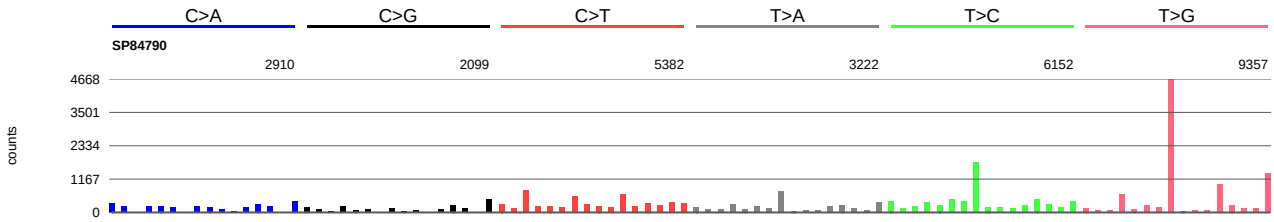
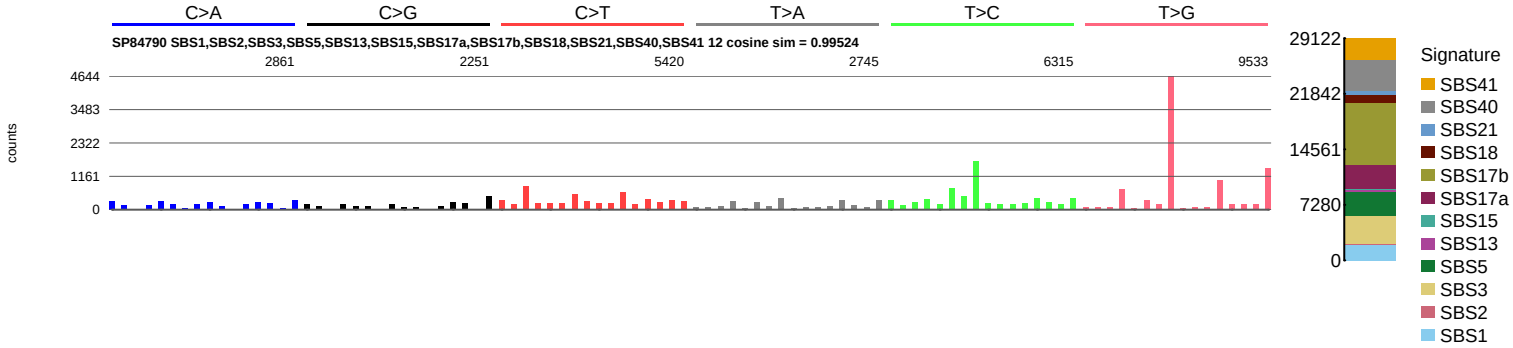
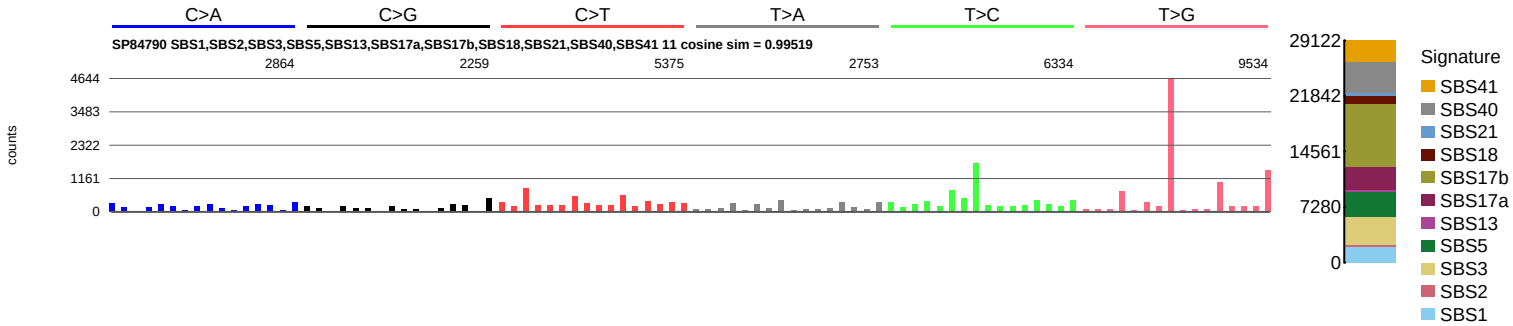
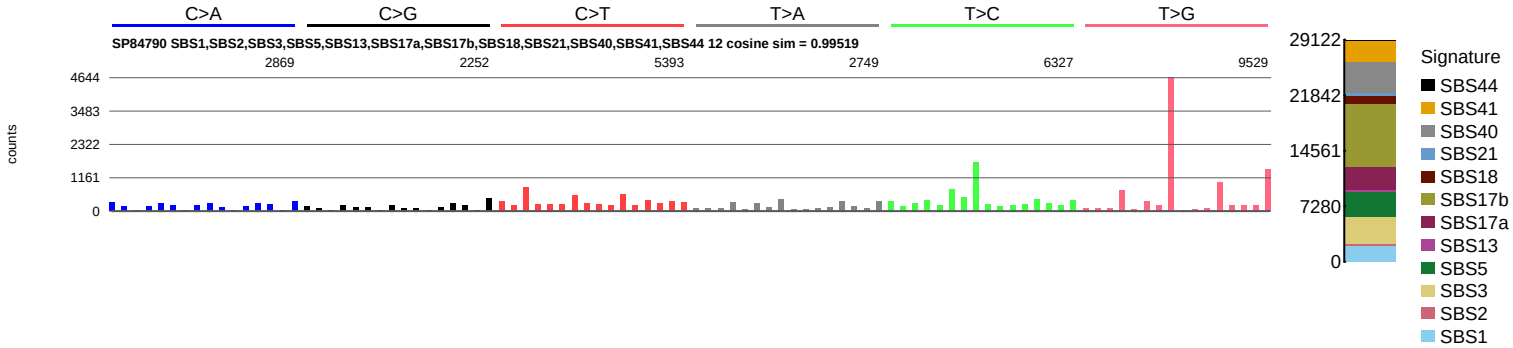
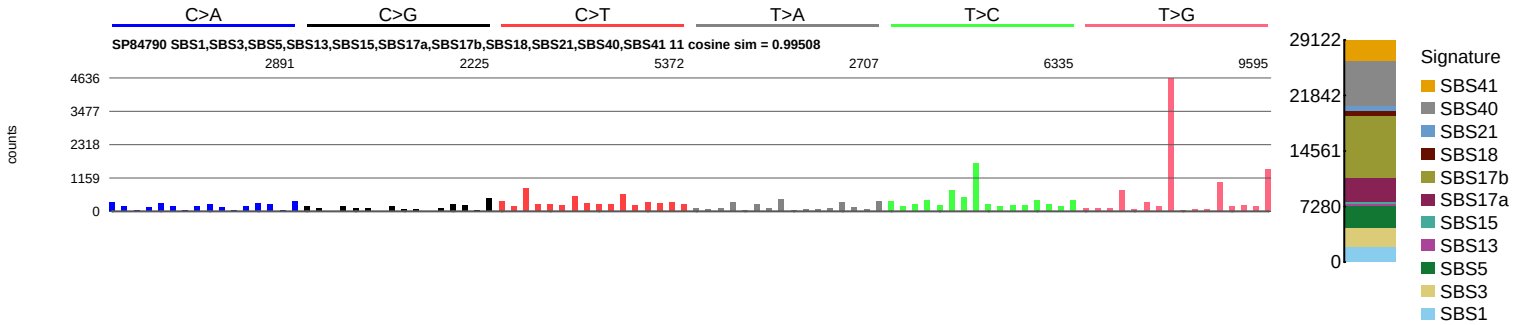
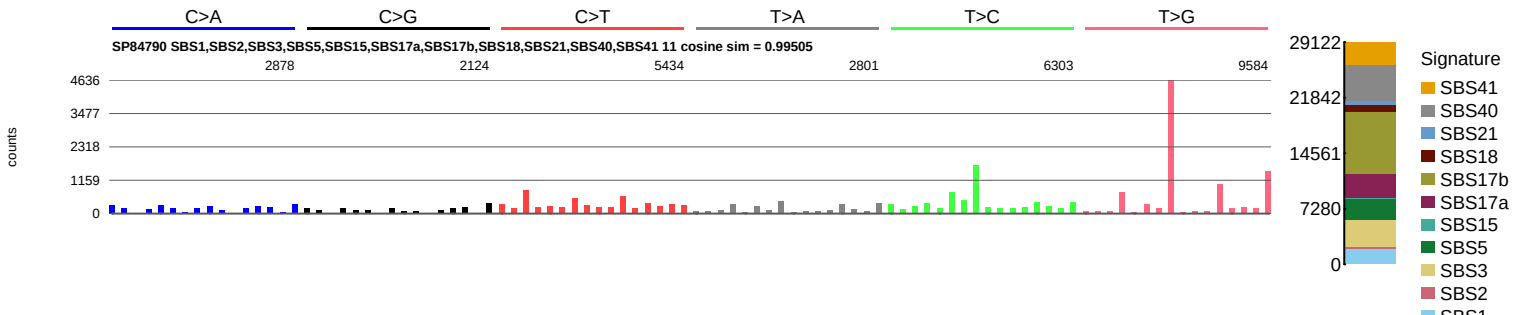

C>A
C>G
C>T
T>A
T>C
T>G
SP84790
2910
2099
5382
3222
6152
9357
4668
3501
2334
1167
0
counts

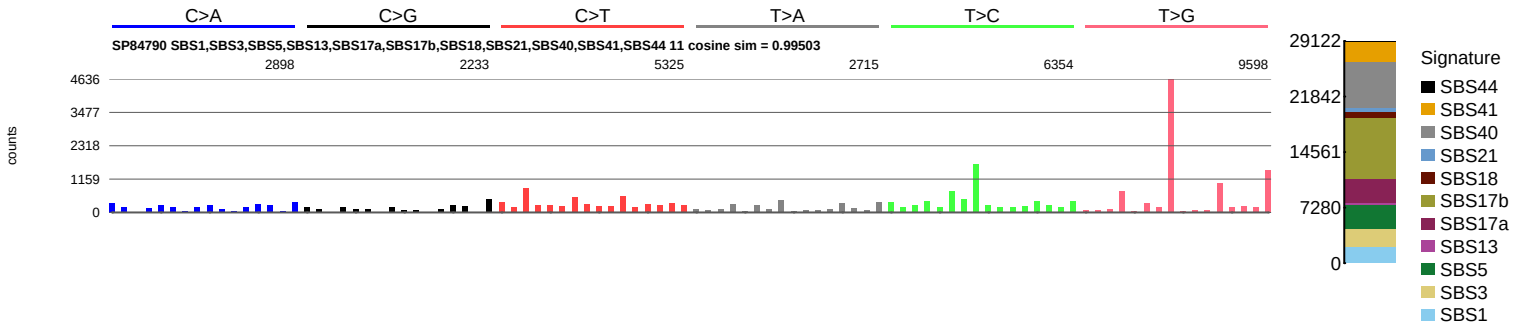

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 12 cosine sim = 0.99524
2861
2251
5420
2745
6315
9533
4644
3483
2322
1161
0
counts
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

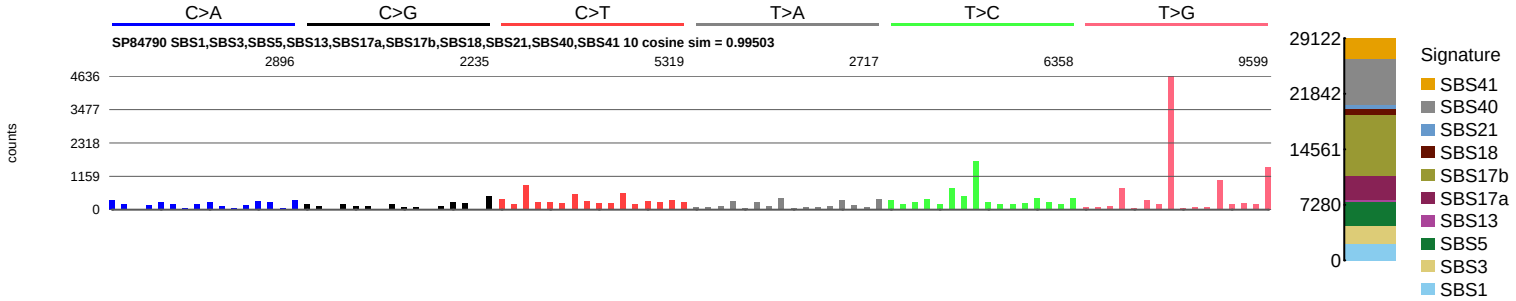

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 11 cosine sim = 0.99519
2864
2259
5375
2753
6334
9534
4644
3483
2322
1161
0
counts
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

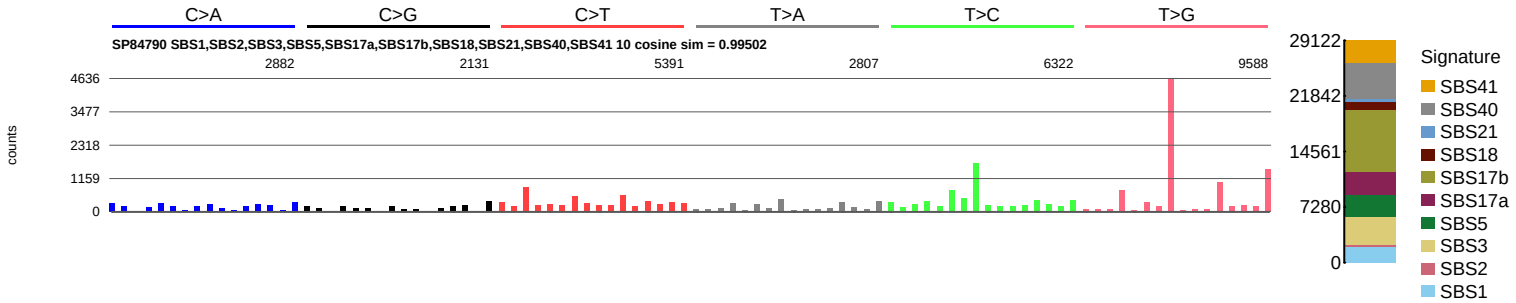

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 12 cosine sim = 0.99519
2869
2252
5393
2749
6327
9529
4644
3483
2322
1161
0
counts
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

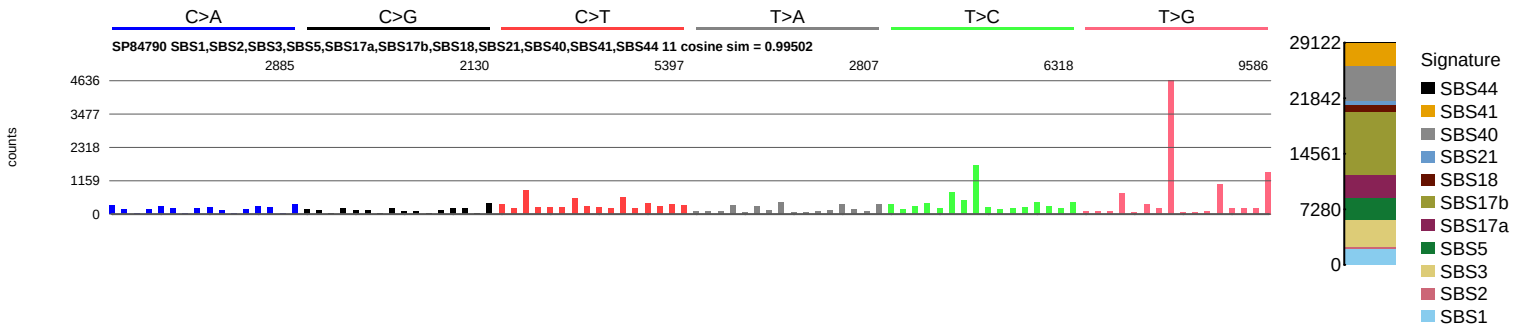

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 11 cosine sim = 0.99508
2891
2225
5372
2707
6335
9595
4636
3477
2318
1159
0
counts
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

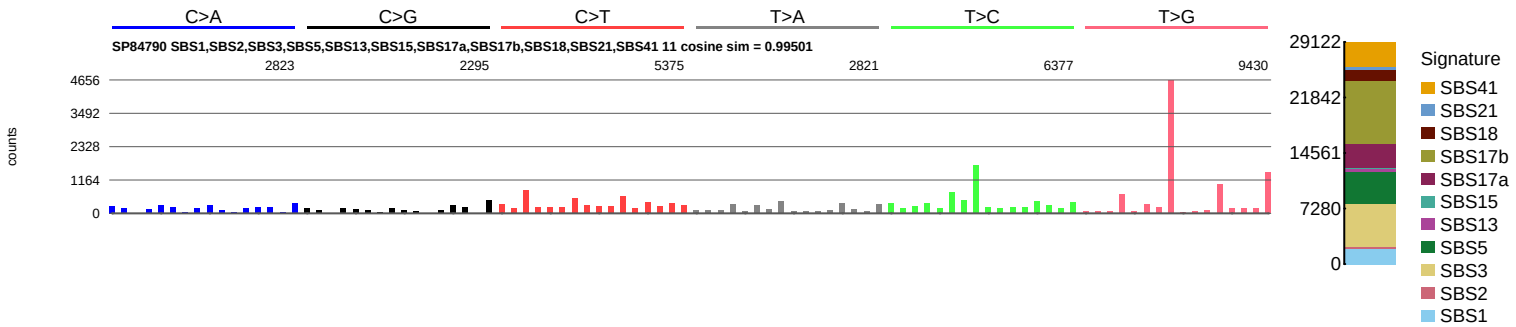

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 11 cosine sim = 0.99505
2878
2124
5434
2801
6303
9584
4636
3477
2318
1159
0
counts
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2

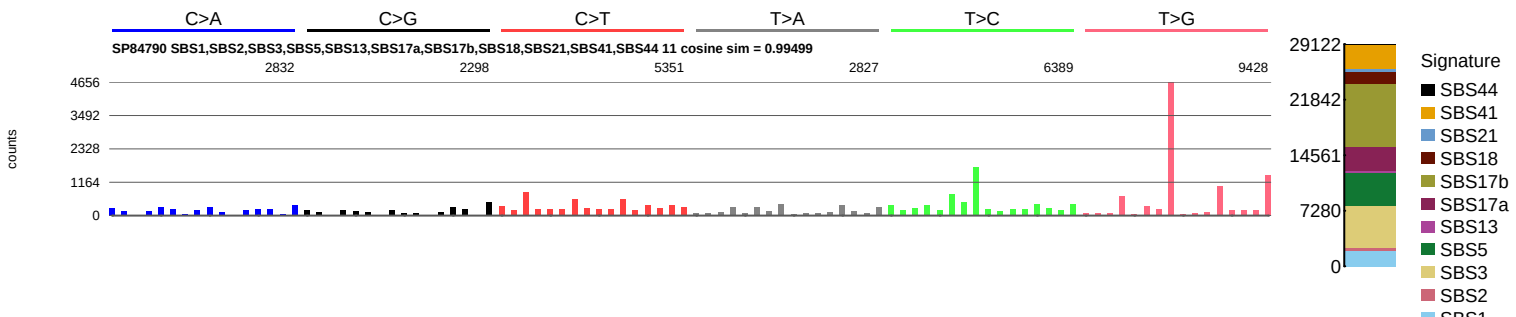

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.99503
2898
2233
5325
2715
6354
9598
counts
4636
3477
2318
1159
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

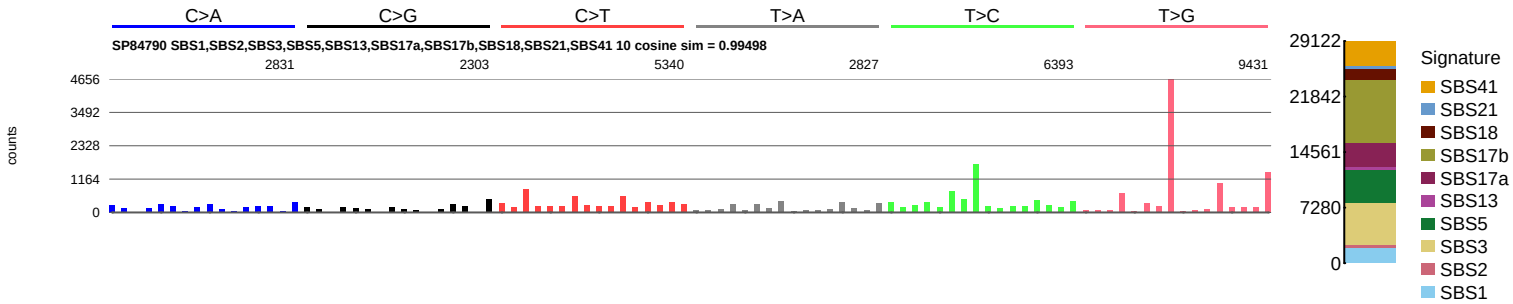

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.99503
2896
2235
5319
2717
6358
9599
counts
4636
3477
2318
1159
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

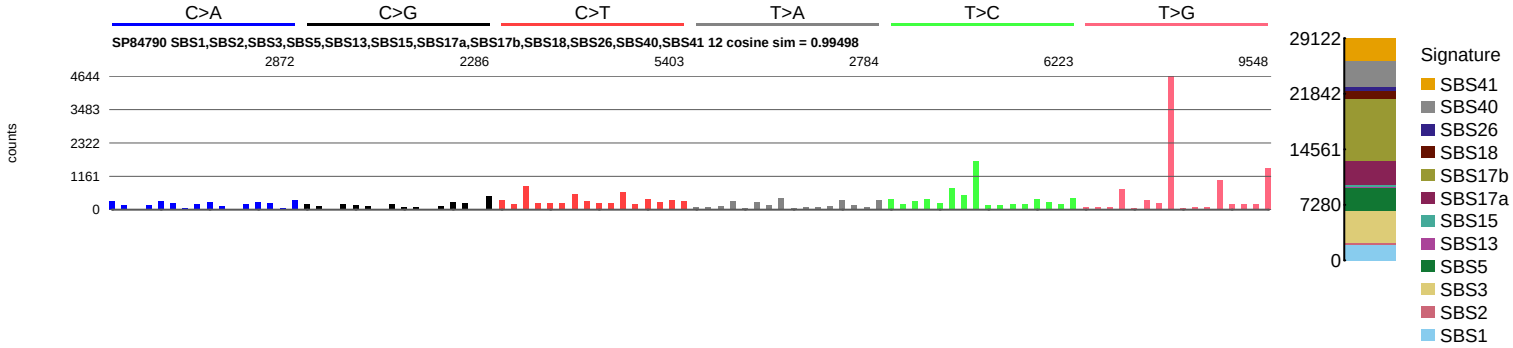

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.99502
2882
2131
5391
2807
6322
9588
counts
4636
3477
2318
1159
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

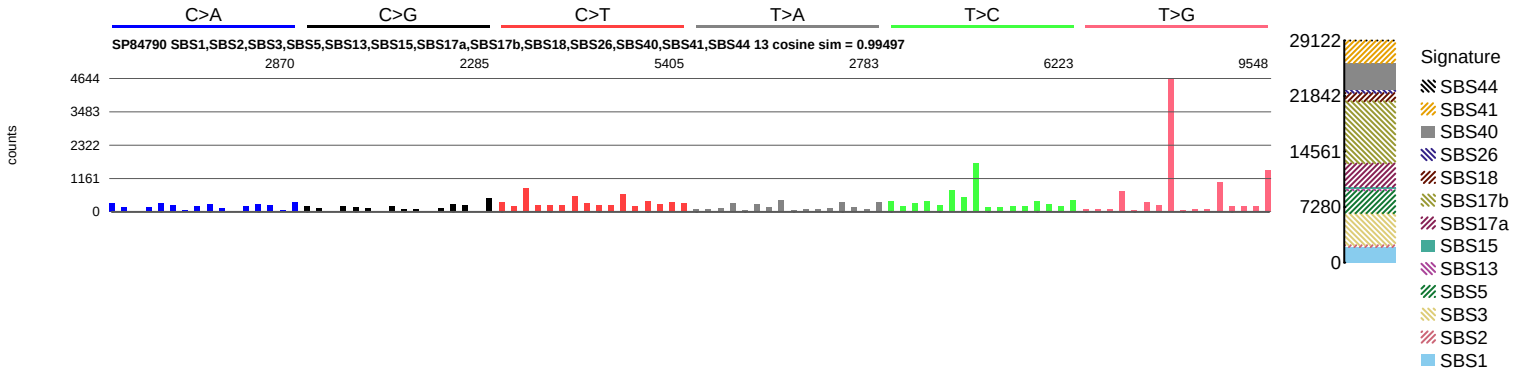

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.99502
2885
2130
5397
2807
6318
9586
counts
4636
3477
2318
1159
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

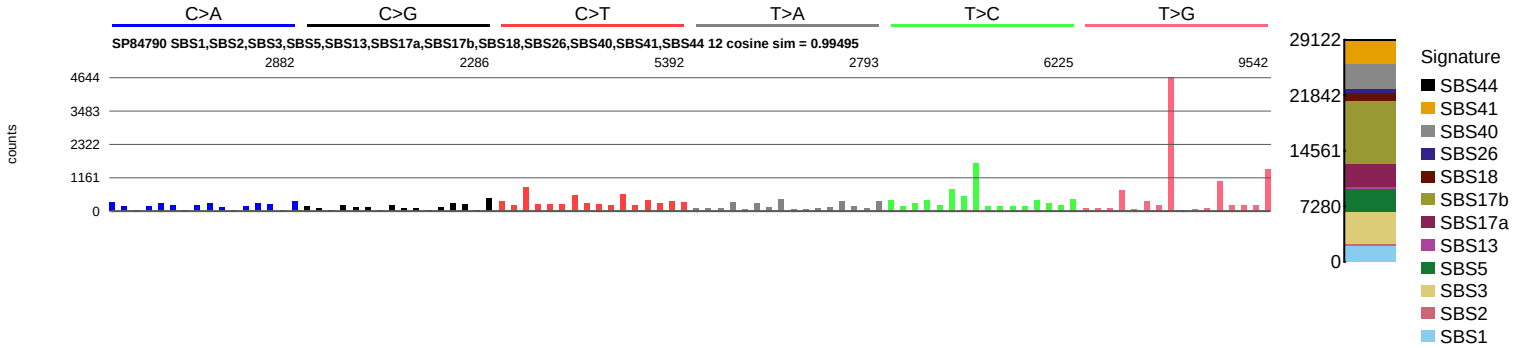

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 11 cosine sim = 0.99501
2823
2295
5375
2821
6377
9430
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

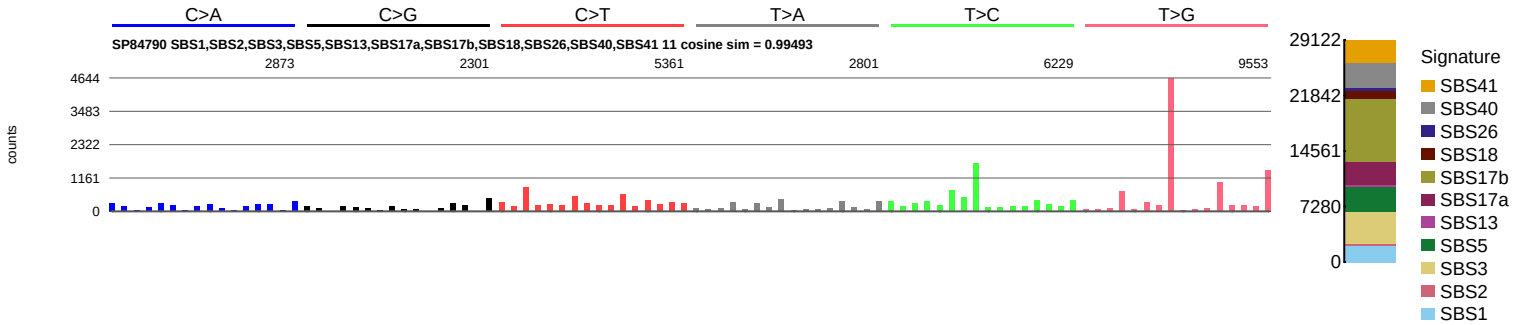

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 11 cosine sim = 0.99499
2832
2298
5351
2827
6389
9428
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2

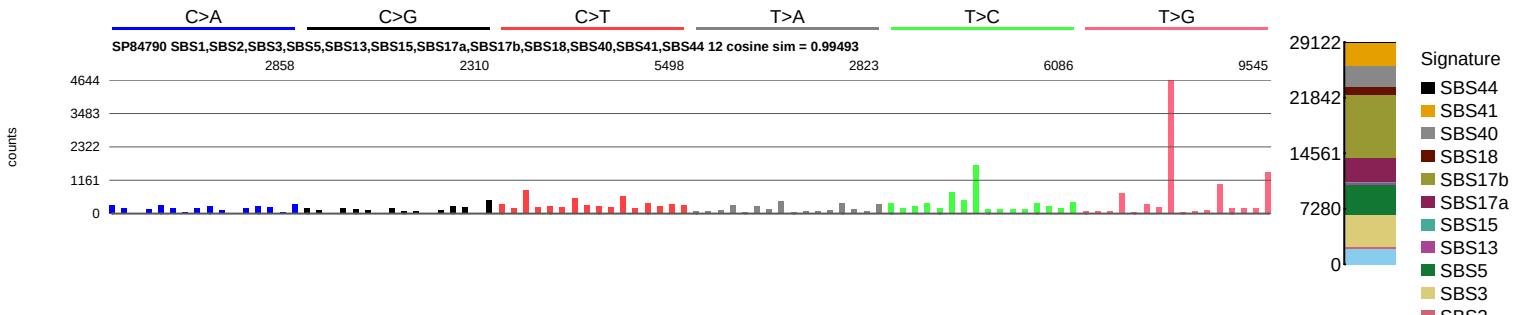

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.99498
2831
2303
5340
2827
6393
9431
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

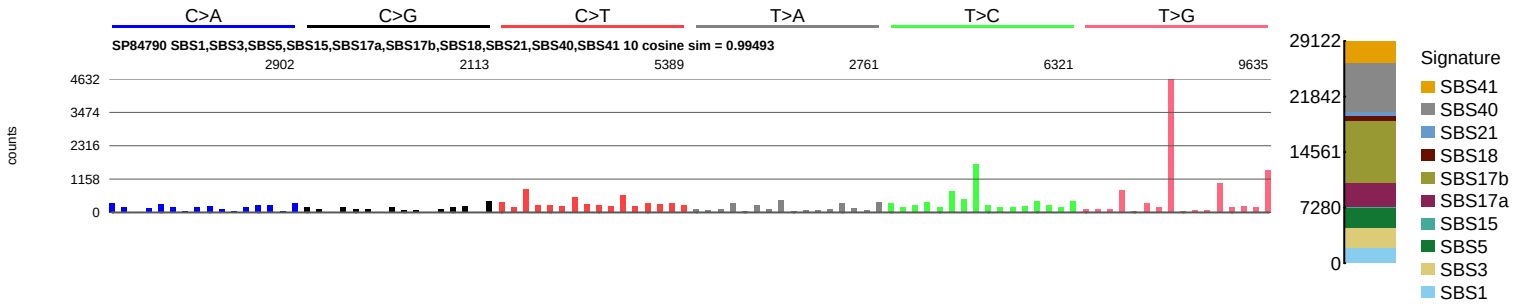

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 12 cosine sim = 0.99498
2872
2286
5403
2784
6223
9548
counts
4644
3483
2322
1161
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

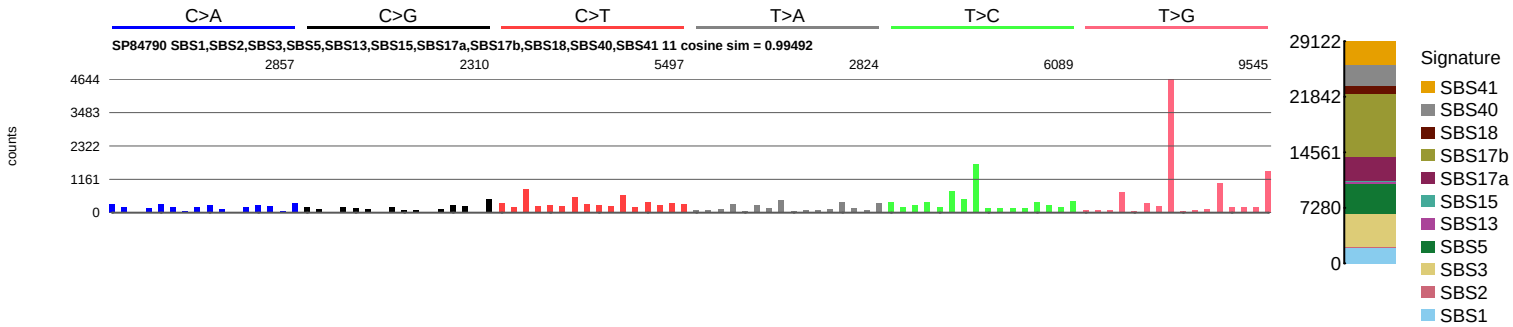

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 13 cosine sim = 0.99497
2870
2285
5405
2783
6223
9548
counts
4644
3483
2322
1161
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

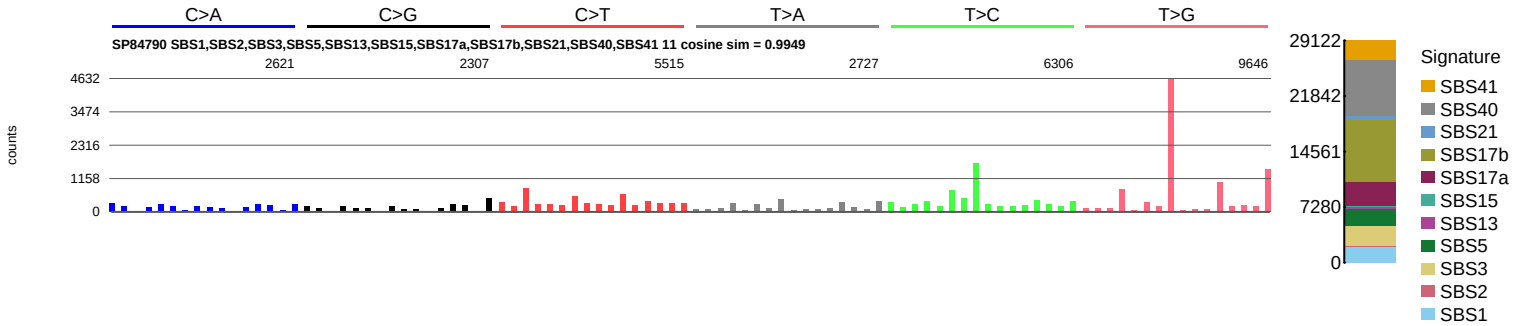

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.99495
2882
2286
5392
2793
6225
9542
counts
4644
3483
2322
1161
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

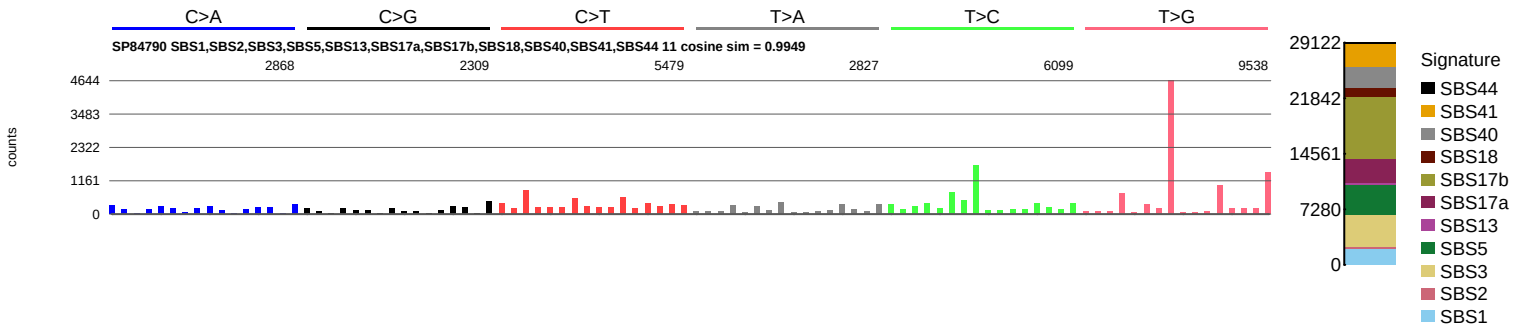

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 11 cosine sim = 0.99493
2873
2301
5361
2801
6229
9553
counts
4644
3483
2322
1161
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

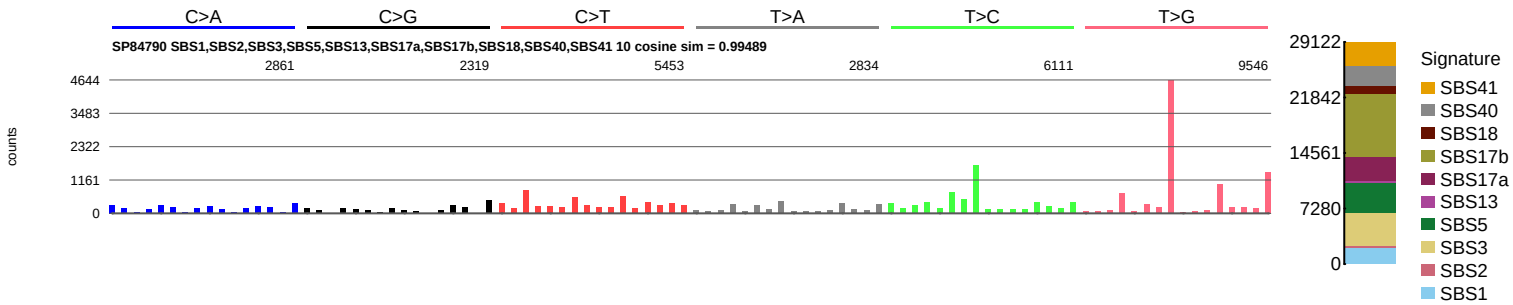

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 12 cosine sim = 0.99493
2858
2310
5498
2823
6086
9545
counts
4644
3483
2322
1161
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3

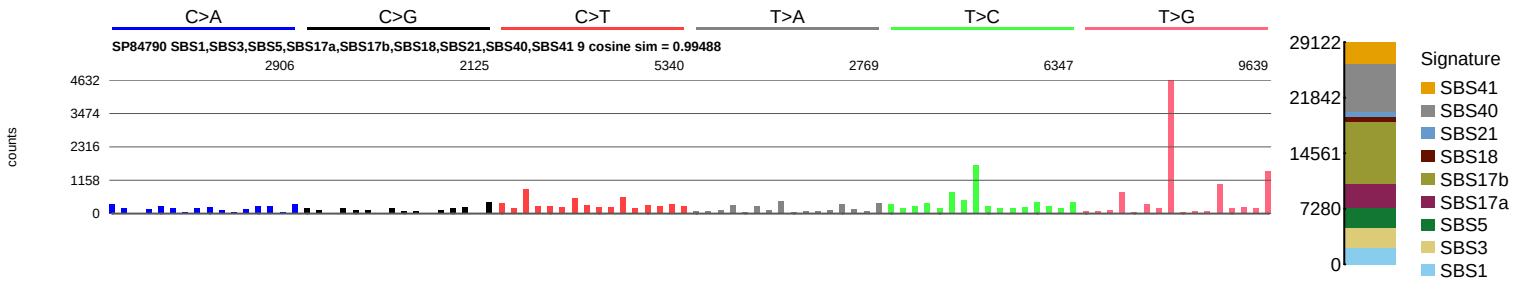

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.99493
2902
2113
5389
2761
6321
9635
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1

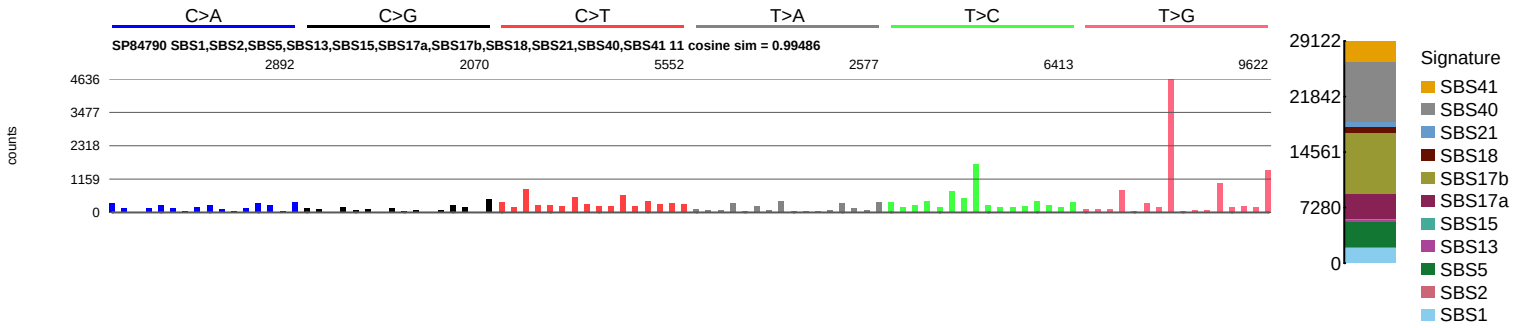

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 11 cosine sim = 0.99492
2857
2310
5497
2824
6089
9545
counts
4644
3483
2322
1161
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

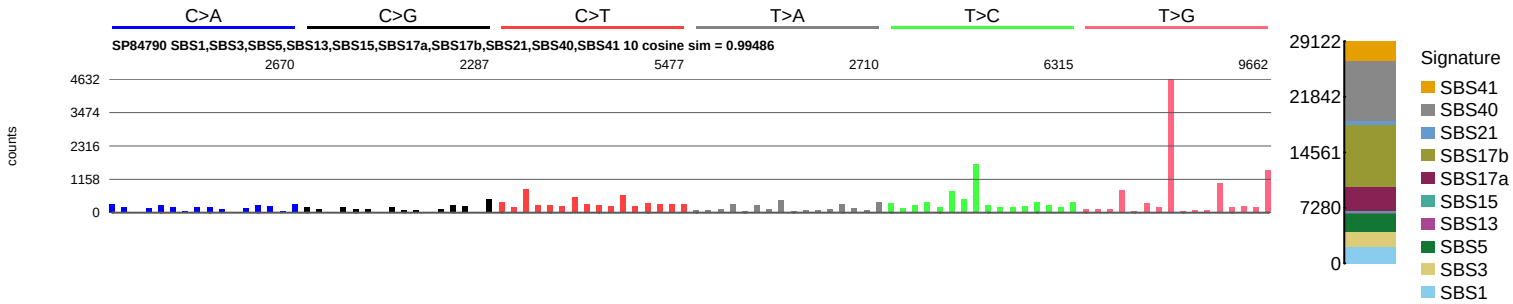

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 11 cosine sim = 0.9949
2621
2307
5515
2727
6306
9646
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

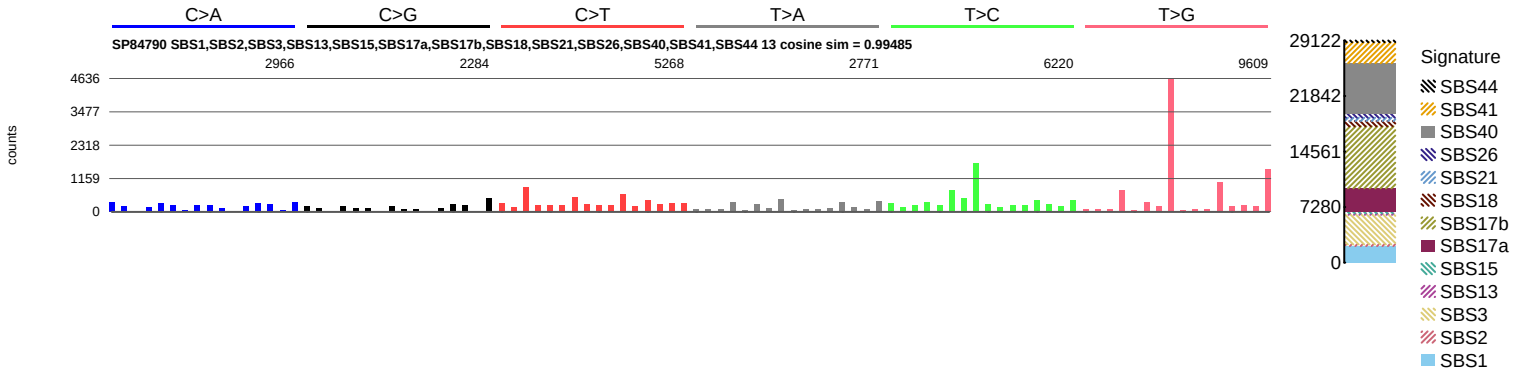

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 11 cosine sim = 0.9949
2868
2309
5479
2827
6099
9538
counts
4644
3483
2322
1161
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

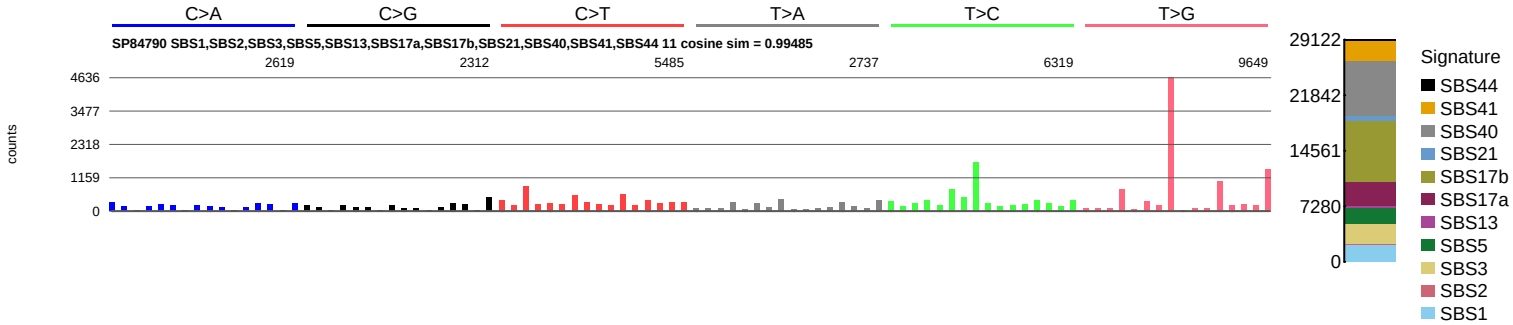

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 10 cosine sim = 0.99489
2861
2319
5453
2834
6111
9546
counts
4644
3483
2322
1161
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

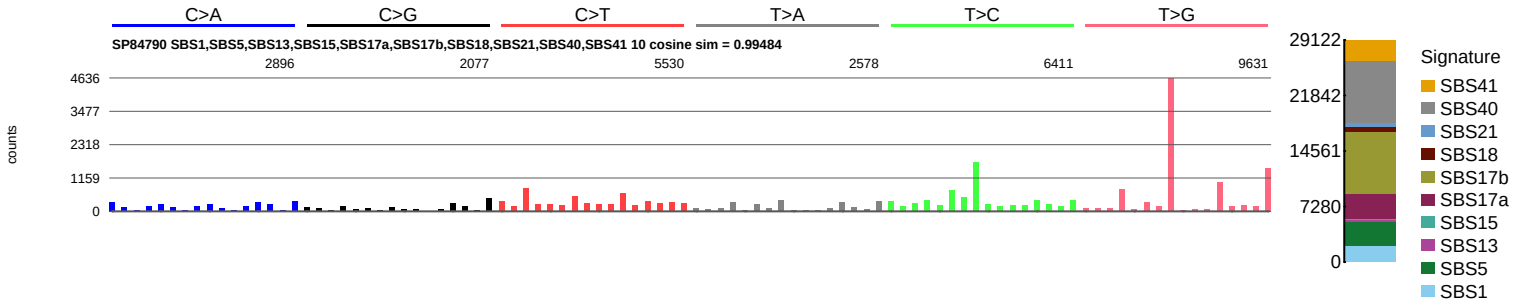

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.99488
2906
2125
5340
2769
6347
9639
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1

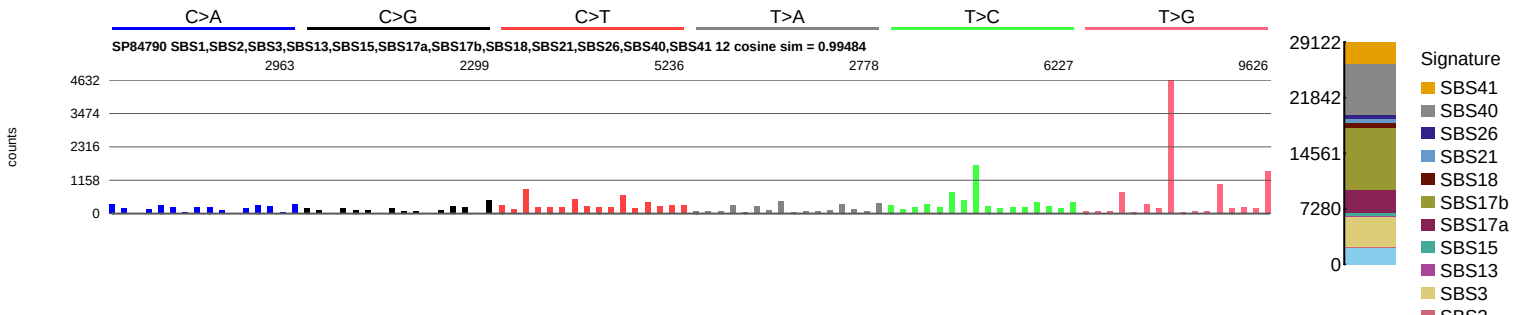

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 11 cosine sim = 0.99486
2892
2070
5552
2577
6413
9622
counts
4636
3477
2318
1159
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

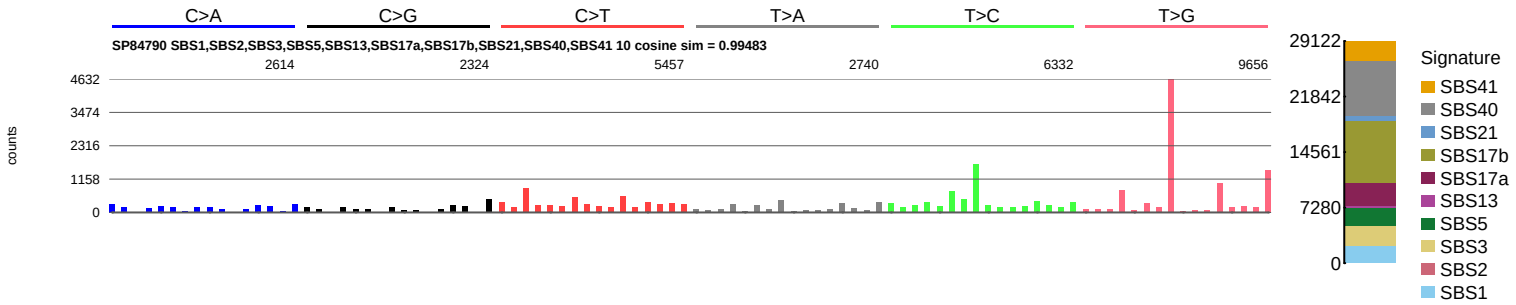

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 10 cosine sim = 0.99486
2670
2287
5477
2710
6315
9662
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

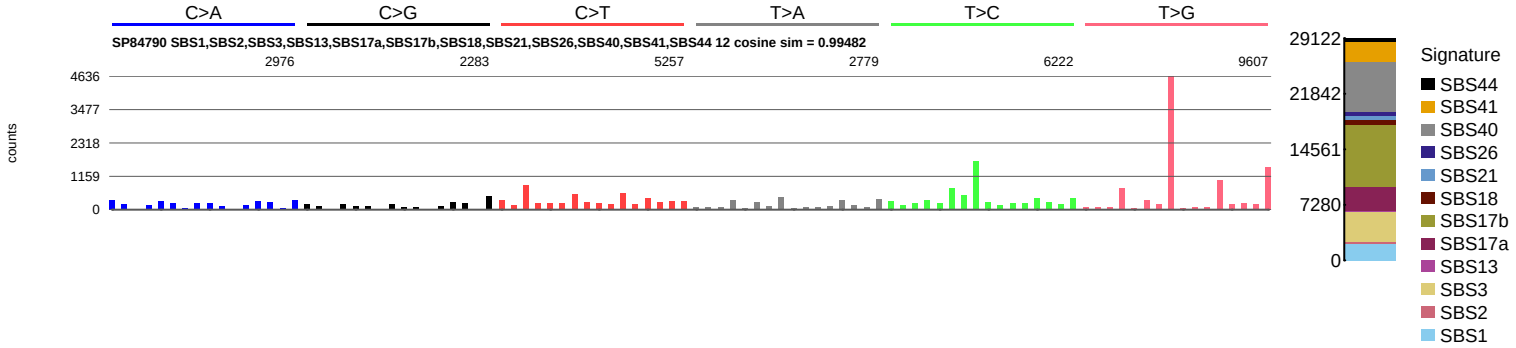

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 13 cosine sim = 0.99485
2966
2284
5268
2771
6220
9609
counts
4636
3477
2318
1159
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

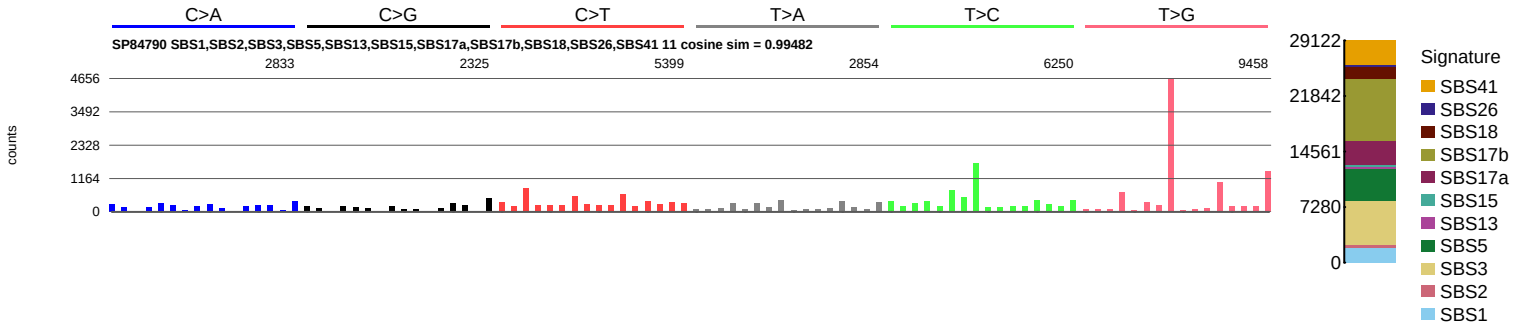

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.99485
2619
2312
5485
2737
6319
9649
counts
4636
3477
2318
1159
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

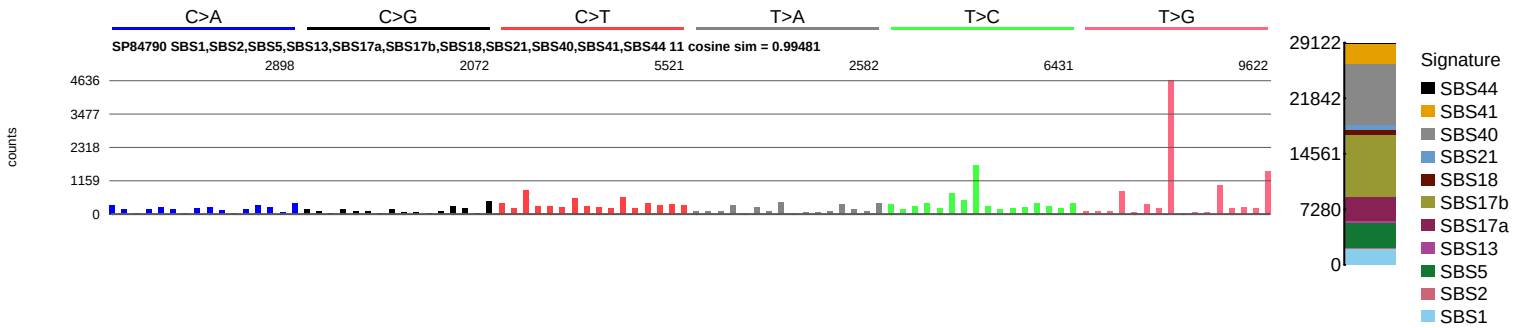

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.99484
2896
2077
5530
2578
6411
9631
counts
4636
3477
2318
1159
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

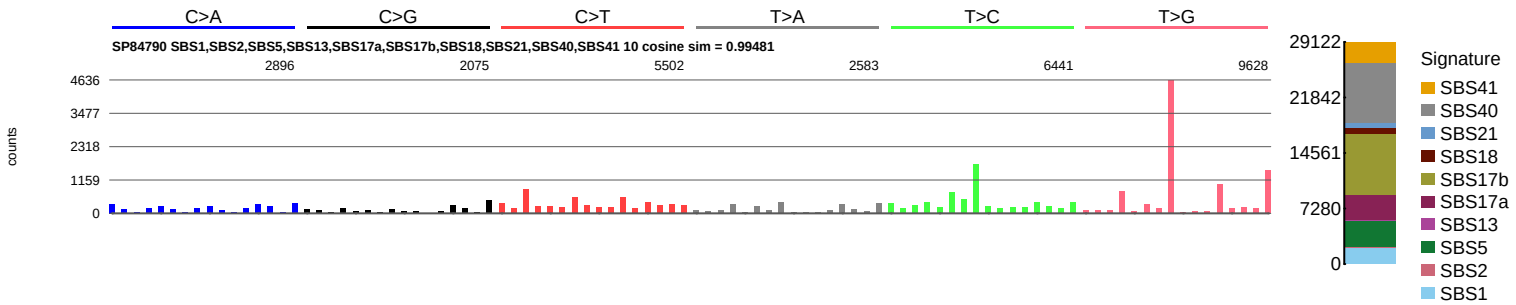

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 12 cosine sim = 0.99484
2963
2299
5236
2778
6227
9626
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3

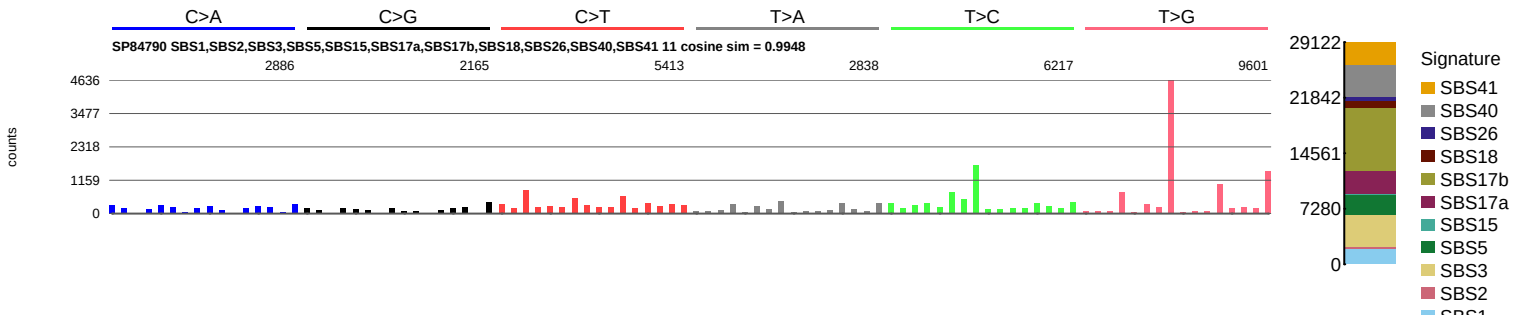

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS41 10 cosine sim = 0.99483
2614
2324
5457
2740
6332
9656
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

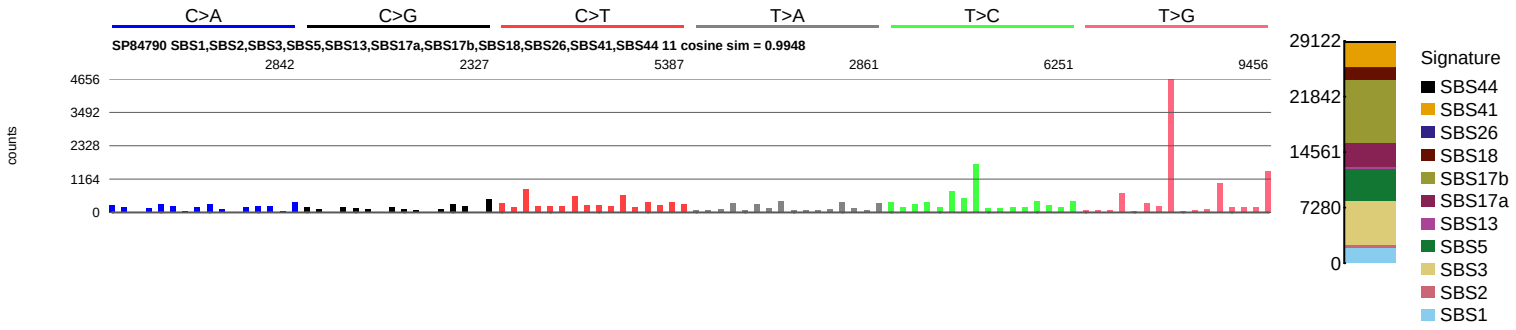

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.99482
2976
2283
5257
2779
6222
9607
counts
4636
3477
2318
1159
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

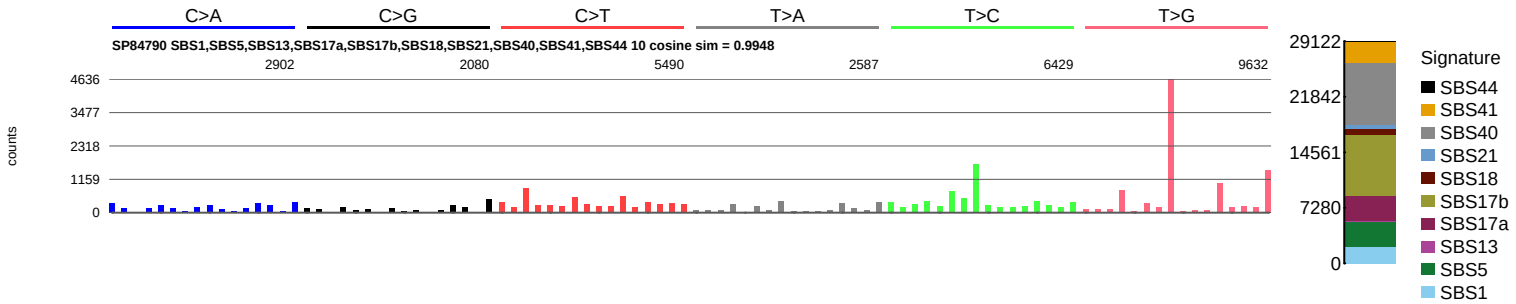

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 11 cosine sim = 0.99482
2833
2325
5399
2854
6250
9458
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

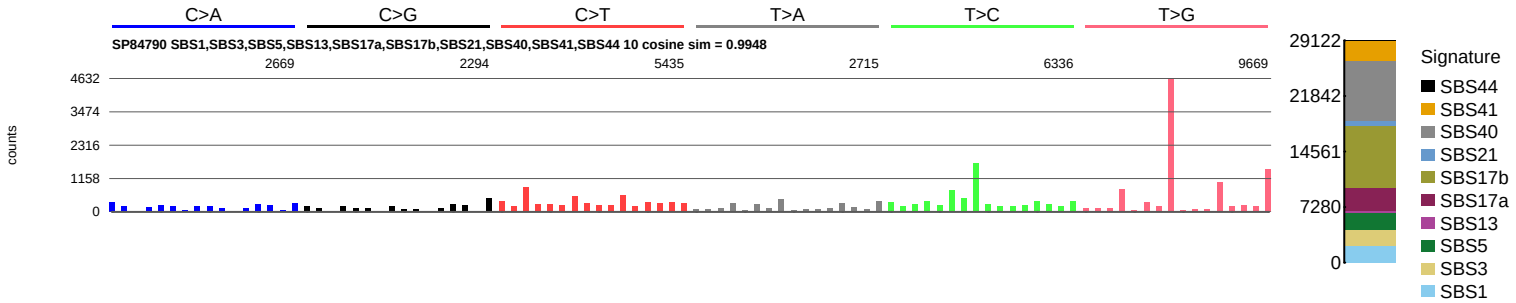

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.99481
2898
2072
5521
2582
6431
9622
counts
4636
3477
2318
1159
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

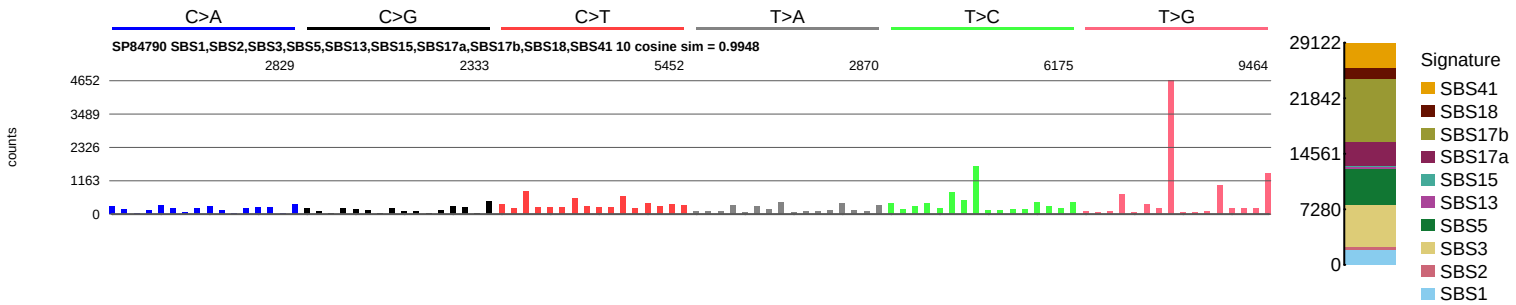

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.99481
2896
2075
5502
2583
6441
9628
counts
4636
3477
2318
1159
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

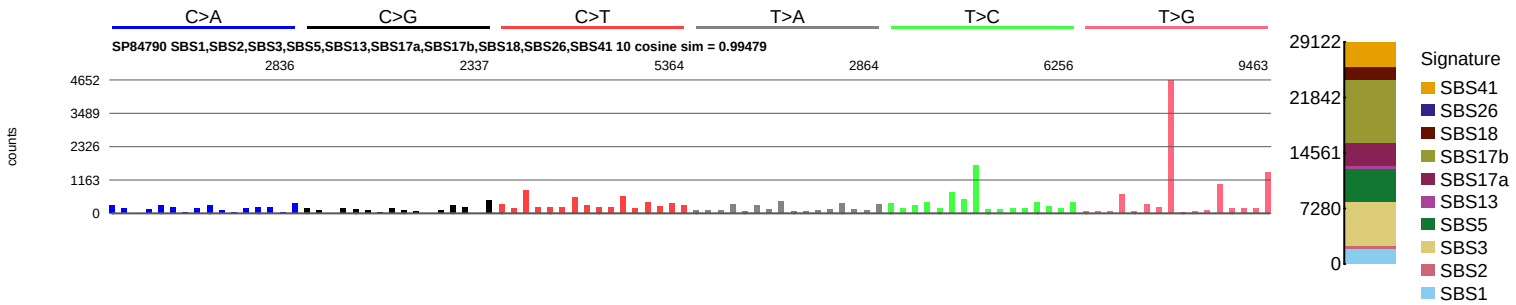

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 11 cosine sim = 0.9948
2886
2165
5413
2838
6217
9601
counts
4636
3477
2318
1159
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2

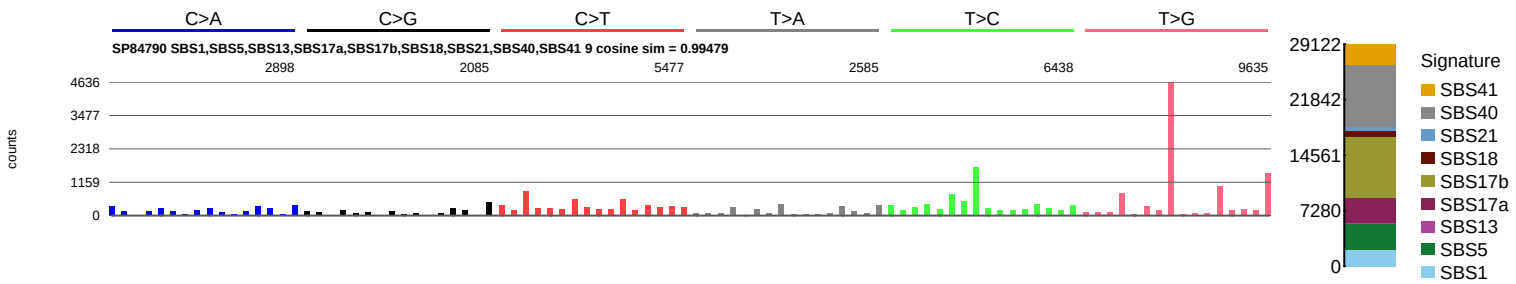

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 11 cosine sim = 0.9948
2842
2327
5387
2861
6251
9456
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

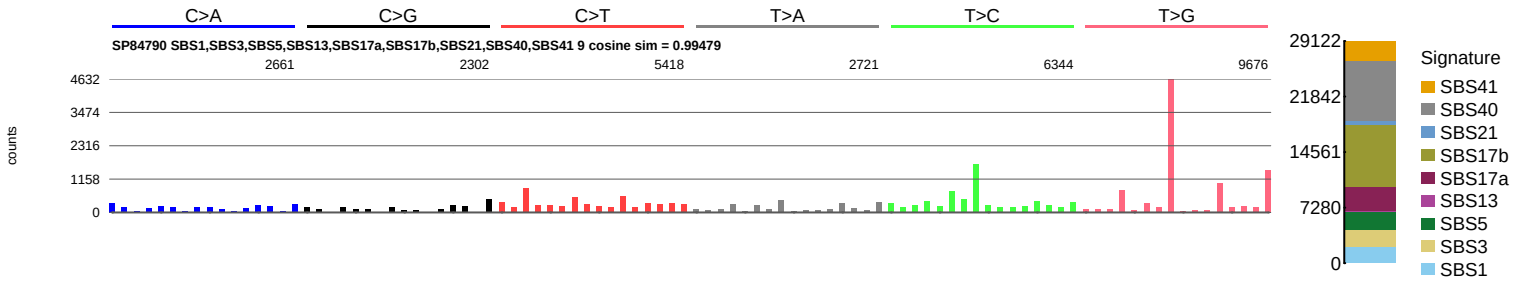

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.9948
2902
2080
5490
2587
6429
9632
counts
4636
3477
2318
1159
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1

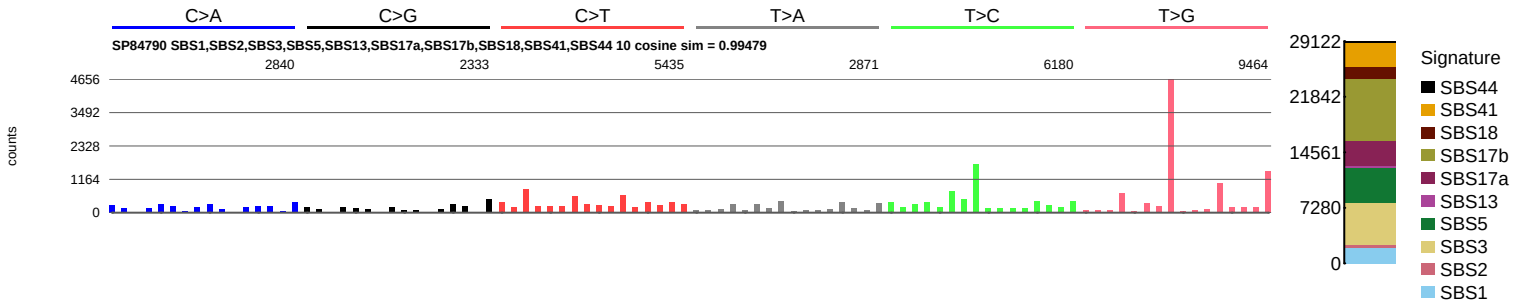

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.9948
2669
2294
5435
2715
6336
9669
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

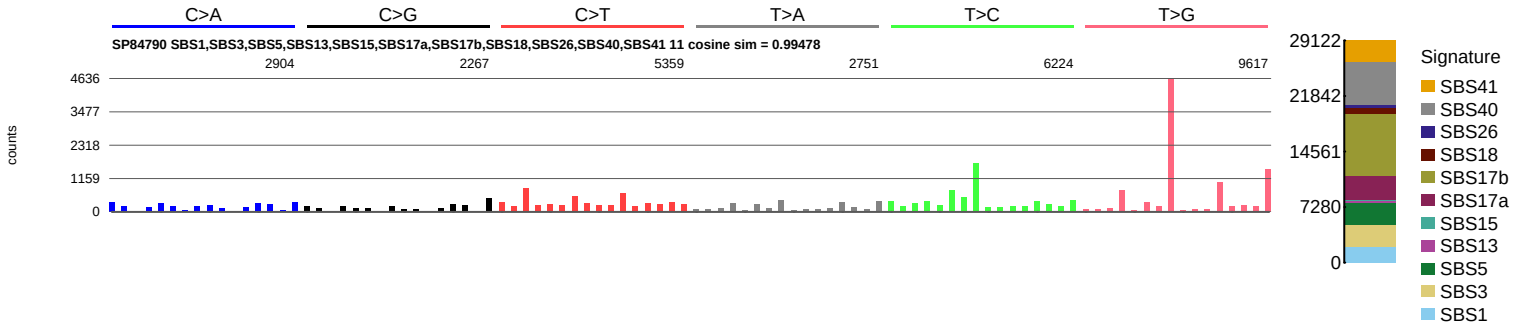

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 10 cosine sim = 0.9948
2829
2333
5452
2870
6175
9464
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

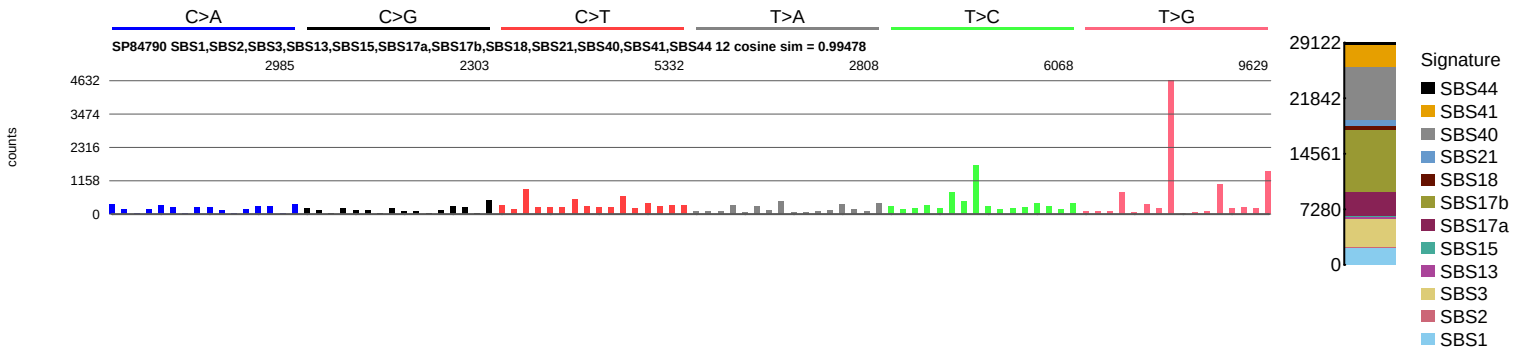

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41 10 cosine sim = 0.99479
2836
2337
5364
2864
6256
9463
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

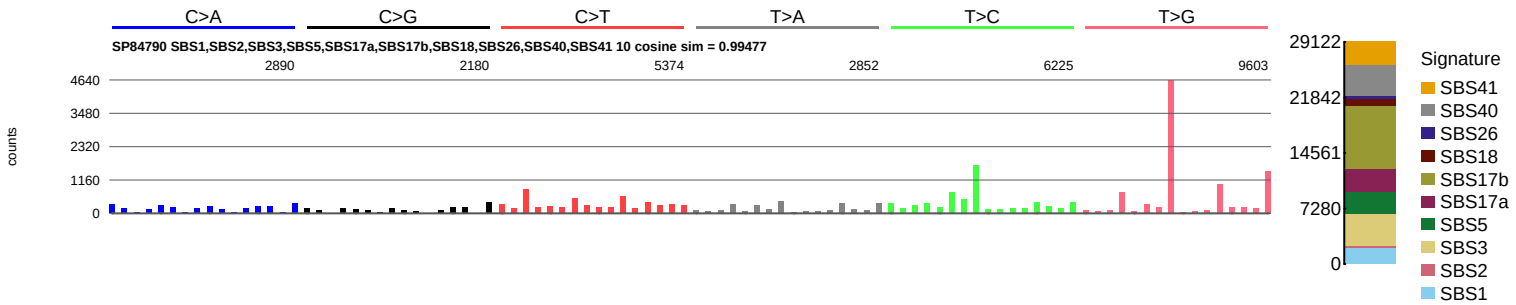

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.99479
2898
2085
5477
2585
6438
9635
counts
4636
3477
2318
1159
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1

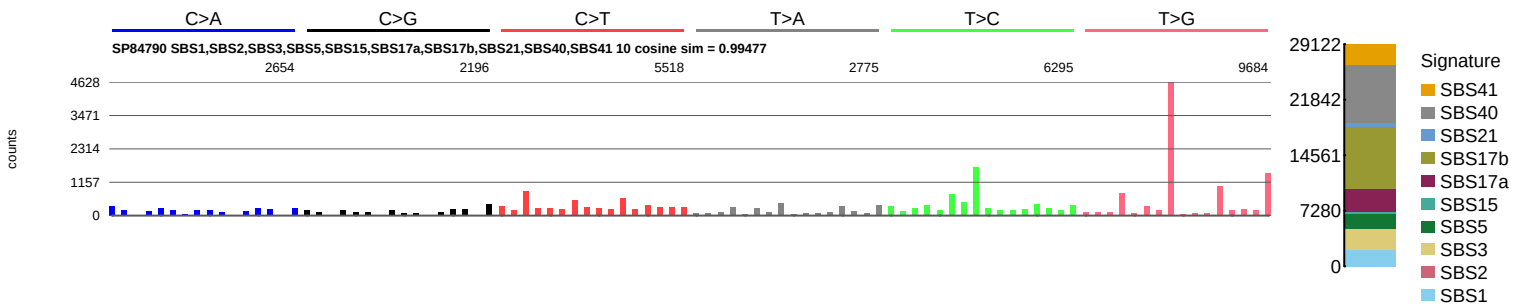

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS41 9 cosine sim = 0.99479
2661
2302
5418
2721
6344
9676
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

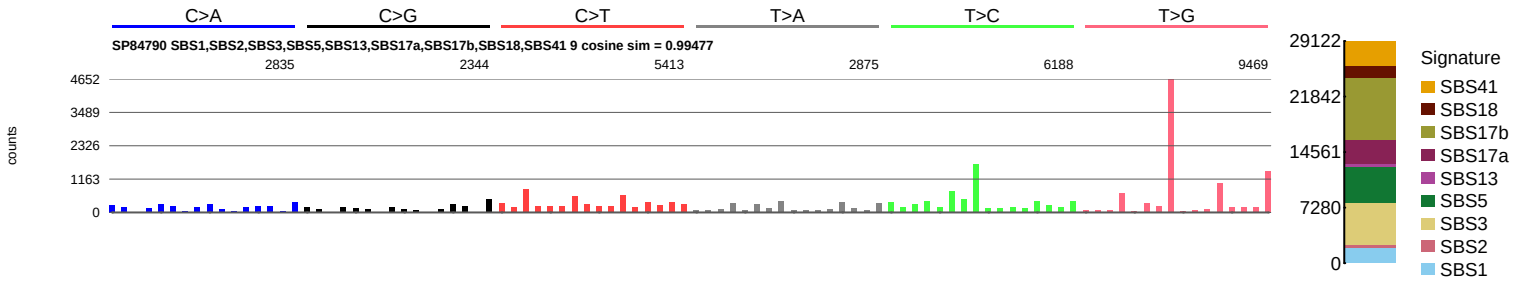

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.99479
2840
2333
5435
2871
6180
9464
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

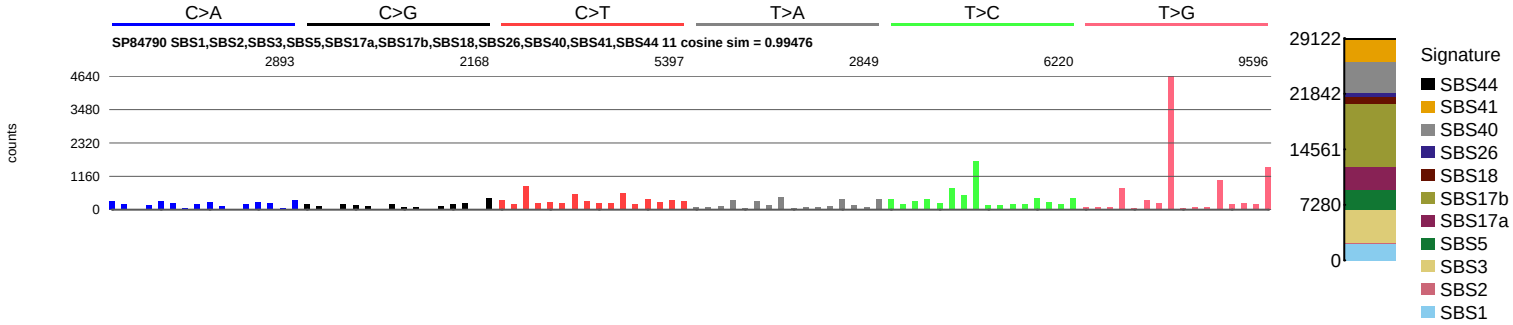

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 11 cosine sim = 0.99478
2904
2267
5359
2751
6224
9617
counts
4636
3477
2318
1159
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

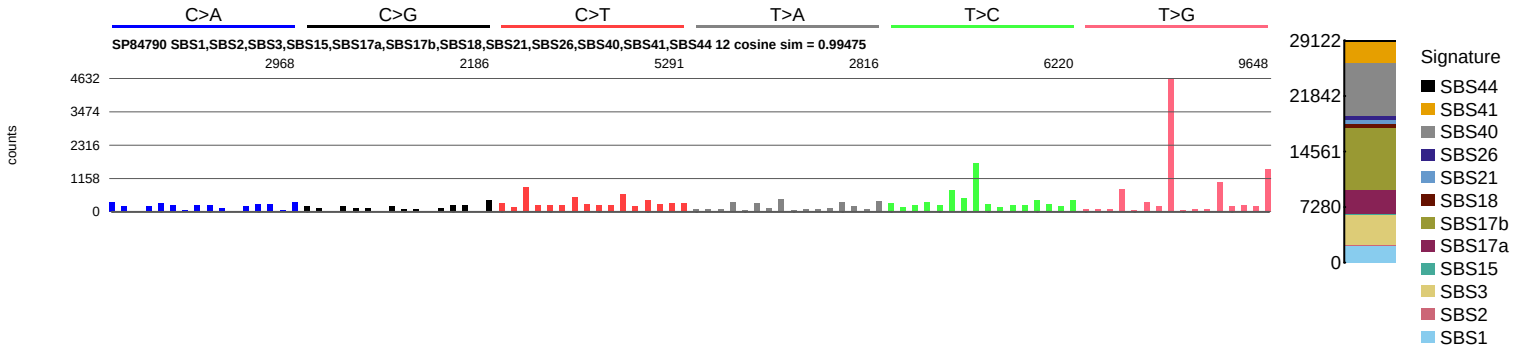

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 12 cosine sim = 0.99478
2985
2303
5332
2808
6068
9629
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

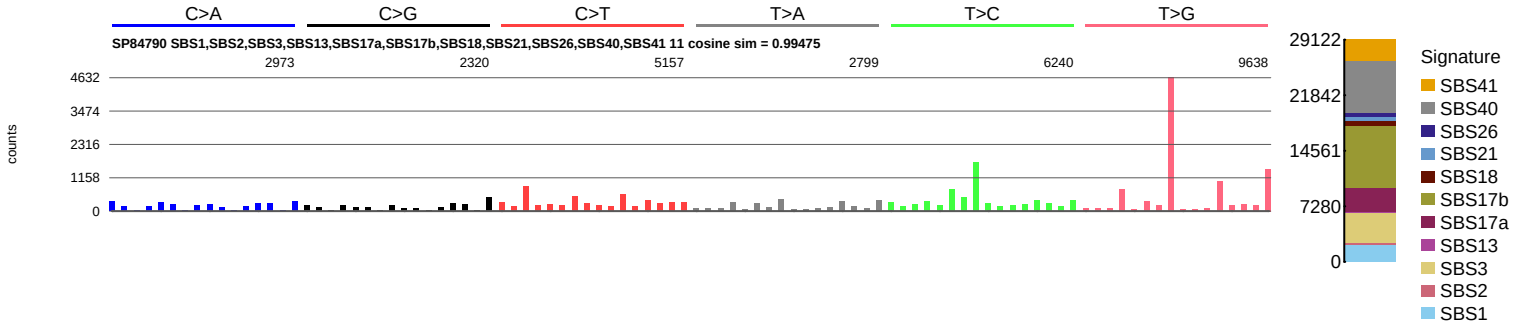

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.99477
2890
2180
5374
2852
6225
9603
counts
4640
3480
2320
1160
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

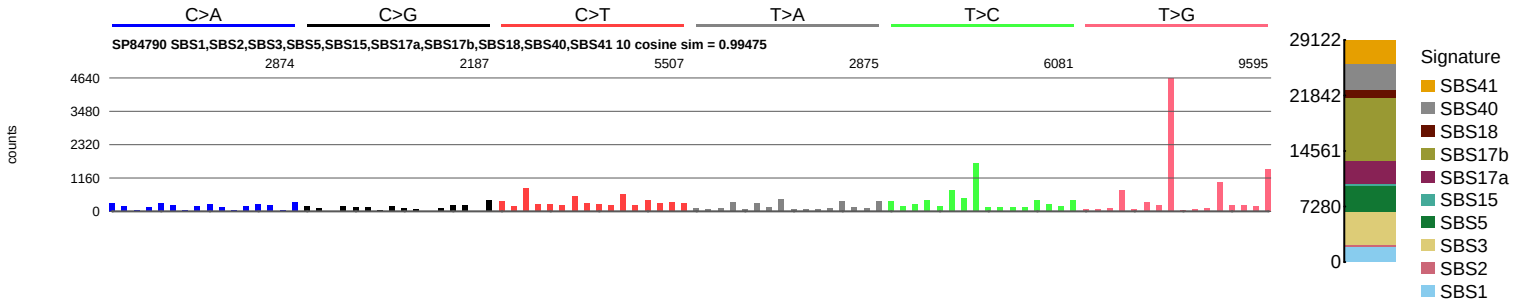

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 10 cosine sim = 0.99477
2654
2196
5518
2775
6295
9684
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

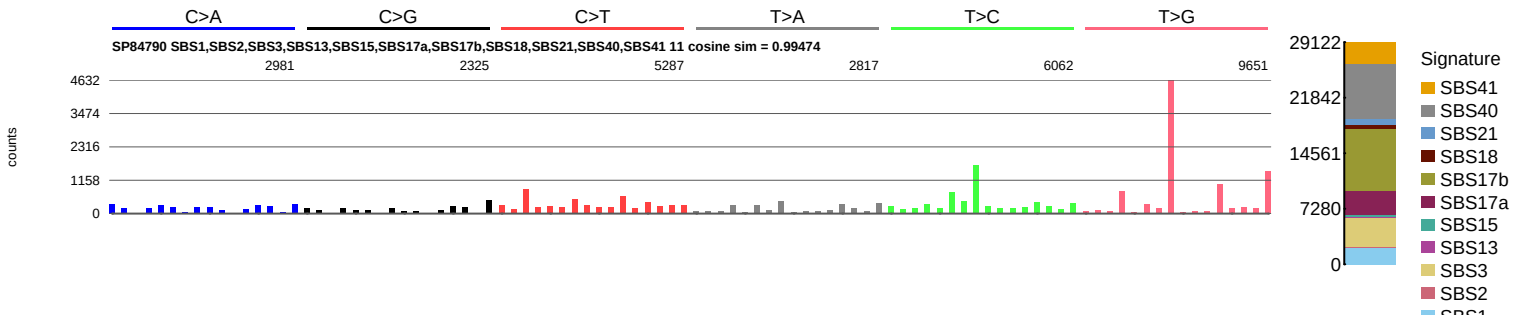

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.99477
2835
2344
5413
2875
6188
9469
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

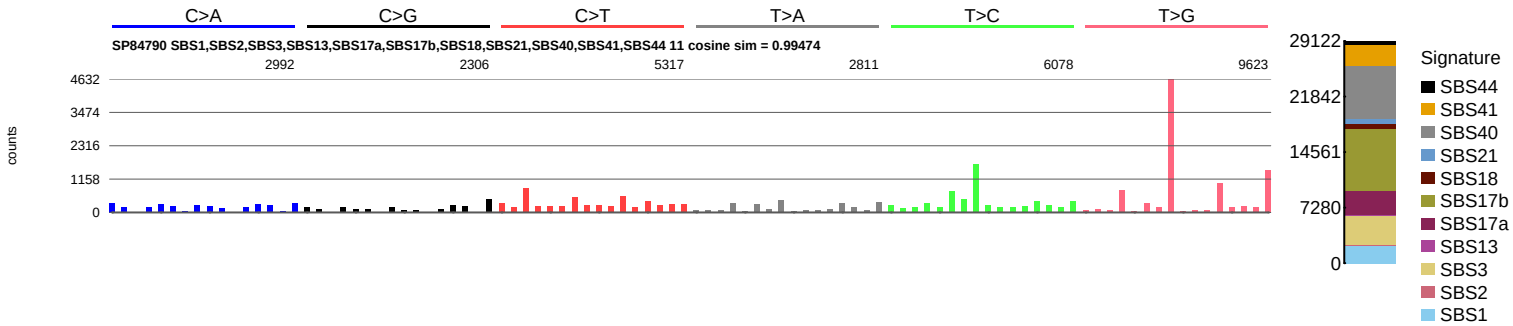

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.99476
2893
2168
5397
2849
6220
9596
counts
4640
3480
2320
1160
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

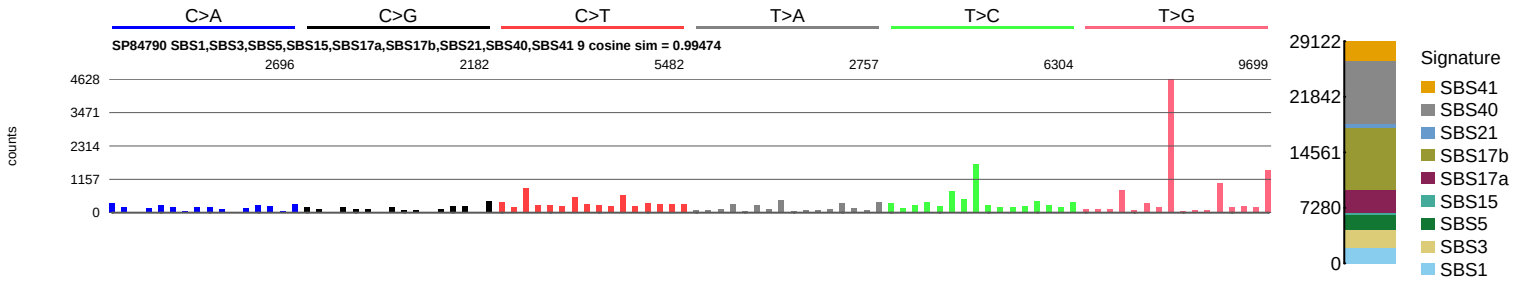

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.99475
2968
2186
5291
2816
6220
9648
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

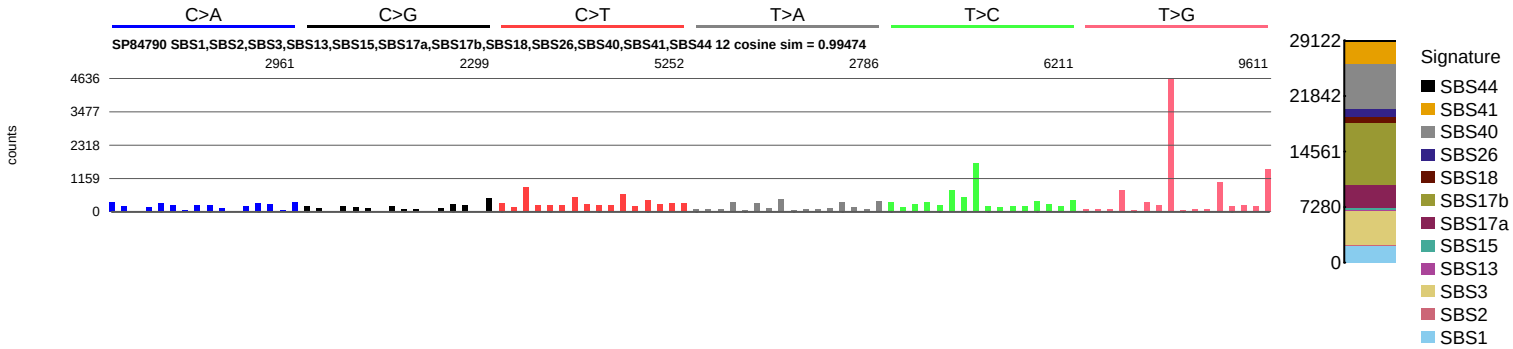

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 11 cosine sim = 0.99475
2973
2320
5157
2799
6240
9638
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

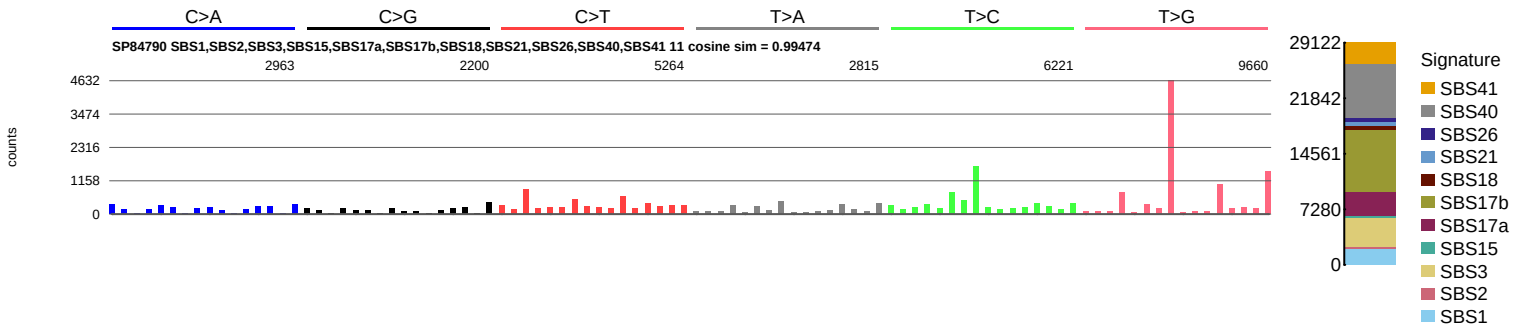

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 10 cosine sim = 0.99475
2874
2187
5507
2875
6081
9595
counts
4640
3480
2320
1160
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

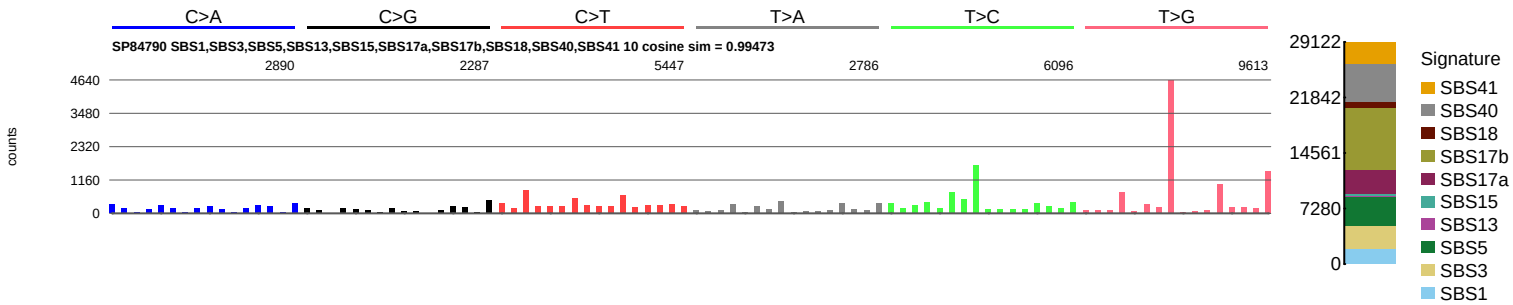

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 11 cosine sim = 0.99474
2981
2325
5287
2817
6062
9651
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2

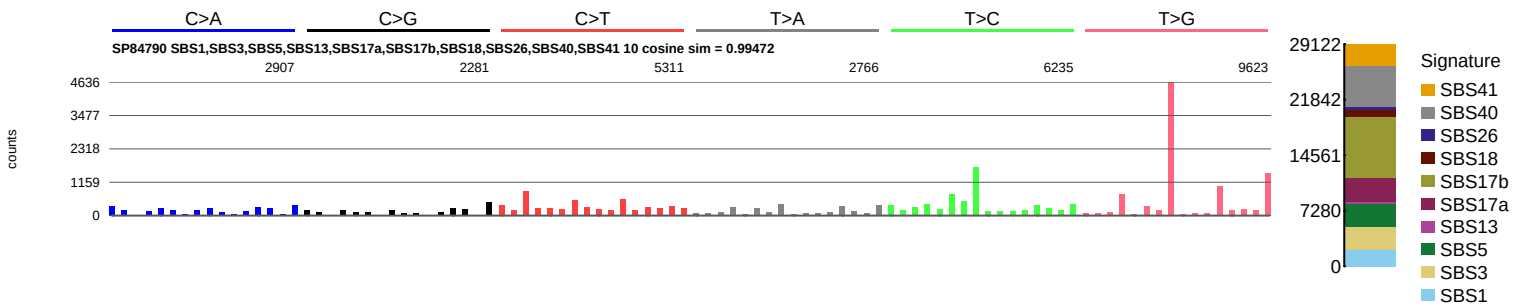

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.99474
2992
2306
5317
2811
6078
9623
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

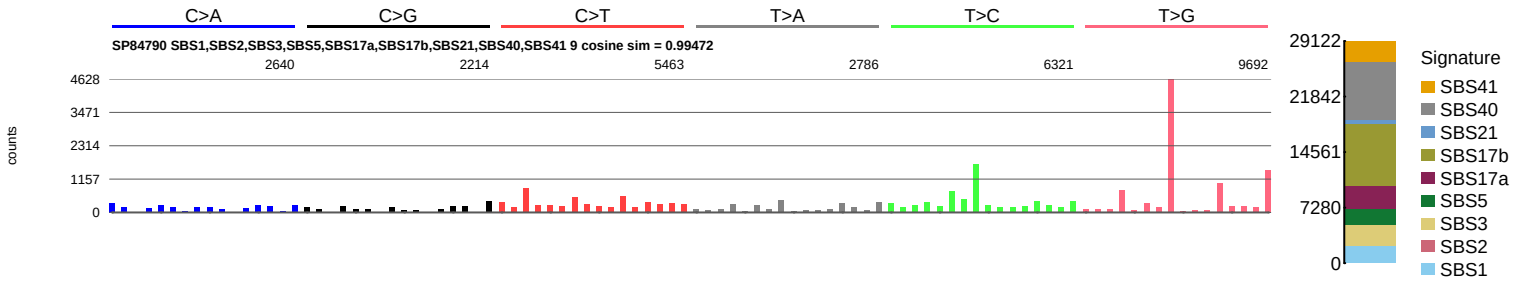

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 9 cosine sim = 0.99474
2696
2182
5482
2757
6304
9699
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1

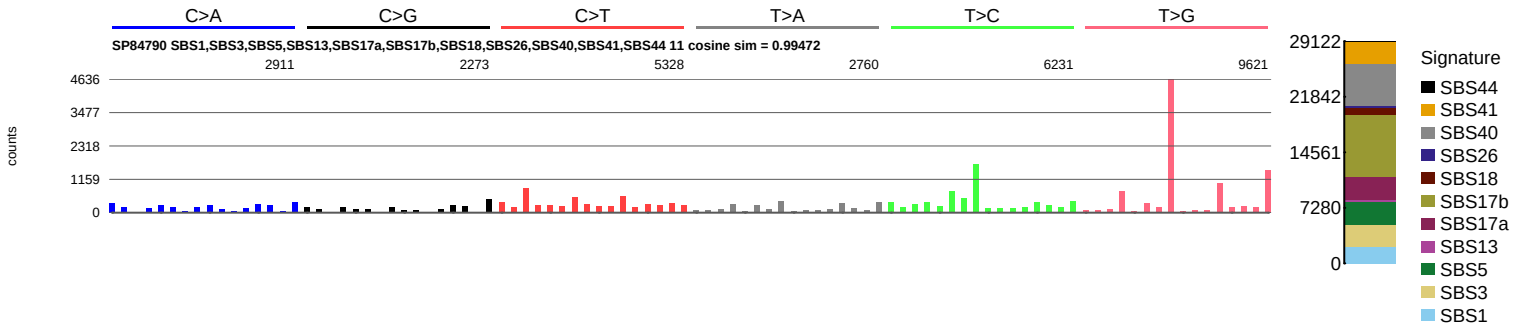

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.99474
2961
2299
5252
2786
6211
9611
counts
4636
3477
2318
1159
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

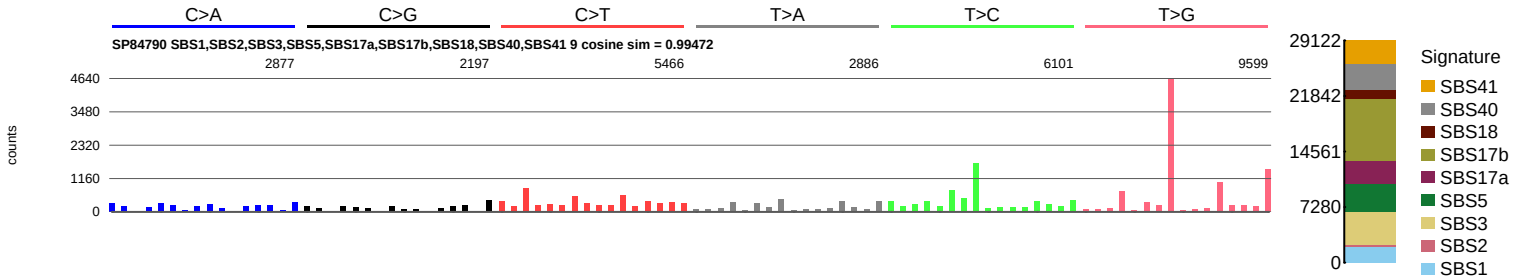

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 11 cosine sim = 0.99474
2963
2200
5264
2815
6221
9660
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

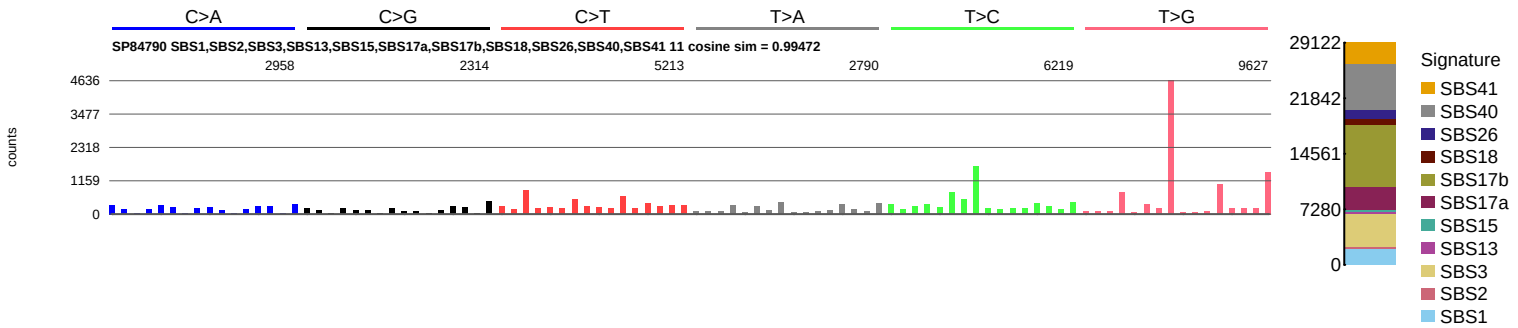

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 10 cosine sim = 0.99473
2890
2287
5447
2786
6096
9613
counts
4640
3480
2320
1160
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

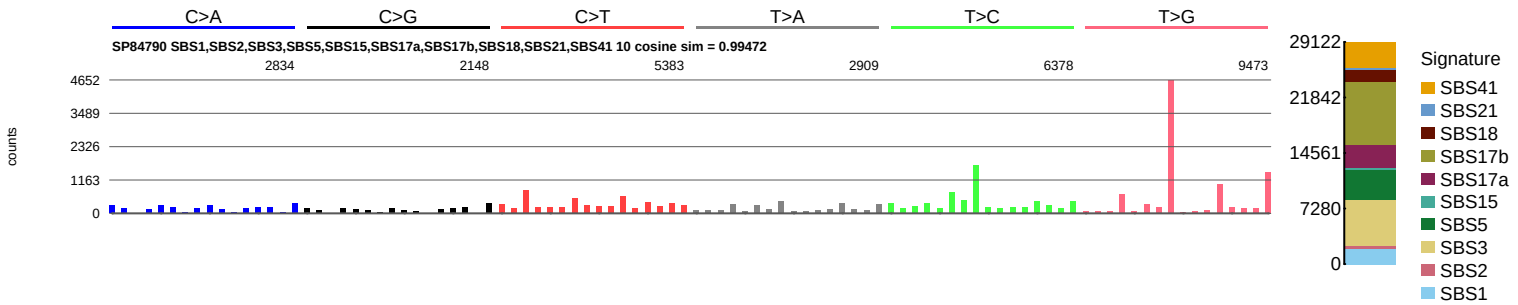

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.99472
2907
2281
5311
2766
6235
9623
counts
4636
3477
2318
1159
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

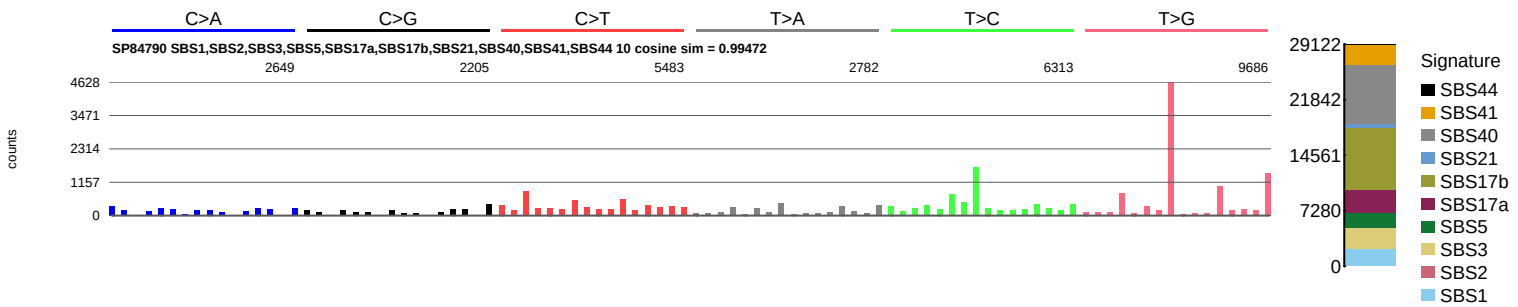

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS41 9 cosine sim = 0.99472
2640
2214
5463
2786
6321
9692
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

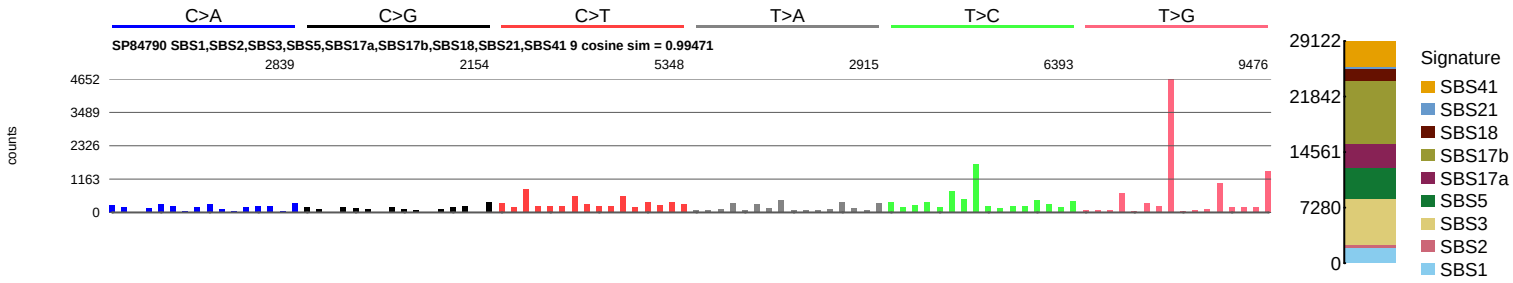

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.99472
2911
2273
5328
2760
6231
9621
counts
4636
3477
2318
1159
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

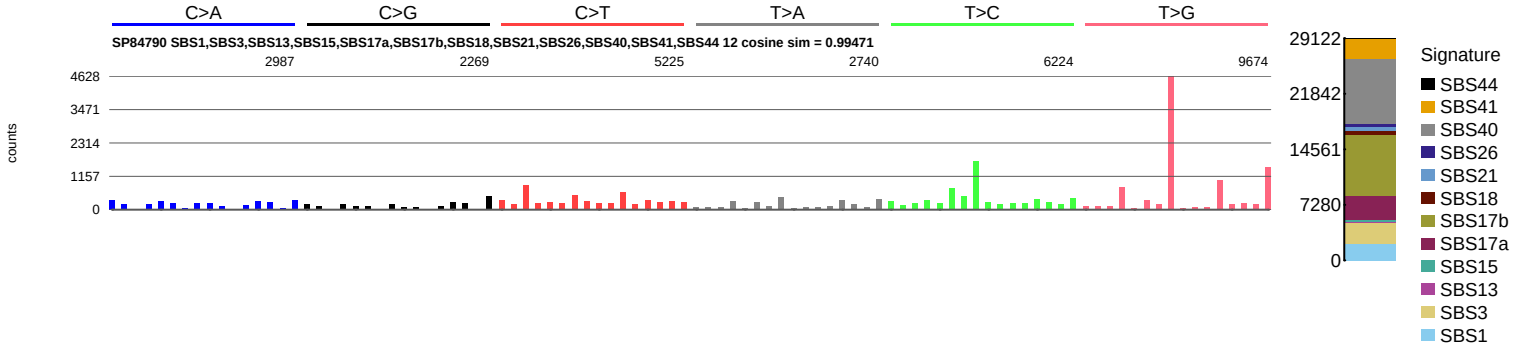

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.99472
2877
2197
5466
2886
6101
9599
counts
4640
3480
2320
1160
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

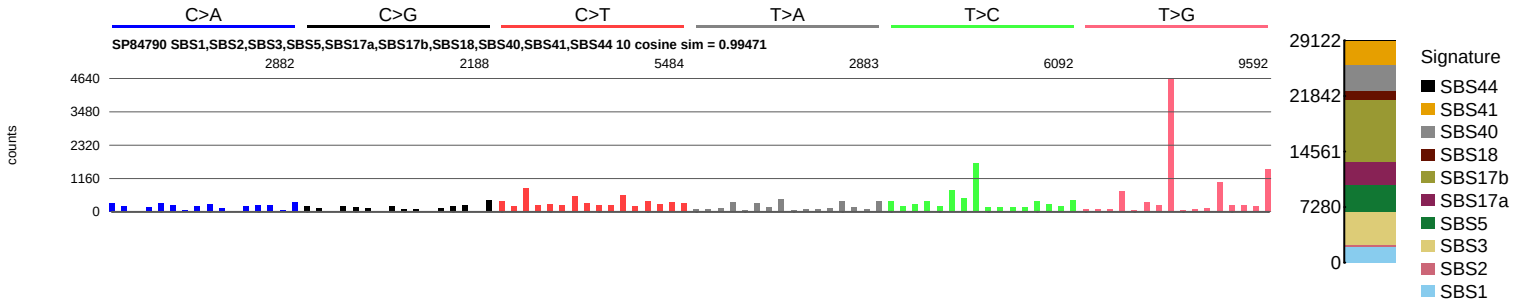

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 11 cosine sim = 0.99472
2958
2314
5213
2790
6219
9627
counts
4636
3477
2318
1159
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

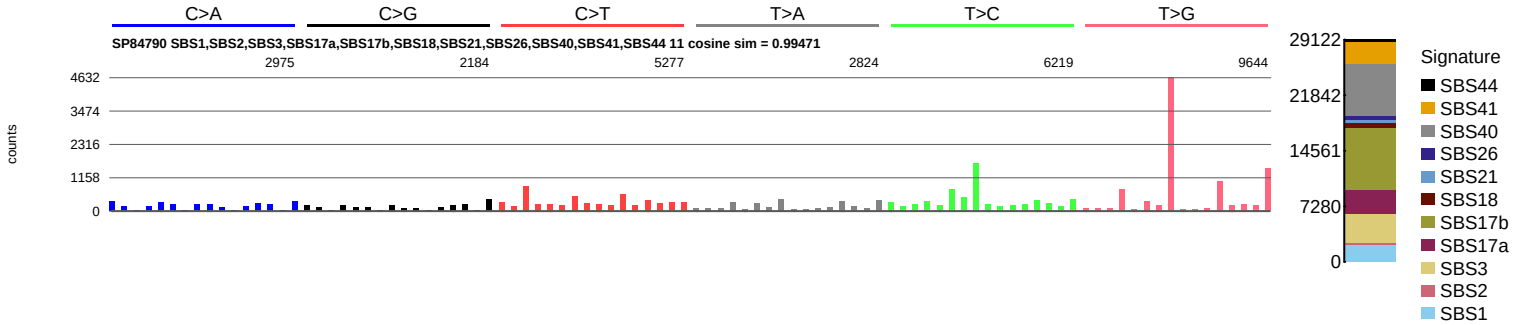

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.99472
2834
2148
5383
2909
6378
9473
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

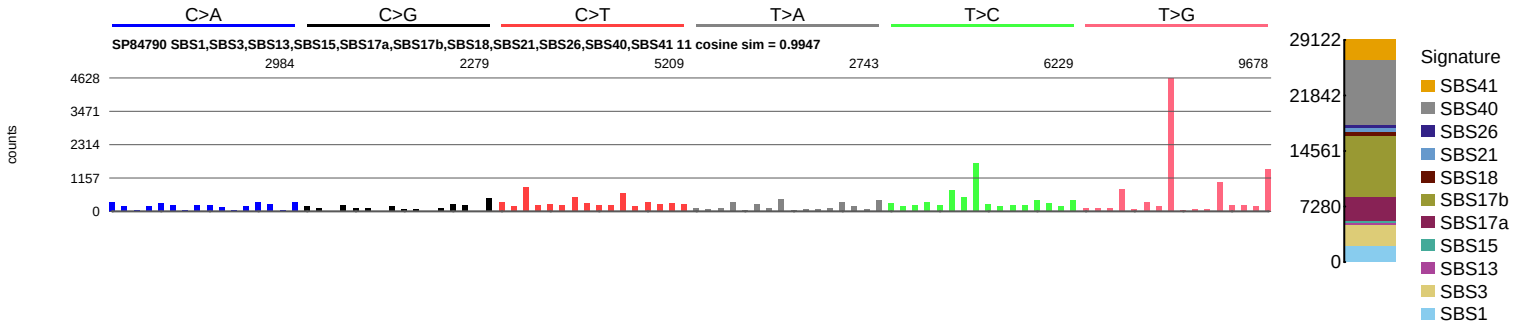

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.99472
2649
2205
5483
2782
6313
9686
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

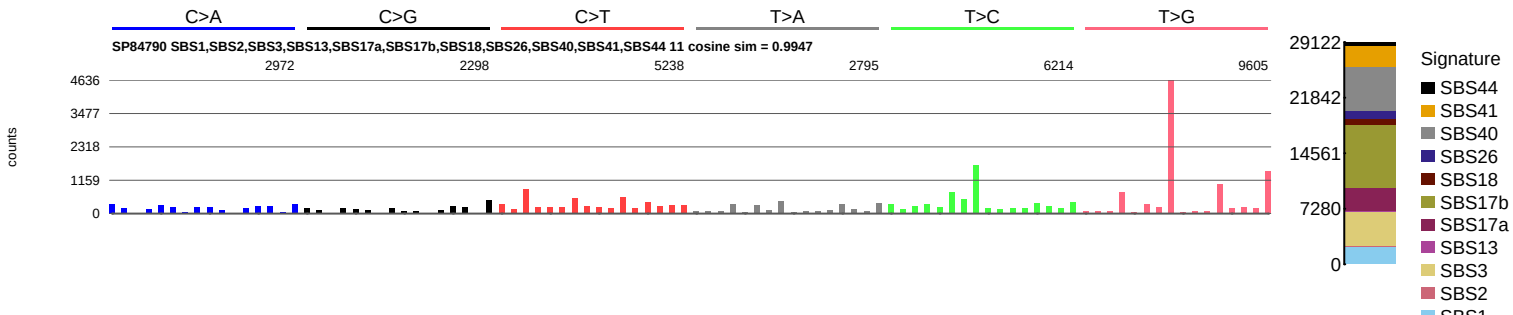

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.99471
2839
2154
5348
2915
6393
9476
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

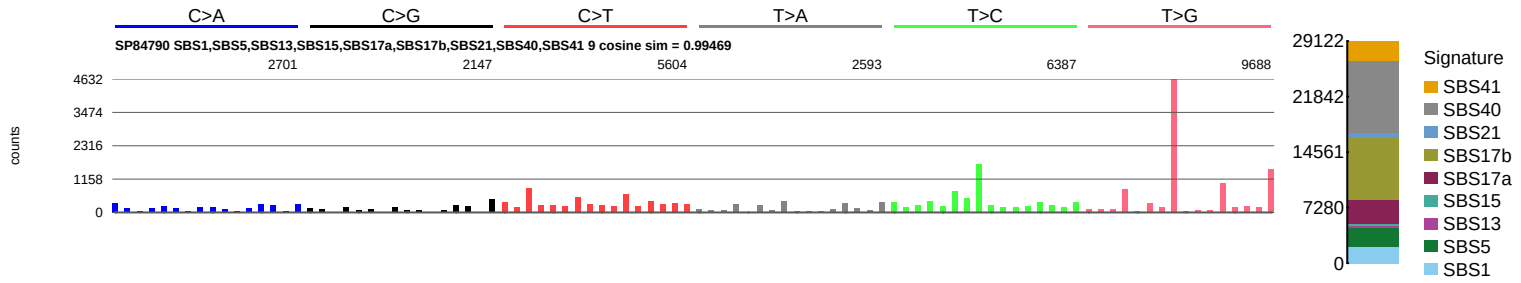

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.99471
2987
2269
5225
2740
6224
9674
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

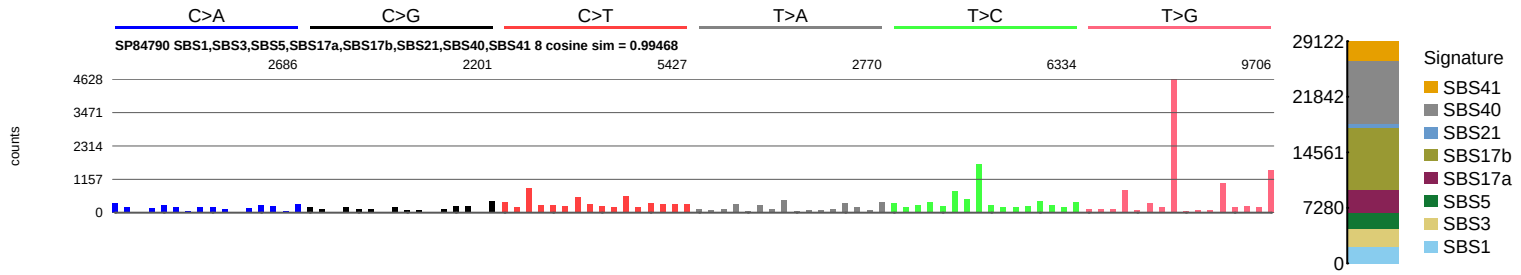

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.99471
2882
2188
5484
2883
6092
9592
counts
4640
3480
2320
1160
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

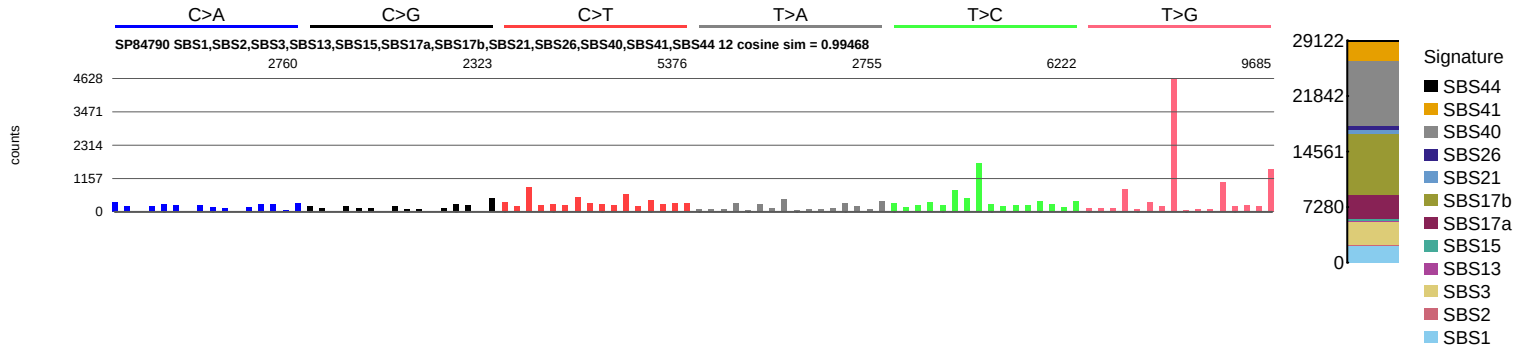

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.99471
2975
2184
5277
2824
6219
9644
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

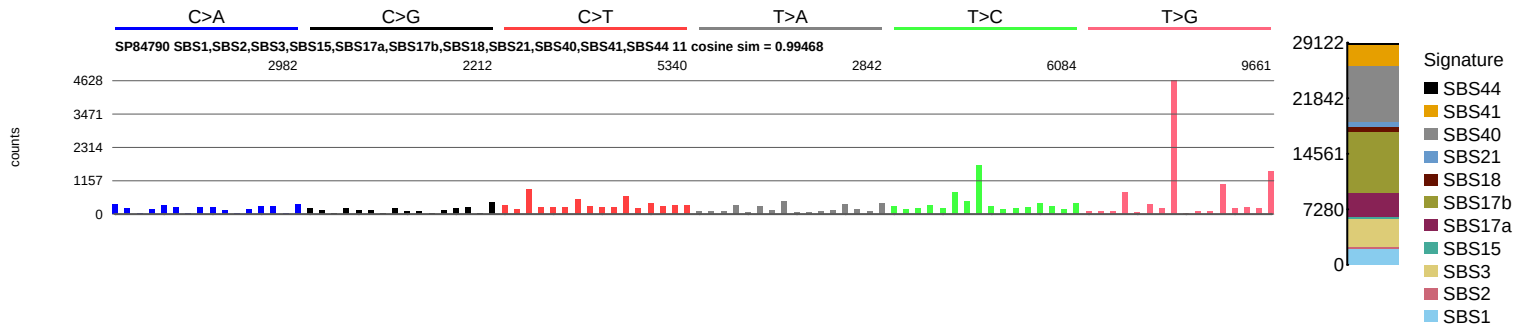

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 11 cosine sim = 0.9947
2984
2279
5209
2743
6229
9678
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

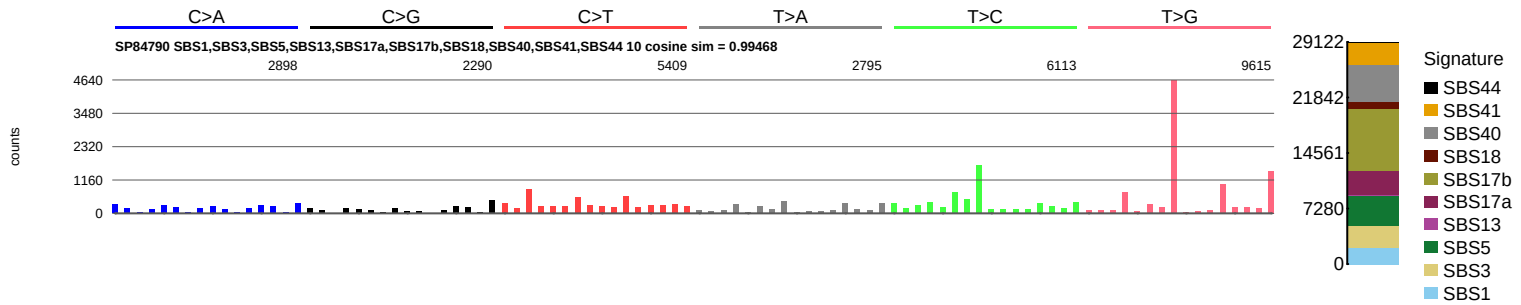

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.9947
2972
2298
5238
2795
6214
9605
counts
4636
3477
2318
1159
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2

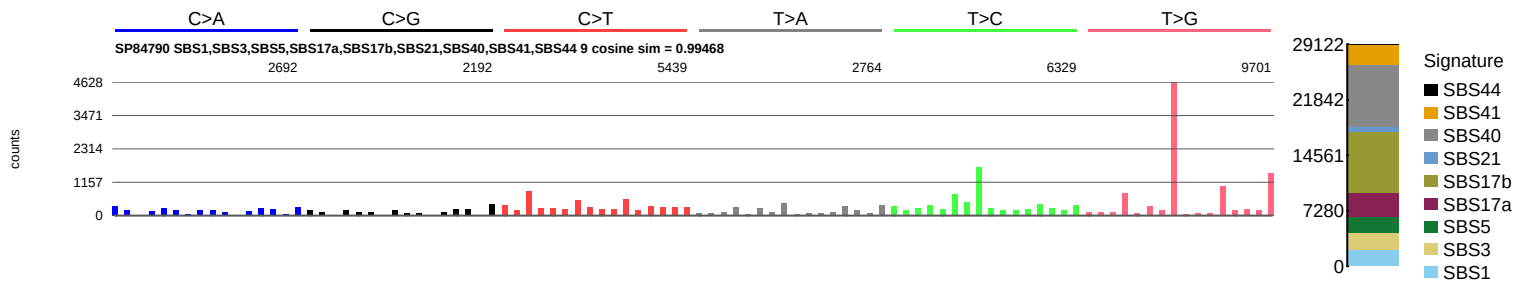

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 9 cosine sim = 0.99469
2701
2147
5604
2593
6387
9688
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

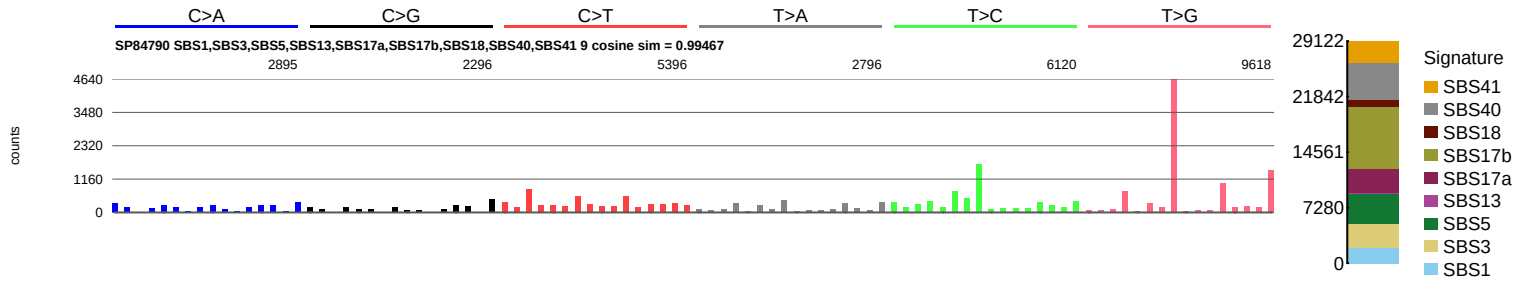

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS41 8 cosine sim = 0.99468
2686
2201
5427
2770
6334
9706
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS1

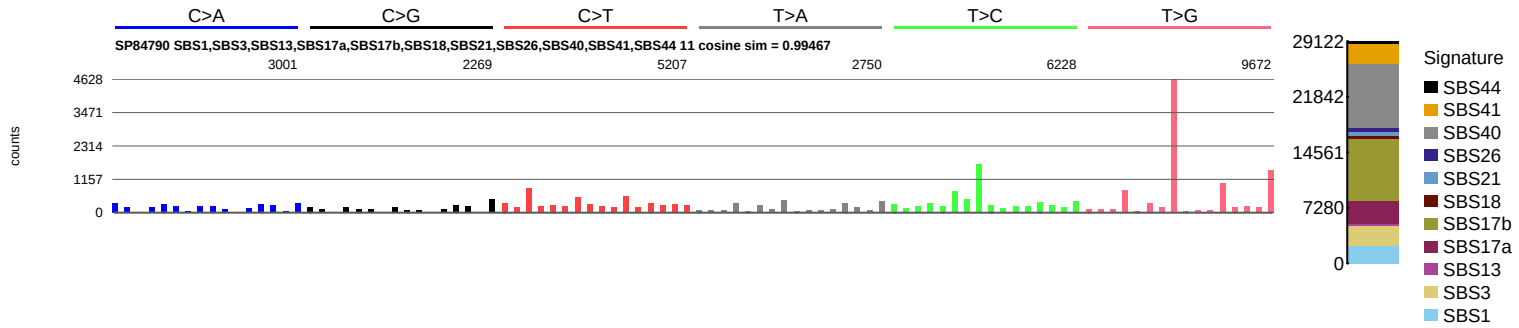

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.99468
2760
2323
5376
2755
6222
9685
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

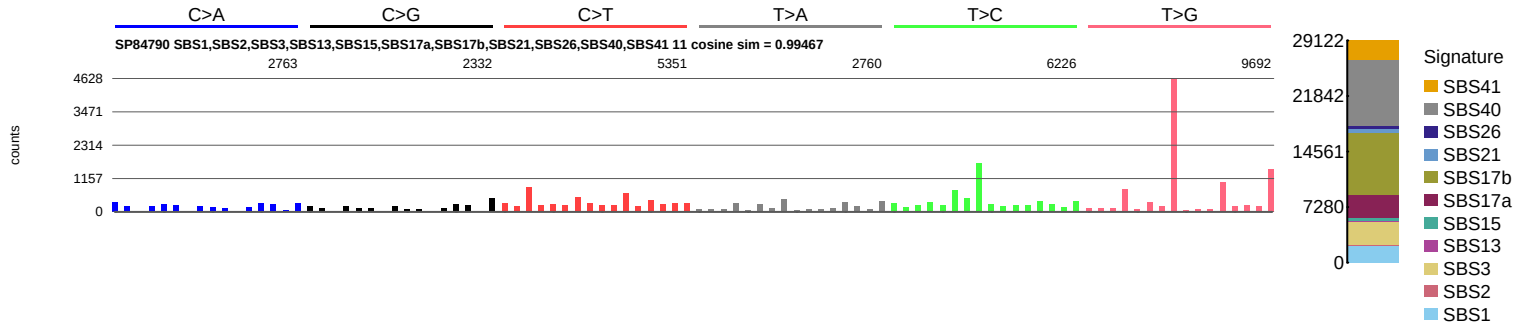

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.99468
2982
2212
5340
2842
6084
9661
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

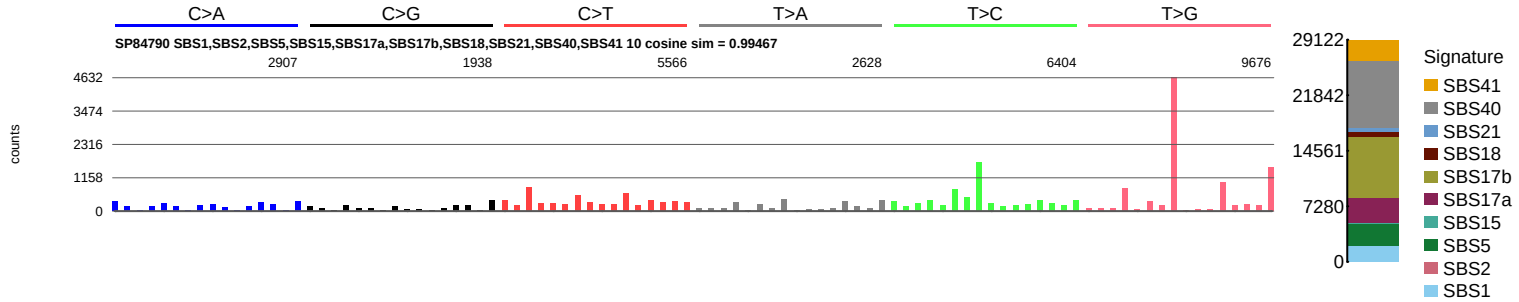

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.99468
2898
2290
5409
2795
6113
9615
counts
4640
3480
2320
1160
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

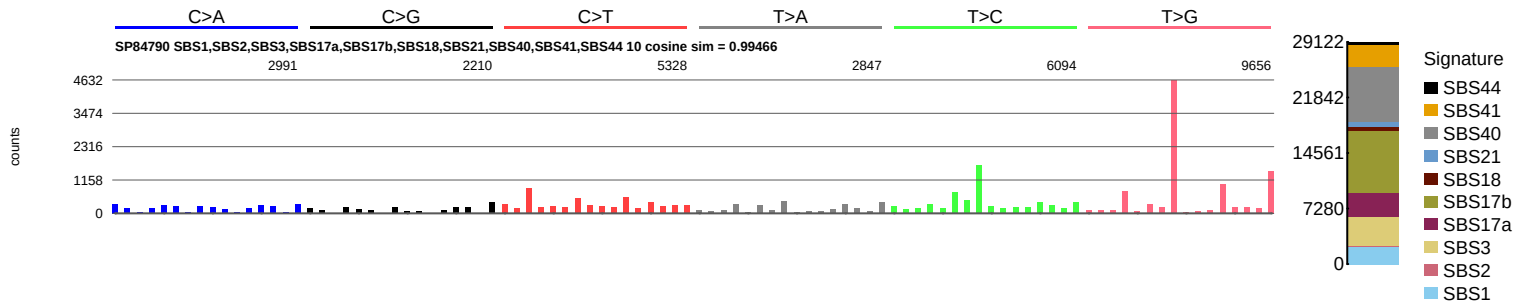

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.99468
2692
2192
5439
2764
6329
9701
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS1

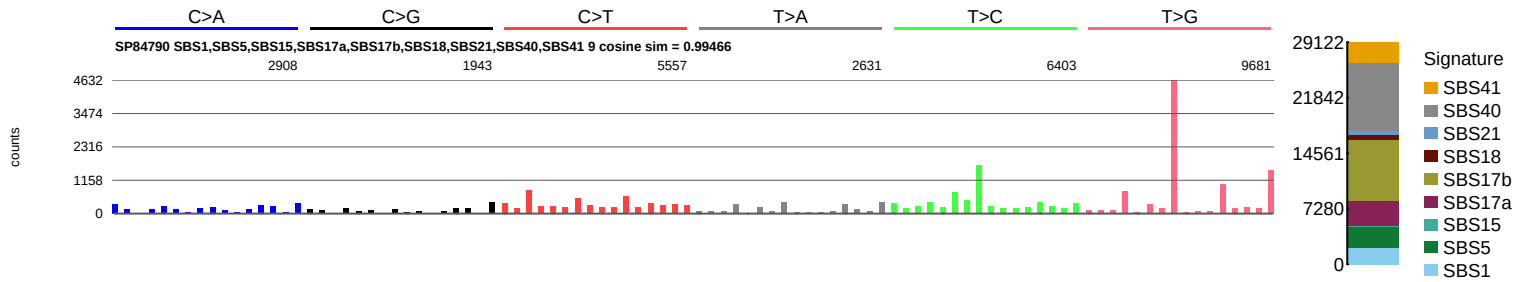

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.99467
2895
2296
5396
2796
6120
9618
counts
4640
3480
2320
1160
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

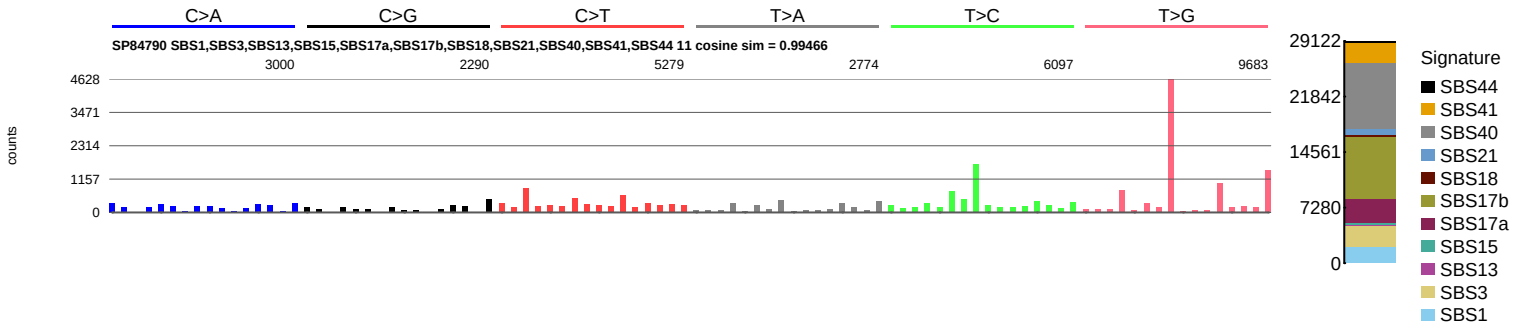

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.99467
3001
2269
5207
2750
6228
9672
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1

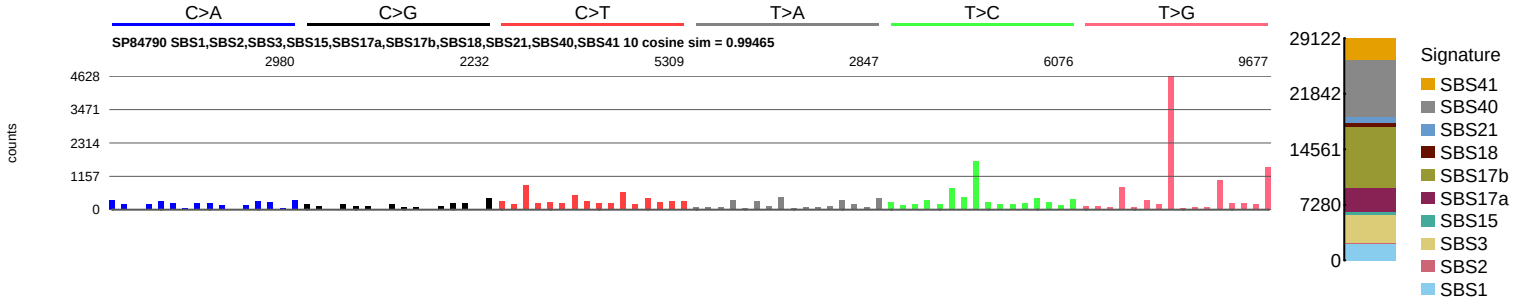

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41 11 cosine sim = 0.99467
2763
2332
5351
2760
6226
9692
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

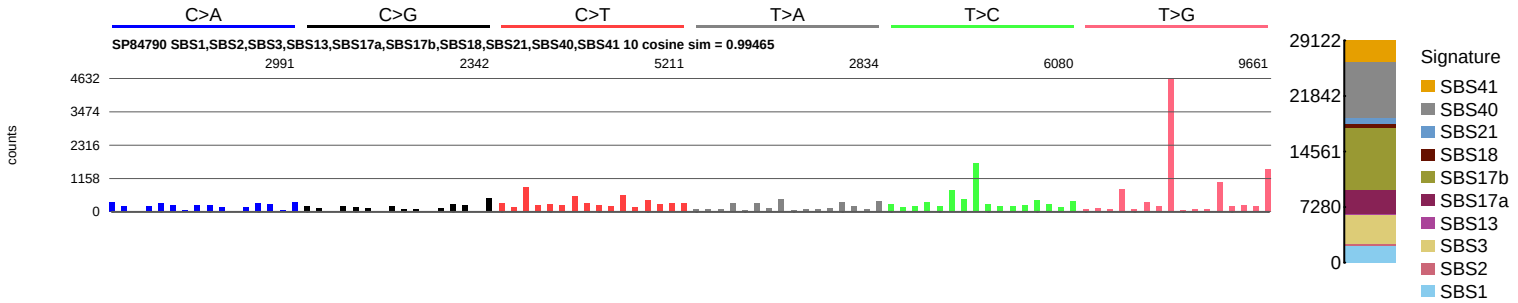

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.99467
2907
1938
5566
2628
6404
9676
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

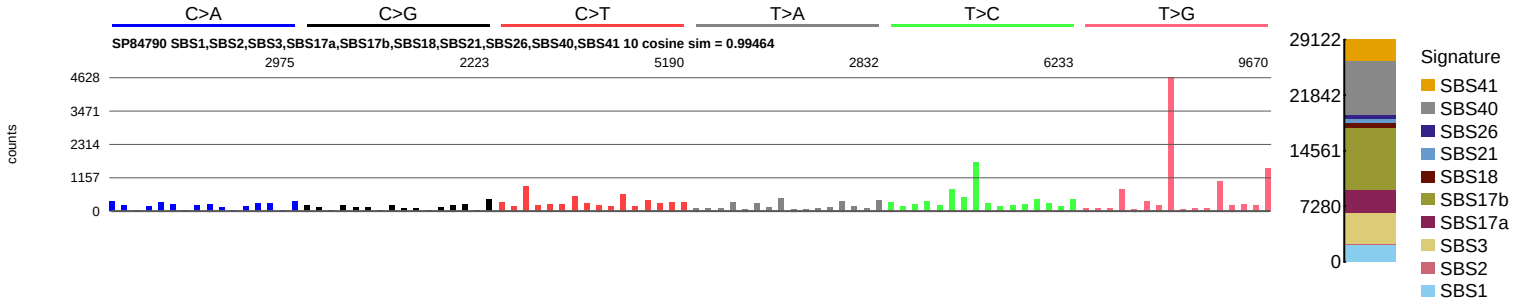

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.99466
2991
2210
5328
2847
6094
9656
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

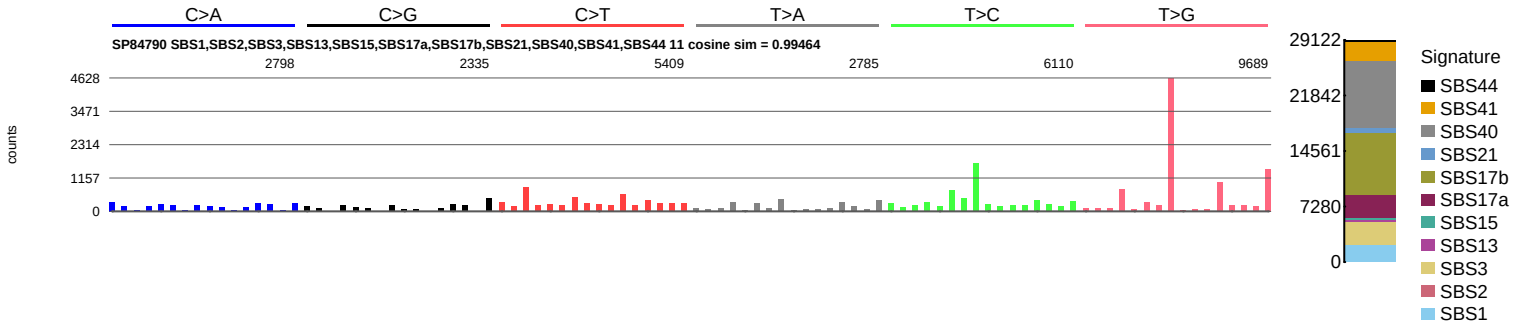

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.99466
2908
1943
5557
2631
6403
9681
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1

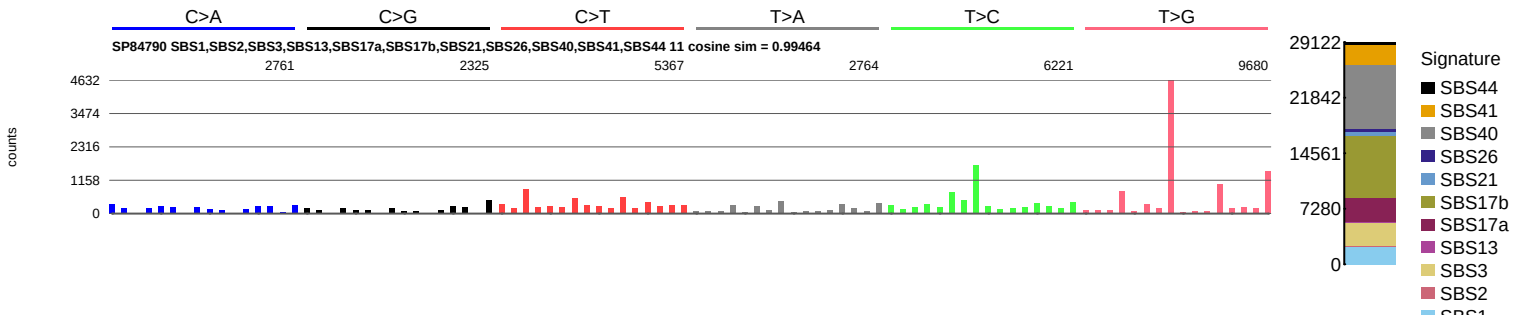

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.99466
3000
2290
5279
2774
6097
9683
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

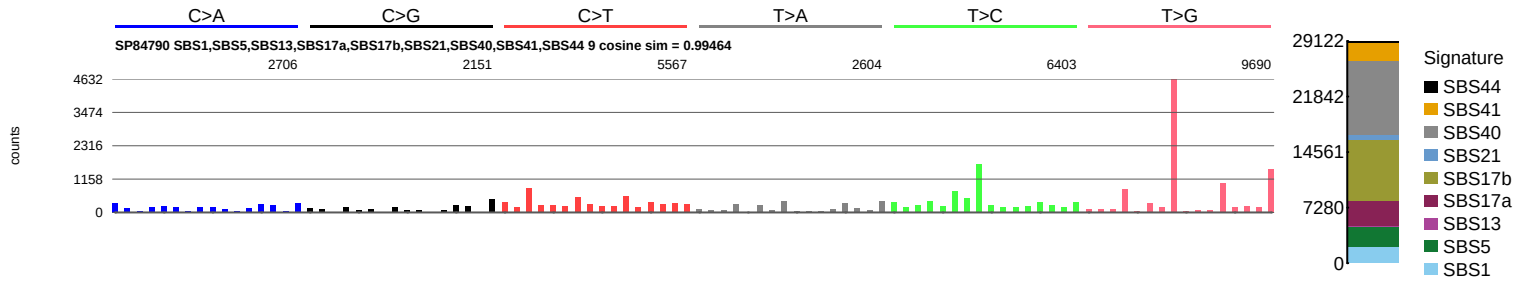

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.99465
2980
2232
5309
2847
6076
9677
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

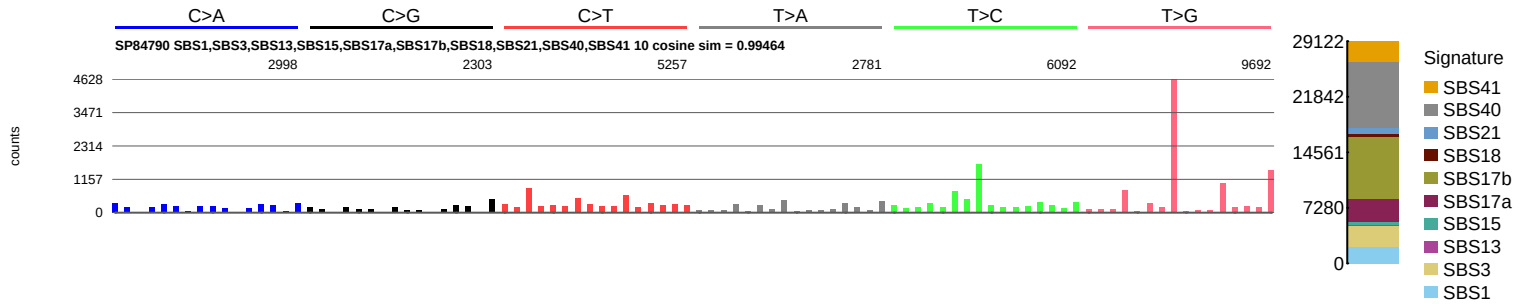

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.99465
2991
2342
5211
2834
6080
9661
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

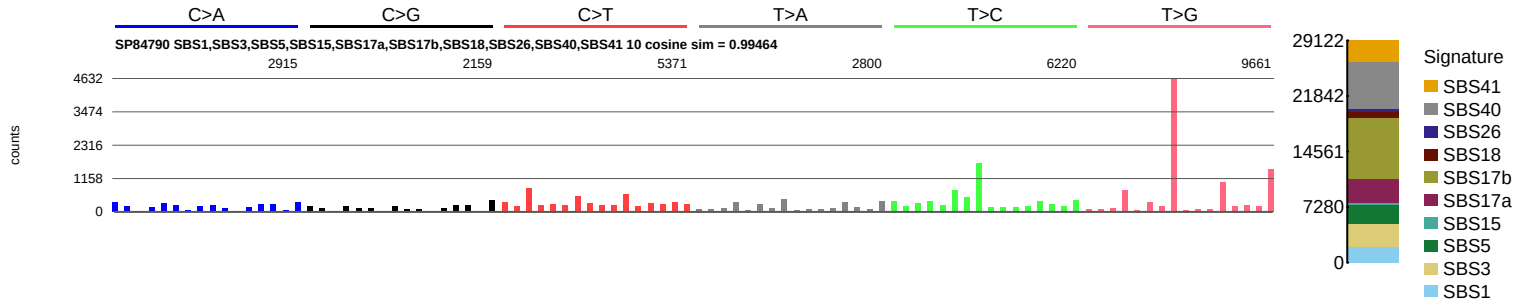

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.99464
2975
2223
5190
2832
6233
9670
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

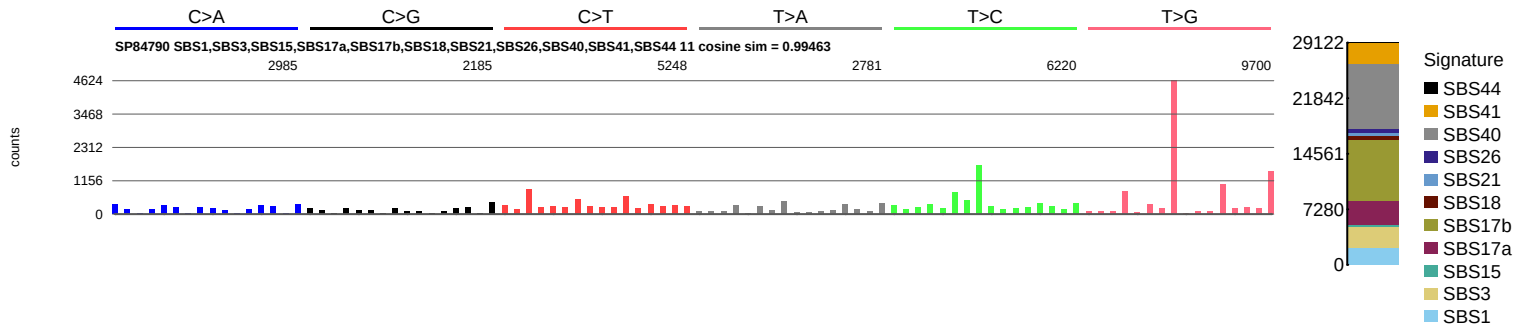

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.99464
2798
2335
5409
2785
6110
9689
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

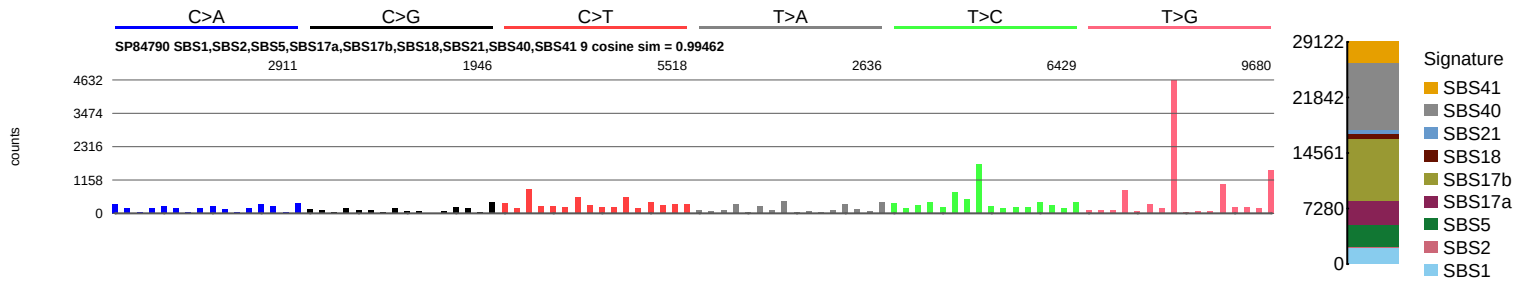

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.99464
2761
2325
5367
2764
6221
9680
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2

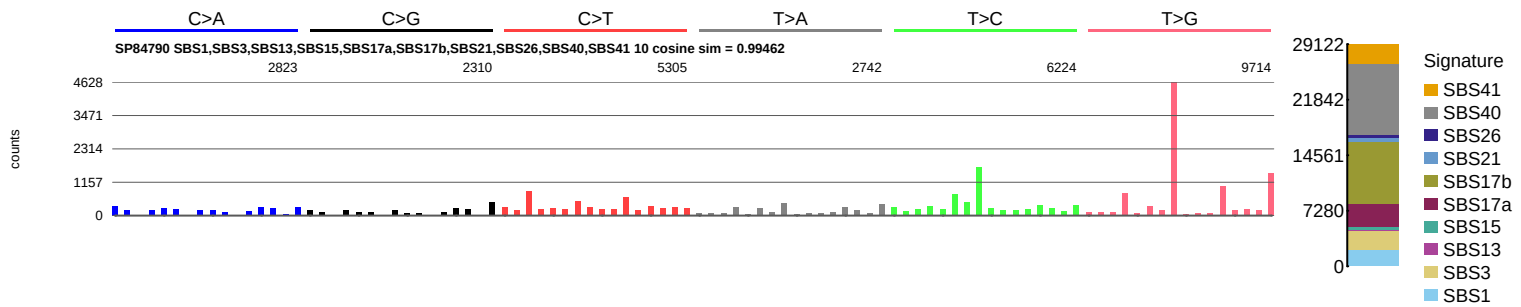

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.99464
2706
2151
5567
2604
6403
9690
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS1

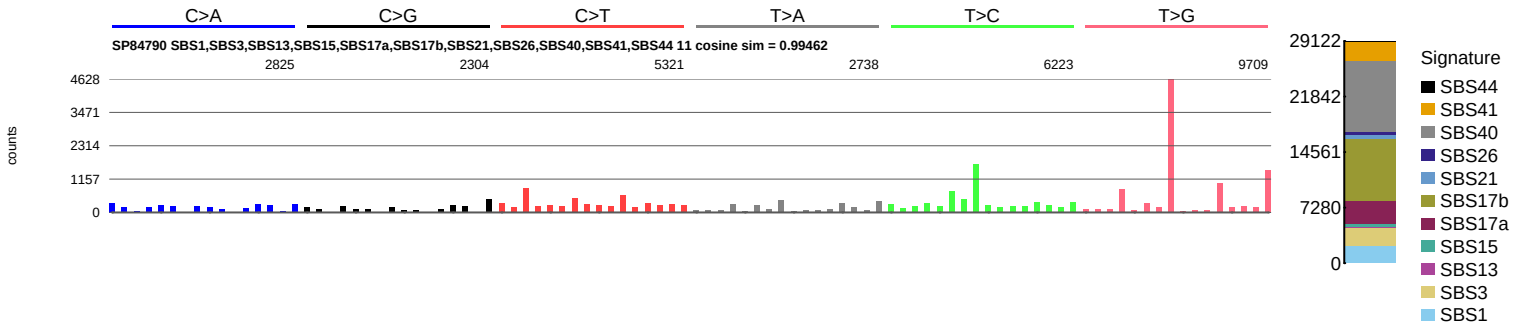

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.99464
2998
2303
5257
2781
6092
9692
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

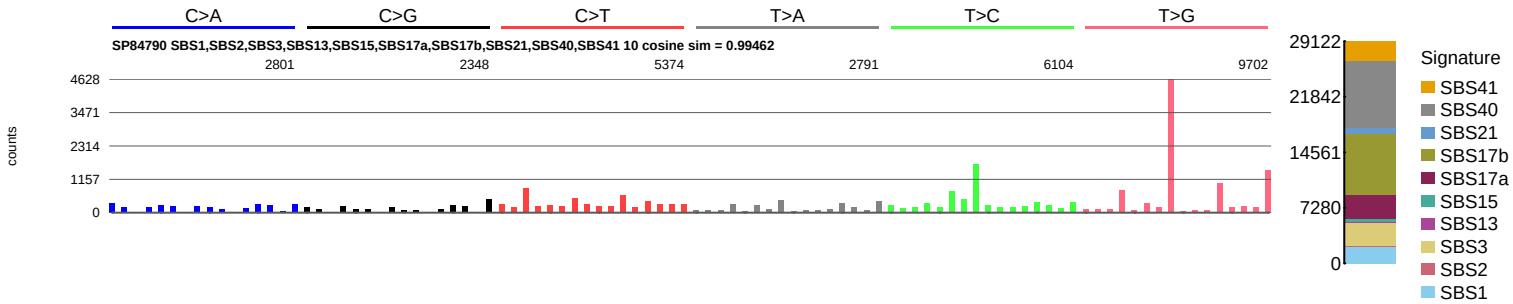

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.99464
2915
2159
5371
2800
6220
9661
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1

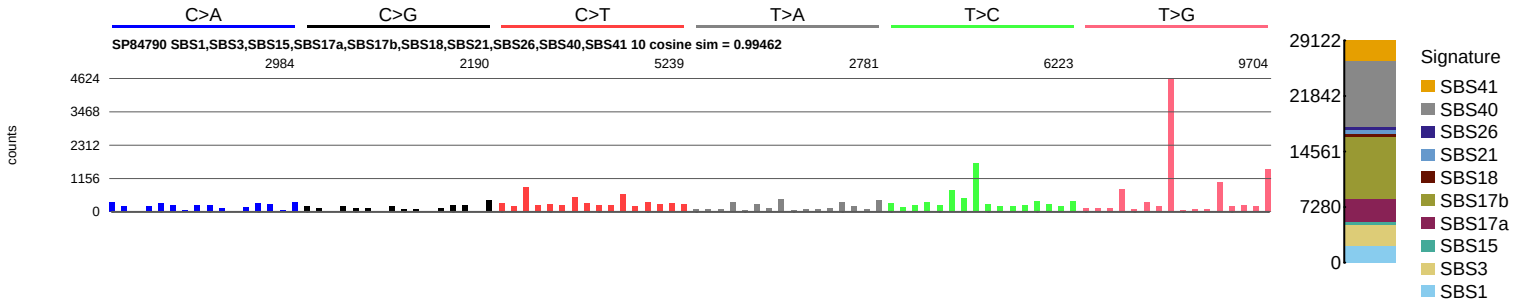

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.99463
2985
2185
5248
2781
6220
9700
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

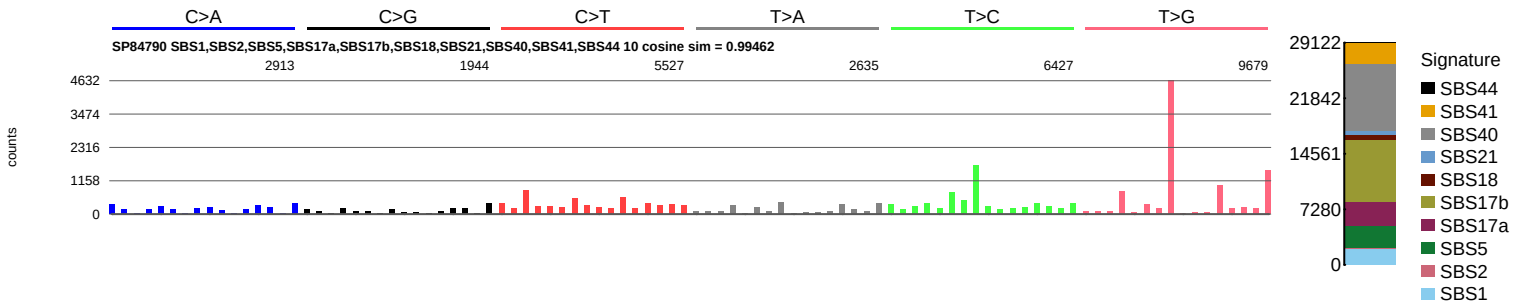

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.99462
2911
1946
5518
2636
6429
9680
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

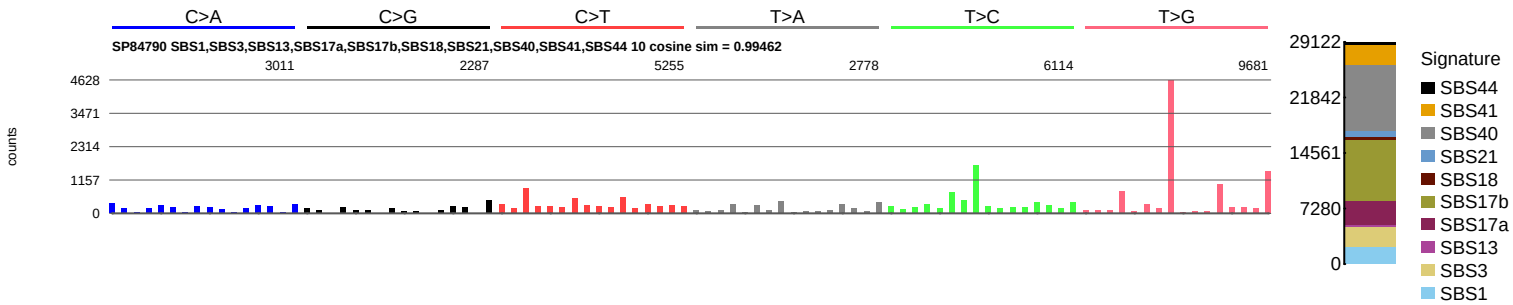

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.99462
2823
2310
5305
2742
6224
9714
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

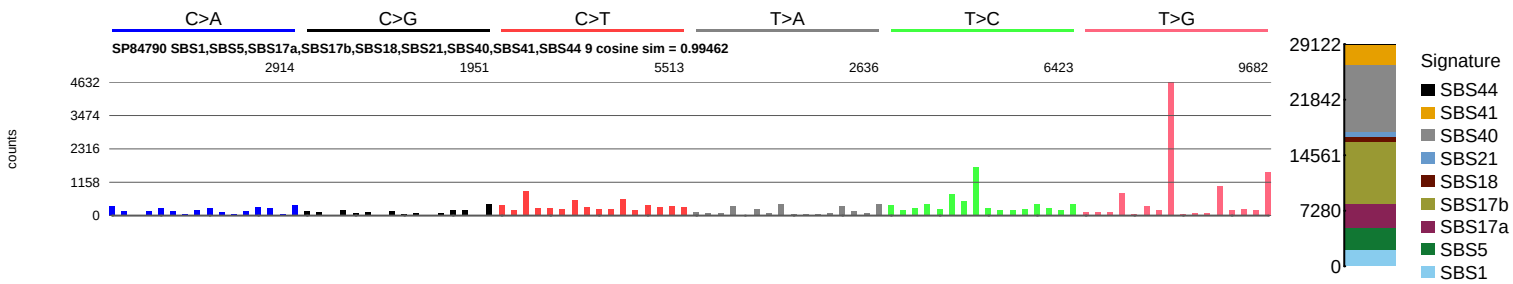

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.99462
2825
2304
5321
2738
6223
9709
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

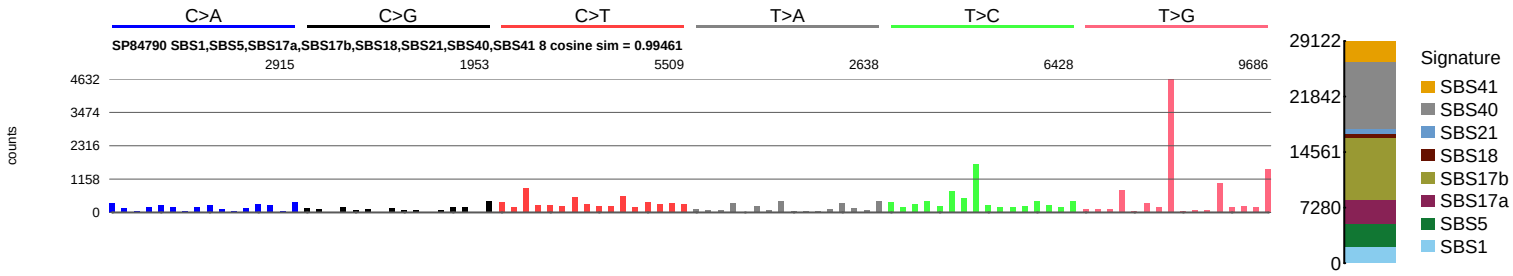

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 10 cosine sim = 0.99462
2801
2348
5374
2791
6104
9702
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

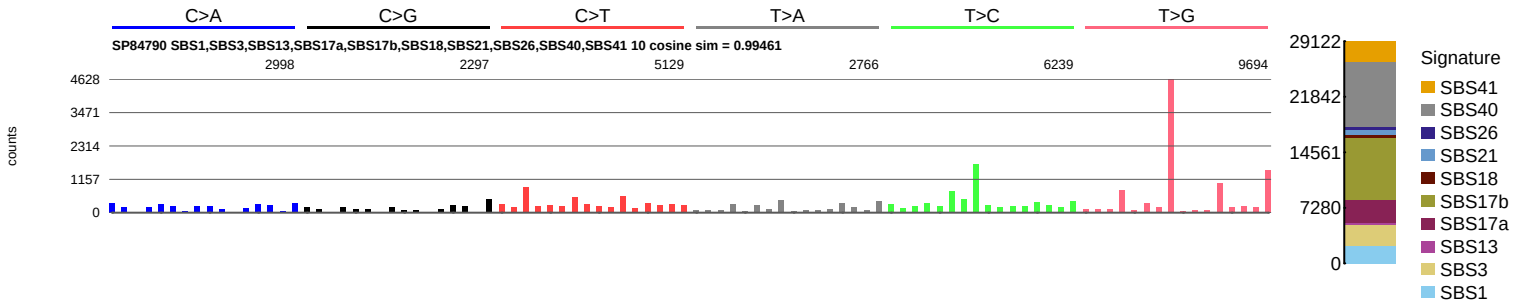

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.99462
2984
2190
5239
2781
6223
9704
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

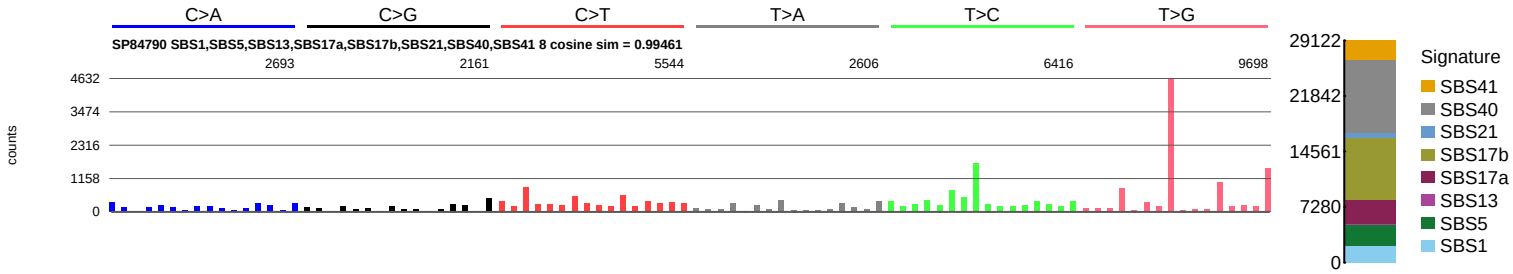

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.99462
2913
1944
5527
2635
6427
9679
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

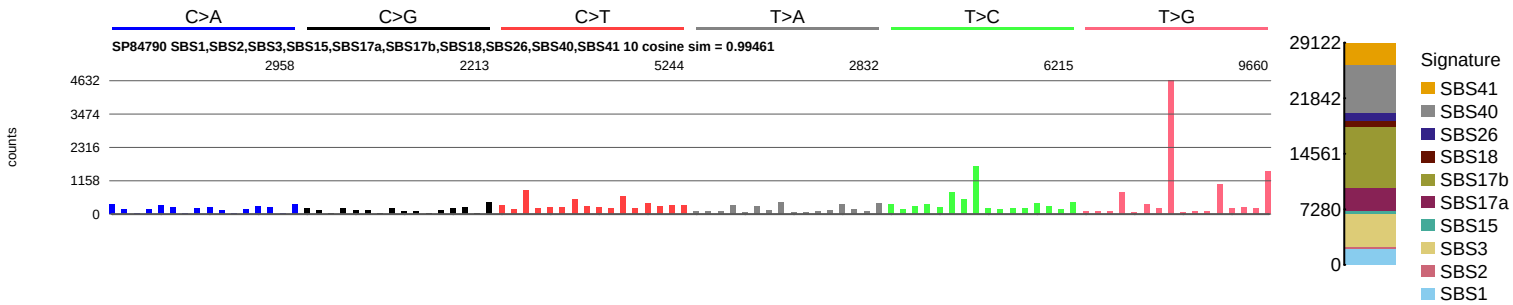

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.99462
3011
2287
5255
2778
6114
9681
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1

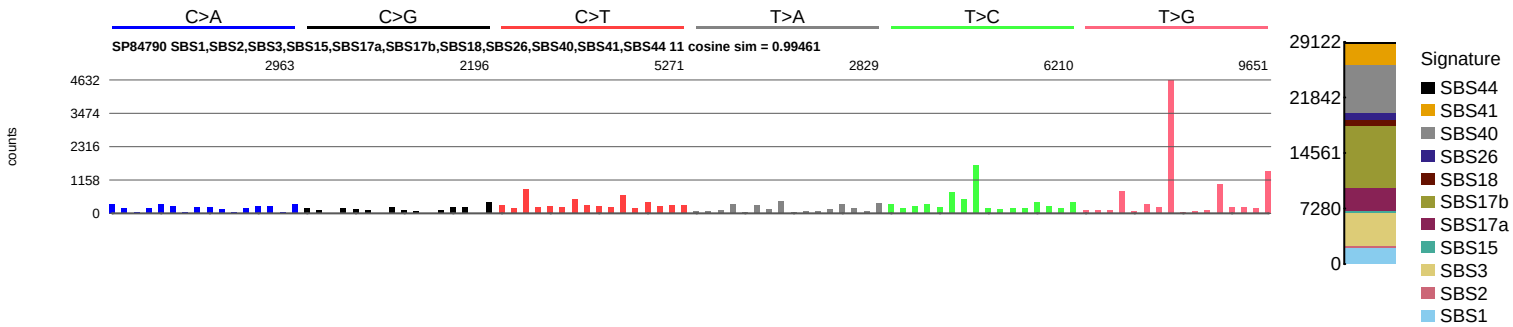

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.99462
2914
1951
5513
2636
6423
9682
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1

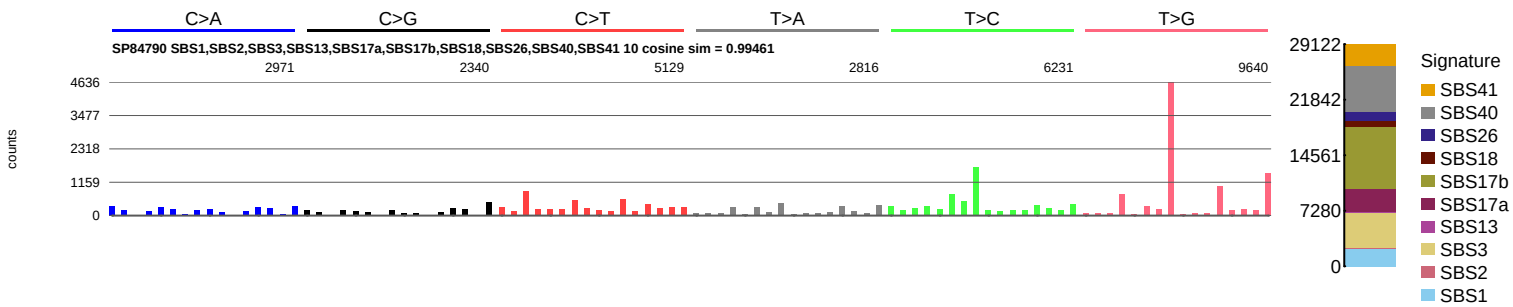

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.99461
2915
1953
5509
2638
6428
9686
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1

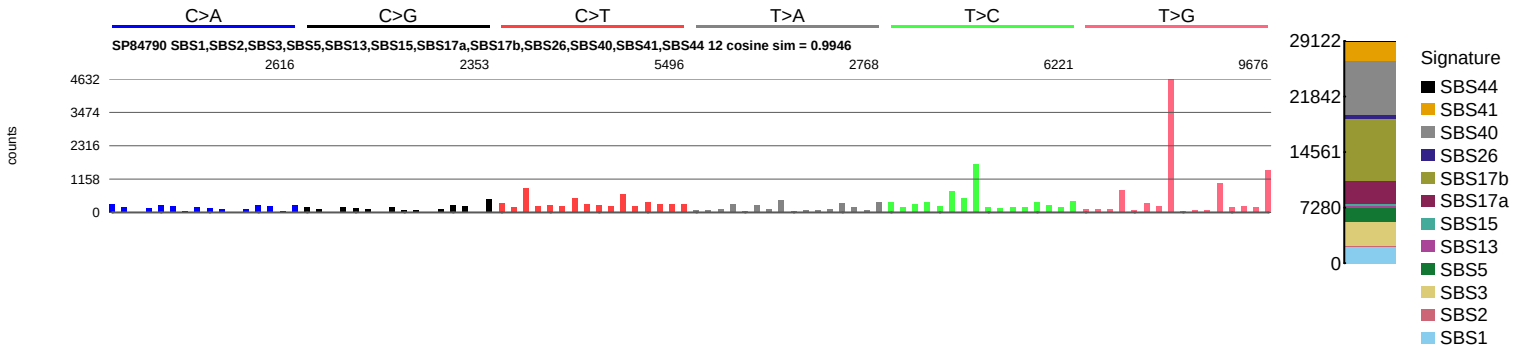

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.99461
2998
2297
5129
2766
6239
9694
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1

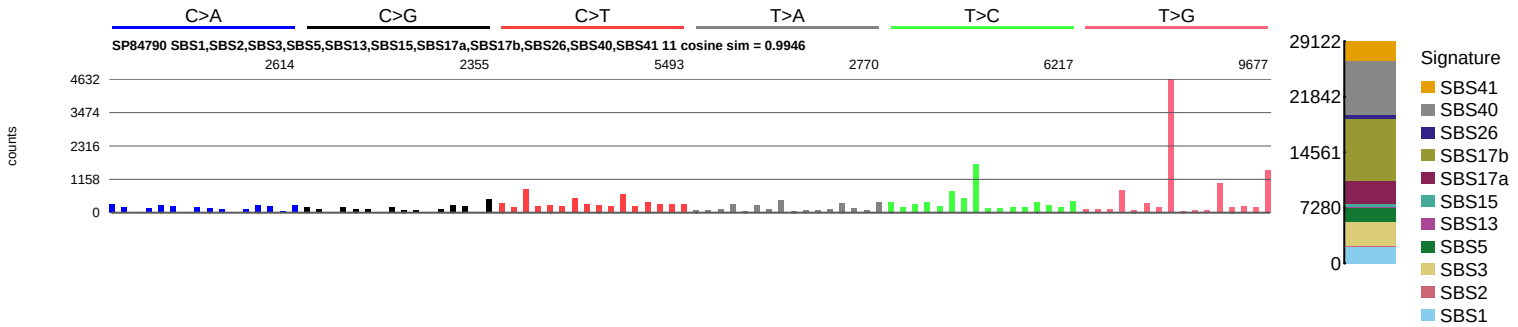

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS41 8 cosine sim = 0.99461
2693
2161
5544
2606
6416
9698
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS1

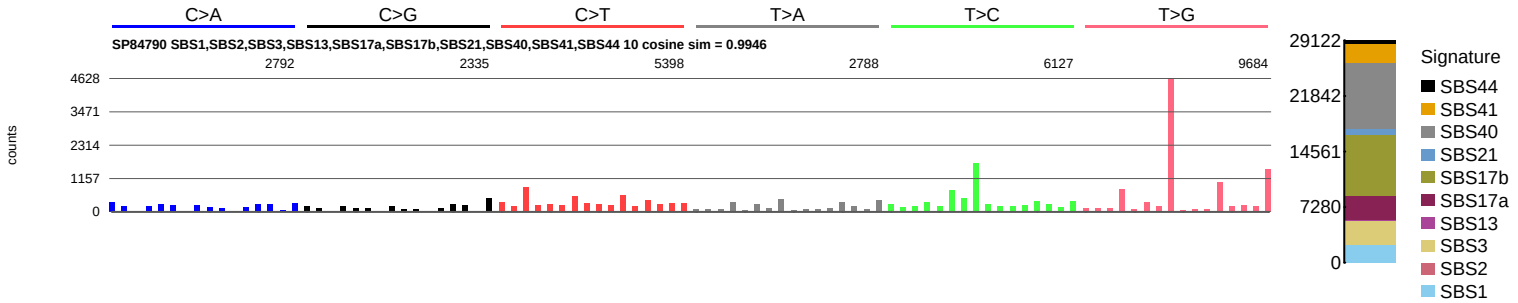

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.99461
2958
2213
5244
2832
6215
9660
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

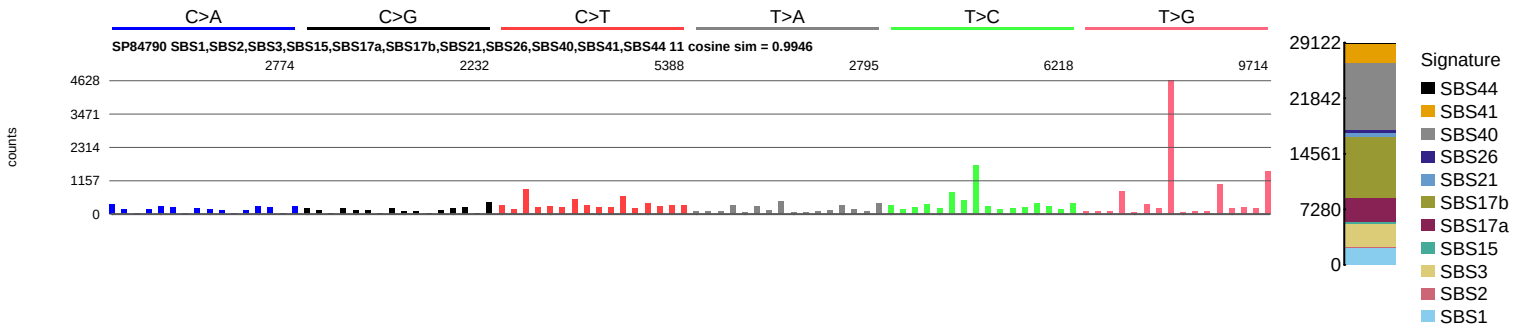

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.99461
2963
2196
5271
2829
6210
9651
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

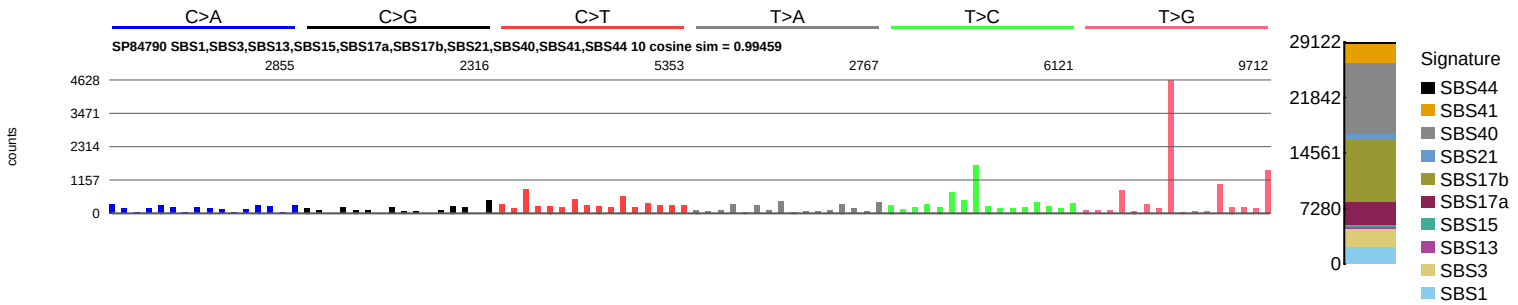

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.99461
2971
2340
5129
2816
6231
9640
counts
4636
3477
2318
1159
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

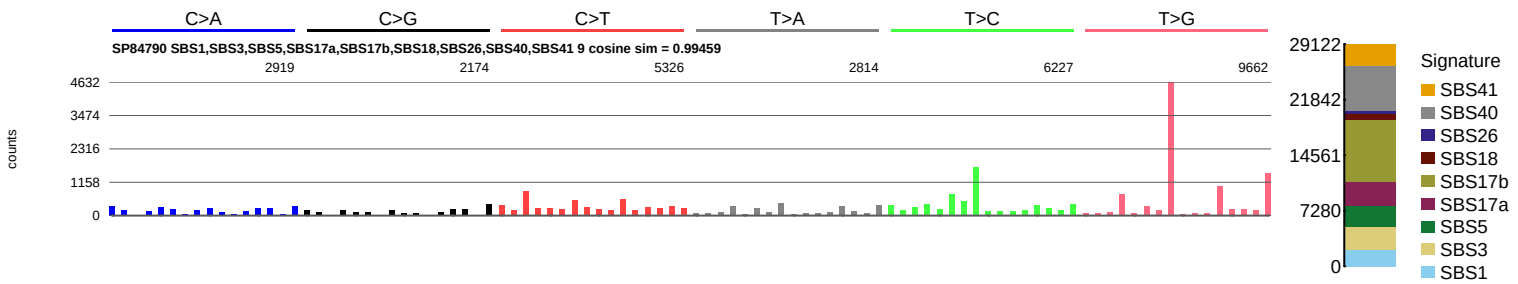

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.9946
2616
2353
5496
2768
6221
9676
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

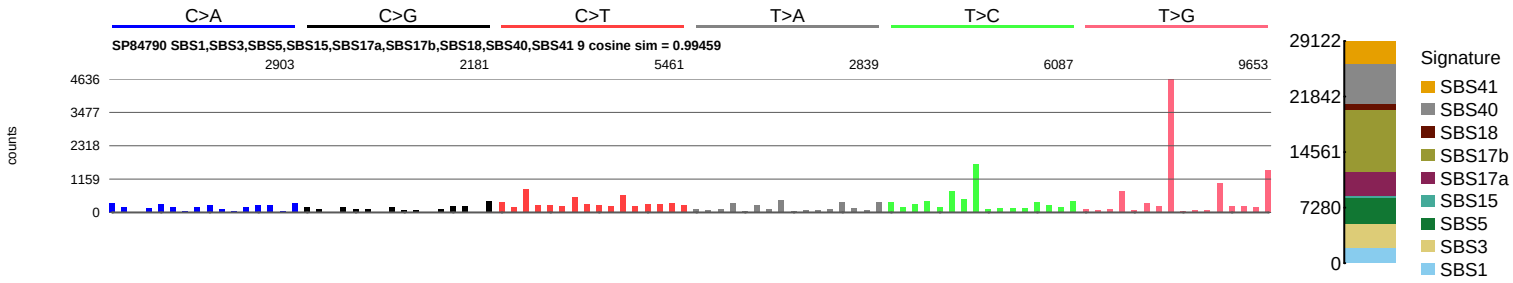

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 11 cosine sim = 0.9946
2614
2355
5493
2770
6217
9677
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

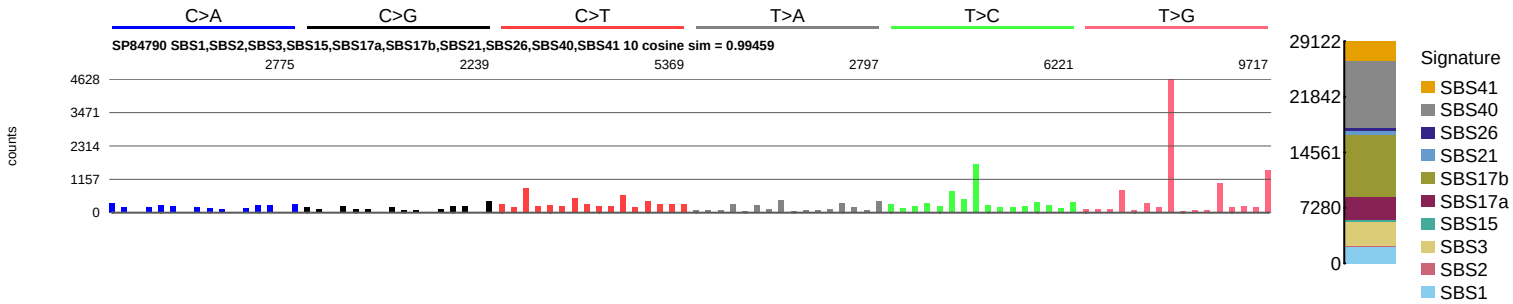

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.9946
2792
2335
5398
2788
6127
9684
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

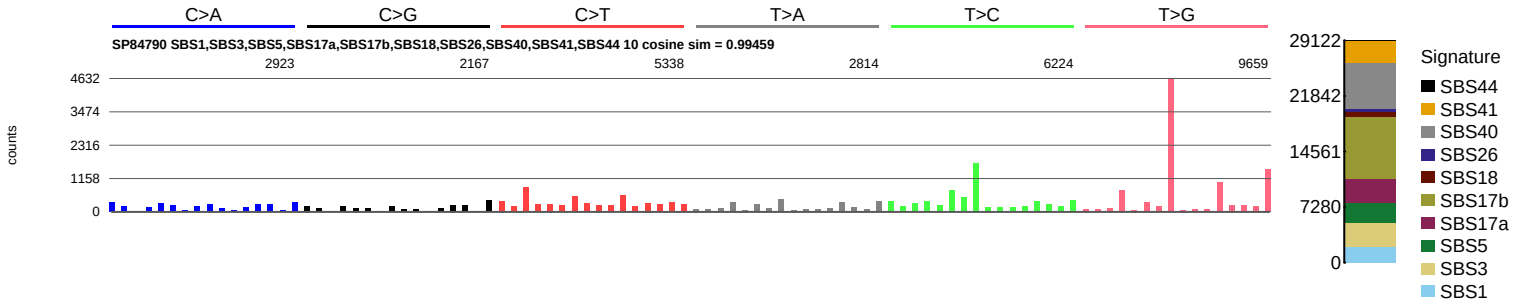

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.9946
2774
2232
5388
2795
6218
9714
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

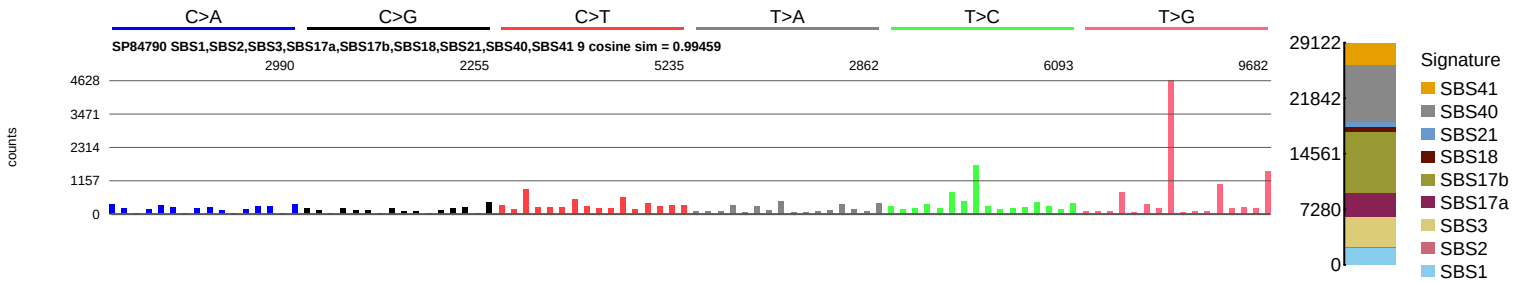

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.99459
2855
2316
5353
2767
6121
9712
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

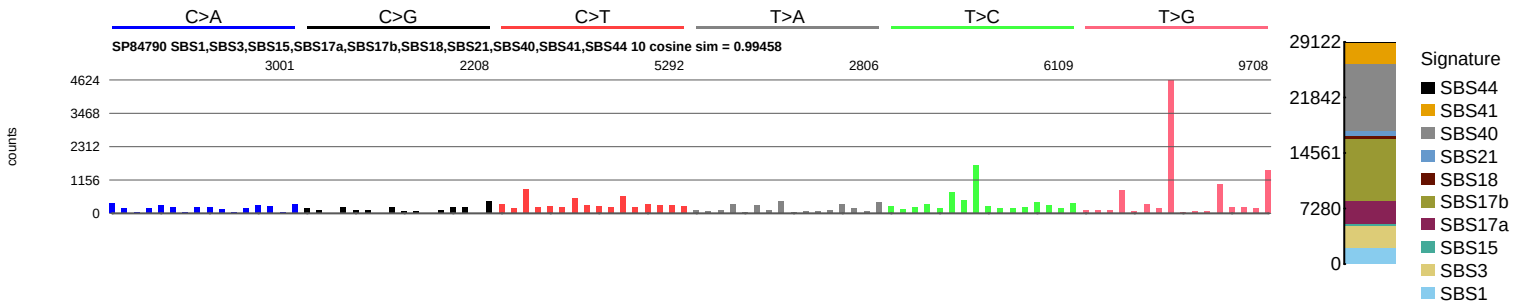

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.99459
2919
2174
5326
2814
6227
9662
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1

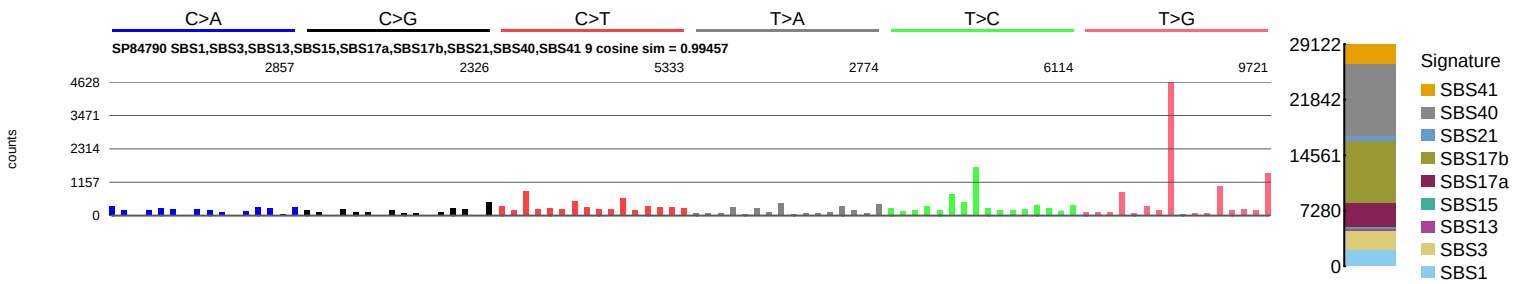

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.99459
2903
2181
5461
2839
6087
9653
4636
3477
2318
1159
0
counts
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1

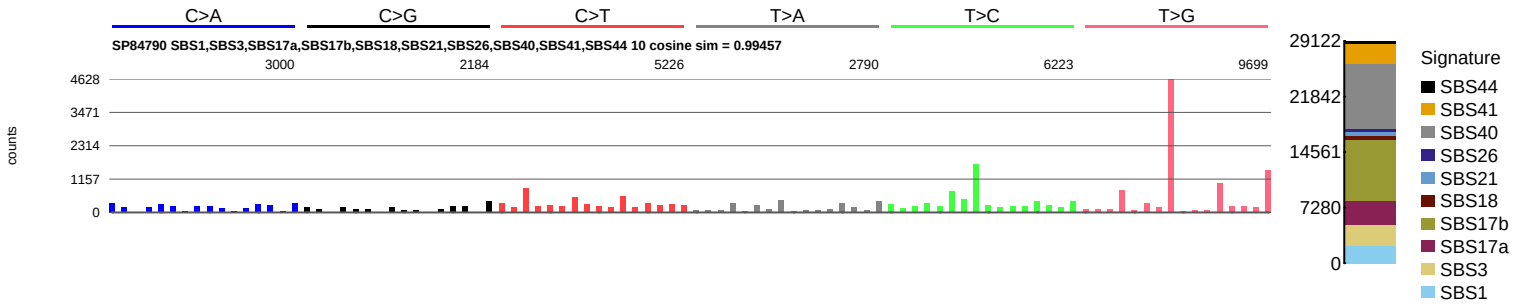

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.99459
2775
2239
5369
2797
6221
9717
4628
3471
2314
1157
0
counts
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

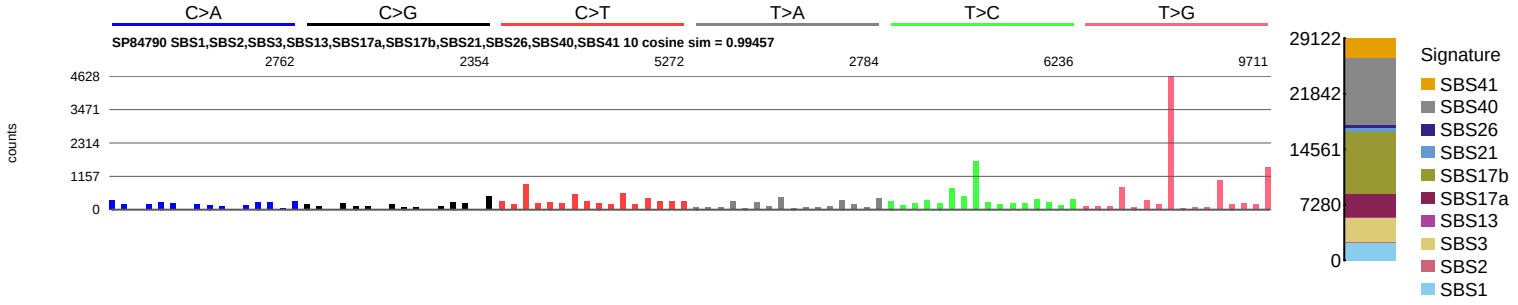

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.99459
2923
2167
5338
2814
6224
9659
4632
3474
2316
1158
0
counts
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1

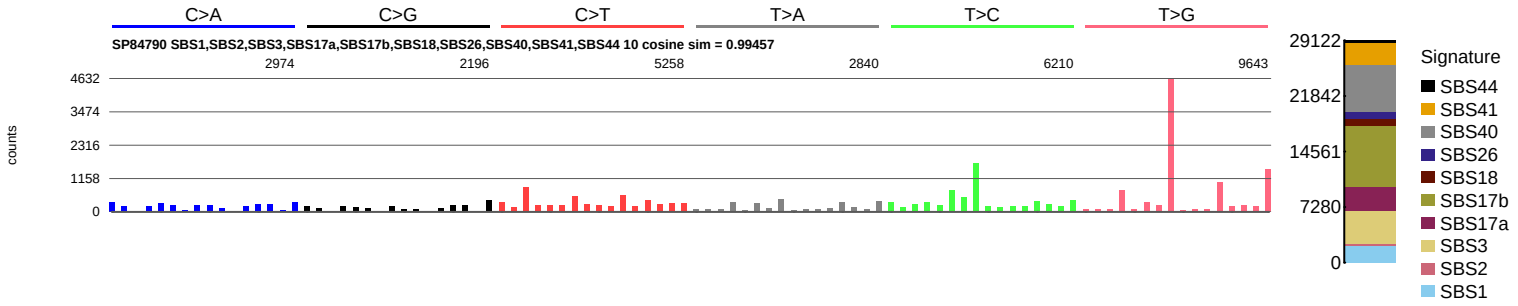

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.99459
2990
2255
5235
2862
6093
9682
4628
3471
2314
1157
0
counts
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

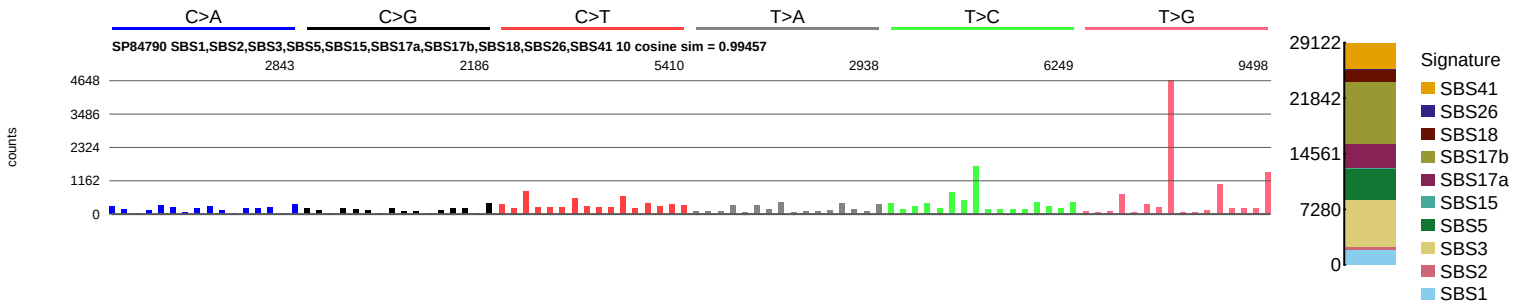

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.99458
3001
2208
5292
2806
6109
9708
4624
3468
2312
1156
0
counts
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

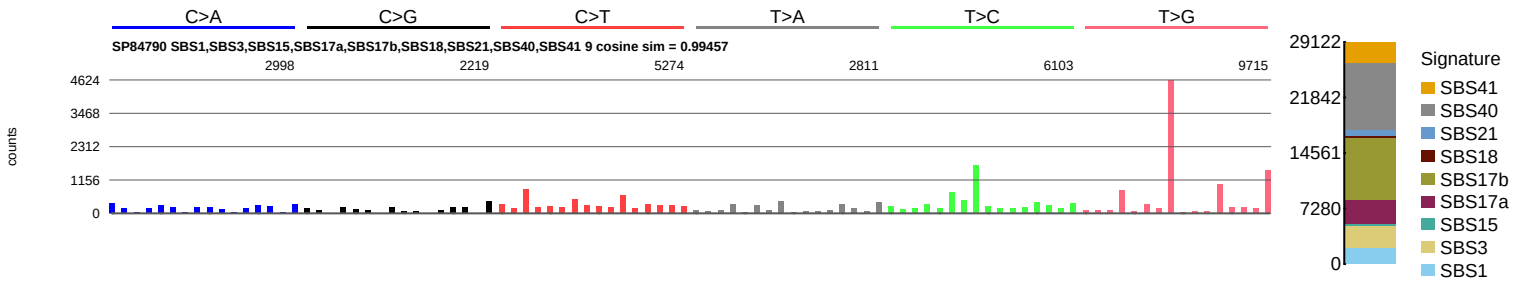

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 9 cosine sim = 0.99457
2857
2326
5333
2774
6114
9721
4628
3471
2314
1157
0
counts
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

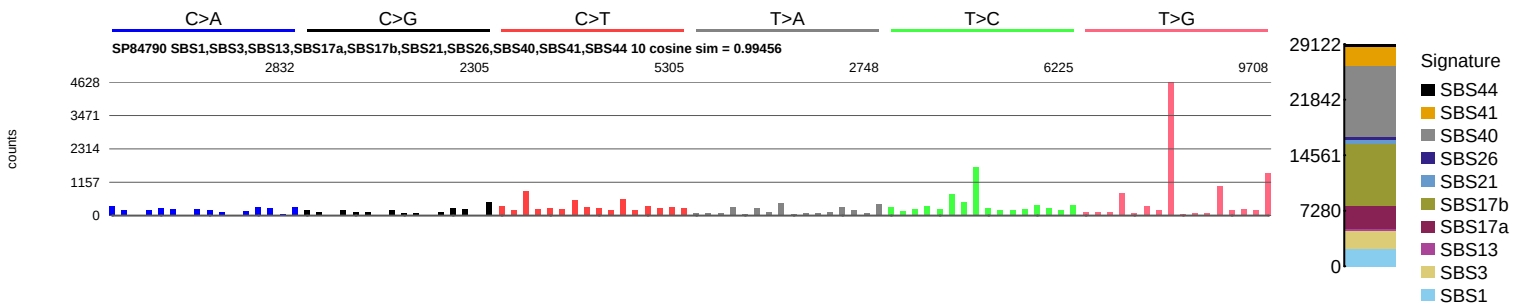

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.99457
3000
2184
5226
2790
6223
9699
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1

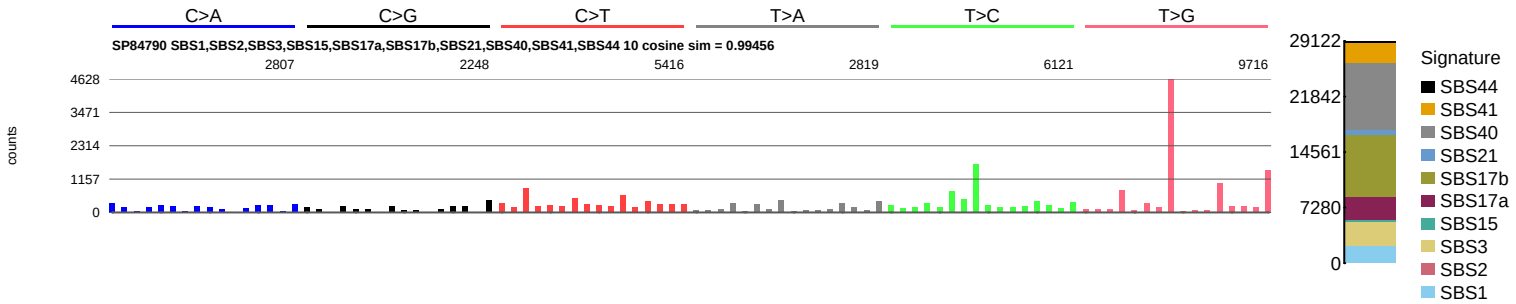

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.99457
2762
2354
5272
2784
6236
9711
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

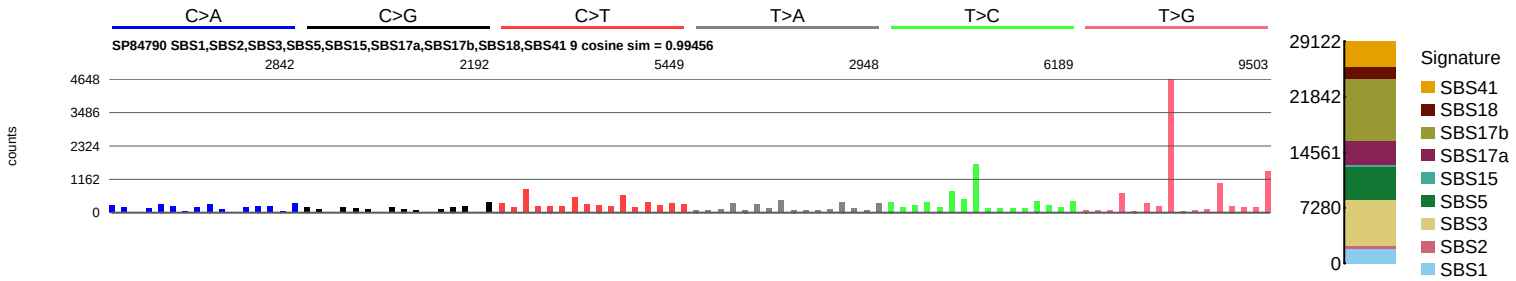

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.99457
2974
2196
5258
2840
6210
9643
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

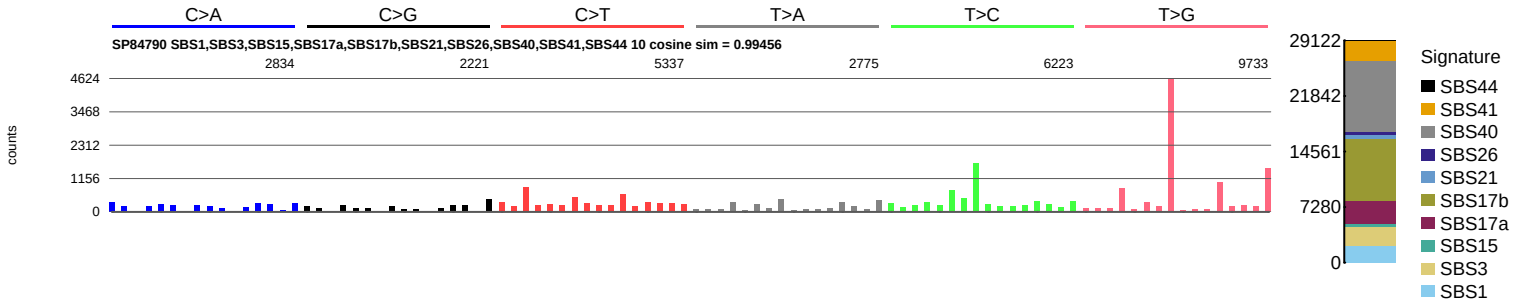

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 10 cosine sim = 0.99457
2843
2186
5410
2938
6249
9498
counts
4648
3486
2324
1162
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

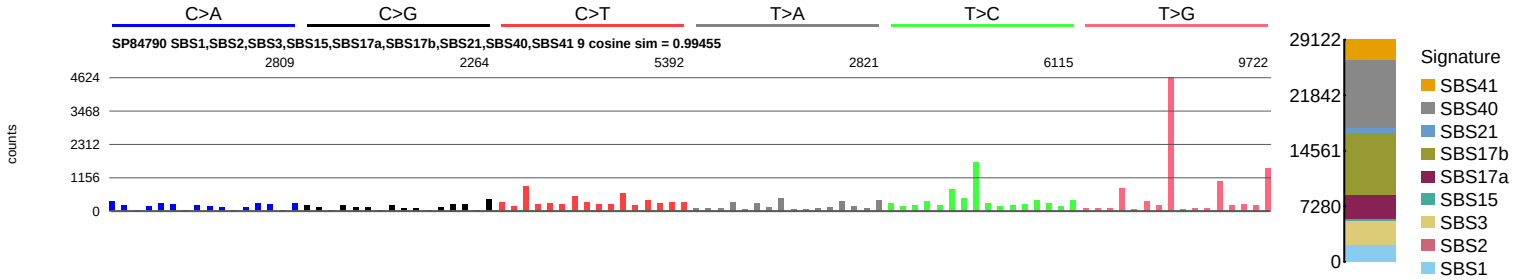

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.99457
2998
2219
5274
2811
6103
9715
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

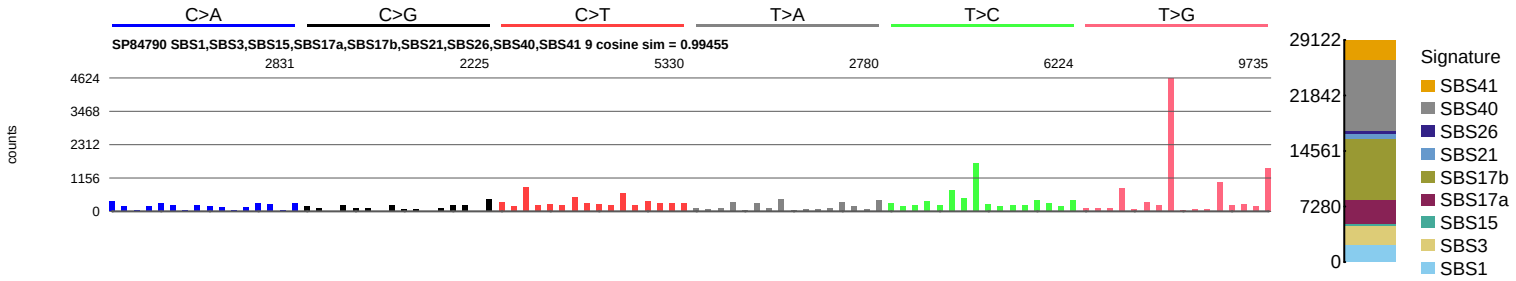

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.99456
2832
2305
5305
2748
6225
9708
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS1

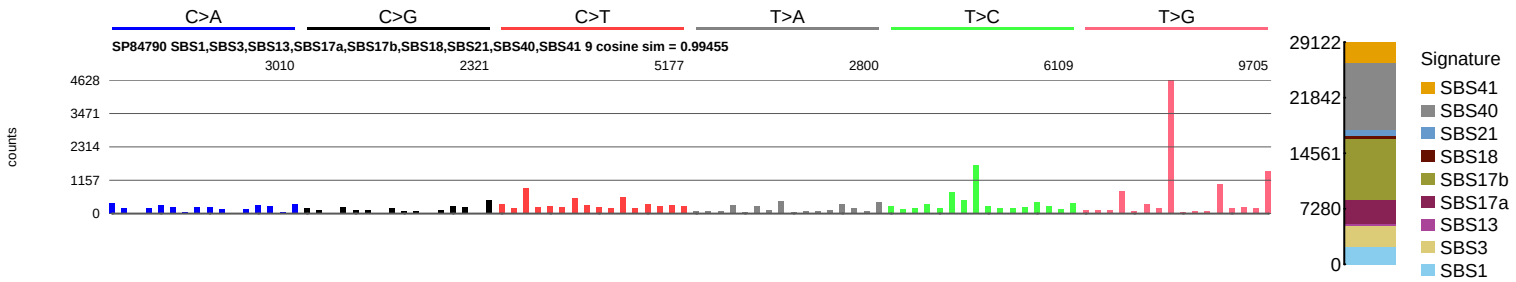

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.99456
2807
2248
5416
2819
6121
9716
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

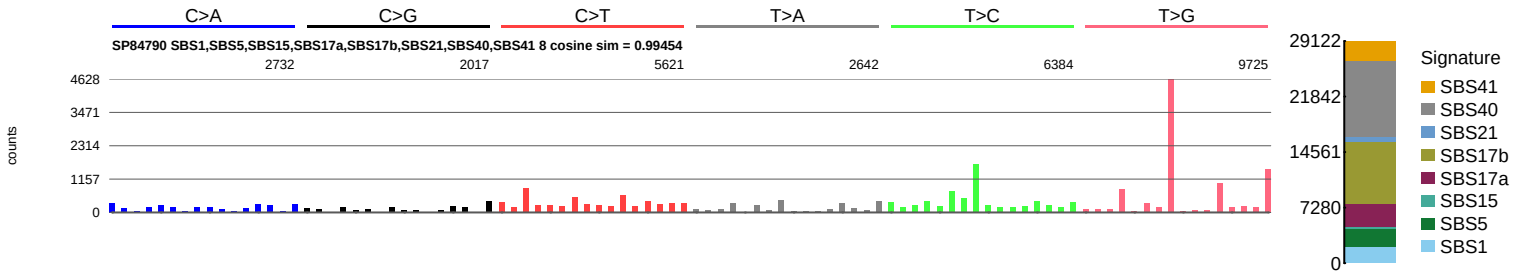

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.99456
2842
2192
5449
2948
6189
9503
counts
4648
3486
2324
1162
0
29122
21842
14561
7280
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

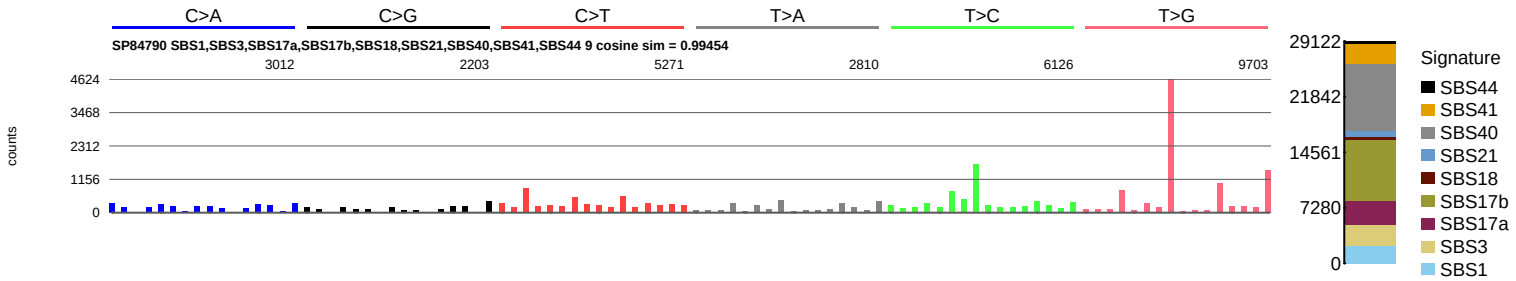

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.99456
2834
2221
5337
2775
6223
9733
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS1

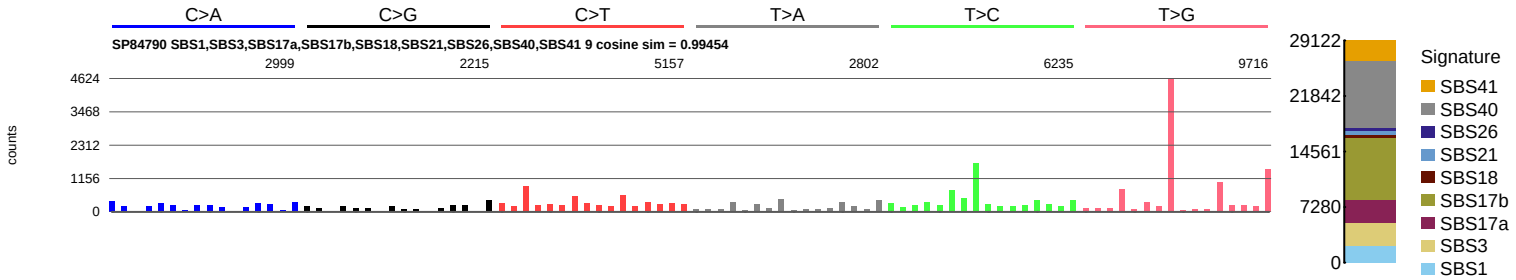

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 9 cosine sim = 0.99455
2809
2264
5392
2821
6115
9722
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

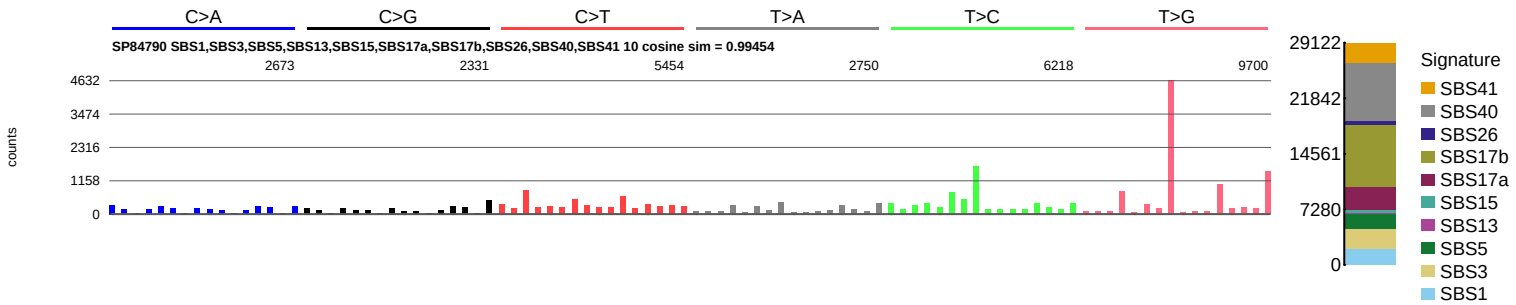

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.99455
2831
2225
5330
2780
6224
9735
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS1

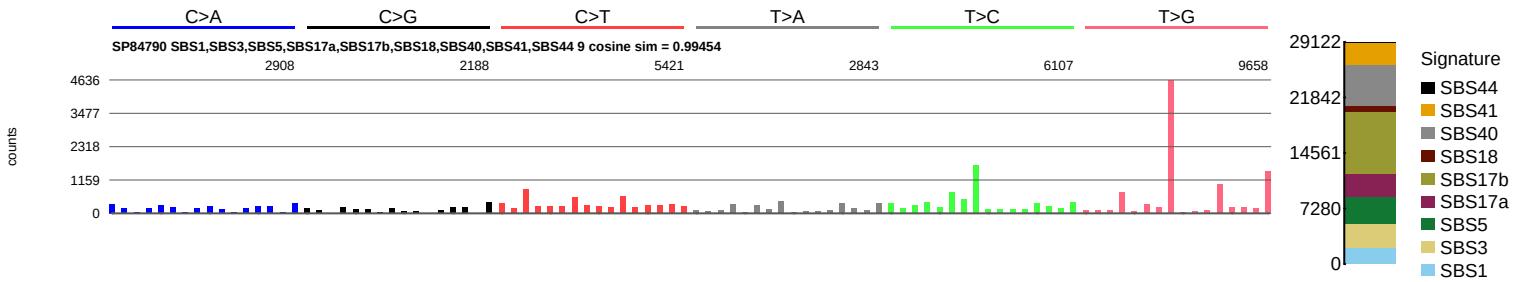

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.99455
3010
2321
5177
2800
6109
9705
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1

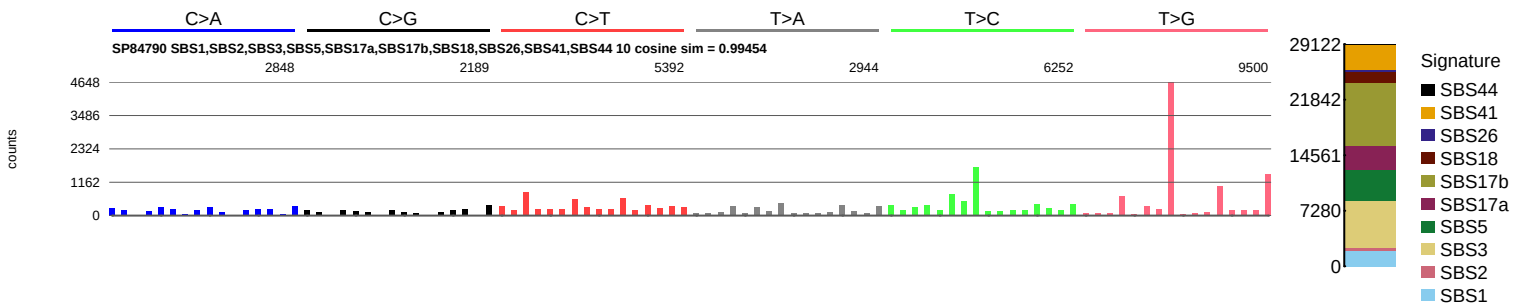

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 8 cosine sim = 0.99454
2732
2017
5621
2642
6384
9725
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS1

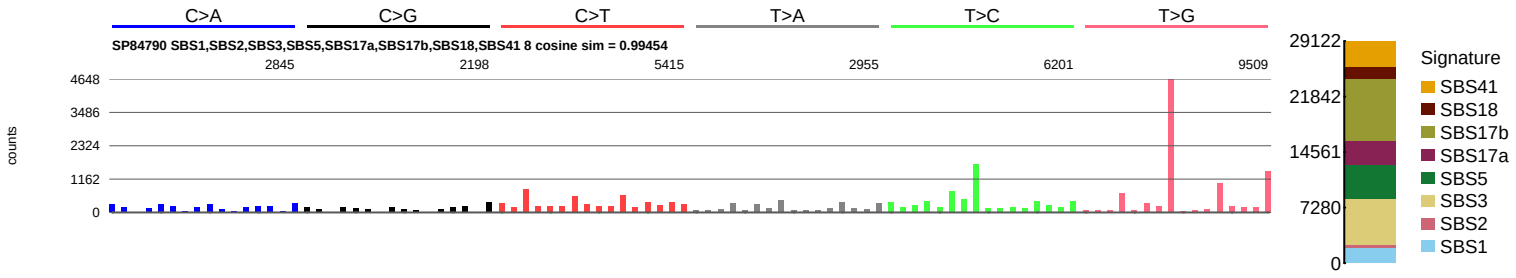

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.99454
3012
2203
5271
2810
6126
9703
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1

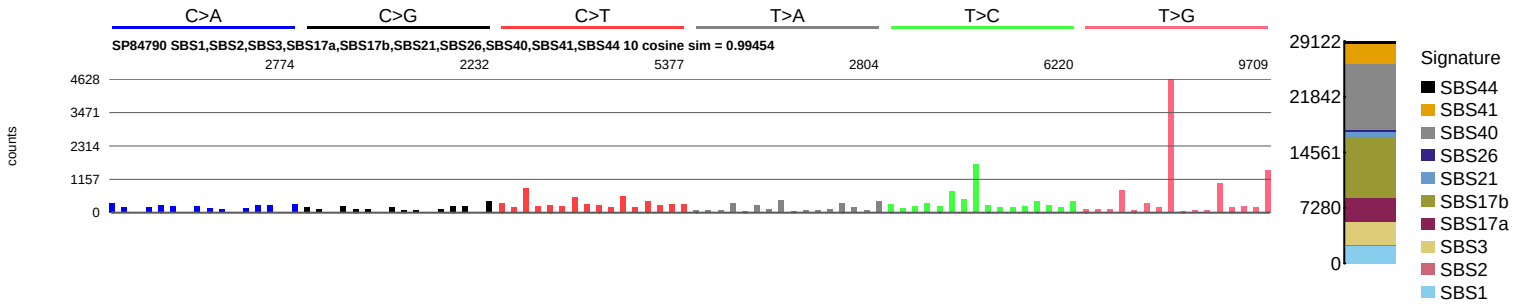

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.99454
2999
2215
5157
2802
6235
9716
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1

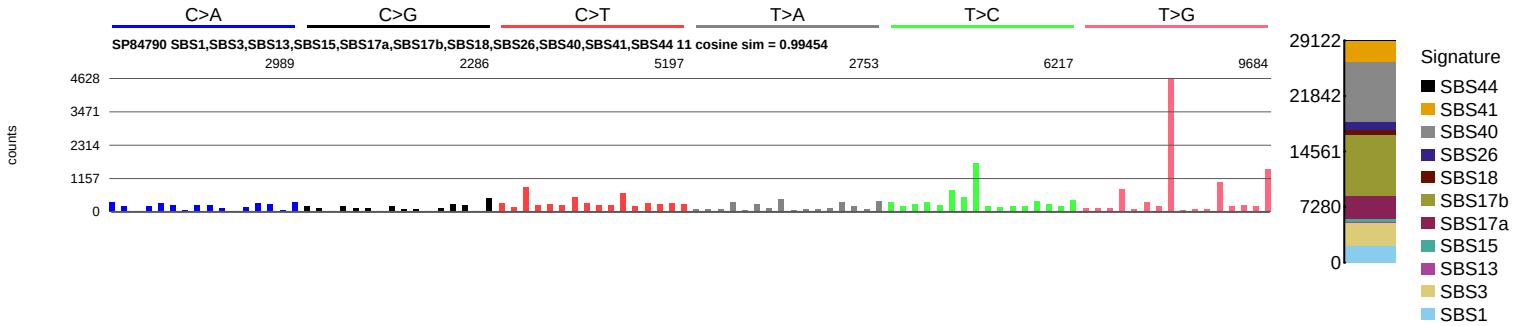

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 10 cosine sim = 0.99454
2673
2331
5454
2750
6218
9700
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

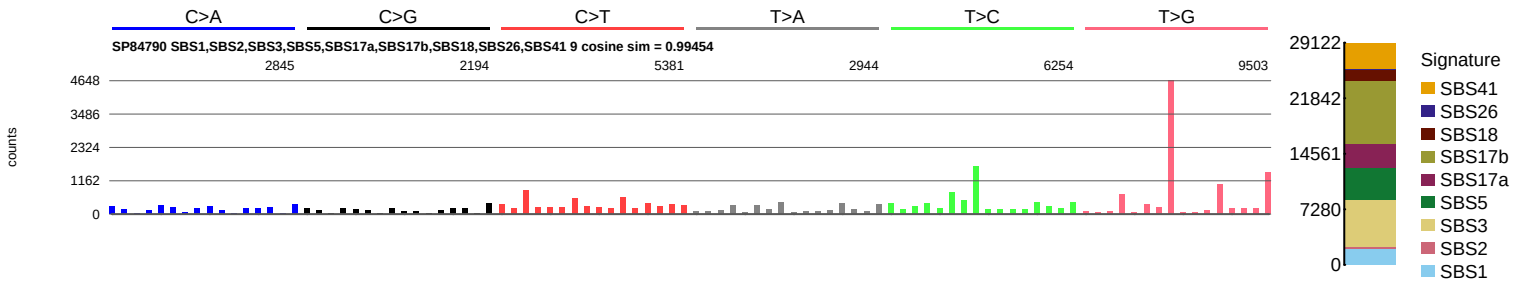

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.99454
2908
2188
5421
2843
6107
9658
counts
4636
3477
2318
1159
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1

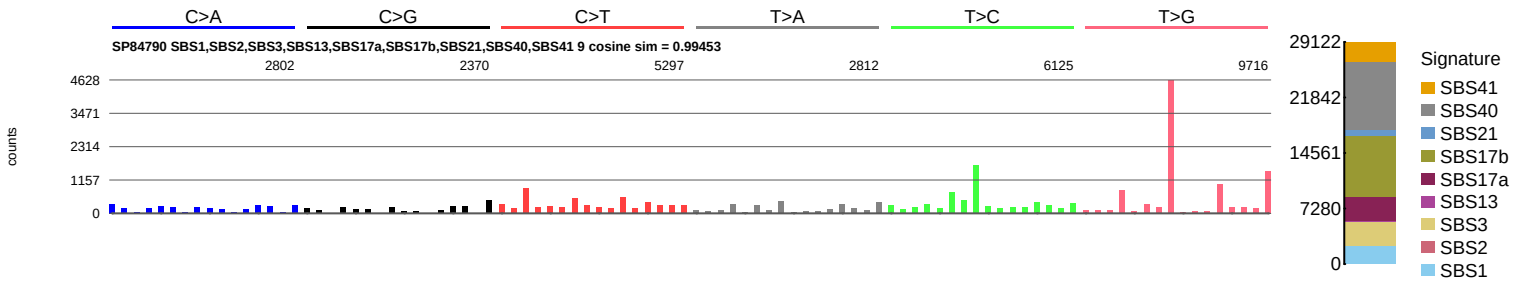

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.99454
2848
2189
5392
2944
6252
9500
counts
4648
3486
2324
1162
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

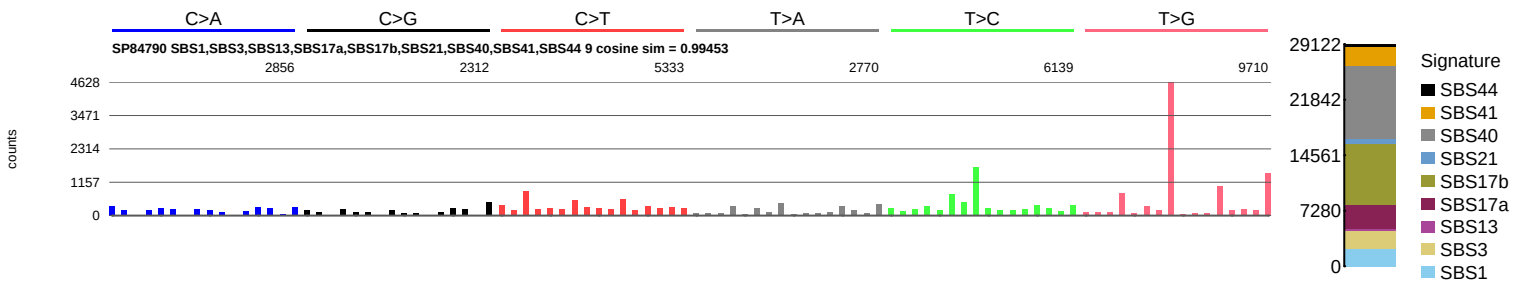

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.99454
2845
2198
5415
2955
6201
9509
counts
4648
3486
2324
1162
0
29122
21842
14561
7280
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

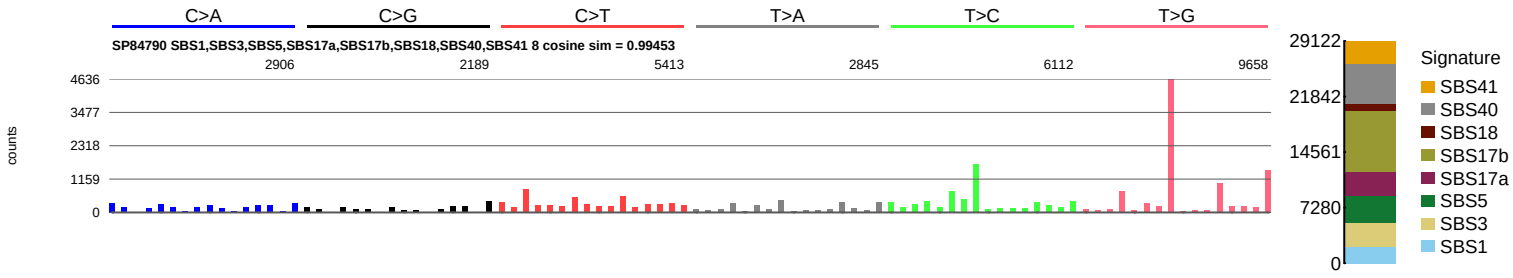

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.99454
2774
2232
5377
2804
6220
9709
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1

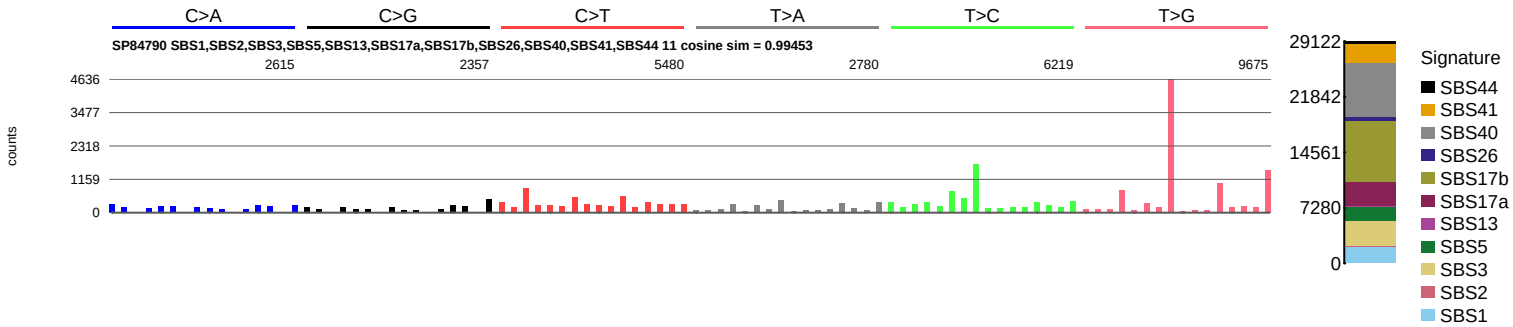

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.99454
2989
2286
5197
2753
6217
9684
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

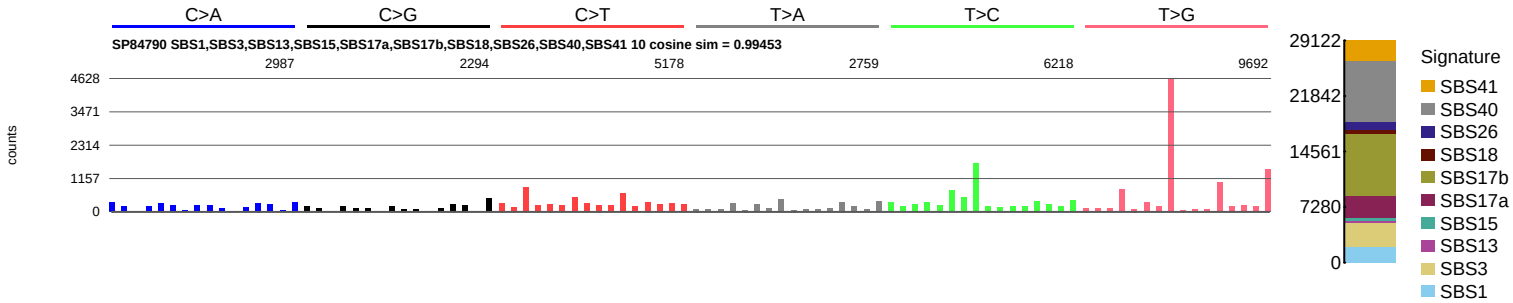

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.99454
2845
2194
5381
2944
6254
9503
counts
4648
3486
2324
1162
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

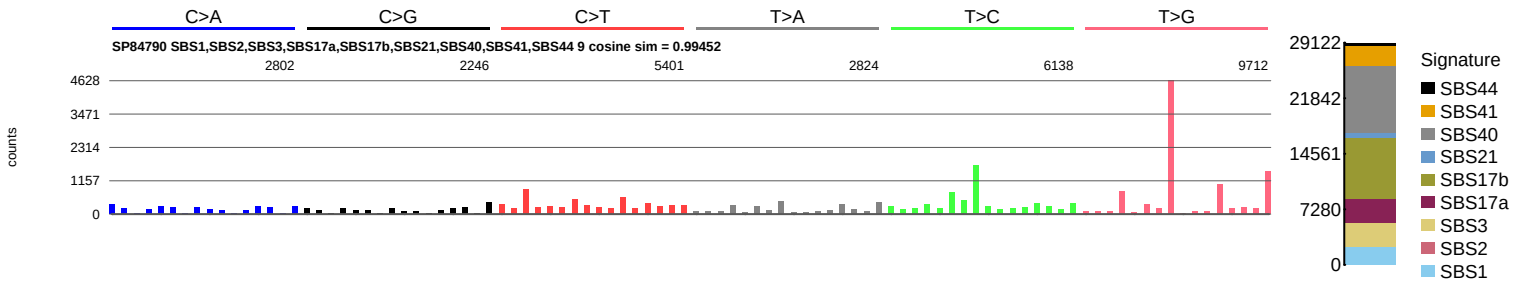

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS41 9 cosine sim = 0.99453
2802
2370
5297
2812
6125
9716
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

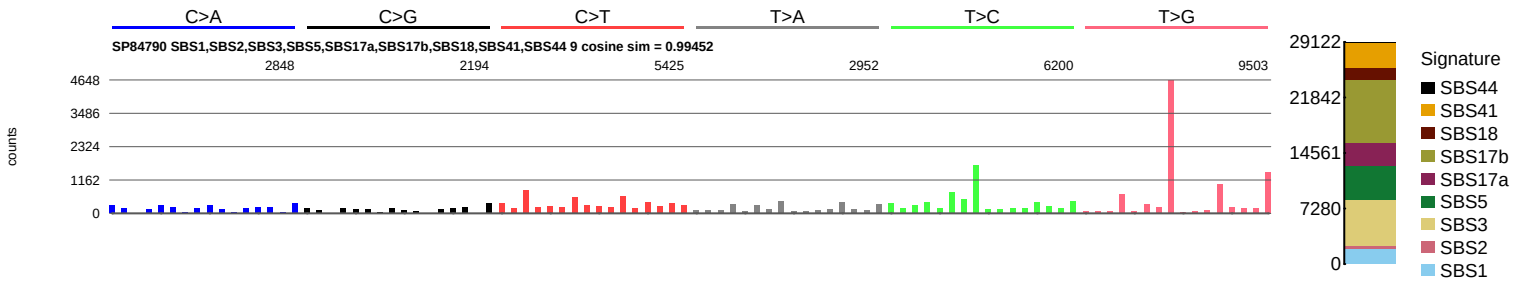

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.99453
2856
2312
5333
2770
6139
9710
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS1

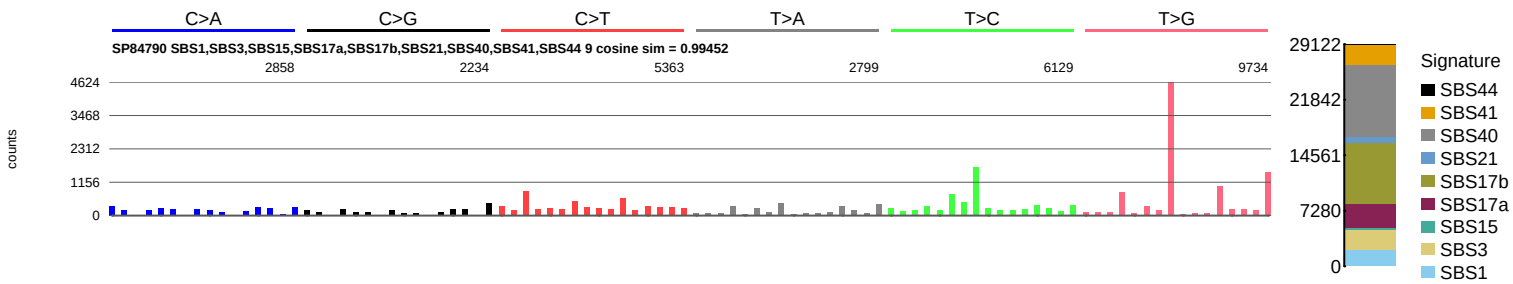

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.99453
2906
2189
5413
2845
6112
9658
counts
4636
3477
2318
1159
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1

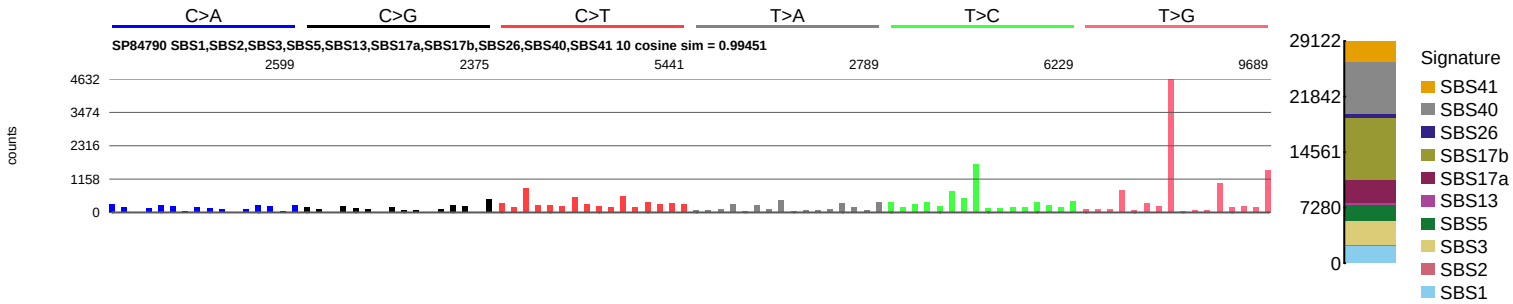

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.99453
2615
2357
5480
2780
6219
9675
counts
4636
3477
2318
1159
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

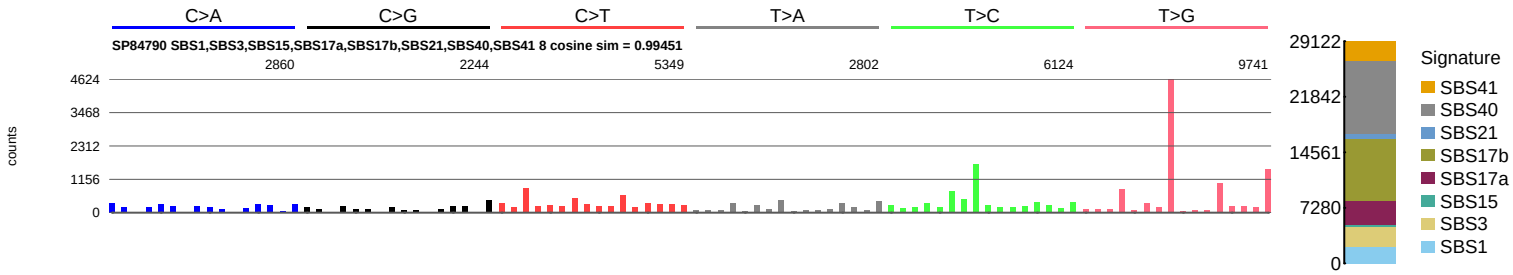

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.99453
2987
2294
5178
2759
6218
9692
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

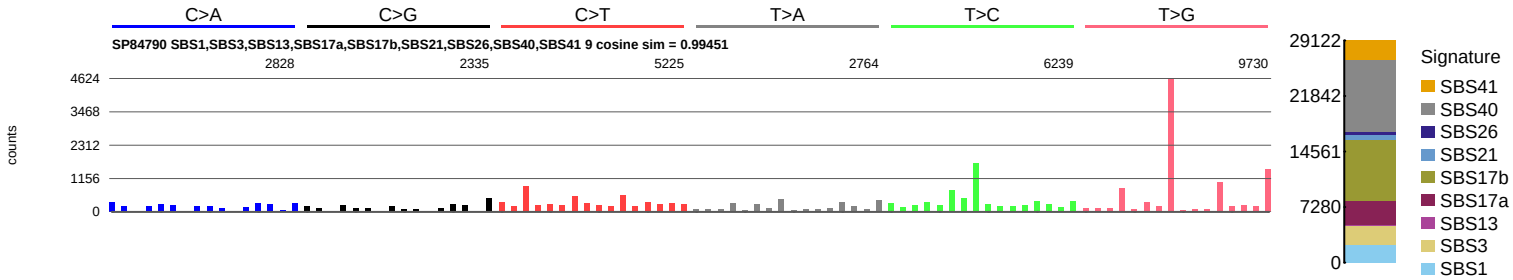

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.99452
2802
2246
5401
2824
6138
9712
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1

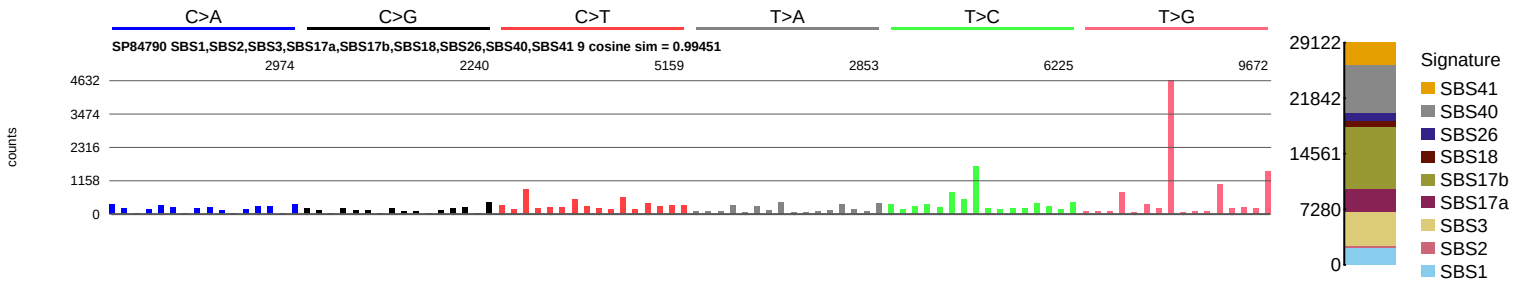

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.99452
2848
2194
5425
2952
6200
9503
counts
4648
3486
2324
1162
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

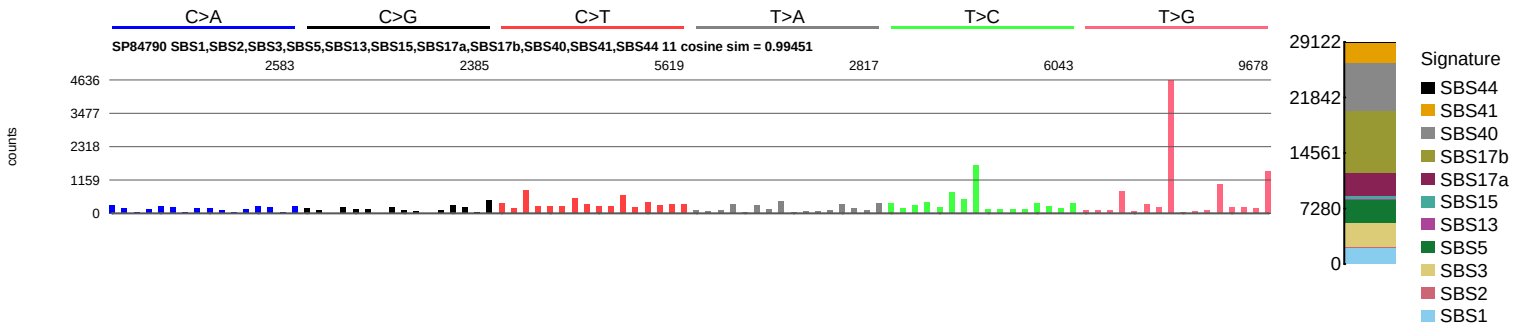

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.99452
2858
2234
5363
2799
6129
9734
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS1

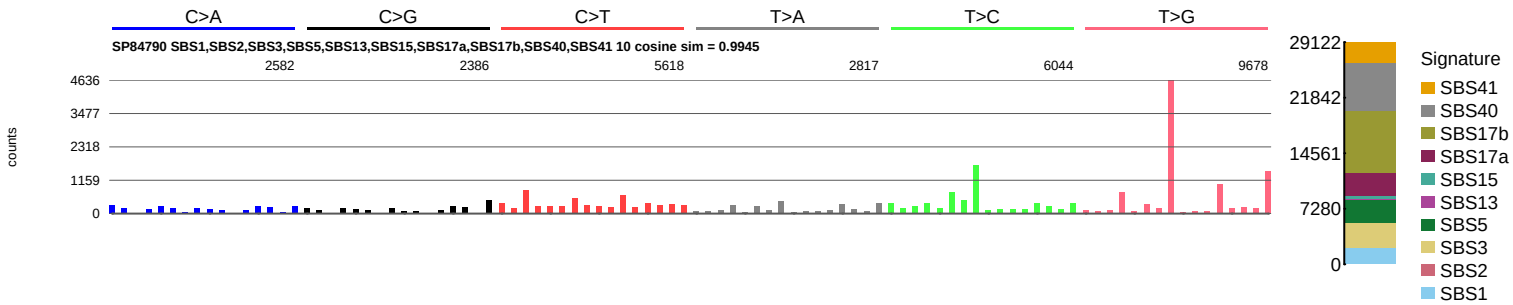

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS41 10 cosine sim = 0.99451
2599
2375
5441
2789
6229
9689
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

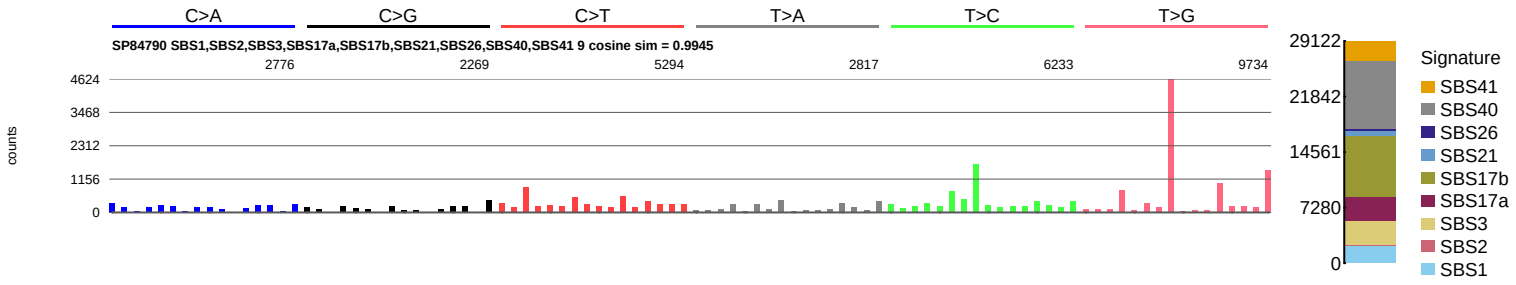

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 8 cosine sim = 0.99451
2860
2244
5349
2802
6124
9741
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS1

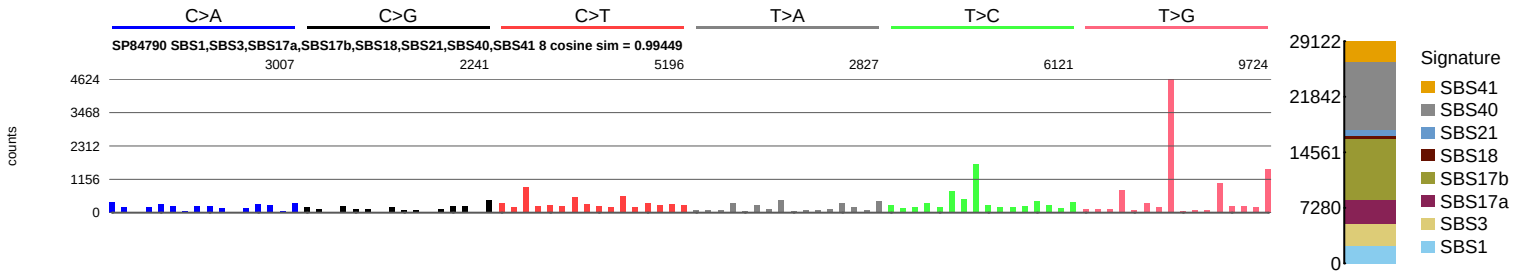

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.99451
2828
2335
5225
2764
6239
9730
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS1

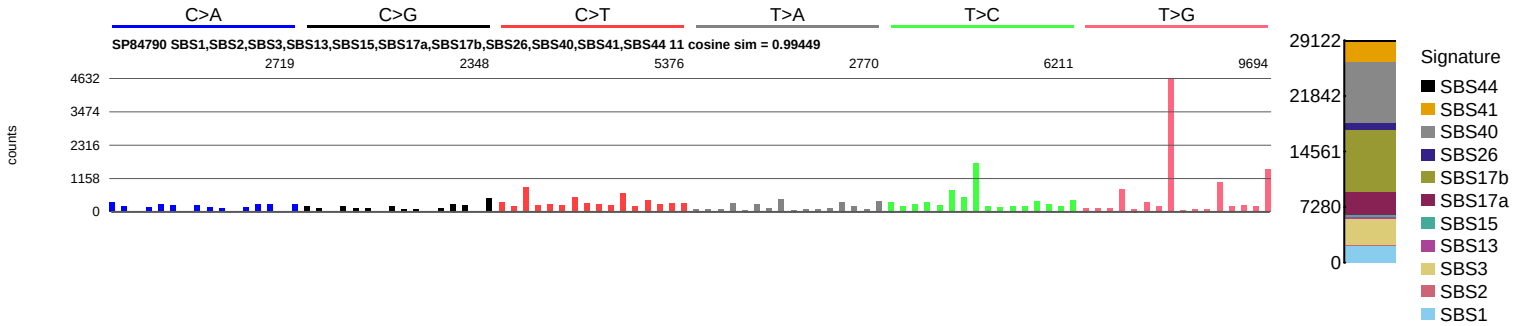

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.99451
2974
2240
5159
2853
6225
9672
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

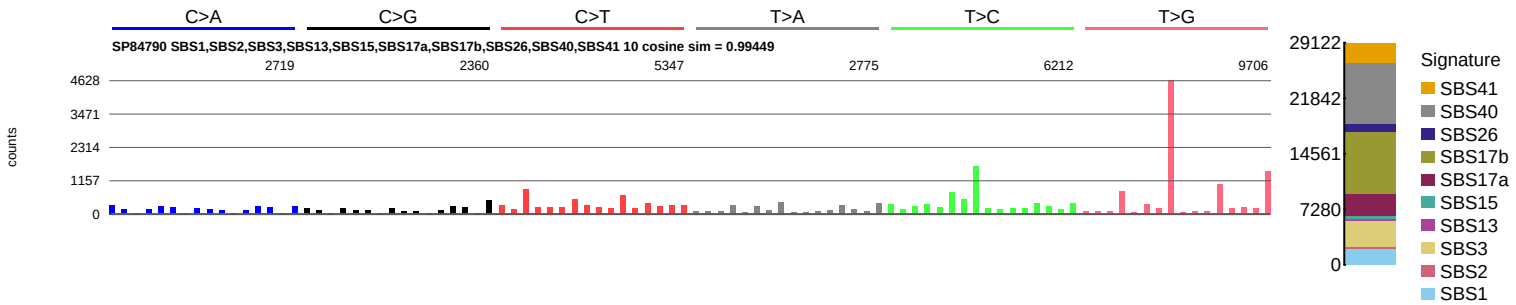

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41,SBS44 11 cosine sim = 0.99451
2583
2385
5619
2817
6043
9678
counts
4636
3477
2318
1159
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

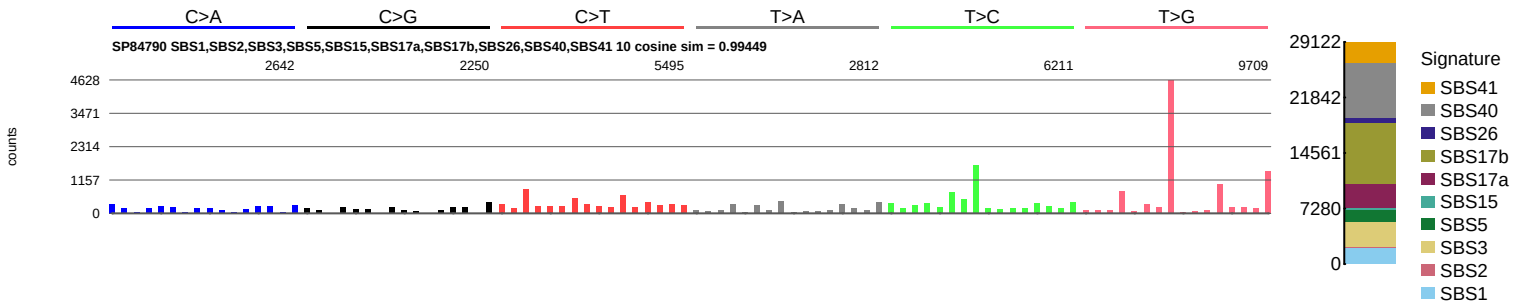

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41 10 cosine sim = 0.9945
2582
2386
5618
2817
6044
9678
counts
4636
3477
2318
1159
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

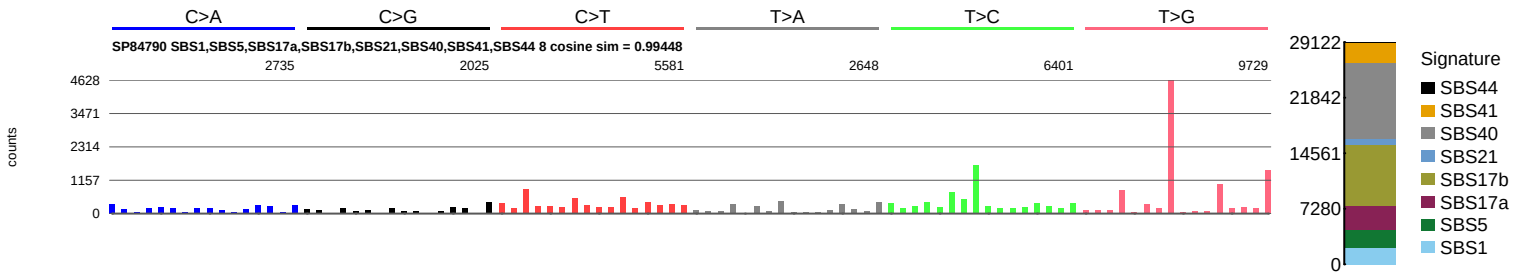

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.9945
2776
2269
5294
2817
6233
9734
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1

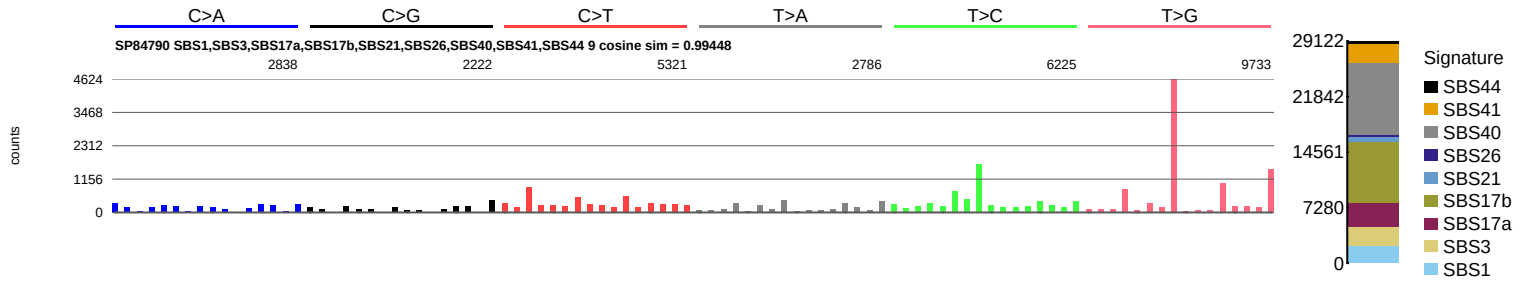

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.99449
3007
2241
5196
2827
6121
9724
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1

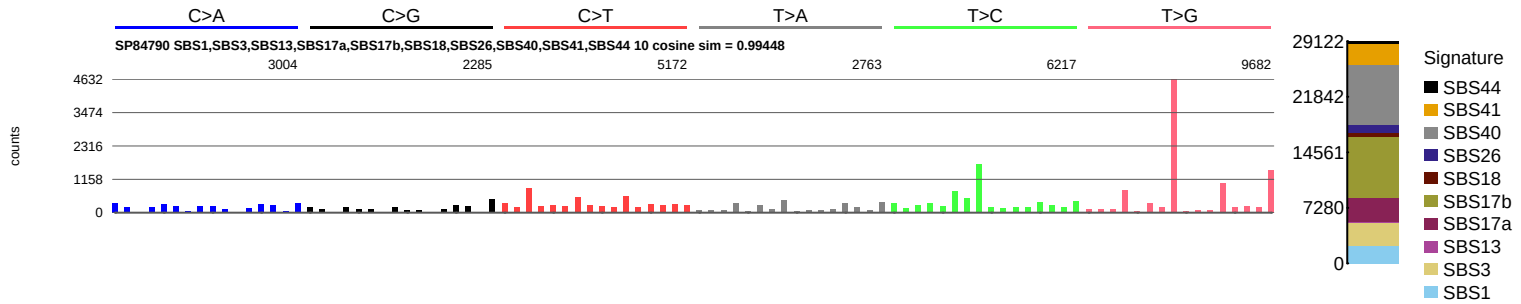

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.99449
2719
2348
5376
2770
6211
9694
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

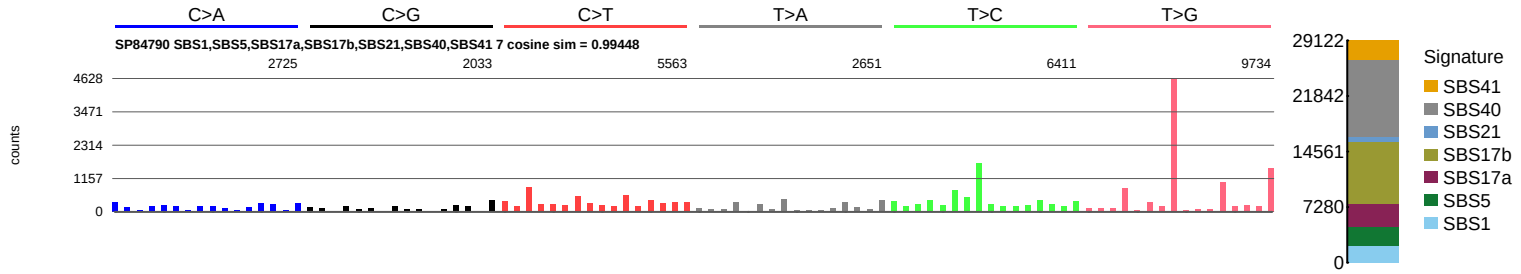

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 10 cosine sim = 0.99449
2719
2360
5347
2775
6212
9706
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

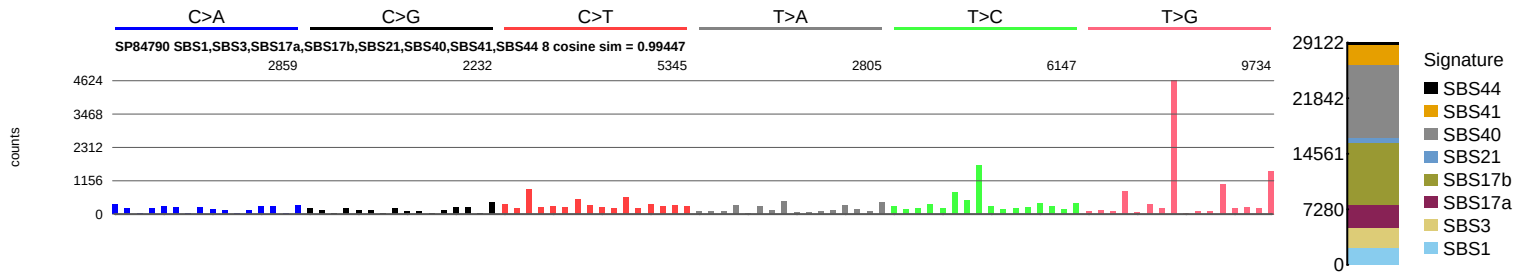

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 10 cosine sim = 0.99449
2642
2250
5495
2812
6211
9709
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

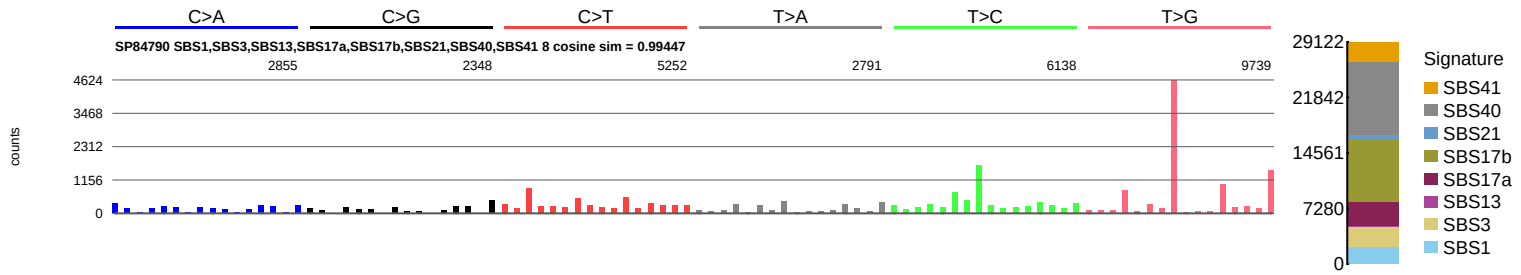

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.99448
2735
2025
5581
2648
6401
9729
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS1

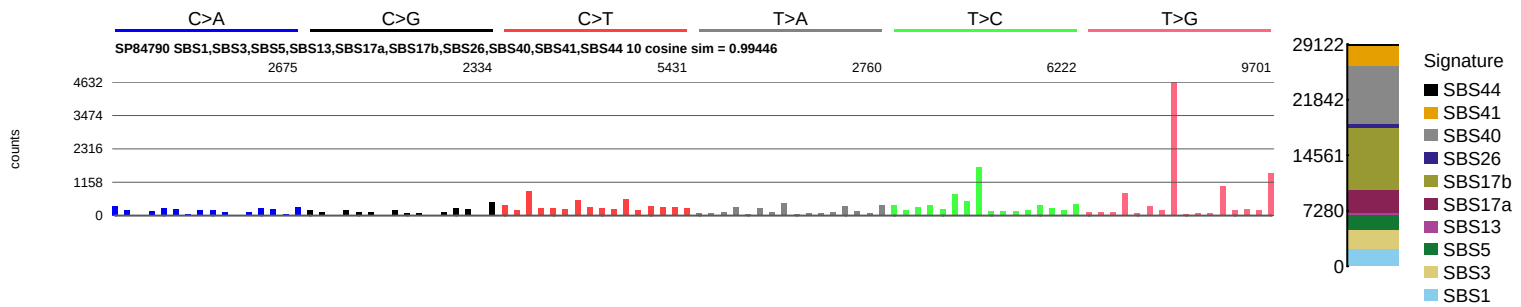

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.99448
2838
2222
5321
2786
6225
9733
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS3
SBS1

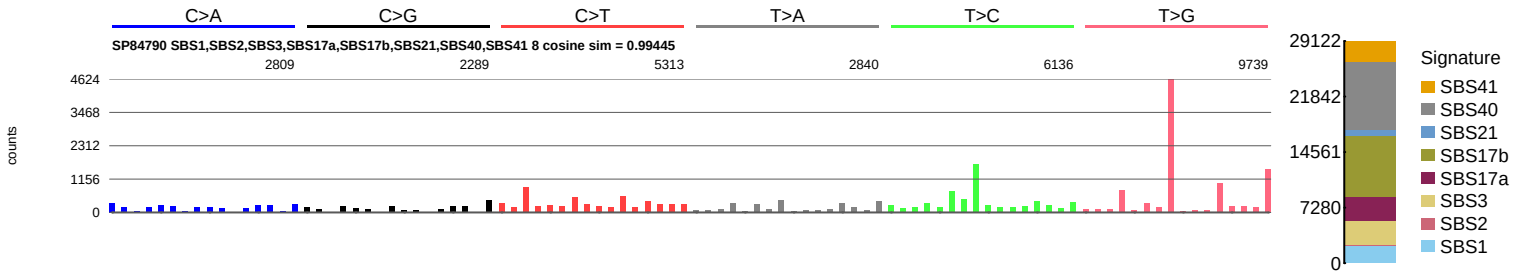

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.99448
3004
2285
5172
2763
6217
9682
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1

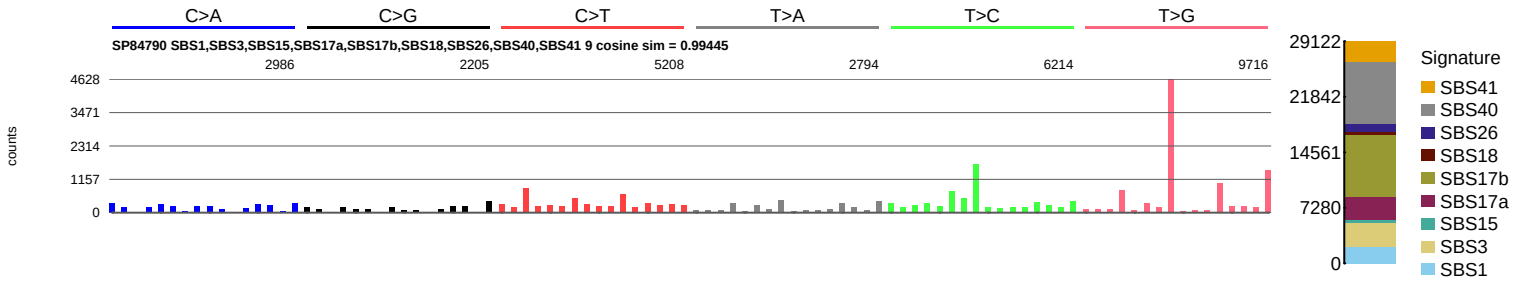

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS41 7 cosine sim = 0.99448
2725
2033
5563
2651
6411
9734
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS1

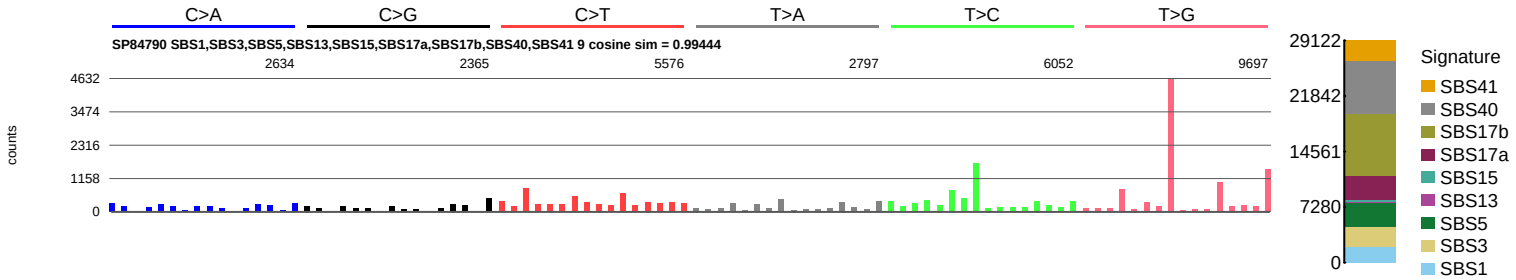

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.99447
2859
2232
5345
2805
6147
9734
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS3
SBS1

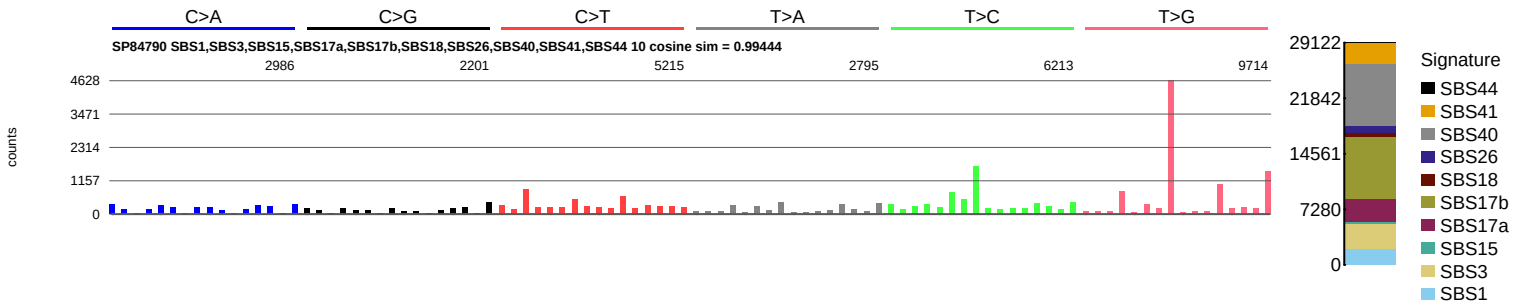

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS41 8 cosine sim = 0.99447
2855
2348
5252
2791
6138
9739
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS1

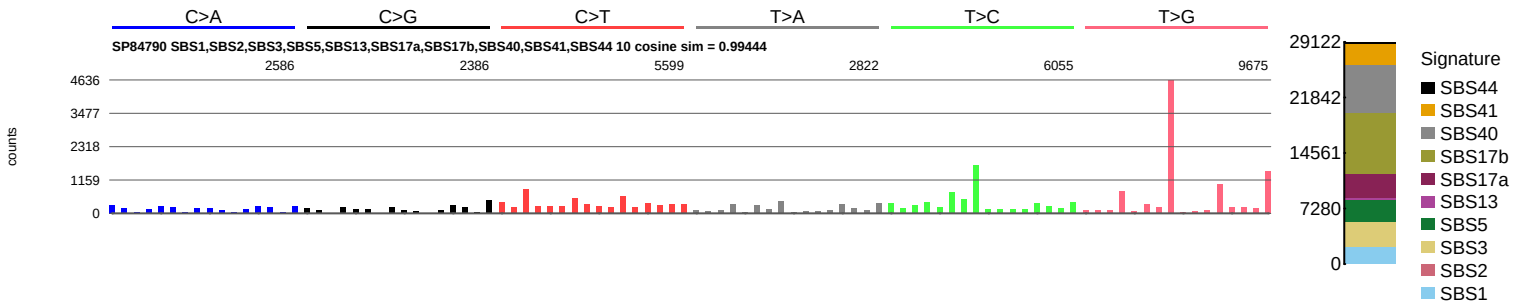

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.99446
2675
2334
5431
2760
6222
9701
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

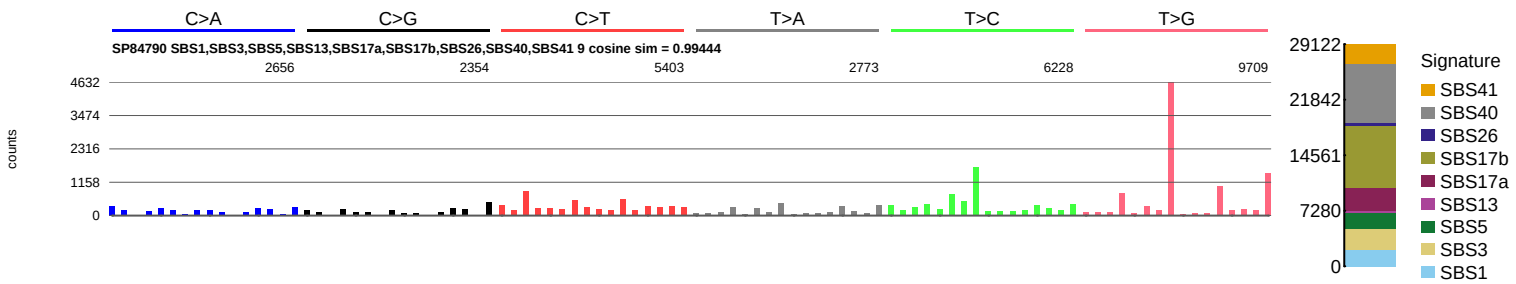

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS40,SBS41 8 cosine sim = 0.99445
2809
2289
5313
2840
6136
9739
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1

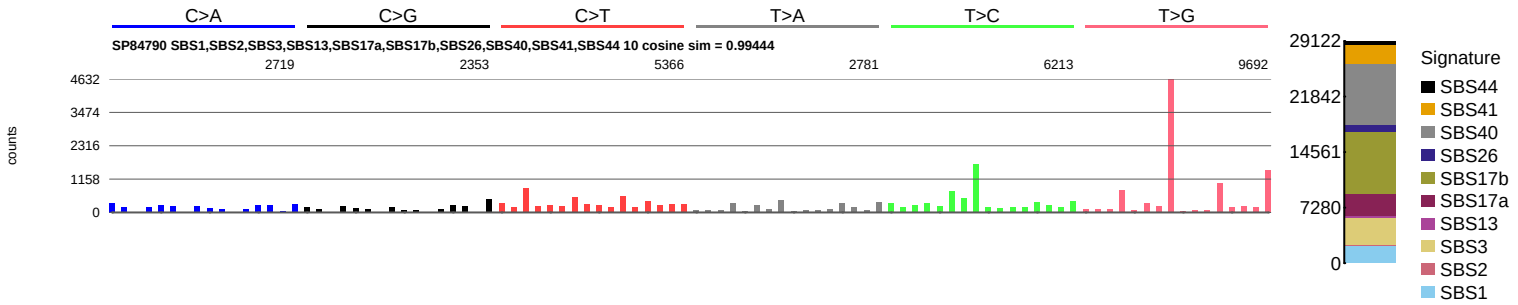

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.99445
2986
2205
5208
2794
6214
9716
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

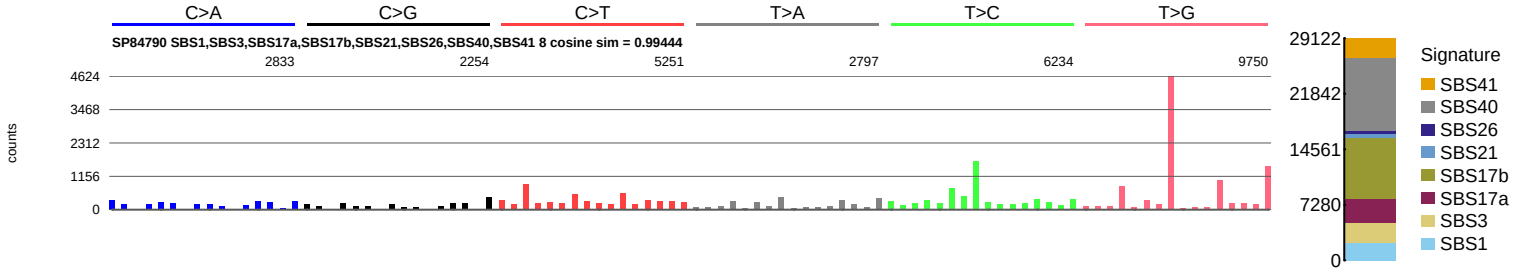

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41 9 cosine sim = 0.99444
2634
2365
5576
2797
6052
9697
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

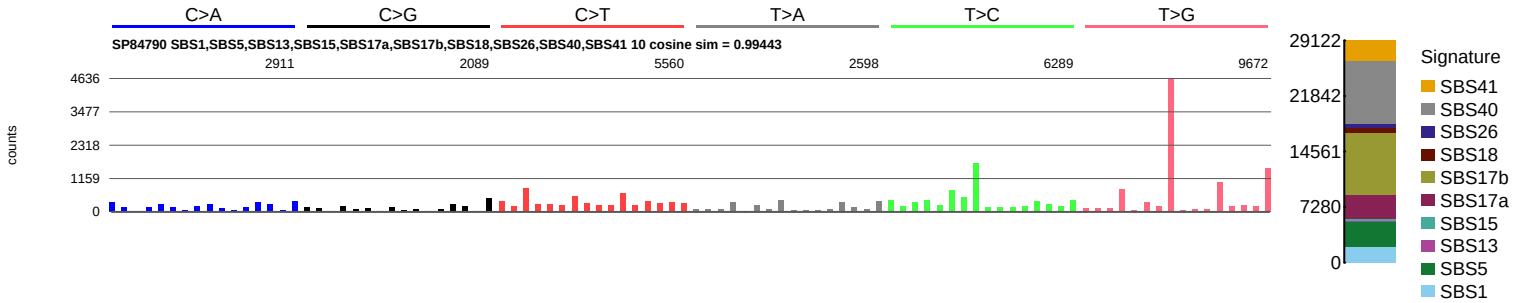

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.99444
2986
2201
5215
2795
6213
9714
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

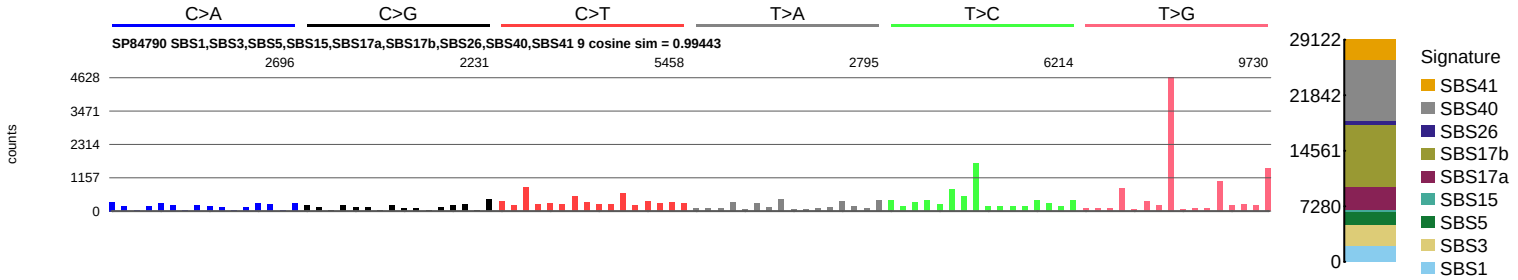

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS40,SBS41,SBS44 10 cosine sim = 0.99444
2586
2386
5599
2822
6055
9675
counts
4636
3477
2318
1159
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

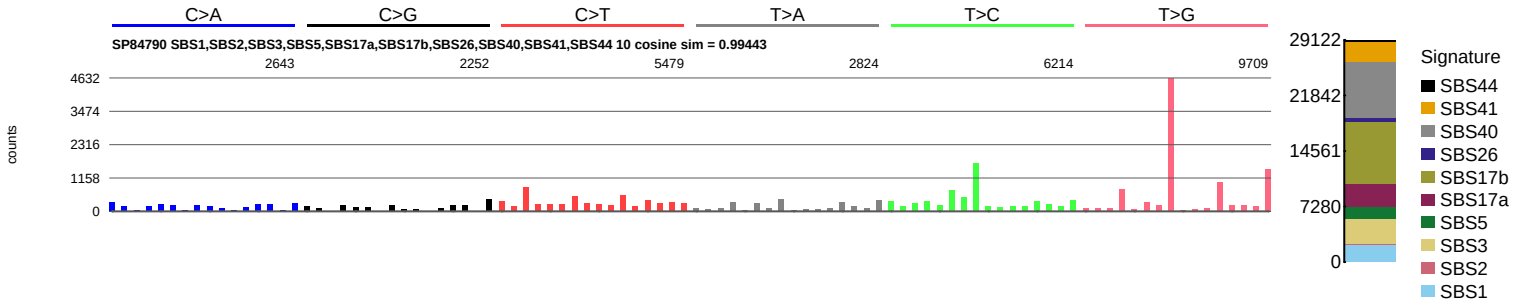

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS41 9 cosine sim = 0.99444
2656
2354
5403
2773
6228
9709
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

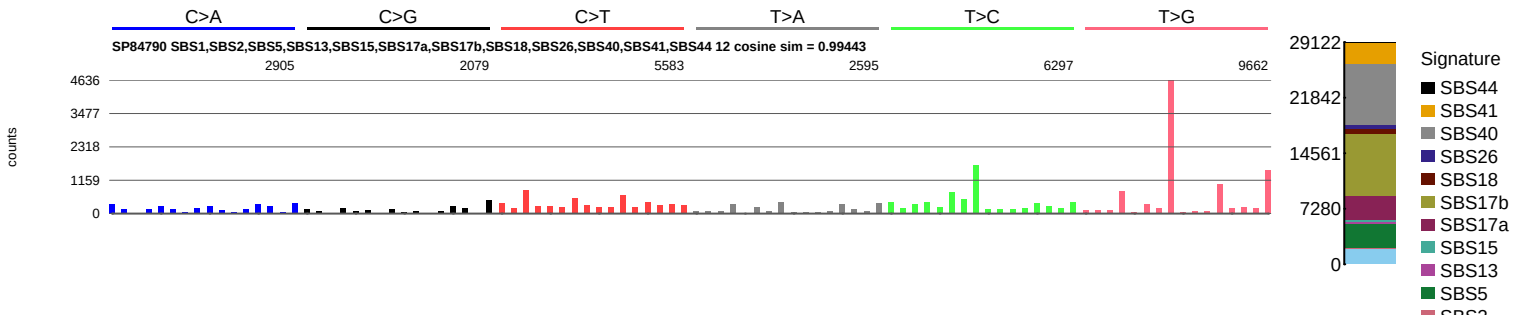

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.99444
2719
2353
5366
2781
6213
9692
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

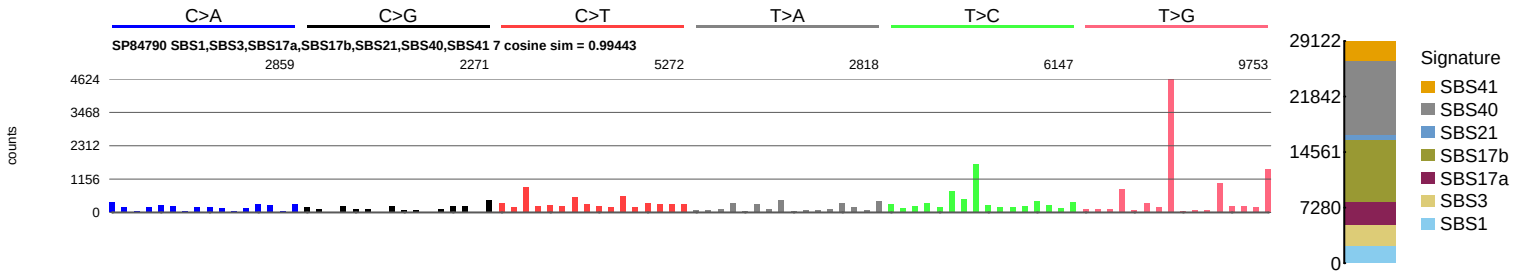

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41 8 cosine sim = 0.99444
2833
2254
5251
2797
6234
9750
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS3
SBS1

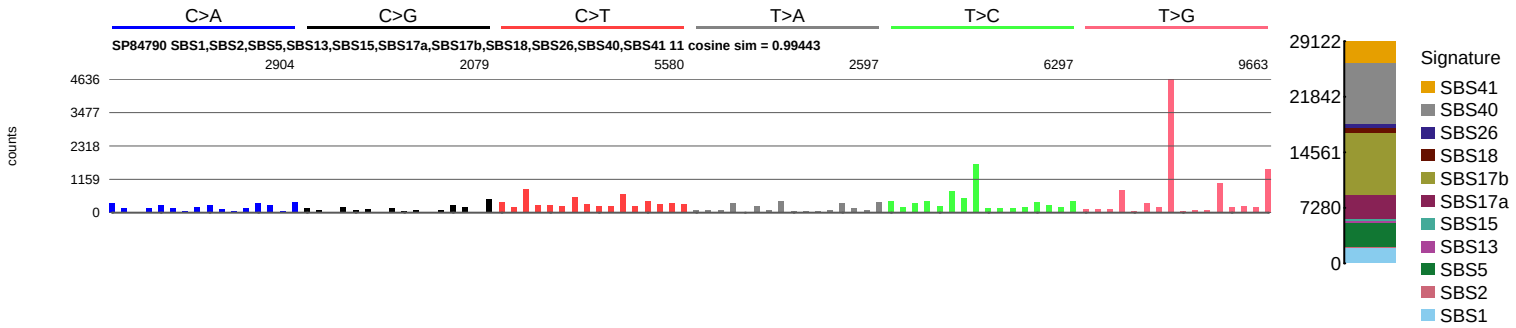

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.99443
2911
2089
5560
2598
6289
9672
counts
4636
3477
2318
1159
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

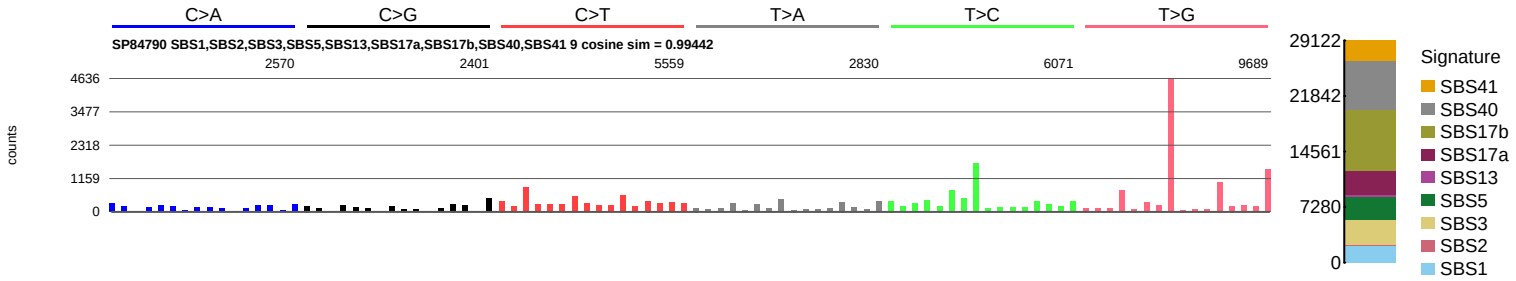

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 9 cosine sim = 0.99443
2696
2231
5458
2795
6214
9730
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1

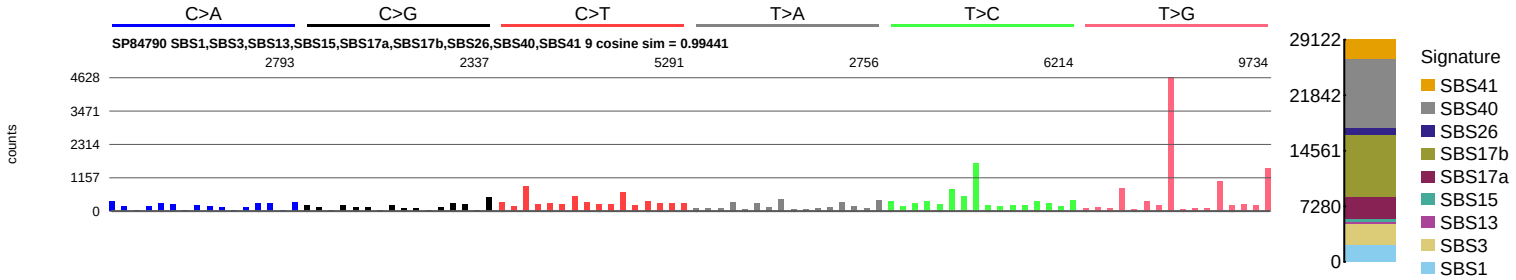

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.99443
2643
2252
5479
2824
6214
9709
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

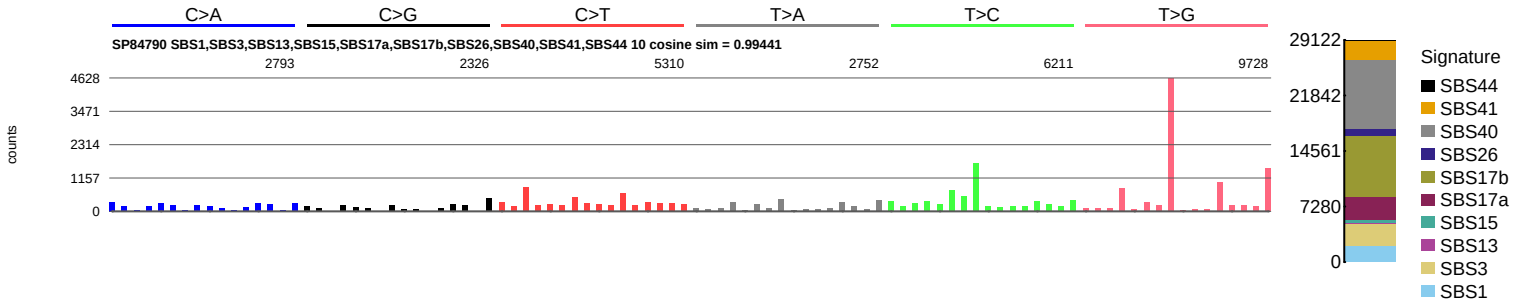

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.99443
2905
2079
5583
2595
6297
9662
counts
4636
3477
2318
1159
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5

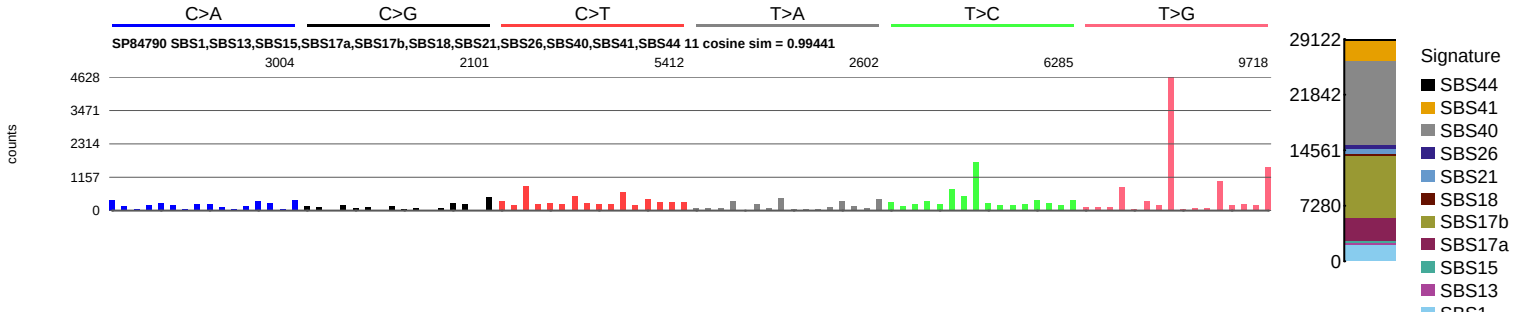

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS21,SBS40,SBS41 7 cosine sim = 0.99443
2859
2271
5272
2818
6147
9753
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS3
SBS1

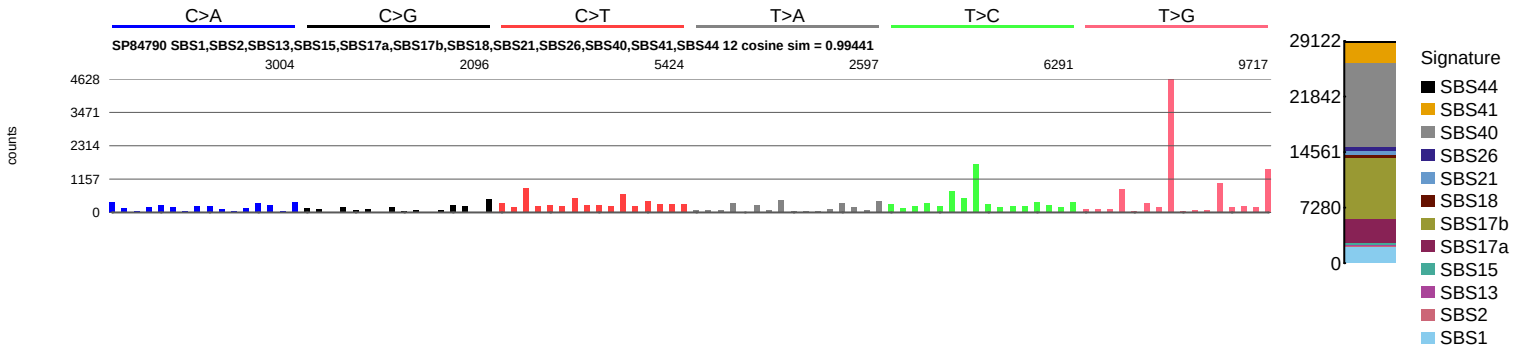

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 11 cosine sim = 0.99443
2904
2079
5580
2597
6297
9663
counts
4636
3477
2318
1159
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

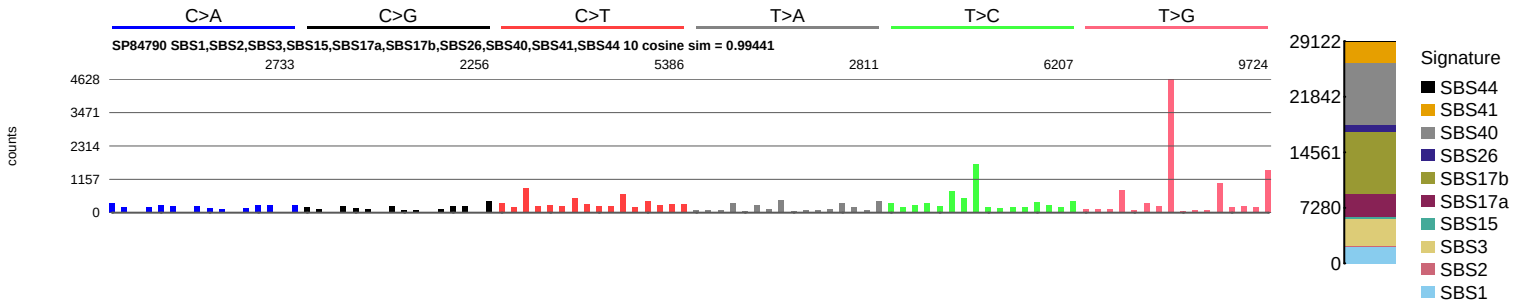

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS40,SBS41 9 cosine sim = 0.99442
2570
2401
5559
2830
6071
9689
counts
4636
3477
2318
1159
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

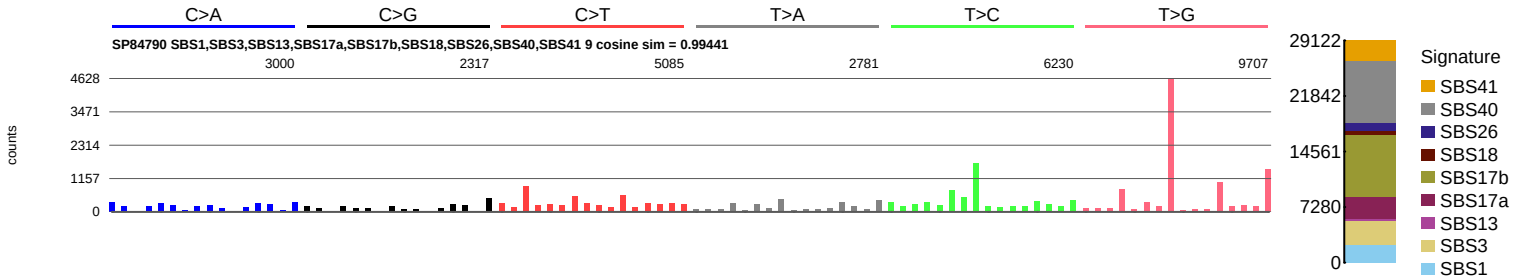

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 9 cosine sim = 0.99441
2793
2337
5291
2756
6214
9734
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

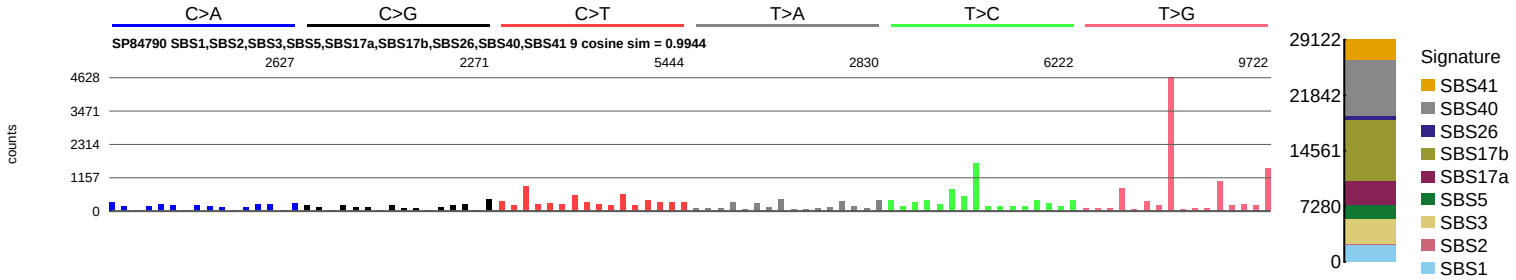

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.99441
2793
2326
5310
2752
6211
9728
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

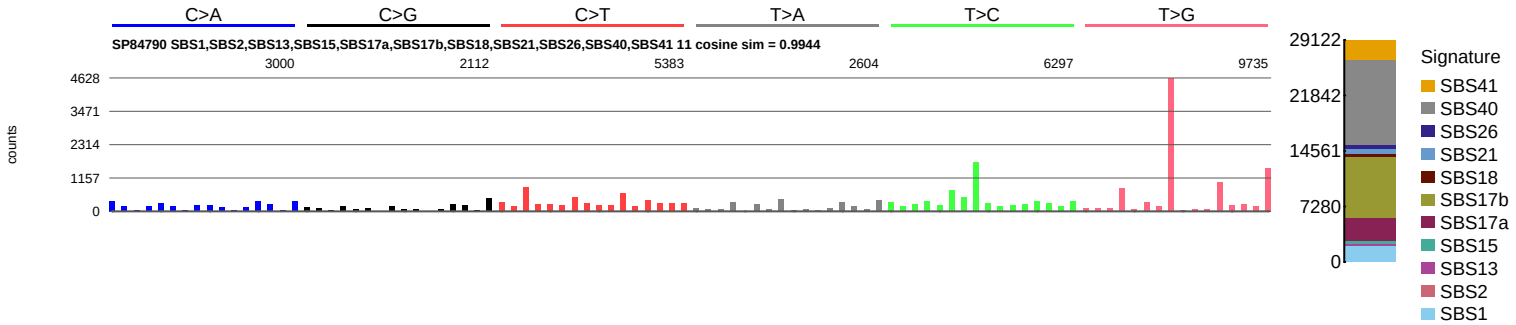

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.99441
3004
2101
5412
2602
6285
9718
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13

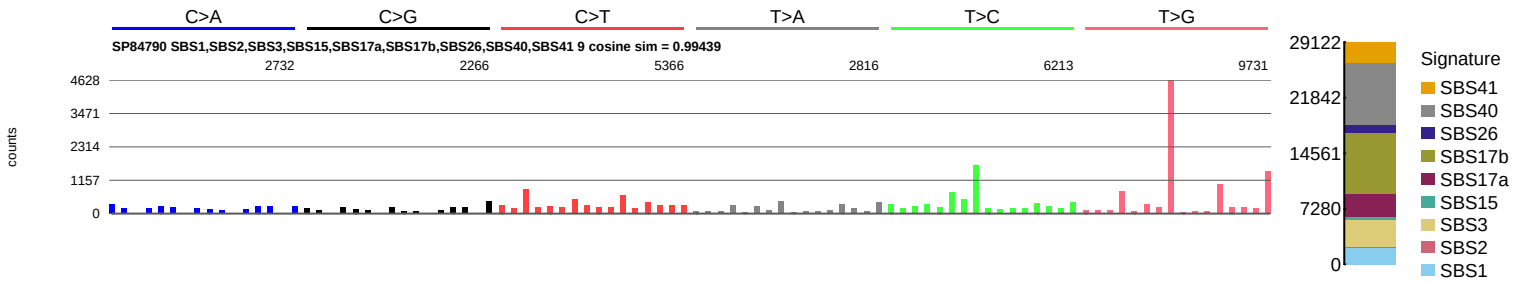

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.99441
3004
2096
5424
2597
6291
9717
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

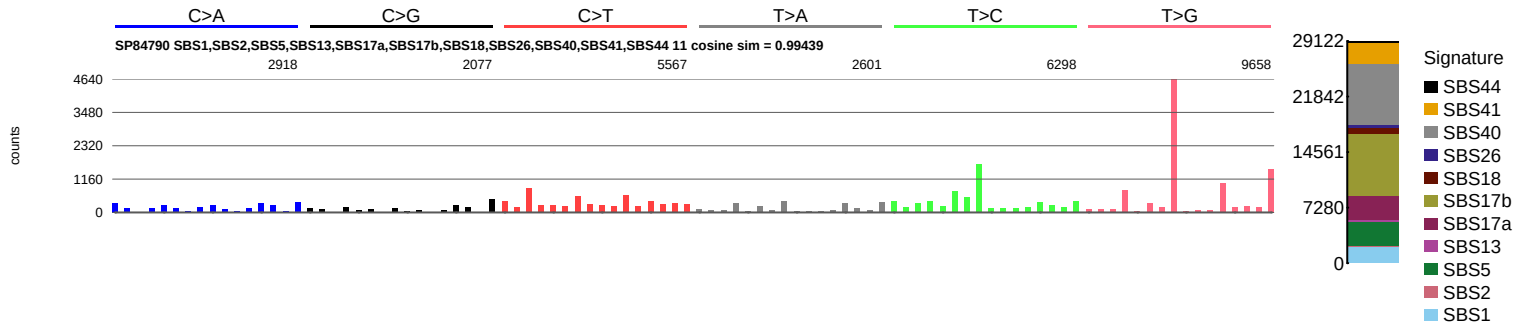

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.99441
2733
2256
5386
2811
6207
9724
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

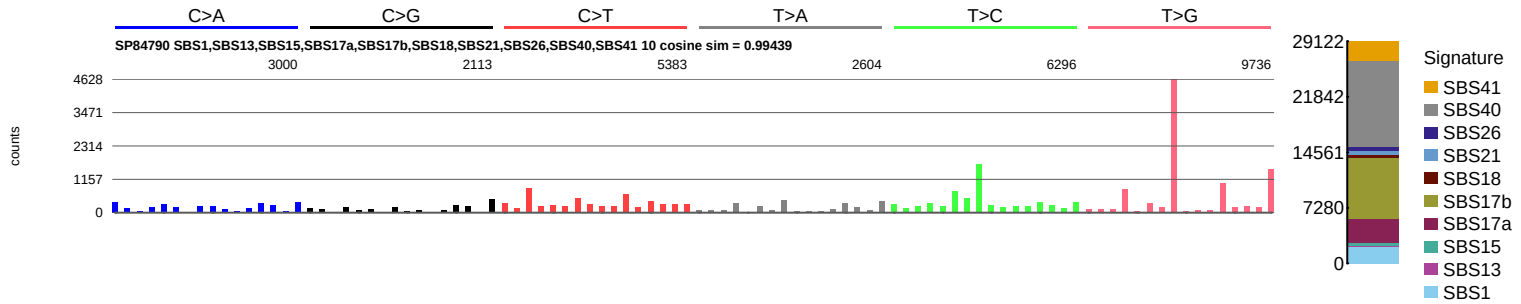

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.99441
3000
2317
5085
2781
6230
9707
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1

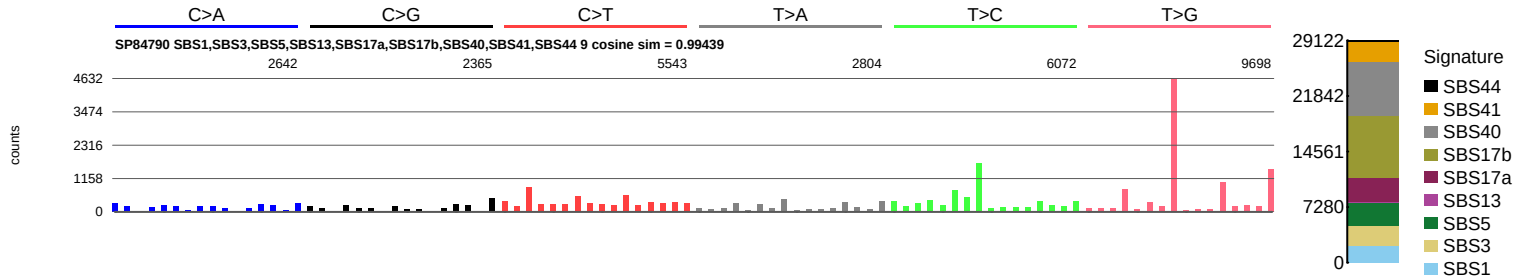

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS26,SBS40,SBS41 9 cosine sim = 0.9944
2627
2271
5444
2830
6222
9722
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

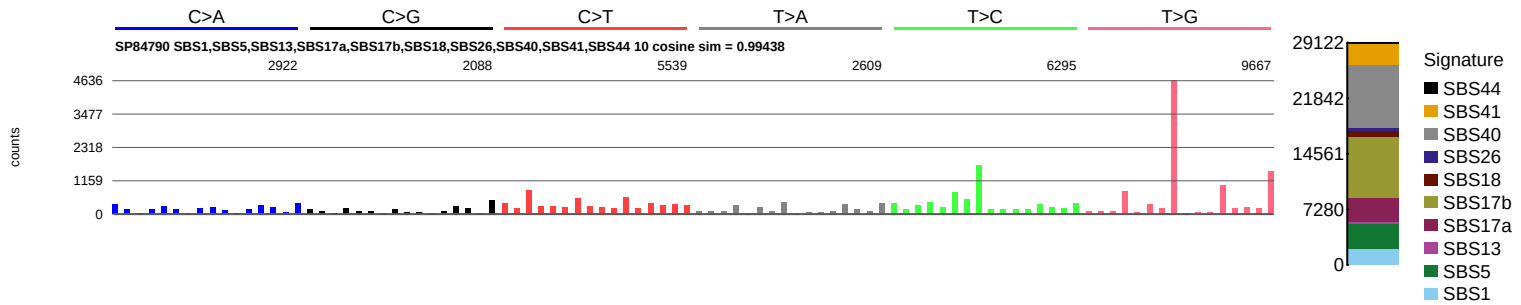

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 11 cosine sim = 0.9944
3000
2112
5383
2604
6297
9735
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

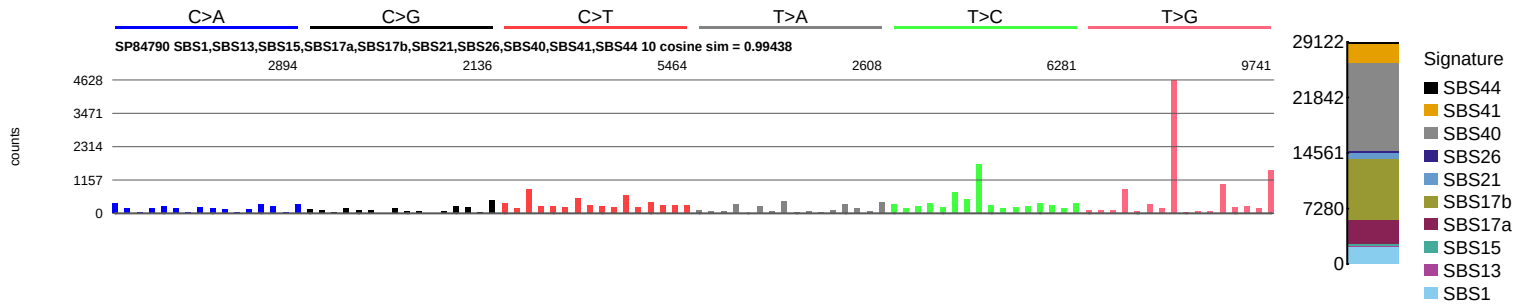

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 9 cosine sim = 0.99439
2732
2266
5366
2816
6213
9731
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

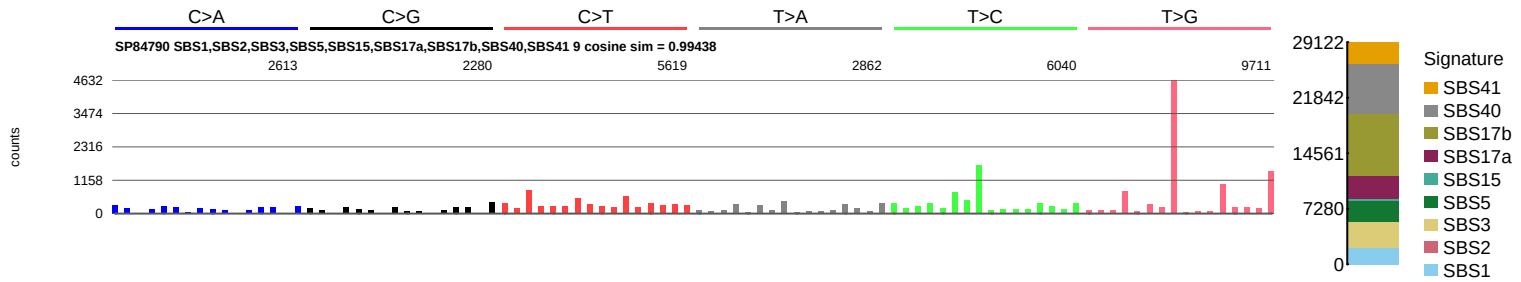

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.99439
2918
2077
5567
2601
6298
9658
counts
4640
3480
2320
1160
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

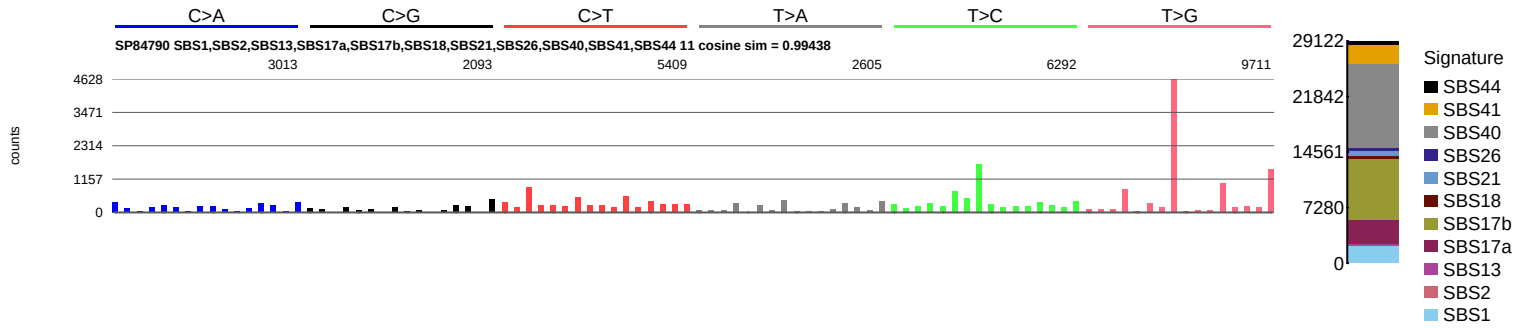

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.99439
3000
2113
5383
2604
6296
9736
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1

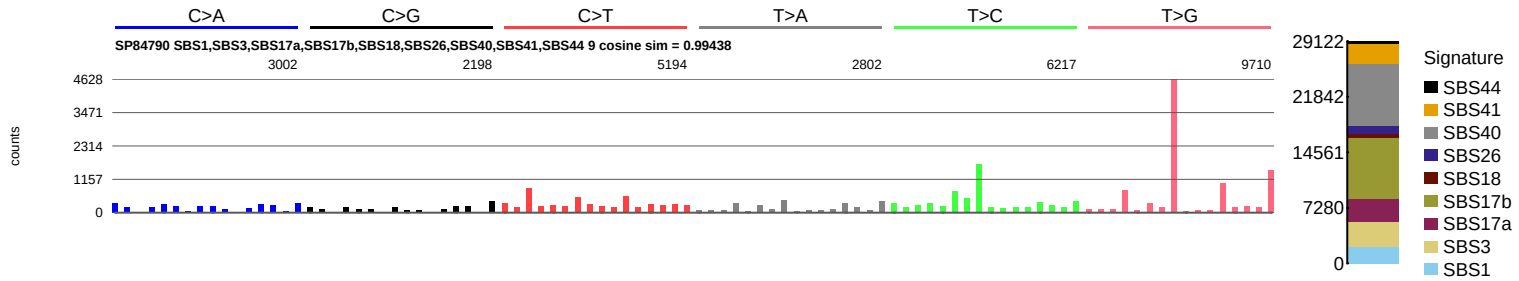

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS40,SBS41,SBS44 9 cosine sim = 0.99439
2642
2365
5543
2804
6072
9698
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

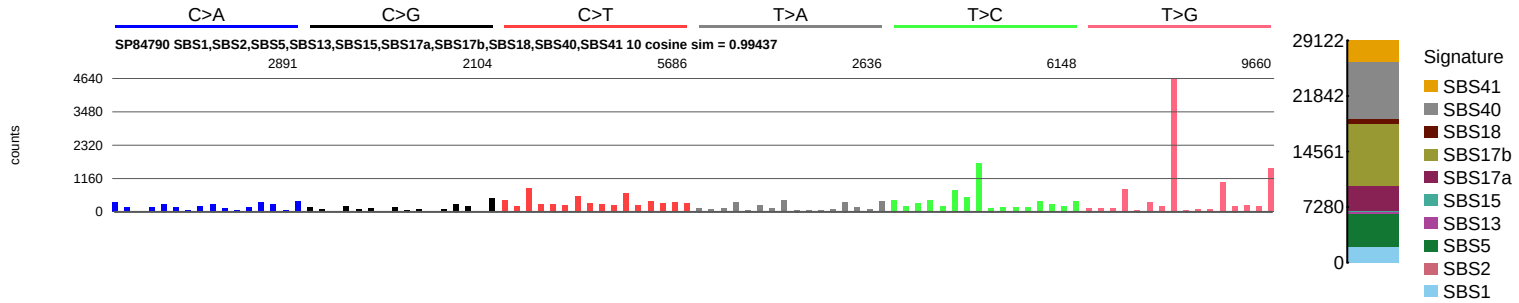

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.99438
2922
2088
5539
2609
6295
9667
counts
4636
3477
2318
1159
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1

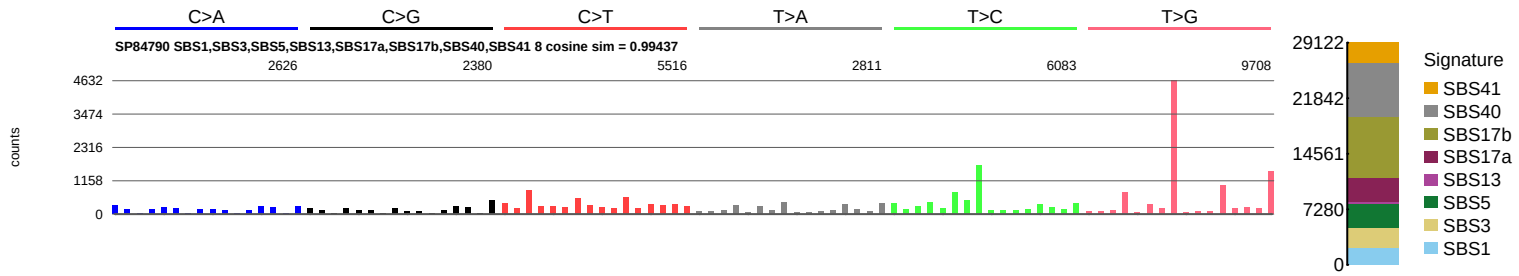

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.99438
2894
2136
5464
2608
6281
9741
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS1

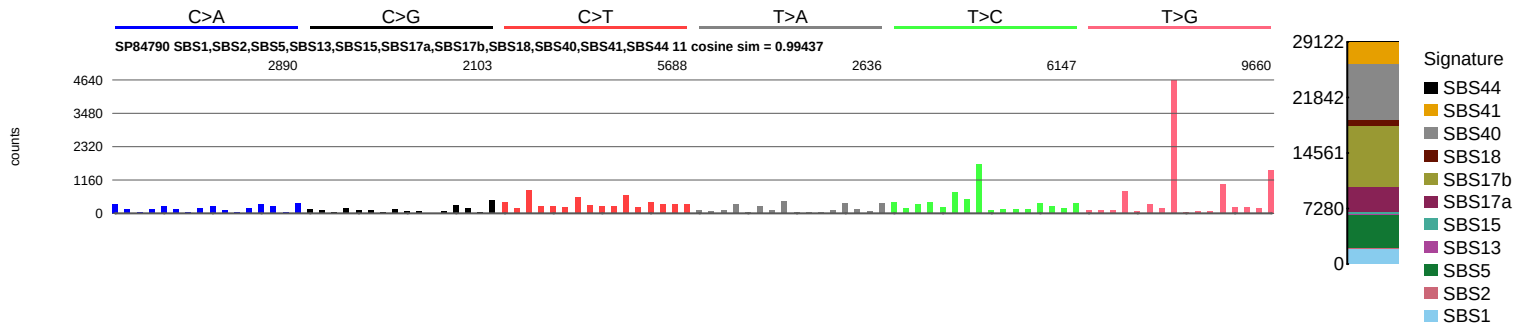

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS40,SBS41 9 cosine sim = 0.99438
2613
2280
5619
2862
6040
9711
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

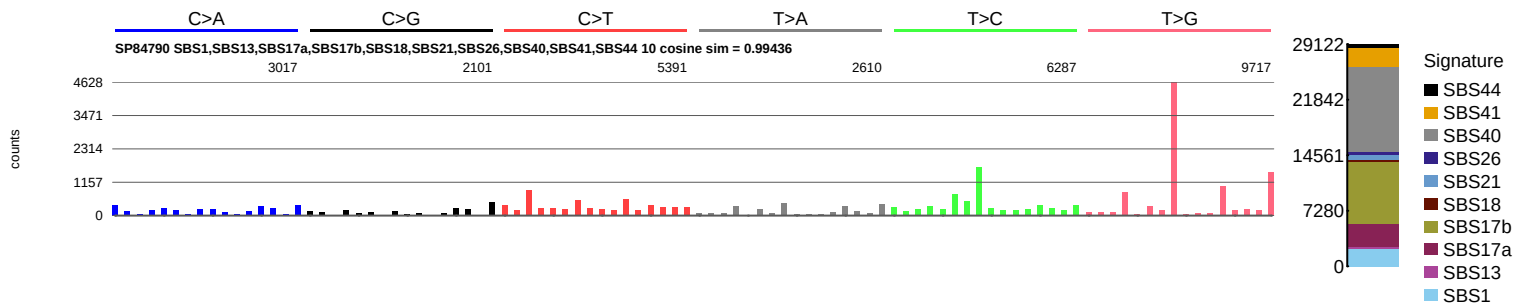

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.99438
3013
2093
5409
2605
6292
9711
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

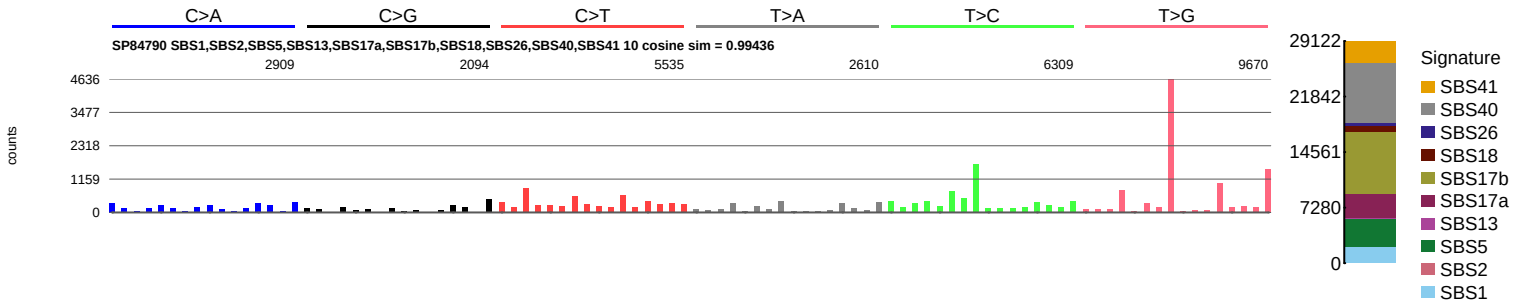

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.99438
3002
2198
5194
2802
6217
9710
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS1

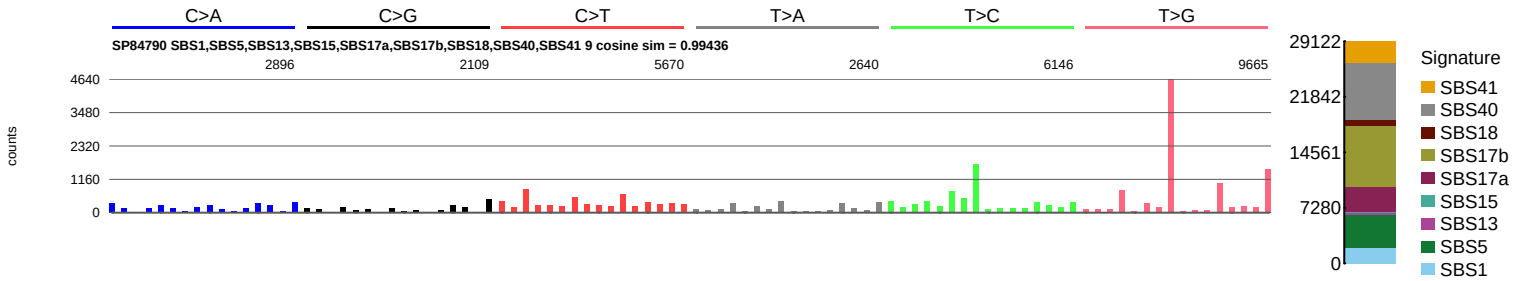

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 10 cosine sim = 0.99437
2891
2104
5686
2636
6148
9660
counts
4640
3480
2320
1160
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

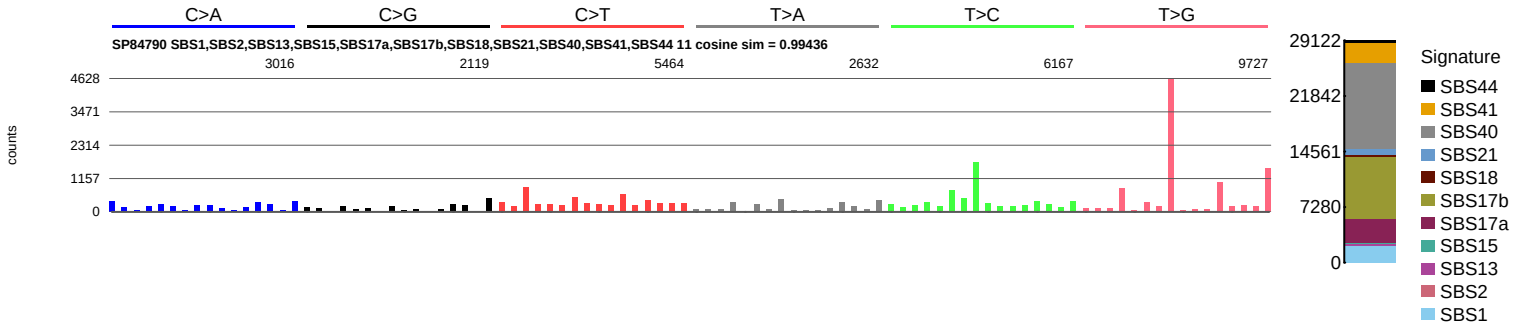

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS40,SBS41 8 cosine sim = 0.99437
2626
2380
5516
2811
6083
9708
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

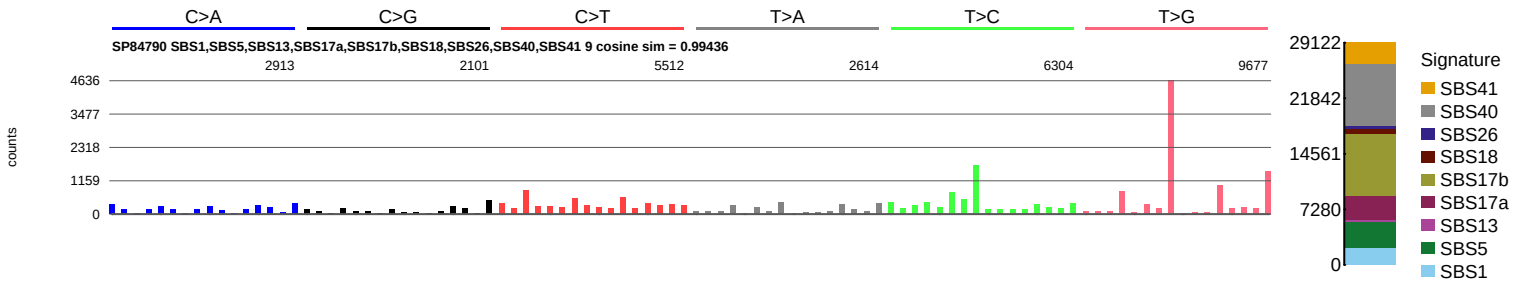

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 11 cosine sim = 0.99437
2890
2103
5688
2636
6147
9660
counts
4640
3480
2320
1160
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

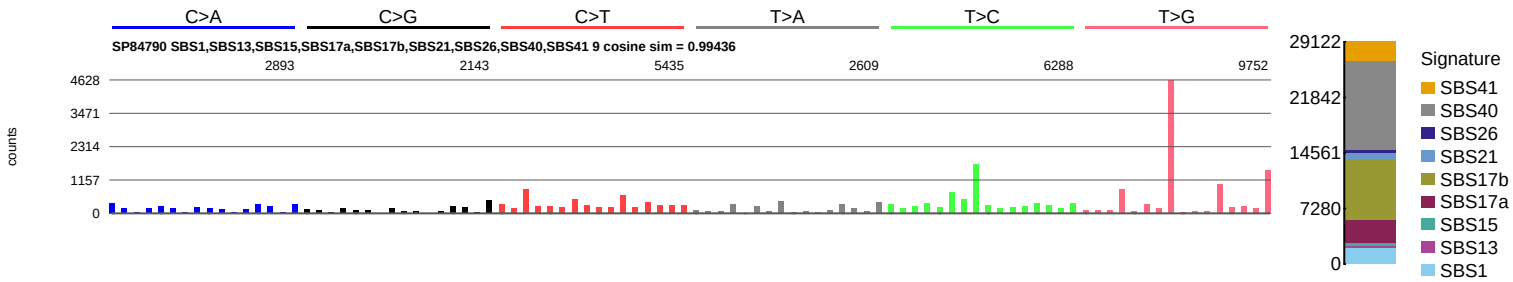

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.99436
3017
2101
5391
2610
6287
9717
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS1

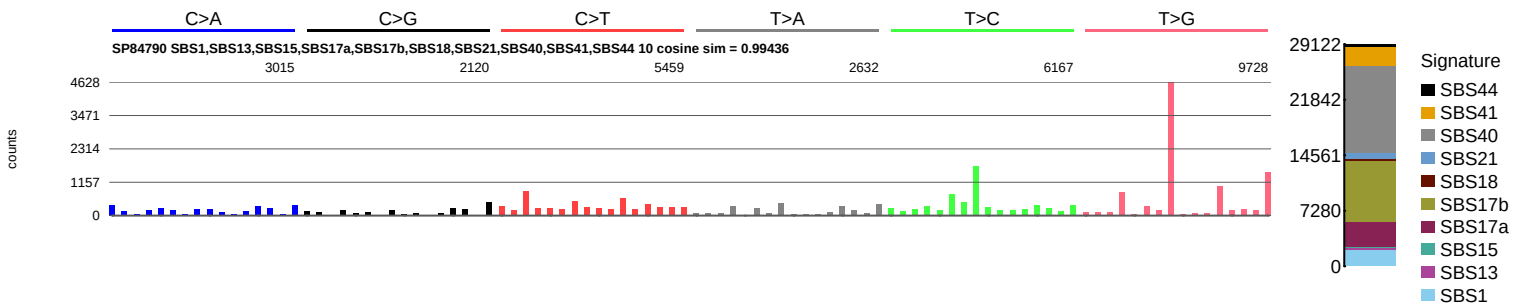

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.99436
2909
2094
5535
2610
6309
9670
counts
4636
3477
2318
1159
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

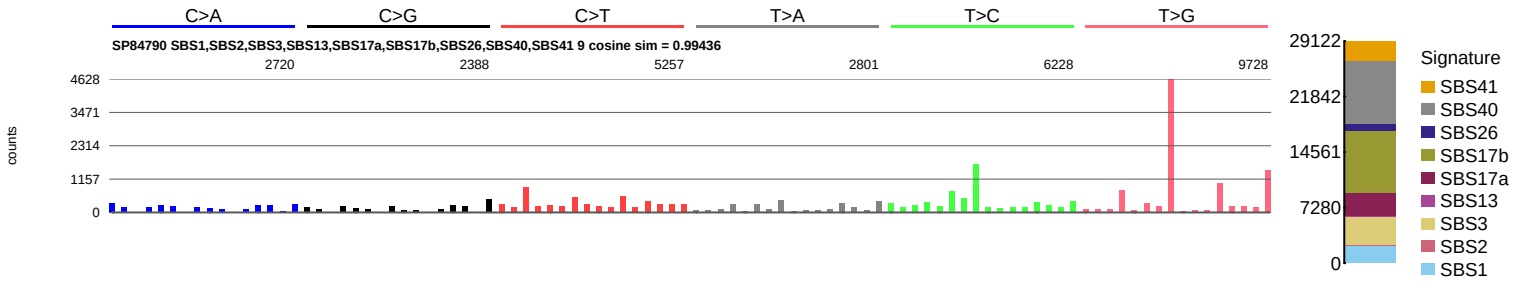

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.99436
2896
2109
5670
2640
6146
9665
counts
4640
3480
2320
1160
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

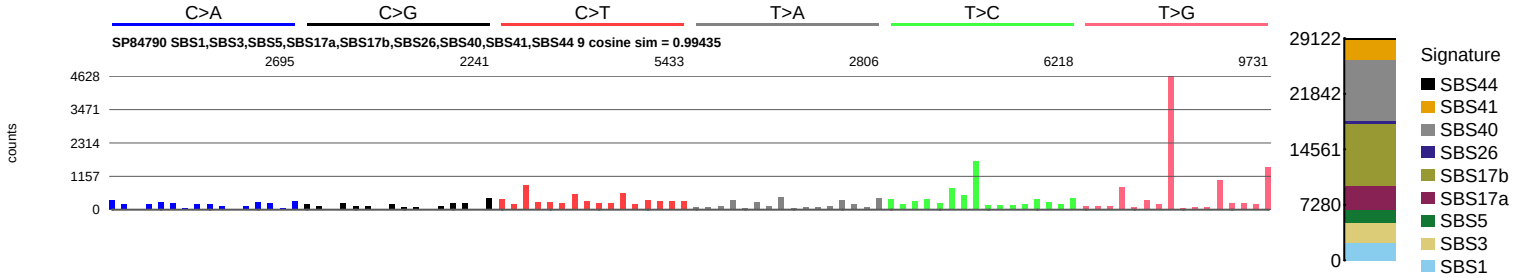

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.99436
3016
2119
5464
2632
6167
9727
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

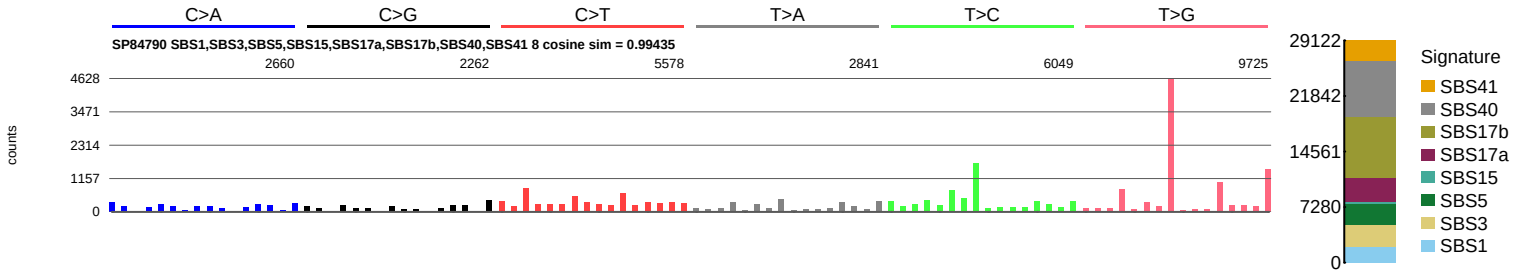

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.99436
2913
2101
5512
2614
6304
9677
counts
4636
3477
2318
1159
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1

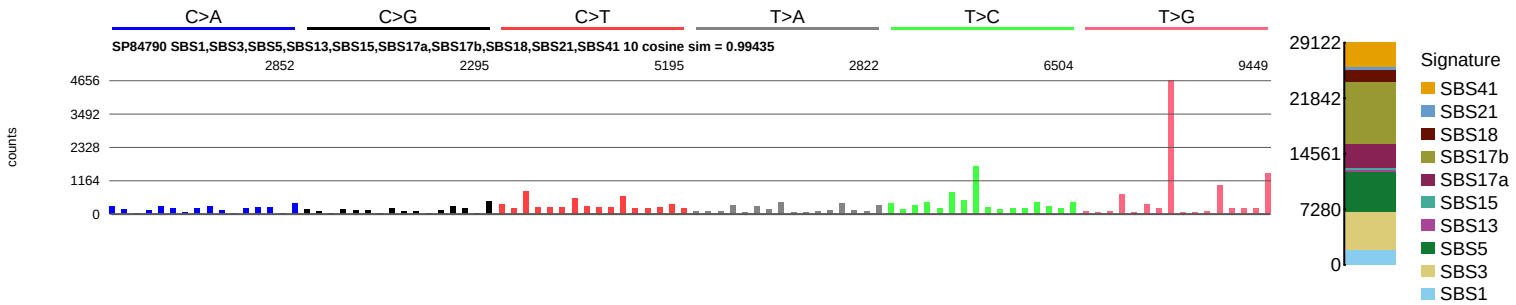

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.99436
2893
2143
5435
2609
6288
9752
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS1

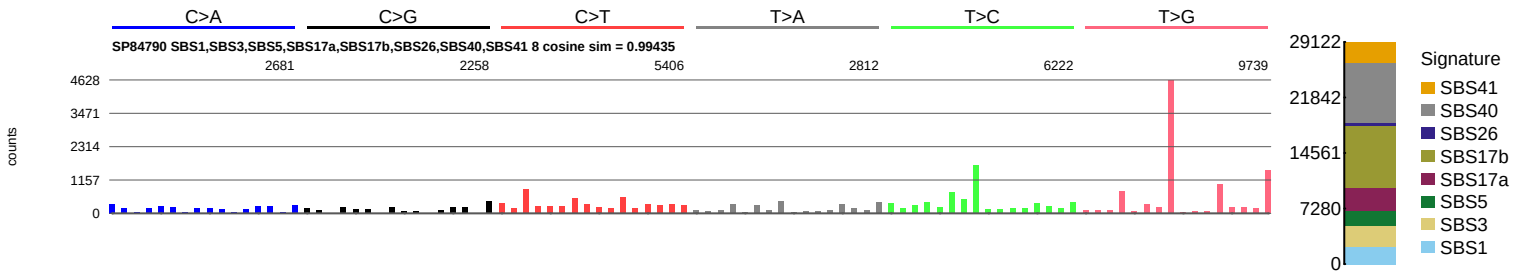

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.99436
3015
2120
5459
2632
6167
9728
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1

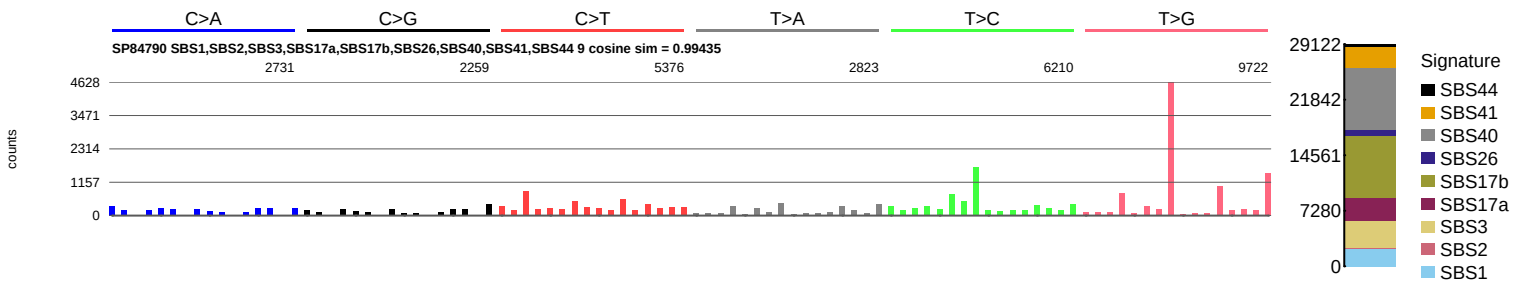

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS41 9 cosine sim = 0.99436
2720
2388
5257
2801
6228
9728
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

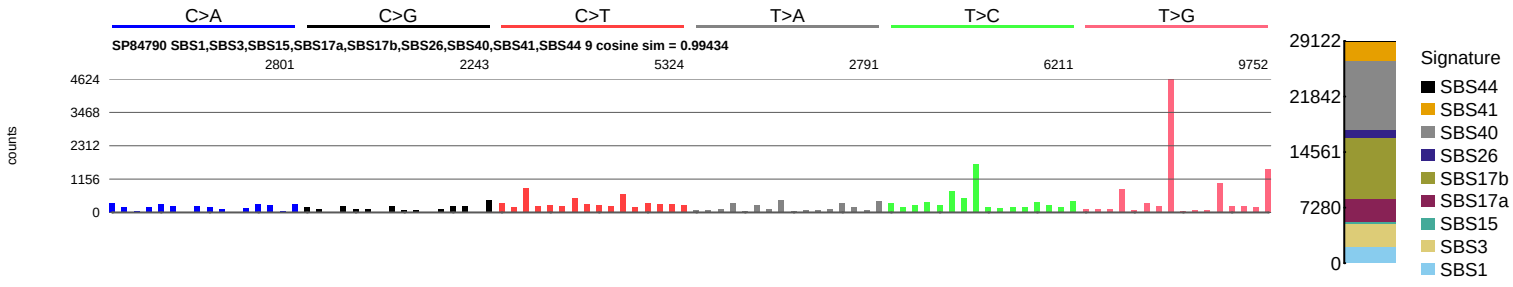

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.99435
2695
2241
5433
2806
6218
9731
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS5
SBS3
SBS1

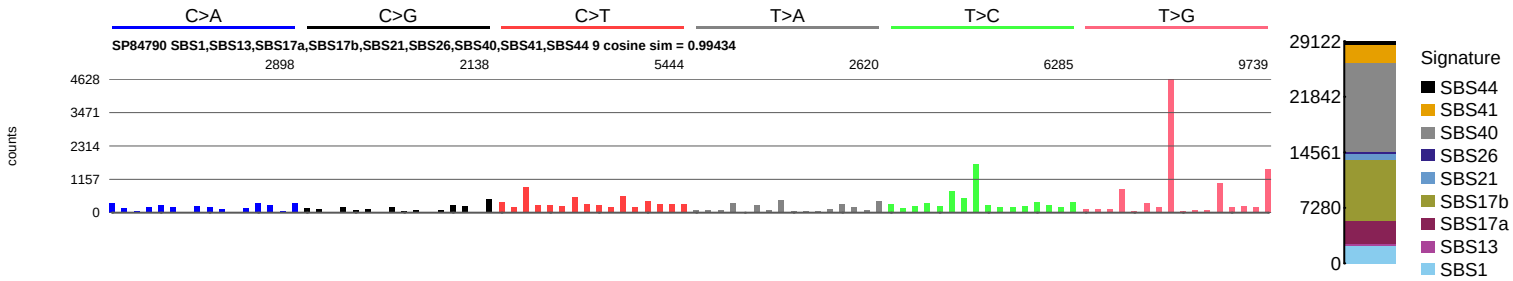

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS40,SBS41 8 cosine sim = 0.99435
2660
2262
5578
2841
6049
9725
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1

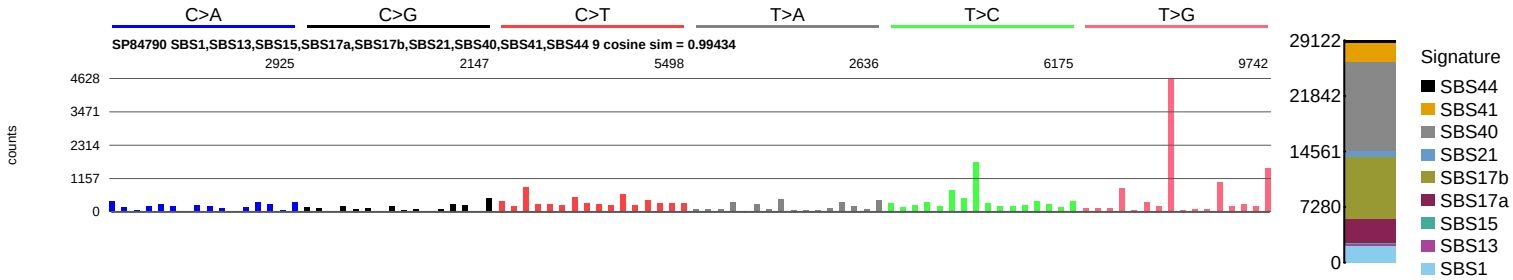

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.99435
2852
2295
5195
2822
6504
9449
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

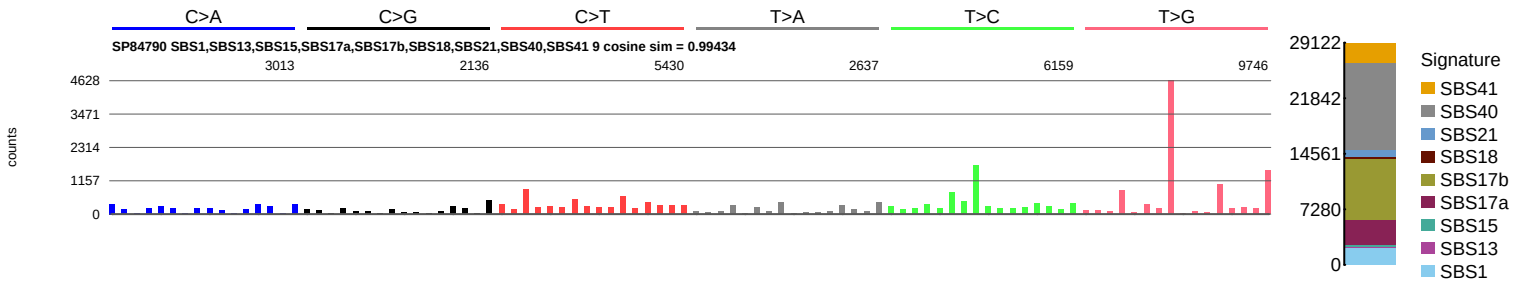

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS26,SBS40,SBS41 8 cosine sim = 0.99435
2681
2258
5406
2812
6222
9739
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS5
SBS3
SBS1

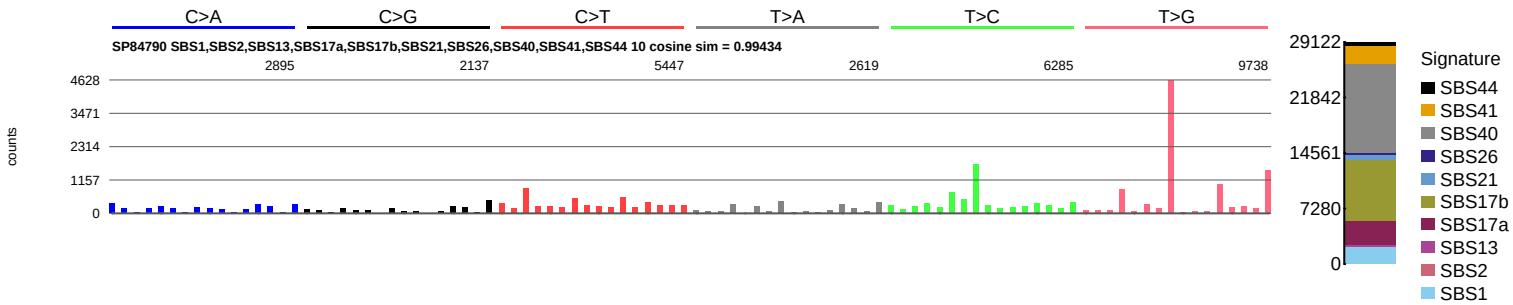

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.99435
2731
2259
5376
2823
6210
9722
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS3
SBS2
SBS1

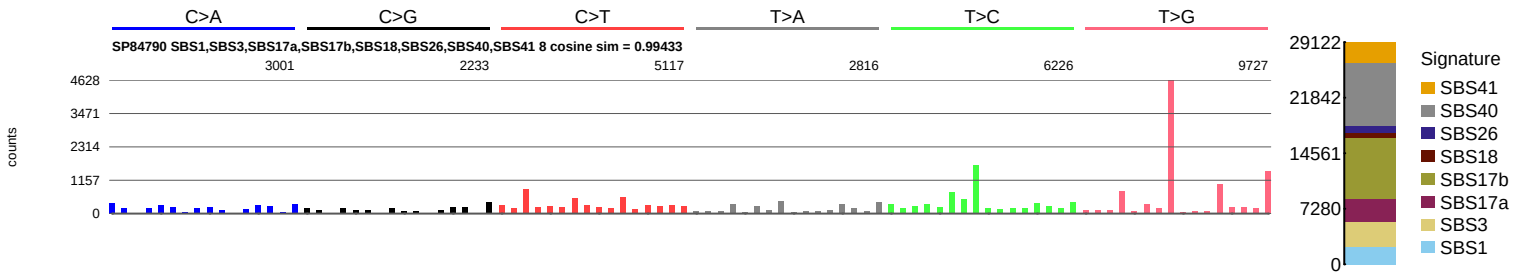

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.99434
2801
2243
5324
2791
6211
9752
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS1

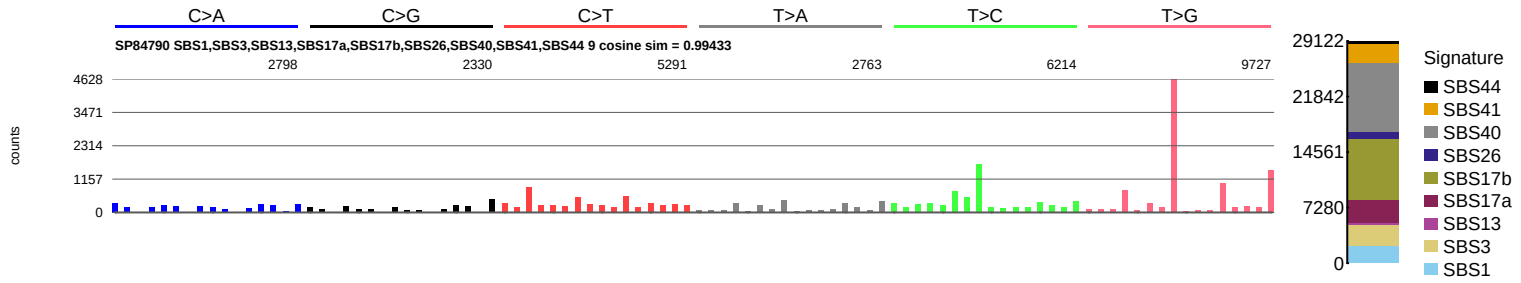

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.99434
2898
2138
5444
2620
6285
9739
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS1

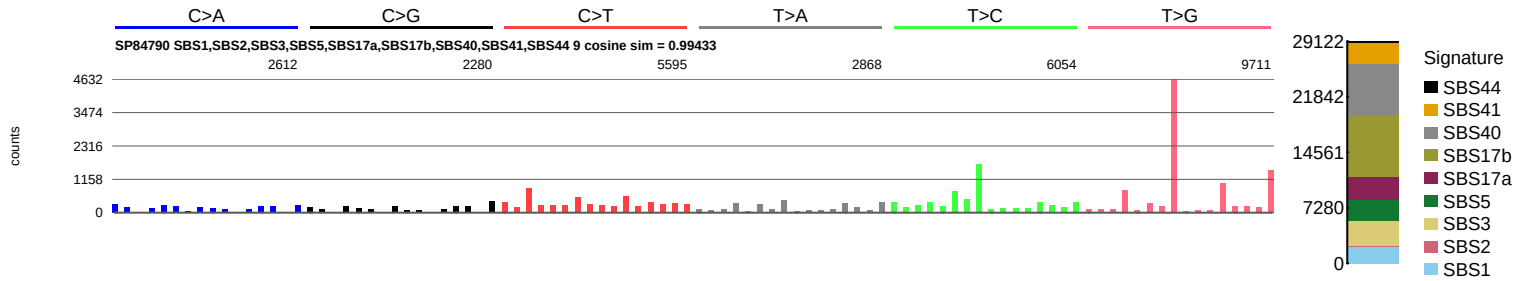

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.99434
2925
2147
5498
2636
6175
9742
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS1

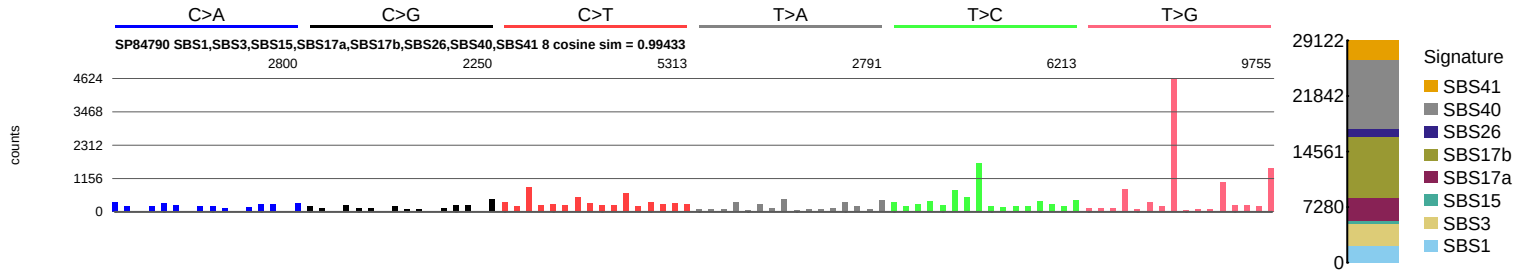

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.99434
3013
2136
5430
2637
6159
9746
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1

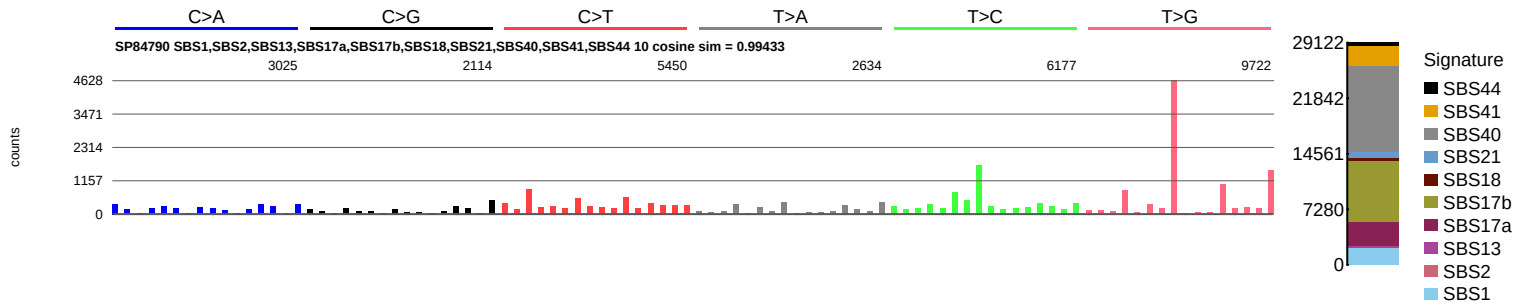

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.99434
2895
2137
5447
2619
6285
9738
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS2
SBS1

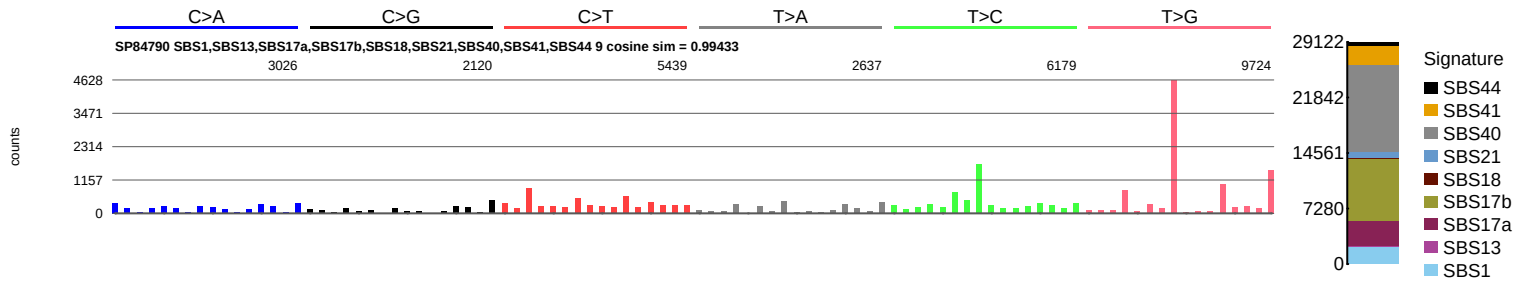

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.99433
3001
2233
5117
2816
6226
9727
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS1

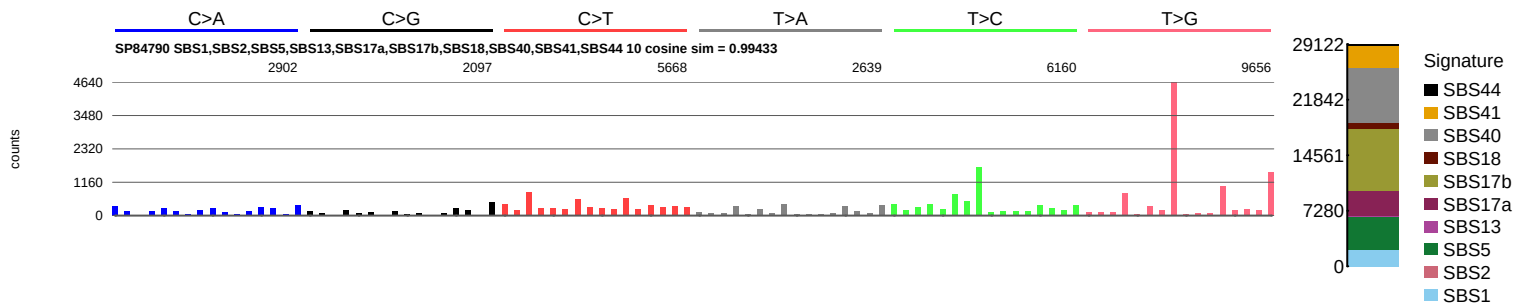

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.99433
2798
2330
5291
2763
6214
9727
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS3
SBS1

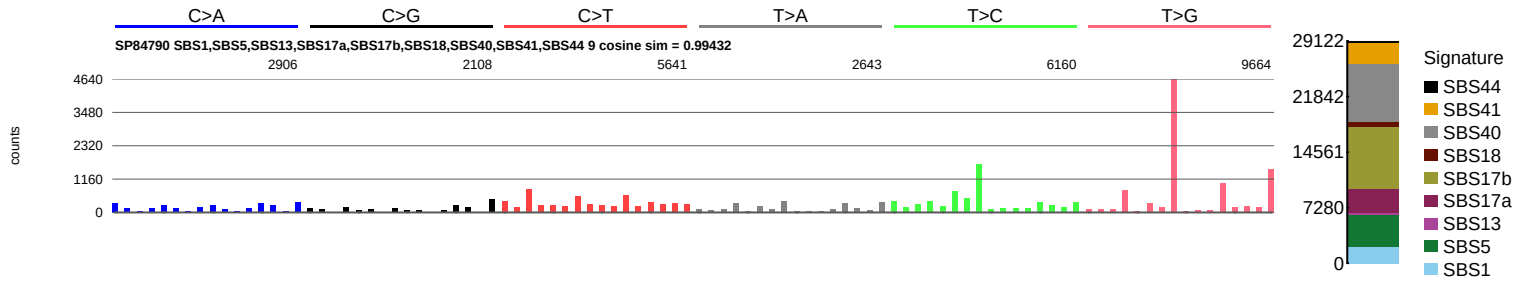

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS40,SBS41,SBS44 9 cosine sim = 0.99433
2612
2280
5595
2868
6054
9711
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

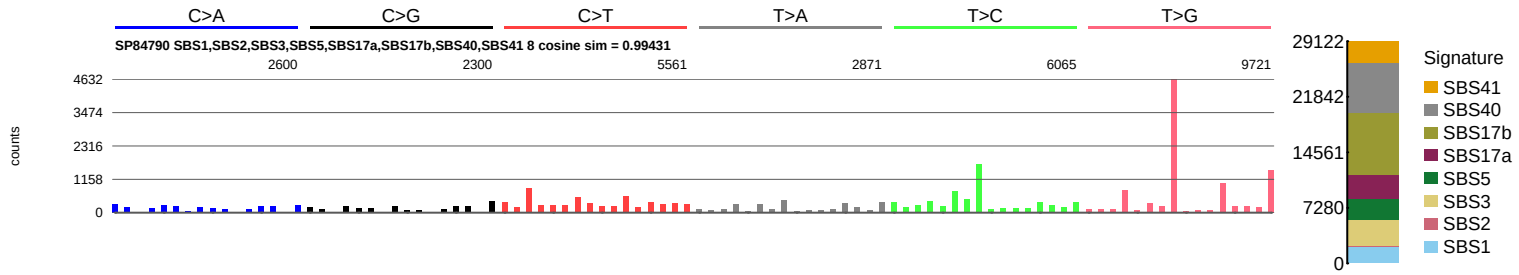

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 8 cosine sim = 0.99433
2800
2250
5313
2791
6213
9755
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS1

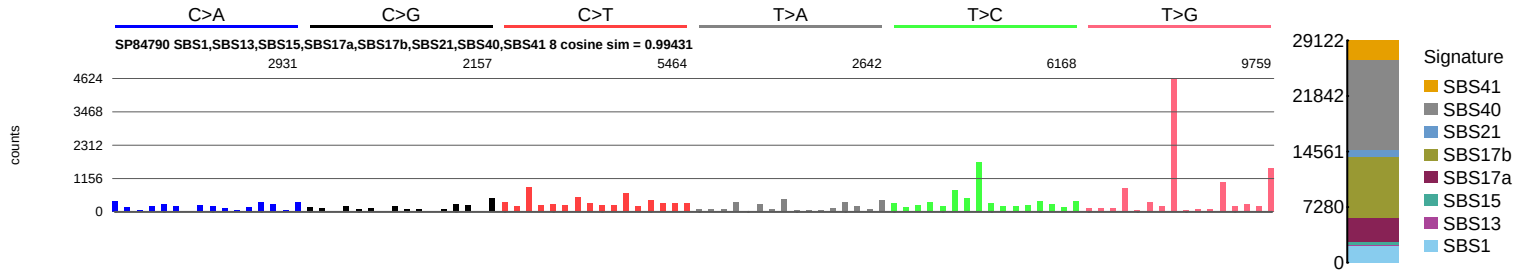

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.99433
3025
2114
5450
2634
6177
9722
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

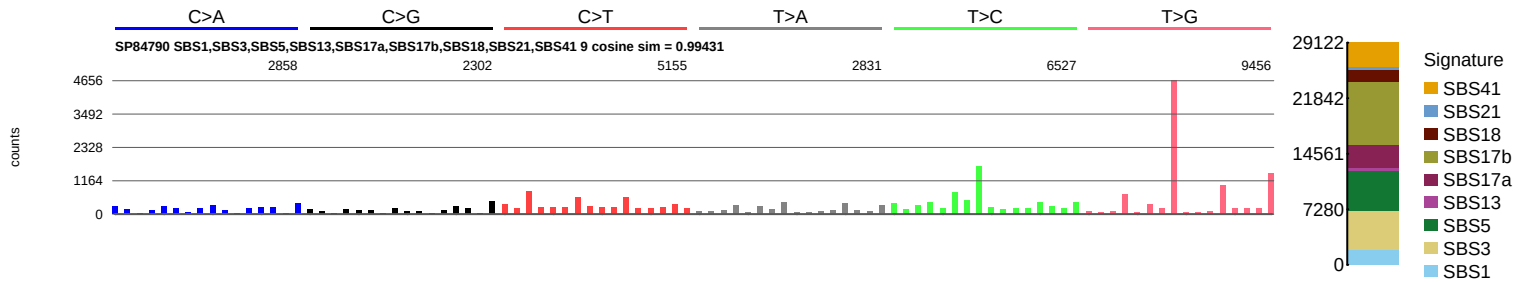

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.99433
3026
2120
5439
2637
6179
9724
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS1

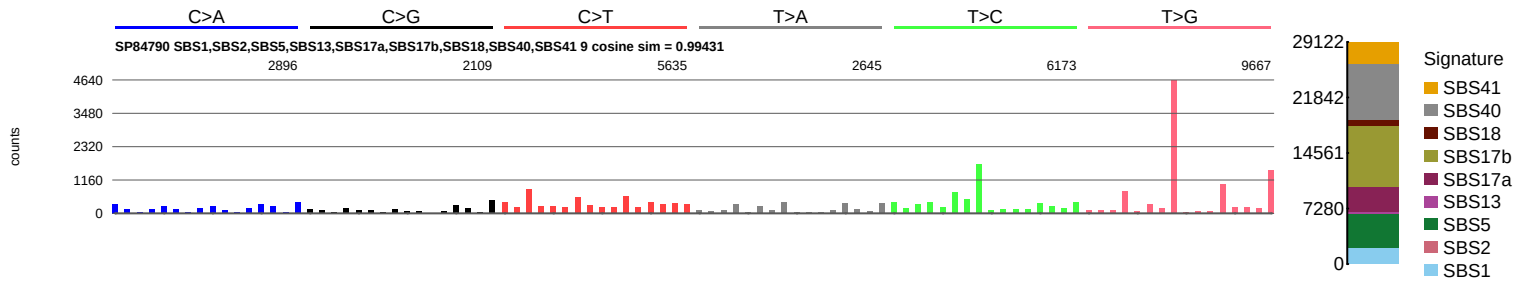

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.99433
2902
2097
5668
2639
6160
9656
counts
4640
3480
2320
1160
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

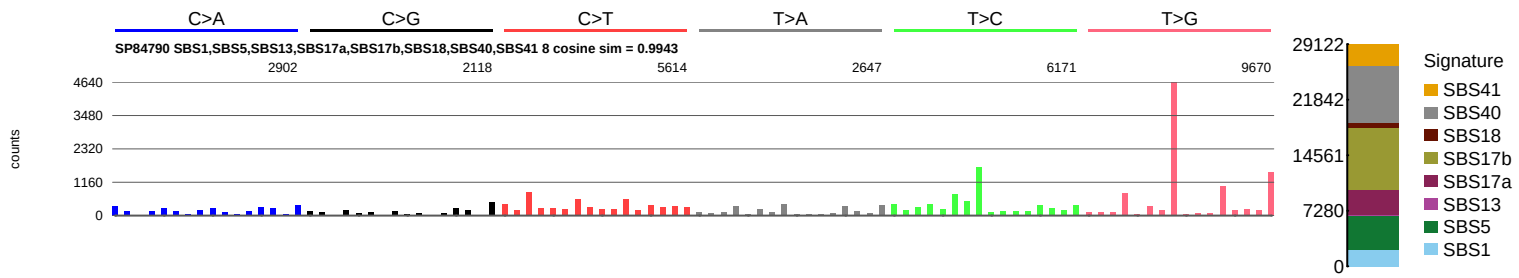

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.99432
2906
2108
5641
2643
6160
9664
counts
4640
3480
2320
1160
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1

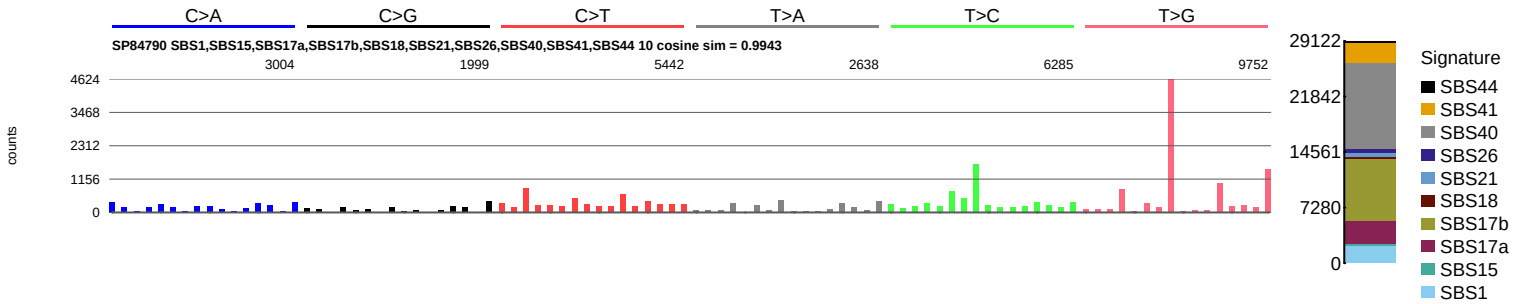

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS40,SBS41 8 cosine sim = 0.99431
2600
2300
5561
2871
6065
9721
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

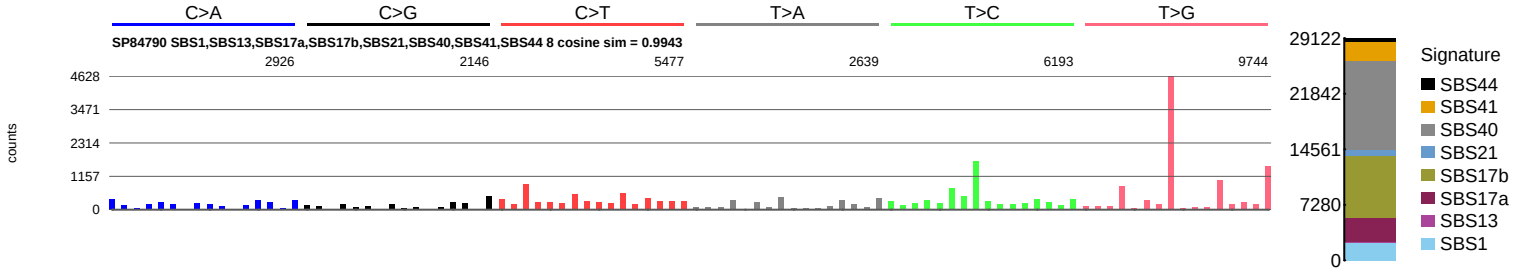

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 8 cosine sim = 0.99431
2931
2157
5464
2642
6168
9759
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS1

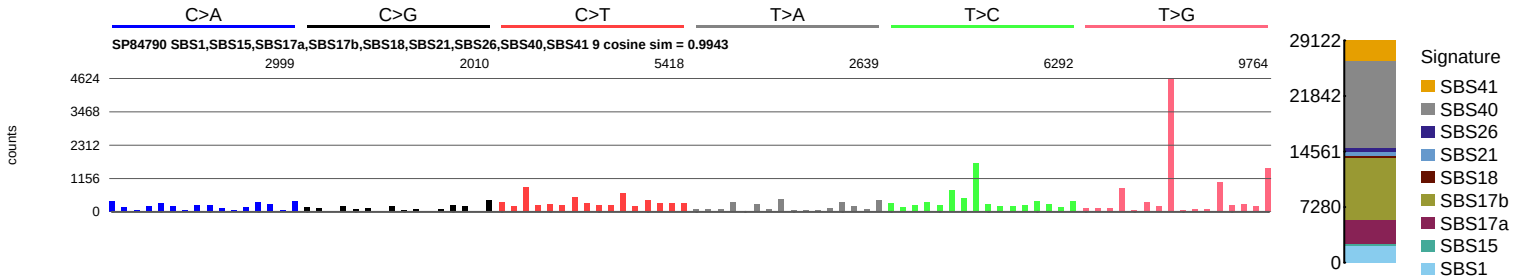

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.99431
2858
2302
5155
2831
6527
9456
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

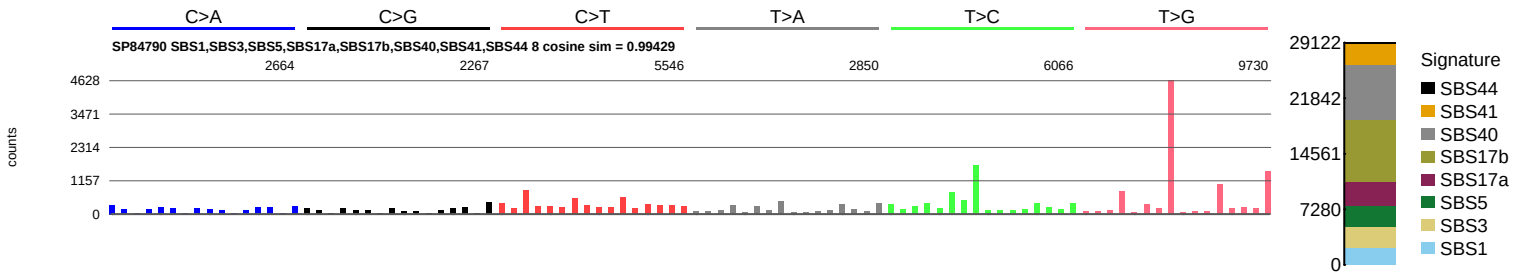

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.99431
2896
2109
5635
2645
6173
9667
counts
4640
3480
2320
1160
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

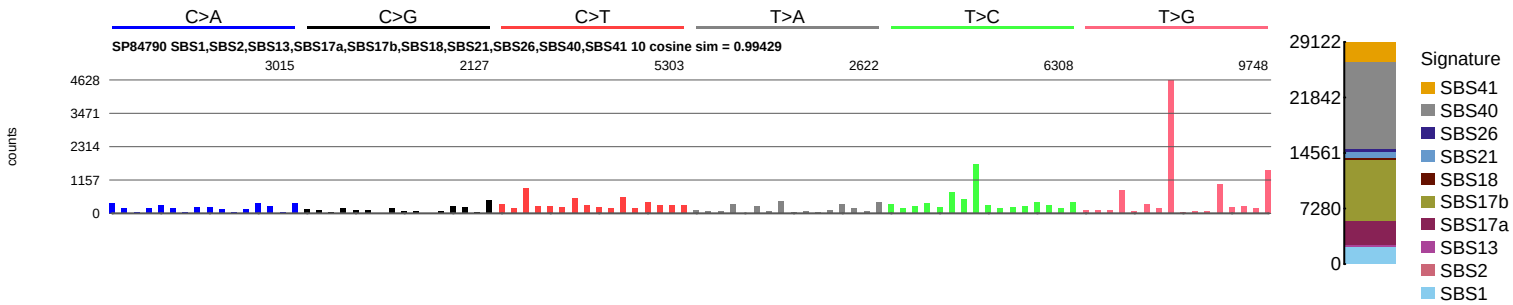

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.9943
2902
2118
5614
2647
6171
9670
counts
4640
3480
2320
1160
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1

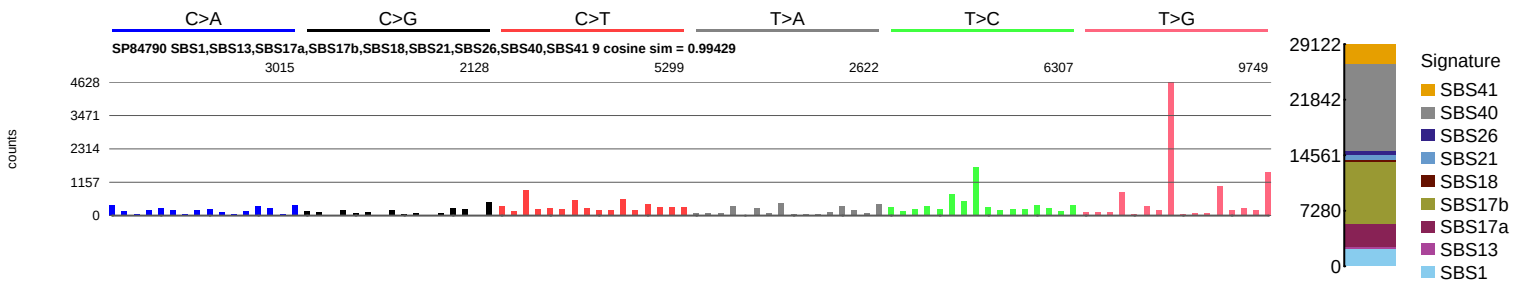

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.9943
3004
1999
5442
2638
6285
9752
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1

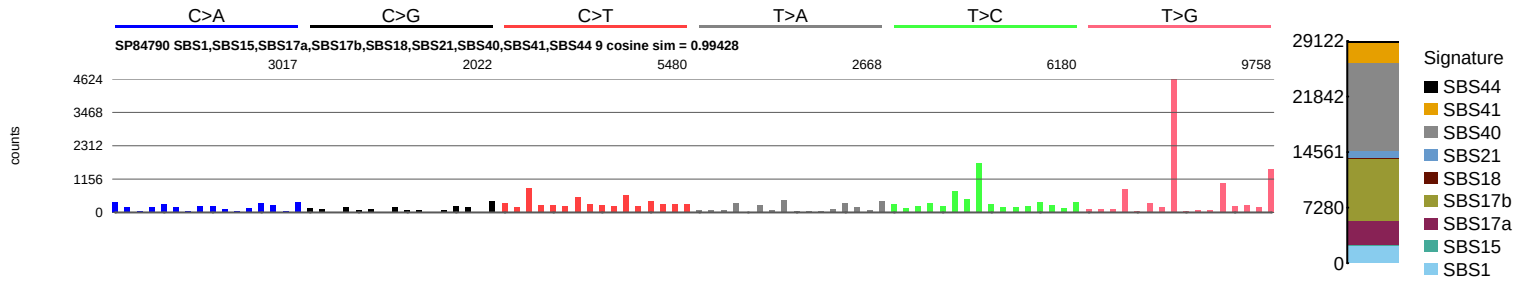

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.9943
2926
2146
5477
2639
6193
9744
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS1

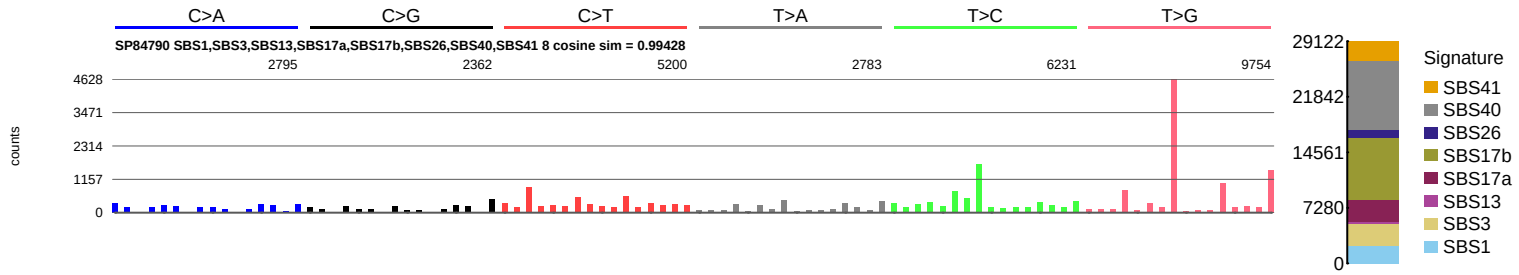

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.9943
2999
2010
5418
2639
6292
9764
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1

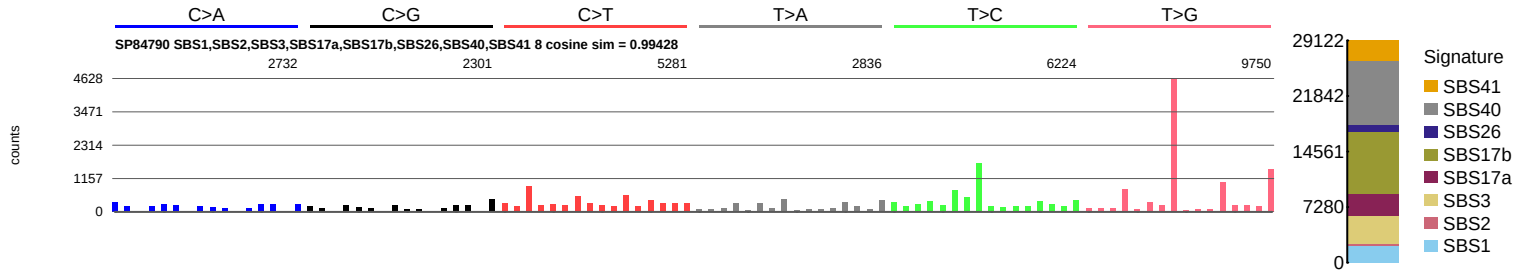

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS40,SBS41,SBS44 8 cosine sim = 0.99429
2664
2267
5546
2850
6066
9730
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS5
SBS3
SBS1

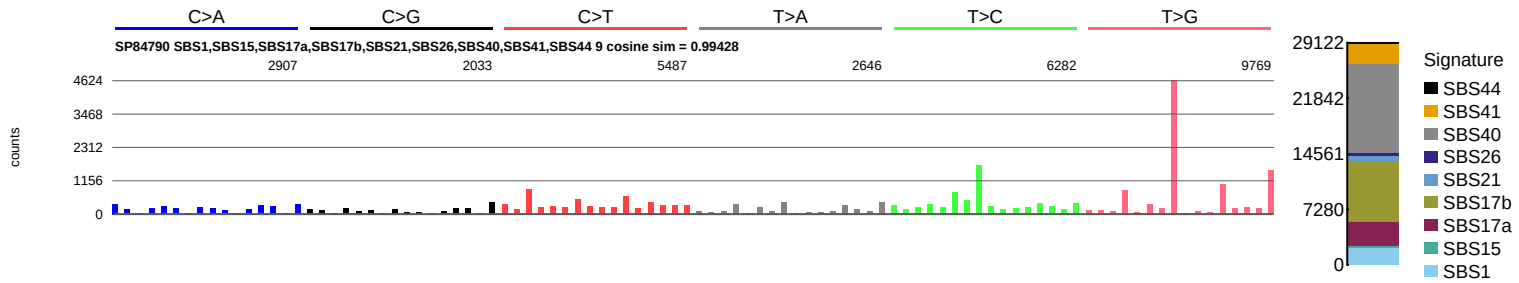

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.99429
3015
2127
5303
2622
6308
9748
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

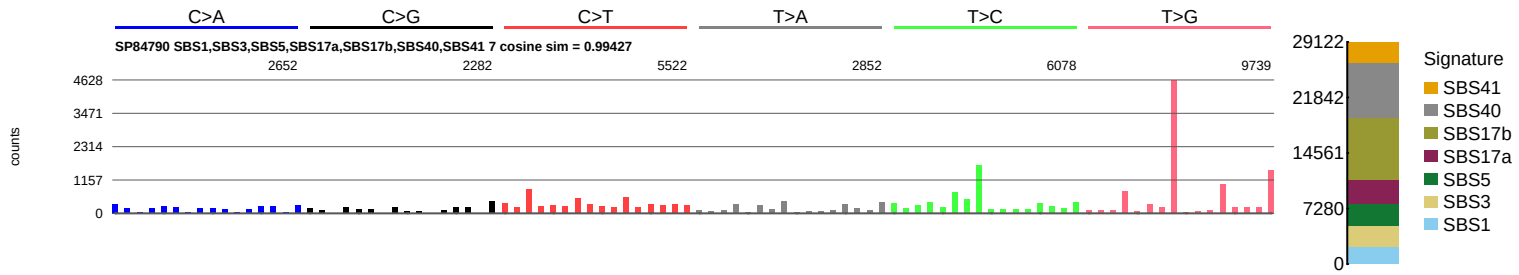

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.99429
3015
2128
5299
2622
6307
9749
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS1

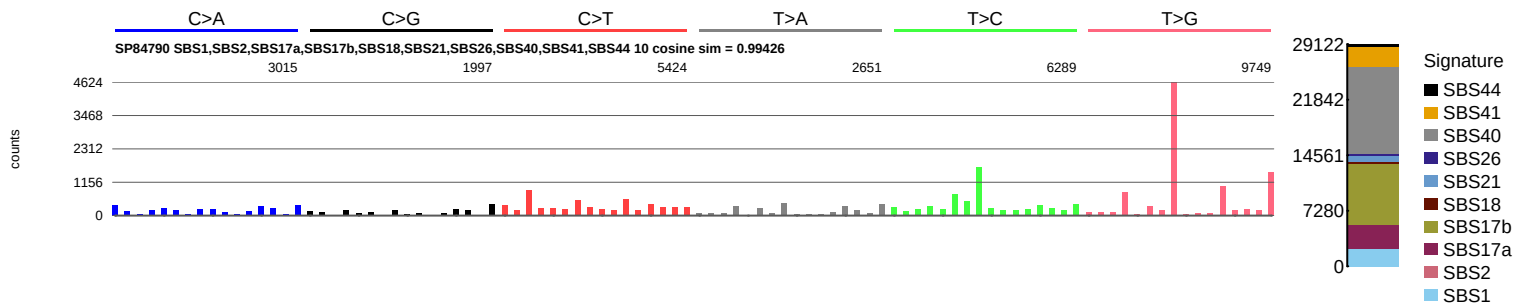

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.99428
3017
2022
5480
2668
6180
9758
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1

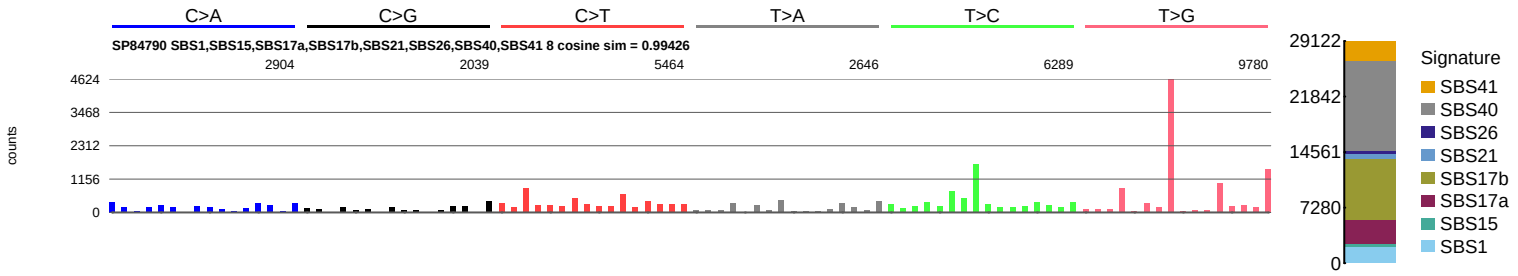

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS41 8 cosine sim = 0.99428
2795
2362
5200
2783
6231
9754
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS3
SBS1

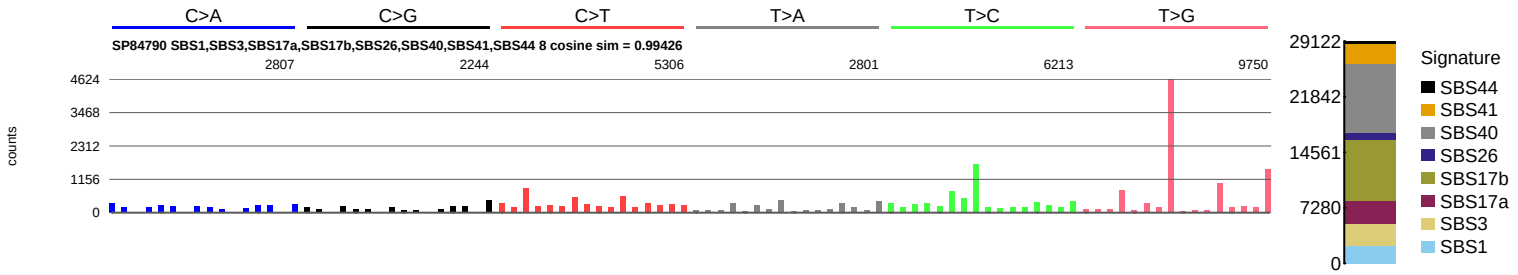

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS26,SBS40,SBS41 8 cosine sim = 0.99428
2732
2301
5281
2836
6224
9750
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS3
SBS2
SBS1

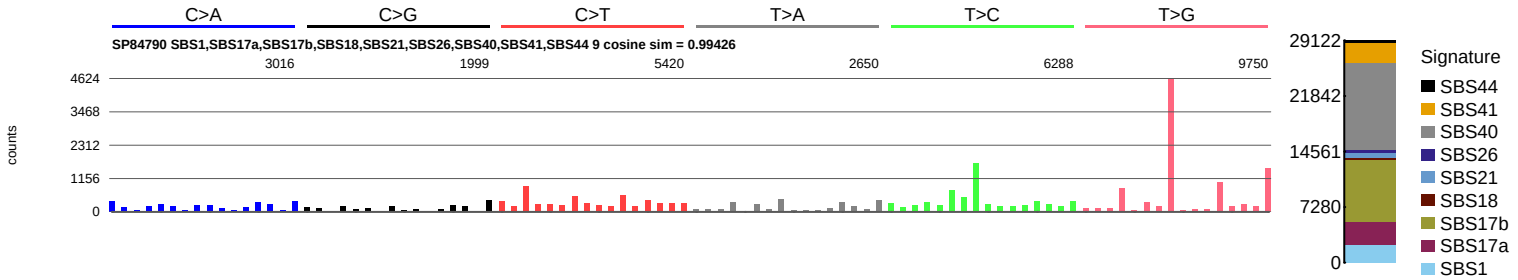

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.99428
2907
2033
5487
2646
6282
9769
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS1

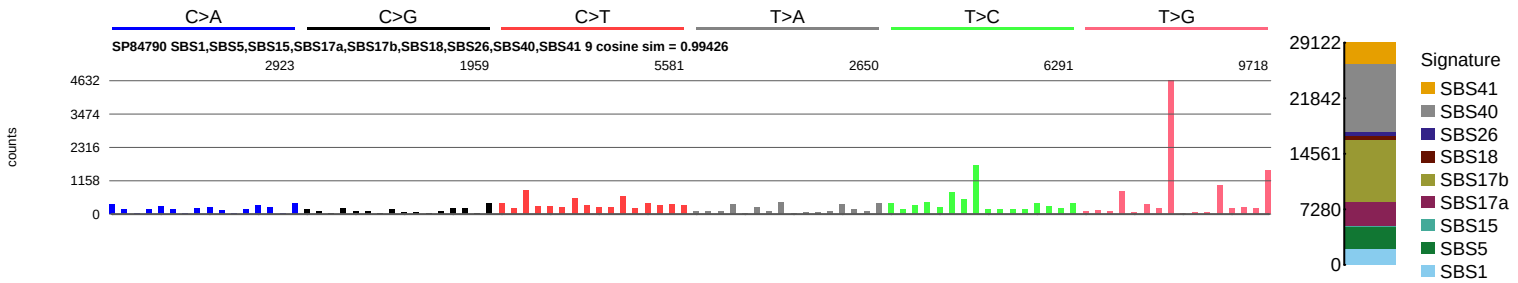

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.99427
2652
2282
5522
2852
6078
9739
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS5
SBS3
SBS1

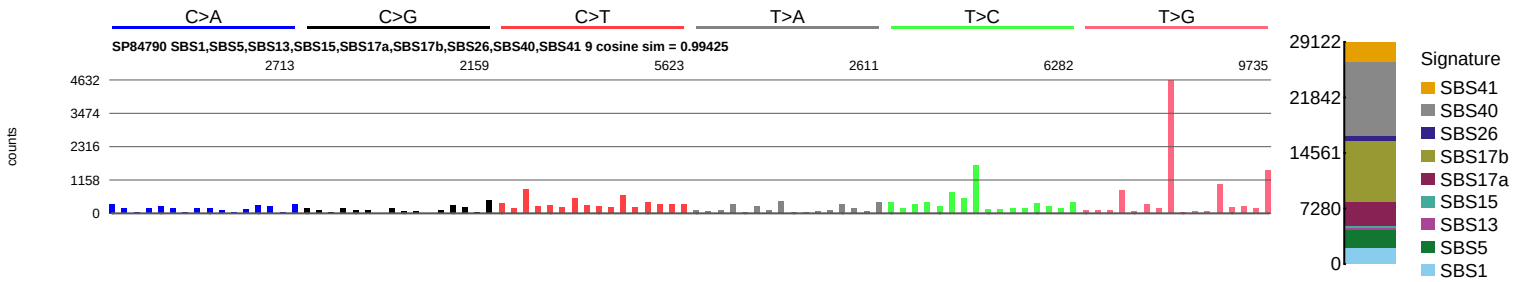

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.99426
3015
1997
5424
2651
6289
9749
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1

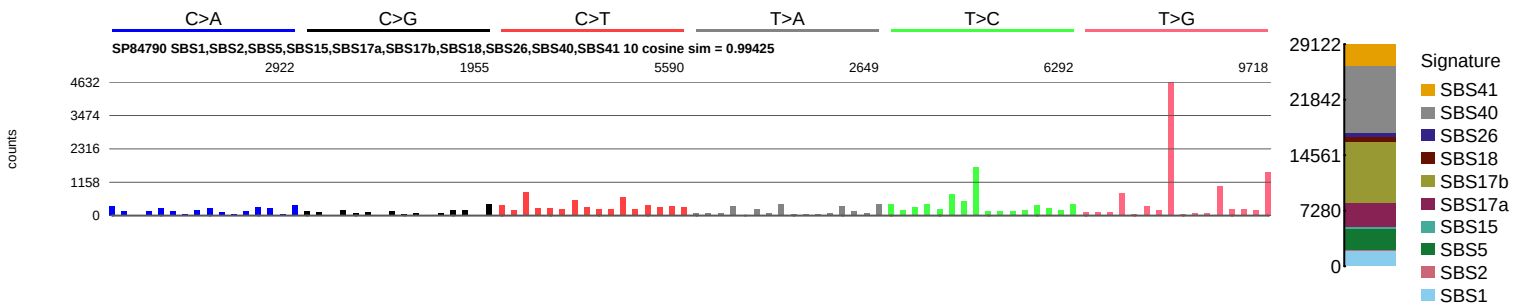

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41 8 cosine sim = 0.99426
2904
2039
5464
2646
6289
9780
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS1

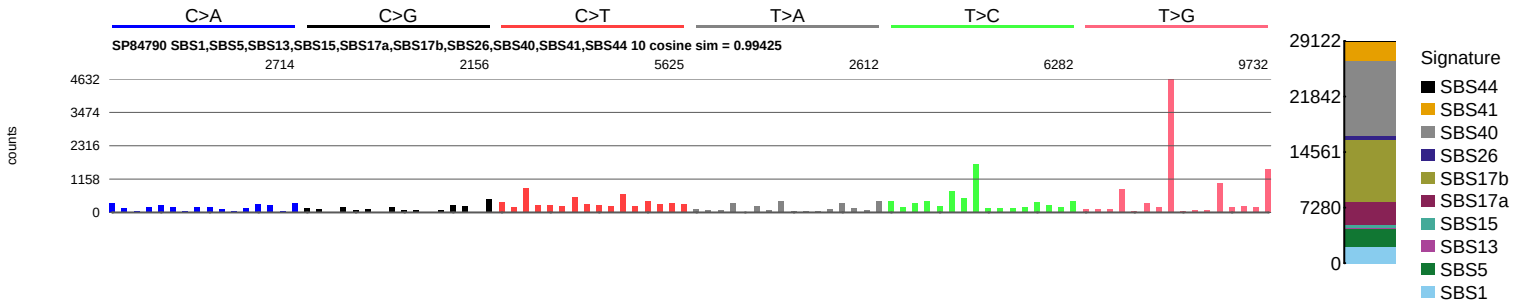

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.99426
2807
2244
5306
2801
6213
9750
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS3
SBS1

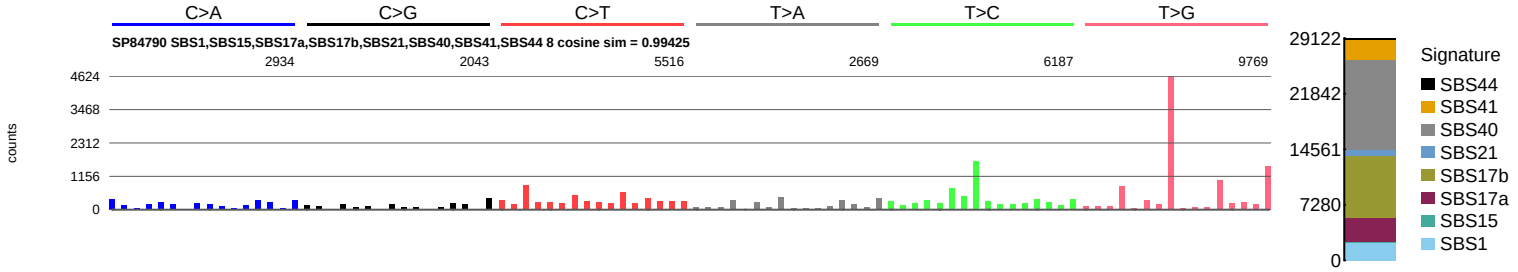

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.99426
3016
1999
5420
2650
6288
9750
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS1

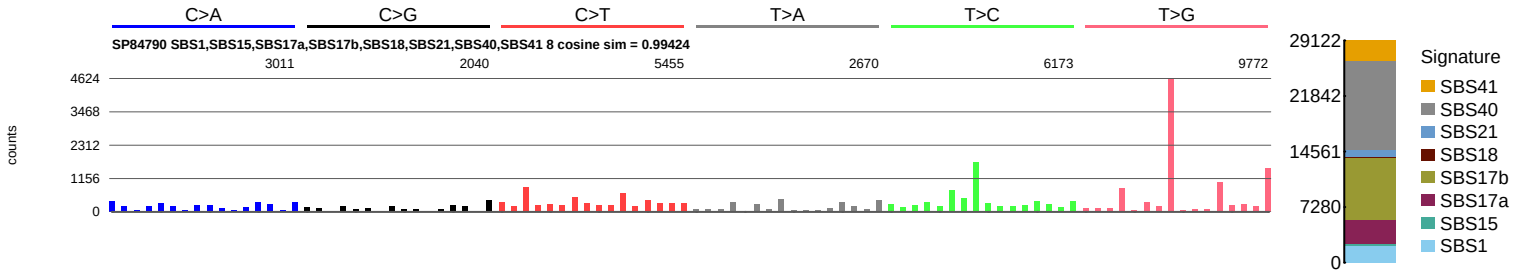

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.99426
2923
1959
5581
2650
6291
9718
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1

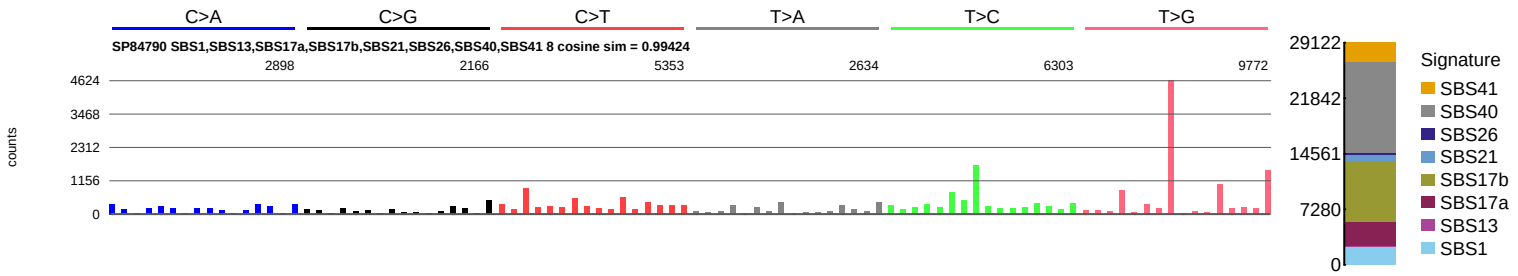

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 9 cosine sim = 0.99425
2713
2159
5623
2611
6282
9735
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

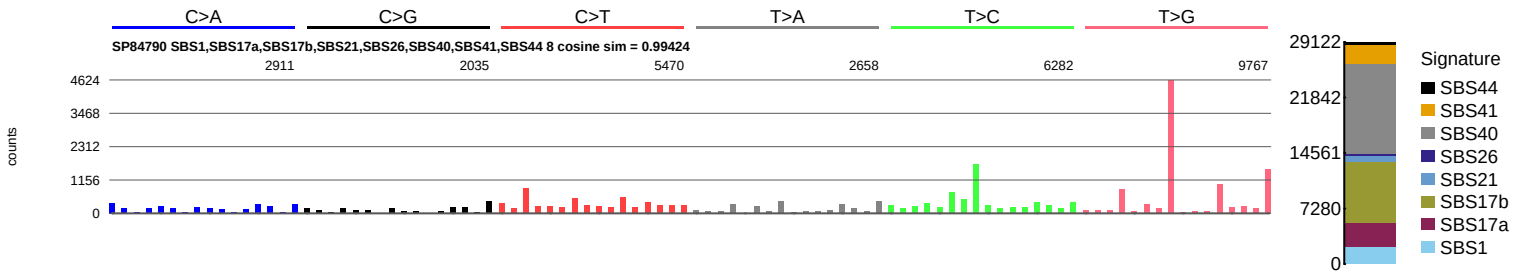

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.99425
2922
1955
5590
2649
6292
9718
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

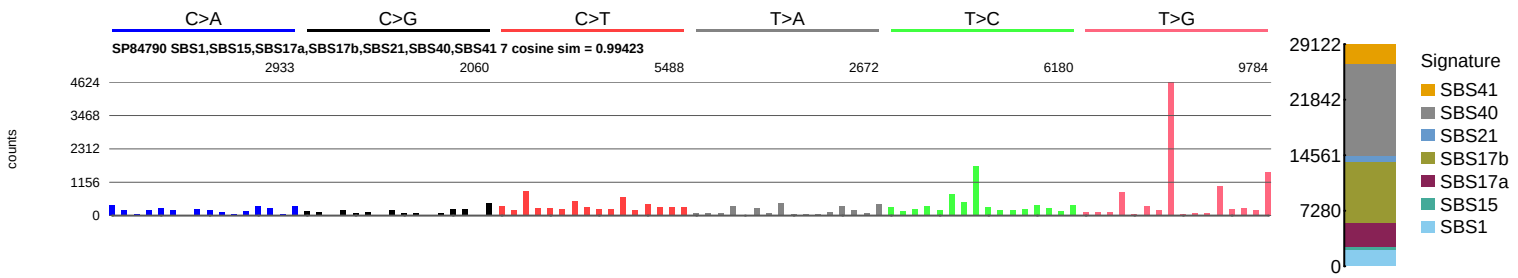

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.99425
2714
2156
5625
2612
6282
9732
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

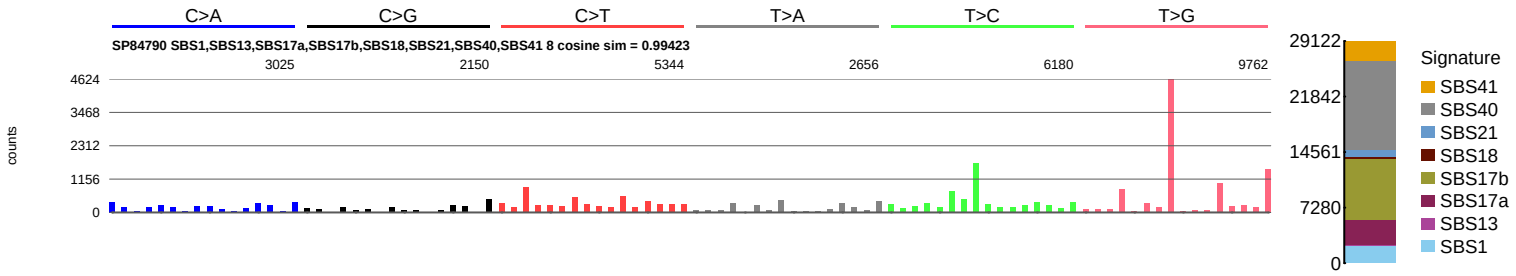

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.99425
2934
2043
5516
2669
6187
9769
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS1

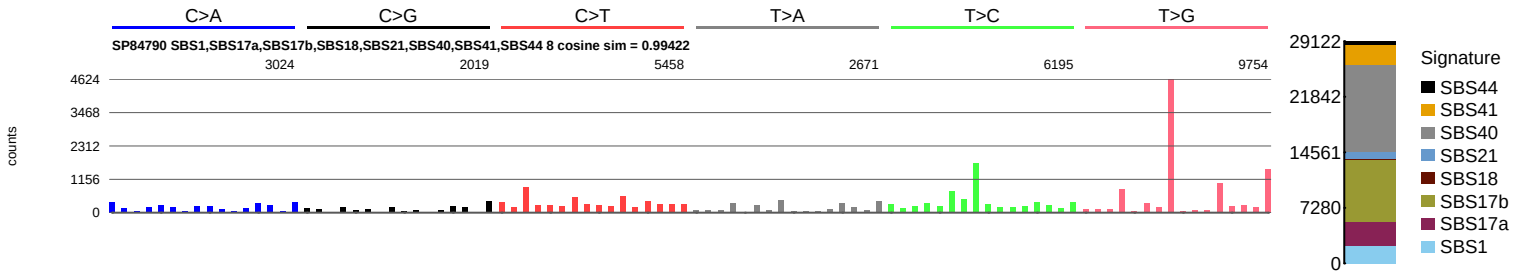

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.99424
3011
2040
5455
2670
6173
9772
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1

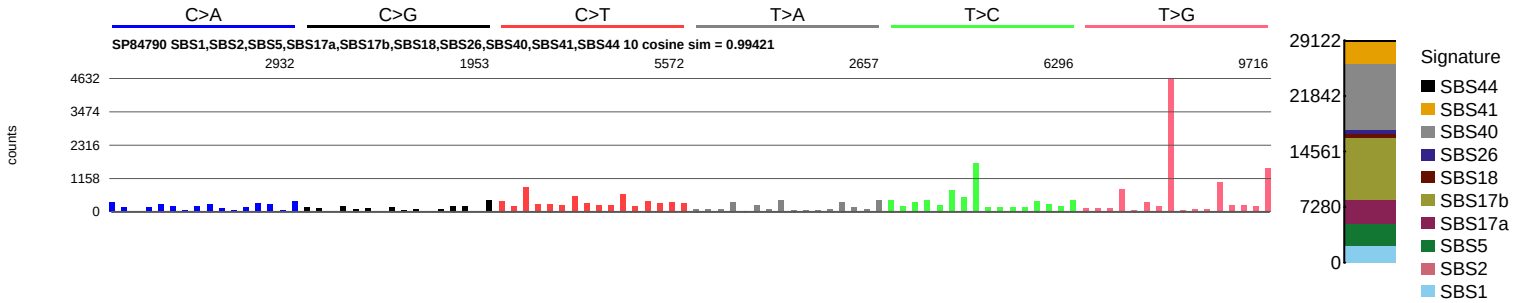

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41 8 cosine sim = 0.99424
2898
2166
5353
2634
6303
9772
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS1

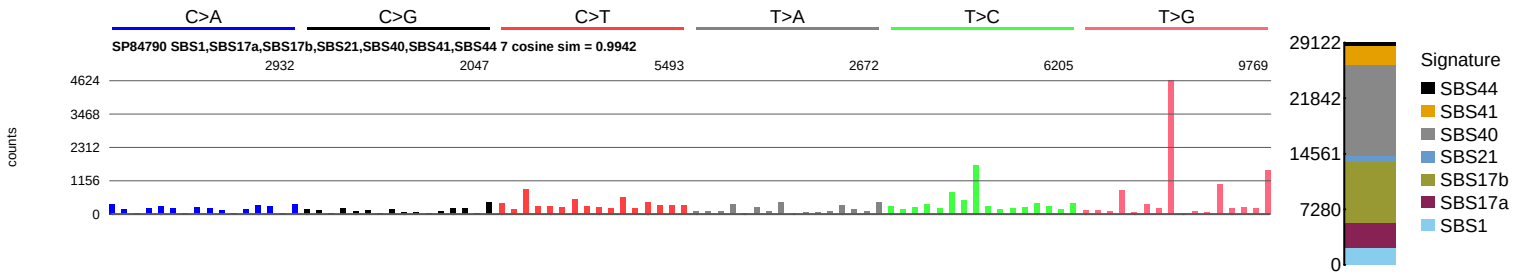

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.99424
2911
2035
5470
2658
6282
9767
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS1

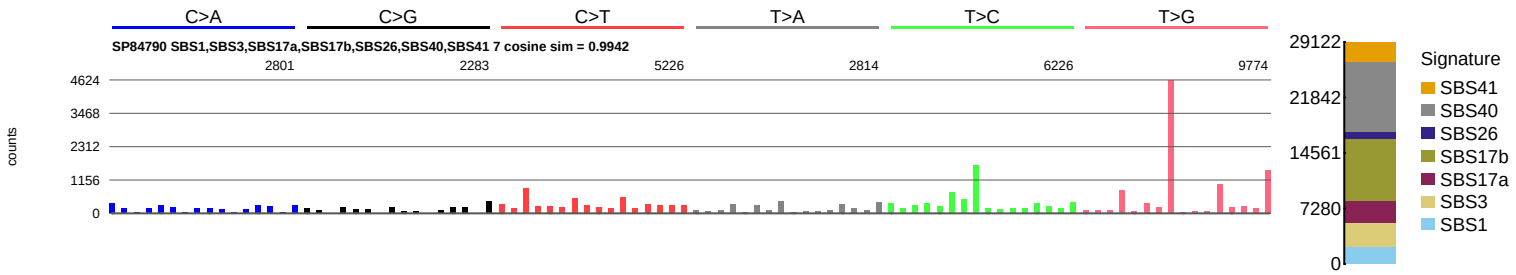

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 7 cosine sim = 0.99423
2933
2060
5488
2672
6180
9784
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS1

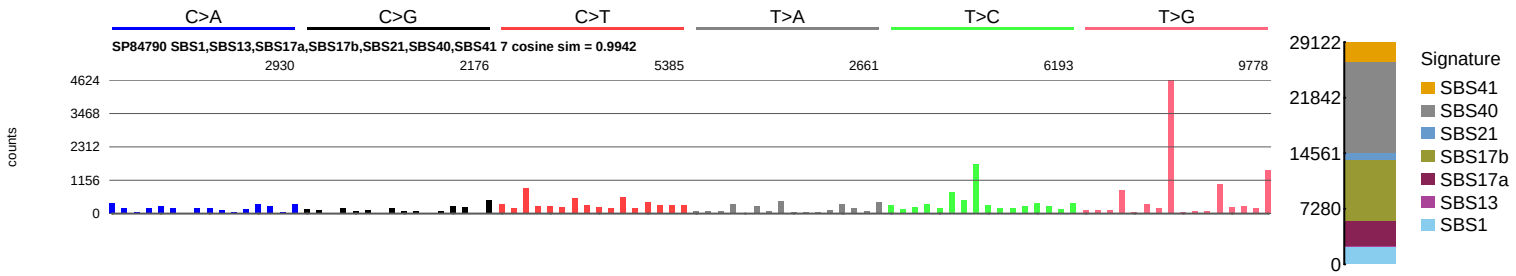

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.99423
3025
2150
5344
2656
6180
9762
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS1

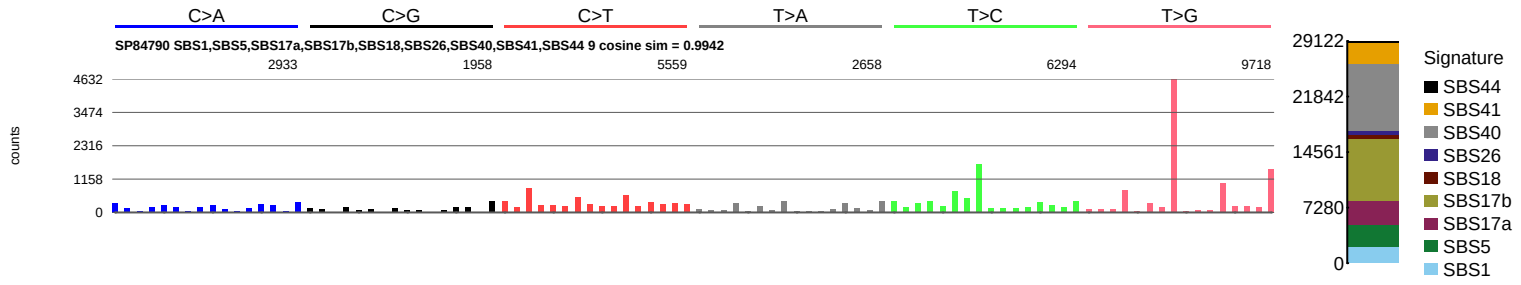

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.99422
3024
2019
5458
2671
6195
9754
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS1

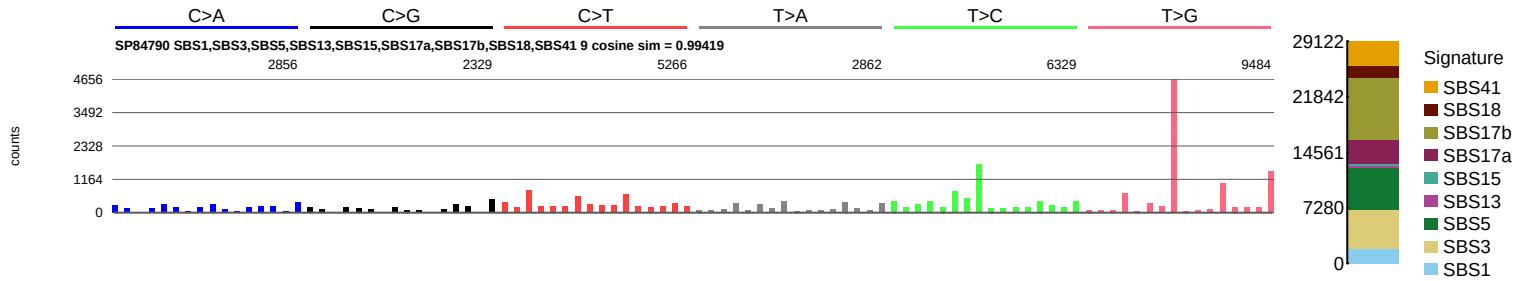

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.99421
2932
1953
5572
2657
6296
9716
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

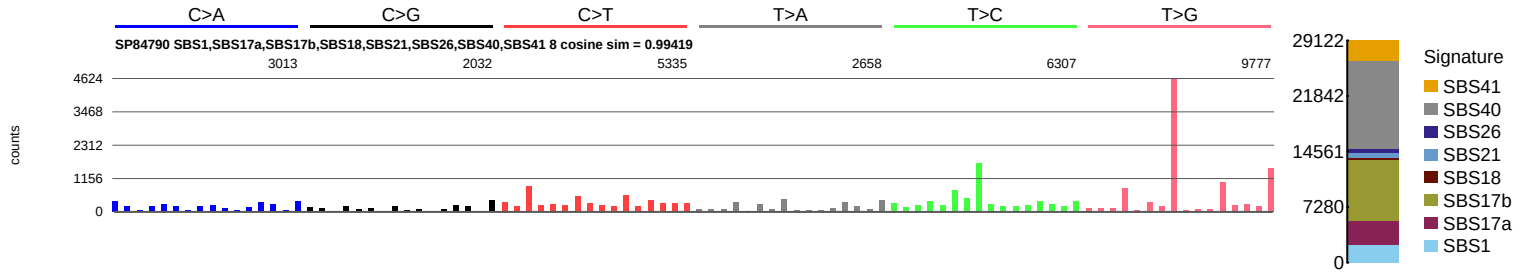

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 7 cosine sim = 0.9942
2932
2047
5493
2672
6205
9769
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS1

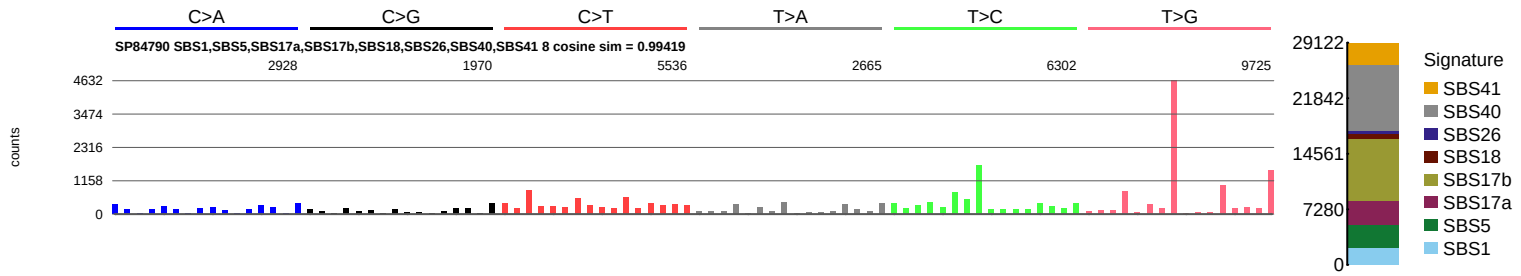

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS26,SBS40,SBS41 7 cosine sim = 0.9942
2801
2283
5226
2814
6226
9774
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS3
SBS1

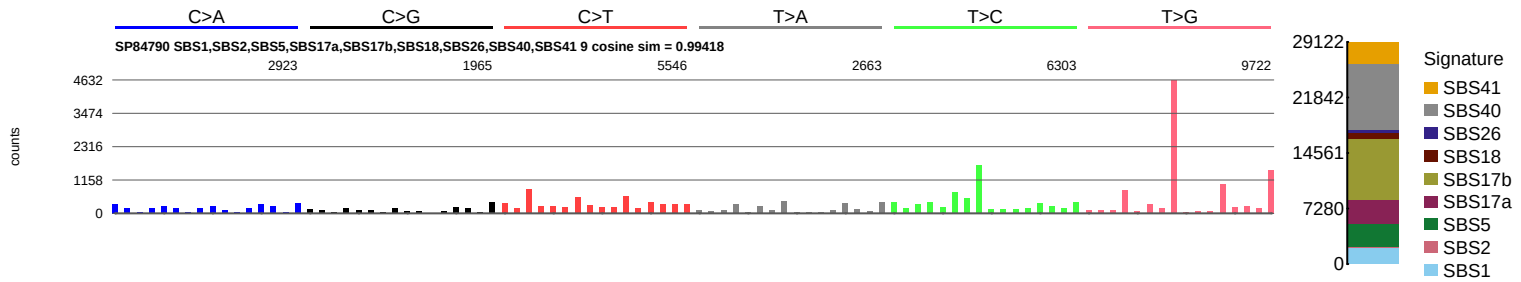

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS41 7 cosine sim = 0.9942
2930
2176
5385
2661
6193
9778
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS1

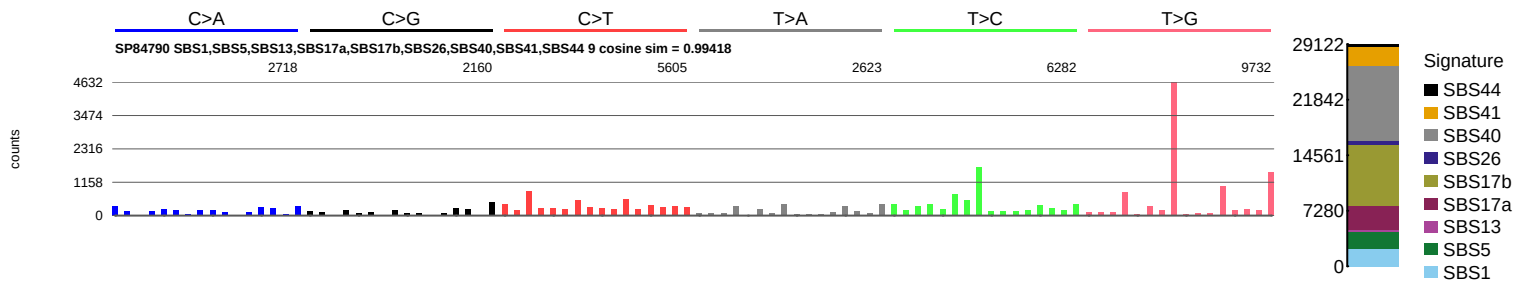

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.9942
2933
1958
5559
2658
6294
9718
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS1

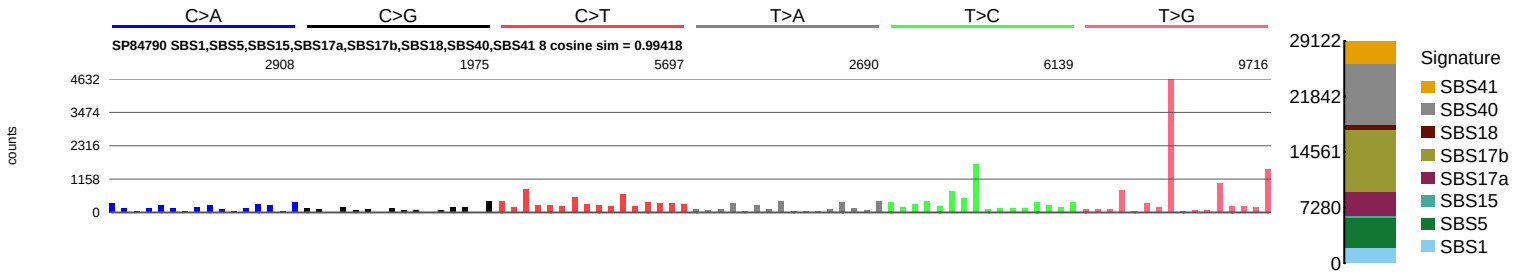

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.99419
2856
2329
5266
2862
6329
9484
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

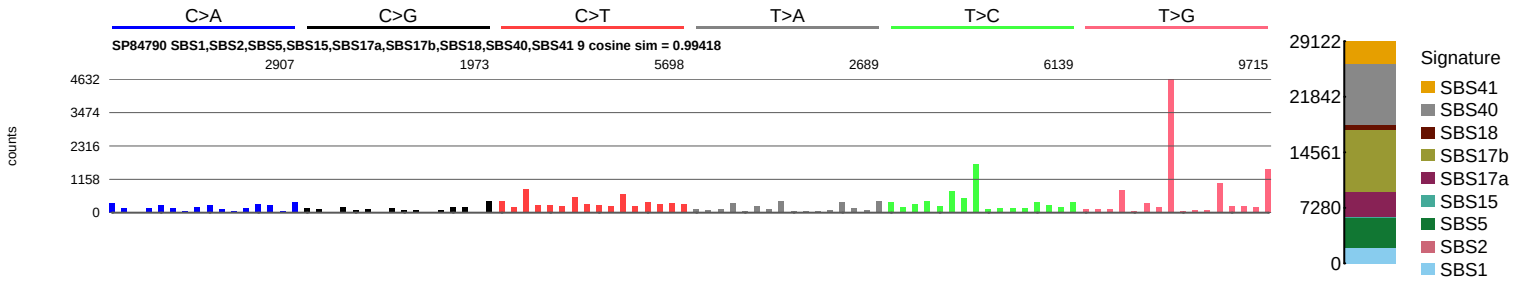

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 8 cosine sim = 0.99419
3013
2032
5335
2658
6307
9777
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS1

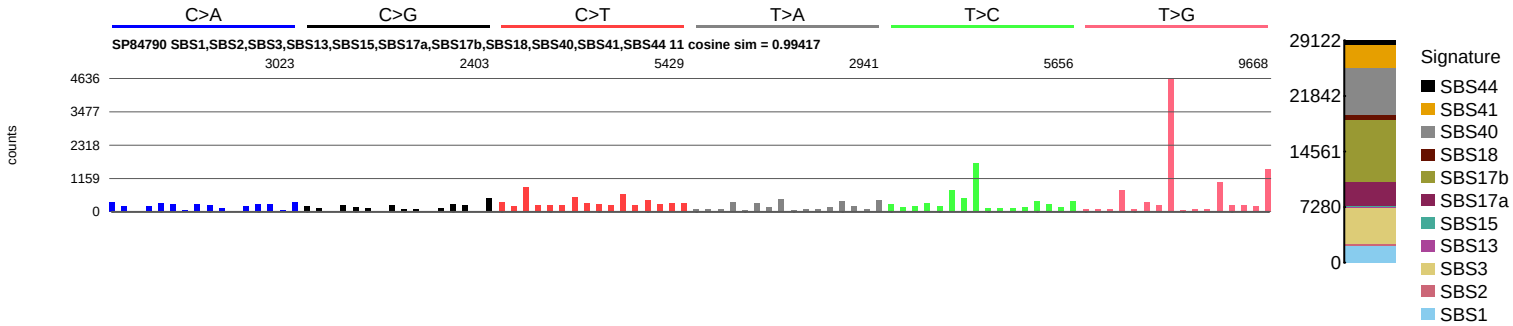

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.99419
2928
1970
5536
2665
6302
9725
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS1

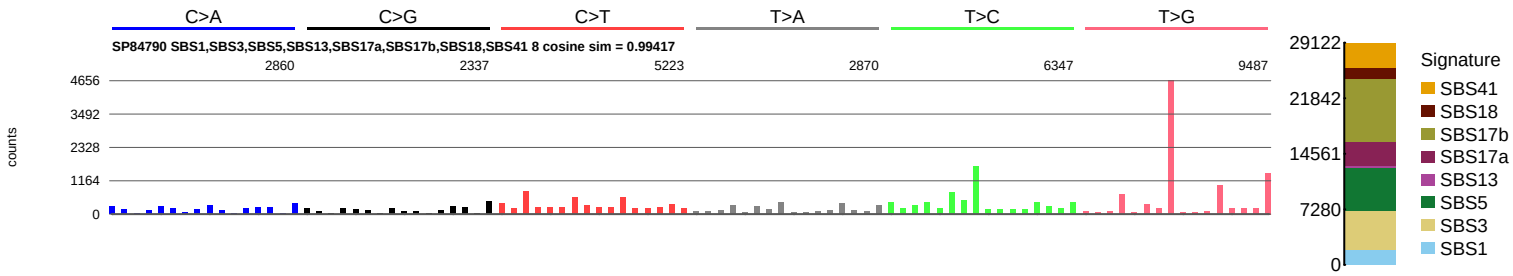

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.99418
2923
1965
5546
2663
6303
9722
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

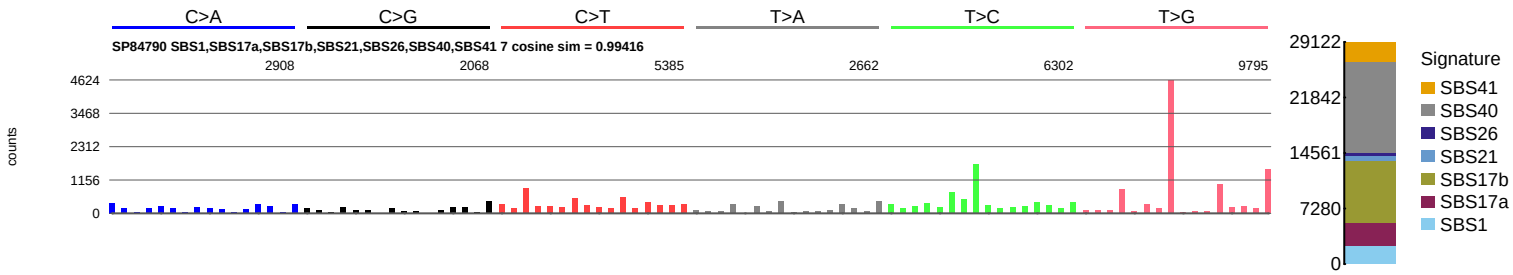

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.99418
2718
2160
5605
2623
6282
9732
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS1

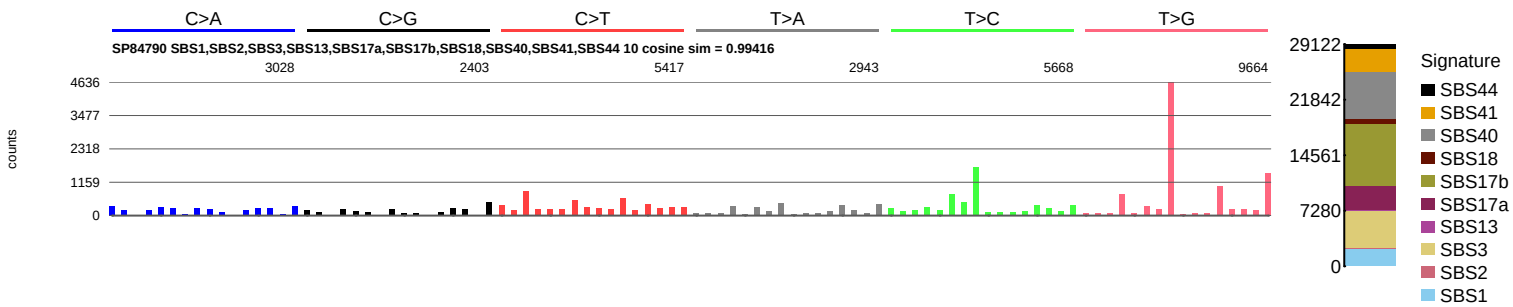

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.99418
2908
1975
5697
2690
6139
9716
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1

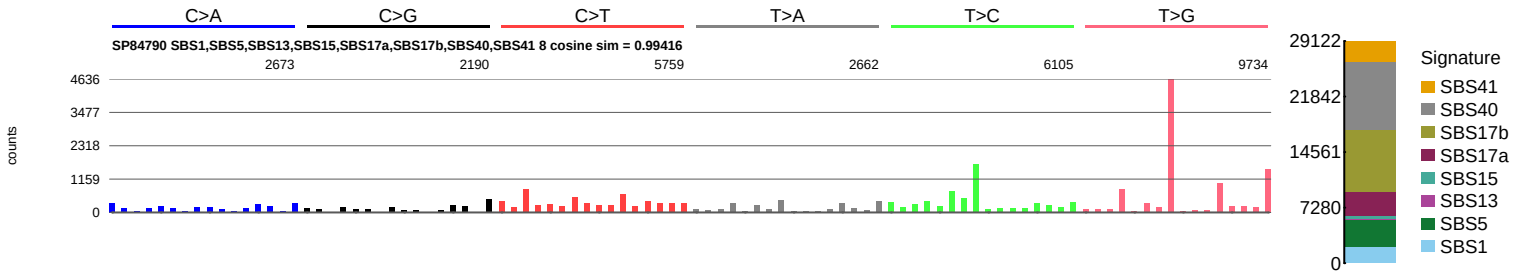

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.99418
2907
1973
5698
2689
6139
9715
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

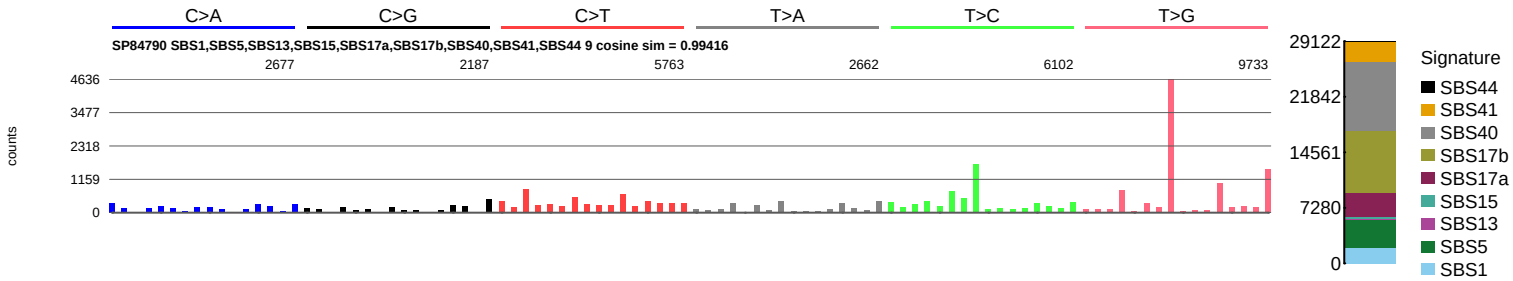

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 11 cosine sim = 0.99417
3023
2403
5429
2941
5656
9668
counts
4636
3477
2318
1159
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

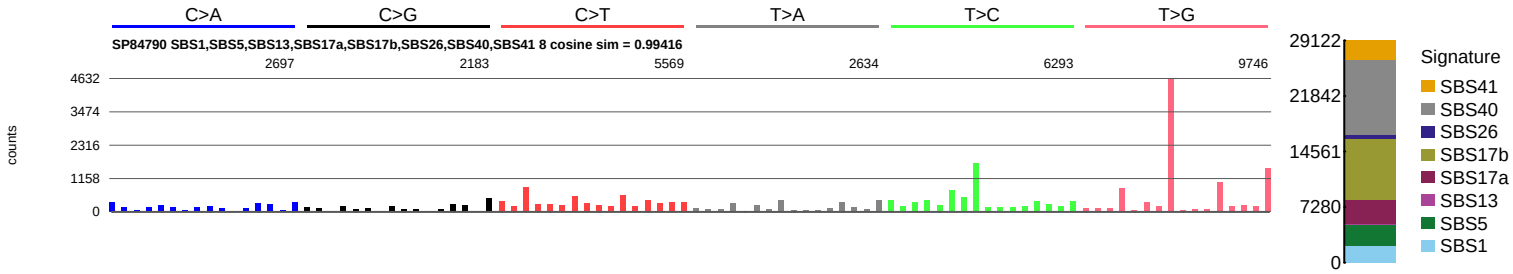

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.99417
2860
2337
5223
2870
6347
9487
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

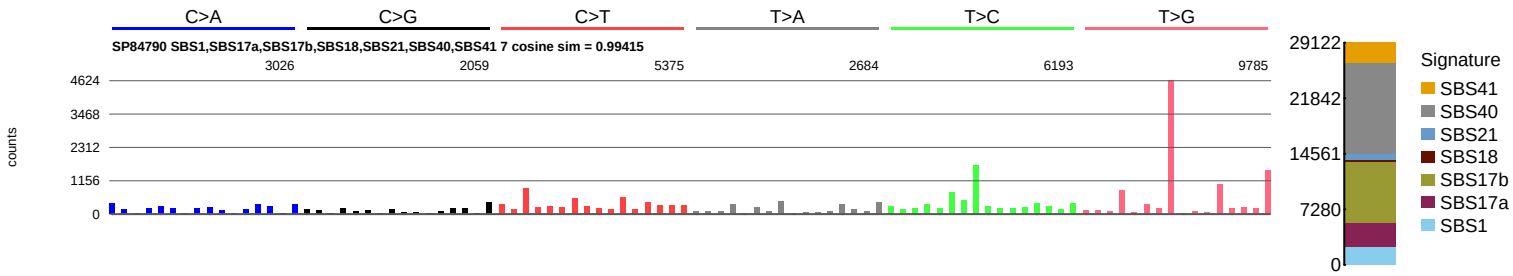

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41 7 cosine sim = 0.99416
2908
2068
5385
2662
6302
9795
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS1

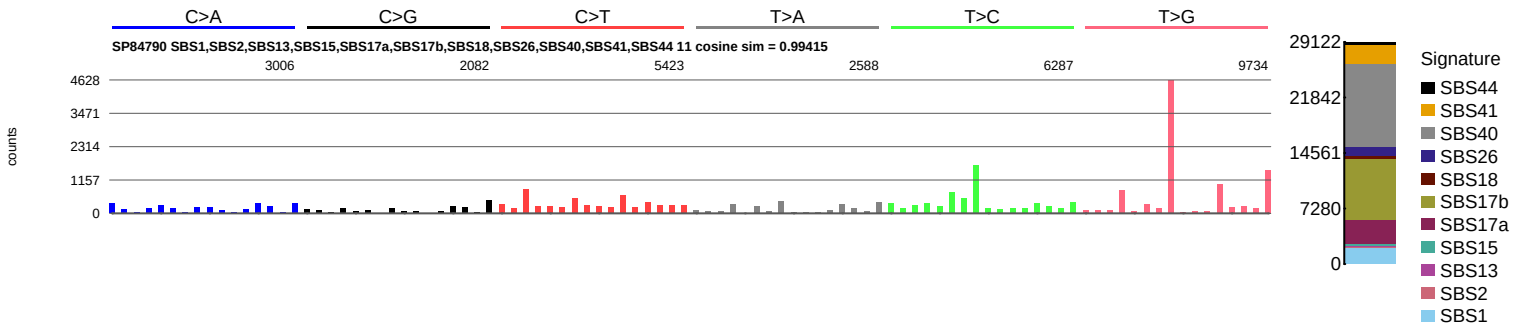

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.99416
3028
2403
5417
2943
5668
9664
counts
4636
3477
2318
1159
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

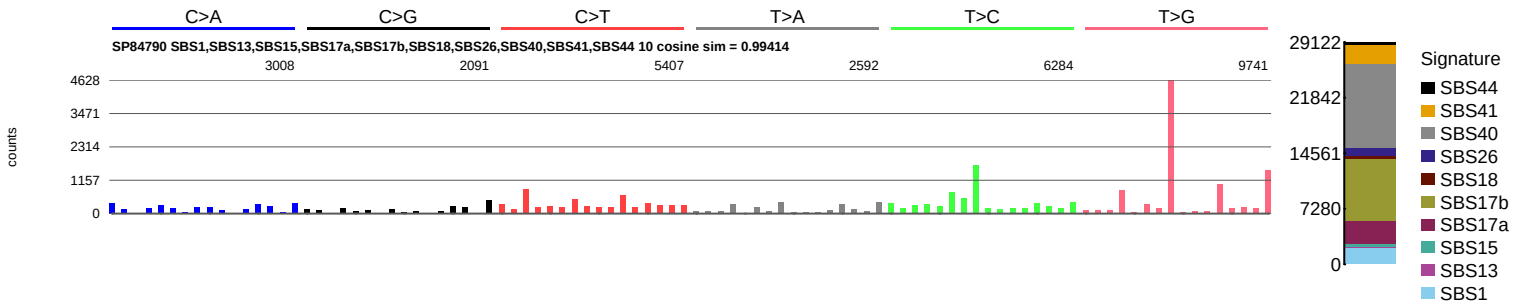

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41 8 cosine sim = 0.99416
2673
2190
5759
2662
6105
9734
counts
4636
3477
2318
1159
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

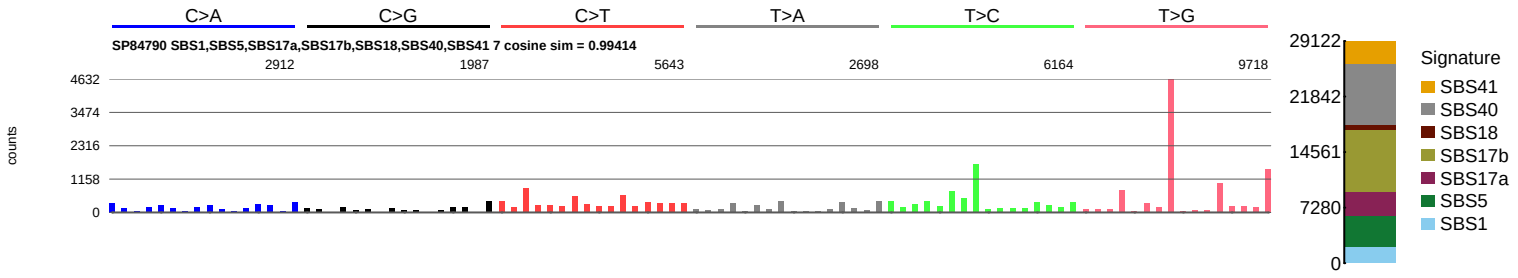

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41,SBS44 9 cosine sim = 0.99416
2677
2187
5763
2662
6102
9733
counts
4636
3477
2318
1159
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

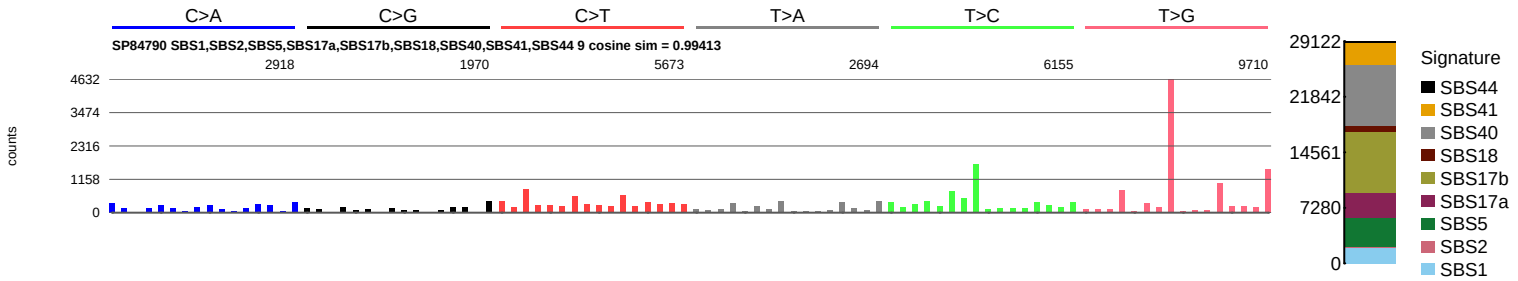

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS41 8 cosine sim = 0.99416
2697
2183
5569
2634
6293
9746
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS1

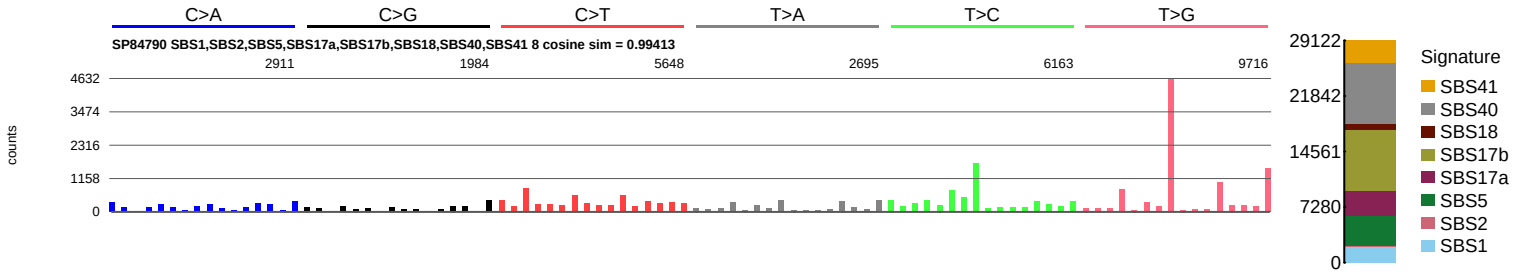

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.99415
3026
2059
5375
2684
6193
9785
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS1

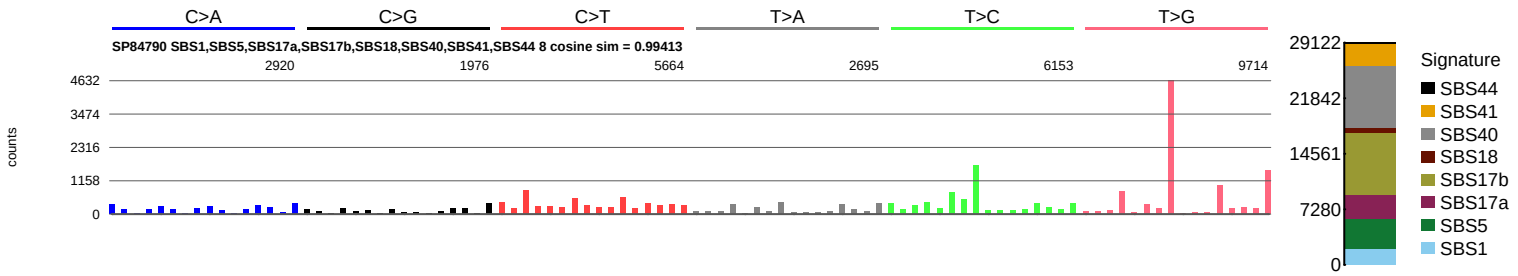

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.99415
3006
2082
5423
2588
6287
9734
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

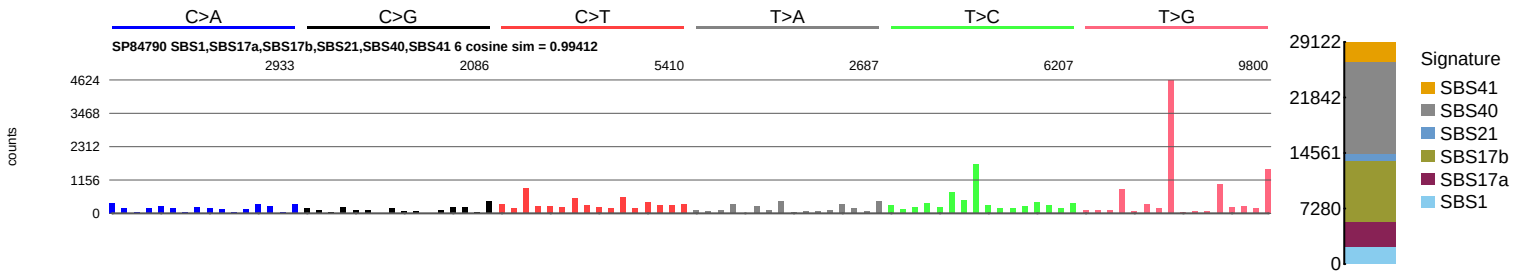

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.99414
3008
2091
5407
2592
6284
9741
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1

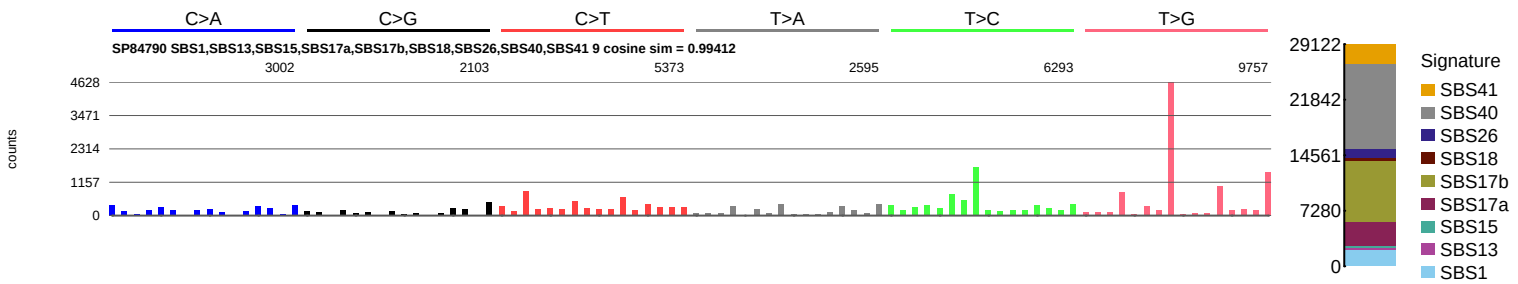

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.99414
2912
1987
5643
2698
6164
9718
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS1

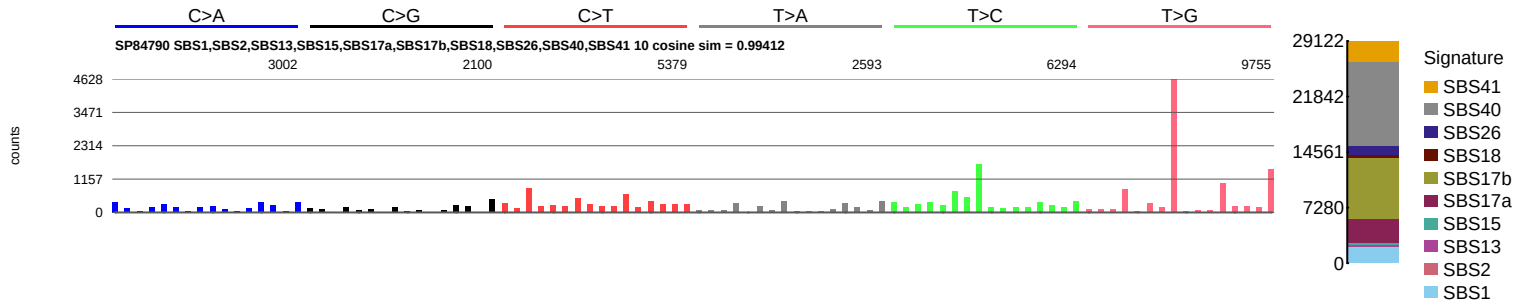

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.99413
2918
1970
5673
2694
6155
9710
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

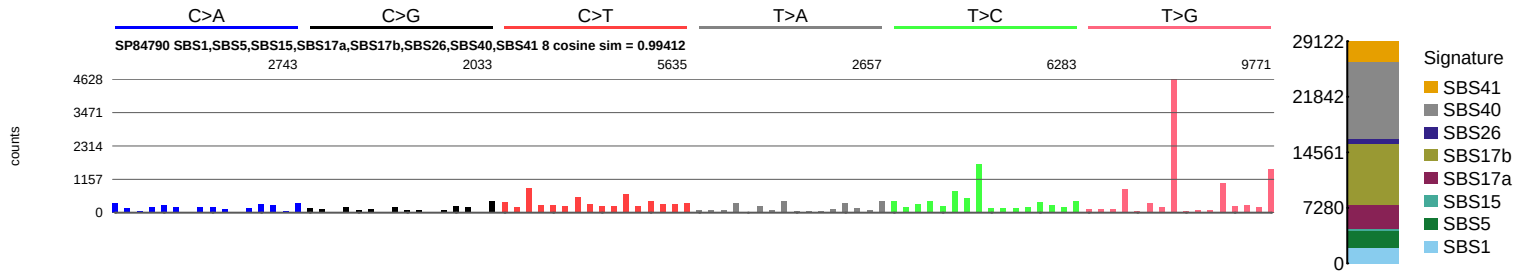

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.99413
2911
1984
5648
2695
6163
9716
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

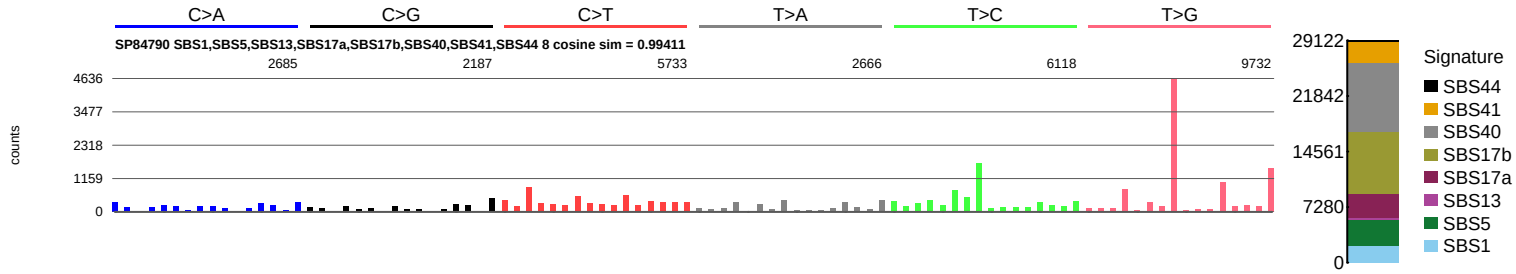

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.99413
2920
1976
5664
2695
6153
9714
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS1

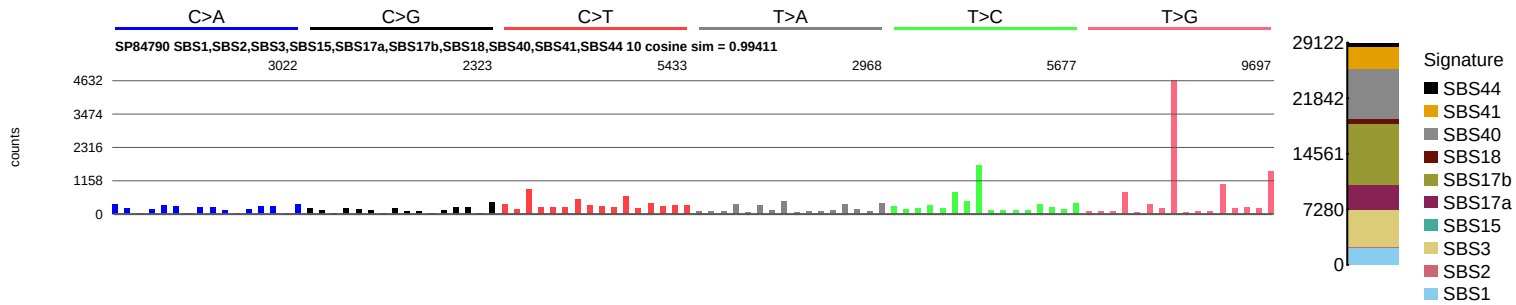

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS17a,SBS17b,SBS21,SBS40,SBS41 6 cosine sim = 0.99412
2933
2086
5410
2687
6207
9800
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS1

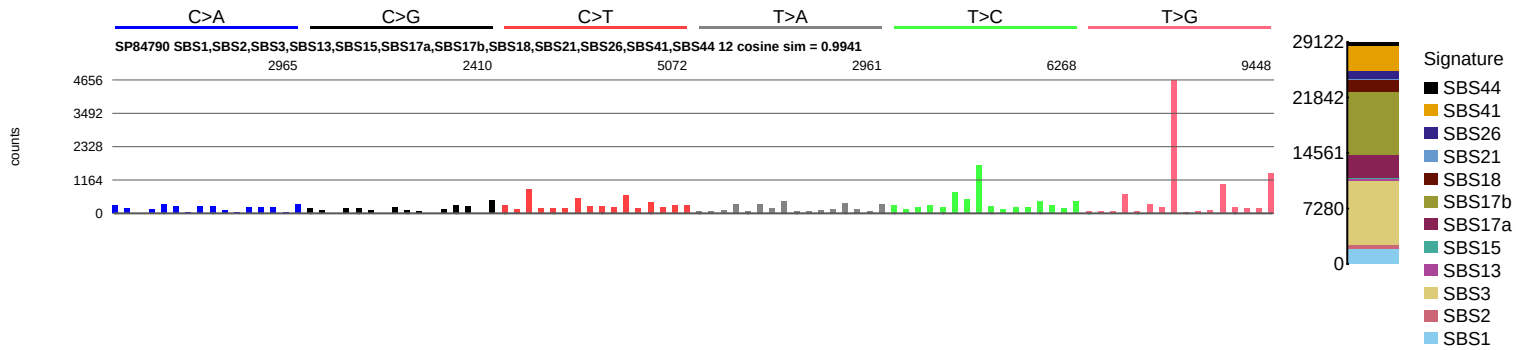

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.99412
3002
2103
5373
2595
6293
9757
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1

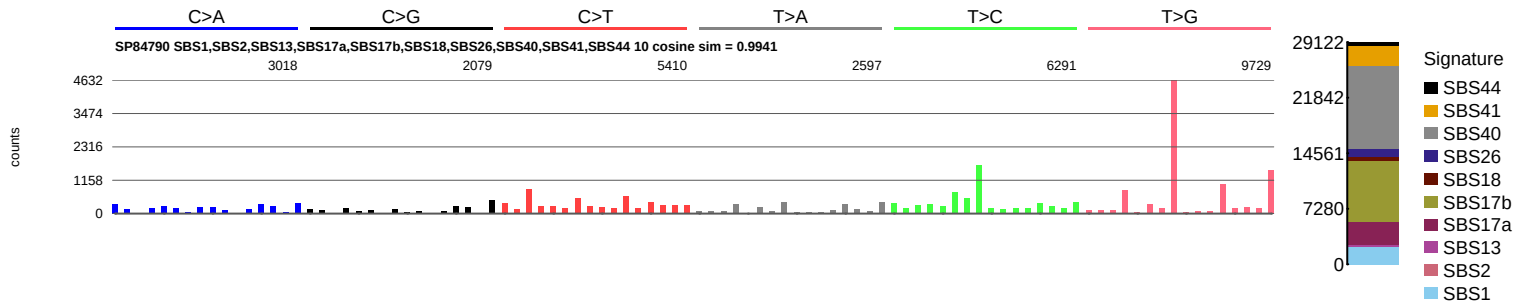

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.99412
3002
2100
5379
2593
6294
9755
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

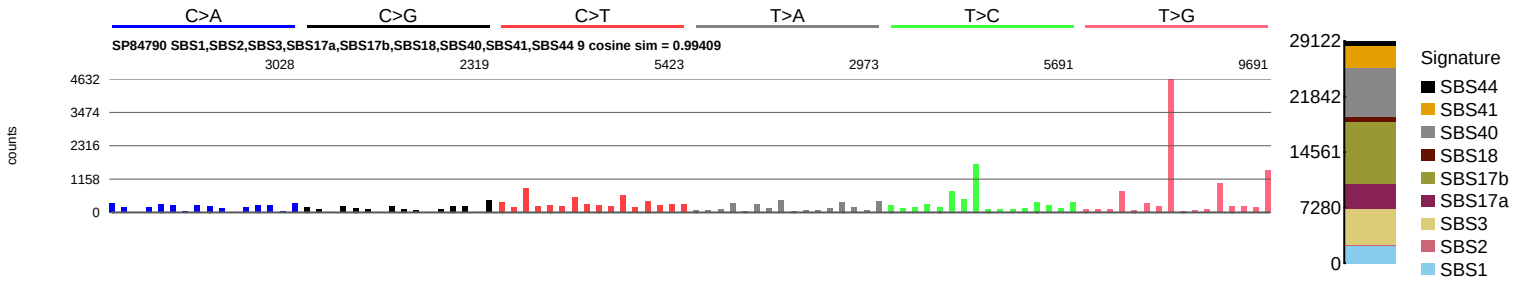

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 8 cosine sim = 0.99412
2743
2033
5635
2657
6283
9771
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS1

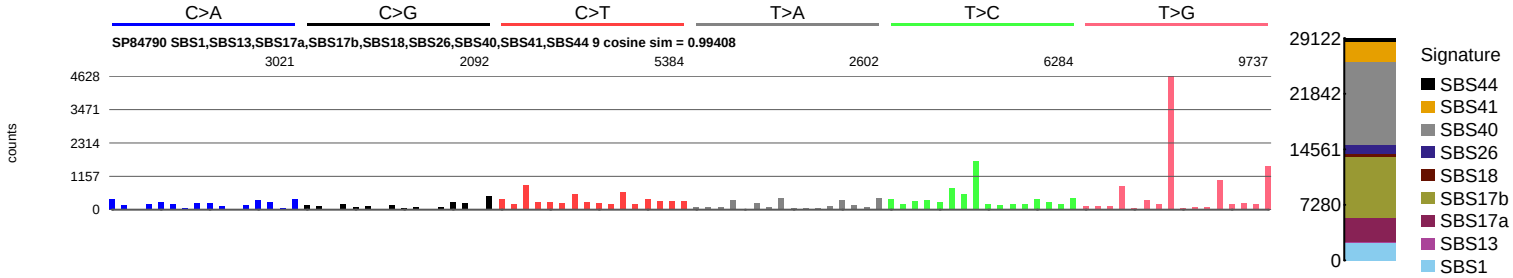

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS40,SBS41,SBS44 8 cosine sim = 0.99411
2685
2187
5733
2666
6118
9732
counts
4636
3477
2318
1159
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS13
SBS5
SBS1

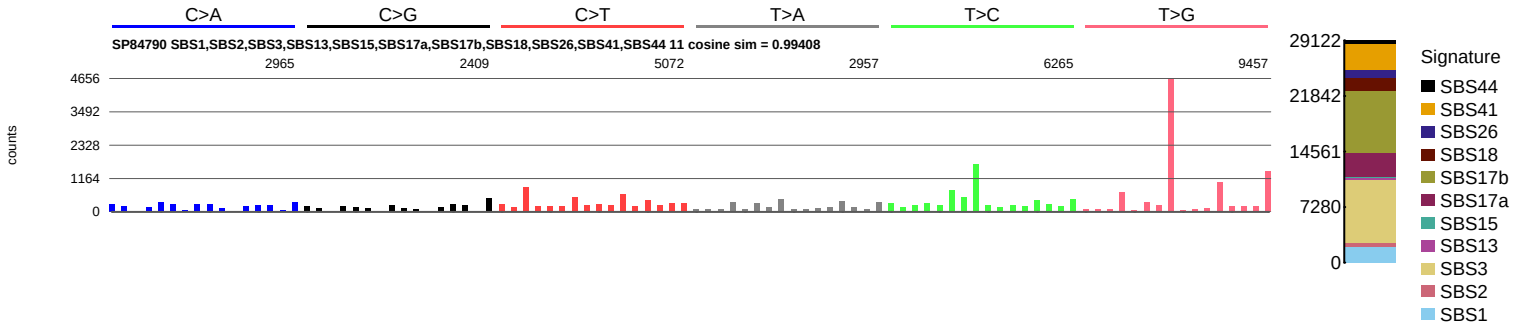

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.99411
3022
2323
5433
2968
5677
9697
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

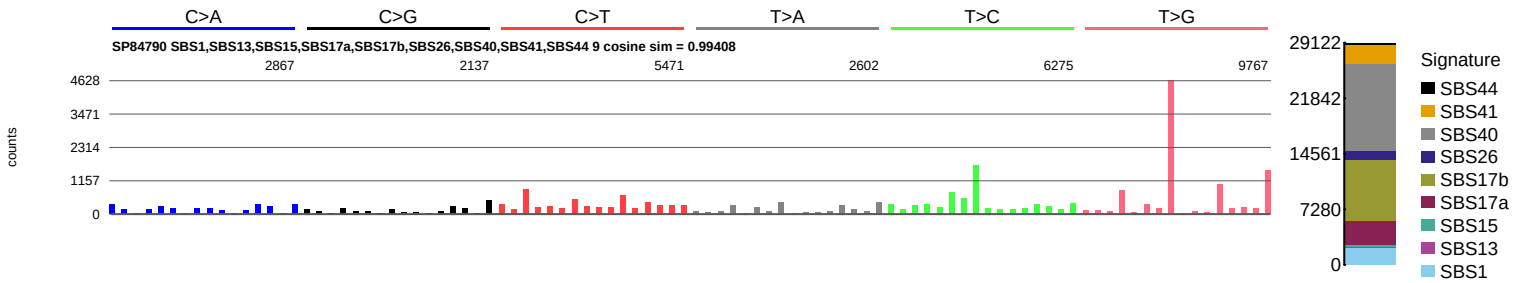

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 12 cosine sim = 0.9941
2965
2410
5072
2961
6268
9448
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

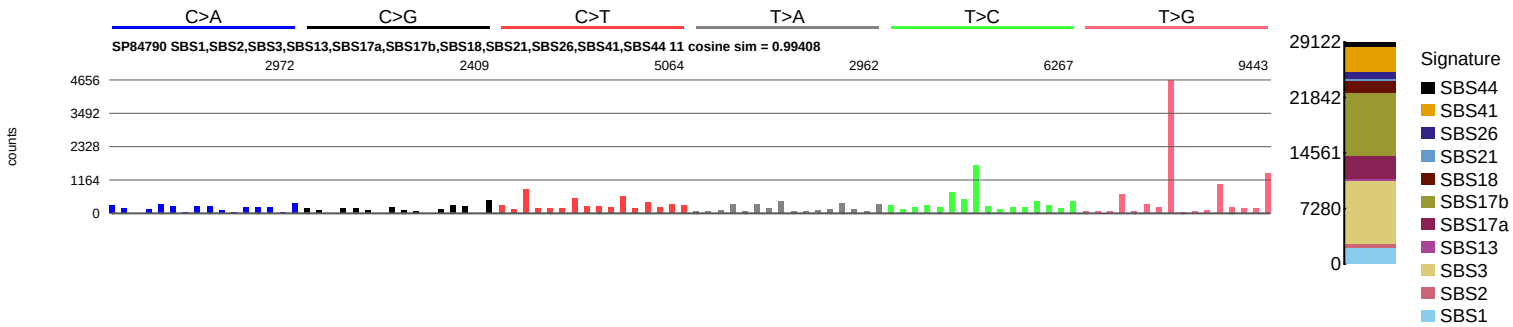

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.9941
3018
2079
5410
2597
6291
9729
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

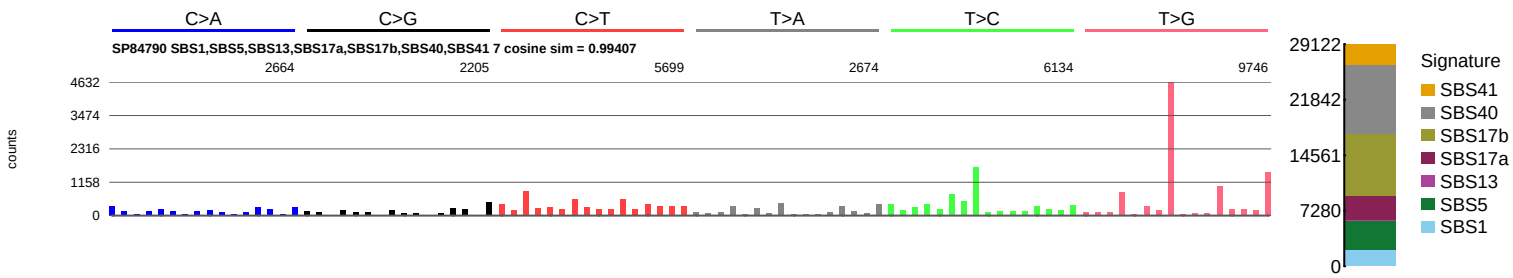

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.99409
3028
2319
5423
2973
5691
9691
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

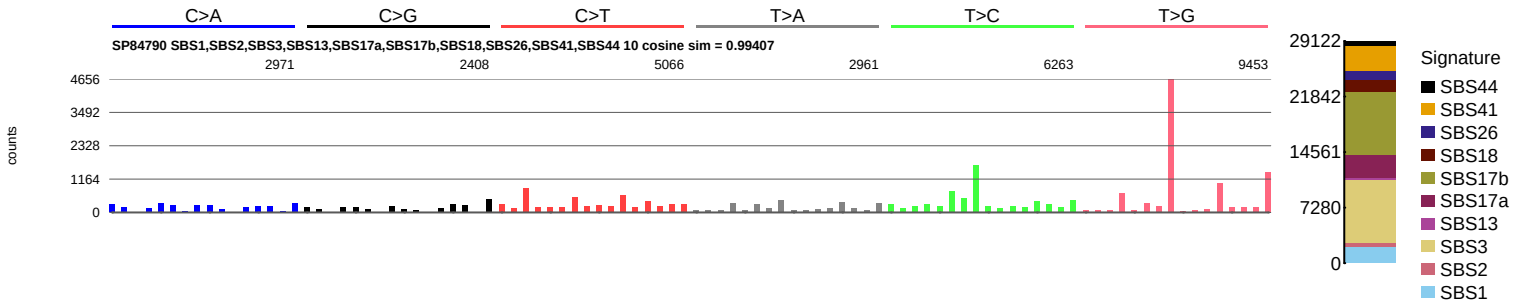

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.99408
3021
2092
5384
2602
6284
9737
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS1

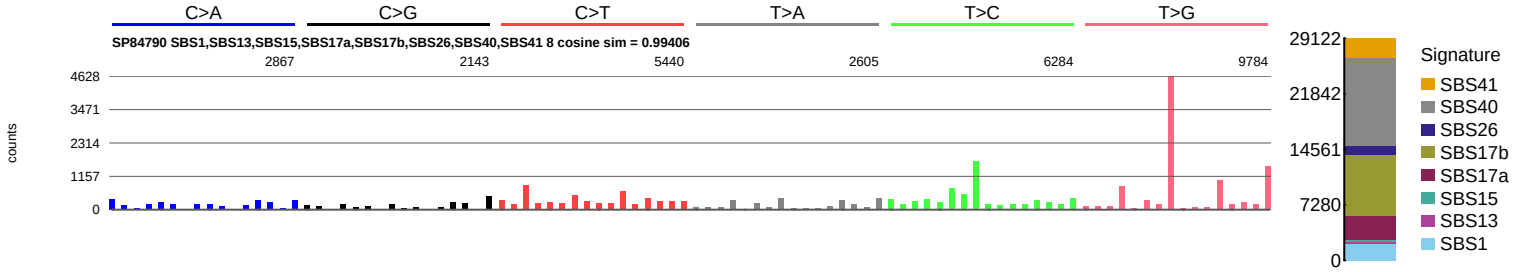

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 11 cosine sim = 0.99408
2965
2409
5072
2957
6265
9457
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

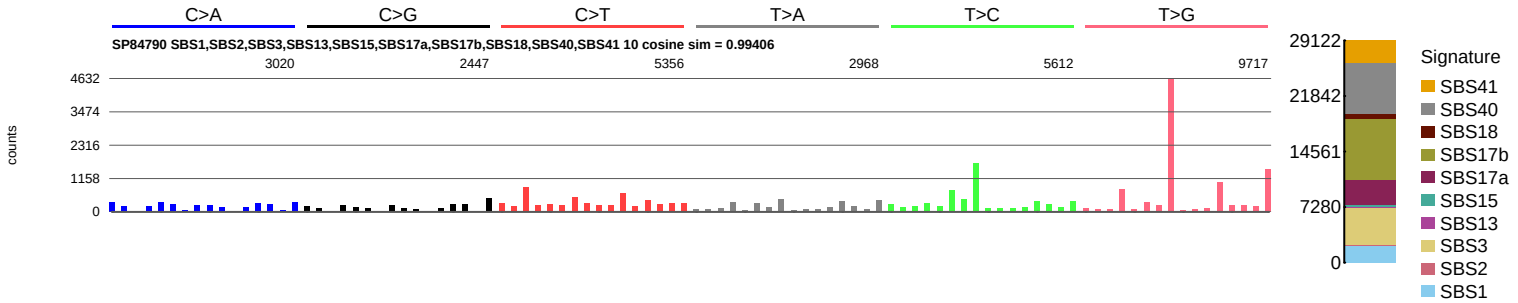

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.99408
2867
2137
5471
2602
6275
9767
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS1

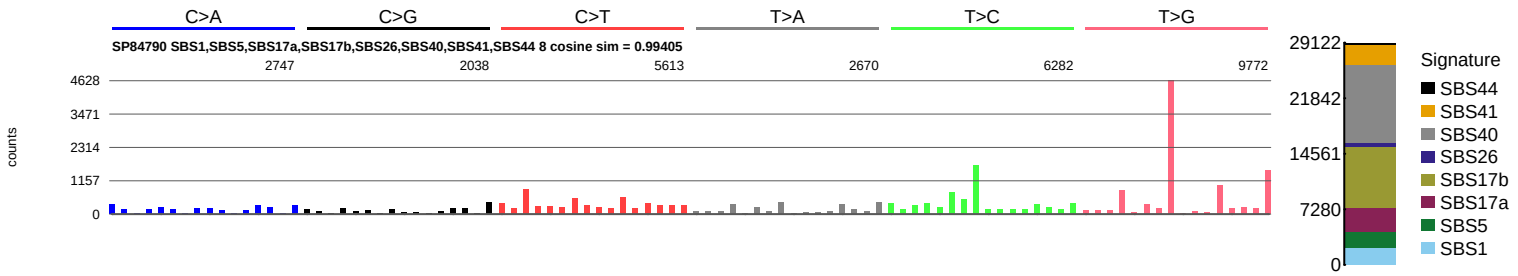

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 11 cosine sim = 0.99408
2972
2409
5064
2962
6267
9443
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

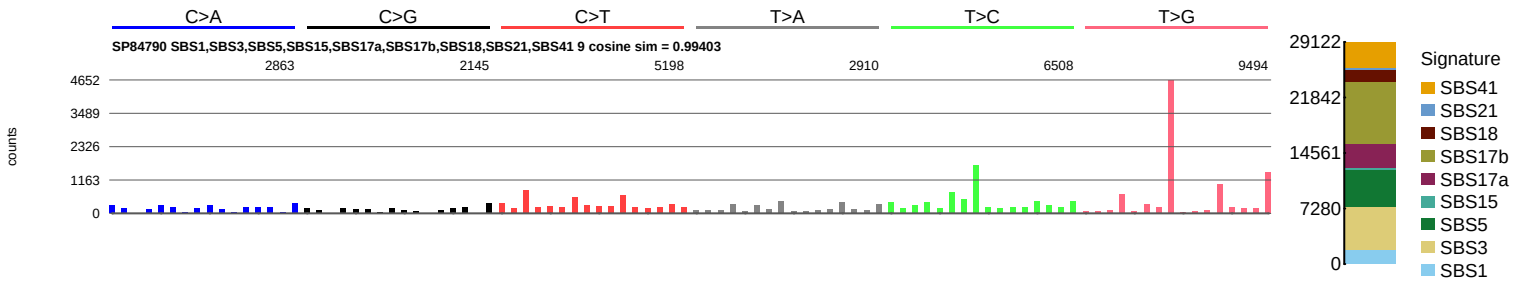

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.99407
2664
2205
5699
2674
6134
9746
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS13
SBS5
SBS1

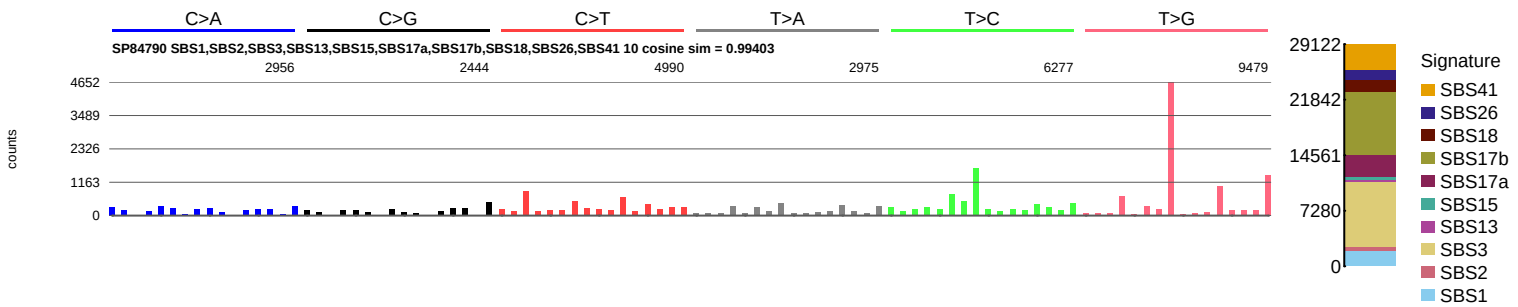

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.99407
2971
2408
5066
2961
6263
9453
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

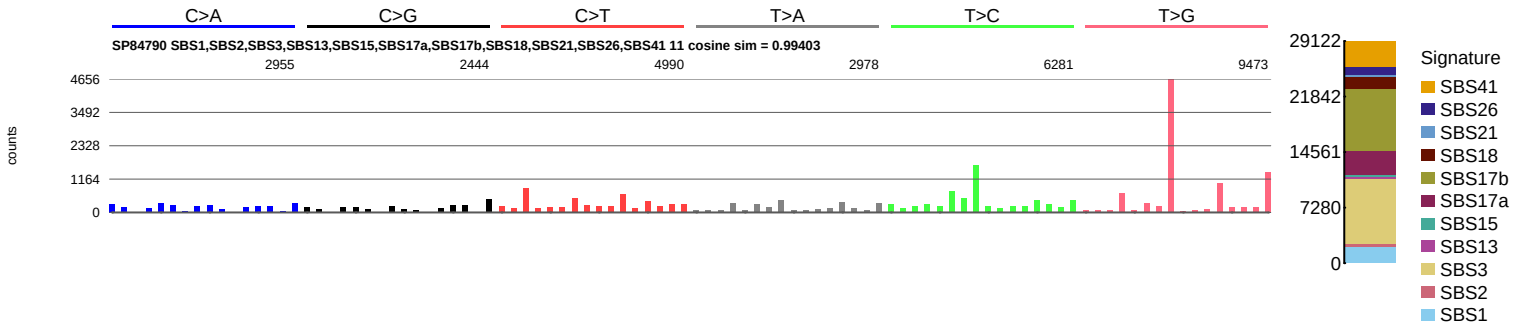

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 8 cosine sim = 0.99406
2867
2143
5440
2605
6284
9784
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS1

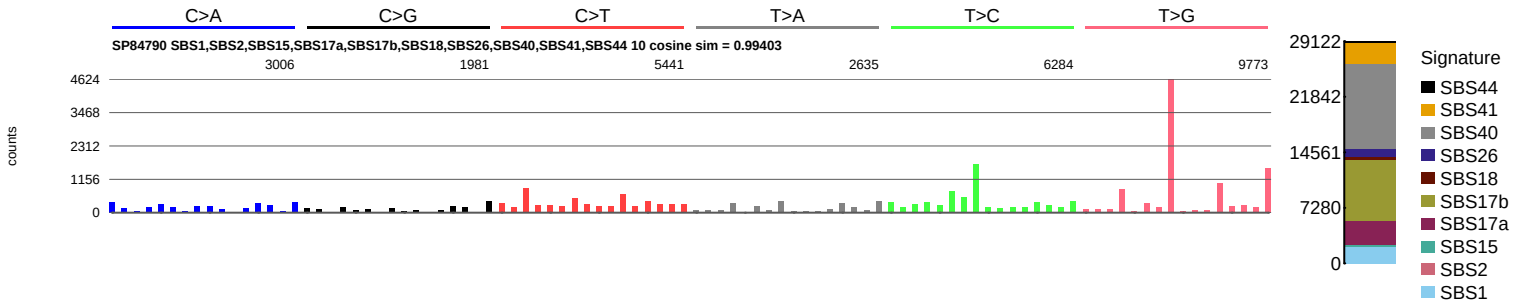

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 10 cosine sim = 0.99406
3020
2447
5356
2968
5612
9717
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

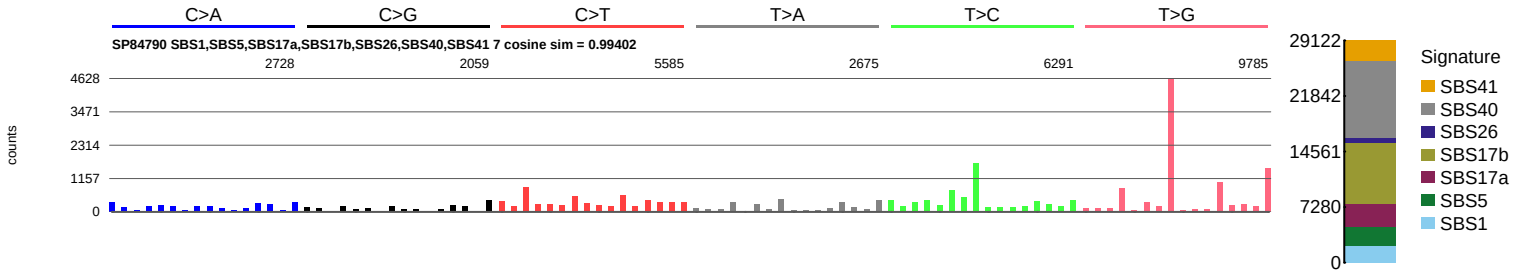

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.99405
2747
2038
5613
2670
6282
9772
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS5
SBS1

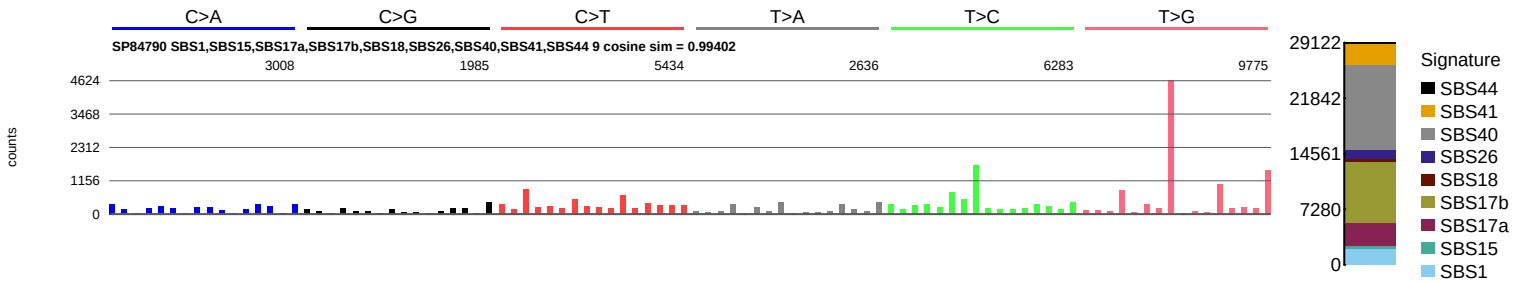

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.99403
2863
2145
5198
2910
6508
9494
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1

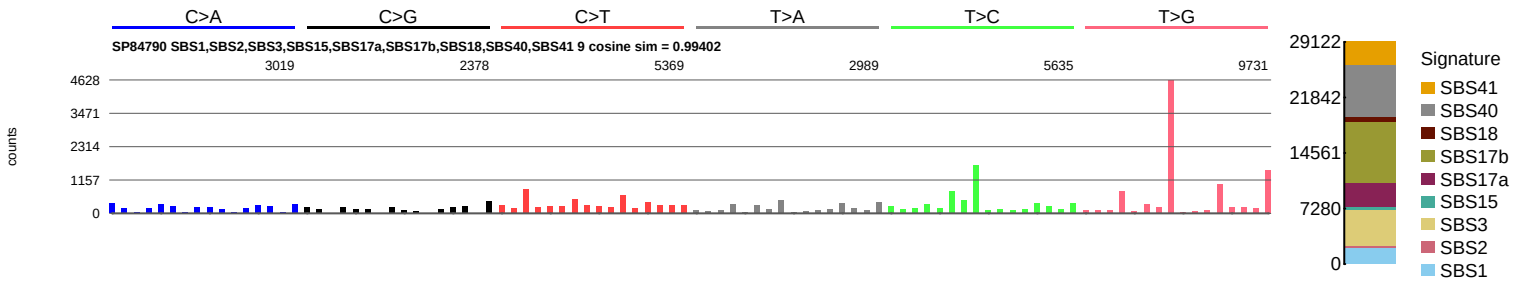

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 10 cosine sim = 0.99403
2956
2444
4990
2975
6277
9479
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

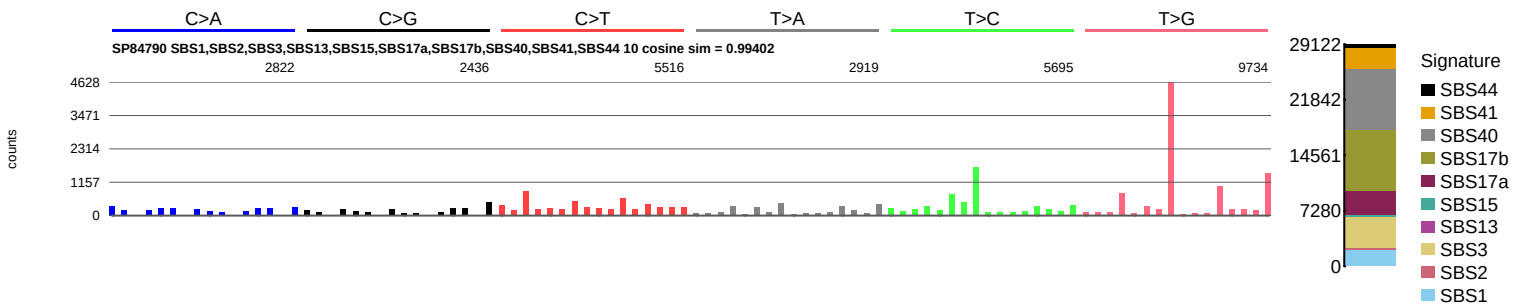

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 11 cosine sim = 0.99403
2955
2444
4990
2978
6281
9473
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

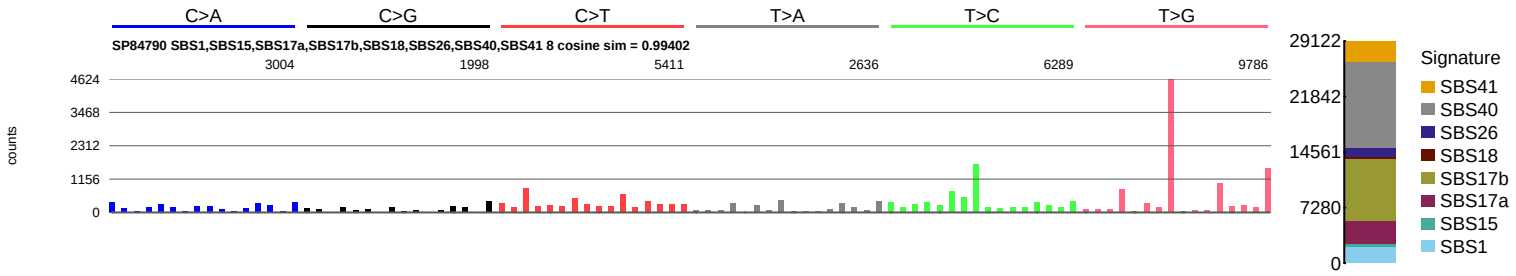

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.99403
3006
1981
5441
2635
6284
9773
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

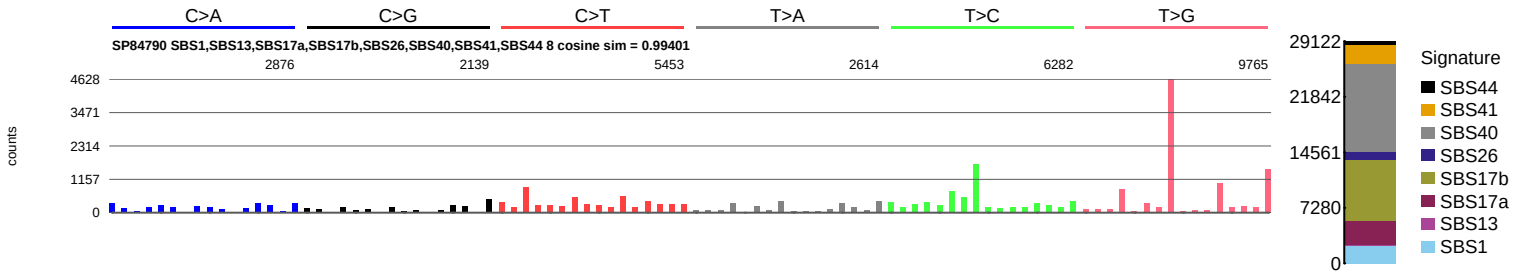

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS17a,SBS17b,SBS26,SBS40,SBS41 7 cosine sim = 0.99402
2728
2059
5585
2675
6291
9785
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS5
SBS1

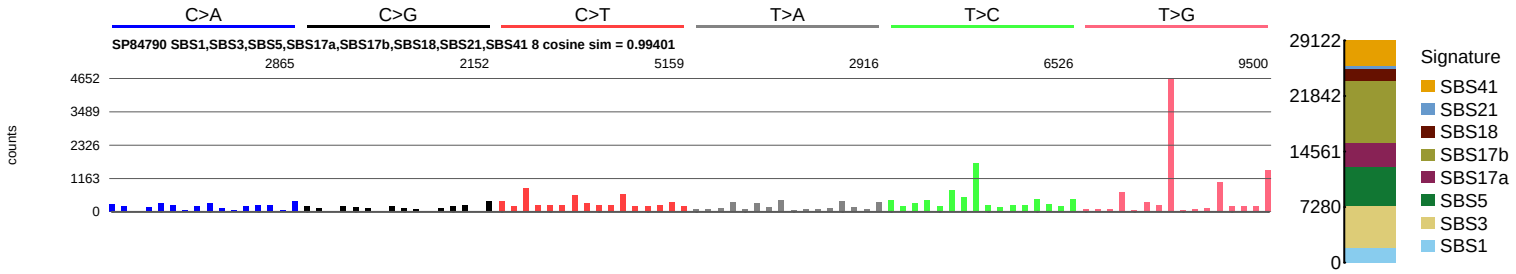

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.99402
3008
1985
5434
2636
6283
9775
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1

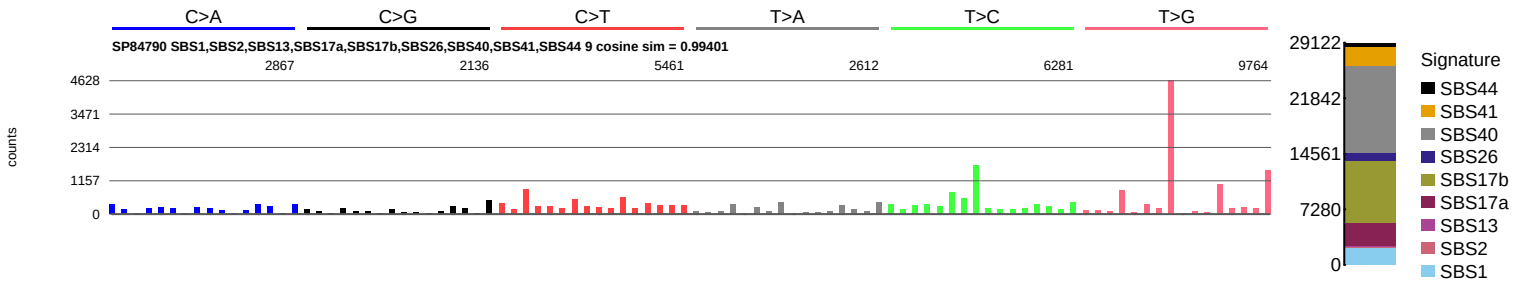

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.99402
3019
2378
5369
2989
5635
9731
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

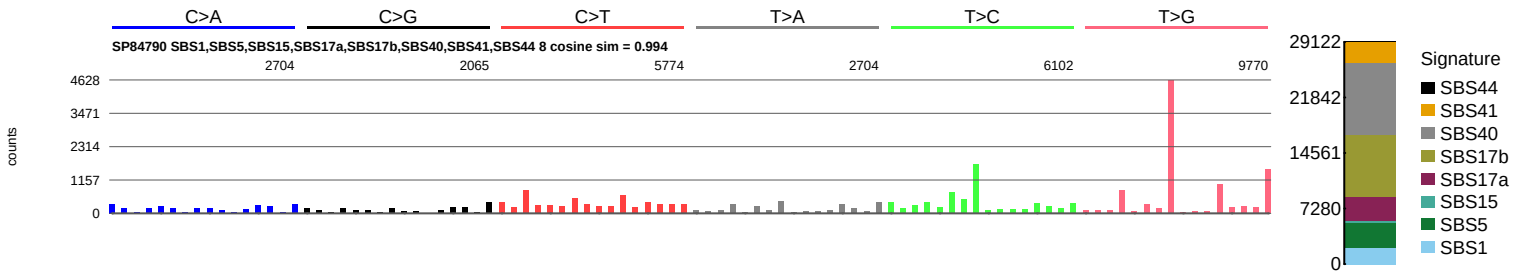

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41,SBS44 10 cosine sim = 0.99402
2822
2436
5516
2919
5695
9734
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

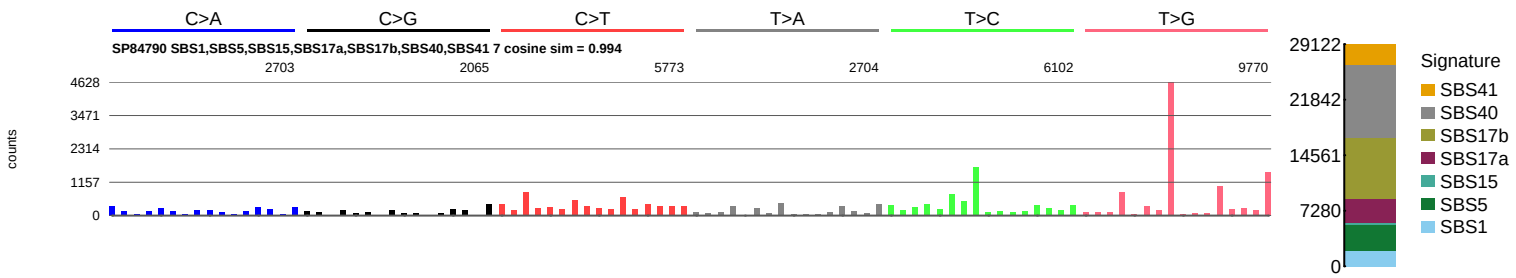

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.99402
3004
1998
5411
2636
6289
9786
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1

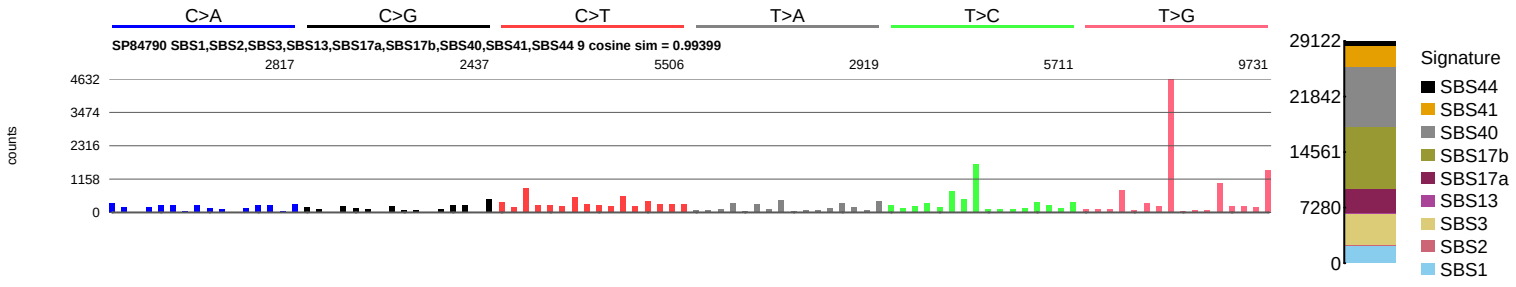

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.99401
2876
2139
5453
2614
6282
9765
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS1

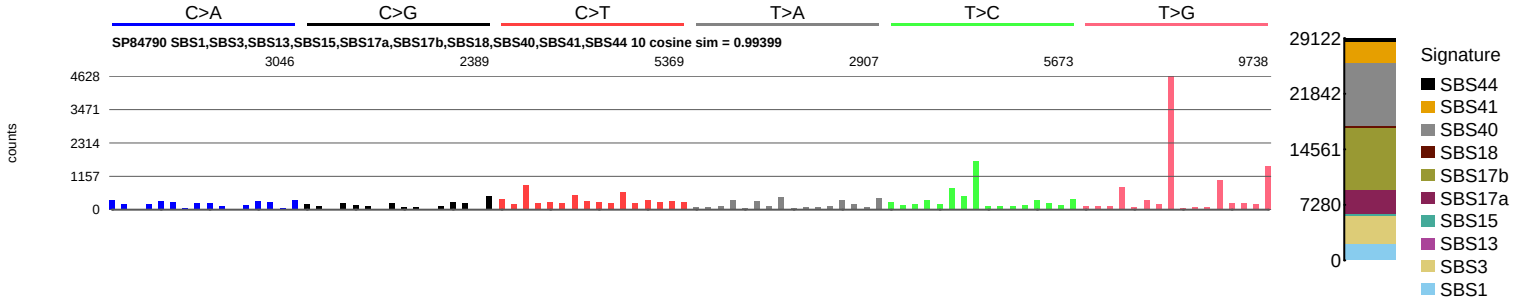

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.99401
2865
2152
5159
2916
6526
9500
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1

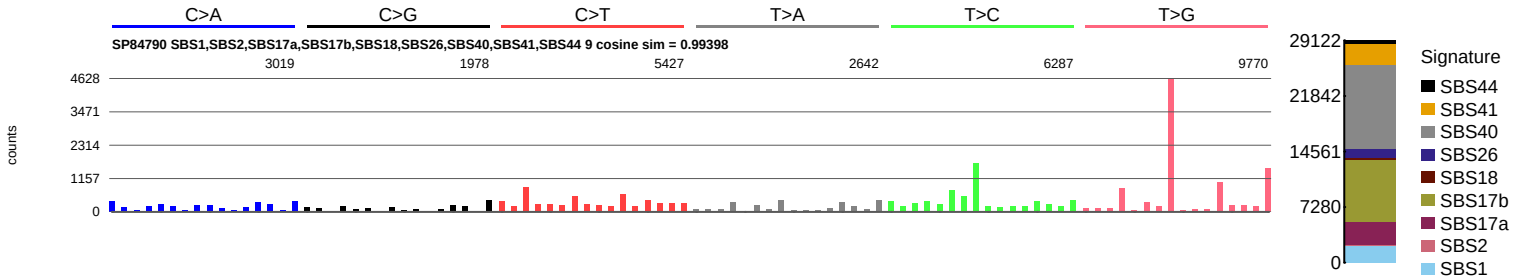

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.99401
2867
2136
5461
2612
6281
9764
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS2
SBS1

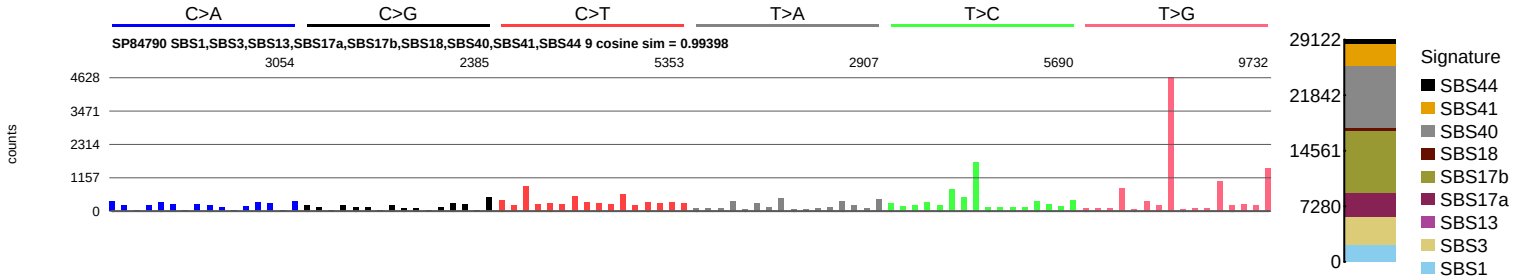

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS40,SBS41,SBS44 8 cosine sim = 0.994
2704
2065
5774
2704
6102
9770
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS1

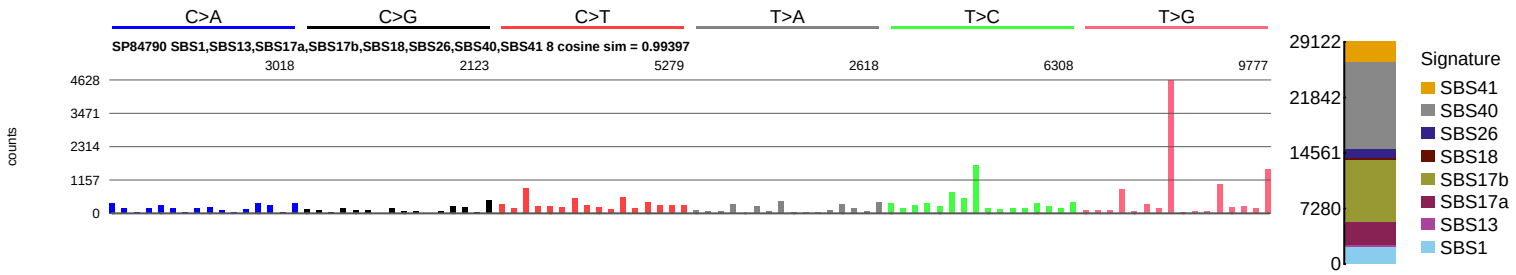

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.994
2703
2065
5773
2704
6102
9770
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS1

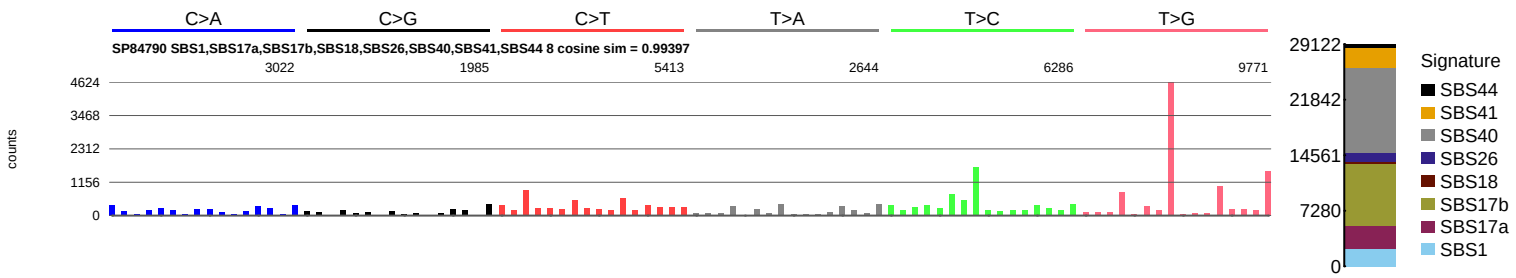

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS40,SBS41,SBS44 9 cosine sim = 0.99399
2817
2437
5506
2919
5711
9731
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

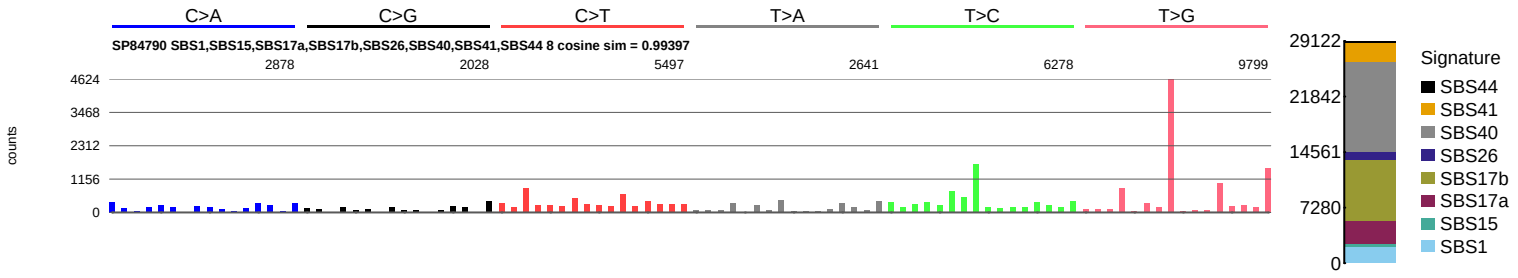

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.99399
3046
2389
5369
2907
5673
9738
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

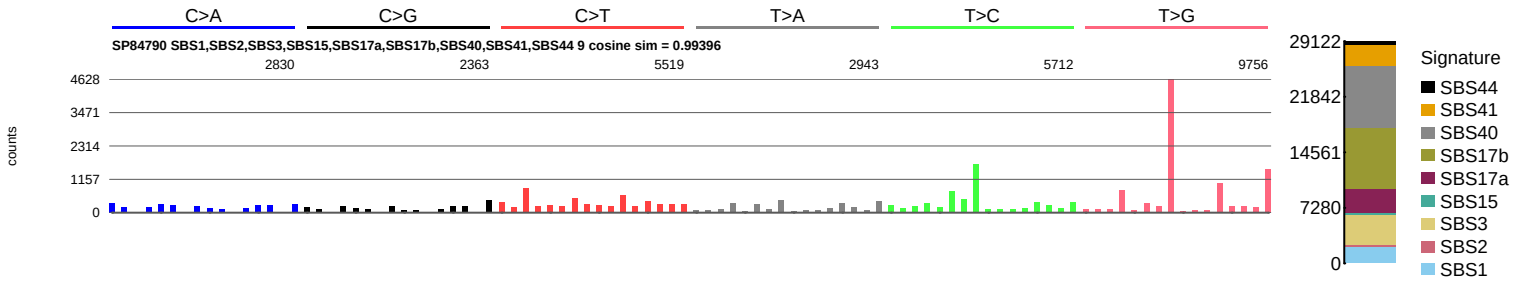

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.99398
3019
1978
5427
2642
6287
9770
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1

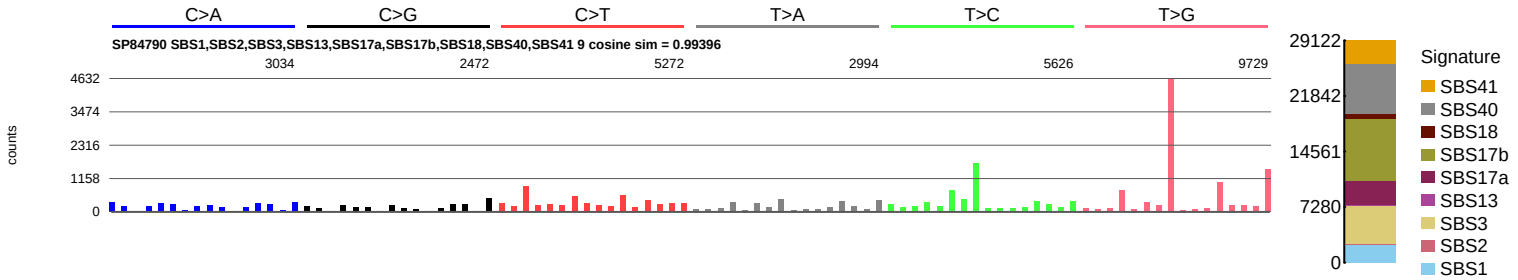

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.99398
3054
2385
5353
2907
5690
9732
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1

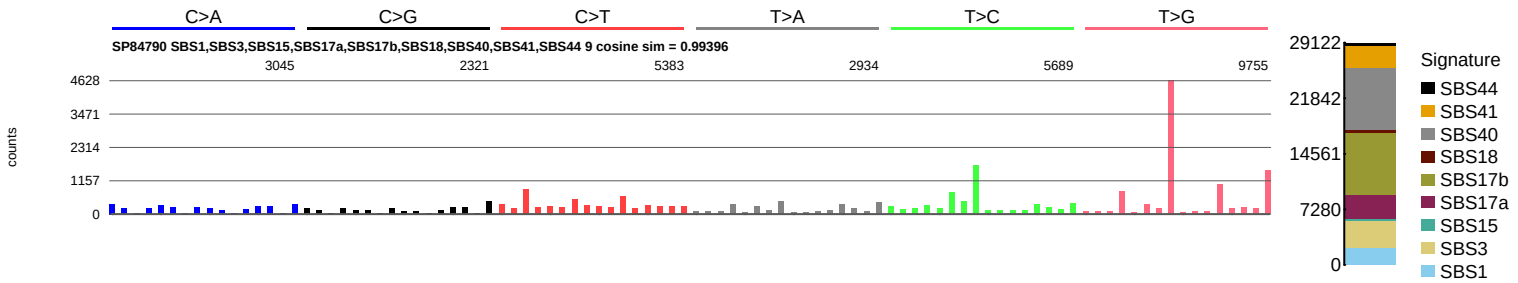

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.99397
3018
2123
5279
2618
6308
9777
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS1

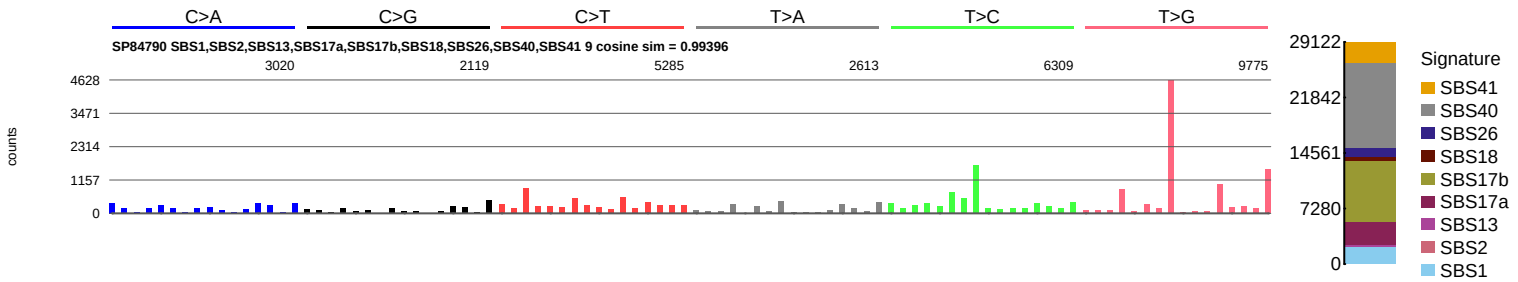

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.99397
3022
1985
5413
2644
6286
9771
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS1

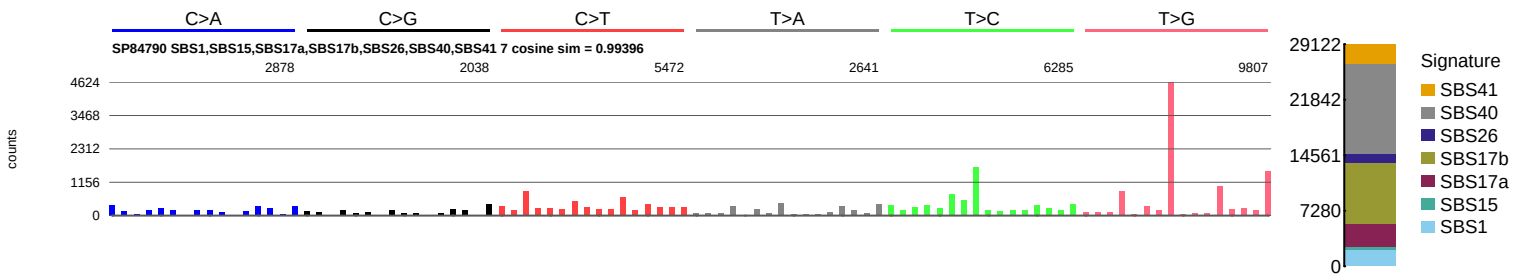

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.99397
2878
2028
5497
2641
6278
9799
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS1

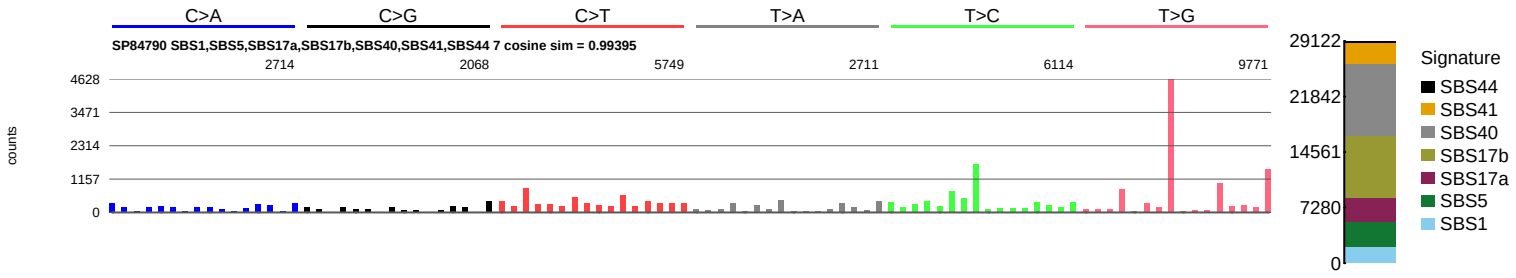

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS40,SBS41,SBS44 9 cosine sim = 0.99396
2830
2363
5519
2943
5712
9756
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

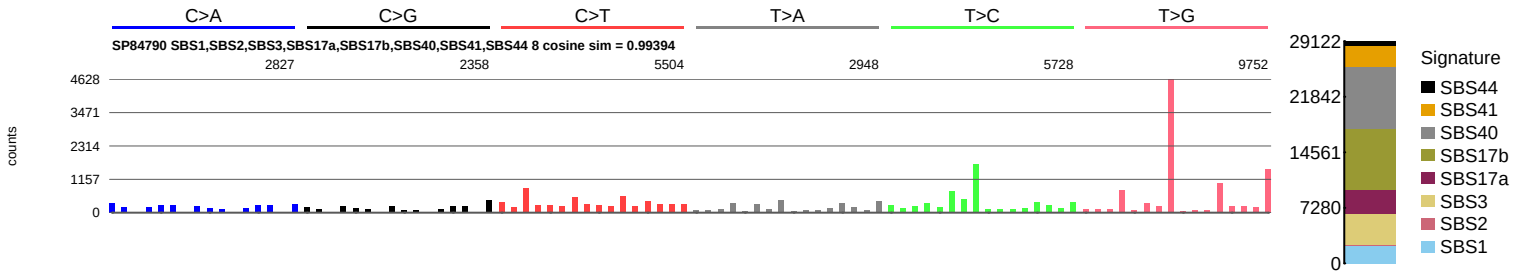

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.99396
3034
2472
5272
2994
5626
9729
counts
4632
3474
2316
1158
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

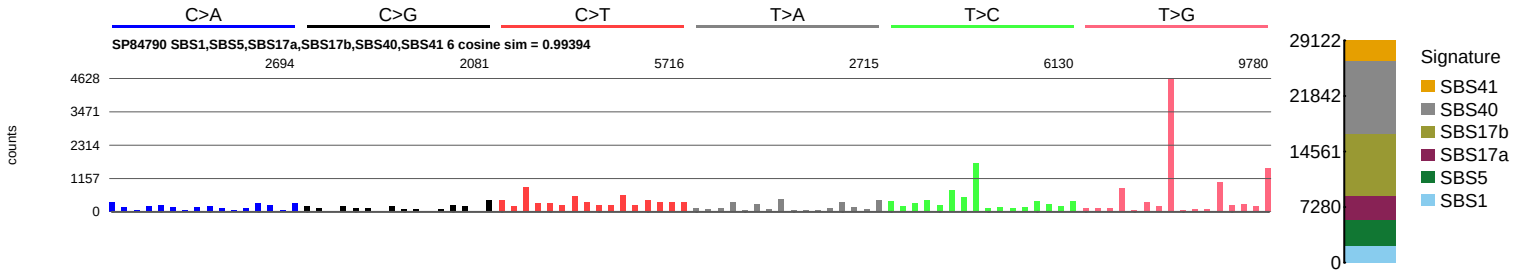

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.99396
3045
2321
5383
2934
5689
9755
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

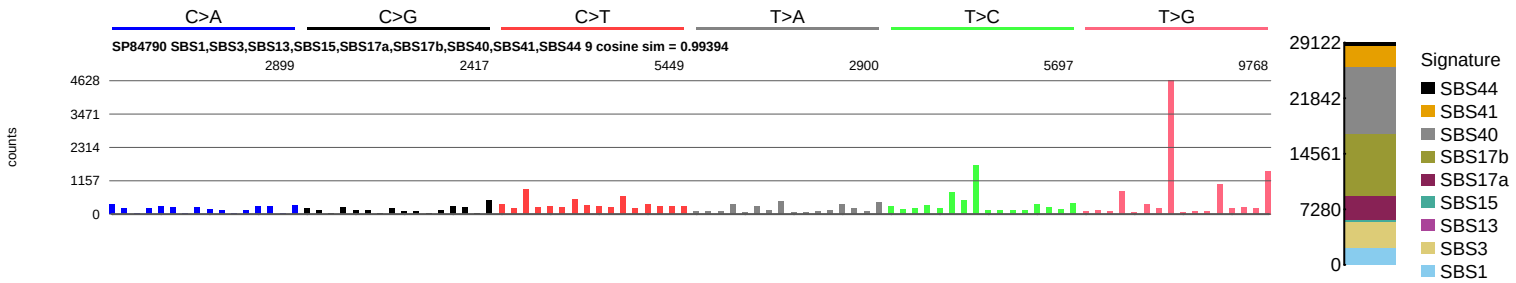

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.99396
3020
2119
5285
2613
6309
9775
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

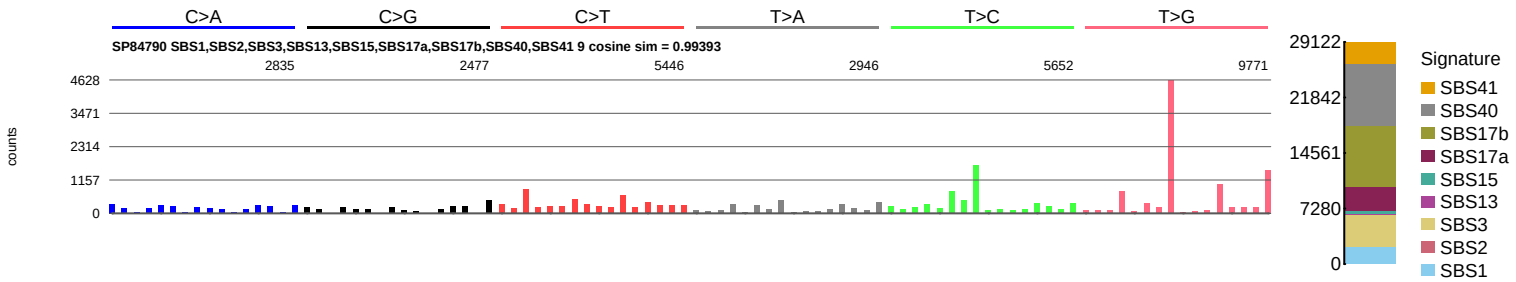

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 7 cosine sim = 0.99396
2878
2038
5472
2641
6285
9807
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS1

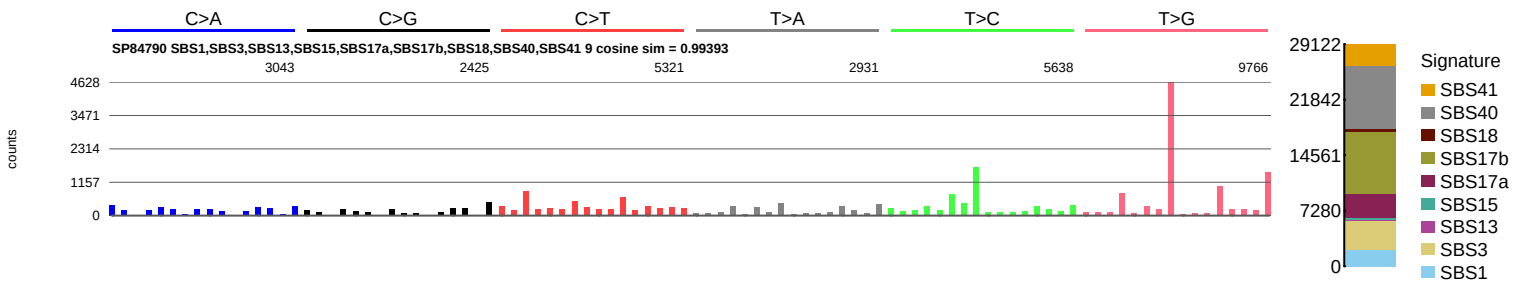

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS17a,SBS17b,SBS40,SBS41,SBS44 7 cosine sim = 0.99395
2714
2068
5749
2711
6114
9771
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS5
SBS1

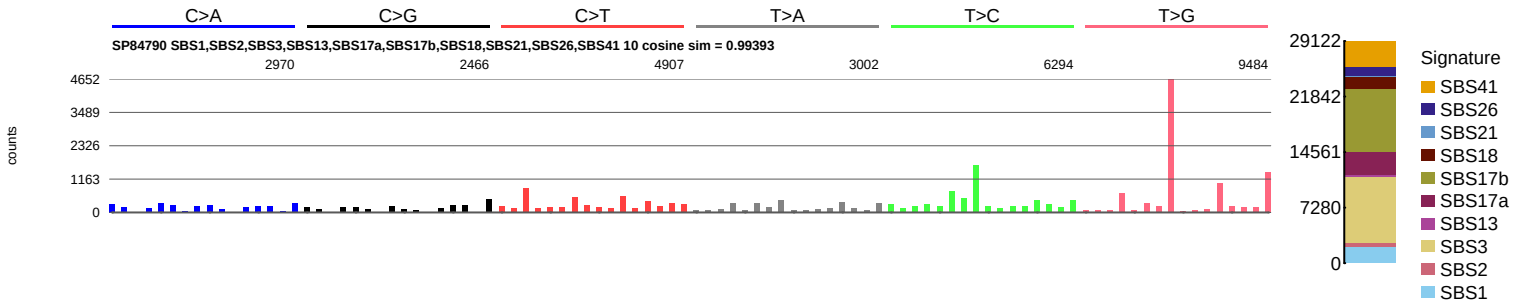

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS40,SBS41,SBS44 8 cosine sim = 0.99394
2827
2358
5504
2948
5728
9752
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS3
SBS2
SBS1

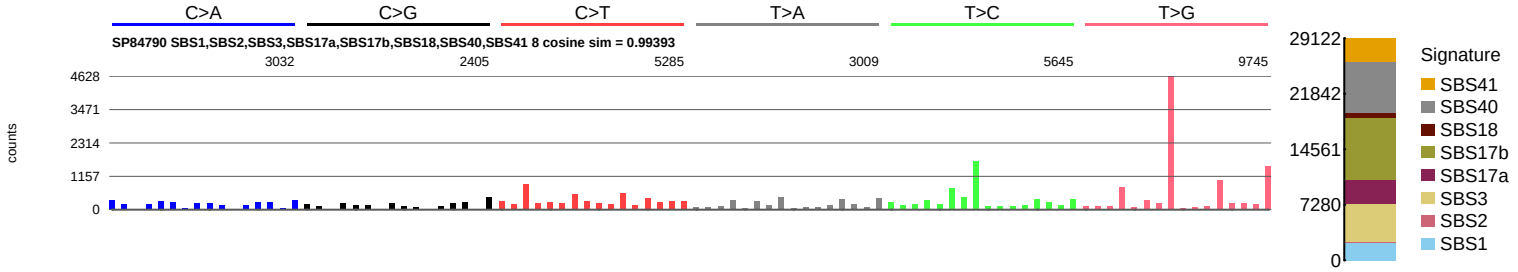

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.99394
2694
2081
5716
2715
6130
9780
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS5
SBS1

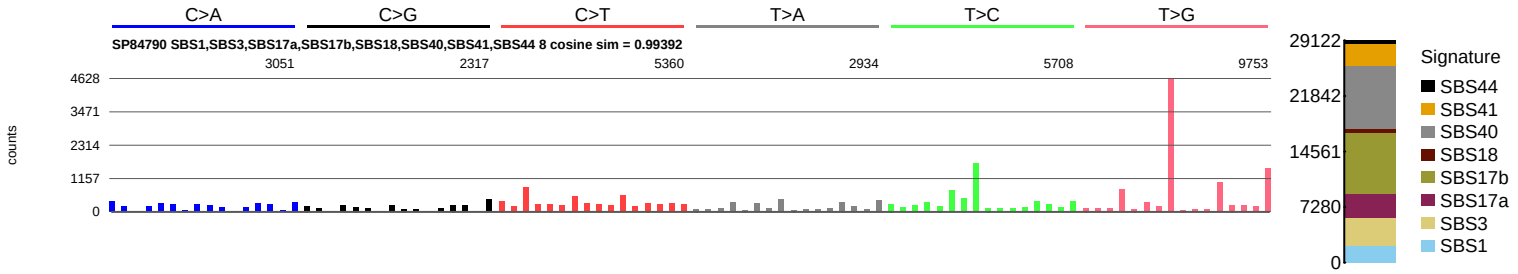

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41,SBS44 9 cosine sim = 0.99394
2899
2417
5449
2900
5697
9768
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

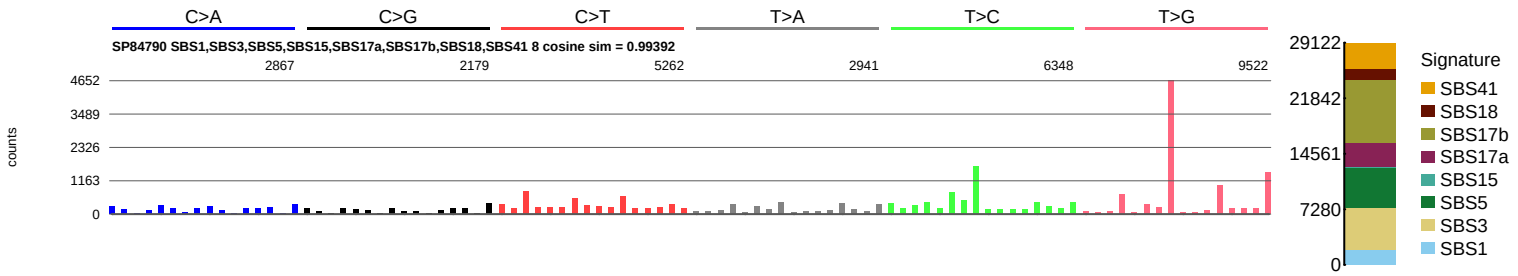

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41 9 cosine sim = 0.99393
2835
2477
5446
2946
5652
9771
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

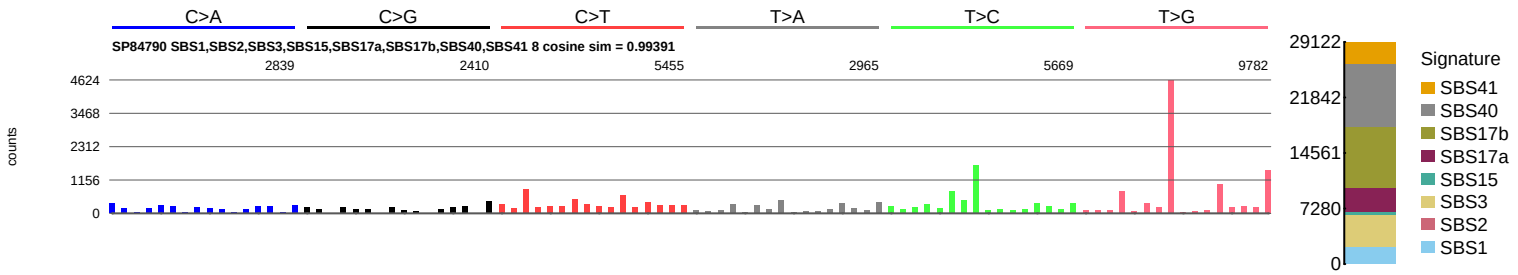

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.99393
3043
2425
5321
2931
5638
9766
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

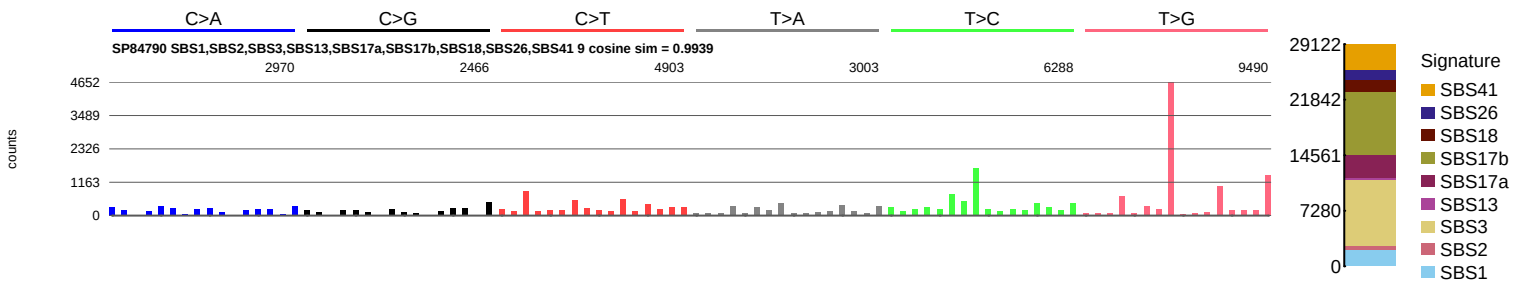

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 10 cosine sim = 0.99393
2970
2466
4907
3002
6294
9484
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

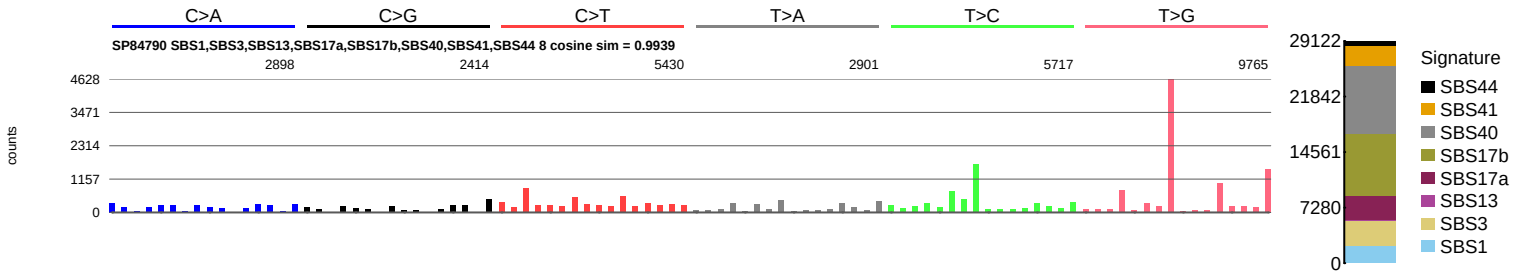

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.99393
3032
2405
5285
3009
5645
9745
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

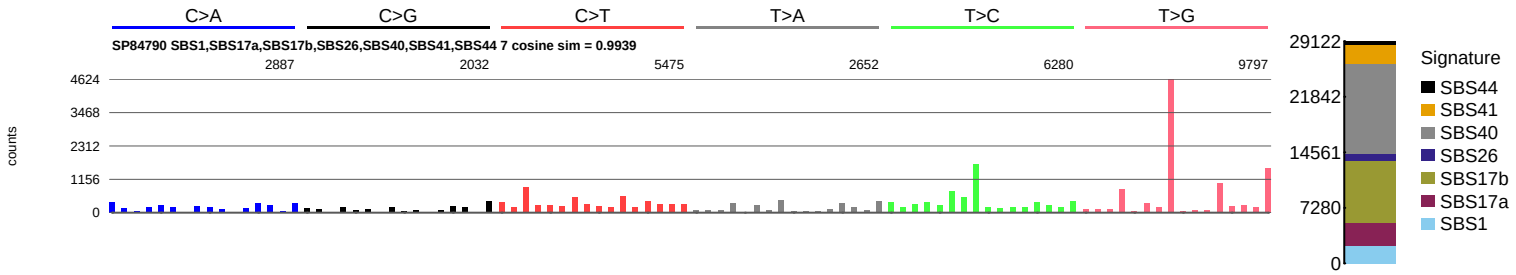

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.99392
3051
2317
5360
2934
5708
9753
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS3
SBS1

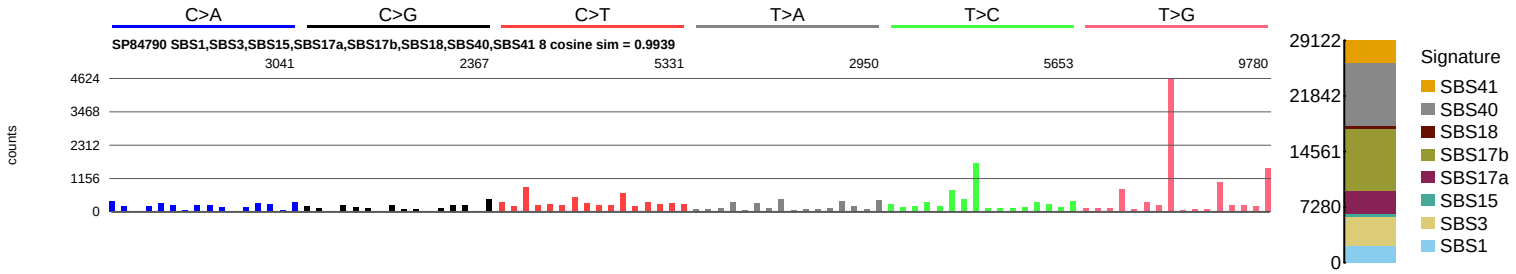

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.99392
2867
2179
5262
2941
6348
9522
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1

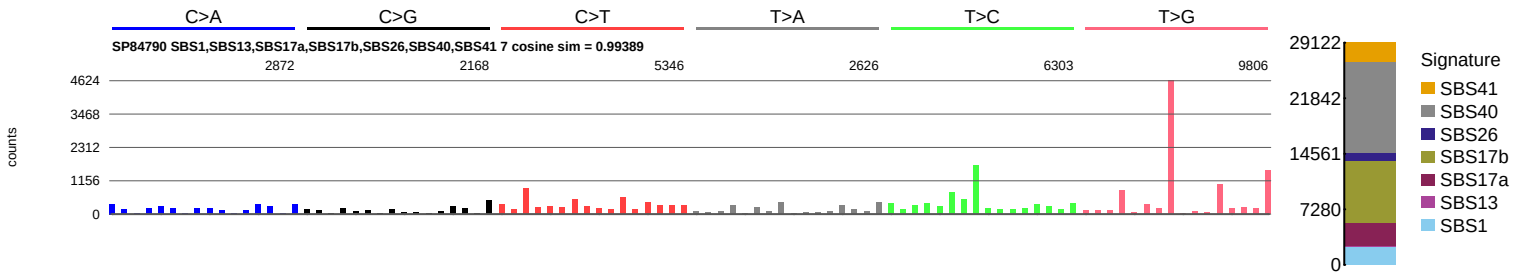

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS40,SBS41 8 cosine sim = 0.99391
2839
2410
5455
2965
5669
9782
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

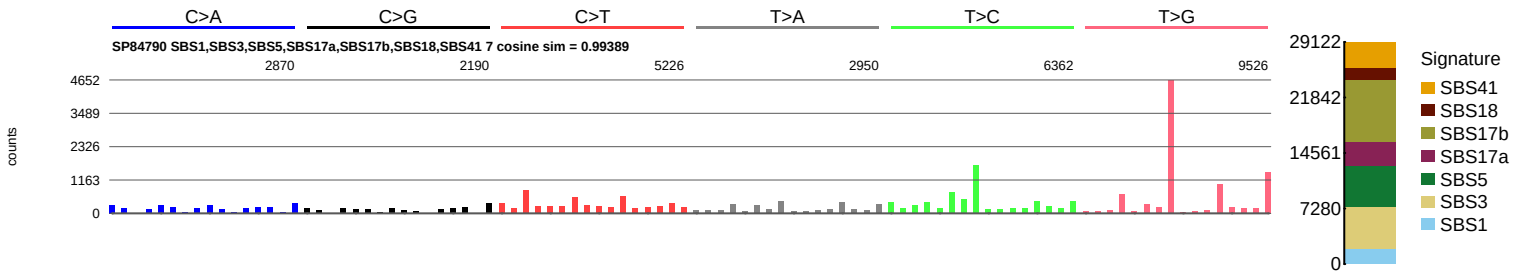

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.9939
2970
2466
4903
3003
6288
9490
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

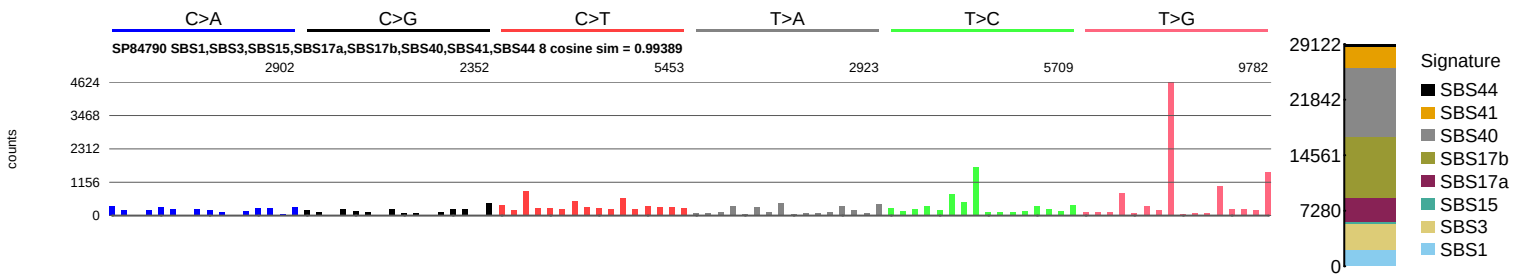

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS40,SBS41,SBS44 8 cosine sim = 0.9939
2898
2414
5430
2901
5717
9765
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS13
SBS3
SBS1

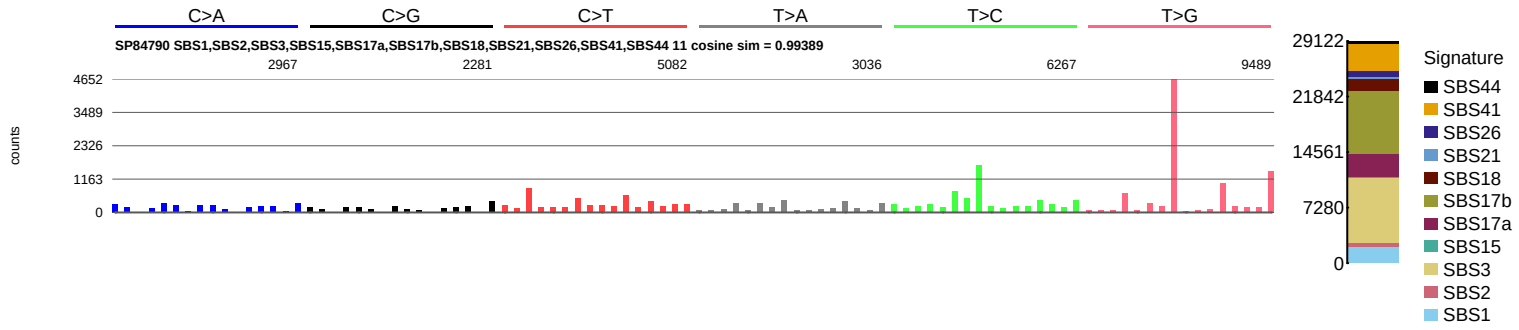

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 7 cosine sim = 0.9939
2887
2032
5475
2652
6280
9797
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS1

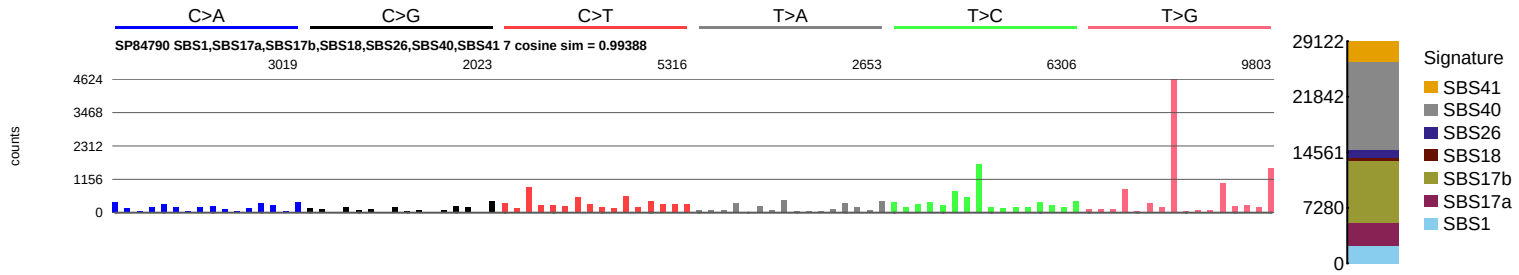

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.9939
3041
2367
5331
2950
5653
9780
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

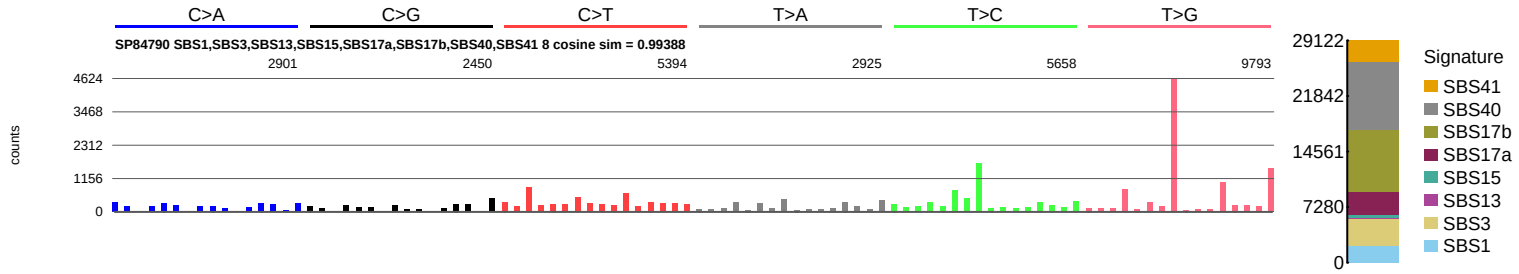

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS41 7 cosine sim = 0.99389
2872
2168
5346
2626
6303
9806
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS1

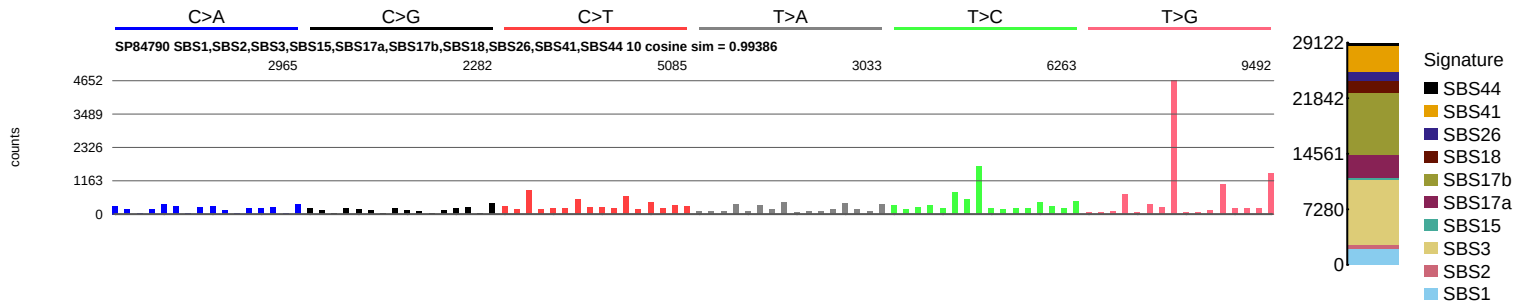

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.99389
2870
2190
5226
2950
6362
9526
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1

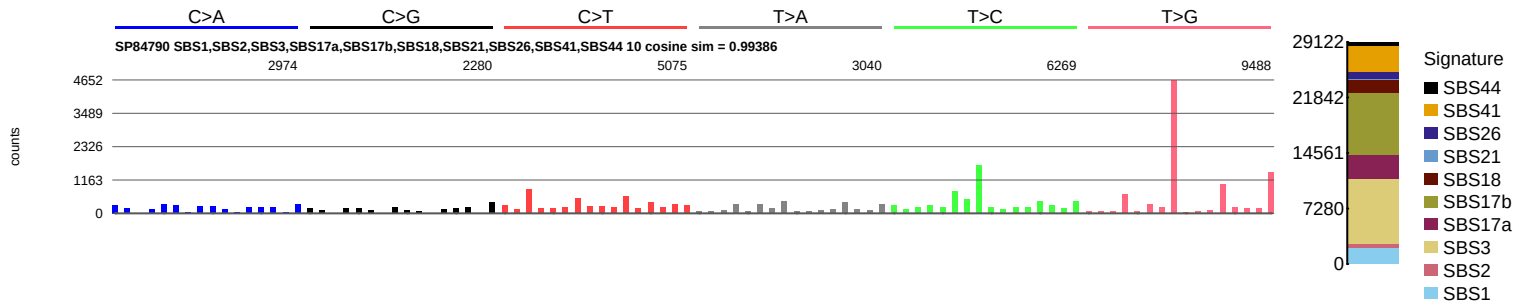

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS40,SBS41,SBS44 8 cosine sim = 0.99389
2902
2352
5453
2923
5709
9782
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS3
SBS1

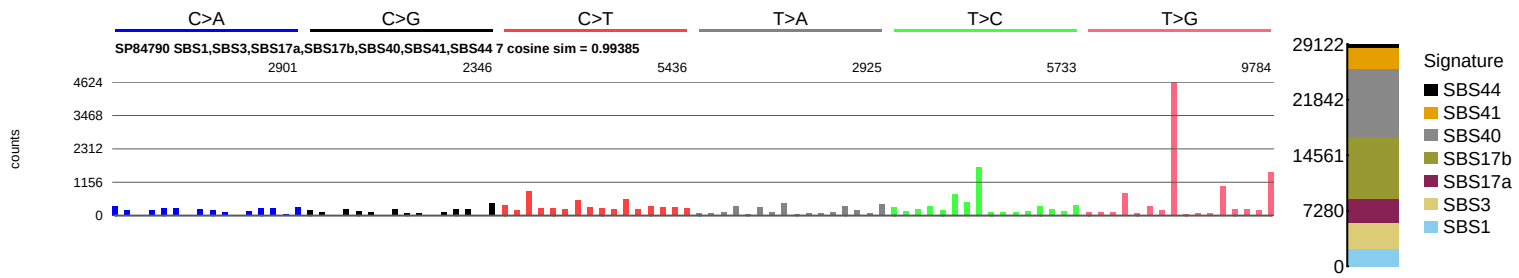

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 11 cosine sim = 0.99389
2967
2281
5082
3036
6267
9489
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

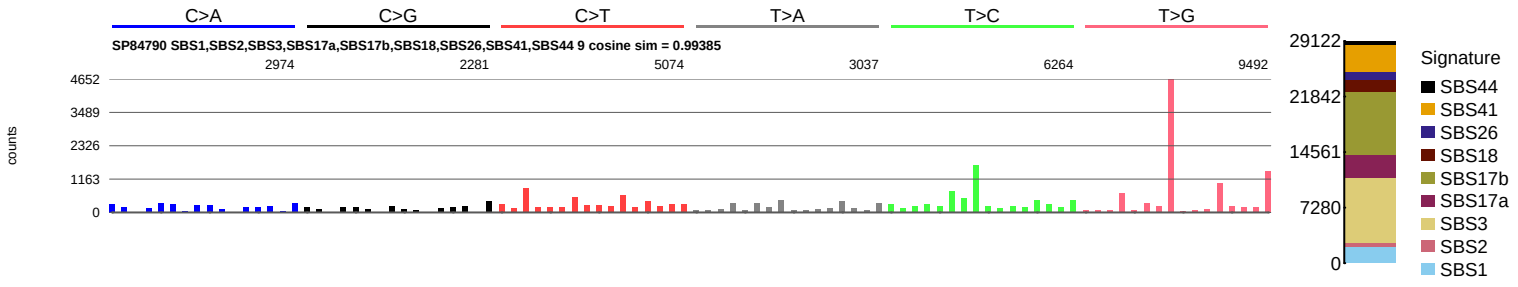

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.99388
3019
2023
5316
2653
6306
9803
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS1

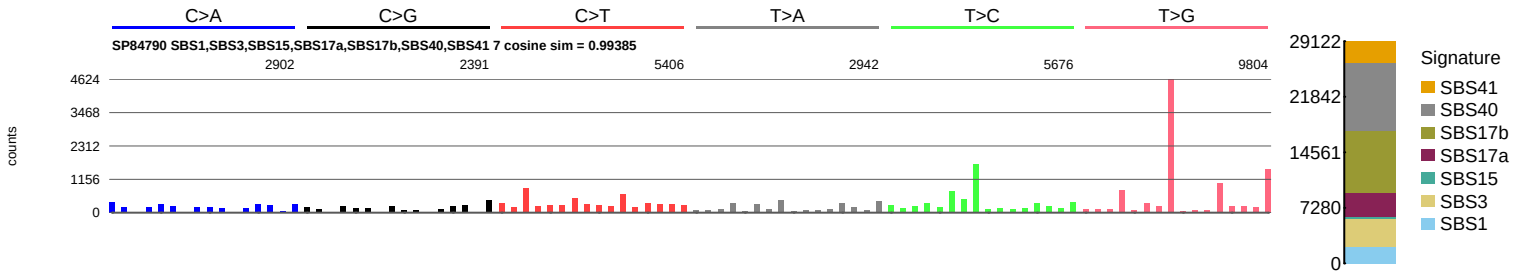

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41 8 cosine sim = 0.99388
2901
2450
5394
2925
5658
9793
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

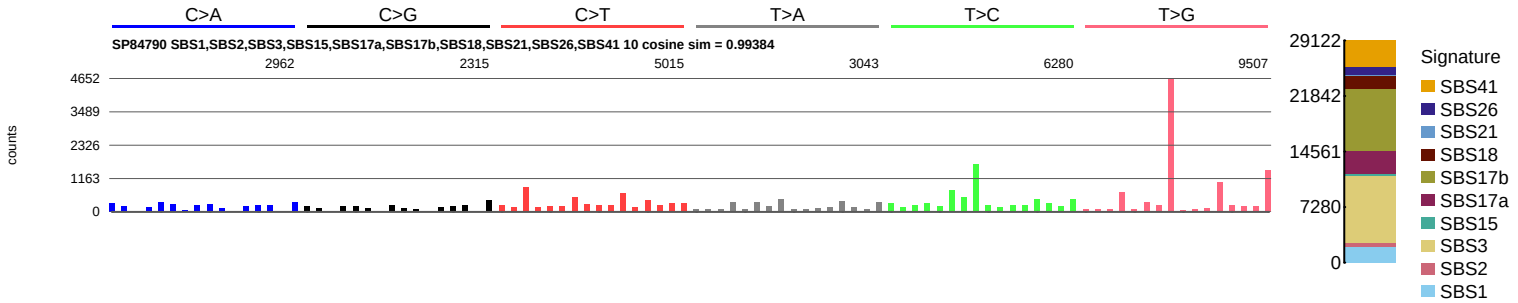

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.99386
2965
2282
5085
3033
6263
9492
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

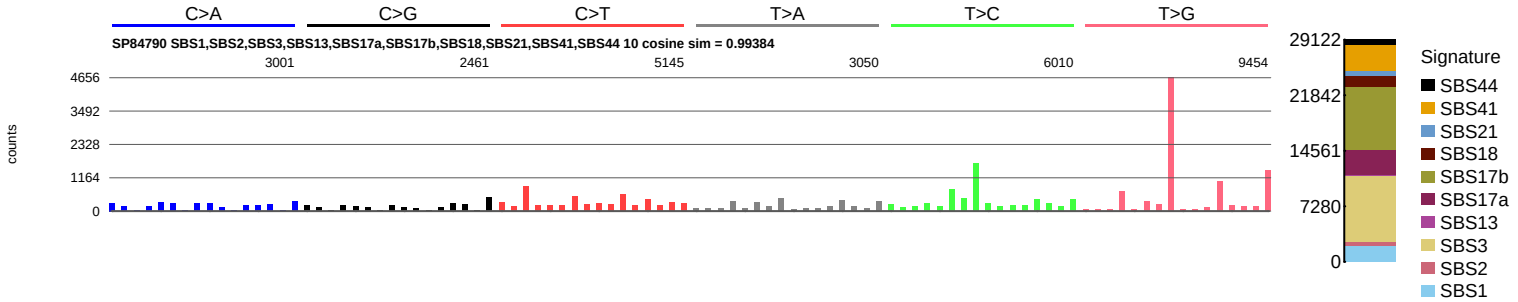

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 10 cosine sim = 0.99386
2974
2280
5075
3040
6269
9488
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

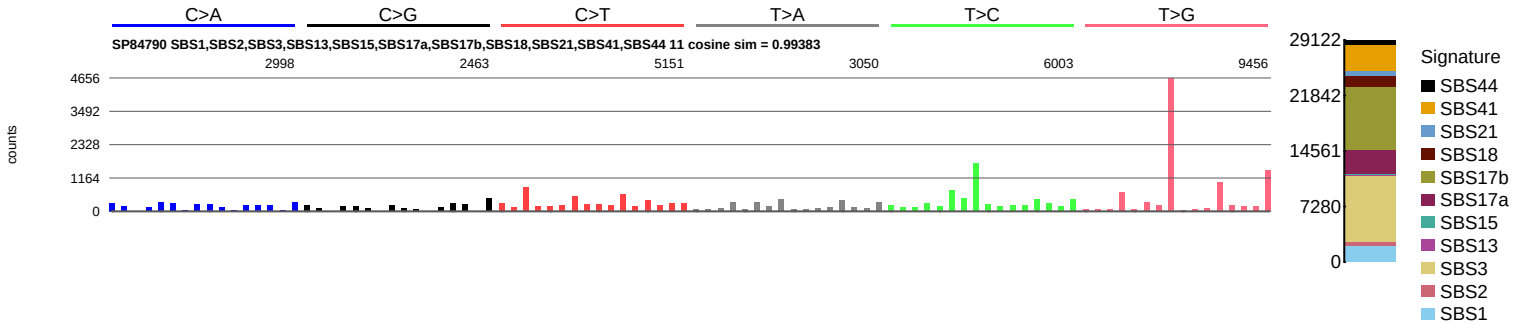

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS40,SBS41,SBS44 7 cosine sim = 0.99385
2901
2346
5436
2925
5733
9784
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS3
SBS1

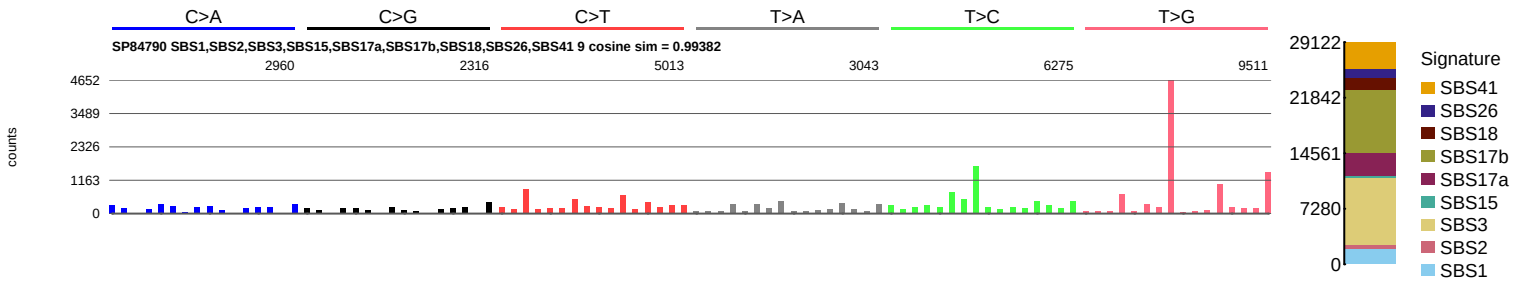

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.99385
2974
2281
5074
3037
6264
9492
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

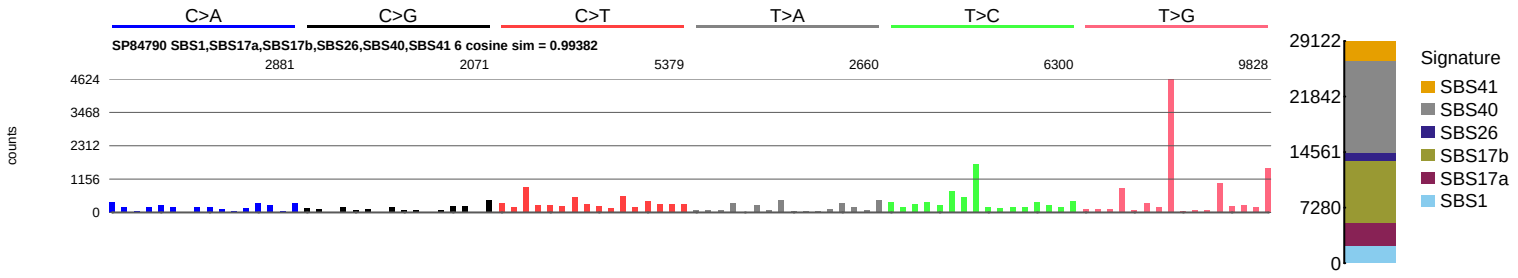

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.99385
2902
2391
5406
2942
5676
9804
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS3
SBS1

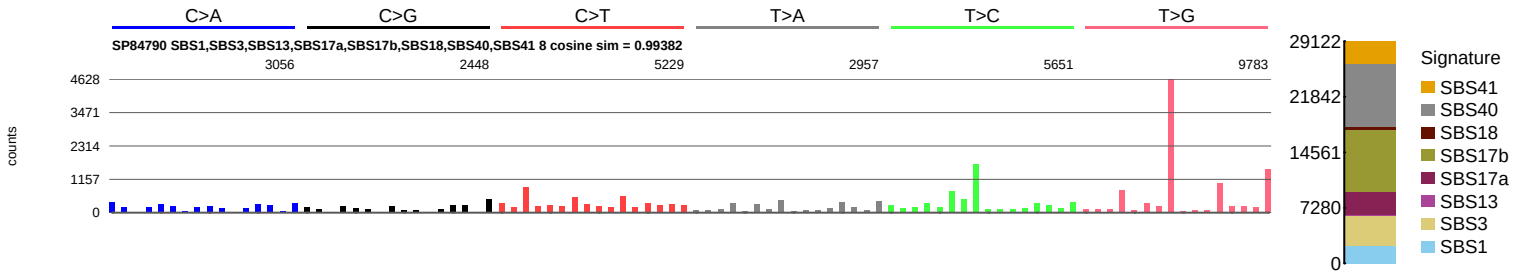

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 10 cosine sim = 0.99384
2962
2315
5015
3043
6280
9507
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

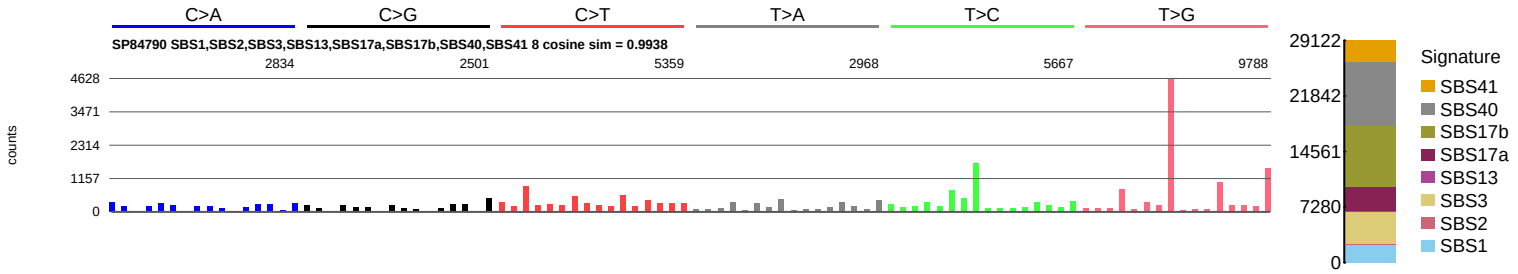

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.99384
3001
2461
5145
3050
6010
9454
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

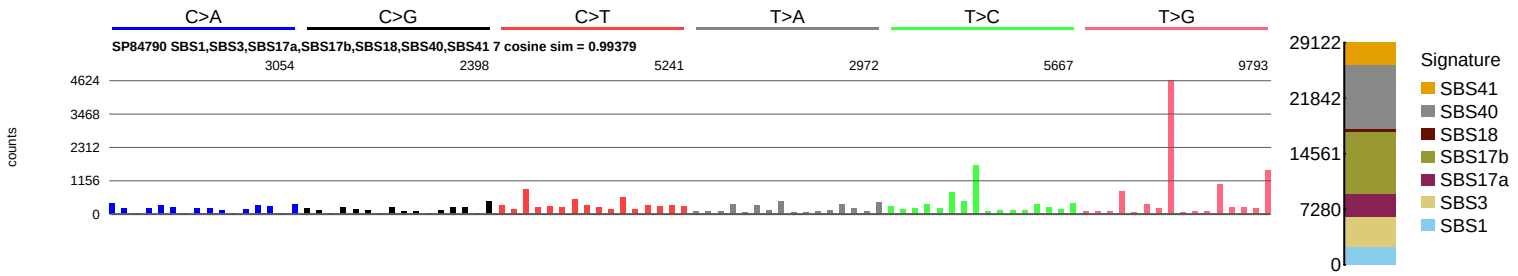

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 11 cosine sim = 0.99383
2998
2463
5151
3050
6003
9456
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

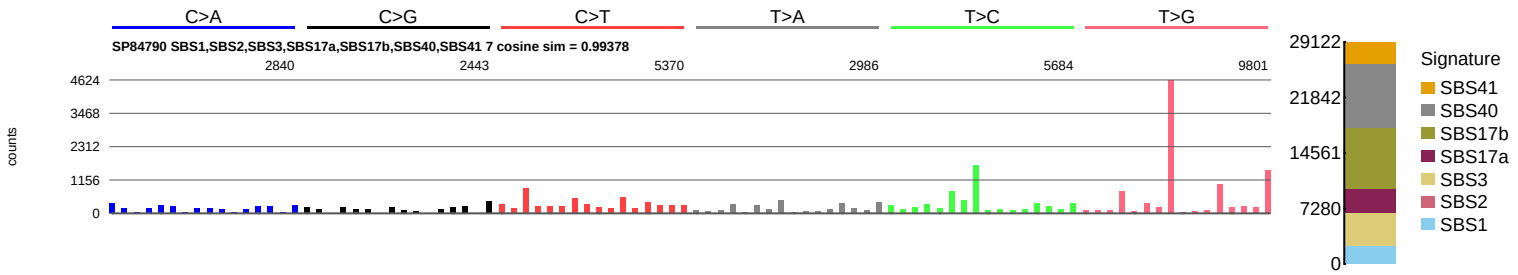

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.99382
2960
2316
5013
3043
6275
9511
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

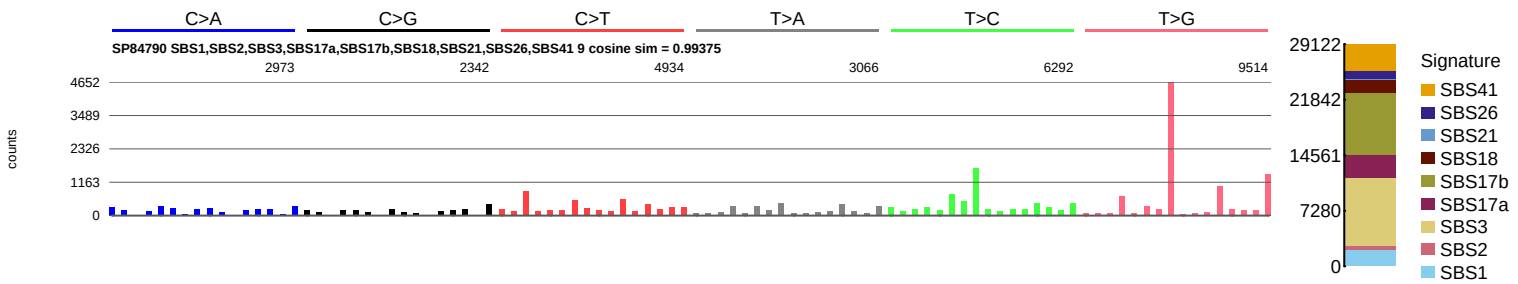

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS17a,SBS17b,SBS26,SBS40,SBS41 6 cosine sim = 0.99382
2881
2071
5379
2660
6300
9828
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS1

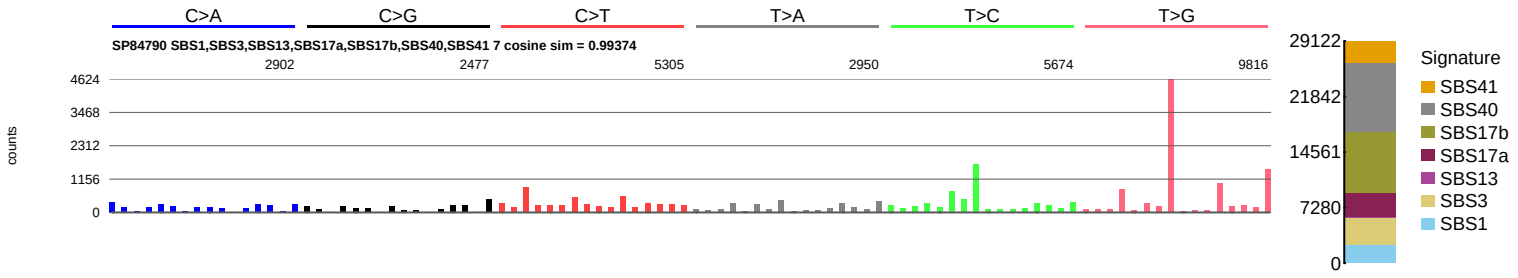

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.99382
3056
2448
5229
2957
5651
9783
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1

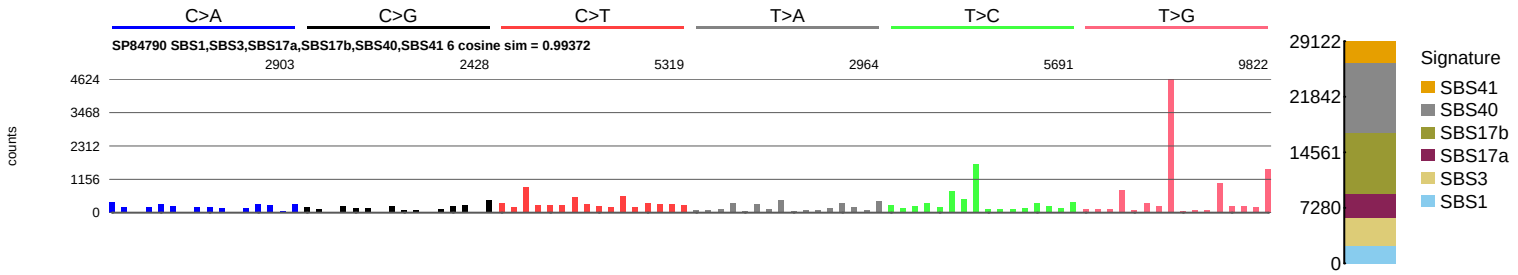

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS40,SBS41 8 cosine sim = 0.9938
2834
2501
5359
2968
5667
9788
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

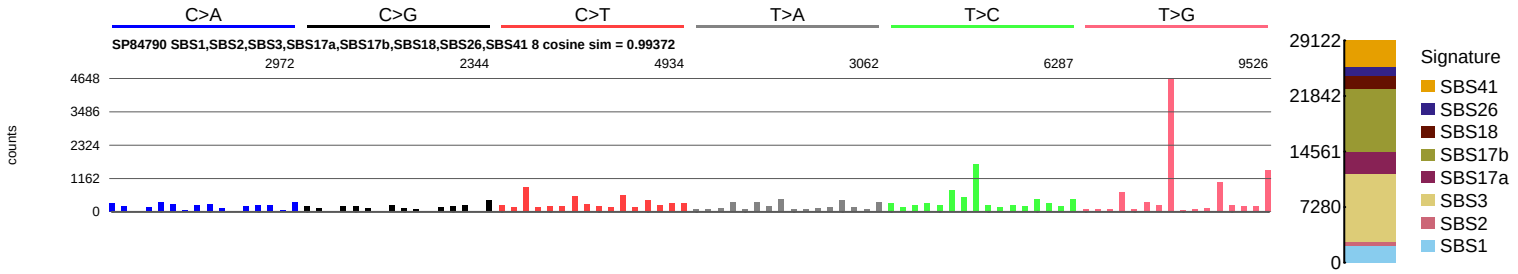

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.99379
3054
2398
5241
2972
5667
9793
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS3
SBS1

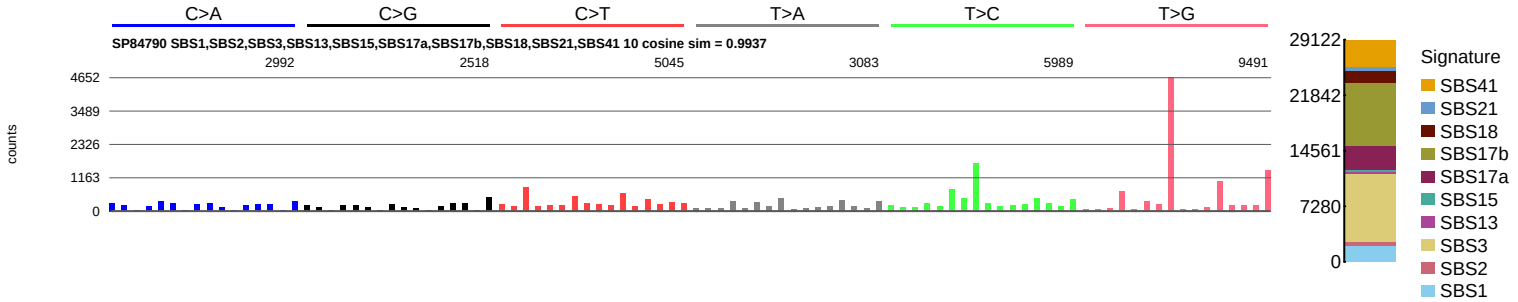

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.99378
2840
2443
5370
2986
5684
9801
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS3
SBS2
SBS1

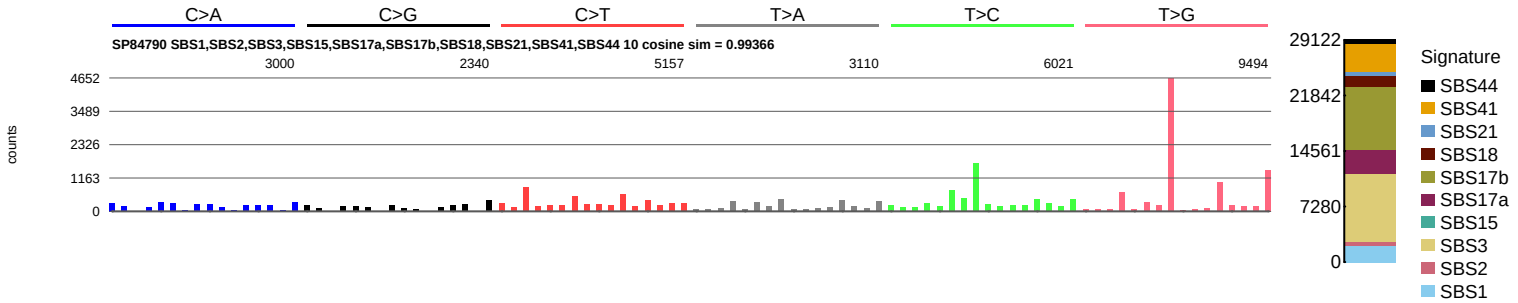

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.99375
2973
2342
4934
3066
6292
9514
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

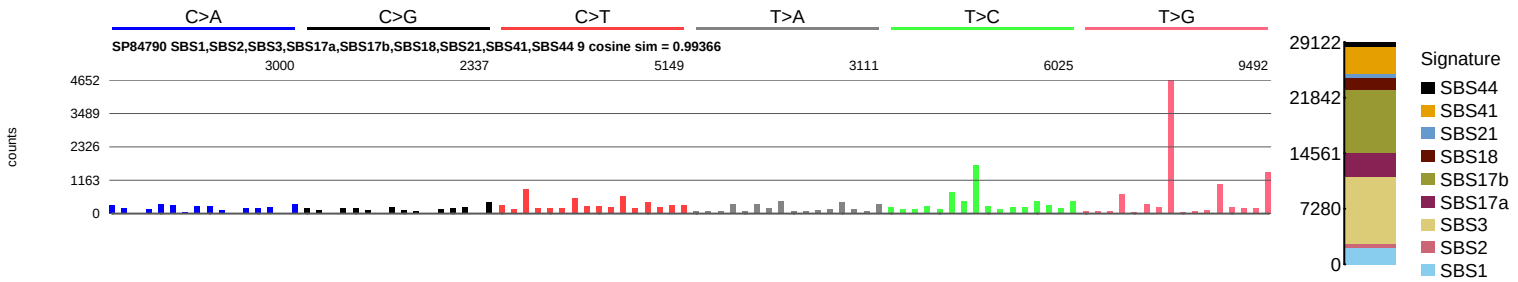

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.99374
2902
2477
5305
2950
5674
9816
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS13
SBS3
SBS1

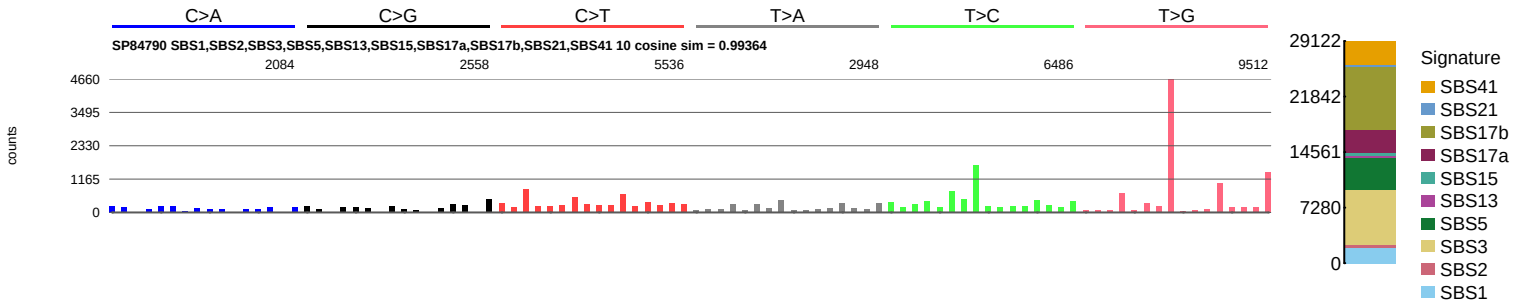

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.99372
2903
2428
5319
2964
5691
9822
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS3
SBS1

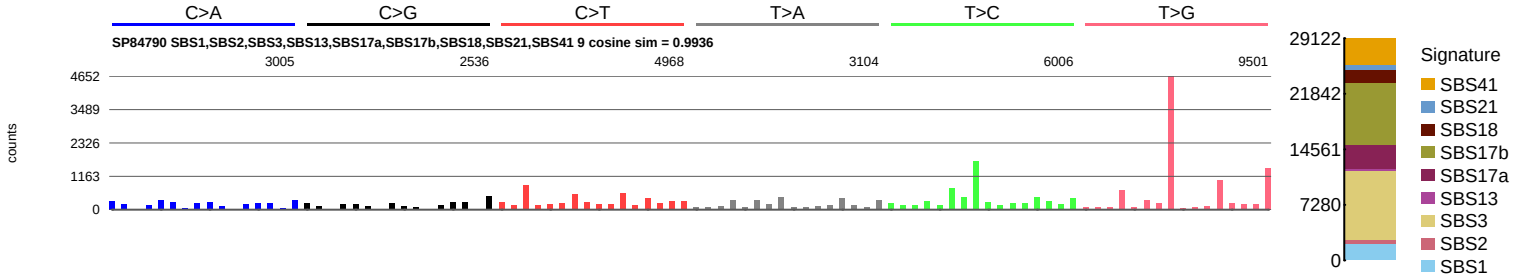

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.99372
2972
2344
4934
3062
6287
9526
counts
4648
3486
2324
1162
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

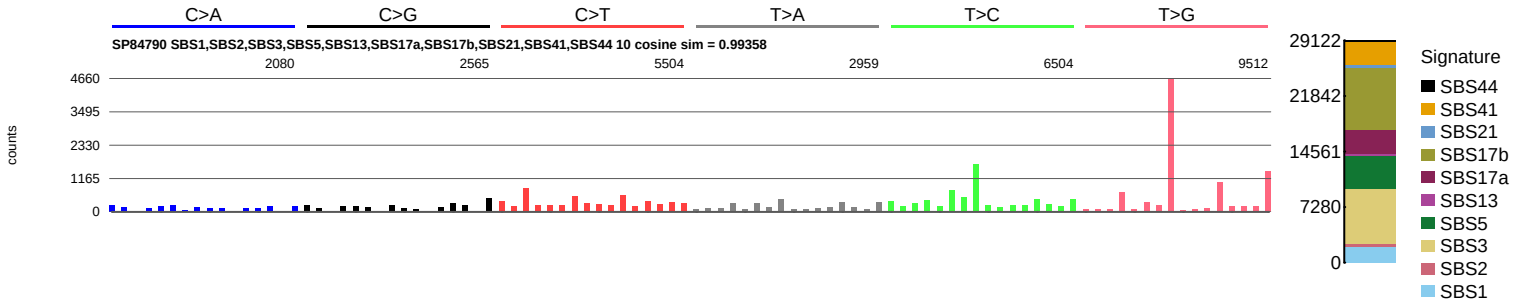

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.9937
2992
2518
5045
3083
5989
9491
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

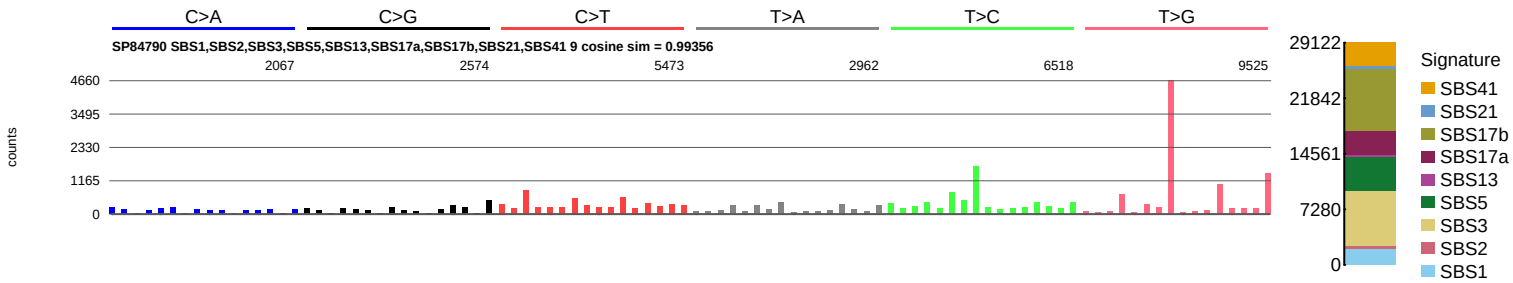

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.99366
3000
2340
5157
3110
6021
9494
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

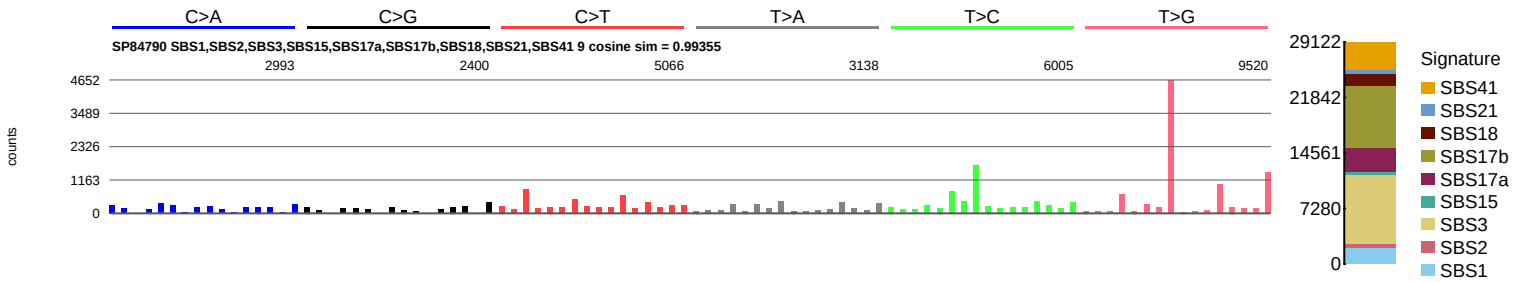

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.99366
3000
2337
5149
3111
6025
9492
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

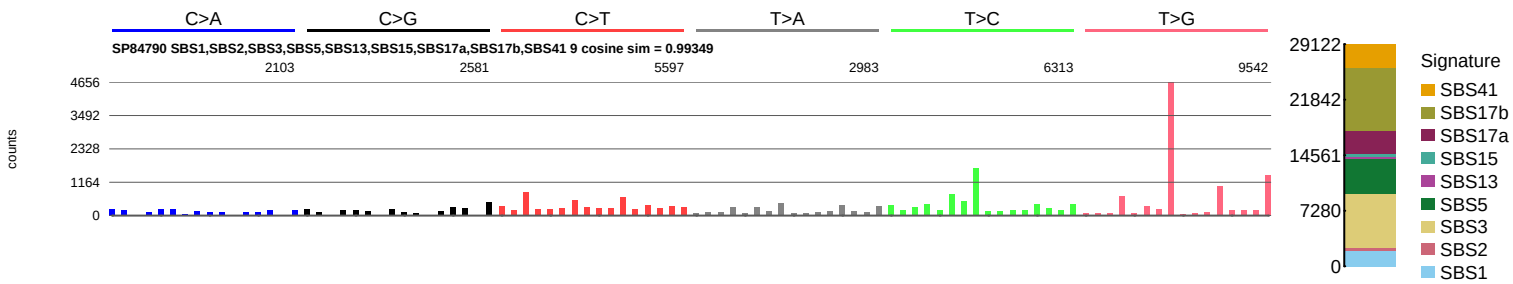

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 10 cosine sim = 0.99364
2084
2558
5536
2948
6486
9512
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

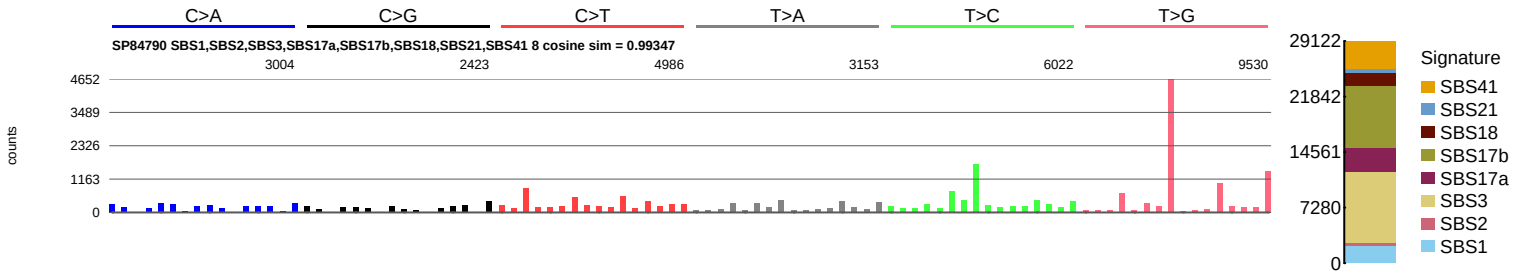

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.9936
3005
2536
4968
3104
6006
9501
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

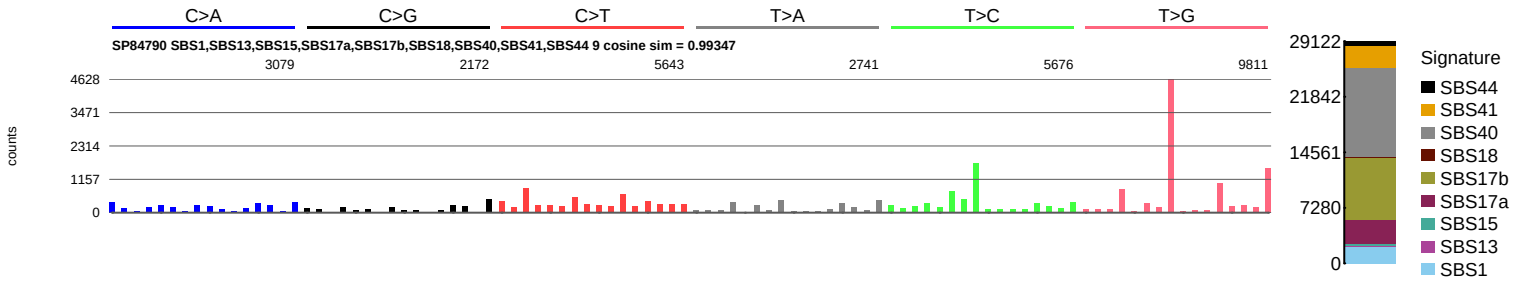

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS41,SBS44 10 cosine sim = 0.99358
2080
2565
5504
2959
6504
9512
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

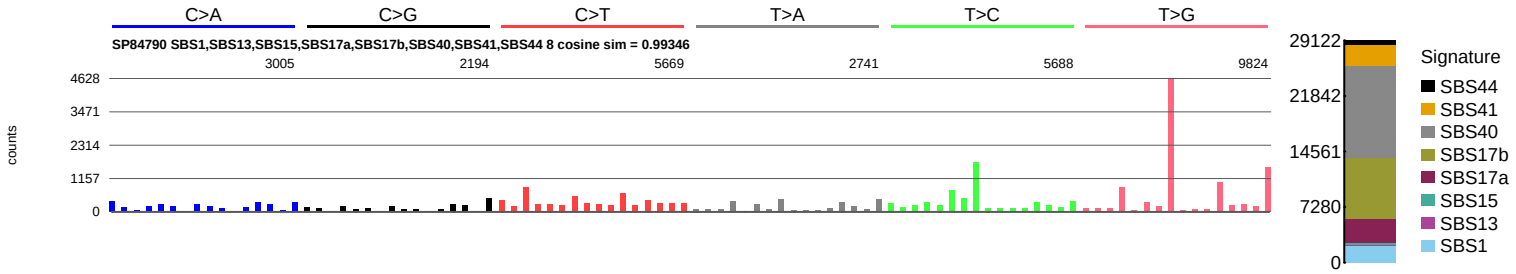

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS41 9 cosine sim = 0.99356
2067
2574
5473
2962
6518
9525
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

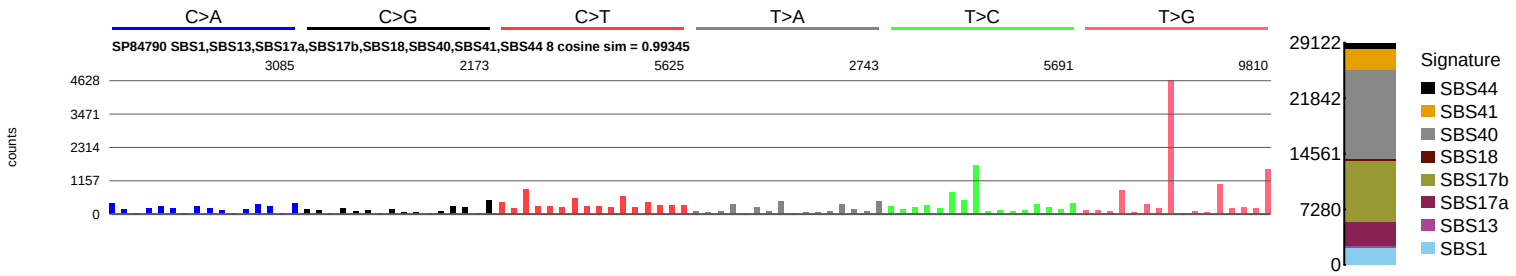

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.99355
2993
2400
5066
3138
6005
9520
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

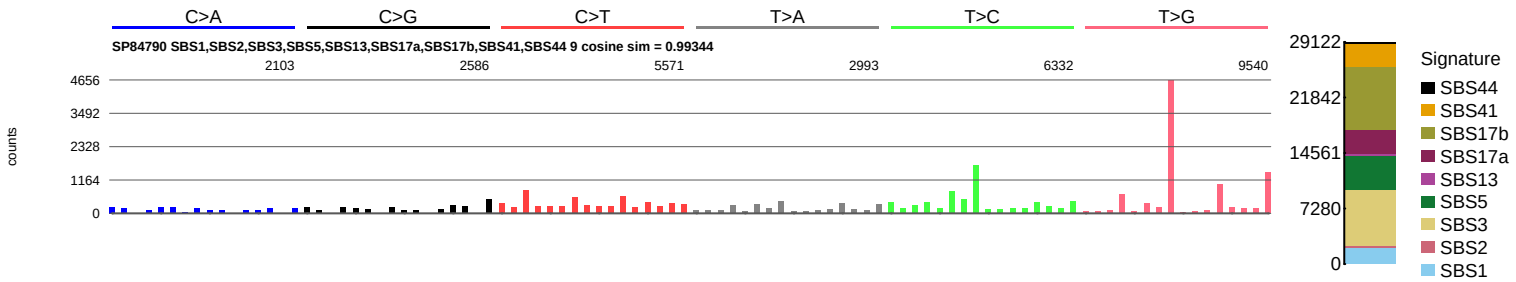

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS41 9 cosine sim = 0.99349
2103
2581
5597
2983
6313
9542
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

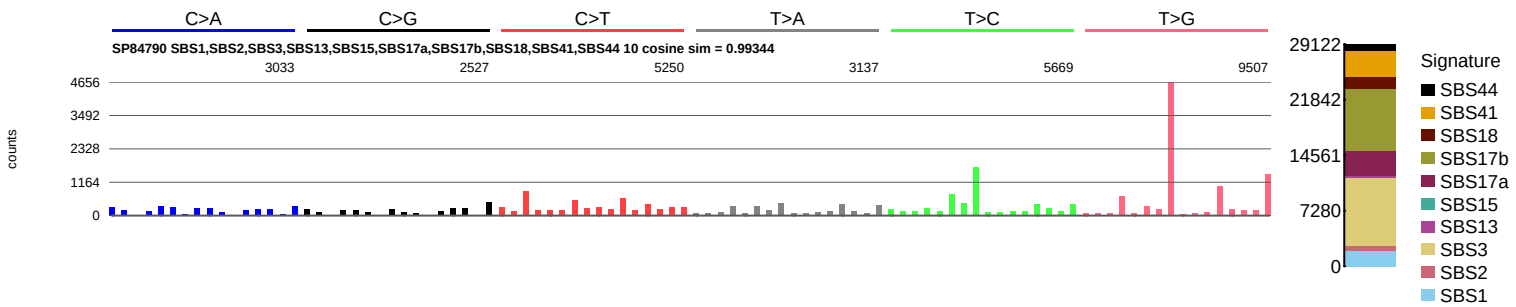

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.99347
3004
2423
4986
3153
6022
9530
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

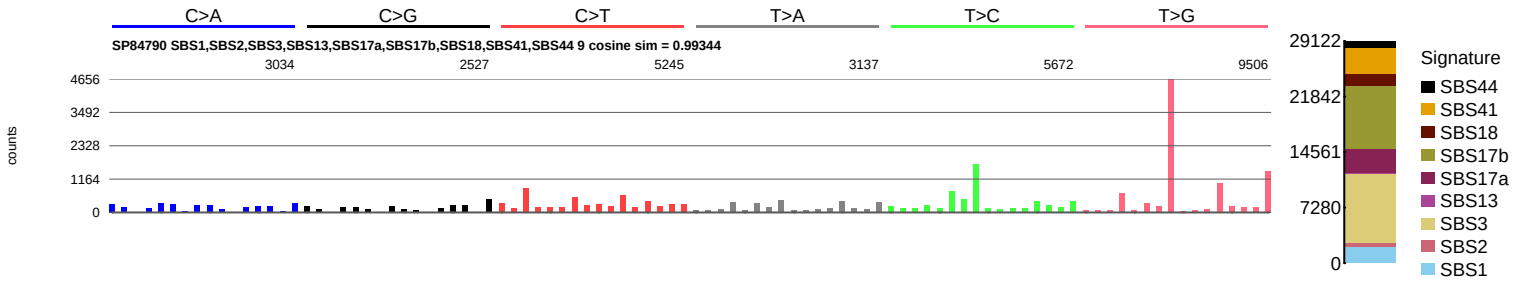

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.99347
3079
2172
5643
2741
5676
9811
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1

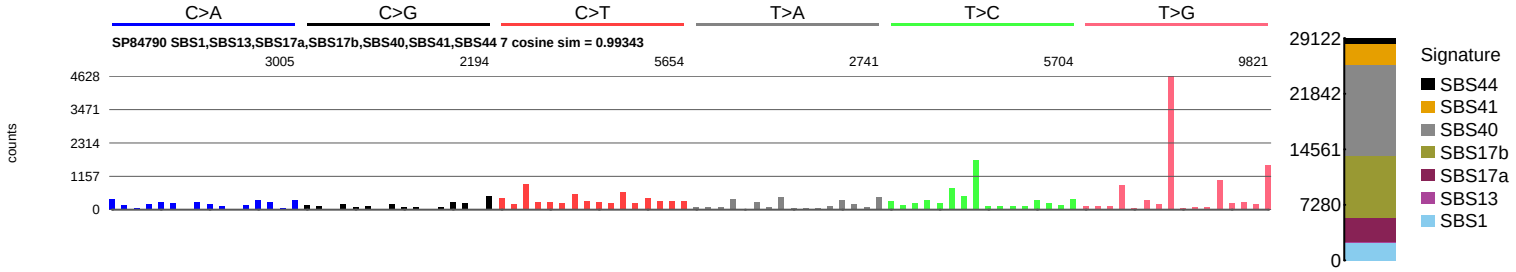

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41,SBS44 8 cosine sim = 0.99346
3005
2194
5669
2741
5688
9824
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS1

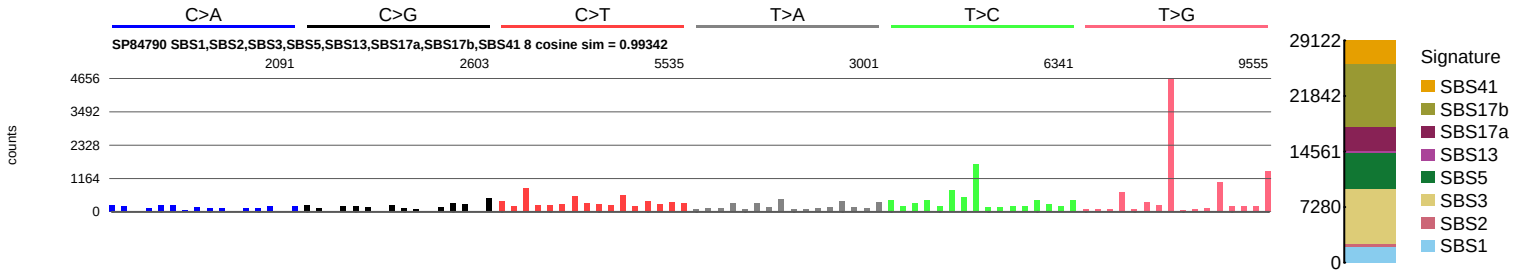

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.99345
3085
2173
5625
2743
5691
9810
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS1

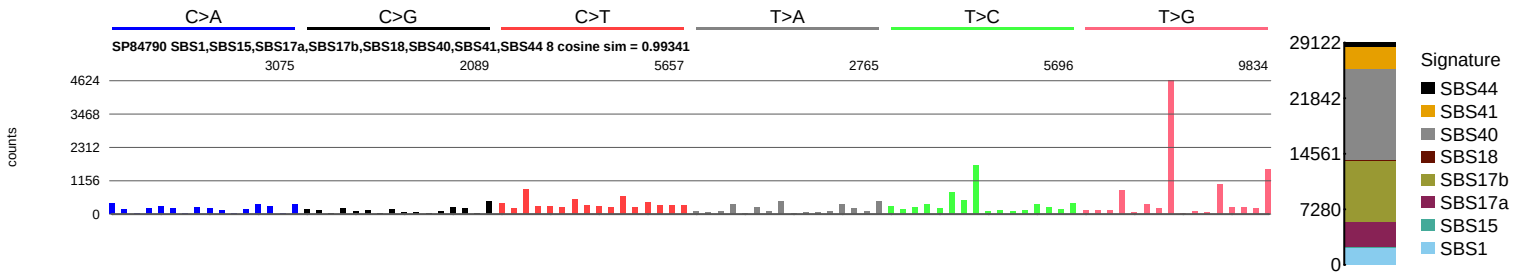

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS41,SBS44 9 cosine sim = 0.99344
2103
2586
5571
2993
6332
9540
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

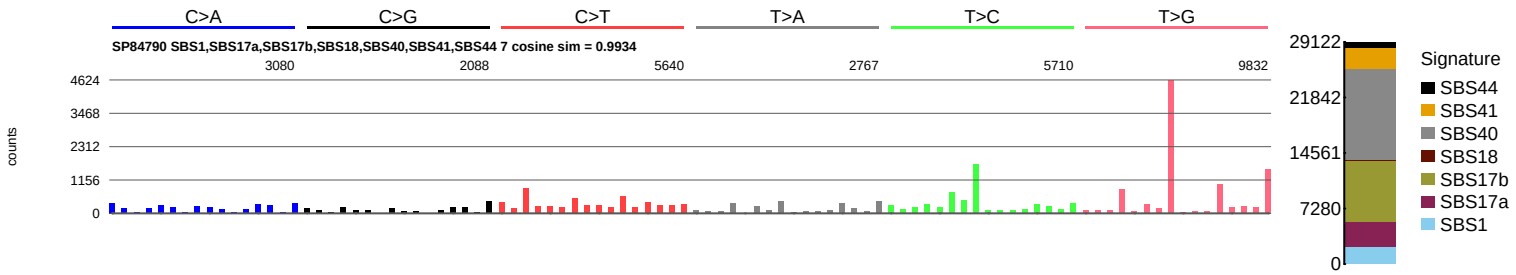

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.99344
3033
2527
5250
3137
5669
9507
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

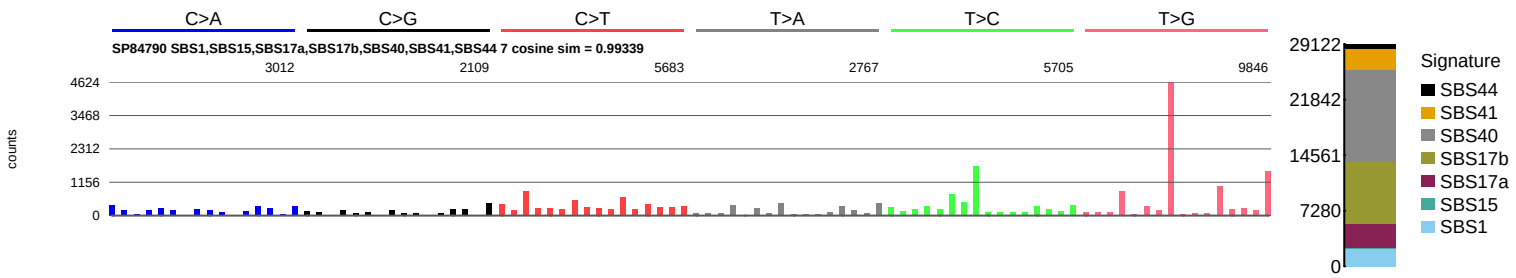

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.99344
3034
2527
5245
3137
5672
9506
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

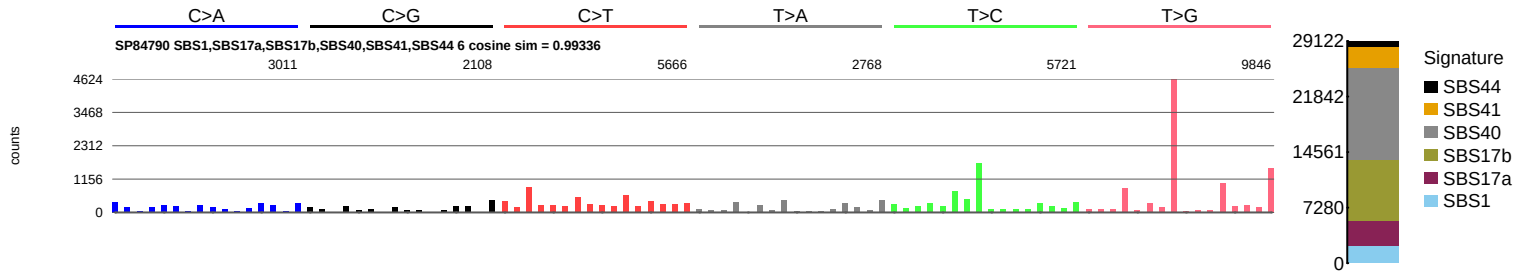

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS17a,SBS17b,SBS40,SBS41,SBS44 7 cosine sim = 0.99343
3005
2194
5654
2741
5704
9821
counts
4628
3471
2314
1157
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS13
SBS1

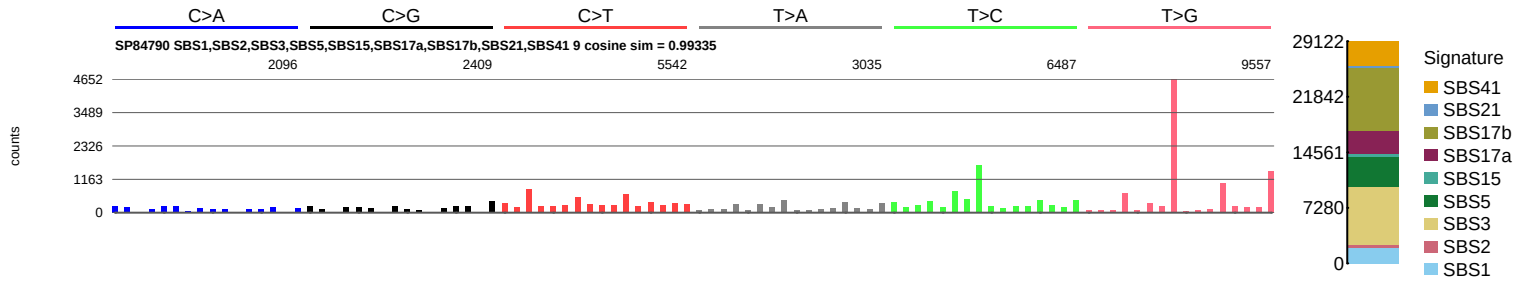

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS41 8 cosine sim = 0.99342
2091
2603
5535
3001
6341
9555
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS41
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

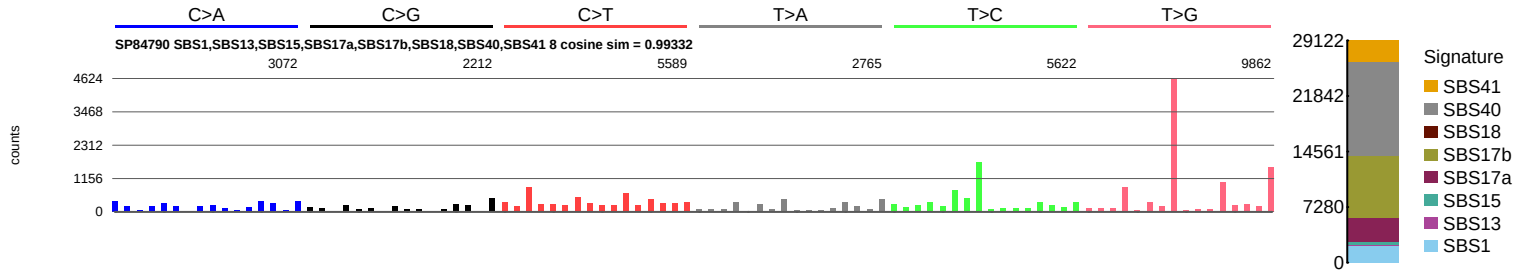

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.99341
3075
2089
5657
2765
5696
9834
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS1

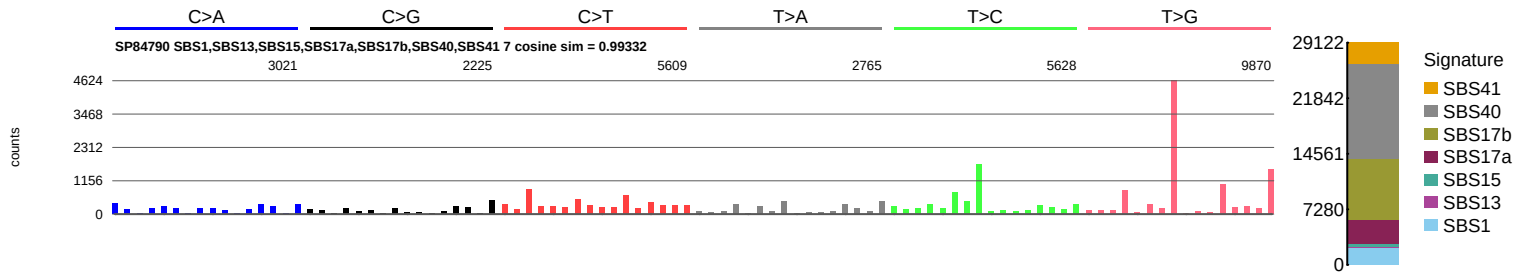

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 7 cosine sim = 0.9934
3080
2088
5640
2767
5710
9832
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS1

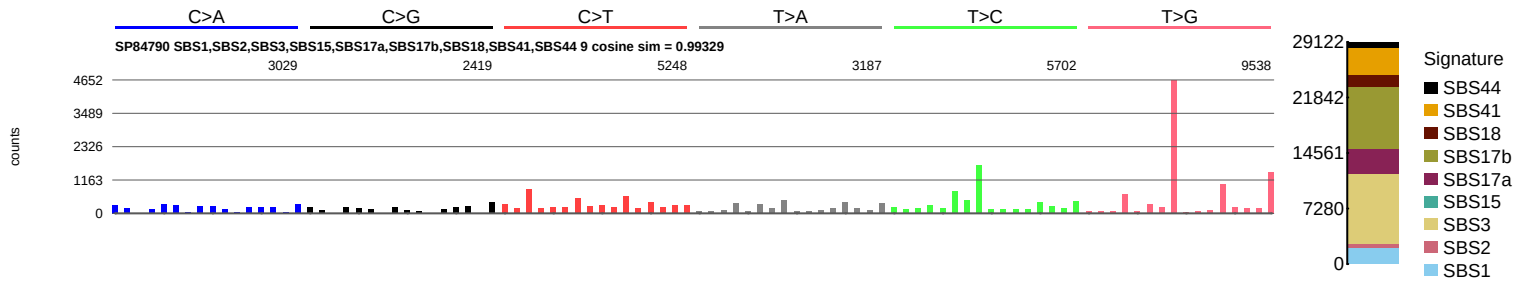

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS15,SBS17a,SBS17b,SBS40,SBS41,SBS44 7 cosine sim = 0.99339
3012
2109
5683
2767
5705
9846
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS1

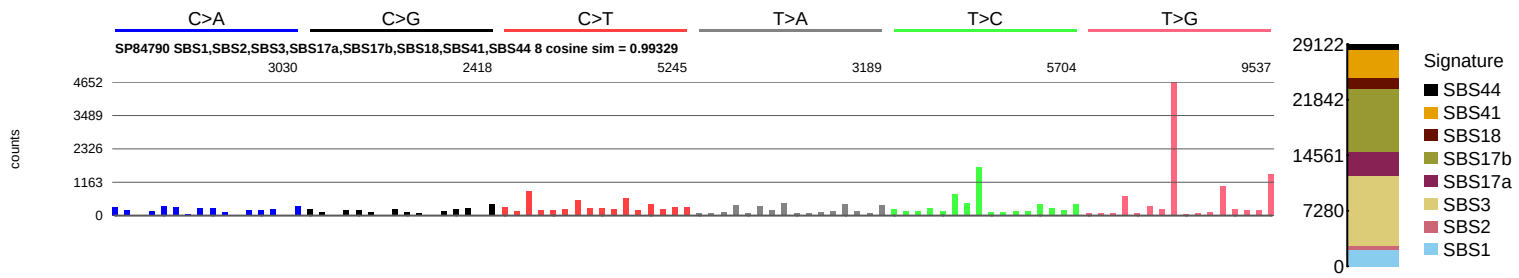

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS17a,SBS17b,SBS40,SBS41,SBS44 6 cosine sim = 0.99336
3011
2108
5666
2768
5721
9846
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS1

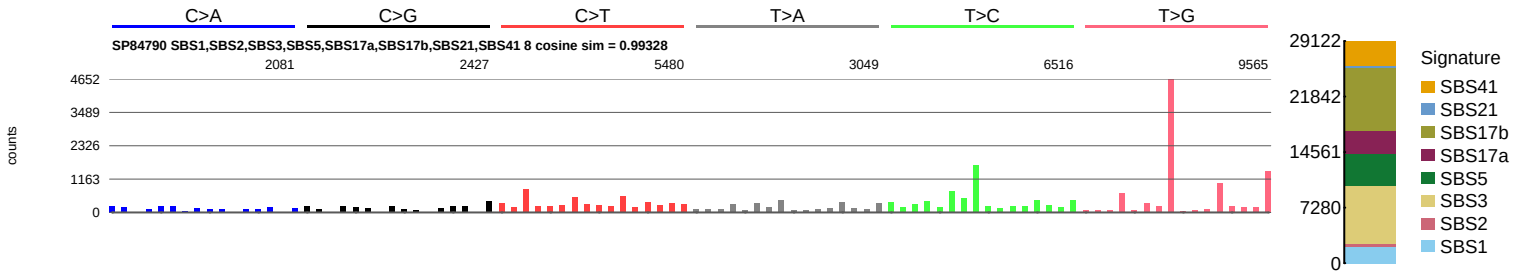

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 9 cosine sim = 0.99335
2096
2409
5542
3035
6487
9557
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

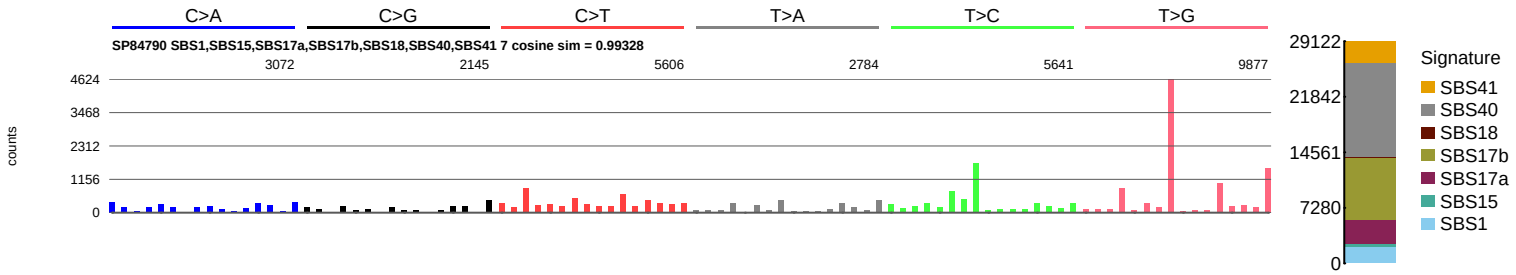

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.99332
3072
2212
5589
2765
5622
9862
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1

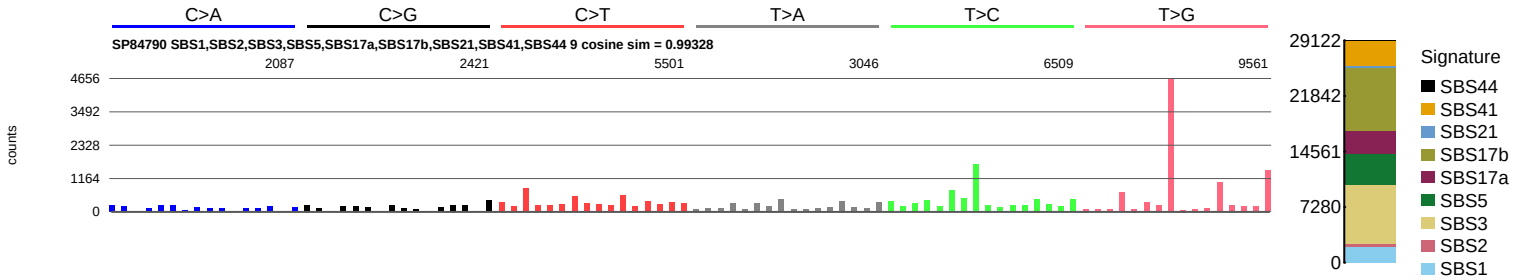

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.99332
3021
2225
5609
2765
5628
9870
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS1

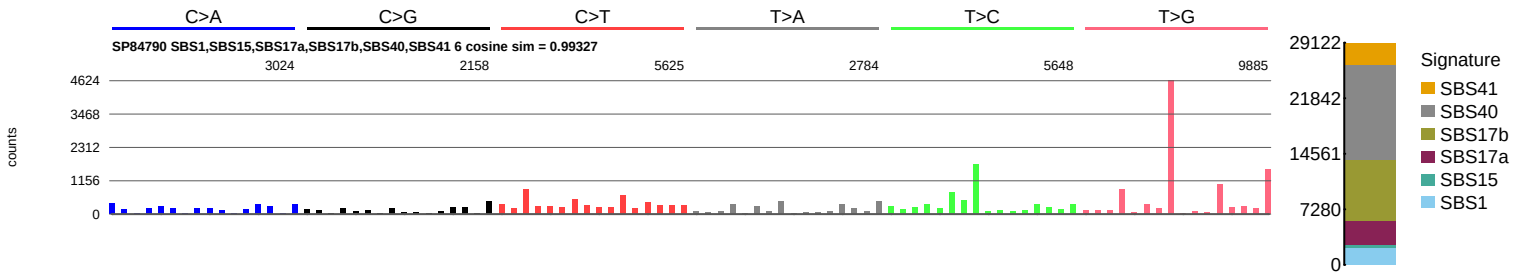

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.99329
3029
2419
5248
3187
5702
9538
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

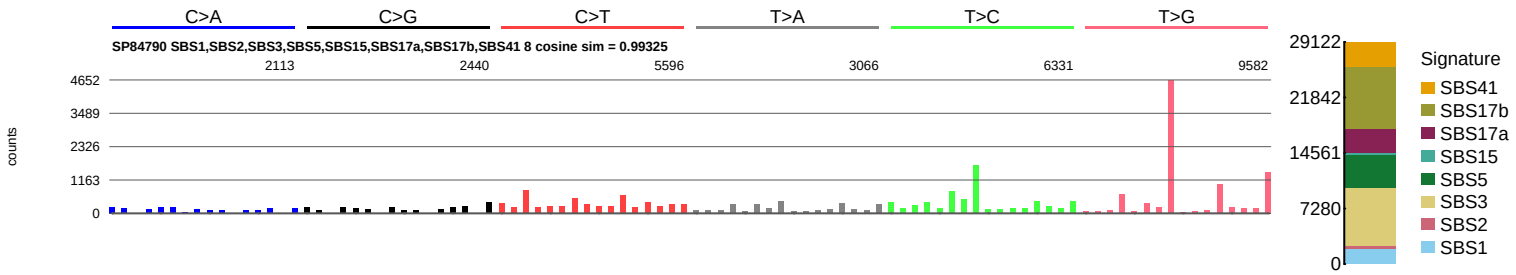

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.99329
3030
2418
5245
3189
5704
9537
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

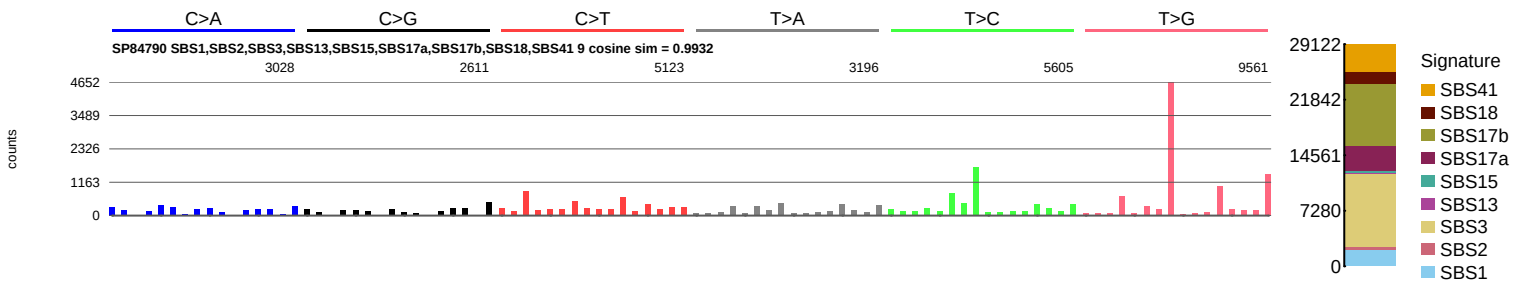

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.99328
2081
2427
5480
3049
6516
9565
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

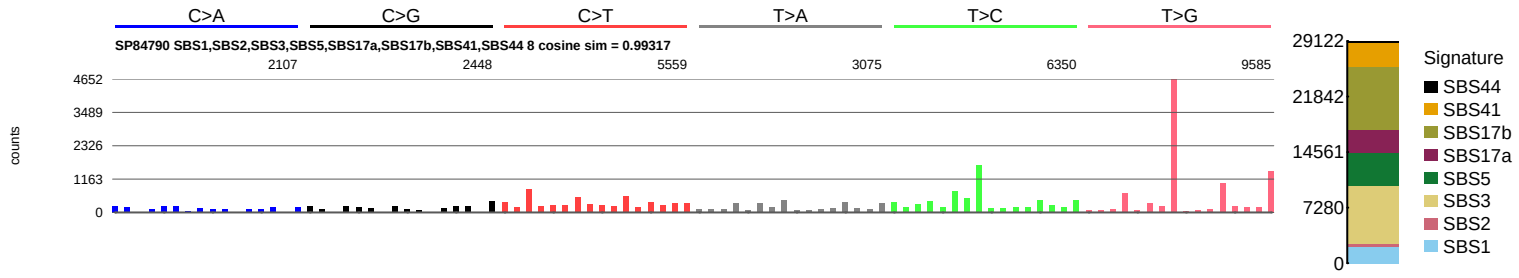

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.99328
3072
2145
5606
2784
5641
9877
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS1

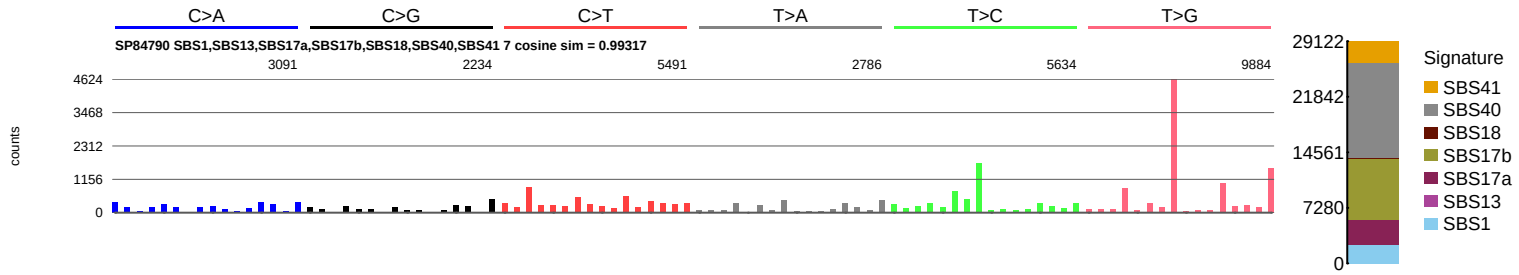

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS41,SBS44 9 cosine sim = 0.99328
2087
2421
5501
3046
6509
9561
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

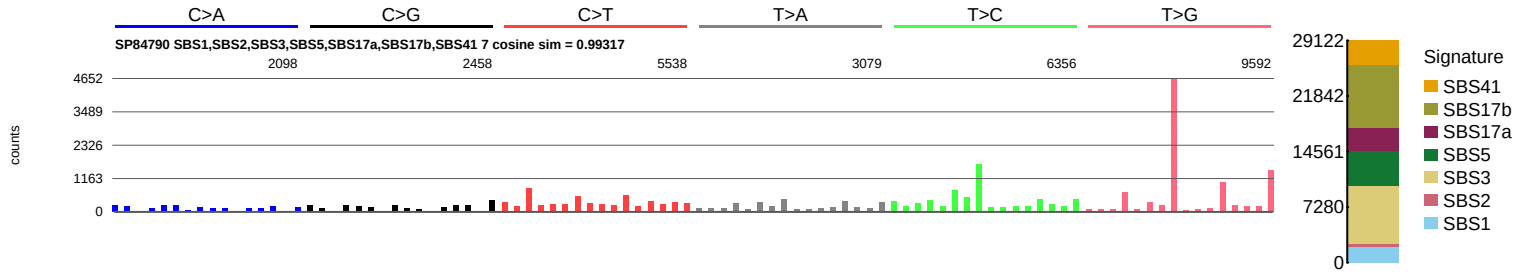

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS15,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.99327
3024
2158
5625
2784
5648
9885
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS1

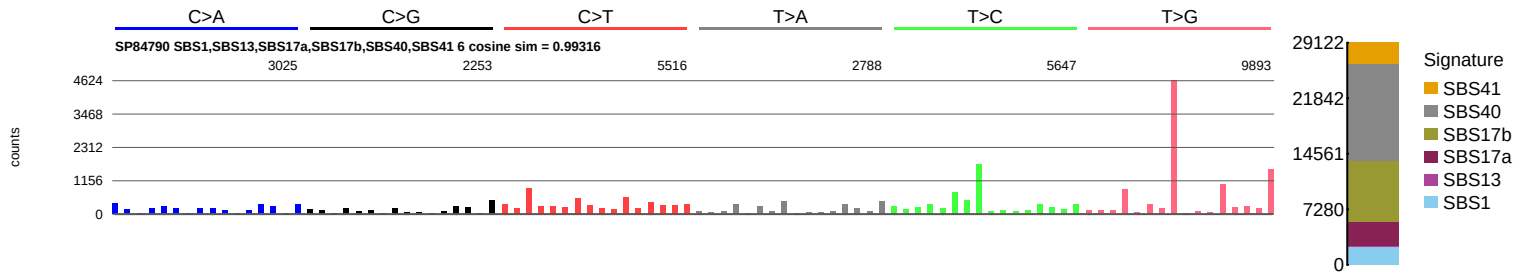

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS41 8 cosine sim = 0.99325
2113
2440
5596
3066
6331
9582
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

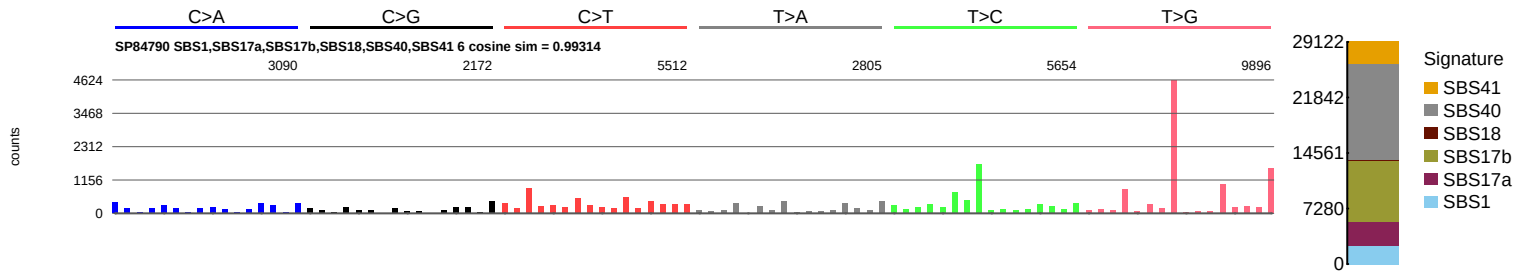

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.9932
3028
2611
5123
3196
5605
9561
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

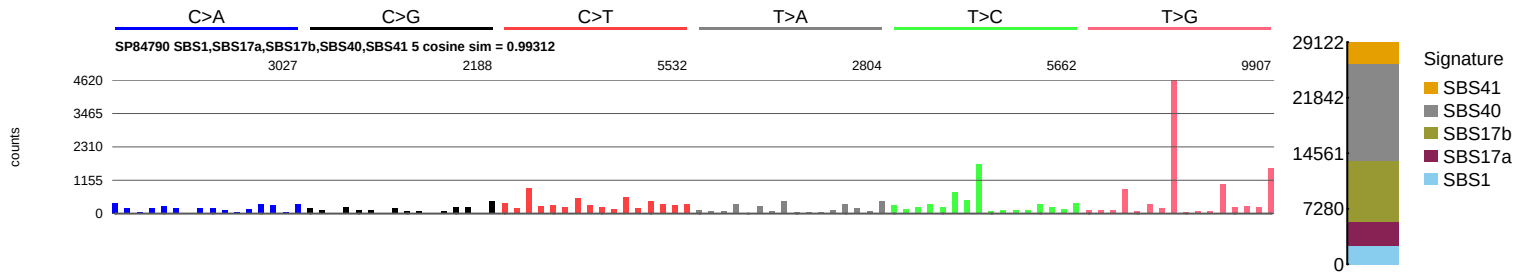

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS41,SBS44 8 cosine sim = 0.99317
2107
2448
5559
3075
6350
9585
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

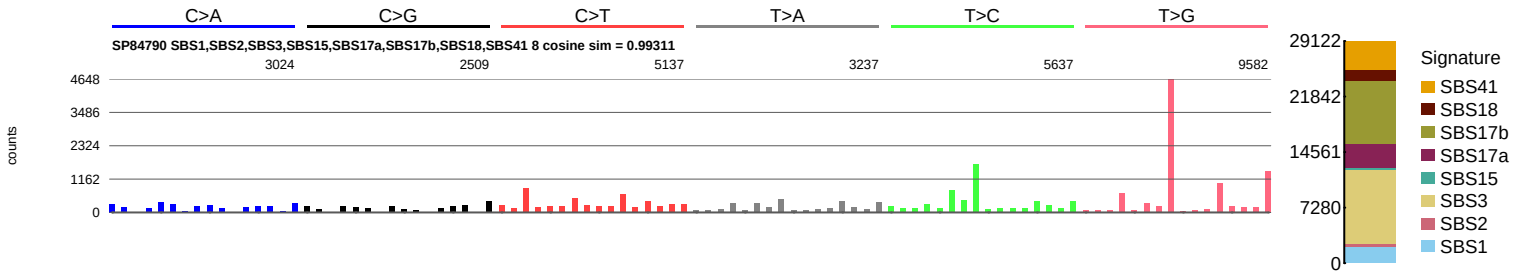

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.99317
3091
2234
5491
2786
5634
9884
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS1

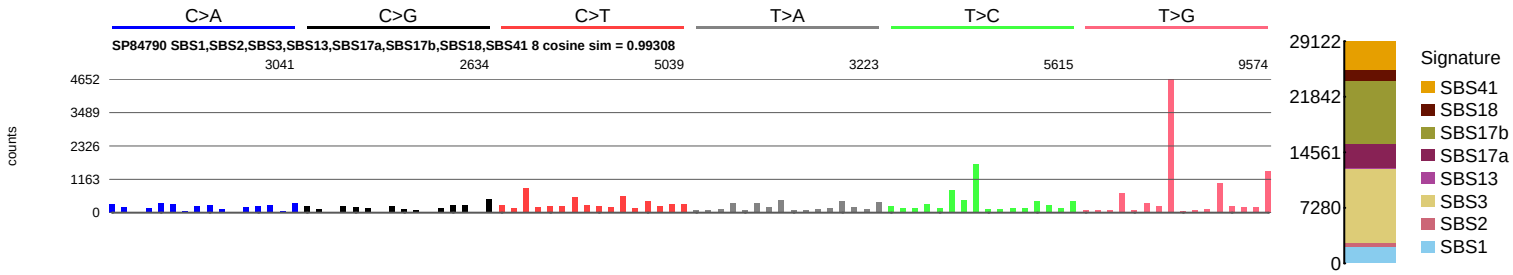

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS41 7 cosine sim = 0.99317
2098
2458
5538
3079
6356
9592
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS41
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

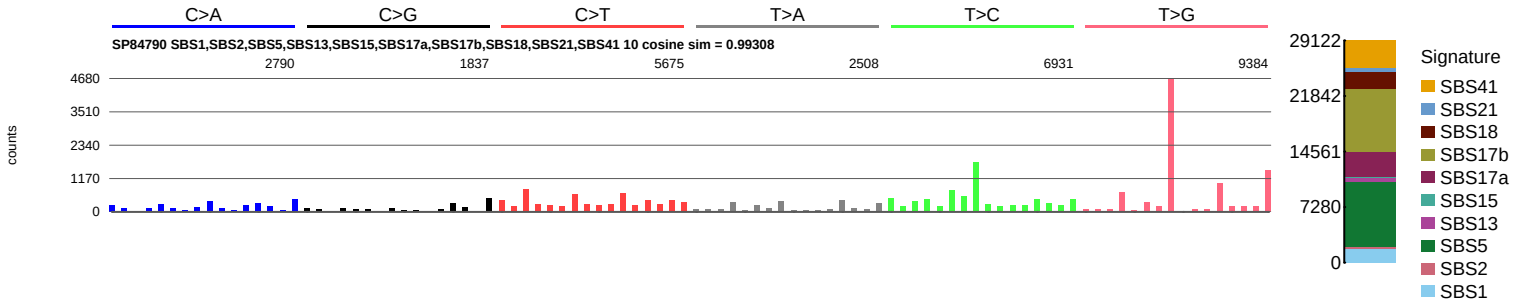

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.99316
3025
2253
5516
2788
5647
9893
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS13
SBS1

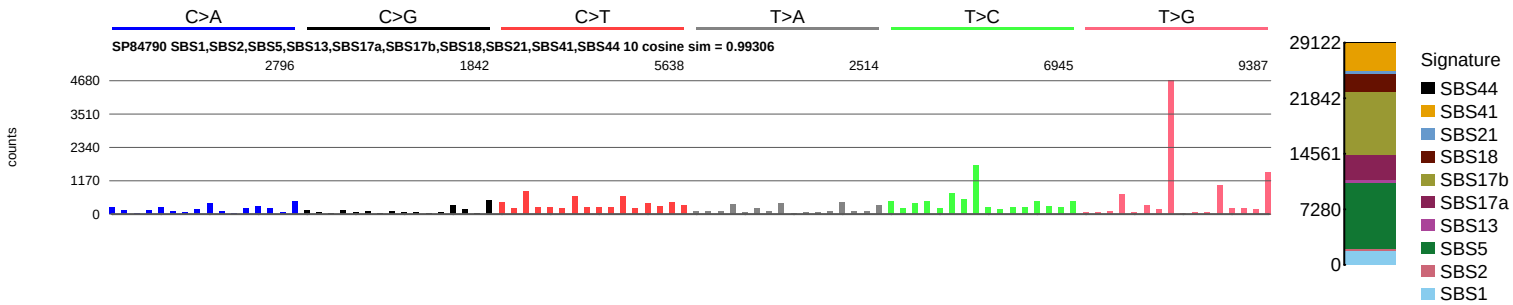

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS17a,SBS17b,SBS18,SBS40,SBS41 6 cosine sim = 0.99314
3090
2172
5512
2805
5654
9896
counts
4624
3468
2312
1156
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS1

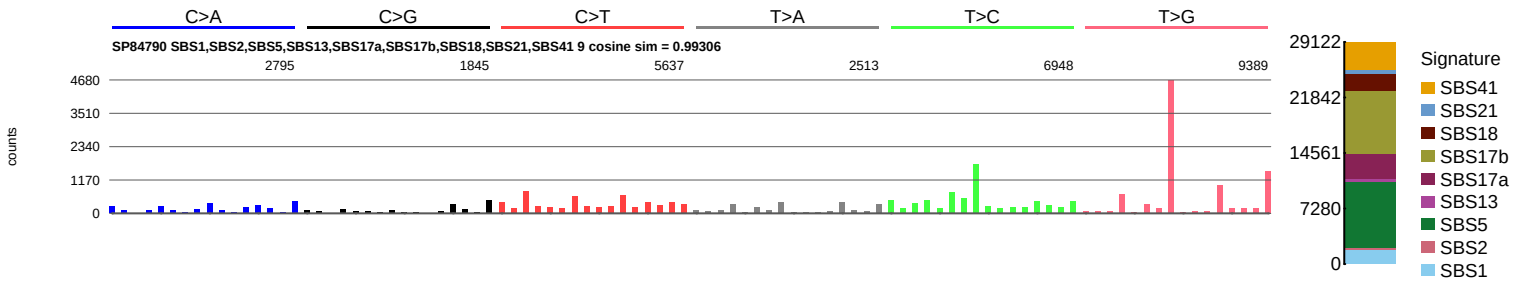

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS17a,SBS17b,SBS40,SBS41 5 cosine sim = 0.99312
3027
2188
5532
2804
5662
9907
counts
4620
3465
2310
1155
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS1

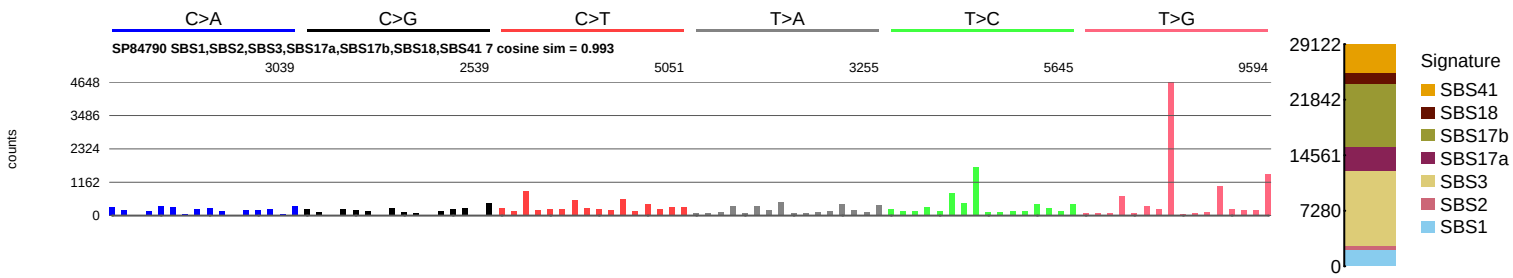

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.99311
3024
2509
5137
3237
5637
9582
counts
4648
3486
2324
1162
0
29122
21842
14561
7280
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

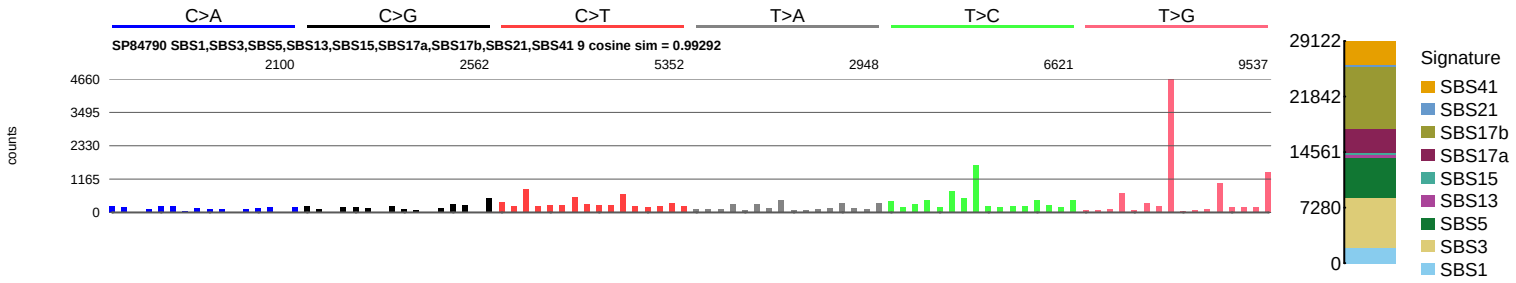

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.99308
3041
2634
5039
3223
5615
9574
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

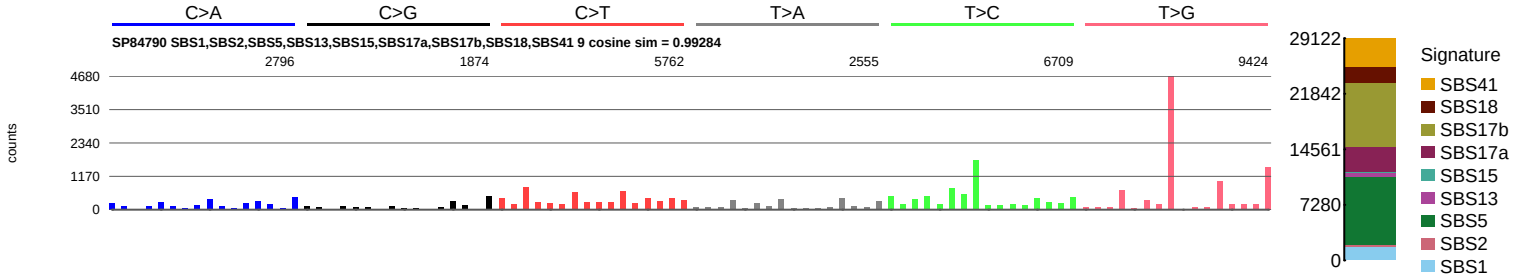

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.99308
2790
1837
5675
2508
6931
9384
counts
4680
3510
2340
1170
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

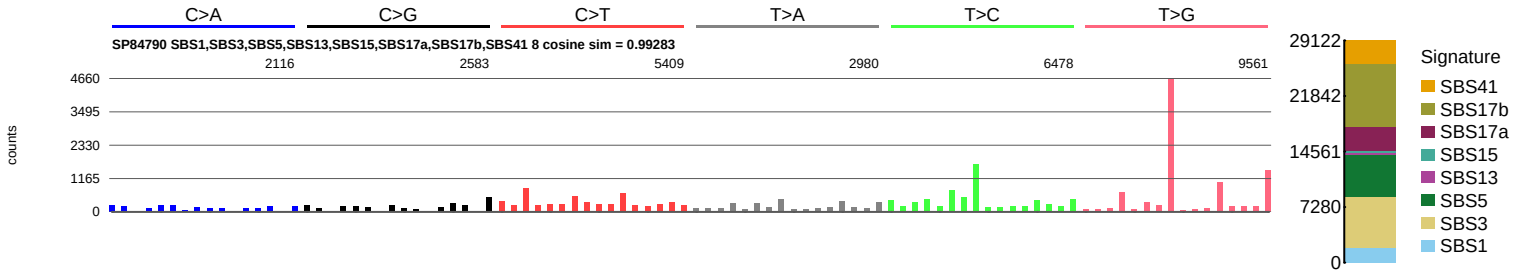

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.99306
2796
1842
5638
2514
6945
9387
counts
4680
3510
2340
1170
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

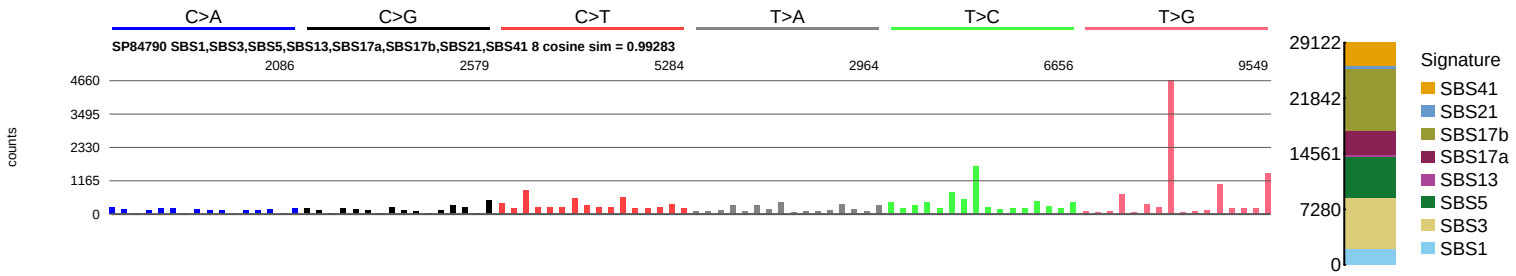

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.99306
2795
1845
5637
2513
6948
9389
counts
4680
3510
2340
1170
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

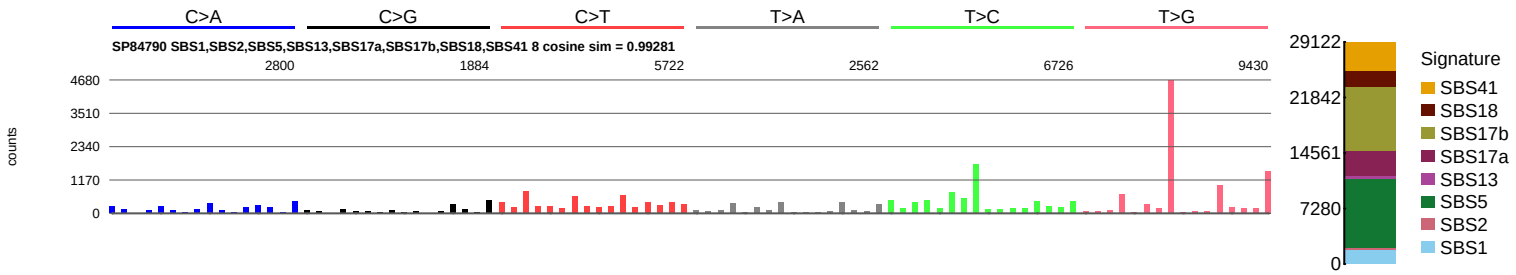

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.993
3039
2539
5051
3255
5645
9594
counts
4648
3486
2324
1162
0
29122
21842
14561
7280
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

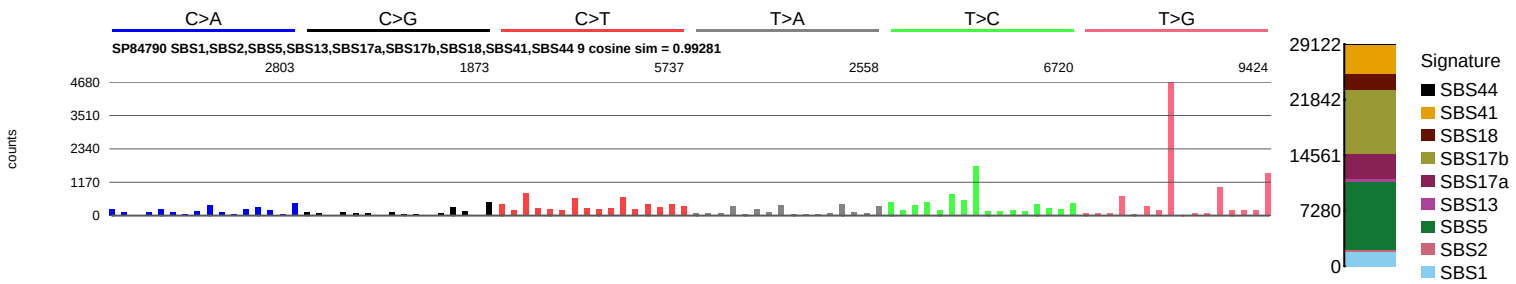

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 9 cosine sim = 0.99292
2100
2562
5352
2948
6621
9537
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

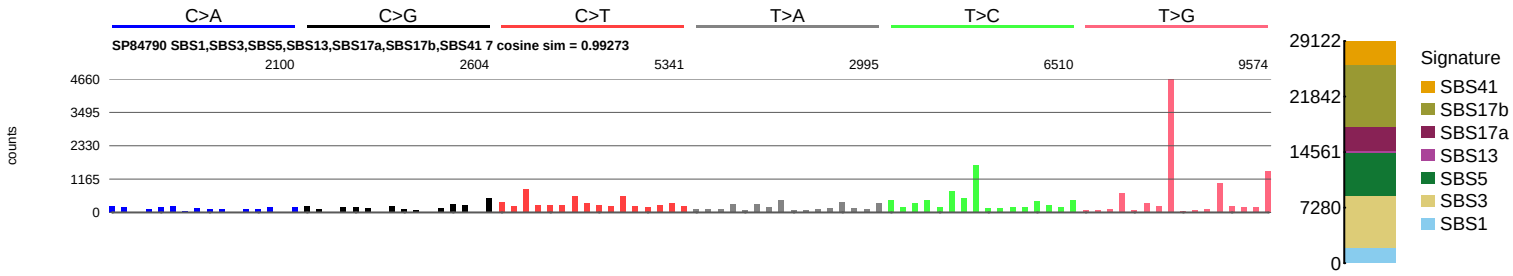

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.99284
2796
1874
5762
2555
6709
9424
counts
4680
3510
2340
1170
0
29122
21842
14561
7280
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

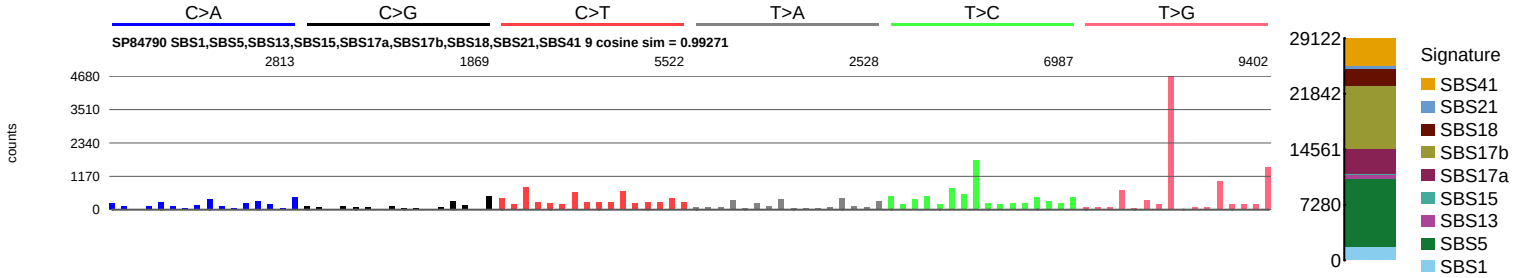

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS41 8 cosine sim = 0.99283
2116
2583
5409
2980
6478
9561
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

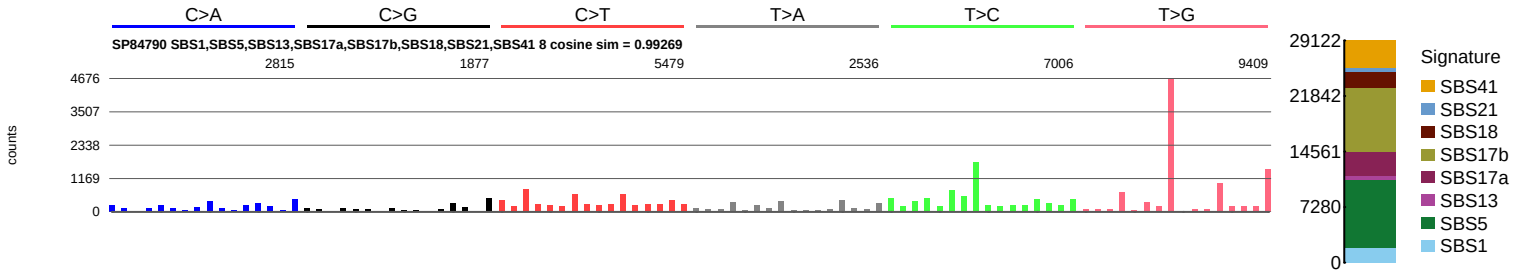

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.99283
2086
2579
5284
2964
6656
9549
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

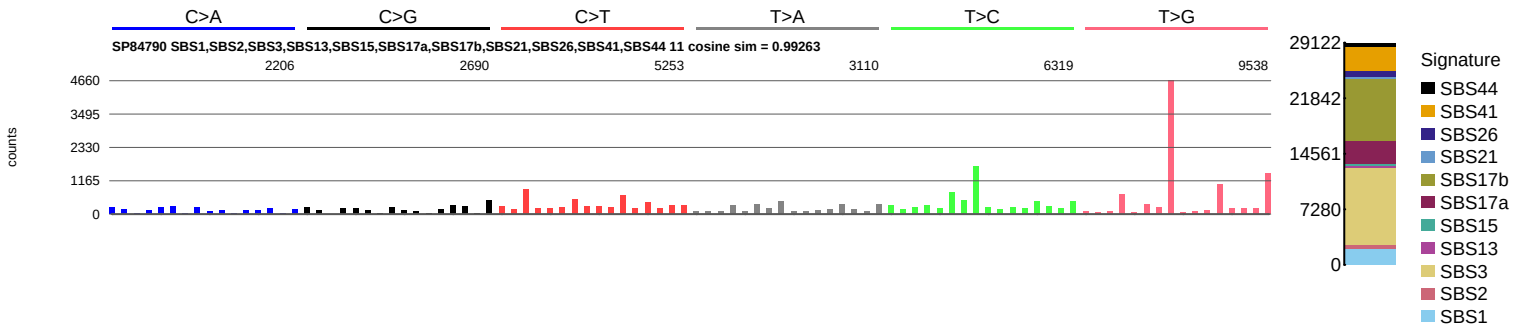

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.99281
2800
1884
5722
2562
6726
9430
counts
4680
3510
2340
1170
0
29122
21842
14561
7280
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

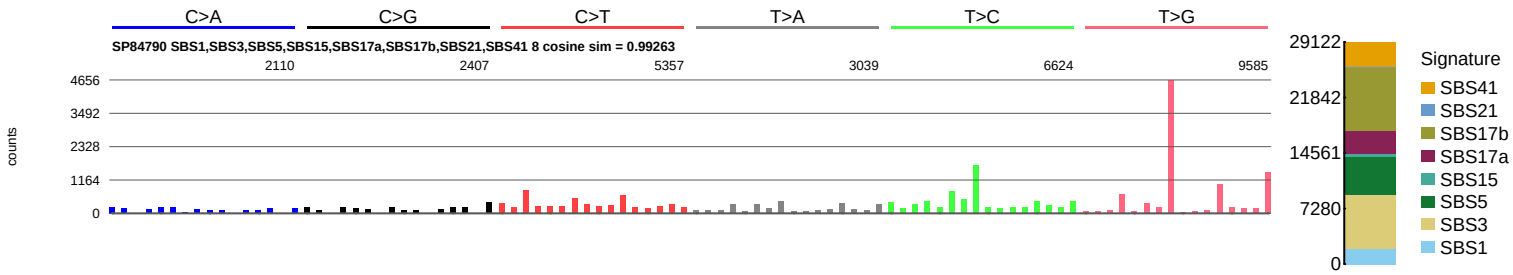

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.99281
2803
1873
5737
2558
6720
9424
counts
4680
3510
2340
1170
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

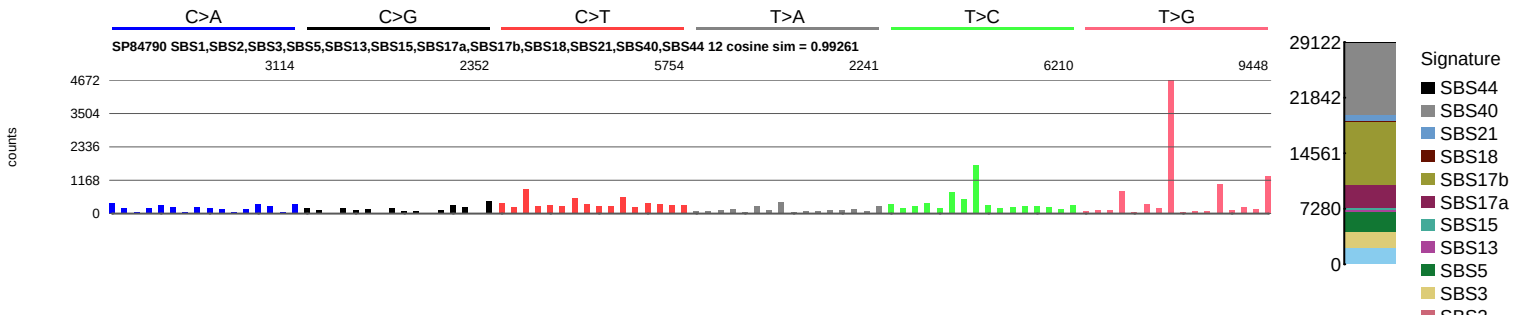

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS41 7 cosine sim = 0.99273
2100
2604
5341
2995
6510
9574
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS41
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

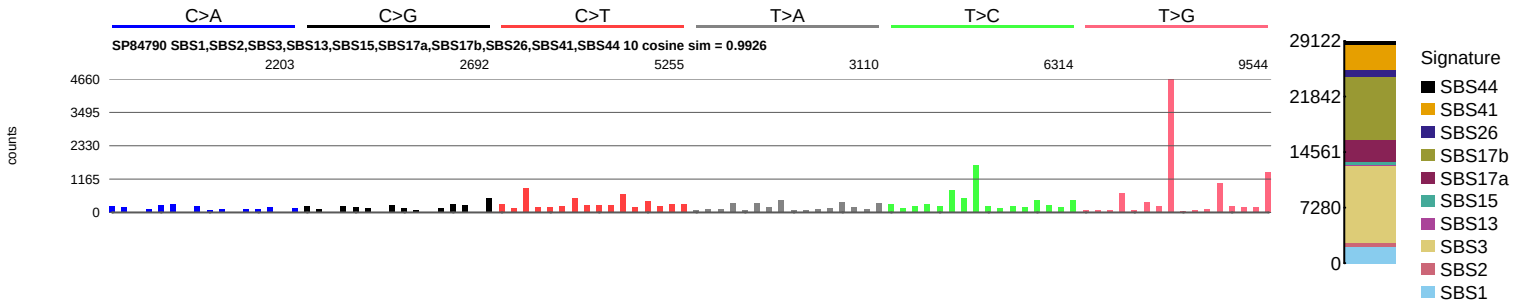

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.99271
2813
1869
5522
2528
6987
9402
counts
4680
3510
2340
1170
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

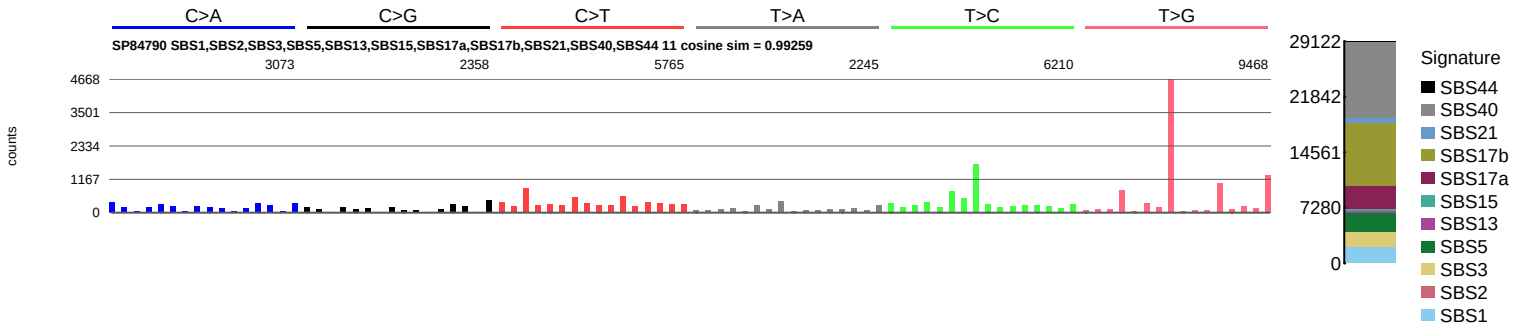

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.99269
2815
1877
5479
2536
7006
9409
counts
4676
3507
2338
1169
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1

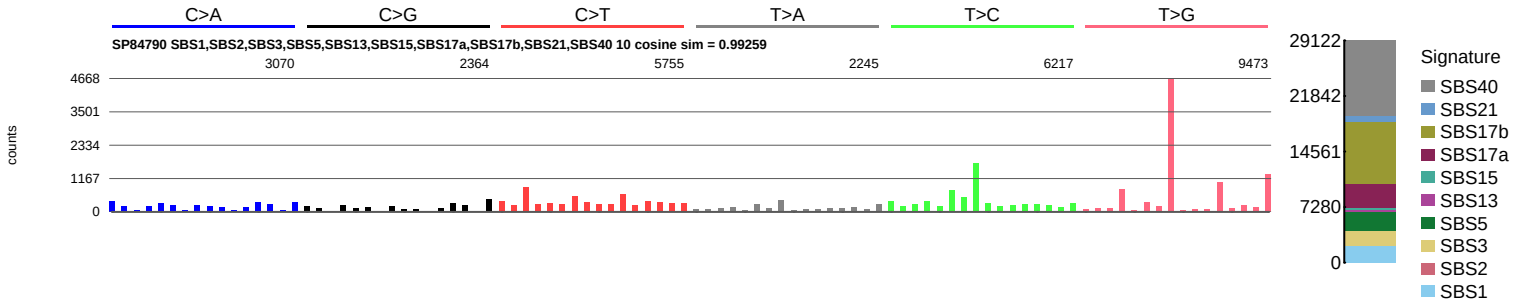

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 11 cosine sim = 0.99263
2206
2690
5253
3110
6319
9538
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

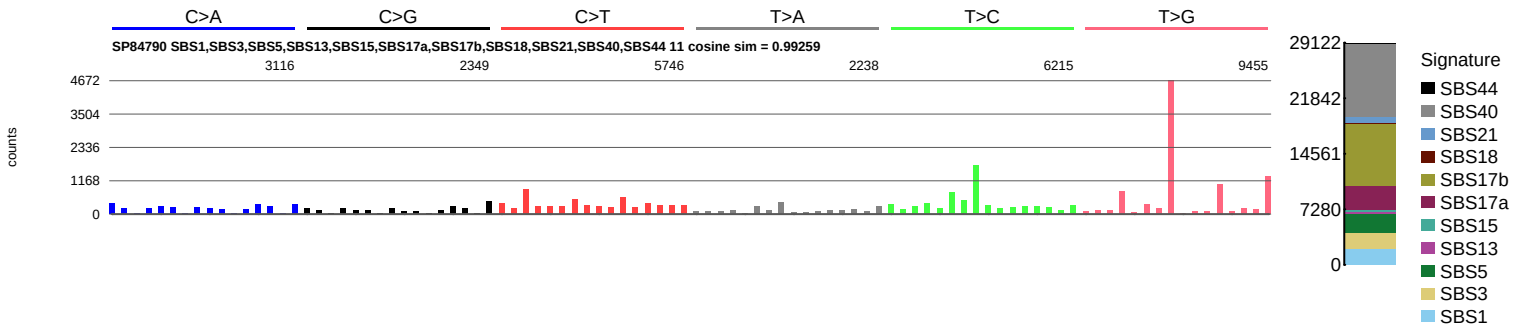

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.99263
2110
2407
5357
3039
6624
9585
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1

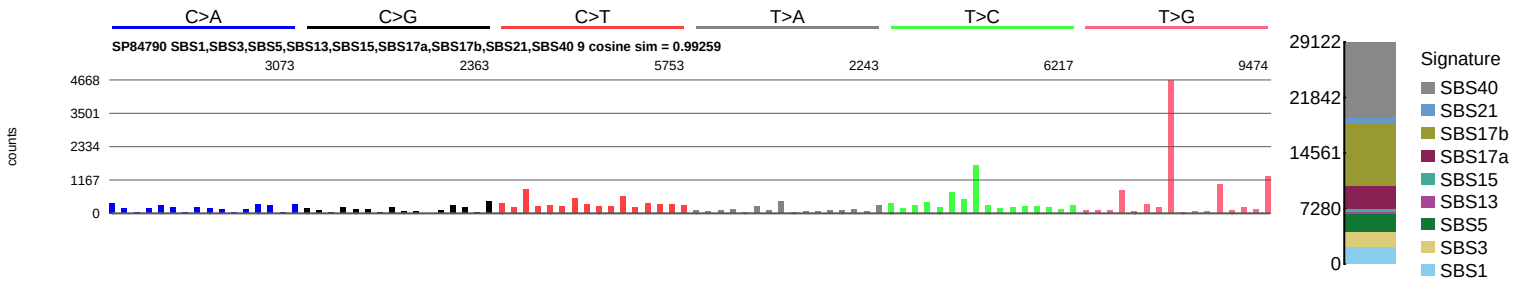

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 12 cosine sim = 0.99261
3114
2352
5754
2241
6210
9448
counts
4672
3504
2336
1168
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3

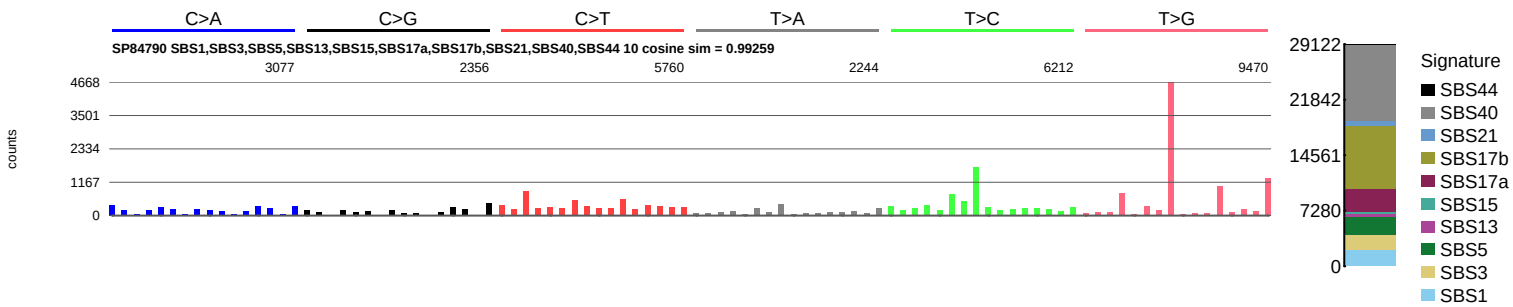

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41,SBS44 10 cosine sim = 0.9926
2203
2692
5255
3110
6314
9544
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

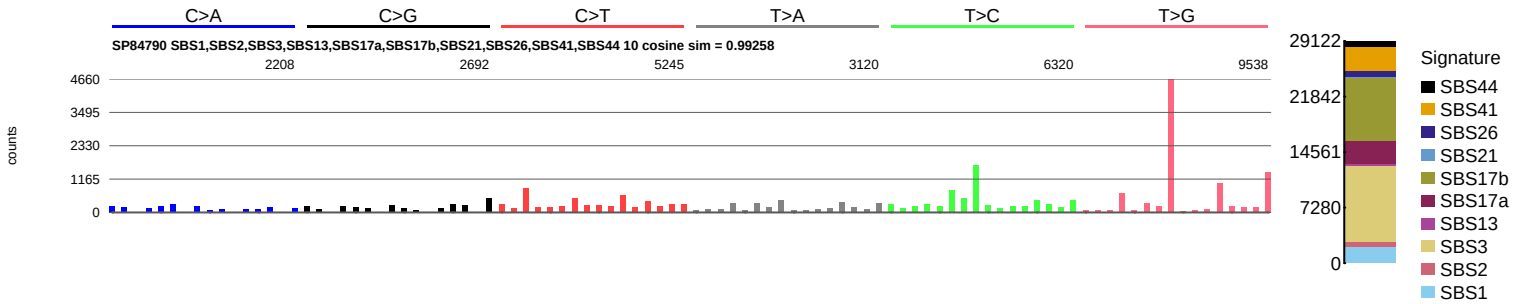

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS44 11 cosine sim = 0.99259
3073
2358
5765
2245
6210
9468
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

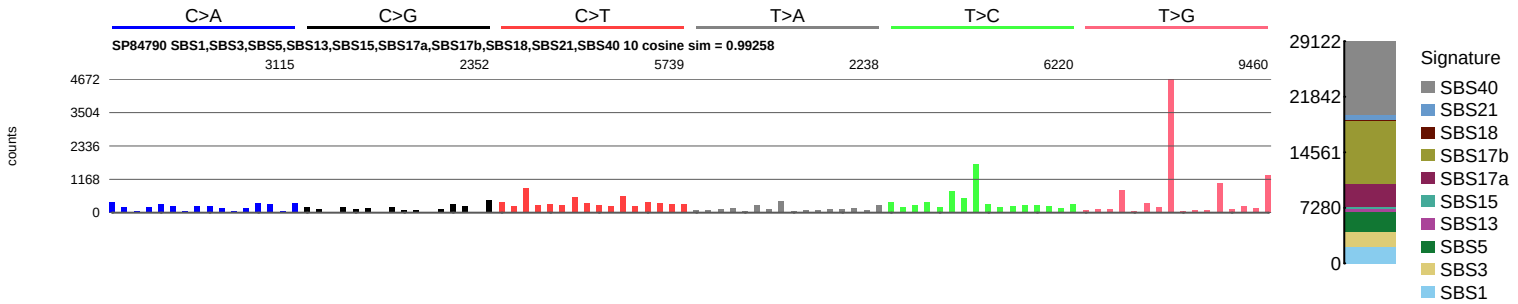

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40 10 cosine sim = 0.99259
3070
2364
5755
2245
6217
9473
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

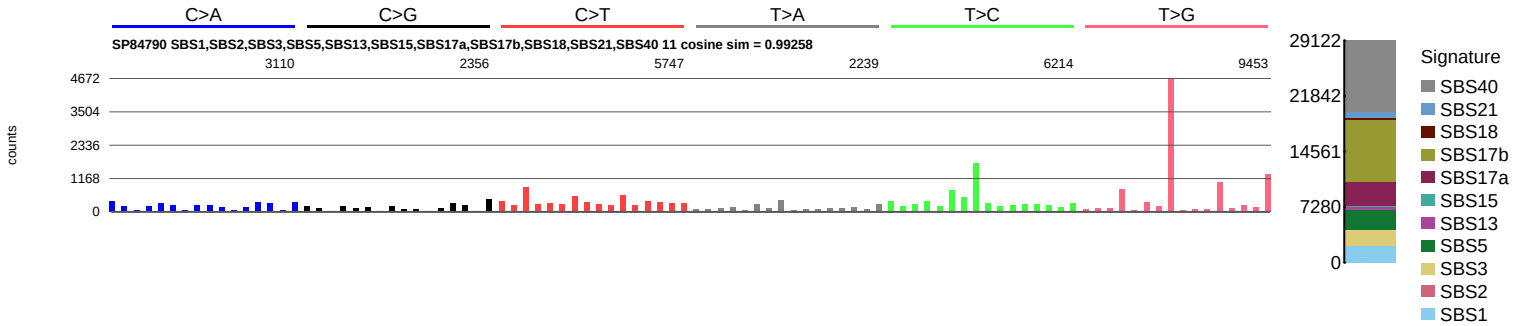

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 11 cosine sim = 0.99259
3116
2349
5746
2238
6215
9455
counts
4672
3504
2336
1168
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

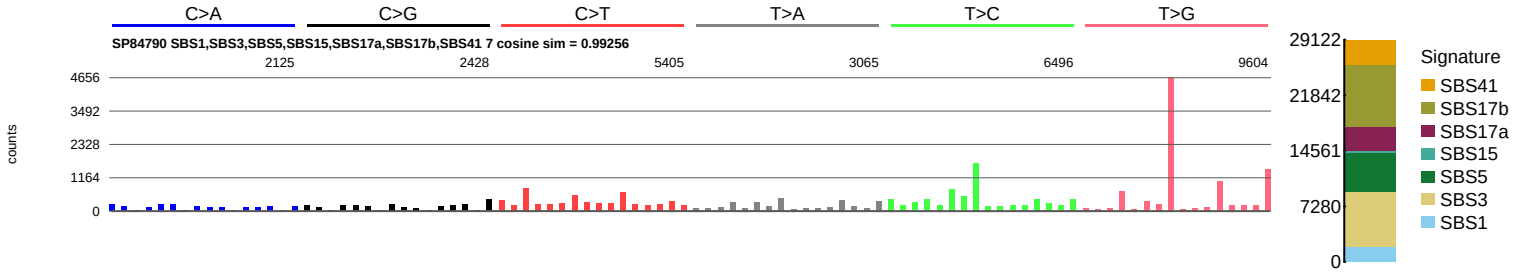

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40 9 cosine sim = 0.99259
3073
2363
5753
2243
6217
9474
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

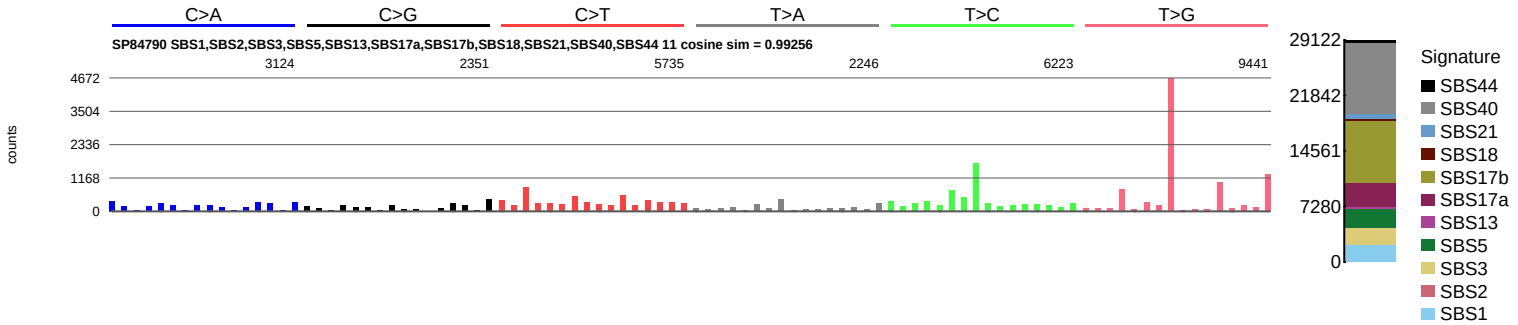

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS44 10 cosine sim = 0.99259
3077
2356
5760
2244
6212
9470
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

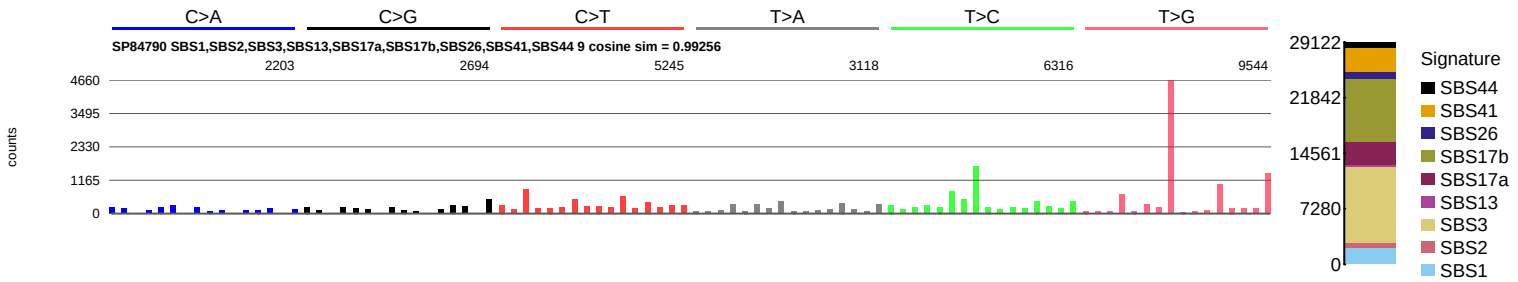

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 10 cosine sim = 0.99258
2208
2692
5245
3120
6320
9538
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

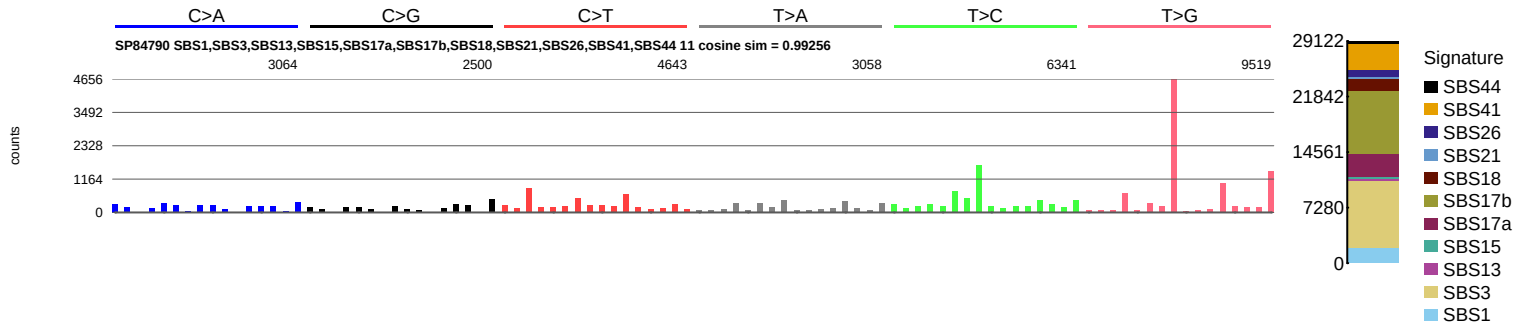

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 10 cosine sim = 0.99258
3115
2352
5739
2238
6220
9460
counts
4672
3504
2336
1168
0
29122
21842
14561
7280
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

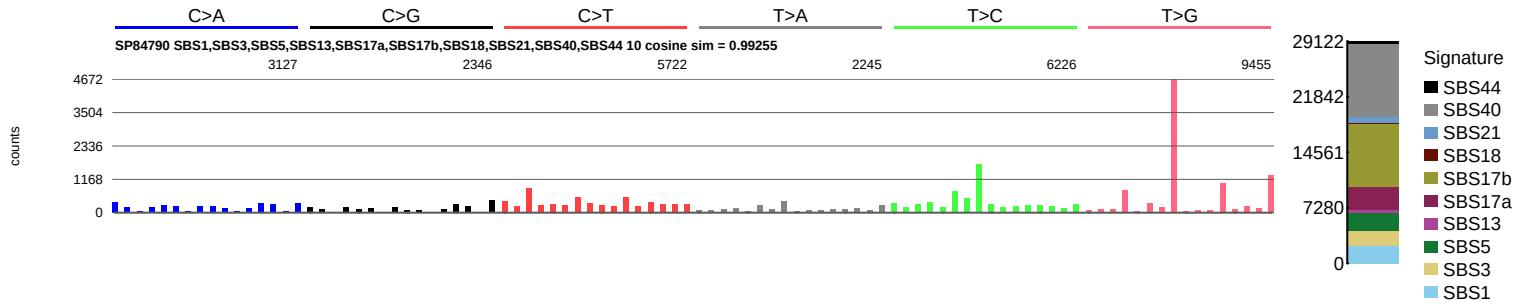

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 11 cosine sim = 0.99258
3110
2356
5747
2239
6214
9453
counts
4672
3504
2336
1168
0
29122
21842
14561
7280
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

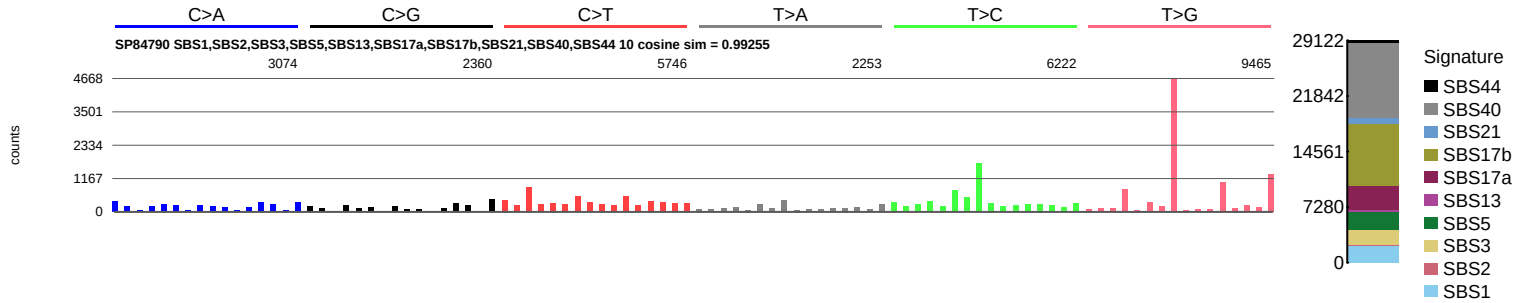

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.99256
2125
2428
5405
3065
6496
9604
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1

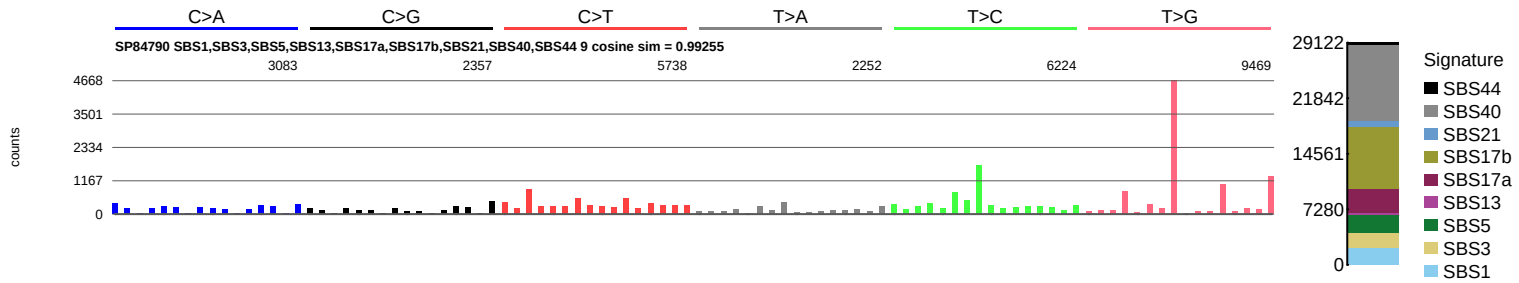

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 11 cosine sim = 0.99256
3124
2351
5735
2246
6223
9441
counts
4672
3504
2336
1168
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

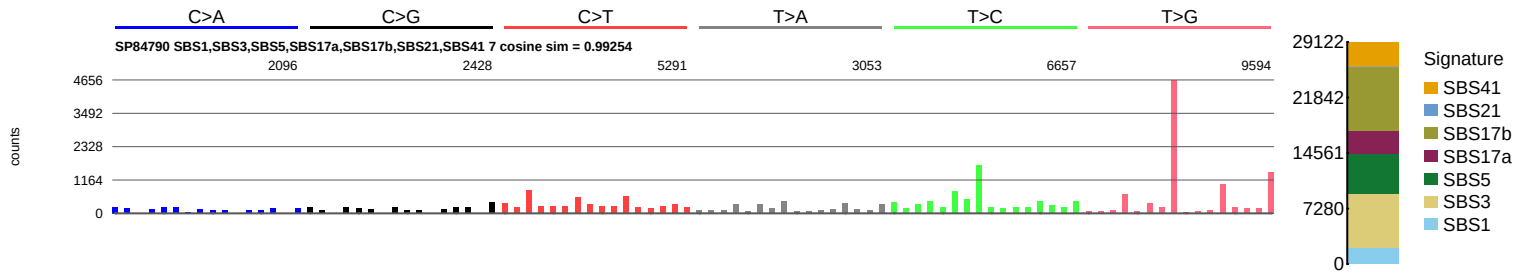

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS26,SBS41,SBS44 9 cosine sim = 0.99256
2203
2694
5245
3118
6316
9544
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

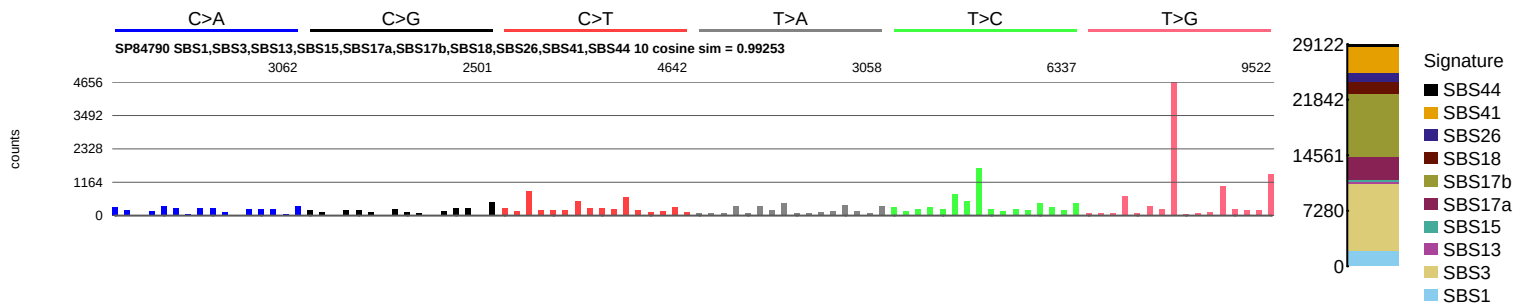

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 11 cosine sim = 0.99256
3064
2500
4643
3058
6341
9519
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

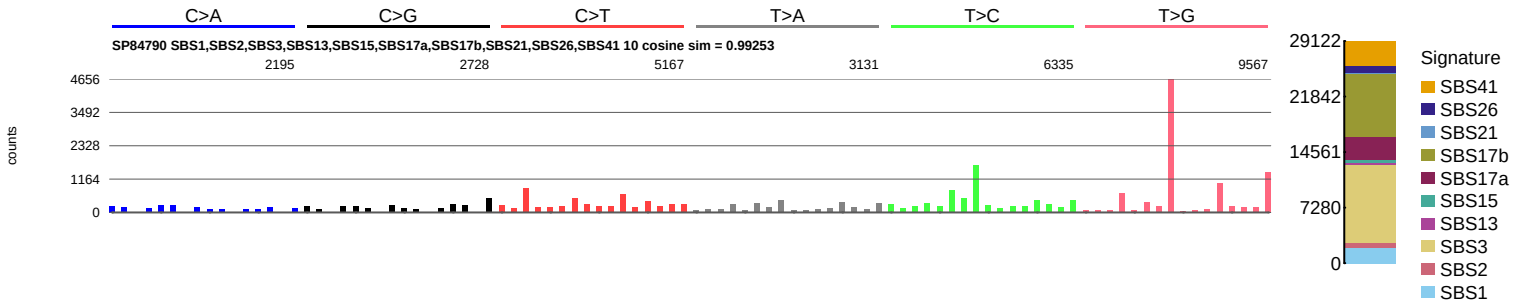

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.99255
3127
2346
5722
2245
6226
9455
counts
4672
3504
2336
1168
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

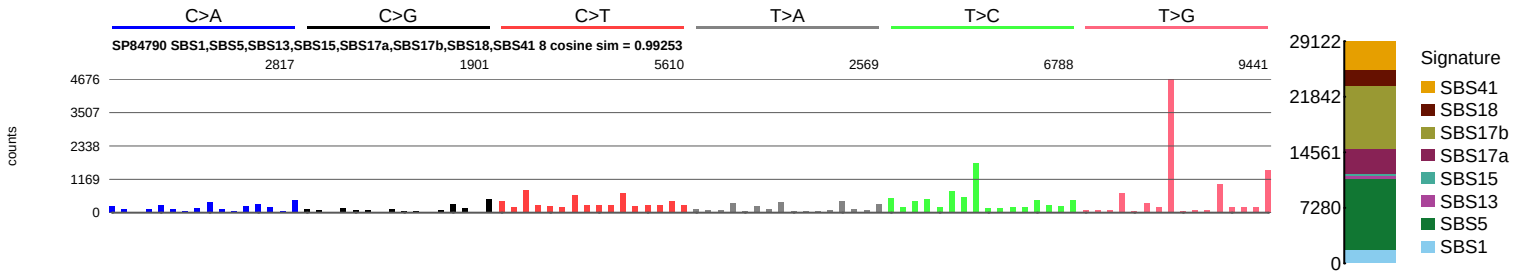

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS44 10 cosine sim = 0.99255
3074
2360
5746
2253
6222
9465
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

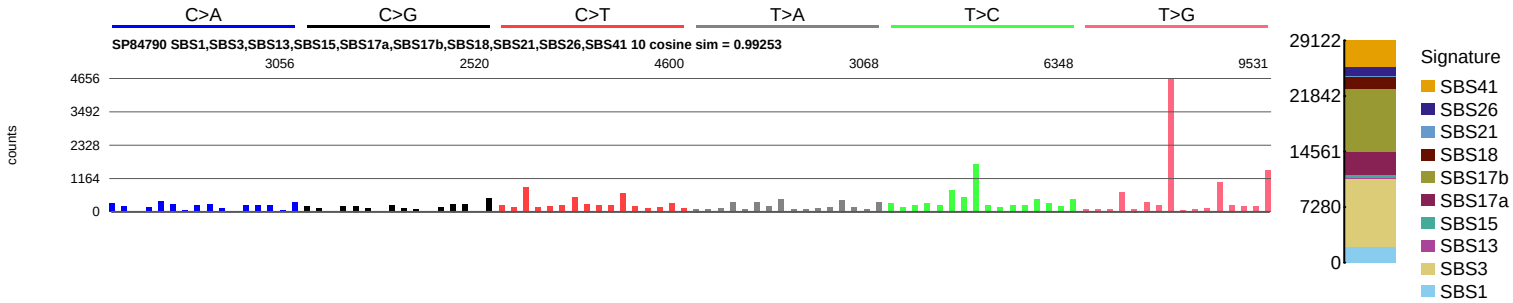

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS44 9 cosine sim = 0.99255
3083
2357
5738
2252
6224
9469
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

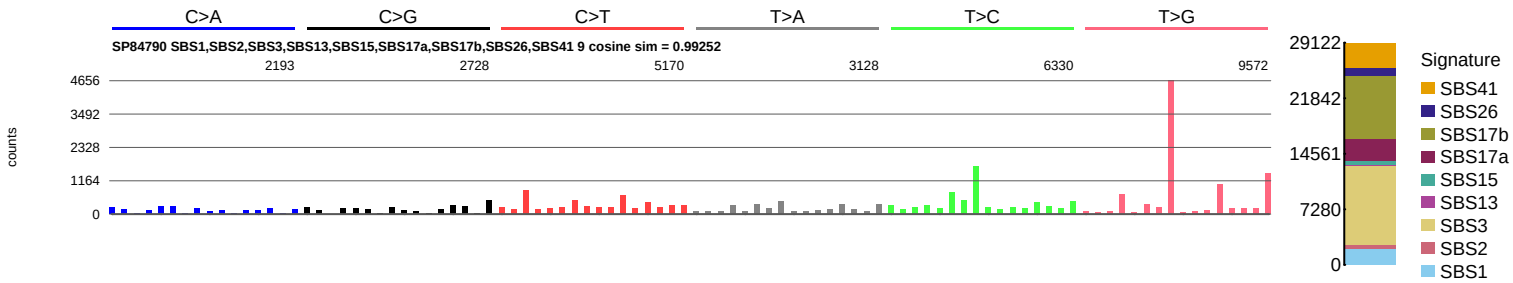

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.99254
2096
2428
5291
3053
6657
9594
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS1

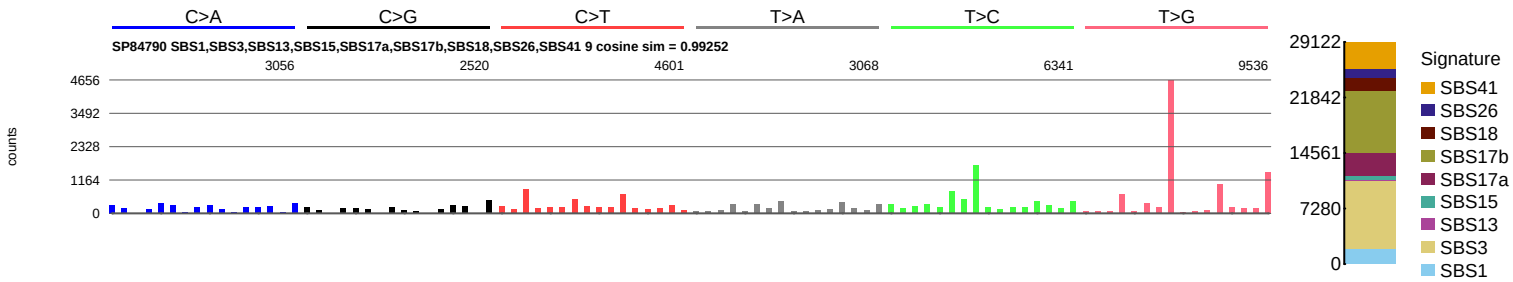

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.99253
3062
2501
4642
3058
6337
9522
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

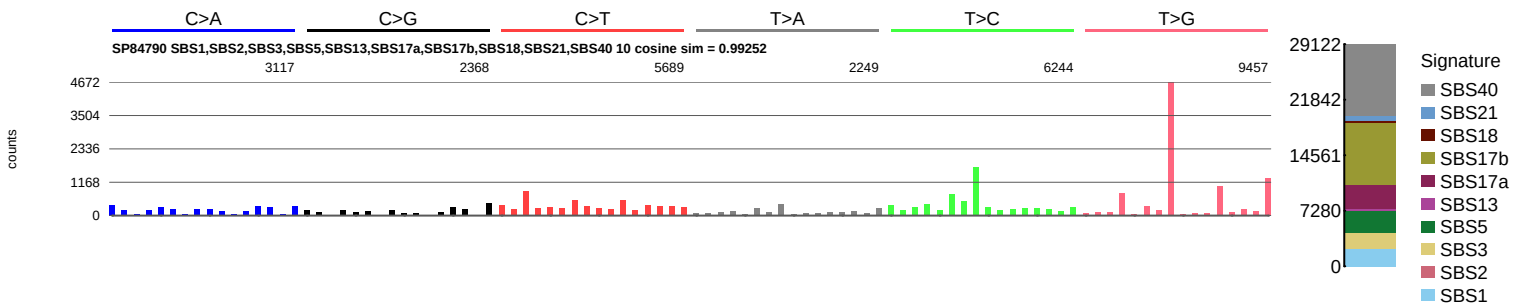

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41 10 cosine sim = 0.99253
2195
2728
5167
3131
6335
9567
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

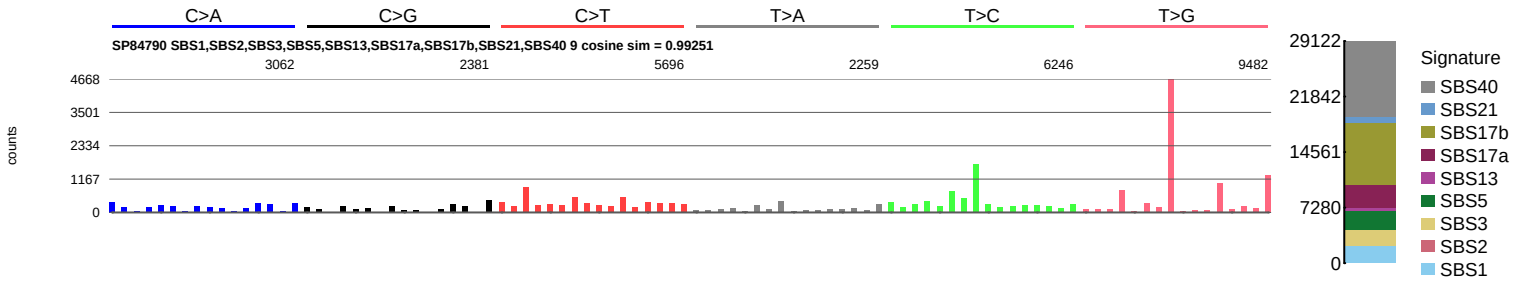

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.99253
2817
1901
5610
2569
6788
9441
counts
4676
3507
2338
1169
0
29122
21842
14561
7280
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

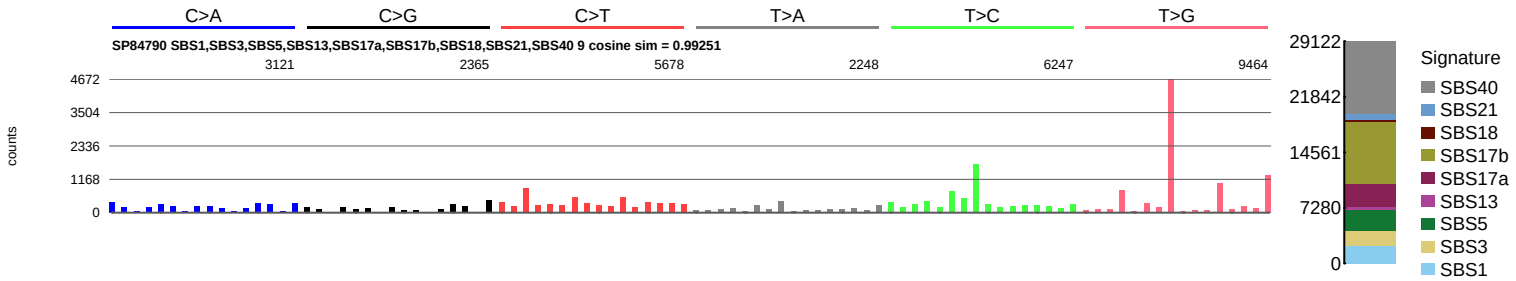

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 10 cosine sim = 0.99253
3056
2520
4600
3068
6348
9531
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

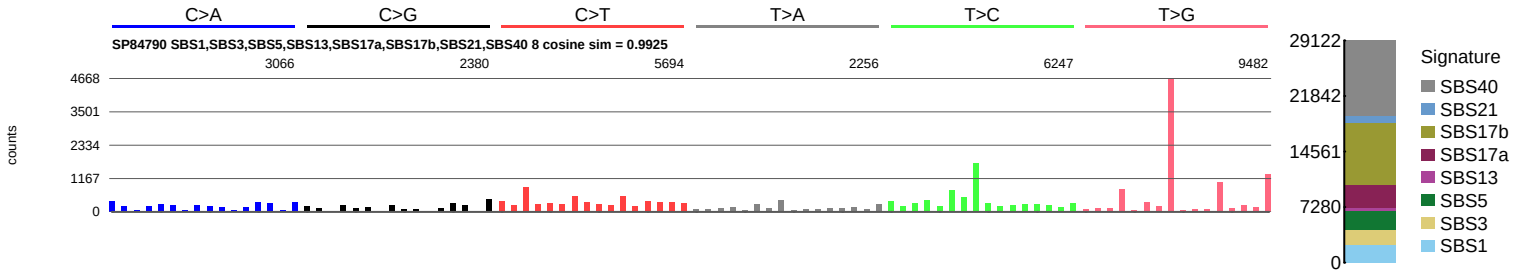

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41 9 cosine sim = 0.99252
2193
2728
5170
3128
6330
9572
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

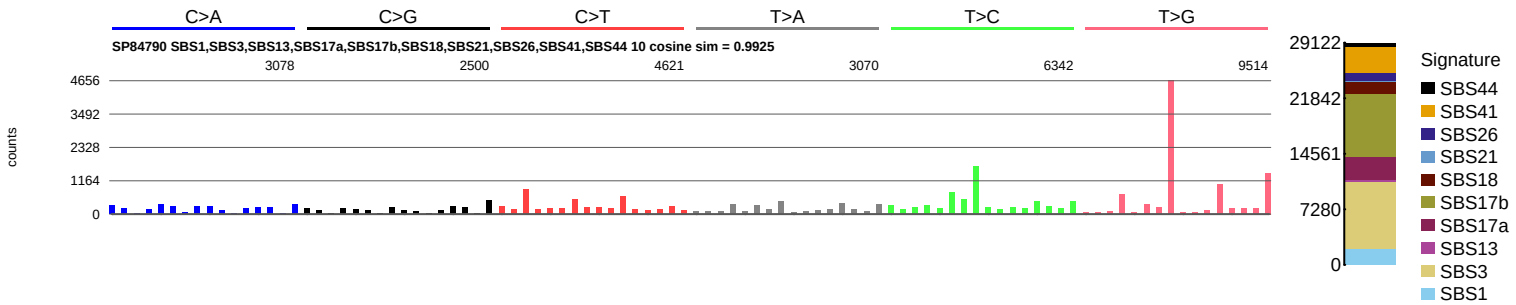

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.99252
3056
2520
4601
3068
6341
9536
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

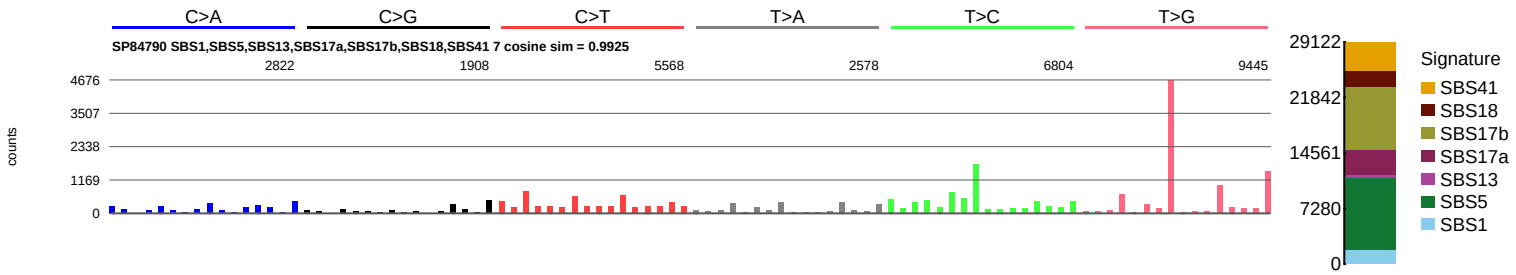

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 10 cosine sim = 0.99252
3117
2368
5689
2249
6244
9457
counts
4672
3504
2336
1168
0
29122
21842
14561
7280
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

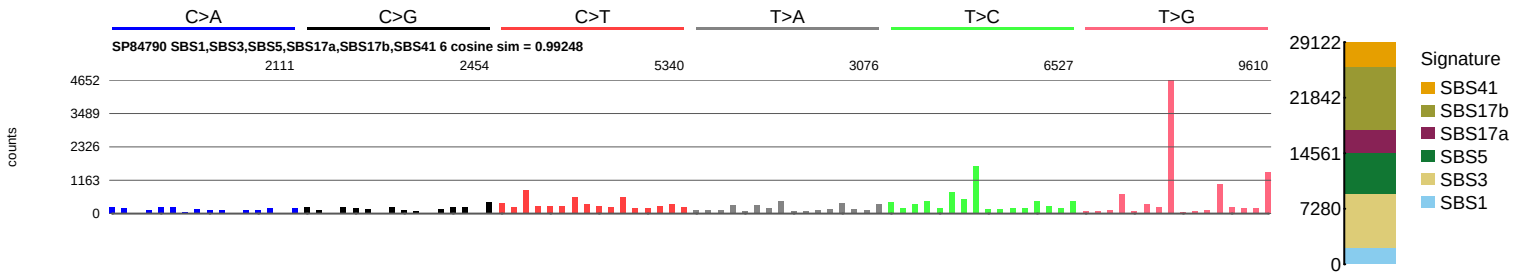

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS40 9 cosine sim = 0.99251
3062
2381
5696
2259
6246
9482
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

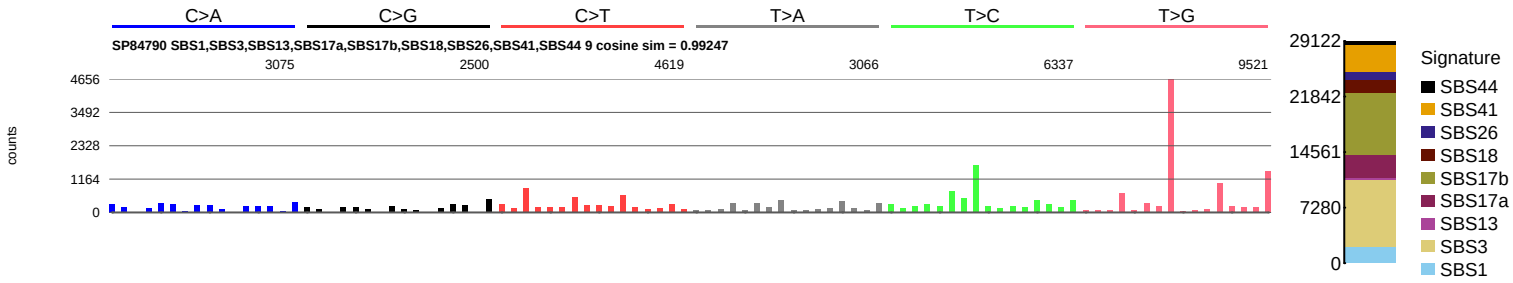

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.99251
3121
2365
5678
2248
6247
9464
counts
4672
3504
2336
1168
0
29122
21842
14561
7280
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

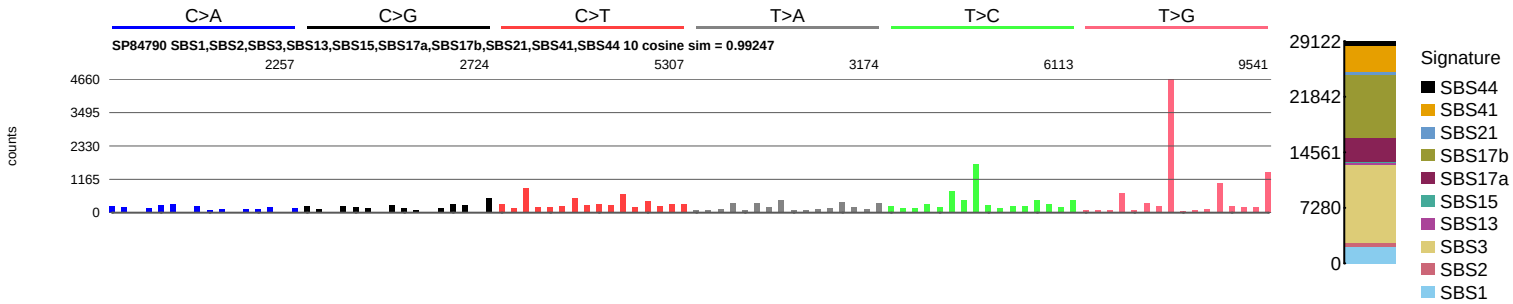

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS40 8 cosine sim = 0.9925
3066
2380
5694
2256
6247
9482
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

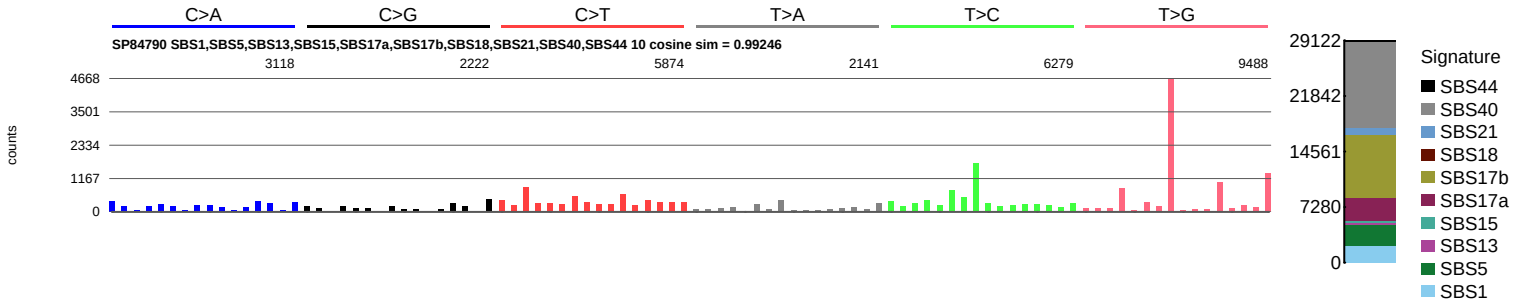

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 10 cosine sim = 0.9925
3078
2500
4621
3070
6342
9514
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1

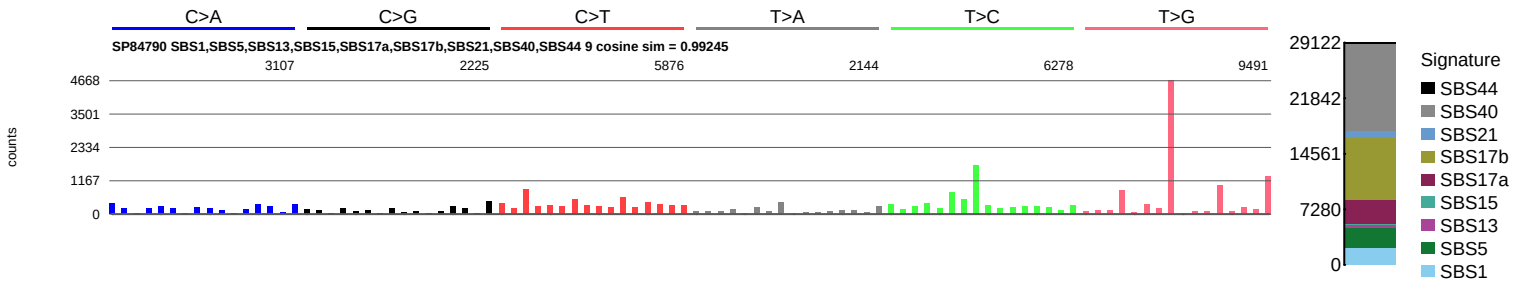

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.9925
2822
1908
5568
2578
6804
9445
counts
4676
3507
2338
1169
0
29122
21842
14561
7280
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1

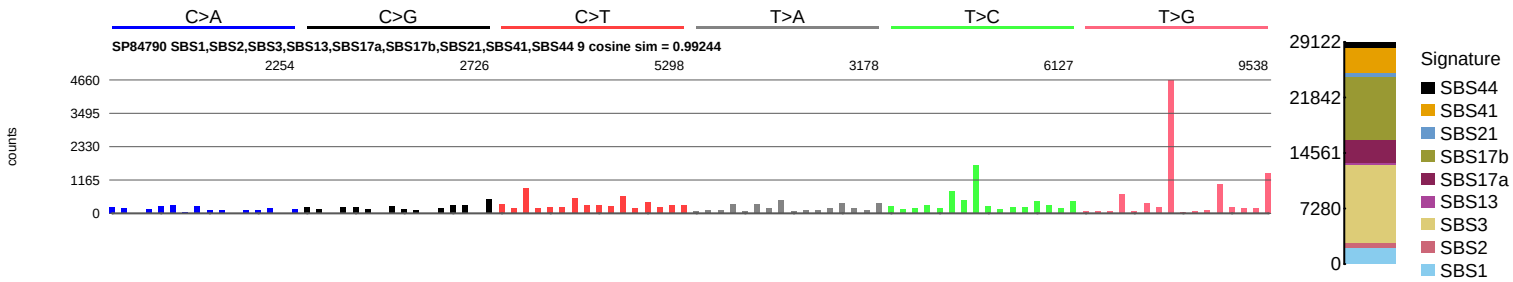

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS41 6 cosine sim = 0.99248
2111
2454
5340
3076
6527
9610
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS41
SBS17b
SBS17a
SBS5
SBS3
SBS1

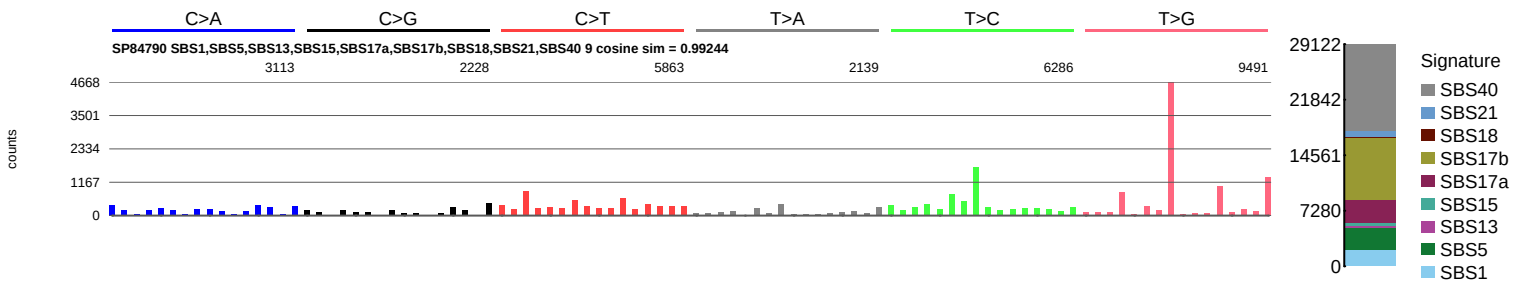

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.99247
3075
2500
4619
3066
6337
9521
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1

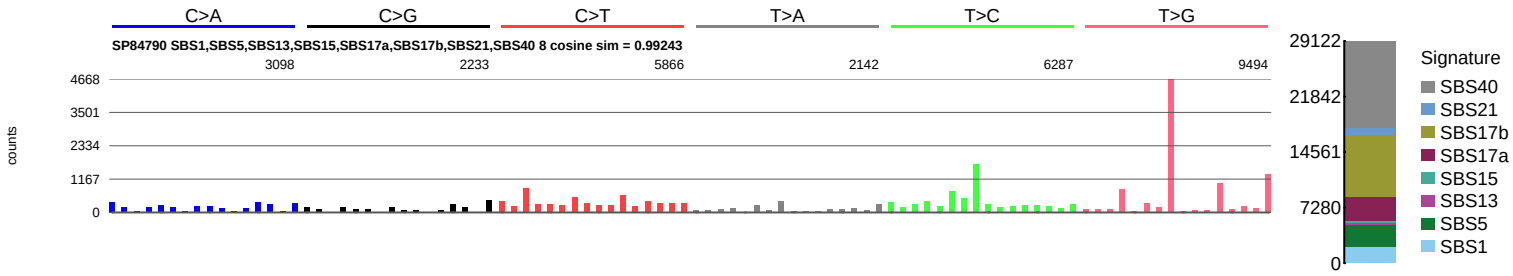

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41,SBS44 10 cosine sim = 0.99247
2257
2724
5307
3174
6113
9541
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

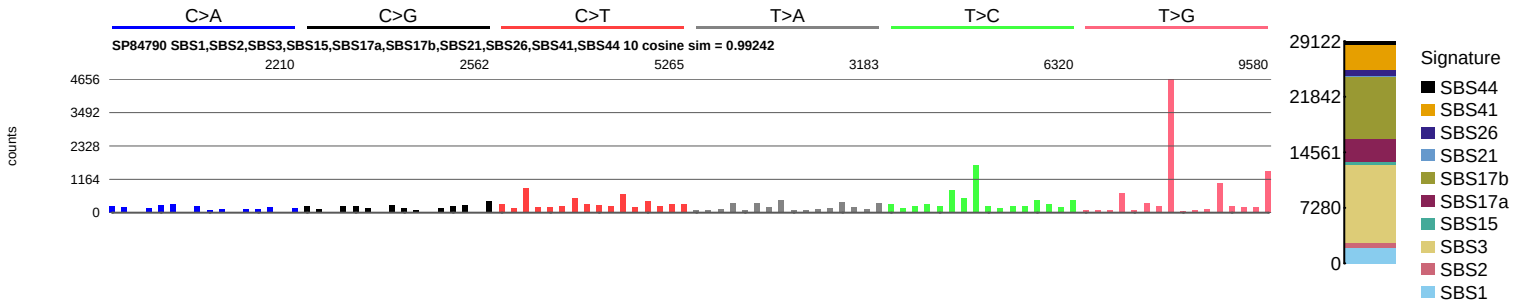

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.99246
3118
2222
5874
2141
6279
9488
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

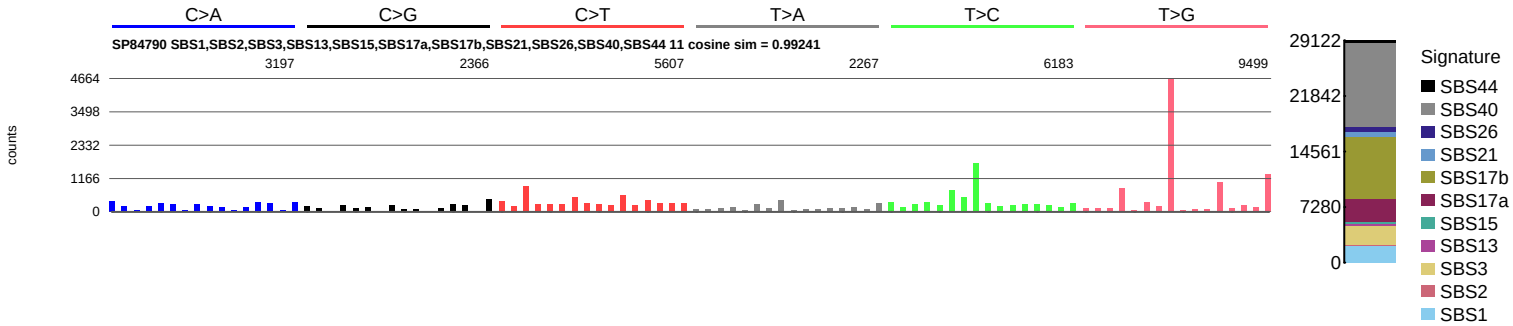

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS44 9 cosine sim = 0.99245
3107
2225
5876
2144
6278
9491
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

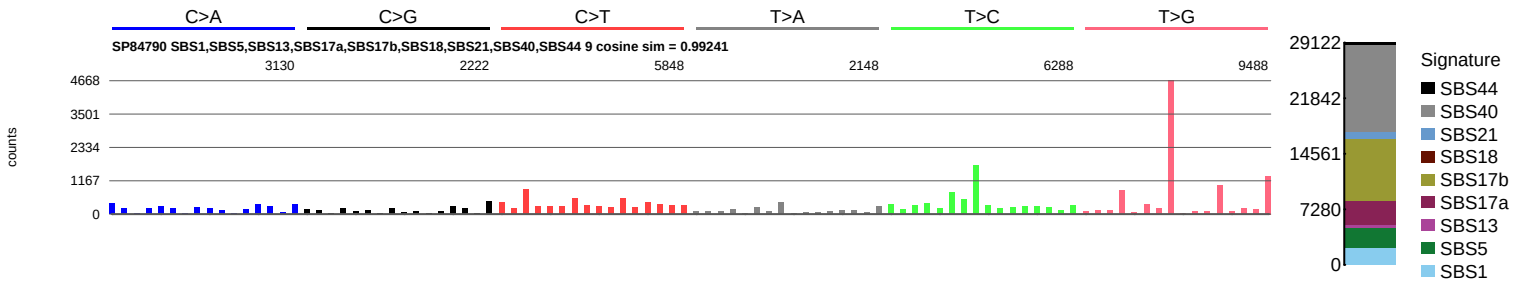

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS41,SBS44 9 cosine sim = 0.99244
2254
2726
5298
3178
6127
9538
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

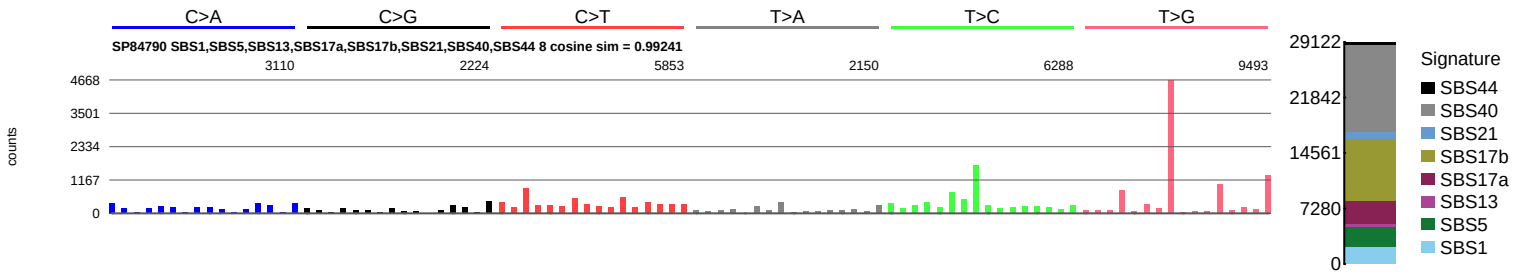

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.99244
3113
2228
5863
2139
6286
9491
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

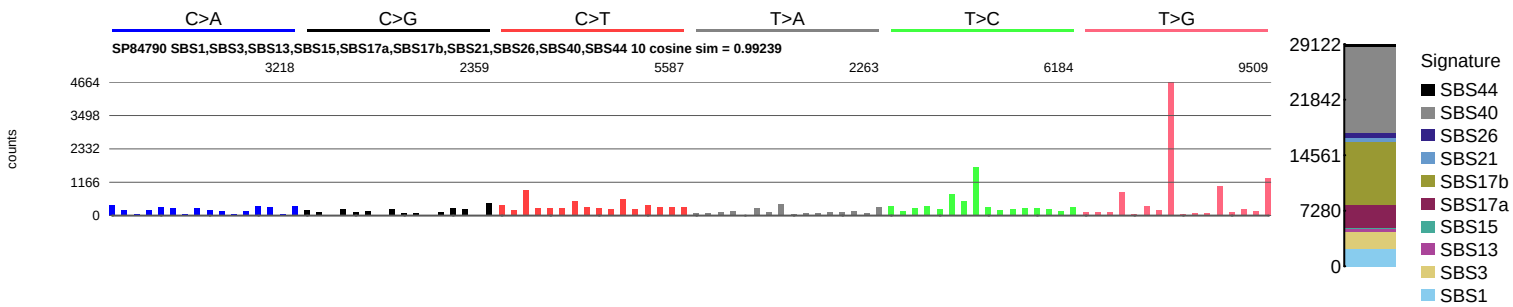

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40 8 cosine sim = 0.99243
3098
2233
5866
2142
6287
9494
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

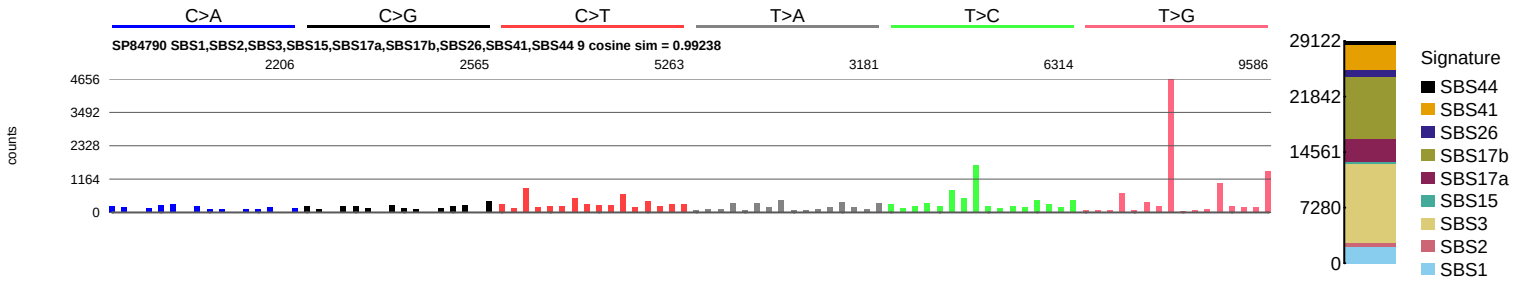

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 10 cosine sim = 0.99242
2210
2562
5265
3183
6320
9580
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

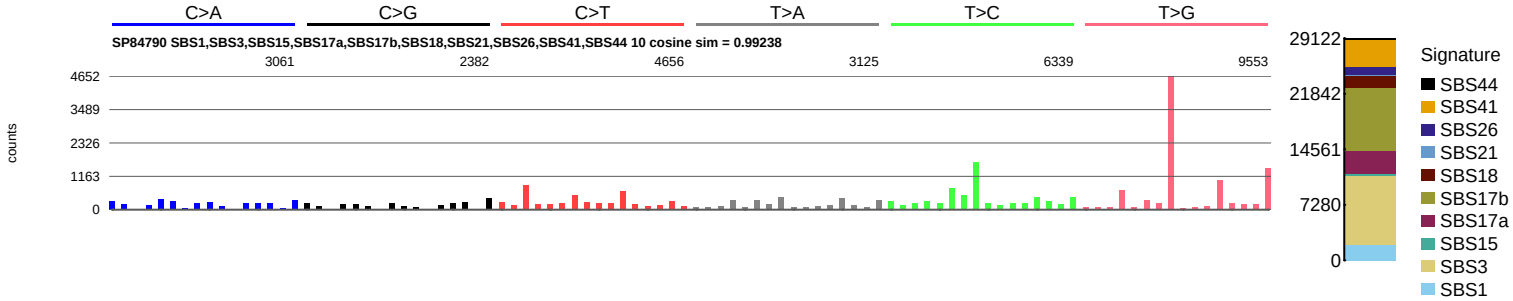

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS44 11 cosine sim = 0.99241
3197
2366
5607
2267
6183
9499
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

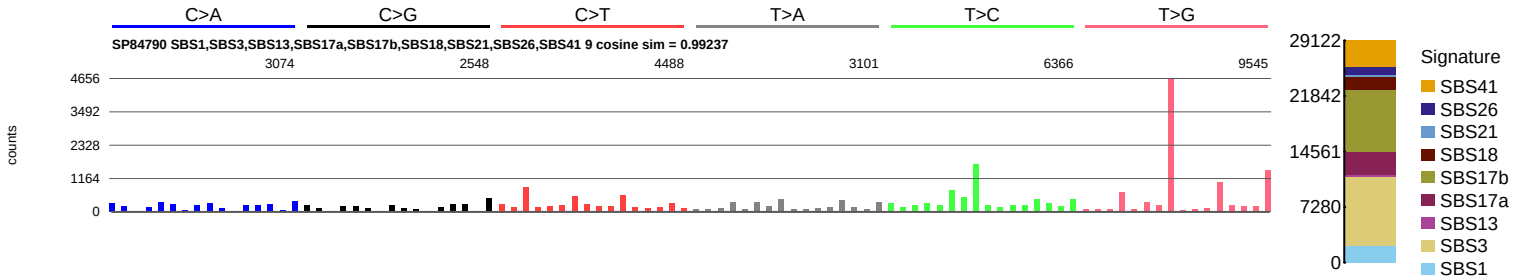

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.99241
3130
2222
5848
2148
6288
9488
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1

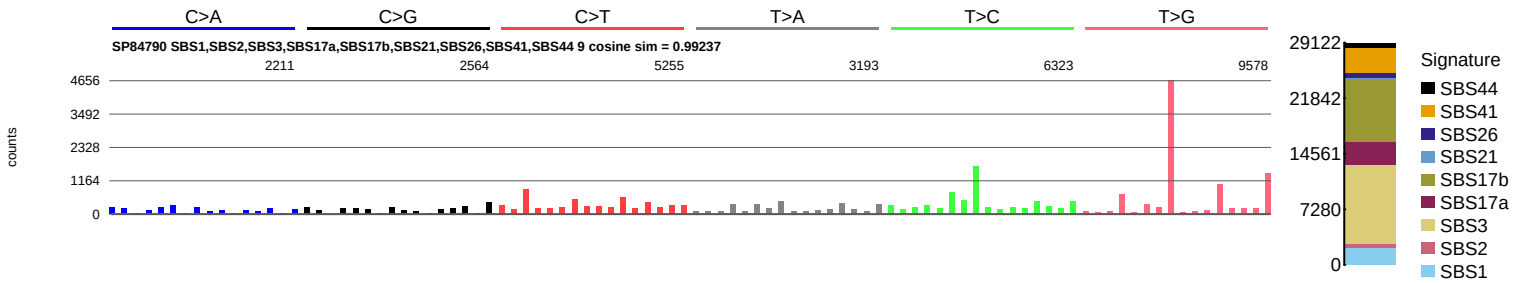

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS44 8 cosine sim = 0.99241
3110
2224
5853
2150
6288
9493
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS1

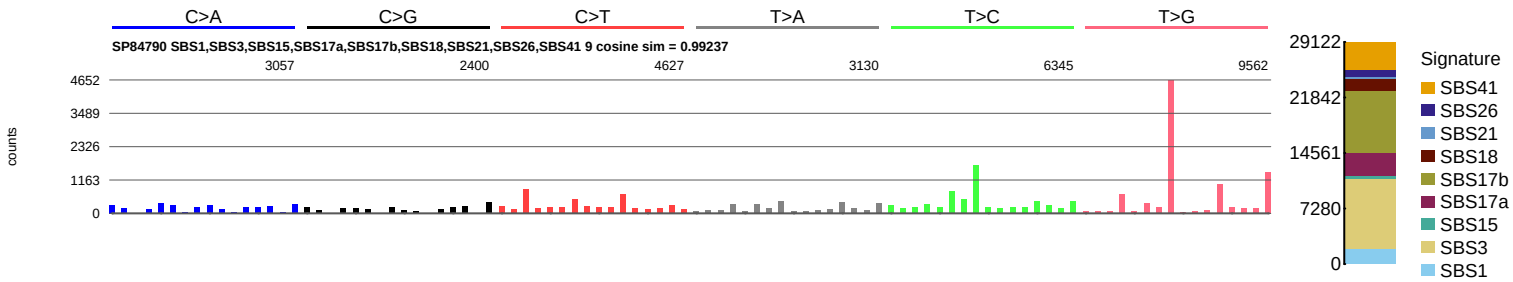

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.99239
3218
2359
5587
2263
6184
9509
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

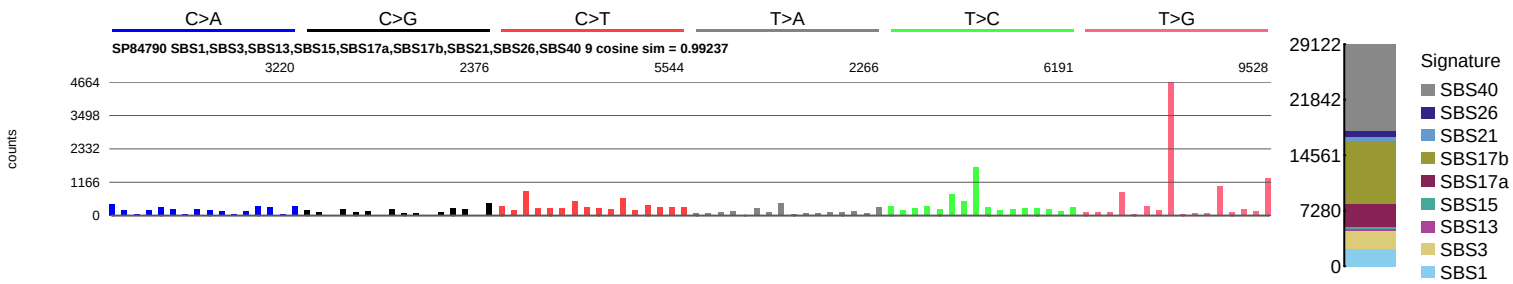

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS41,SBS44 9 cosine sim = 0.99238
2206
2565
5263
3181
6314
9586
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

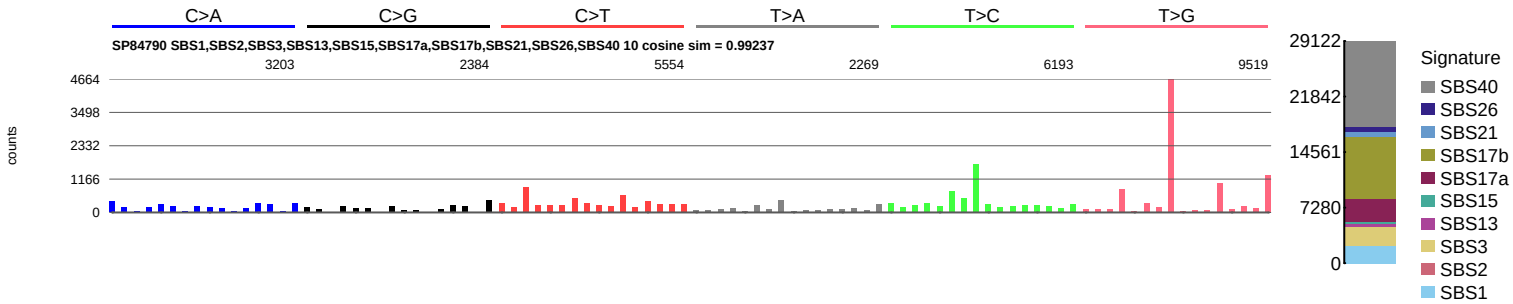

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 10 cosine sim = 0.99238
3061
2382
4656
3125
6339
9553
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

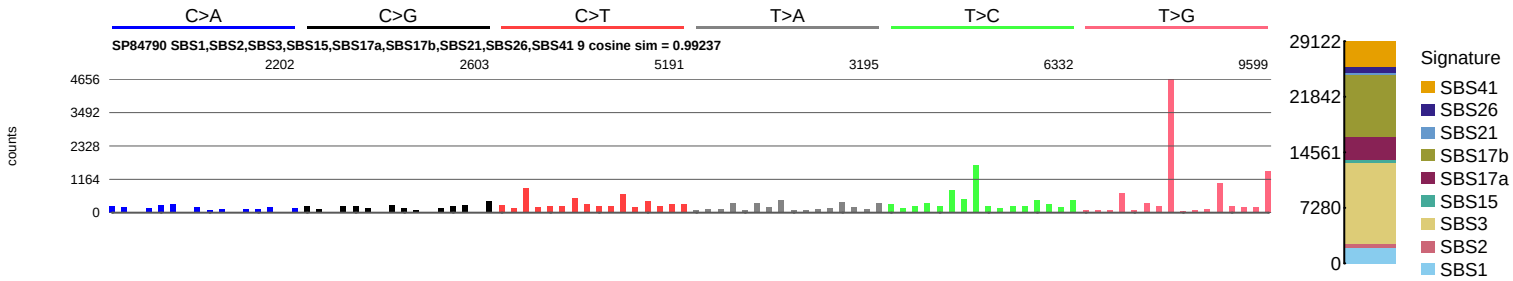

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.99237
3074
2548
4488
3101
6366
9545
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1

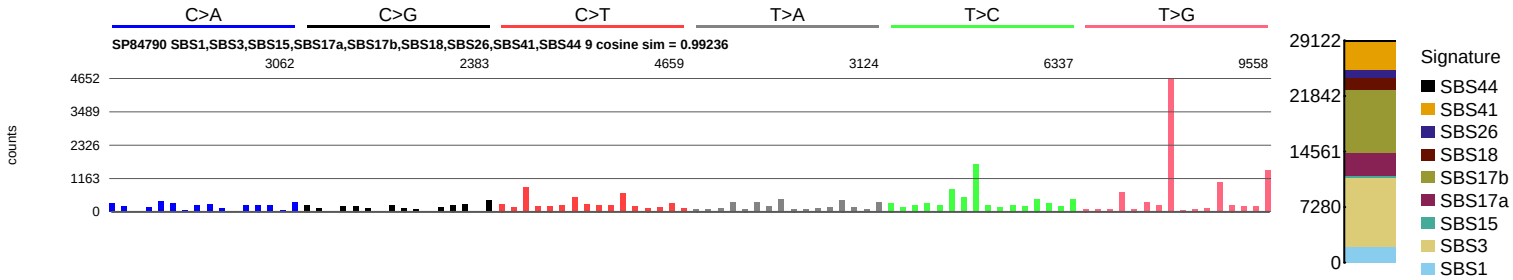

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 9 cosine sim = 0.99237
2211
2564
5255
3193
6323
9578
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1

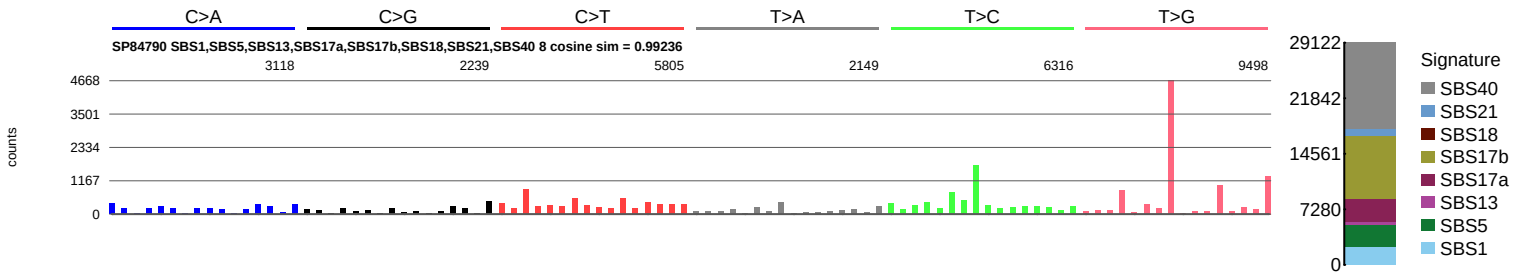

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.99237
3057
2400
4627
3130
6345
9562
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

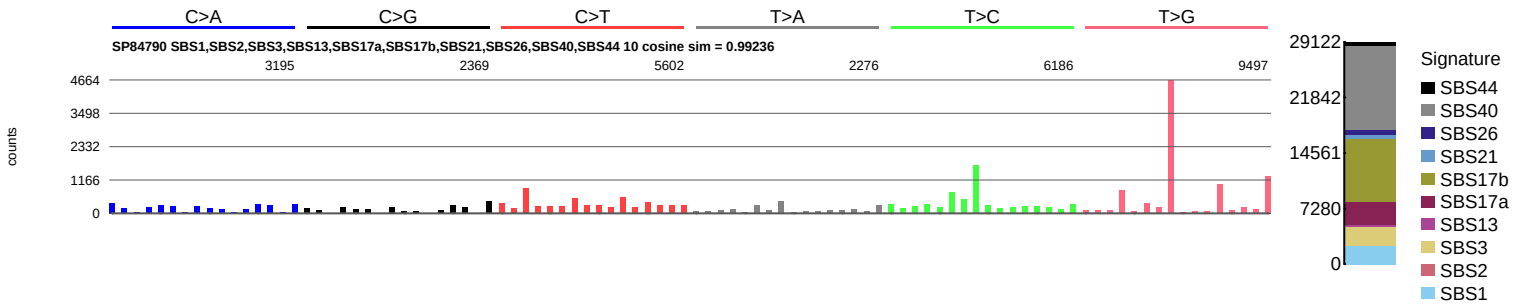

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40 9 cosine sim = 0.99237
3220
2376
5544
2266
6191
9528
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

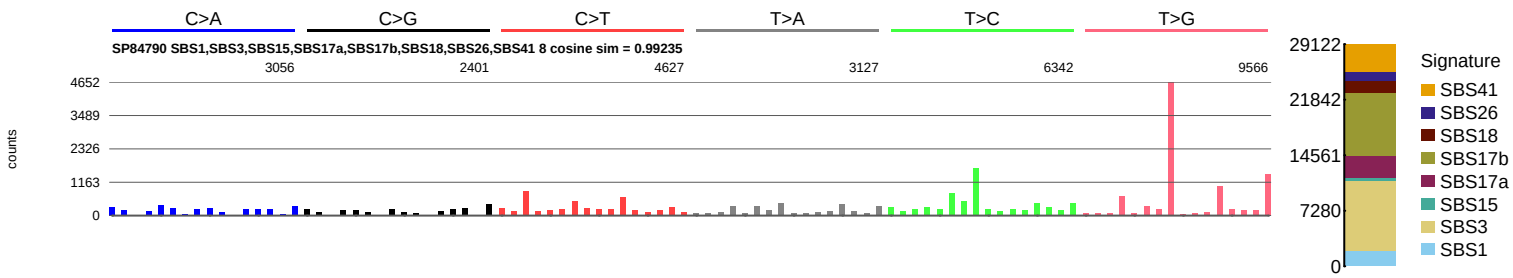

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40 10 cosine sim = 0.99237
3203
2384
5554
2269
6193
9519
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

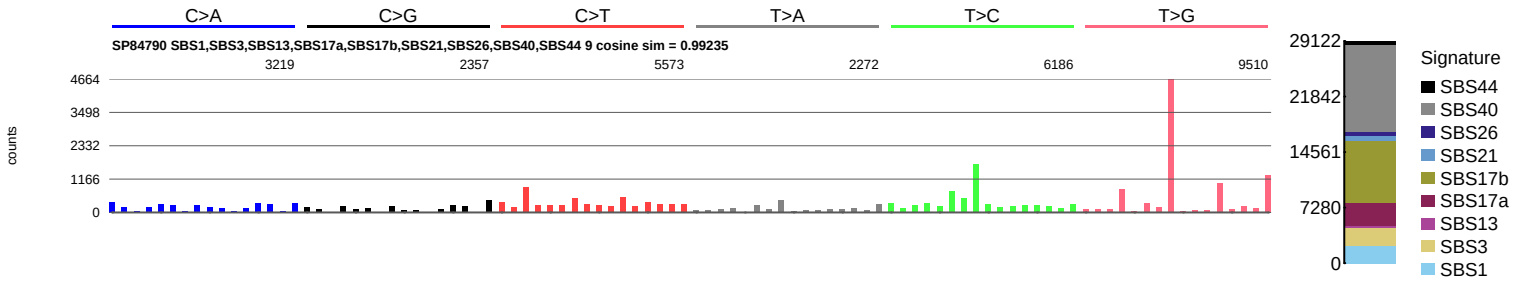

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41 9 cosine sim = 0.99237
2202
2603
5191
3195
6332
9599
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

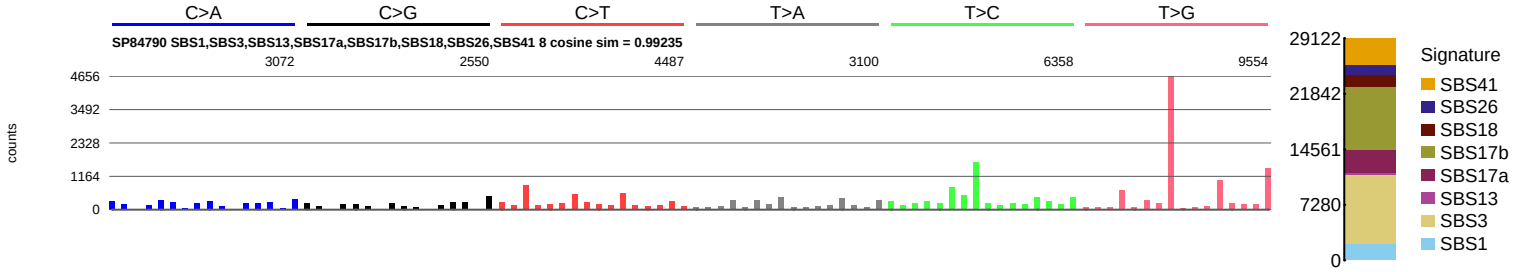

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.99236
3062
2383
4659
3124
6337
9558
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

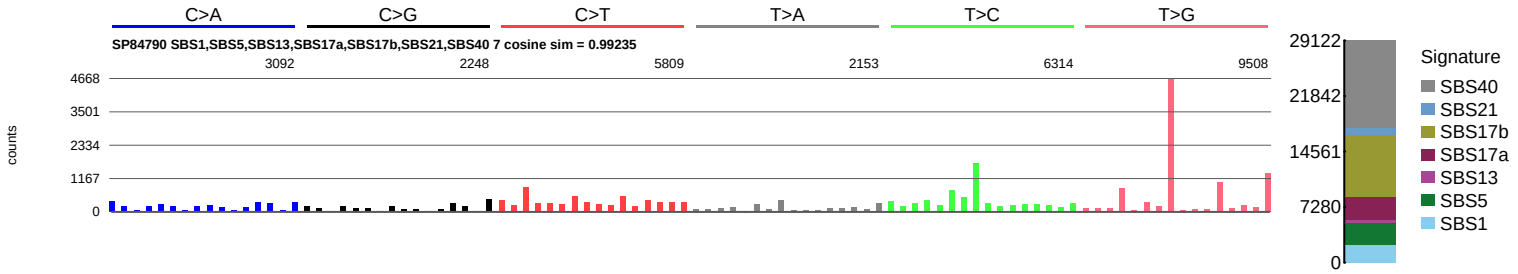

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.99236
3118
2239
5805
2149
6316
9498
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1

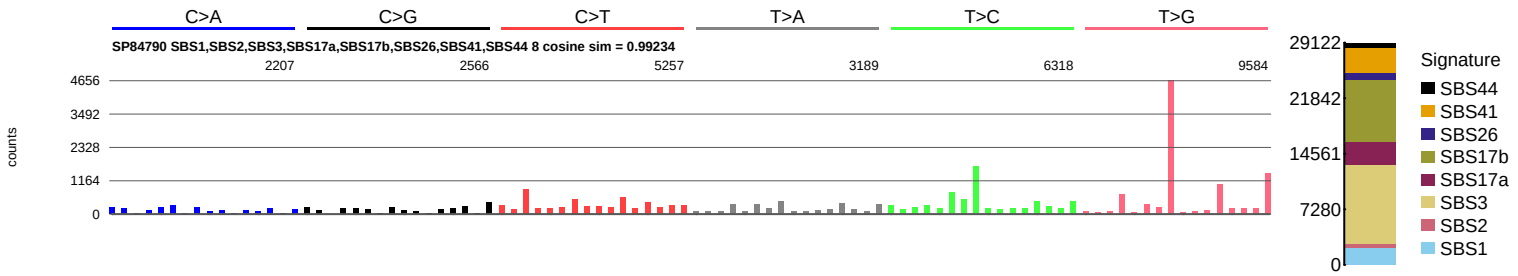

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.99236
3195
2369
5602
2276
6186
9497
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

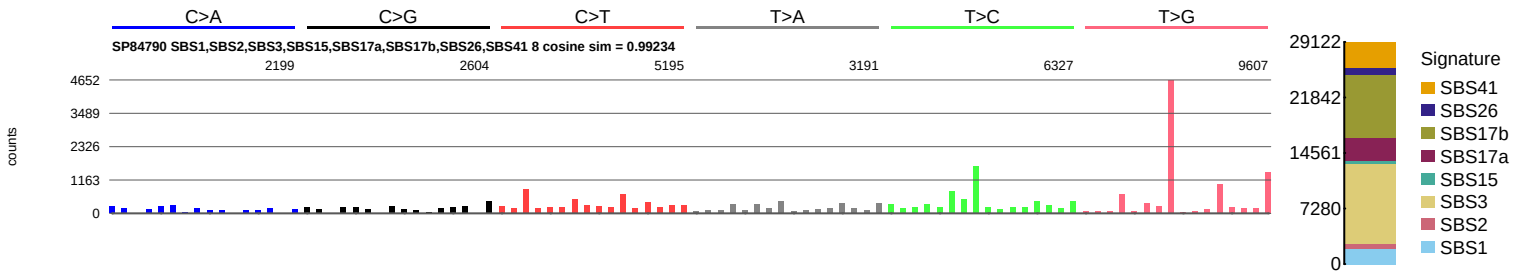

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.99235
3056
2401
4627
3127
6342
9566
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

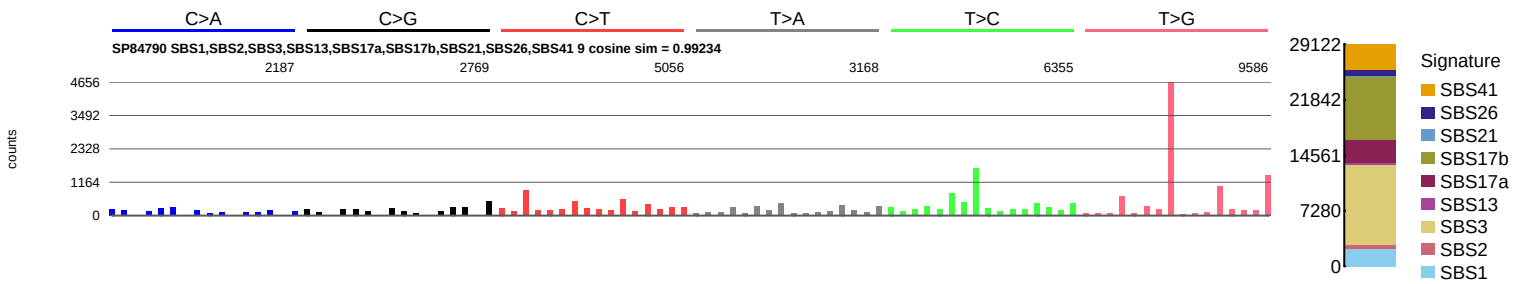

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.99235
3219
2357
5573
2272
6186
9510
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS1

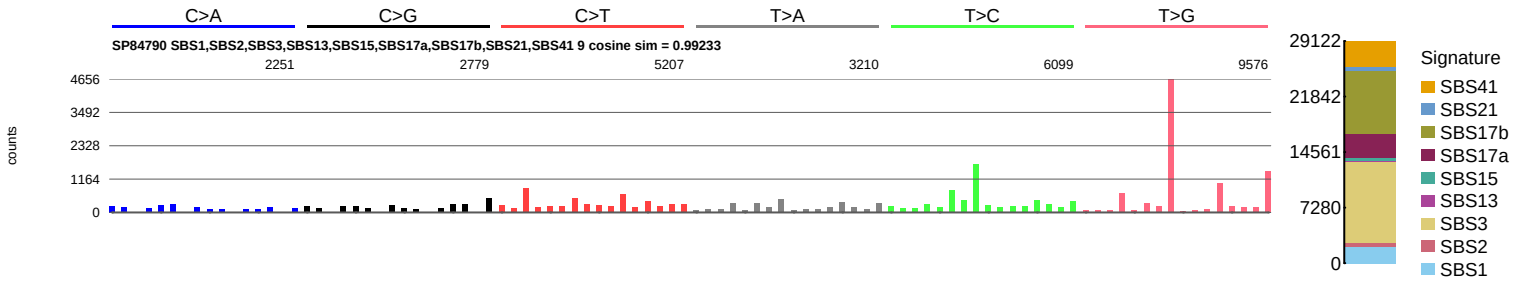

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.99235
3072
2550
4487
3100
6358
9554
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1

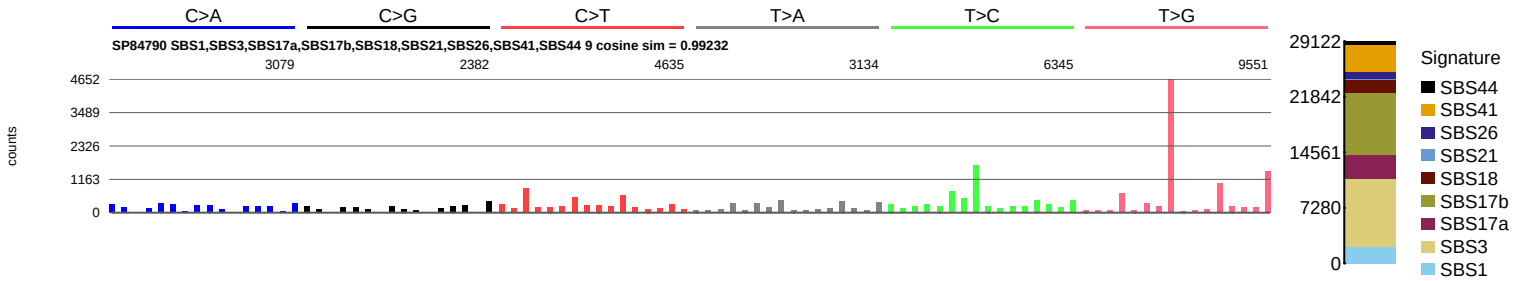

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS40 7 cosine sim = 0.99235
3092
2248
5809
2153
6314
9508
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS1

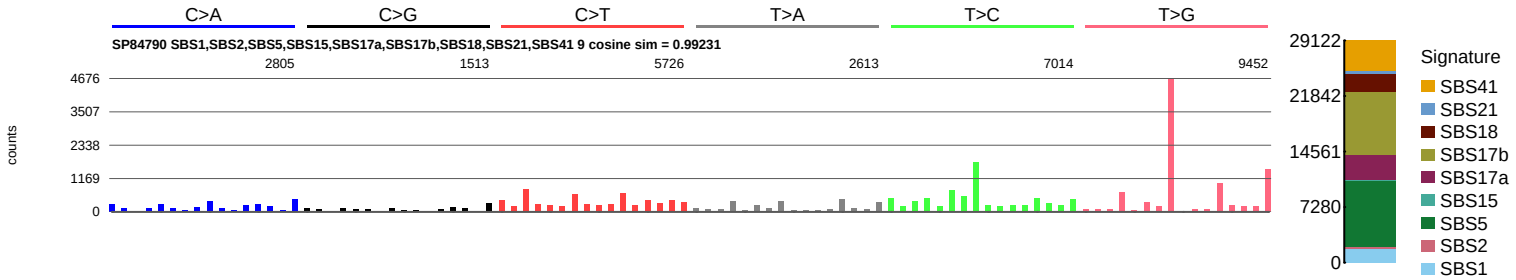

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS26,SBS41,SBS44 8 cosine sim = 0.99234
2207
2566
5257
3189
6318
9584
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS3
SBS2
SBS1

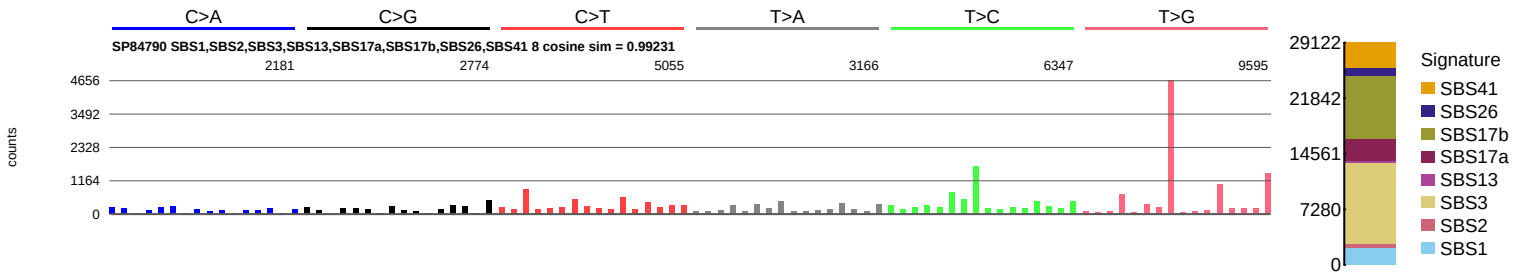

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS41 8 cosine sim = 0.99234
2199
2604
5195
3191
6327
9607
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

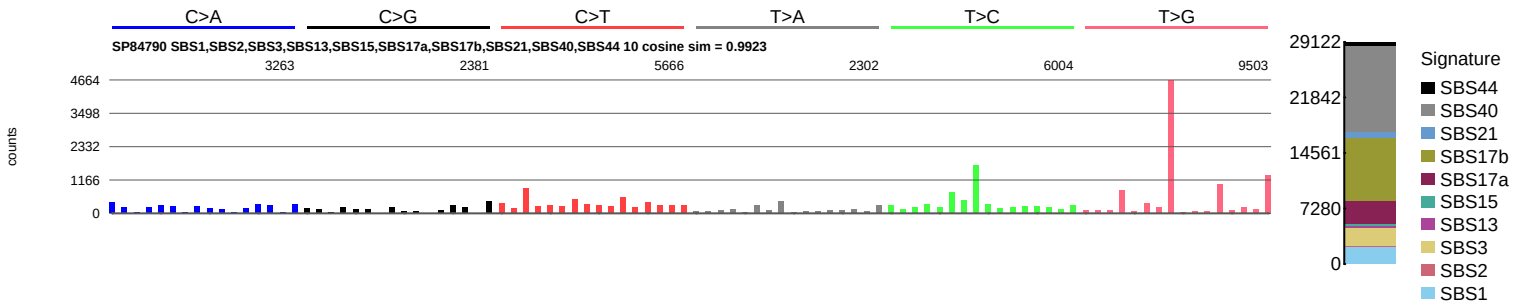

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS41 9 cosine sim = 0.99234
2187
2769
5056
3168
6355
9586
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

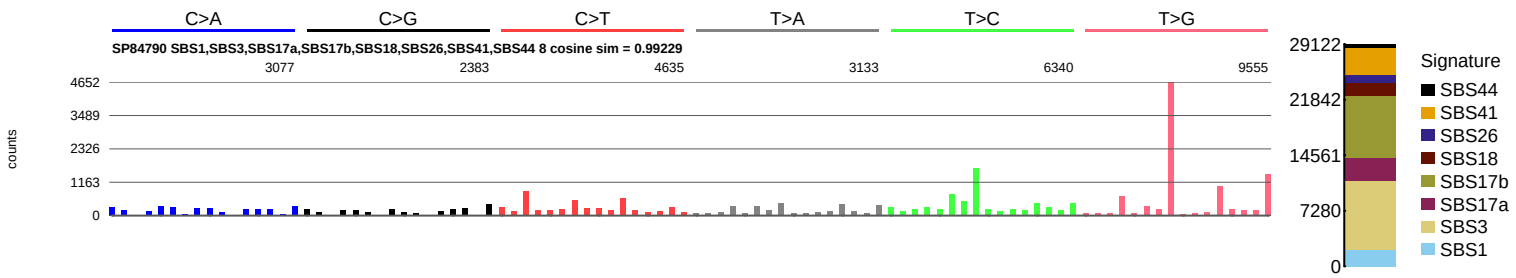

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 9 cosine sim = 0.99233
2251
2779
5207
3210
6099
9576
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

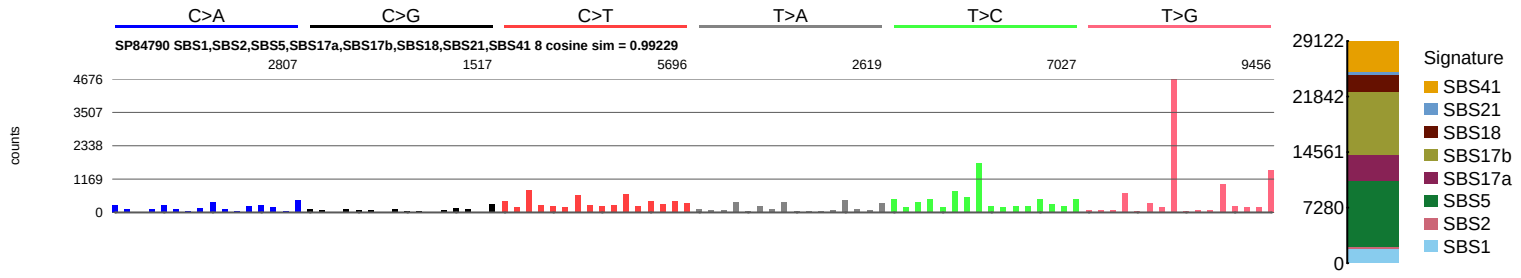

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 9 cosine sim = 0.99232
3079
2382
4635
3134
6345
9551
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1

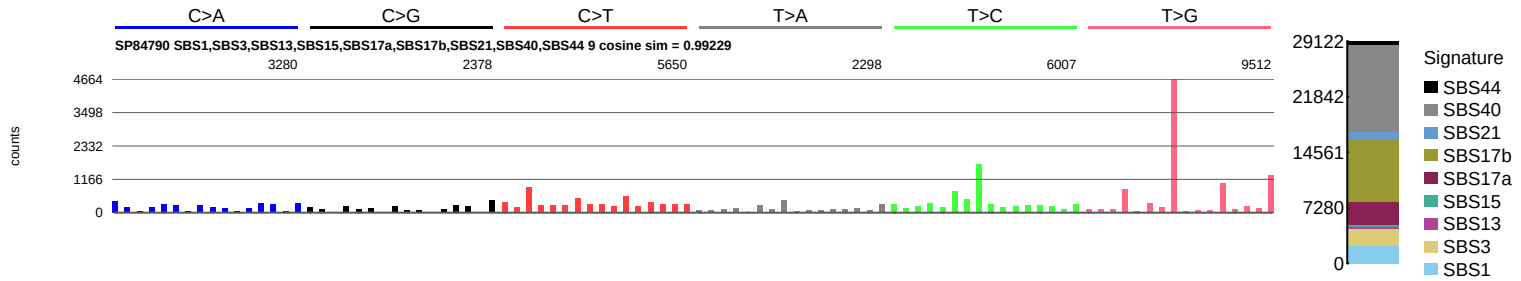

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.99231
2805
1513
5726
2613
7014
9452
counts
4676
3507
2338
1169
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

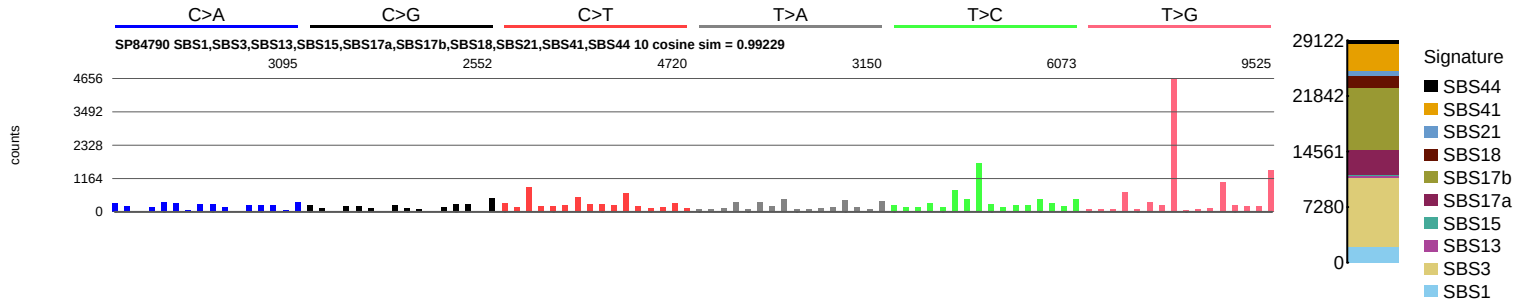

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS26,SBS41 8 cosine sim = 0.99231
2181
2774
5055
3166
6347
9595
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

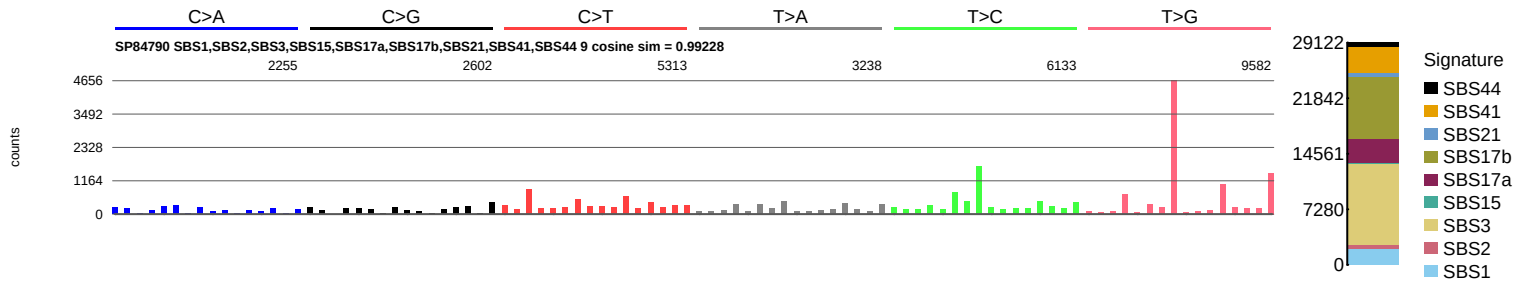

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS44 10 cosine sim = 0.9923
3263
2381
5666
2302
6004
9503
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

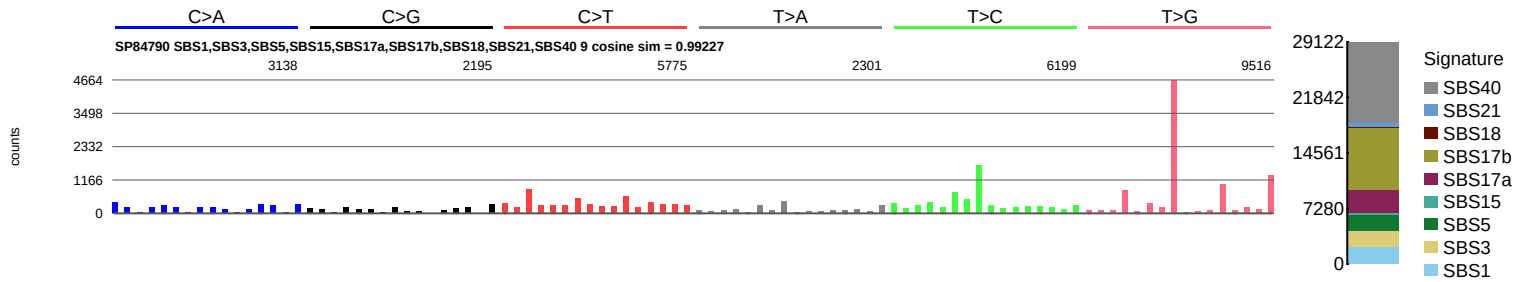

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.99229
3077
2383
4635
3133
6340
9555
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS1

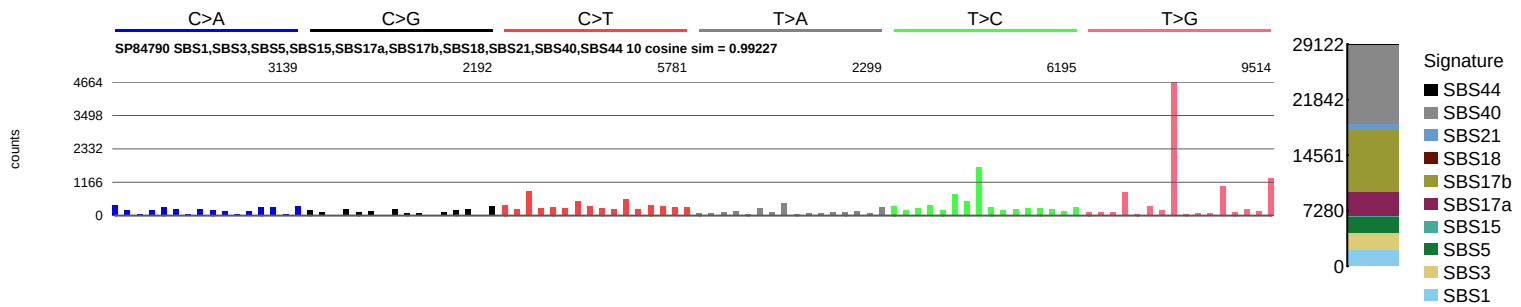

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.99229
2807
1517
5696
2619
7027
9456
counts
4676
3507
2338
1169
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

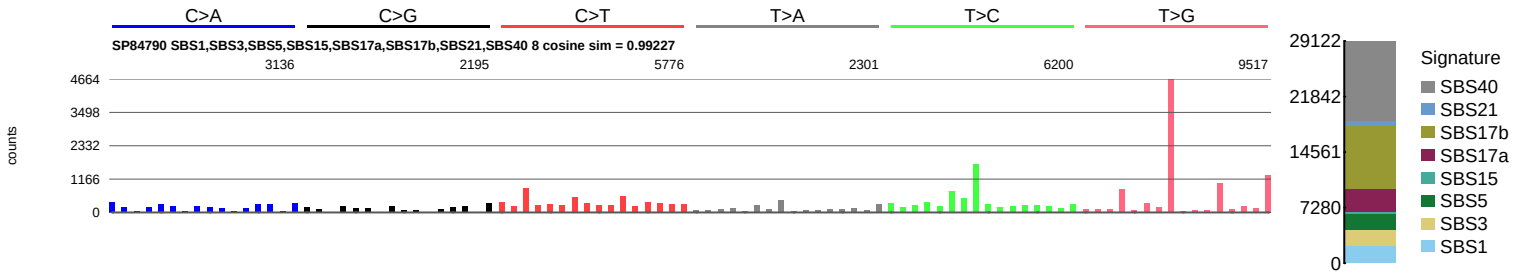

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS44 9 cosine sim = 0.99229
3280
2378
5650
2298
6007
9512
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

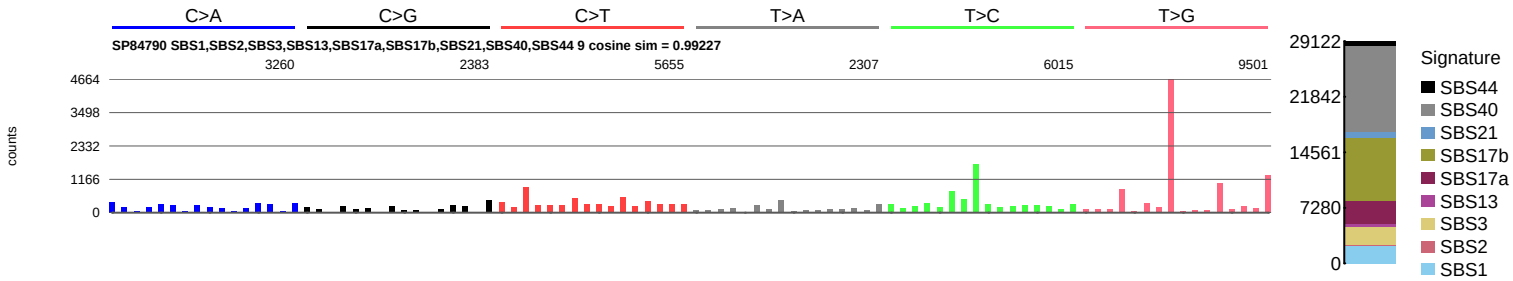

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.99229
3095
2552
4720
3150
6073
9525
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

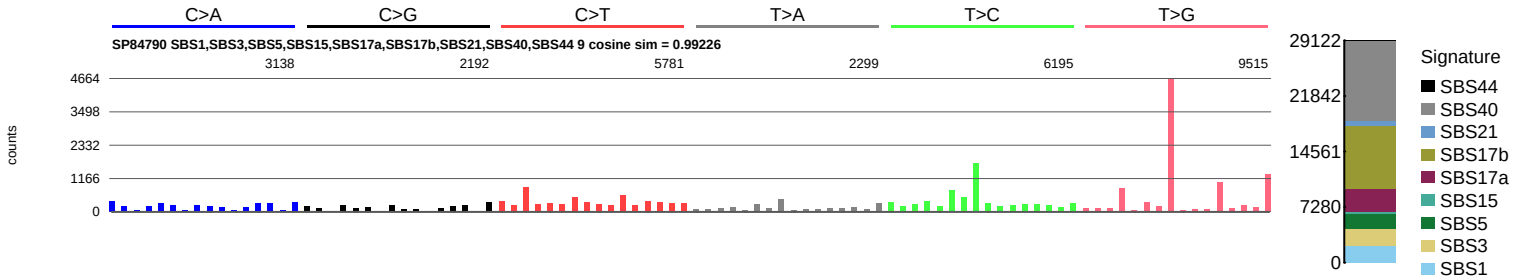

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS41,SBS44 9 cosine sim = 0.99228
2255
2602
5313
3238
6133
9582
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

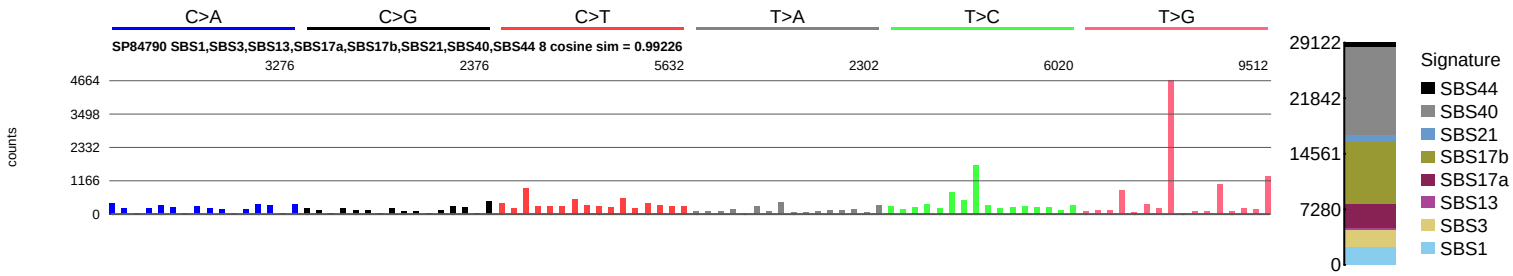

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.99227
3138
2195
5775
2301
6199
9516
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1

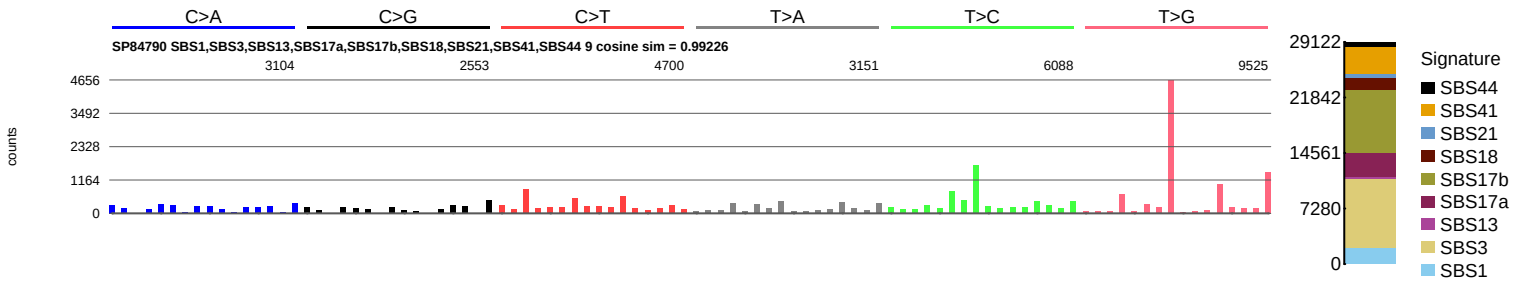

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.99227
3139
2192
5781
2299
6195
9514
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1

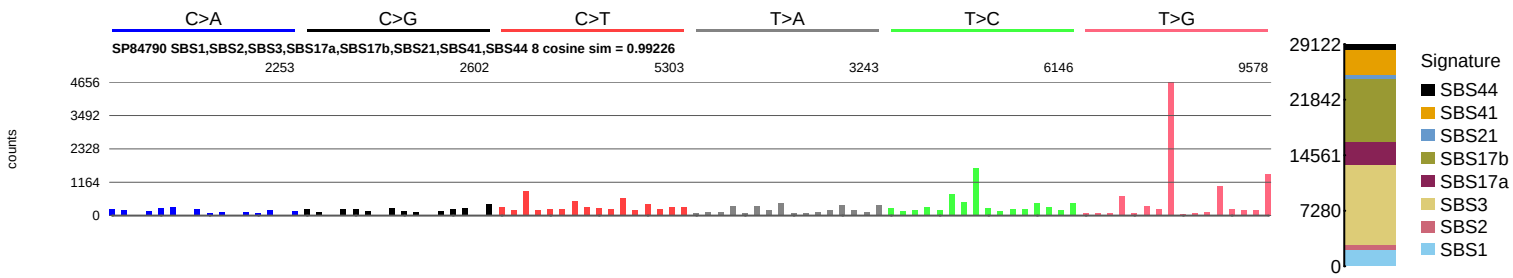

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40 8 cosine sim = 0.99227
3136
2195
5776
2301
6200
9517
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1

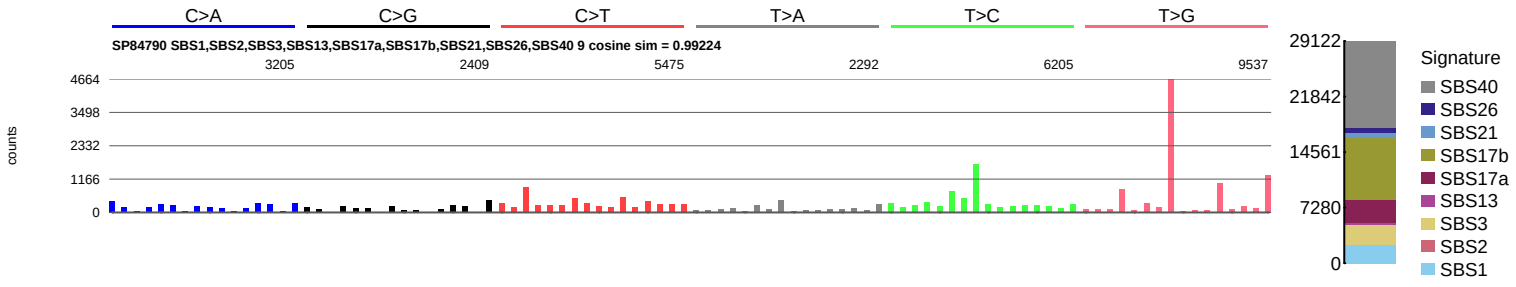

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS44 9 cosine sim = 0.99227
3260
2383
5655
2307
6015
9501
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

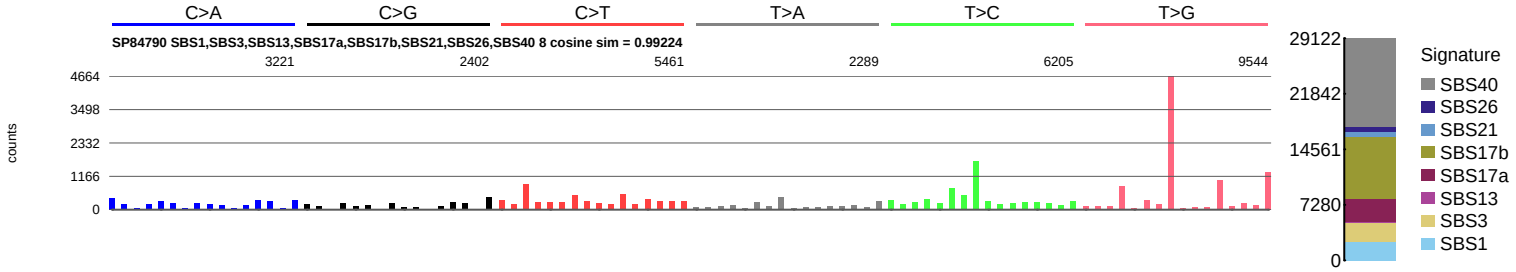

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS44 9 cosine sim = 0.99226
3138
2192
5781
2299
6195
9515
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1

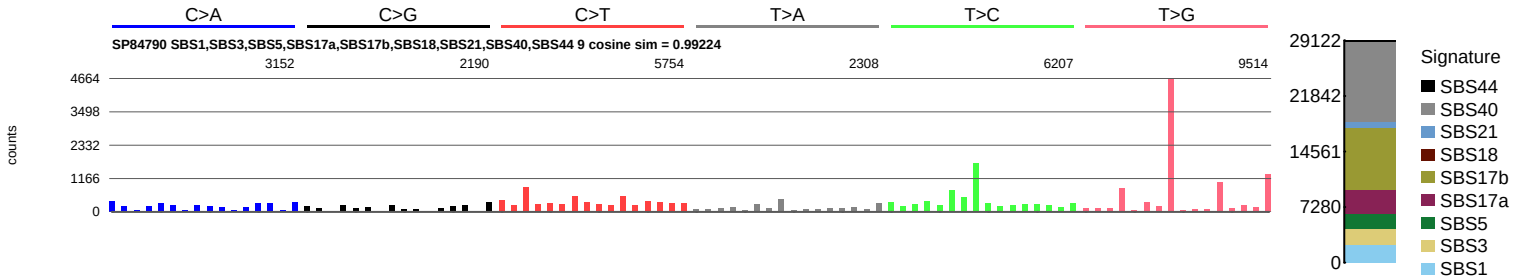

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS44 8 cosine sim = 0.99226
3276
2376
5632
2302
6020
9512
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS1

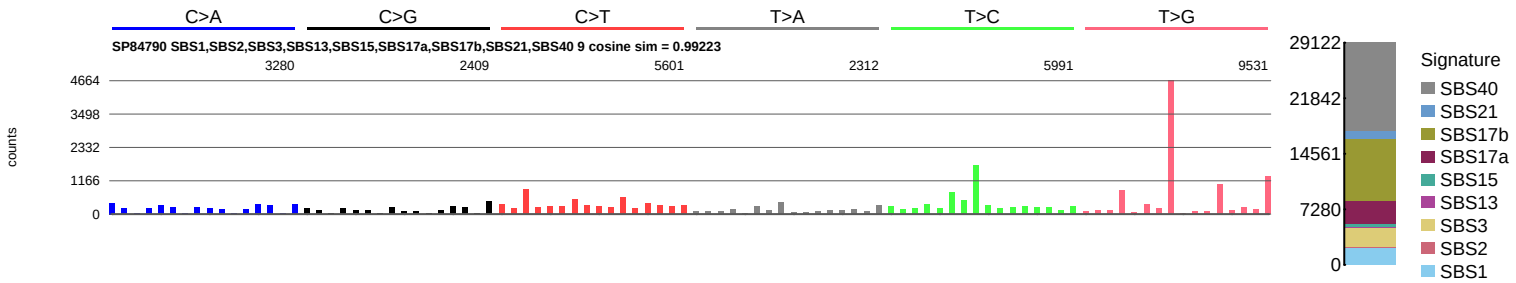

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.99226
3104
2553
4700
3151
6088
9525
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1

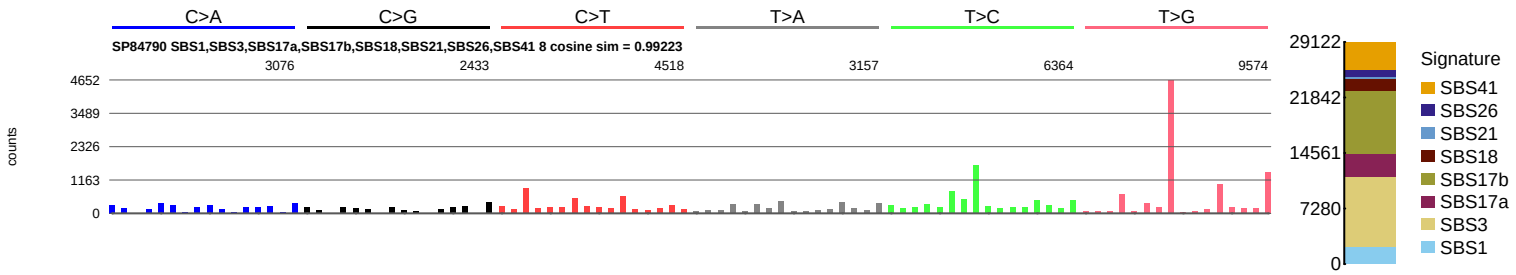

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS41,SBS44 8 cosine sim = 0.99226
2253
2602
5303
3243
6146
9578
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1

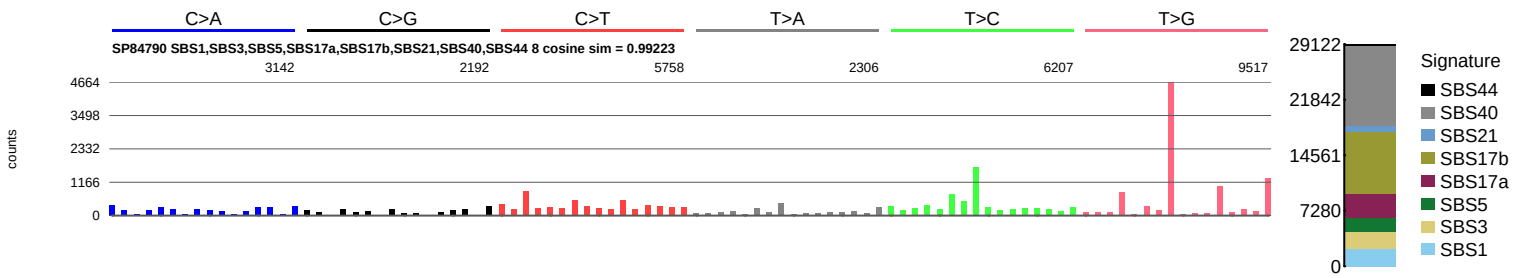

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS40 9 cosine sim = 0.99224
3205
2409
5475
2292
6205
9537
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

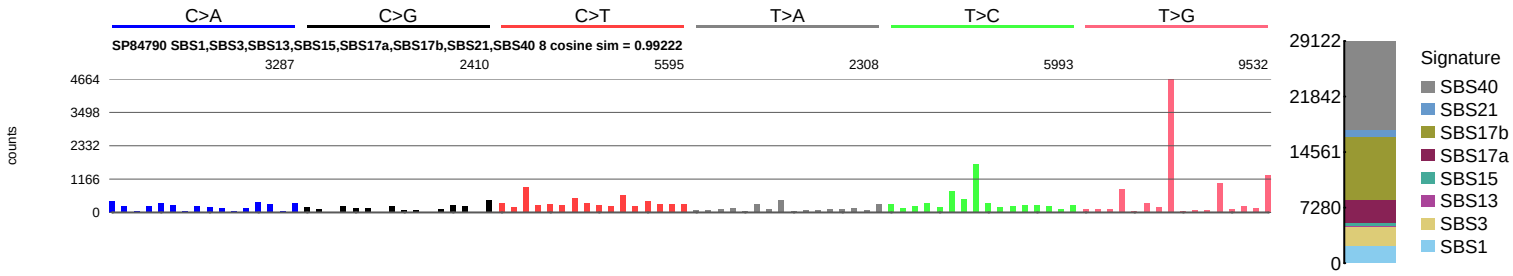

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS40 8 cosine sim = 0.99224
3221
2402
5461
2289
6205
9544
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS1

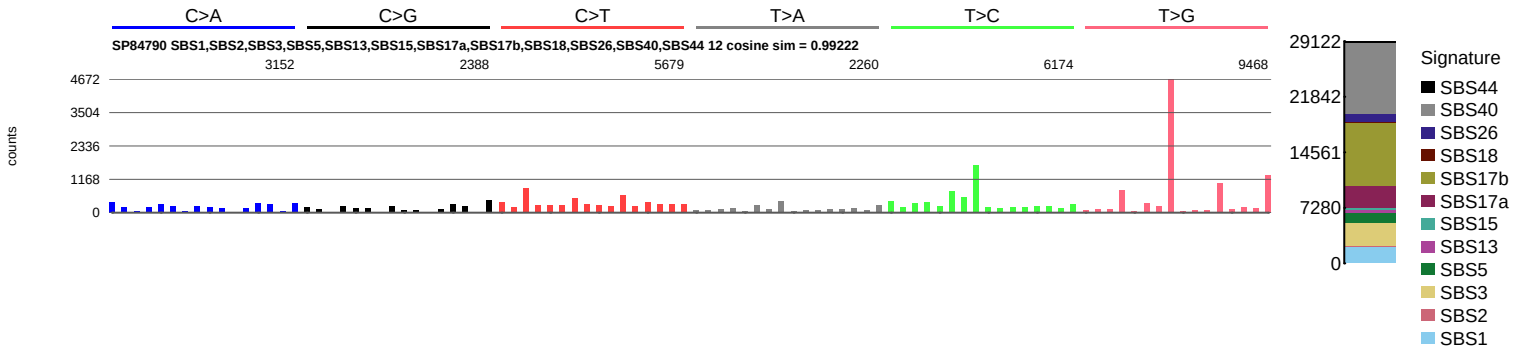

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.99224
3152
2190
5754
2308
6207
9514
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1

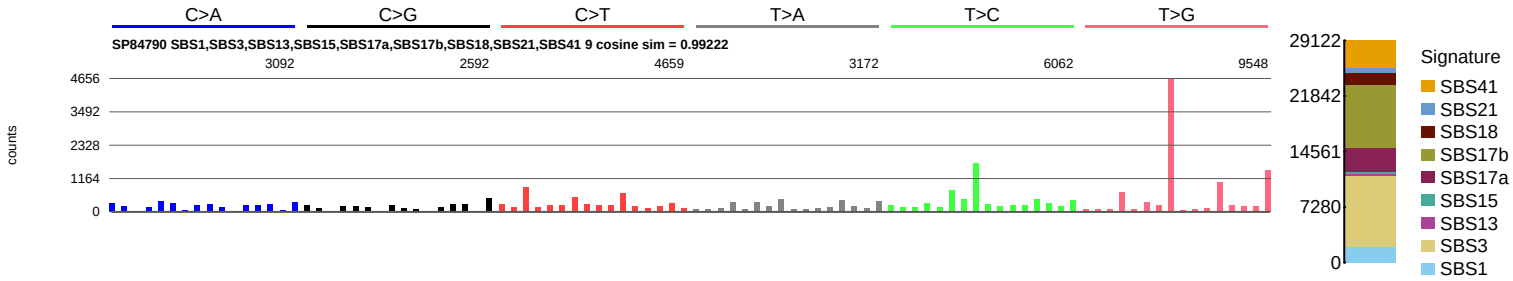

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40 9 cosine sim = 0.99223
3280
2409
5601
2312
5991
9531
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

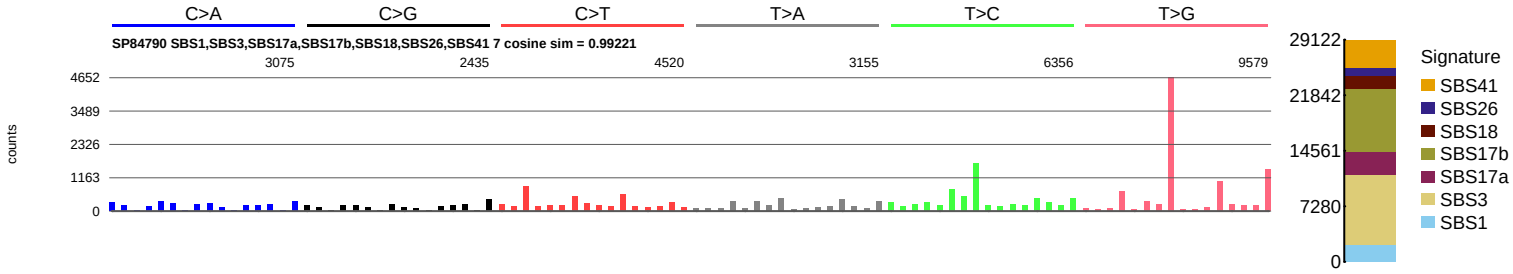

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 8 cosine sim = 0.99223
3076
2433
4518
3157
6364
9574
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1

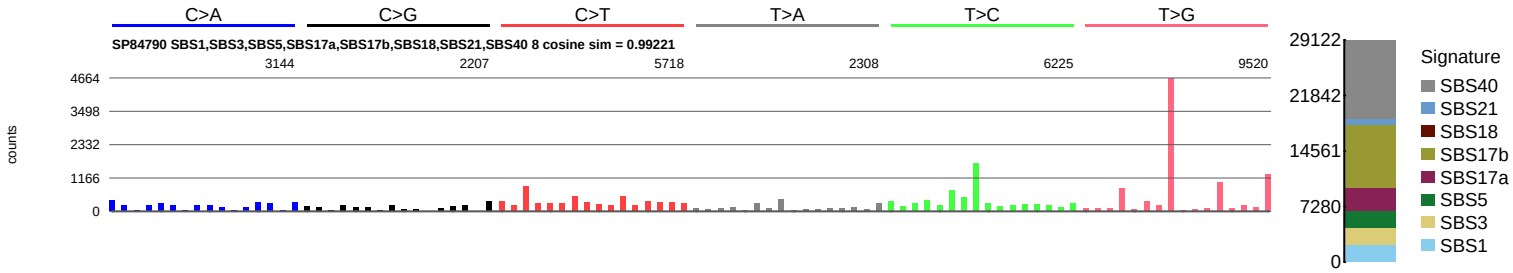

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS44 8 cosine sim = 0.99223
3142
2192
5758
2306
6207
9517
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS1

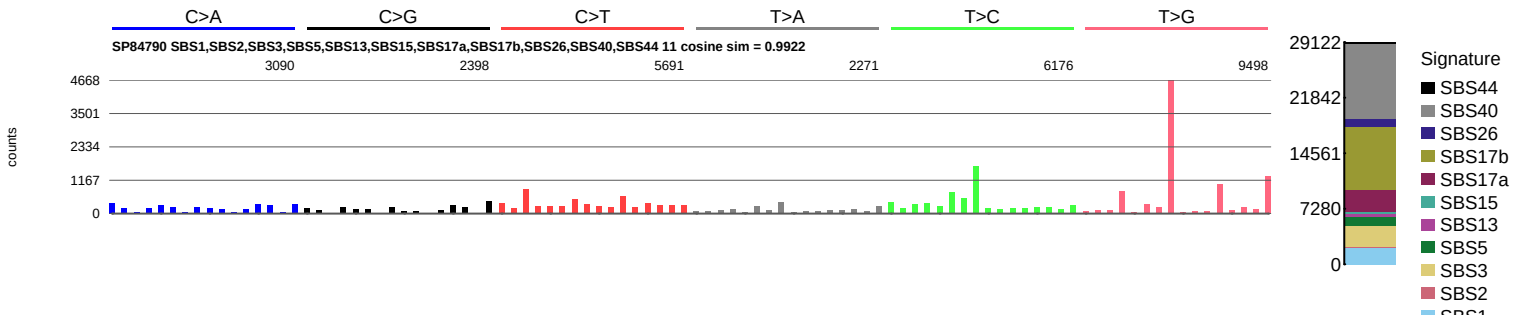

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40 8 cosine sim = 0.99222
3287
2410
5595
2308
5993
9532
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

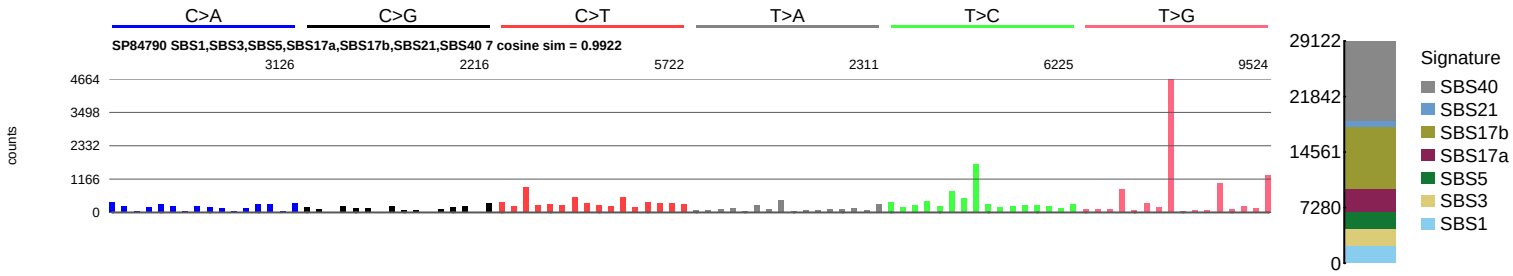

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 12 cosine sim = 0.99222
3152
2388
5679
2260
6174
9468
counts
4672
3504
2336
1168
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

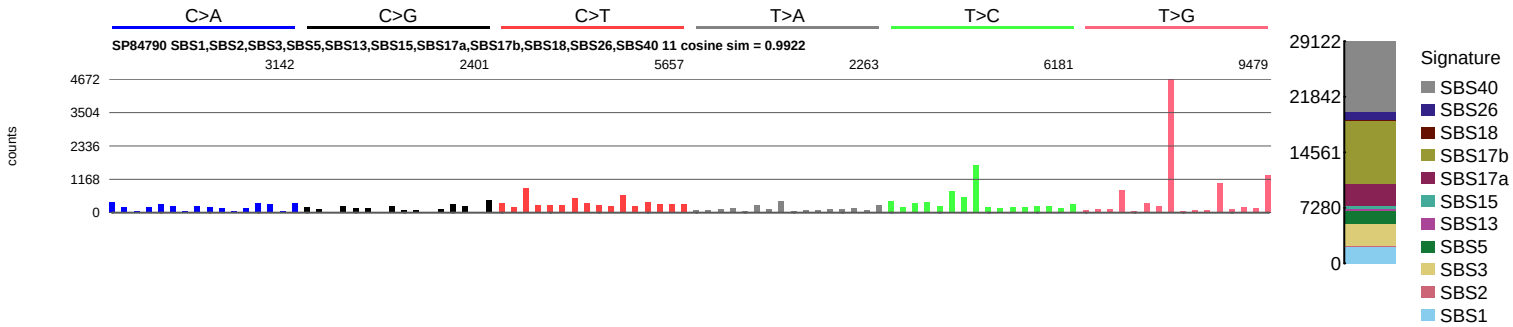

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.99222
3092
2592
4659
3172
6062
9548
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

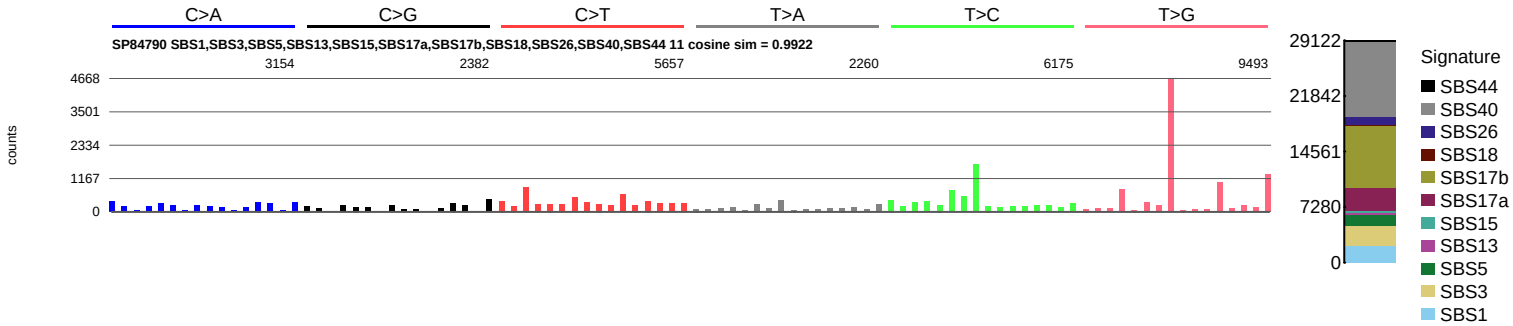

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.99221
3075
2435
4520
3155
6356
9579
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS1

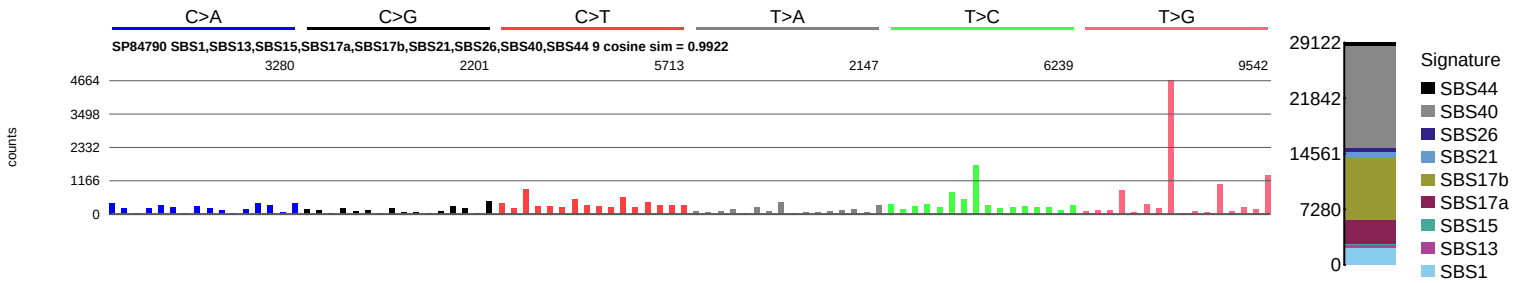

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.99221
3144
2207
5718
2308
6225
9520
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1

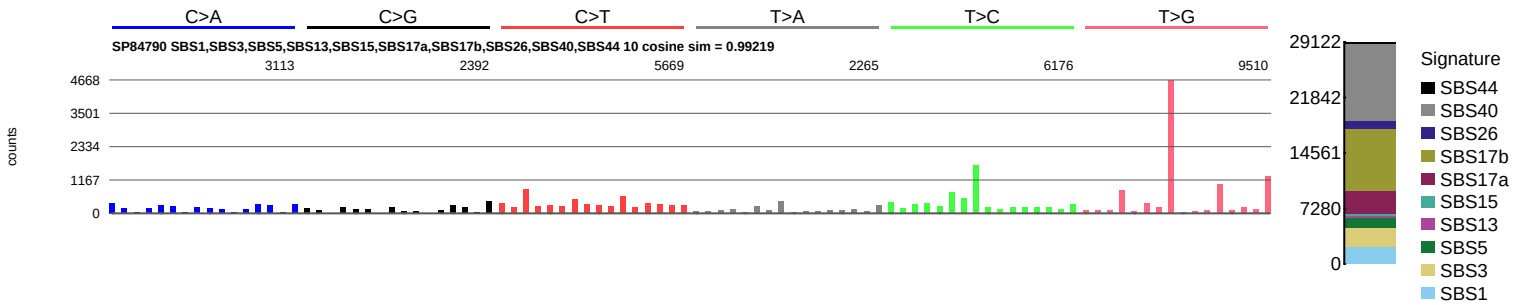

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS44 11 cosine sim = 0.9922
3090
2398
5691
2271
6176
9498
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2

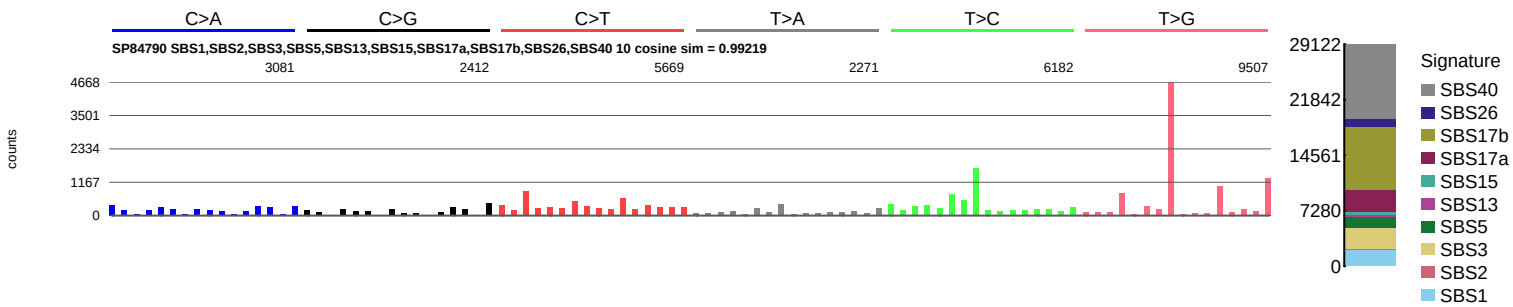

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS40 7 cosine sim = 0.9922
3126
2216
5722
2311
6225
9524
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS1

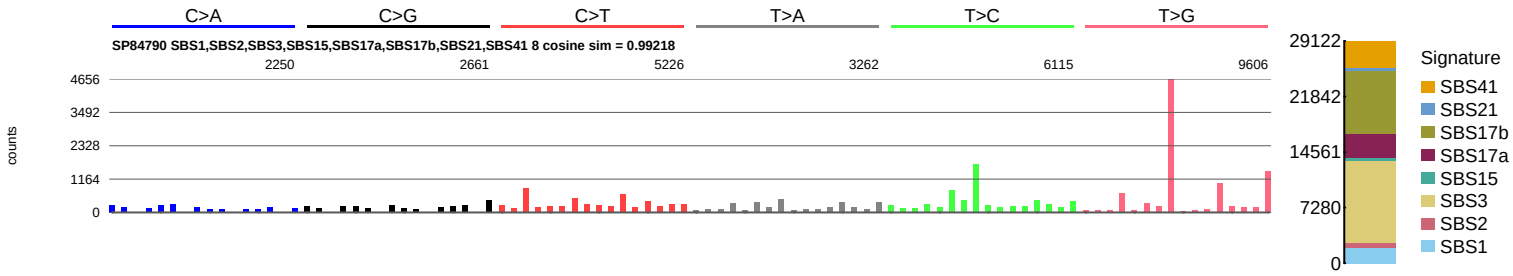

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 11 cosine sim = 0.9922
3142
2401
5657
2263
6181
9479
counts
4672
3504
2336
1168
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

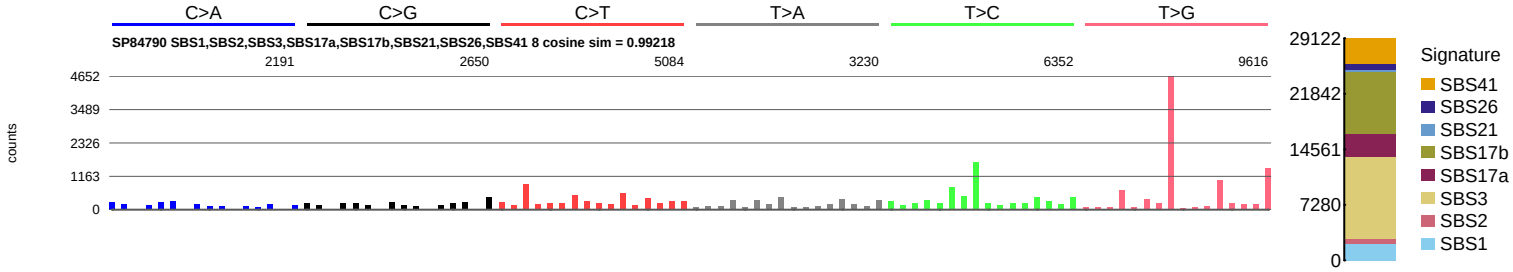

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 11 cosine sim = 0.9922
3154
2382
5657
2260
6175
9493
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

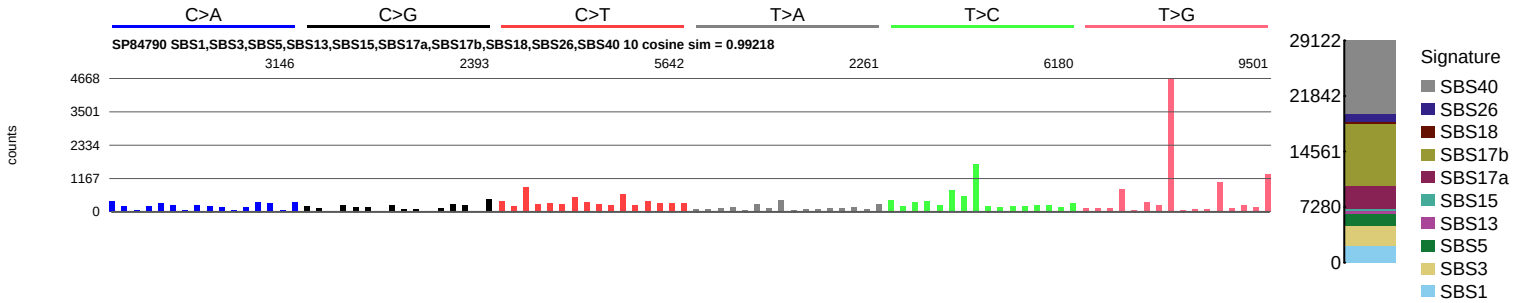

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.9922
3280
2201
5713
2147
6239
9542
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS1

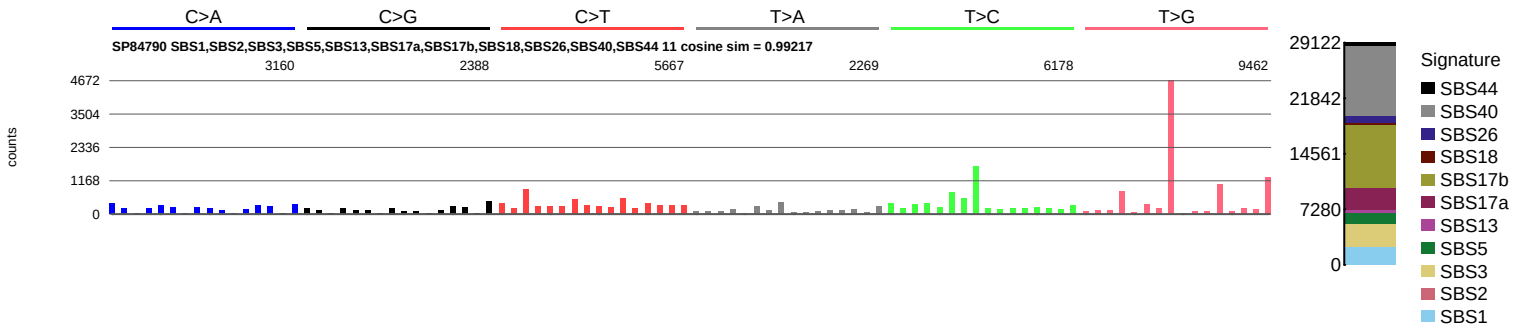

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS44 10 cosine sim = 0.99219
3113
2392
5669
2265
6176
9510
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

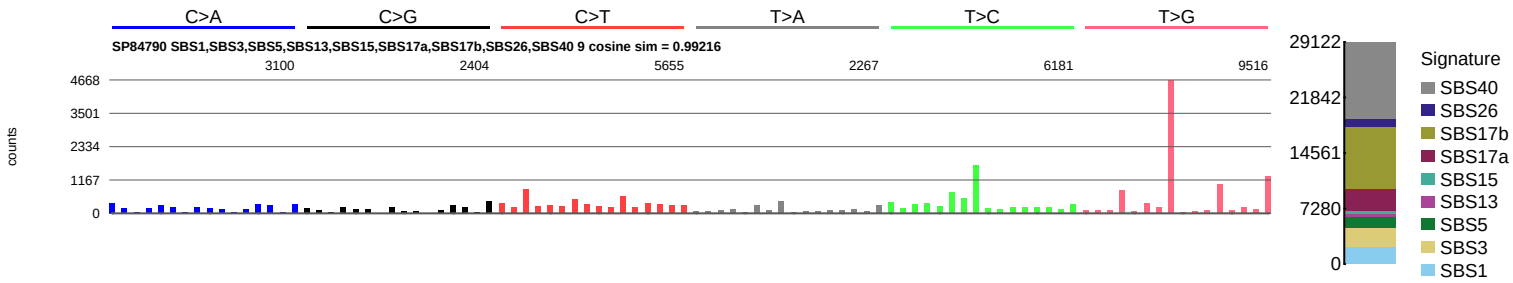

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40 10 cosine sim = 0.99219
3081
2412
5669
2271
6182
9507
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

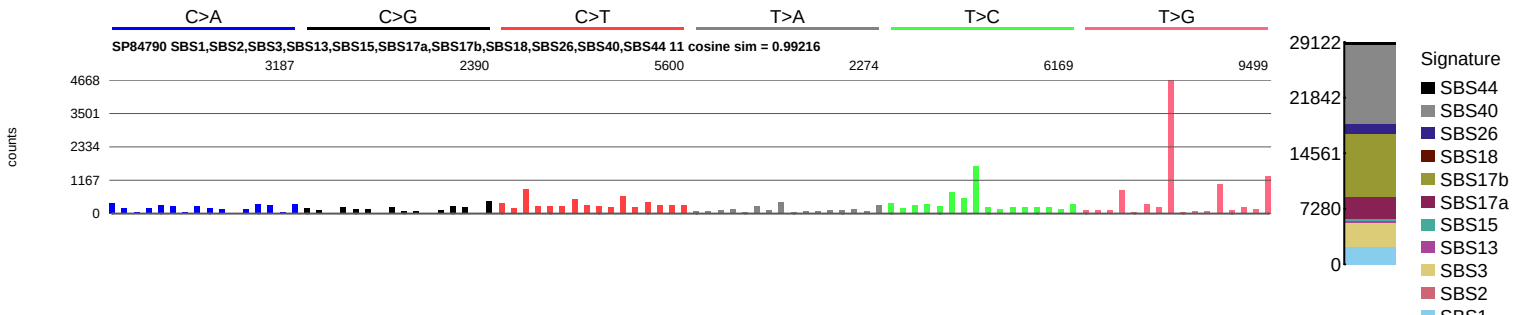

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.99218
2250
2661
5226
3262
6115
9606
4656
3492
2328
1164
0
counts
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

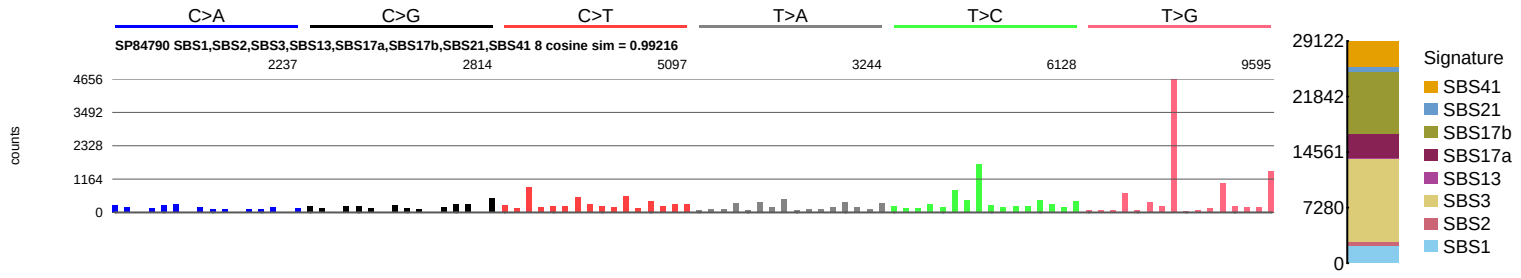

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS26,SBS41 8 cosine sim = 0.99218
2191
2650
5084
3230
6352
9616
4652
3489
2326
1163
0
counts
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1

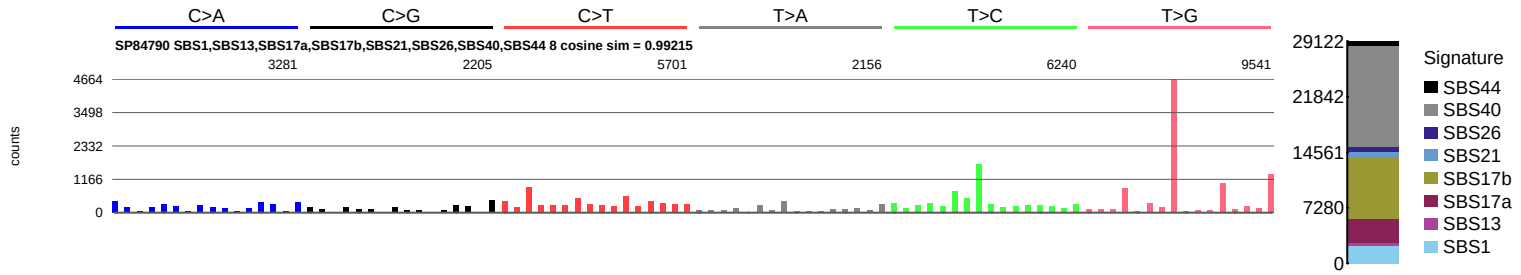

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 10 cosine sim = 0.99218
3146
2393
5642
2261
6180
9501
4668
3501
2334
1167
0
counts
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

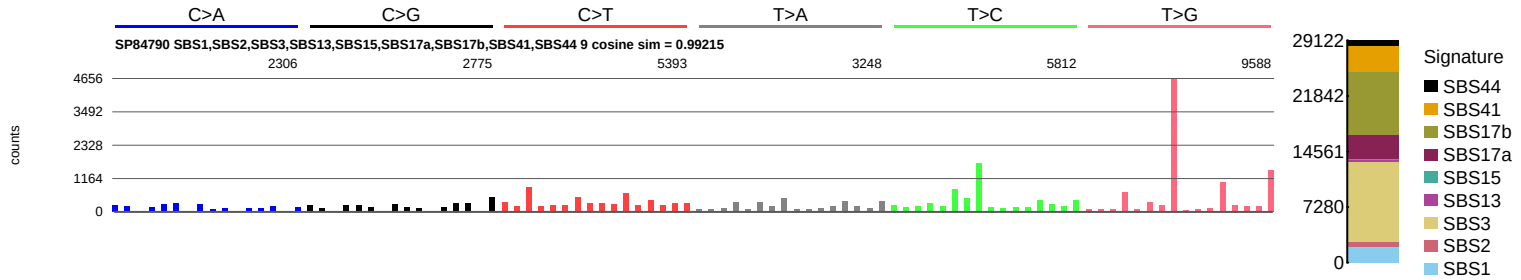

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 11 cosine sim = 0.99217
3160
2388
5667
2269
6178
9462
4672
3504
2336
1168
0
counts
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

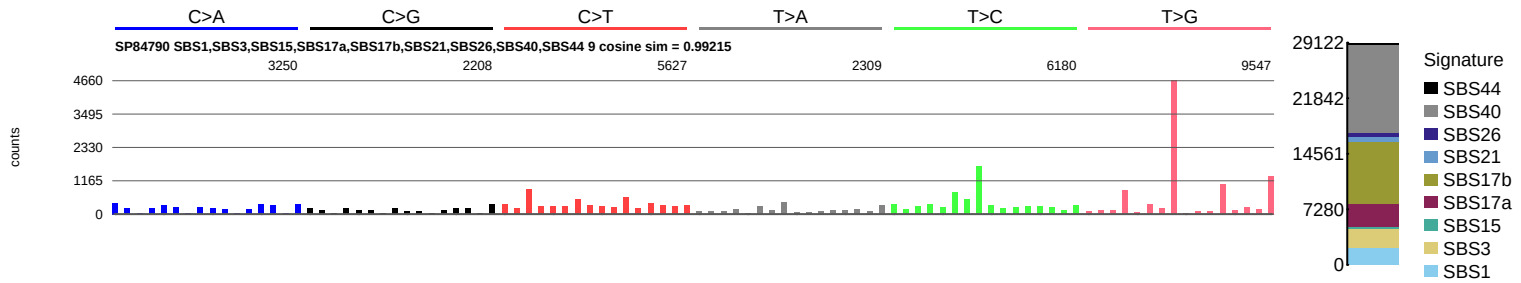

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40 9 cosine sim = 0.99216
3100
2404
5655
2267
6181
9516
4668
3501
2334
1167
0
counts
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

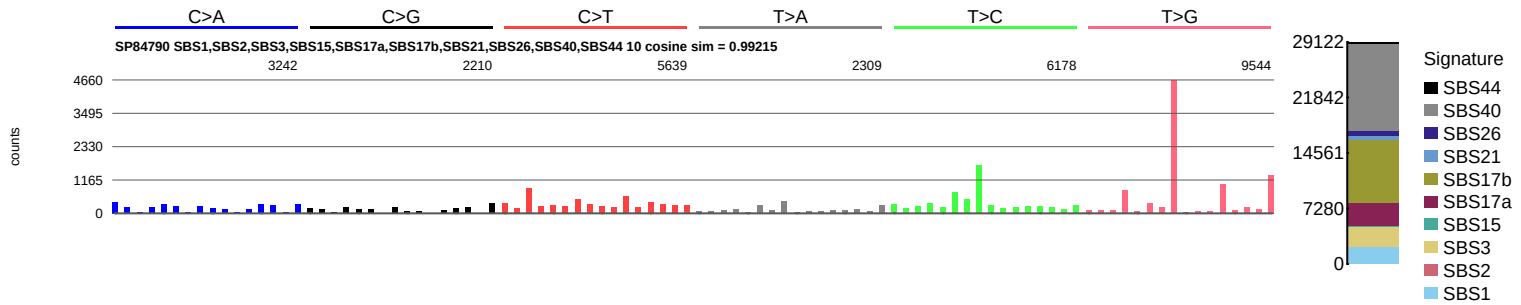

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 11 cosine sim = 0.99216
3187
2390
5600
2274
6169
9499
4668
3501
2334
1167
0
counts
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2

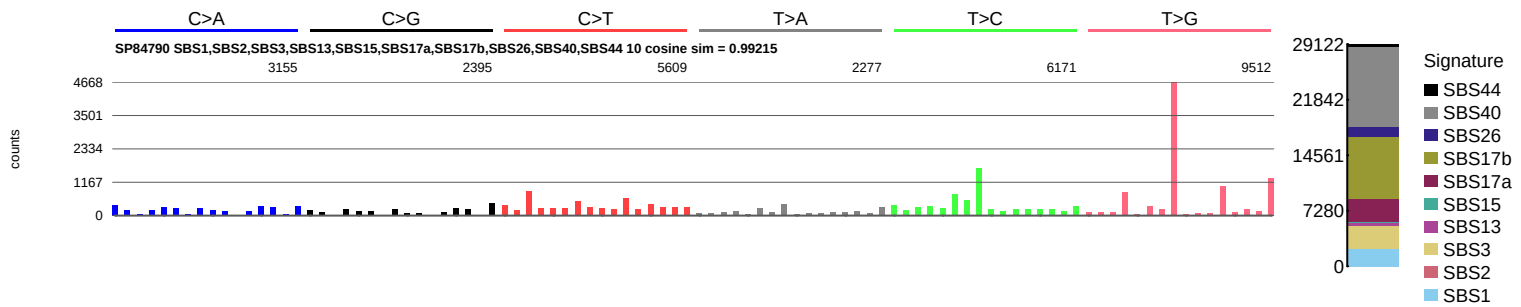

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.99216
2237
2814
5097
3244
6128
9595
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

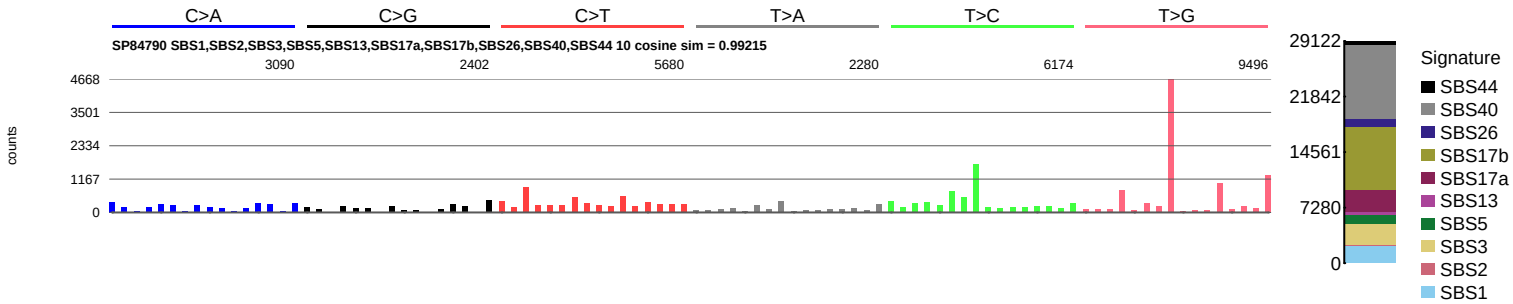

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS44 8 cosine sim = 0.99215
3281
2205
5701
2156
6240
9541
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS1

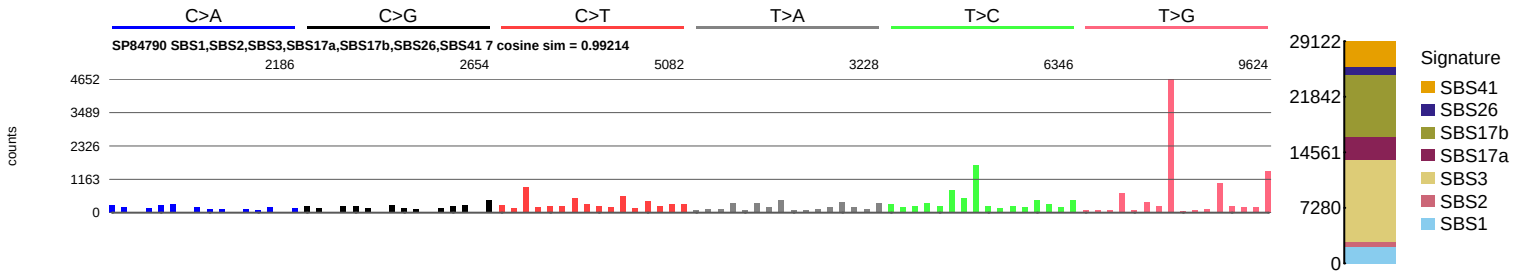

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS41,SBS44 9 cosine sim = 0.99215
2306
2775
5393
3248
5812
9588
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

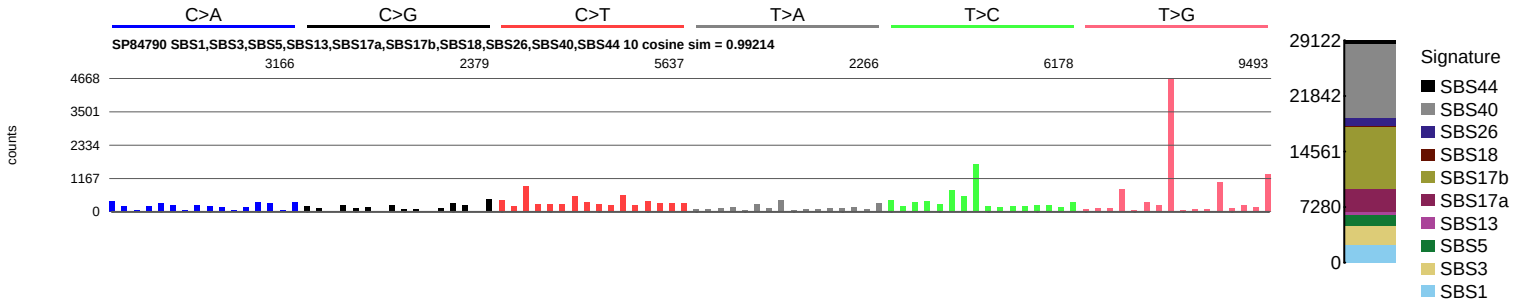

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.99215
3250
2208
5627
2309
6180
9547
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS1

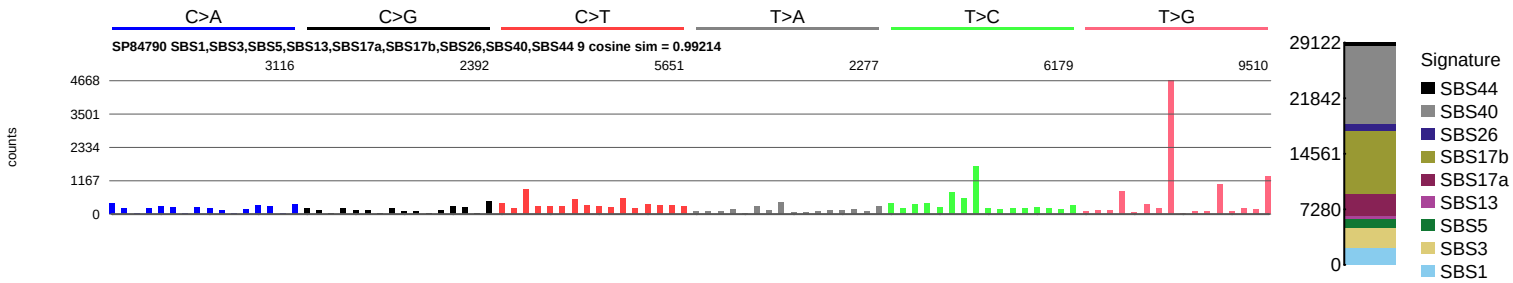

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.99215
3242
2210
5639
2309
6178
9544
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

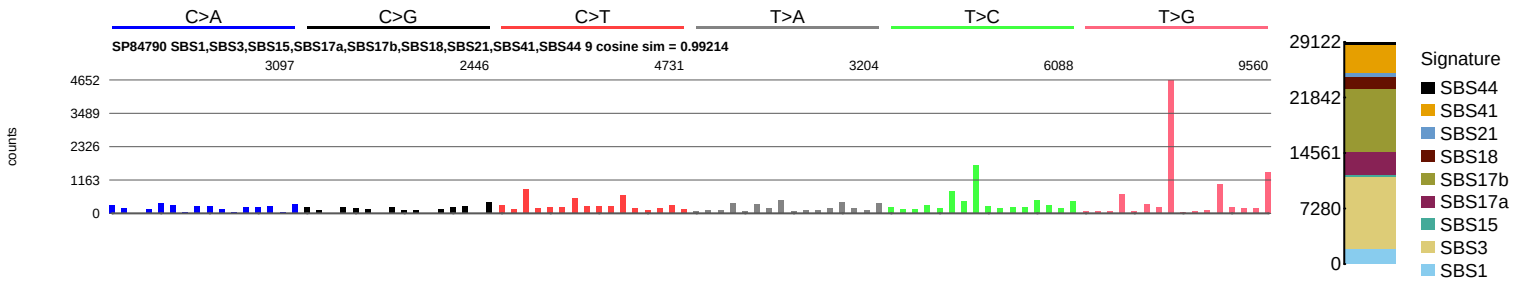

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS44 10 cosine sim = 0.99215
3155
2395
5609
2277
6171
9512
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

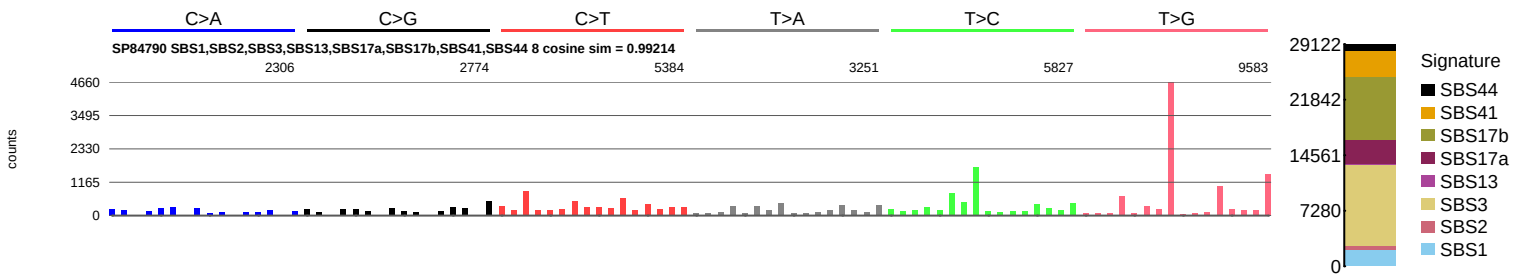

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS44 10 cosine sim = 0.99215
3090
2402
5680
2280
6174
9496
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

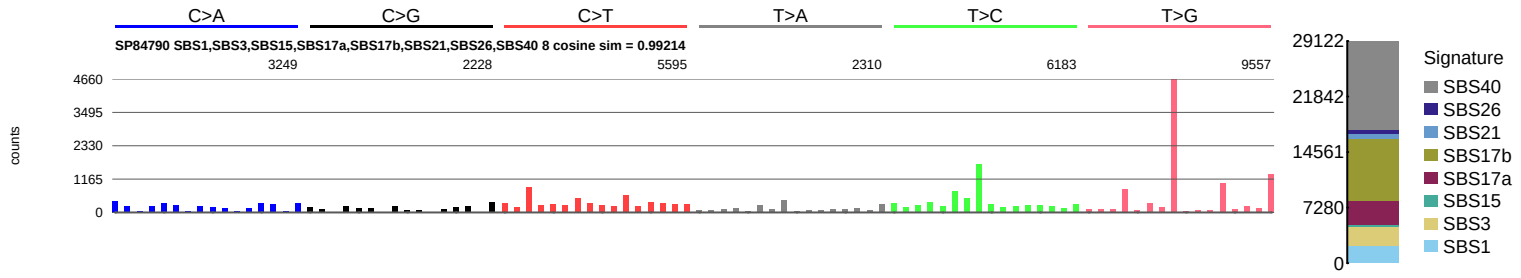

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS26,SBS41 7 cosine sim = 0.99214
2186
2654
5082
3228
6346
9624
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS3
SBS2
SBS1

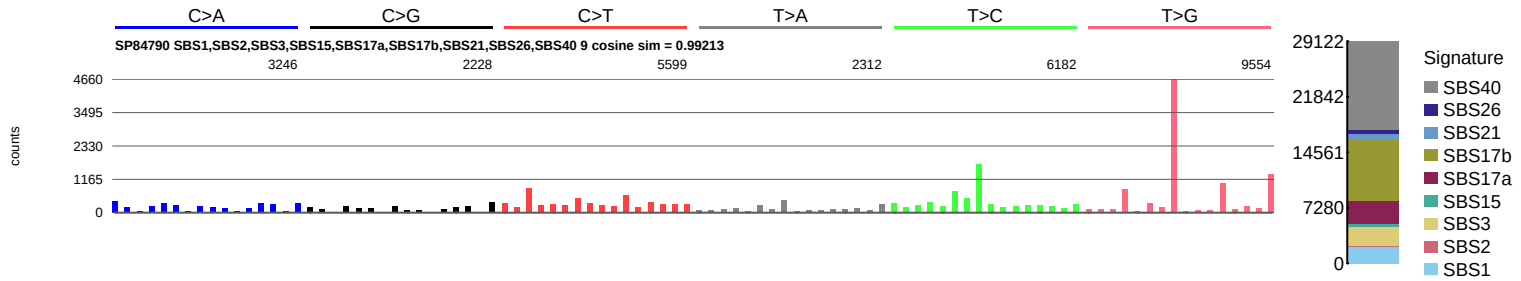

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.99214
3166
2379
5637
2266
6178
9493
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

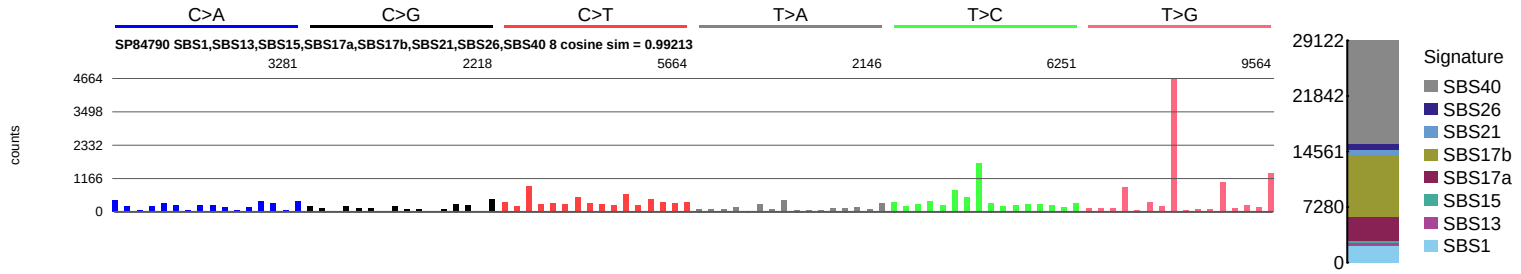

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS44 9 cosine sim = 0.99214
3116
2392
5651
2277
6179
9510
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

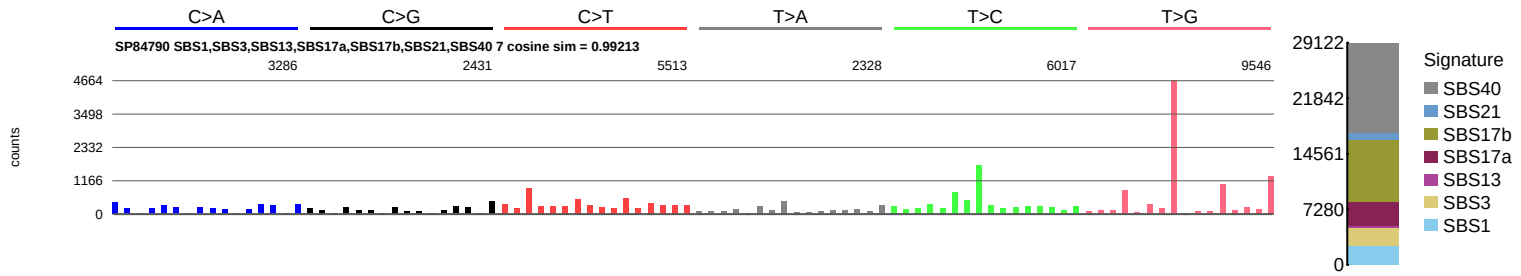

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.99214
3097
2446
4731
3204
6088
9560
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

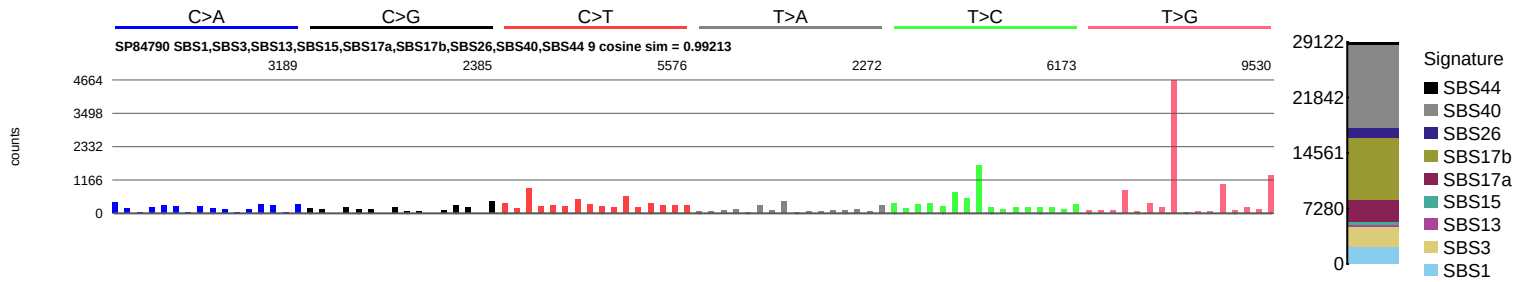

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS41,SBS44 8 cosine sim = 0.99214
2306
2774
5384
3251
5827
9583
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

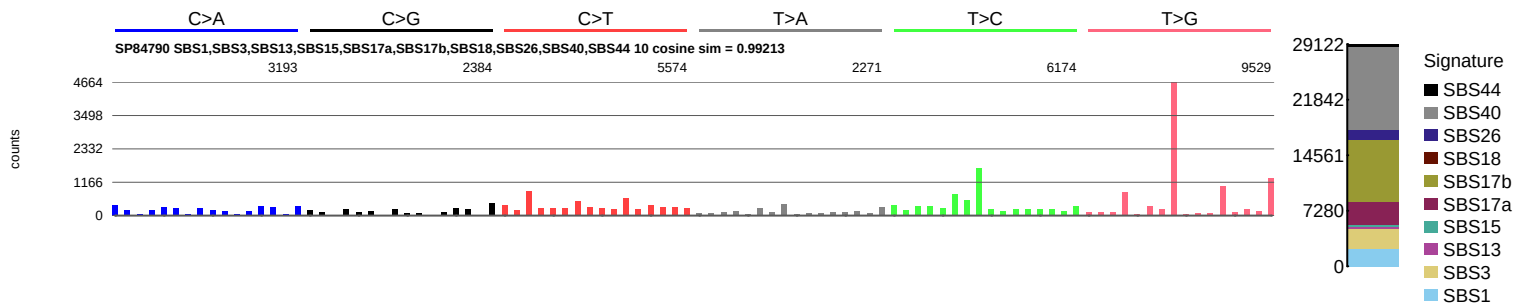

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40 8 cosine sim = 0.99214
3249
2228
5595
2310
6183
9557
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS1

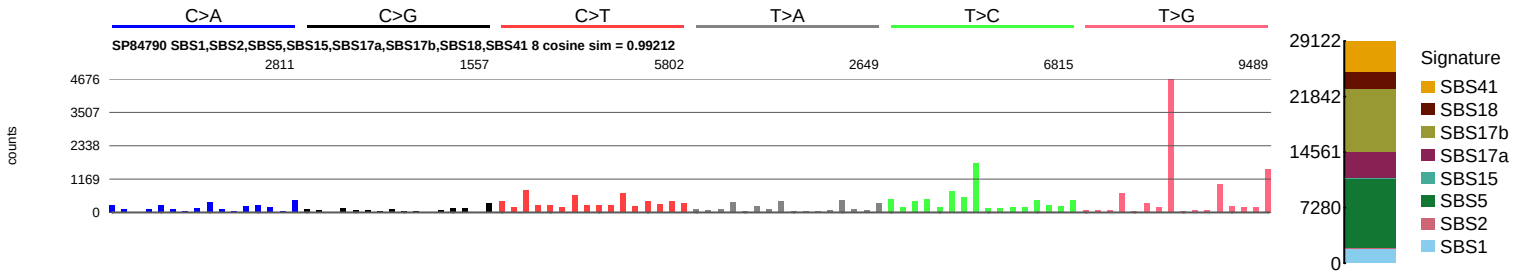

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40 9 cosine sim = 0.99213
3246
2228
5599
2312
6182
9554
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

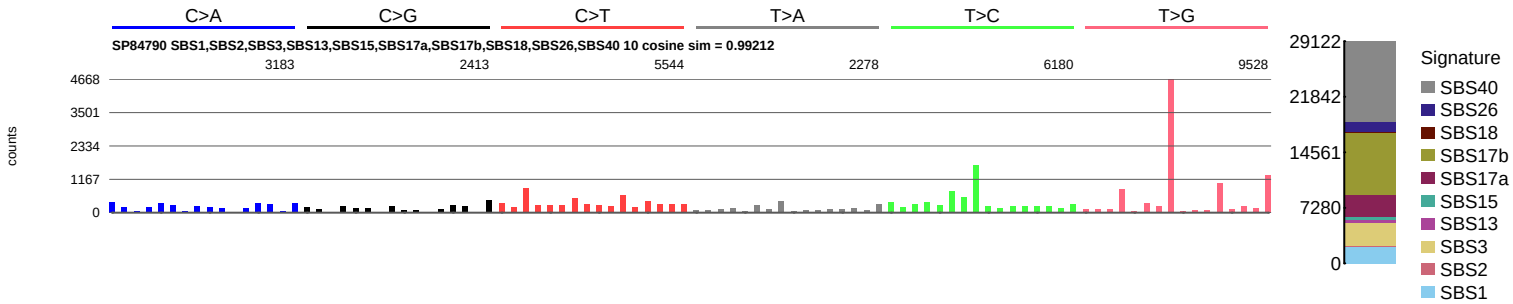

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40 8 cosine sim = 0.99213
3281
2218
5664
2146
6251
9564
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS1

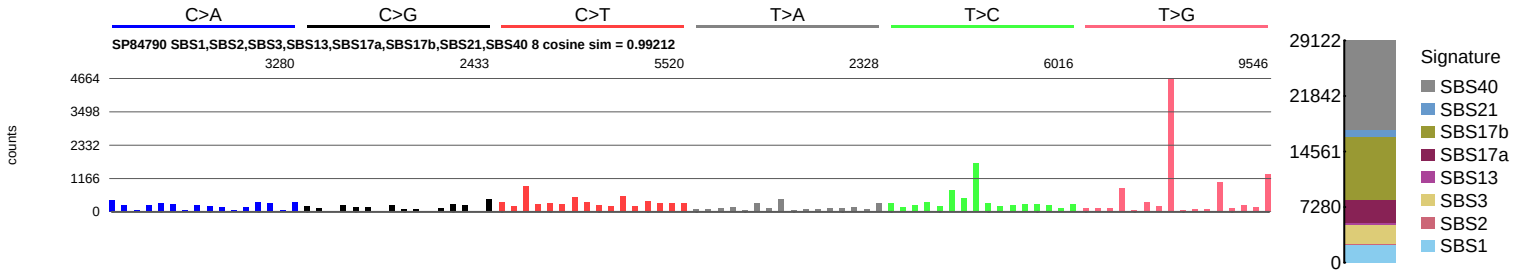

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS40 7 cosine sim = 0.99213
3286
2431
5513
2328
6017
9546
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS1

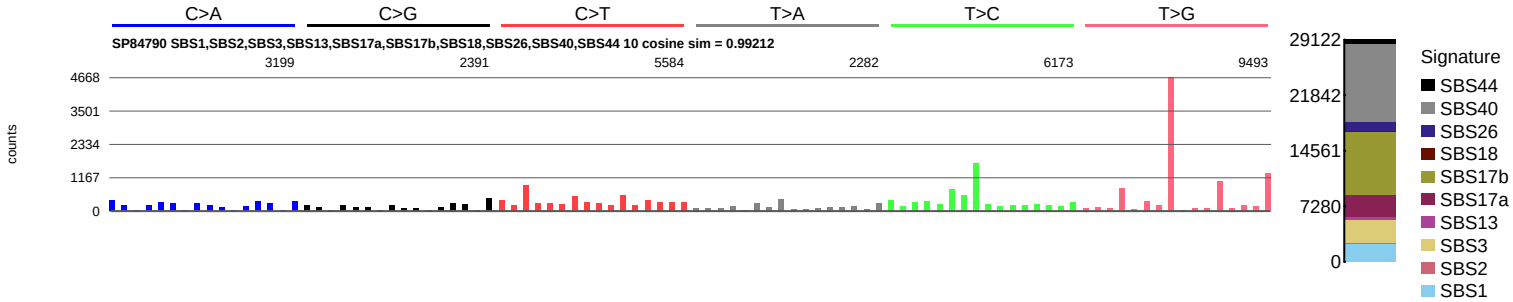

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS44 9 cosine sim = 0.99213
3189
2385
5576
2272
6173
9530
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

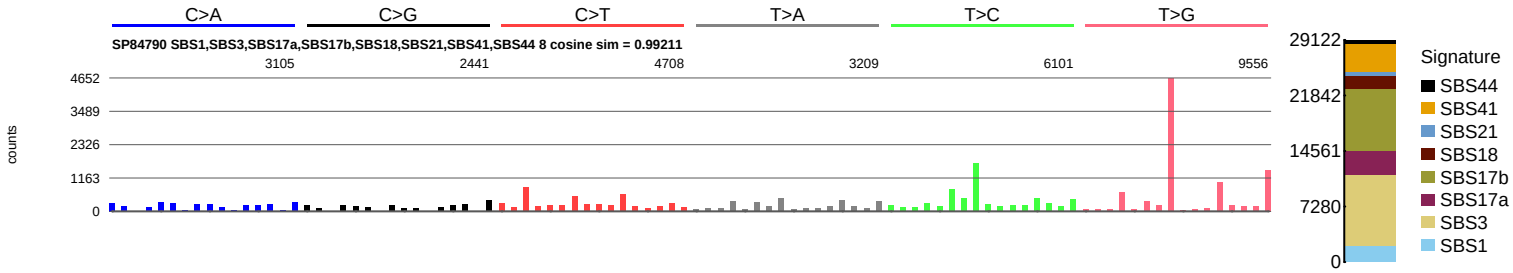

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.99213
3193
2384
5574
2271
6174
9529
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

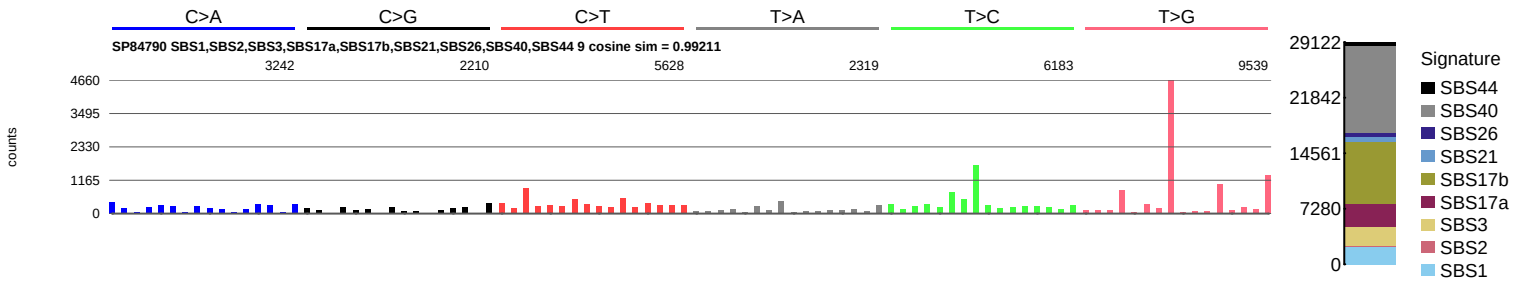

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.99212
2811
1557
5802
2649
6815
9489
counts
4676
3507
2338
1169
0
29122
21842
14561
7280
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

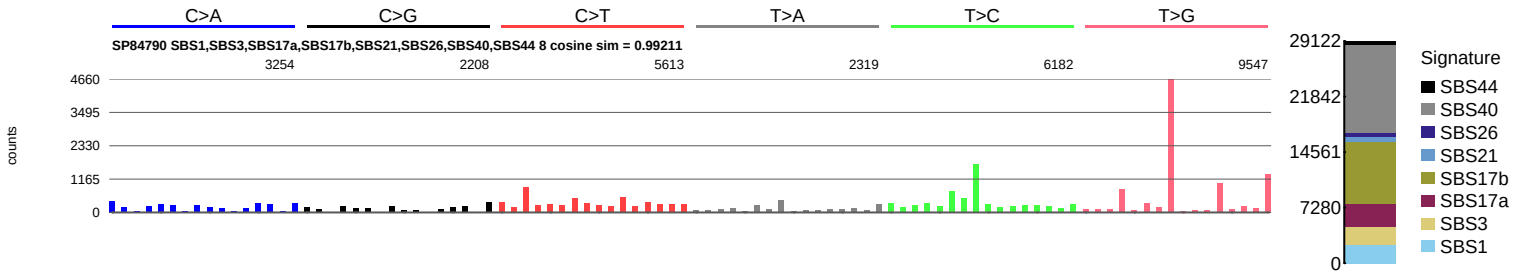

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 10 cosine sim = 0.99212
3183
2413
5544
2278
6180
9528
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

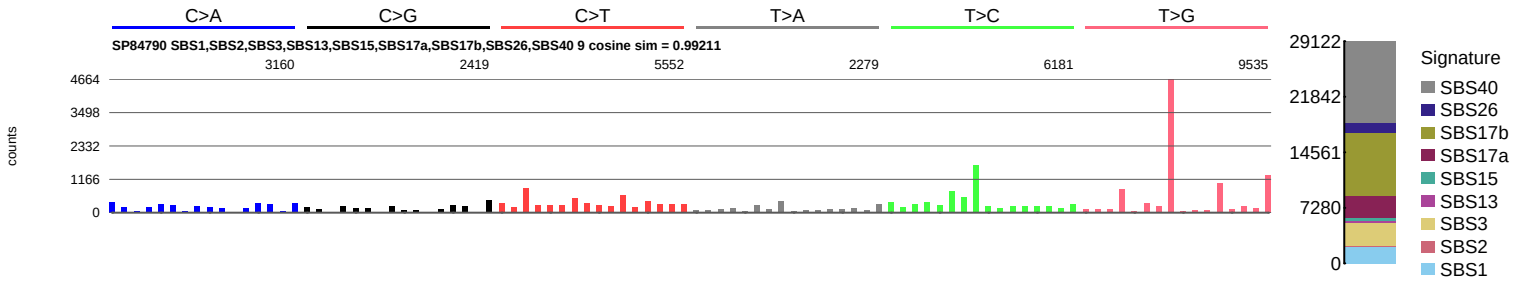

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS40 8 cosine sim = 0.99212
3280
2433
5520
2328
6016
9546
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

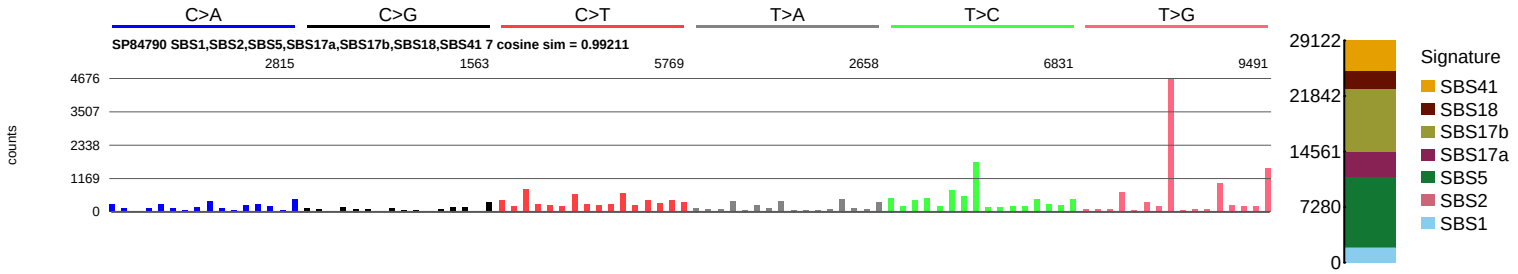

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.99212
3199
2391
5584
2282
6173
9493
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

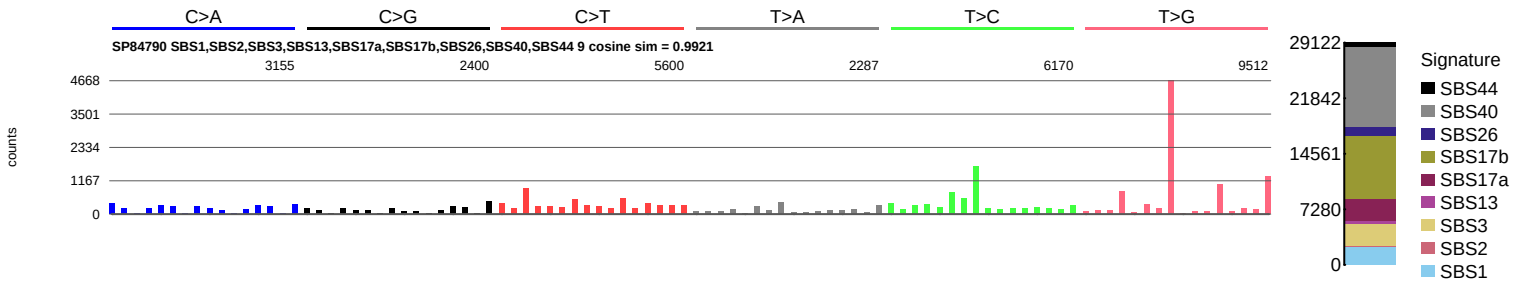

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.99211
3105
2441
4708
3209
6101
9556
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1

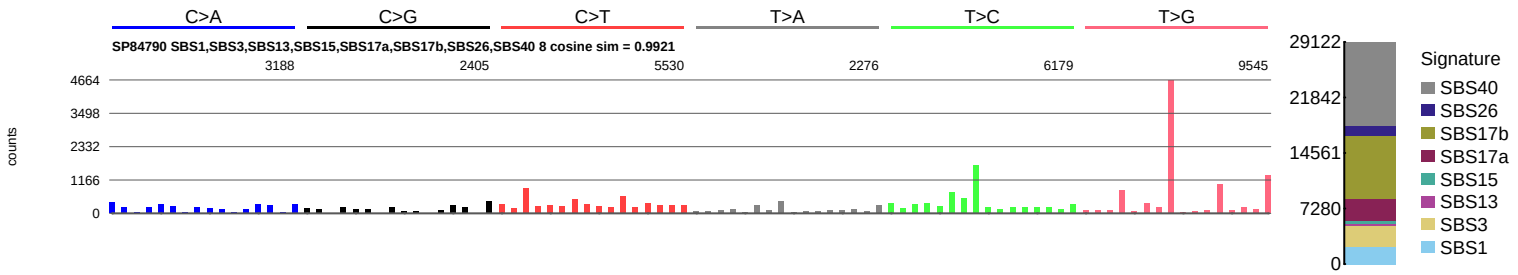

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.99211
3242
2210
5628
2319
6183
9539
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1

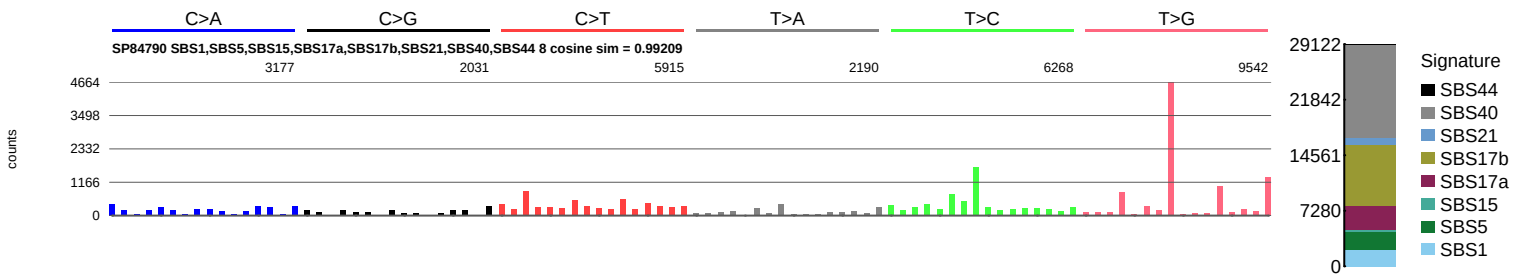

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS44 8 cosine sim = 0.99211
3254
2208
5613
2319
6182
9547
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS3
SBS1

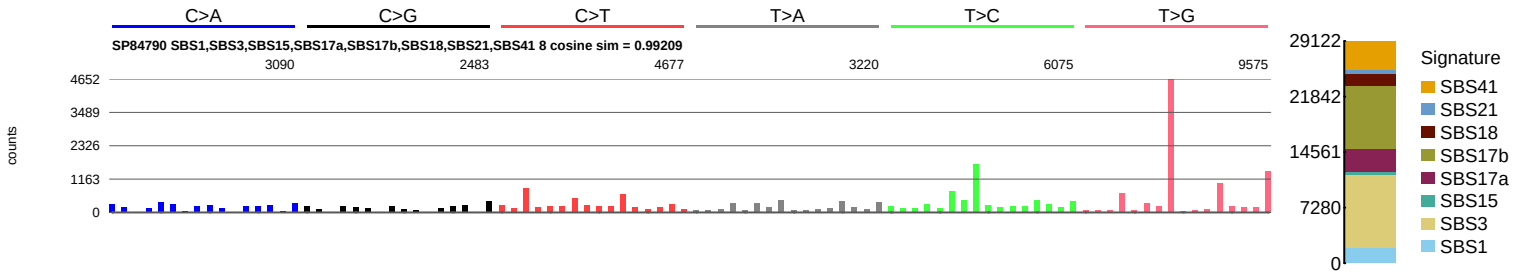

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40 9 cosine sim = 0.99211
3160
2419
5552
2279
6181
9535
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

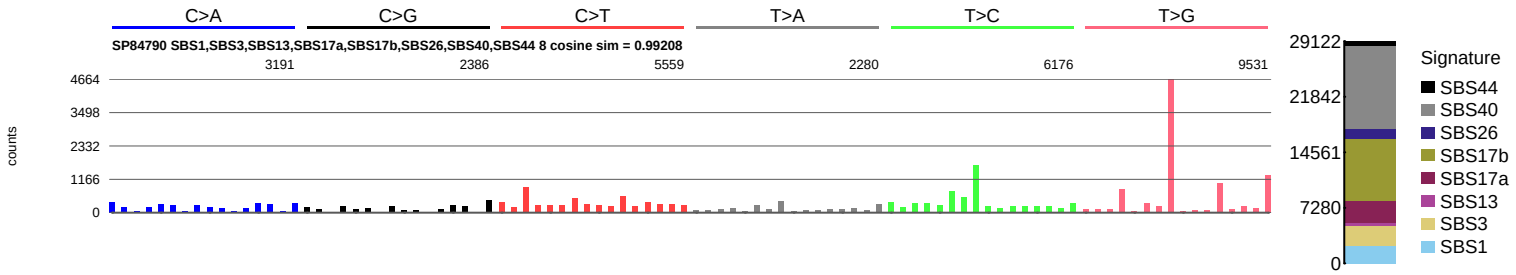

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.99211
2815
1563
5769
2658
6831
9491
counts
4676
3507
2338
1169
0
29122
21842
14561
7280
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

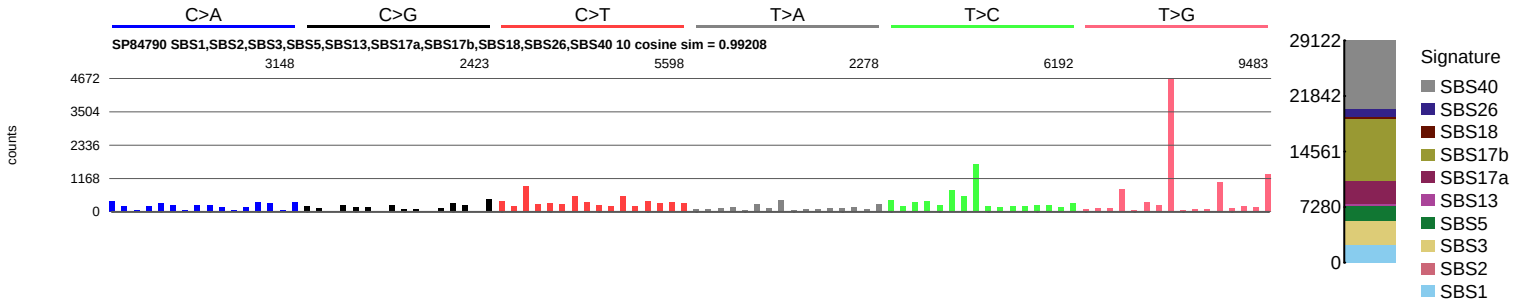

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS44 9 cosine sim = 0.9921
3155
2400
5600
2287
6170
9512
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

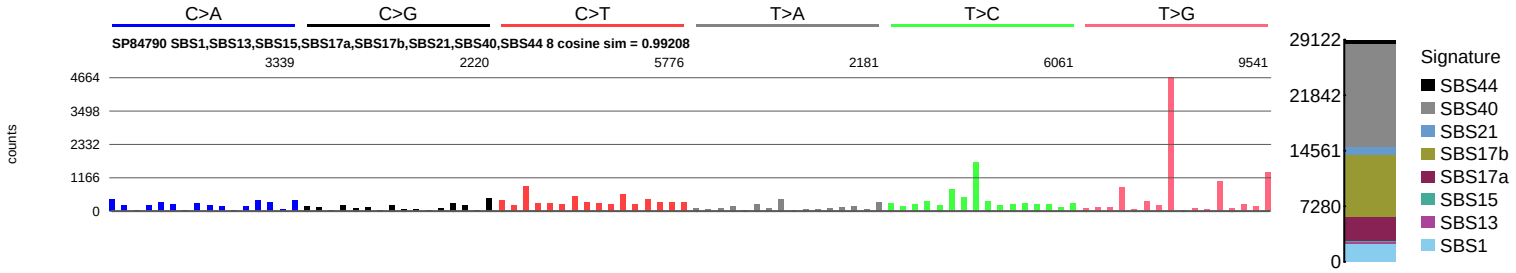

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40 8 cosine sim = 0.9921
3188
2405
5530
2276
6179
9545
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

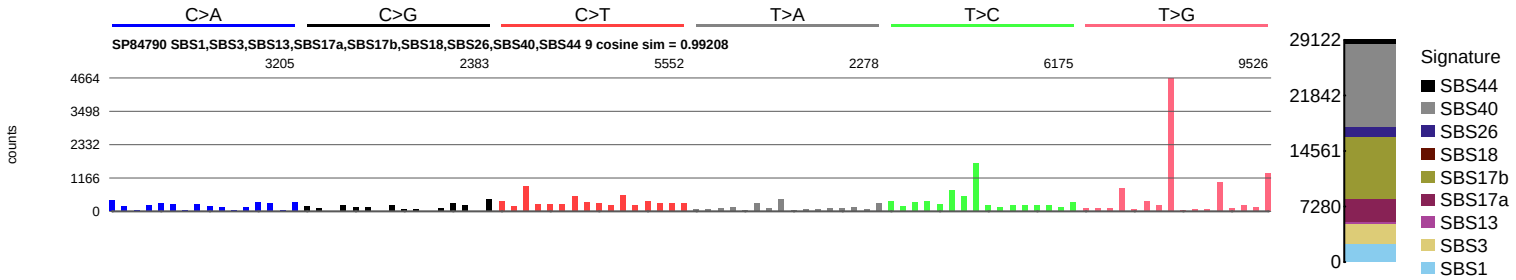

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS44 8 cosine sim = 0.99209
3177
2031
5915
2190
6268
9542
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS1

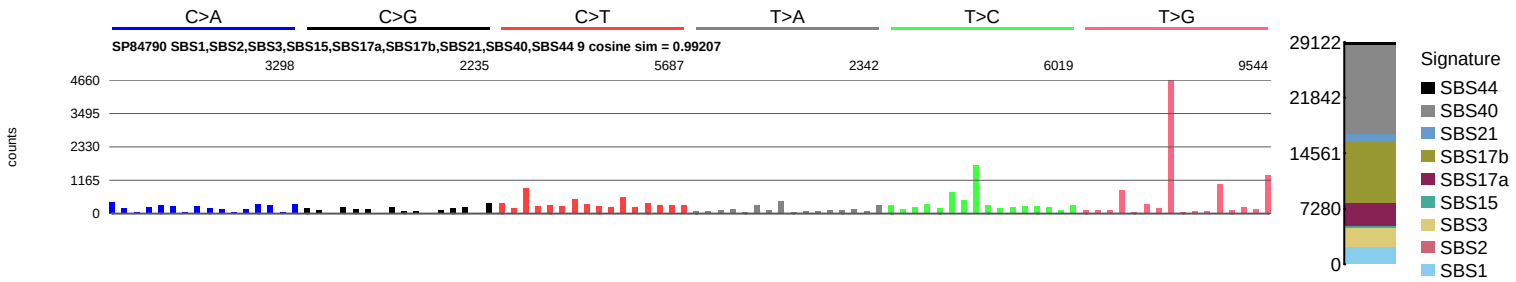

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.99209
3090
2483
4677
3220
6075
9575
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

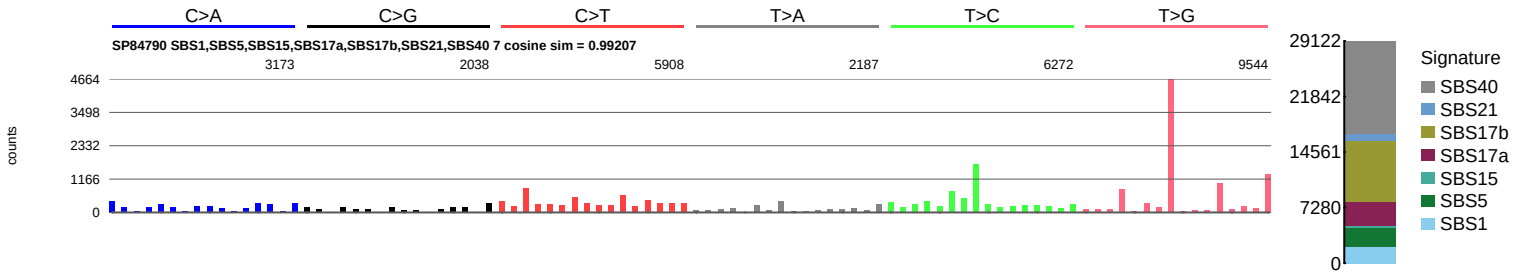

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS44 8 cosine sim = 0.99208
3191
2386
5559
2280
6176
9531
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS3
SBS1

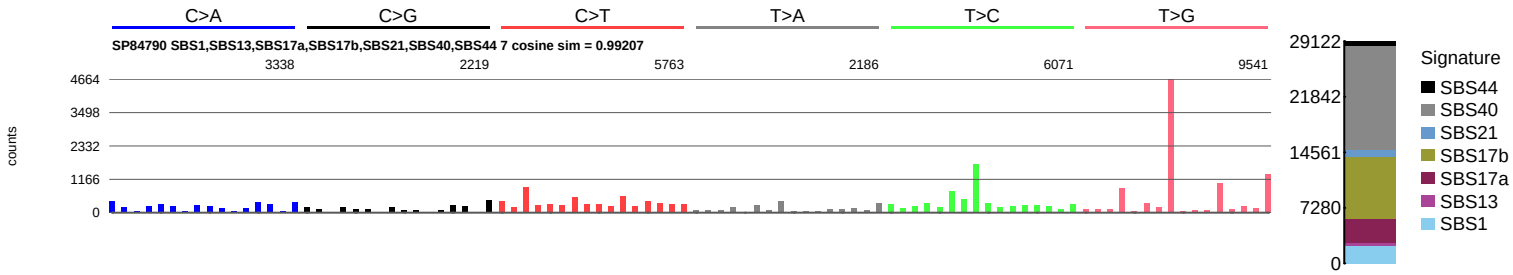

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40 10 cosine sim = 0.99208
3148
2423
5598
2278
6192
9483
counts
4672
3504
2336
1168
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

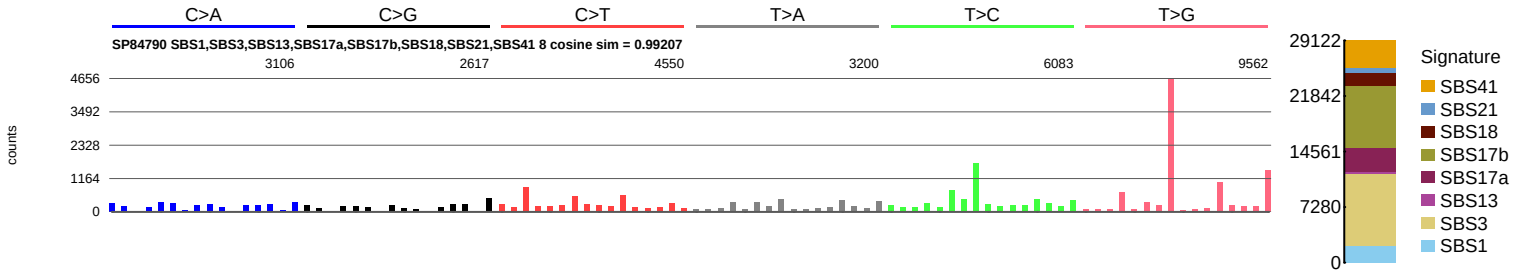

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS44 8 cosine sim = 0.99208
3339
2220
5776
2181
6061
9541
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS1

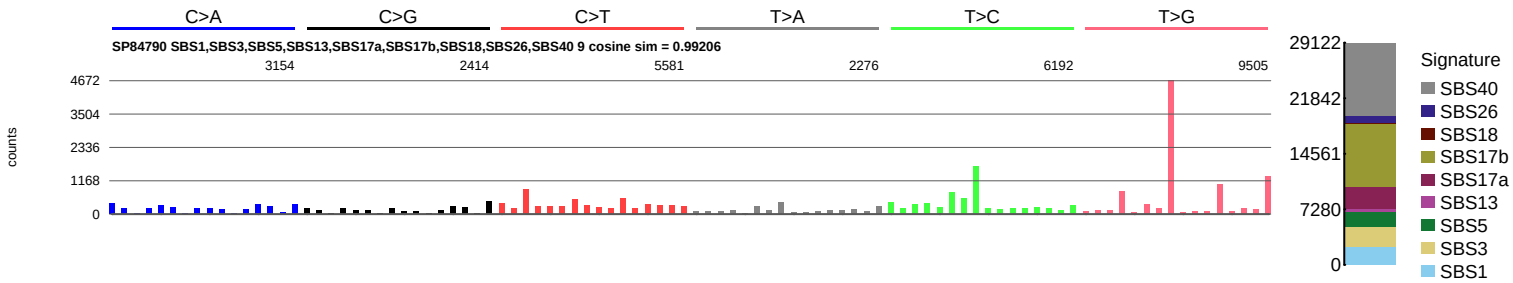

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.99208
3205
2383
5552
2278
6175
9526
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1

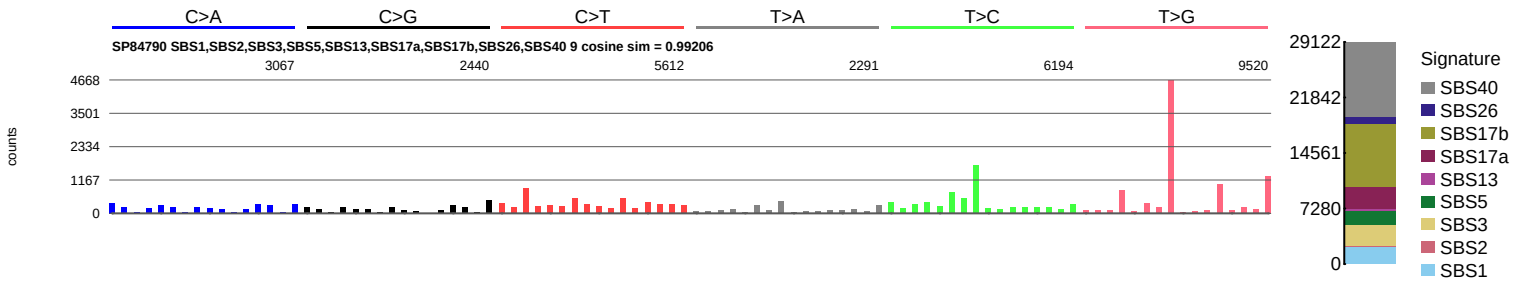

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS44 9 cosine sim = 0.99207
3298
2235
5687
2342
6019
9544
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

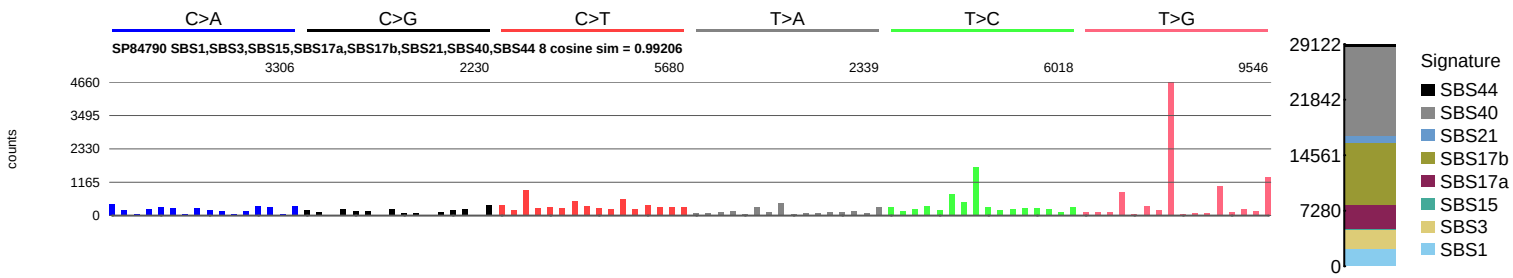

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40 7 cosine sim = 0.99207
3173
2038
5908
2187
6272
9544
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS1

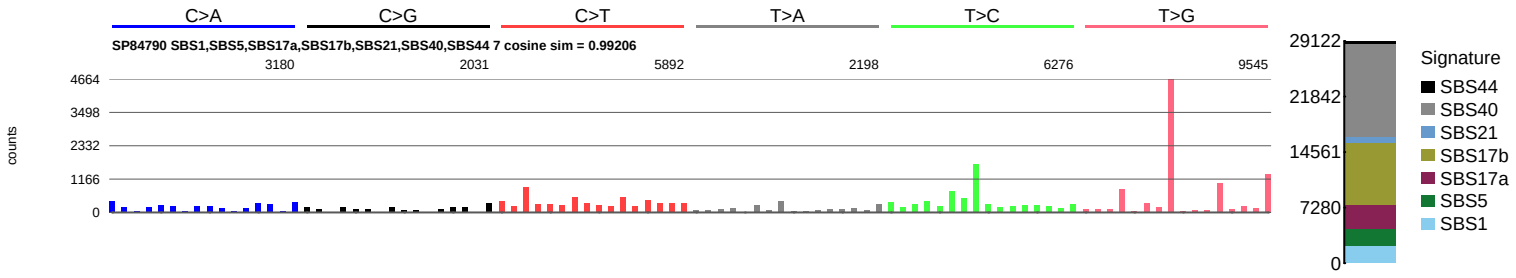

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS44 7 cosine sim = 0.99207
3338
2219
5763
2186
6071
9541
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS1

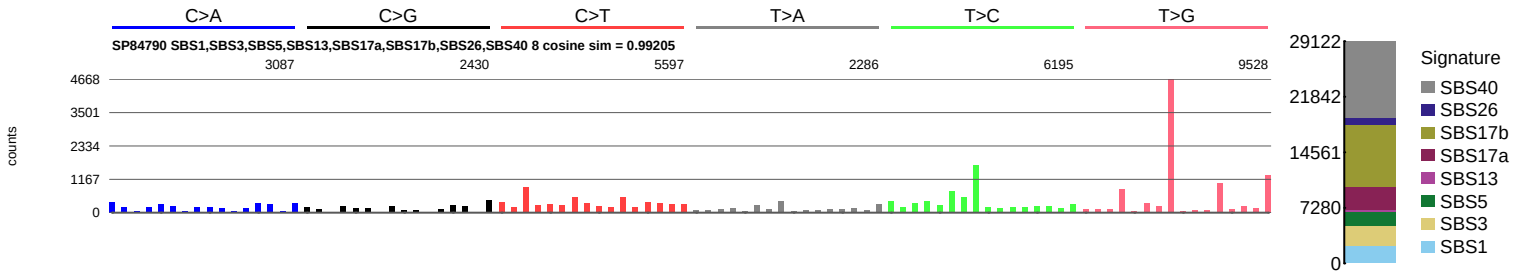

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.99207
3106
2617
4550
3200
6083
9562
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1

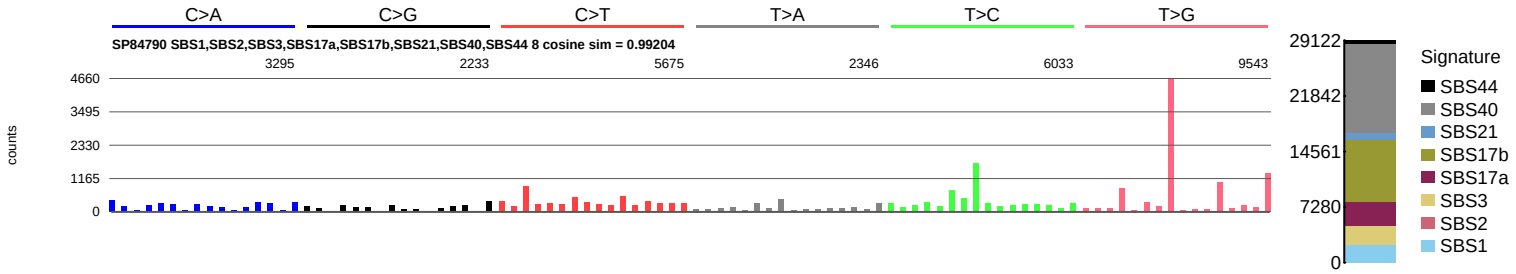

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.99206
3154
2414
5581
2276
6192
9505
counts
4672
3504
2336
1168
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

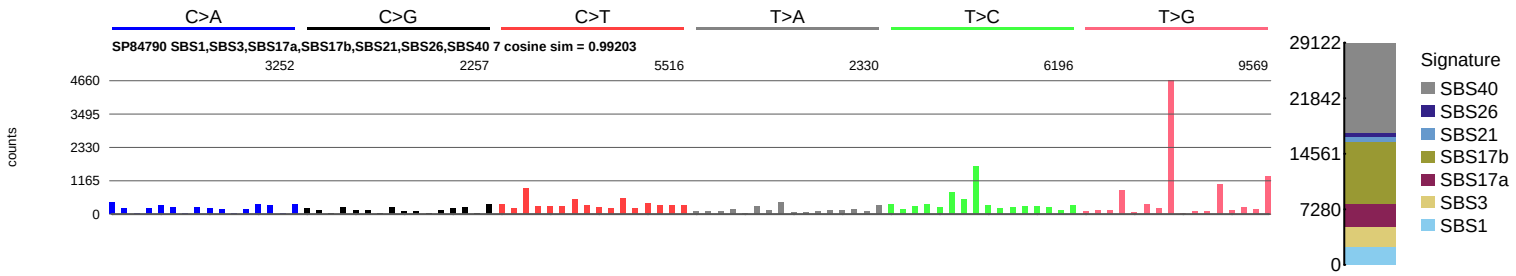

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS26,SBS40 9 cosine sim = 0.99206
3067
2440
5612
2291
6194
9520
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

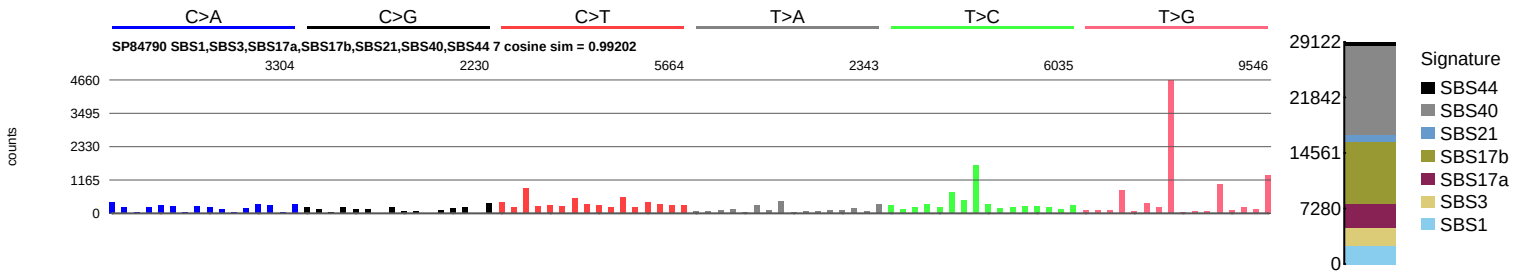

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS44 8 cosine sim = 0.99206
3306
2230
5680
2339
6018
9546
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS1

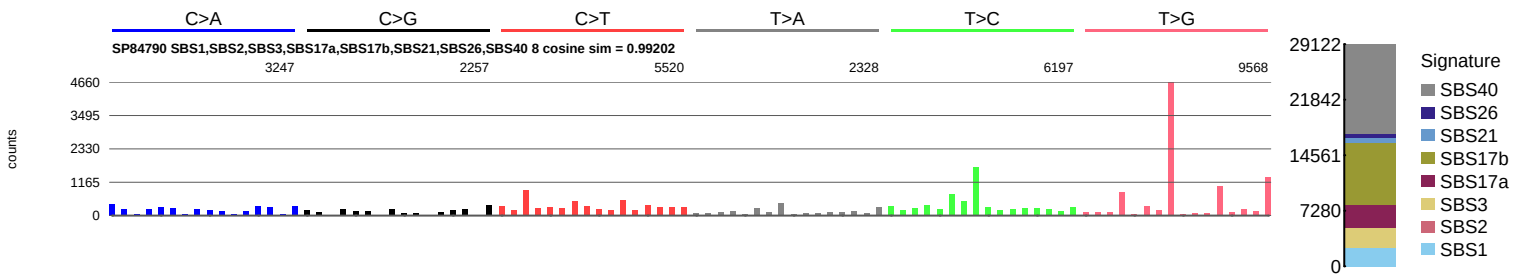

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS44 7 cosine sim = 0.99206
3180
2031
5892
2198
6276
9545
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS1

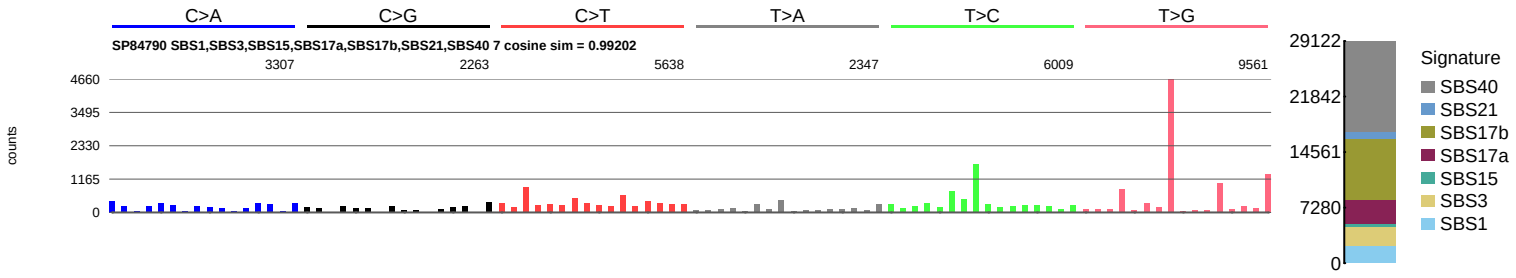

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS26,SBS40 8 cosine sim = 0.99205
3087
2430
5597
2286
6195
9528
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

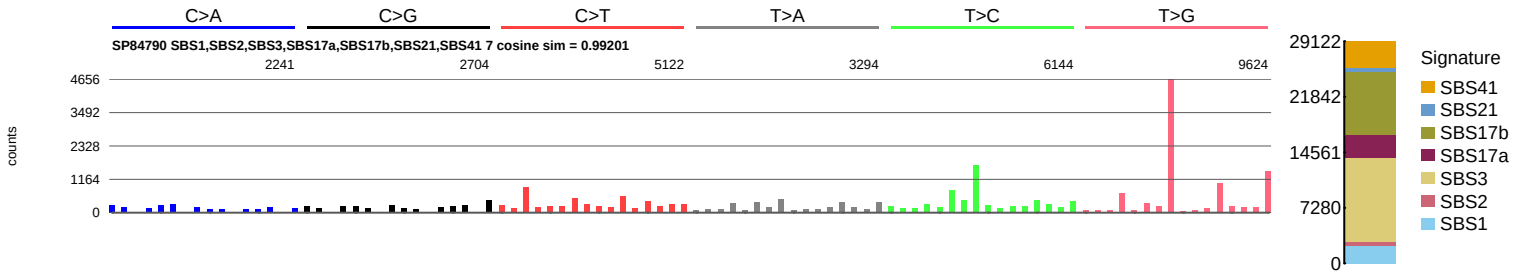

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS40,SBS44 8 cosine sim = 0.99204
3295
2233
5675
2346
6033
9543
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1

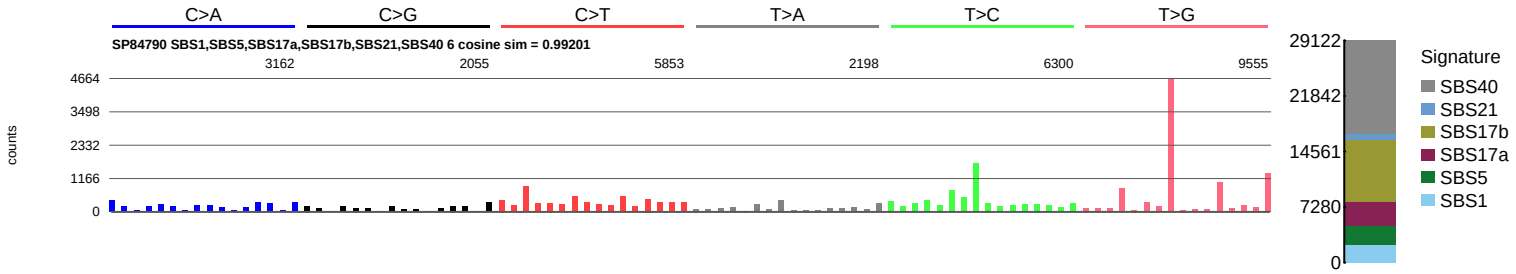

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS21,SBS26,SBS40 7 cosine sim = 0.99203
3252
2257
5516
2330
6196
9569
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS3
SBS1

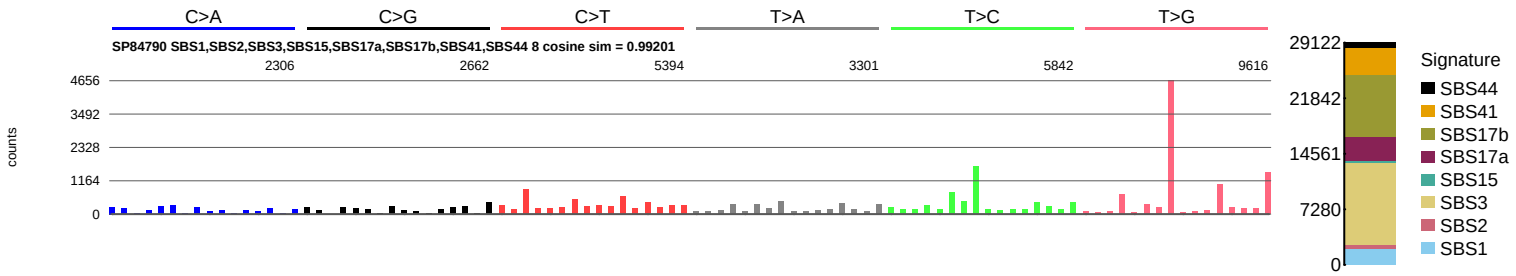

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS21,SBS40,SBS44 7 cosine sim = 0.99202
3304
2230
5664
2343
6035
9546
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS3
SBS1

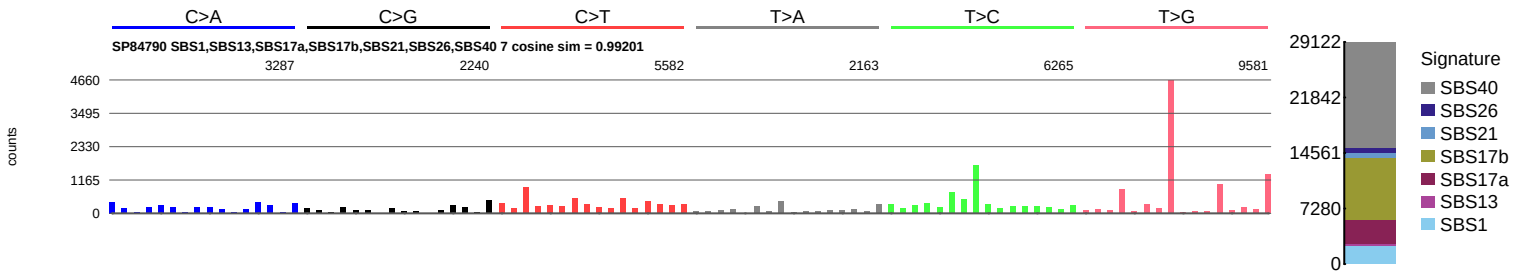

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS26,SBS40 8 cosine sim = 0.99202
3247
2257
5520
2328
6197
9568
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1

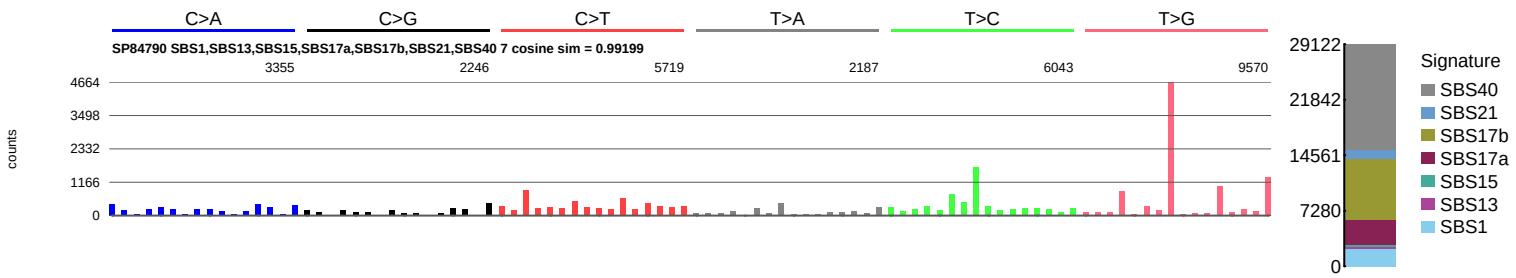

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS40 7 cosine sim = 0.99202
3307
2263
5638
2347
6009
9561
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS1

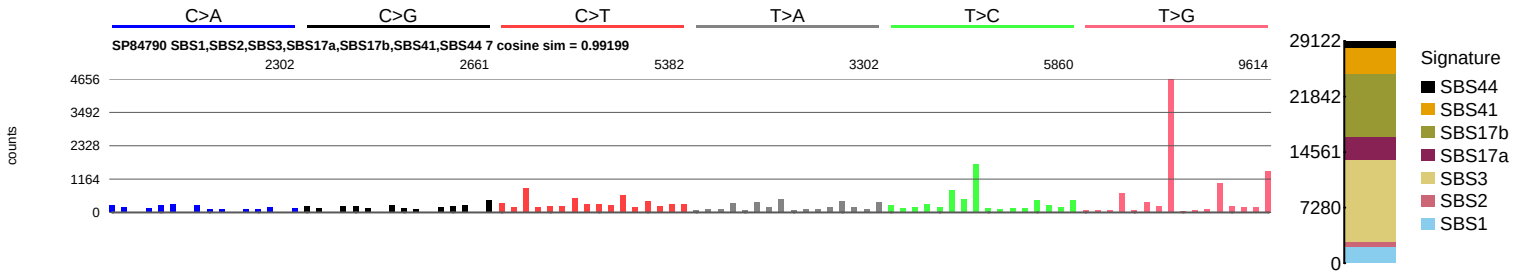

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.99201
2241
2704
5122
3294
6144
9624
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1

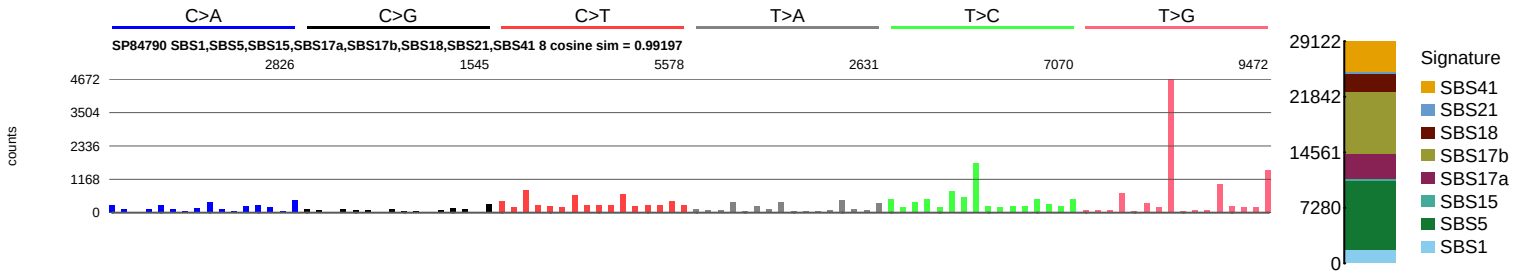

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS17a,SBS17b,SBS21,SBS40 6 cosine sim = 0.99201
3162
2055
5853
2198
6300
9555
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS1

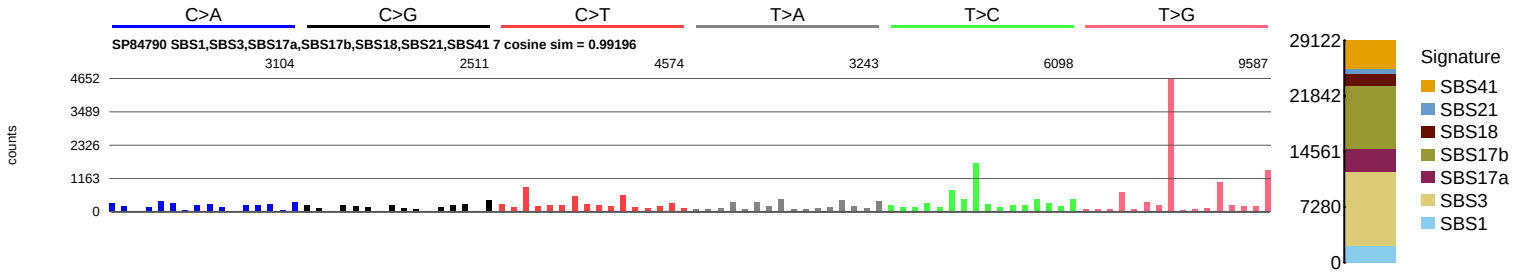

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS41,SBS44 8 cosine sim = 0.99201
2306
2662
5394
3301
5842
9616
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

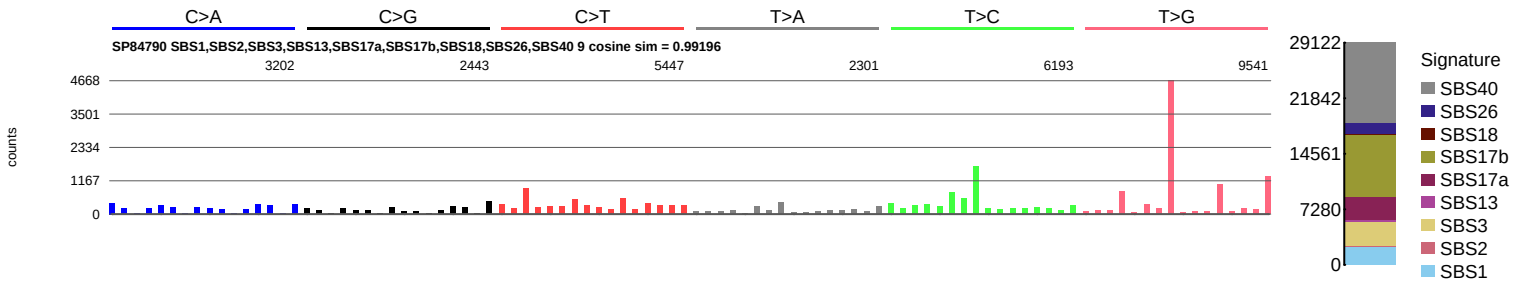

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS40 7 cosine sim = 0.99201
3287
2240
5582
2163
6265
9581
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS1

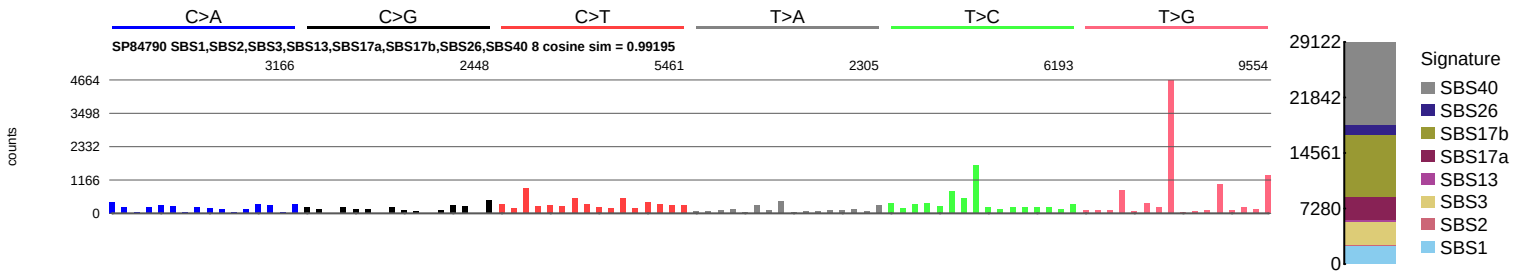

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40 7 cosine sim = 0.99199
3355
2246
5719
2187
6043
9570
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS1

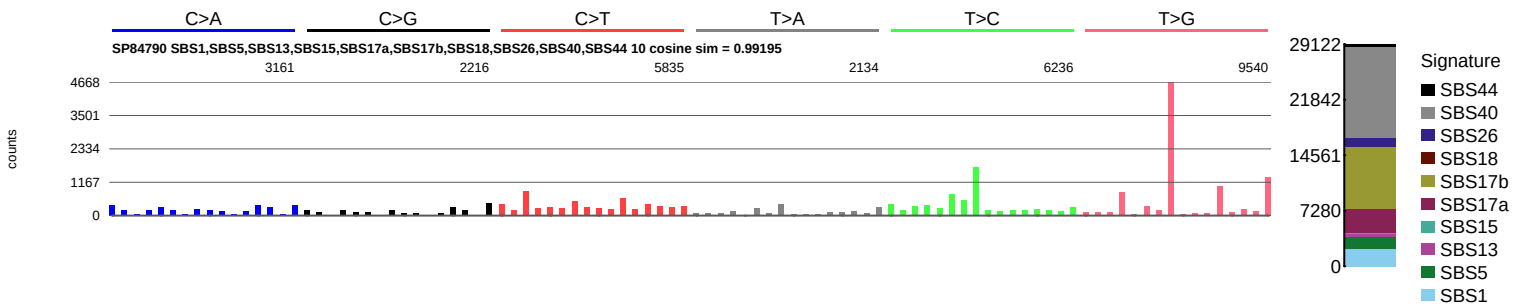

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS41,SBS44 7 cosine sim = 0.99199
2302
2661
5382
3302
5860
9614
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS3
SBS2
SBS1

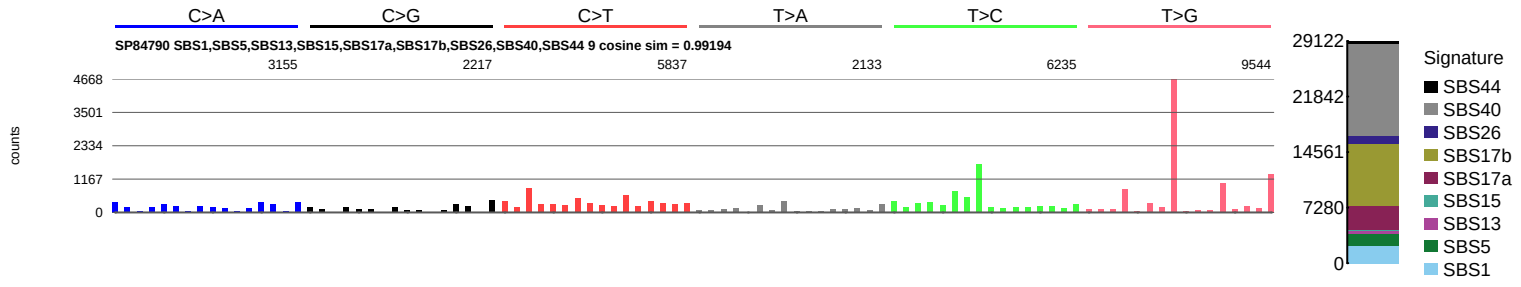

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.99197
2826
1545
5578
2631
7070
9472
counts
4672
3504
2336
1168
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1

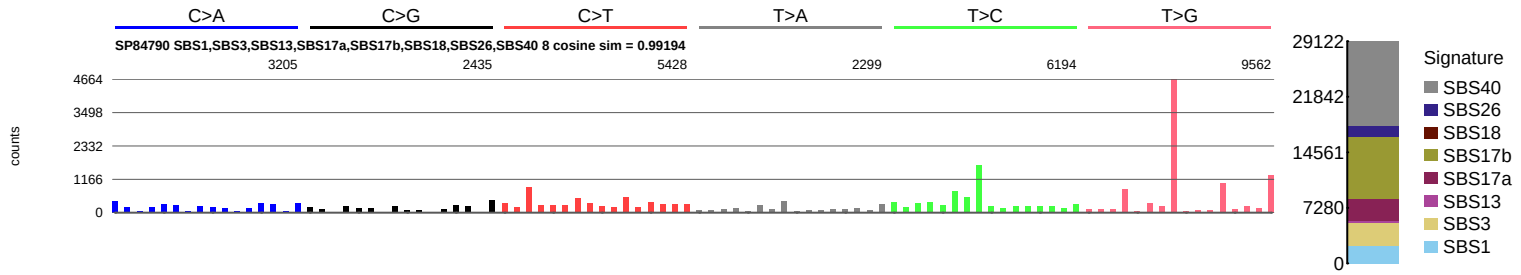

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.99196
3104
2511
4574
3243
6098
9587
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1

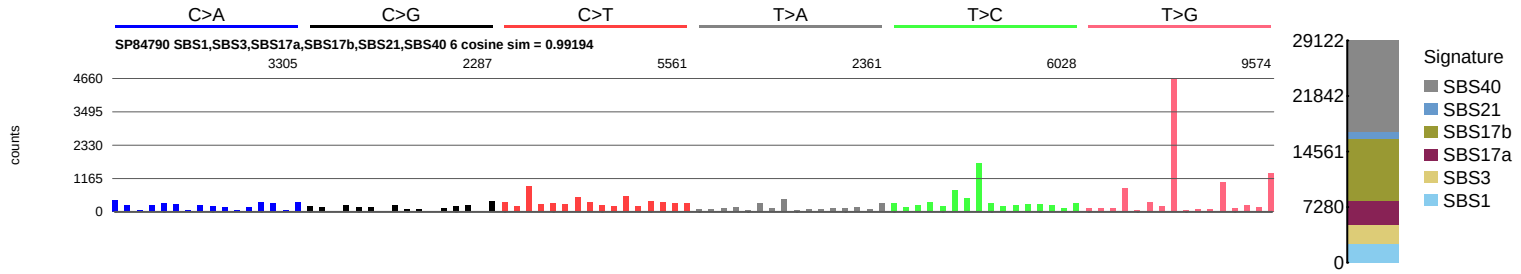

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.99196
3202
2443
5447
2301
6193
9541
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

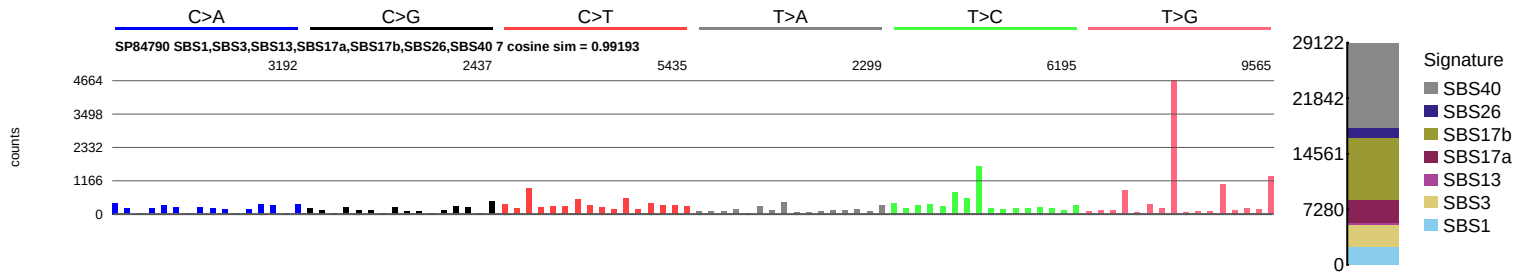

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS26,SBS40 8 cosine sim = 0.99195
3166
2448
5461
2305
6193
9554
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

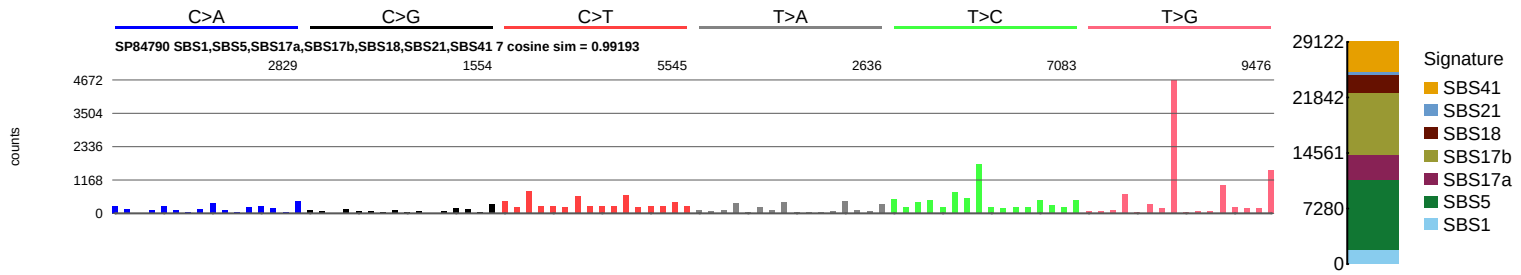

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.99195
3161
2216
5835
2134
6236
9540
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

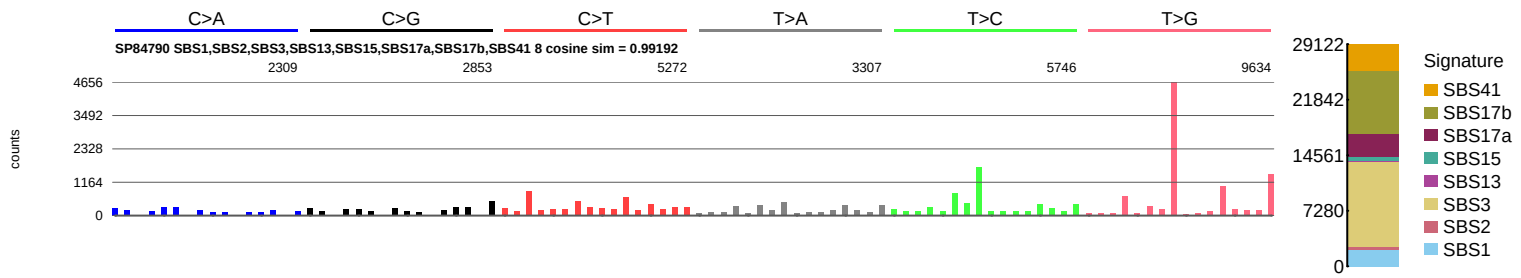

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS44 9 cosine sim = 0.99194
3155
2217
5837
2133
6235
9544
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

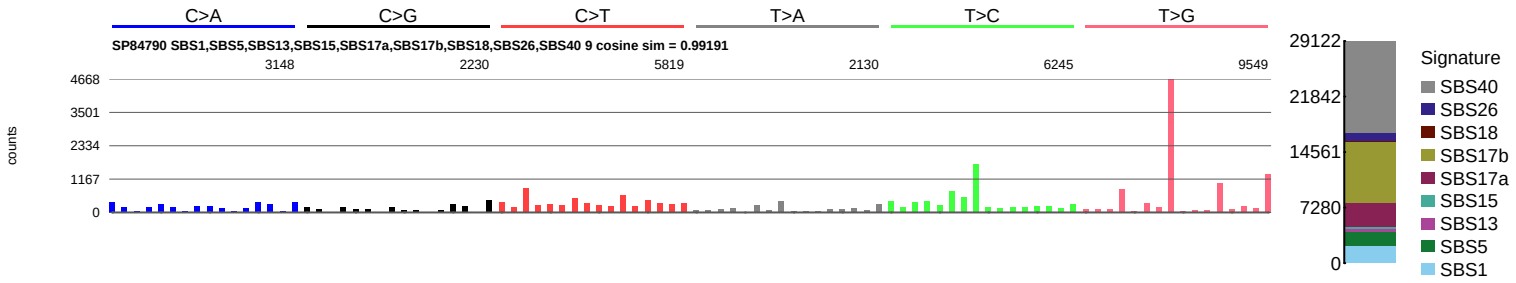

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.99194
3205
2435
5428
2299
6194
9562
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1

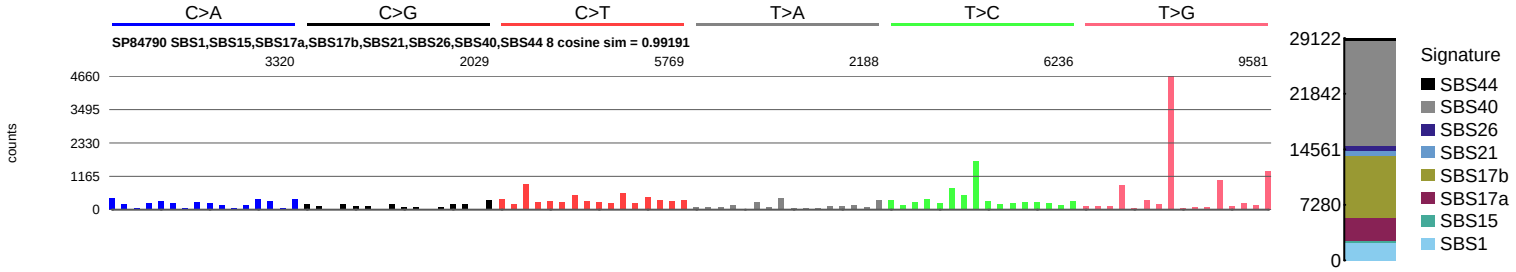

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS21,SBS40 6 cosine sim = 0.99194
3305
2287
5561
2361
6028
9574
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS3
SBS1

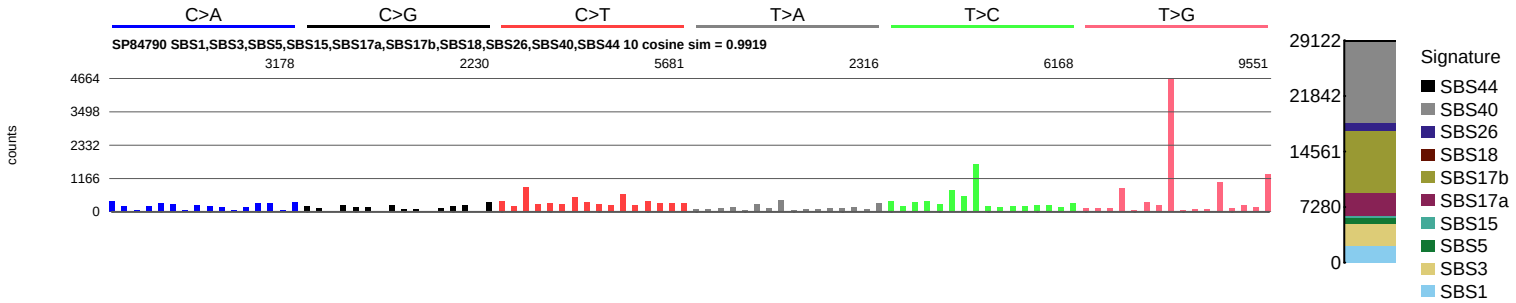

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS26,SBS40 7 cosine sim = 0.99193
3192
2437
5435
2299
6195
9565
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS3
SBS1

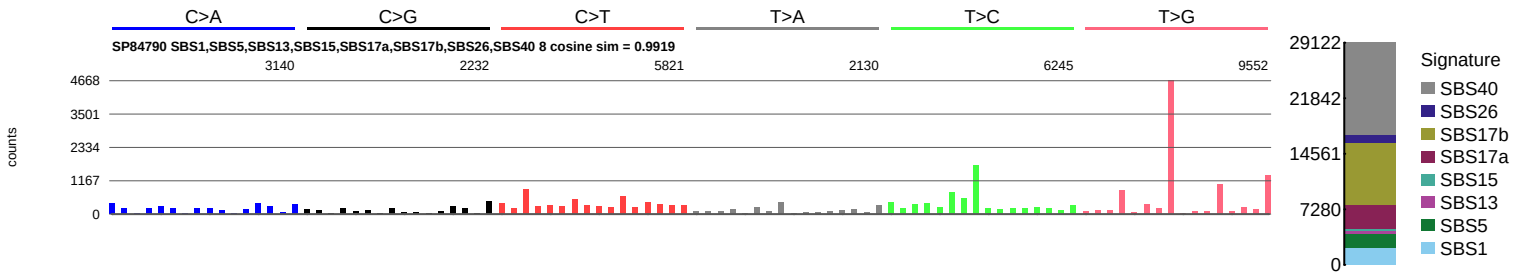

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.99193
2829
1554
5545
2636
7083
9476
counts
4672
3504
2336
1168
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1

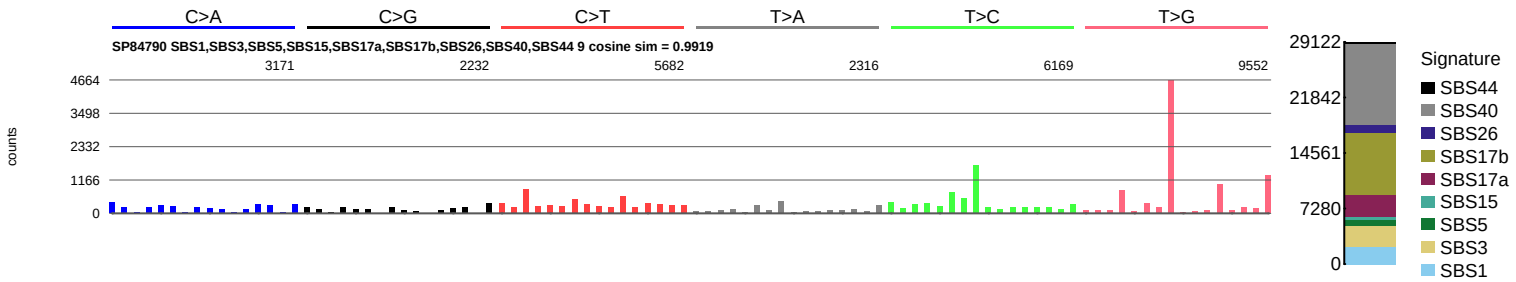

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS41 8 cosine sim = 0.99192
2309
2853
5272
3307
5746
9634
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

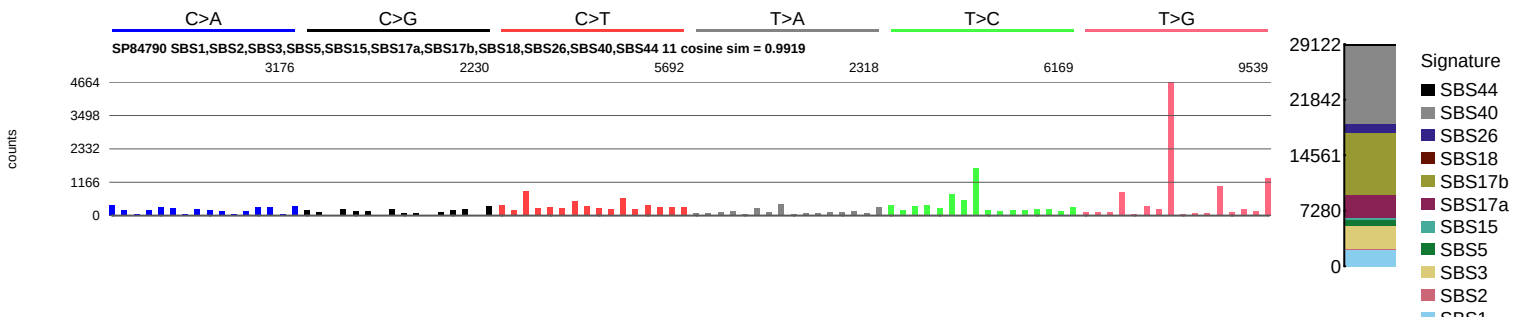

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.99191
3148
2230
5819
2130
6245
9549
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

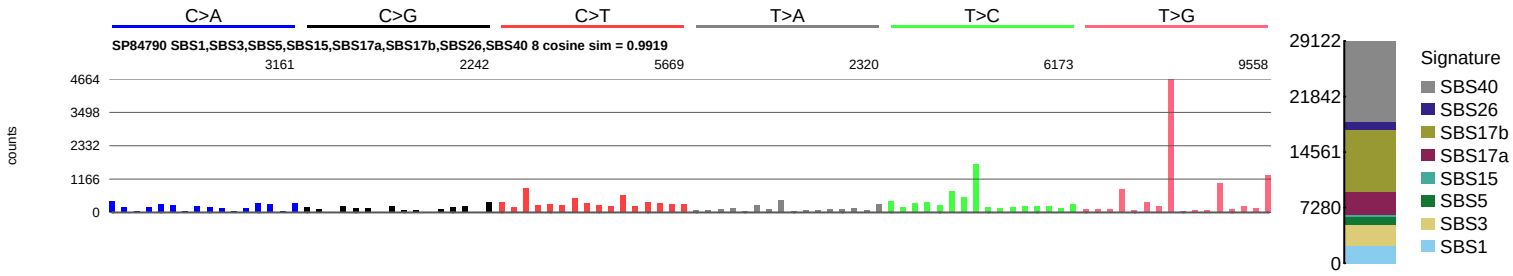

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS44 8 cosine sim = 0.99191
3320
2029
5769
2188
6236
9581
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS1

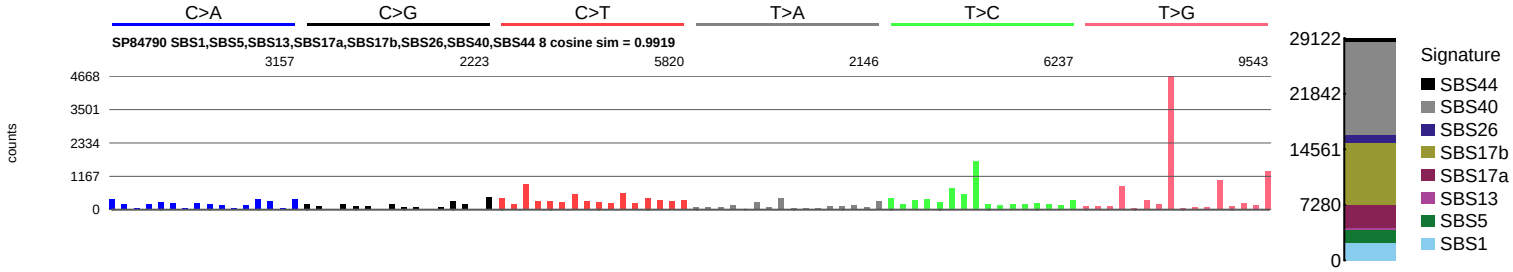

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.9919
3178
2230
5681
2316
6168
9551
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1

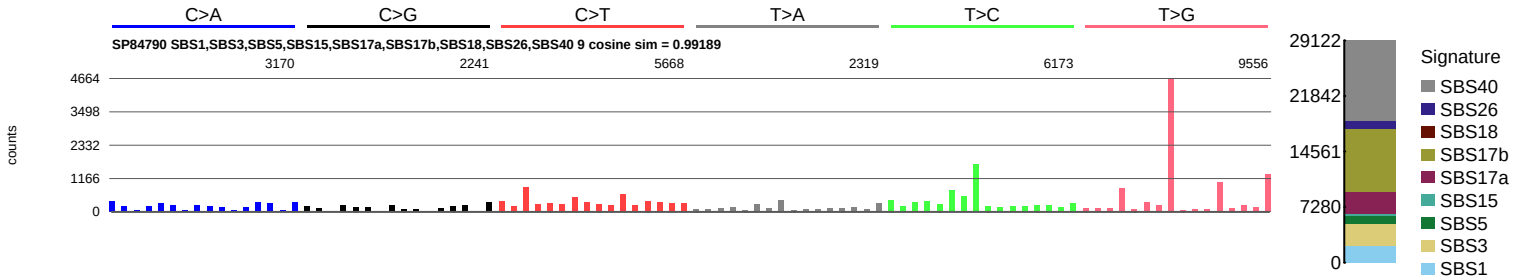

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40 8 cosine sim = 0.9919
3140
2232
5821
2130
6245
9552
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

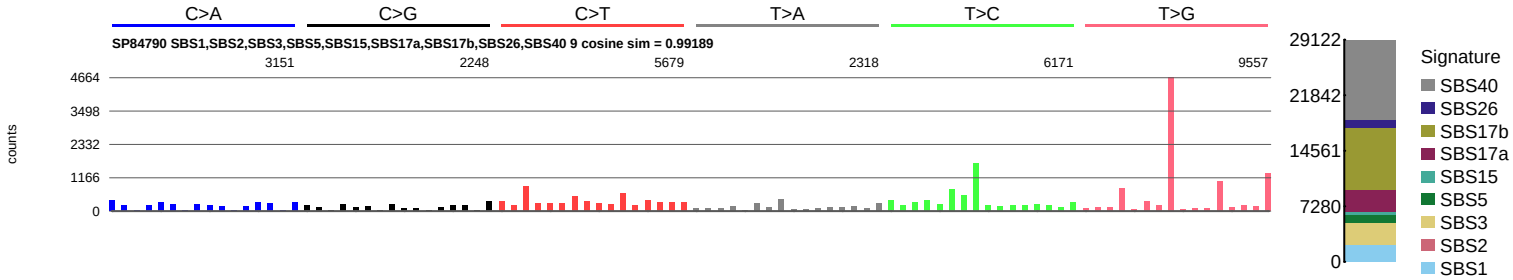

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS44 9 cosine sim = 0.9919
3171
2232
5682
2316
6169
9552
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1

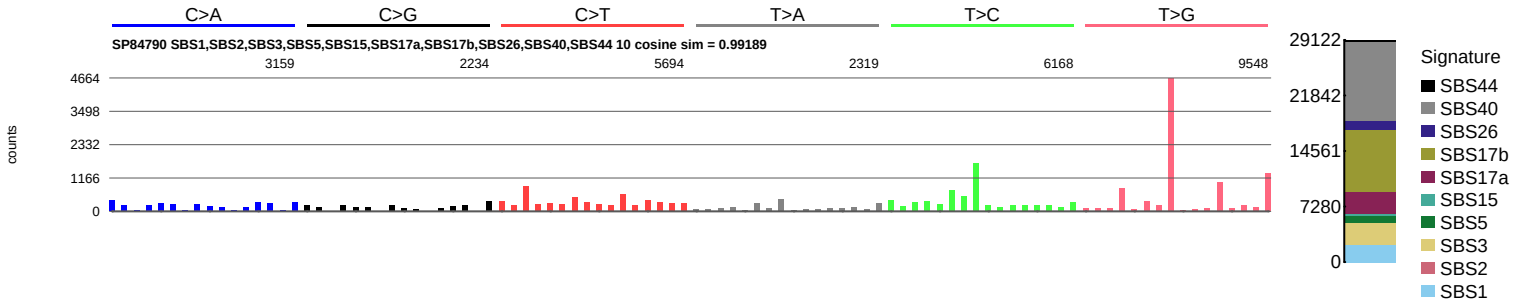

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 11 cosine sim = 0.9919
3176
2230
5692
2318
6169
9539
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2

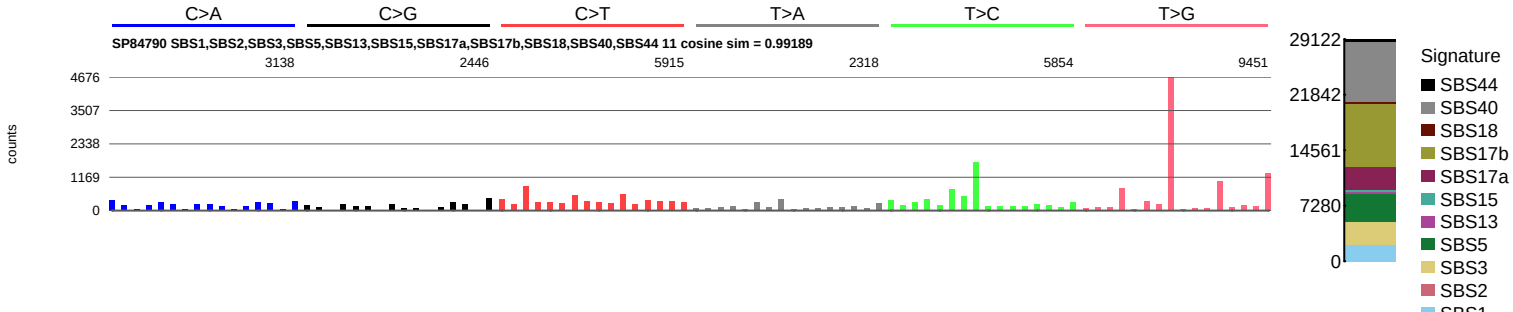

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS40 8 cosine sim = 0.9919
3161
2242
5669
2320
6173
9558
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1

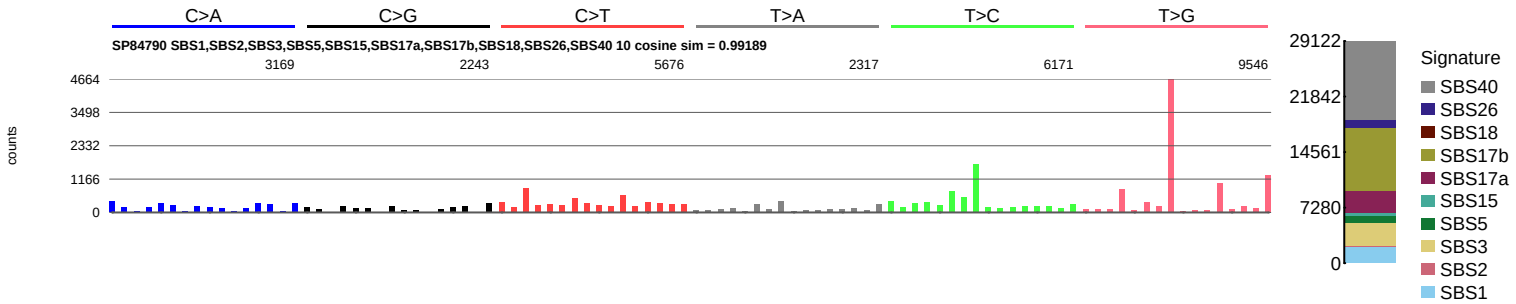

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS44 8 cosine sim = 0.9919
3157
2223
5820
2146
6237
9543
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS1

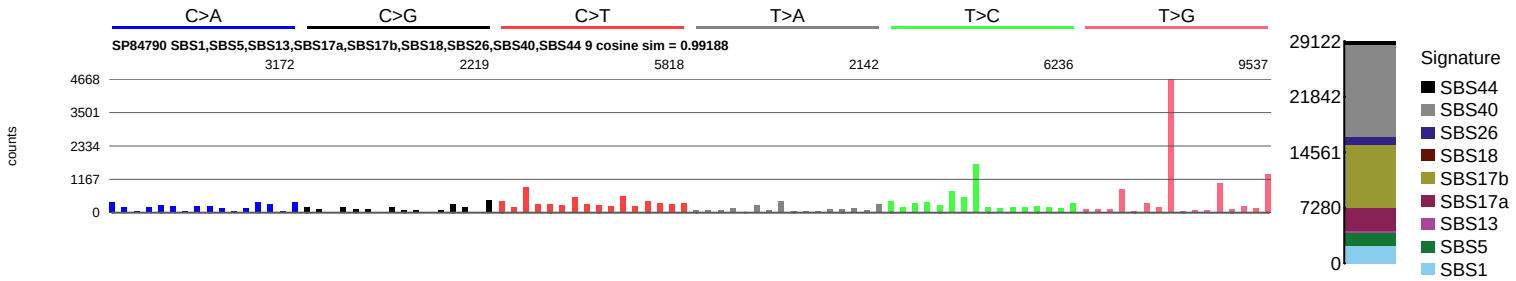

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.99189
3170
2241
5668
2319
6173
9556
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1

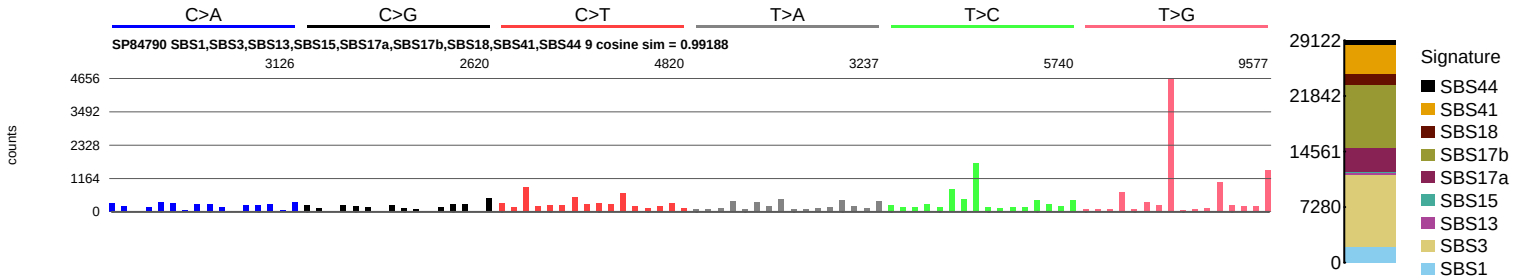

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS40 9 cosine sim = 0.99189
3151
2248
5679
2318
6171
9557
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

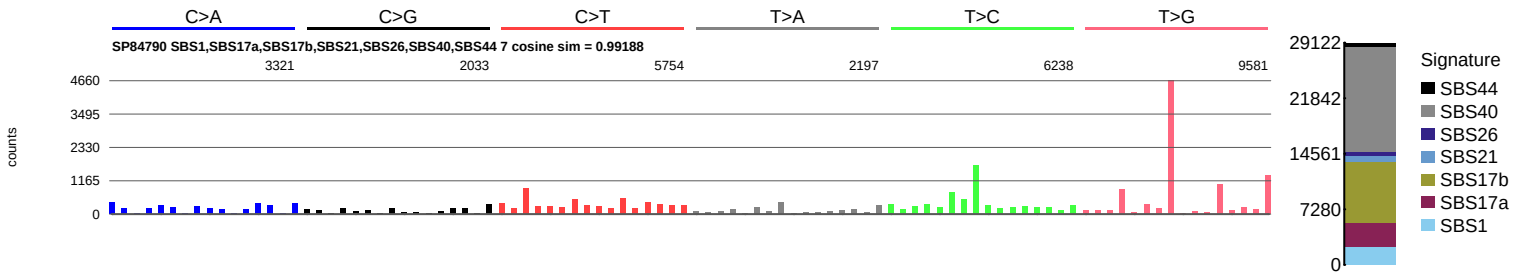

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS44 10 cosine sim = 0.99189
3159
2234
5694
2319
6168
9548
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

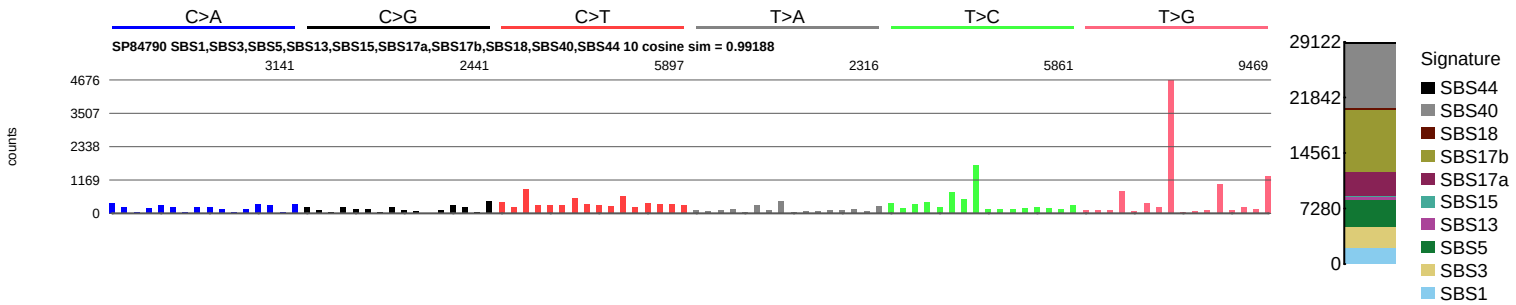

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 11 cosine sim = 0.99189
3138
2446
5915
2318
5854
9451
counts
4676
3507
2338
1169
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2

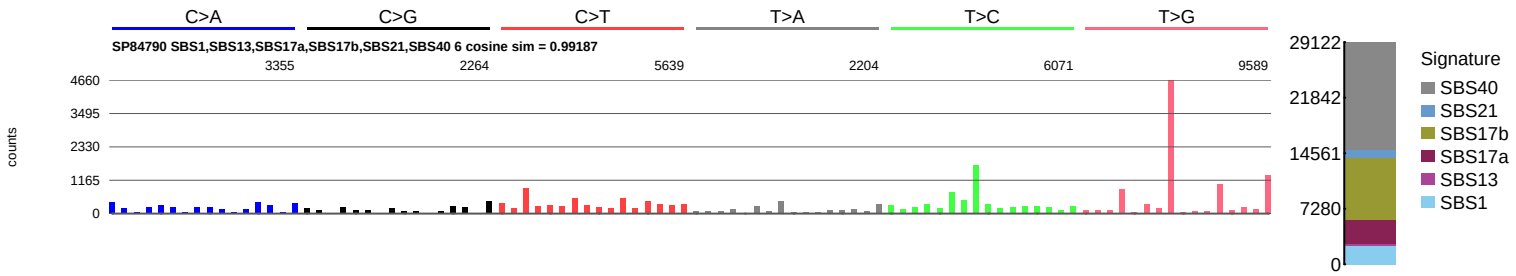

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 10 cosine sim = 0.99189
3169
2243
5676
2317
6171
9546
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

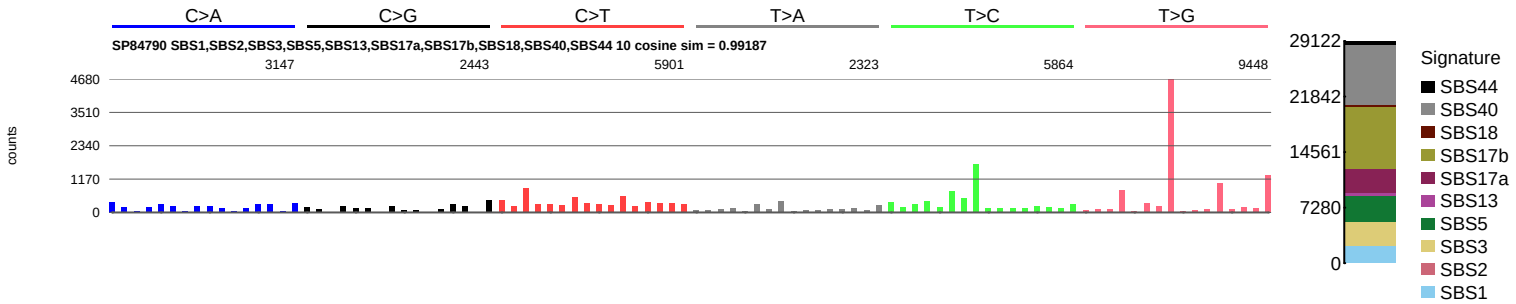

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.99188
3172
2219
5818
2142
6236
9537
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1

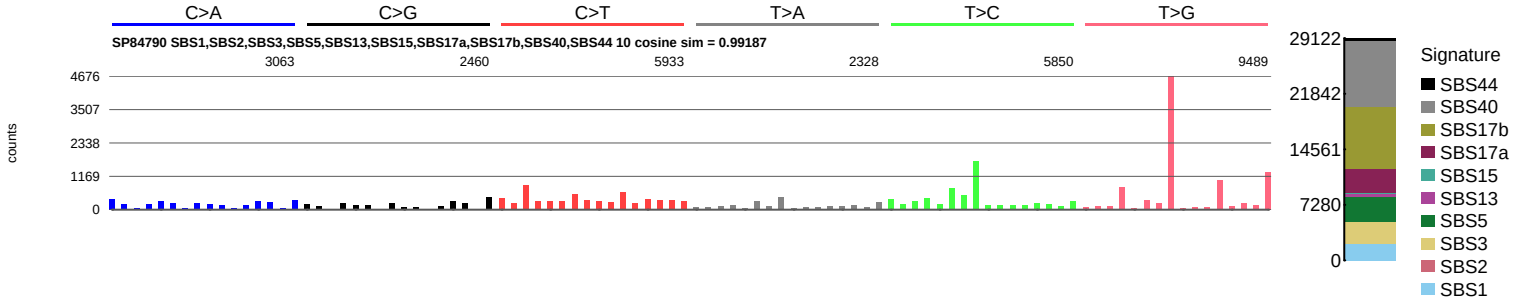

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.99188
3126
2620
4820
3237
5740
9577
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

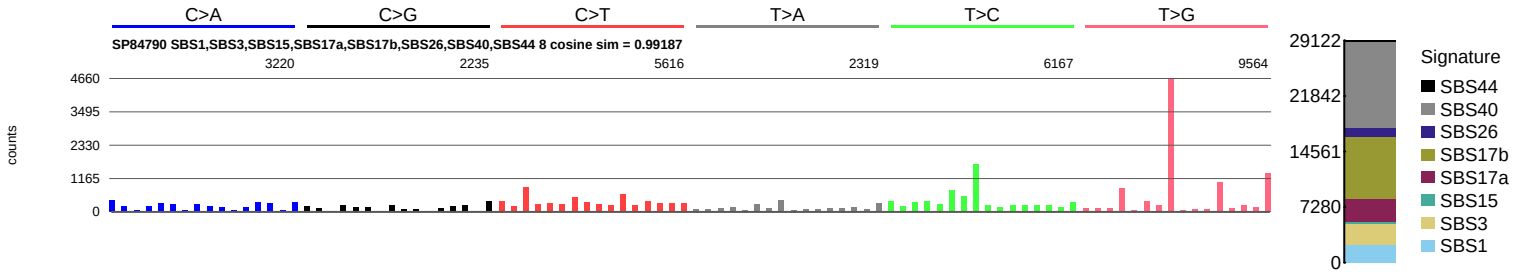

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS44 7 cosine sim = 0.99188
3321
2033
5754
2197
6238
9581
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS1

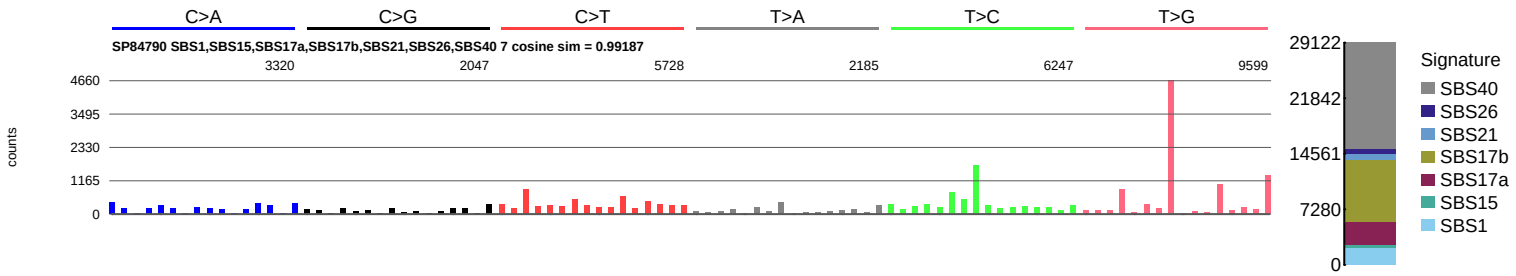

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 10 cosine sim = 0.99188
3141
2441
5897
2316
5861
9469
counts
4676
3507
2338
1169
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

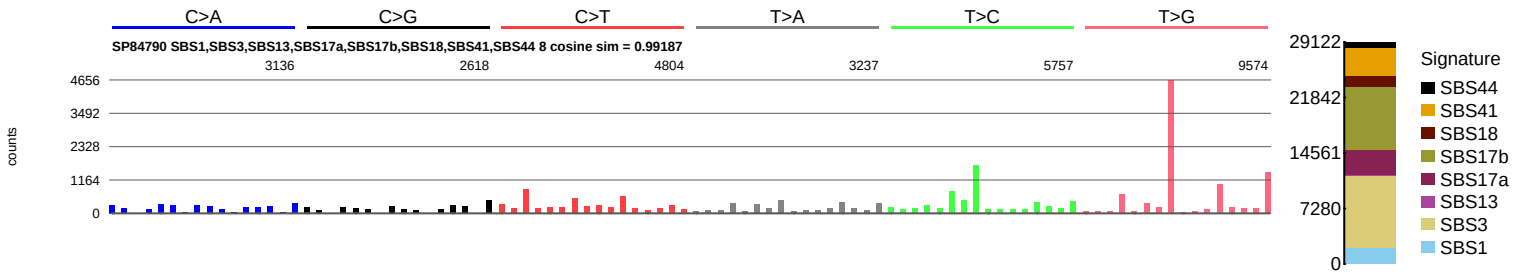

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS17a,SBS17b,SBS21,SBS40 6 cosine sim = 0.99187
3355
2264
5639
2204
6071
9589
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS1

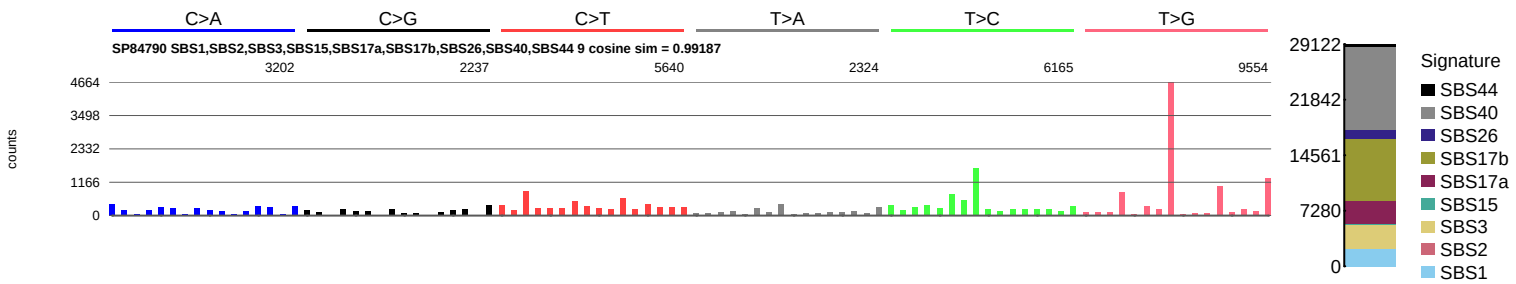

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 10 cosine sim = 0.99187
3147
2443
5901
2323
5864
9448
counts
4680
3510
2340
1170
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

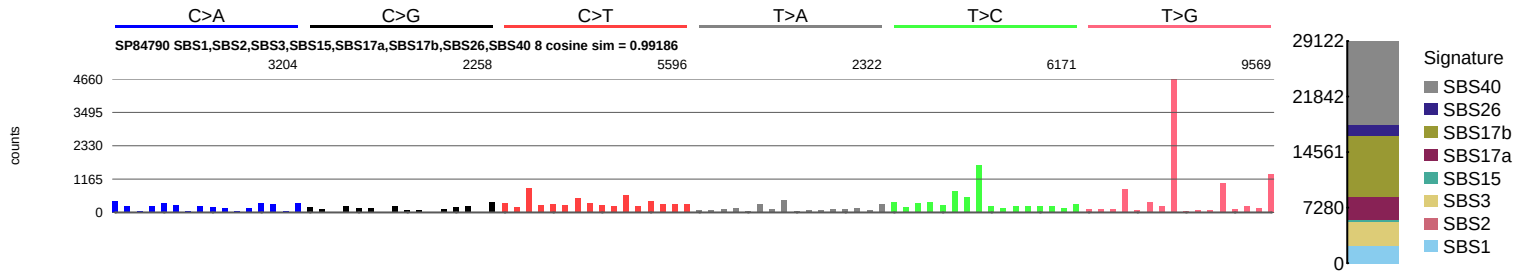

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS44 10 cosine sim = 0.99187
3063
2460
5933
2328
5850
9489
counts
4676
3507
2338
1169
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

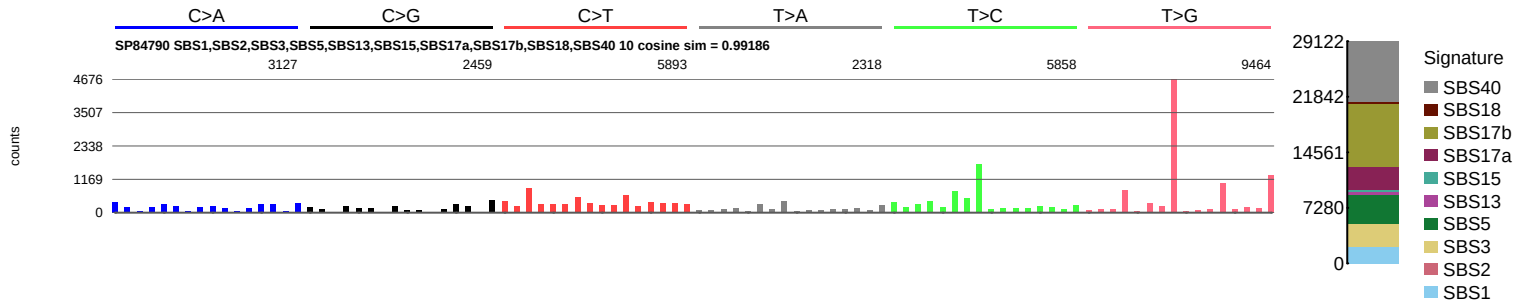

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS44 8 cosine sim = 0.99187
3220
2235
5616
2319
6167
9564
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS1

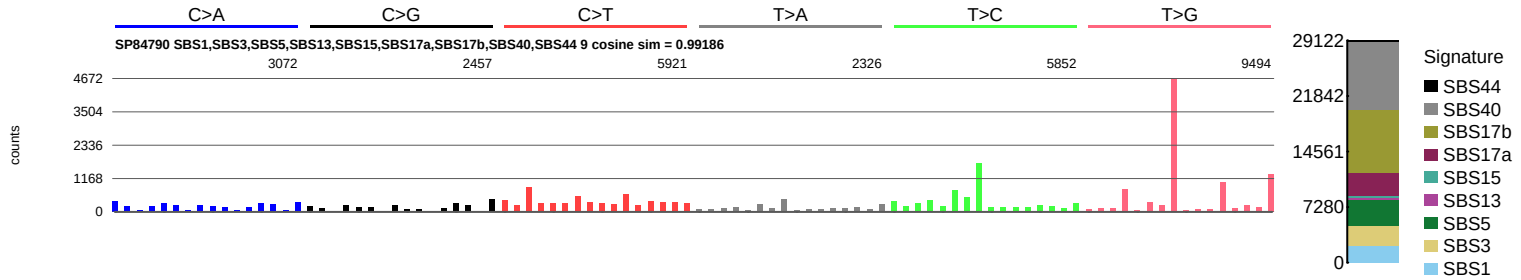

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40 7 cosine sim = 0.99187
3320
2047
5728
2185
6247
9599
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS1

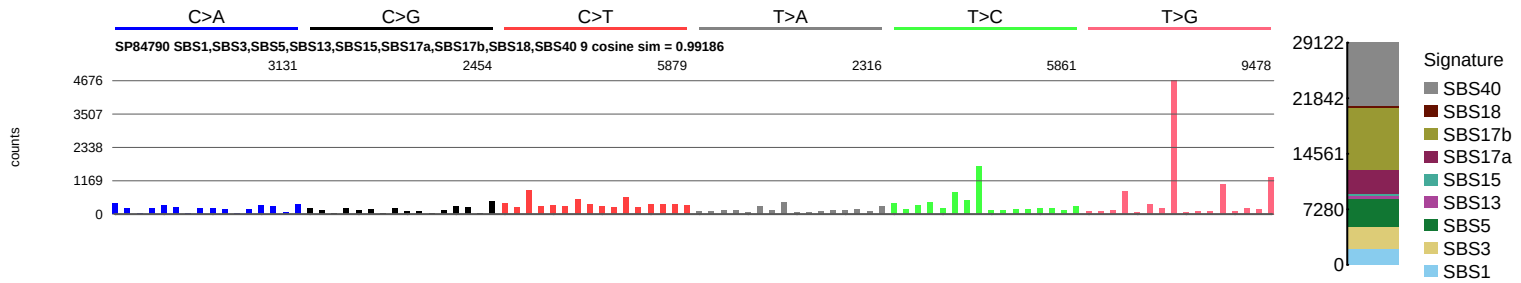

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.99187
3136
2618
4804
3237
5757
9574
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1

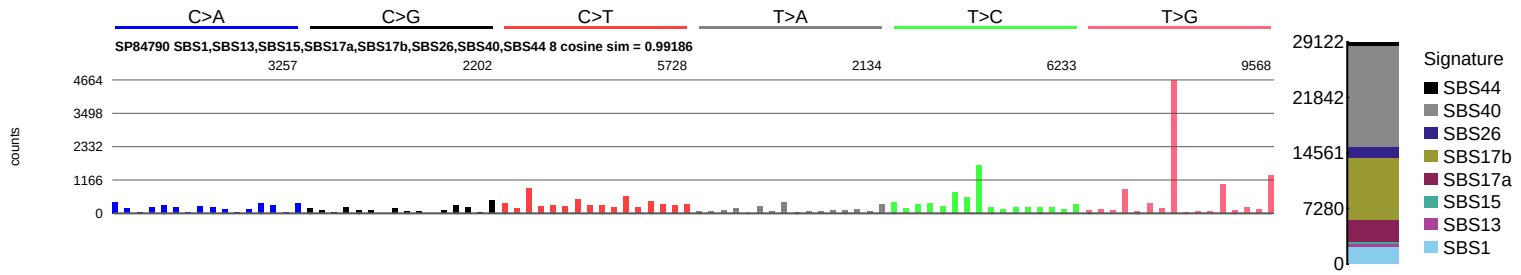

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS44 9 cosine sim = 0.99187
3202
2237
5640
2324
6165
9554
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

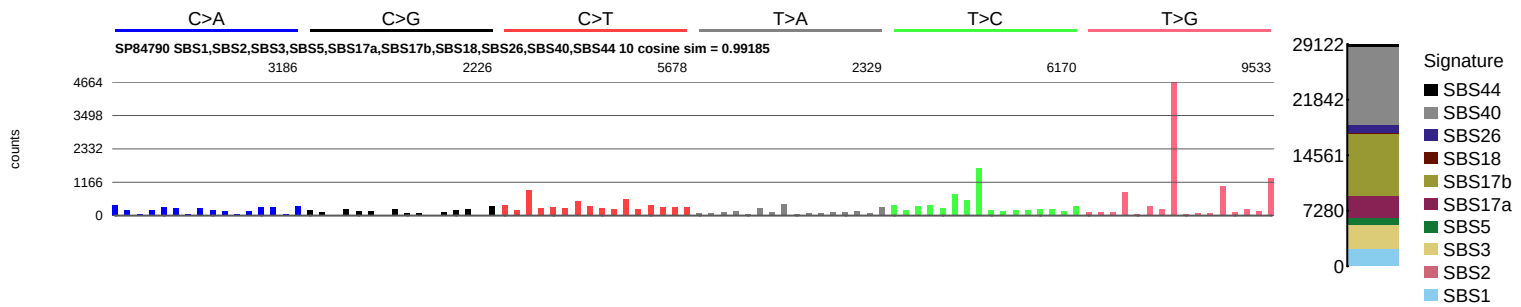

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS40 8 cosine sim = 0.99186
3204
2258
5596
2322
6171
9569
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

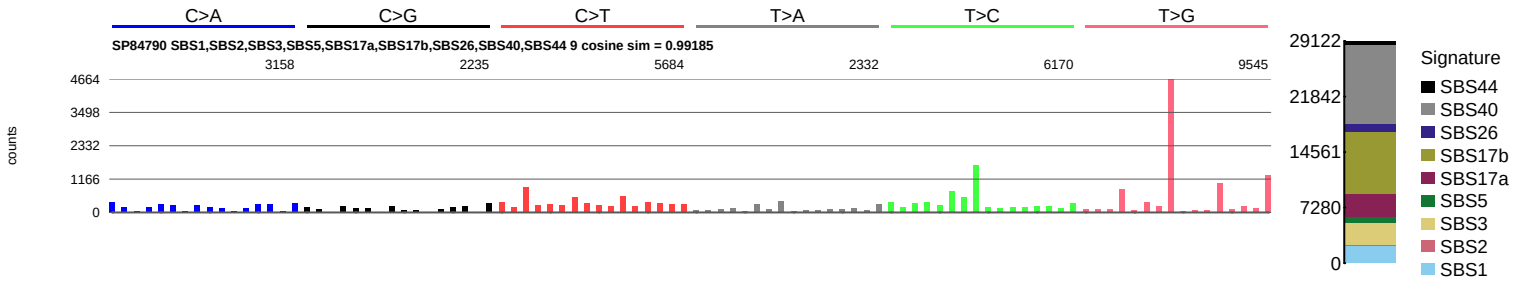

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 10 cosine sim = 0.99186
3127
2459
5893
2318
5858
9464
counts
4676
3507
2338
1169
0
29122
21842
14561
7280
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

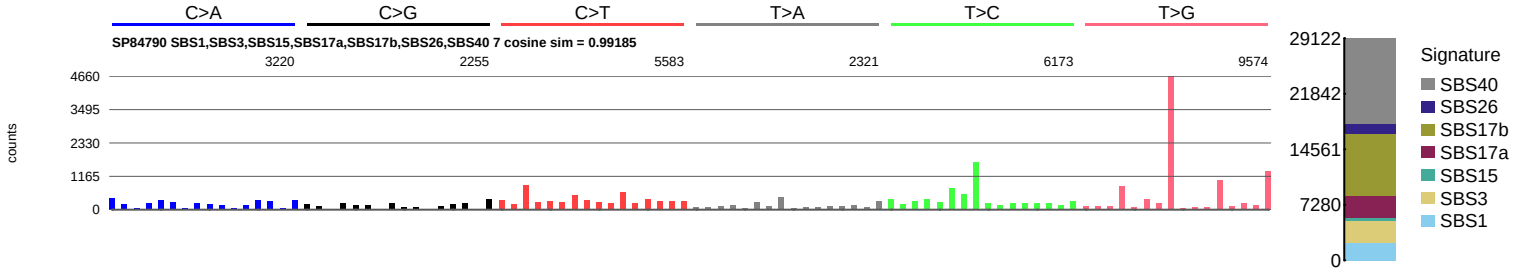

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS44 9 cosine sim = 0.99186
3072
2457
5921
2326
5852
9494
counts
4672
3504
2336
1168
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

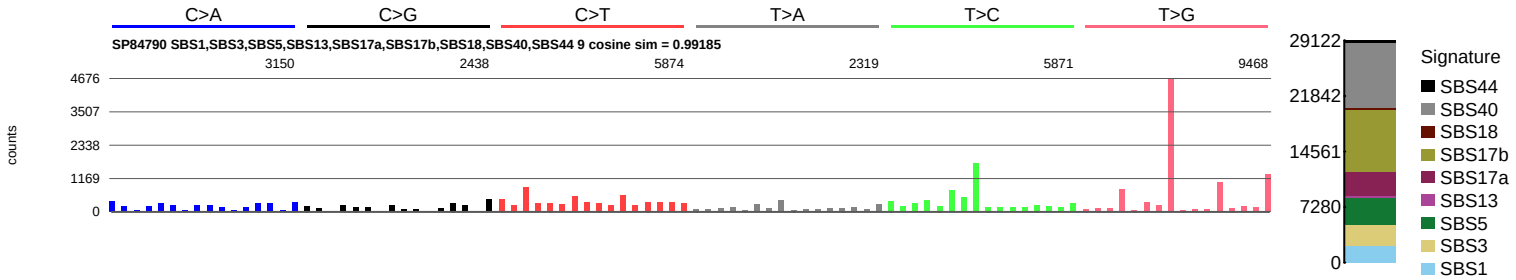

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 9 cosine sim = 0.99186
3131
2454
5879
2316
5861
9478
counts
4676
3507
2338
1169
0
29122
21842
14561
7280
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

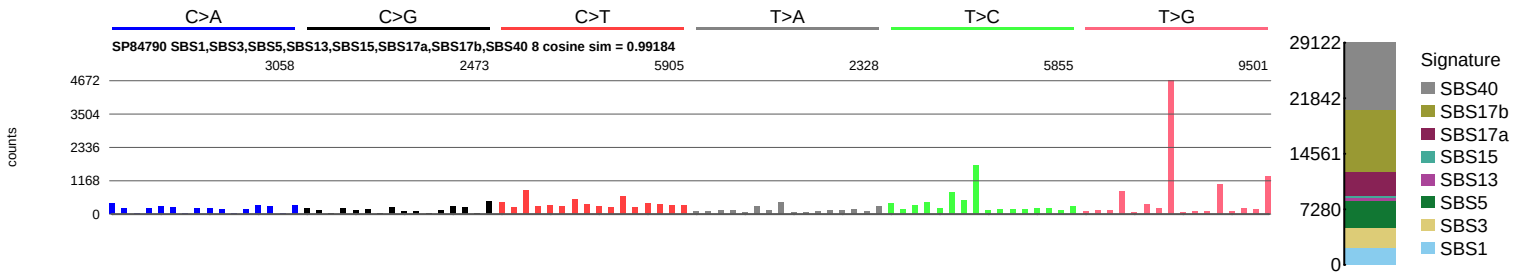

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS44 8 cosine sim = 0.99186
3257
2202
5728
2134
6233
9568
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS1

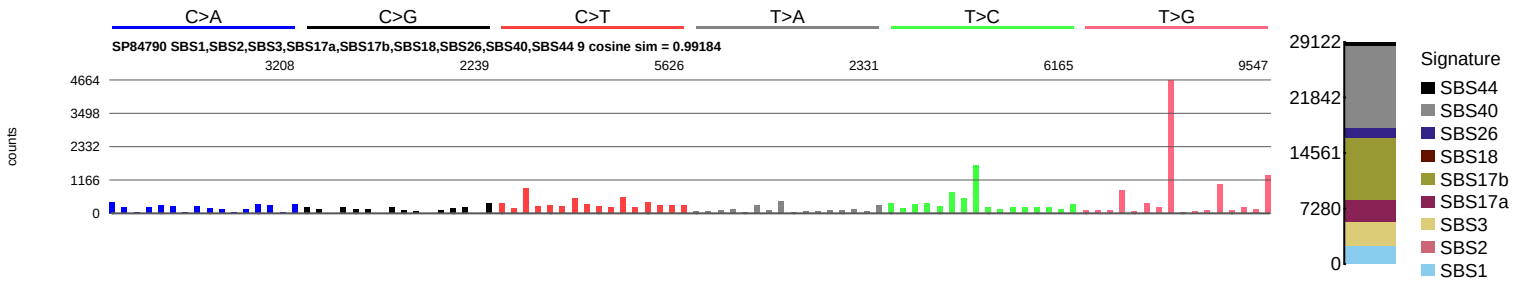

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.99185
3186
2226
5678
2329
6170
9533
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

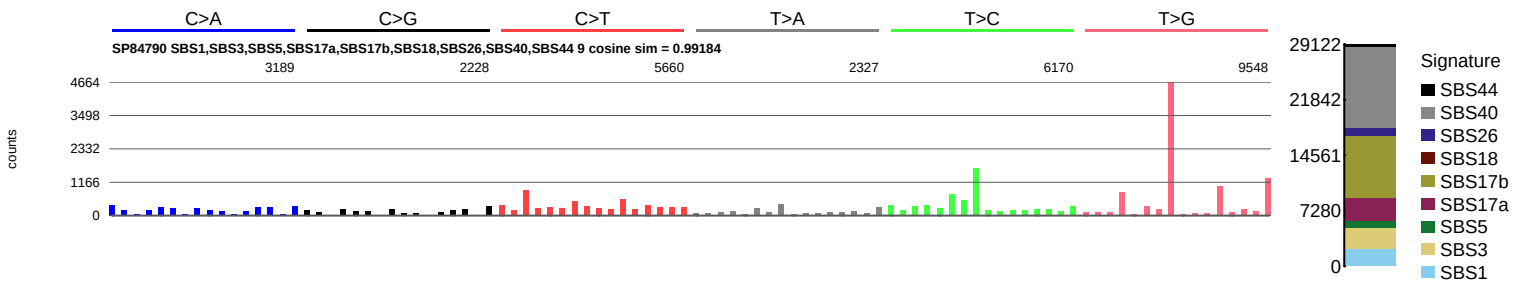

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS26,SBS40,SBS44 9 cosine sim = 0.99185
3158
2235
5684
2332
6170
9545
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

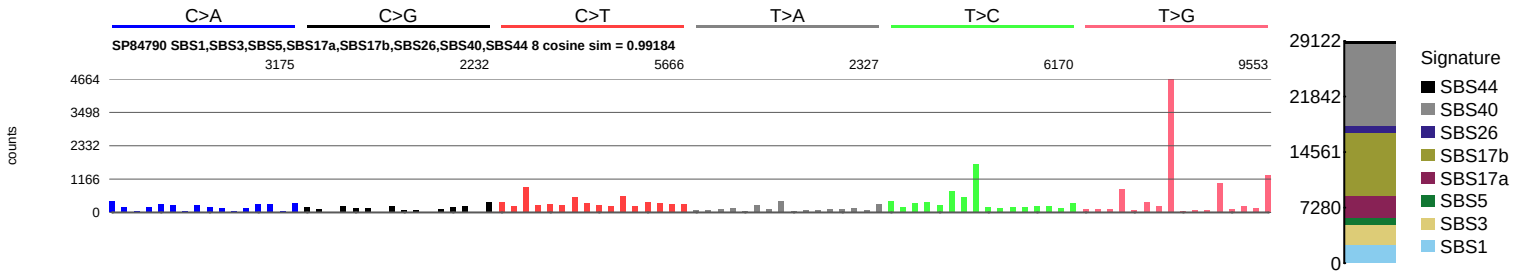

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS40 7 cosine sim = 0.99185
3220
2255
5583
2321
6173
9574
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS1

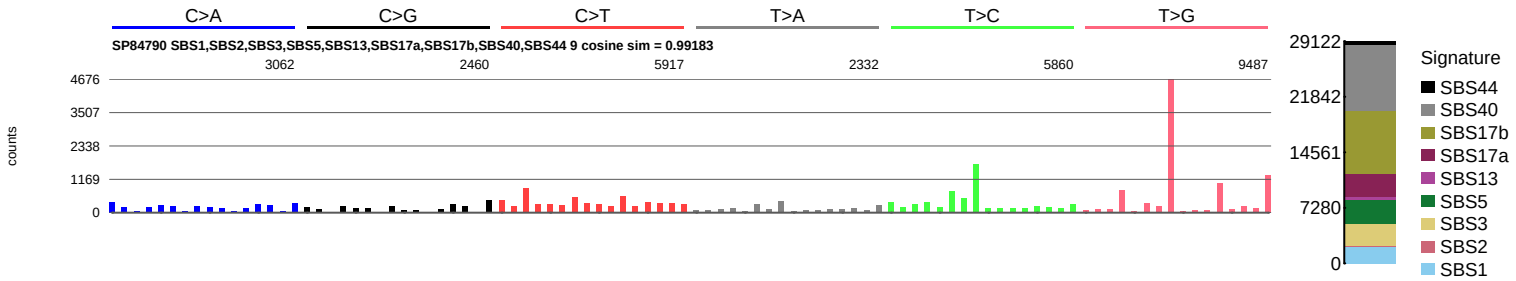

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.99185
3150
2438
5874
2319
5871
9468
counts
4676
3507
2338
1169
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

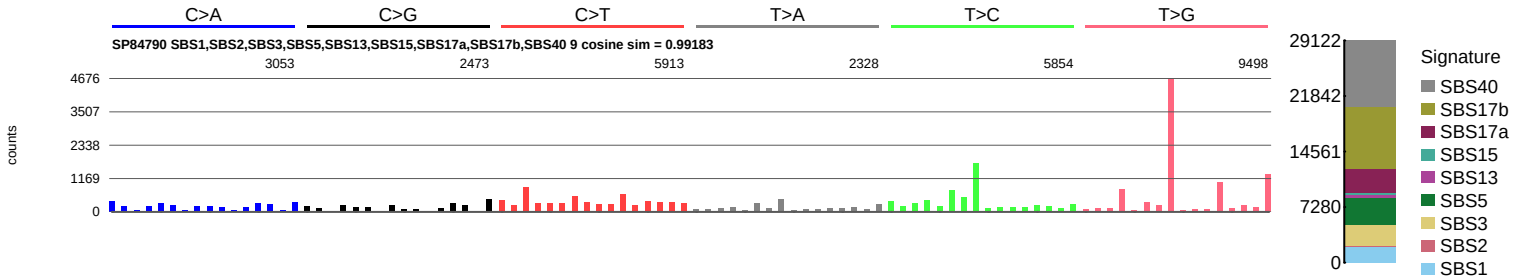

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40 8 cosine sim = 0.99184
3058
2473
5905
2328
5855
9501
counts
4672
3504
2336
1168
0
29122
21842
14561
7280
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

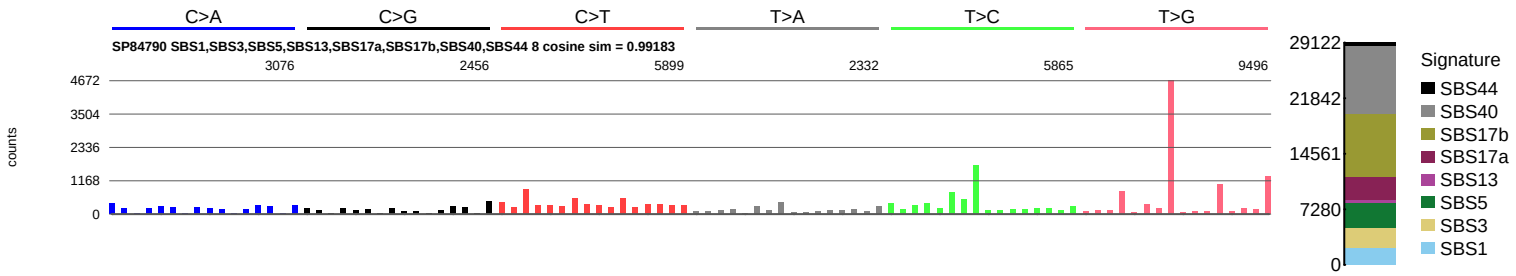

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.99184
3208
2239
5626
2331
6165
9547
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

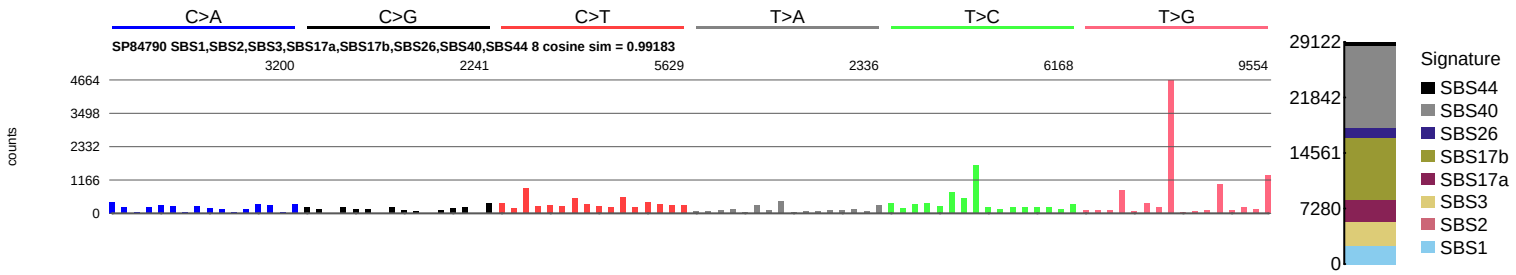

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.99184
3189
2228
5660
2327
6170
9548
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1

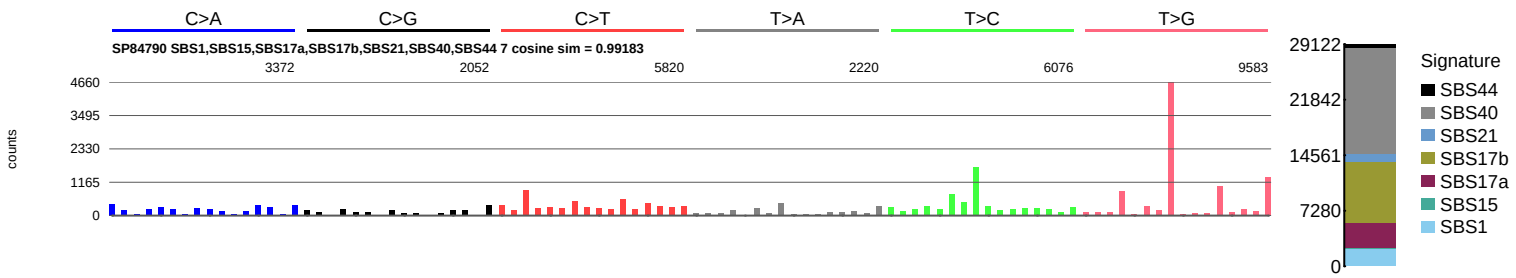

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS26,SBS40,SBS44 8 cosine sim = 0.99184
3175
2232
5666
2327
6170
9553
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS5
SBS3
SBS1

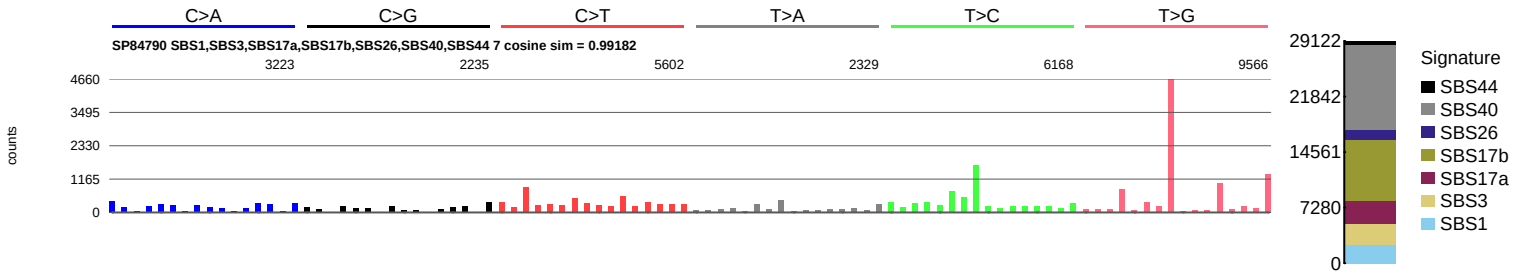

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS40,SBS44 9 cosine sim = 0.99183
3062
2460
5917
2332
5860
9487
counts
4676
3507
2338
1169
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

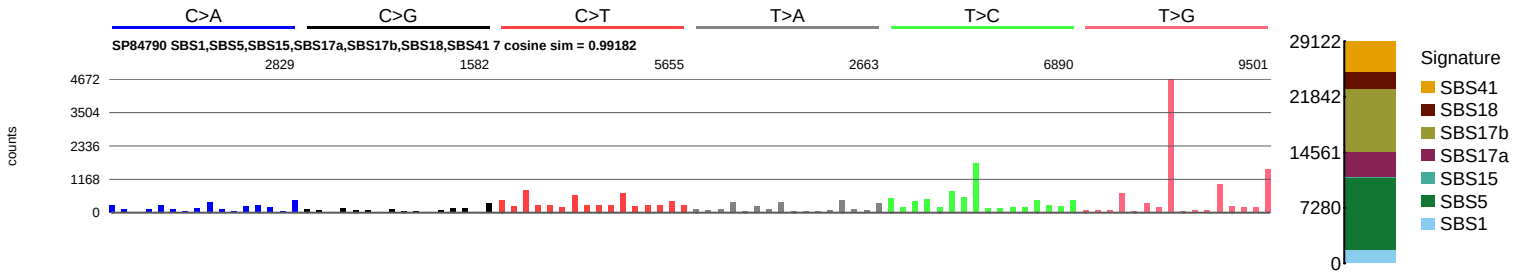

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40 9 cosine sim = 0.99183
3053
2473
5913
2328
5854
9498
counts
4676
3507
2338
1169
0
29122
21842
14561
7280
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

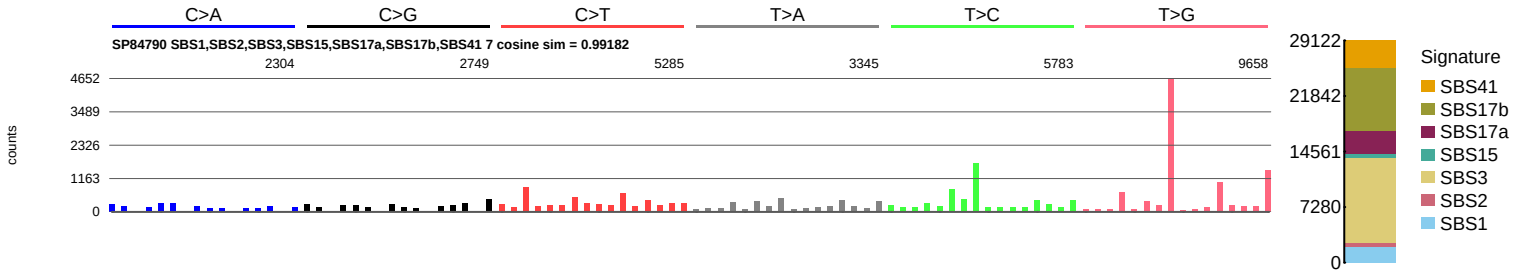

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS40,SBS44 8 cosine sim = 0.99183
3076
2456
5899
2332
5865
9496
counts
4672
3504
2336
1168
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

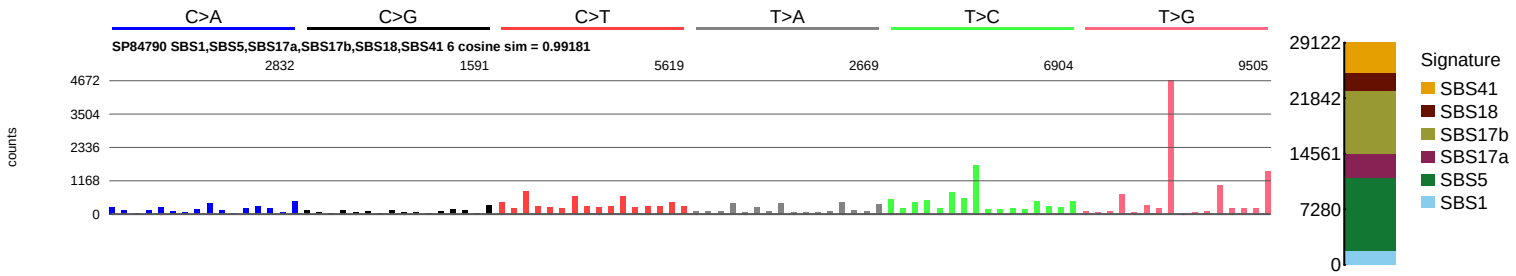

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS26,SBS40,SBS44 8 cosine sim = 0.99183
3200
2241
5629
2336
6168
9554
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS3
SBS2
SBS1

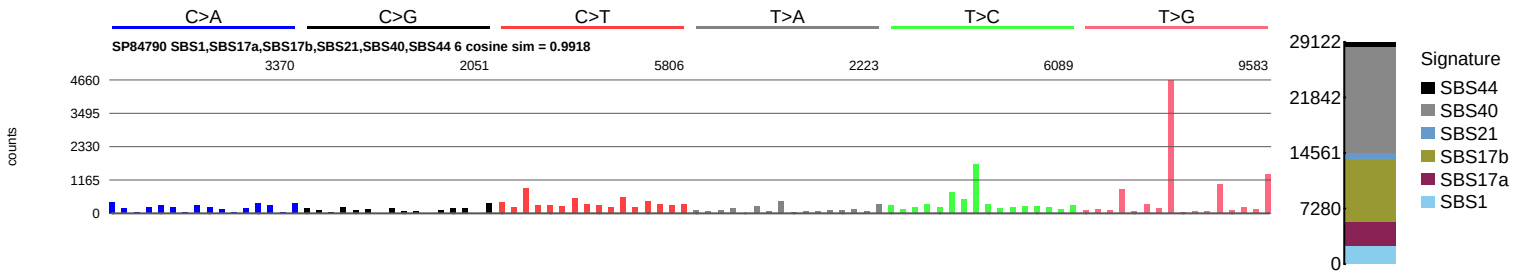

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS44 7 cosine sim = 0.99183
3372
2052
5820
2220
6076
9583
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS1

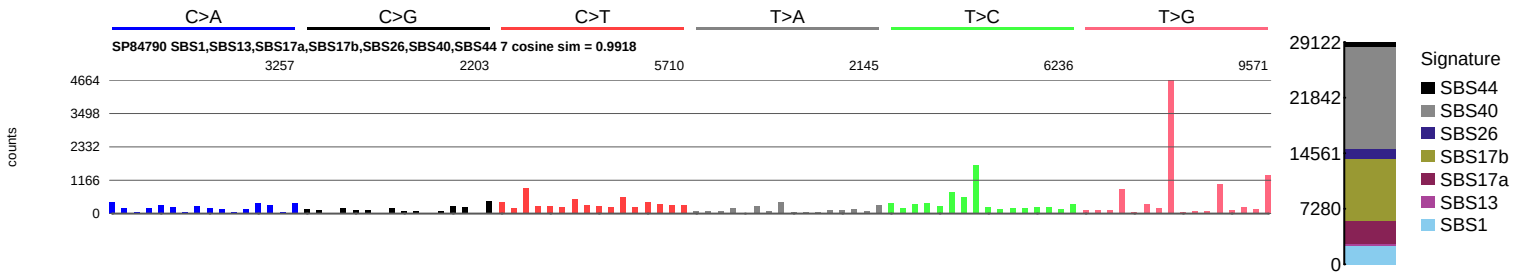

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS26,SBS40,SBS44 7 cosine sim = 0.99182
3223
2235
5602
2329
6168
9566
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS3
SBS1

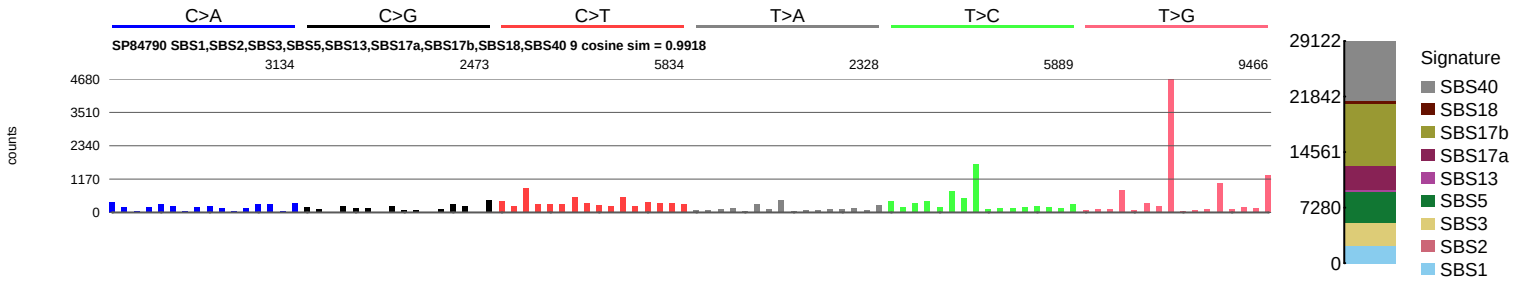

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.99182
2829
1582
5655
2663
6890
9501
counts
4672
3504
2336
1168
0
29122
21842
14561
7280
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1

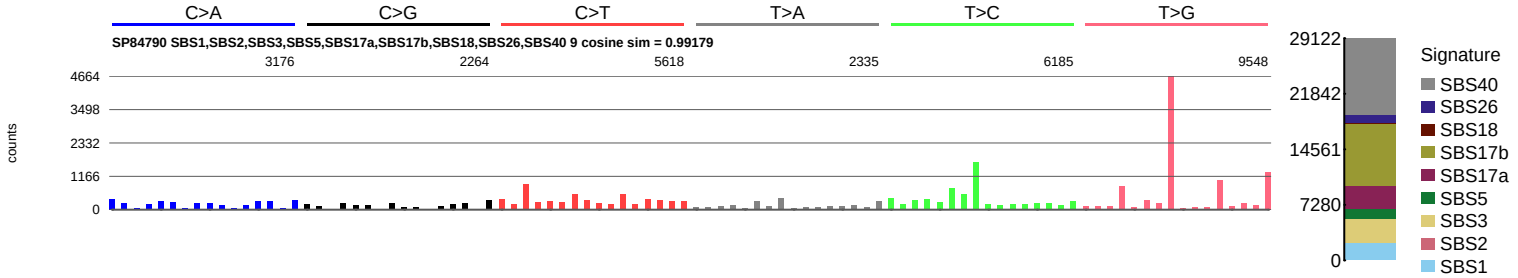

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.99182
2304
2749
5285
3345
5783
9658
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

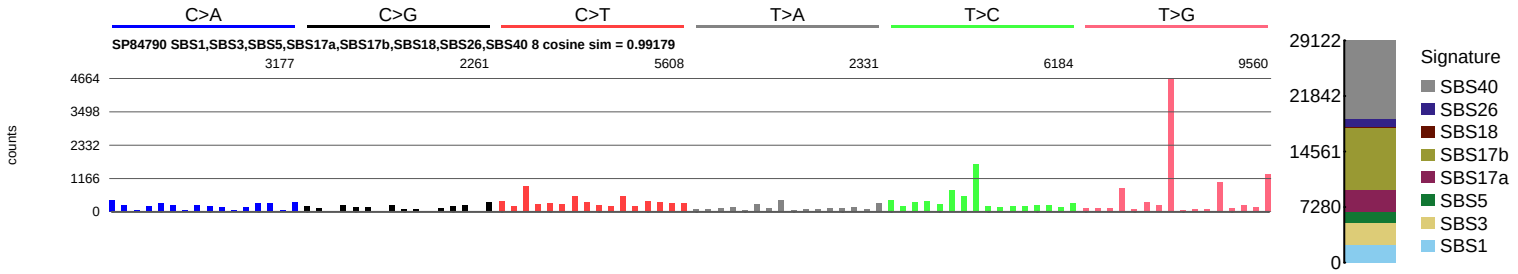

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.99181
2832
1591
5619
2669
6904
9505
counts
4672
3504
2336
1168
0
29122
21842
14561
7280
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS1

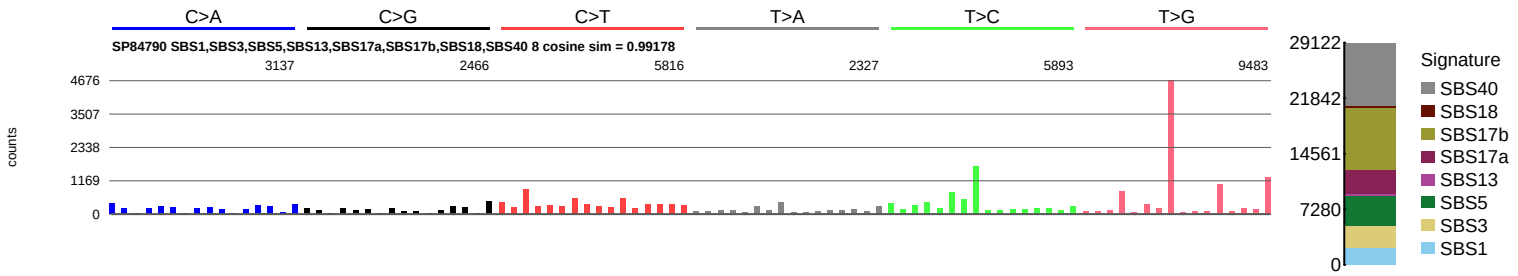

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS17a,SBS17b,SBS21,SBS40,SBS44 6 cosine sim = 0.9918
3370
2051
5806
2223
6089
9583
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS1

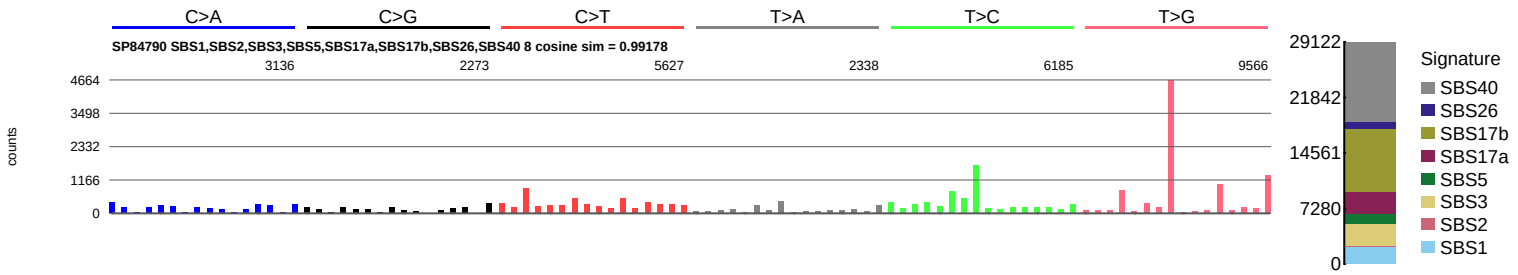

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS44 7 cosine sim = 0.9918
3257
2203
5710
2145
6236
9571
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS1

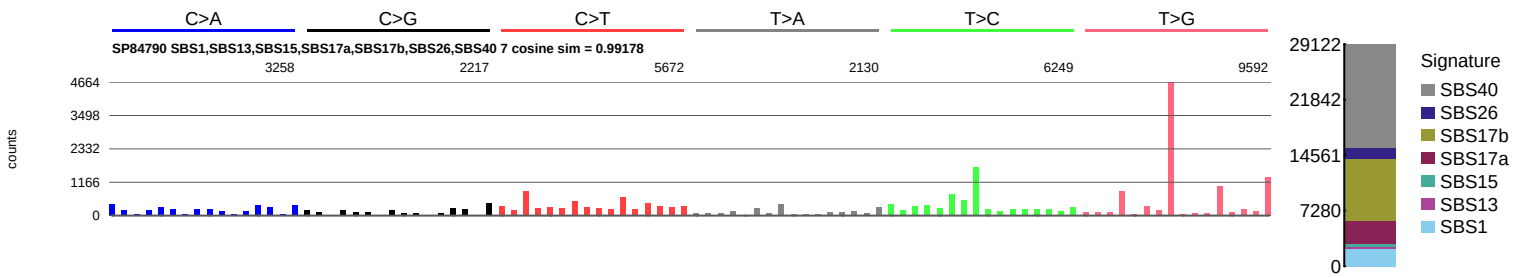

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40 9 cosine sim = 0.9918
3134
2473
5834
2328
5889
9466
counts
4680
3510
2340
1170
0
29122
21842
14561
7280
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

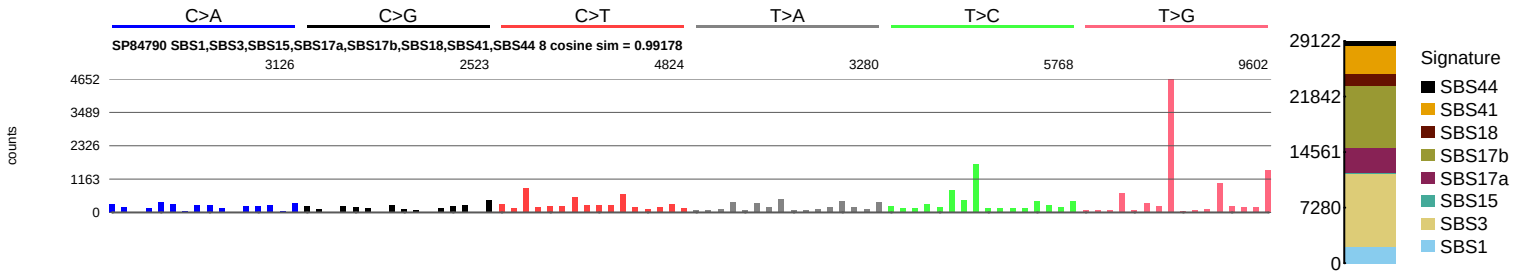

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.99179
3176
2264
5618
2335
6185
9548
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

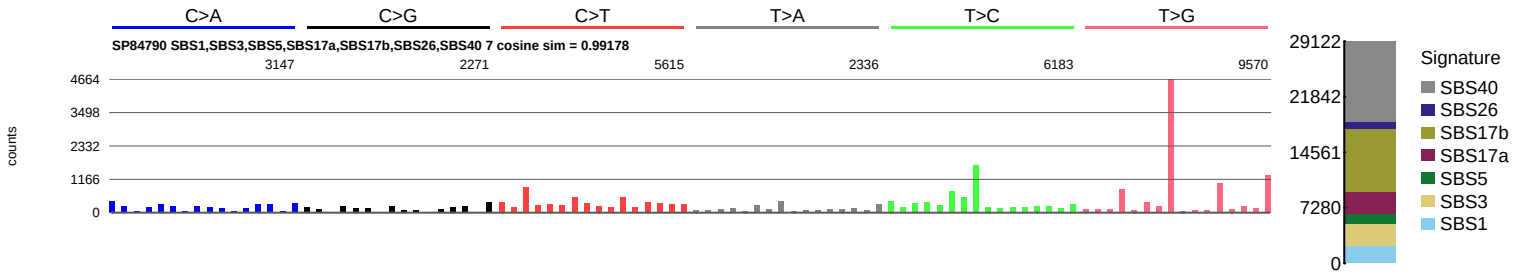

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.99179
3177
2261
5608
2331
6184
9560
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1

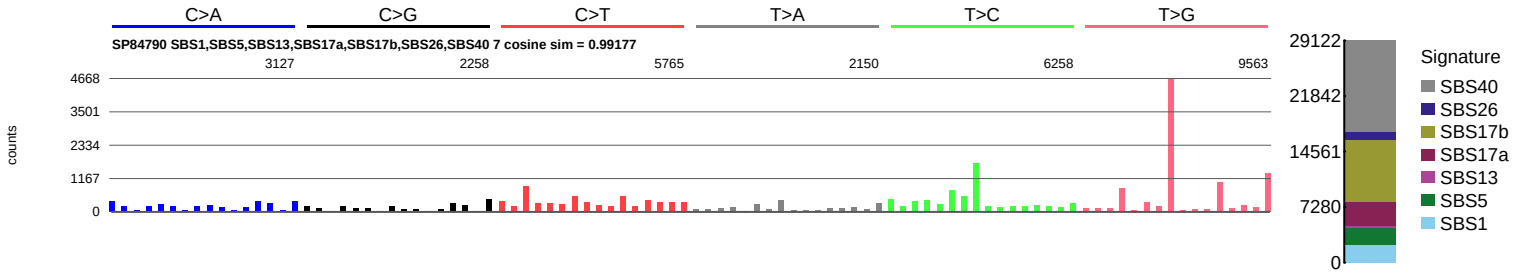

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.99178
3137
2466
5816
2327
5893
9483
counts
4676
3507
2338
1169
0
29122
21842
14561
7280
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

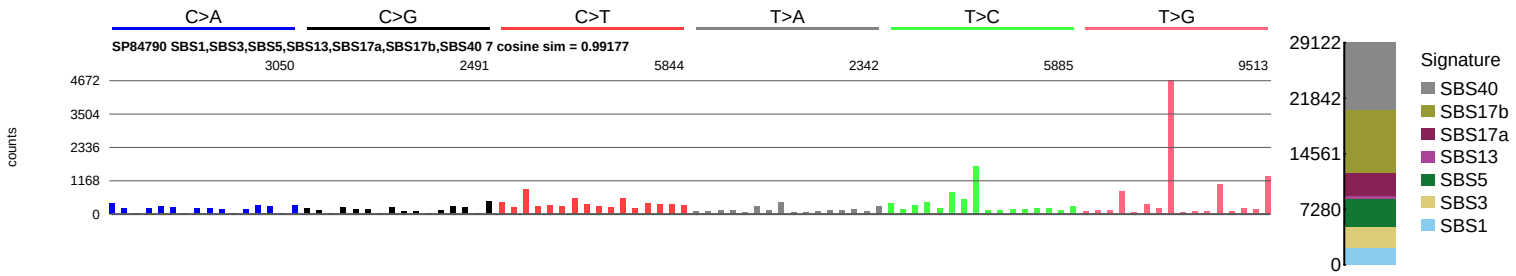

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS26,SBS40 8 cosine sim = 0.99178
3136
2273
5627
2338
6185
9566
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

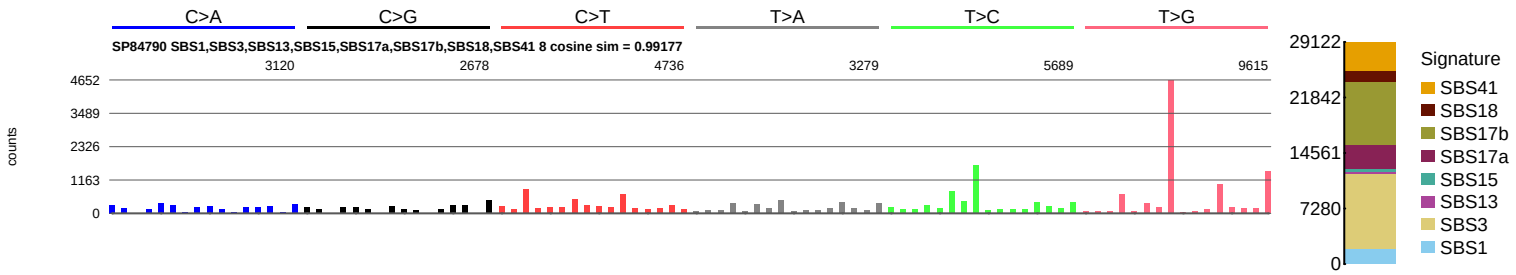

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40 7 cosine sim = 0.99178
3258
2217
5672
2130
6249
9592
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS1

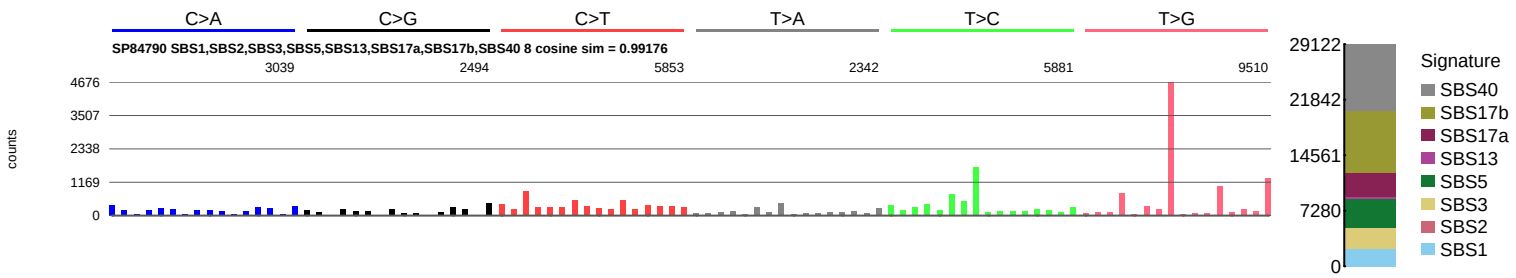

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.99178
3126
2523
4824
3280
5768
9602
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

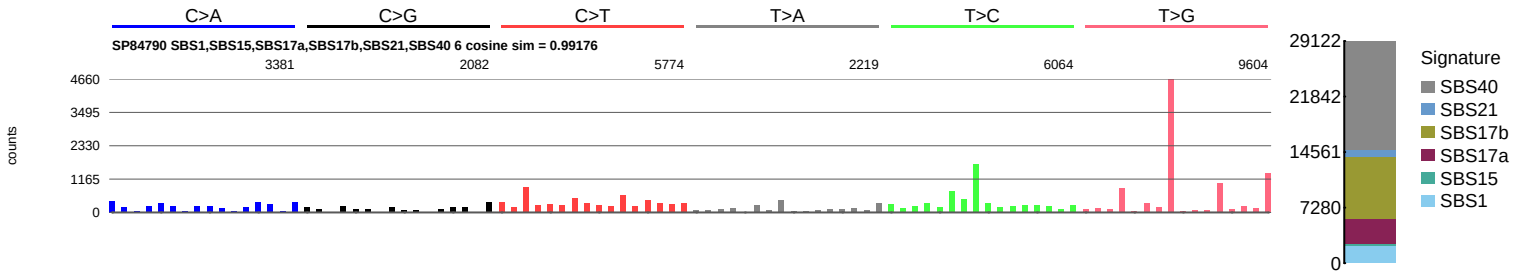

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS26,SBS40 7 cosine sim = 0.99178
3147
2271
5615
2336
6183
9570
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS5
SBS3
SBS1

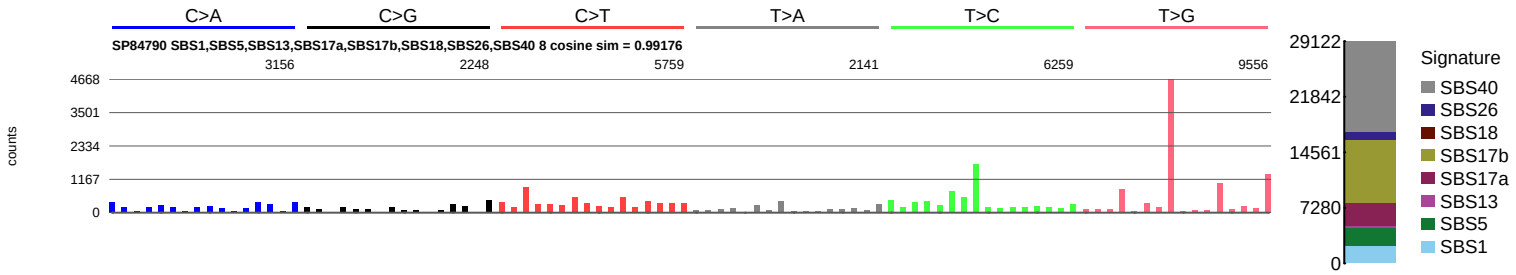

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS26,SBS40 7 cosine sim = 0.99177
3127
2258
5765
2150
6258
9563
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS1

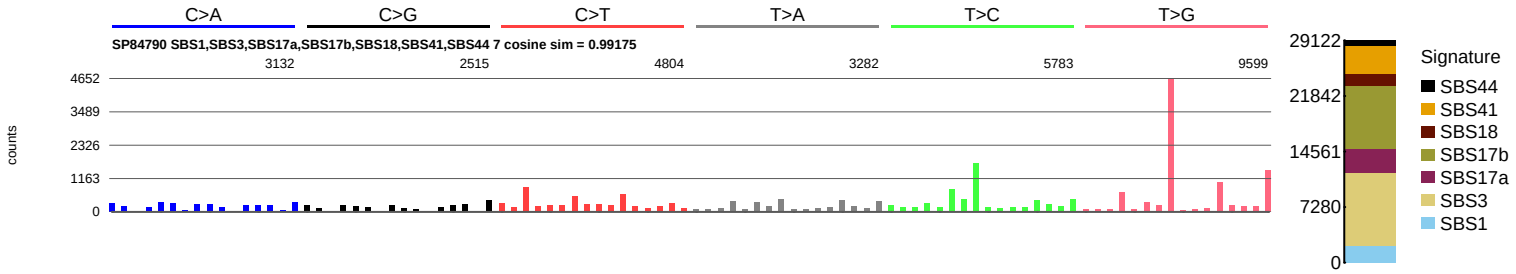

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS40 7 cosine sim = 0.99177
3050
2491
5844
2342
5885
9513
counts
4672
3504
2336
1168
0
29122
21842
14561
7280
0
Signature
SBS40
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

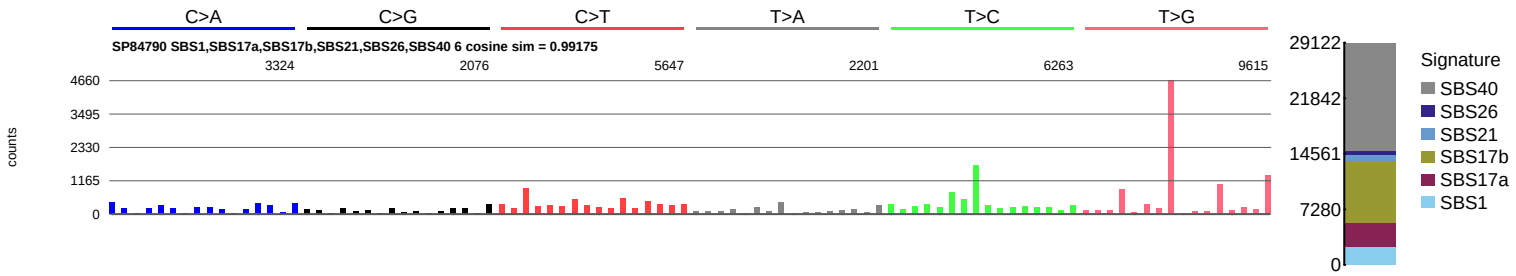

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.99177
3120
2678
4736
3279
5689
9615
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

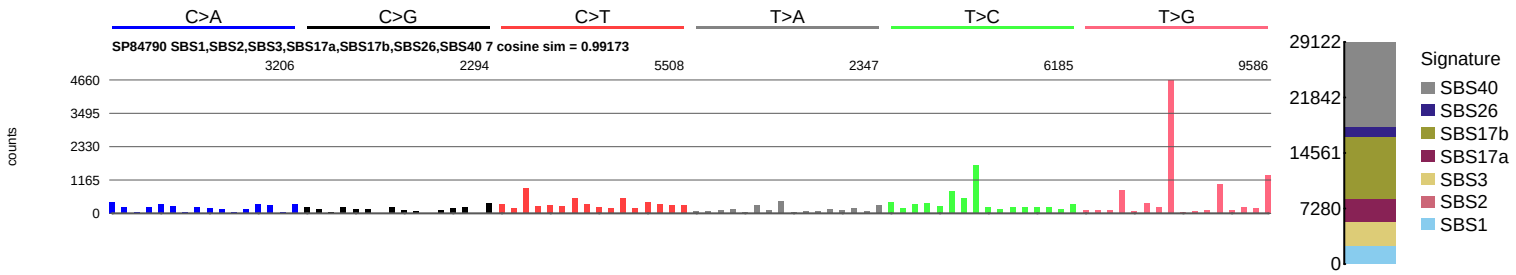

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS40 8 cosine sim = 0.99176
3039
2494
5853
2342
5881
9510
counts
4676
3507
2338
1169
0
29122
21842
14561
7280
0
Signature
SBS40
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

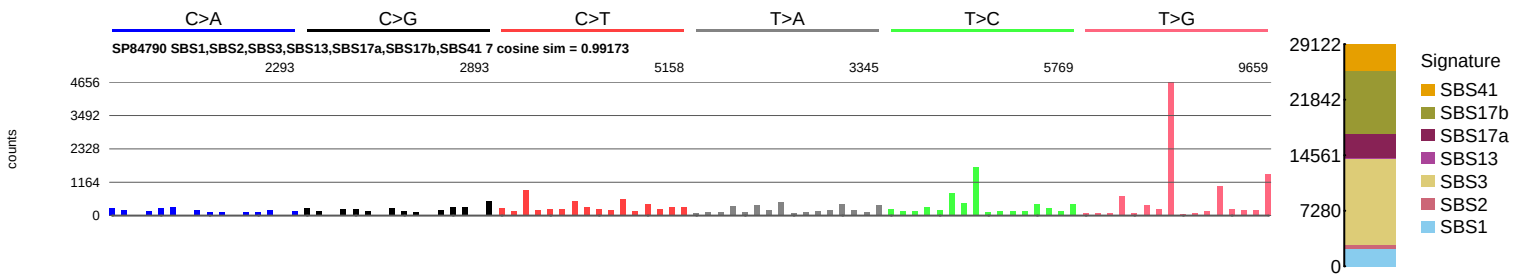

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS40 6 cosine sim = 0.99176
3381
2082
5774
2219
6064
9604
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS1

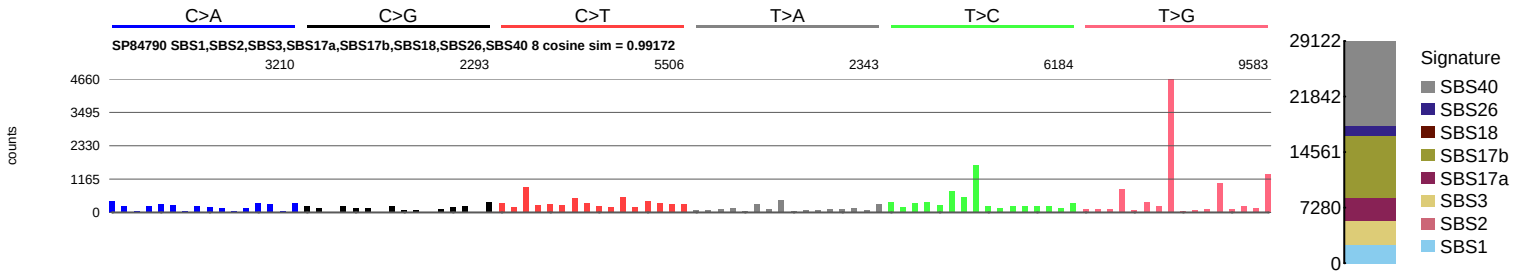

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.99176
3156
2248
5759
2141
6259
9556
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1

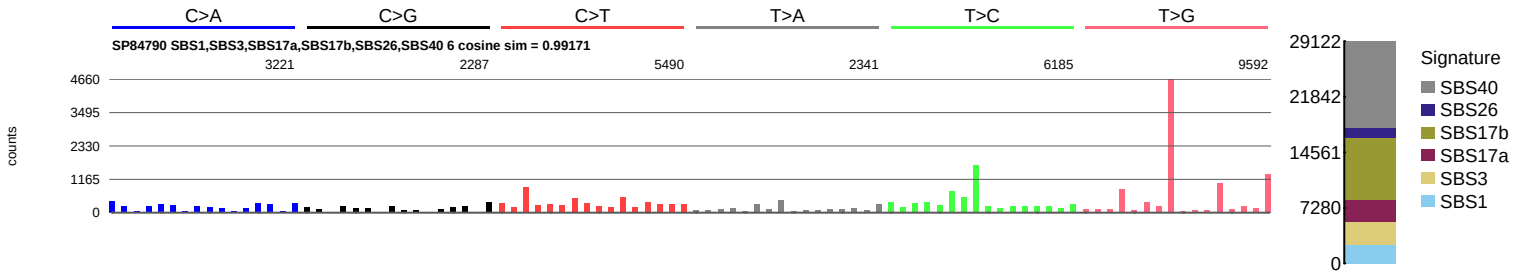

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.99175
3132
2515
4804
3282
5783
9599
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS1

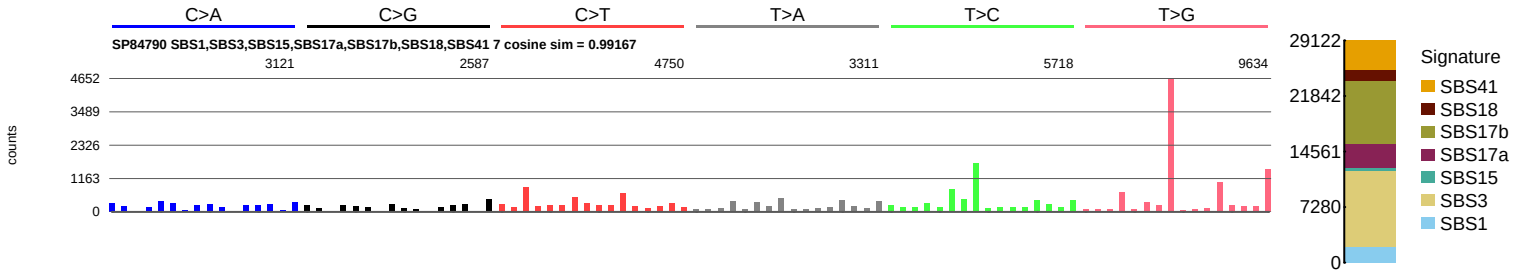

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS17a,SBS17b,SBS21,SBS26,SBS40 6 cosine sim = 0.99175
3324
2076
5647
2201
6263
9615
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS1

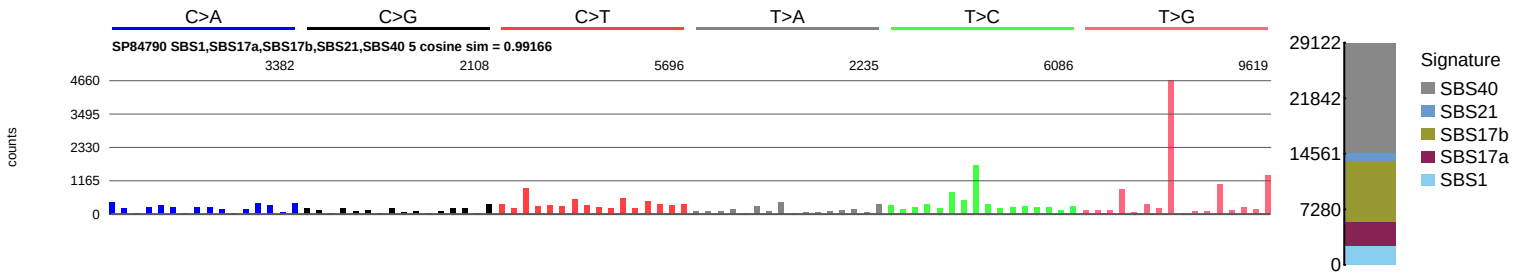

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS26,SBS40 7 cosine sim = 0.99173
3206
2294
5508
2347
6185
9586
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS3
SBS2
SBS1

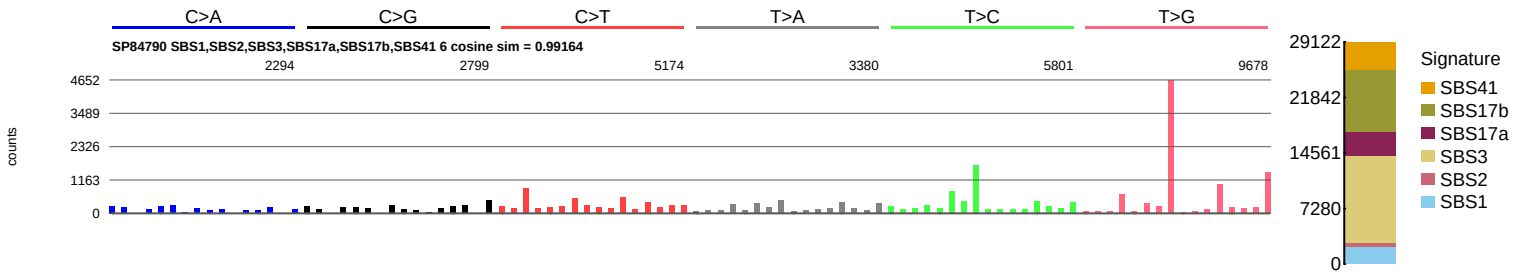

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS41 7 cosine sim = 0.99173
2293
2893
5158
3345
5769
9659
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS41
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

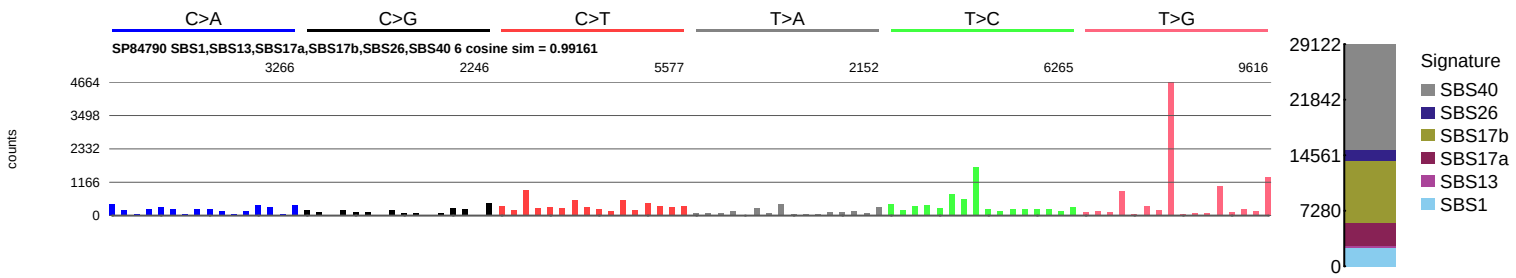

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.99172
3210
2293
5506
2343
6184
9583
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

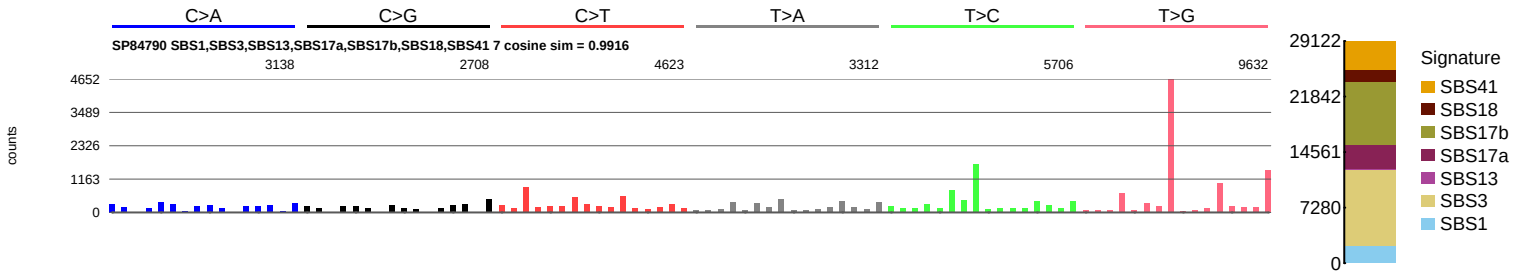

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS26,SBS40 6 cosine sim = 0.99171
3221
2287
5490
2341
6185
9592
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS3
SBS1

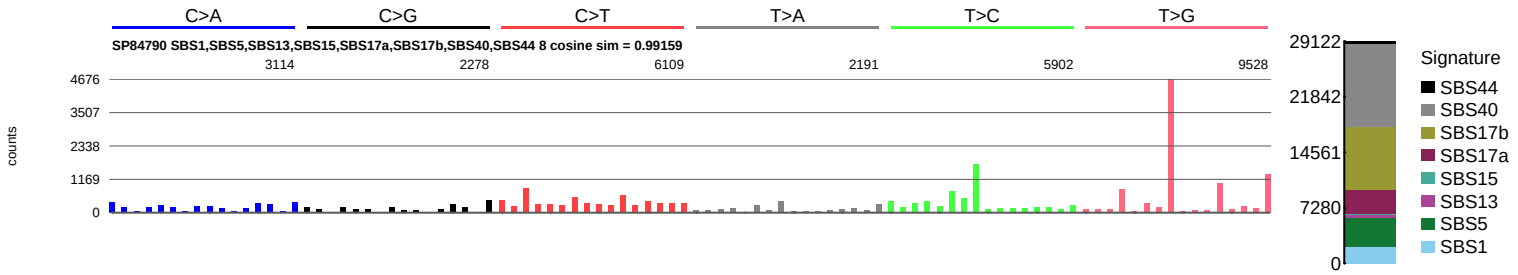

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.99167
3121
2587
4750
3311
5718
9634
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

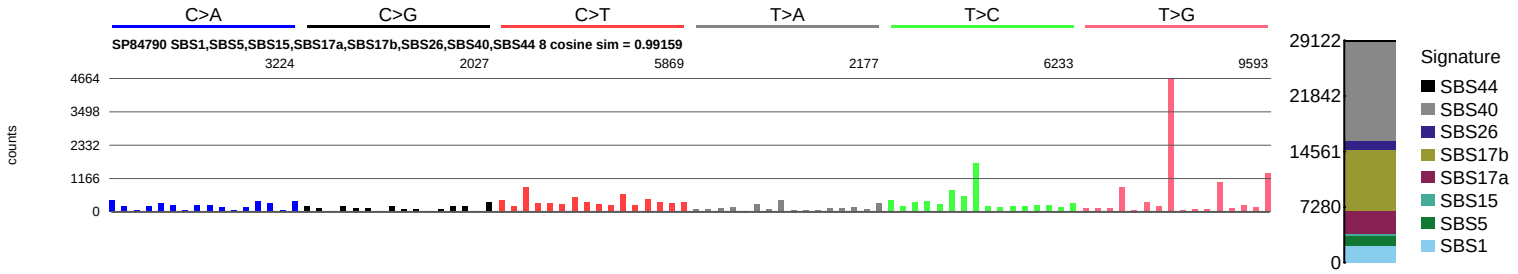

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS17a,SBS17b,SBS21,SBS40 5 cosine sim = 0.99166
3382
2108
5696
2235
6086
9619
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS1

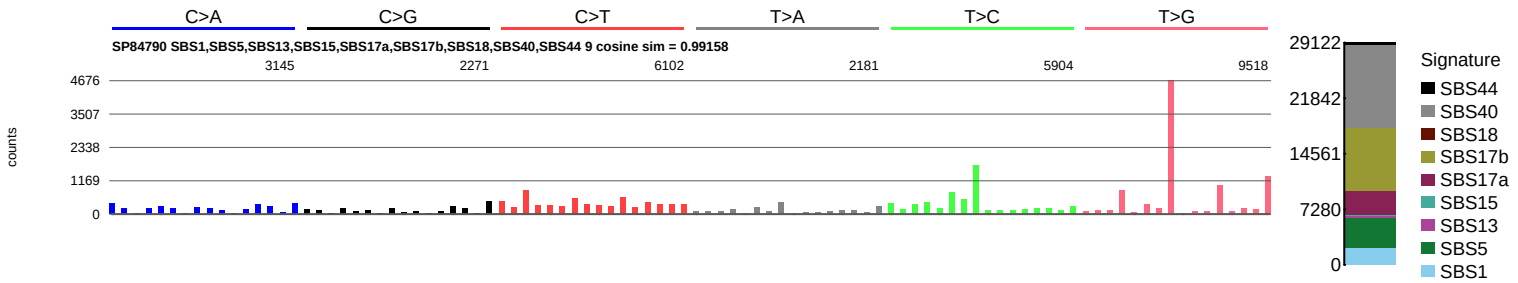

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS41 6 cosine sim = 0.99164
2294
2799
5174
3380
5801
9678
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS41
SBS17b
SBS17a
SBS3
SBS2
SBS1

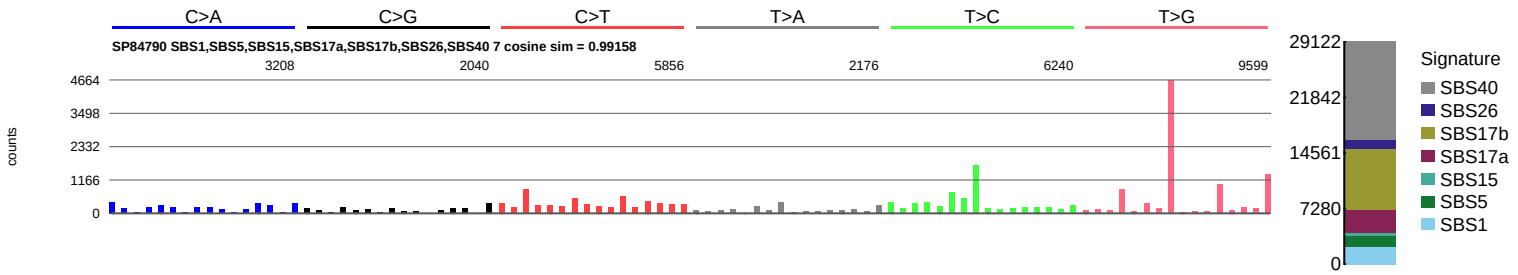

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS17a,SBS17b,SBS26,SBS40 6 cosine sim = 0.99161
3266
2246
5577
2152
6265
9616
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS1

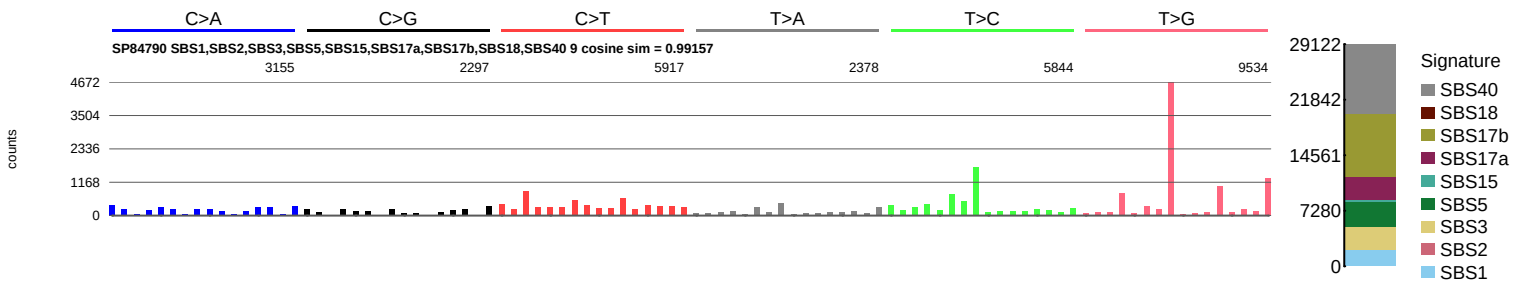

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.9916
3138
2708
4623
3312
5706
9632
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1

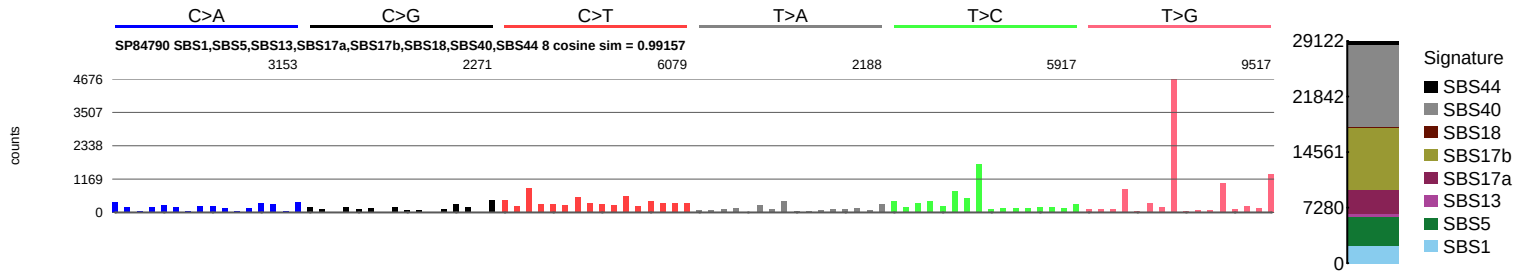

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS44 8 cosine sim = 0.99159
3114
2278
6109
2191
5902
9528
counts
4676
3507
2338
1169
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

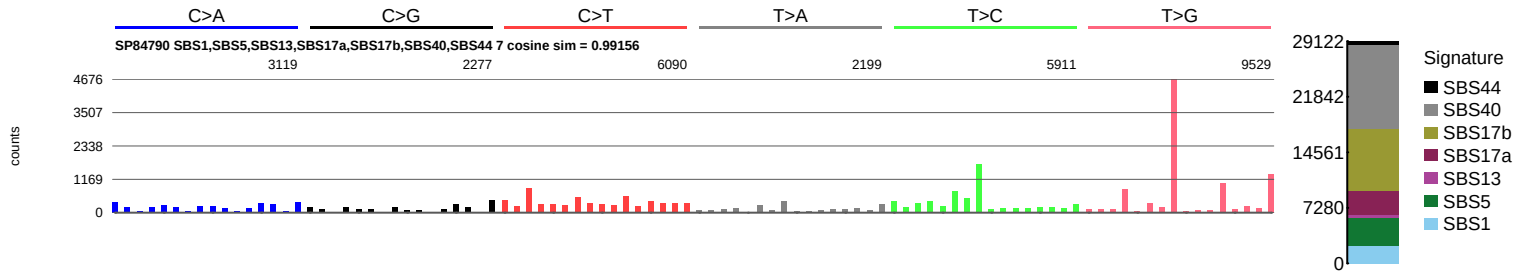

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS44 8 cosine sim = 0.99159
3224
2027
5869
2177
6233
9593
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS1

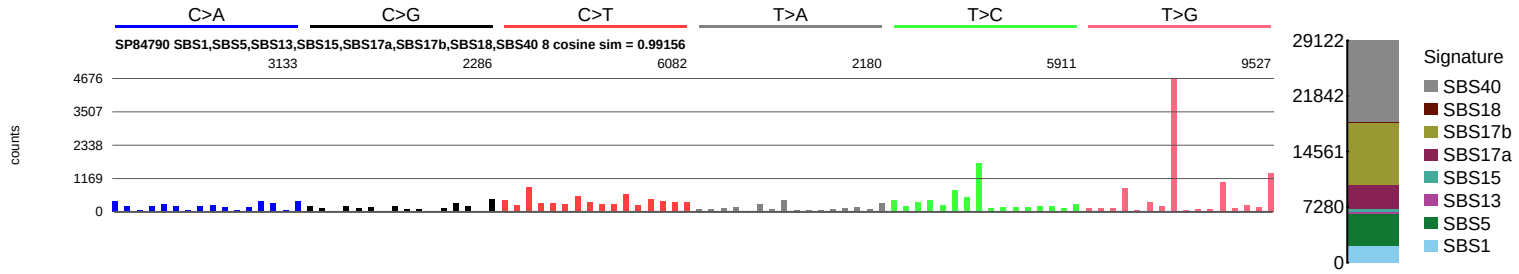

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.99158
3145
2271
6102
2181
5904
9518
counts
4676
3507
2338
1169
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

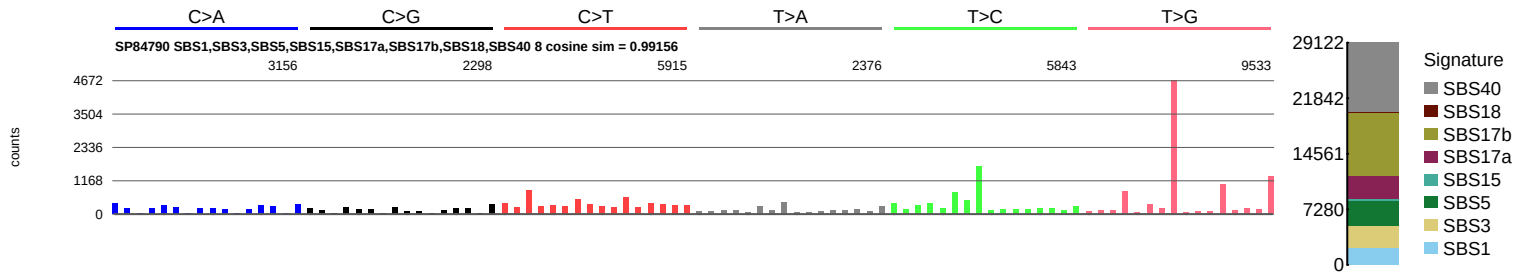

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS40 7 cosine sim = 0.99158
3208
2040
5856
2176
6240
9599
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS1

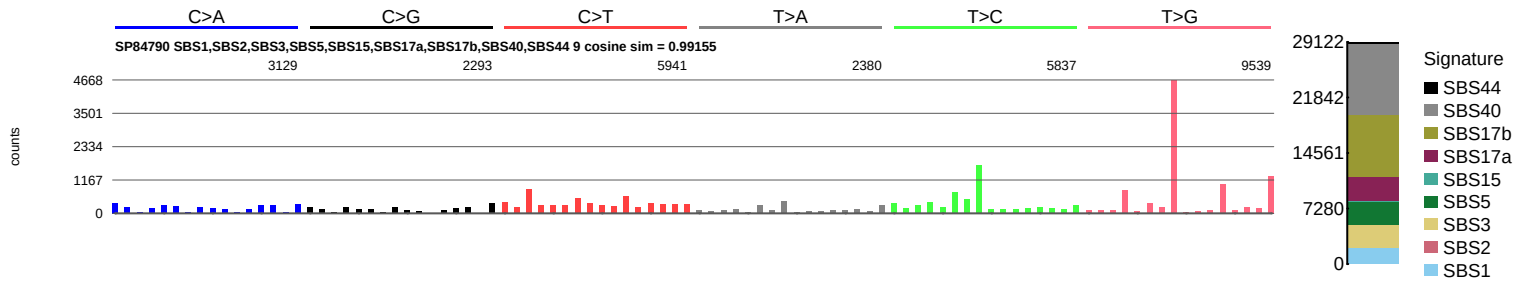

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40 9 cosine sim = 0.99157
3155
2297
5917
2378
5844
9534
counts
4672
3504
2336
1168
0
29122
21842
14561
7280
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

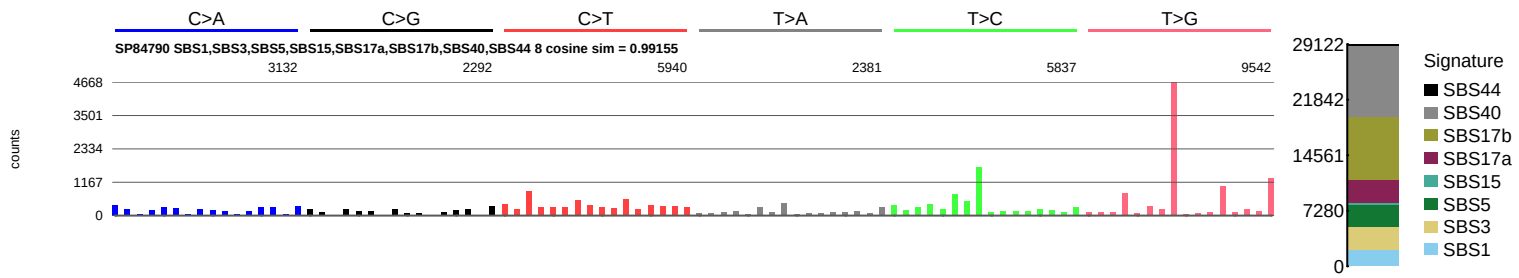

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.99157
3153
2271
6079
2188
5917
9517
counts
4676
3507
2338
1169
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1

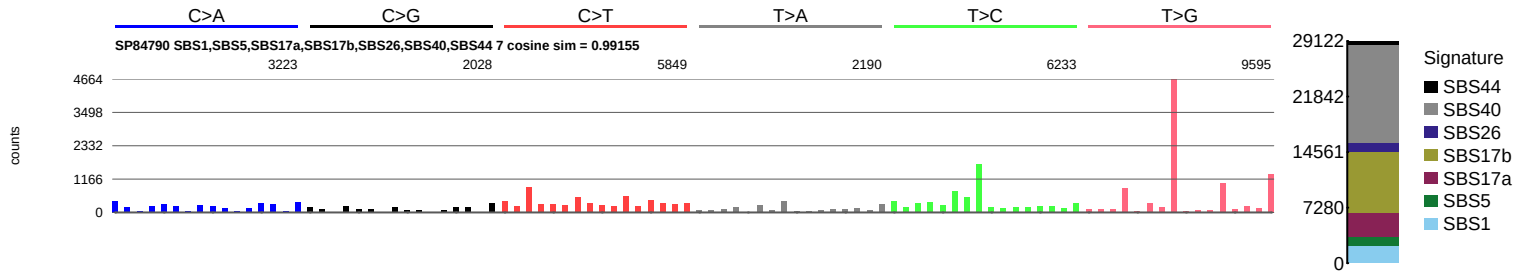

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS40,SBS44 7 cosine sim = 0.99156
3119
2277
6090
2199
5911
9529
counts
4676
3507
2338
1169
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS13
SBS5
SBS1

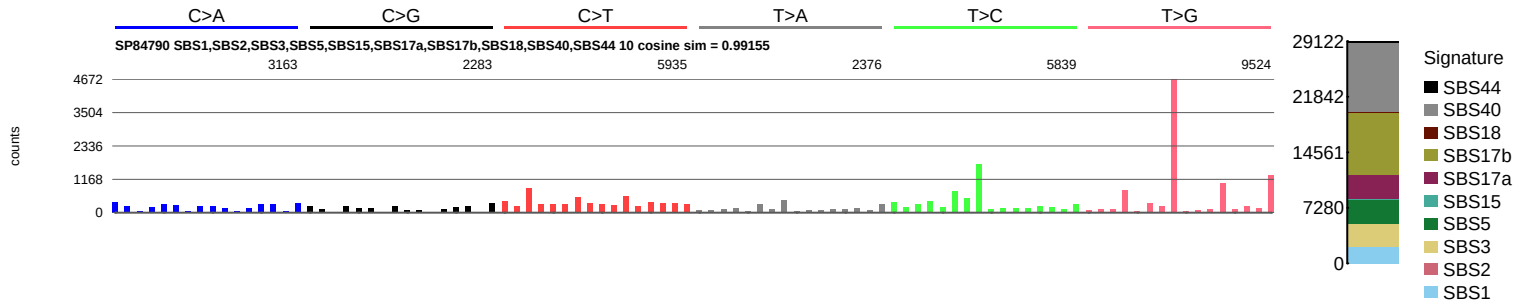

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.99156
3133
2286
6082
2180
5911
9527
counts
4676
3507
2338
1169
0
29122
21842
14561
7280
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

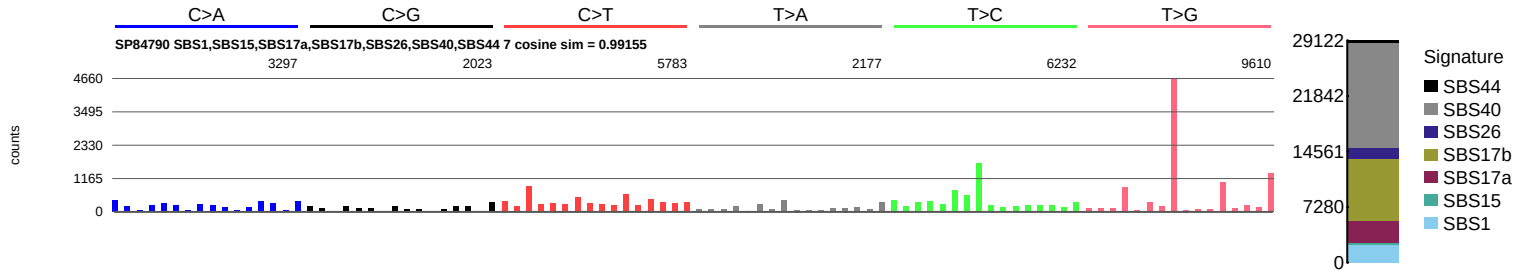

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.99156
3156
2298
5915
2376
5843
9533
counts
4672
3504
2336
1168
0
29122
21842
14561
7280
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1

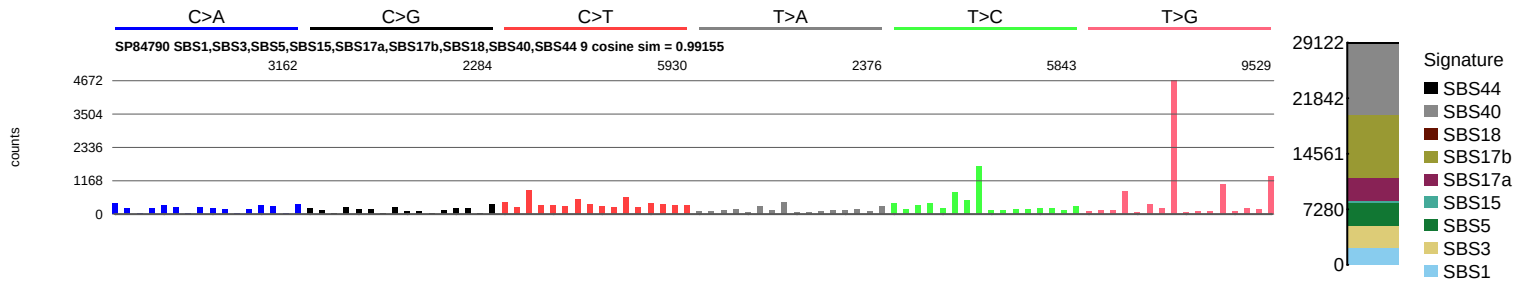

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS40,SBS44 9 cosine sim = 0.99155
3129
2293
5941
2380
5837
9539
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

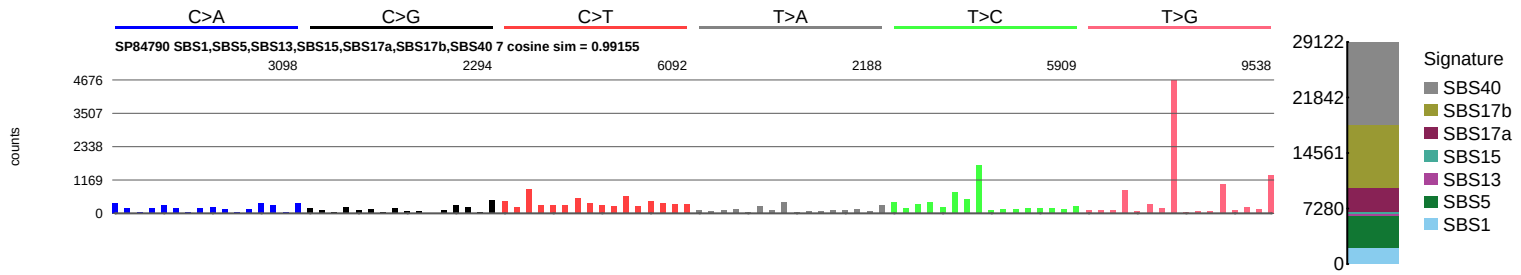

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS40,SBS44 8 cosine sim = 0.99155
3132
2292
5940
2381
5837
9542
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1

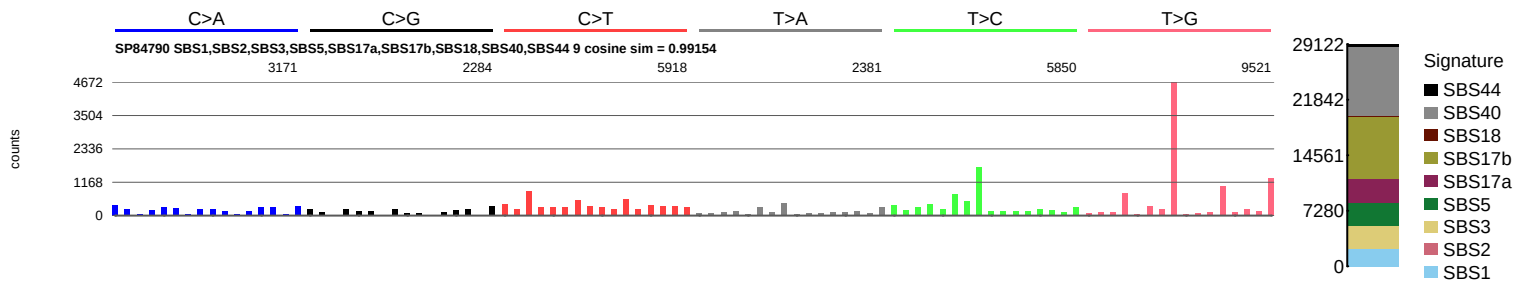

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS17a,SBS17b,SBS26,SBS40,SBS44 7 cosine sim = 0.99155
3223
2028
5849
2190
6233
9595
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS5
SBS1

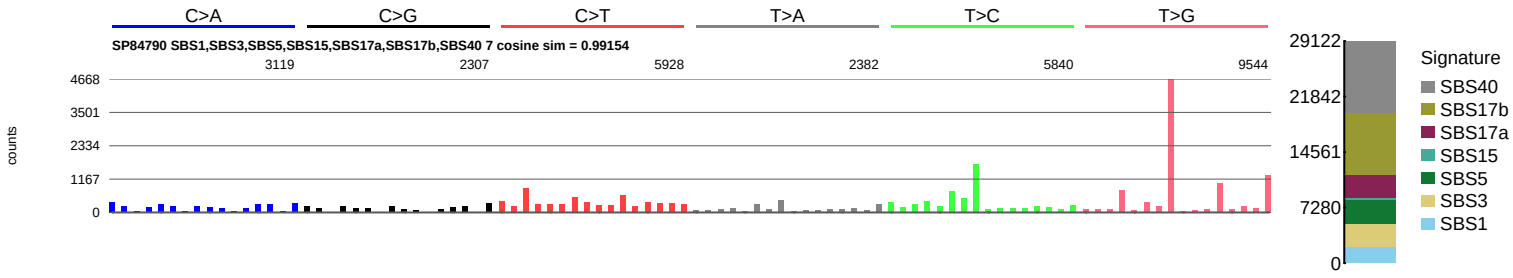

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 10 cosine sim = 0.99155
3163
2283
5935
2376
5839
9524
counts
4672
3504
2336
1168
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

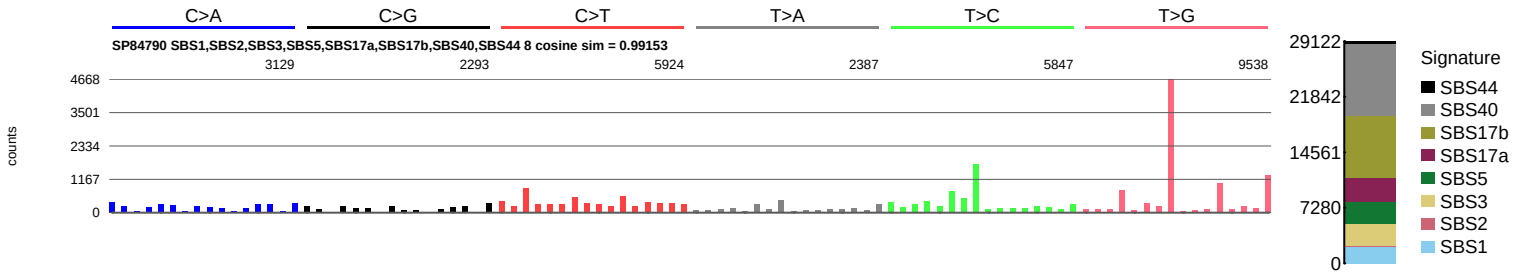

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS44 7 cosine sim = 0.99155
3297
2023
5783
2177
6232
9610
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS1

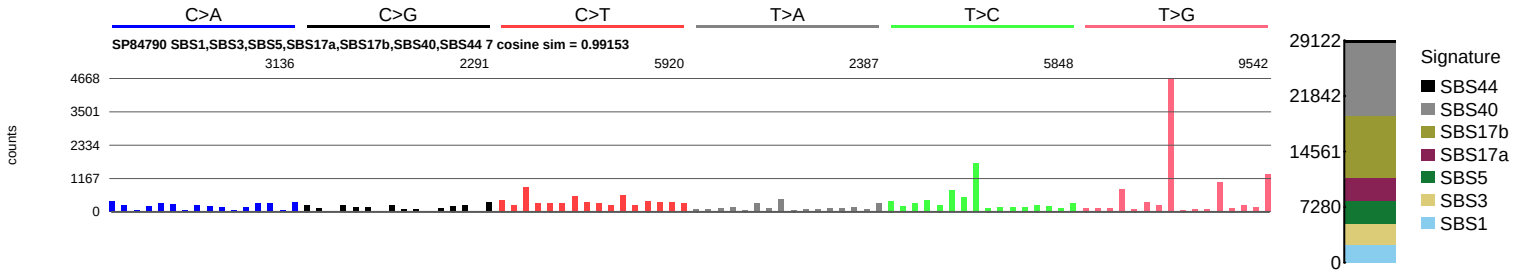

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.99155
3162
2284
5930
2376
5843
9529
counts
4672
3504
2336
1168
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1

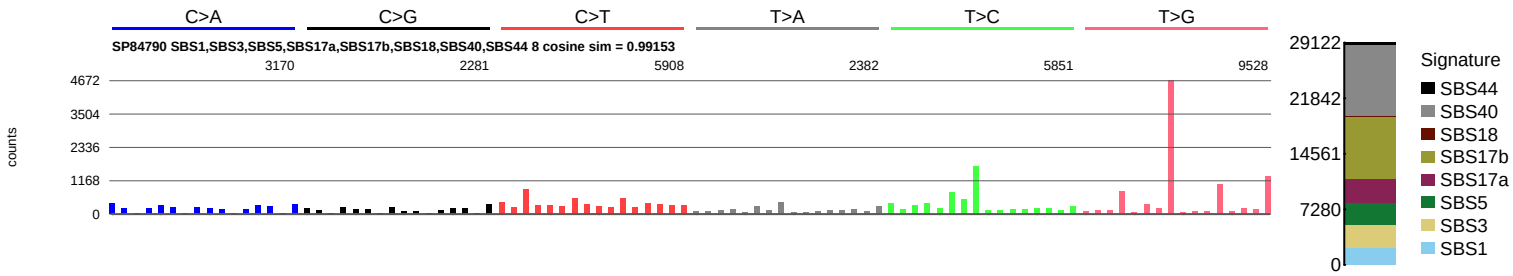

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40 7 cosine sim = 0.99155
3098
2294
6092
2188
5909
9538
counts
4676
3507
2338
1169
0
29122
21842
14561
7280
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

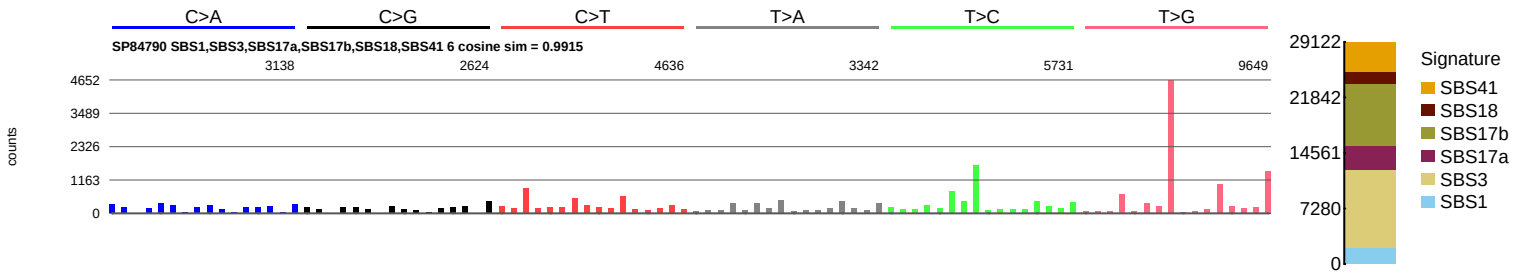

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.99154
3171
2284
5918
2381
5850
9521
counts
4672
3504
2336
1168
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

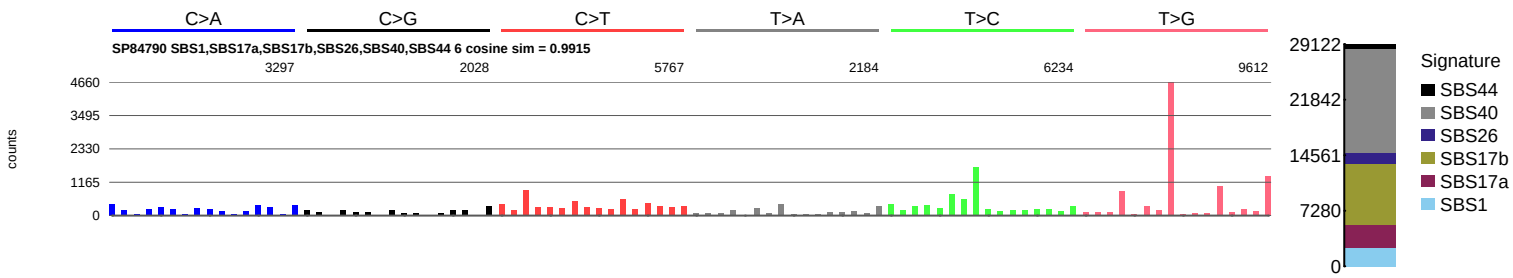

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS40 7 cosine sim = 0.99154
3119
2307
5928
2382
5840
9544
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1

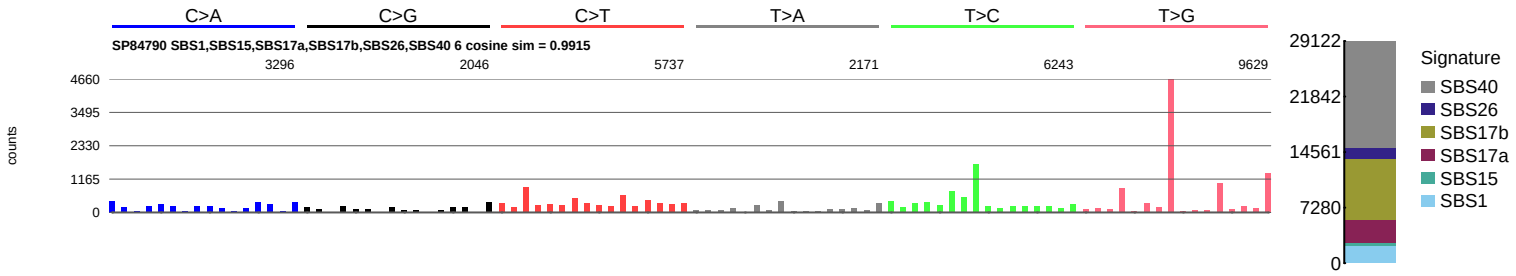

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS40,SBS44 8 cosine sim = 0.99153
3129
2293
5924
2387
5847
9538
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

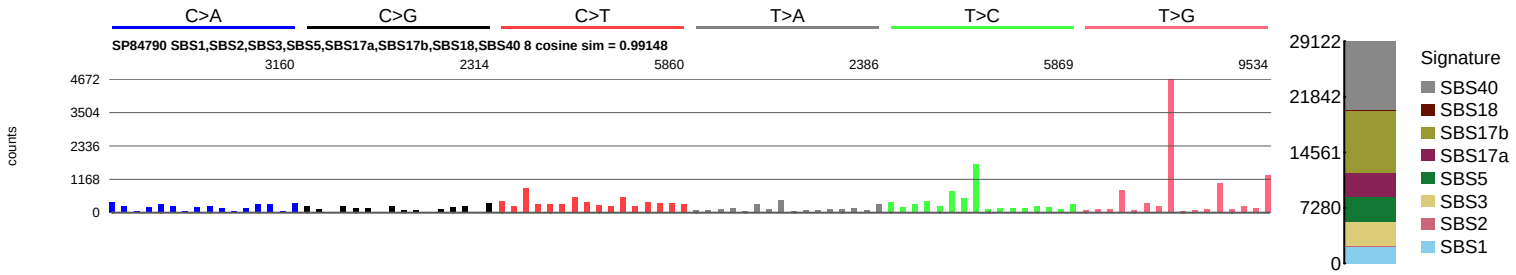

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS40,SBS44 7 cosine sim = 0.99153
3136
2291
5920
2387
5848
9542
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS5
SBS3
SBS1

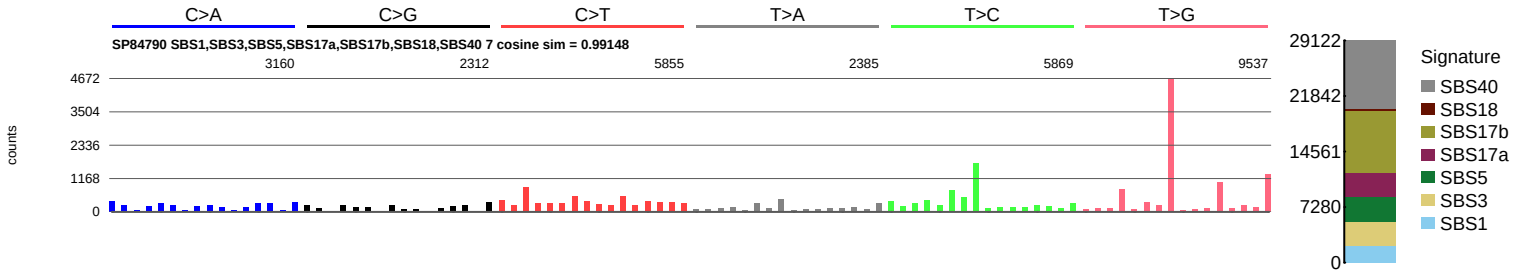

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.99153
3170
2281
5908
2382
5851
9528
counts
4672
3504
2336
1168
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1

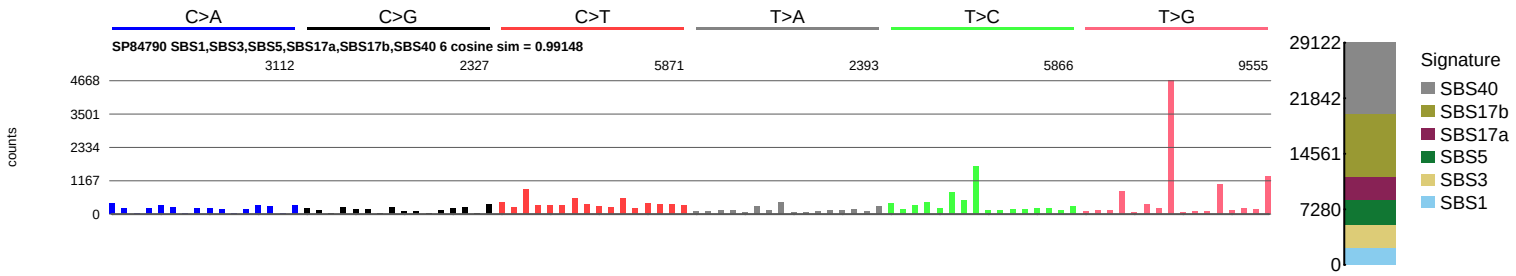

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.9915
3138
2624
4636
3342
5731
9649
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS1

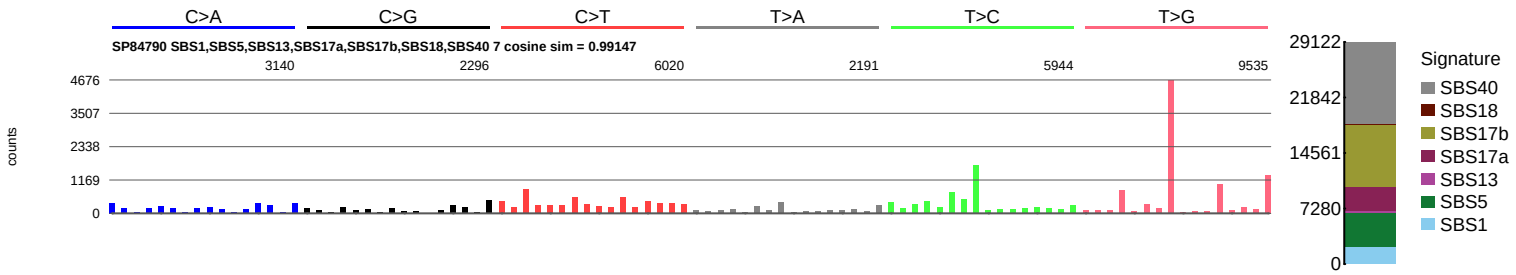

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS17a,SBS17b,SBS26,SBS40,SBS44 6 cosine sim = 0.9915
3297
2028
5767
2184
6234
9612
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS1

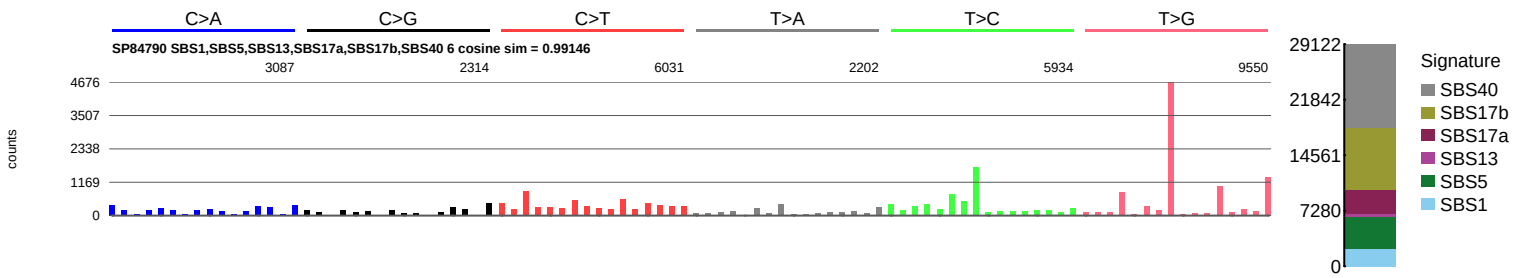

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS15,SBS17a,SBS17b,SBS26,SBS40 6 cosine sim = 0.9915
3296
2046
5737
2171
6243
9629
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS1

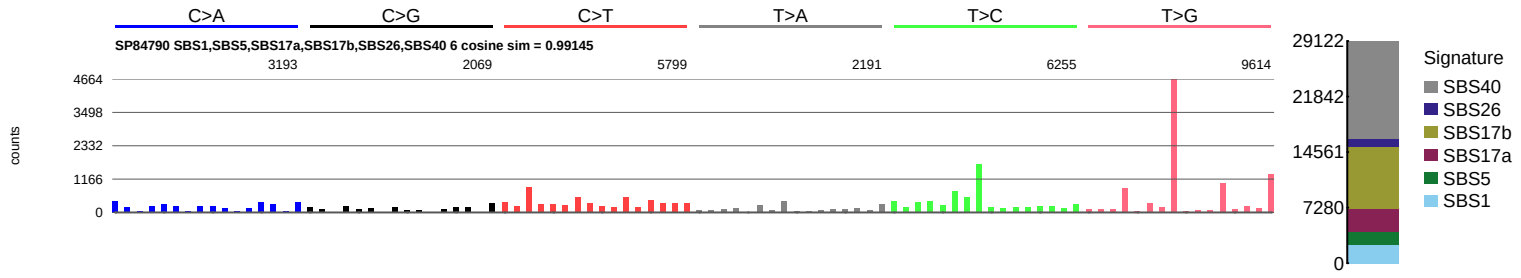

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.99148
3160
2314
5860
2386
5869
9534
counts
4672
3504
2336
1168
0
29122
21842
14561
7280
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

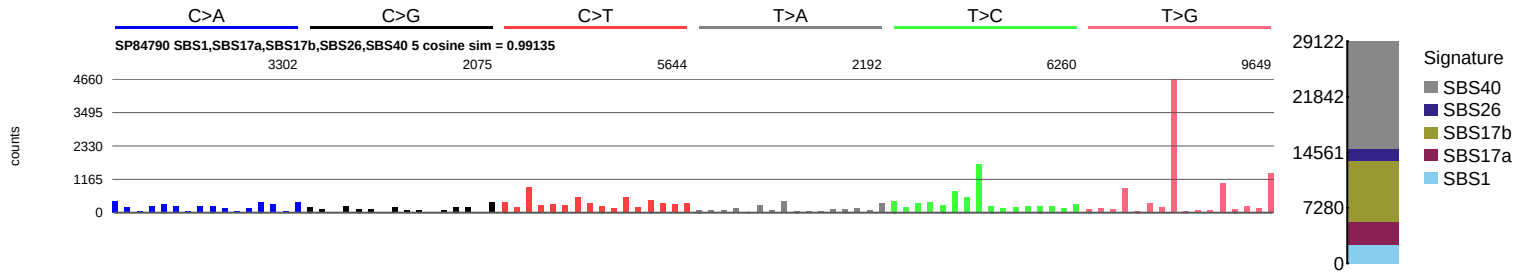

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.99148
3160
2312
5855
2385
5869
9537
counts
4672
3504
2336
1168
0
29122
21842
14561
7280
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1

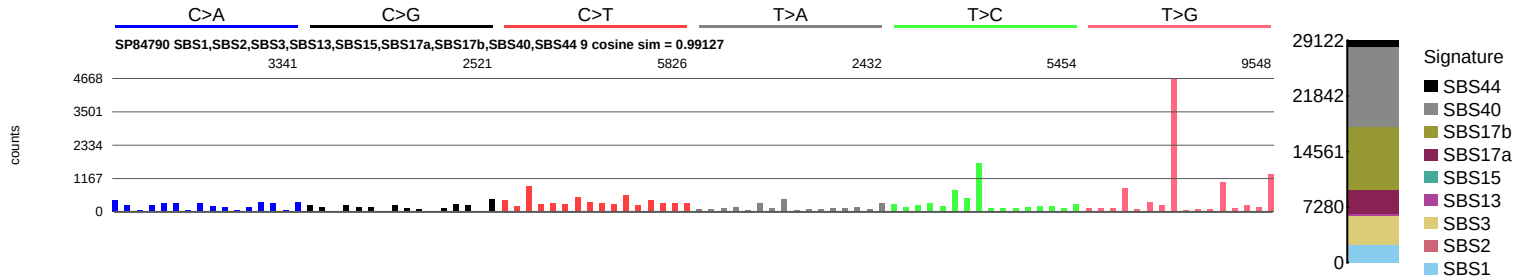

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS40 6 cosine sim = 0.99148
3112
2327
5871
2393
5866
9555
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS40
SBS17b
SBS17a
SBS5
SBS3
SBS1

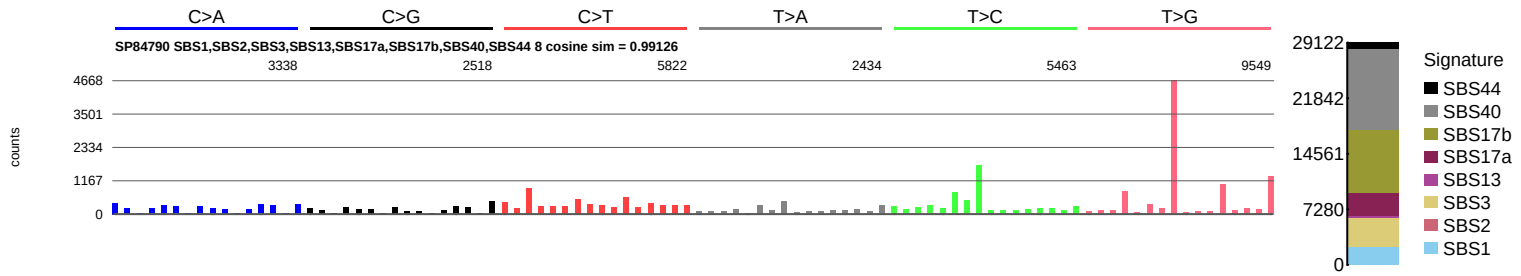

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.99147
3140
2296
6020
2191
5944
9535
counts
4676
3507
2338
1169
0
29122
21842
14561
7280
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1

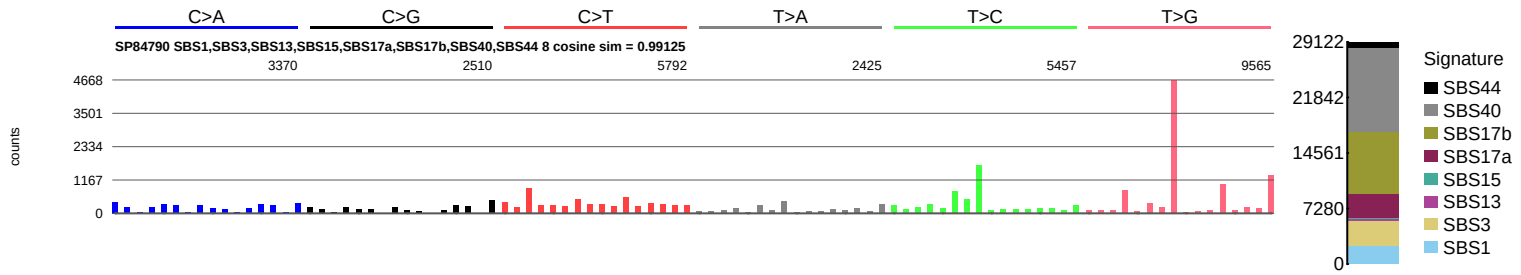

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS40 6 cosine sim = 0.99146
3087
2314
6031
2202
5934
9550
counts
4676
3507
2338
1169
0
29122
21842
14561
7280
0
Signature
SBS40
SBS17b
SBS17a
SBS13
SBS5
SBS1

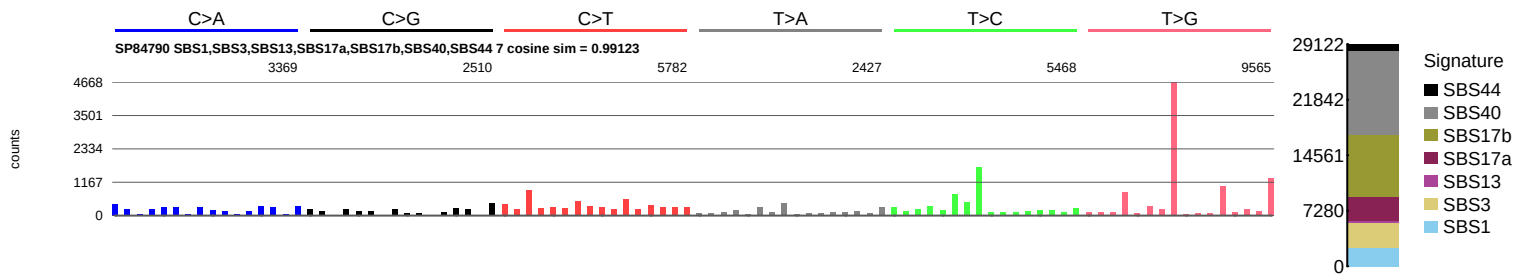

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS17a,SBS17b,SBS26,SBS40 6 cosine sim = 0.99145
3193
2069
5799
2191
6255
9614
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS5
SBS1

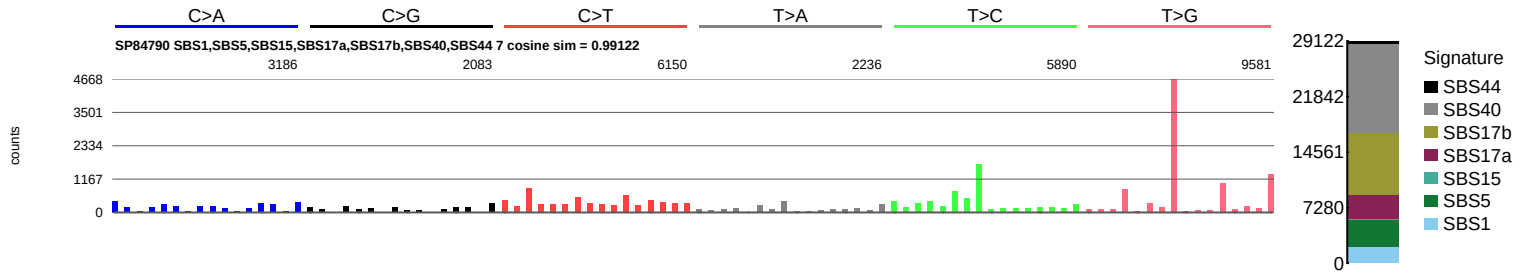

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS17a,SBS17b,SBS26,SBS40 5 cosine sim = 0.99135
3302
2075
5644
2192
6260
9649
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS1

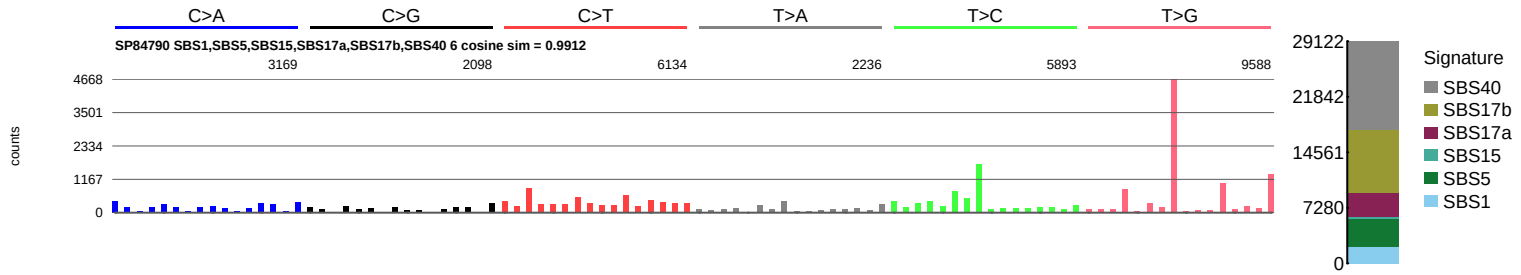

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS44 9 cosine sim = 0.99127
3341
2521
5826
2432
5454
9548
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

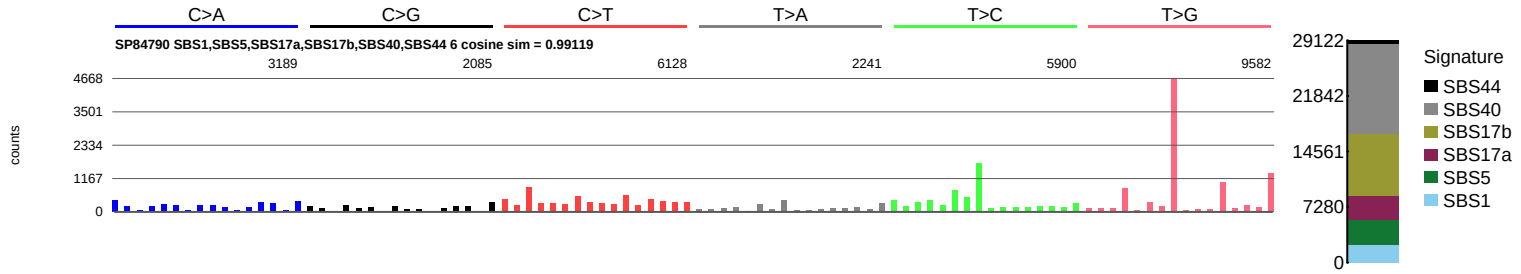

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS40,SBS44 8 cosine sim = 0.99126
3338
2518
5822
2434
5463
9549
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

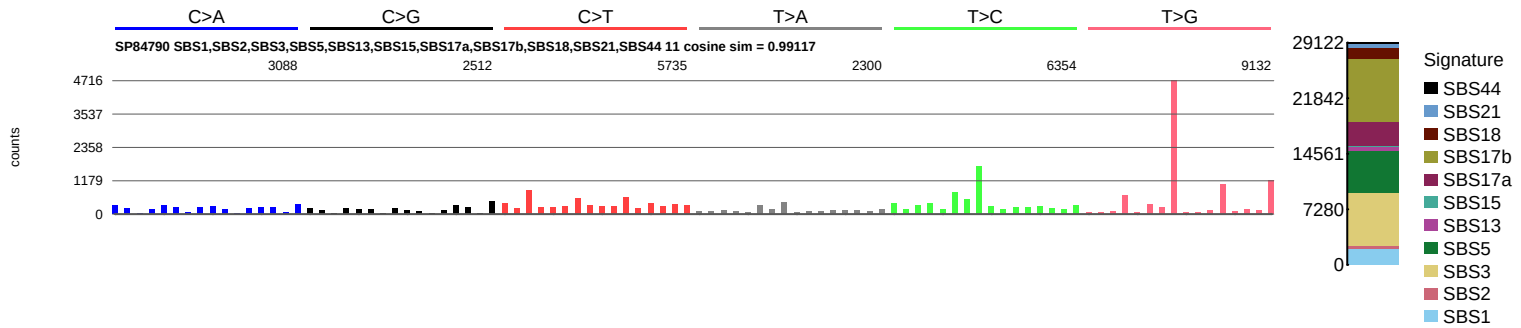

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS44 8 cosine sim = 0.99125
3370
2510
5792
2425
5457
9565
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

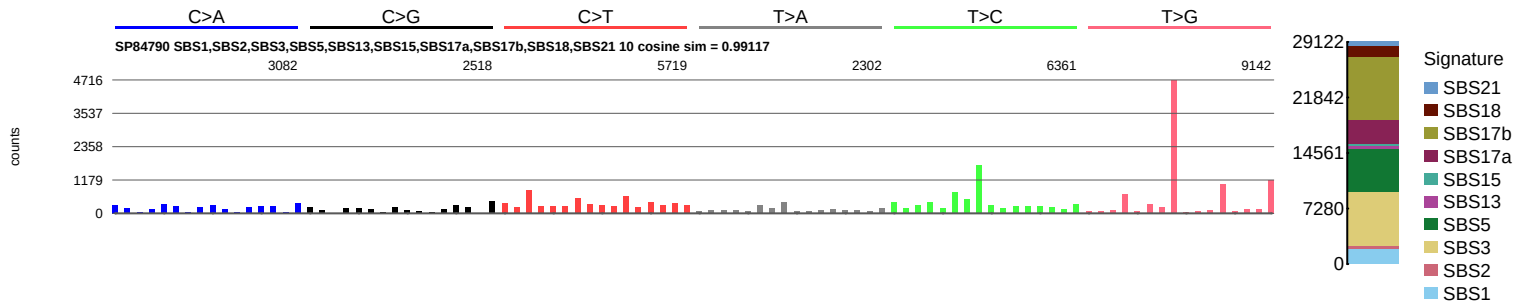

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS40,SBS44 7 cosine sim = 0.99123
3369
2510
5782
2427
5468
9565
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS13
SBS3
SBS1

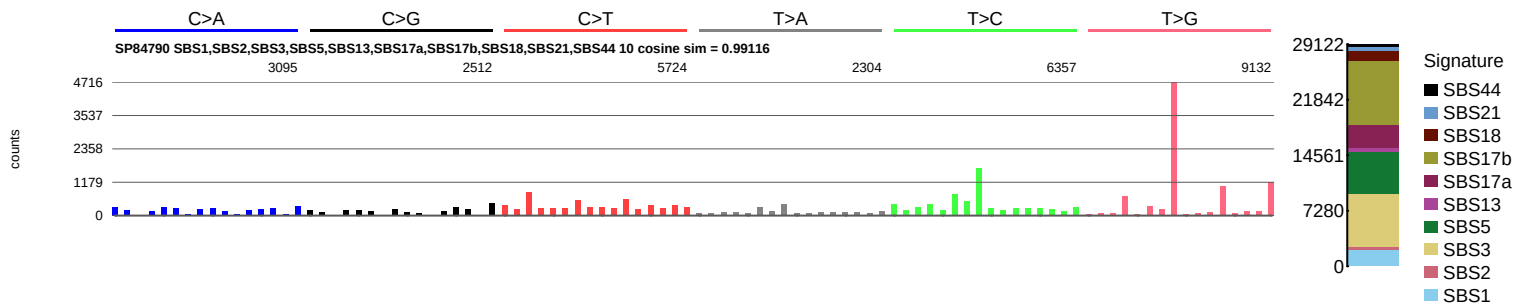

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS40,SBS44 7 cosine sim = 0.99122
3186
2083
6150
2236
5890
9581
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS1

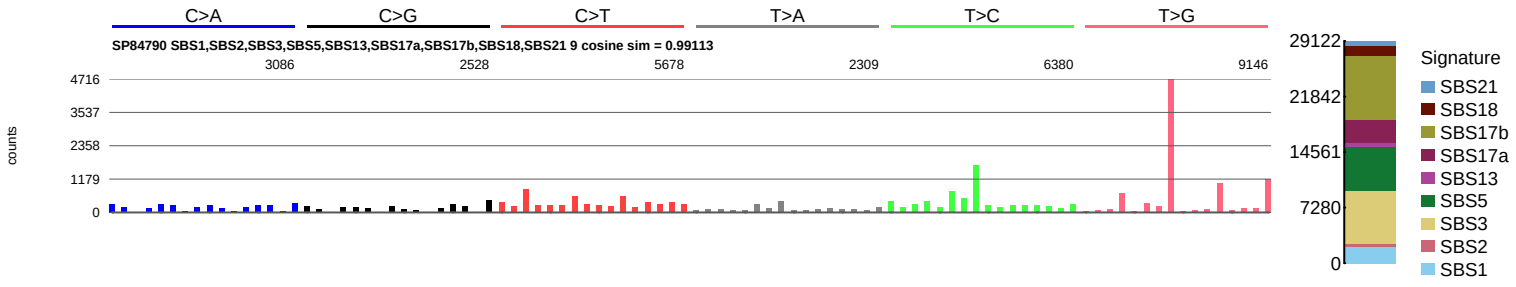

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS40 6 cosine sim = 0.9912
3169
2098
6134
2236
5893
9588
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS1

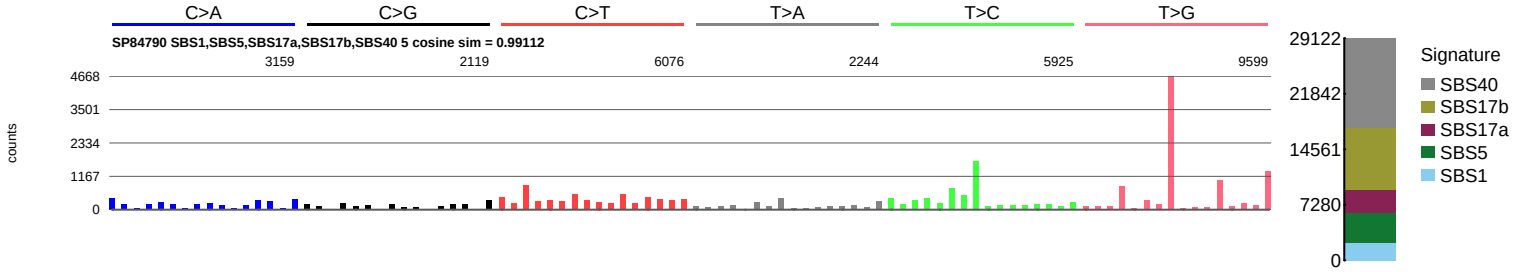

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS17a,SBS17b,SBS40,SBS44 6 cosine sim = 0.99119
3189
2085
6128
2241
5900
9582
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS5
SBS1

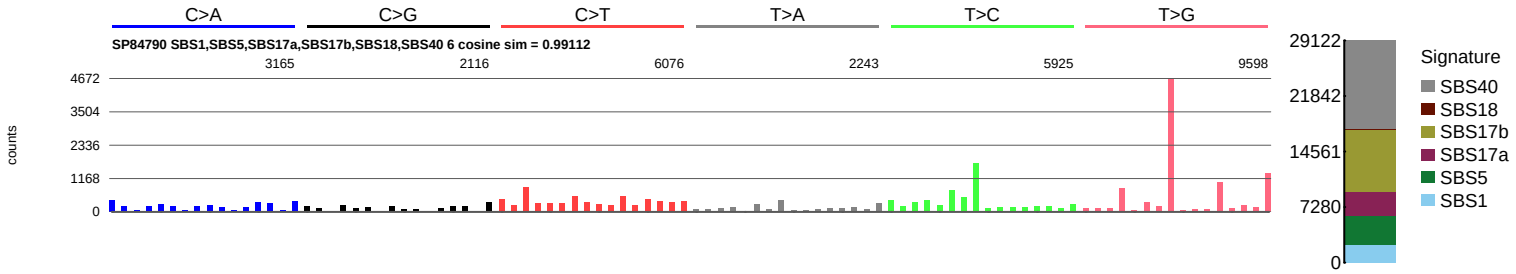

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 11 cosine sim = 0.99117
3088
2512
5735
2300
6354
9132
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

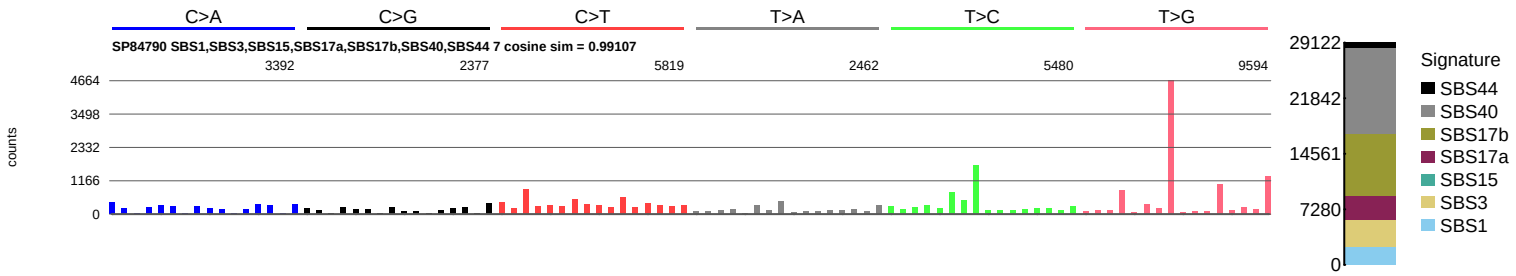

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 10 cosine sim = 0.99117
3082
2518
5719
2302
6361
9142
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

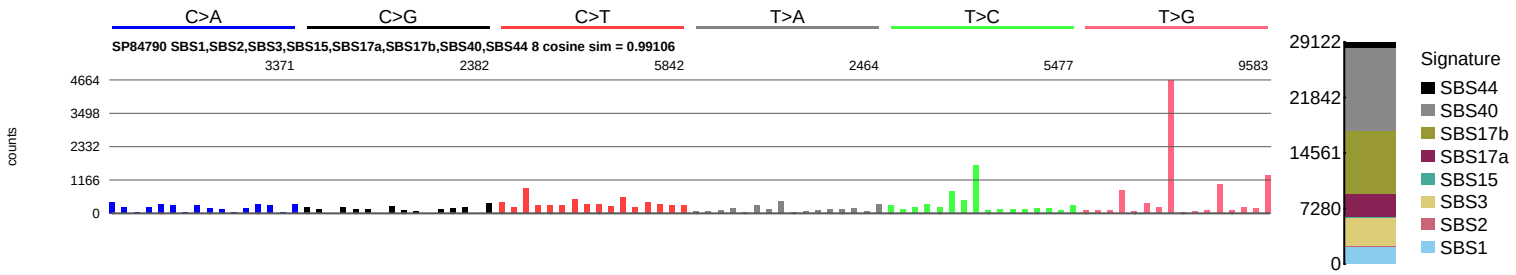

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.99116
3095
2512
5724
2304
6357
9132
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

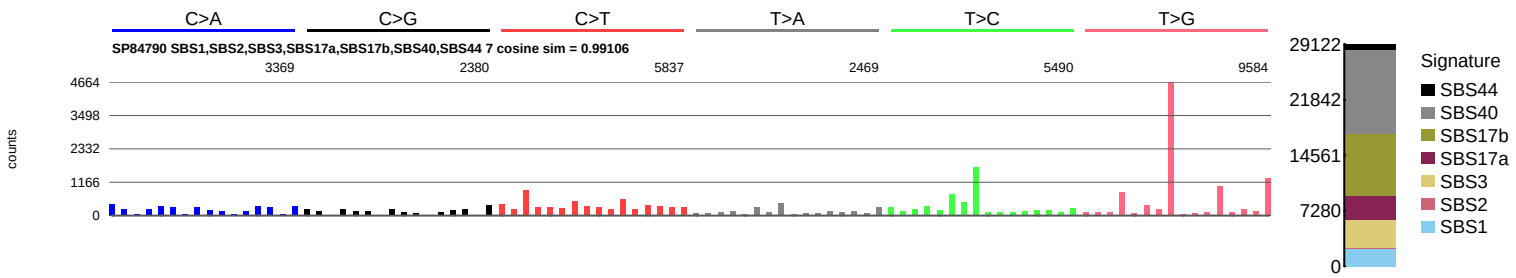

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.99113
3086
2528
5678
2309
6380
9146
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

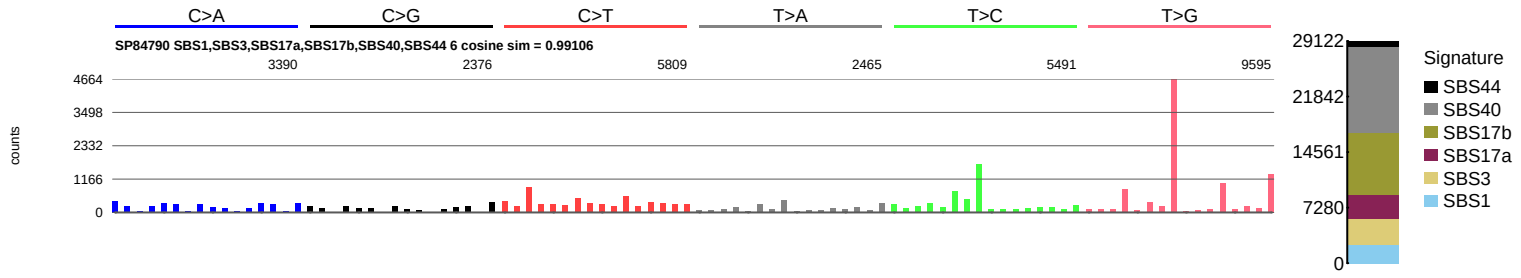

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS17a,SBS17b,SBS40 5 cosine sim = 0.99112
3159
2119
6076
2244
5925
9599
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS40
SBS17b
SBS17a
SBS5
SBS1

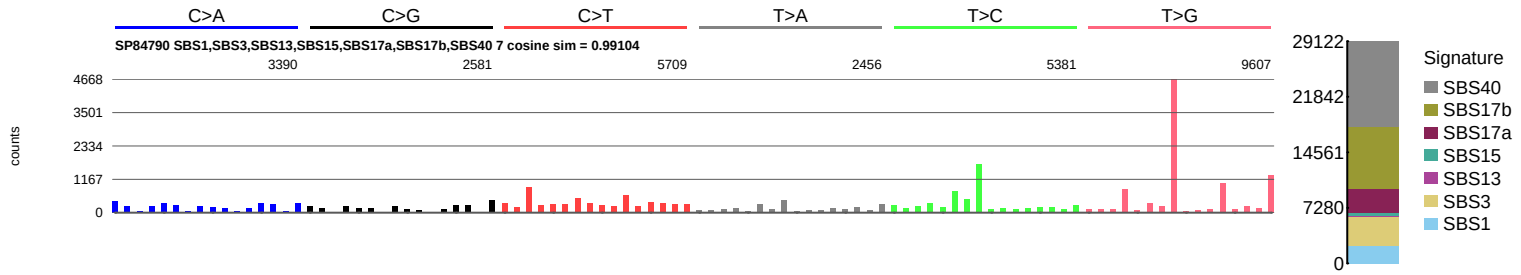

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS40 6 cosine sim = 0.99112
3165
2116
6076
2243
5925
9598
counts
4672
3504
2336
1168
0
29122
21842
14561
7280
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS1

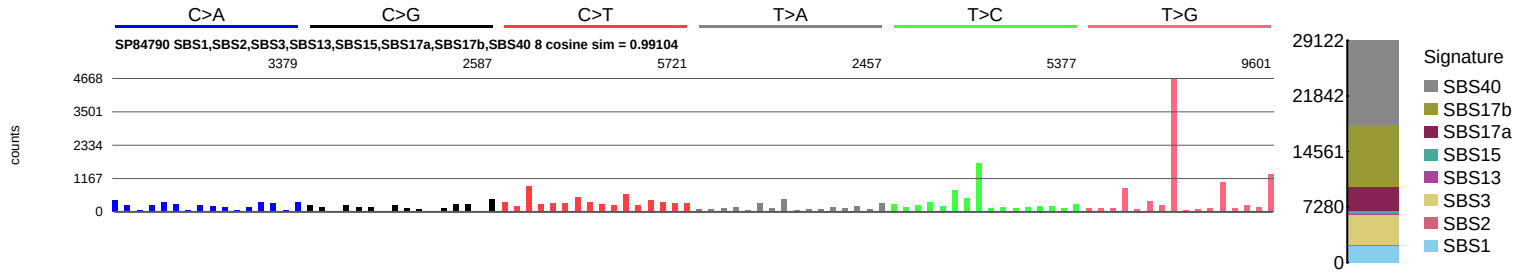

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS40,SBS44 7 cosine sim = 0.99107
3392
2377
5819
2462
5480
9594
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS15
SBS3
SBS1

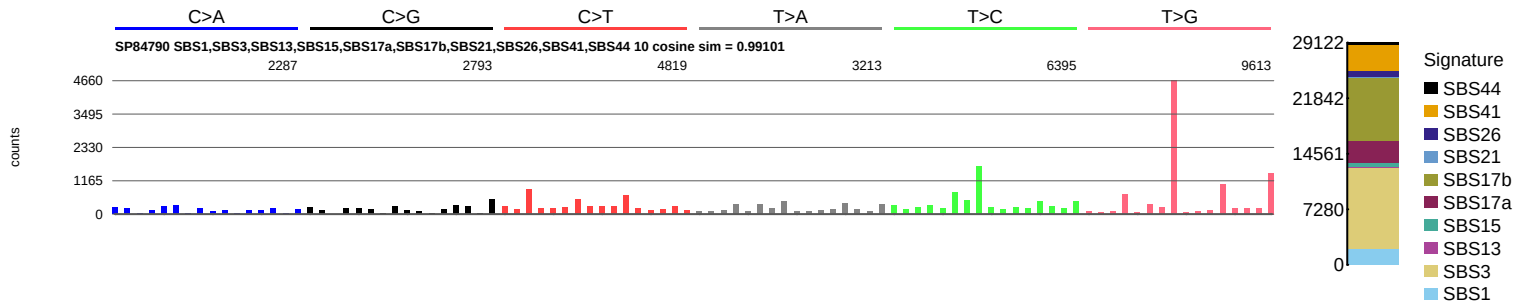

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS40,SBS44 8 cosine sim = 0.99106
3371
2382
5842
2464
5477
9583
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

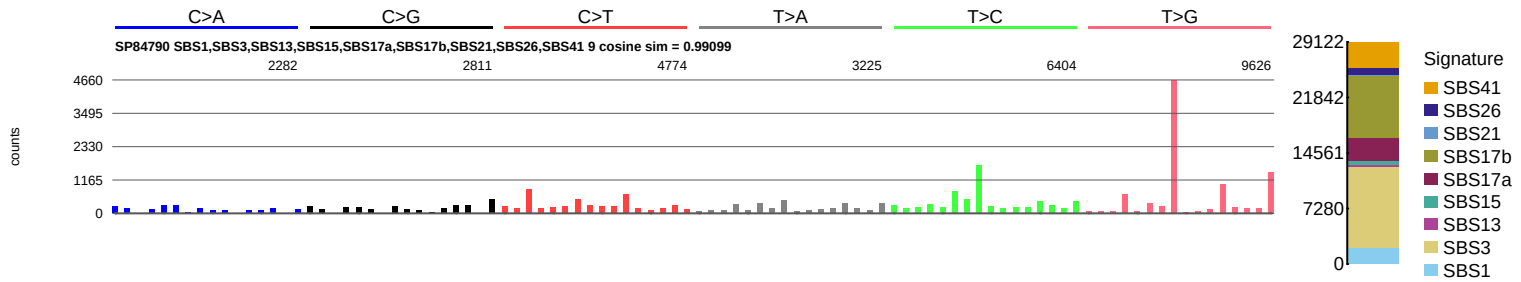

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS40,SBS44 7 cosine sim = 0.99106
3369
2380
5837
2469
5490
9584
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS3
SBS2
SBS1

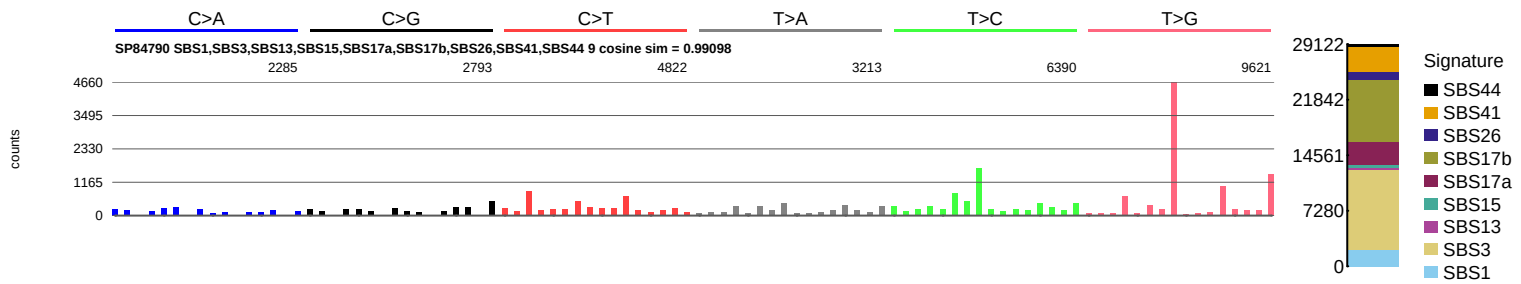

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS40,SBS44 6 cosine sim = 0.99106
3390
2376
5809
2465
5491
9595
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS3
SBS1

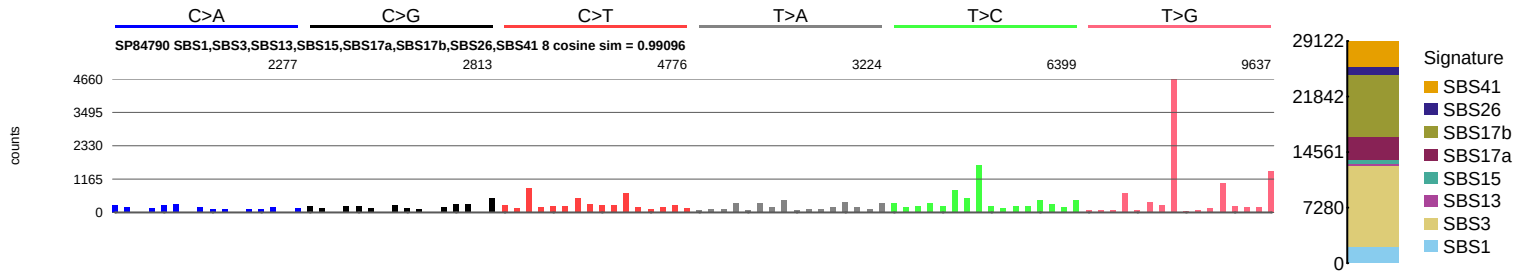

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS40 7 cosine sim = 0.99104
3390
2581
5709
2456
5381
9607
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

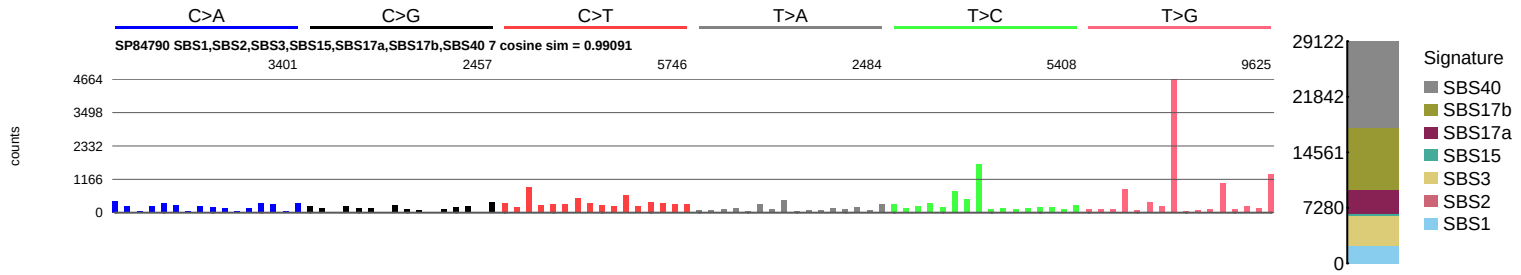

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS40 8 cosine sim = 0.99104
3379
2587
5721
2457
5377
9601
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

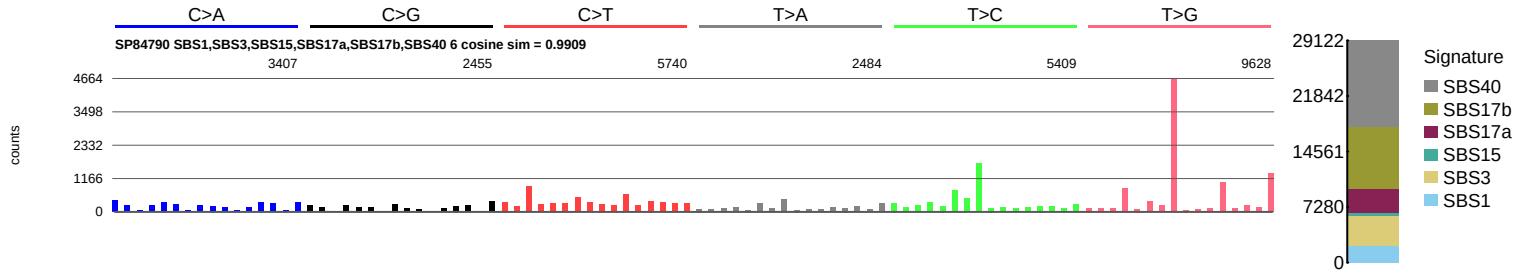

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 10 cosine sim = 0.99101
2287
2793
4819
3213
6395
9613
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

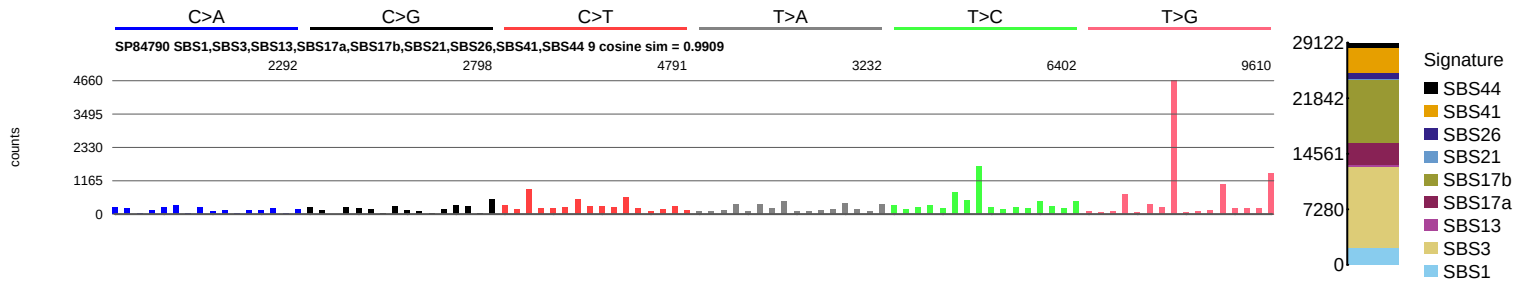

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41 9 cosine sim = 0.99099
2282
2811
4774
3225
6404
9626
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

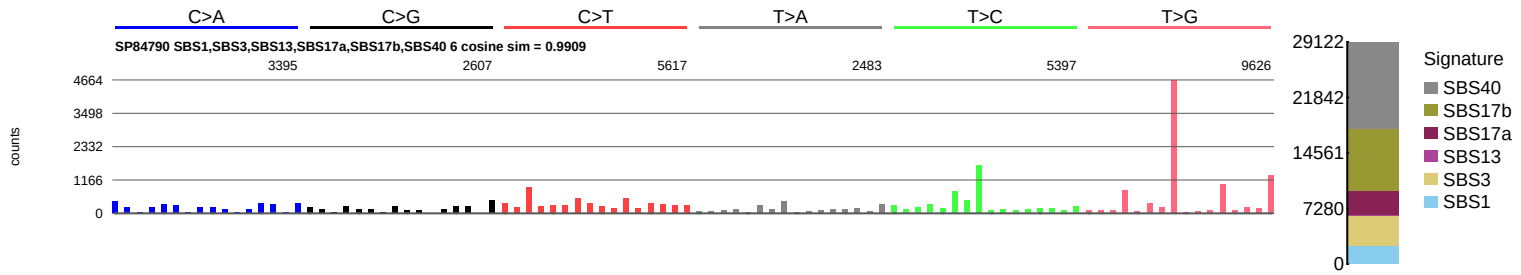

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41,SBS44 9 cosine sim = 0.99098
2285
2793
4822
3213
6390
9621
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

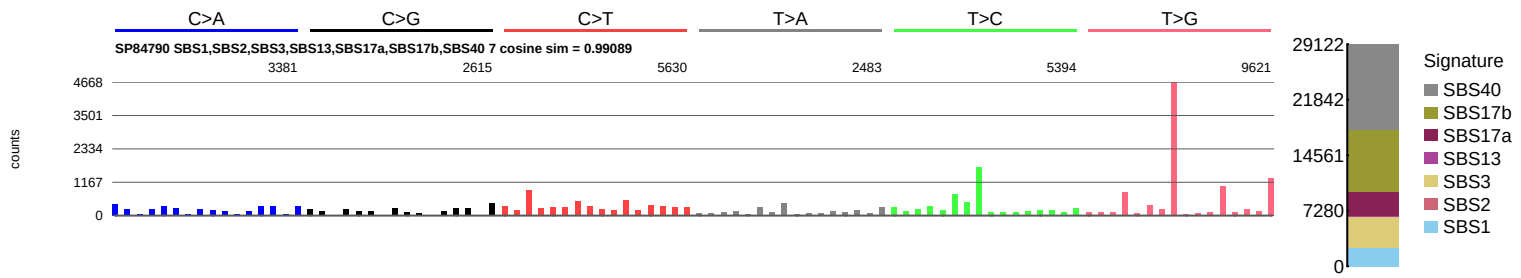

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41 8 cosine sim = 0.99096
2277
2813
4776
3224
6399
9637
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

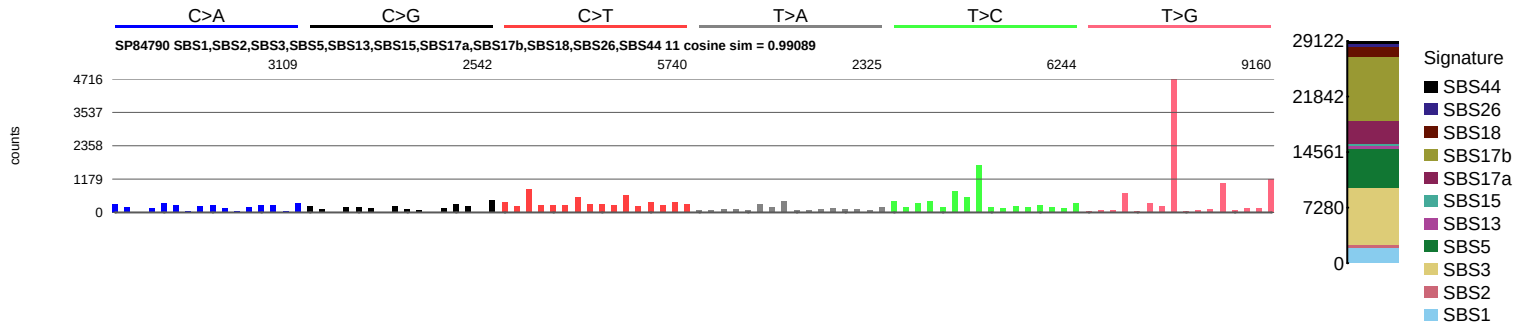

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS40 7 cosine sim = 0.99091
3401
2457
5746
2484
5408
9625
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

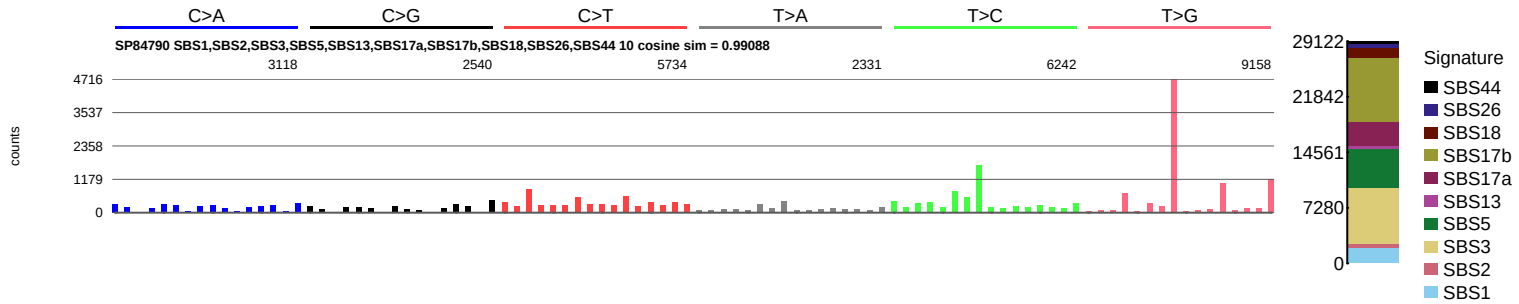

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS40 6 cosine sim = 0.9909
3407
2455
5740
2484
5409
9628
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS3
SBS1

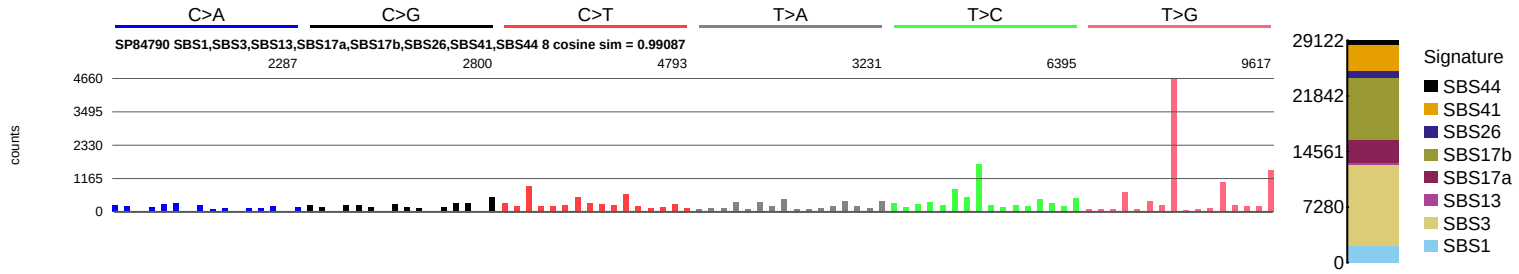

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 9 cosine sim = 0.9909
2292
2798
4791
3232
6402
9610
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS1

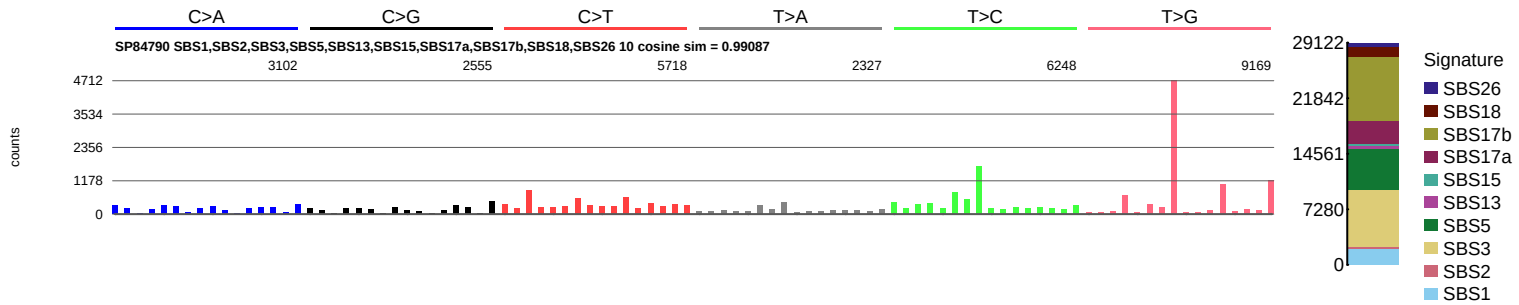

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS40 6 cosine sim = 0.9909
3395
2607
5617
2483
5397
9626
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS40
SBS17b
SBS17a
SBS13
SBS3
SBS1

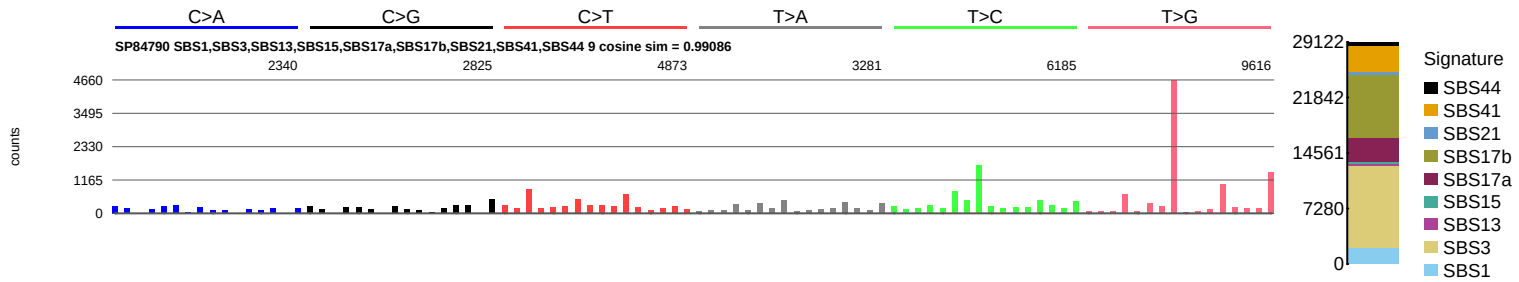

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS40 7 cosine sim = 0.99089
3381
2615
5630
2483
5394
9621
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS40
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

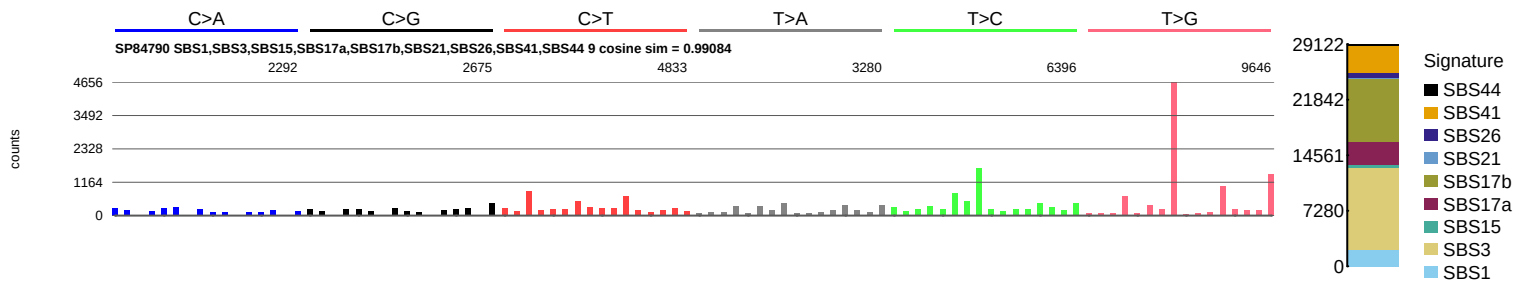

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 11 cosine sim = 0.99089
3109
2542
5740
2325
6244
9160
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

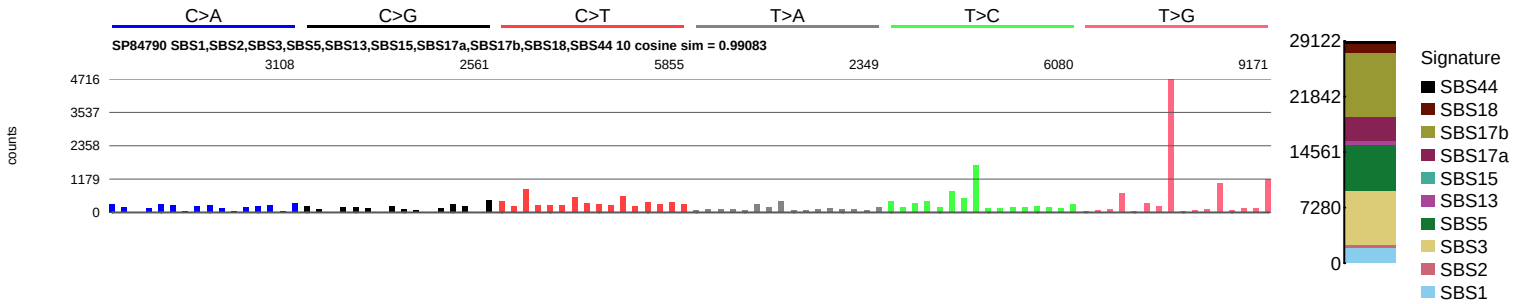

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 10 cosine sim = 0.99088
3118
2540
5734
2331
6242
9158
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

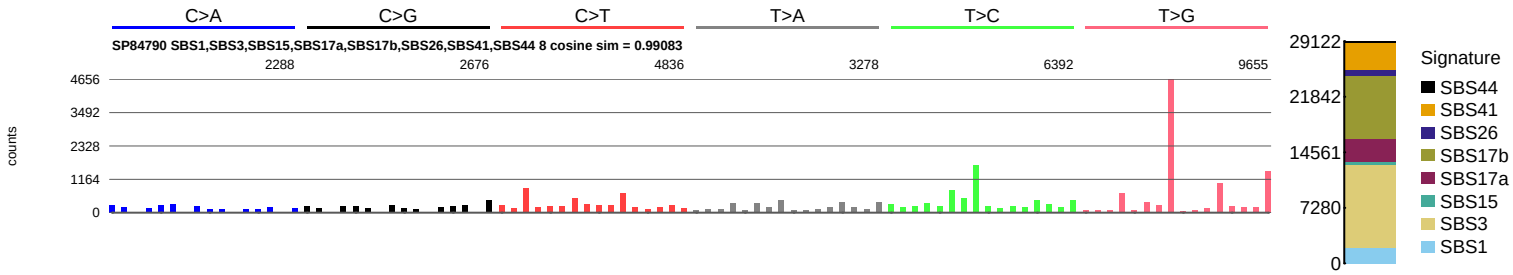

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS26,SBS41,SBS44 8 cosine sim = 0.99087
2287
2800
4793
3231
6395
9617
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS13
SBS3
SBS1

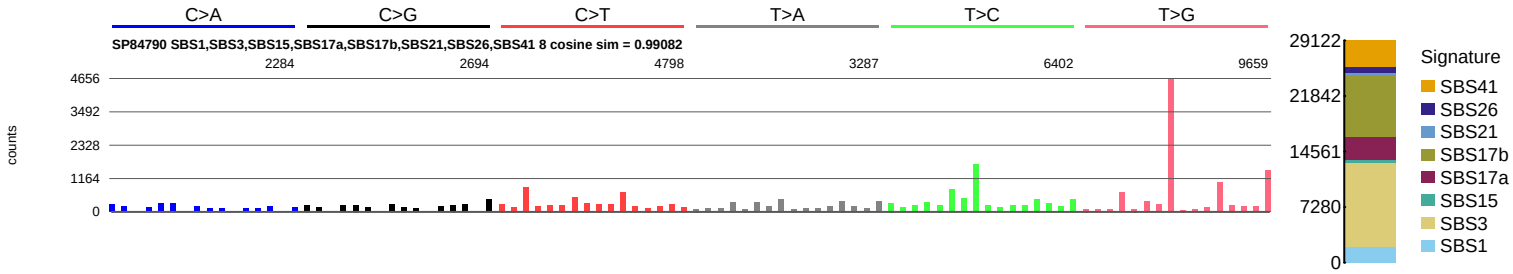

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 10 cosine sim = 0.99087
3102
2555
5718
2327
6248
9169
counts
4712
3534
2356
1178
0
29122
21842
14561
7280
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

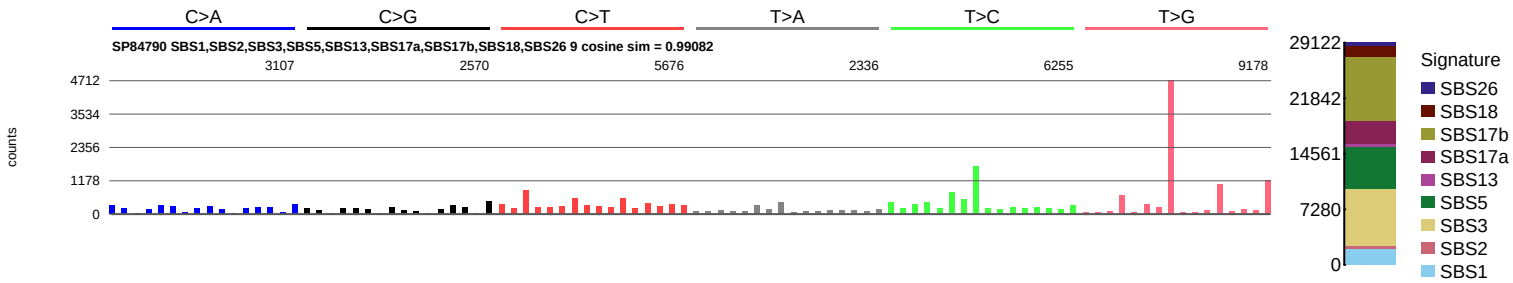

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41,SBS44 9 cosine sim = 0.99086
2340
2825
4873
3281
6185
9616
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

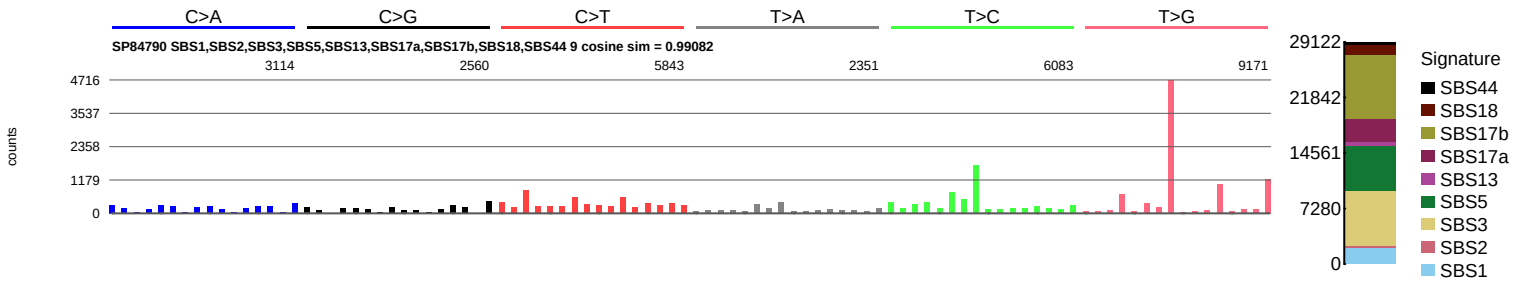

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 9 cosine sim = 0.99084
2292
2675
4833
3280
6396
9646
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS1

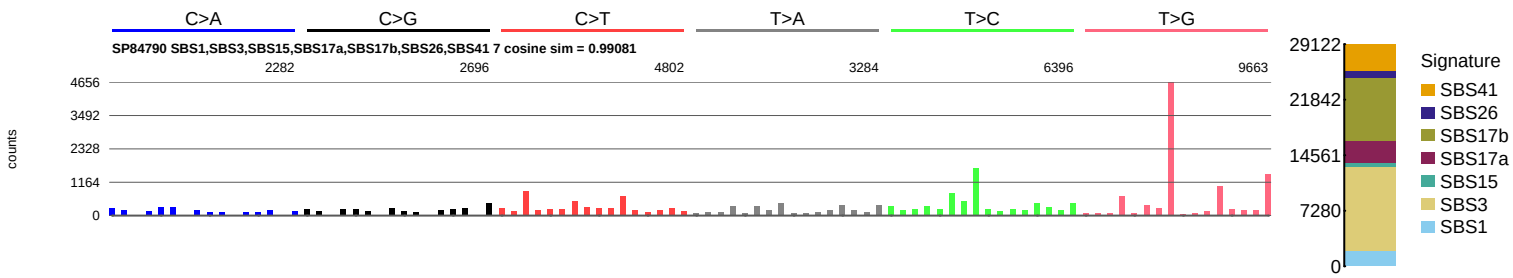

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS44 10 cosine sim = 0.99083
3108
2561
5855
2349
6080
9171
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

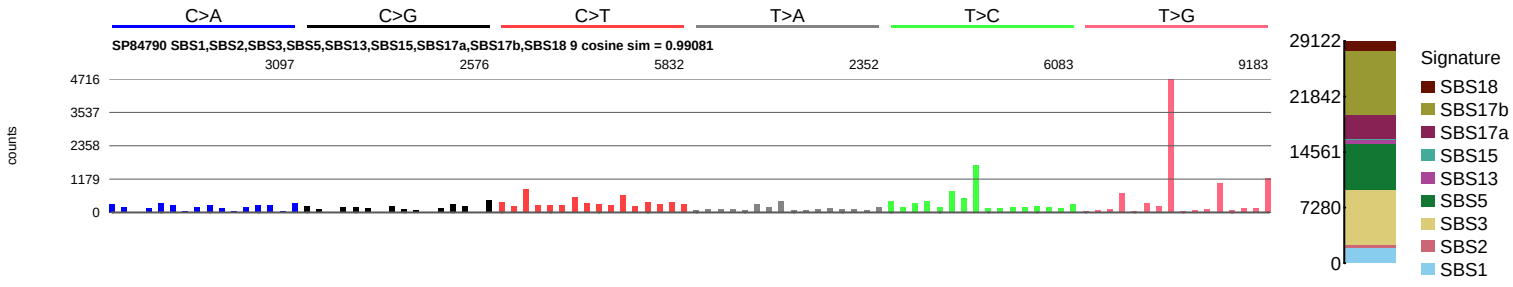

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS41,SBS44 8 cosine sim = 0.99083
2288
2676
4836
3278
6392
9655
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS1

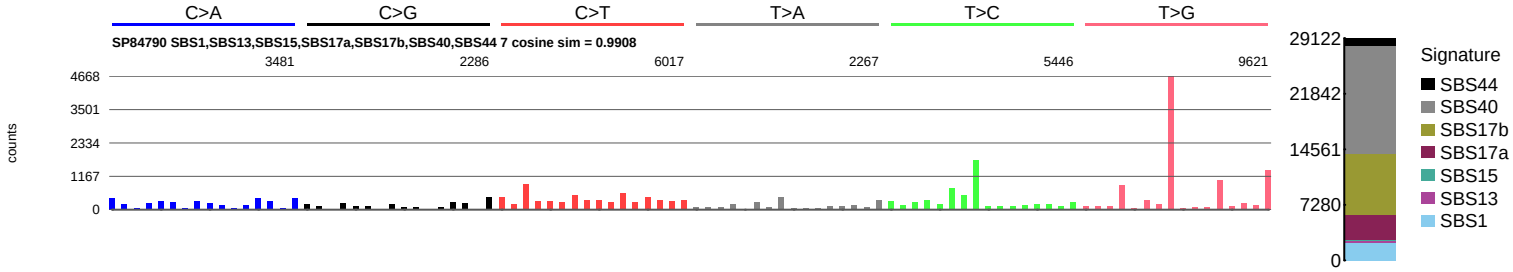

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41 8 cosine sim = 0.99082
2284
2694
4798
3287
6402
9659
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS1

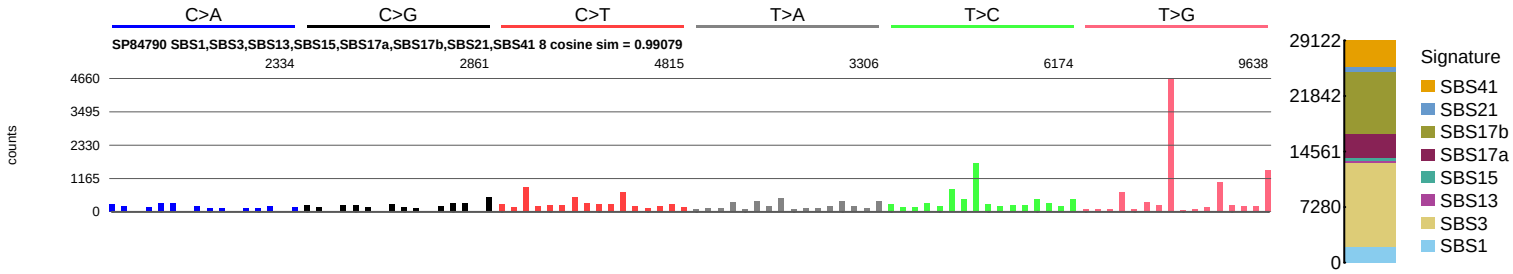

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26 9 cosine sim = 0.99082
3107
2570
5676
2336
6255
9178
counts
4712
3534
2356
1178
0
29122
21842
14561
7280
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

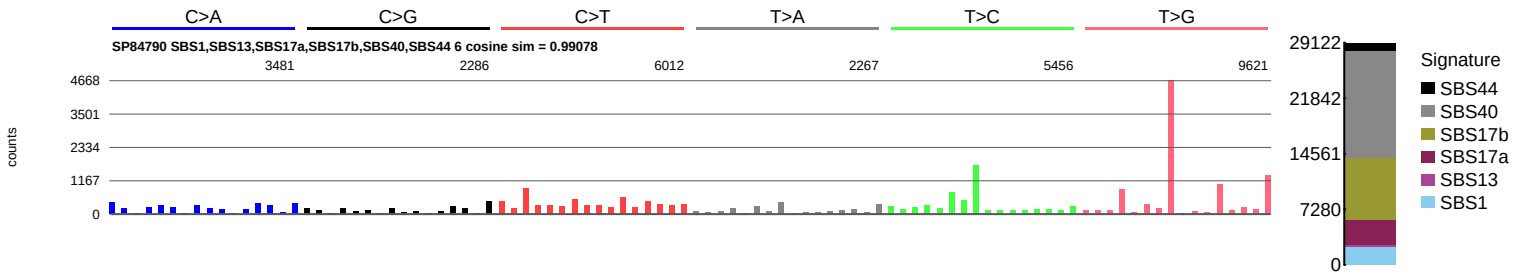

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 9 cosine sim = 0.99082
3114
2560
5843
2351
6083
9171
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

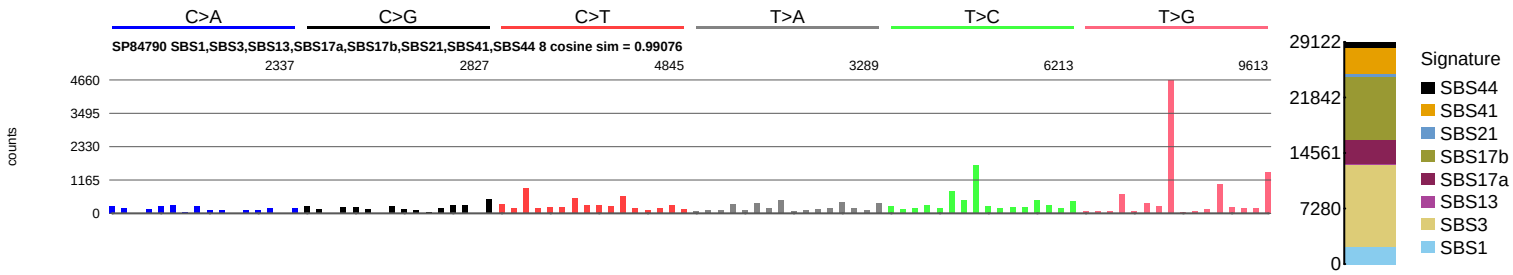

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS41 7 cosine sim = 0.99081
2282
2696
4802
3284
6396
9663
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS1

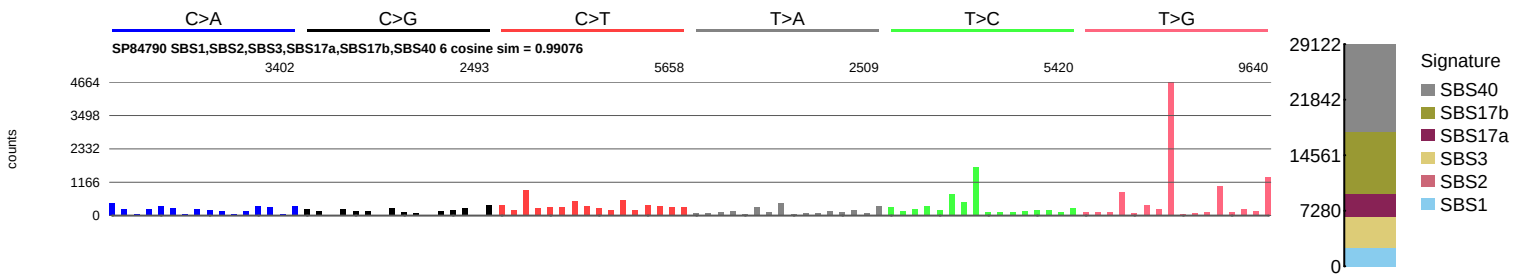

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 9 cosine sim = 0.99081
3097
2576
5832
2352
6083
9183
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

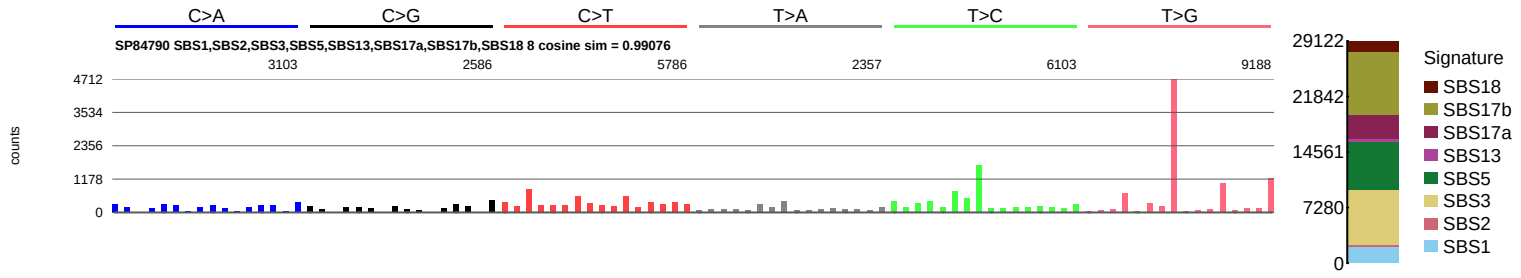

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS44 7 cosine sim = 0.9908
3481
2286
6017
2267
5446
9621
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS1

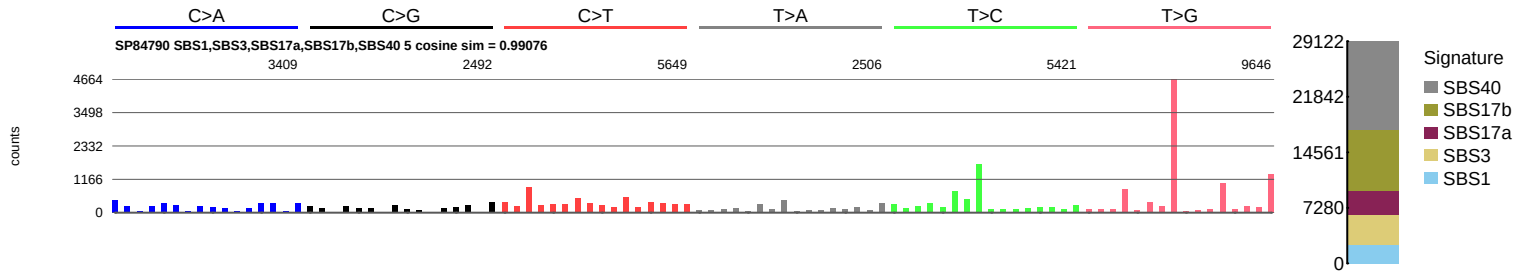

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.99079
2334
2861
4815
3306
6174
9638
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

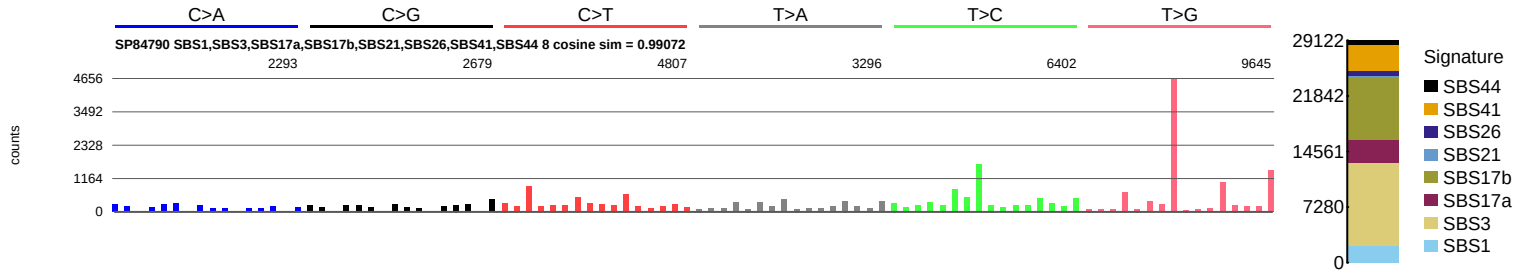

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS17a,SBS17b,SBS40,SBS44 6 cosine sim = 0.99078
3481
2286
6012
2267
5456
9621
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS13
SBS1

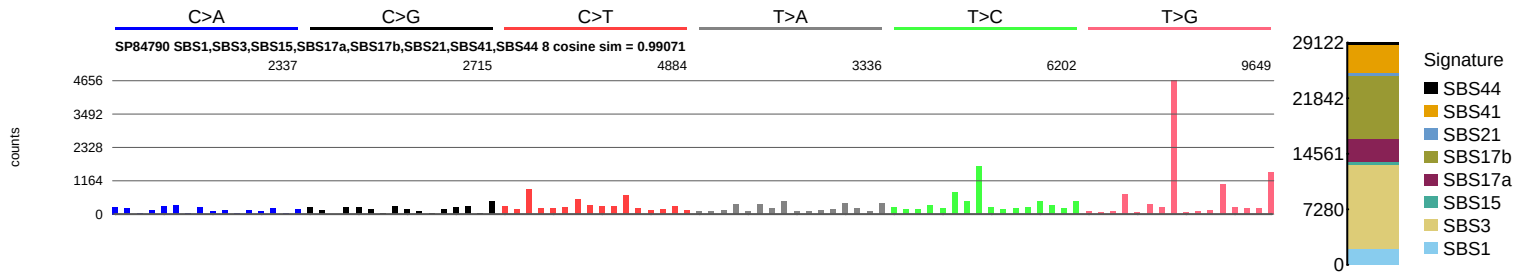

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS41,SBS44 8 cosine sim = 0.99076
2337
2827
4845
3289
6213
9613
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS1

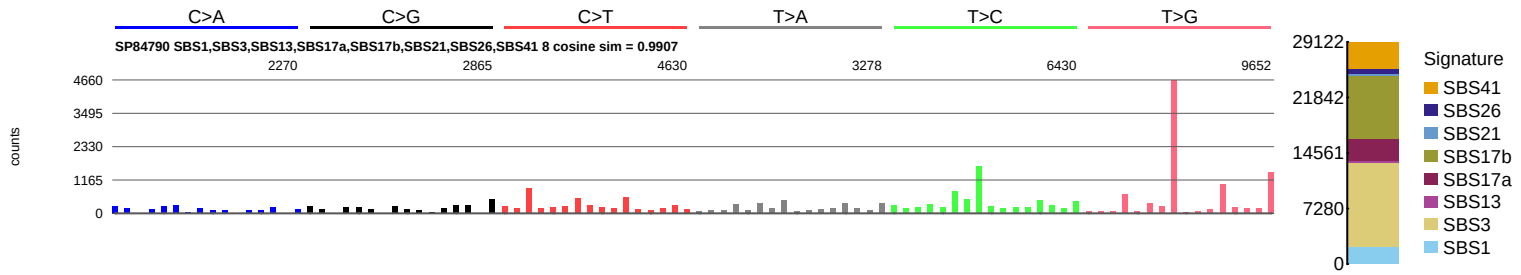

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS40 6 cosine sim = 0.99076
3402
2493
5658
2509
5420
9640
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS40
SBS17b
SBS17a
SBS3
SBS2
SBS1

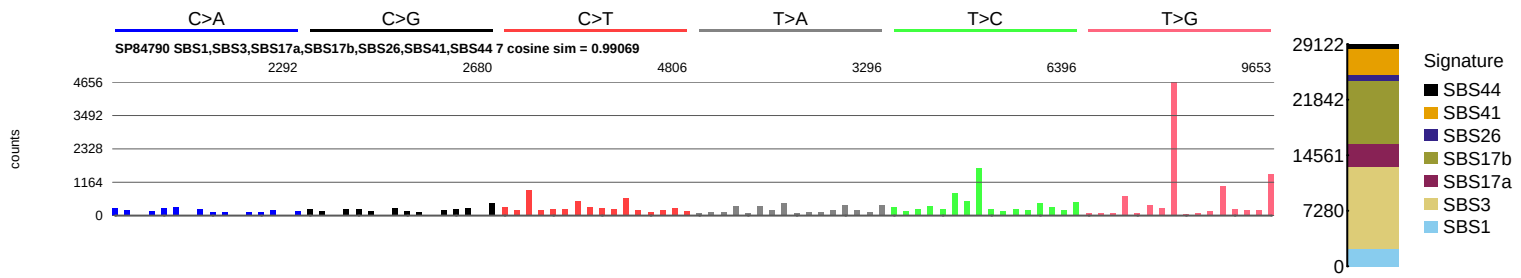

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18 8 cosine sim = 0.99076
3103
2586
5786
2357
6103
9188
4712
3534
2356
1178
0
counts
29122
21842
14561
7280
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

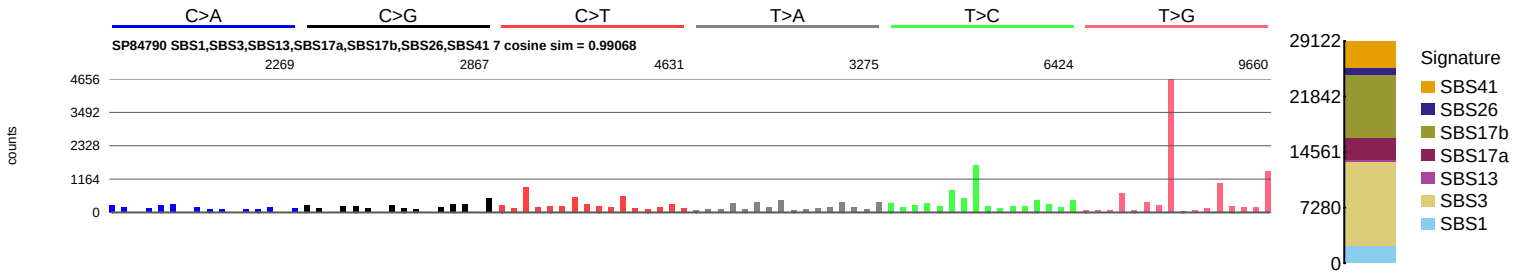

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS40 5 cosine sim = 0.99076
3409
2492
5649
2506
5421
9646
4664
3498
2332
1166
0
counts
29122
21842
14561
7280
0
Signature
SBS40
SBS17b
SBS17a
SBS3
SBS1

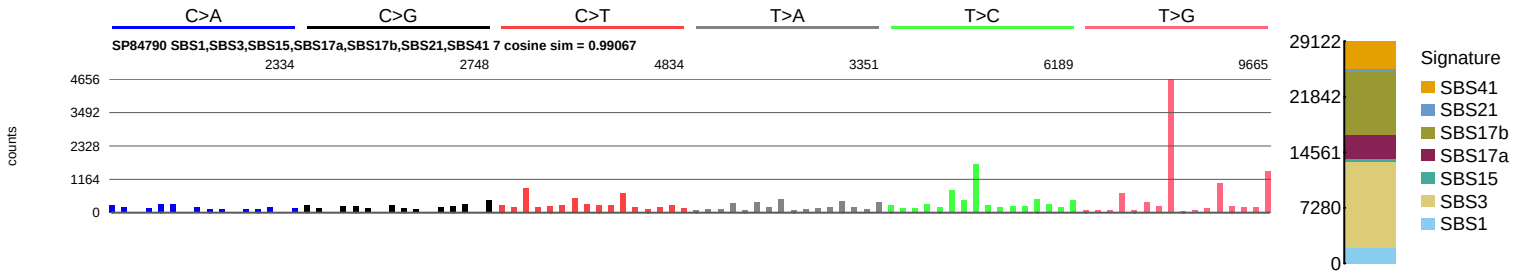

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 8 cosine sim = 0.99072
2293
2679
4807
3296
6402
9645
4656
3492
2328
1164
0
counts
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS3
SBS1

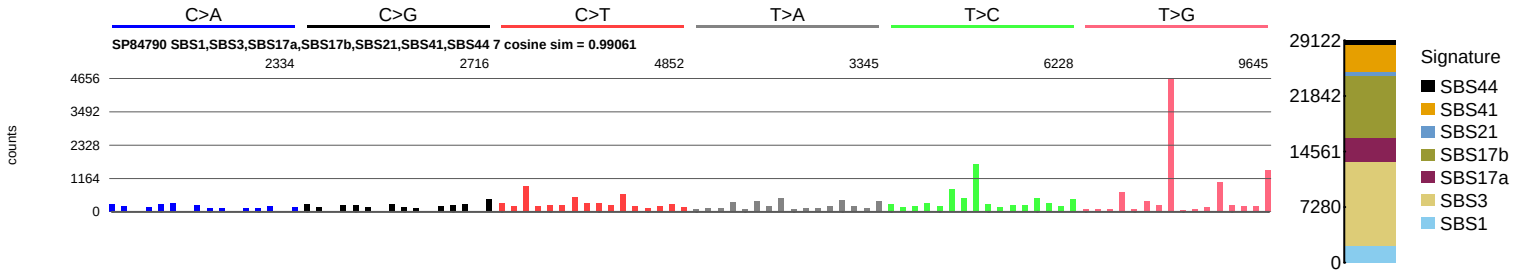

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS41,SBS44 8 cosine sim = 0.99071
2337
2715
4884
3336
6202
9649
4656
3492
2328
1164
0
counts
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS1

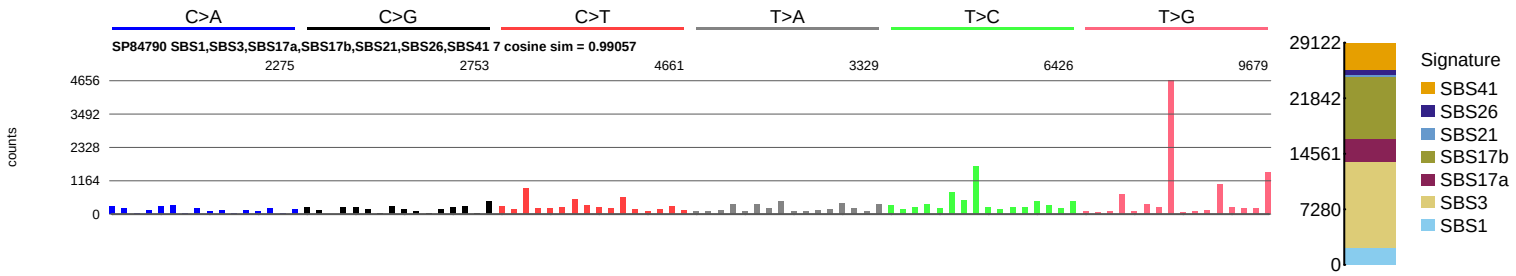

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS41 8 cosine sim = 0.9907
2270
2865
4630
3278
6430
9652
4660
3495
2330
1165
0
counts
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS1

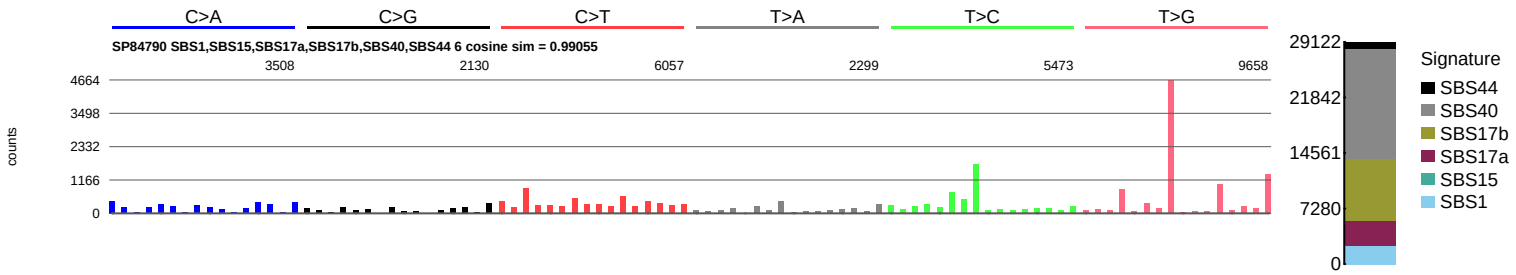

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS26,SBS41,SBS44 7 cosine sim = 0.99069
2292
2680
4806
3296
6396
9653
4656
3492
2328
1164
0
counts
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS3
SBS1

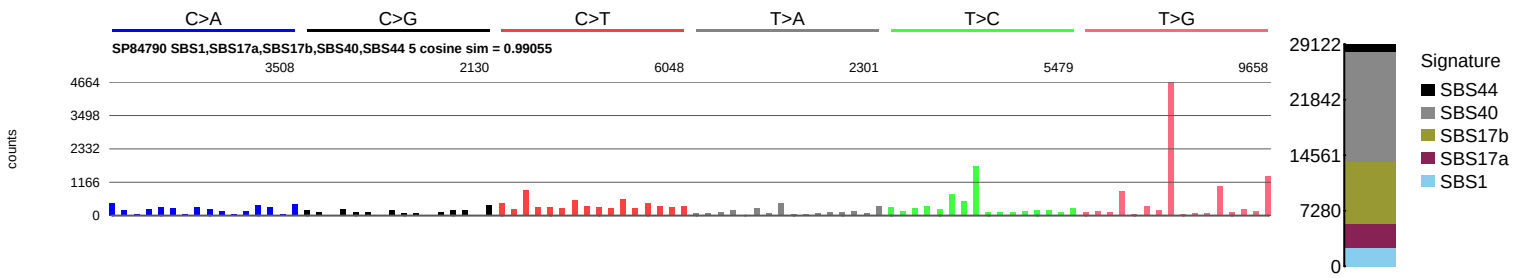

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS26,SBS41 7 cosine sim = 0.99068
2269
2867
4631
3275
6424
9660
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS13
SBS3
SBS1

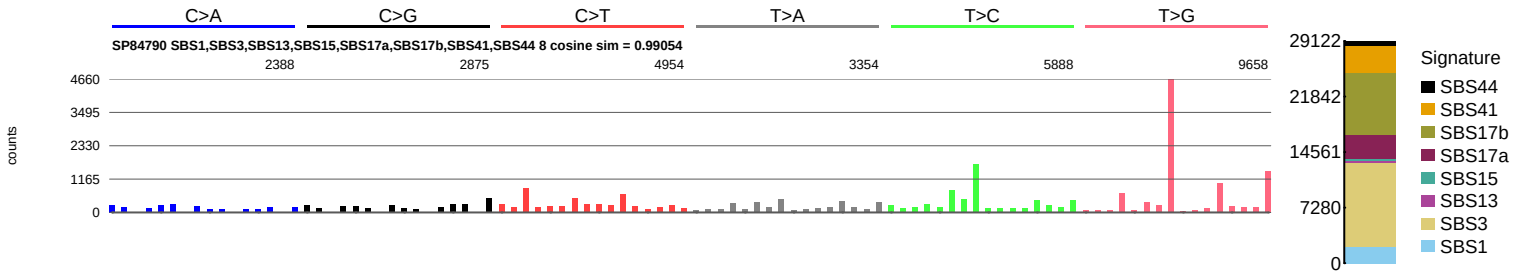

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.99067
2334
2748
4834
3351
6189
9665
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS1

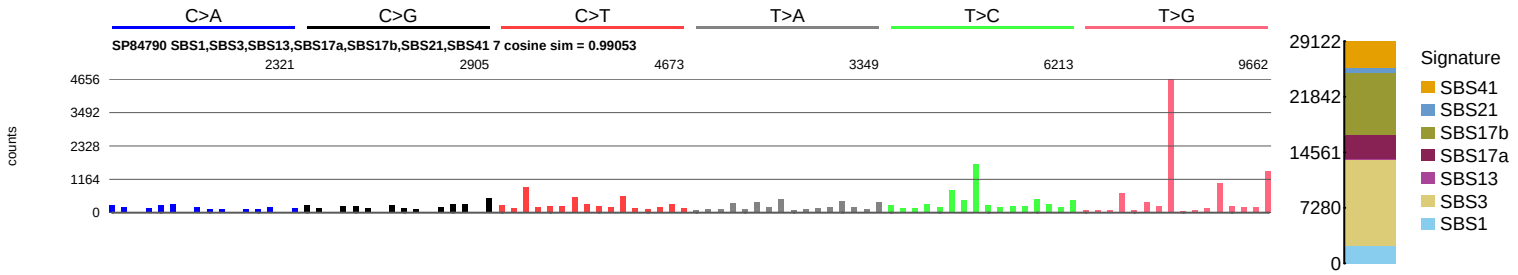

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS21,SBS41,SBS44 7 cosine sim = 0.99061
2334
2716
4852
3345
6228
9645
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS3
SBS1

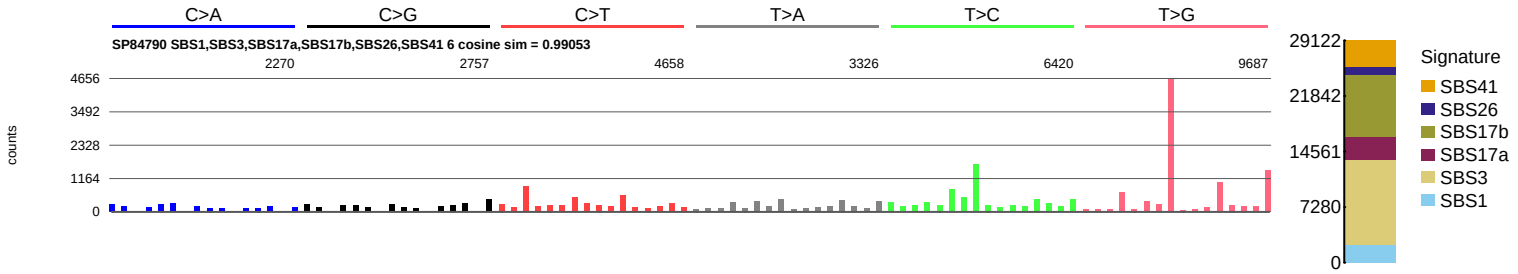

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS21,SBS26,SBS41 7 cosine sim = 0.99057
2275
2753
4661
3329
6426
9679
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS3
SBS1

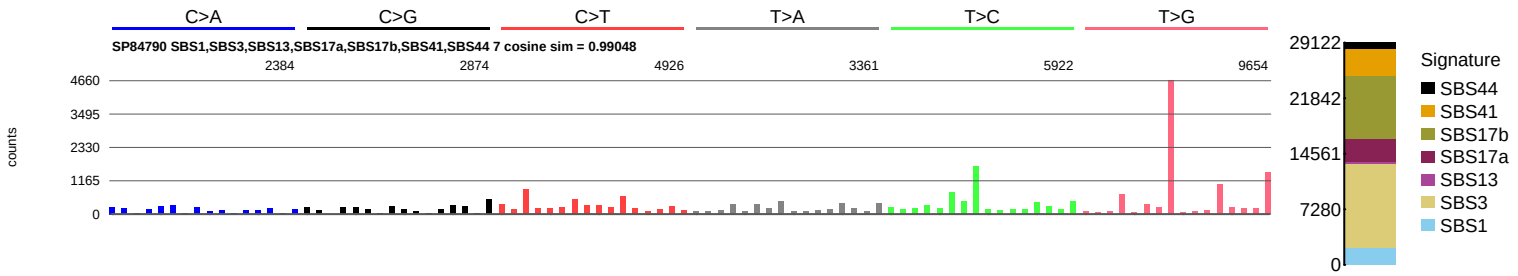

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS15,SBS17a,SBS17b,SBS40,SBS44 6 cosine sim = 0.99055
3508
2130
6057
2299
5473
9658
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS15
SBS1

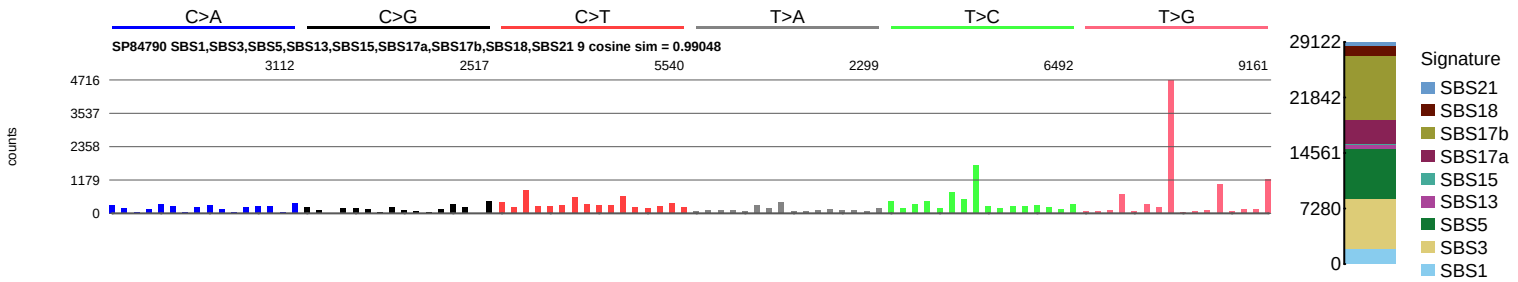

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS17a,SBS17b,SBS40,SBS44 5 cosine sim = 0.99055
3508
2130
6048
2301
5479
9658
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS1

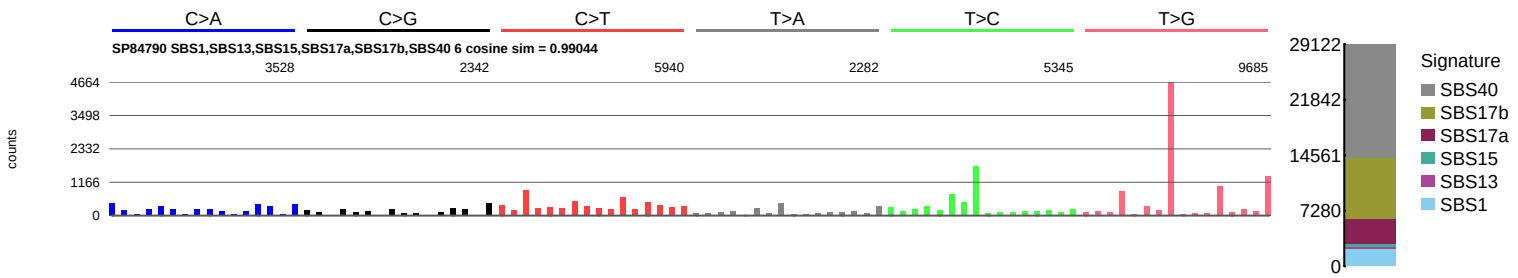

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS41,SBS44 8 cosine sim = 0.99054
2388
2875
4954
3354
5888
9658
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

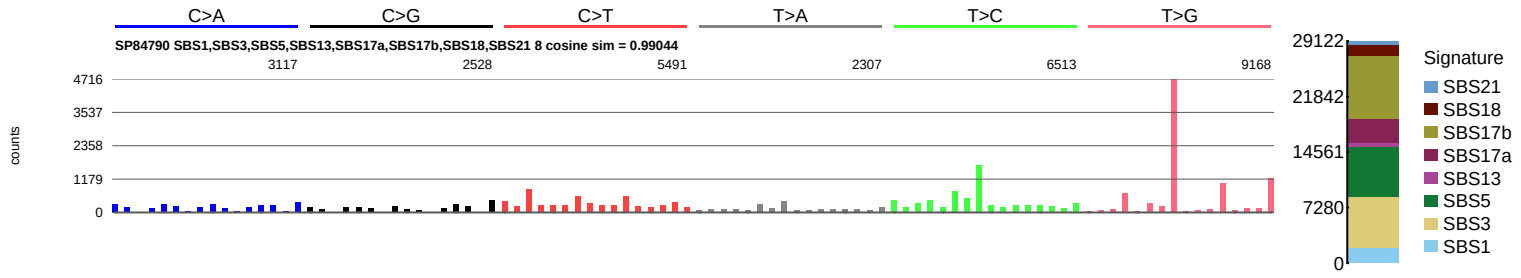

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.99053
2321
2905
4673
3349
6213
9662
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS1

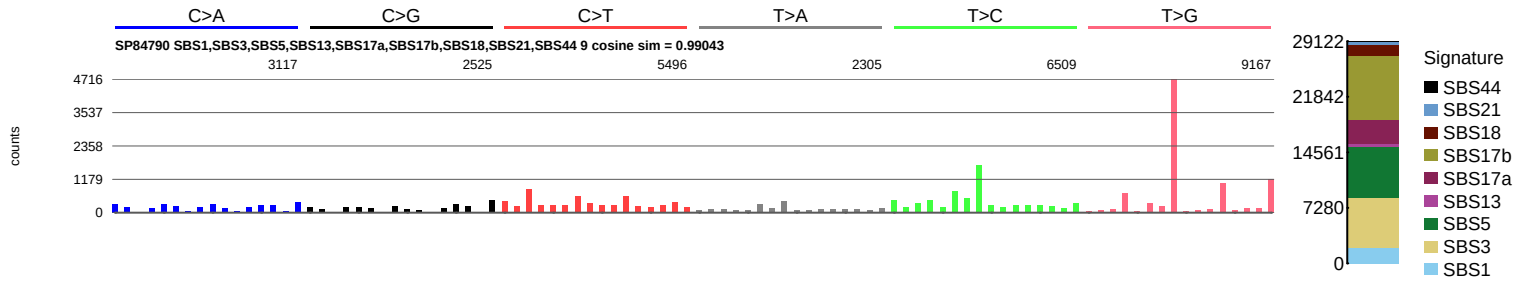

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS26,SBS41 6 cosine sim = 0.99053
2270
2757
4658
3326
6420
9687
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS3
SBS1

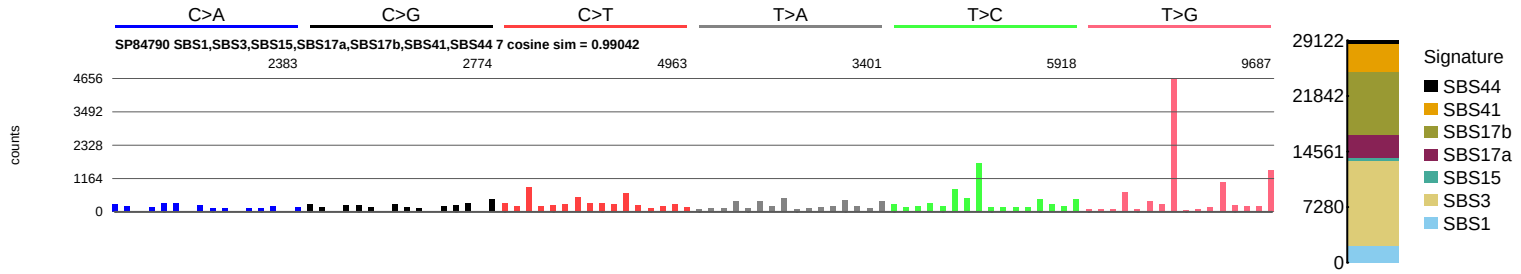

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS41,SBS44 7 cosine sim = 0.99048
2384
2874
4926
3361
5922
9654
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS13
SBS3
SBS1

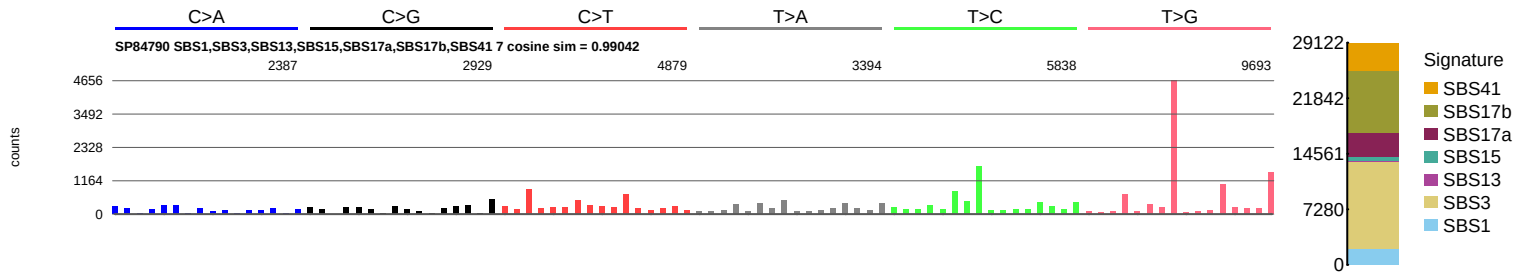

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.99048
3112
2517
5540
2299
6492
9161
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

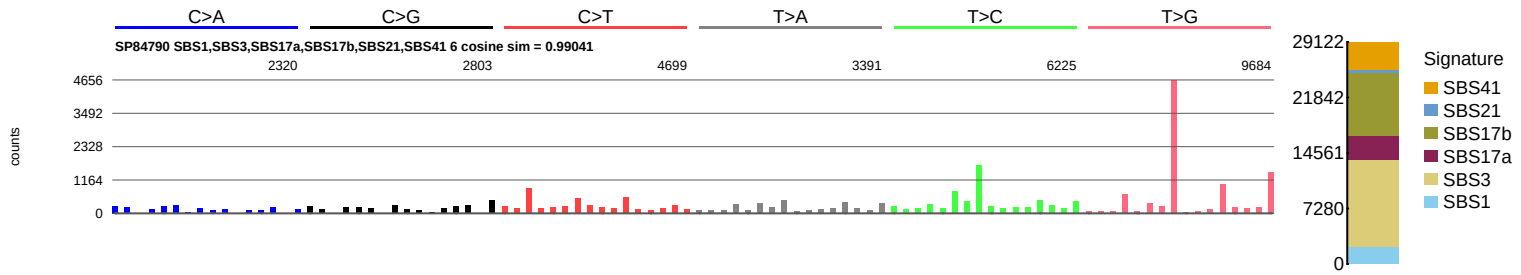

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS40 6 cosine sim = 0.99044
3528
2342
5940
2282
5345
9685
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS1

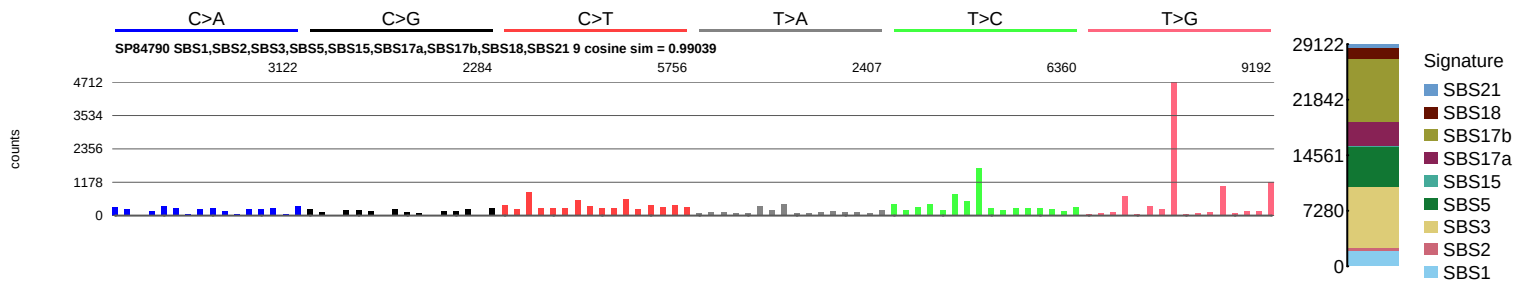

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.99044
3117
2528
5491
2307
6513
9168
4716
3537
2358
1179
0
counts
29122
21842
14561
7280
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

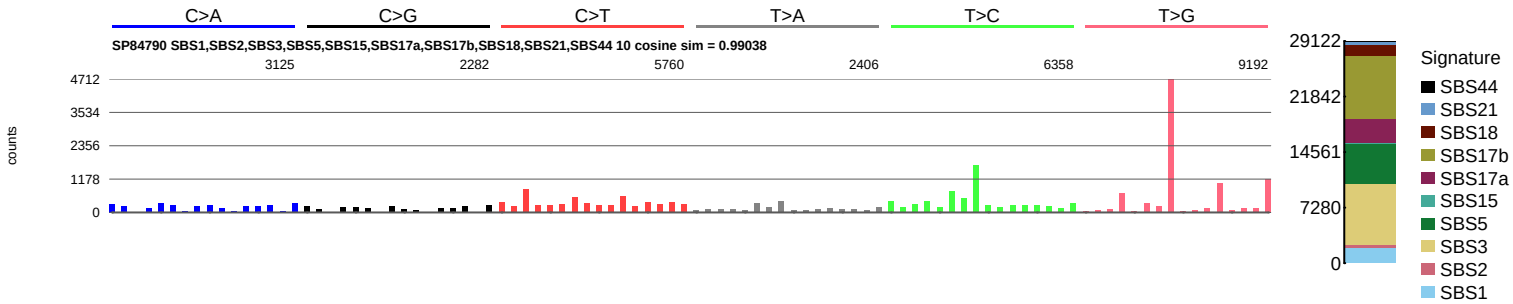

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.99043
3117
2525
5496
2305
6509
9167
4716
3537
2358
1179
0
counts
29122
21842
14561
7280
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

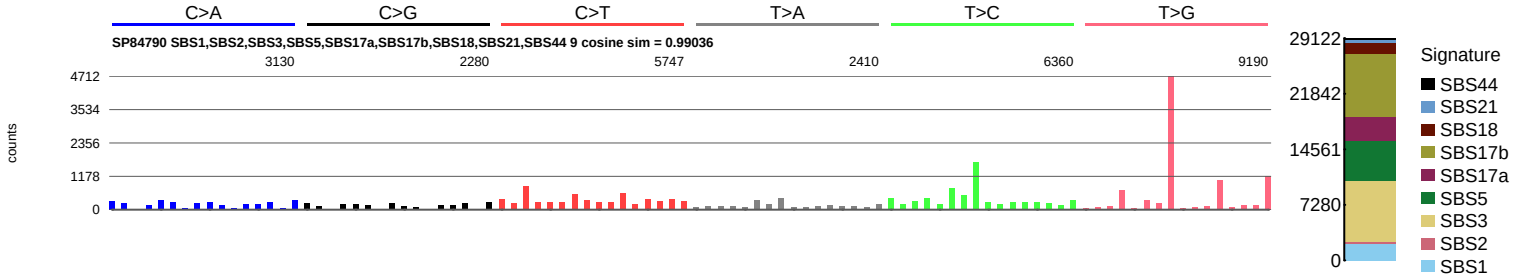

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS41,SBS44 7 cosine sim = 0.99042
2383
2774
4963
3401
5918
9687
4656
3492
2328
1164
0
counts
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS15
SBS3
SBS1

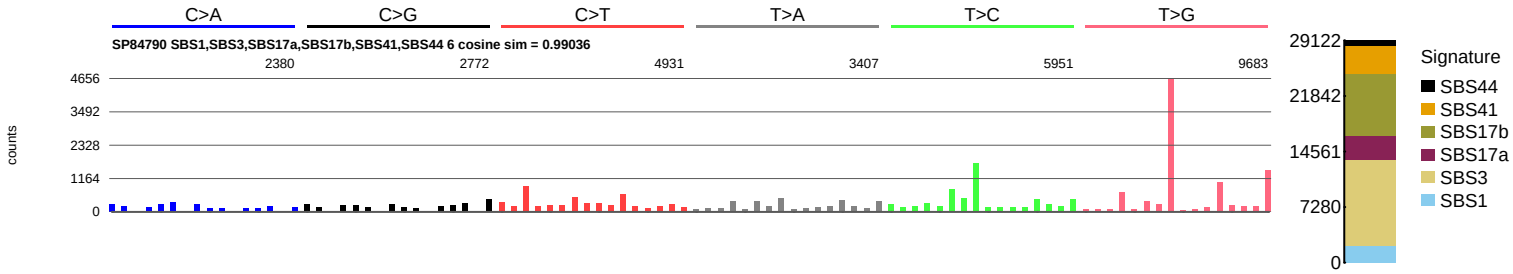

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.99042
2387
2929
4879
3394
5838
9693
4656
3492
2328
1164
0
counts
29122
21842
14561
7280
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

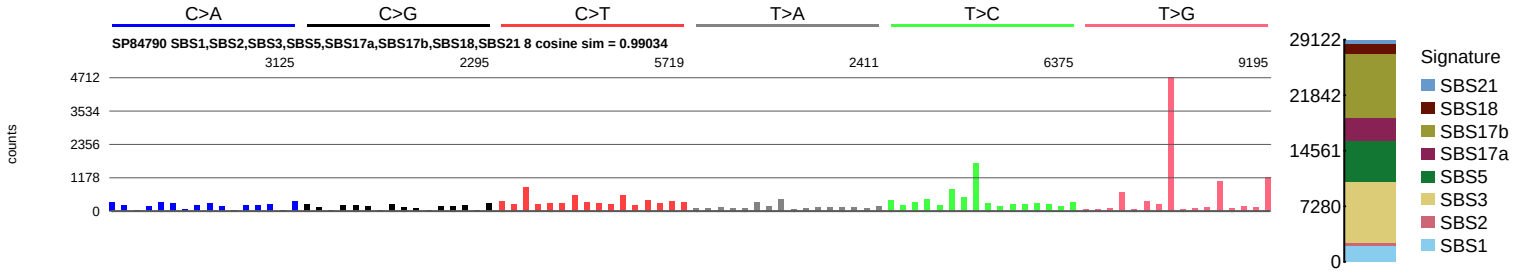

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.99041
2320
2803
4699
3391
6225
9684
4656
3492
2328
1164
0
counts
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS3
SBS1

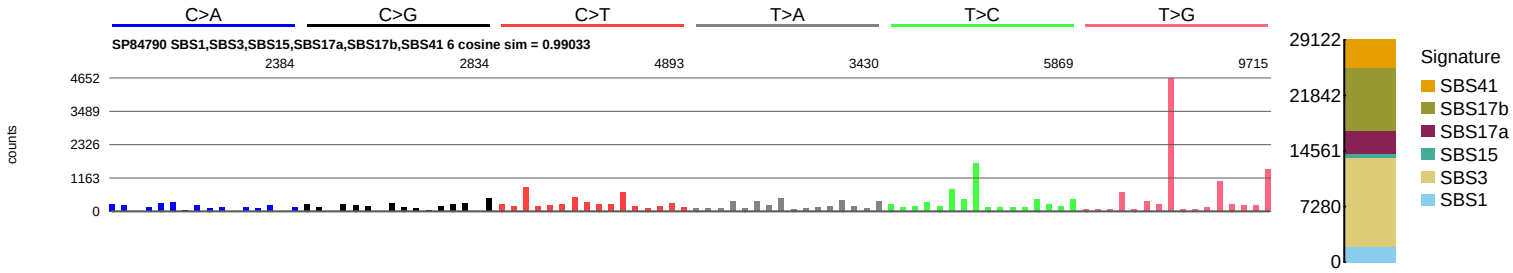

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.99039
3122
2284
5756
2407
6360
9192
4712
3534
2356
1178
0
counts
29122
21842
14561
7280
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

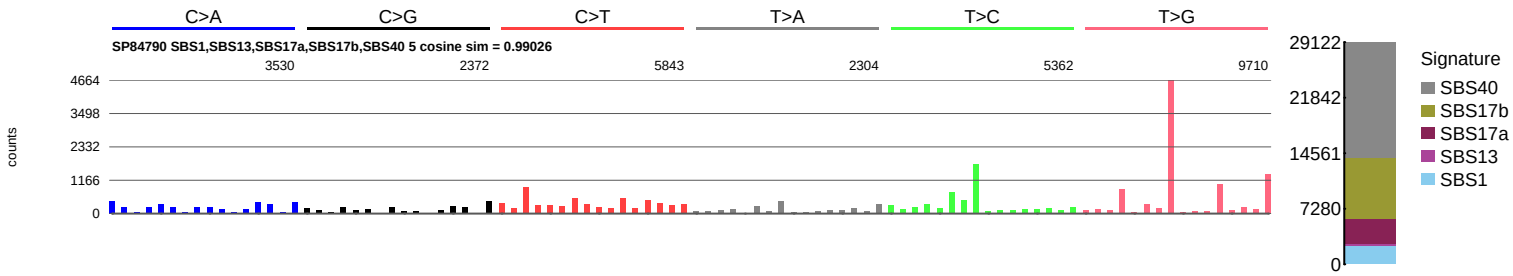

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.99038
3125
2282
5760
2406
6358
9192
counts
4712
3534
2356
1178
0
29122
21842
14561
7280
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

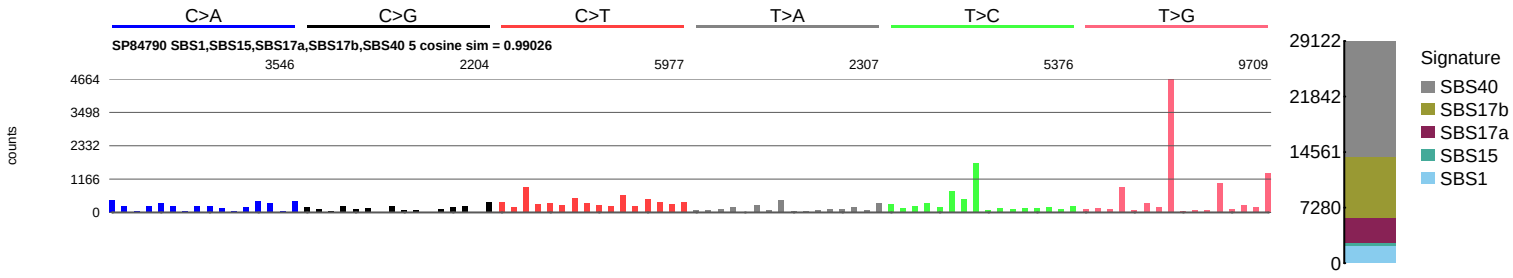

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.99036
3130
2280
5747
2410
6360
9190
counts
4712
3534
2356
1178
0
29122
21842
14561
7280
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

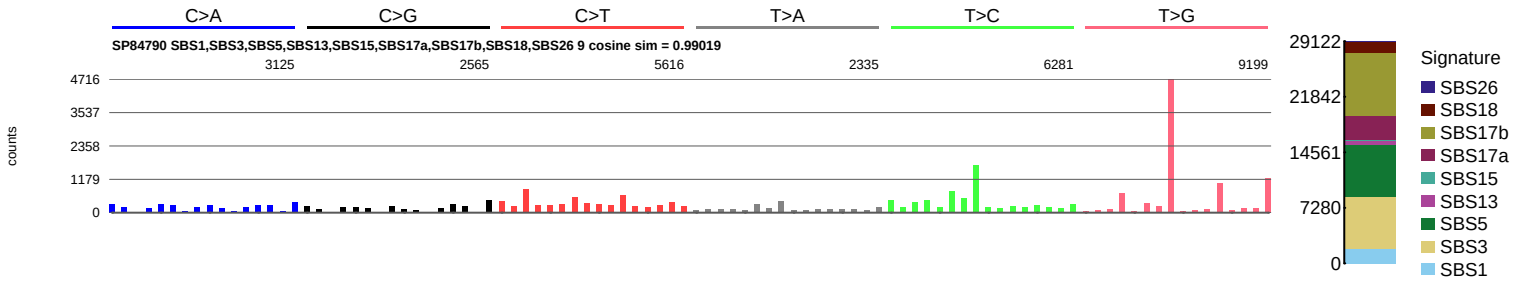

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS41,SBS44 6 cosine sim = 0.99036
2380
2772
4931
3407
5951
9683
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS3
SBS1

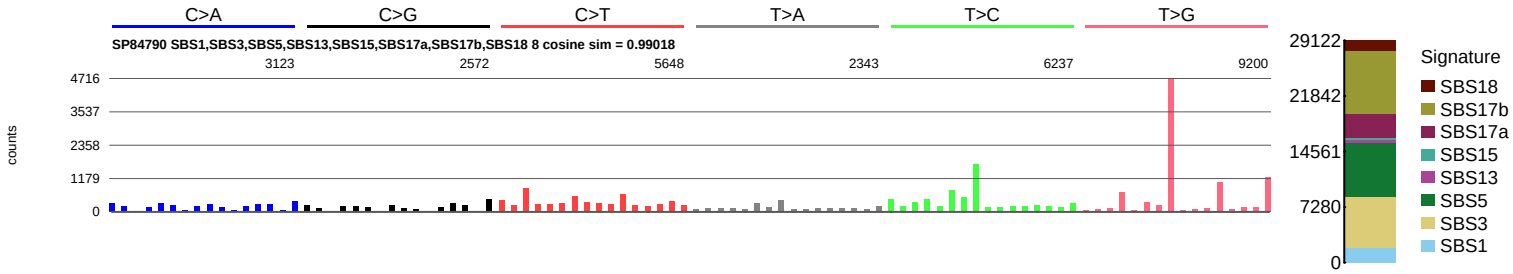

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.99034
3125
2295
5719
2411
6375
9195
counts
4712
3534
2356
1178
0
29122
21842
14561
7280
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

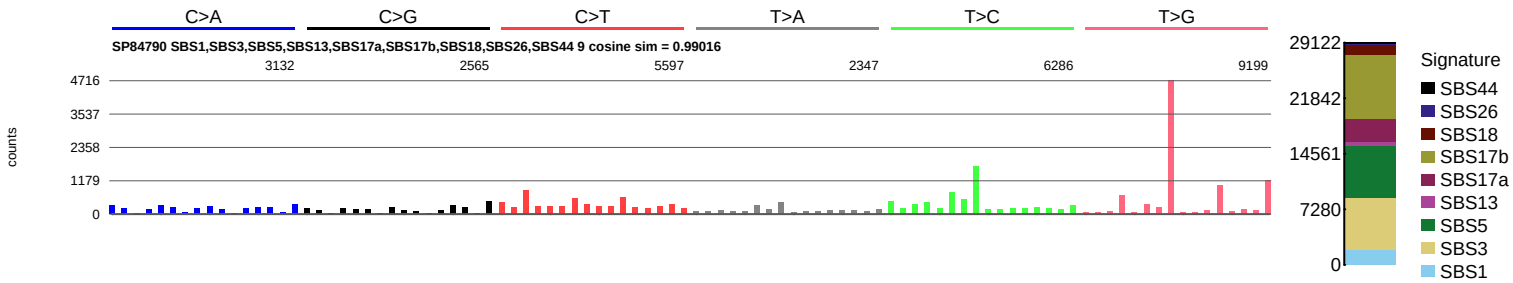

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS41 6 cosine sim = 0.99033
2384
2834
4893
3430
5869
9715
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS3
SBS1

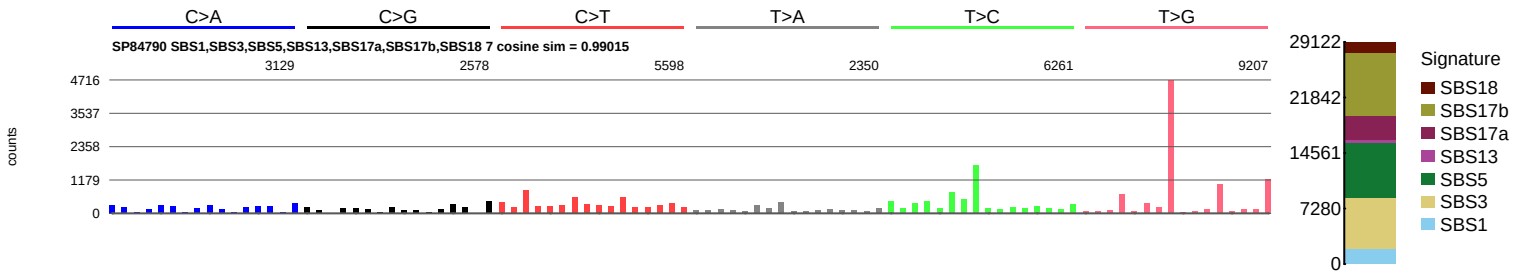

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS17a,SBS17b,SBS40 5 cosine sim = 0.99026
3530
2372
5843
2304
5362
9710
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS40
SBS17b
SBS17a
SBS13
SBS1

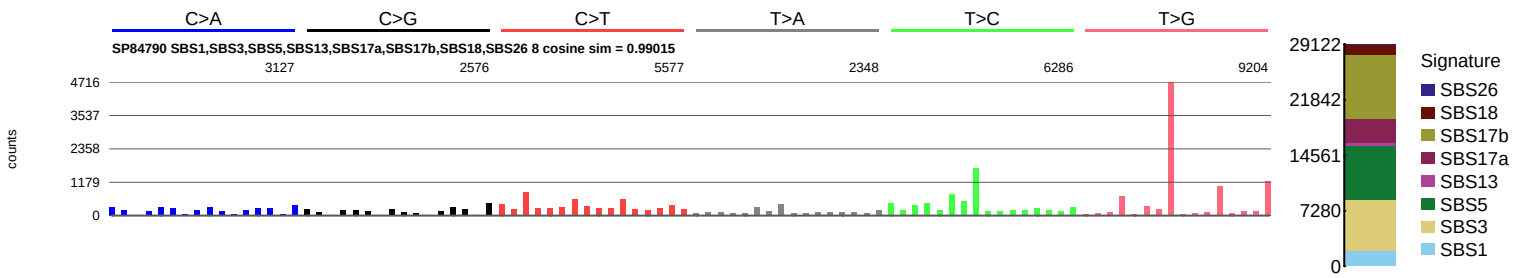

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS15,SBS17a,SBS17b,SBS40 5 cosine sim = 0.99026
3546
2204
5977
2307
5376
9709
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS1

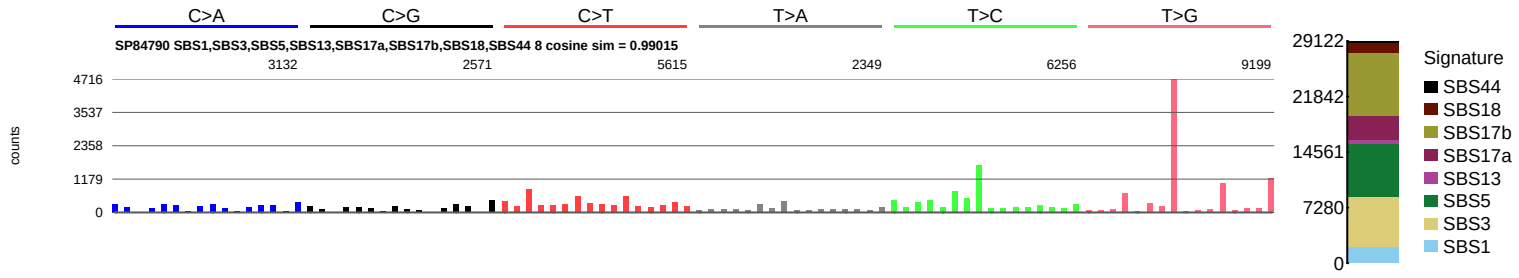

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 9 cosine sim = 0.99019
3125
2565
5616
2335
6281
9199
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

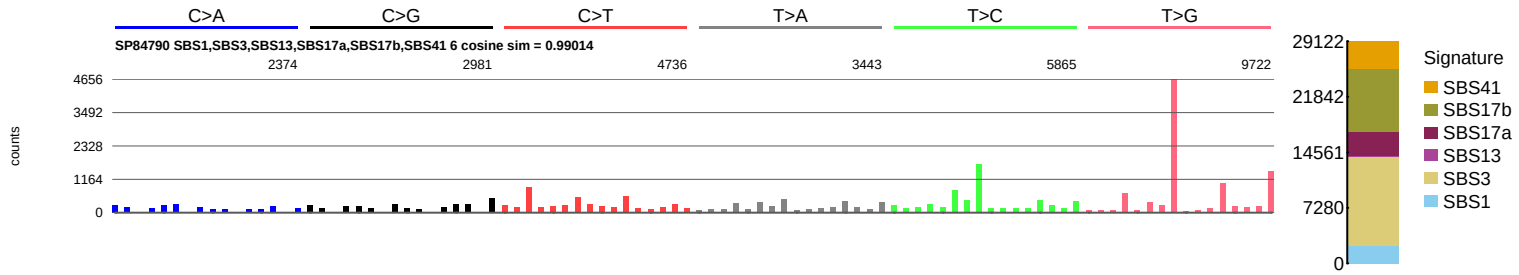

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.99018
3123
2572
5648
2343
6237
9200
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

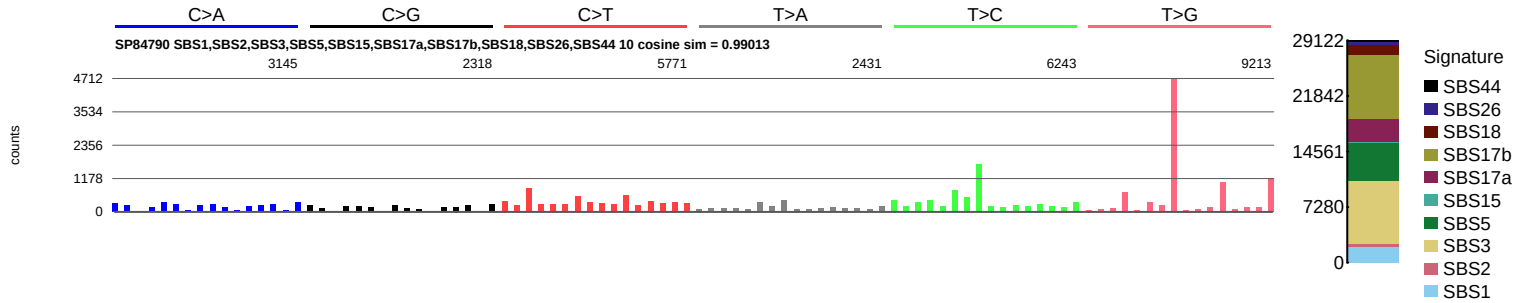

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.99016
3132
2565
5597
2347
6286
9199
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

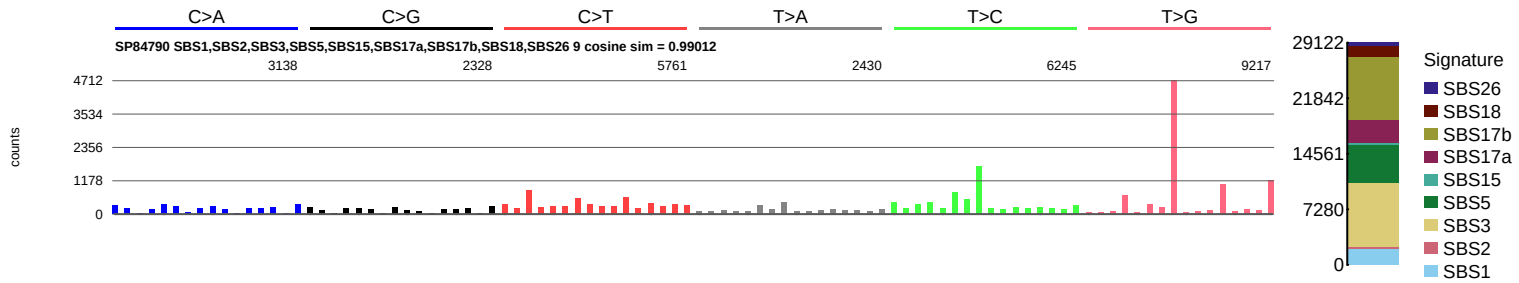

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18 7 cosine sim = 0.99015
3129
2578
5598
2350
6261
9207
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

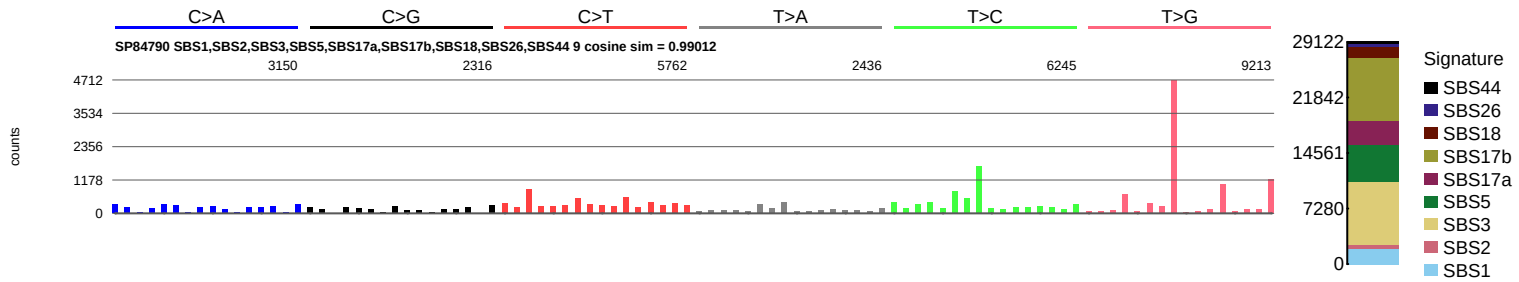

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.99015
3127
2576
5577
2348
6286
9204
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

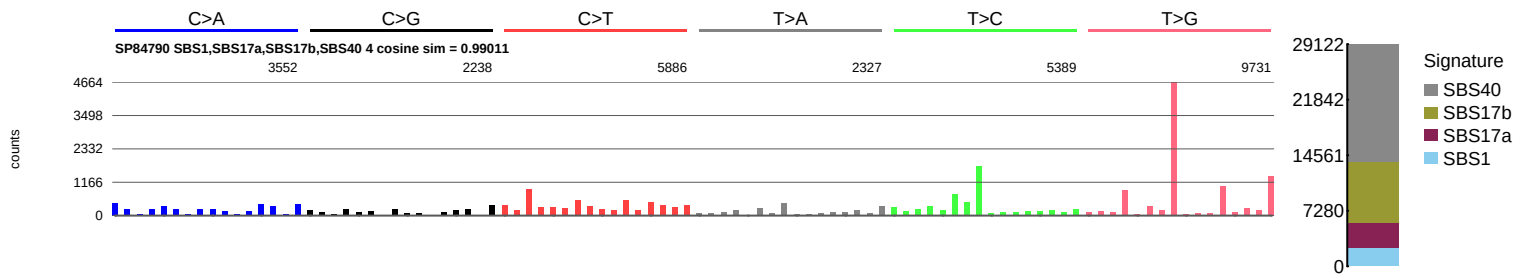

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.99015
3132
2571
5615
2349
6256
9199
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

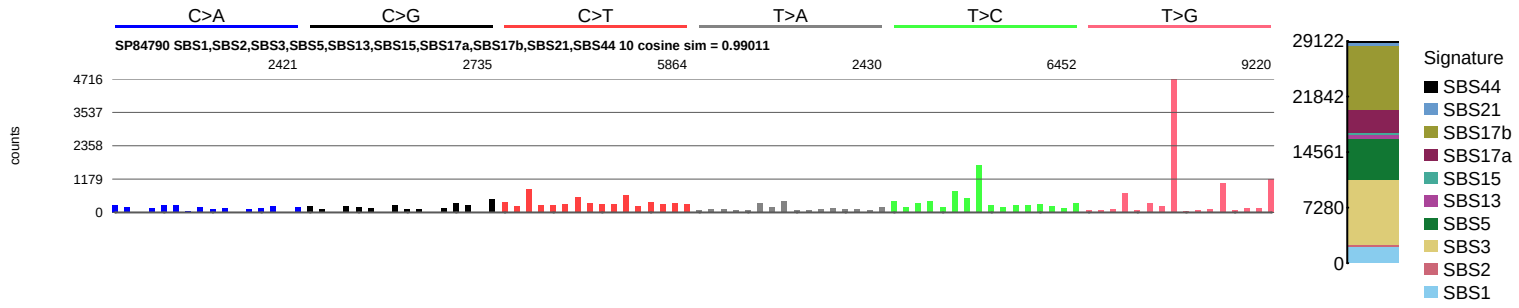

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS41 6 cosine sim = 0.99014
2374
2981
4736
3443
5865
9722
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS41
SBS17b
SBS17a
SBS13
SBS3
SBS1

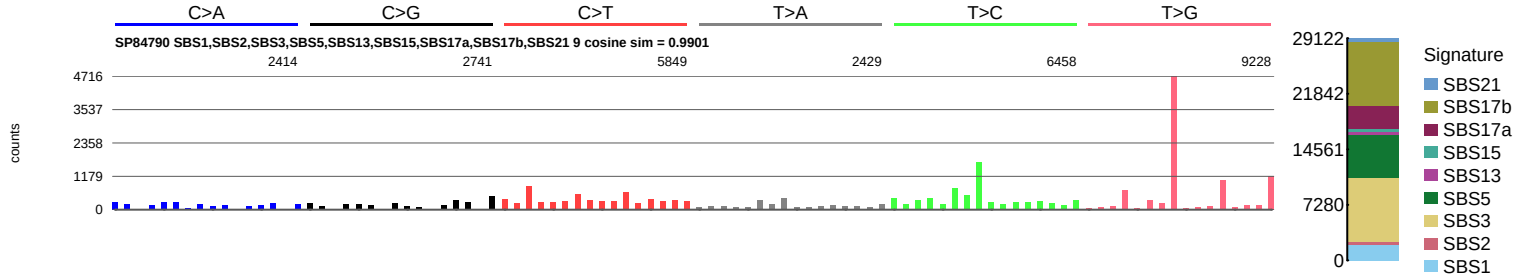

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 10 cosine sim = 0.99013
3145
2318
5771
2431
6243
9213
counts
4712
3534
2356
1178
0
29122
21842
14561
7280
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

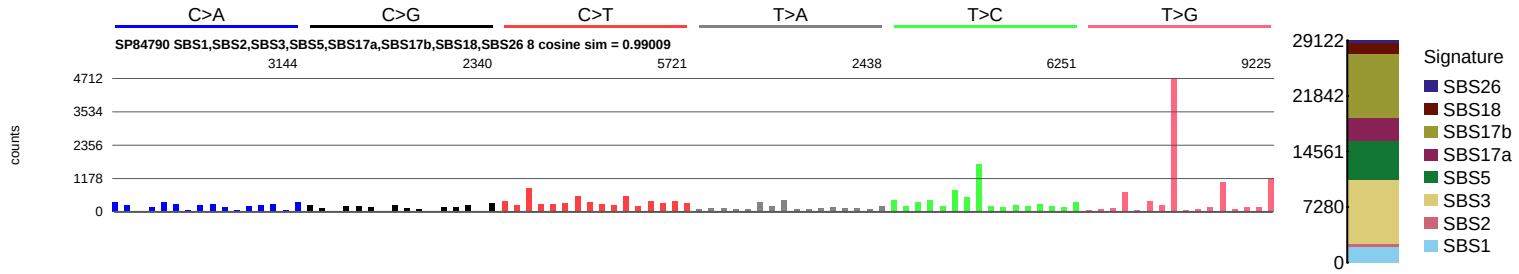

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26 9 cosine sim = 0.99012
3138
2328
5761
2430
6245
9217
counts
4712
3534
2356
1178
0
29122
21842
14561
7280
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

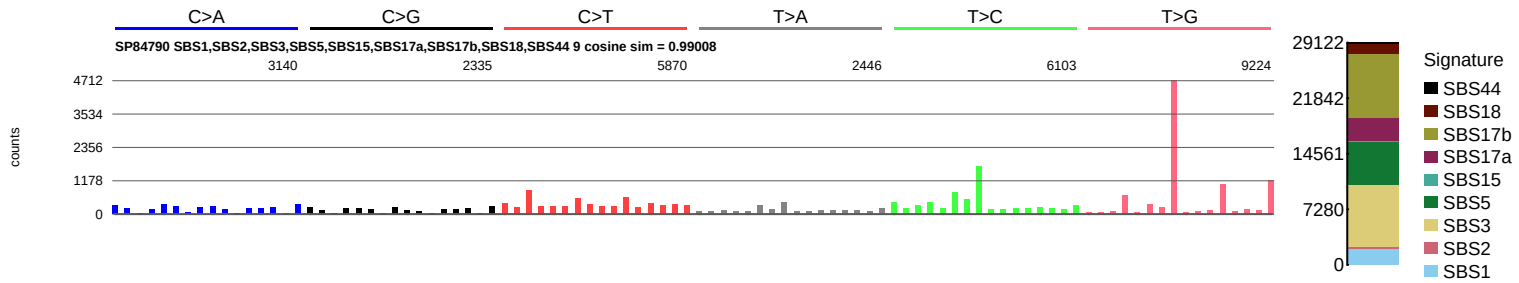

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.99012
3150
2316
5762
2436
6245
9213
counts
4712
3534
2356
1178
0
29122
21842
14561
7280
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

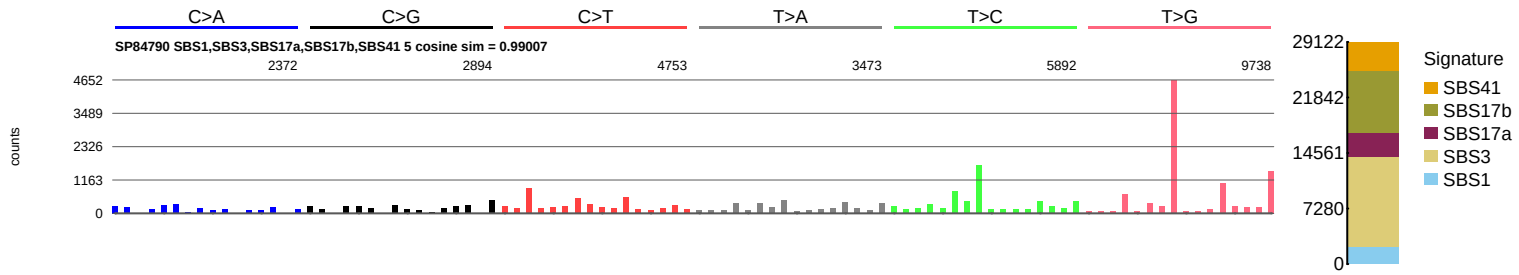

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS17a,SBS17b,SBS40 4 cosine sim = 0.99011
3552
2238
5886
2327
5389
9731
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS40
SBS17b
SBS17a
SBS1

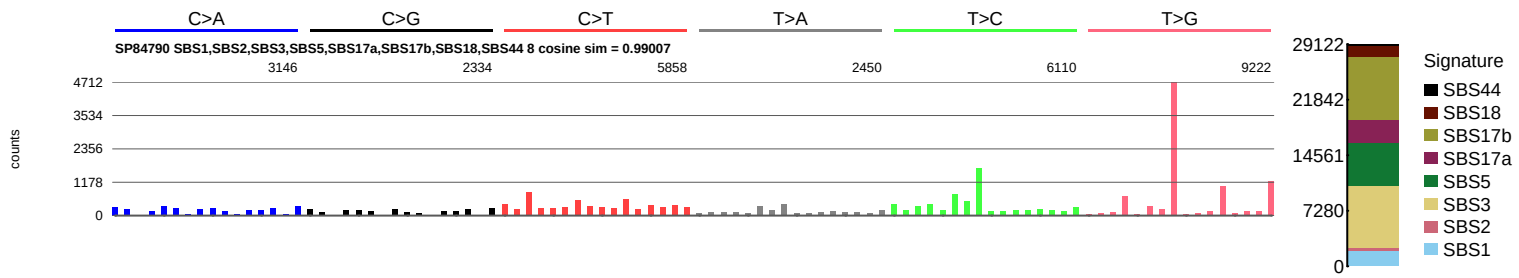

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS44 10 cosine sim = 0.99011
2421
2735
5864
2430
6452
9220
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

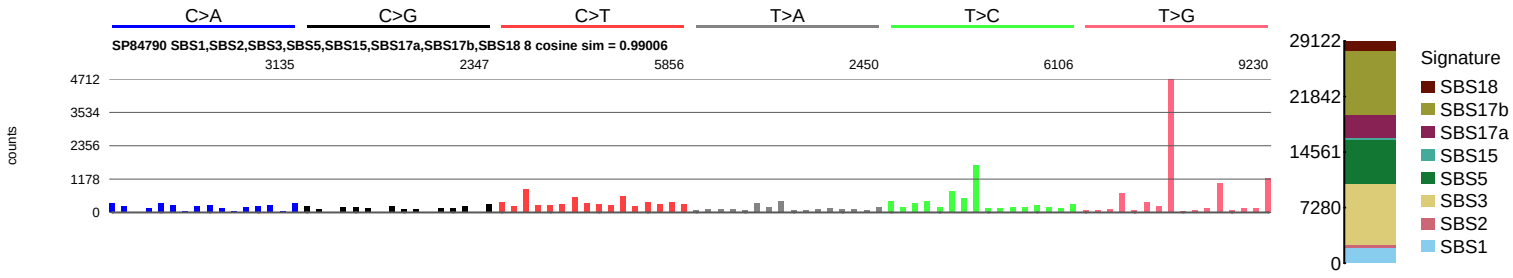

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21 9 cosine sim = 0.9901
2414
2741
5849
2429
6458
9228
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

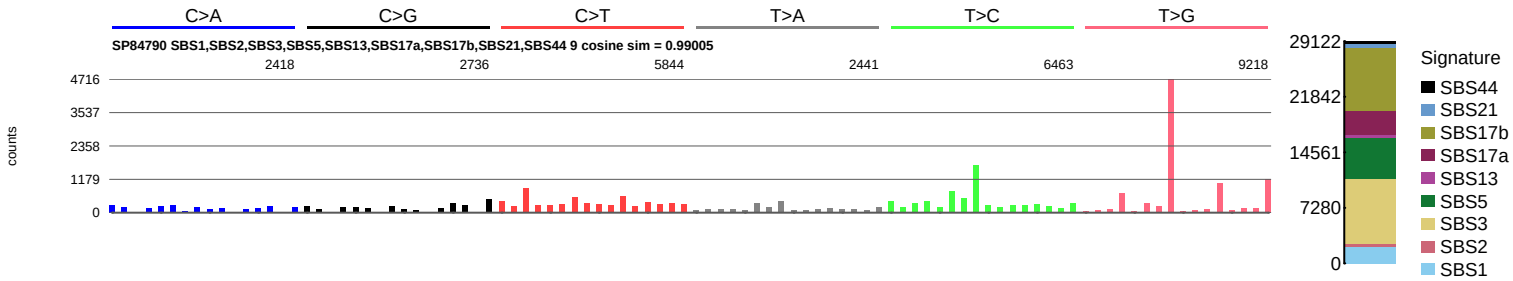

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.99009
3144
2340
5721
2438
6251
9225
counts
4712
3534
2356
1178
0
29122
21842
14561
7280
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

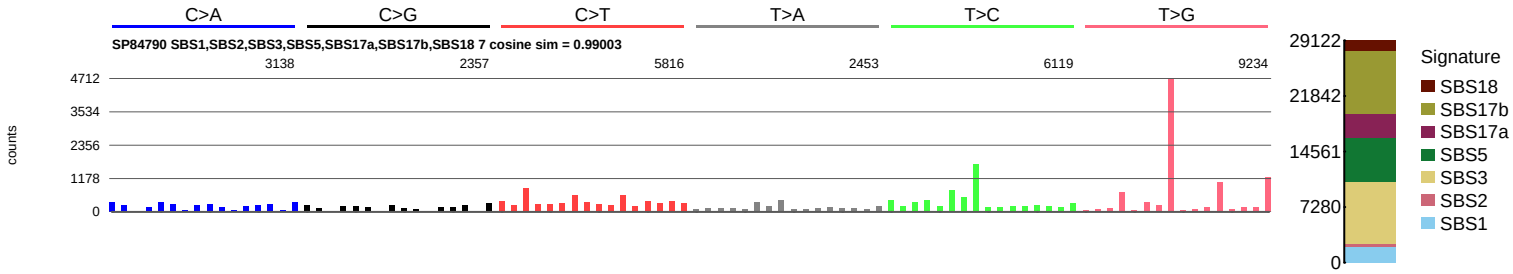

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS44 9 cosine sim = 0.99008
3140
2335
5870
2446
6103
9224
counts
4712
3534
2356
1178
0
29122
21842
14561
7280
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

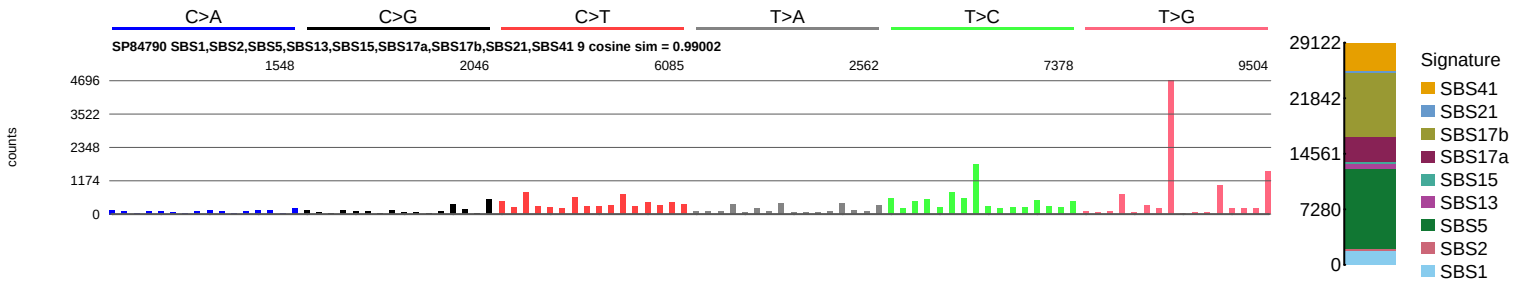

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS41 5 cosine sim = 0.99007
2372
2894
4753
3473
5892
9738
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS41
SBS17b
SBS17a
SBS3
SBS1

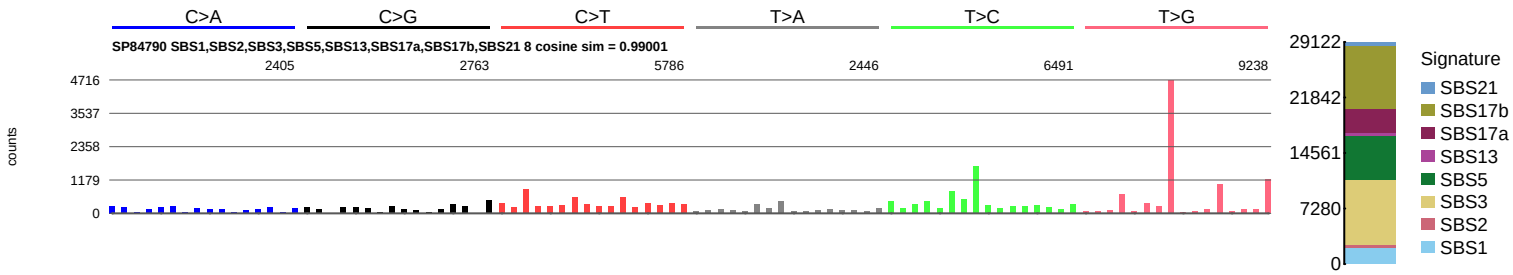

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.99007
3146
2334
5858
2450
6110
9222
counts
4712
3534
2356
1178
0
29122
21842
14561
7280
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

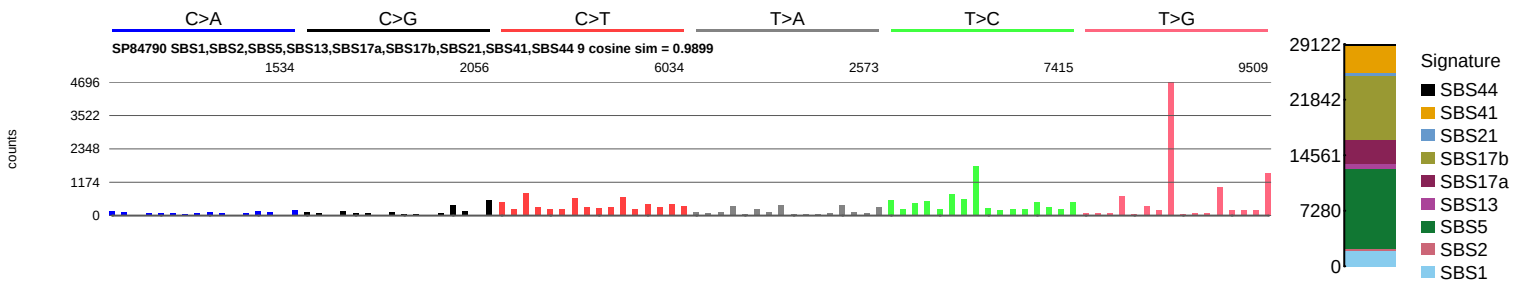

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.99006
3135
2347
5856
2450
6106
9230
counts
4712
3534
2356
1178
0
29122
21842
14561
7280
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

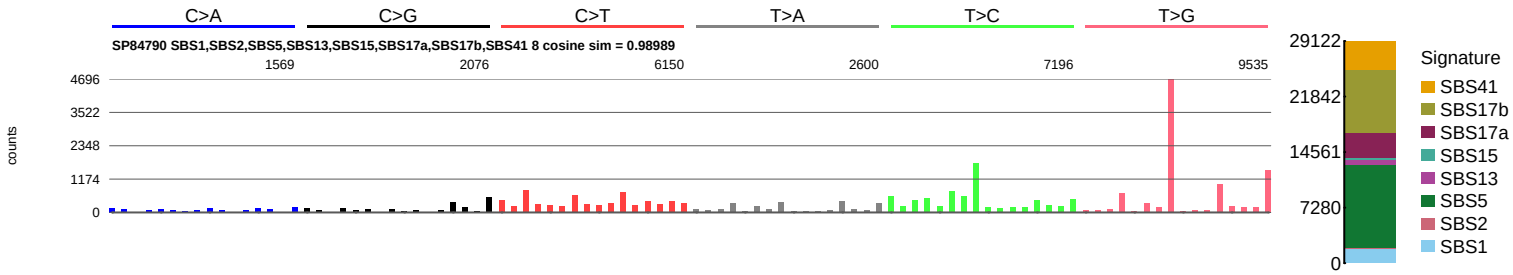

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS44 9 cosine sim = 0.99005
2418
2736
5844
2441
6463
9218
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

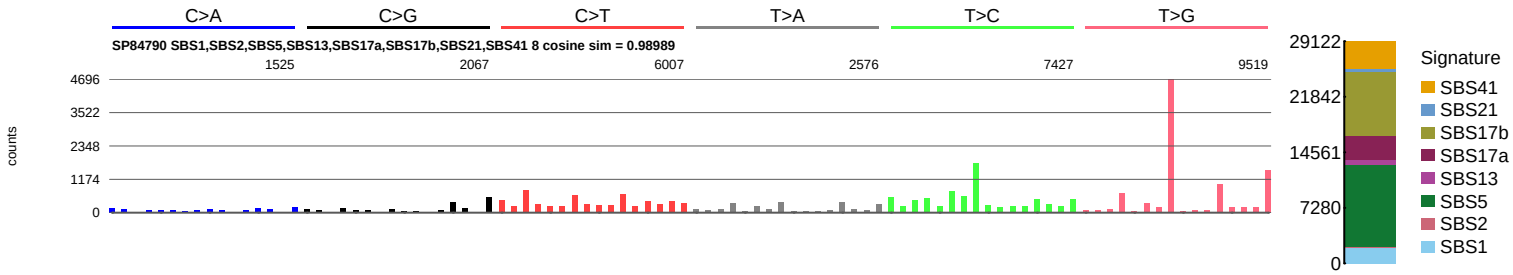

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18 7 cosine sim = 0.99003
3138
2357
5816
2453
6119
9234
counts
4712
3534
2356
1178
0
29122
21842
14561
7280
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

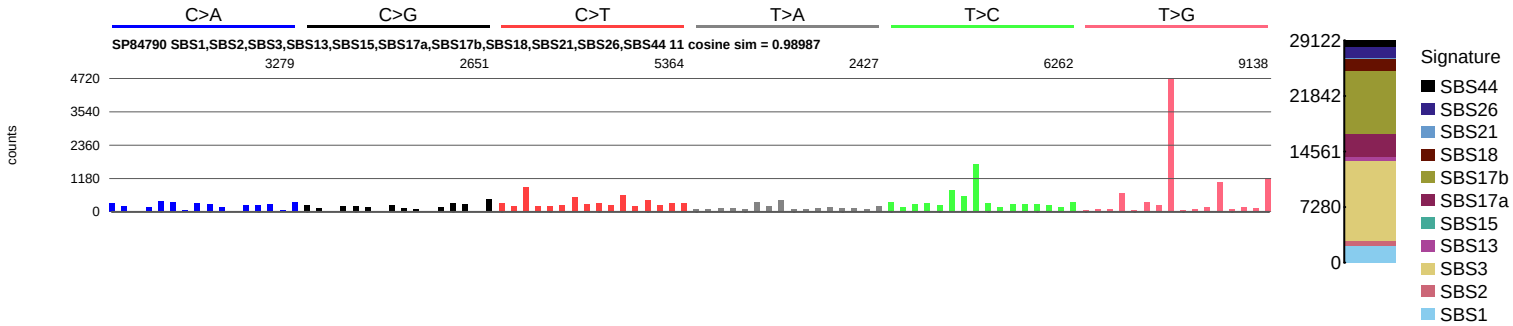

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 9 cosine sim = 0.99002
1548
2046
6085
2562
7378
9504
counts
4696
3522
2348
1174
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

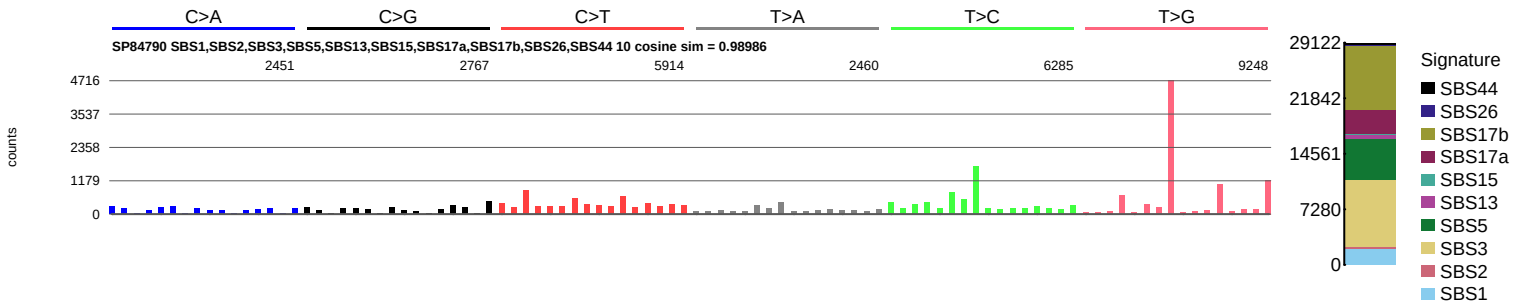

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21 8 cosine sim = 0.99001
2405
2763
5786
2446
6491
9238
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

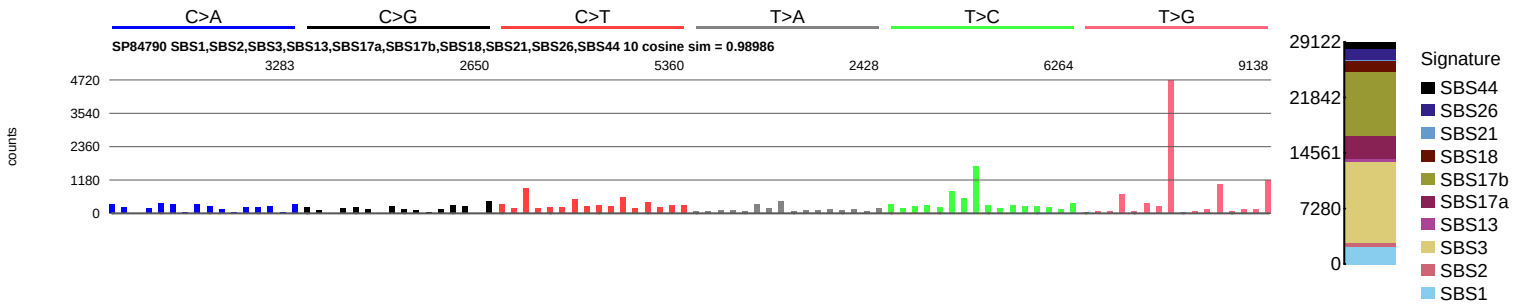

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS41,SBS44 9 cosine sim = 0.9899
1534
2056
6034
2573
7415
9509
counts
4696
3522
2348
1174
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

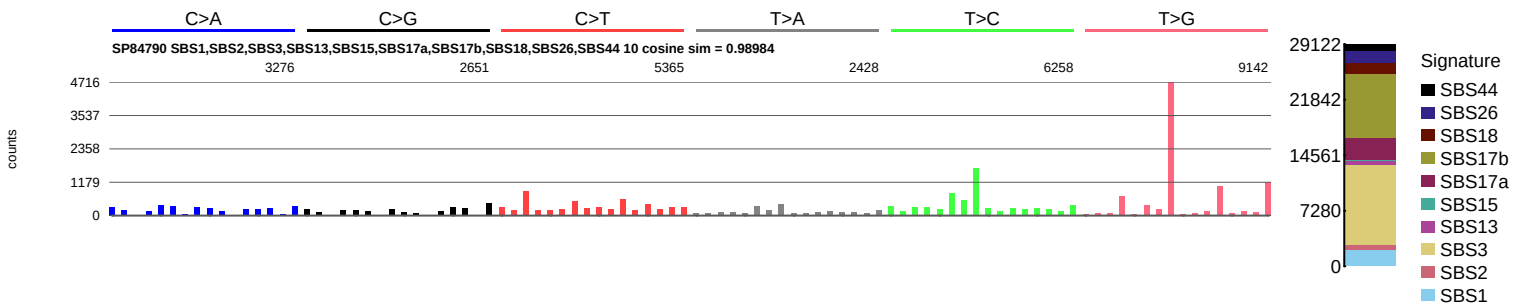

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS41 8 cosine sim = 0.98989
1569
2076
6150
2600
7196
9535
counts
4696
3522
2348
1174
0
29122
21842
14561
7280
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

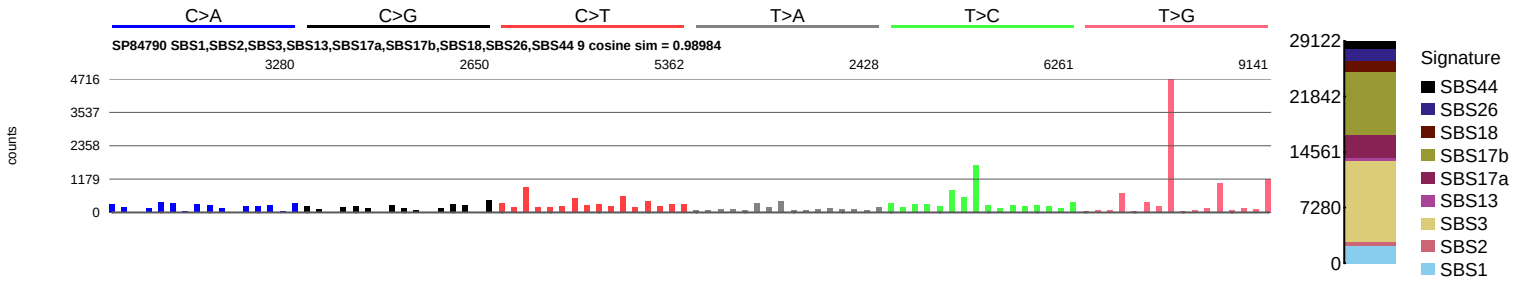

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.98989
1525
2067
6007
2576
7427
9519
counts
4696
3522
2348
1174
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

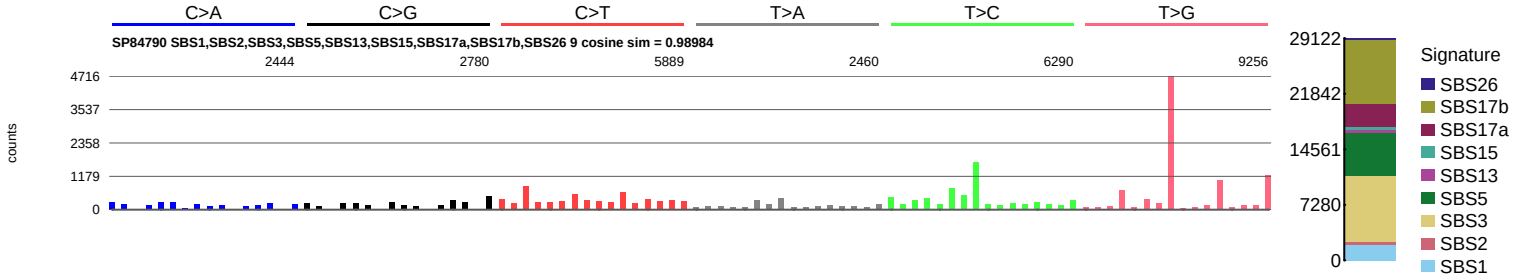

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 11 cosine sim = 0.98987
3279
2651
5364
2427
6262
9138
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

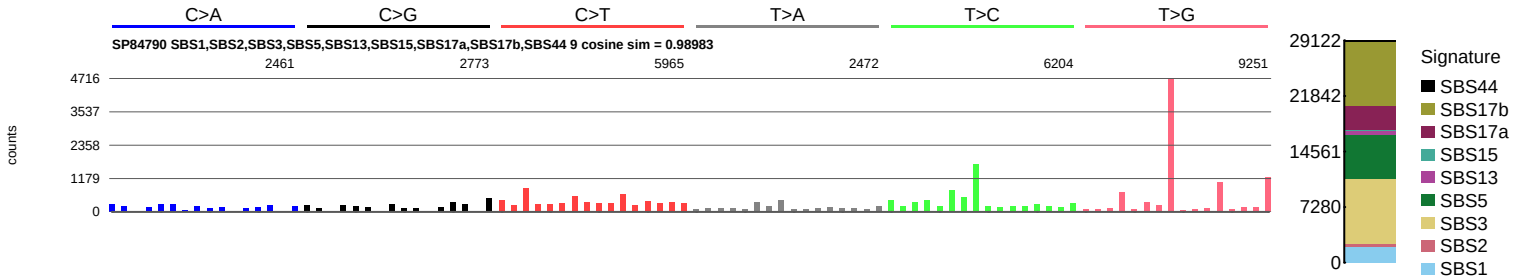

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS44 10 cosine sim = 0.98986
2451
2767
5914
2460
6285
9248
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

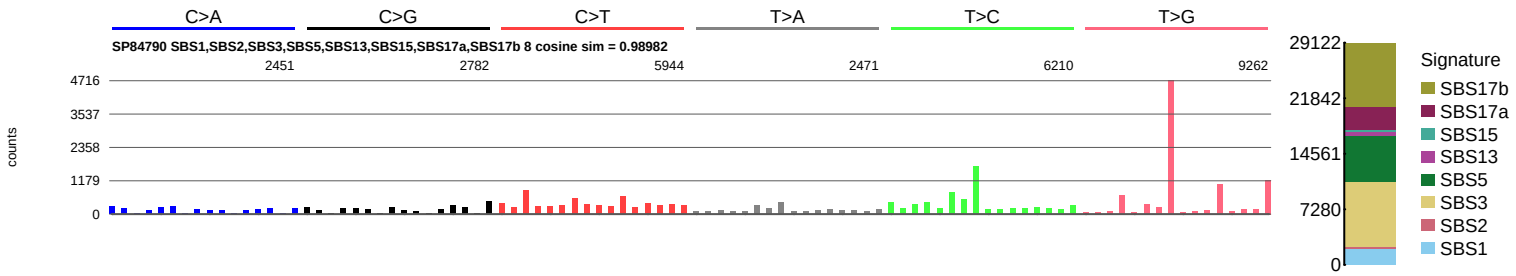

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 10 cosine sim = 0.98986
3283
2650
5360
2428
6264
9138
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

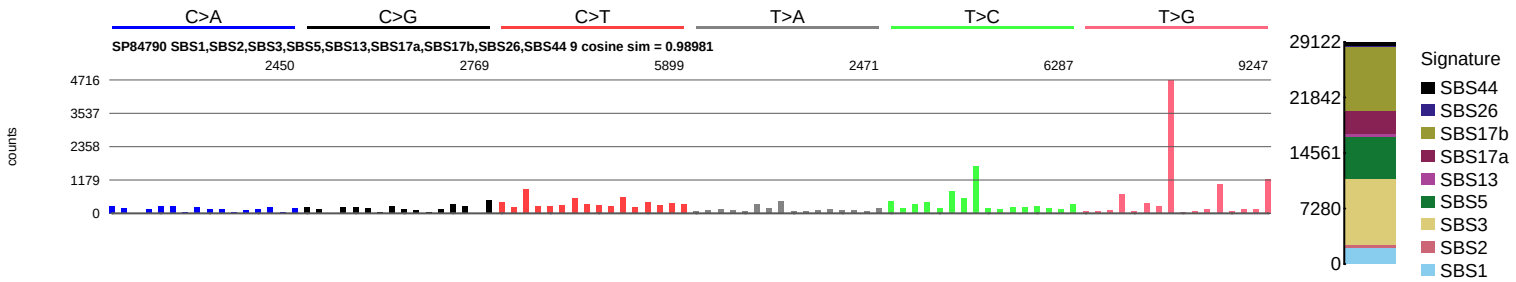

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 10 cosine sim = 0.98984
3276
2651
5365
2428
6258
9142
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

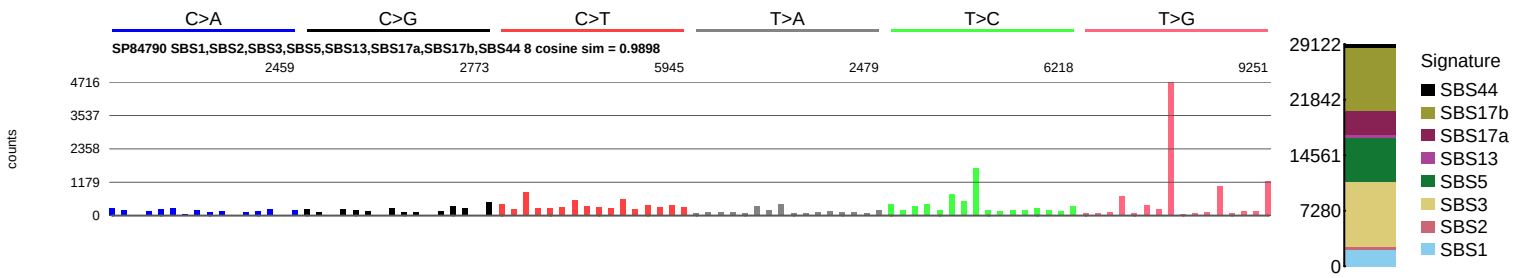

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.98984
3280
2650
5362
2428
6261
9141
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

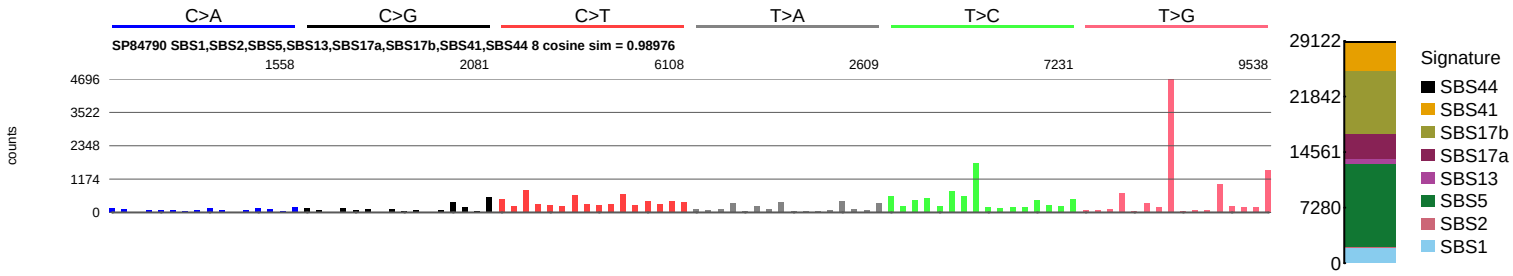

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26 9 cosine sim = 0.98984
2444
2780
5889
2460
6290
9256
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

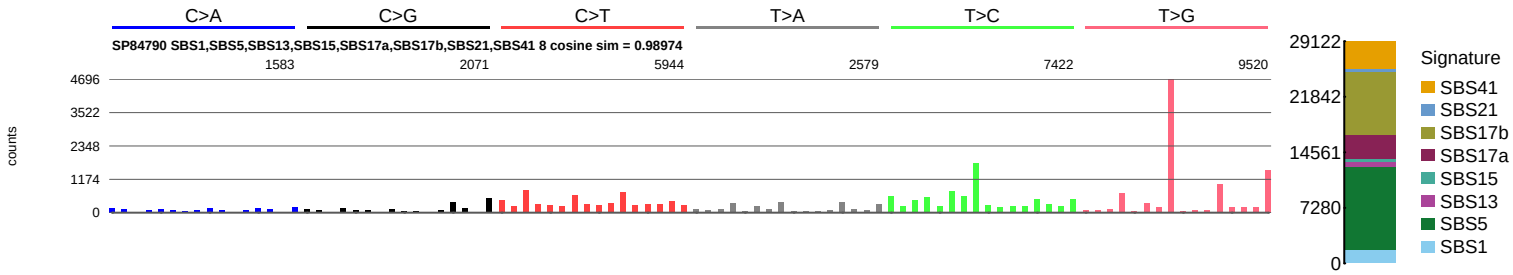

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS44 9 cosine sim = 0.98983
2461
2773
5965
2472
6204
9251
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS44
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

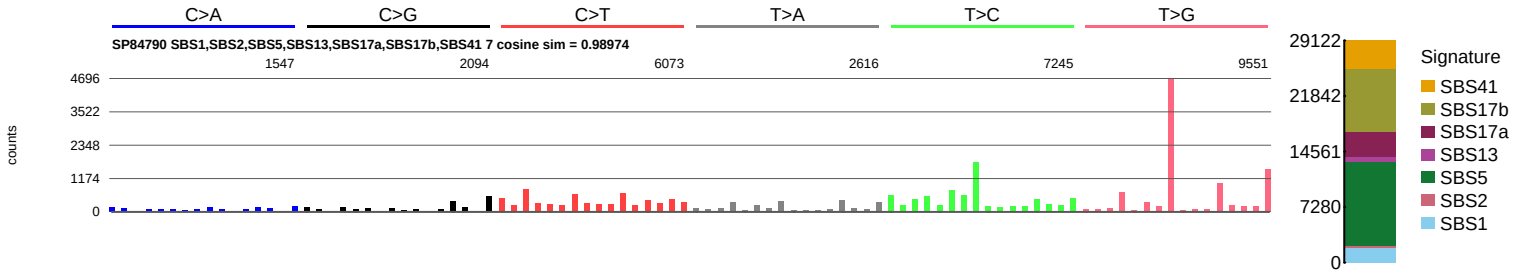

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b 8 cosine sim = 0.98982
2451
2782
5944
2471
6210
9262
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

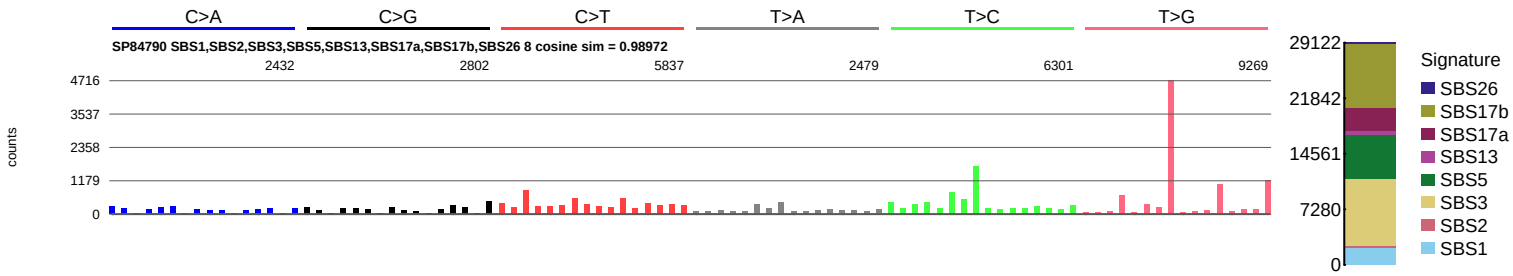

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS26,SBS44 9 cosine sim = 0.98981
2450
2769
5899
2471
6287
9247
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

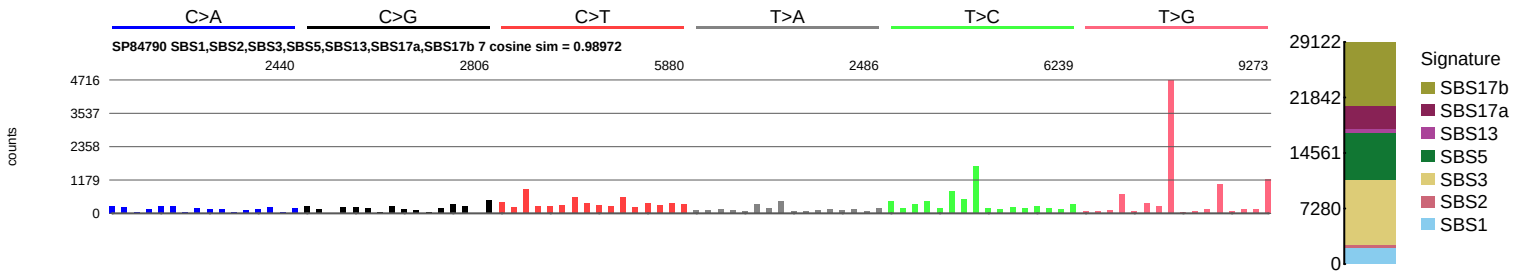

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS44 8 cosine sim = 0.9898
2459
2773
5945
2479
6218
9251
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS44
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

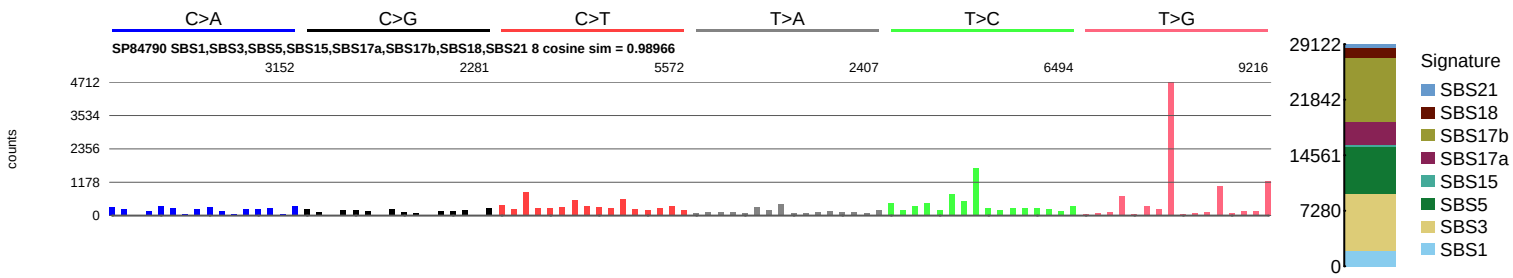

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS41,SBS44 8 cosine sim = 0.98976
1558
2081
6108
2609
7231
9538
counts
4696
3522
2348
1174
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

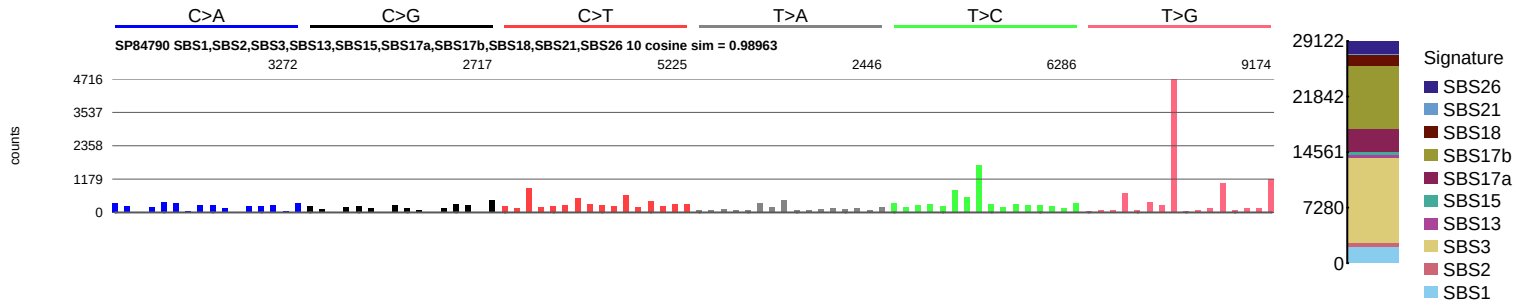

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.98974
1583
2071
5944
2579
7422
9520
counts
4696
3522
2348
1174
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

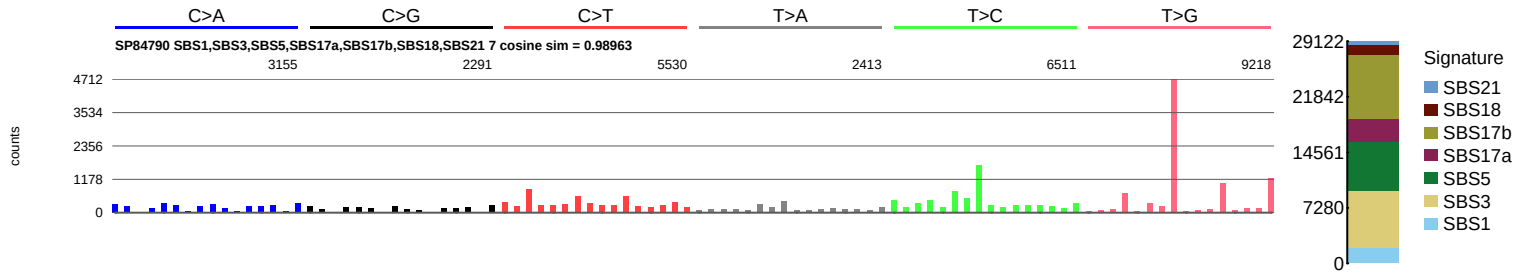

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98974
1547
2094
6073
2616
7245
9551
counts
4696
3522
2348
1174
0
29122
21842
14561
7280
0
Signature
SBS41
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

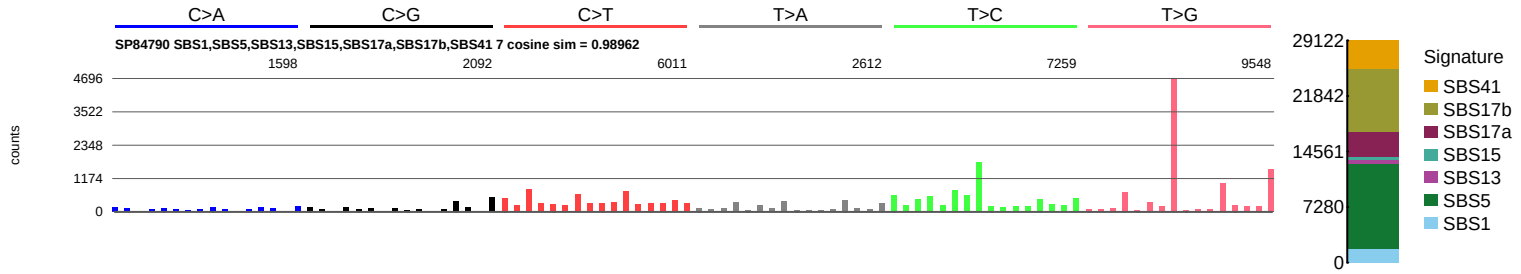

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS26 8 cosine sim = 0.98972
2432
2802
5837
2479
6301
9269
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

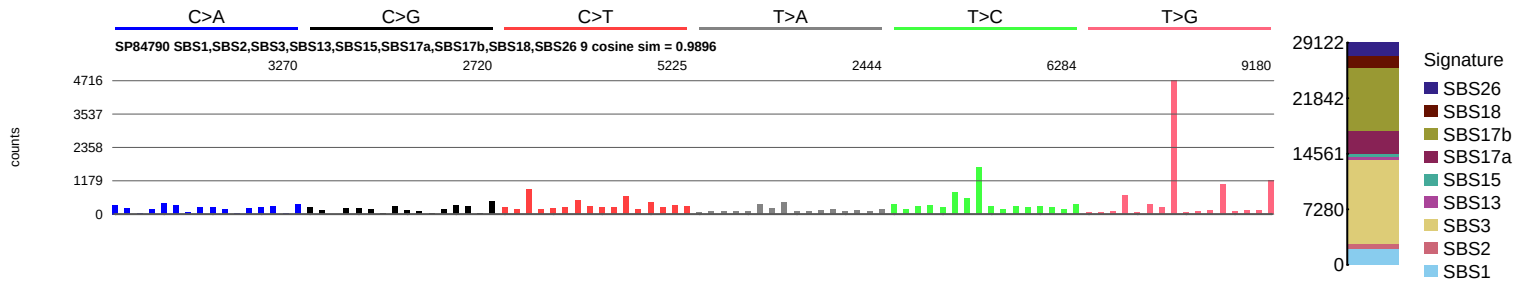

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b 7 cosine sim = 0.98972
2440
2806
5880
2486
6239
9273
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

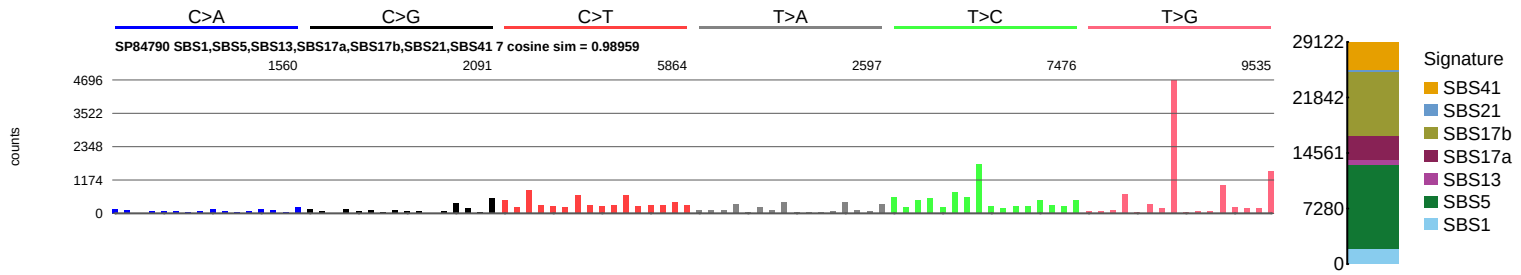

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98966
3152
2281
5572
2407
6494
9216
counts
4712
3534
2356
1178
0
29122
21842
14561
7280
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1

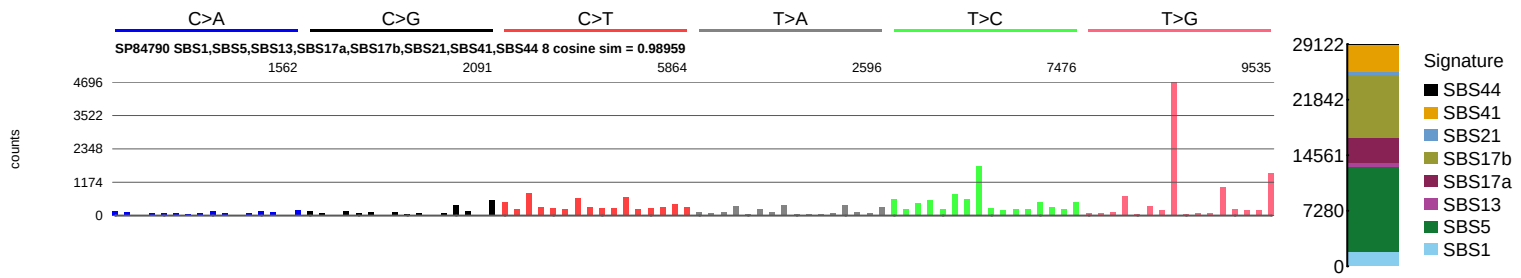

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26 10 cosine sim = 0.98963
3272
2717
5225
2446
6286
9174
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

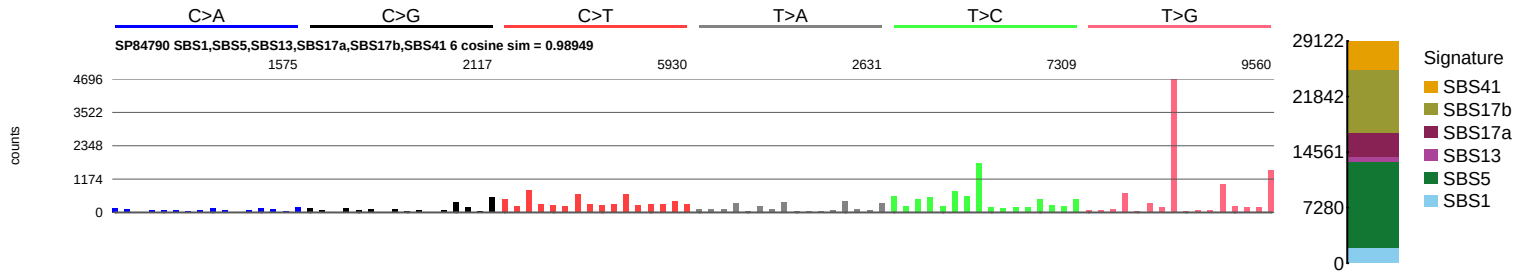

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98963
3155
2291
5530
2413
6511
9218
counts
4712
3534
2356
1178
0
29122
21842
14561
7280
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1

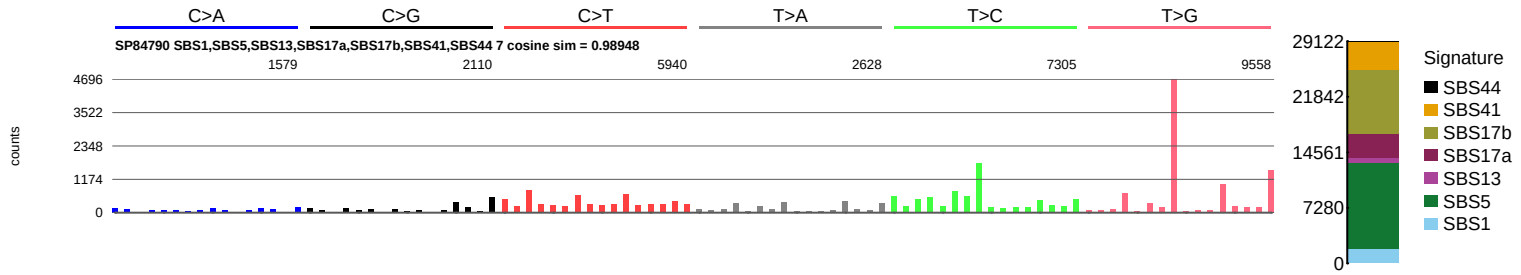

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98962
1598
2092
6011
2612
7259
9548
counts
4696
3522
2348
1174
0
29122
21842
14561
7280
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

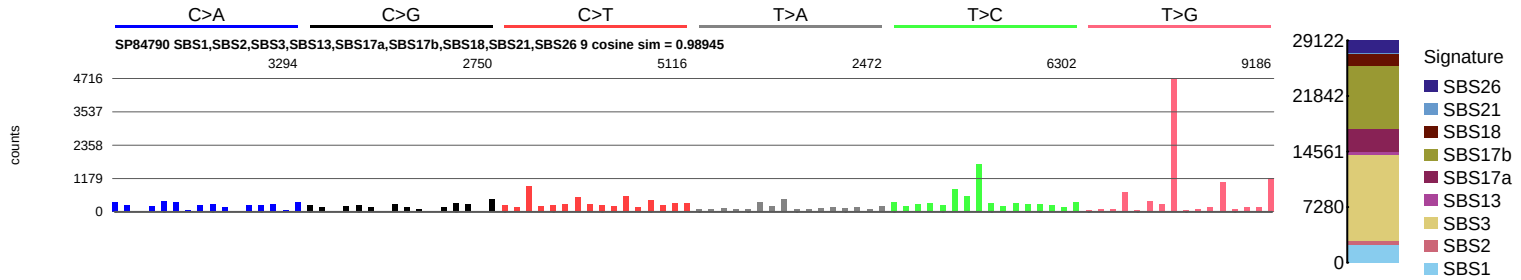

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 9 cosine sim = 0.9896
3270
2720
5225
2444
6284
9180
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

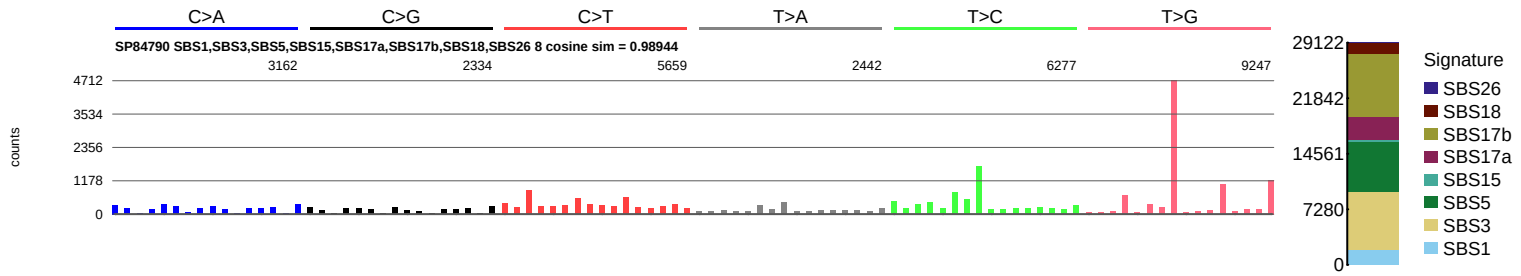

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98959
1560
2091
5864
2597
7476
9535
counts
4696
3522
2348
1174
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS1

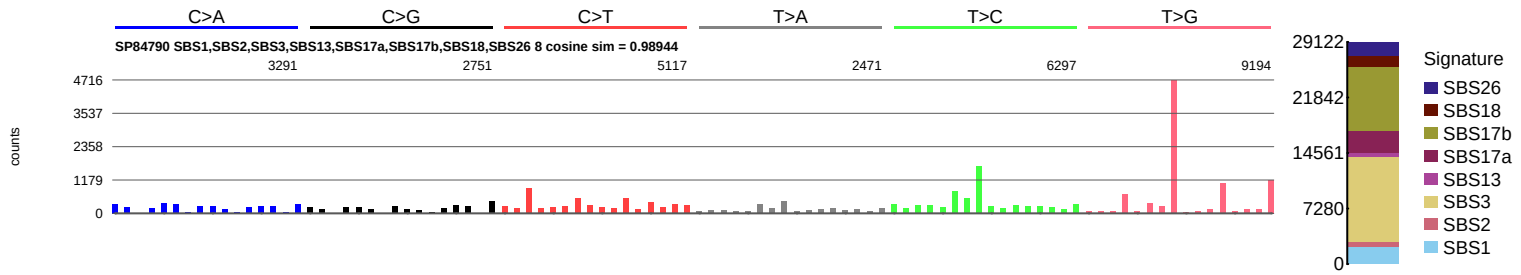

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS41,SBS44 8 cosine sim = 0.98959
1562
2091
5864
2596
7476
9535
counts
4696
3522
2348
1174
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS1

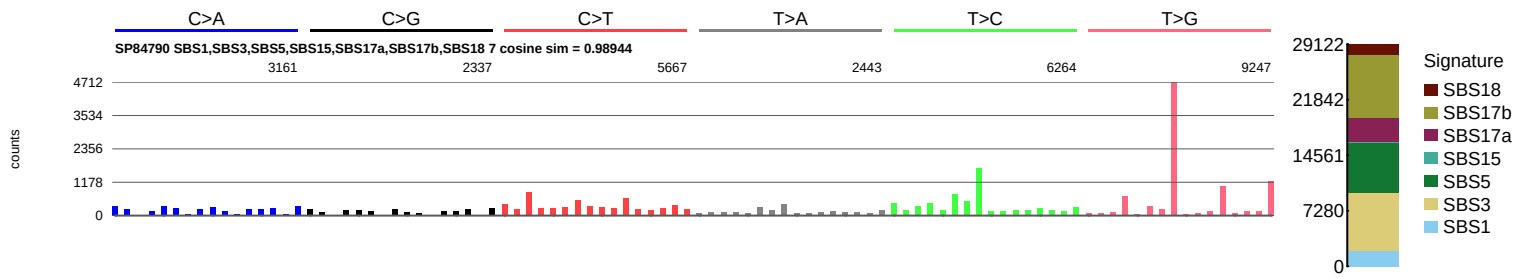

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98949
1575
2117
5930
2631
7309
9560
counts
4696
3522
2348
1174
0
29122
21842
14561
7280
0
Signature
SBS41
SBS17b
SBS17a
SBS13
SBS5
SBS1

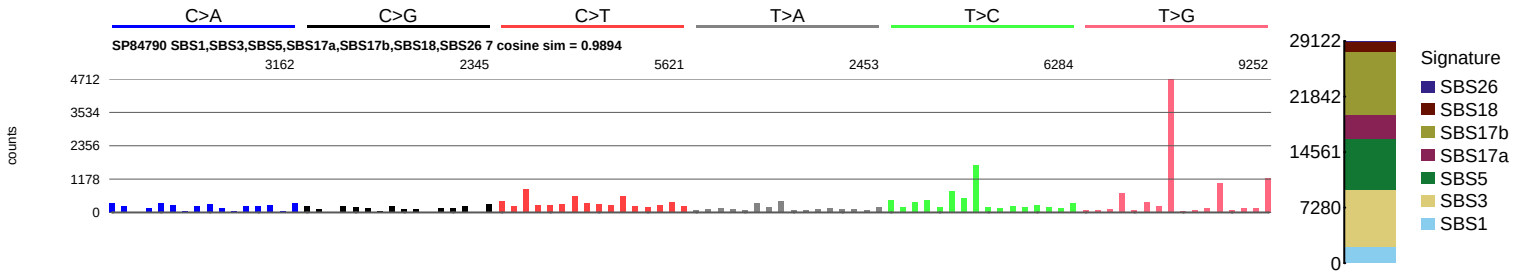

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS41,SBS44 7 cosine sim = 0.98948
1579
2110
5940
2628
7305
9558
counts
4696
3522
2348
1174
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS13
SBS5
SBS1

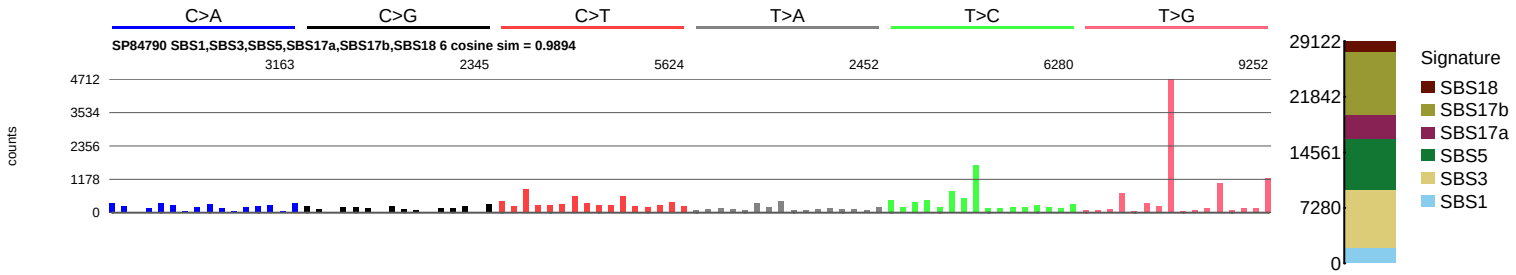

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26 9 cosine sim = 0.98945
3294
2750
5116
2472
6302
9186
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

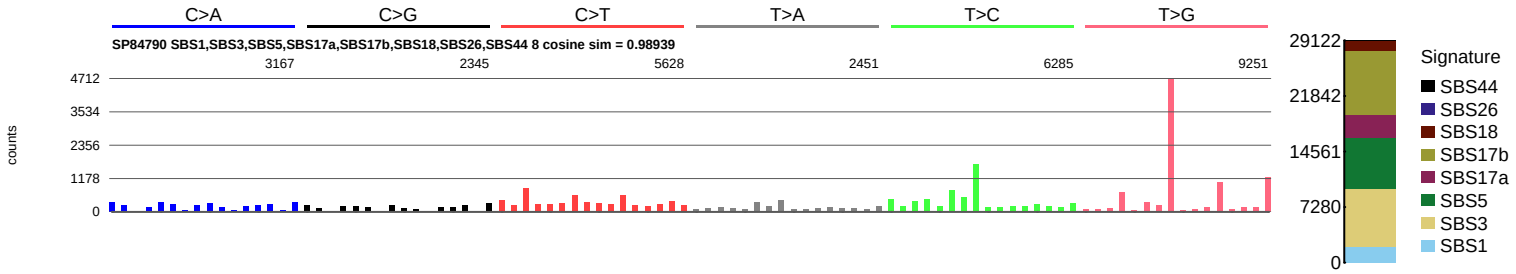

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98944
3162
2334
5659
2442
6277
9247
counts
4712
3534
2356
1178
0
29122
21842
14561
7280
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1

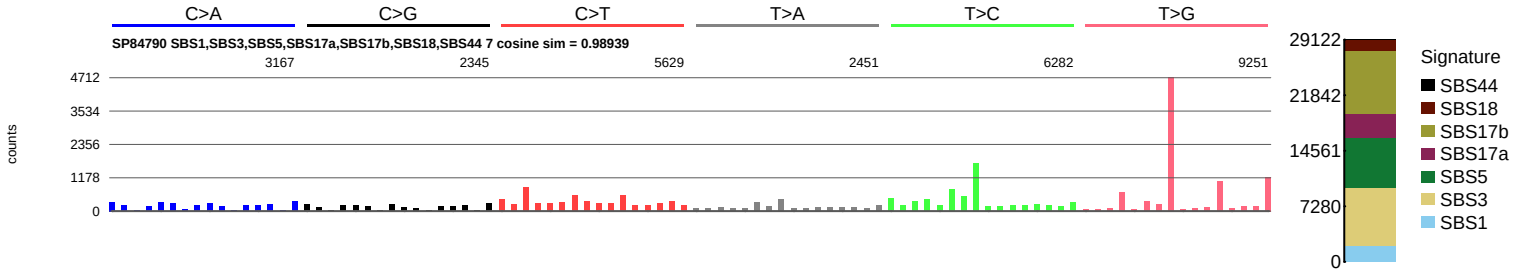

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98944
3291
2751
5117
2471
6297
9194
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

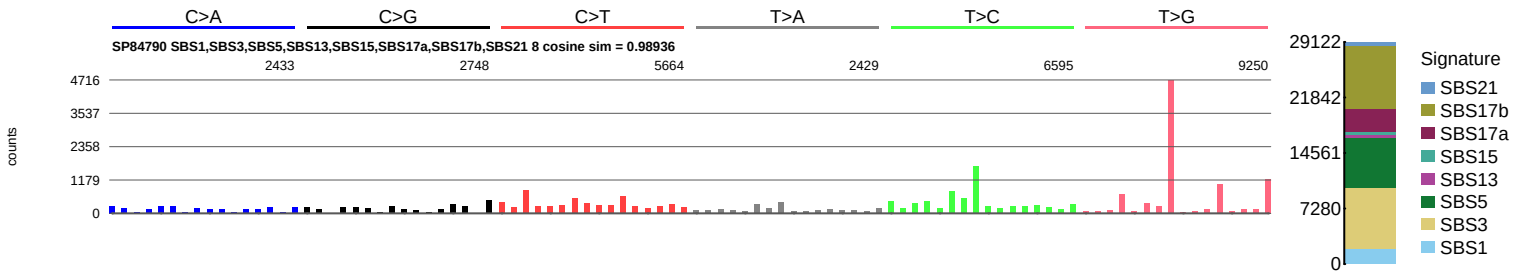

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98944
3161
2337
5667
2443
6264
9247
counts
4712
3534
2356
1178
0
29122
21842
14561
7280
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1

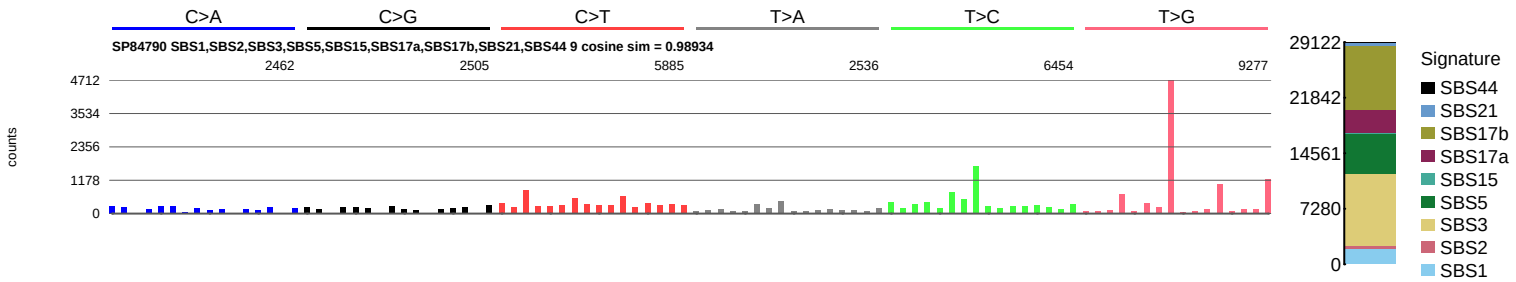

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.9894
3162
2345
5621
2453
6284
9252
4712
3534
2356
1178
0
counts
29122
21842
14561
7280
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1

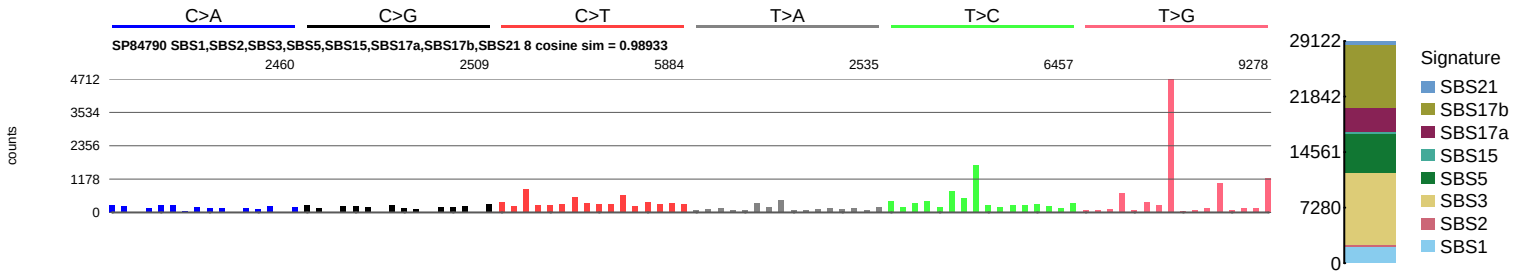

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18 6 cosine sim = 0.9894
3163
2345
5624
2452
6280
9252
4712
3534
2356
1178
0
counts
29122
21842
14561
7280
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1

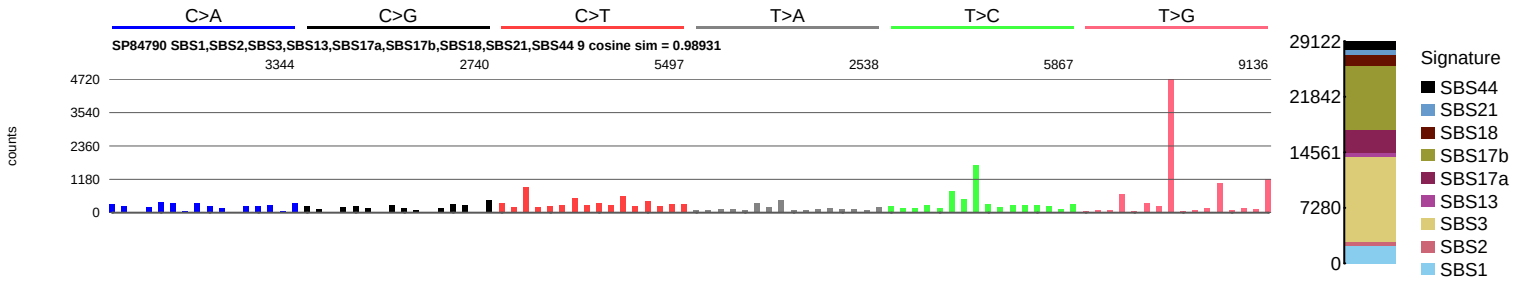

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98939
3167
2345
5628
2451
6285
9251
4712
3534
2356
1178
0
counts
29122
21842
14561
7280
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1

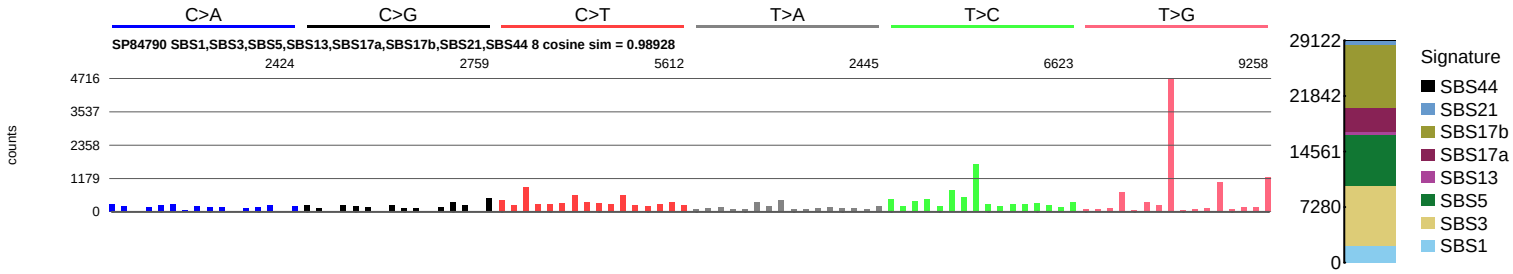

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.98939
3167
2345
5629
2451
6282
9251
4712
3534
2356
1178
0
counts
29122
21842
14561
7280
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1

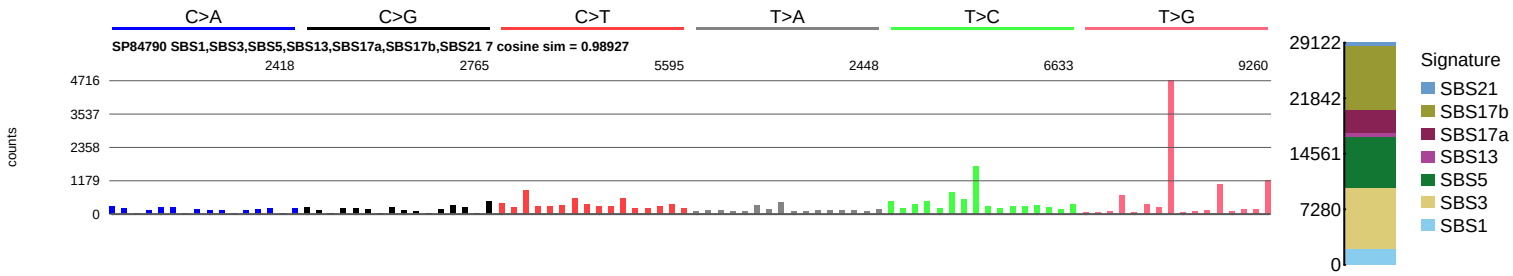

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21 8 cosine sim = 0.98936
2433
2748
5664
2429
6595
9250
4716
3537
2358
1179
0
counts
29122
21842
14561
7280
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

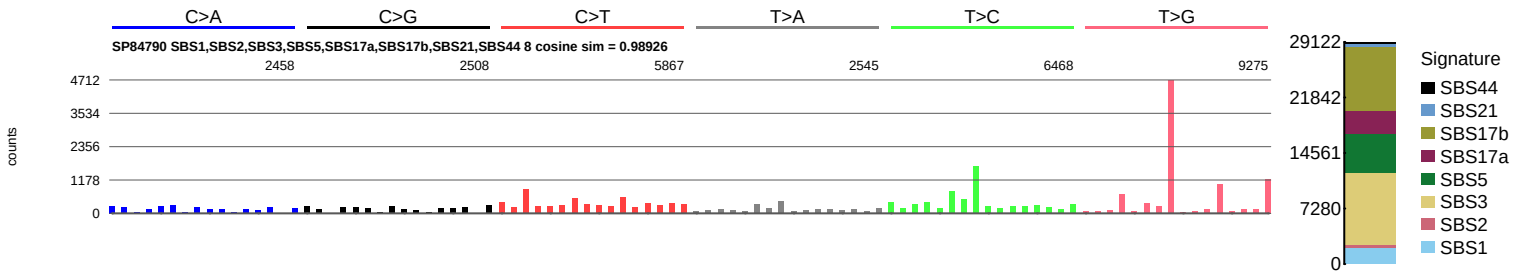

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS44 9 cosine sim = 0.98934
2462
2505
5885
2536
6454
9277
4712
3534
2356
1178
0
counts
29122
21842
14561
7280
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

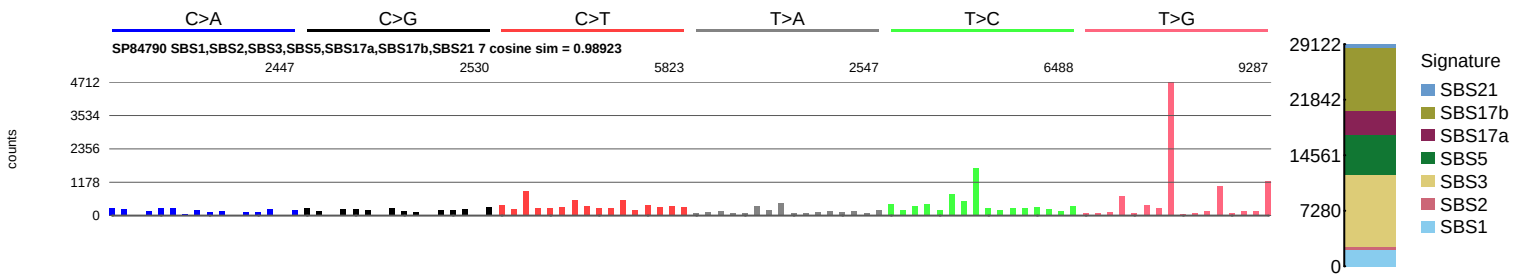

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21 8 cosine sim = 0.98933
2460
2509
5884
2535
6457
9278
counts
4712
3534
2356
1178
0
29122
21842
14561
7280
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

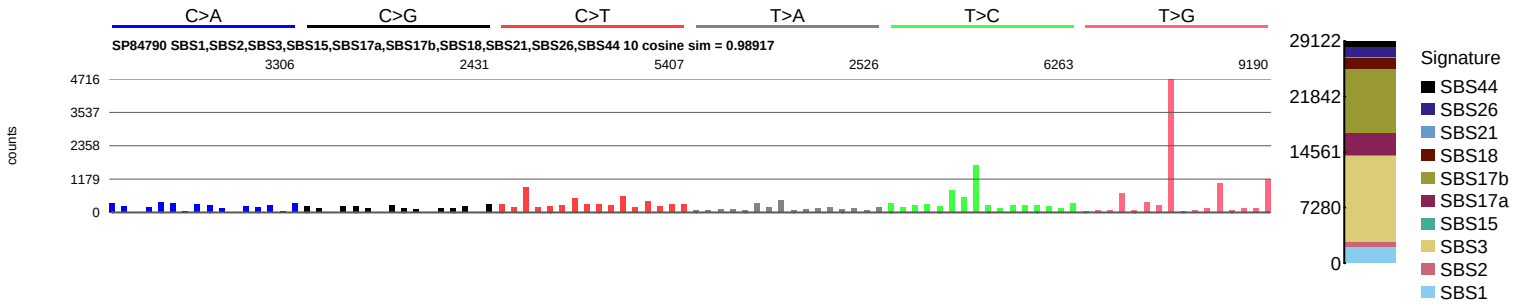

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98931
3344
2740
5497
2538
5867
9136
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

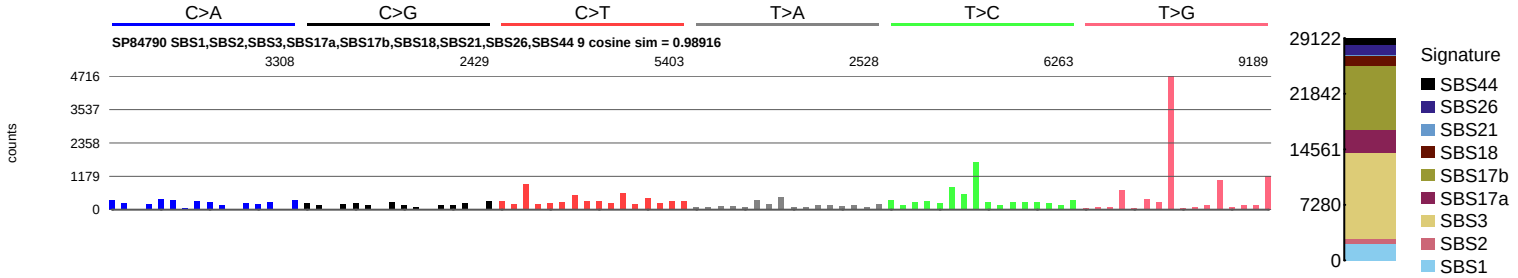

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS44 8 cosine sim = 0.98928
2424
2759
5612
2445
6623
9258
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

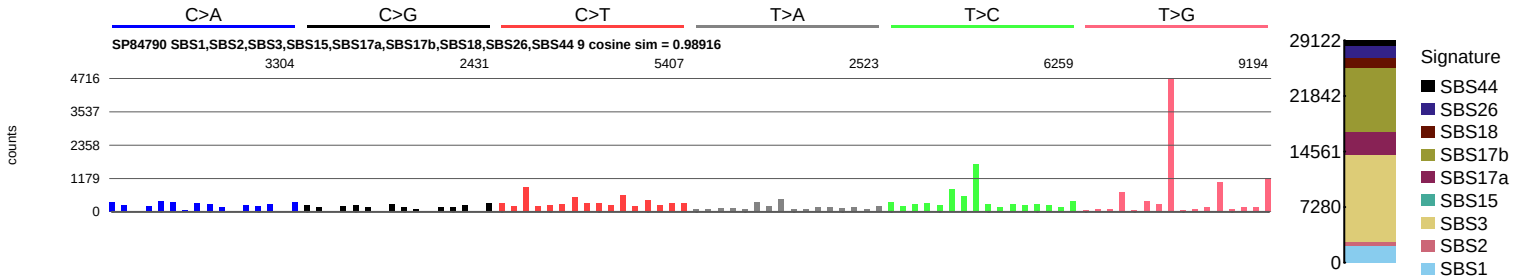

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98927
2418
2765
5595
2448
6633
9260
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

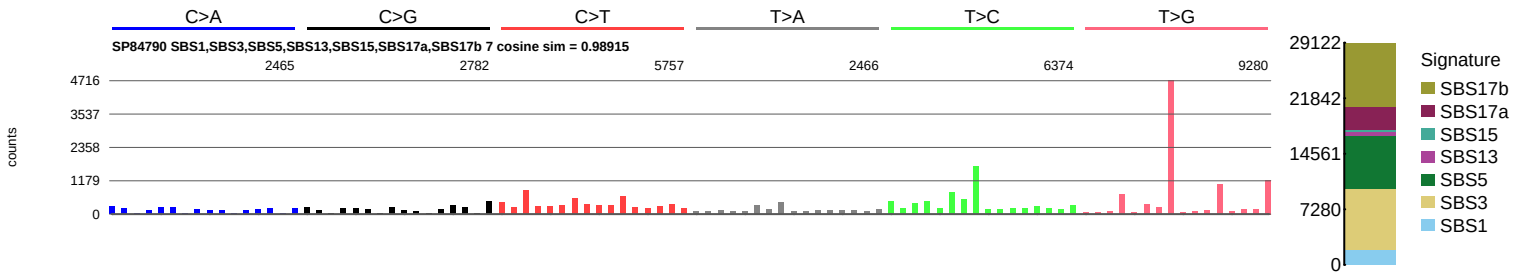

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS44 8 cosine sim = 0.98926
2458
2508
5867
2545
6468
9275
counts
4712
3534
2356
1178
0
29122
21842
14561
7280
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

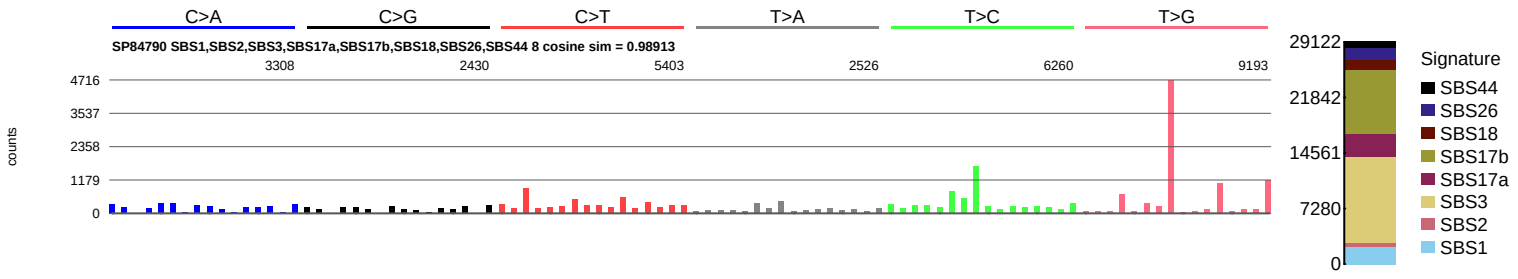

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98923
2447
2530
5823
2547
6488
9287
counts
4712
3534
2356
1178
0
29122
21842
14561
7280
0
Signature
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

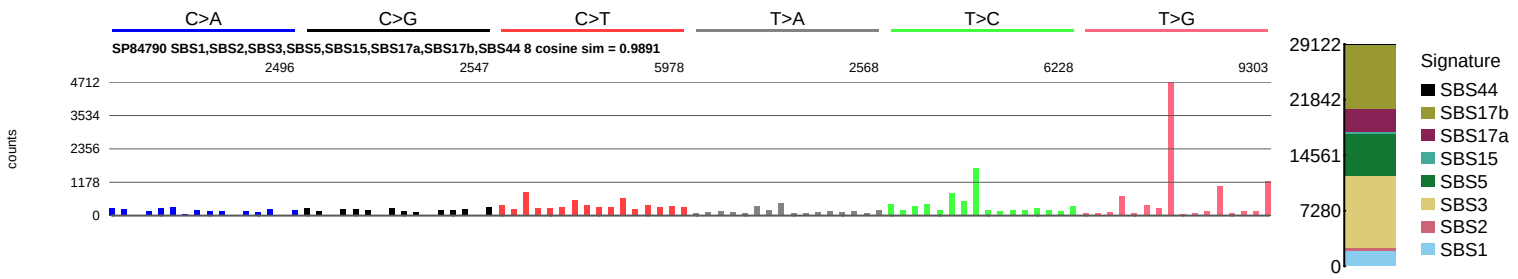

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 10 cosine sim = 0.98917
3306
2431
5407
2526
6263
9190
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

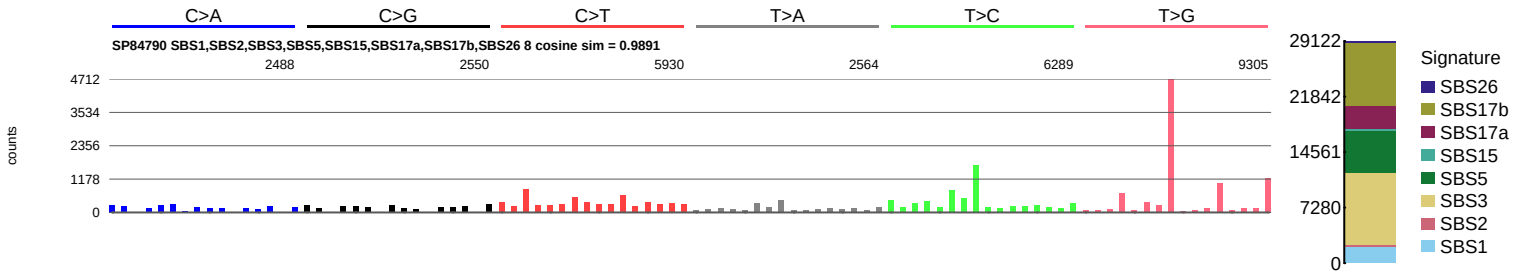

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 9 cosine sim = 0.98916
3308
2429
5403
2528
6263
9189
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

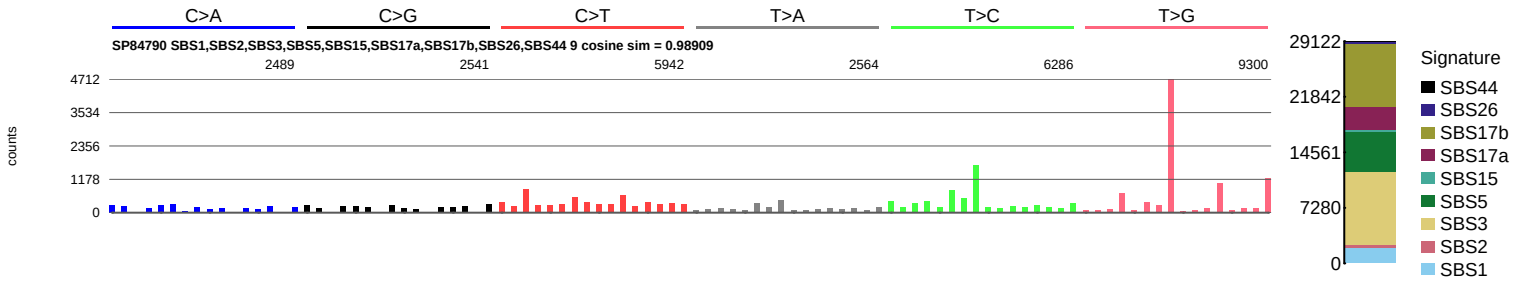

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.98916
3304
2431
5407
2523
6259
9194
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

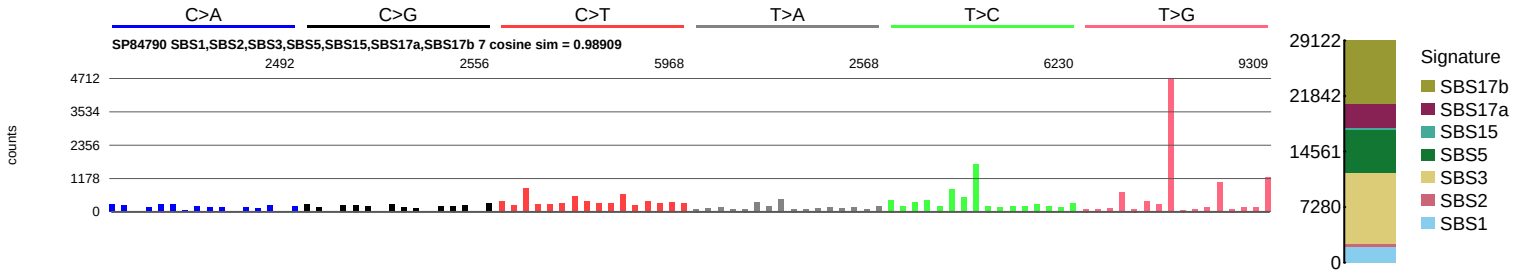

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b 7 cosine sim = 0.98915
2465
2782
5757
2466
6374
9280
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

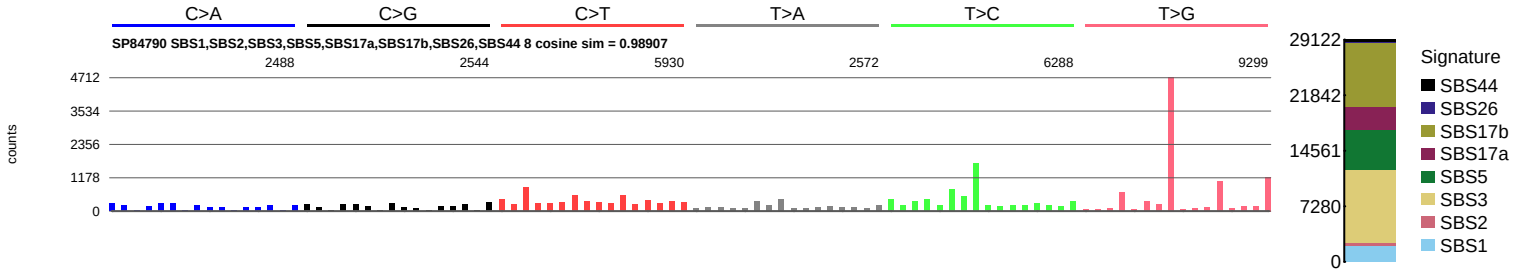

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98913
3308
2430
5403
2526
6260
9193
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

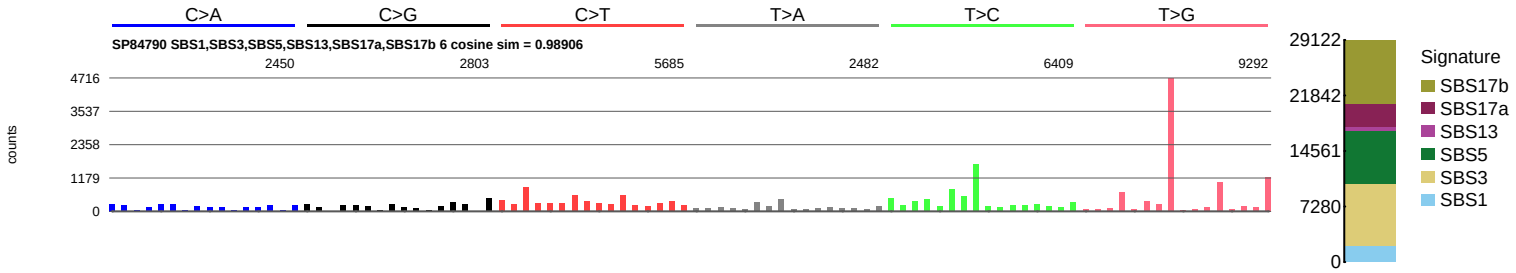

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS44 8 cosine sim = 0.9891
2496
2547
5978
2568
6228
9303
counts
4712
3534
2356
1178
0
29122
21842
14561
7280
0
Signature
SBS44
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

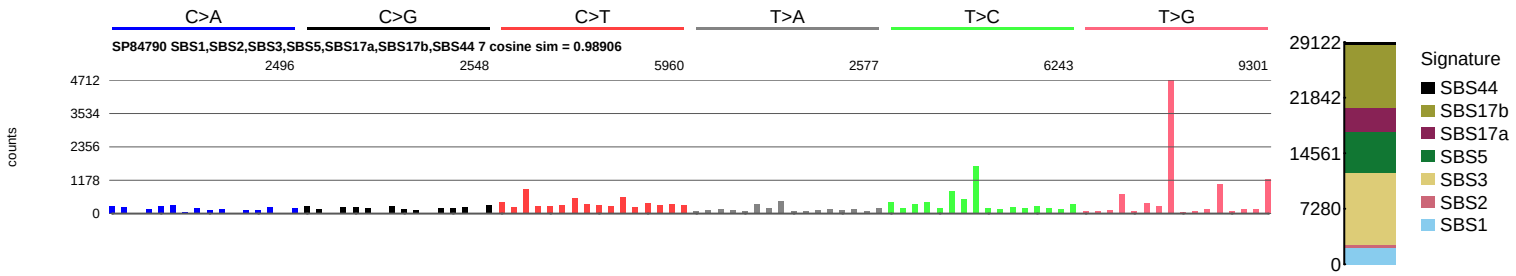

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS26 8 cosine sim = 0.9891
2488
2550
5930
2564
6289
9305
counts
4712
3534
2356
1178
0
29122
21842
14561
7280
0
Signature
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

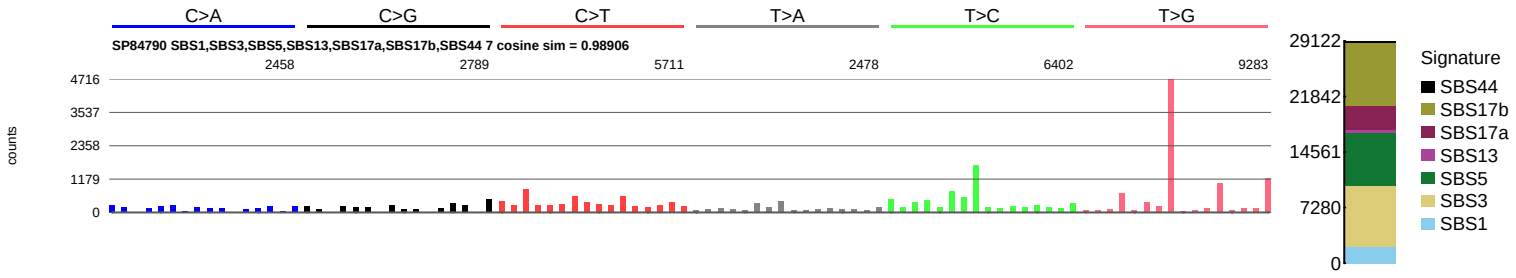

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS44 9 cosine sim = 0.98909
2489
2541
5942
2564
6286
9300
counts
4712
3534
2356
1178
0
29122
21842
14561
7280
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

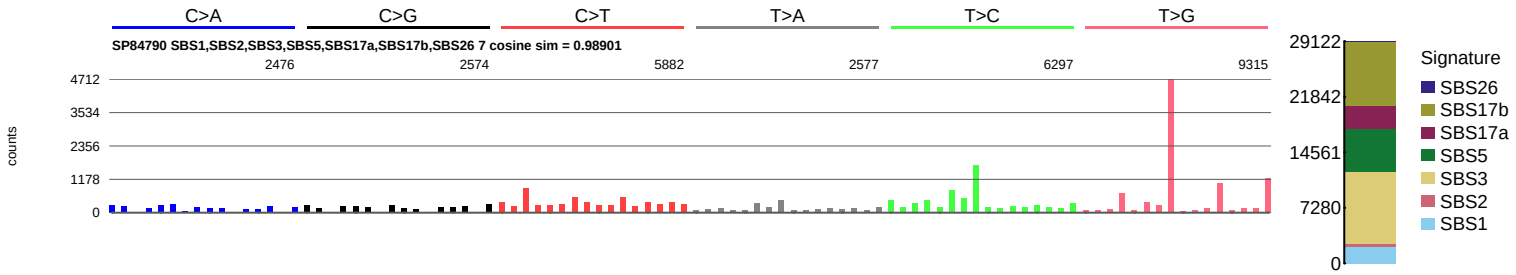

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b 7 cosine sim = 0.98909
2492
2556
5968
2568
6230
9309
counts
4712
3534
2356
1178
0
29122
21842
14561
7280
0
Signature
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

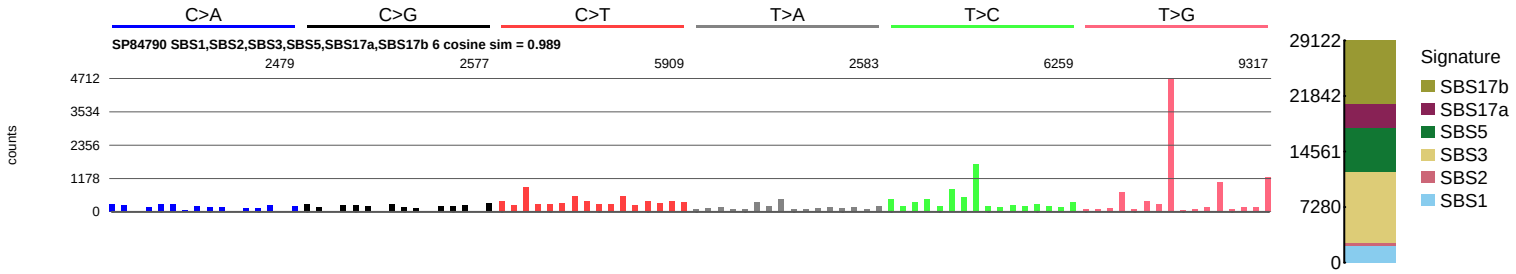

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS26,SBS44 8 cosine sim = 0.98907
2488
2544
5930
2572
6288
9299
counts
4712
3534
2356
1178
0
29122
21842
14561
7280
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

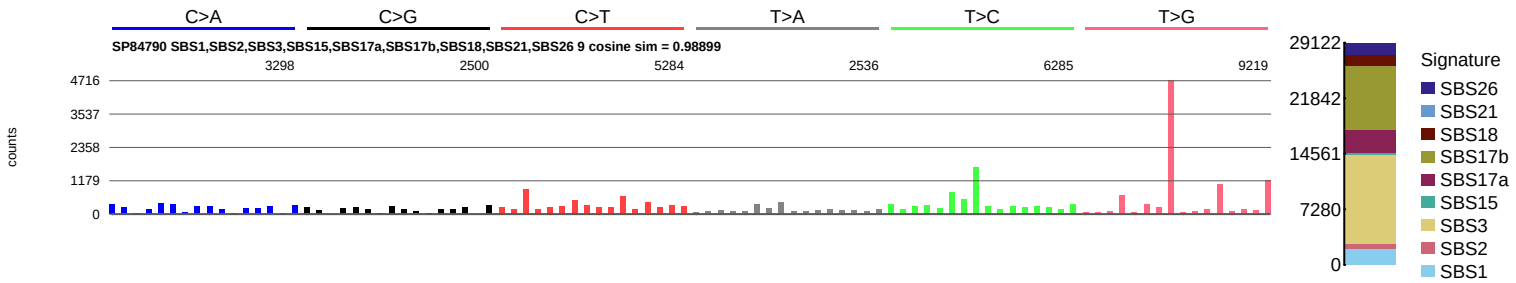

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b 6 cosine sim = 0.98906
2450
2803
5685
2482
6409
9292
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

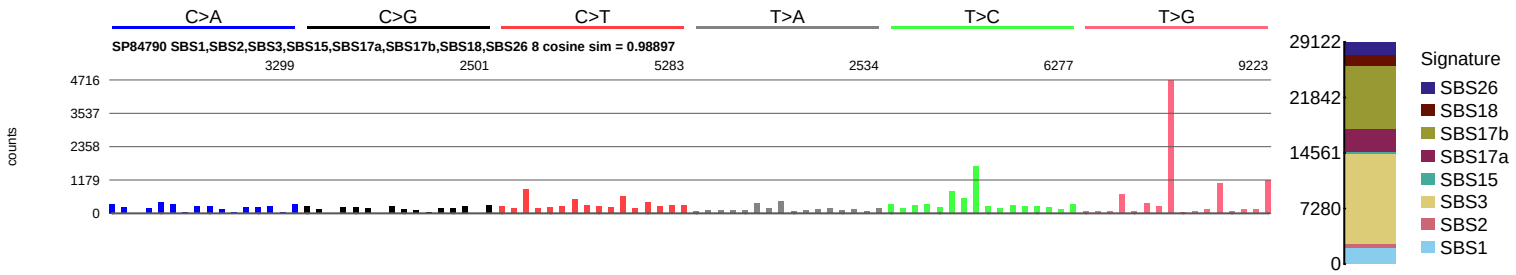

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS44 7 cosine sim = 0.98906
2496
2548
5960
2577
6243
9301
counts
4712
3534
2356
1178
0
29122
21842
14561
7280
0
Signature
SBS44
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

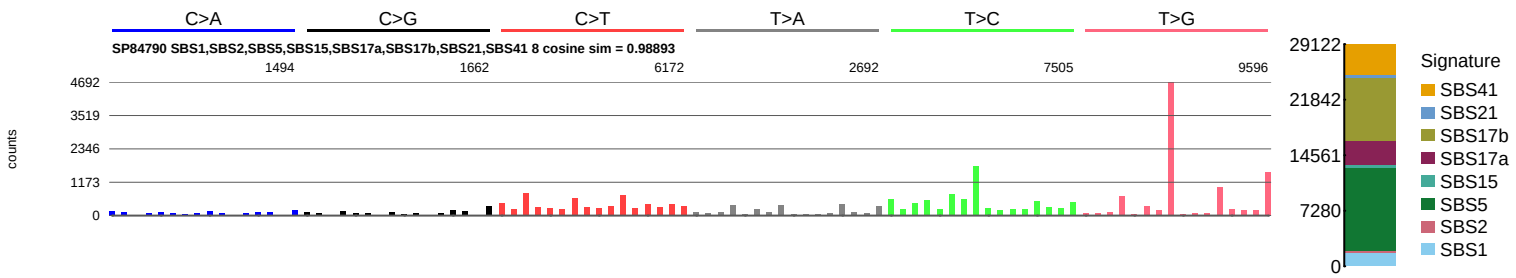

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS44 7 cosine sim = 0.98906
2458
2789
5711
2478
6402
9283
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS44
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

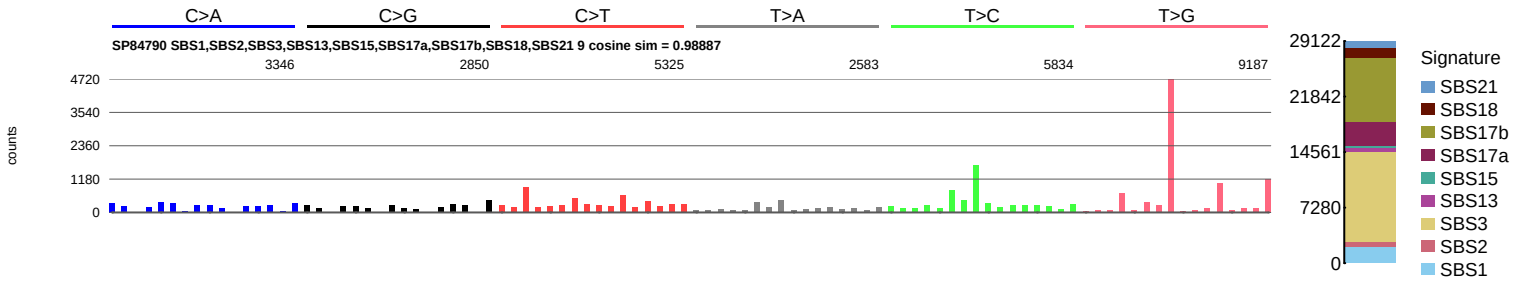

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS26 7 cosine sim = 0.98901
2476
2574
5882
2577
6297
9315
counts
4712
3534
2356
1178
0
29122
21842
14561
7280
0
Signature
SBS26
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

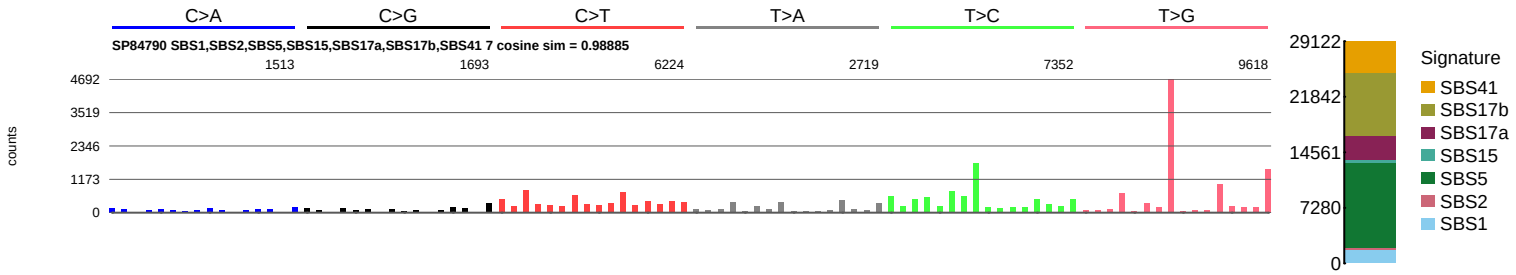

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b 6 cosine sim = 0.989
2479
2577
5909
2583
6259
9317
counts
4712
3534
2356
1178
0
29122
21842
14561
7280
0
Signature
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

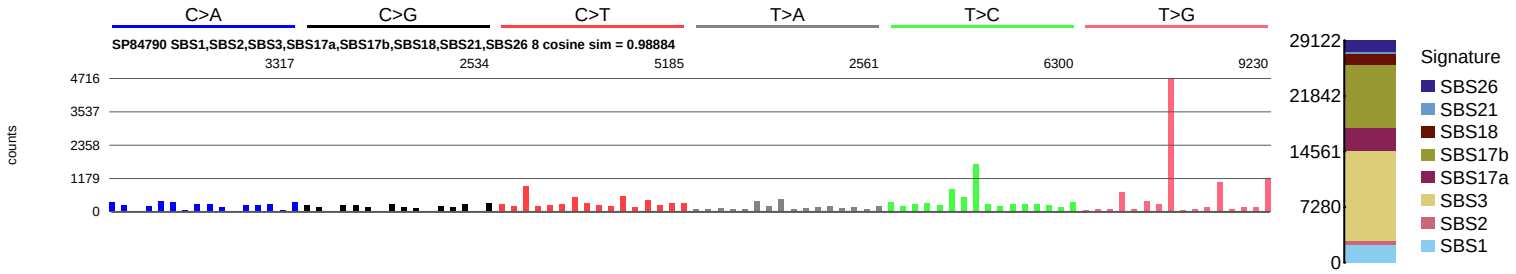

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26 9 cosine sim = 0.98899
3298
2500
5284
2536
6285
9219
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

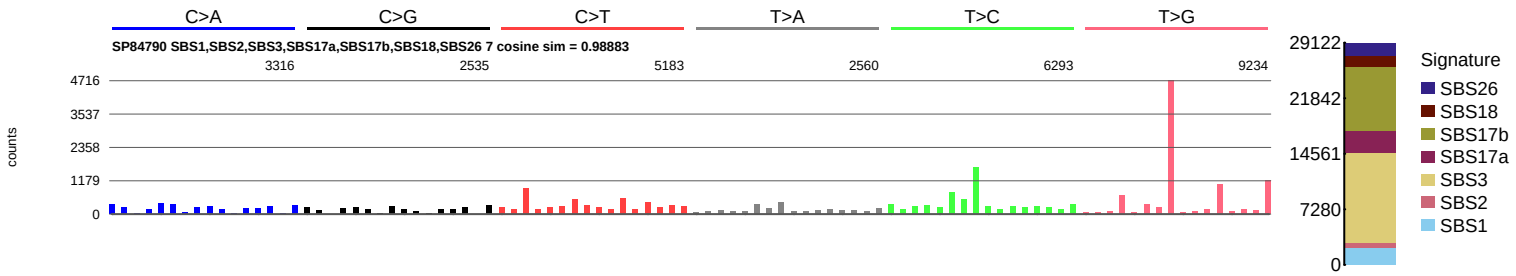

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98897
3299
2501
5283
2534
6277
9223
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

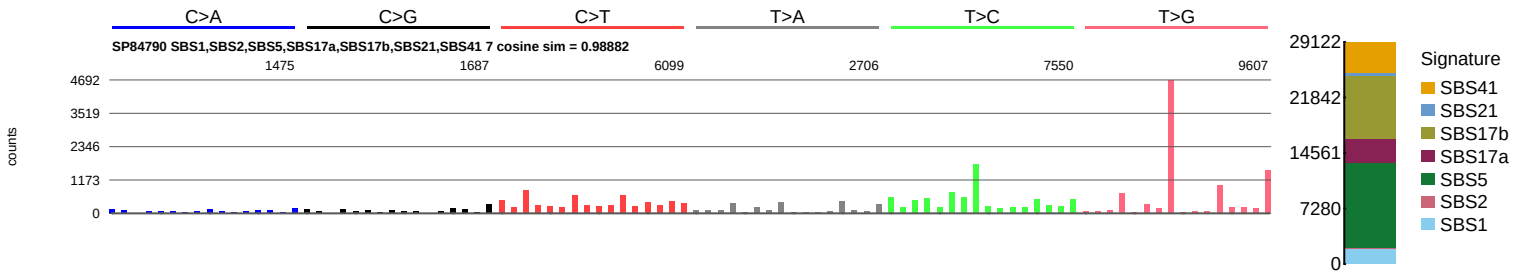

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.98893
1494
1662
6172
2692
7505
9596
counts
4692
3519
2346
1173
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

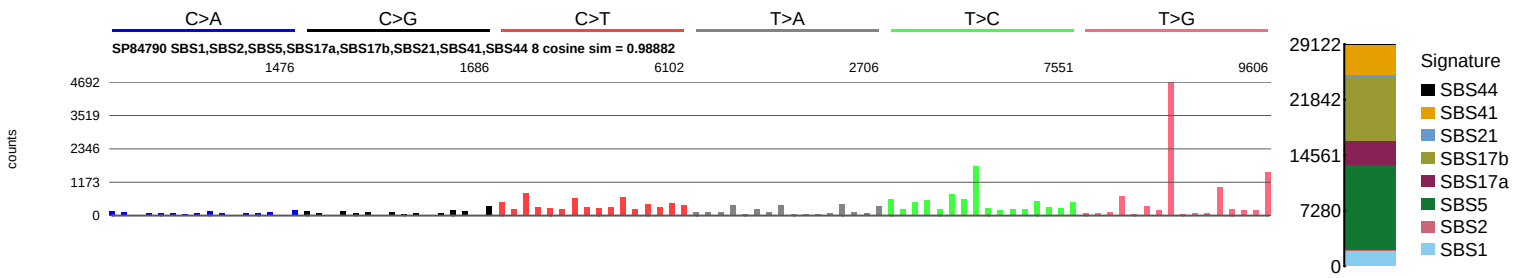

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98887
3346
2850
5325
2583
5834
9187
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

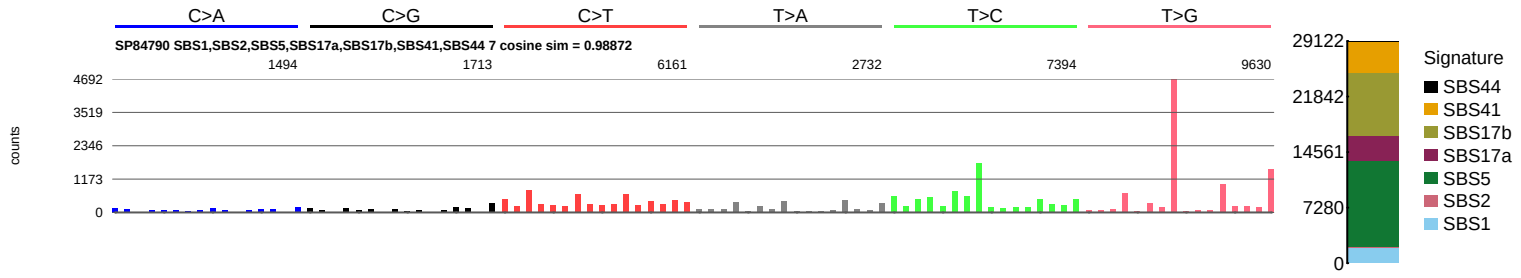

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98885
1513
1693
6224
2719
7352
9618
counts
4692
3519
2346
1173
0
29122
21842
14561
7280
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

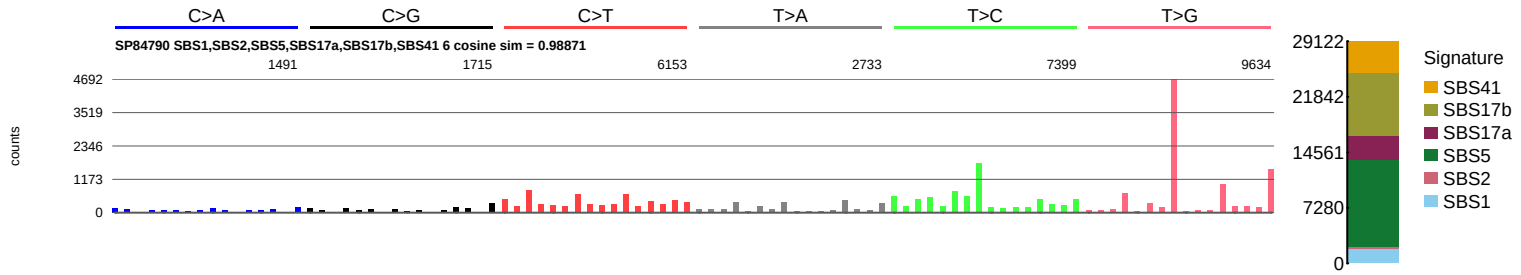

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26 8 cosine sim = 0.98884
3317
2534
5185
2561
6300
9230
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

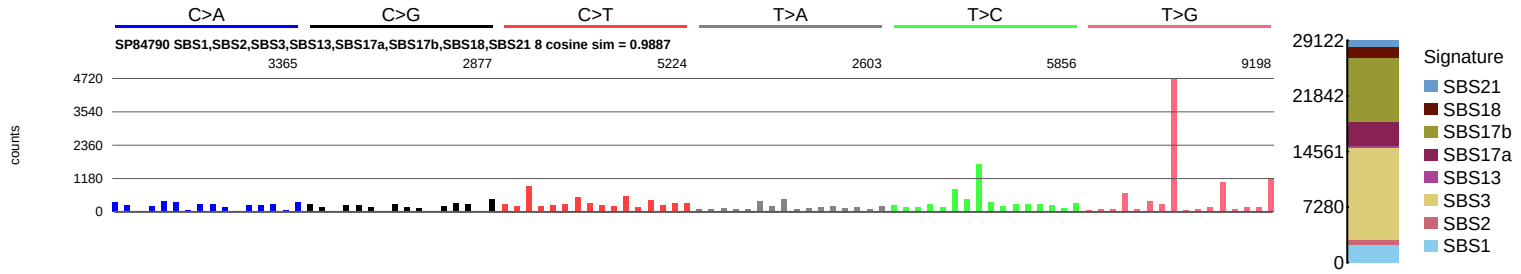

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98883
3316
2535
5183
2560
6293
9234
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

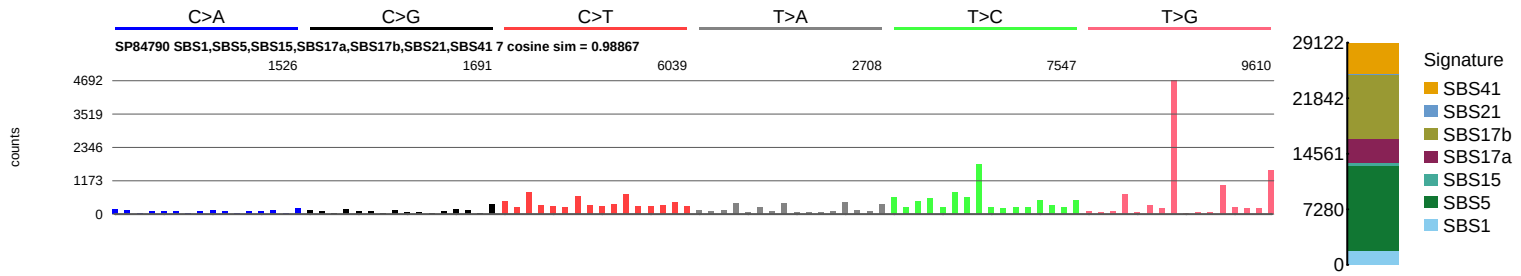

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98882
1475
1687
6099
2706
7550
9607
counts
4692
3519
2346
1173
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS2
SBS1

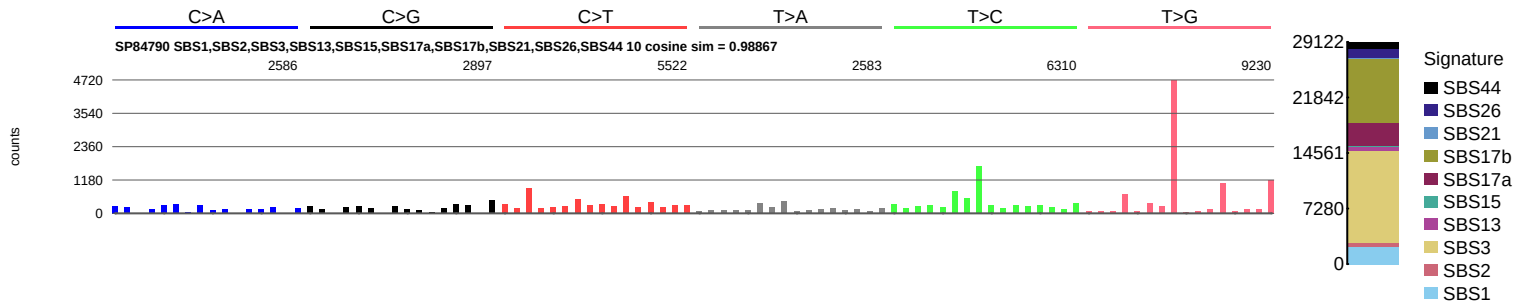

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS41,SBS44 8 cosine sim = 0.98882
1476
1686
6102
2706
7551
9606
counts
4692
3519
2346
1173
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS2
SBS1

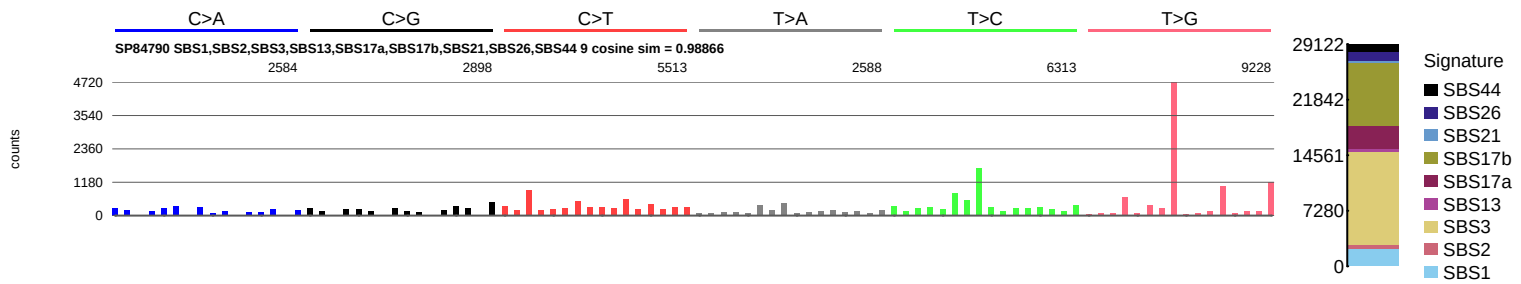

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS41,SBS44 7 cosine sim = 0.98872
1494
1713
6161
2732
7394
9630
counts
4692
3519
2346
1173
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS5
SBS2
SBS1

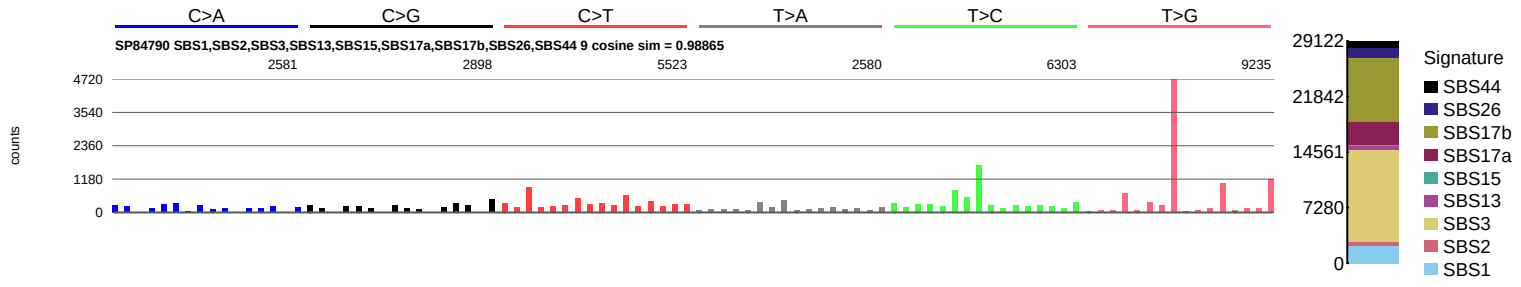

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98871
1491
1715
6153
2733
7399
9634
counts
4692
3519
2346
1173
0
29122
21842
14561
7280
0
Signature
SBS41
SBS17b
SBS17a
SBS5
SBS2
SBS1

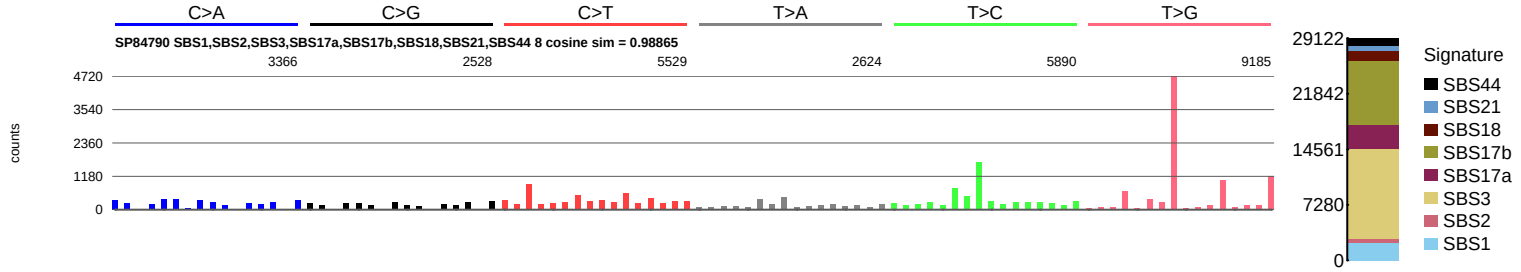

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.9887
3365
2877
5224
2603
5856
9198
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

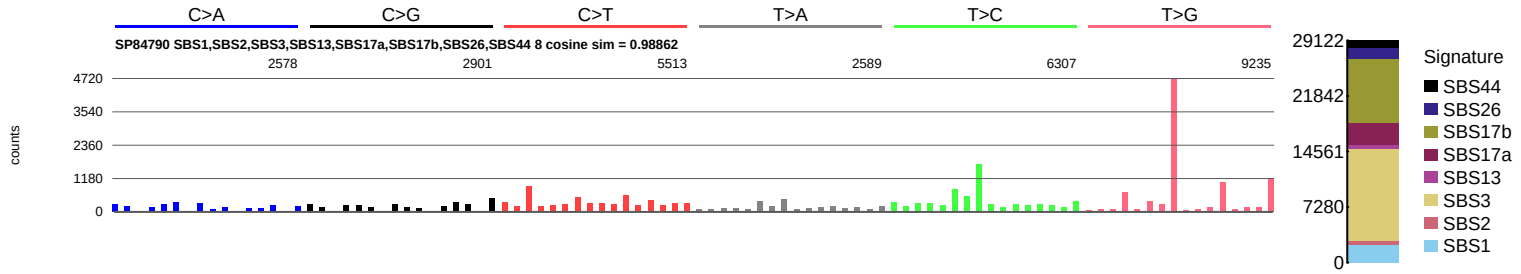

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98867
1526
1691
6039
2708
7547
9610
counts
4692
3519
2346
1173
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS1

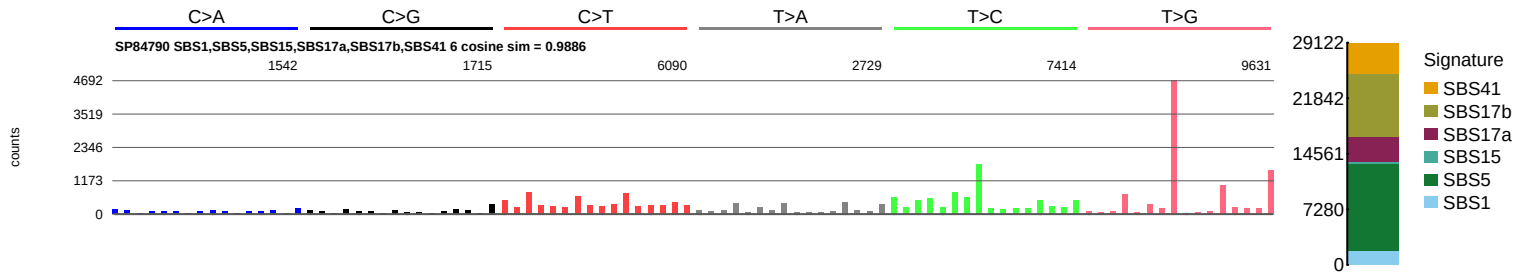

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS44 10 cosine sim = 0.98867
2586
2897
5522
2583
6310
9230
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS44
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

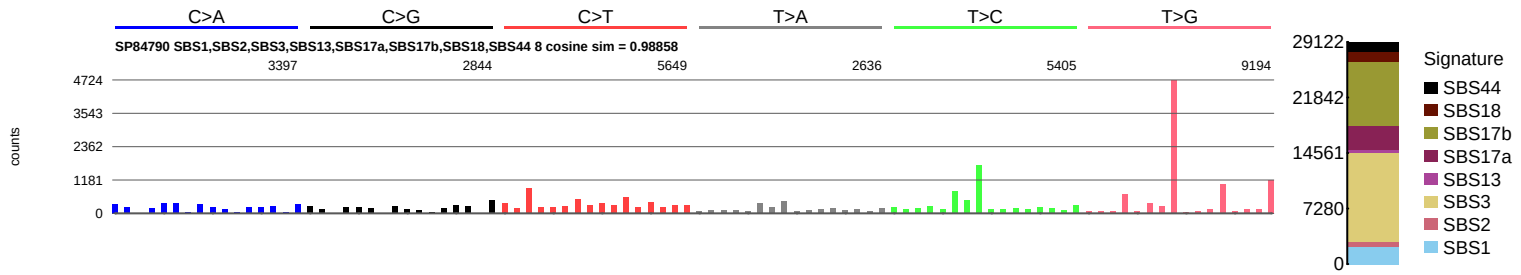

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS44 9 cosine sim = 0.98866
2584
2898
5513
2588
6313
9228
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS44
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

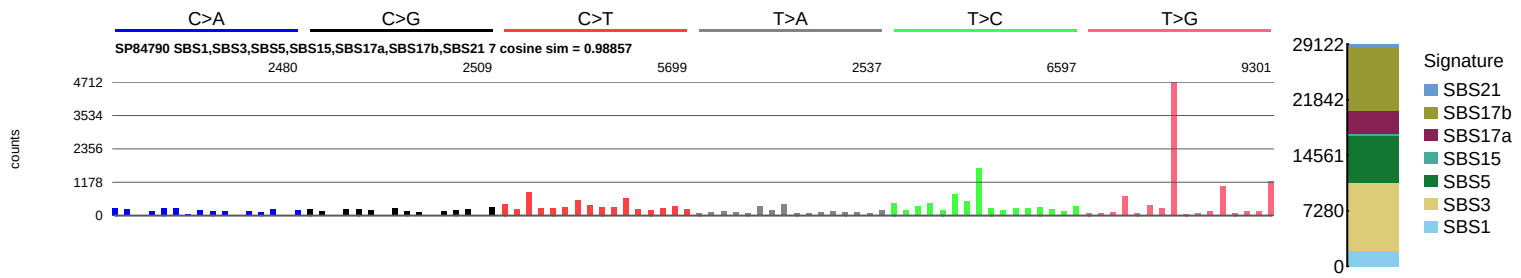

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS44 9 cosine sim = 0.98865
2581
2898
5523
2580
6303
9235
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

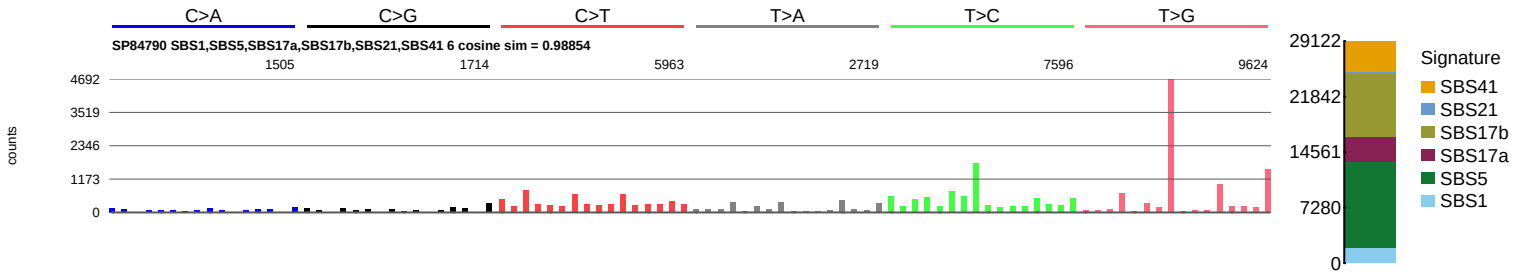

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98865
3366
2528
5529
2624
5890
9185
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

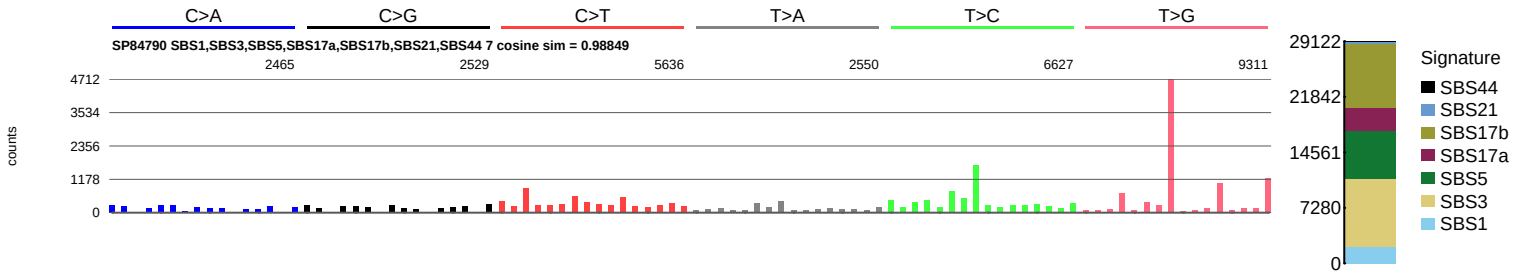

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS26,SBS44 8 cosine sim = 0.98862
2578
2901
5513
2589
6307
9235
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

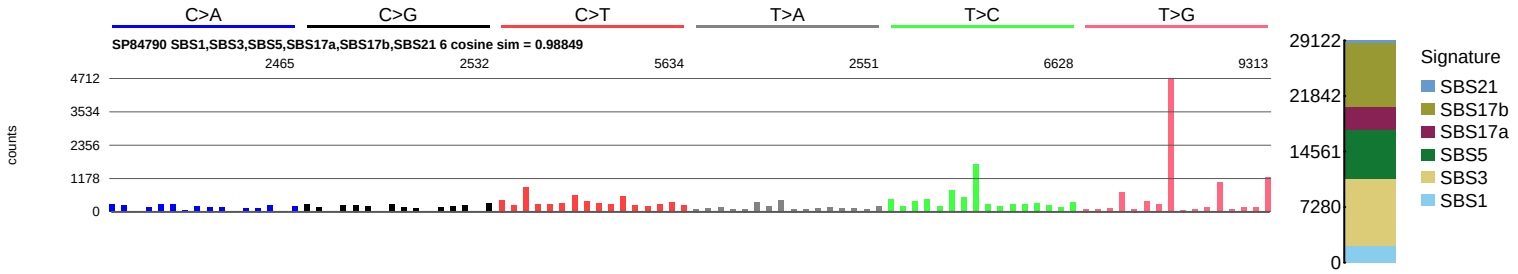

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS41 6 cosine sim = 0.9886
1542
1715
6090
2729
7414
9631
counts
4692
3519
2346
1173
0
29122
21842
14561
7280
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS1

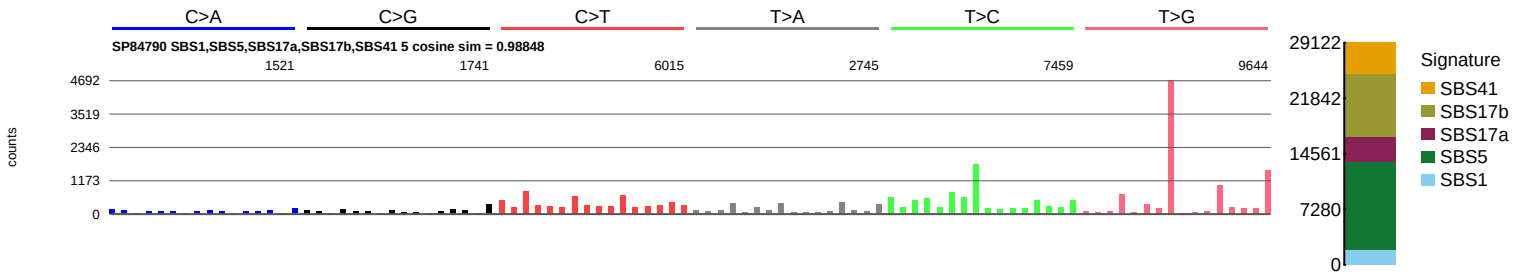

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98858
3397
2844
5649
2636
5405
9194
counts
4724
3543
2362
1181
0
29122
21842
14561
7280
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

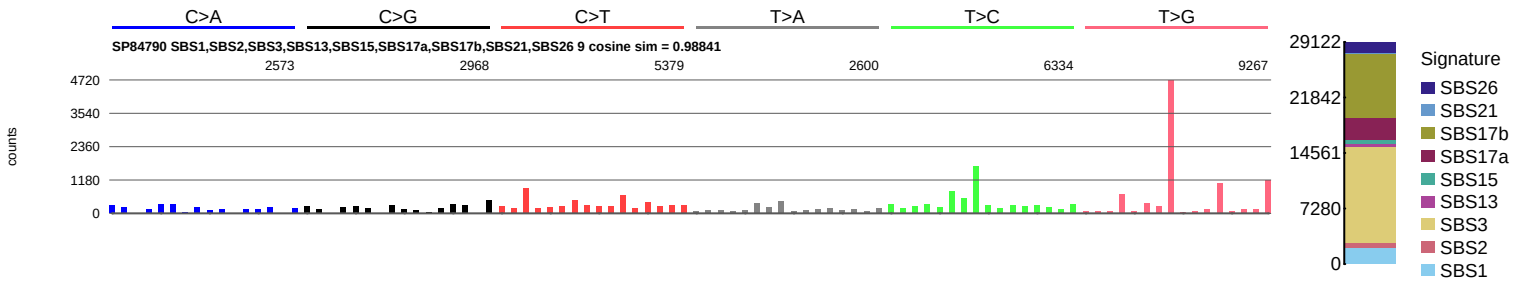

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98857
2480
2509
5699
2537
6597
9301
counts
4712
3534
2356
1178
0
29122
21842
14561
7280
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1

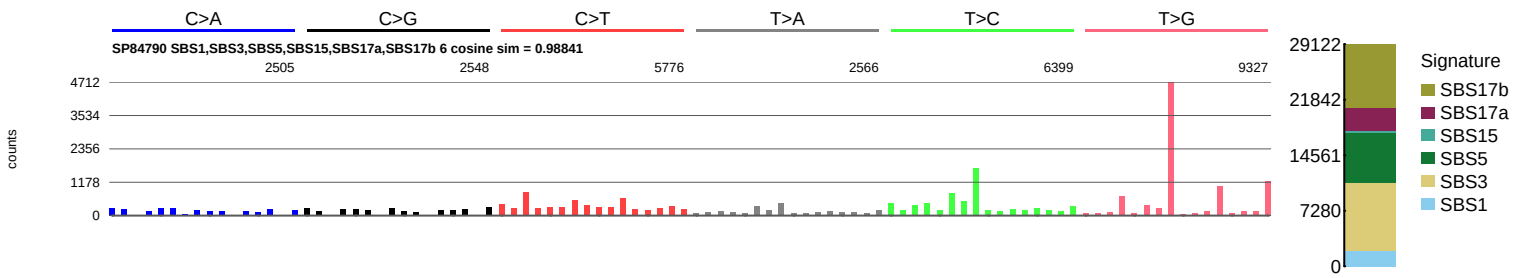

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.98854
1505
1714
5963
2719
7596
9624
counts
4692
3519
2346
1173
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS1

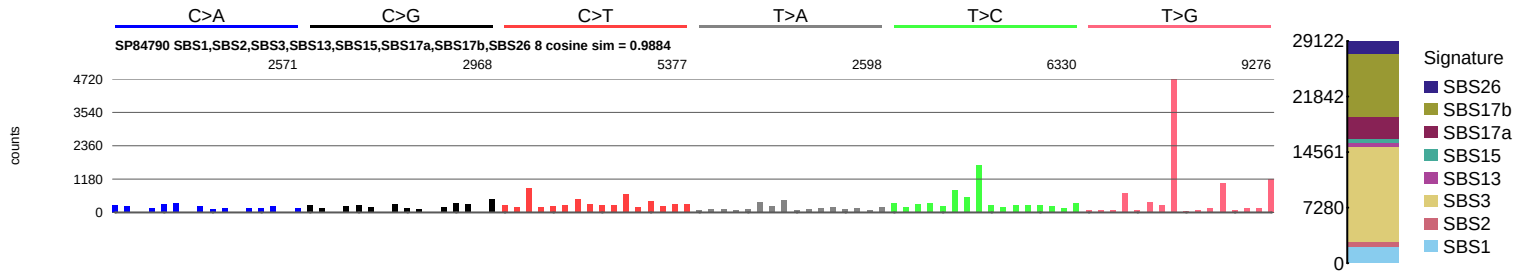

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS44 7 cosine sim = 0.98849
2465
2529
5636
2550
6627
9311
counts
4712
3534
2356
1178
0
29122
21842
14561
7280
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS1

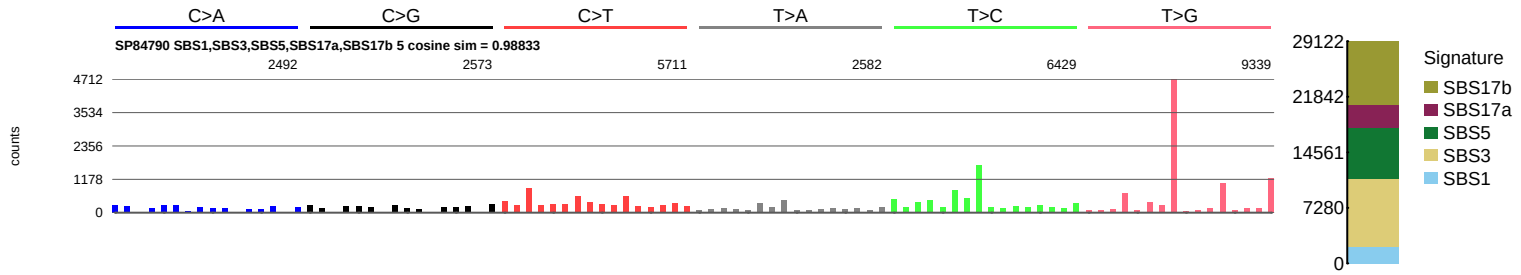

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS21 6 cosine sim = 0.98849
2465
2532
5634
2551
6628
9313
counts
4712
3534
2356
1178
0
29122
21842
14561
7280
0
Signature
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS1

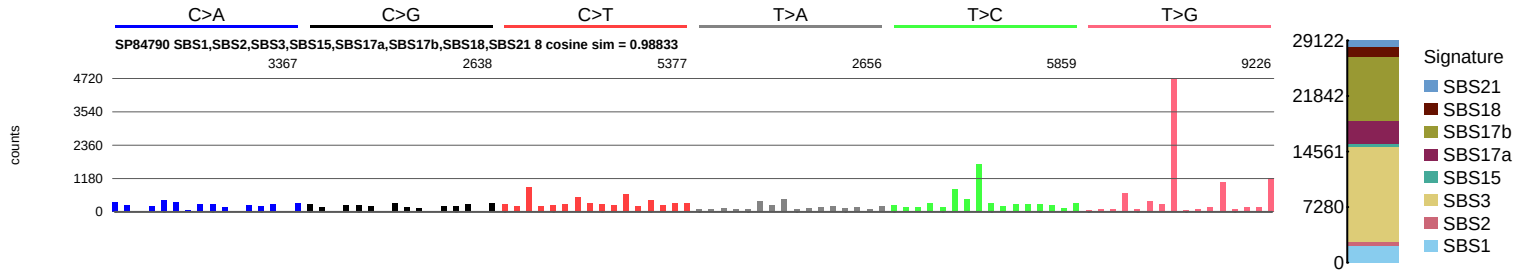

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS17a,SBS17b,SBS41 5 cosine sim = 0.98848
1521
1741
6015
2745
7459
9644
counts
4692
3519
2346
1173
0
29122
21842
14561
7280
0
Signature
SBS41
SBS17b
SBS17a
SBS5
SBS1

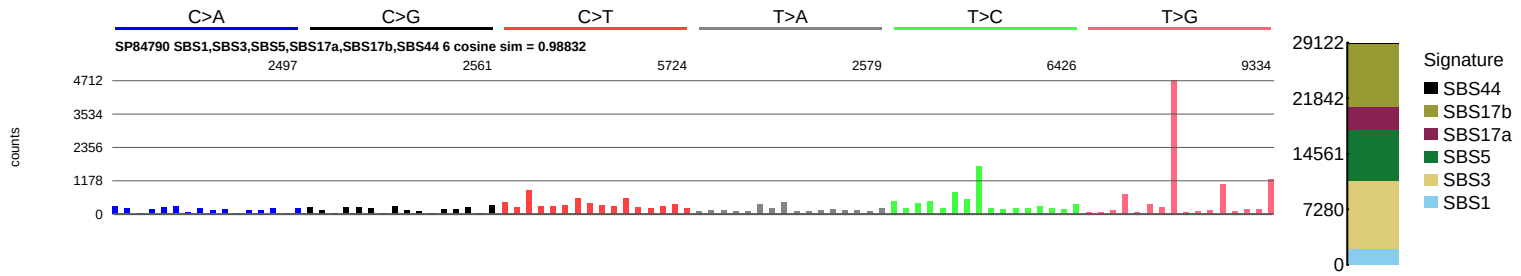

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26 9 cosine sim = 0.98841
2573
2968
5379
2600
6334
9267
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

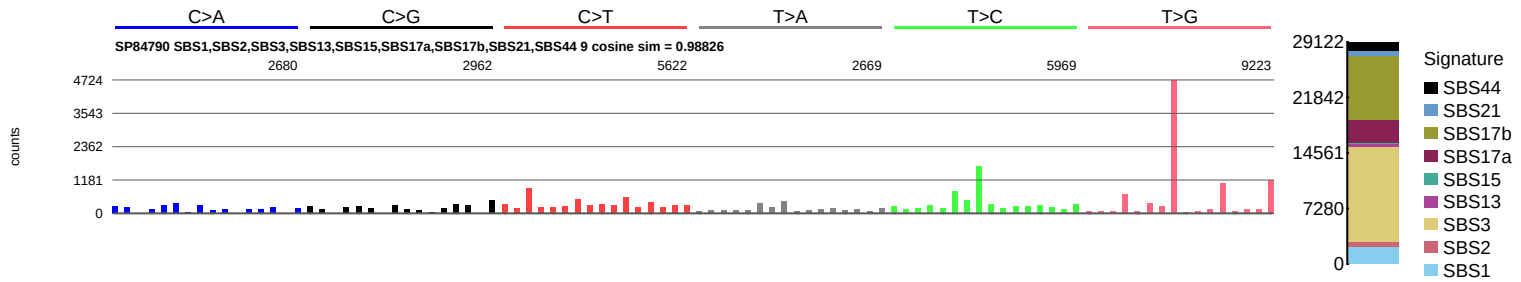

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b 6 cosine sim = 0.98841
2505
2548
5776
2566
6399
9327
counts
4712
3534
2356
1178
0
29122
21842
14561
7280
0
Signature
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1

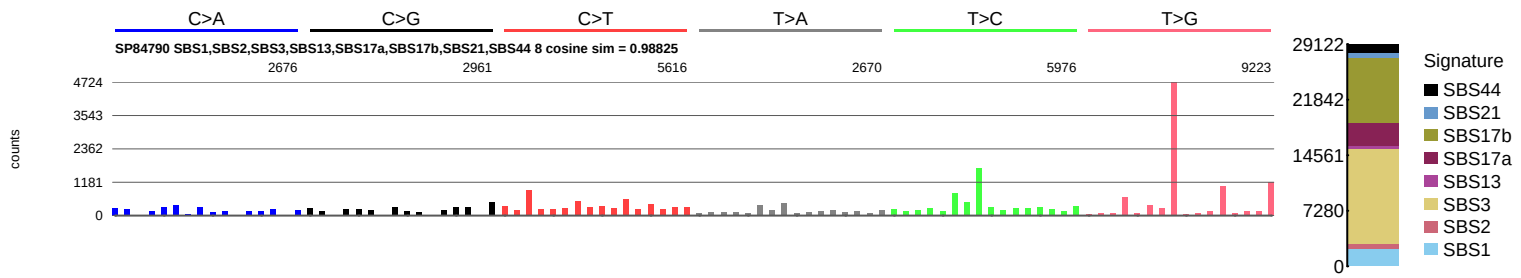

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26 8 cosine sim = 0.9884
2571
2968
5377
2598
6330
9276
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

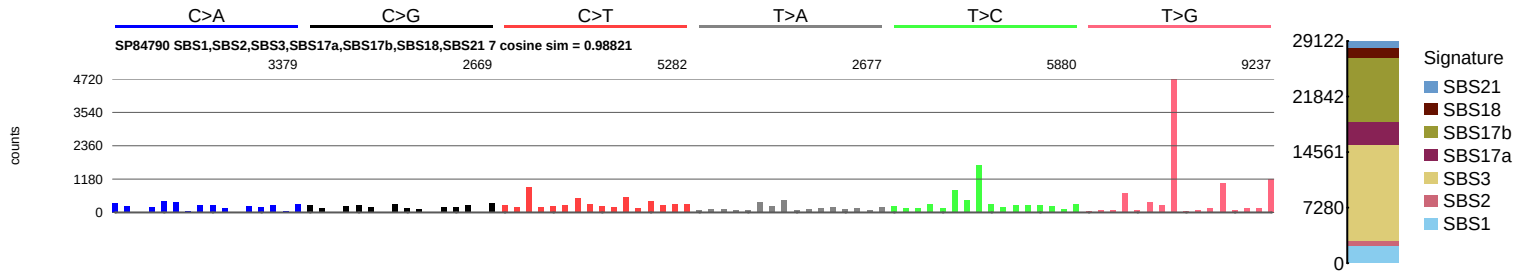

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS17a,SBS17b 5 cosine sim = 0.98833
2492
2573
5711
2582
6429
9339
counts
4712
3534
2356
1178
0
29122
21842
14561
7280
0
Signature
SBS17b
SBS17a
SBS5
SBS3
SBS1

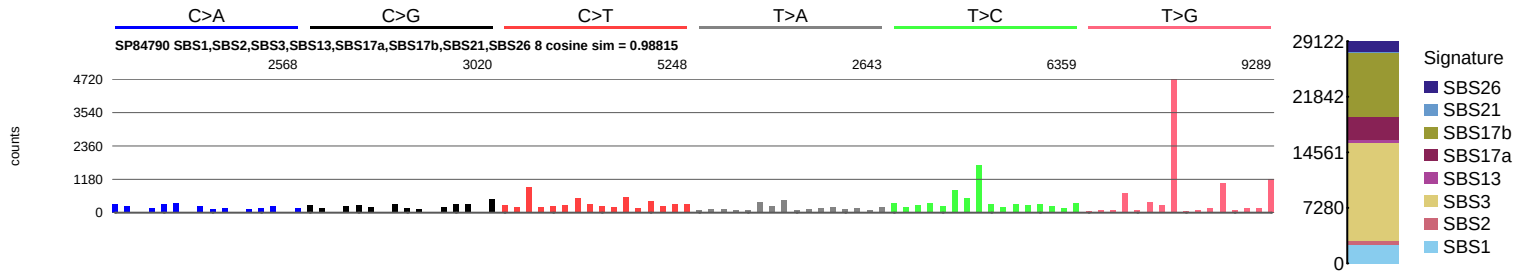

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98833
3367
2638
5377
2656
5859
9226
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

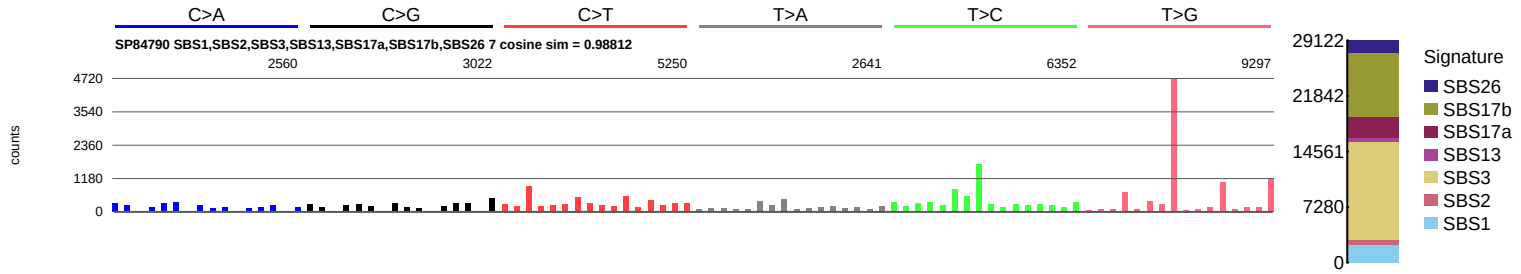

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS44 6 cosine sim = 0.98832
2497
2561
5724
2579
6426
9334
counts
4712
3534
2356
1178
0
29122
21842
14561
7280
0
Signature
SBS44
SBS17b
SBS17a
SBS5
SBS3
SBS1

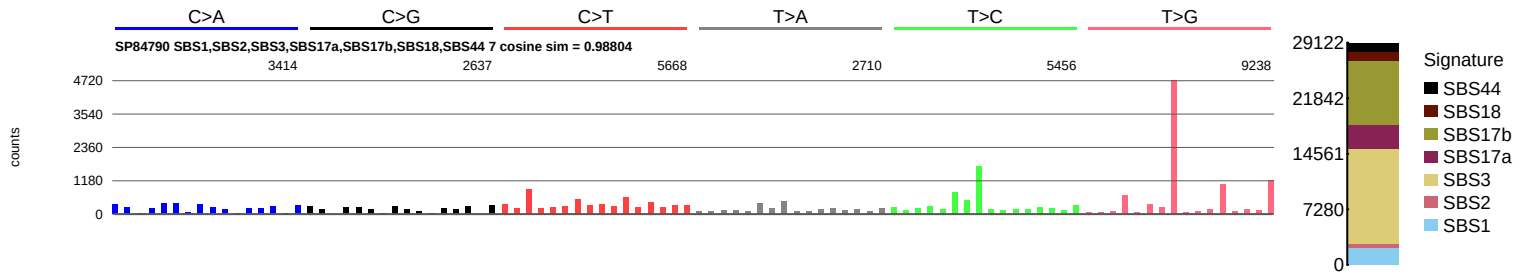

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS44 9 cosine sim = 0.98826
2680
2962
5622
2669
5969
9223
counts
4724
3543
2362
1181
0
29122
21842
14561
7280
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

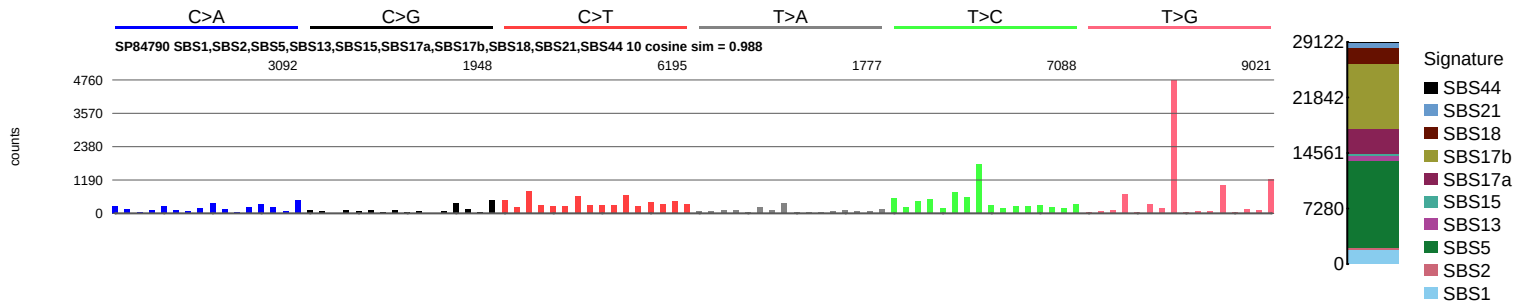

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS44 8 cosine sim = 0.98825
2676
2961
5616
2670
5976
9223
counts
4724
3543
2362
1181
0
29122
21842
14561
7280
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

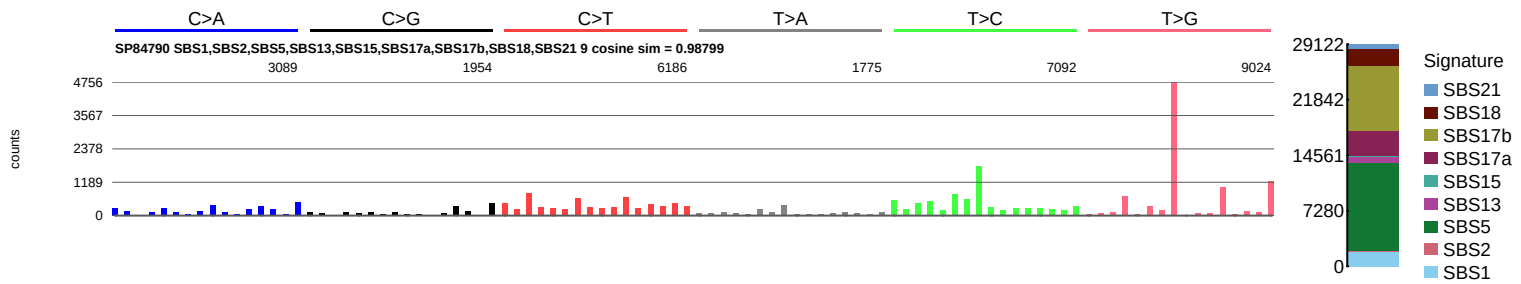

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98821
3379
2669
5282
2677
5880
9237
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

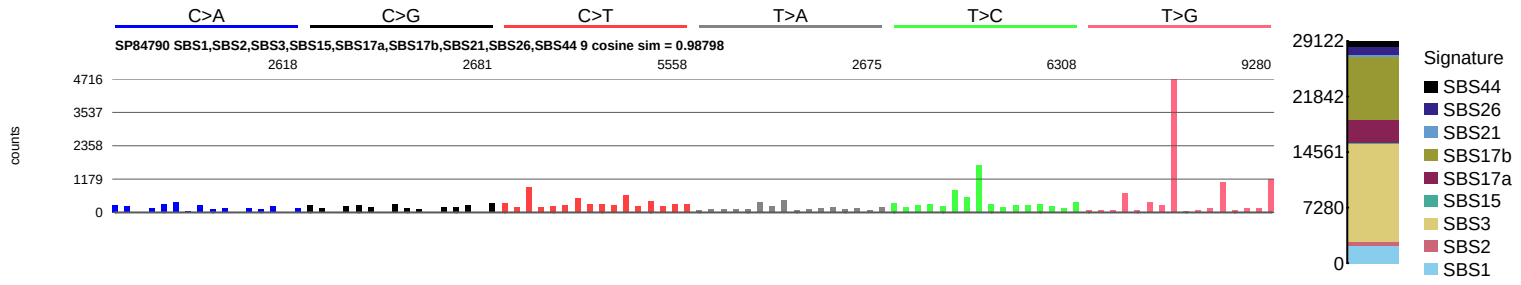

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS26 8 cosine sim = 0.98815
2568
3020
5248
2643
6359
9289
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

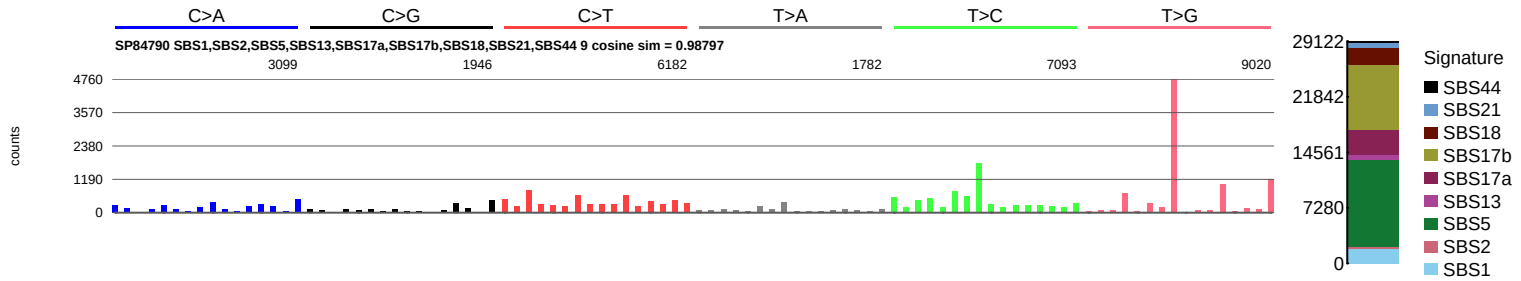

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS26 7 cosine sim = 0.98812
2560
3022
5250
2641
6352
9297
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS26
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

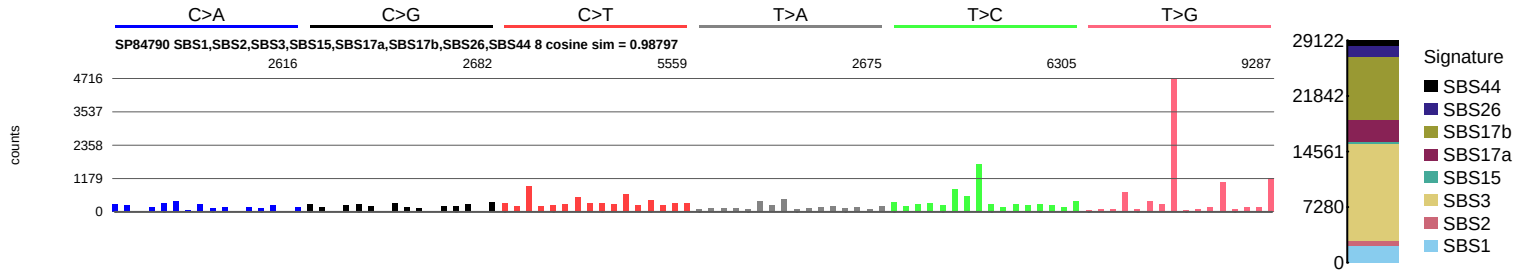

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.98804
3414
2637
5668
2710
5456
9238
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

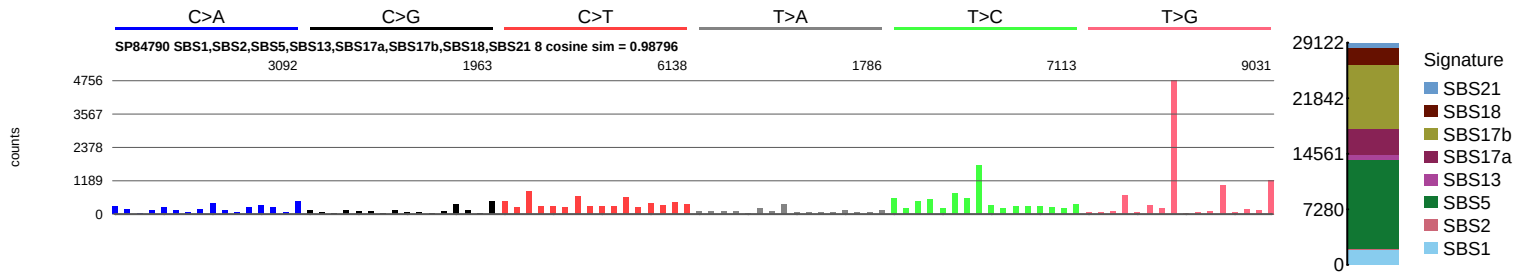

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.988
3092
1948
6195
1777
7088
9021
counts
4760
3570
2380
1190
0
29122
21842
14561
7280
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

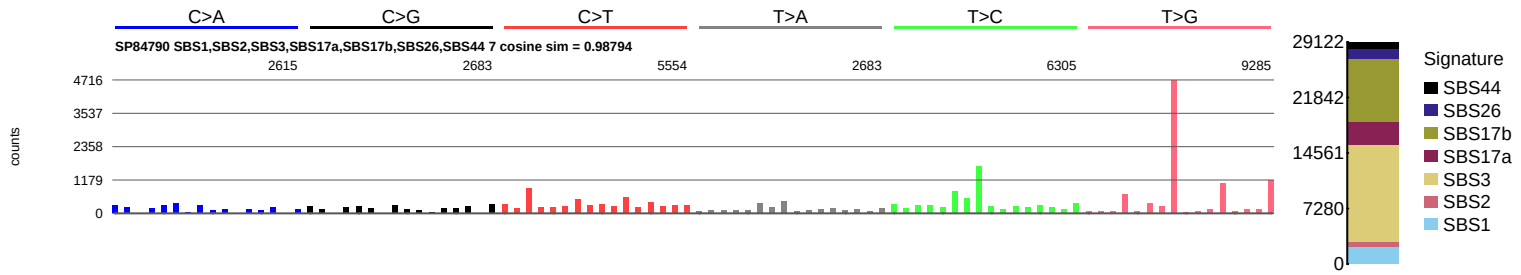

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98799
3089
1954
6186
1775
7092
9024
counts
4756
3567
2378
1189
0
29122
21842
14561
7280
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

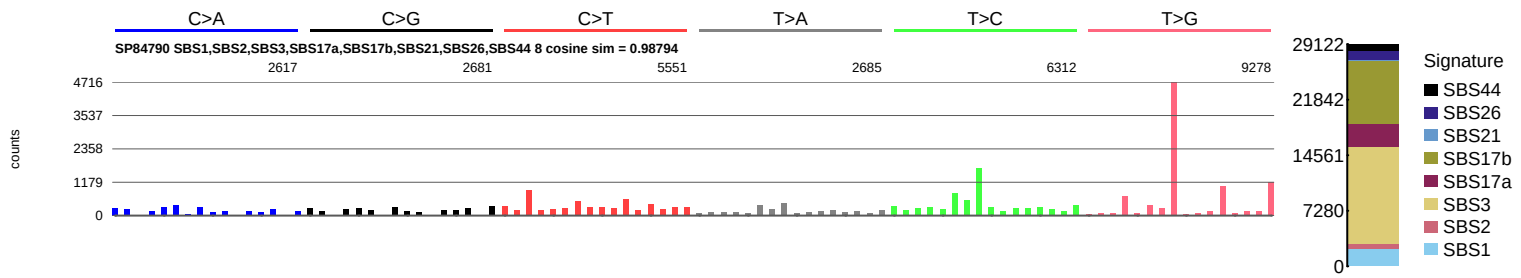

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS44 9 cosine sim = 0.98798
2618
2681
5558
2675
6308
9280
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS44
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

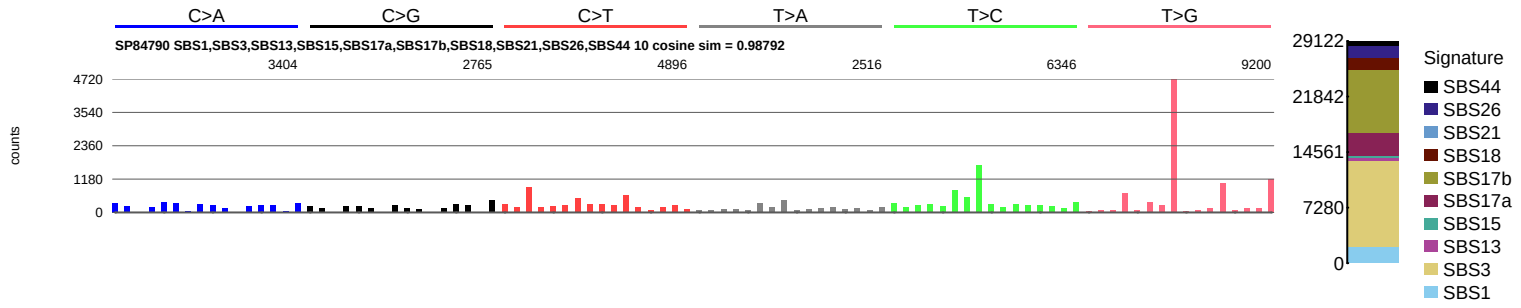

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98797
3099
1946
6182
1782
7093
9020
counts
4760
3570
2380
1190
0
29122
21842
14561
7280
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

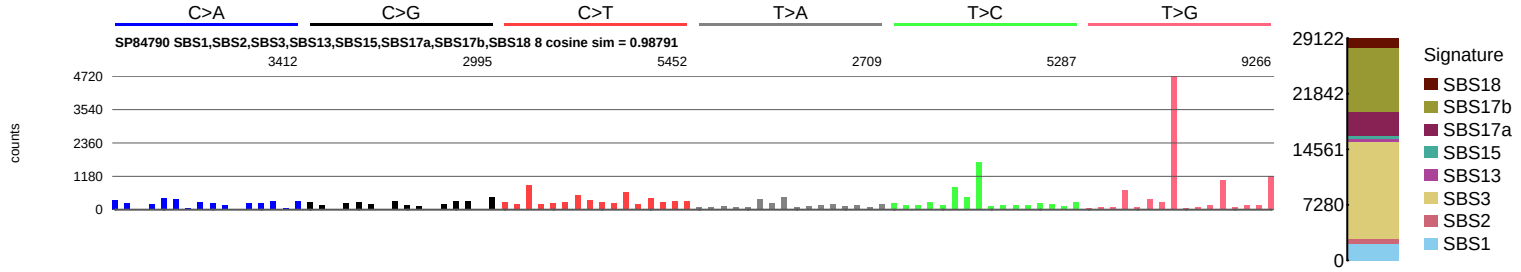

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS44 8 cosine sim = 0.98797
2616
2682
5559
2675
6305
9287
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

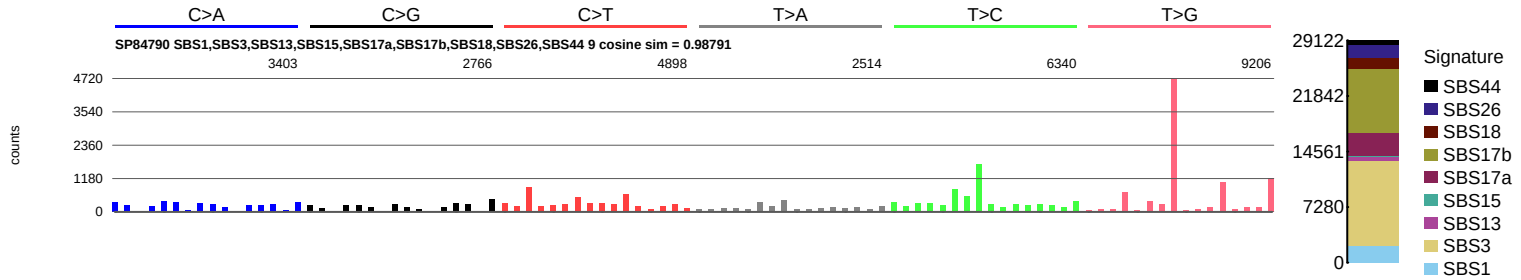

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98796
3092
1963
6138
1786
7113
9031
counts
4756
3567
2378
1189
0
29122
21842
14561
7280
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

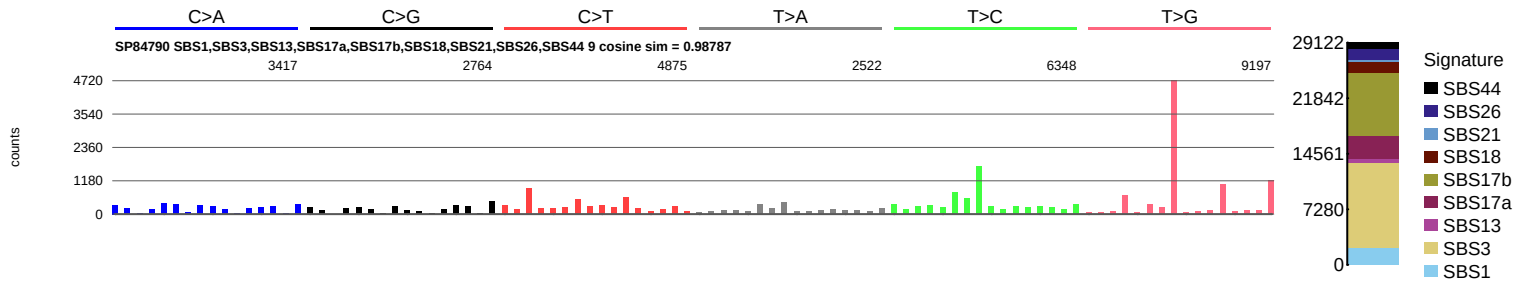

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS26,SBS44 7 cosine sim = 0.98794
2615
2683
5554
2683
6305
9285
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS3
SBS2
SBS1

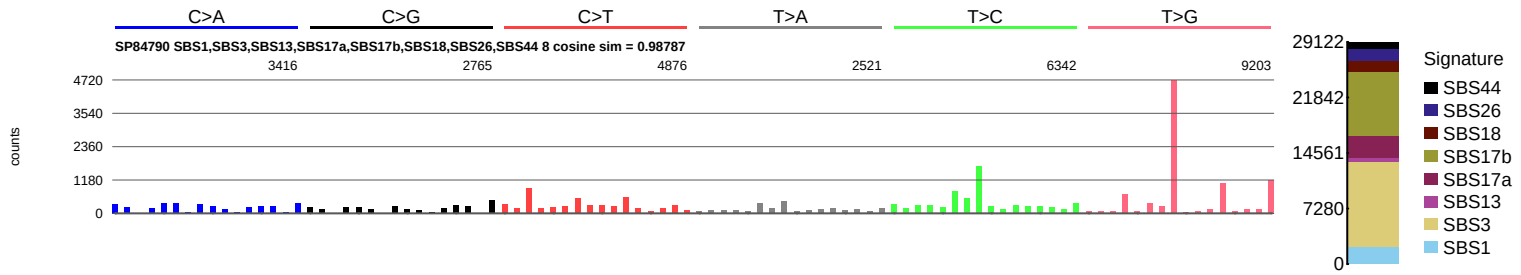

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS26,SBS44 8 cosine sim = 0.98794
2617
2681
5551
2685
6312
9278
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS44
SBS26
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1

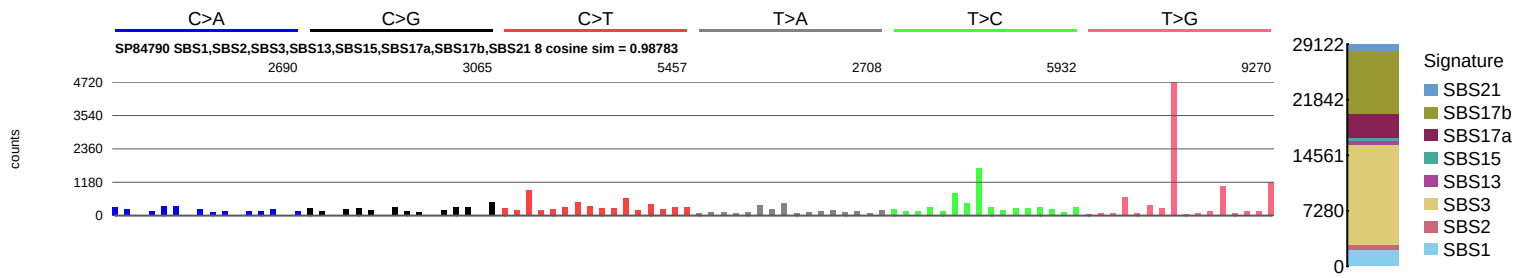

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 10 cosine sim = 0.98792
3404
2765
4896
2516
6346
9200
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

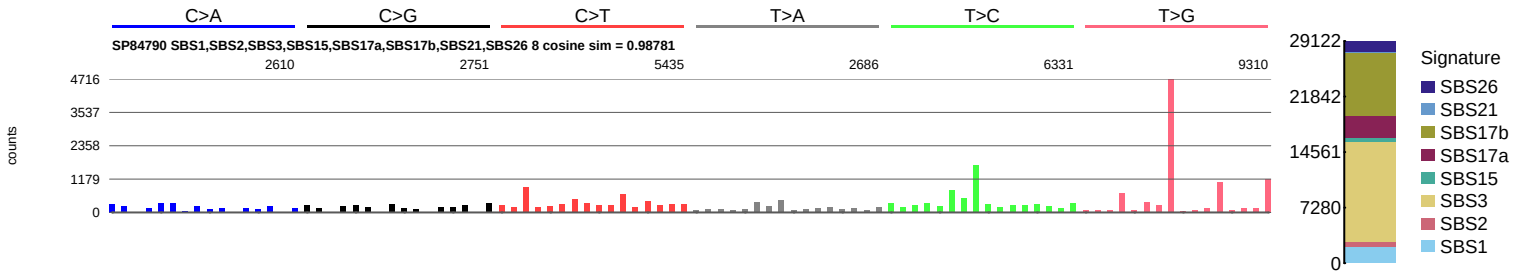

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98791
3412
2995
5452
2709
5287
9266
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

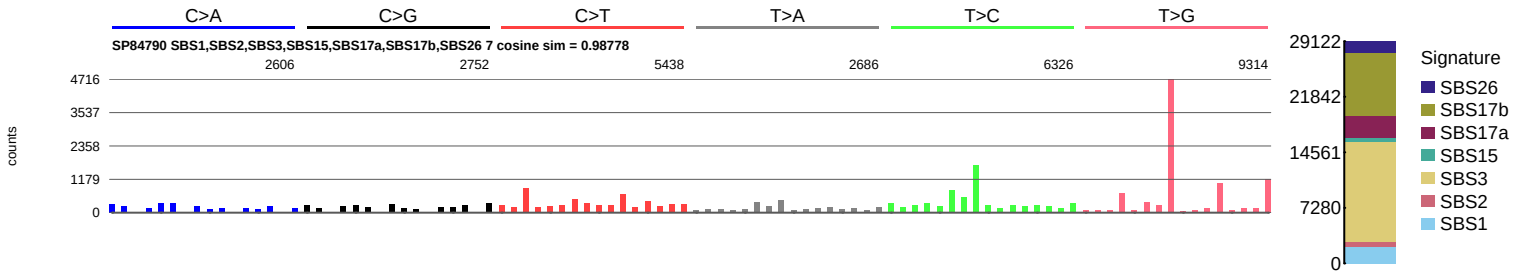

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.98791
3403
2766
4898
2514
6340
9206
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

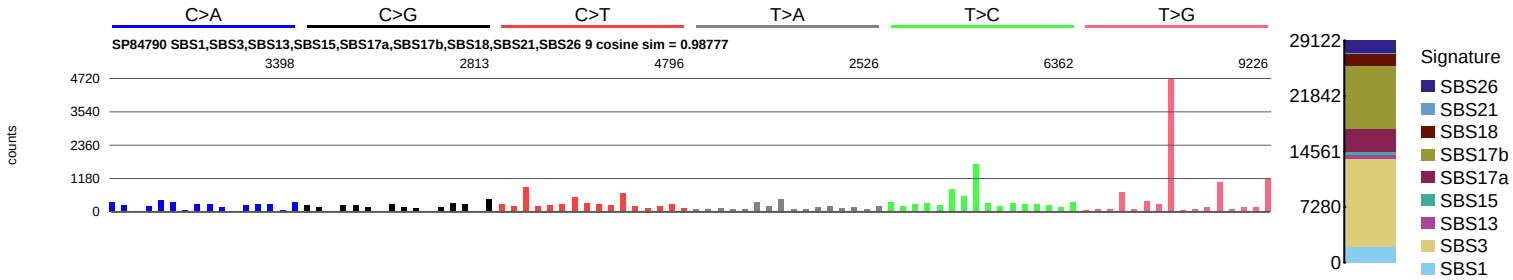

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 9 cosine sim = 0.98787
3417
2764
4875
2522
6348
9197
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1

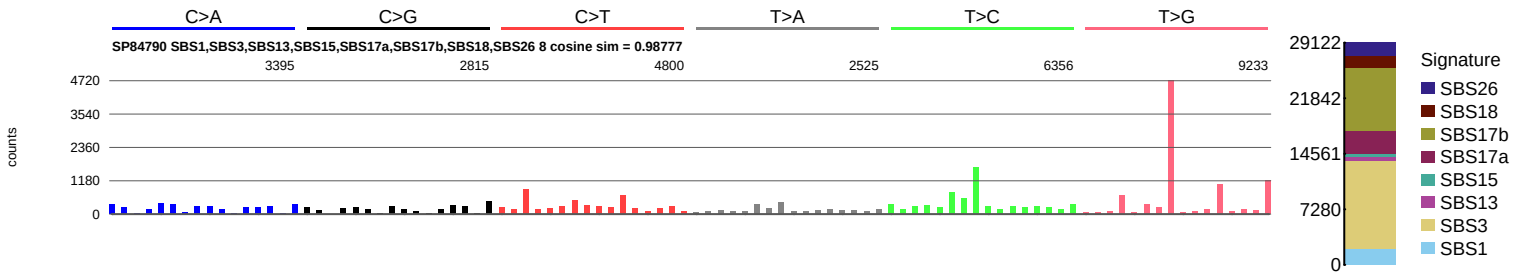

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98787
3416
2765
4876
2521
6342
9203
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1

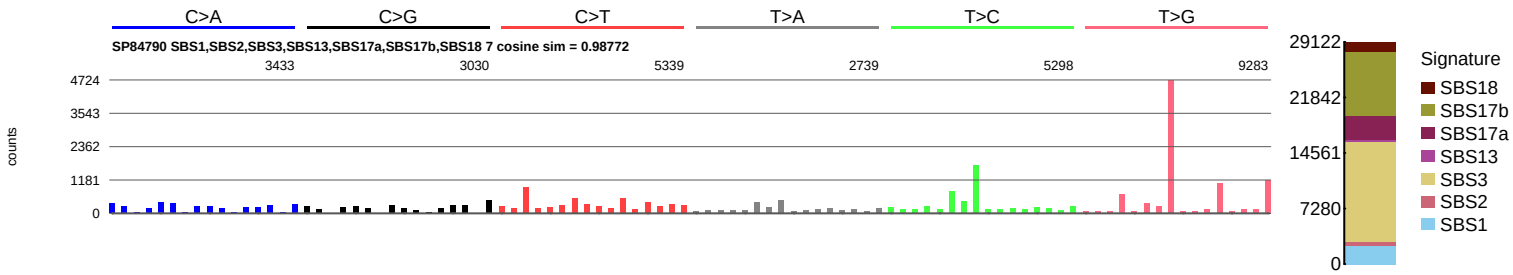

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21 8 cosine sim = 0.98783
2690
3065
5457
2708
5932
9270
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

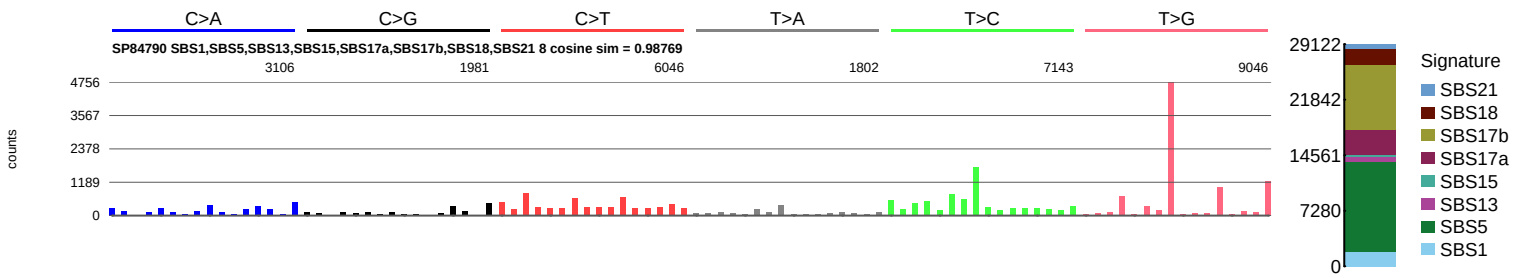

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS26 8 cosine sim = 0.98781
2610
2751
5435
2686
6331
9310
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

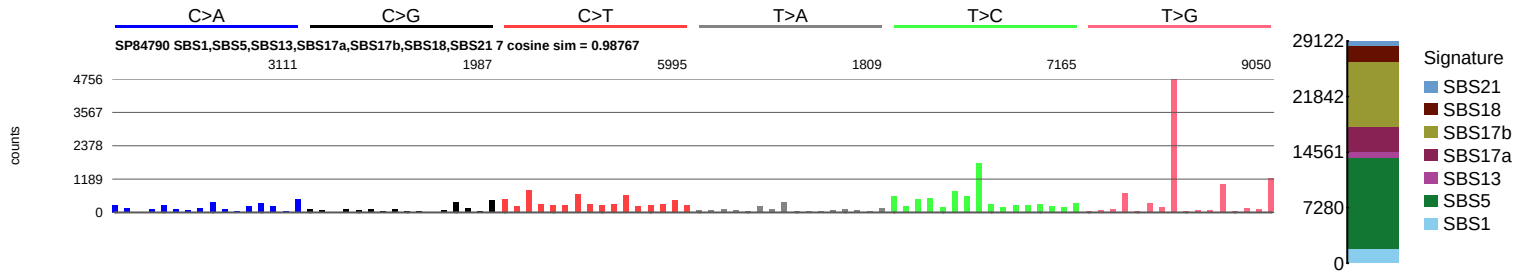

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26 7 cosine sim = 0.98778
2606
2752
5438
2686
6326
9314
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

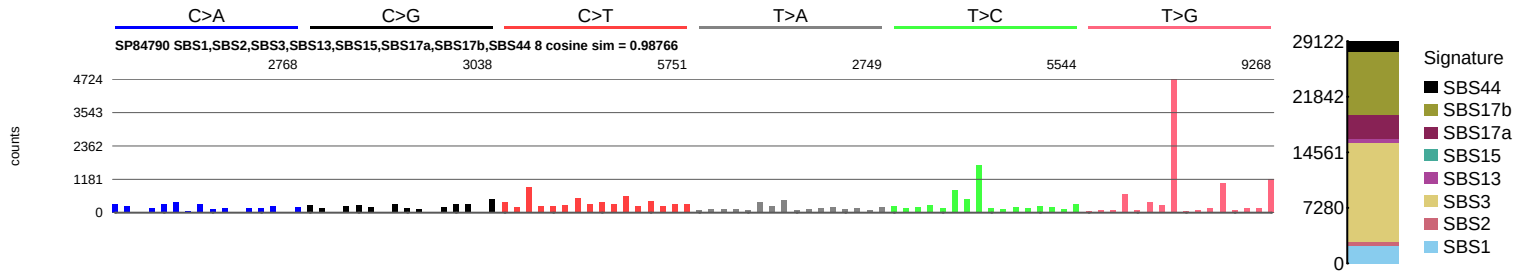

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26 9 cosine sim = 0.98777
3398
2813
4796
2526
6362
9226
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

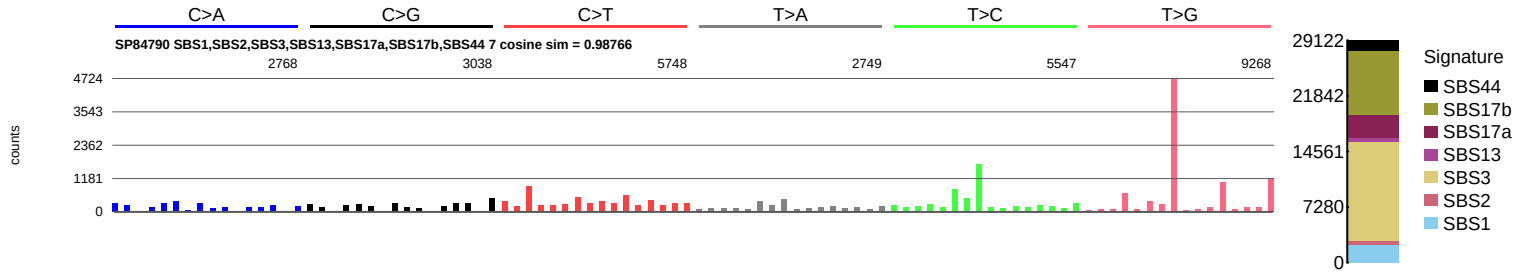

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98777
3395
2815
4800
2525
6356
9233
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

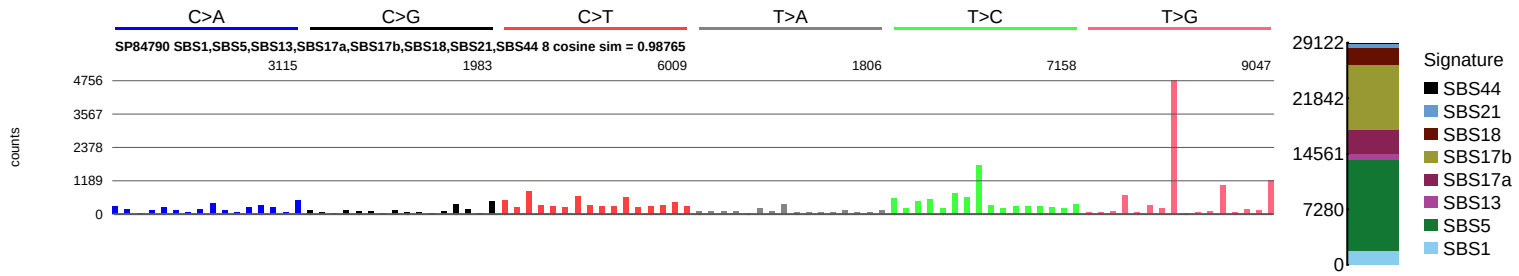

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98772
3433
3030
5339
2739
5298
9283
counts
4724
3543
2362
1181
0
29122
21842
14561
7280
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

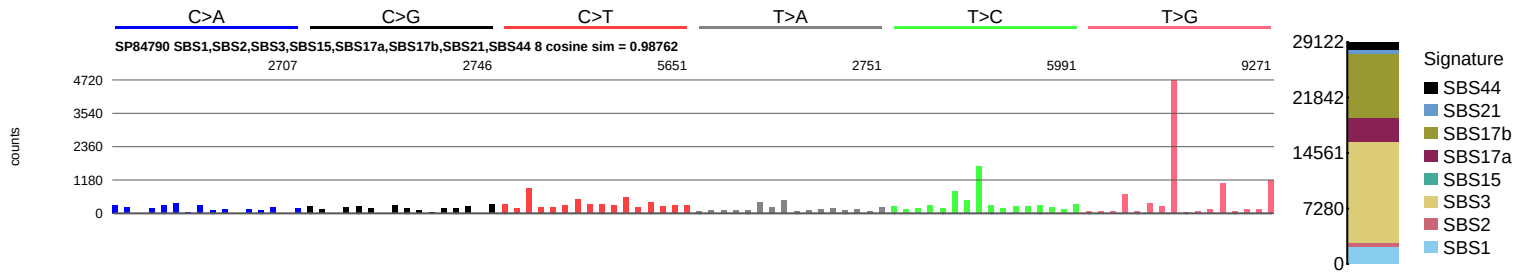

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98769
3106
1981
6046
1802
7143
9046
counts
4756
3567
2378
1189
0
29122
21842
14561
7280
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

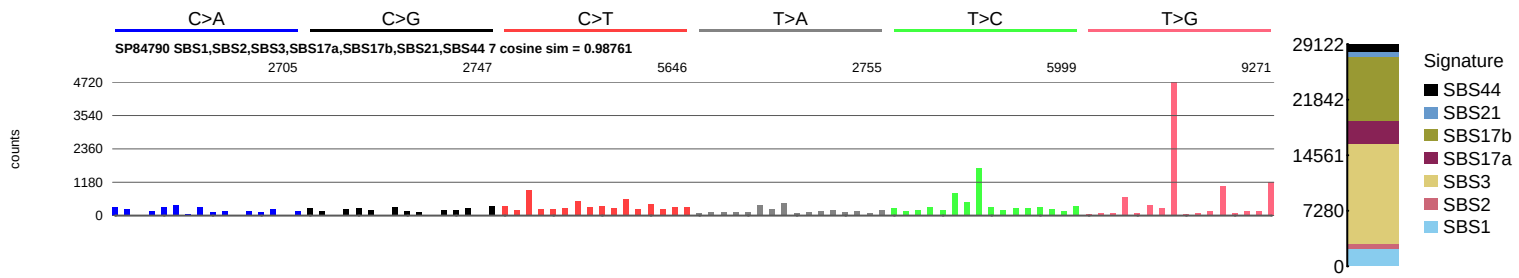

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98767
3111
1987
5995
1809
7165
9050
counts
4756
3567
2378
1189
0
29122
21842
14561
7280
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1

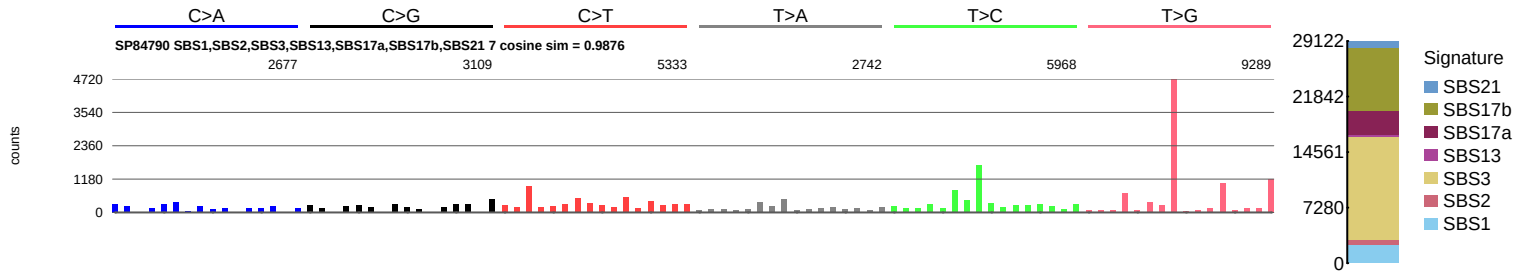

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS44 8 cosine sim = 0.98766
2768
3038
5751
2749
5544
9268
counts
4724
3543
2362
1181
0
29122
21842
14561
7280
0
Signature
SBS44
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

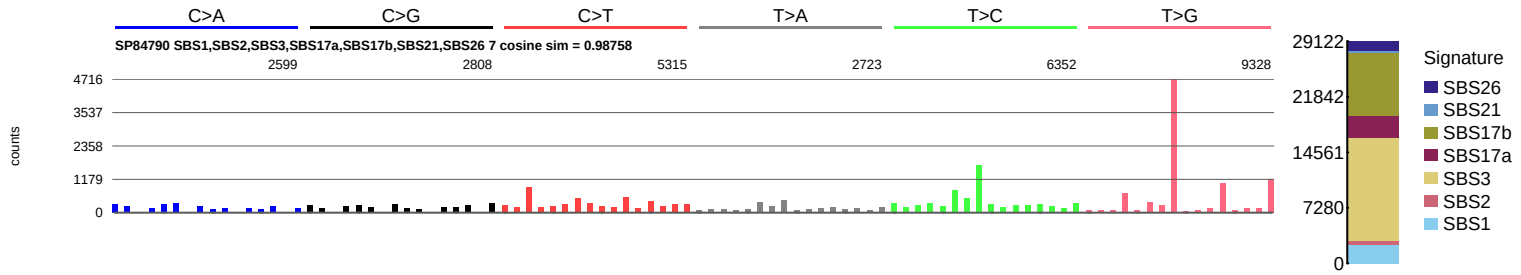

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS44 7 cosine sim = 0.98766
2768
3038
5748
2749
5547
9268
counts
4724
3543
2362
1181
0
29122
21842
14561
7280
0
Signature
SBS44
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

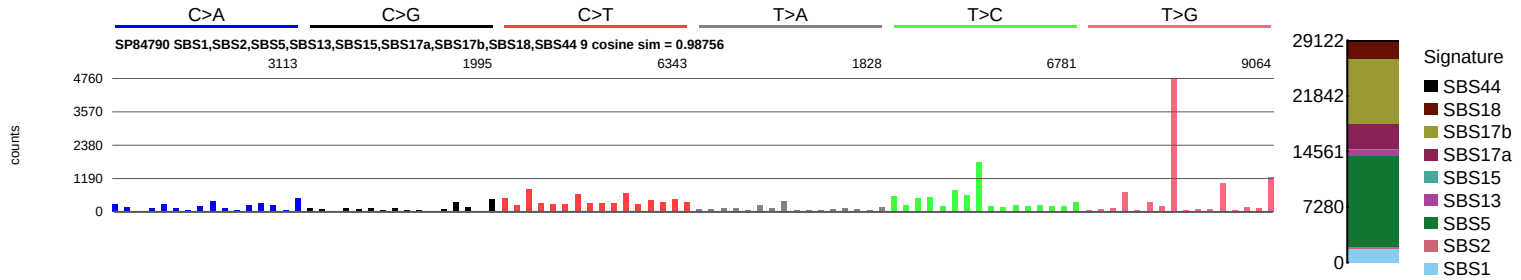

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98765
3115
1983
6009
1806
7158
9047
counts
4756
3567
2378
1189
0
29122
21842
14561
7280
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1

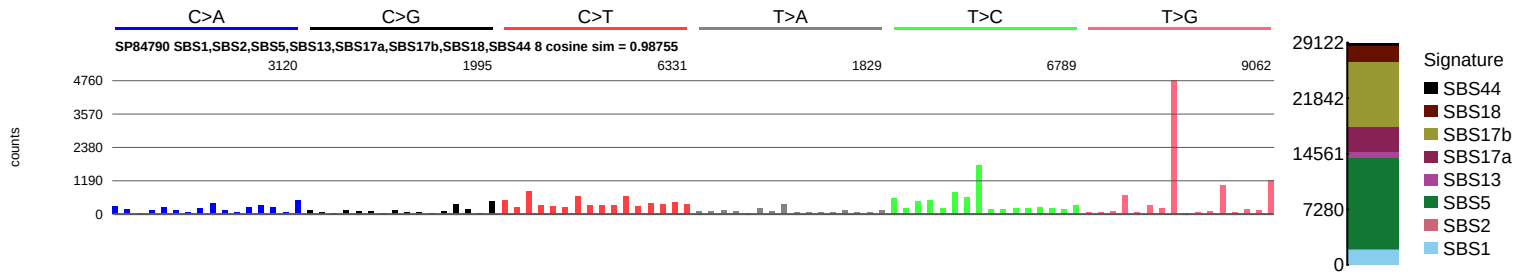

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS44 8 cosine sim = 0.98762
2707
2746
5651
2751
5991
9271
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

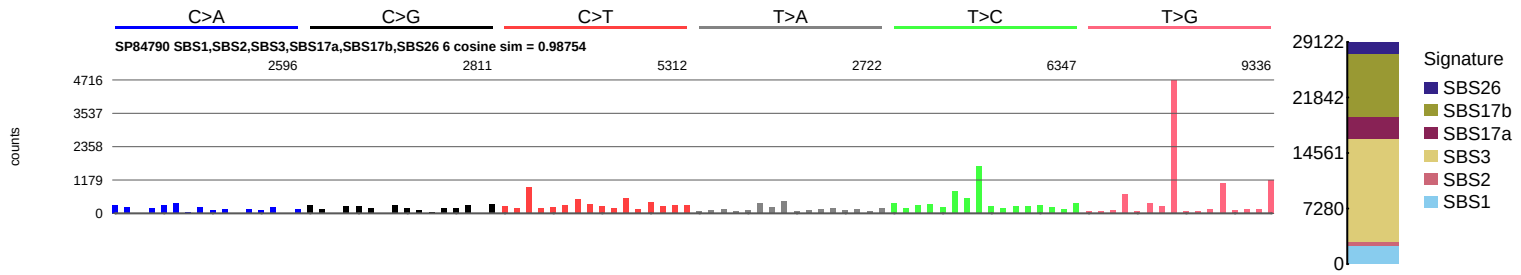

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS44 7 cosine sim = 0.98761
2705
2747
5646
2755
5999
9271
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1

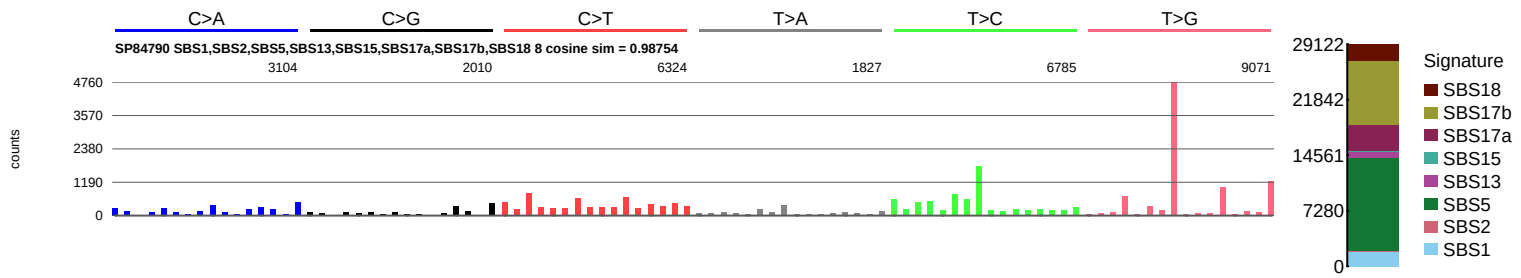

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21 7 cosine sim = 0.9876
2677
3109
5333
2742
5968
9289
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

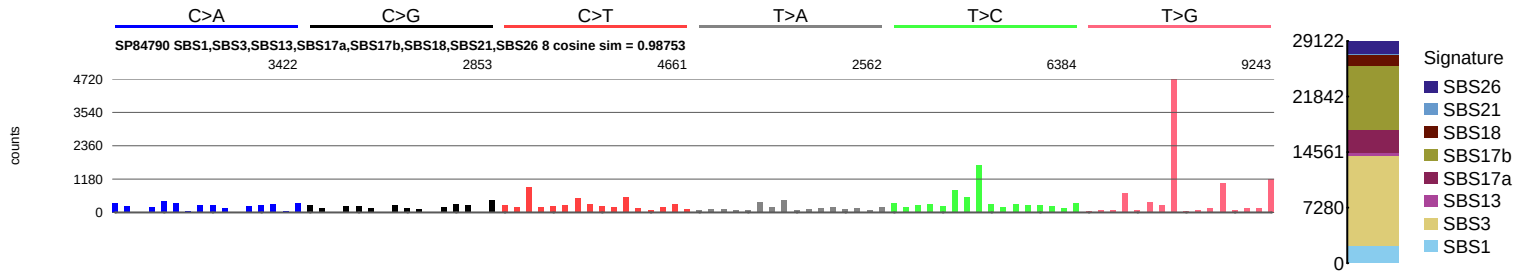

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS26 7 cosine sim = 0.98758
2599
2808
5315
2723
6352
9328
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS26
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1

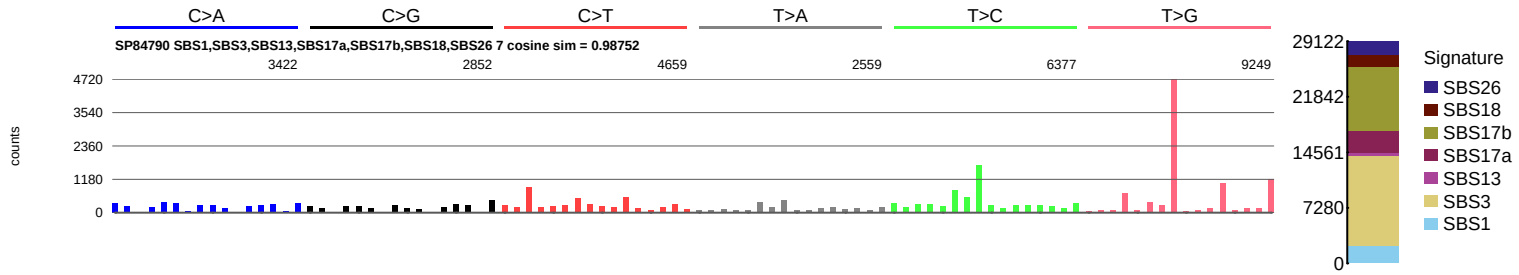

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS44 9 cosine sim = 0.98756
3113
1995
6343
1828
6781
9064
counts
4760
3570
2380
1190
0
29122
21842
14561
7280
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

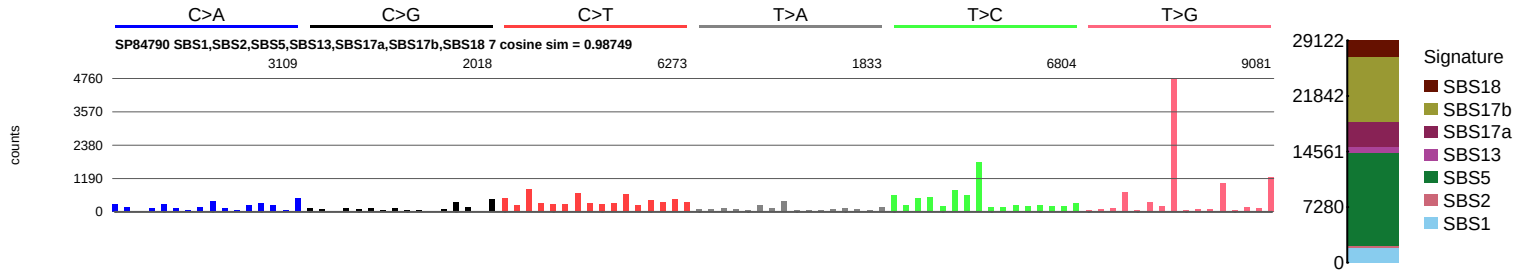

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98755
3120
1995
6331
1829
6789
9062
counts
4760
3570
2380
1190
0
29122
21842
14561
7280
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

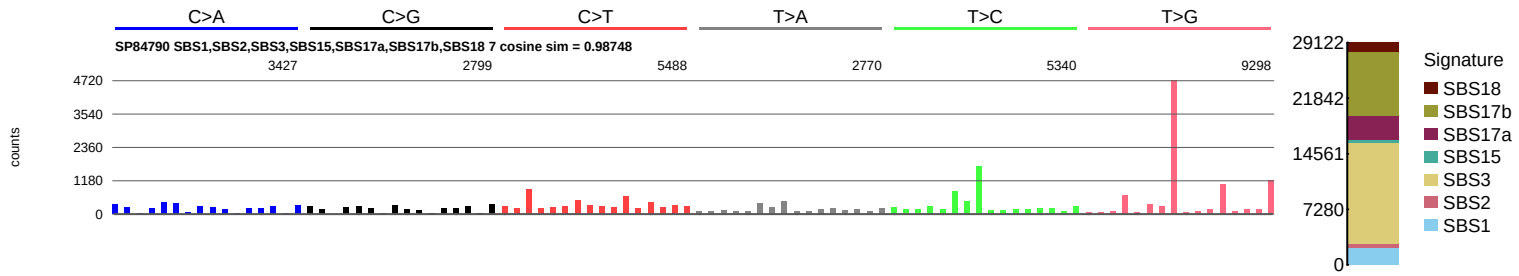

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS26 6 cosine sim = 0.98754
2596
2811
5312
2722
6347
9336
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS26
SBS17b
SBS17a
SBS3
SBS2
SBS1

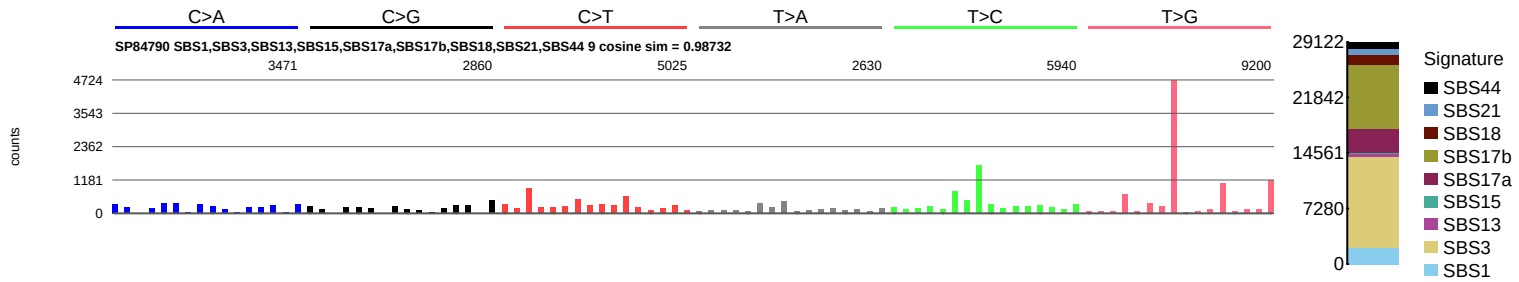

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98754
3104
2010
6324
1827
6785
9071
counts
4760
3570
2380
1190
0
29122
21842
14561
7280
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

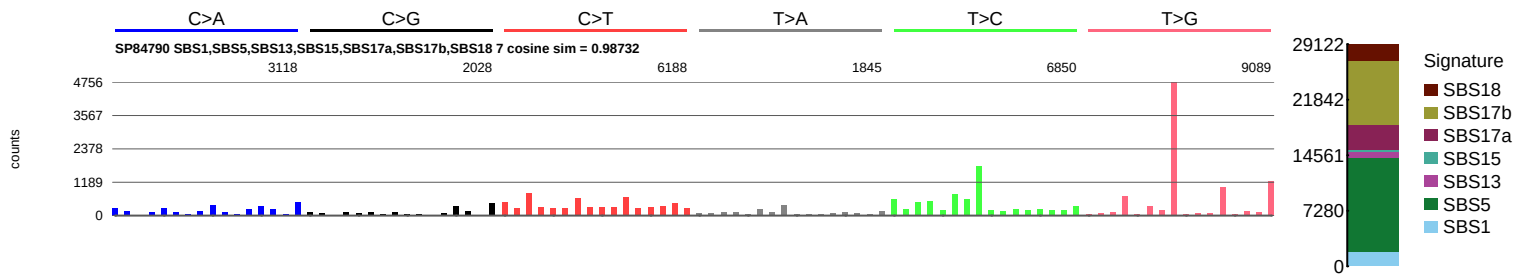

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26 8 cosine sim = 0.98753
3422
2853
4661
2562
6384
9243
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1

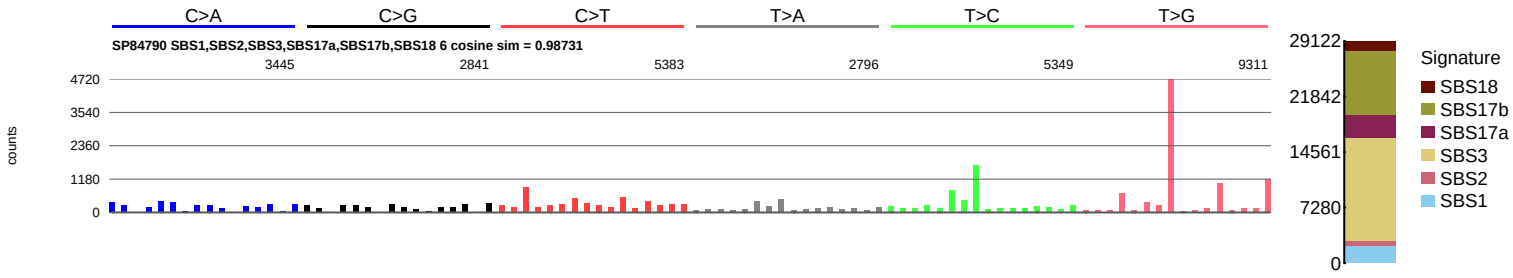

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98752
3422
2852
4659
2559
6377
9249
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1

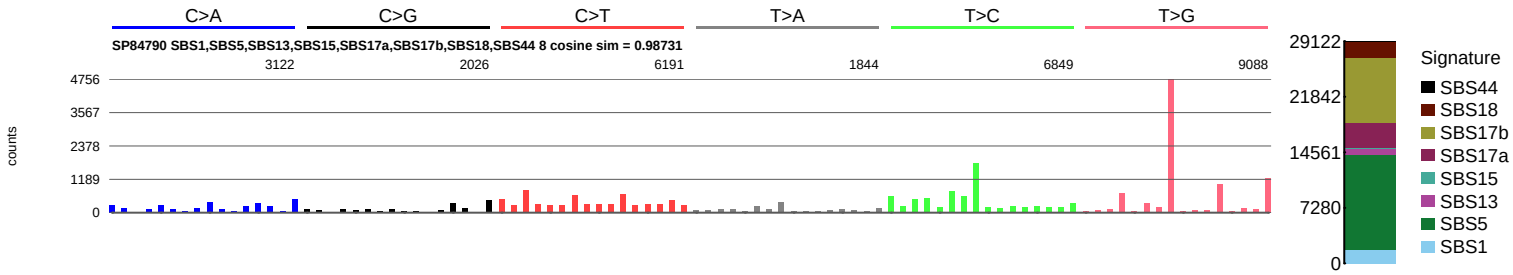

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98749
3109
2018
6273
1833
6804
9081
counts
4760
3570
2380
1190
0
29122
21842
14561
7280
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

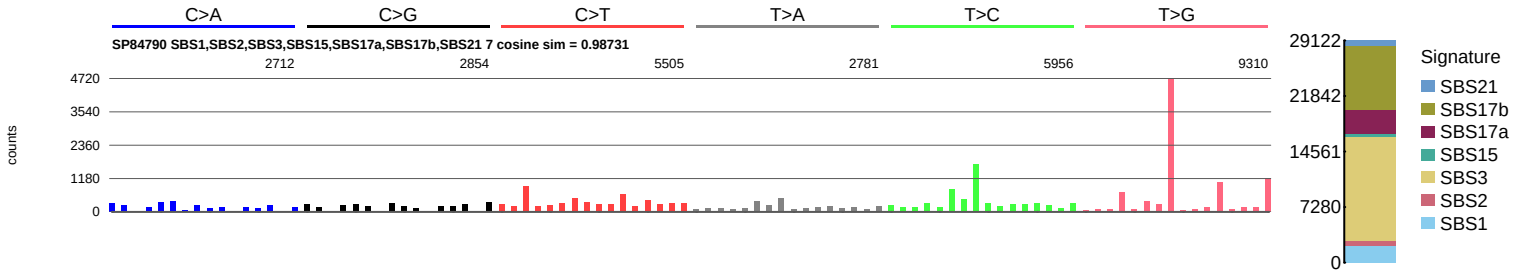

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98748
3427
2799
5488
2770
5340
9298
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

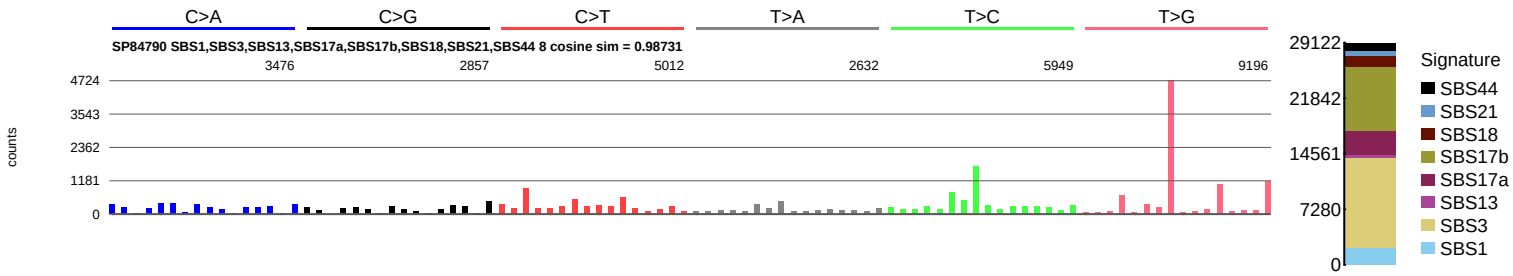

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98732
3471
2860
5025
2630
5940
9200
counts
4724
3543
2362
1181
0
29122
21842
14561
7280
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

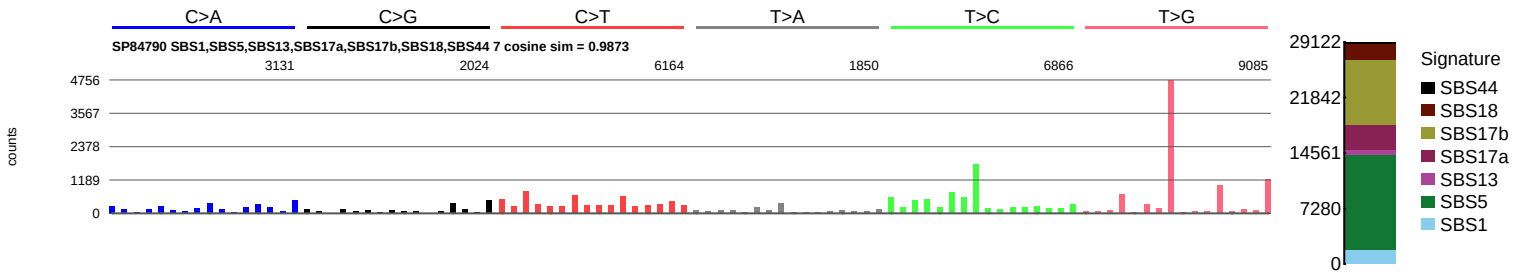

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98732
3118
2028
6188
1845
6850
9089
counts
4756
3567
2378
1189
0
29122
21842
14561
7280
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

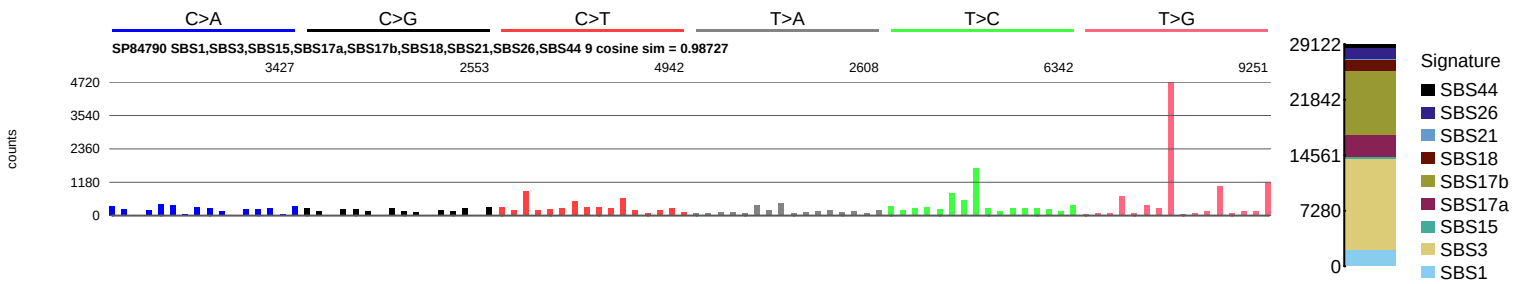

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98731
3445
2841
5383
2796
5349
9311
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

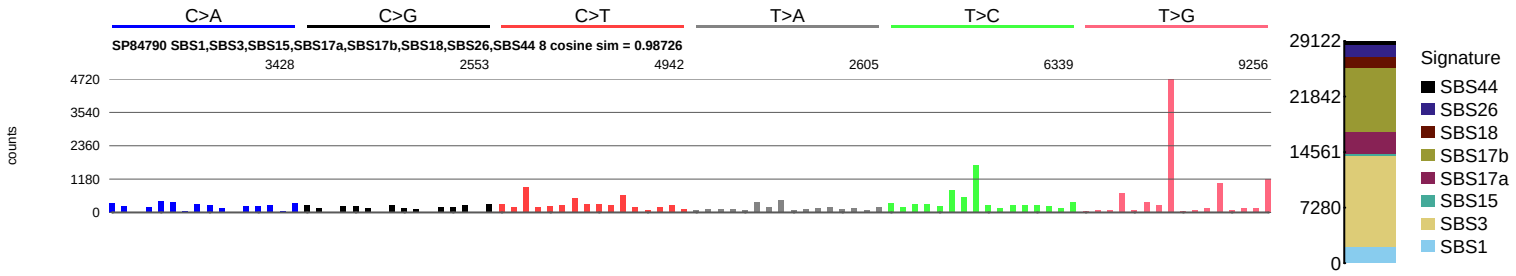

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98731
3122
2026
6191
1844
6849
9088
counts
4756
3567
2378
1189
0
29122
21842
14561
7280
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

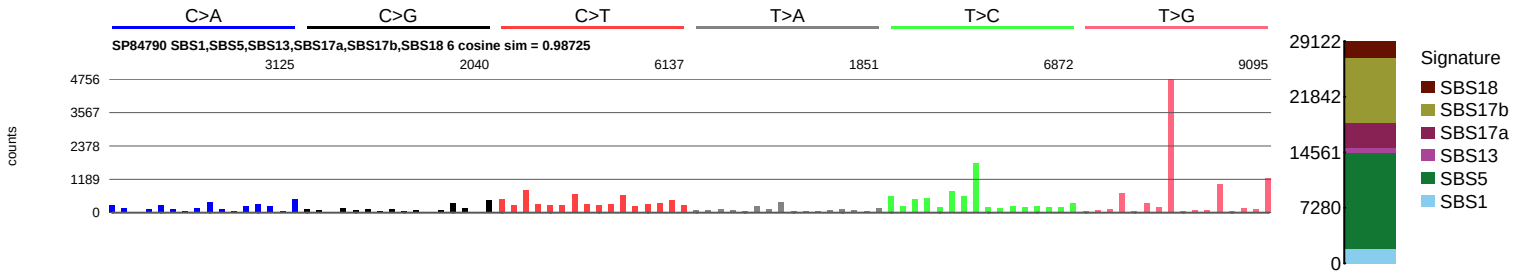

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98731
2712
2854
5505
2781
5956
9310
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

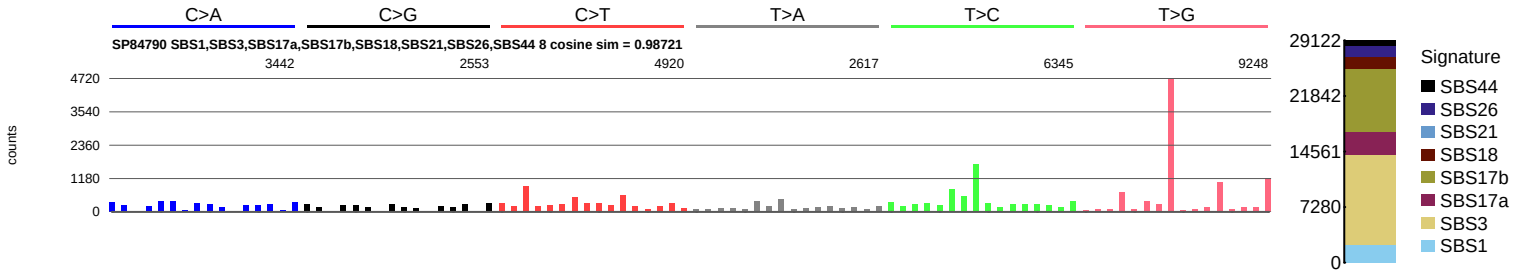

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98731
3476
2857
5012
2632
5949
9196
counts
4724
3543
2362
1181
0
29122
21842
14561
7280
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1

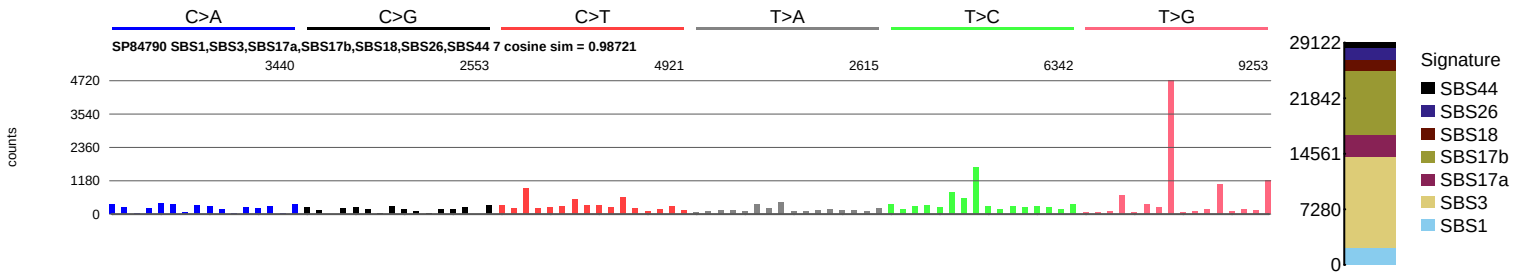

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.9873
3131
2024
6164
1850
6866
9085
counts
4756
3567
2378
1189
0
29122
21842
14561
7280
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1

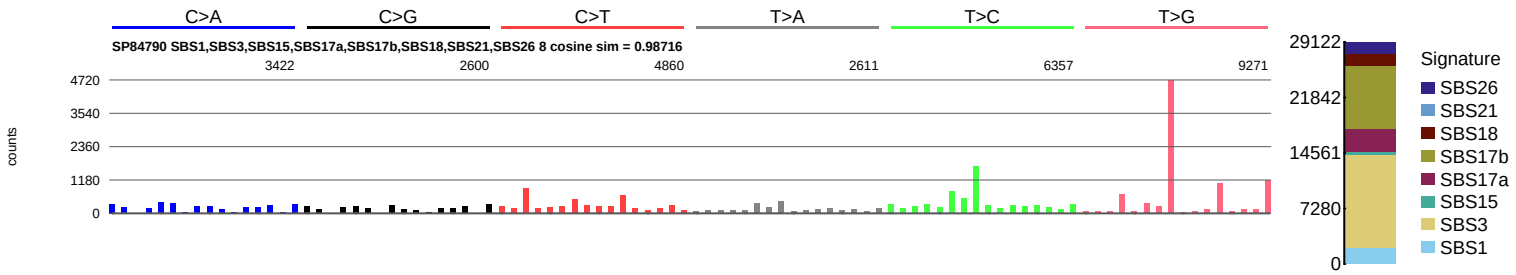

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 9 cosine sim = 0.98727
3427
2553
4942
2608
6342
9251
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

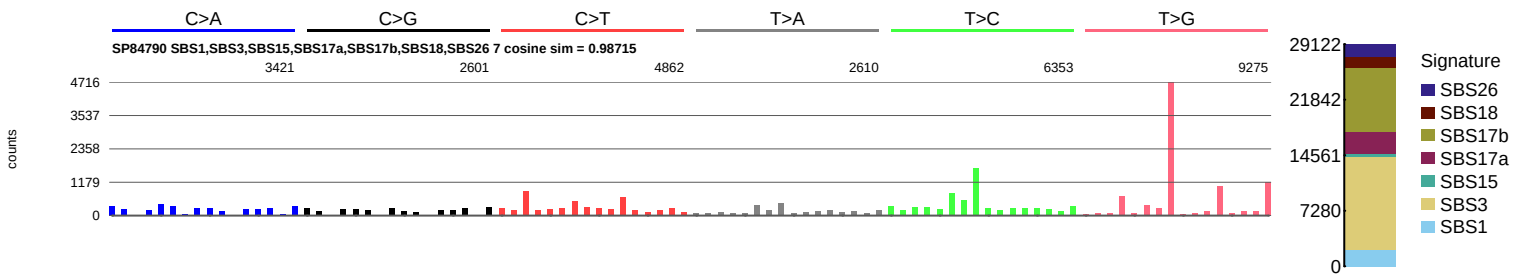

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98726
3428
2553
4942
2605
6339
9256
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

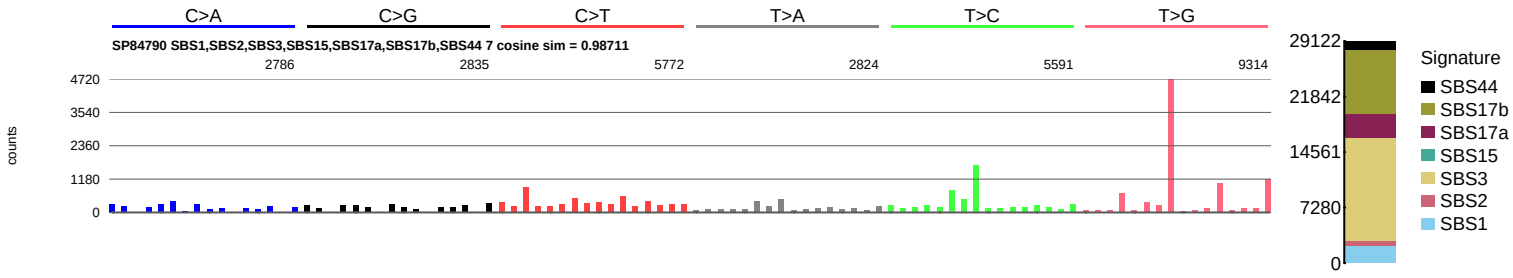

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98725
3125
2040
6137
1851
6872
9095
counts
4756
3567
2378
1189
0
29122
21842
14561
7280
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1

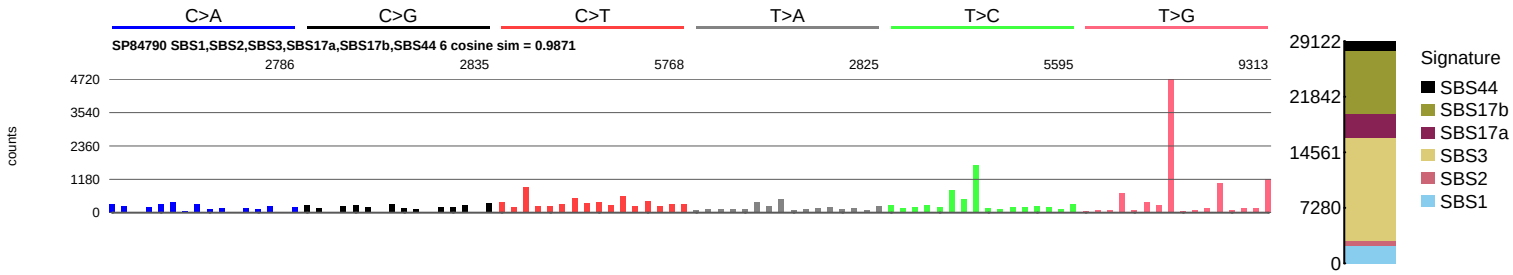

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 8 cosine sim = 0.98721
3442
2553
4920
2617
6345
9248
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1

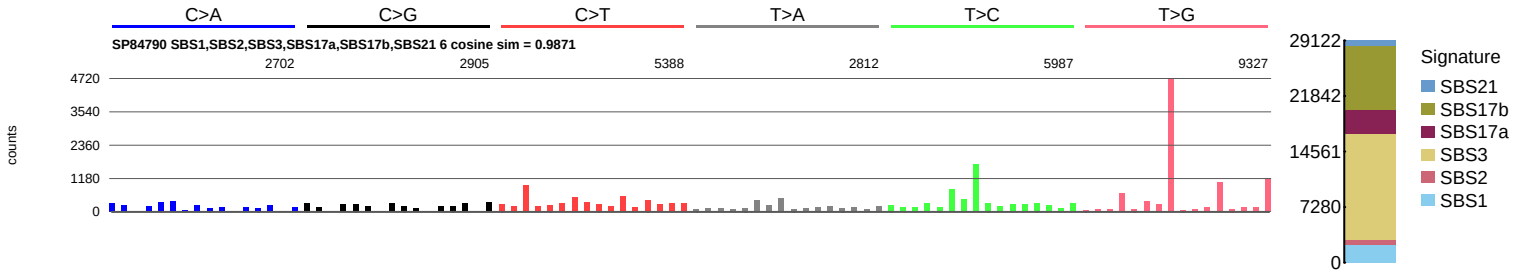

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS44 7 cosine sim = 0.98721
3440
2553
4921
2615
6342
9253
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS1

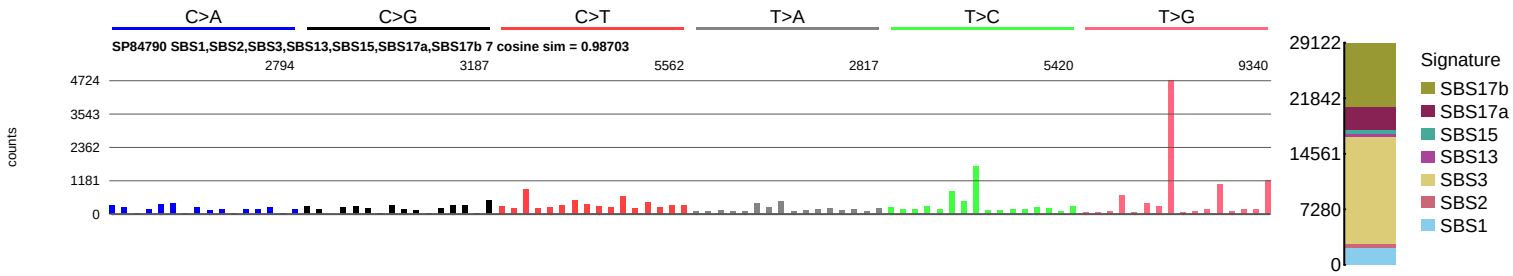

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26 8 cosine sim = 0.98716
3422
2600
4860
2611
6357
9271
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

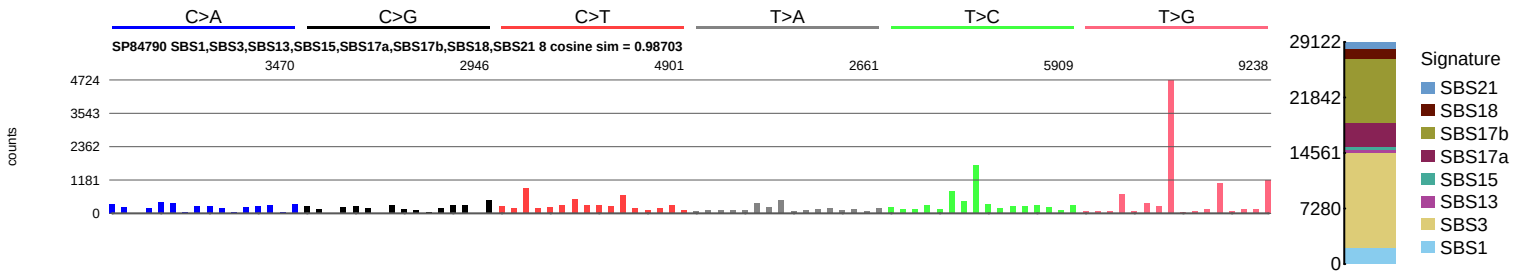

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98715
3421
2601
4862
2610
6353
9275
counts
4716
3537
2358
1179
0
29122
21842
14561
7280
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

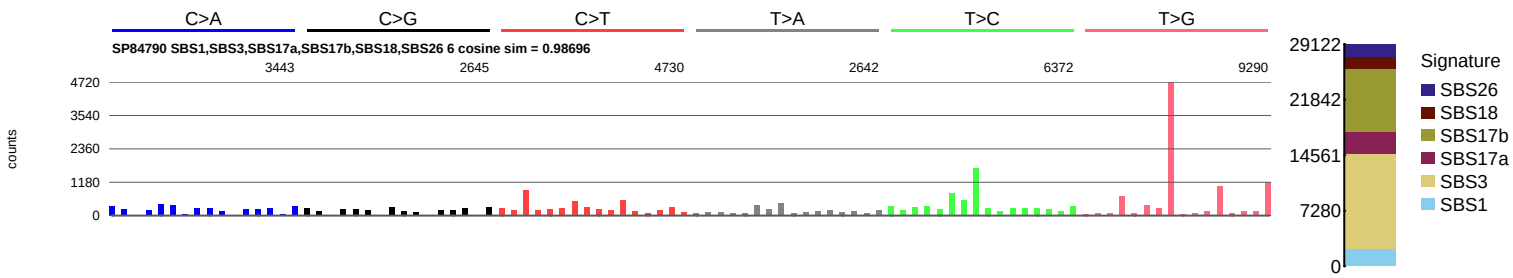

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS44 7 cosine sim = 0.98711
2786
2835
5772
2824
5591
9314
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS44
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

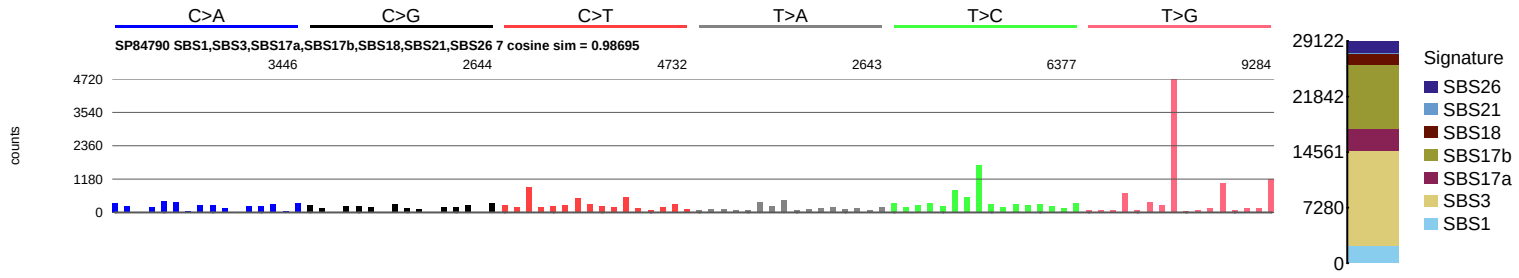

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS44 6 cosine sim = 0.9871
2786
2835
5768
2825
5595
9313
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS44
SBS17b
SBS17a
SBS3
SBS2
SBS1

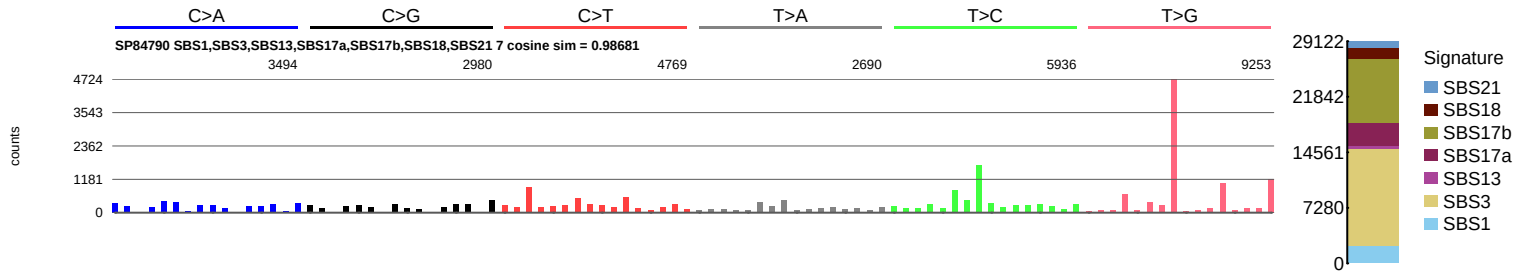

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21 6 cosine sim = 0.9871
2702
2905
5388
2812
5987
9327
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1

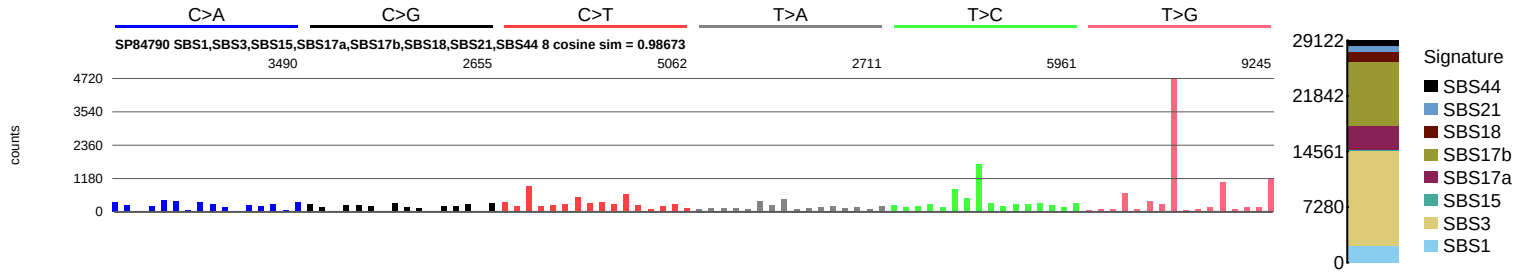

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b 7 cosine sim = 0.98703
2794
3187
5562
2817
5420
9340
counts
4724
3543
2362
1181
0
29122
21842
14561
7280
0
Signature
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

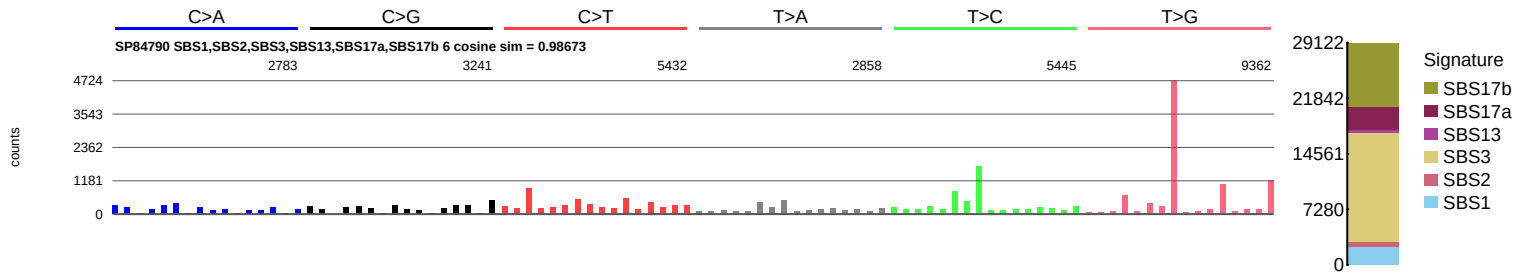

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98703
3470
2946
4901
2661
5909
9238
counts
4724
3543
2362
1181
0
29122
21842
14561
7280
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

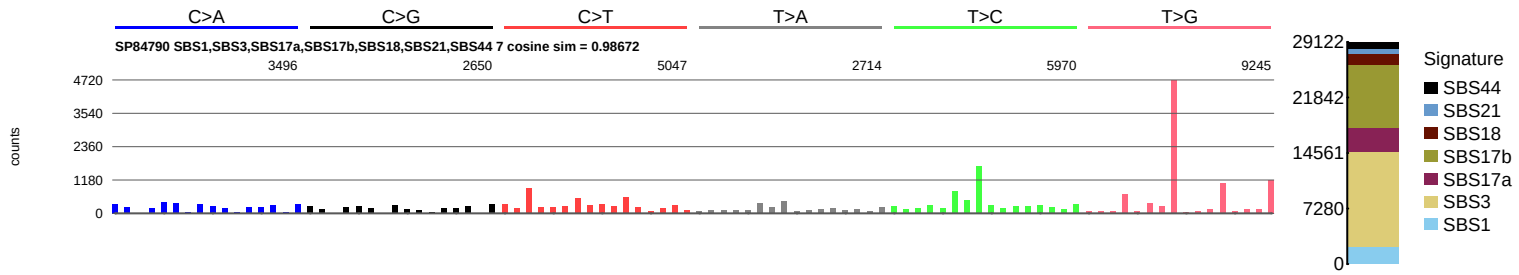

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS26 6 cosine sim = 0.98696
3443
2645
4730
2642
6372
9290
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS1

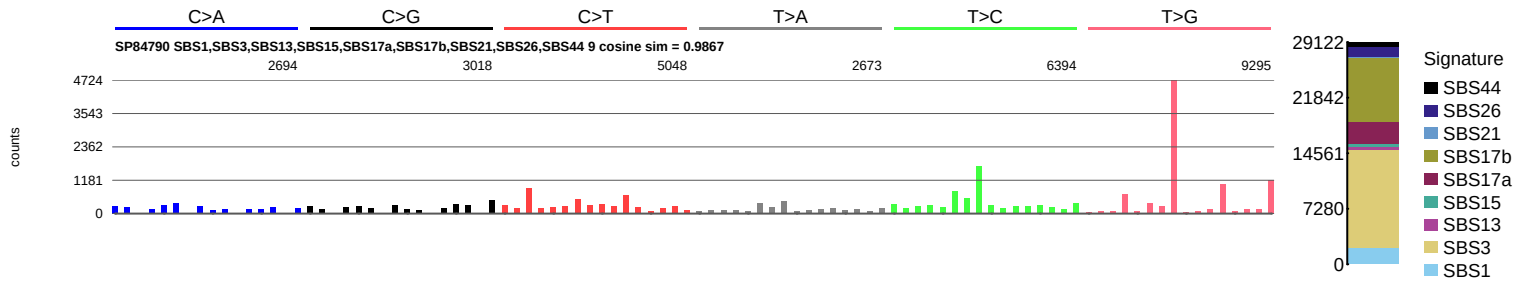

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26 7 cosine sim = 0.98695
3446
2644
4732
2643
6377
9284
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1

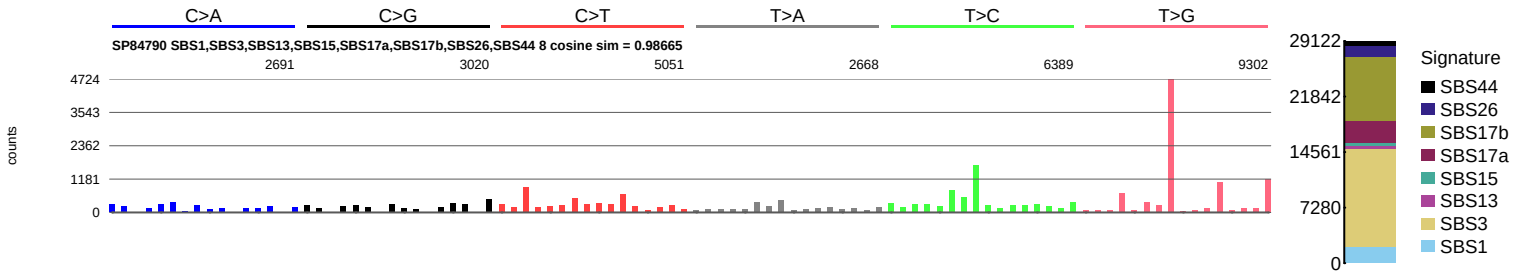

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98681
3494
2980
4769
2690
5936
9253
counts
4724
3543
2362
1181
0
29122
21842
14561
7280
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1

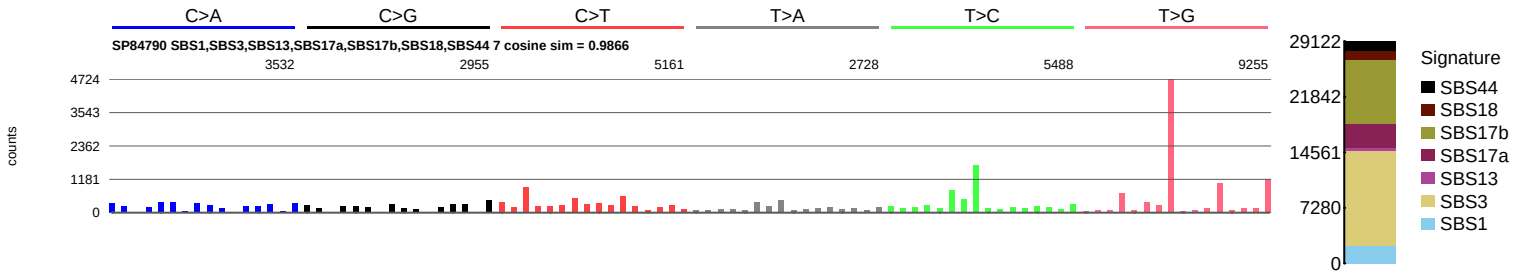

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98673
3490
2655
5062
2711
5961
9245
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

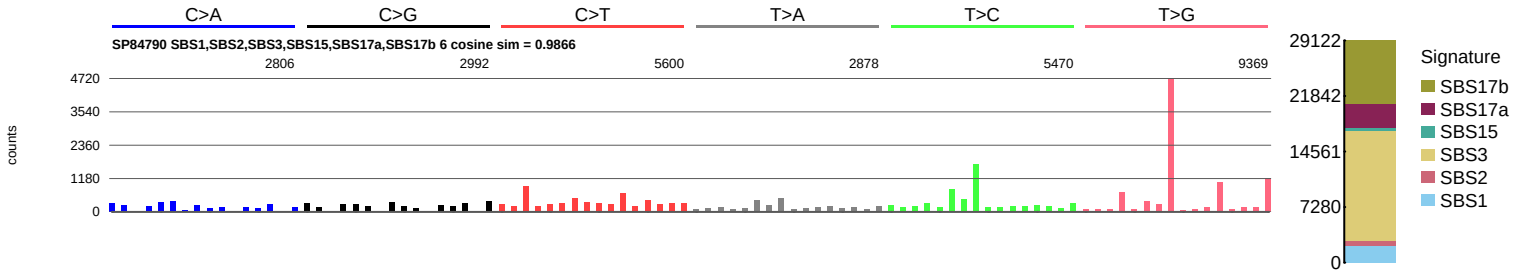

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b 6 cosine sim = 0.98673
2783
3241
5432
2858
5445
9362
counts
4724
3543
2362
1181
0
29122
21842
14561
7280
0
Signature
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

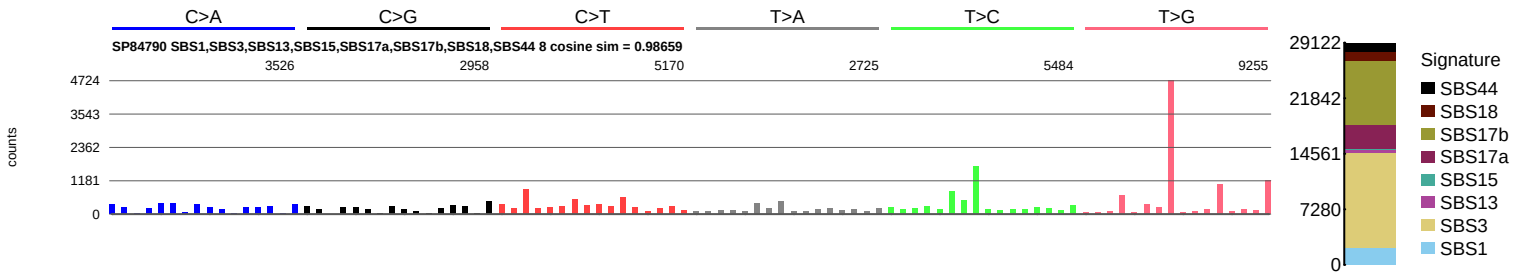

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS44 7 cosine sim = 0.98672
3496
2650
5047
2714
5970
9245
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1

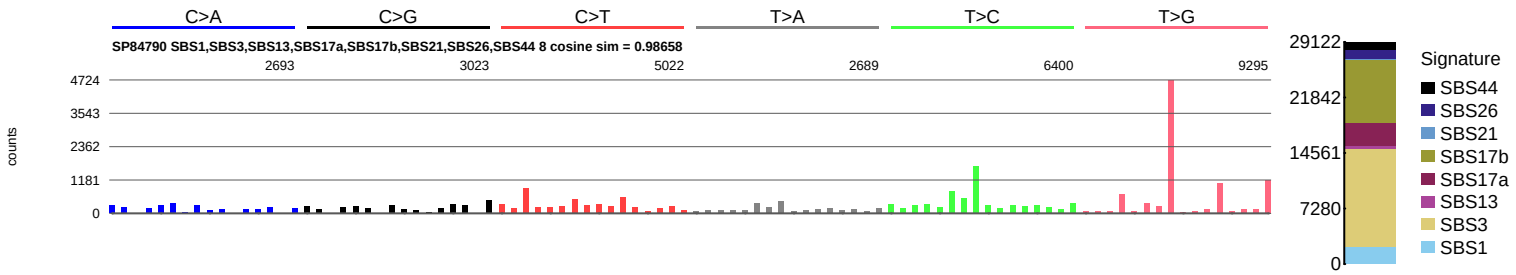

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS44 9 cosine sim = 0.9867
2694
3018
5048
2673
6394
9295
counts
4724
3543
2362
1181
0
29122
21842
14561
7280
0
Signature
SBS44
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

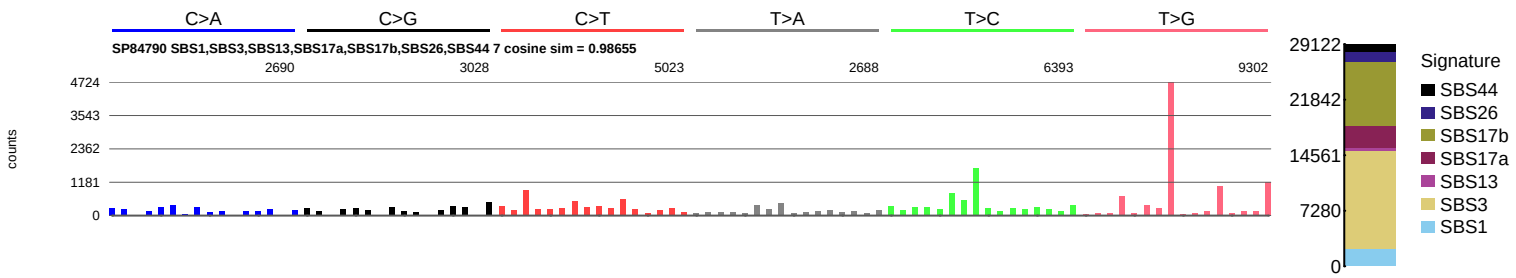

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS44 8 cosine sim = 0.98665
2691
3020
5051
2668
6389
9302
counts
4724
3543
2362
1181
0
29122
21842
14561
7280
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

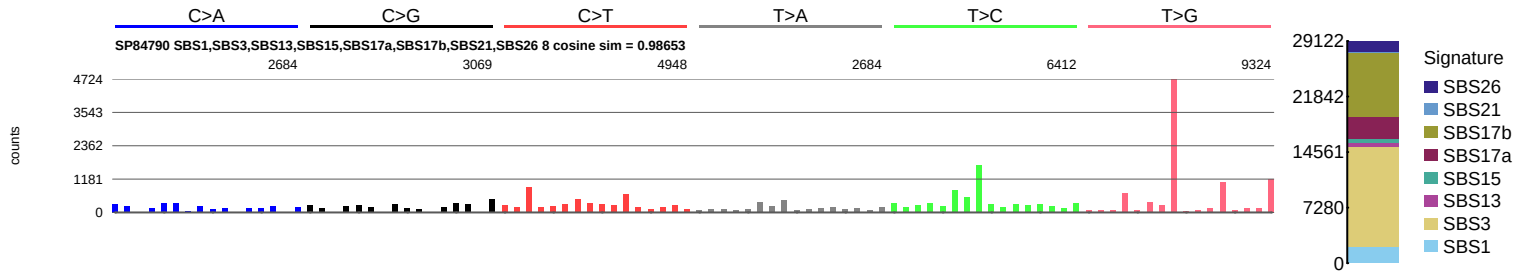

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.9866
3532
2955
5161
2728
5488
9255
counts
4724
3543
2362
1181
0
29122
21842
14561
7280
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1

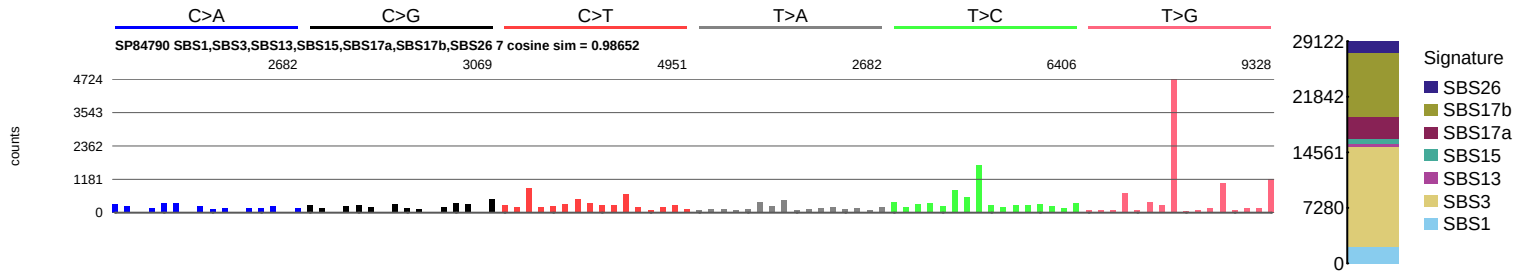

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b 6 cosine sim = 0.9866
2806
2992
5600
2878
5470
9369
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

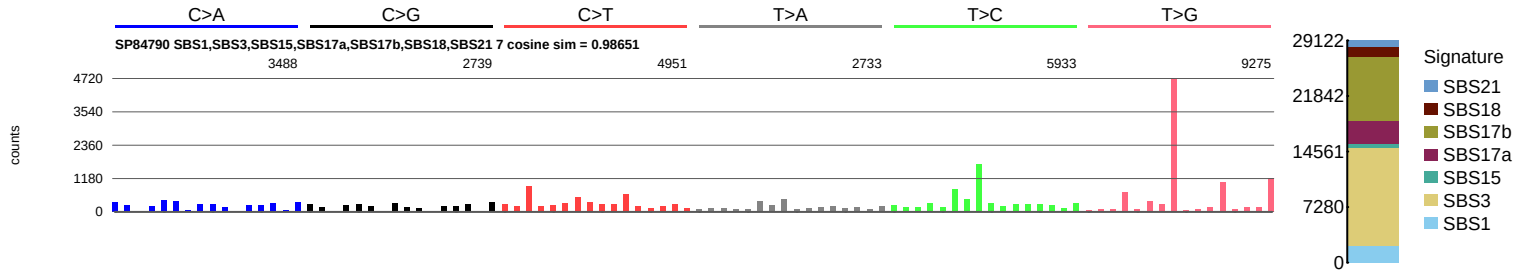

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98659
3526
2958
5170
2725
5484
9255
counts
4724
3543
2362
1181
0
29122
21842
14561
7280
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

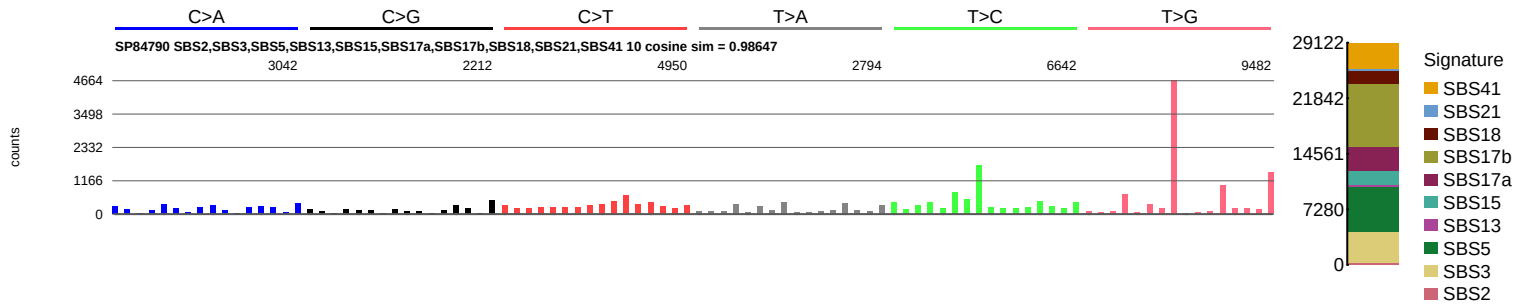

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS44 8 cosine sim = 0.98658
2693
3023
5022
2689
6400
9295
counts
4724
3543
2362
1181
0
29122
21842
14561
7280
0
Signature
SBS44
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS1

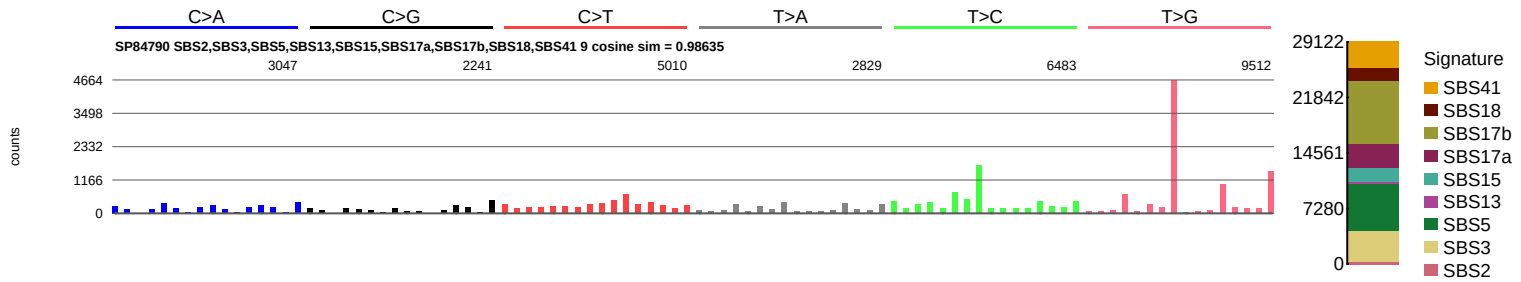

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS26,SBS44 7 cosine sim = 0.98655
2690
3028
5023
2688
6393
9302
counts
4724
3543
2362
1181
0
29122
21842
14561
7280
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS13
SBS3
SBS1

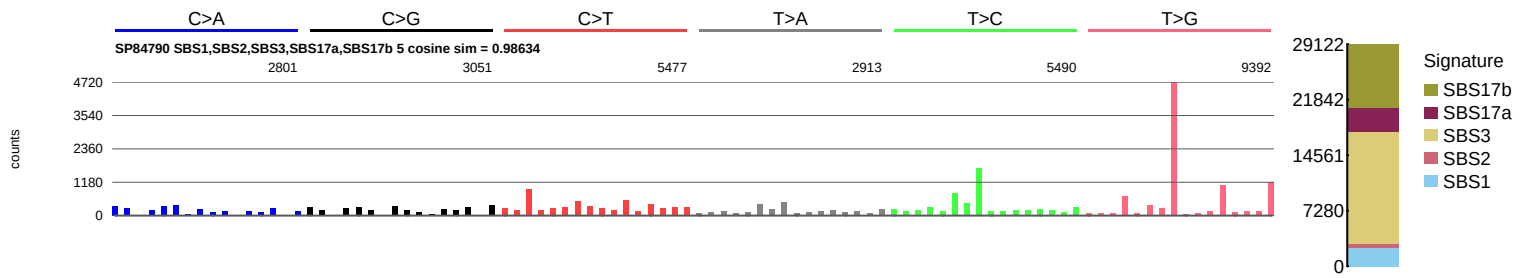

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26 8 cosine sim = 0.98653
2684
3069
4948
2684
6412
9324
counts
4724
3543
2362
1181
0
29122
21842
14561
7280
0
Signature
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

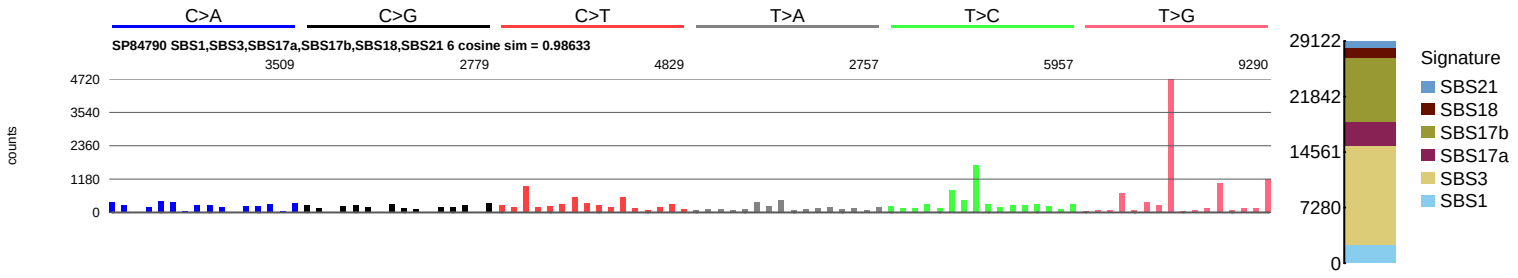

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26 7 cosine sim = 0.98652
2682
3069
4951
2682
6406
9328
counts
4724
3543
2362
1181
0
29122
21842
14561
7280
0
Signature
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

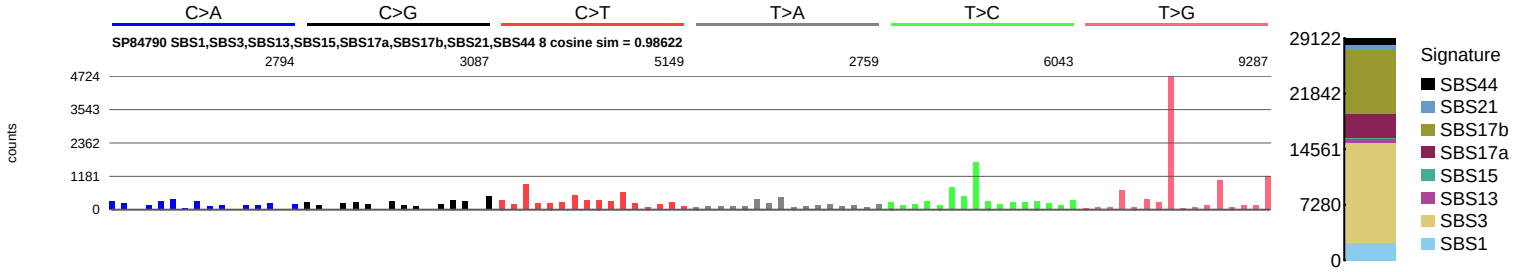

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98651
3488
2739
4951
2733
5933
9275
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

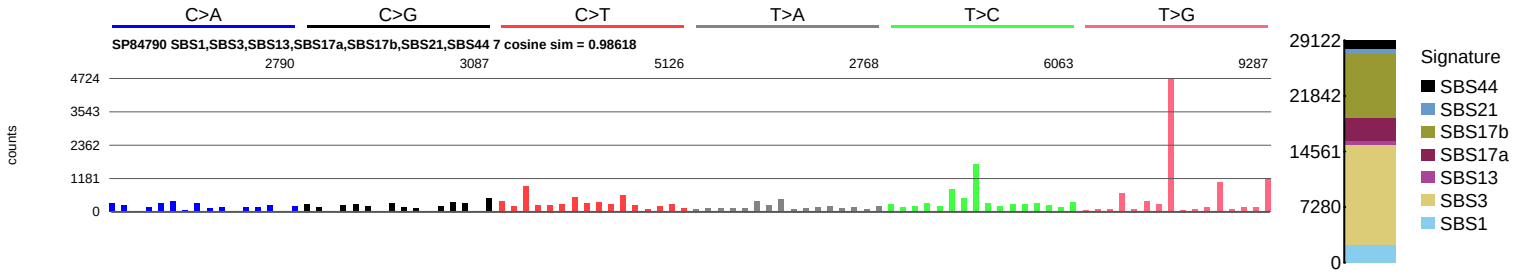

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98647
3042
2212
4950
2794
6642
9482
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2

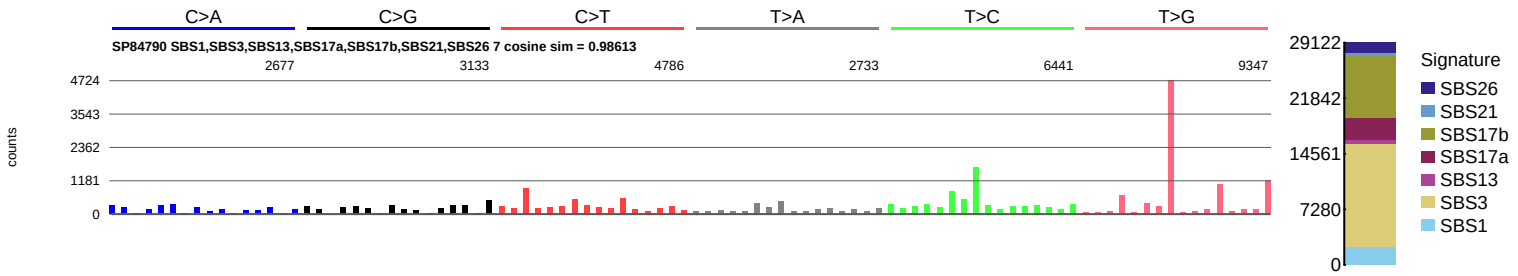

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.98635
3047
2241
5010
2829
6483
9512
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2

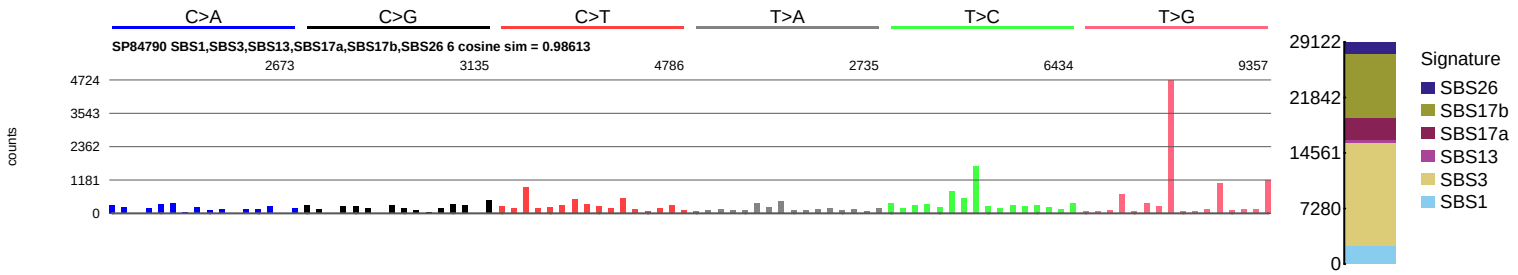

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS3,SBS17a,SBS17b 5 cosine sim = 0.98634
2801
3051
5477
2913
5490
9392
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS17b
SBS17a
SBS3
SBS2
SBS1

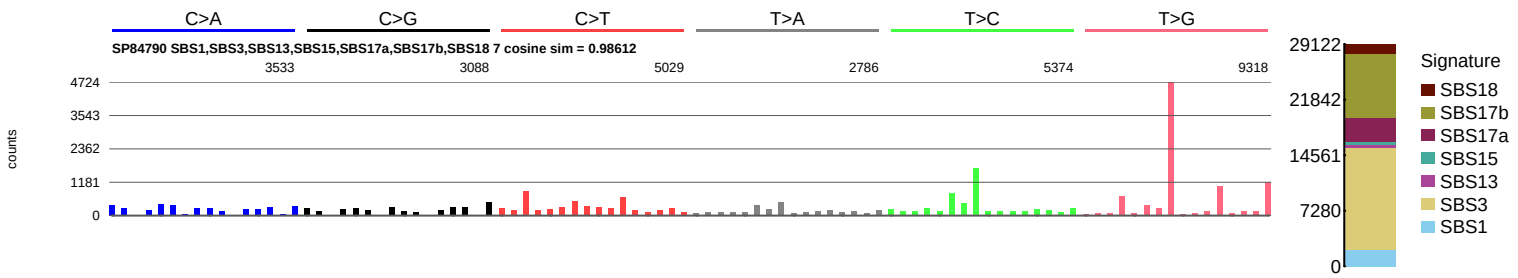

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21 6 cosine sim = 0.98633
3509
2779
4829
2757
5957
9290
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1

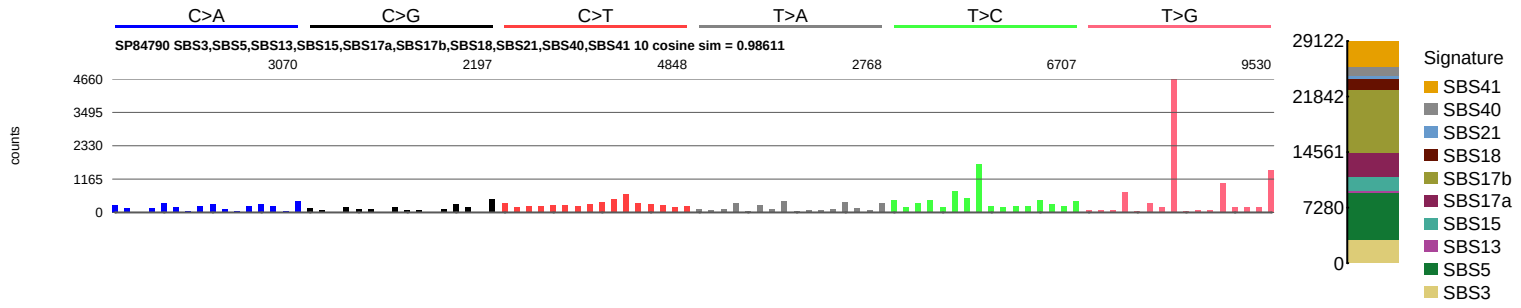

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS44 8 cosine sim = 0.98622
2794
3087
5149
2759
6043
9287
counts
4724
3543
2362
1181
0
29122
21842
14561
7280
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

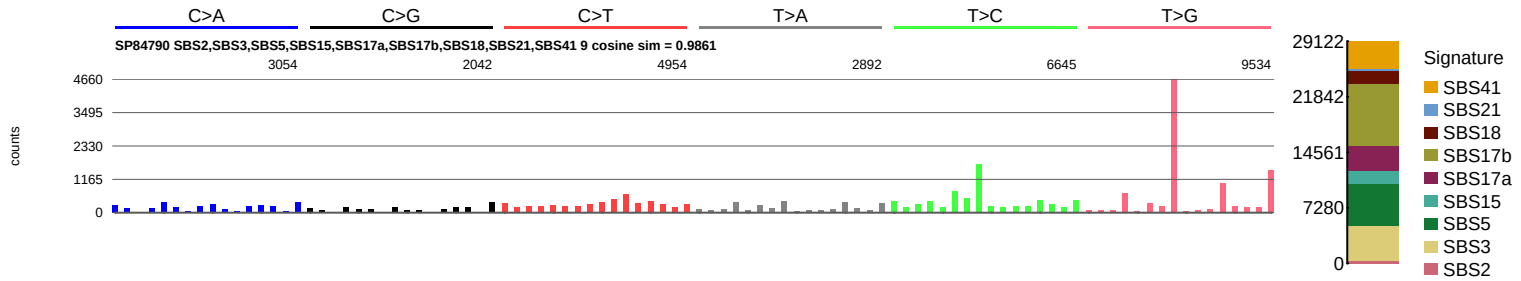

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS44 7 cosine sim = 0.98618
2790
3087
5126
2768
6063
9287
counts
4724
3543
2362
1181
0
29122
21842
14561
7280
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS1

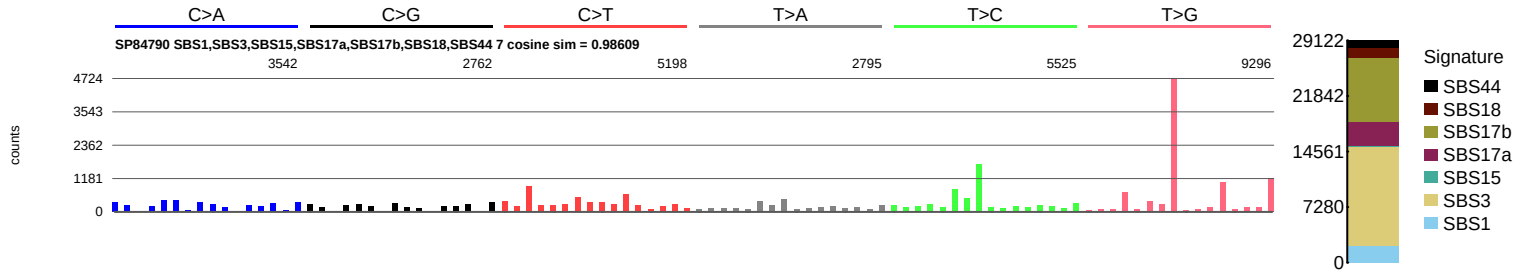

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS26 7 cosine sim = 0.98613
2677
3133
4786
2733
6441
9347
counts
4724
3543
2362
1181
0
29122
21842
14561
7280
0
Signature
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS1

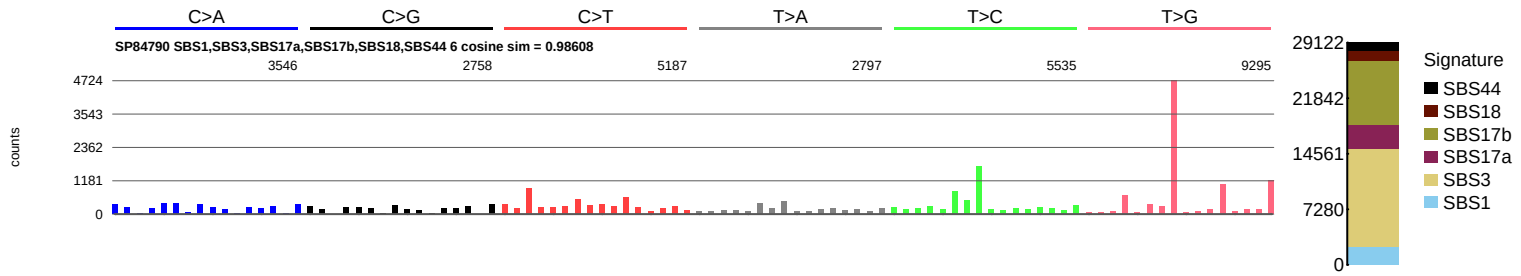

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS26 6 cosine sim = 0.98613
2673
3135
4786
2735
6434
9357
counts
4724
3543
2362
1181
0
29122
21842
14561
7280
0
Signature
SBS26
SBS17b
SBS17a
SBS13
SBS3
SBS1

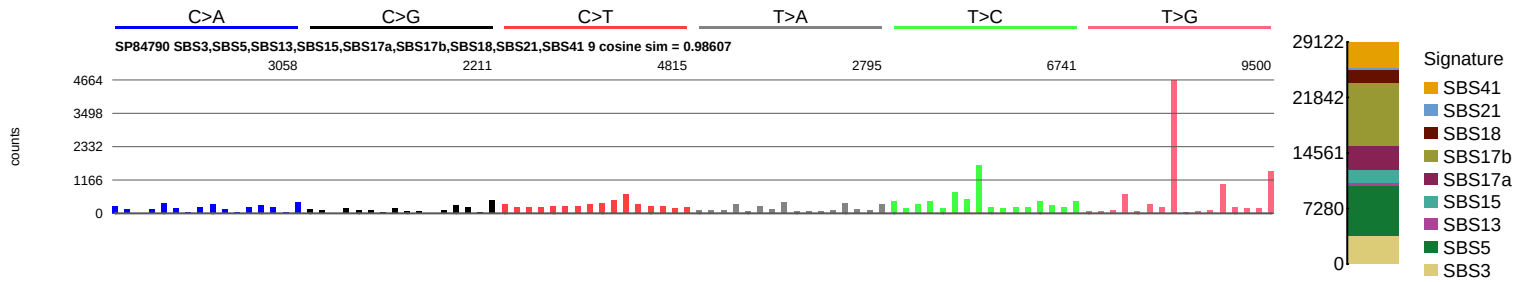

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98612
3533
3088
5029
2786
5374
9318
counts
4724
3543
2362
1181
0
29122
21842
14561
7280
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

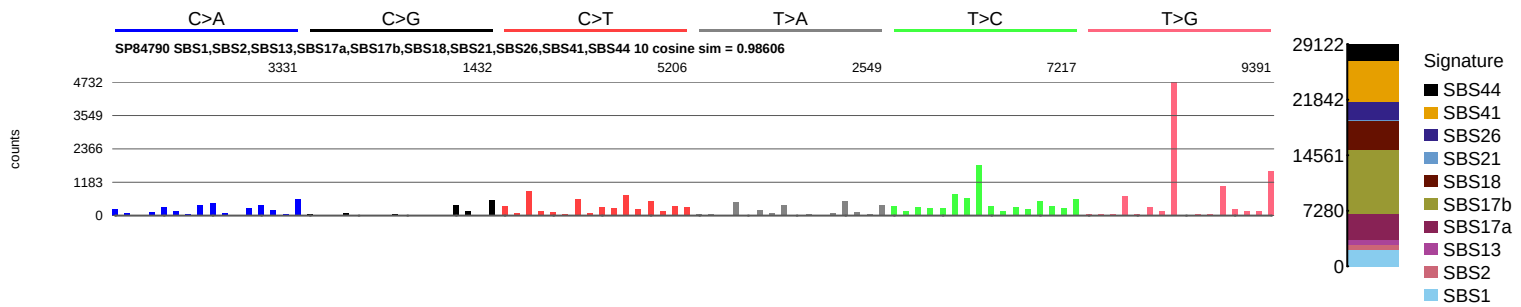

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98611
3070
2197
4848
2768
6707
9530
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3

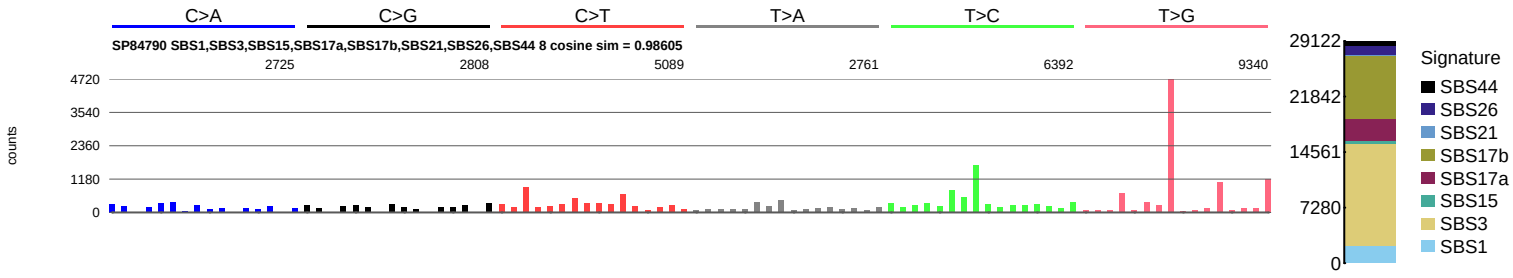

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.9861
3054
2042
4954
2892
6645
9534
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2

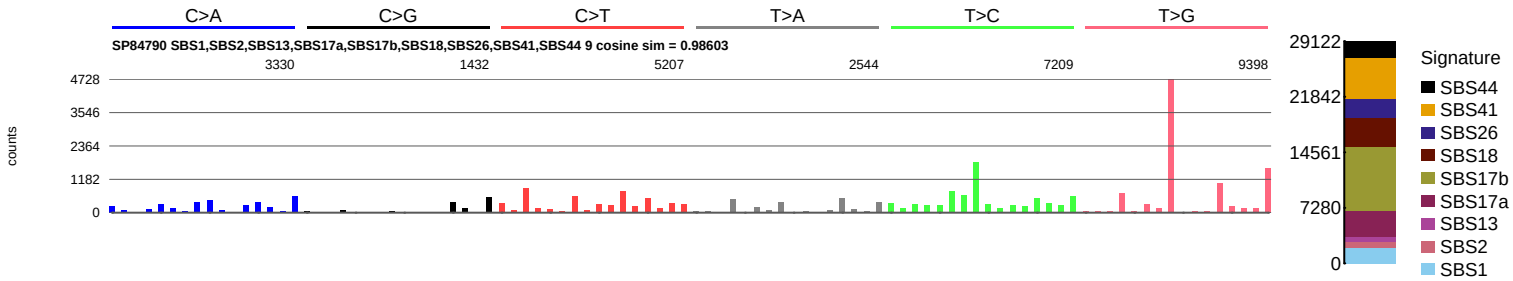

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.98609
3542
2762
5198
2795
5525
9296
counts
4724
3543
2362
1181
0
29122
21842
14561
7280
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

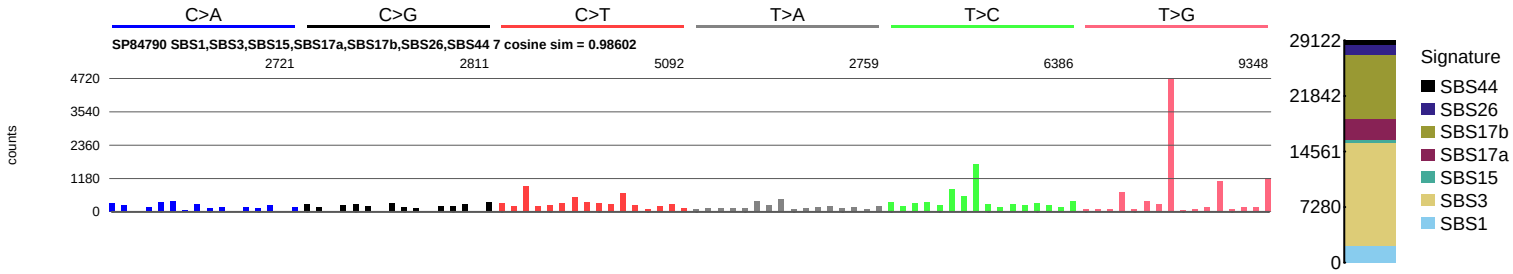

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS44 6 cosine sim = 0.98608
3546
2758
5187
2797
5535
9295
counts
4724
3543
2362
1181
0
29122
21842
14561
7280
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS3
SBS1

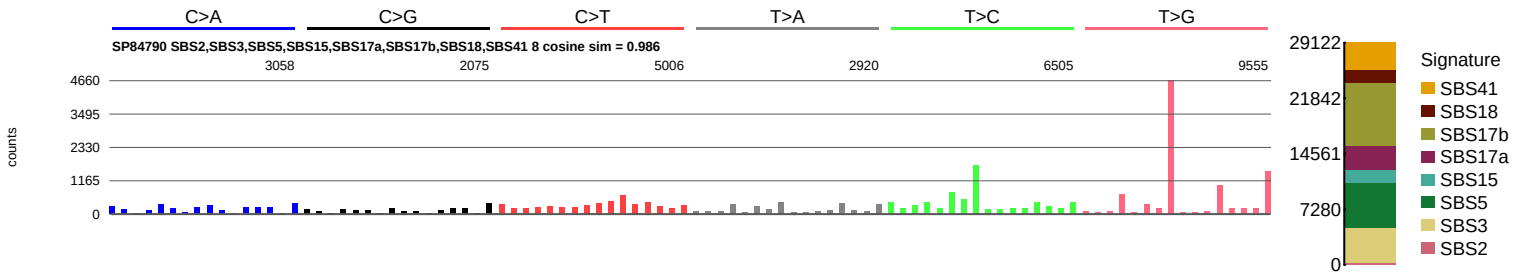

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98607
3058
2211
4815
2795
6741
9500
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3

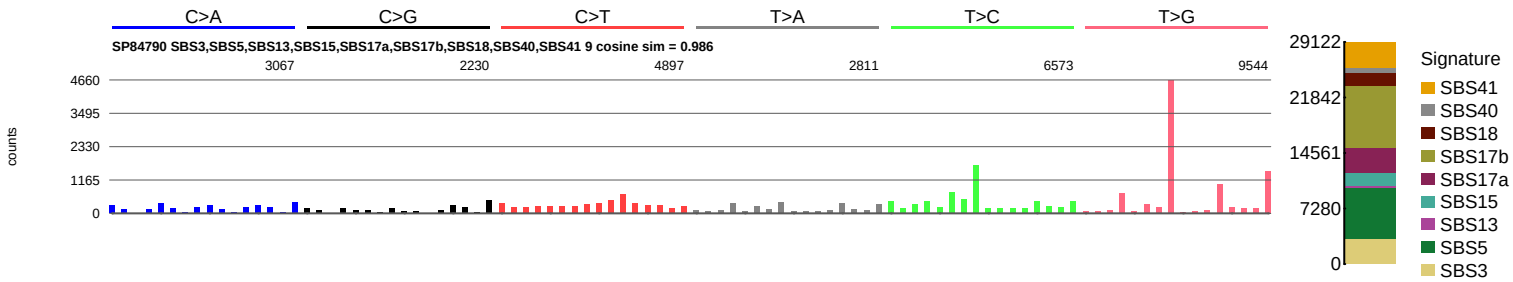

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 10 cosine sim = 0.98606
3331
1432
5206
2549
7217
9391
counts
4732
3549
2366
1183
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

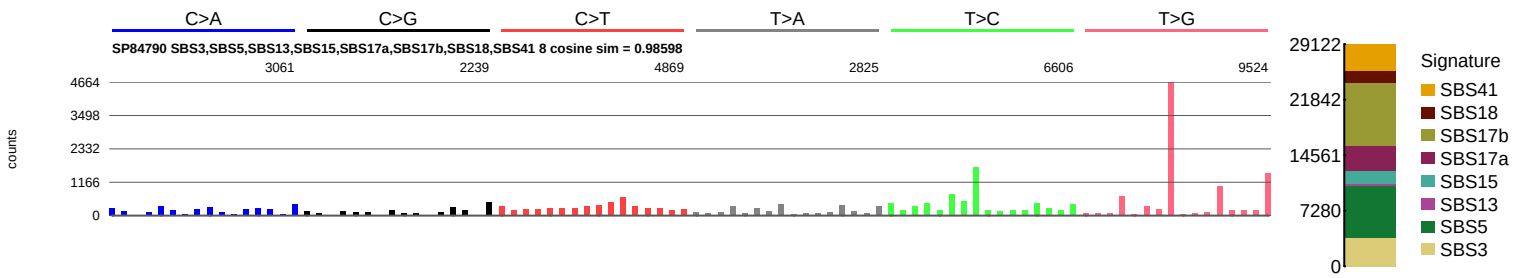

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS44 8 cosine sim = 0.98605
2725
2808
5089
2761
6392
9340
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS44
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS1

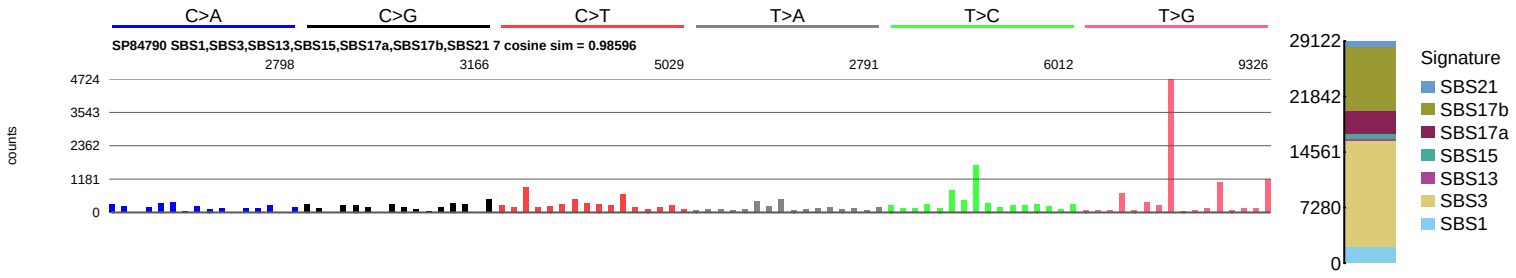

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98603
3330
1432
5207
2544
7209
9398
counts
4728
3546
2364
1182
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

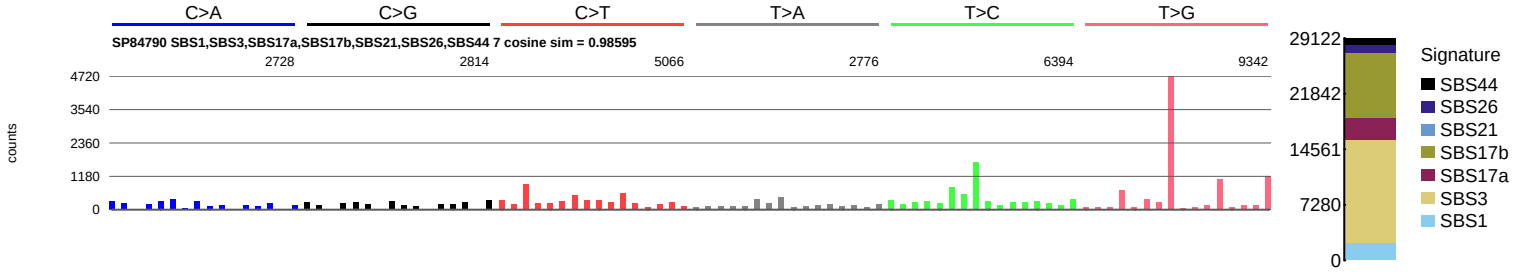

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS44 7 cosine sim = 0.98602
2721
2811
5092
2759
6386
9348
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS1

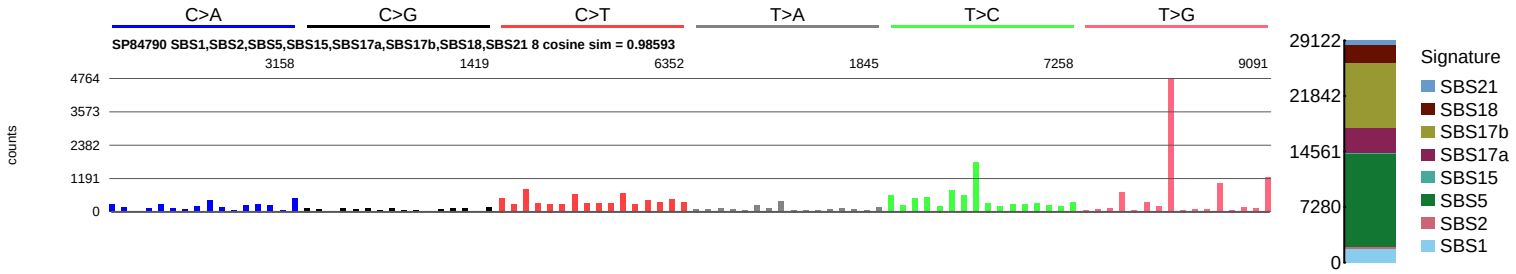

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.986
3058
2075
5006
2920
6505
9555
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2

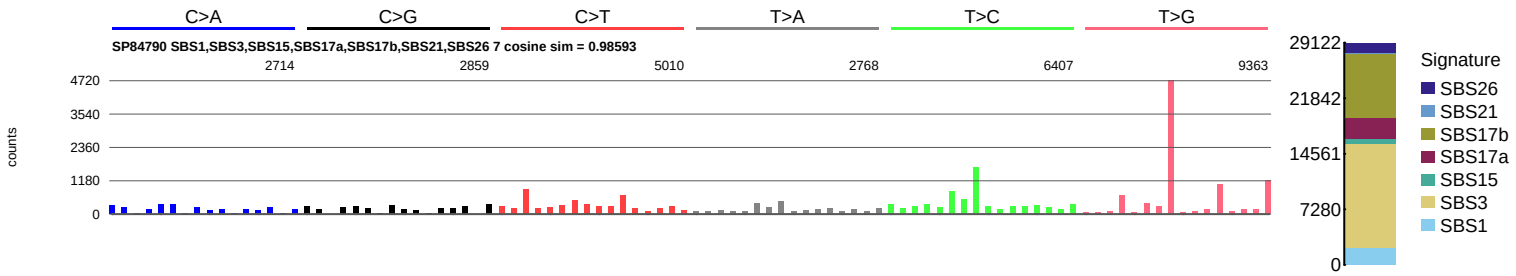

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.986
3067
2230
4897
2811
6573
9544
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3

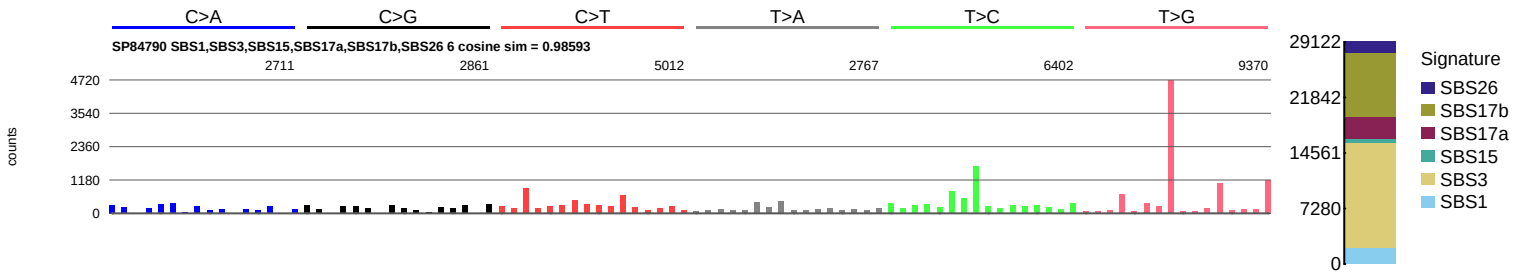

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98598
3061
2239
4869
2825
6606
9524
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3

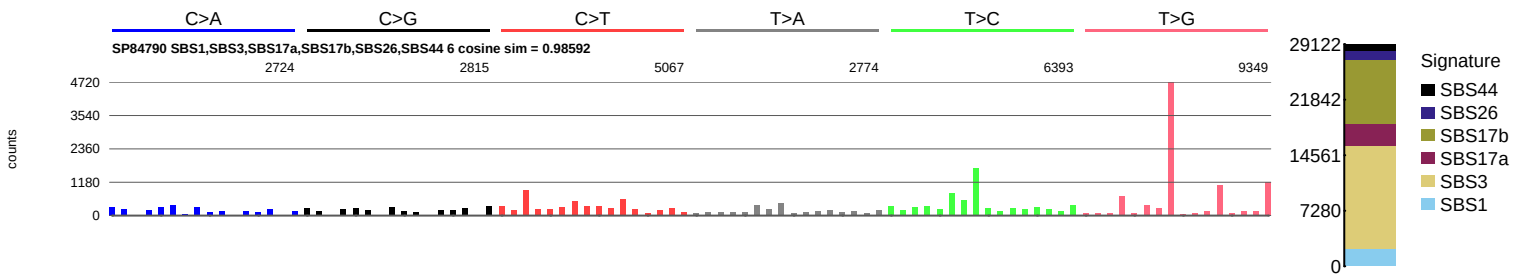

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98596
2798
3166
5029
2791
6012
9326
counts
4724
3543
2362
1181
0
29122
21842
14561
7280
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

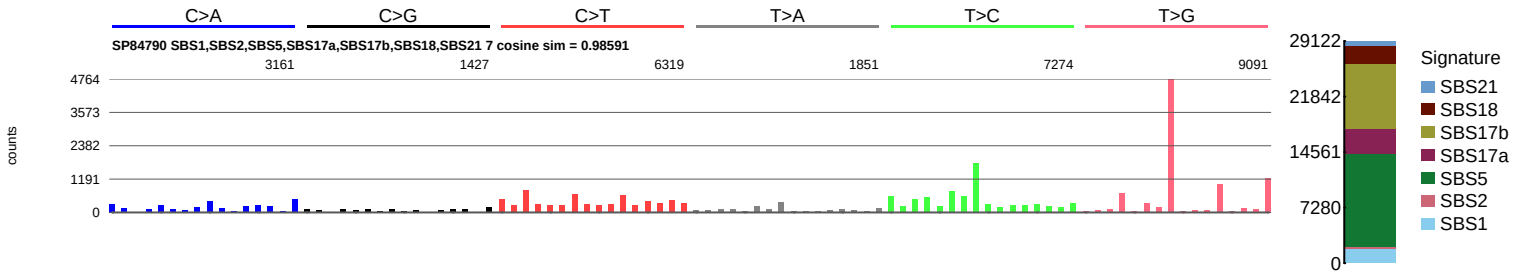

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS21,SBS26,SBS44 7 cosine sim = 0.98595
2728
2814
5066
2776
6394
9342
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS44
SBS26
SBS21
SBS17b
SBS17a
SBS3
SBS1

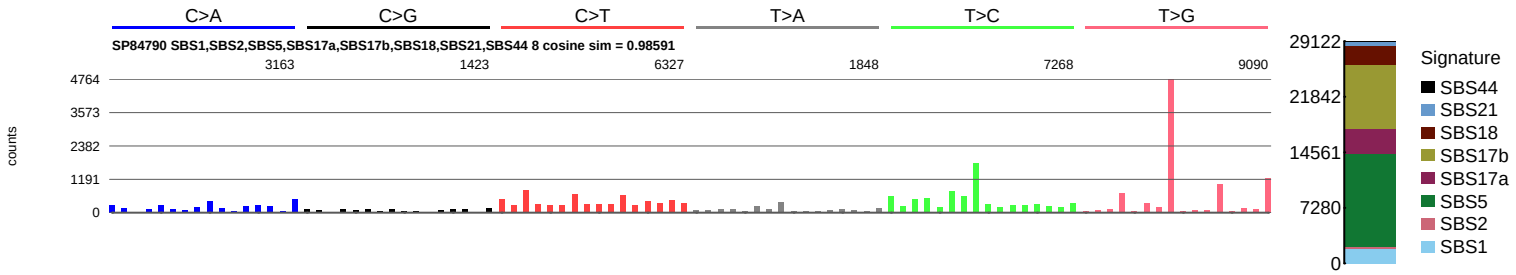

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98593
3158
1419
6352
1845
7258
9091
counts
4764
3573
2382
1191
0
29122
21842
14561
7280
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

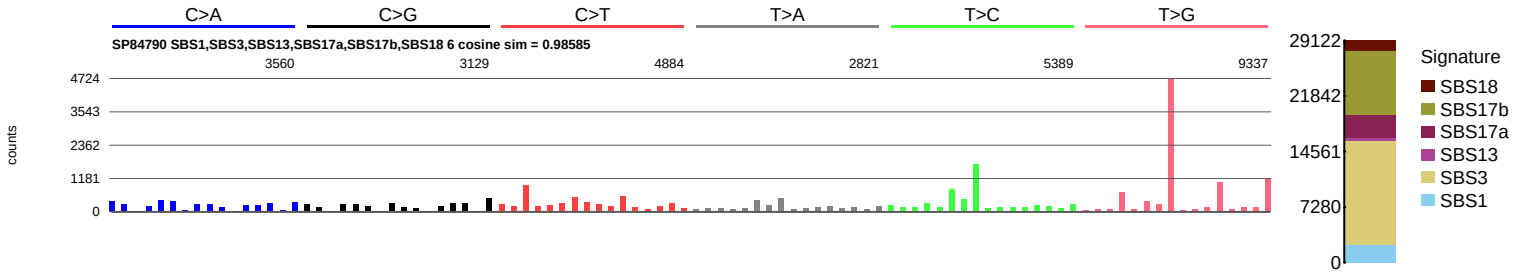

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS26 7 cosine sim = 0.98593
2714
2859
5010
2768
6407
9363
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS1

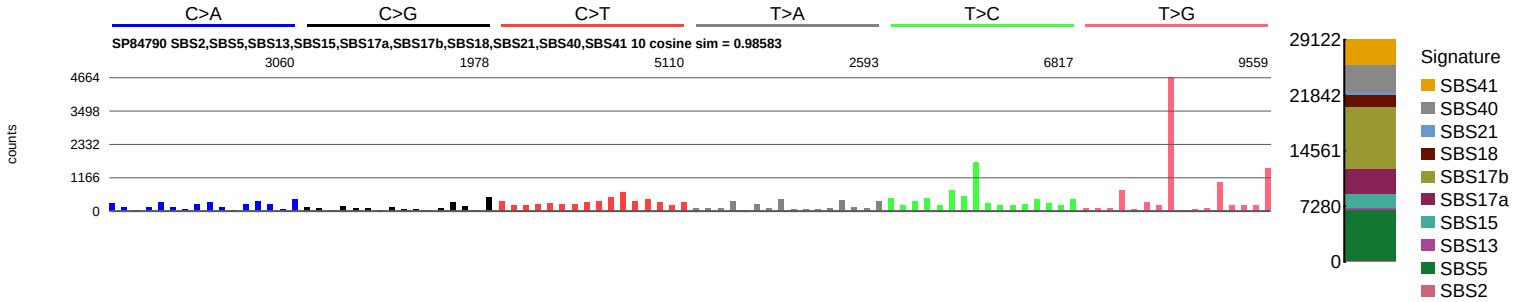

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS26 6 cosine sim = 0.98593
2711
2861
5012
2767
6402
9370
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS1

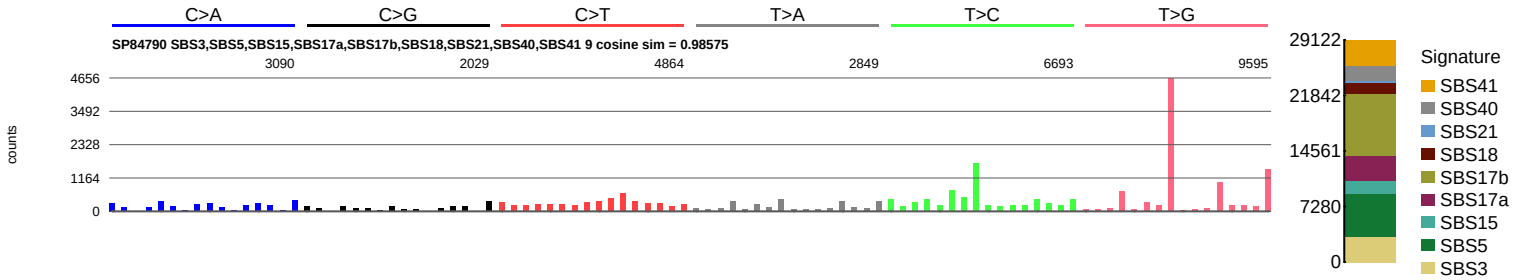

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS26,SBS44 6 cosine sim = 0.98592
2724
2815
5067
2774
6393
9349
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS3
SBS1

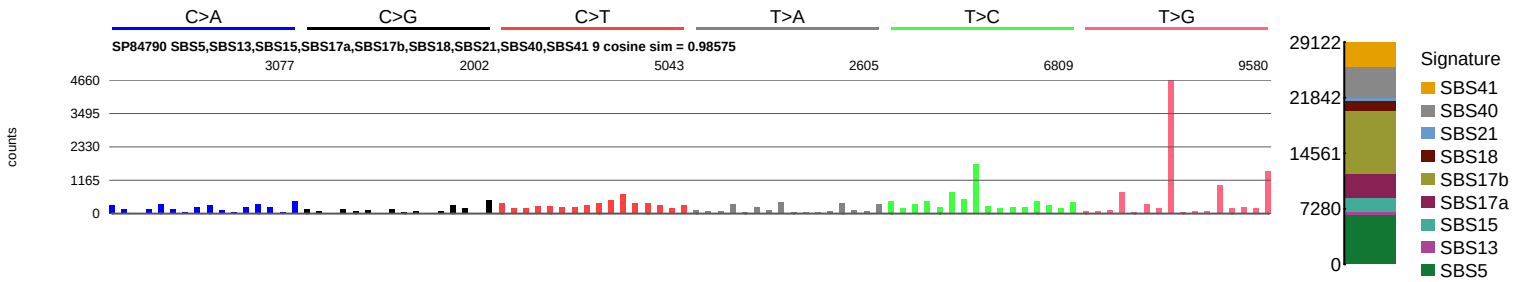

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98591
3161
1427
6319
1851
7274
9091
4764
3573
2382
1191
0
counts
29122
21842
14561
7280
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

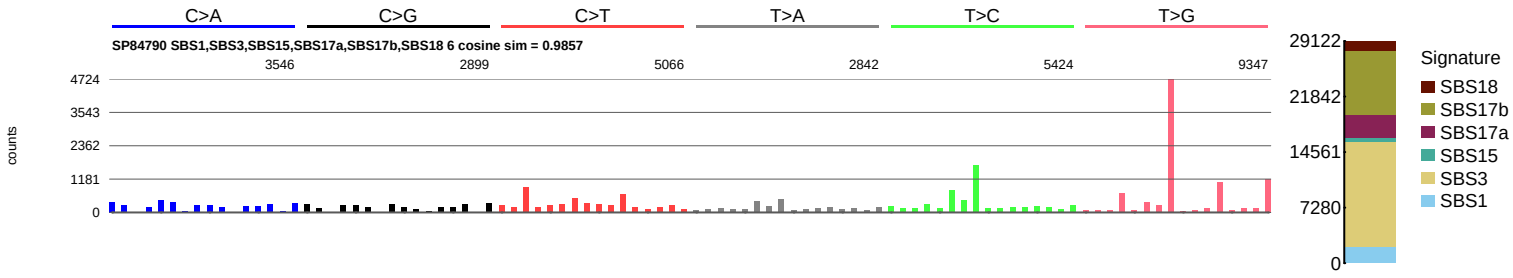

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98591
3163
1423
6327
1848
7268
9090
4764
3573
2382
1191
0
counts
29122
21842
14561
7280
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

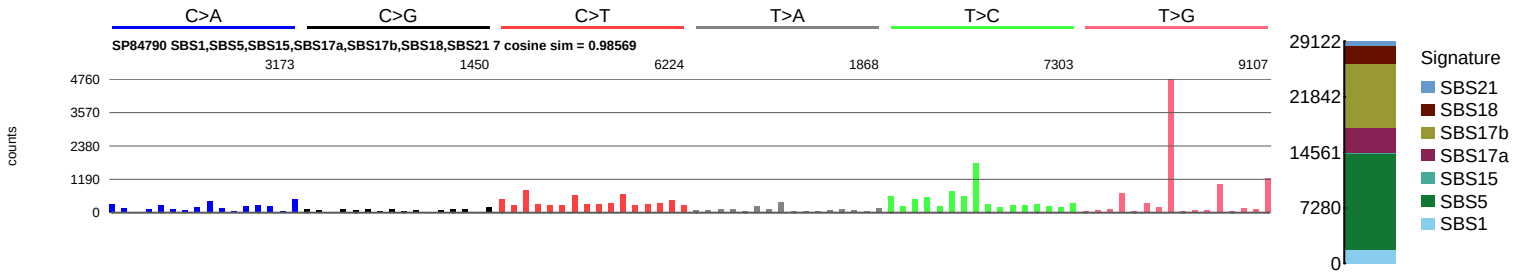

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98585
3560
3129
4884
2821
5389
9337
4724
3543
2362
1181
0
counts
29122
21842
14561
7280
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1

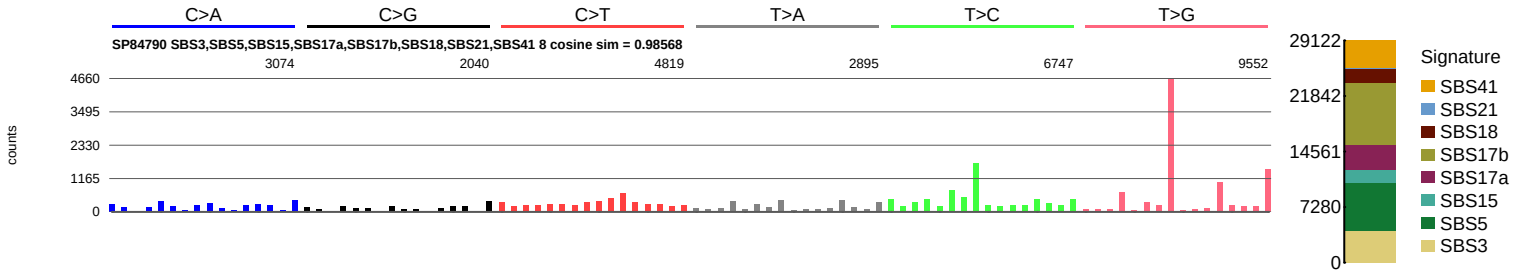

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98583
3060
1978
5110
2593
6817
9559
4664
3498
2332
1166
0
counts
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2

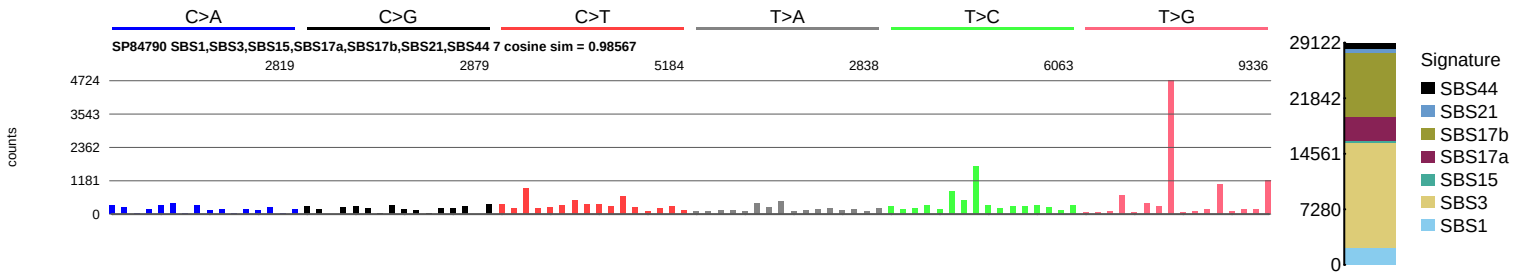

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98575
3090
2029
4864
2849
6693
9595
4656
3492
2328
1164
0
counts
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3

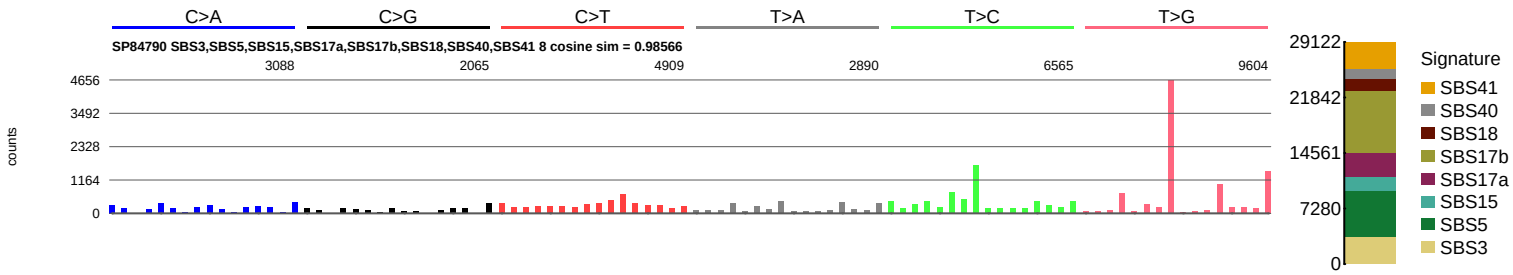

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98575
3077
2002
5043
2605
6809
9580
4660
3495
2330
1165
0
counts
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5

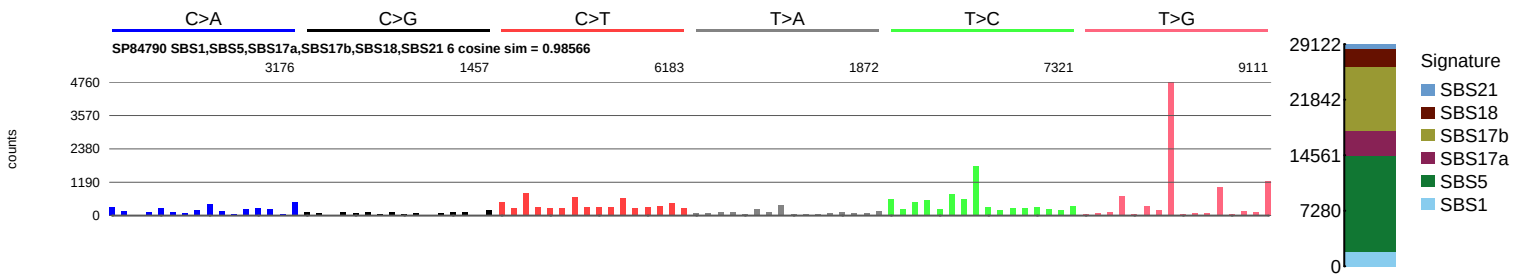

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18 6 cosine sim = 0.9857
3546
2899
5066
2842
5424
9347
counts
4724
3543
2362
1181
0
29122
21842
14561
7280
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

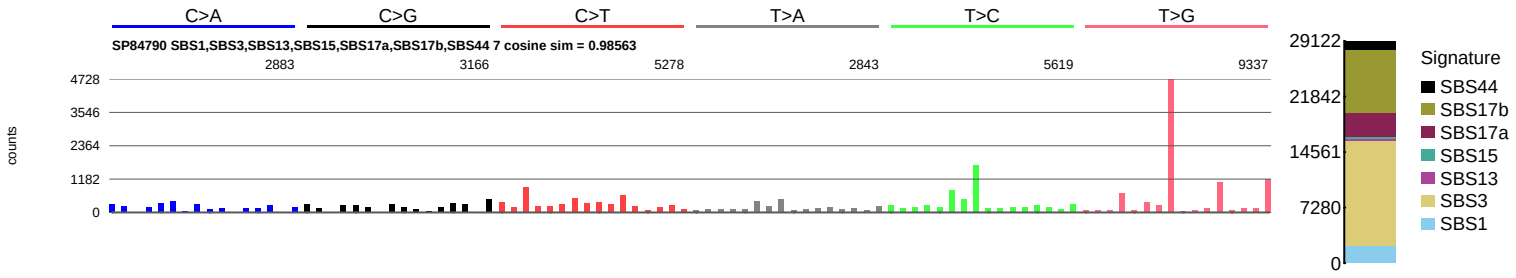

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98569
3173
1450
6224
1868
7303
9107
counts
4760
3570
2380
1190
0
29122
21842
14561
7280
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1

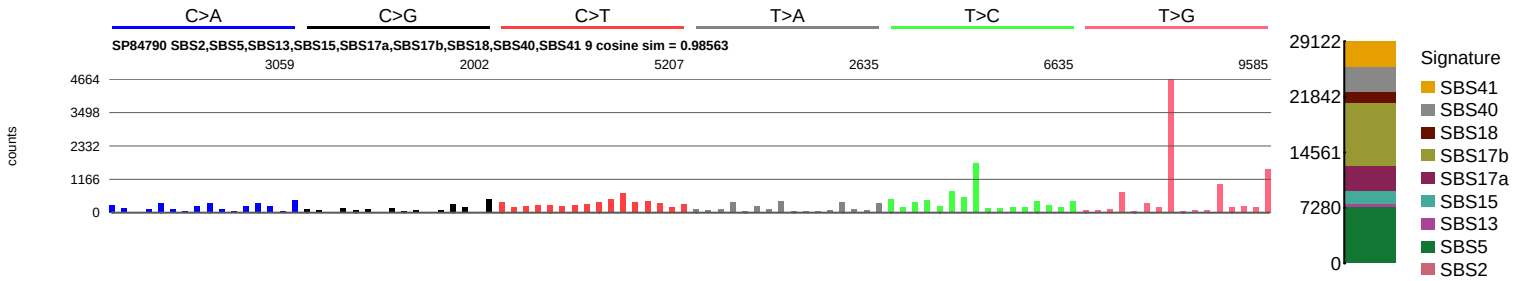

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98568
3074
2040
4819
2895
6747
9552
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3

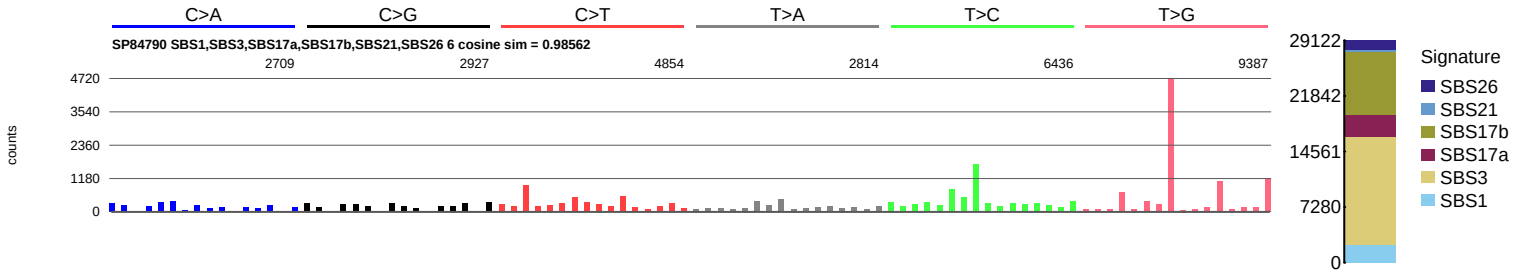

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS44 7 cosine sim = 0.98567
2819
2879
5184
2838
6063
9336
counts
4724
3543
2362
1181
0
29122
21842
14561
7280
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS1

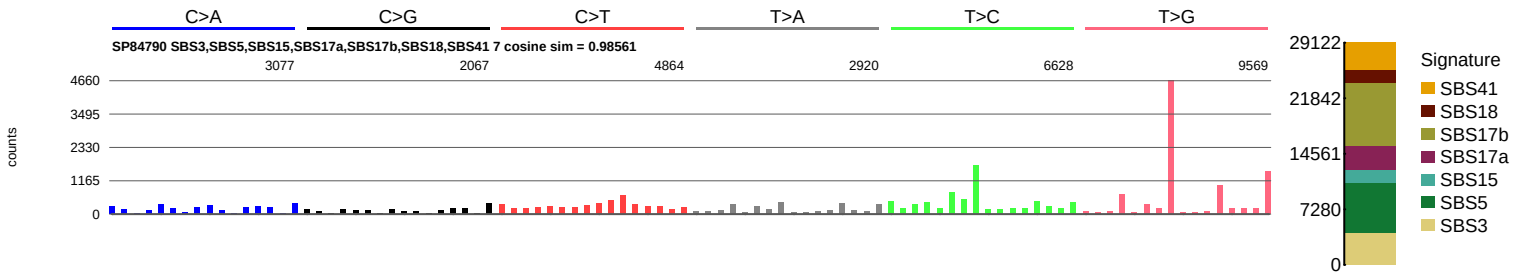

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98566
3088
2065
4909
2890
6565
9604
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3

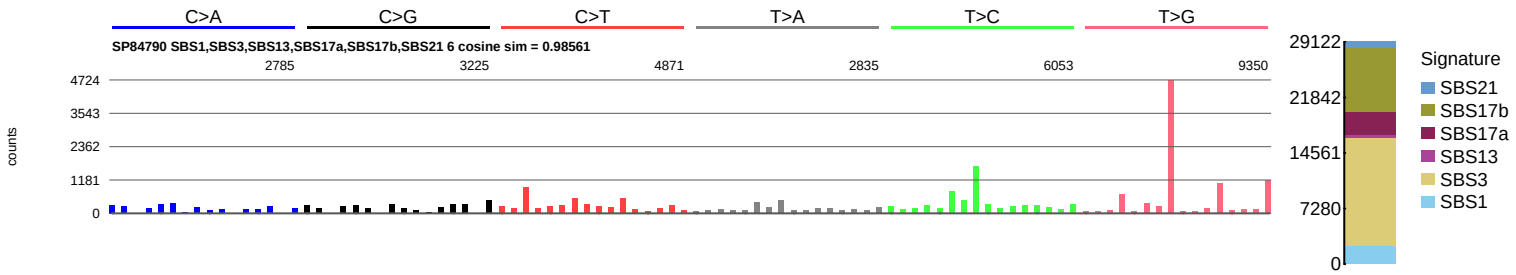

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21 6 cosine sim = 0.98566
3176
1457
6183
1872
7321
9111
counts
4760
3570
2380
1190
0
29122
21842
14561
7280
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1

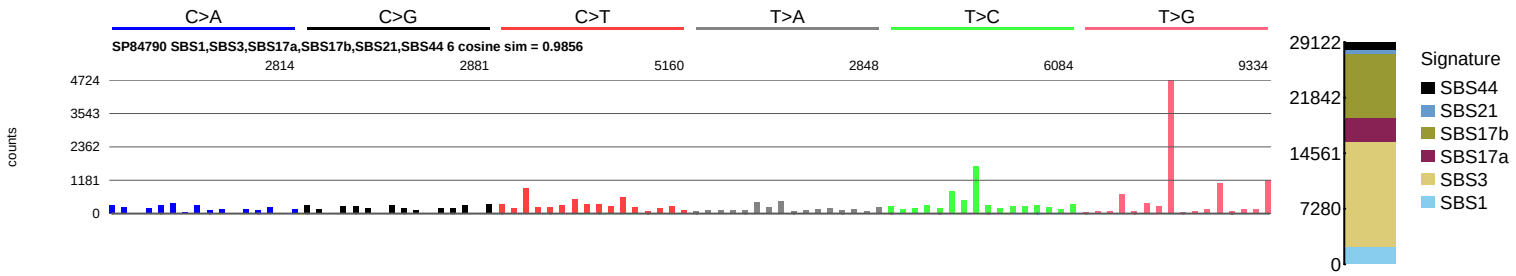

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS44 7 cosine sim = 0.98563
2883
3166
5278
2843
5619
9337
counts
4728
3546
2364
1182
0
29122
21842
14561
7280
0
Signature
SBS44
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

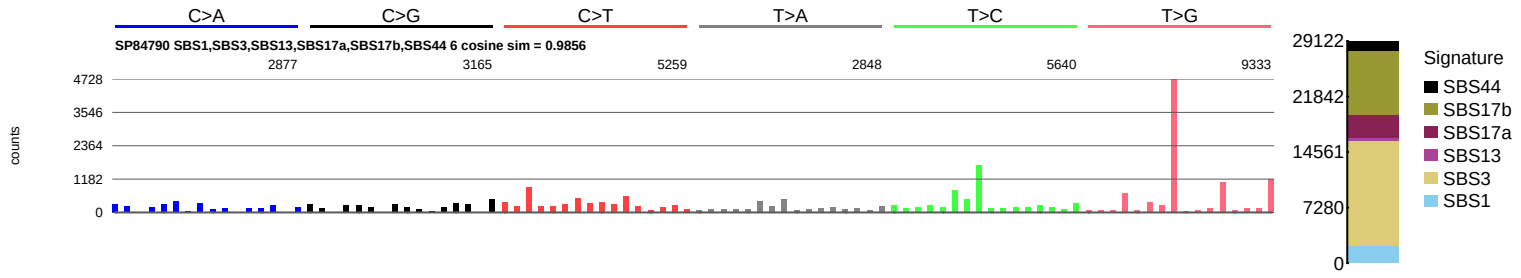

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98563
3059
2002
5207
2635
6635
9585
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2

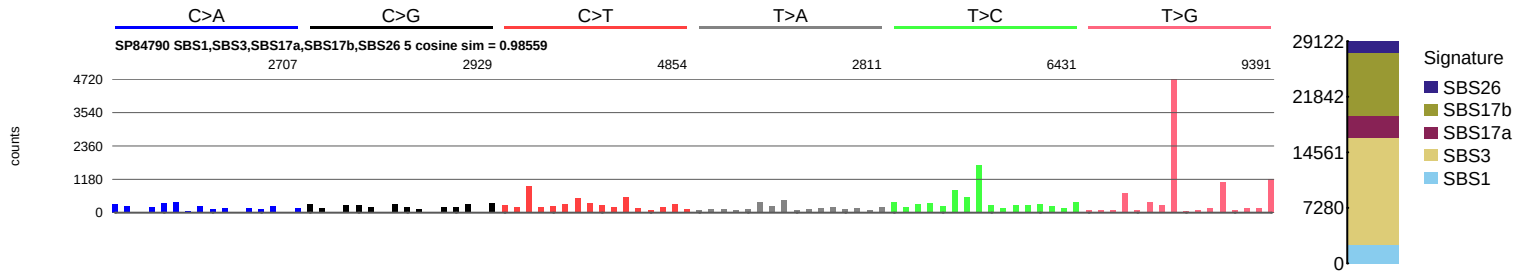

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS21,SBS26 6 cosine sim = 0.98562
2709
2927
4854
2814
6436
9387
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS26
SBS21
SBS17b
SBS17a
SBS3
SBS1

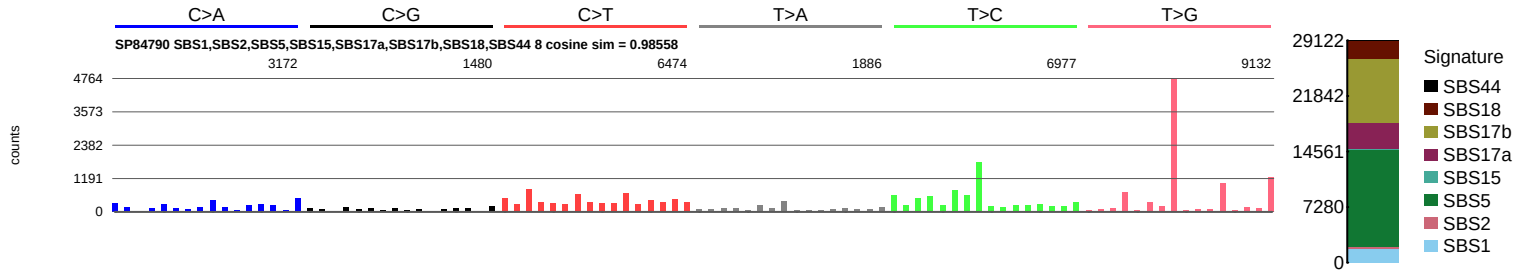

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98561
3077
2067
4864
2920
6628
9569
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3

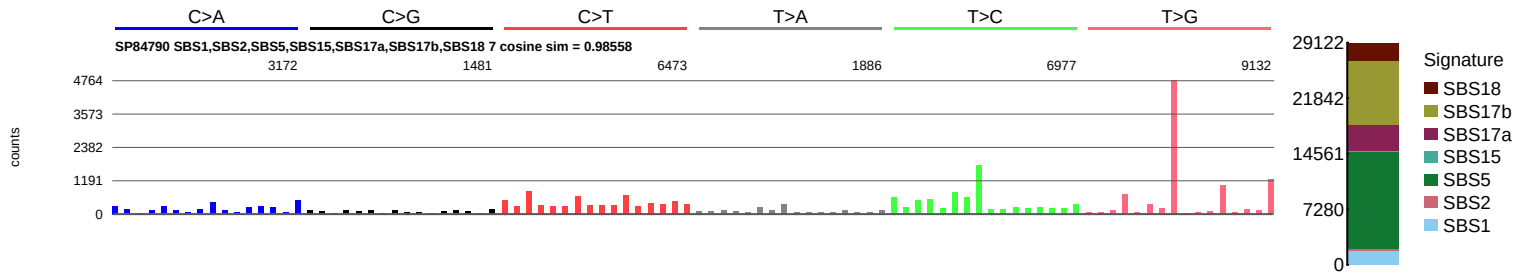

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS21 6 cosine sim = 0.98561
2785
3225
4871
2835
6053
9350
counts
4724
3543
2362
1181
0
29122
21842
14561
7280
0
Signature
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS1

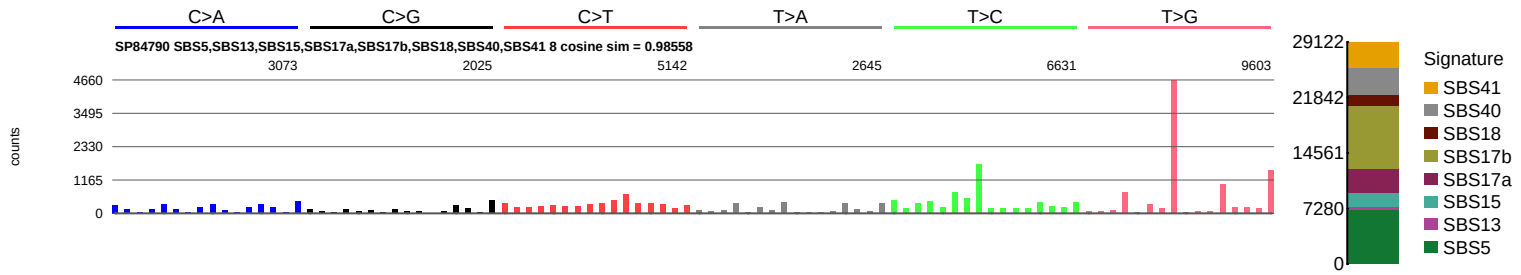

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS21,SBS44 6 cosine sim = 0.9856
2814
2881
5160
2848
6084
9334
counts
4724
3543
2362
1181
0
29122
21842
14561
7280
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS3
SBS1

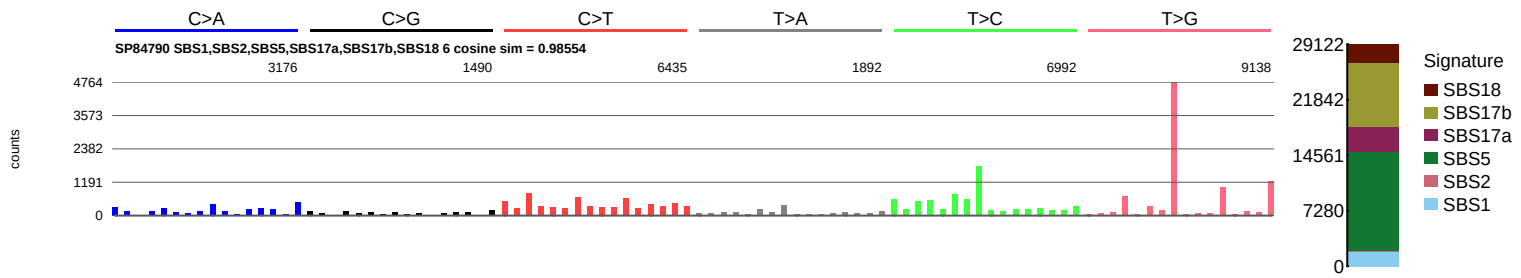

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS44 6 cosine sim = 0.9856
2877
3165
5259
2848
5640
9333
counts
4728
3546
2364
1182
0
29122
21842
14561
7280
0
Signature
SBS44
SBS17b
SBS17a
SBS13
SBS3
SBS1

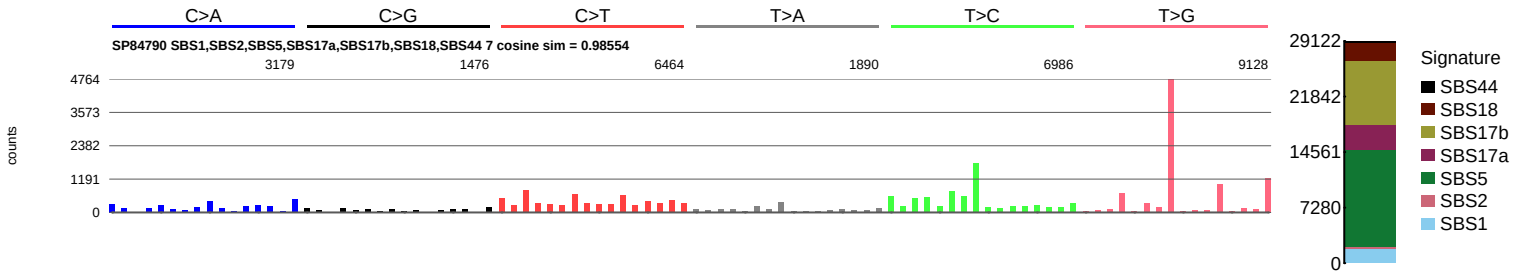

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS26 5 cosine sim = 0.98559
2707
2929
4854
2811
6431
9391
counts
4720
3540
2360
1180
0
29122
21842
14561
7280
0
Signature
SBS26
SBS17b
SBS17a
SBS3
SBS1

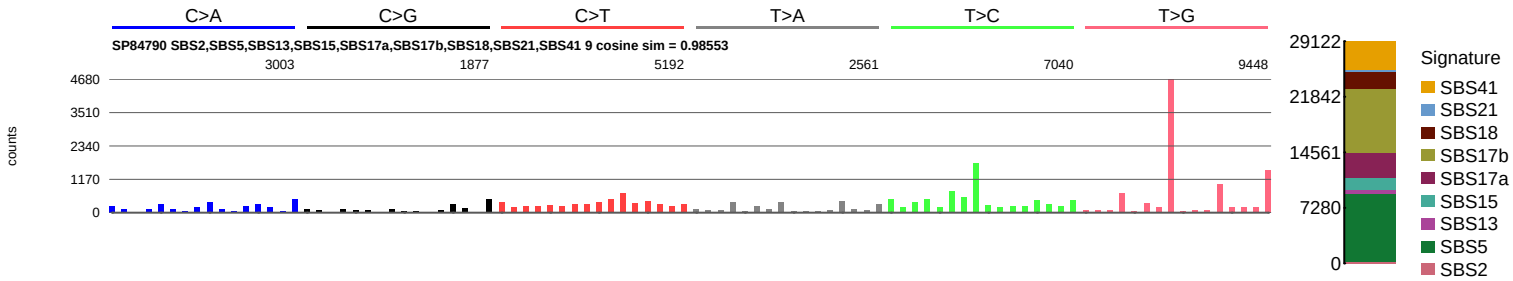

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98558
3172
1480
6474
1886
6977
9132
counts
4764
3573
2382
1191
0
29122
21842
14561
7280
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

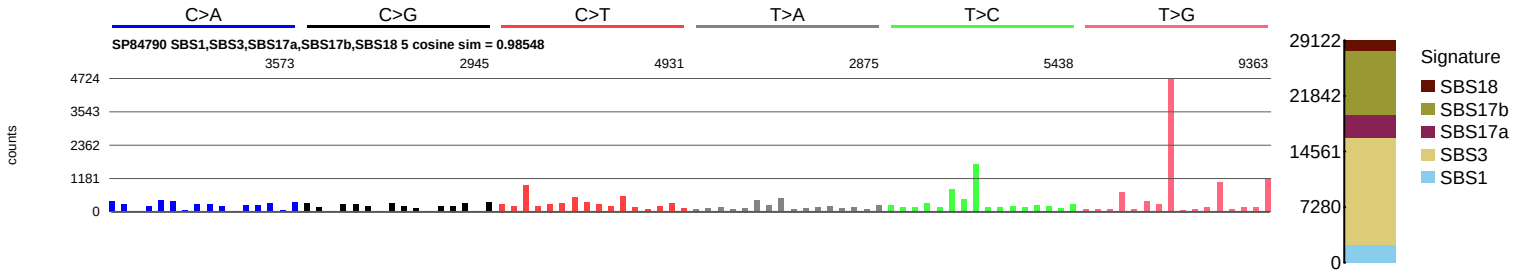

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98558
3172
1481
6473
1886
6977
9132
counts
4764
3573
2382
1191
0
29122
21842
14561
7280
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

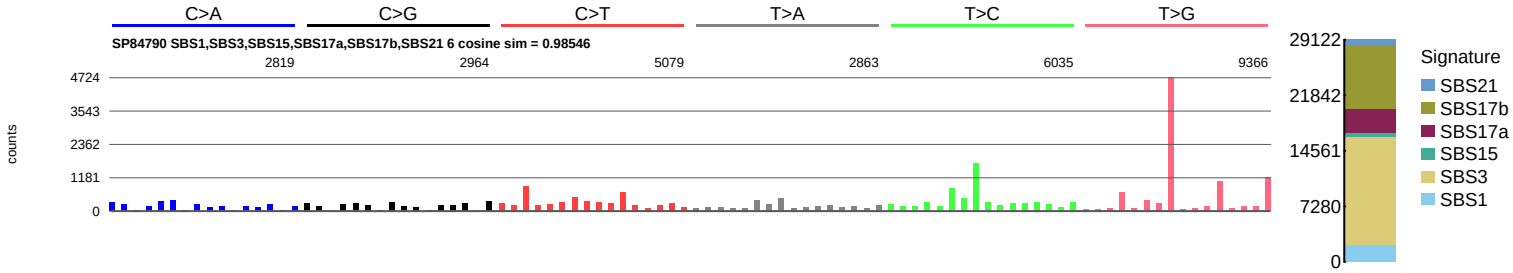

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98558
3073
2025
5142
2645
6631
9603
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5

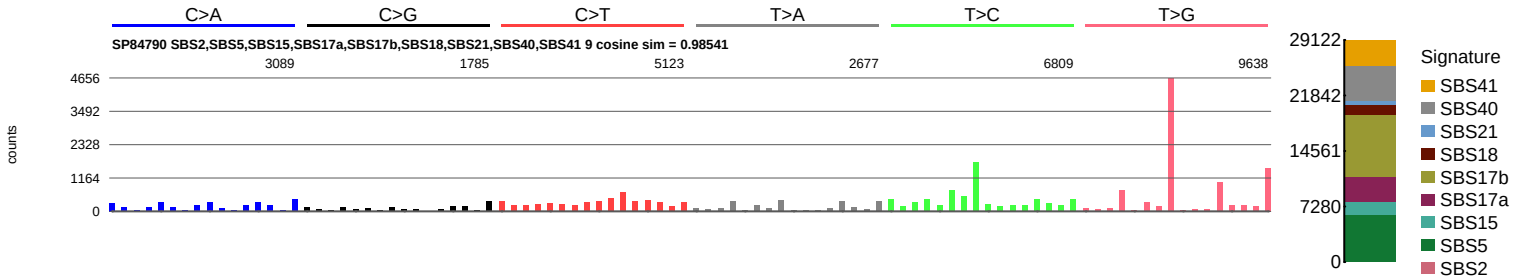

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98554
3176
1490
6435
1892
6992
9138
counts
4764
3573
2382
1191
0
29122
21842
14561
7280
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

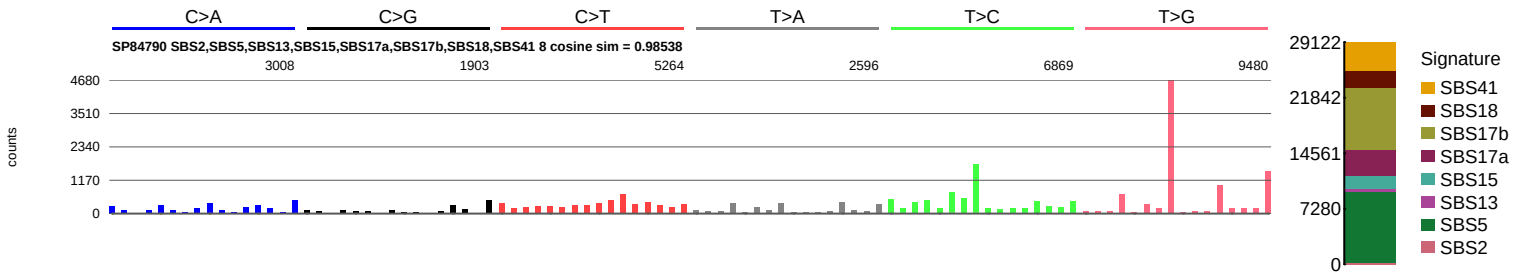

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.98554
3179
1476
6464
1890
6986
9128
counts
4764
3573
2382
1191
0
29122
21842
14561
7280
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

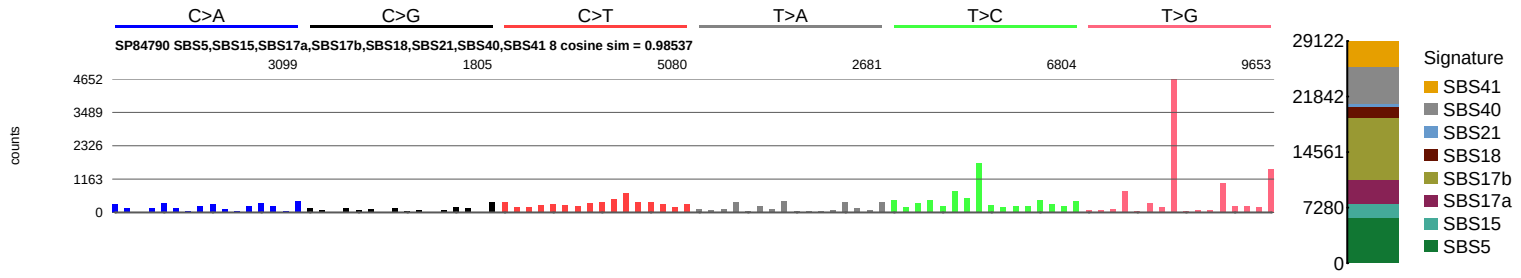

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98553
3003
1877
5192
2561
7040
9448
counts
4680
3510
2340
1170
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2

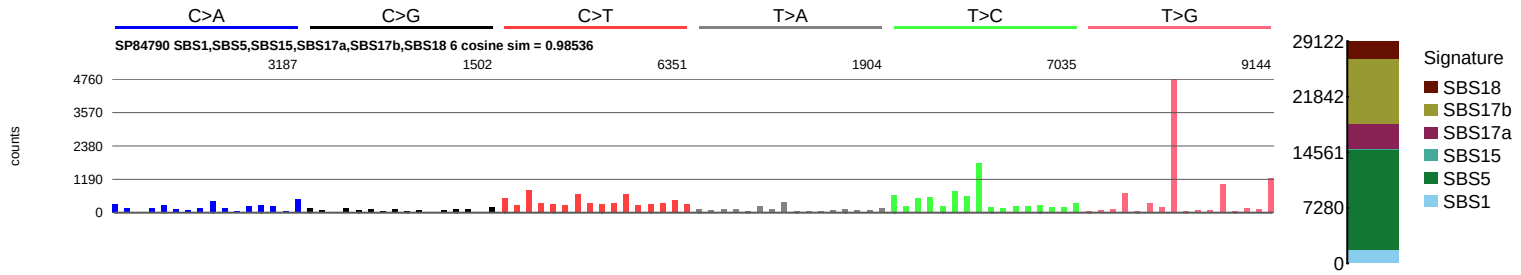

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS18 5 cosine sim = 0.98548
3573
2945
4931
2875
5438
9363
counts
4724
3543
2362
1181
0
29122
21842
14561
7280
0
Signature
SBS18
SBS17b
SBS17a
SBS3
SBS1

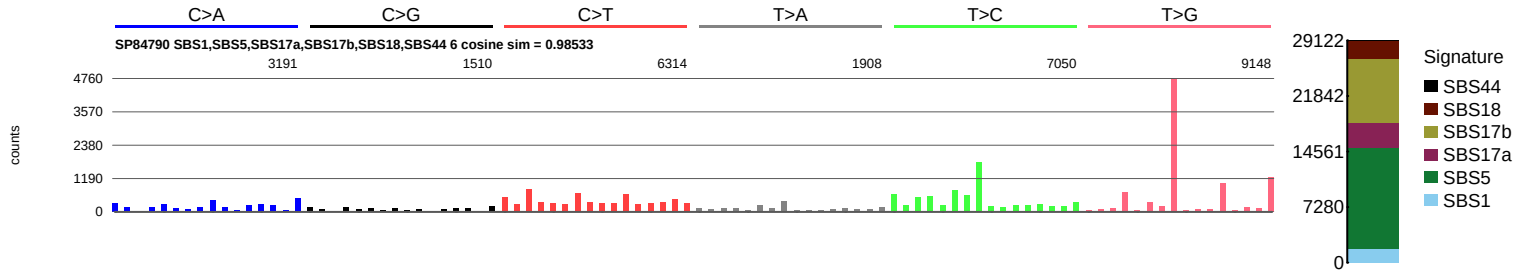

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS21 6 cosine sim = 0.98546
2819
2964
5079
2863
6035
9366
counts
4724
3543
2362
1181
0
29122
21842
14561
7280
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS1

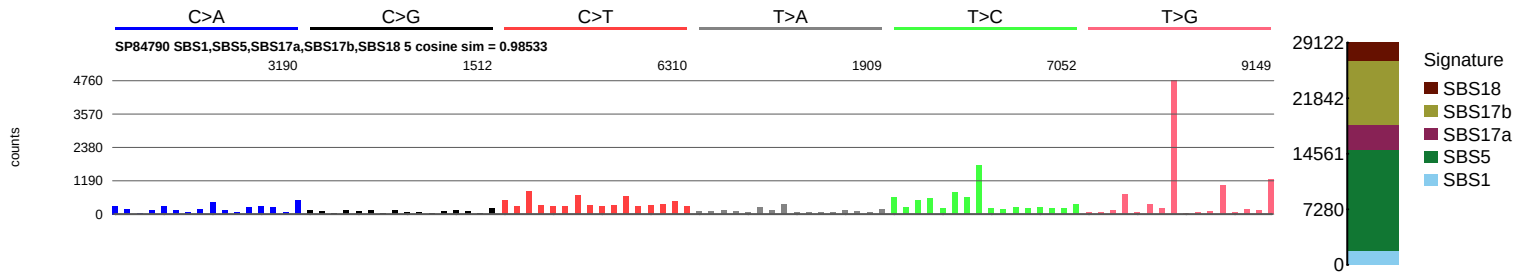

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98541
3089
1785
5123
2677
6809
9638
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2

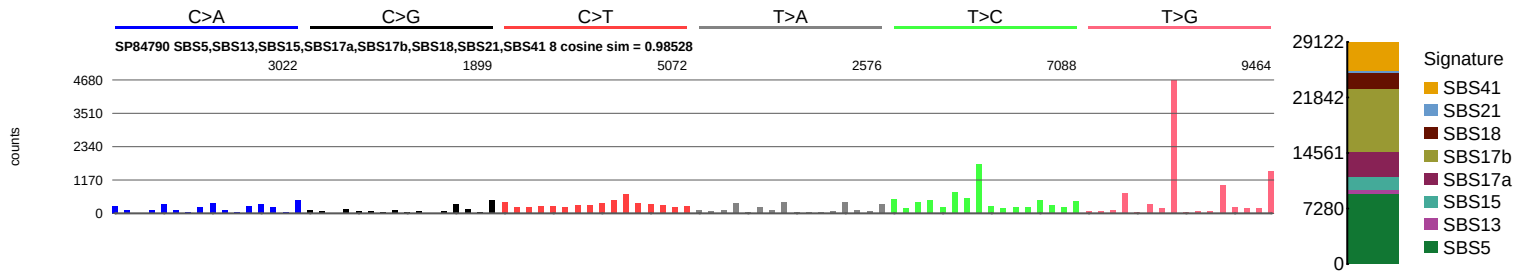

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98538
3008
1903
5264
2596
6869
9480
counts
4680
3510
2340
1170
0
29122
21842
14561
7280
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2

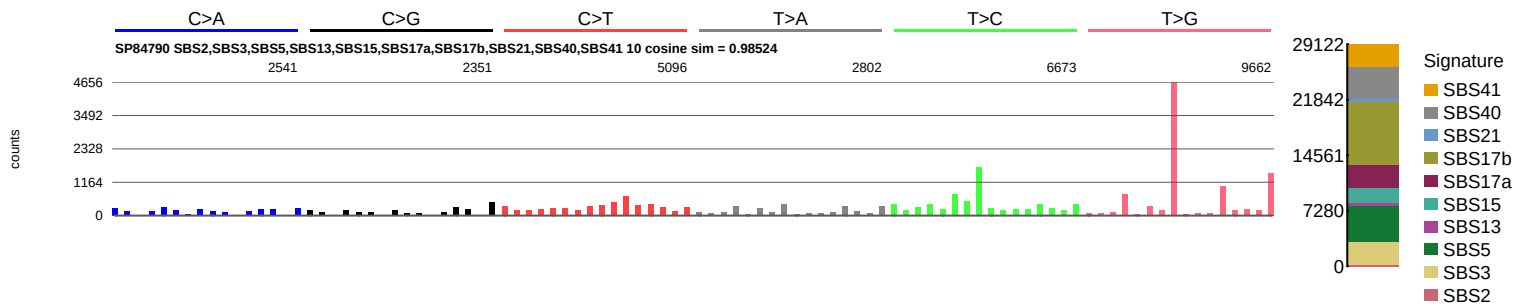

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98537
3099
1805
5080
2681
6804
9653
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5

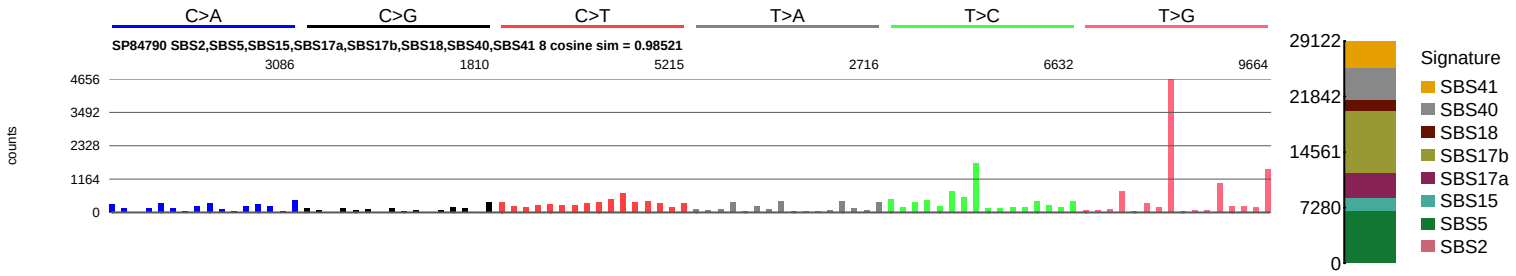

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98536
3187
1502
6351
1904
7035
9144
counts
4760
3570
2380
1190
0
29122
21842
14561
7280
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1

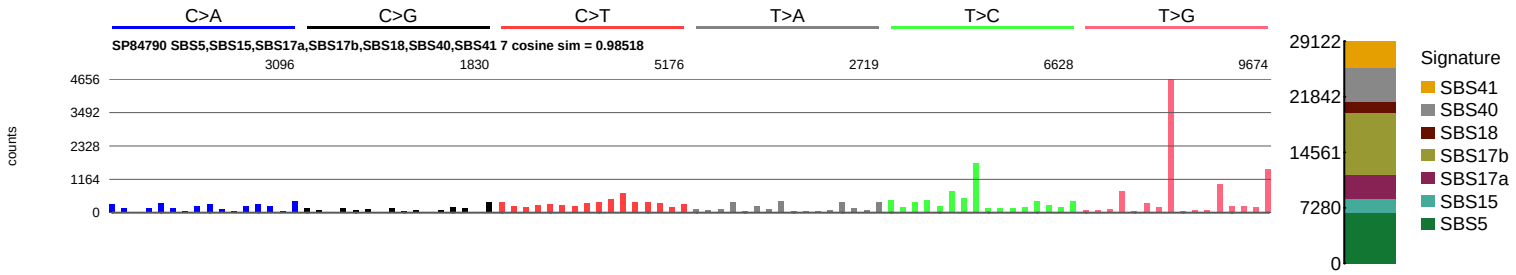

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS44 6 cosine sim = 0.98533
3191
1510
6314
1908
7050
9148
counts
4760
3570
2380
1190
0
29122
21842
14561
7280
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS5
SBS1

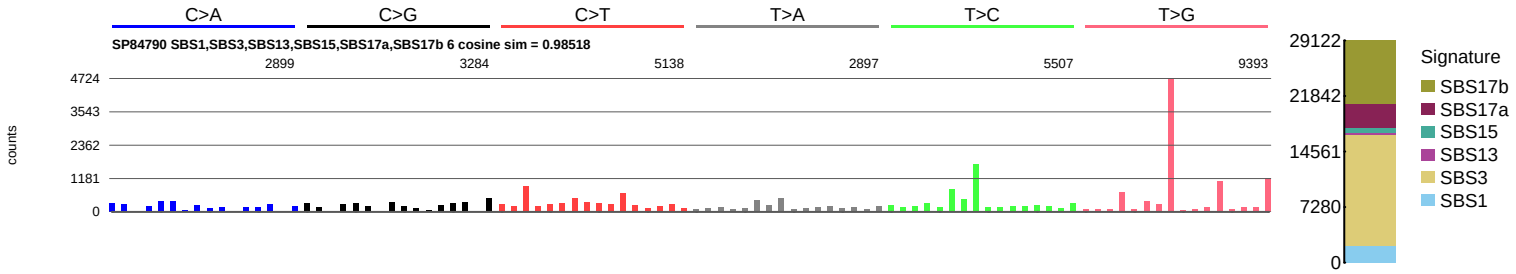

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS17a,SBS17b,SBS18 5 cosine sim = 0.98533
3190
1512
6310
1909
7052
9149
counts
4760
3570
2380
1190
0
29122
21842
14561
7280
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS1

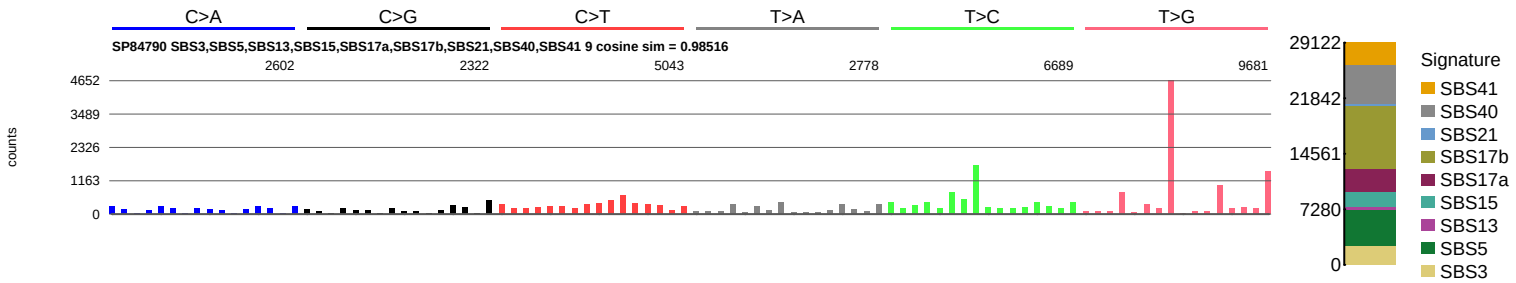

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98528
3022
1899
5072
2576
7088
9464
counts
4680
3510
2340
1170
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5

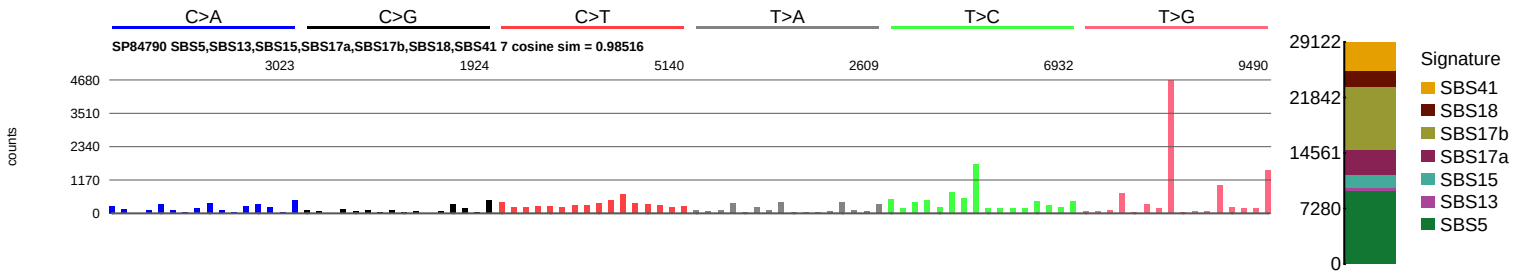

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 10 cosine sim = 0.98524
2541
2351
5096
2802
6673
9662
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2

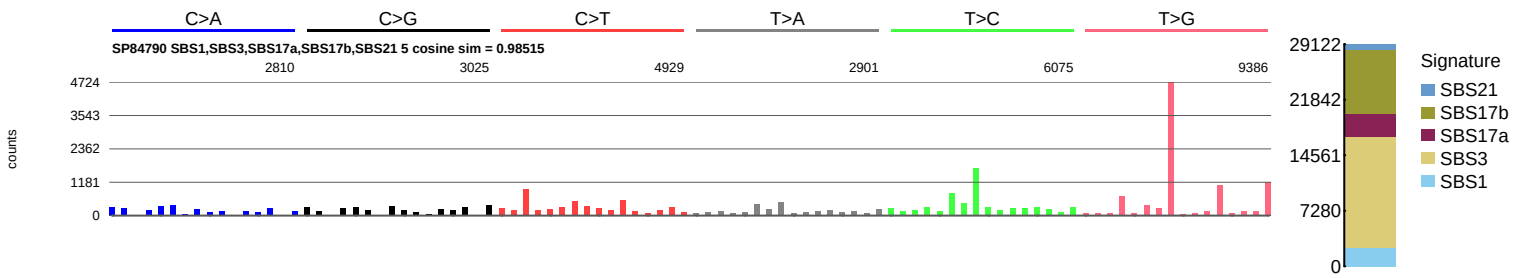

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98521
3086
1810
5215
2716
6632
9664
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2

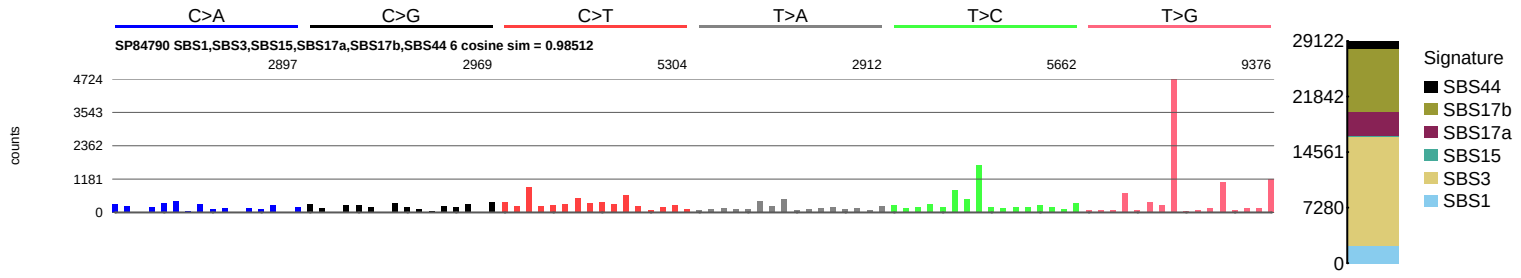

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98518
3096
1830
5176
2719
6628
9674
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5

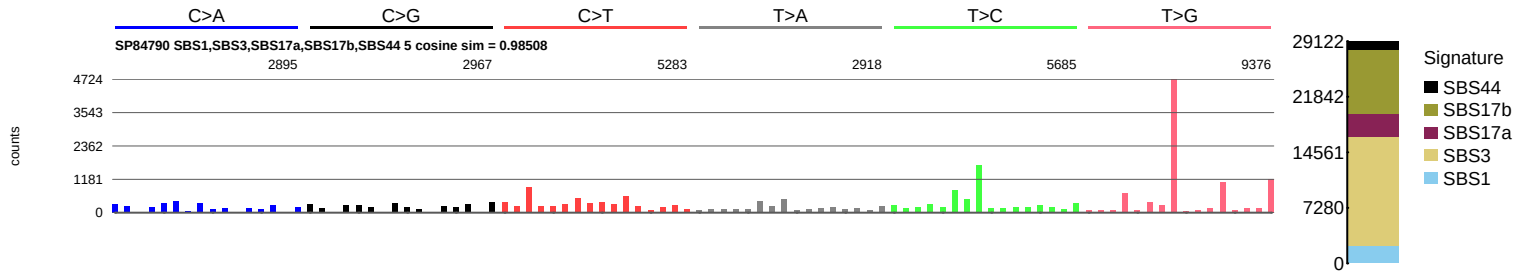

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b 6 cosine sim = 0.98518
2899
3284
5138
2897
5507
9393
counts
4724
3543
2362
1181
0
29122
21842
14561
7280
0
Signature
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

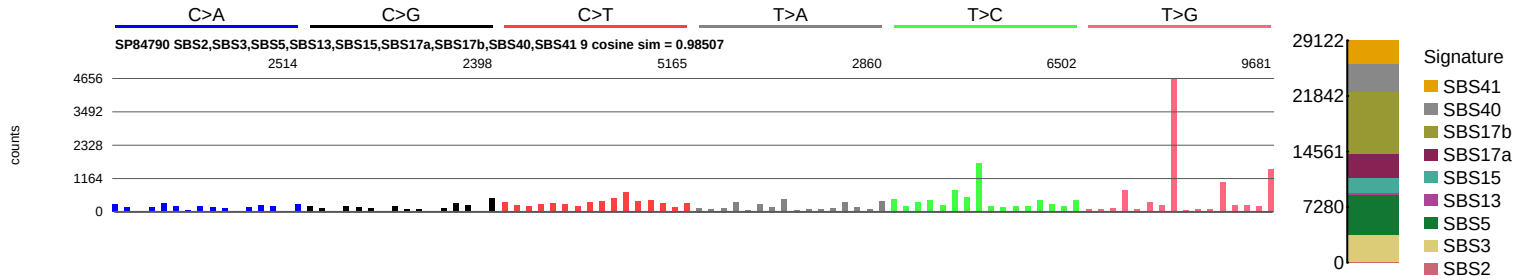

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 9 cosine sim = 0.98516
2602
2322
5043
2778
6689
9681
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3

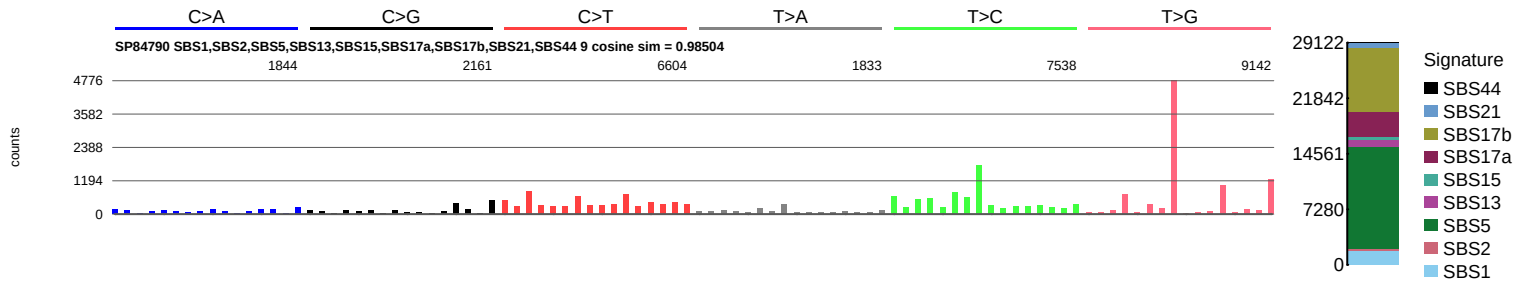

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98516
3023
1924
5140
2609
6932
9490
counts
4680
3510
2340
1170
0
29122
21842
14561
7280
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5

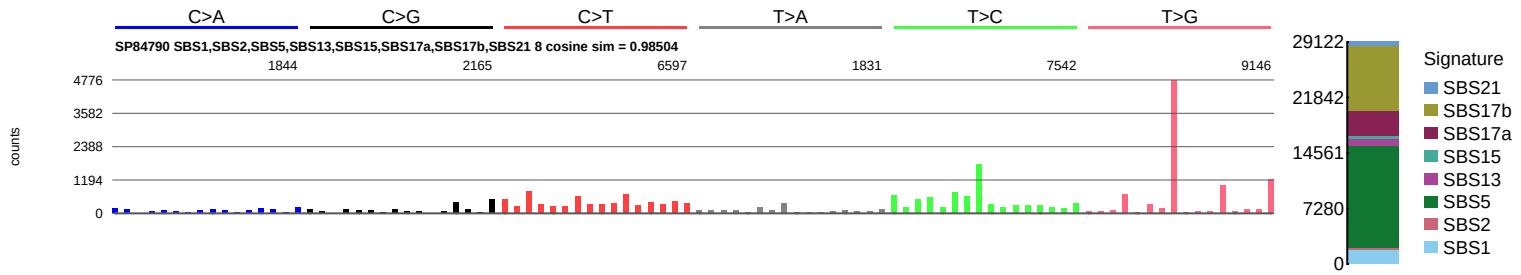

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS21 5 cosine sim = 0.98515
2810
3025
4929
2901
6075
9386
counts
4724
3543
2362
1181
0
29122
21842
14561
7280
0
Signature
SBS21
SBS17b
SBS17a
SBS3
SBS1

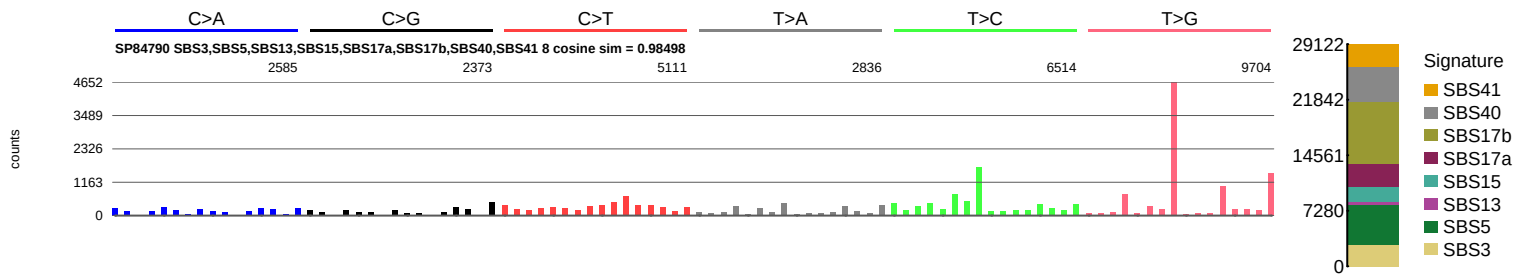

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS44 6 cosine sim = 0.98512
2897
2969
5304
2912
5662
9376
counts
4724
3543
2362
1181
0
29122
21842
14561
7280
0
Signature
SBS44
SBS17b
SBS17a
SBS15
SBS3
SBS1

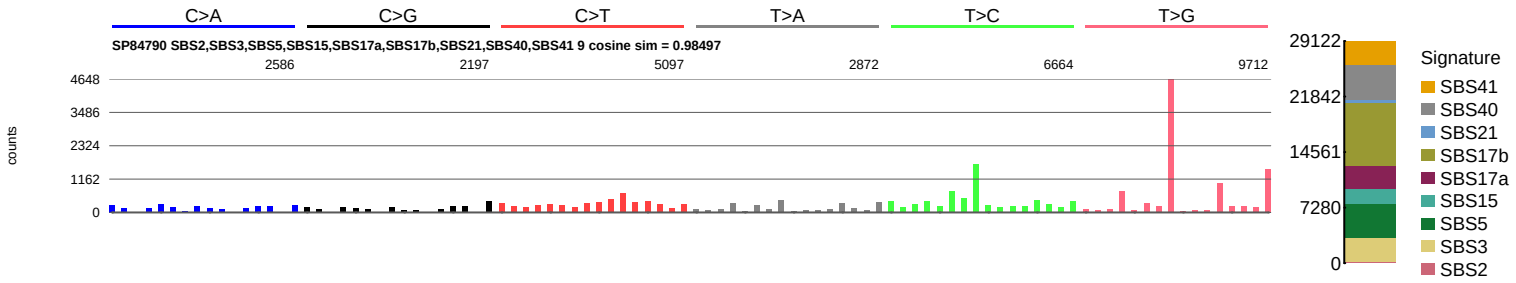

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b,SBS44 5 cosine sim = 0.98508
2895
2967
5283
2918
5685
9376
counts
4724
3543
2362
1181
0
29122
21842
14561
7280
0
Signature
SBS44
SBS17b
SBS17a
SBS3
SBS1

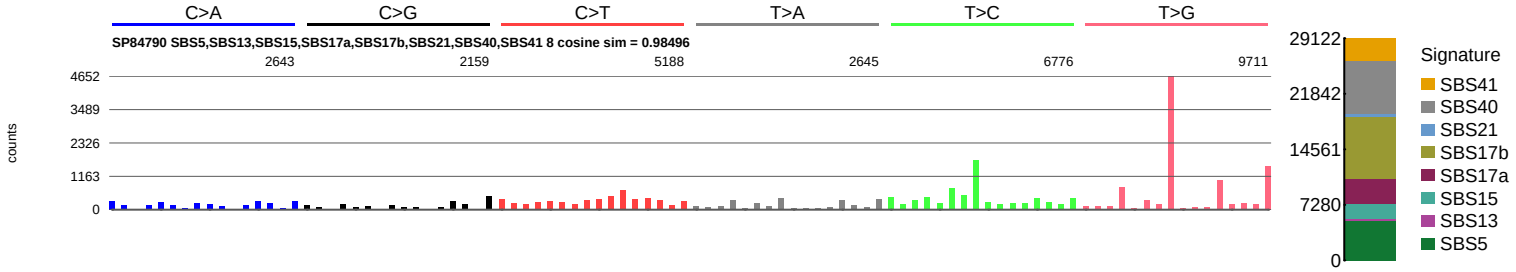

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41 9 cosine sim = 0.98507
2514
2398
5165
2860
6502
9681
counts
4656
3492
2328
1164
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2

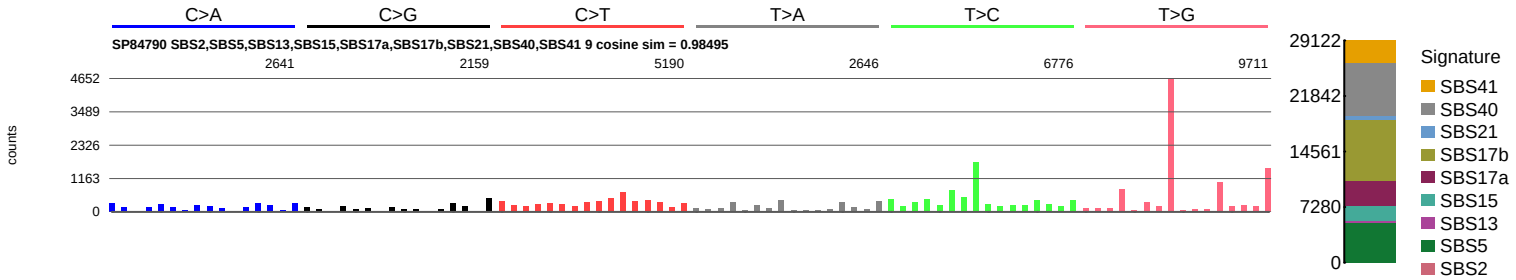

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS44 9 cosine sim = 0.98504
1844
2161
6604
1833
7538
9142
counts
4776
3582
2388
1194
0
29122
21842
14561
7280
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

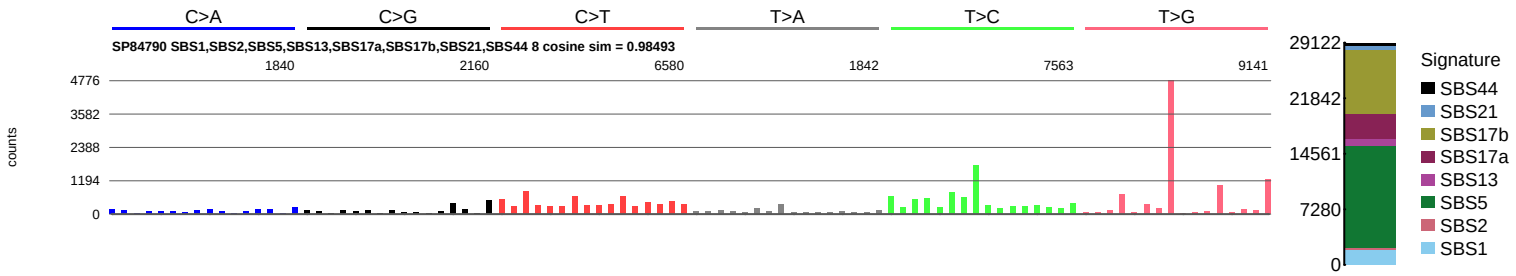

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21 8 cosine sim = 0.98504
1844
2165
6597
1831
7542
9146
counts
4776
3582
2388
1194
0
29122
21842
14561
7280
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

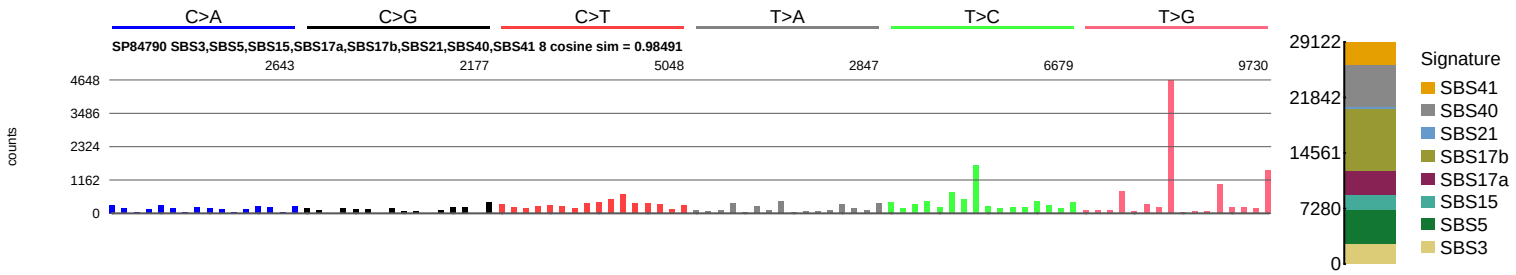

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41 8 cosine sim = 0.98498
2585
2373
5111
2836
6514
9704
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3

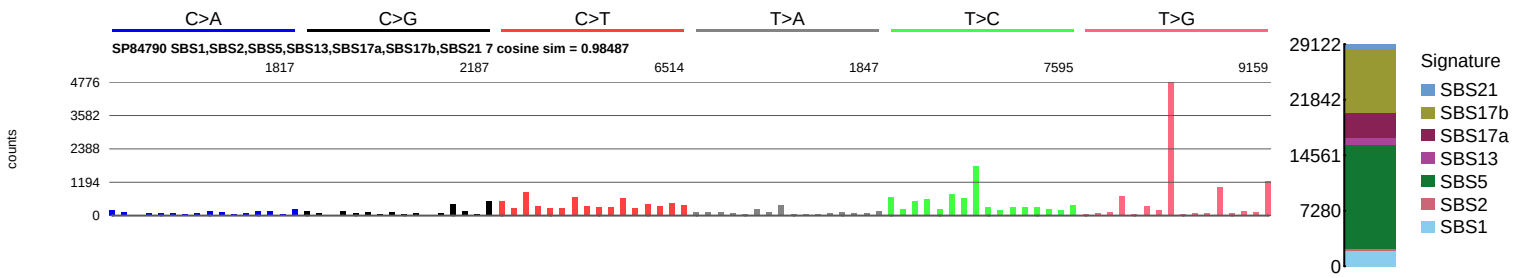

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 9 cosine sim = 0.98497
2586
2197
5097
2872
6664
9712
counts
4648
3486
2324
1162
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2

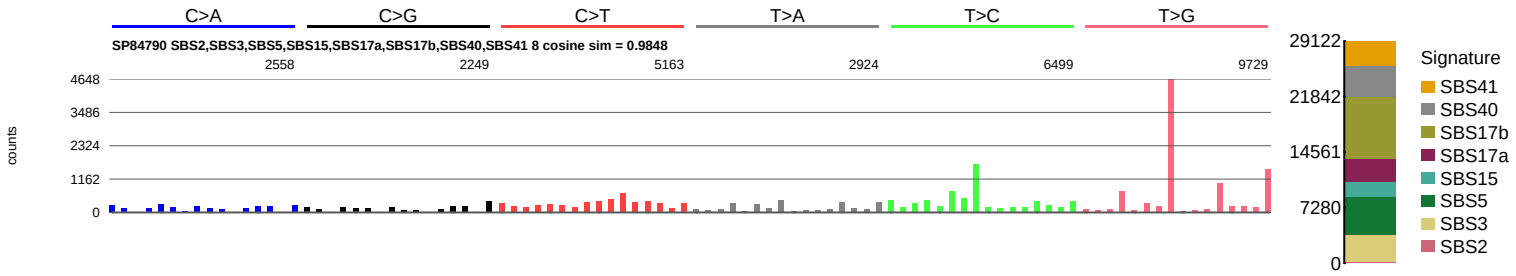

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 8 cosine sim = 0.98496
2643
2159
5188
2645
6776
9711
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5

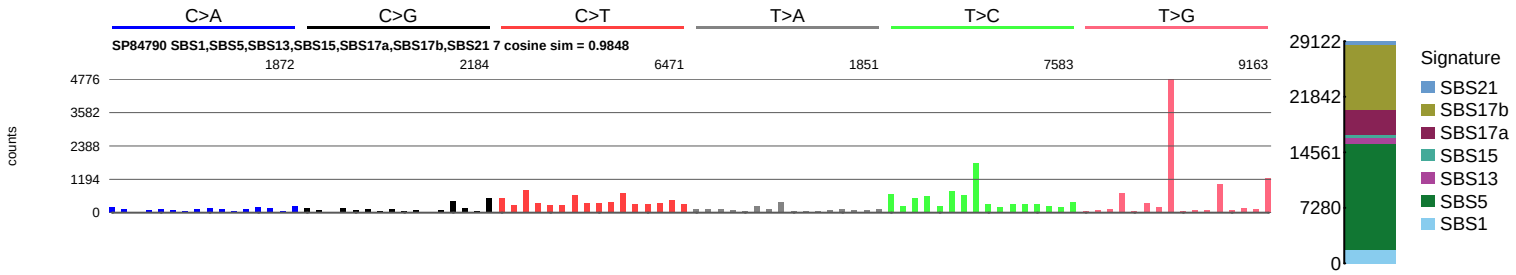

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 9 cosine sim = 0.98495
2641
2159
5190
2646
6776
9711
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2

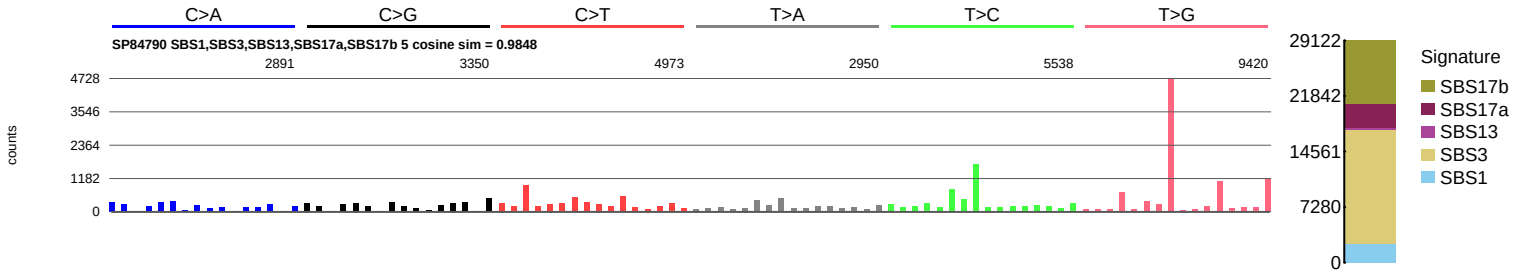

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS44 8 cosine sim = 0.98493
1840
2160
6580
1842
7563
9141
counts
4776
3582
2388
1194
0
29122
21842
14561
7280
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

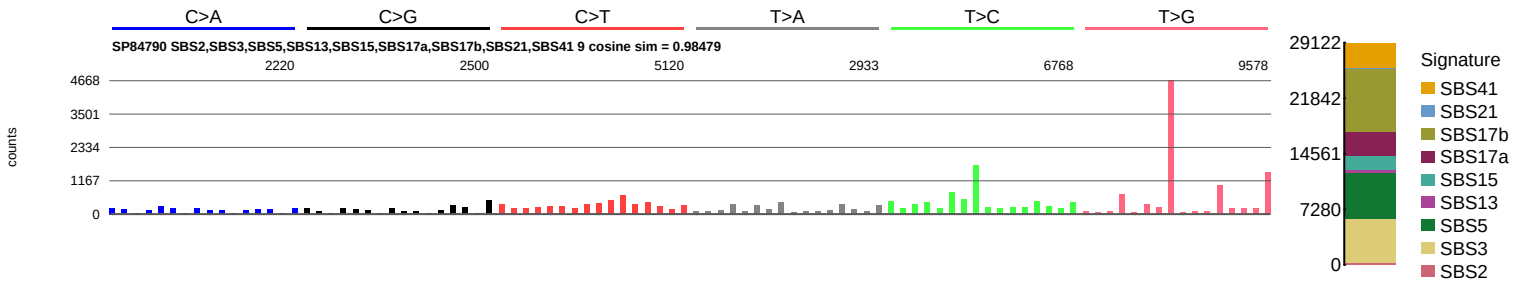

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 8 cosine sim = 0.98491
2643
2177
5048
2847
6679
9730
counts
4648
3486
2324
1162
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3

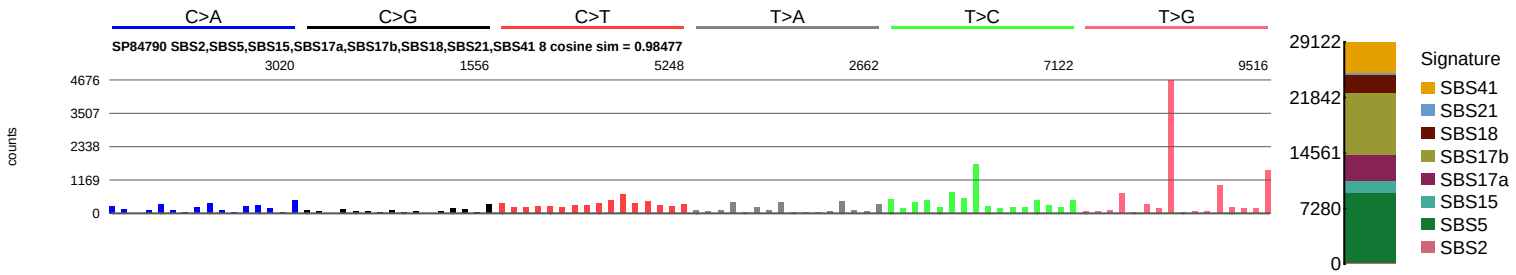

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98487
1817
2187
6514
1847
7595
9159
counts
4776
3582
2388
1194
0
29122
21842
14561
7280
0
Signature
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

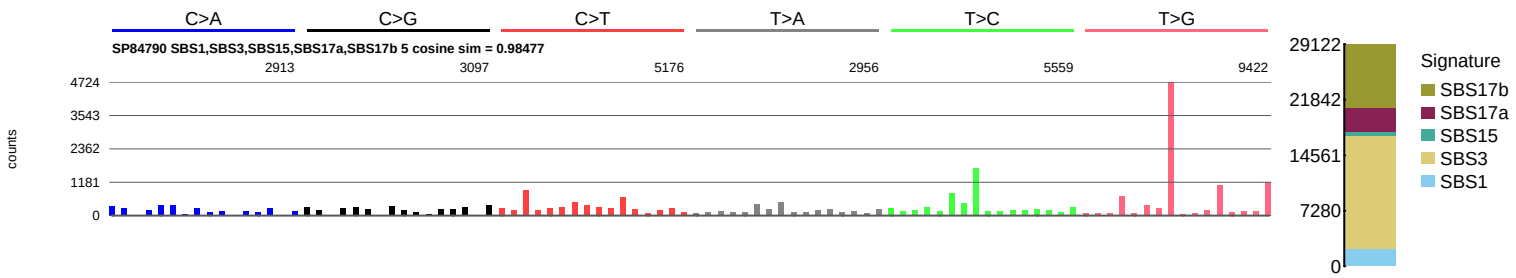

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS40,SBS41 8 cosine sim = 0.9848
2558
2249
5163
2924
6499
9729
counts
4648
3486
2324
1162
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2

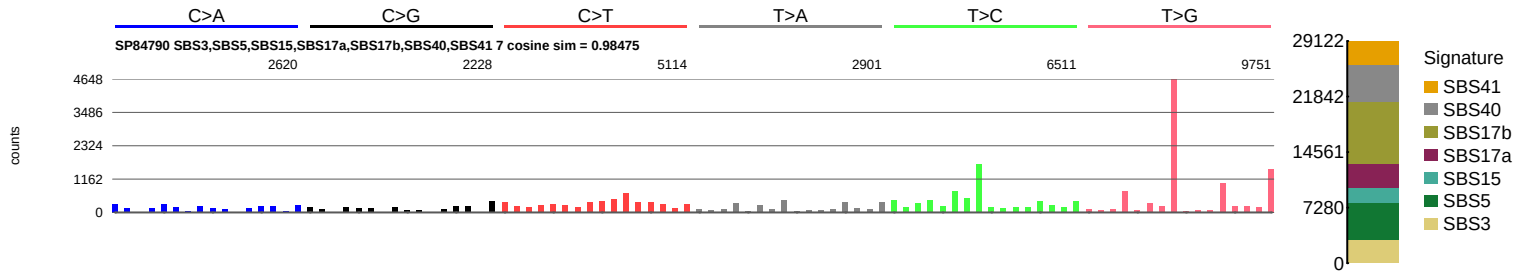

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21 7 cosine sim = 0.9848
1872
2184
6471
1851
7583
9163
counts
4776
3582
2388
1194
0
29122
21842
14561
7280
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

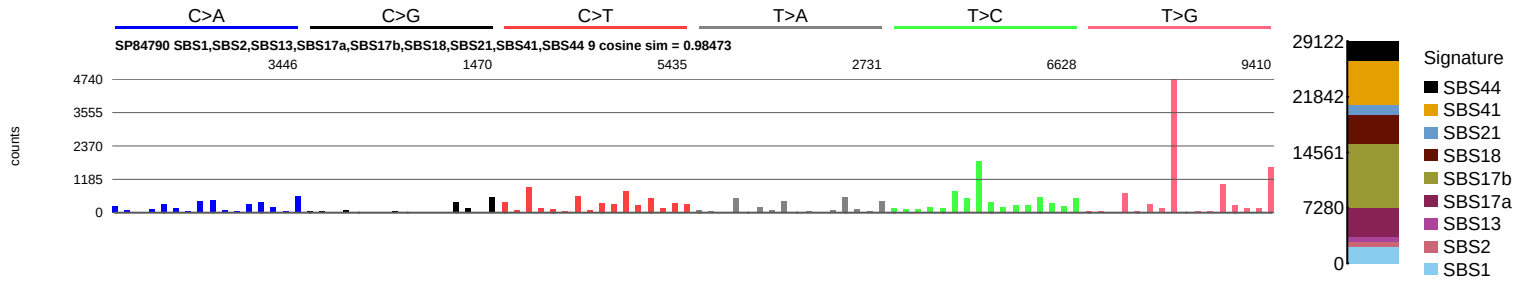

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS13,SBS17a,SBS17b 5 cosine sim = 0.9848
2891
3350
4973
2950
5538
9420
counts
4728
3546
2364
1182
0
29122
21842
14561
7280
0
Signature
SBS17b
SBS17a
SBS13
SBS3
SBS1

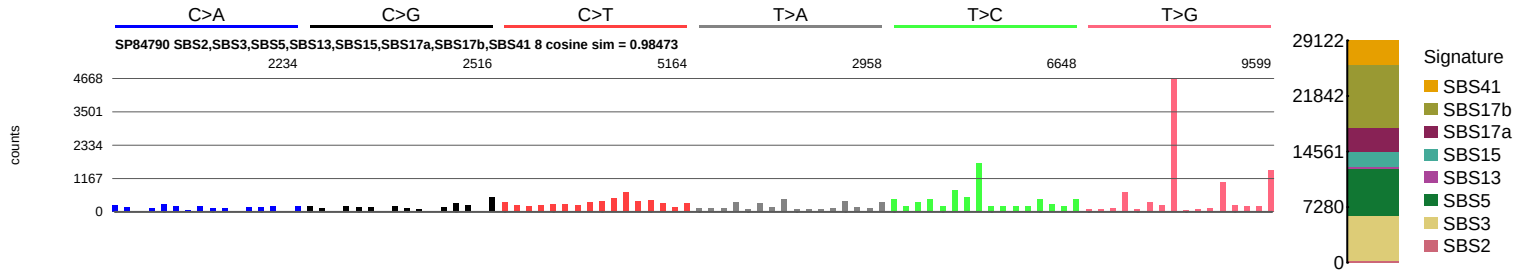

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 9 cosine sim = 0.98479
2220
2500
5120
2933
6768
9578
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2

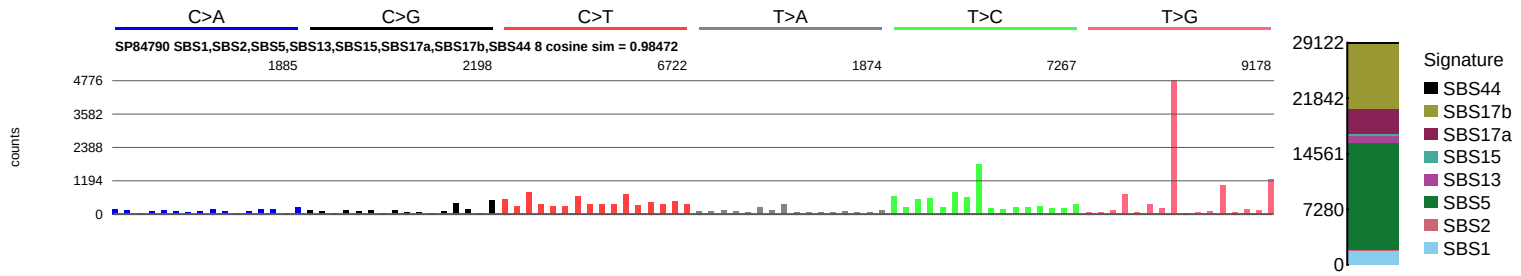

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98477
3020
1556
5248
2662
7122
9516
counts
4676
3507
2338
1169
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2

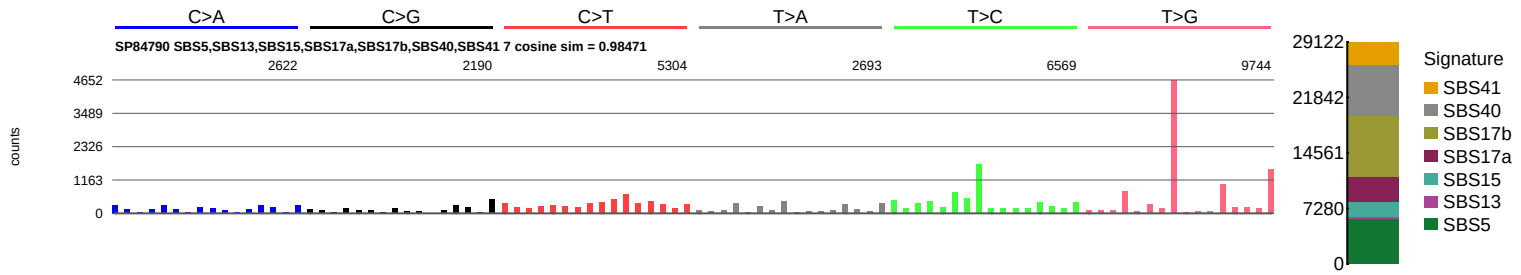

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS15,SBS17a,SBS17b 5 cosine sim = 0.98477
2913
3097
5176
2956
5559
9422
counts
4724
3543
2362
1181
0
29122
21842
14561
7280
0
Signature
SBS17b
SBS17a
SBS15
SBS3
SBS1

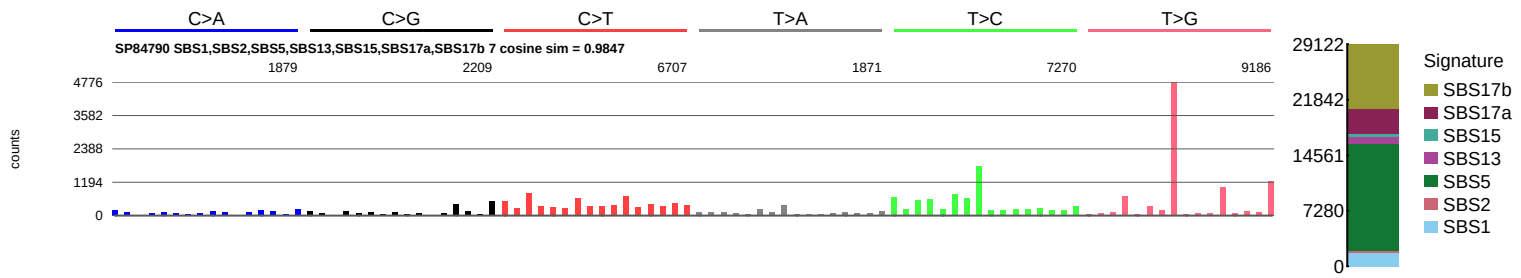

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.98475
2620
2228
5114
2901
6511
9751
counts
4648
3486
2324
1162
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS3

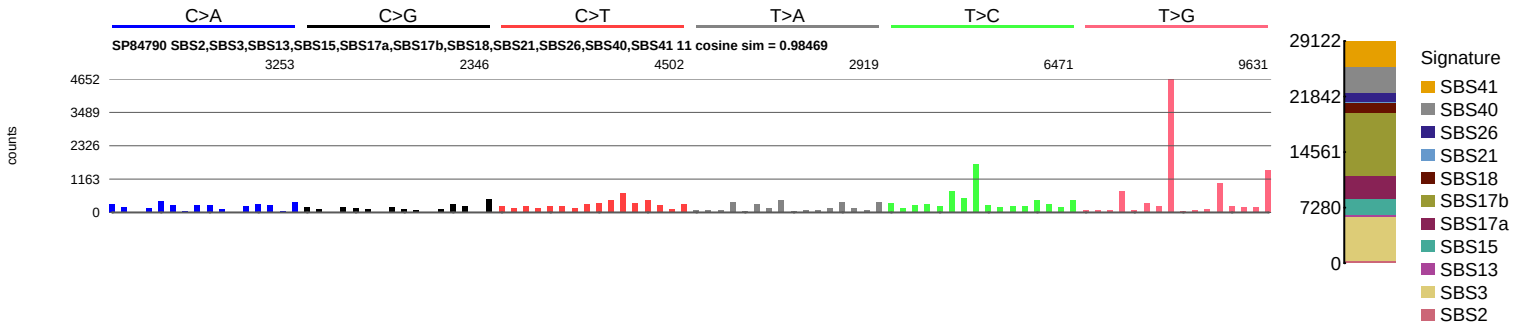

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98473
3446
1470
5435
2731
6628
9410
counts
4740
3555
2370
1185
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

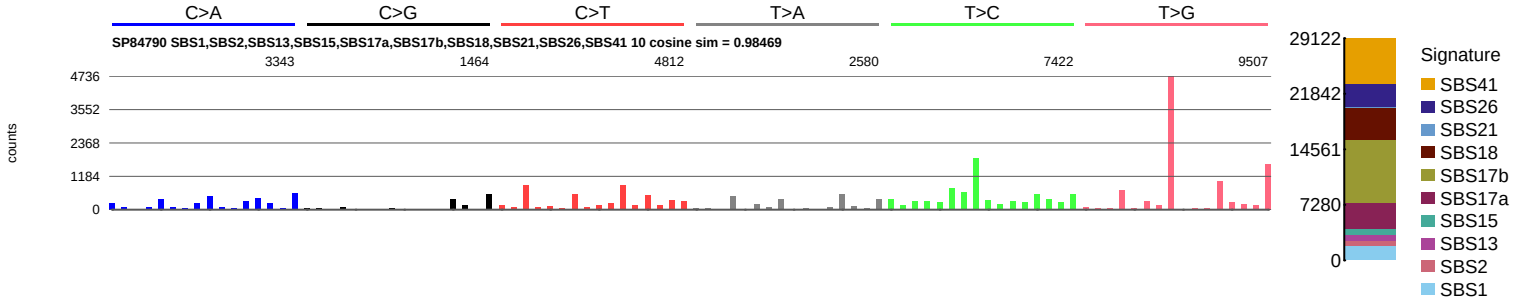

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS41 8 cosine sim = 0.98473
2234
2516
5164
2958
6648
9599
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2

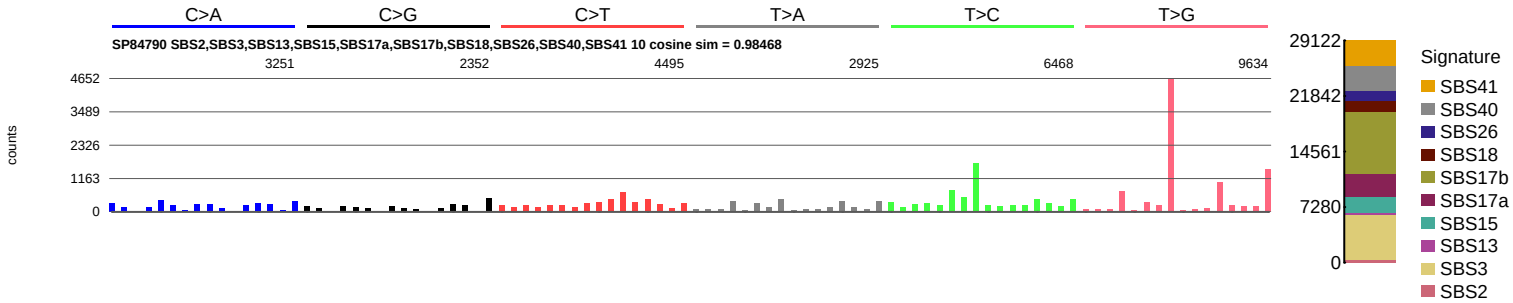

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS44 8 cosine sim = 0.98472
1885
2198
6722
1874
7267
9178
counts
4776
3582
2388
1194
0
29122
21842
14561
7280
0
Signature
SBS44
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

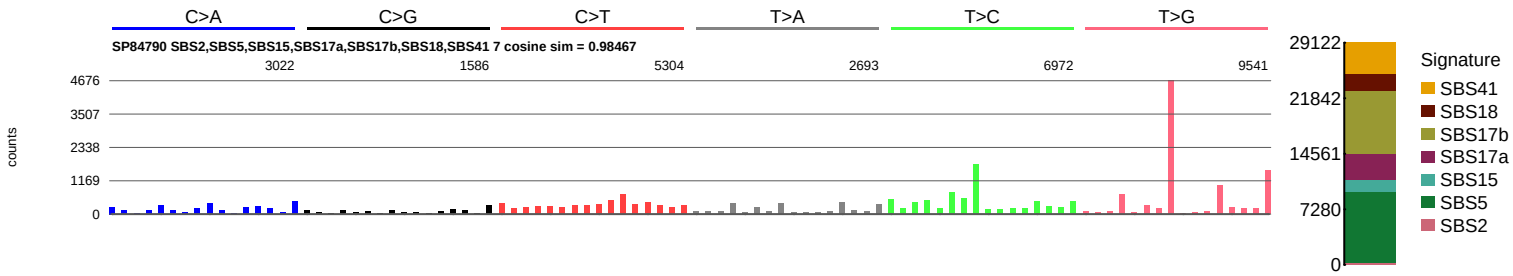

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.98471
2622
2190
5304
2693
6569
9744
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS5

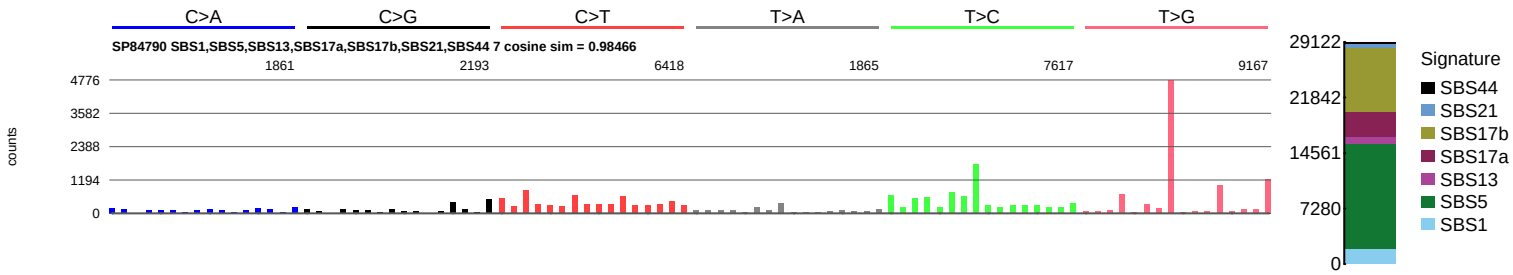

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b 7 cosine sim = 0.9847
1879
2209
6707
1871
7270
9186
counts
4776
3582
2388
1194
0
29122
21842
14561
7280
0
Signature
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

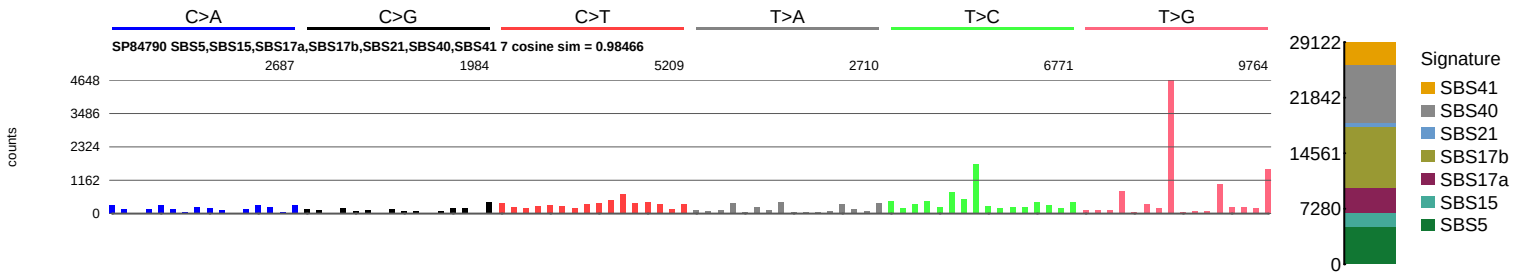

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 11 cosine sim = 0.98469
3253
2346
4502
2919
6471
9631
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2

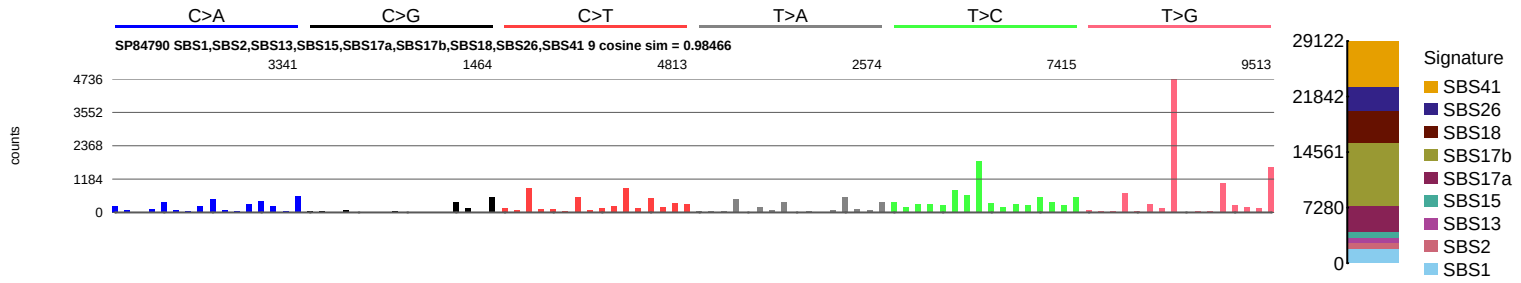

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 10 cosine sim = 0.98469
3343
1464
4812
2580
7422
9507
counts
4736
3552
2368
1184
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

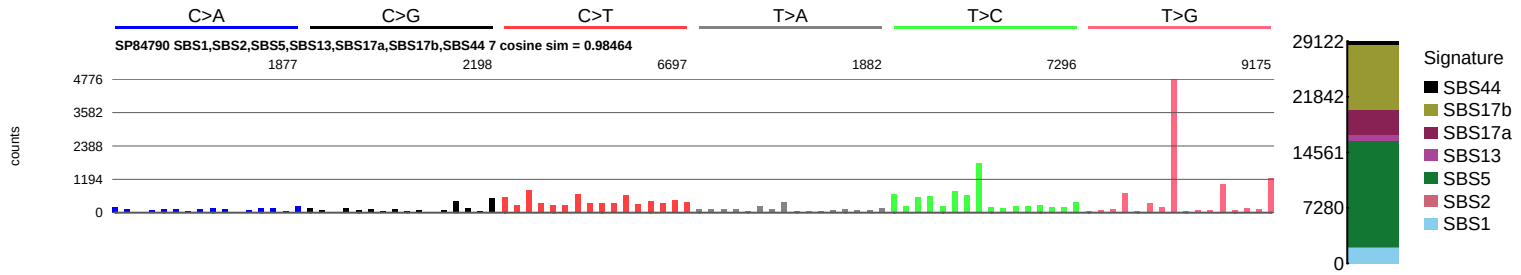

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98468
3251
2352
4495
2925
6468
9634
counts
4652
3489
2326
1163
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2

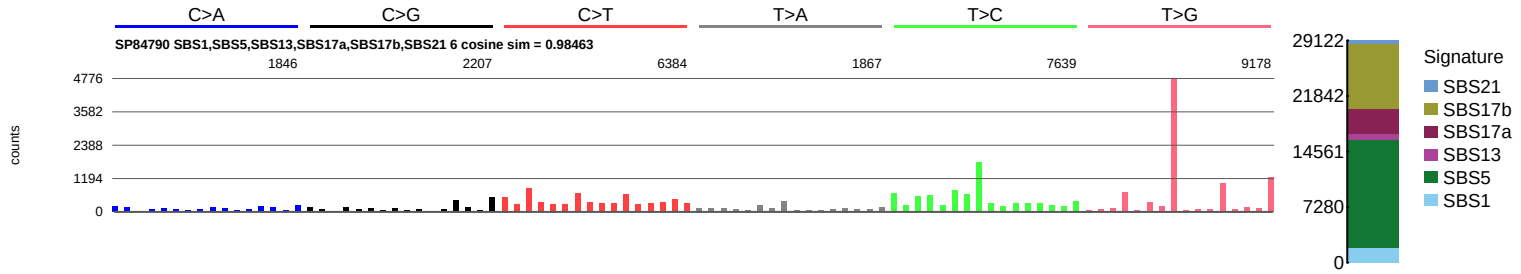

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98467
3022
1586
5304
2693
6972
9541
counts
4676
3507
2338
1169
0
29122
21842
14561
7280
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2

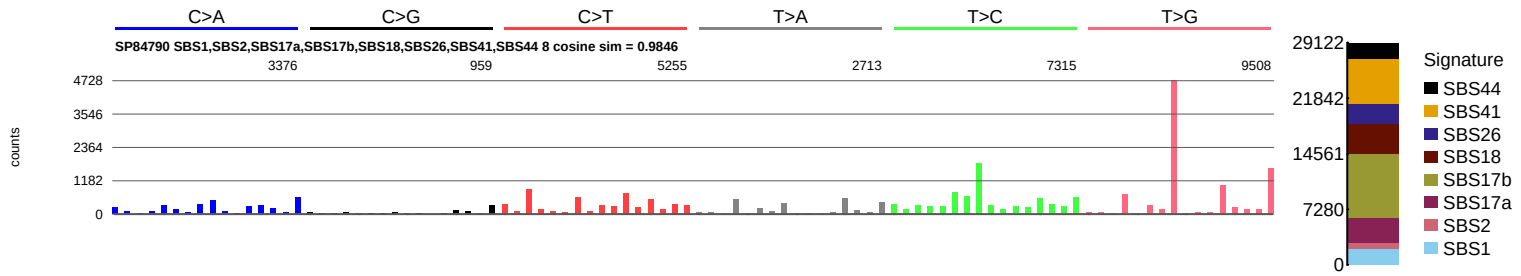

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS44 7 cosine sim = 0.98466
1861
2193
6418
1865
7617
9167
counts
4776
3582
2388
1194
0
29122
21842
14561
7280
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS1

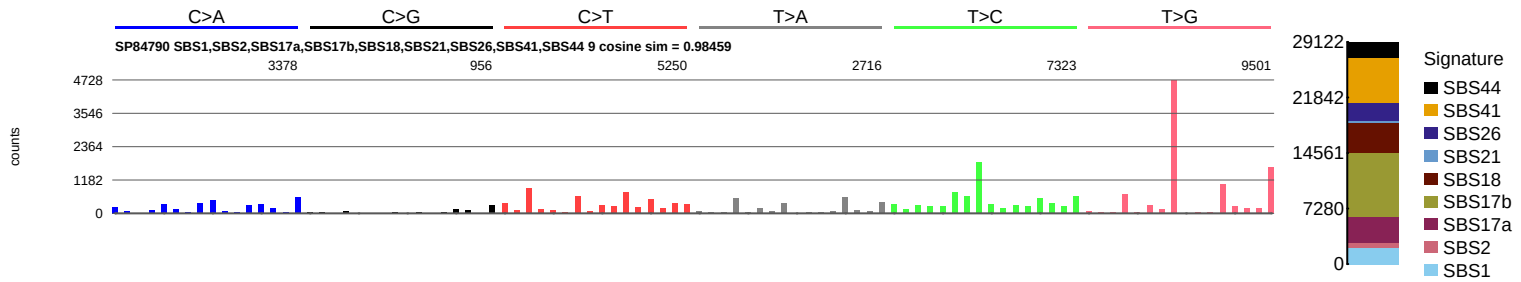

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 7 cosine sim = 0.98466
2687
1984
5209
2710
6771
9764
counts
4648
3486
2324
1162
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5

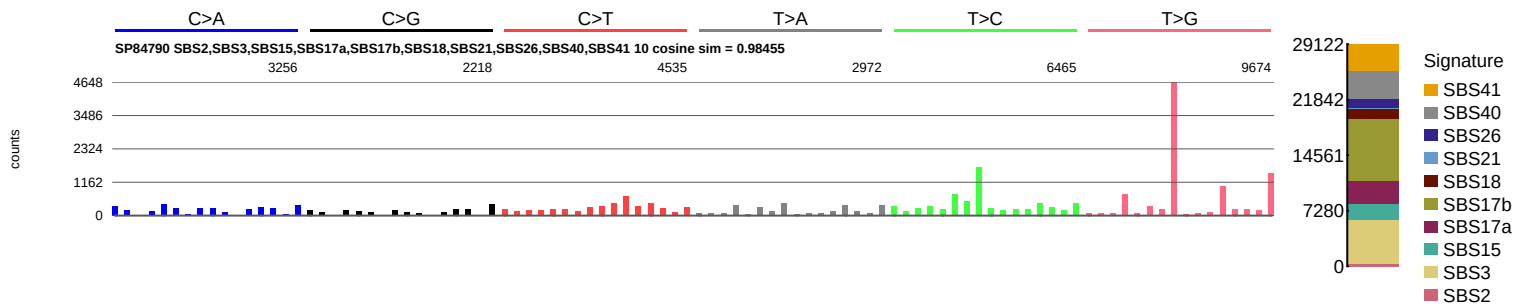

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98466
3341
1464
4813
2574
7415
9513
counts
4736
3552
2368
1184
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

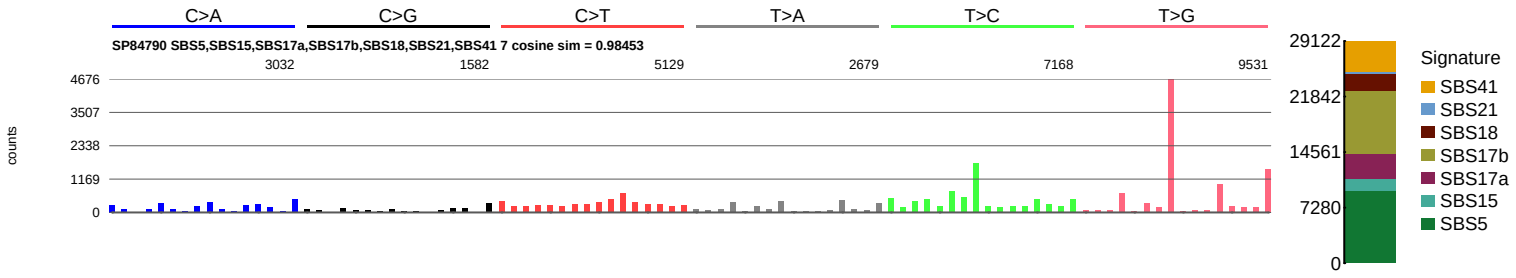

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS44 7 cosine sim = 0.98464
1877
2198
6697
1882
7296
9175
counts
4776
3582
2388
1194
0
29122
21842
14561
7280
0
Signature
SBS44
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

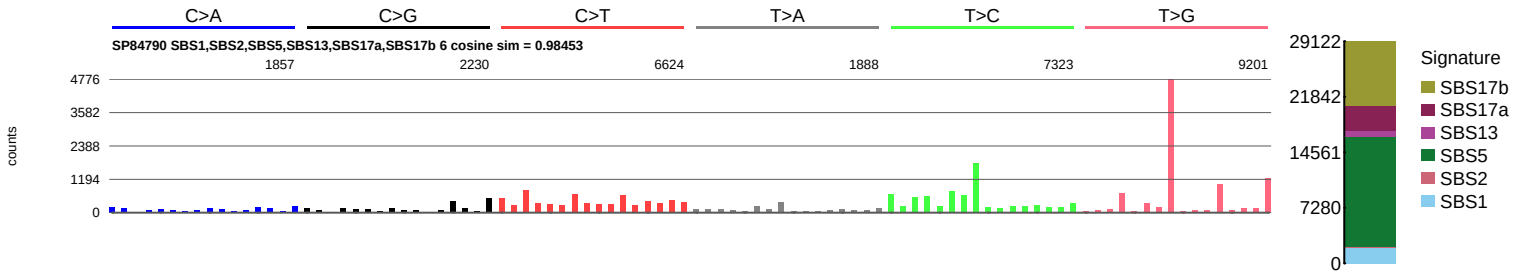

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS21 6 cosine sim = 0.98463
1846
2207
6384
1867
7639
9178
counts
4776
3582
2388
1194
0
29122
21842
14561
7280
0
Signature
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS1

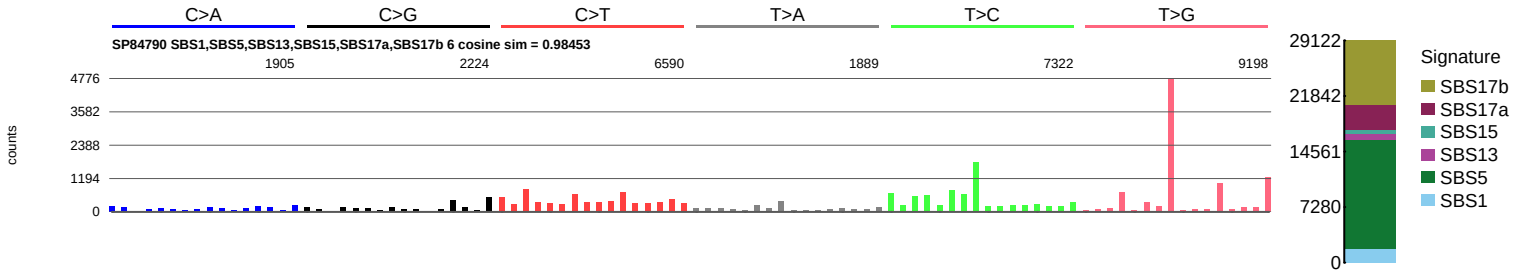

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.9846
3376
959
5255
2713
7315
9508
counts
4728
3546
2364
1182
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1

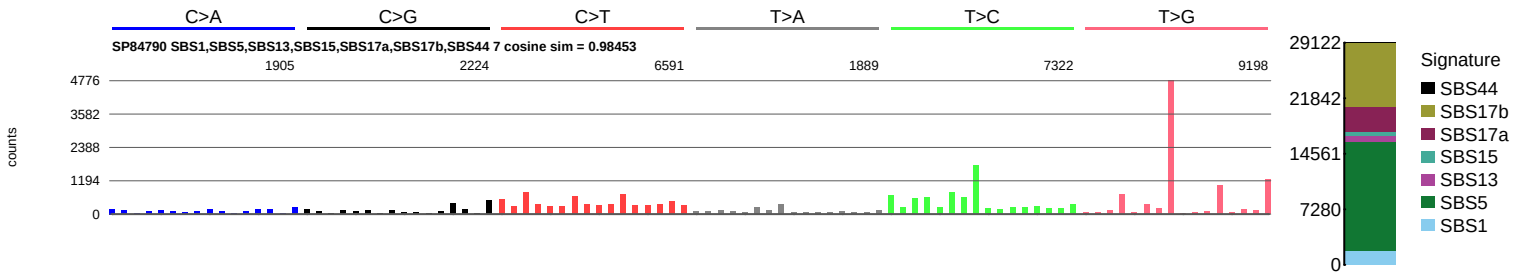

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 9 cosine sim = 0.98459
3378
956
5250
2716
7323
9501
counts
4728
3546
2364
1182
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1

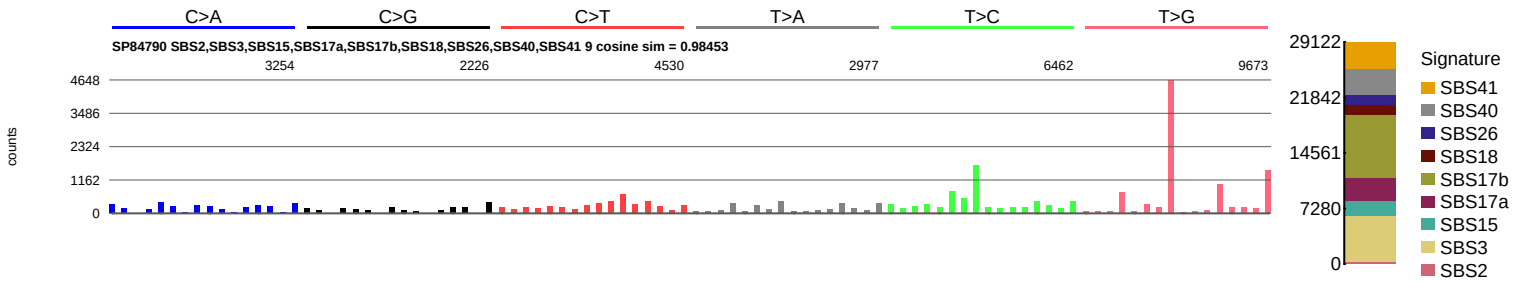

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98455
3256
2218
4535
2972
6465
9674
counts
4648
3486
2324
1162
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2

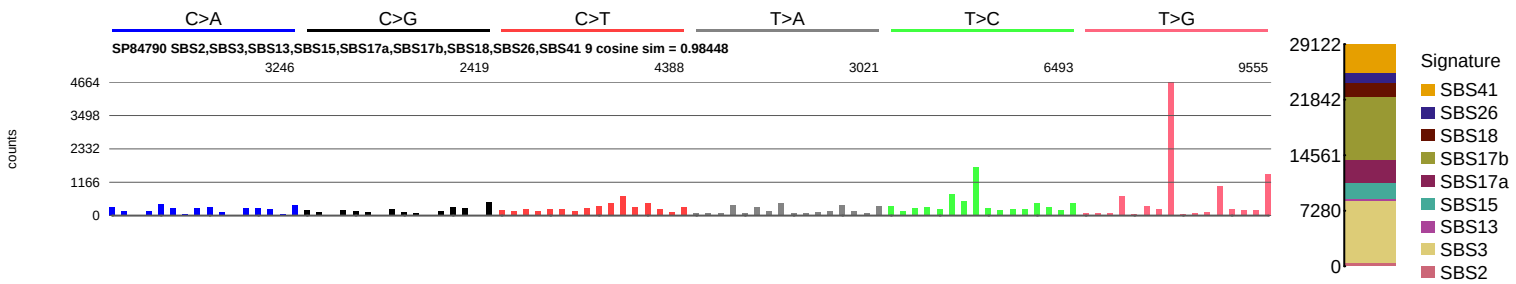

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98453
3032
1582
5129
2679
7168
9531
counts
4676
3507
2338
1169
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5

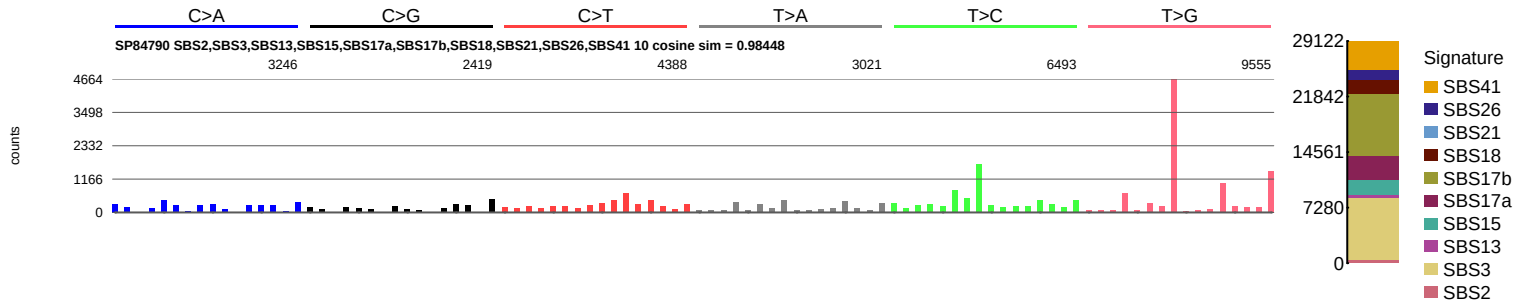

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b 6 cosine sim = 0.98453
1857
2230
6624
1888
7323
9201
counts
4776
3582
2388
1194
0
29122
21842
14561
7280
0
Signature
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

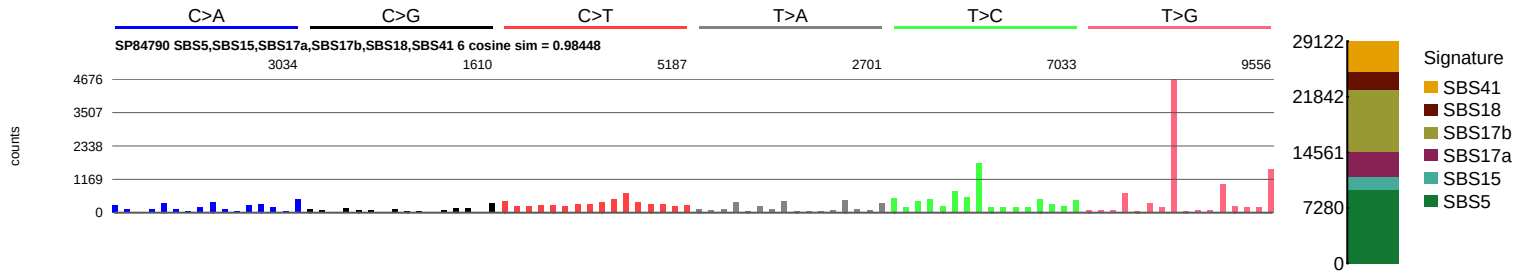

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b 6 cosine sim = 0.98453
1905
2224
6590
1889
7322
9198
counts
4776
3582
2388
1194
0
29122
21842
14561
7280
0
Signature
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

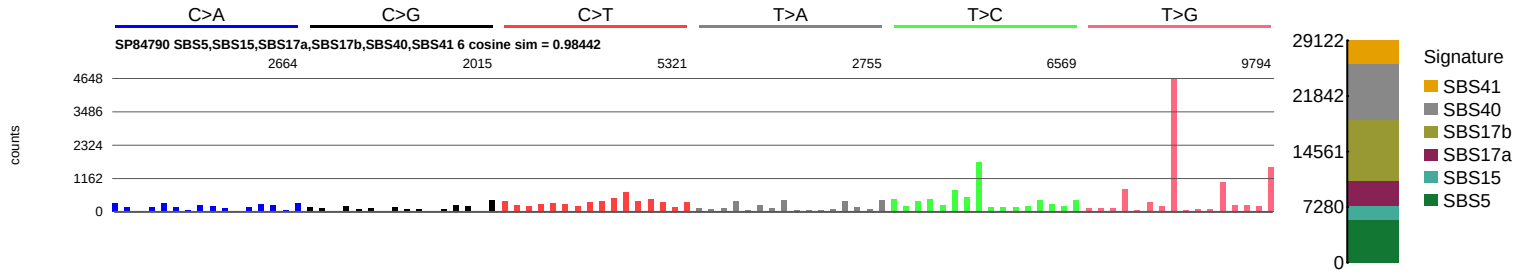

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS44 7 cosine sim = 0.98453
1905
2224
6591
1889
7322
9198
counts
4776
3582
2388
1194
0
29122
21842
14561
7280
0
Signature
SBS44
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

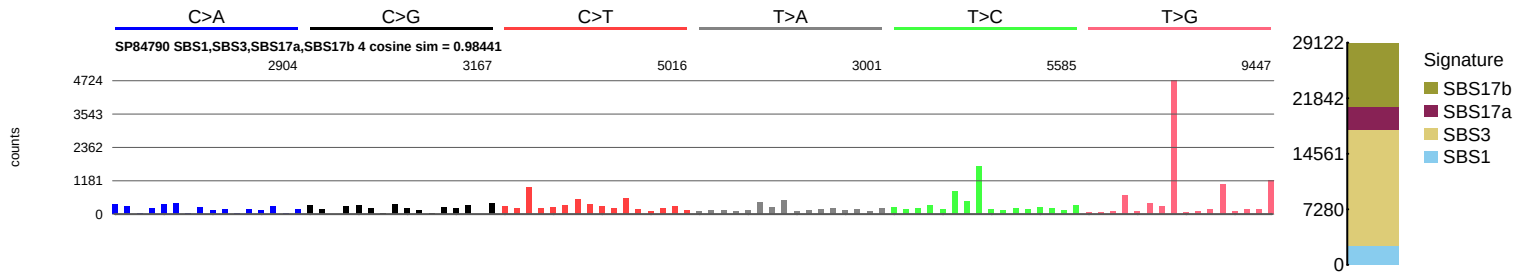

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98453
3254
2226
4530
2977
6462
9673
counts
4648
3486
2324
1162
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2

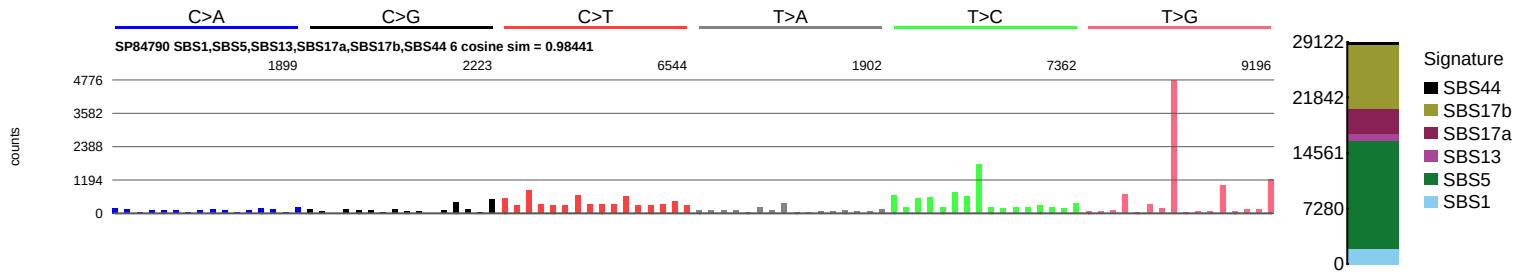

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98448
3246
2419
4388
3021
6493
9555
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2

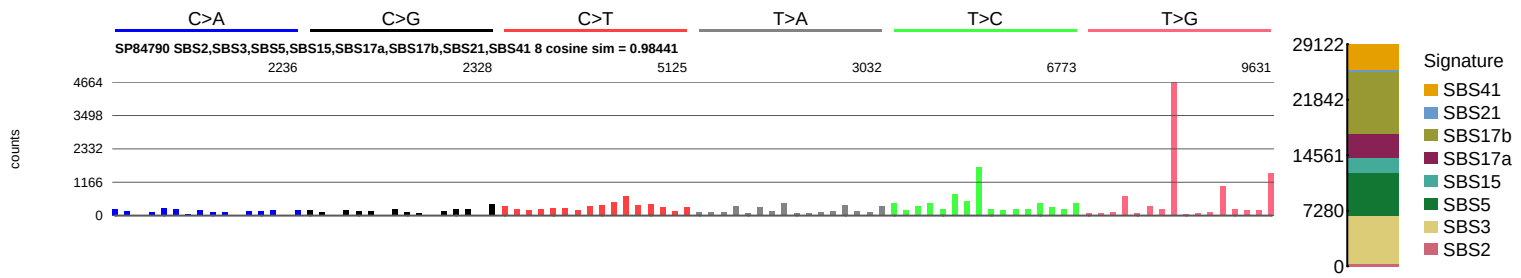

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 10 cosine sim = 0.98448
3246
2419
4388
3021
6493
9555
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2

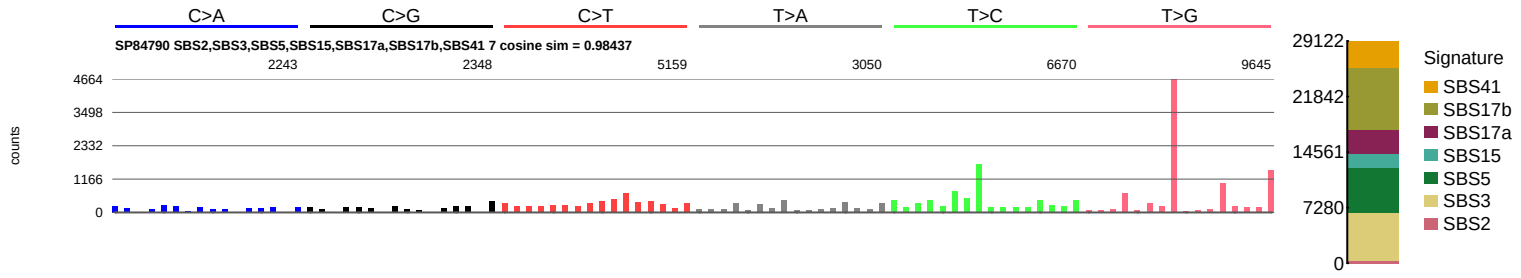

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98448
3034
1610
5187
2701
7033
9556
counts
4676
3507
2338
1169
0
29122
21842
14561
7280
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5

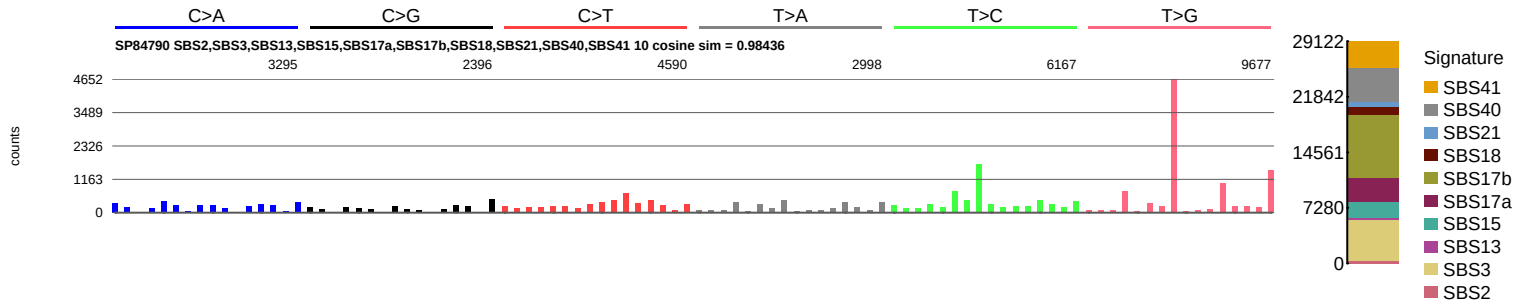

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS5,SBS15,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.98442
2664
2015
5321
2755
6569
9794
counts
4648
3486
2324
1162
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS5

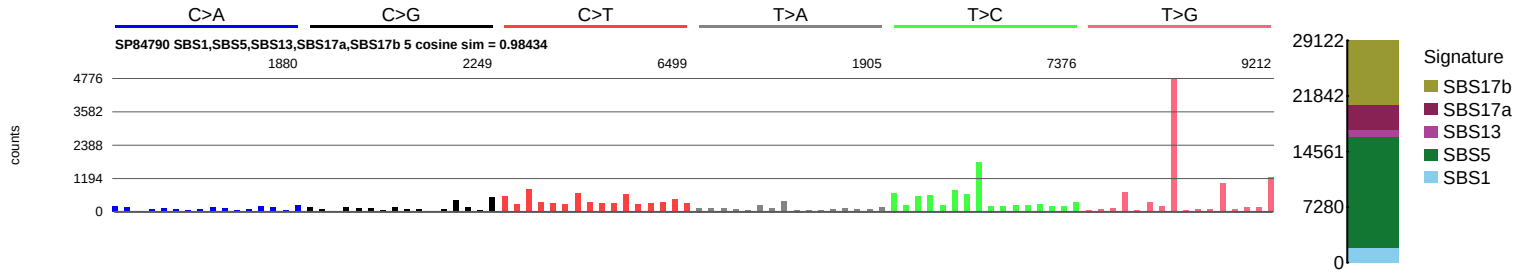

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS3,SBS17a,SBS17b 4 cosine sim = 0.98441
2904
3167
5016
3001
5585
9447
counts
4724
3543
2362
1181
0
29122
21842
14561
7280
0
Signature
SBS17b
SBS17a
SBS3
SBS1

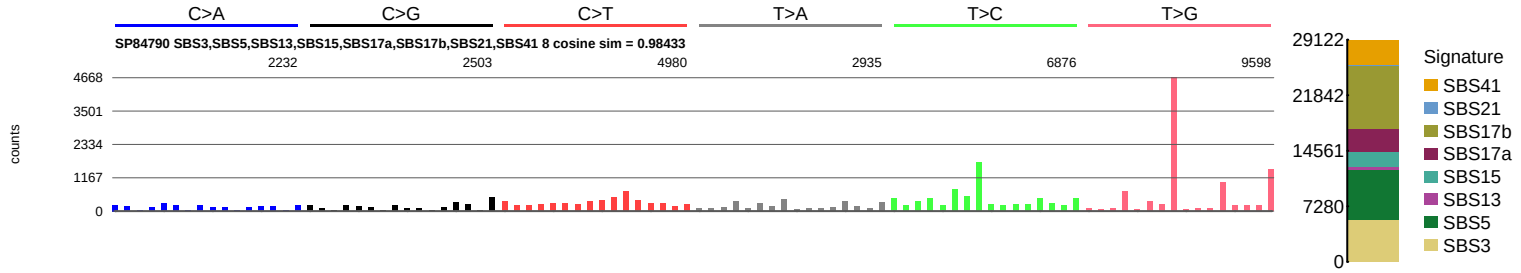

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS44 6 cosine sim = 0.98441
1899
2223
6544
1902
7362
9196
counts
4776
3582
2388
1194
0
29122
21842
14561
7280
0
Signature
SBS44
SBS17b
SBS17a
SBS13
SBS5
SBS1

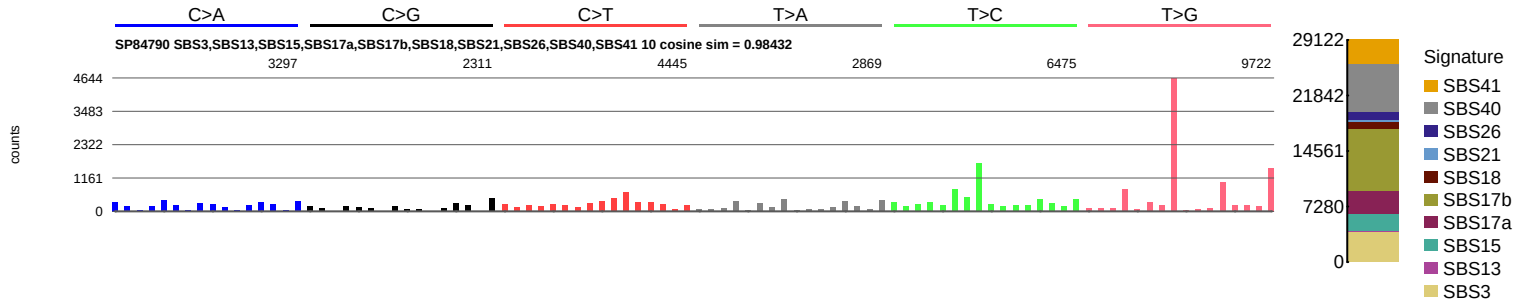

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.98441
2236
2328
5125
3032
6773
9631
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2

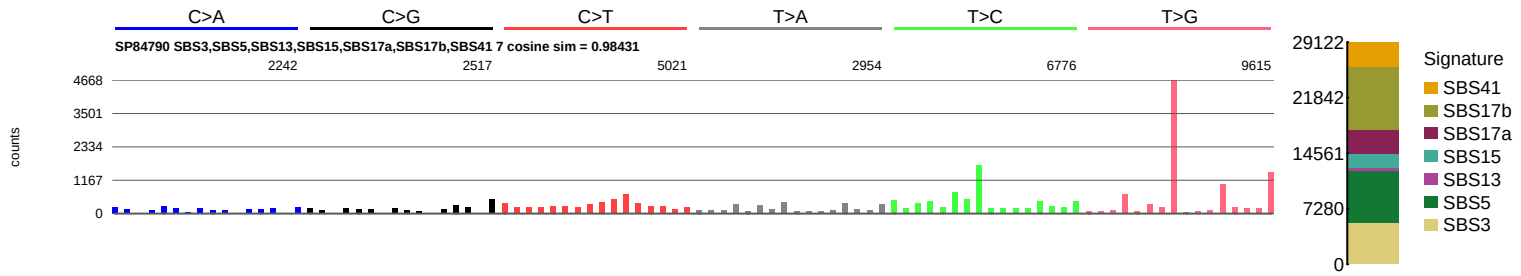

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98437
2243
2348
5159
3050
6670
9645
4664
3498
2332
1166
0
counts
29122
21842
14561
7280
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2

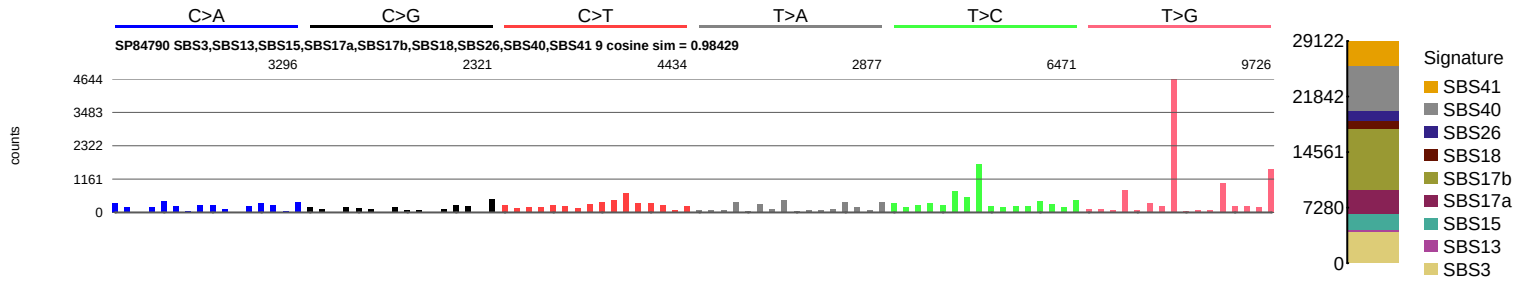

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98436
3295
2396
4590
2998
6167
9677
4652
3489
2326
1163
0
counts
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2

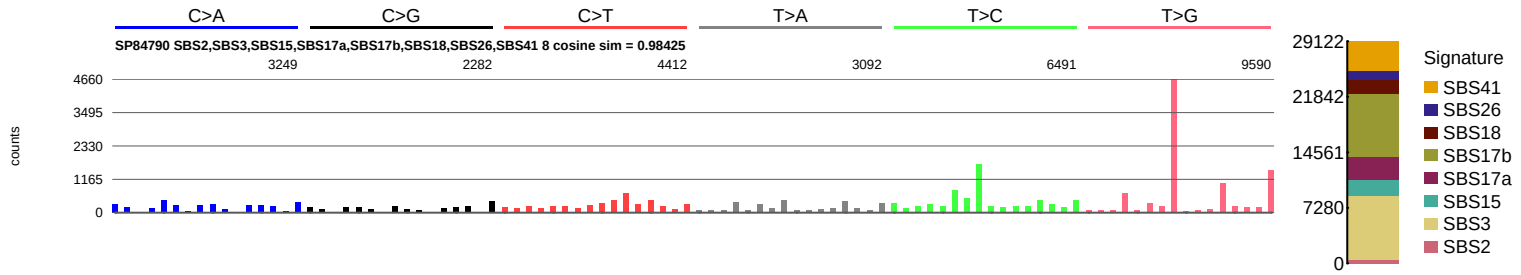

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS13,SBS17a,SBS17b 5 cosine sim = 0.98434
1880
2249
6499
1905
7376
9212
4776
3582
2388
1194
0
counts
29122
21842
14561
7280
0
Signature
SBS17b
SBS17a
SBS13
SBS5
SBS1

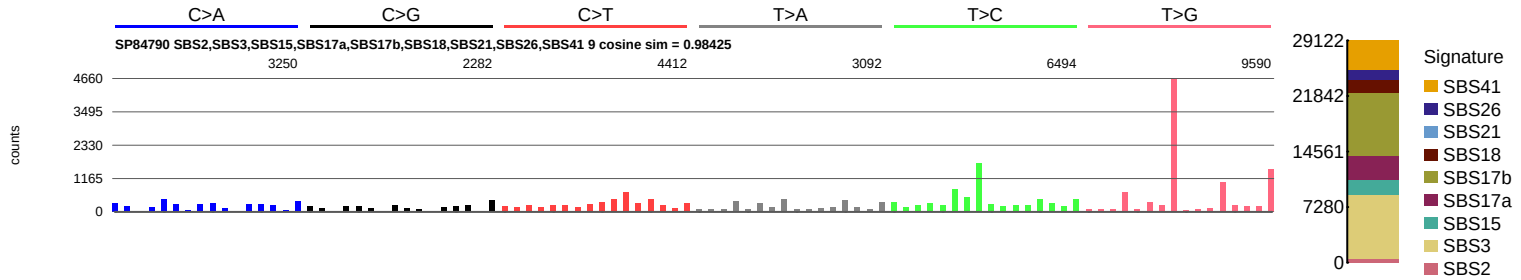

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.98433
2232
2503
4980
2935
6876
9598
4668
3501
2334
1167
0
counts
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3

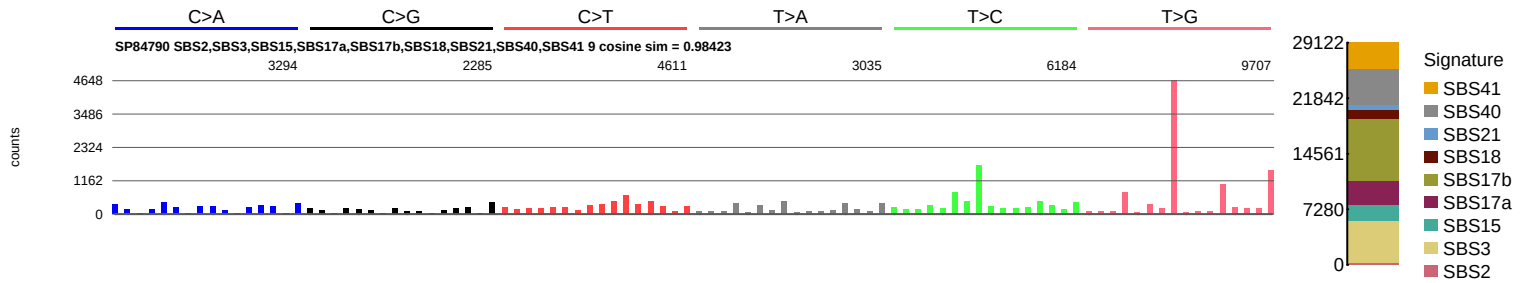

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98432
3297
2311
4445
2869
6475
9722
4644
3483
2322
1161
0
counts
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3

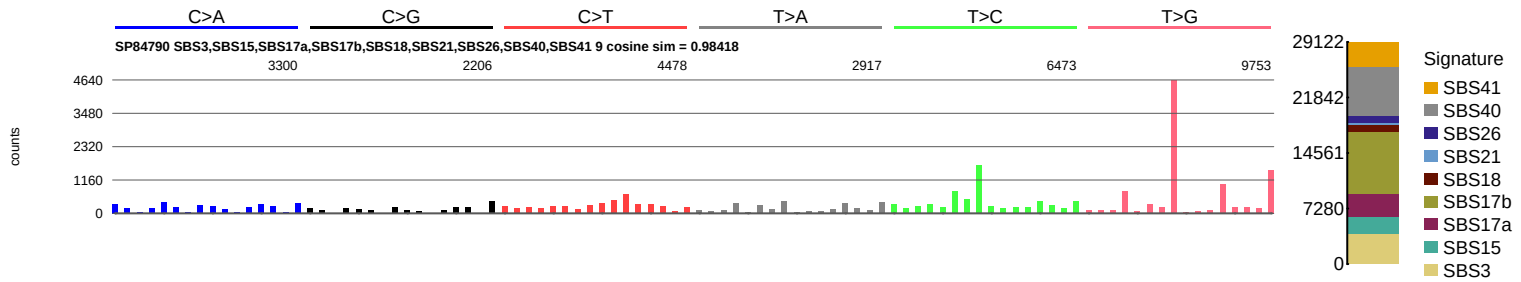

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98431
2242
2517
5021
2954
6776
9615
4668
3501
2334
1167
0
counts
29122
21842
14561
7280
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3

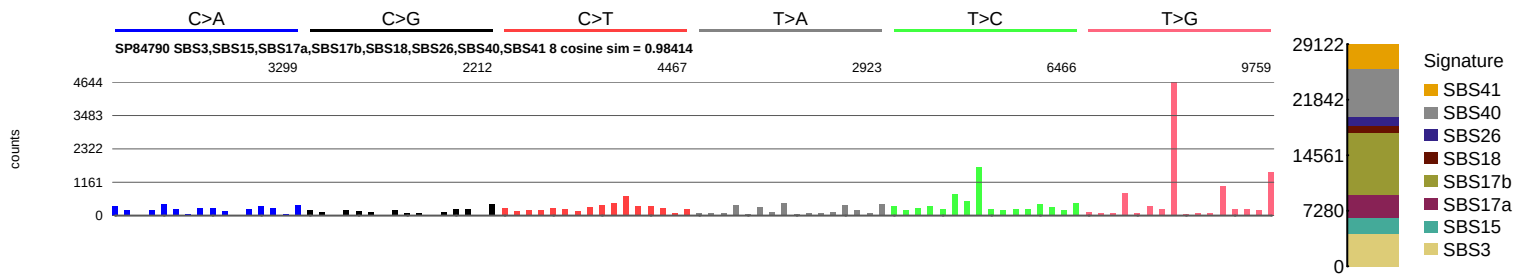

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98429
3296
2321
4434
2877
6471
9726
counts
4644
3483
2322
1161
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3

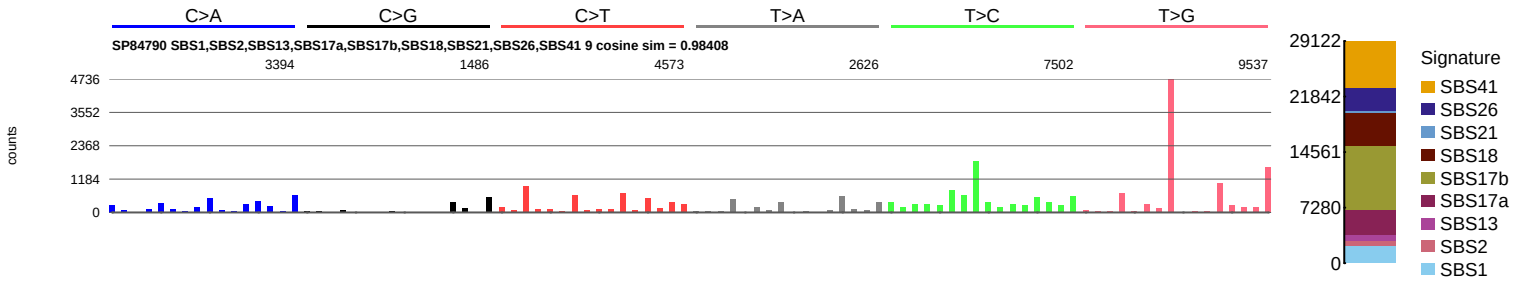

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98425
3249
2282
4412
3092
6491
9590
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2

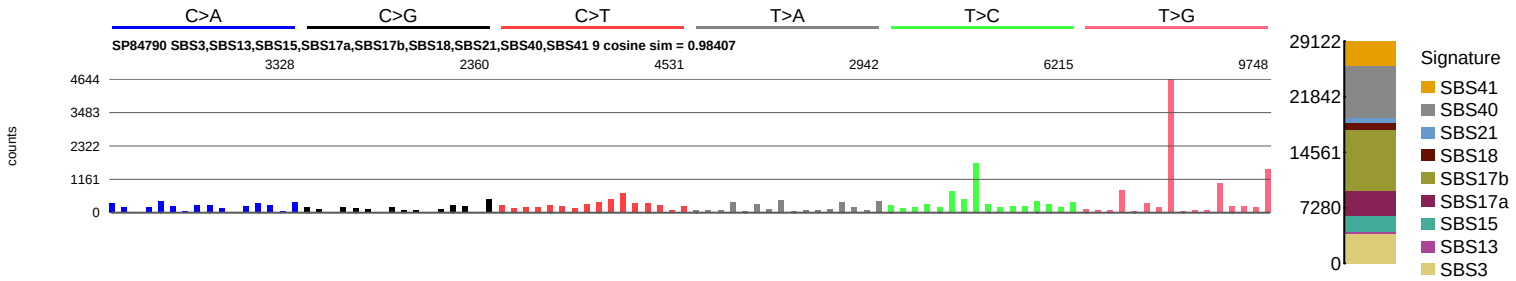

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.98425
3250
2282
4412
3092
6494
9590
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2

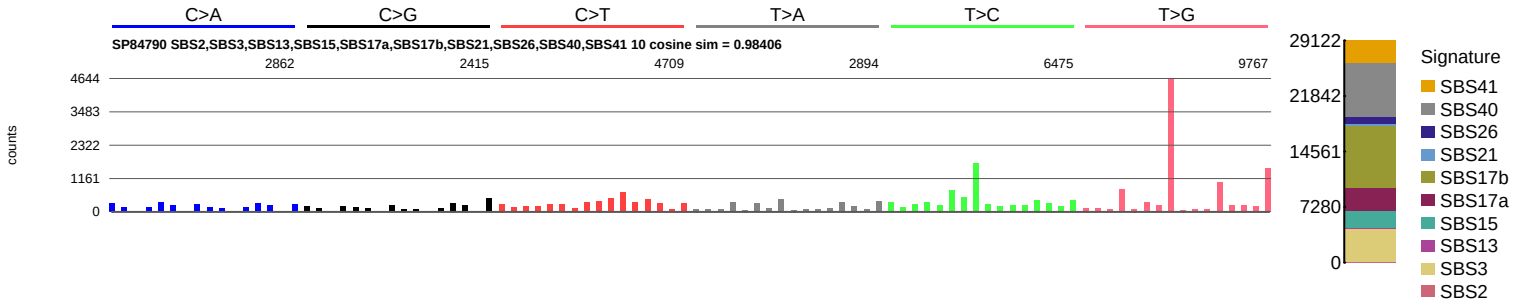

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98423
3294
2285
4611
3035
6184
9707
counts
4648
3486
2324
1162
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2

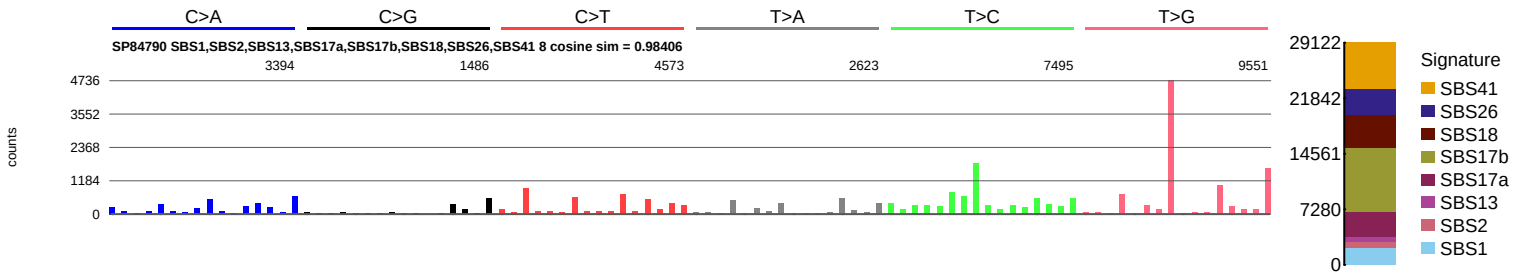

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98418
3300
2206
4478
2917
6473
9753
counts
4640
3480
2320
1160
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3

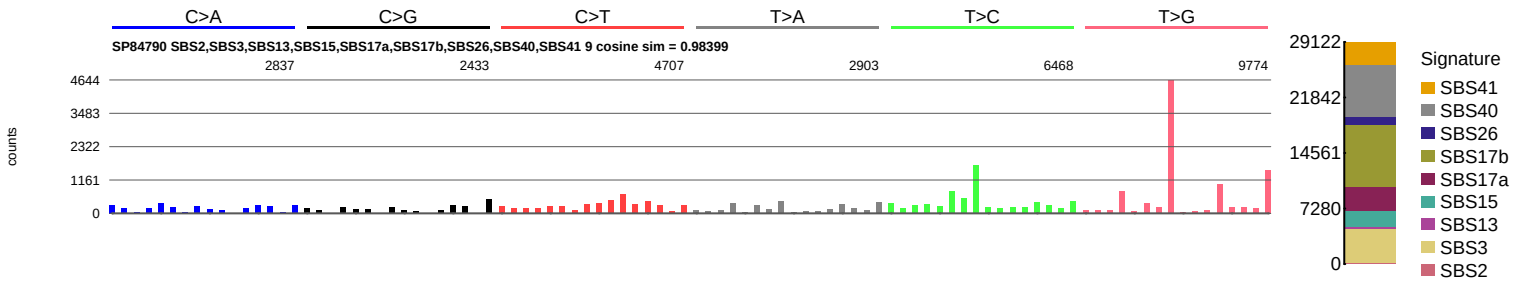

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98414
3299
2212
4467
2923
6466
9759
counts
4644
3483
2322
1161
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3

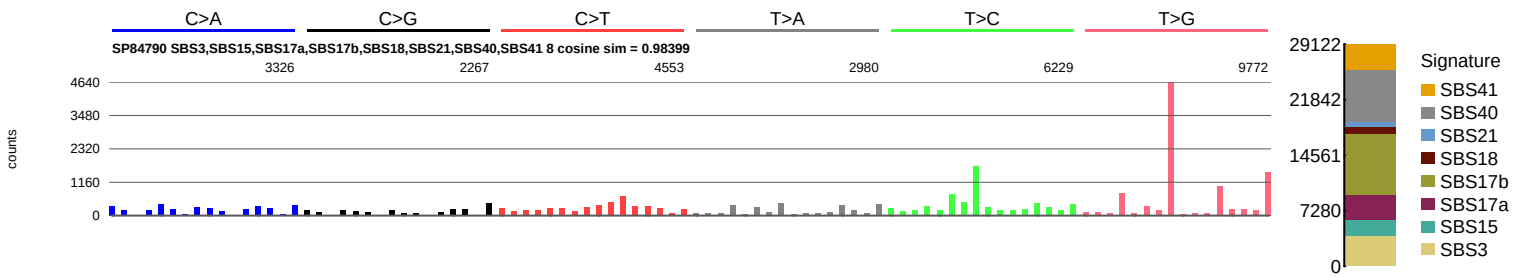

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.98408
3394
1486
4573
2626
7502
9537
counts
4736
3552
2368
1184
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

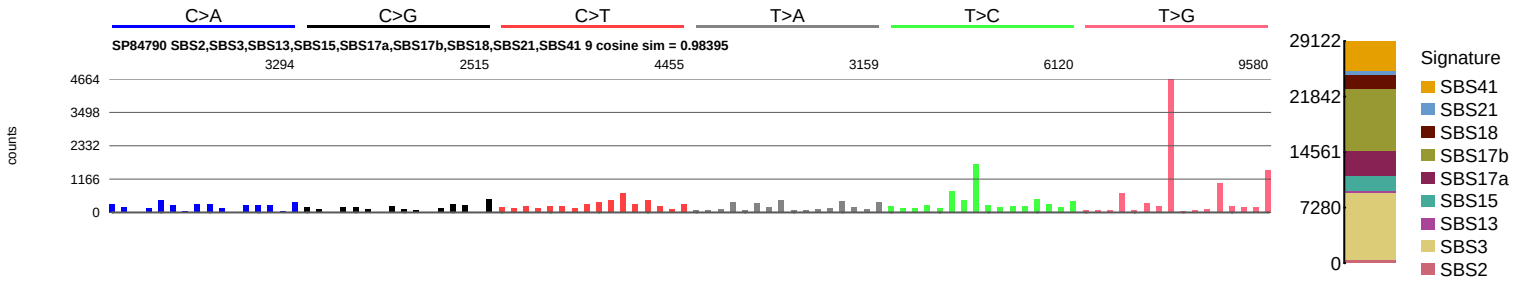

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98407
3328
2360
4531
2942
6215
9748
counts
4644
3483
2322
1161
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3

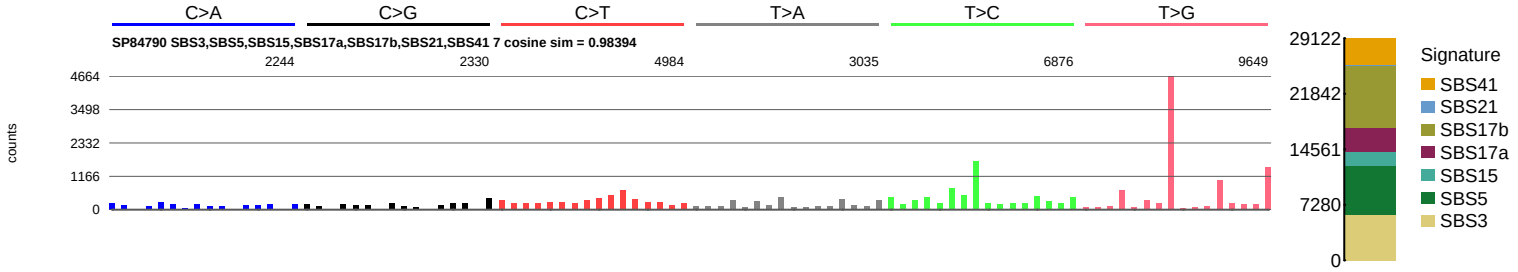

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98406
2862
2415
4709
2894
6475
9767
counts
4644
3483
2322
1161
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2

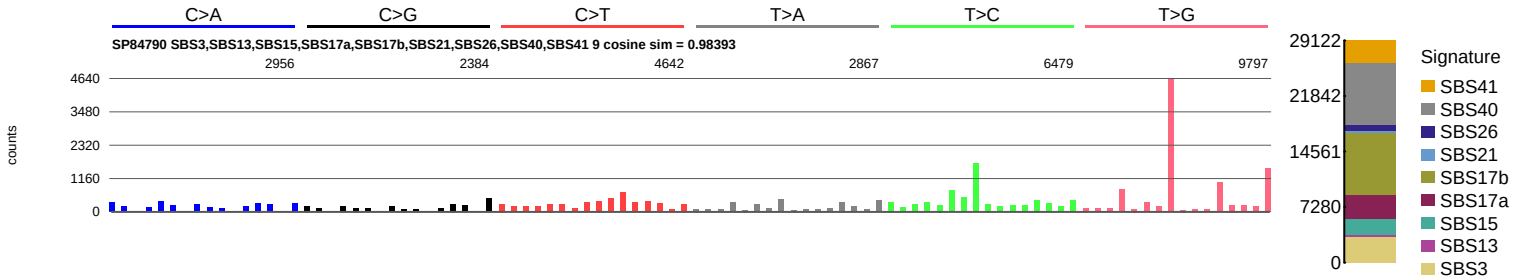

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98406
3394
1486
4573
2623
7495
9551
counts
4736
3552
2368
1184
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

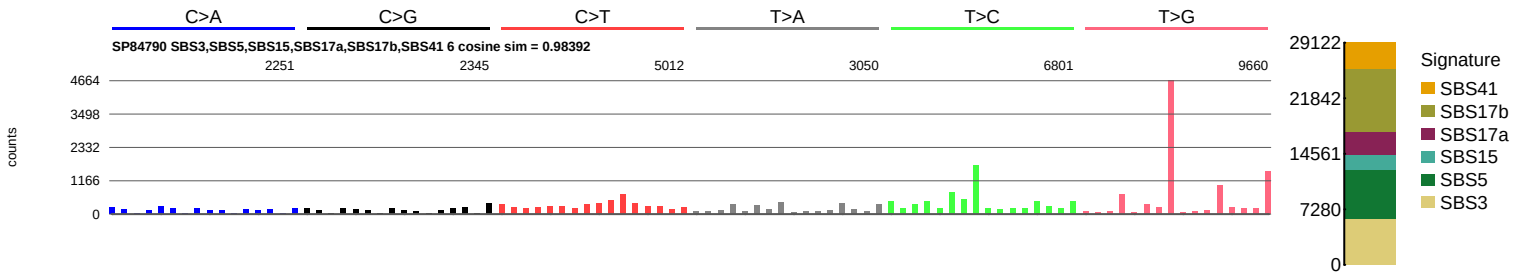

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 9 cosine sim = 0.98399
2837
2433
4707
2903
6468
9774
counts
4644
3483
2322
1161
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2

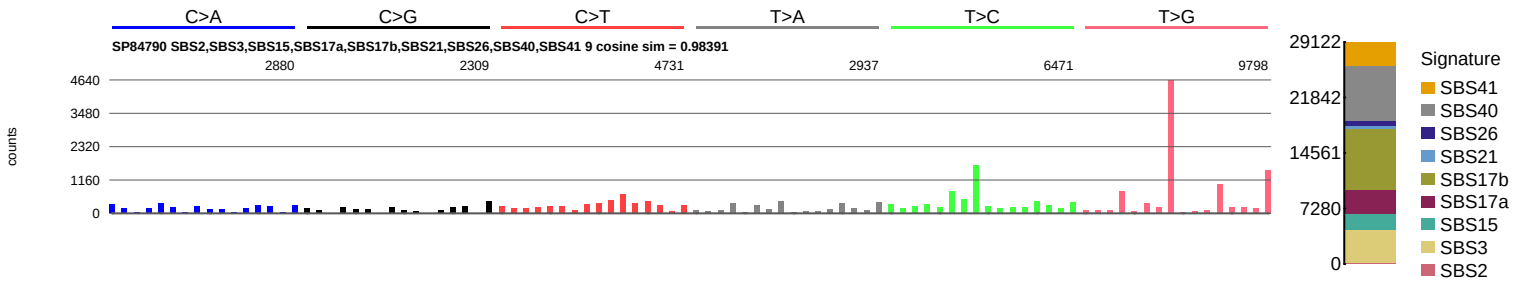

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98399
3326
2267
4553
2980
6229
9772
counts
4640
3480
2320
1160
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3

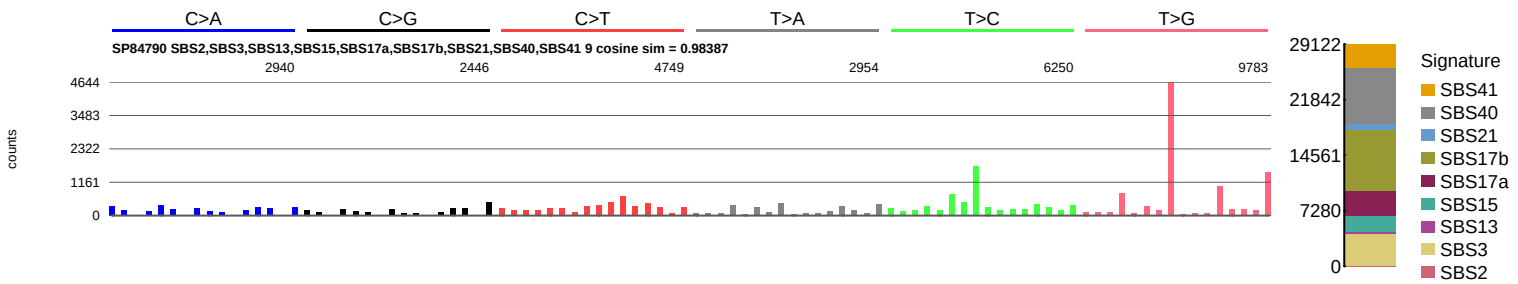

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98395
3294
2515
4455
3159
6120
9580
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2

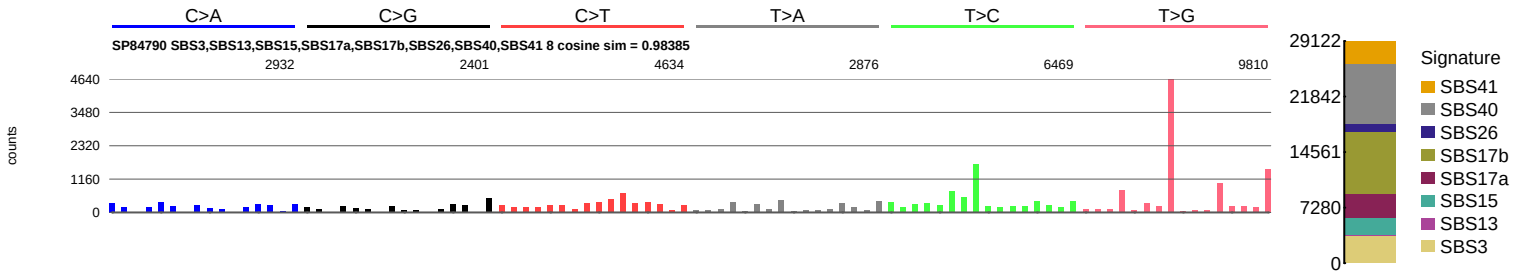

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98394
2244
2330
4984
3035
6876
9649
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3

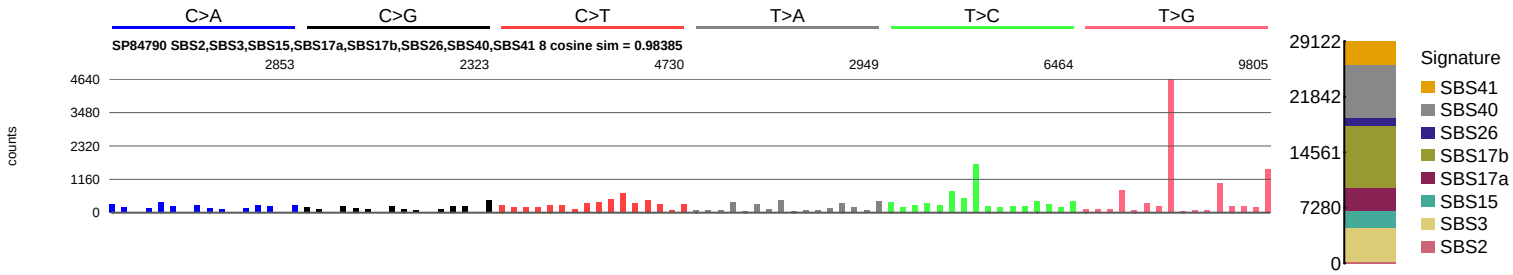

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98393
2956
2384
4642
2867
6479
9797
counts
4640
3480
2320
1160
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3

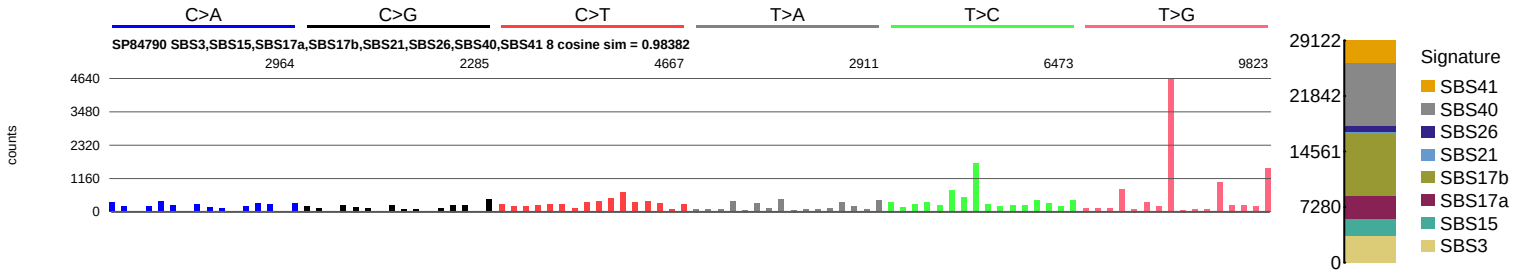

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98392
2251
2345
5012
3050
6801
9660
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS3

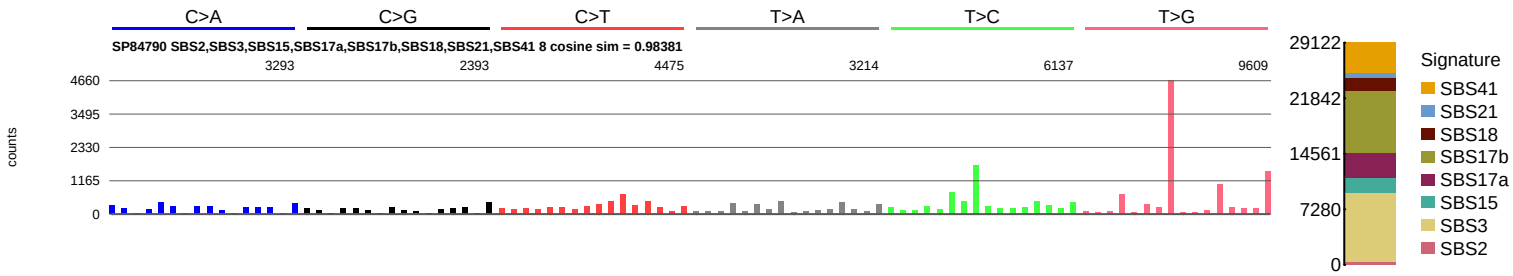

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98391
2880
2309
4731
2937
6471
9798
counts
4640
3480
2320
1160
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2

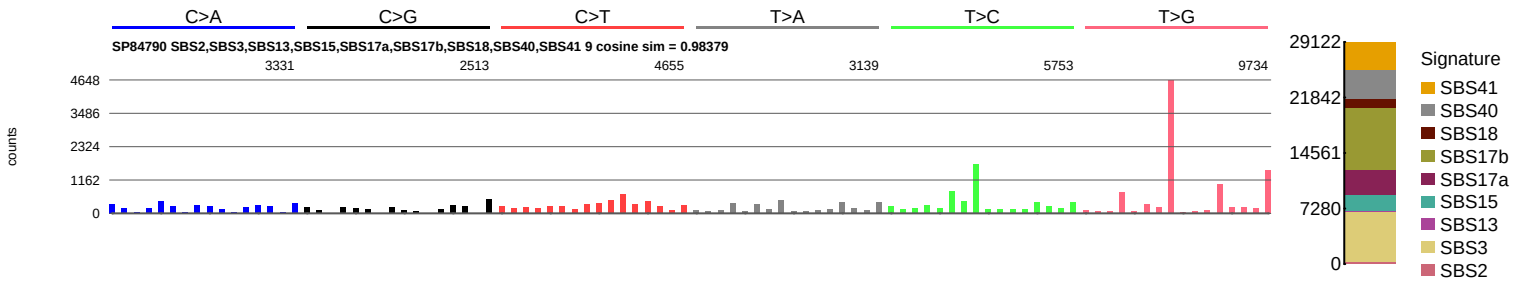

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 9 cosine sim = 0.98387
2940
2446
4749
2954
6250
9783
counts
4644
3483
2322
1161
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2

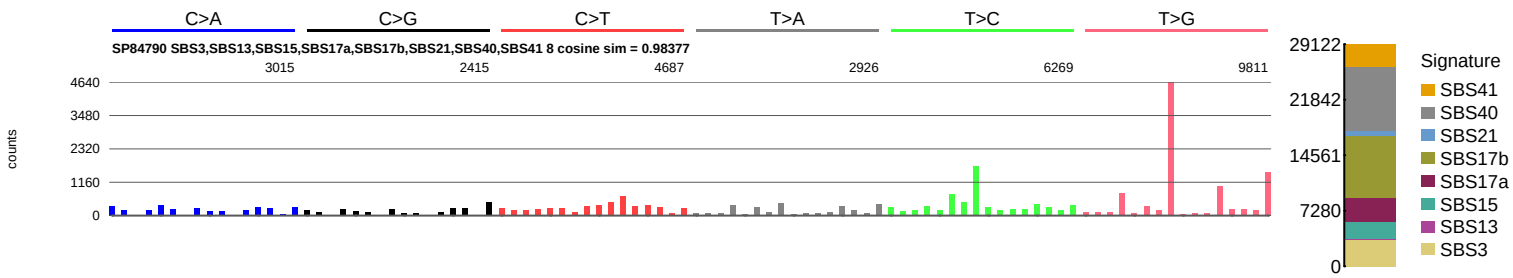

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 8 cosine sim = 0.98385
2932
2401
4634
2876
6469
9810
counts
4640
3480
2320
1160
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3

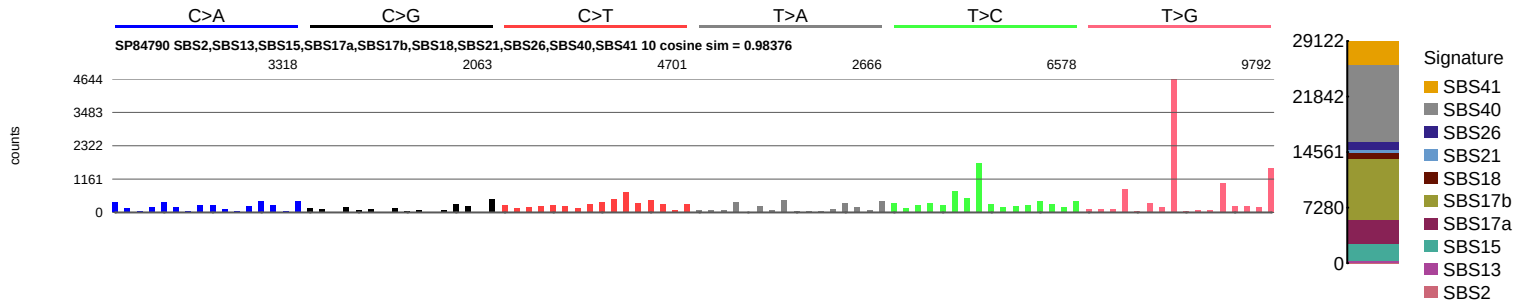

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 8 cosine sim = 0.98385
2853
2323
4730
2949
6464
9805
counts
4640
3480
2320
1160
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS2

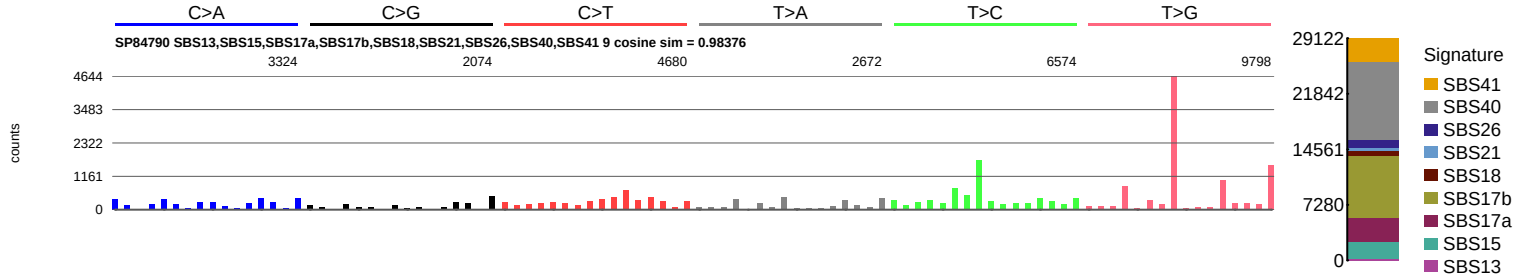

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41 8 cosine sim = 0.98382
2964
2285
4667
2911
6473
9823
counts
4640
3480
2320
1160
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS3

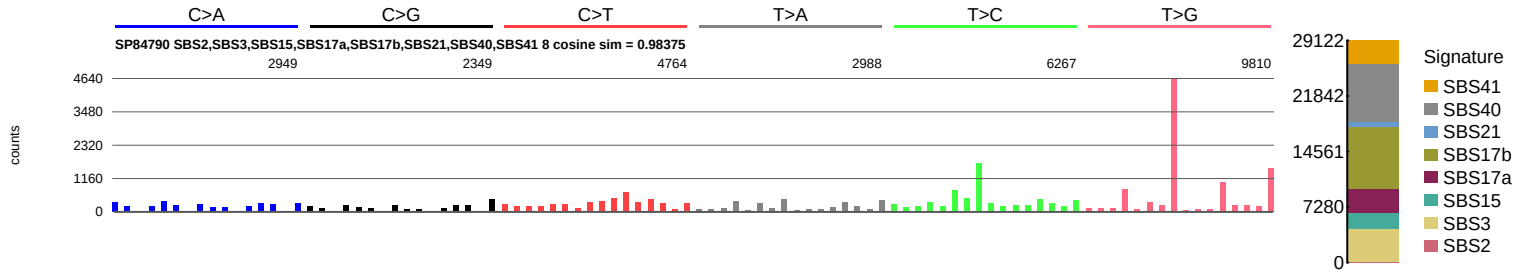

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98381
3293
2393
4475
3214
6137
9609
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2

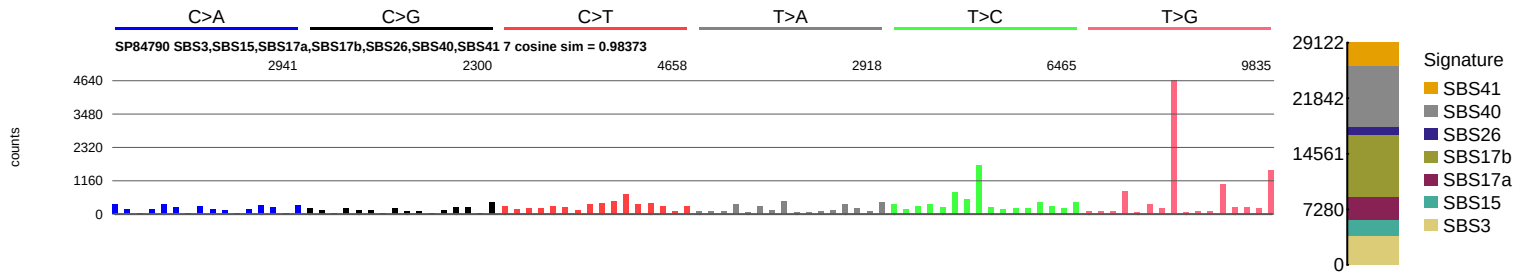

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98379
3331
2513
4655
3139
5753
9734
counts
4648
3486
2324
1162
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2

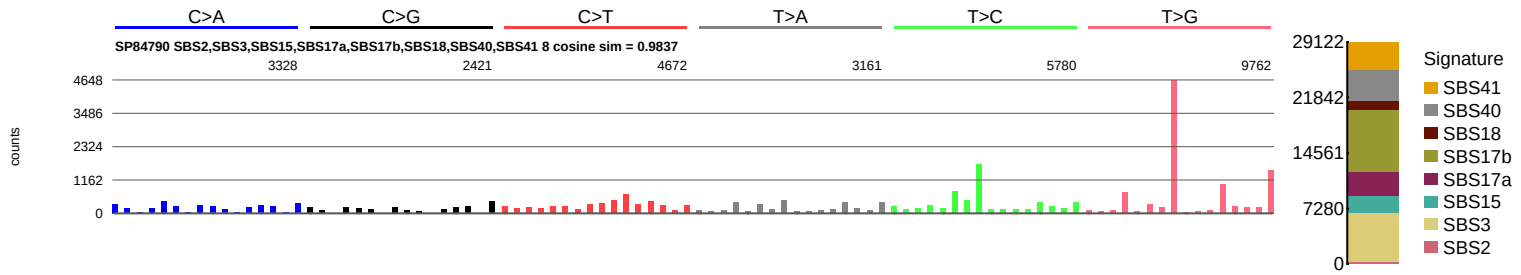

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 8 cosine sim = 0.98377
3015
2415
4687
2926
6269
9811
counts
4640
3480
2320
1160
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3

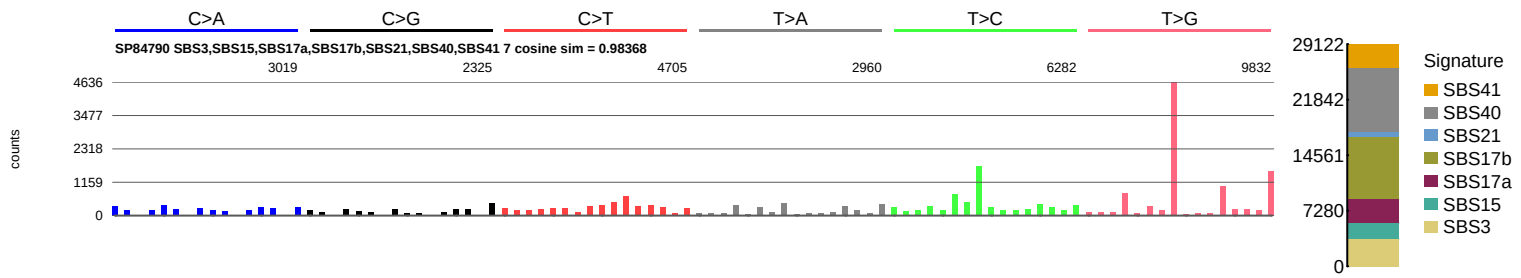

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98376
3318
2063
4701
2666
6578
9792
counts
4644
3483
2322
1161
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2

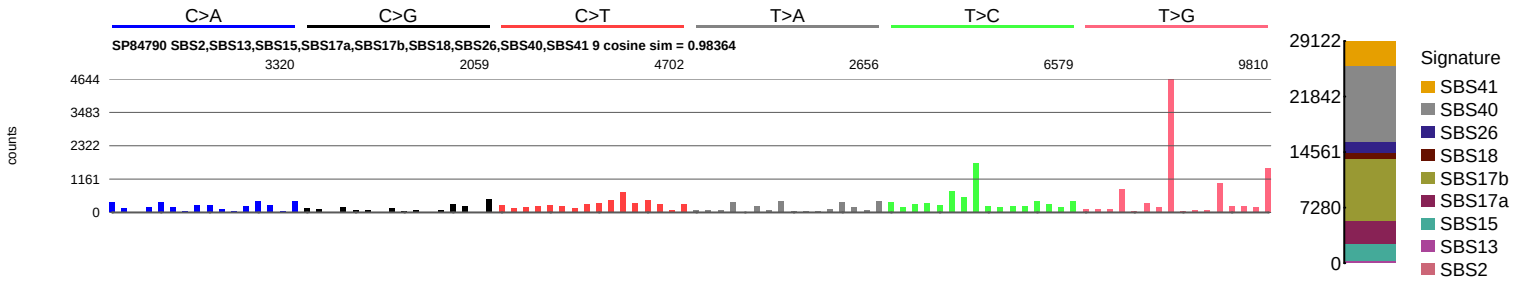

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98376
3324
2074
4680
2672
6574
9798
counts
4644
3483
2322
1161
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13

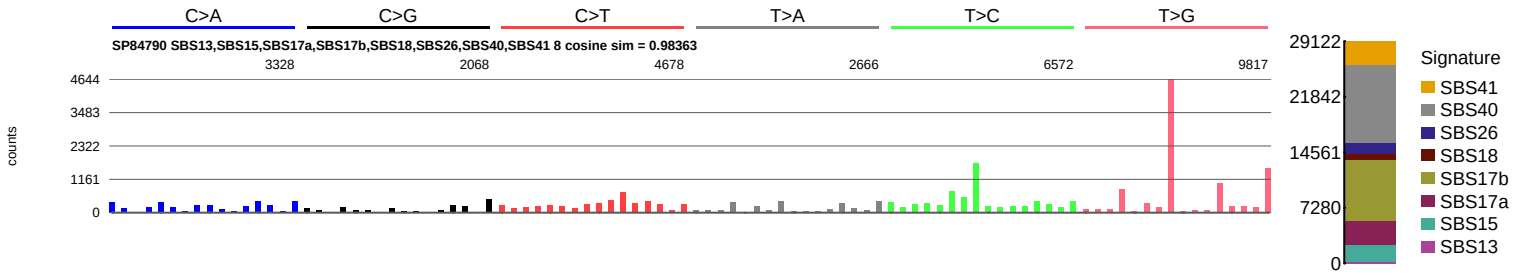

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 8 cosine sim = 0.98375
2949
2349
4764
2988
6267
9810
counts
4640
3480
2320
1160
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2

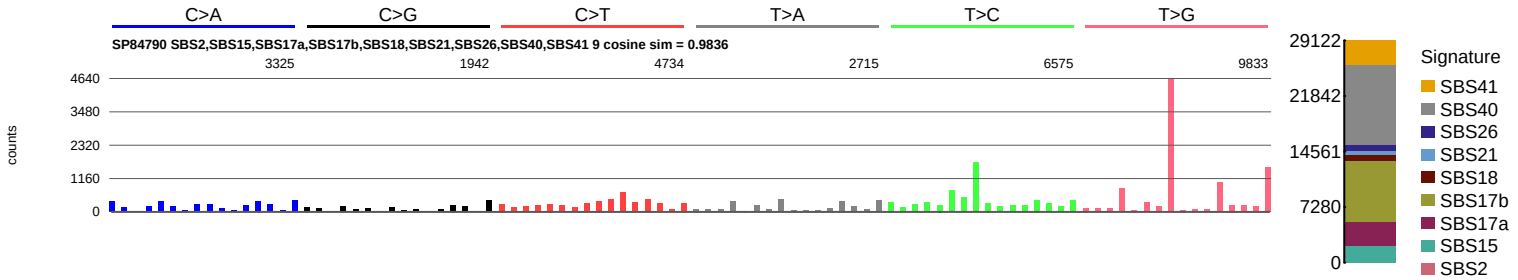

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 7 cosine sim = 0.98373
2941
2300
4658
2918
6465
9835
counts
4640
3480
2320
1160
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS3

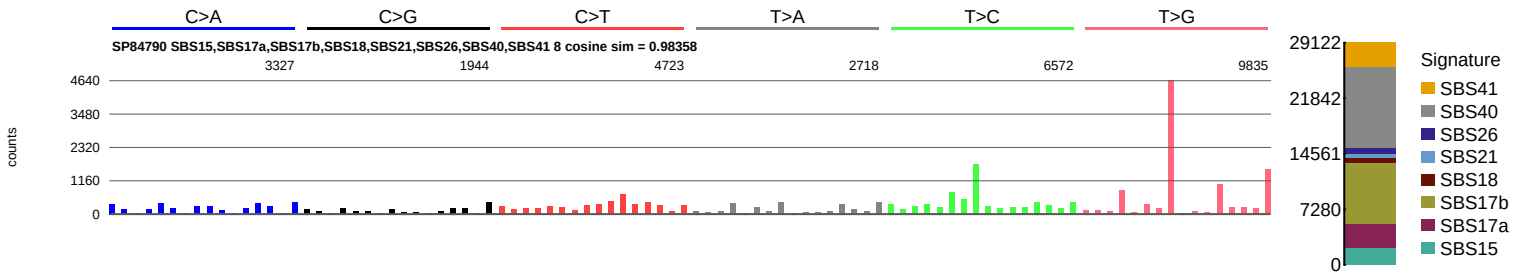

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.9837
3328
2421
4672
3161
5780
9762
counts
4648
3486
2324
1162
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2

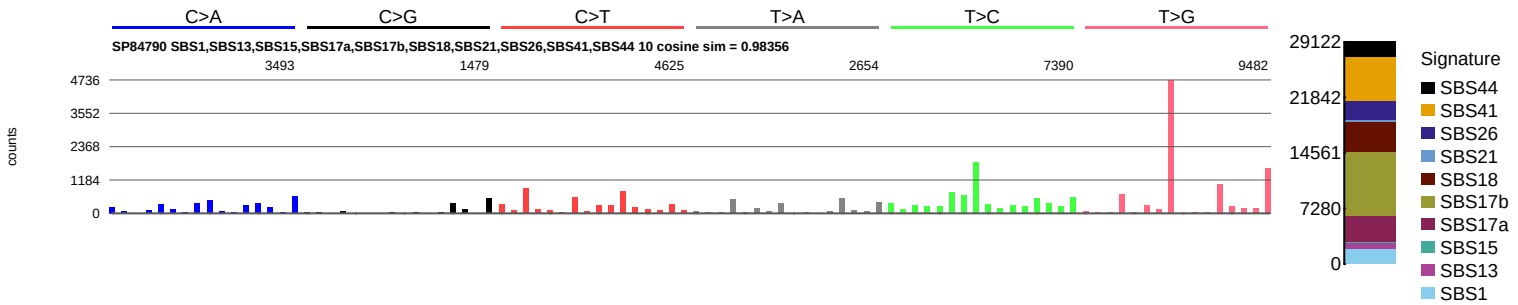

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 7 cosine sim = 0.98368
3019
2325
4705
2960
6282
9832
counts
4636
3477
2318
1159
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS3

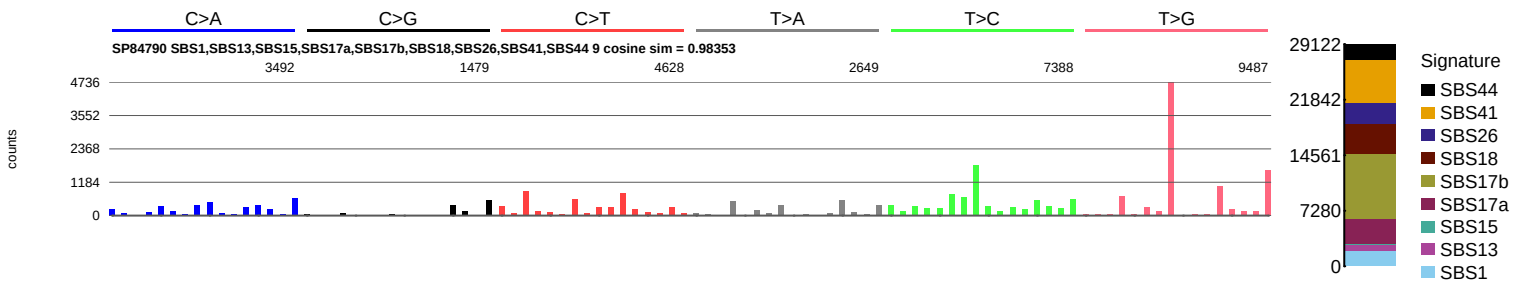

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98364
3320
2059
4702
2656
6579
9810
counts
4644
3483
2322
1161
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2

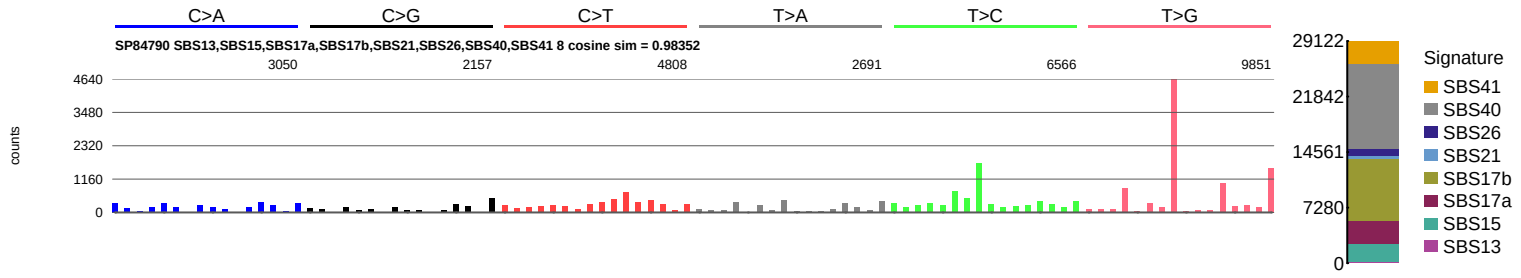

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98363
3328
2068
4678
2666
6572
9817
counts
4644
3483
2322
1161
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13

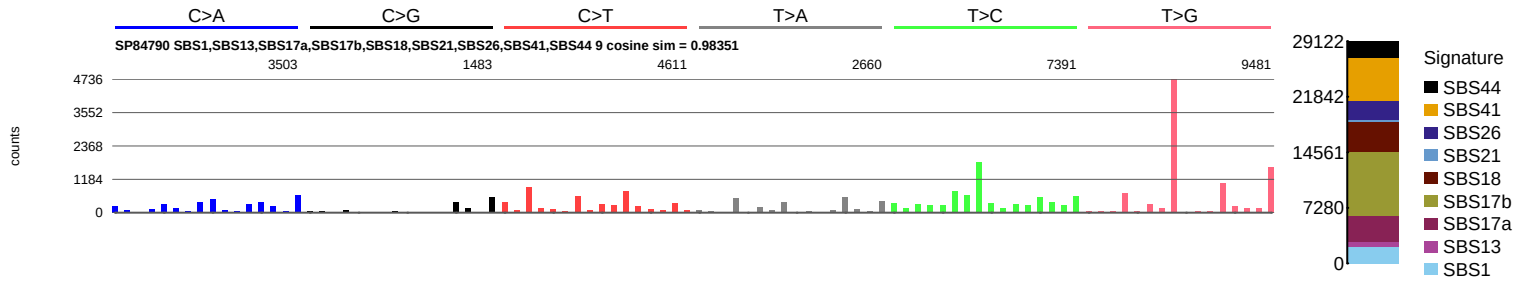

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.9836
3325
1942
4734
2715
6575
9833
counts
4640
3480
2320
1160
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2

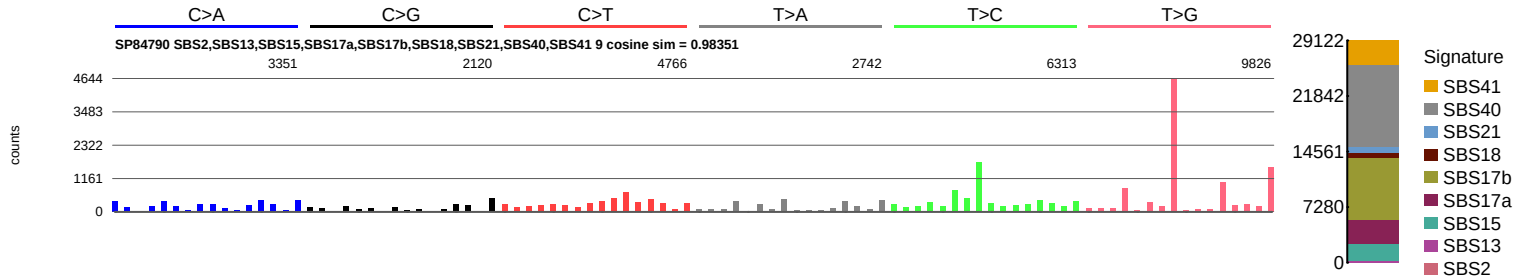

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 8 cosine sim = 0.98358
3327
1944
4723
2718
6572
9835
counts
4640
3480
2320
1160
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15

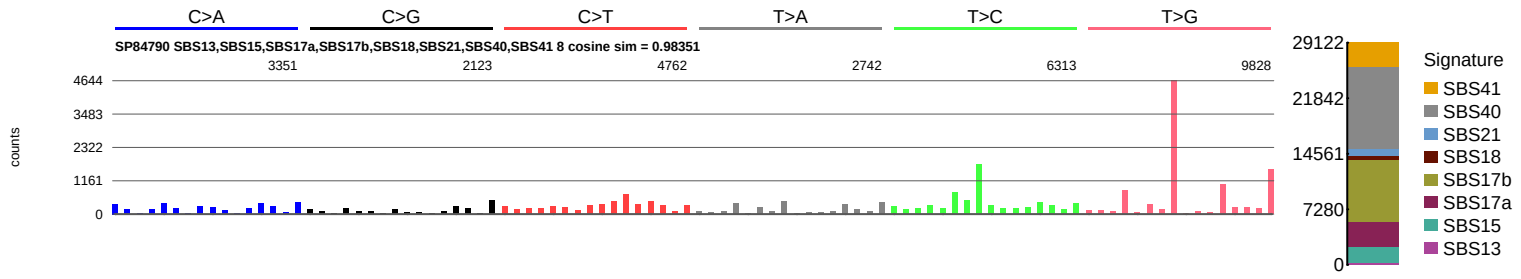

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 10 cosine sim = 0.98356
3493
1479
4625
2654
7390
9482
counts
4736
3552
2368
1184
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1

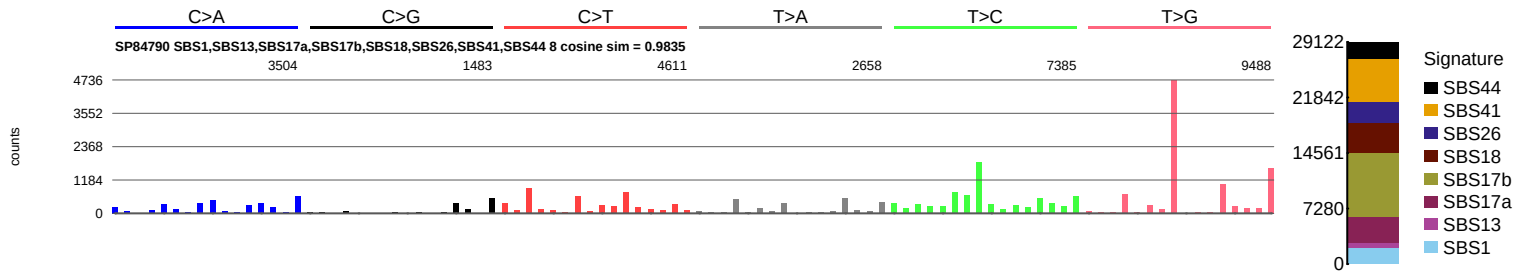

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98353
3492
1479
4628
2649
7388
9487
counts
4736
3552
2368
1184
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1

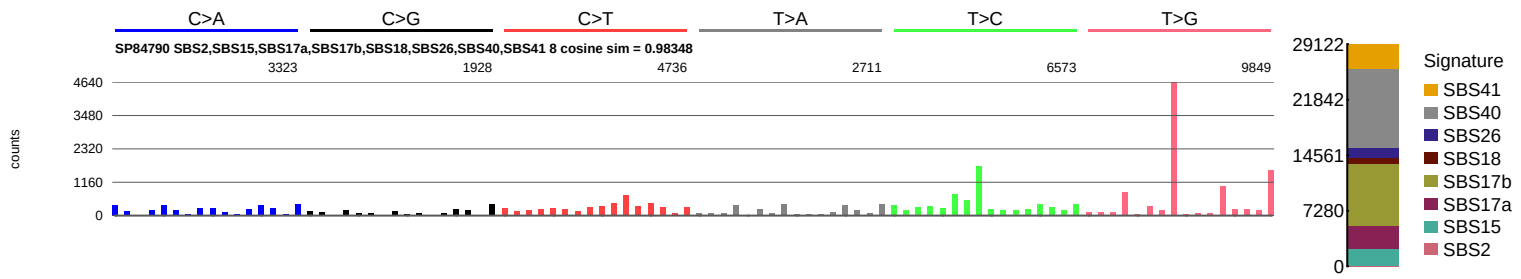

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41 8 cosine sim = 0.98352
3050
2157
4808
2691
6566
9851
counts
4640
3480
2320
1160
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13

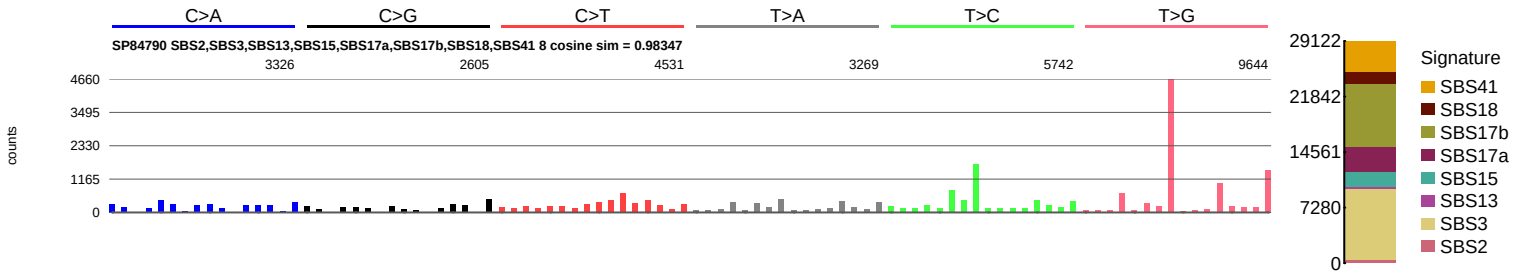

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 9 cosine sim = 0.98351
3503
1483
4611
2660
7391
9481
counts
4736
3552
2368
1184
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS1

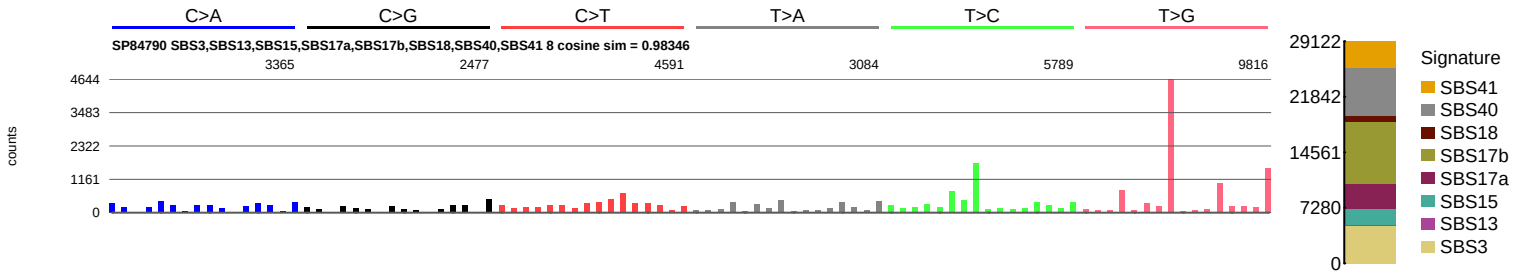

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98351
3351
2120
4766
2742
6313
9826
counts
4644
3483
2322
1161
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2

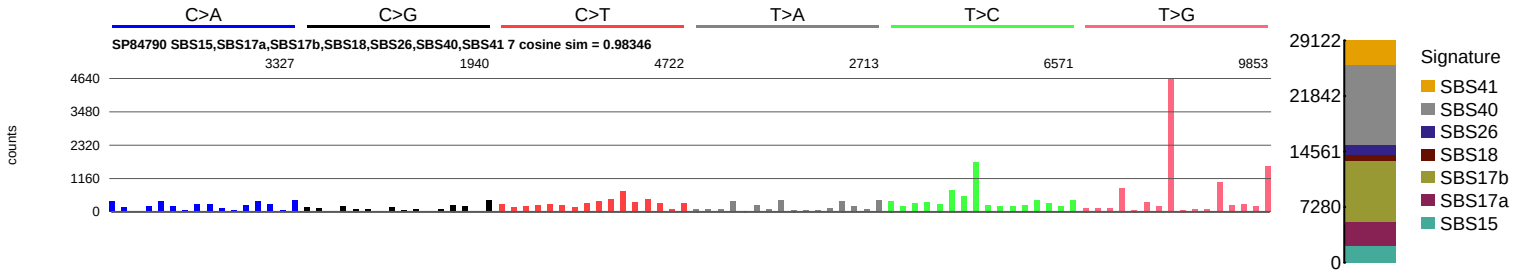

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98351
3351
2123
4762
2742
6313
9828
counts
4644
3483
2322
1161
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13

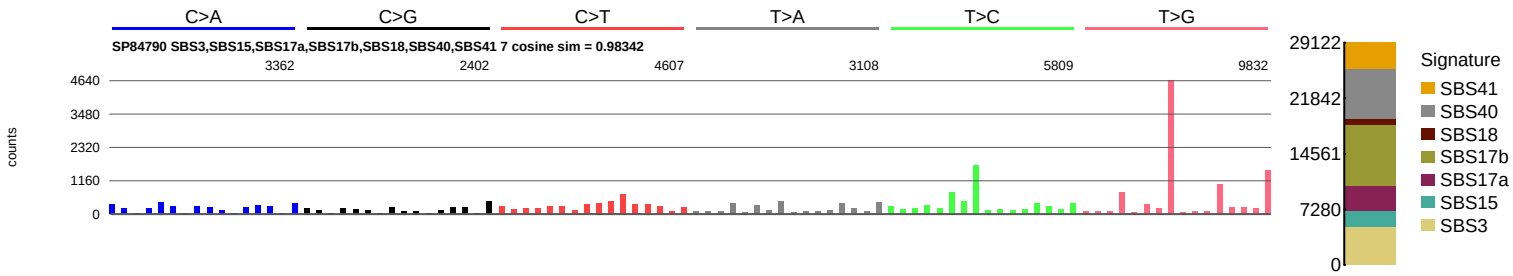

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.9835
3504
1483
4611
2658
7385
9488
counts
4736
3552
2368
1184
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS1

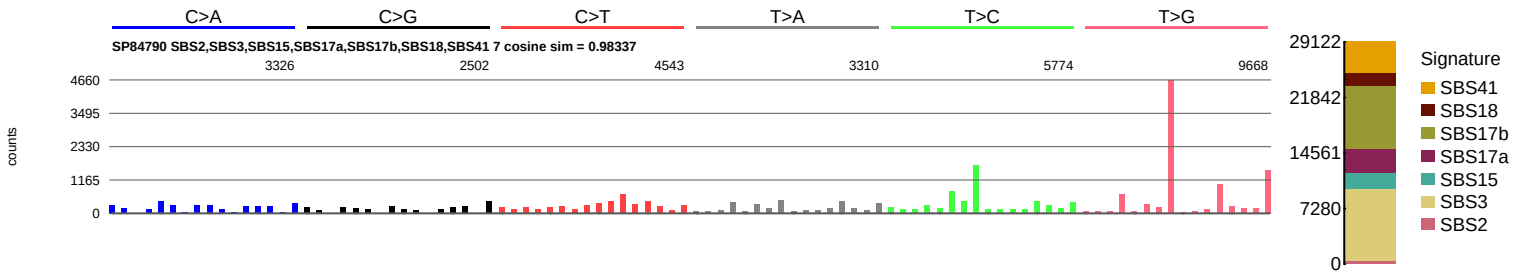

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98348
3323
1928
4736
2711
6573
9849
counts
4640
3480
2320
1160
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2

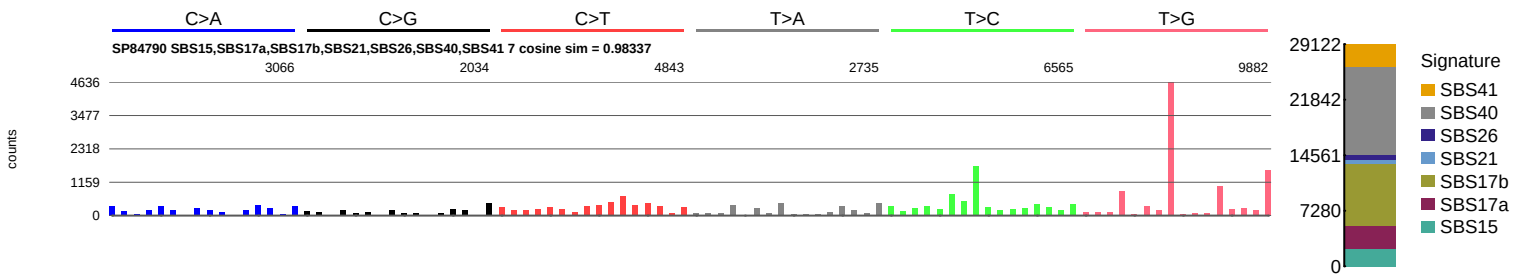

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98347
3326
2605
4531
3269
5742
9644
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2

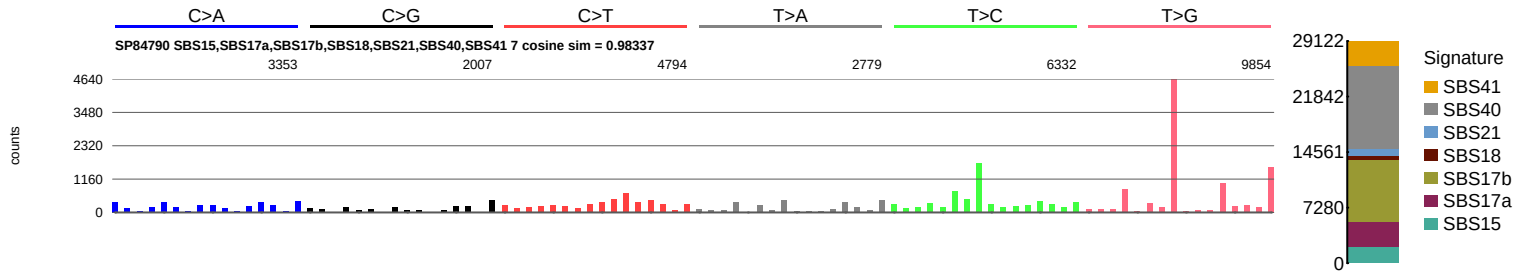

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98346
3365
2477
4591
3084
5789
9816
counts
4644
3483
2322
1161
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3

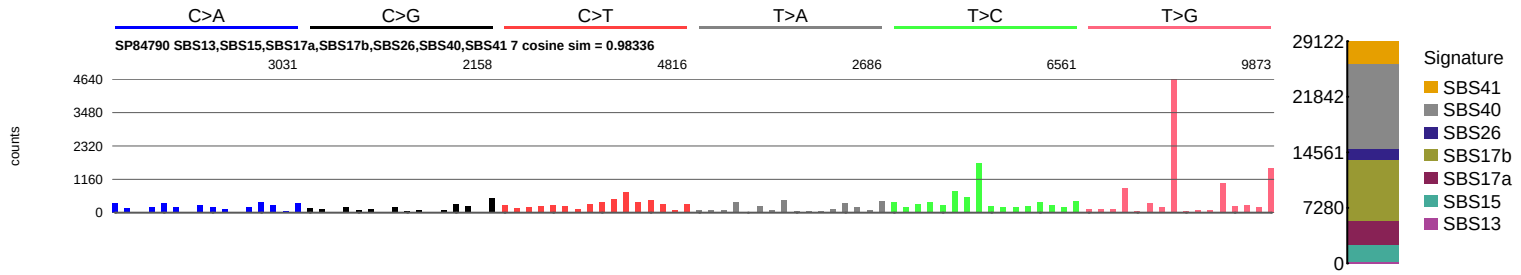

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.98346
3327
1940
4722
2713
6571
9853
counts
4640
3480
2320
1160
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15

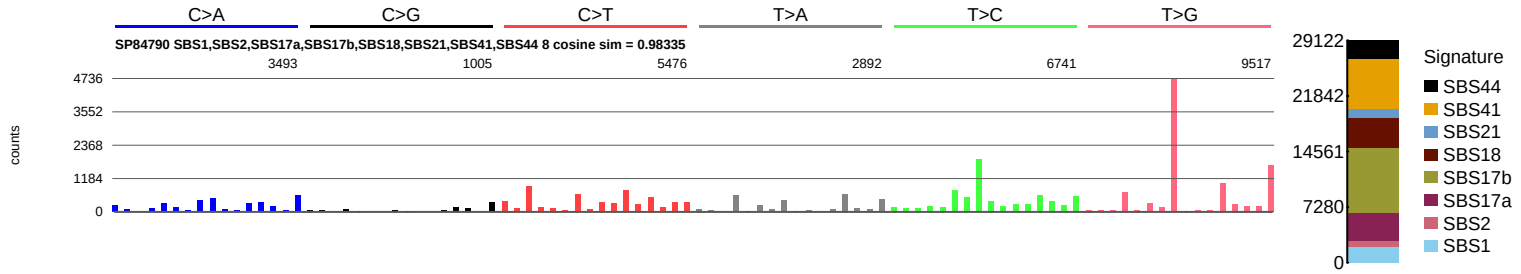

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98342
3362
2402
4607
3108
5809
9832
counts
4640
3480
2320
1160
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS3

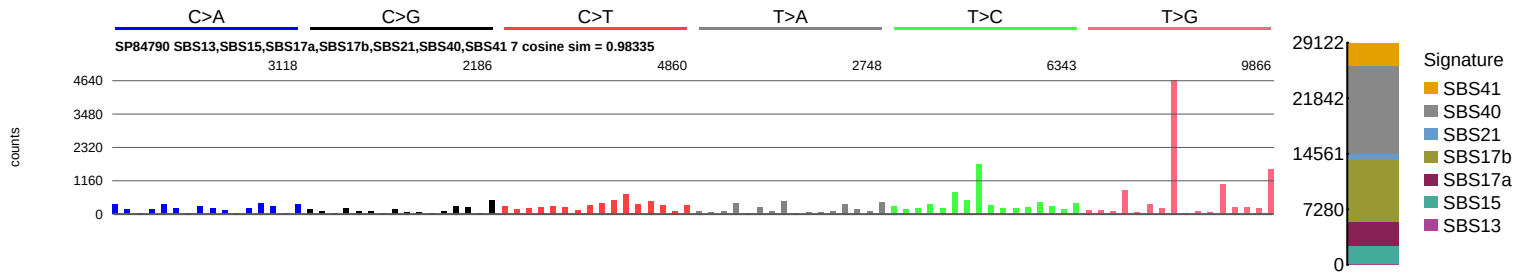

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98337
3326
2502
4543
3310
5774
9668
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2

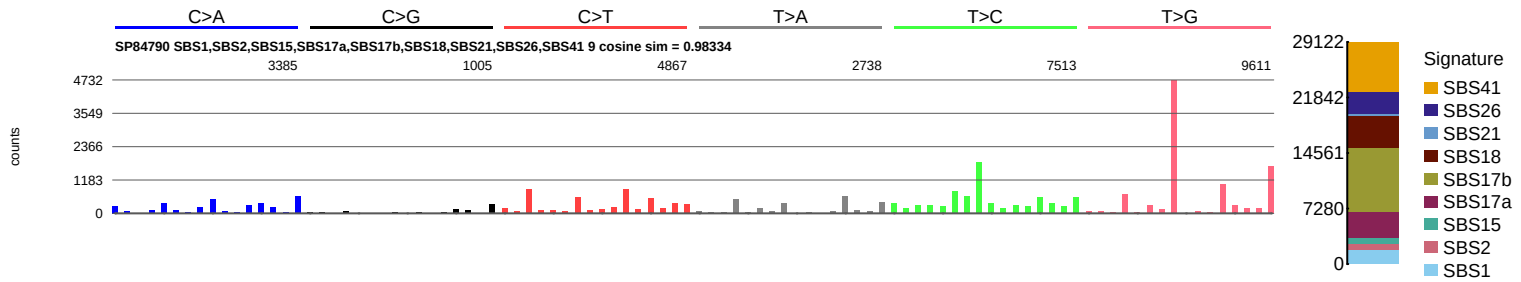

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41 7 cosine sim = 0.98337
3066
2034
4843
2735
6565
9882
counts
4636
3477
2318
1159
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15

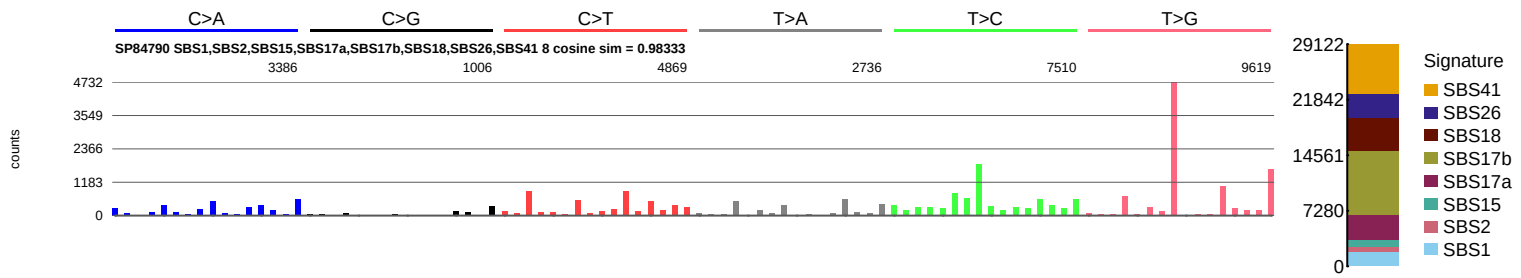

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.98337
3353
2007
4794
2779
6332
9854
counts
4640
3480
2320
1160
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15

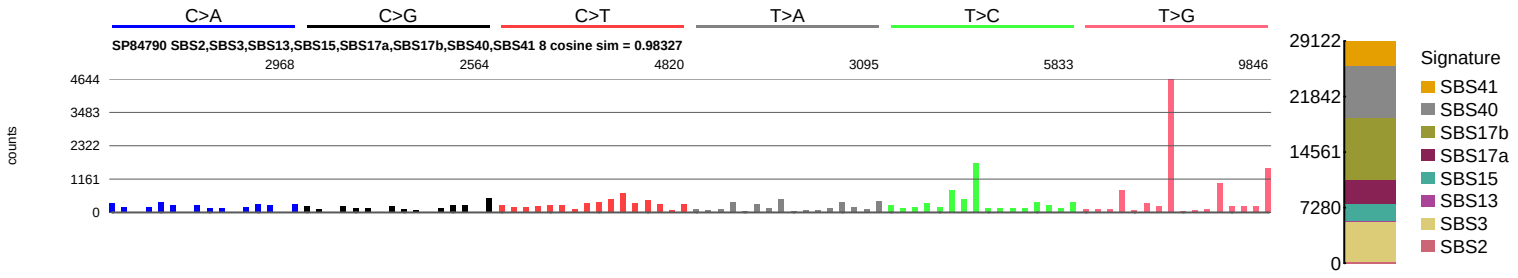

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 7 cosine sim = 0.98336
3031
2158
4816
2686
6561
9873
counts
4640
3480
2320
1160
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13

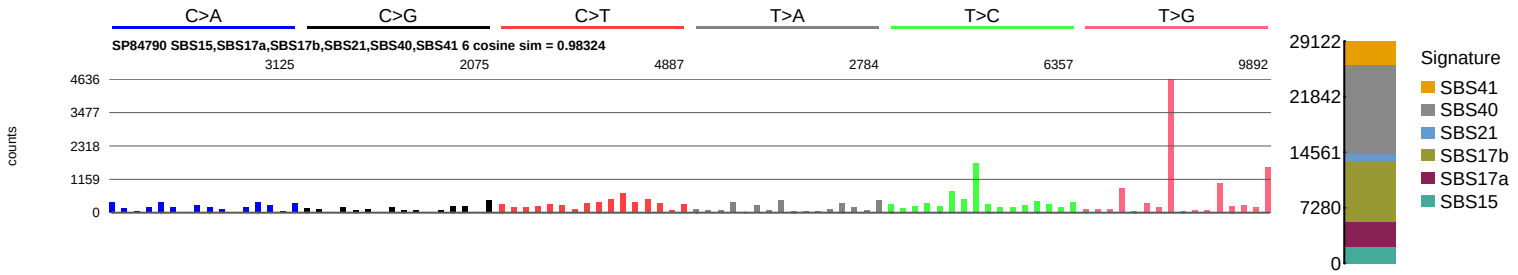

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98335
3493
1005
5476
2892
6741
9517
counts
4736
3552
2368
1184
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1

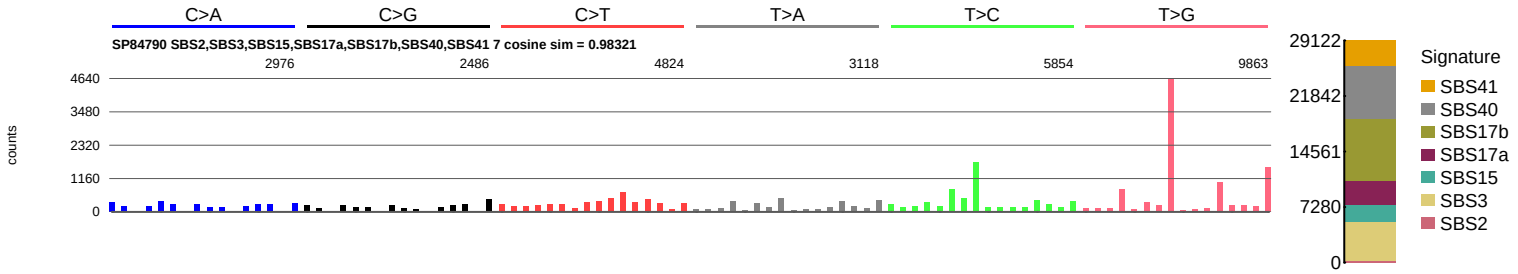

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 7 cosine sim = 0.98335
3118
2186
4860
2748
6343
9866
counts
4640
3480
2320
1160
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13

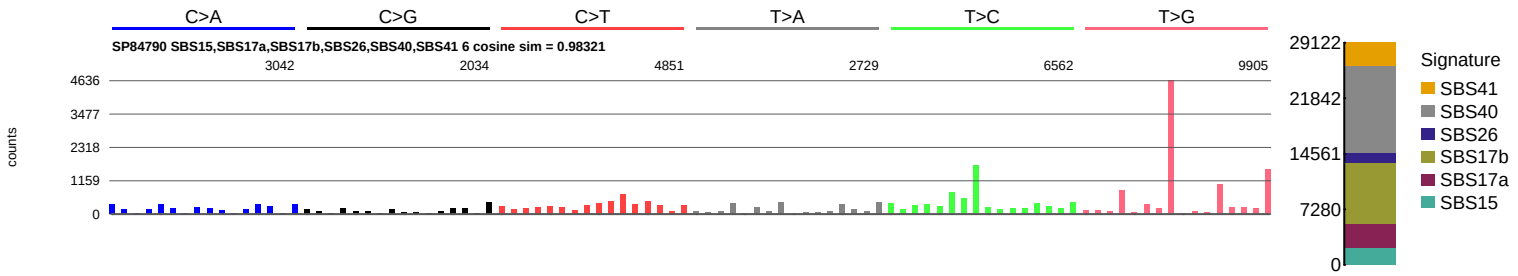

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.98334
3385
1005
4867
2738
7513
9611
counts
4732
3549
2366
1183
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

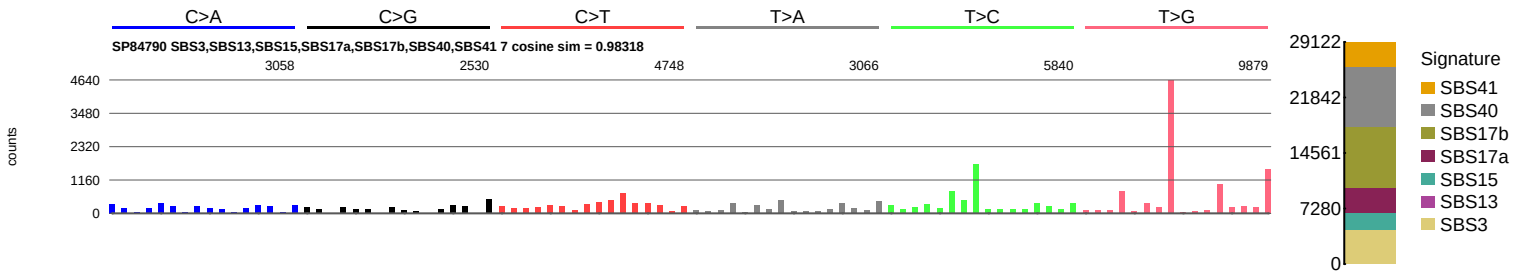

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98333
3386
1006
4869
2736
7510
9619
counts
4732
3549
2366
1183
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

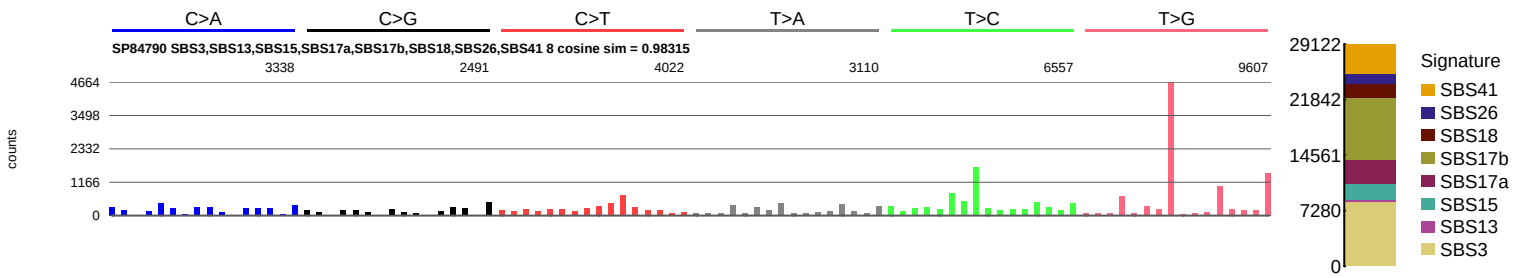

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41 8 cosine sim = 0.98327
2968
2564
4820
3095
5833
9846
counts
4644
3483
2322
1161
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2

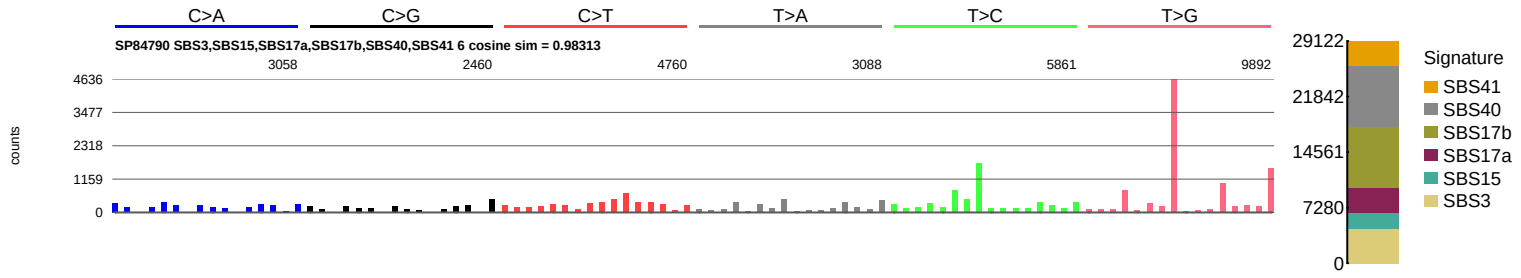

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 6 cosine sim = 0.98324
3125
2075
4887
2784
6357
9892
counts
4636
3477
2318
1159
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15

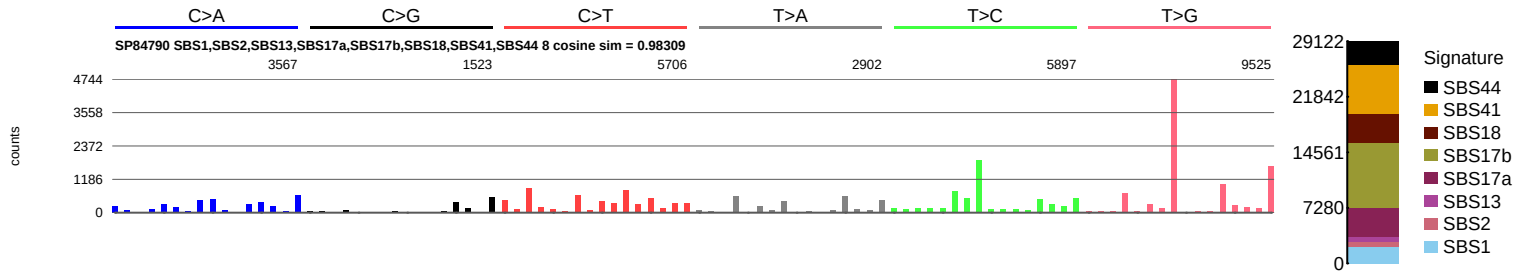

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.98321
2976
2486
4824
3118
5854
9863
counts
4640
3480
2320
1160
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS3
SBS2

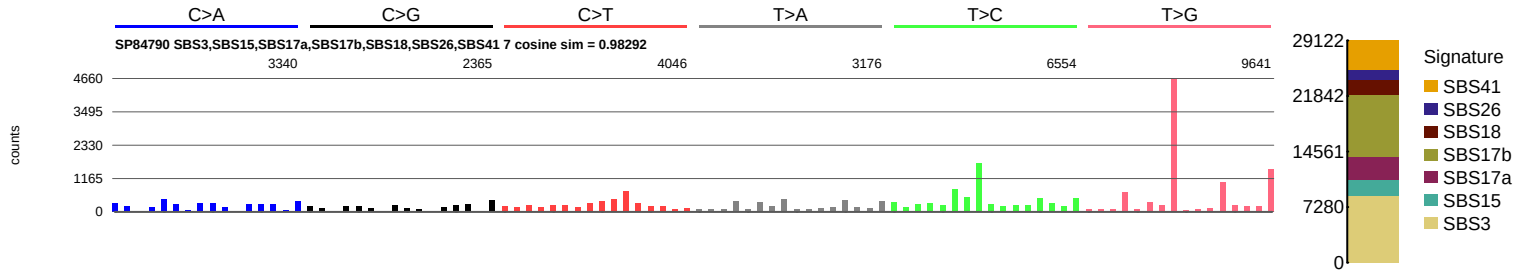

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 6 cosine sim = 0.98321
3042
2034
4851
2729
6562
9905
counts
4636
3477
2318
1159
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15

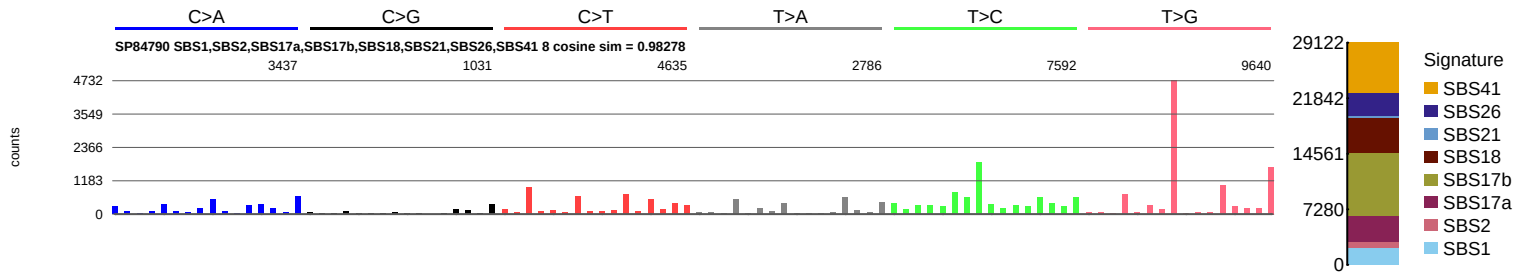

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.98318
3058
2530
4748
3066
5840
9879
counts
4640
3480
2320
1160
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS3

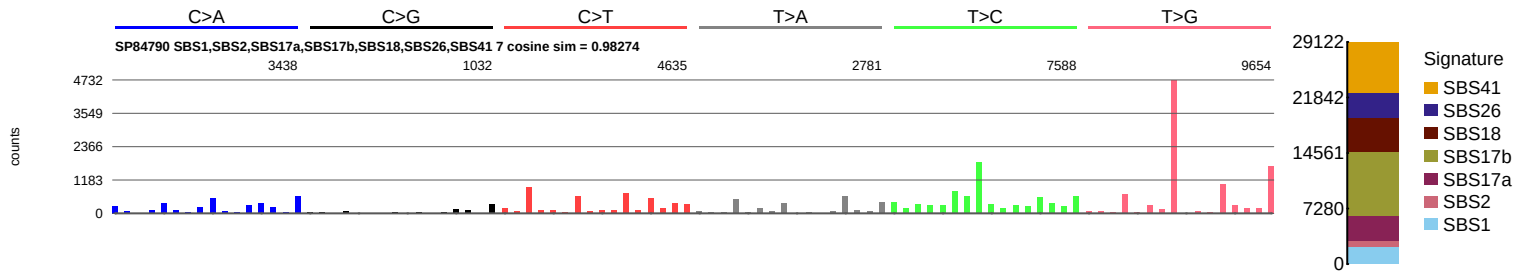

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98315
3338
2491
4022
3110
6557
9607
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3

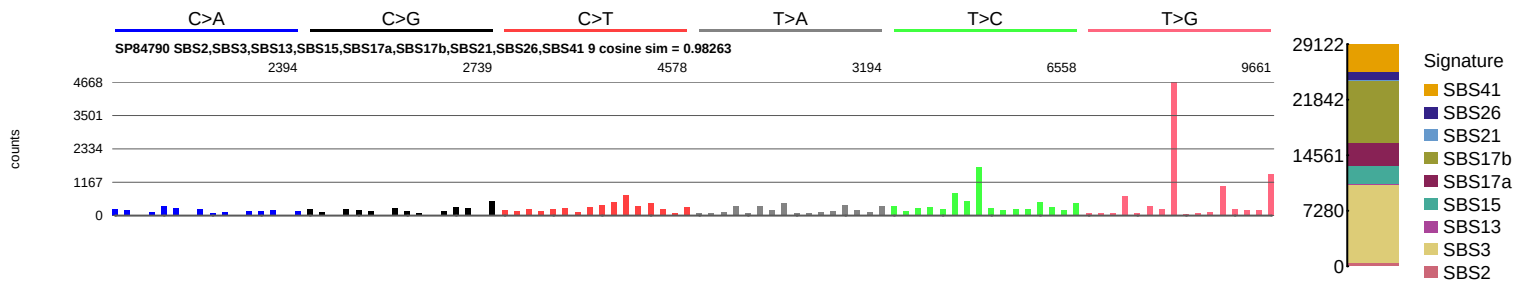

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS15,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.98313
3058
2460
4760
3088
5861
9892
counts
4636
3477
2318
1159
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS3

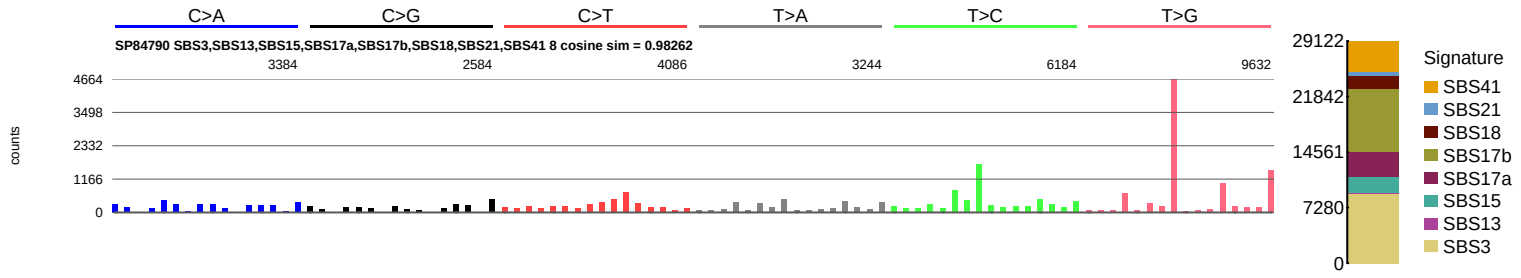

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98309
3567
1523
5706
2902
5897
9525
counts
4744
3558
2372
1186
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

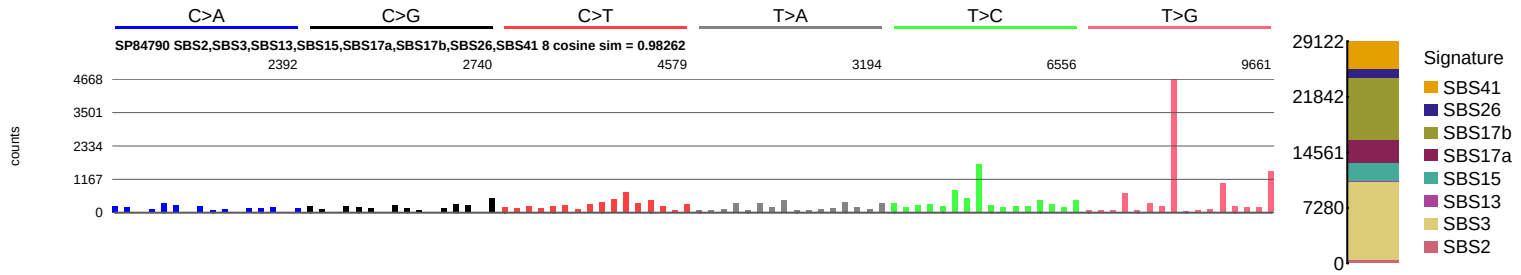

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.98292
3340
2365
4046
3176
6554
9641
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3

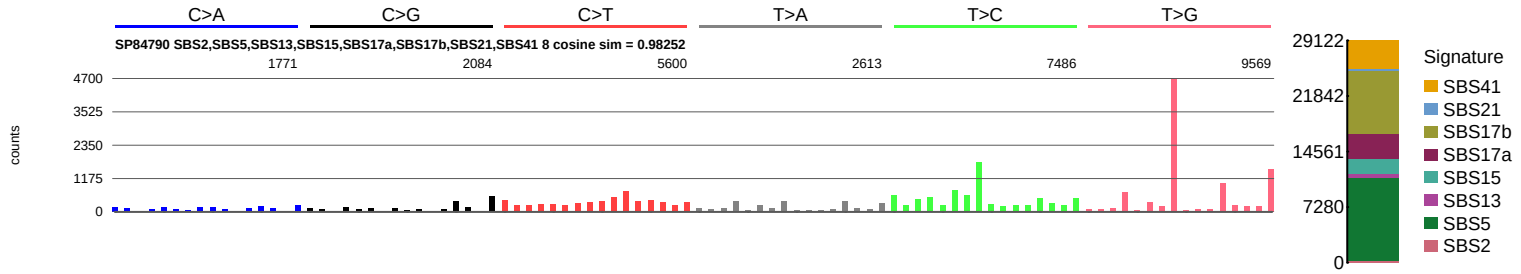

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 8 cosine sim = 0.98278
3437
1031
4635
2786
7592
9640
counts
4732
3549
2366
1183
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1

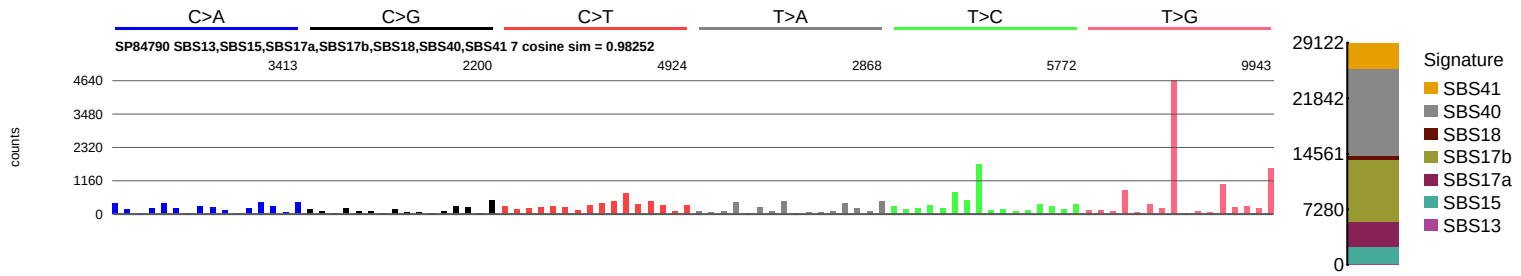

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.98274
3438
1032
4635
2781
7588
9654
counts
4732
3549
2366
1183
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1

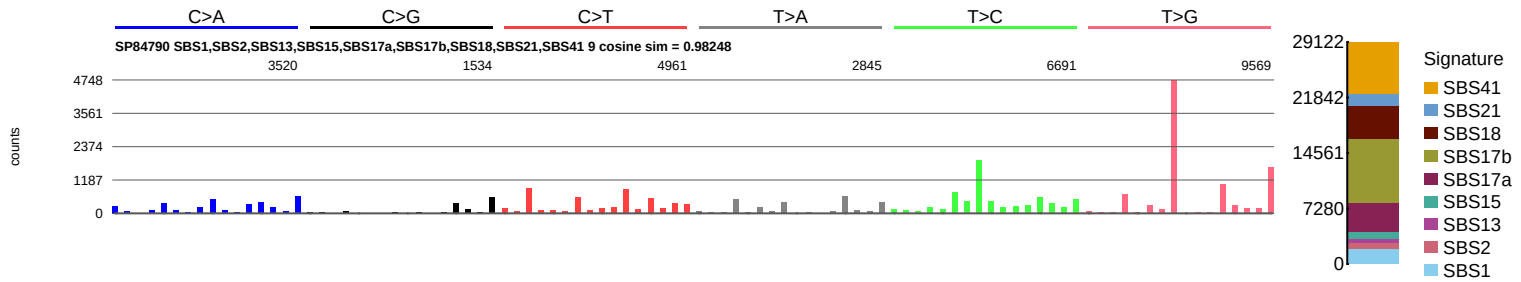

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41 9 cosine sim = 0.98263
2394
2739
4578
3194
6558
9661
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2

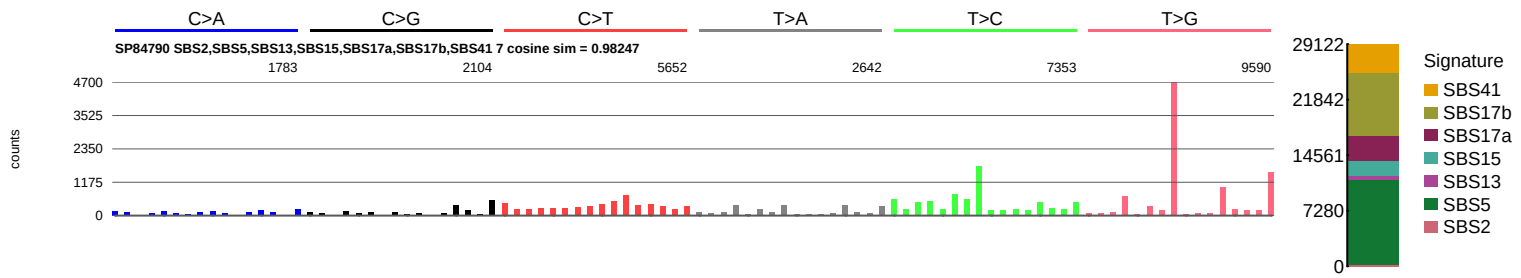

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98262
3384
2584
4086
3244
6184
9632
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3

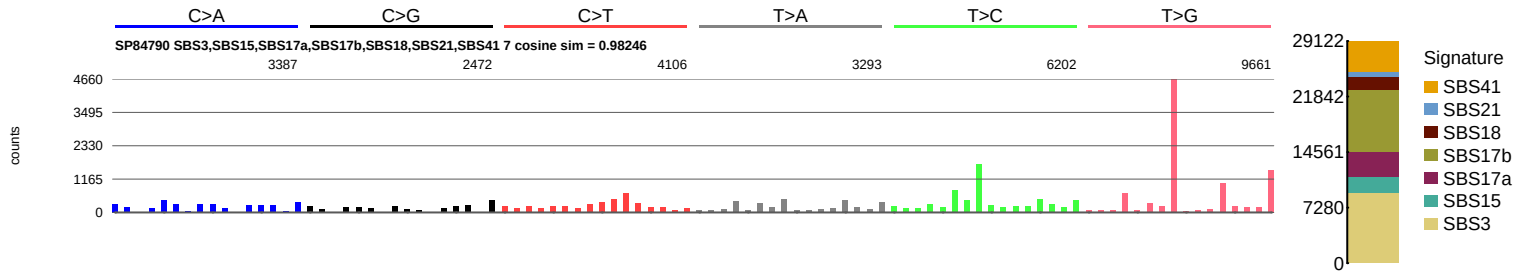

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41 8 cosine sim = 0.98262
2392
2740
4579
3194
6556
9661
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2

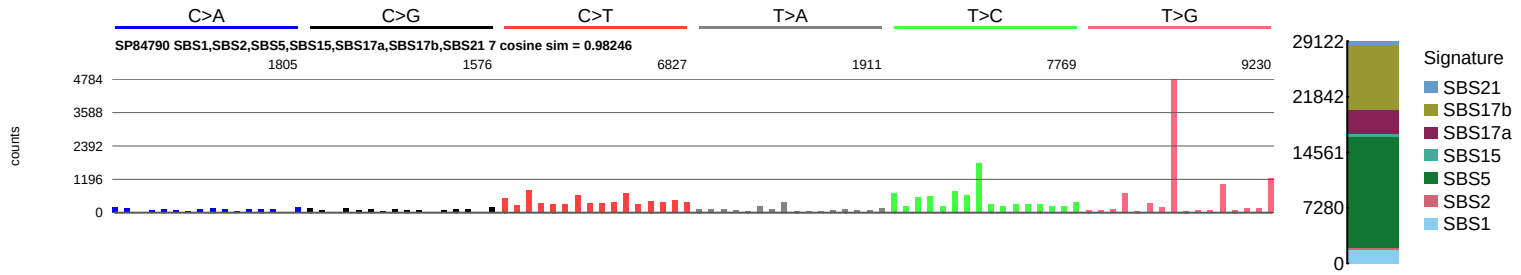

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.98252
1771
2084
5600
2613
7486
9569
counts
4700
3525
2350
1175
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2

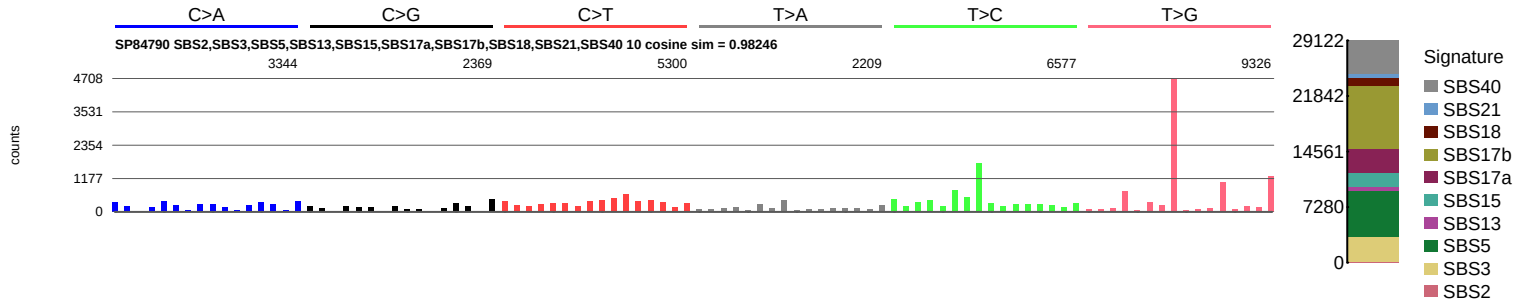

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98252
3413
2200
4924
2868
5772
9943
counts
4640
3480
2320
1160
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13

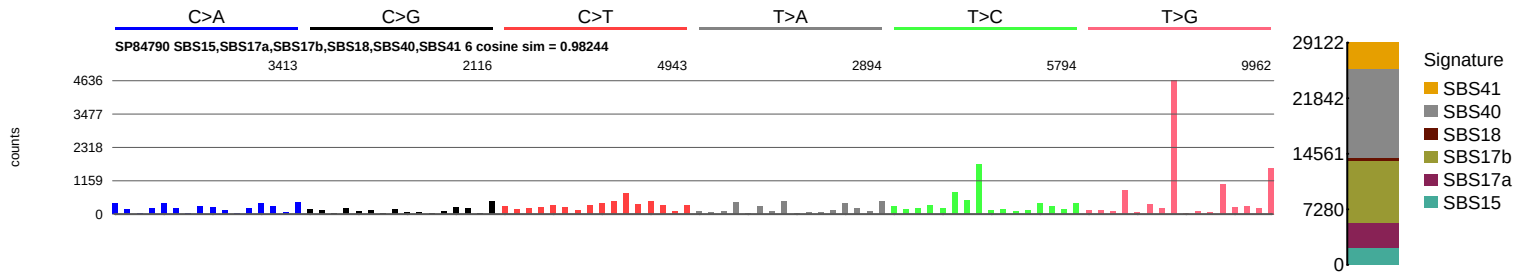

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98248
3520
1534
4961
2845
6691
9569
counts
4748
3561
2374
1187
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

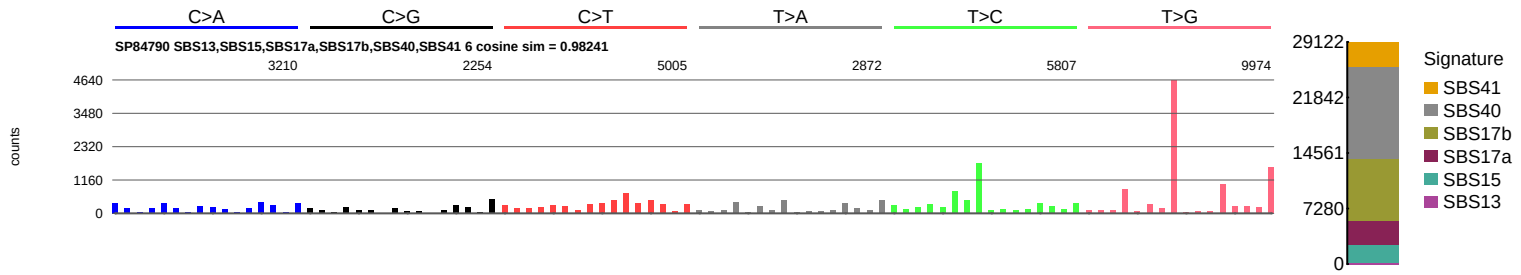

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98247
1783
2104
5652
2642
7353
9590
counts
4700
3525
2350
1175
0
29122
21842
14561
7280
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2

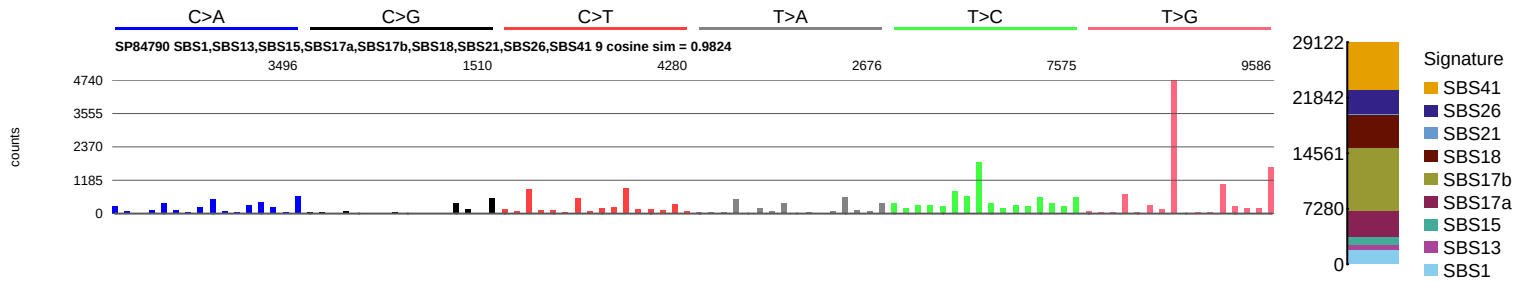

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98246
3387
2472
4106
3293
6202
9661
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3

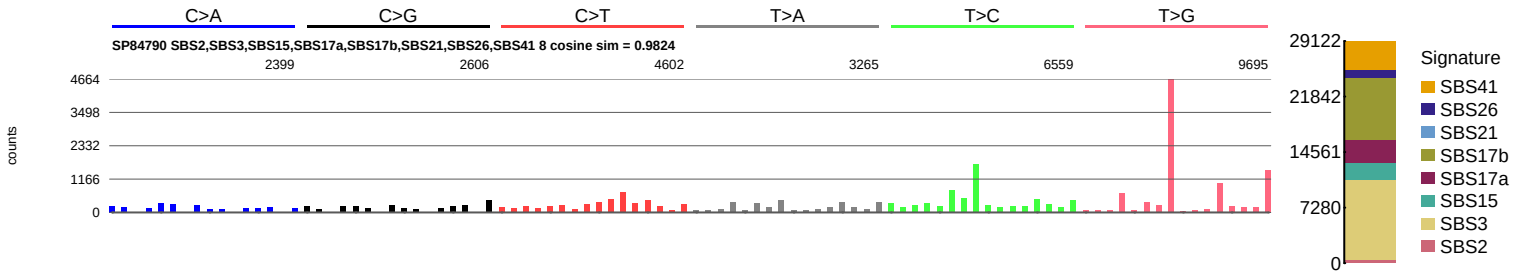

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98246
1805
1576
6827
1911
7769
9230
counts
4784
3588
2392
1196
0
29122
21842
14561
7280
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

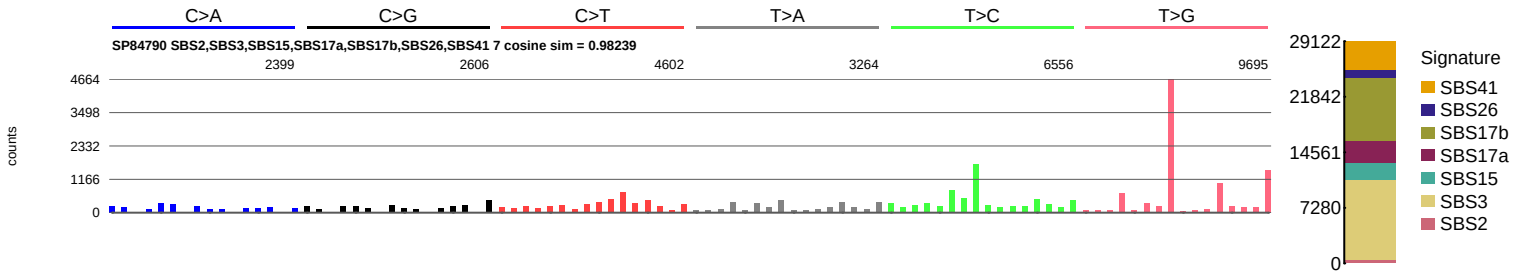

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 10 cosine sim = 0.98246
3344
2369
5300
2209
6577
9326
counts
4708
3531
2354
1177
0
29122
21842
14561
7280
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2

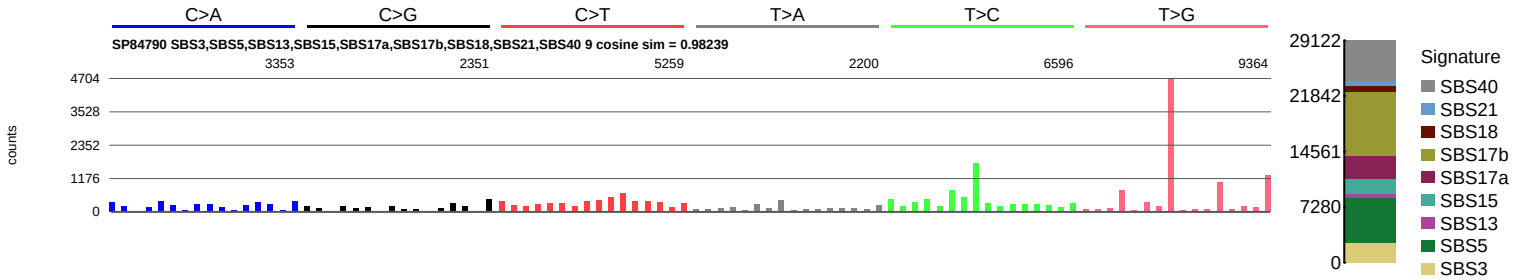

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 6 cosine sim = 0.98244
3413
2116
4943
2894
5794
9962
counts
4636
3477
2318
1159
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15

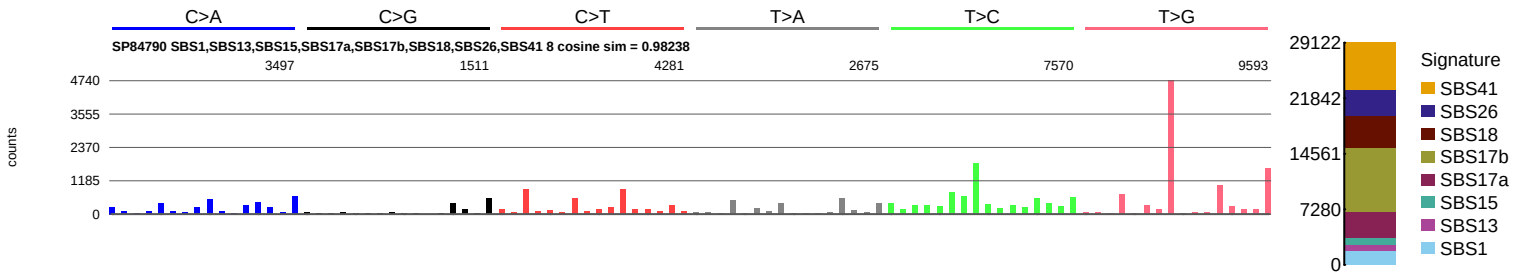

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.98241
3210
2254
5005
2872
5807
9974
counts
4640
3480
2320
1160
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS13

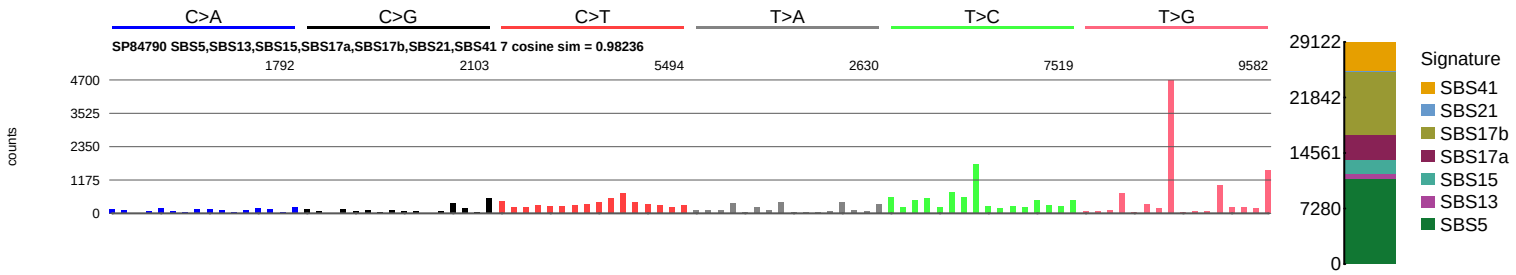

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.9824
3496
1510
4280
2676
7575
9586
counts
4740
3555
2370
1185
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1

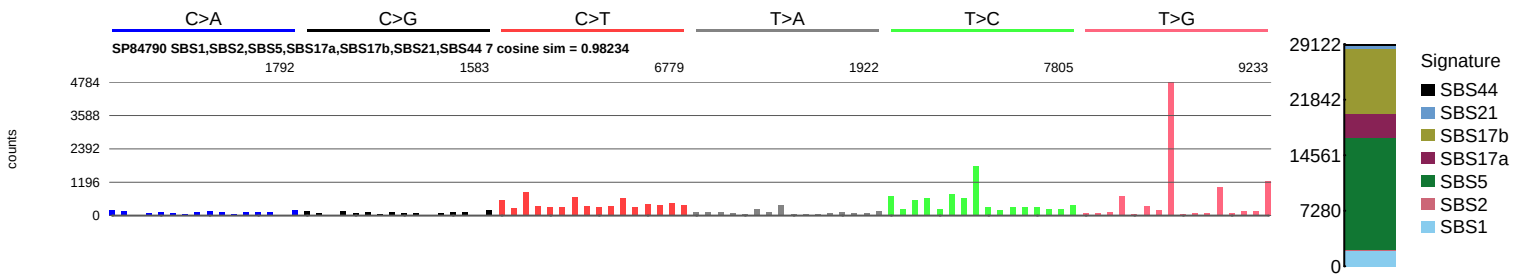

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41 8 cosine sim = 0.9824
2399
2606
4602
3265
6559
9695
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2

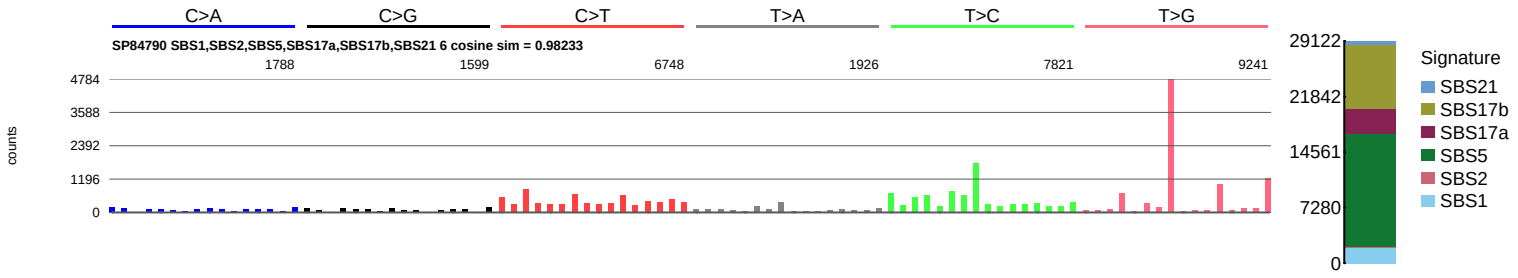

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS41 7 cosine sim = 0.98239
2399
2606
4602
3264
6556
9695
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS2

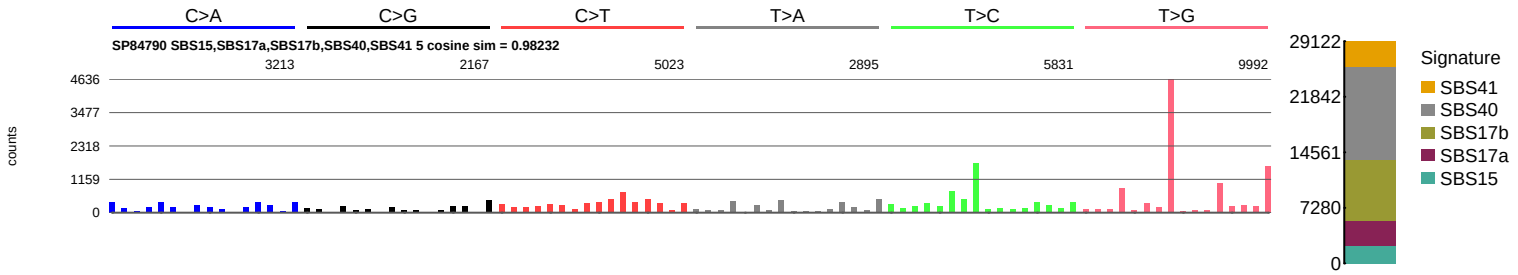

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98239
3353
2351
5259
2200
6596
9364
counts
4704
3528
2352
1176
0
29122
21842
14561
7280
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3

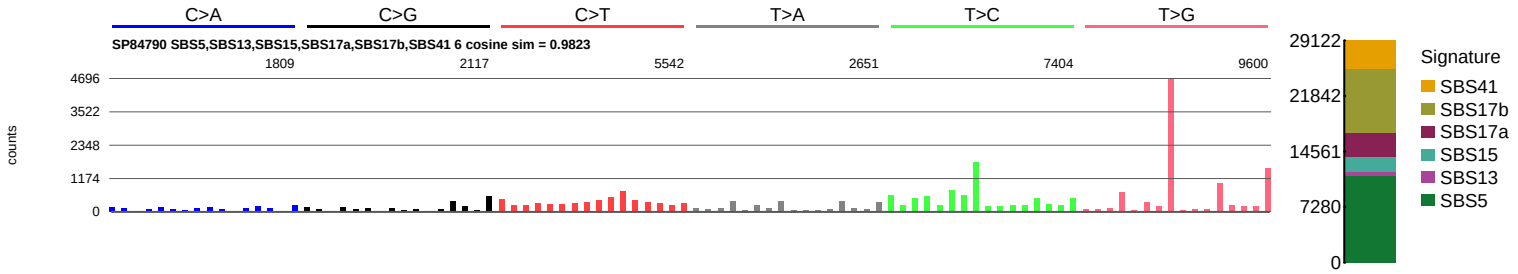

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98238
3497
1511
4281
2675
7570
9593
counts
4740
3555
2370
1185
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1

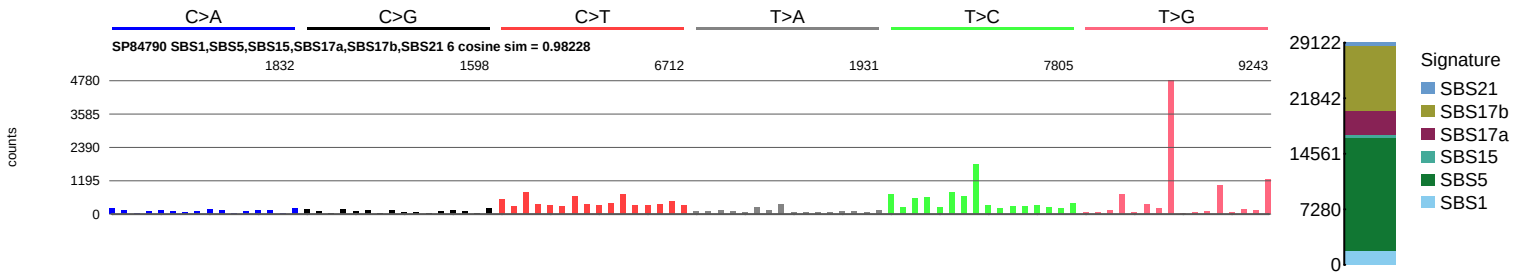

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98236
1792
2103
5494
2630
7519
9582
counts
4700
3525
2350
1175
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5

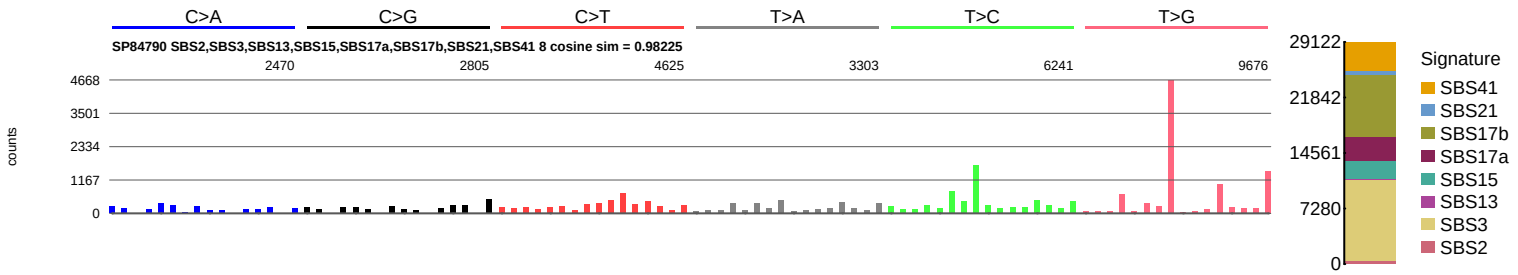

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS44 7 cosine sim = 0.98234
1792
1583
6779
1922
7805
9233
counts
4784
3588
2392
1196
0
29122
21842
14561
7280
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS5
SBS2
SBS1

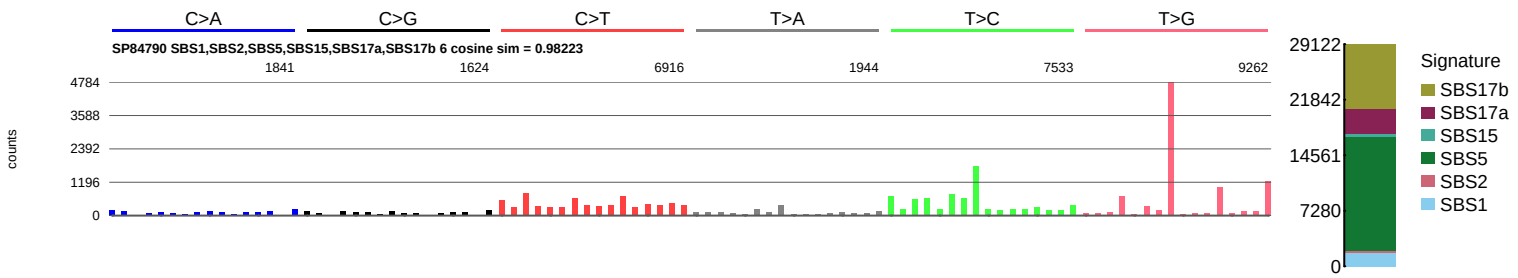

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS21 6 cosine sim = 0.98233
1788
1599
6748
1926
7821
9241
counts
4784
3588
2392
1196
0
29122
21842
14561
7280
0
Signature
SBS21
SBS17b
SBS17a
SBS5
SBS2
SBS1

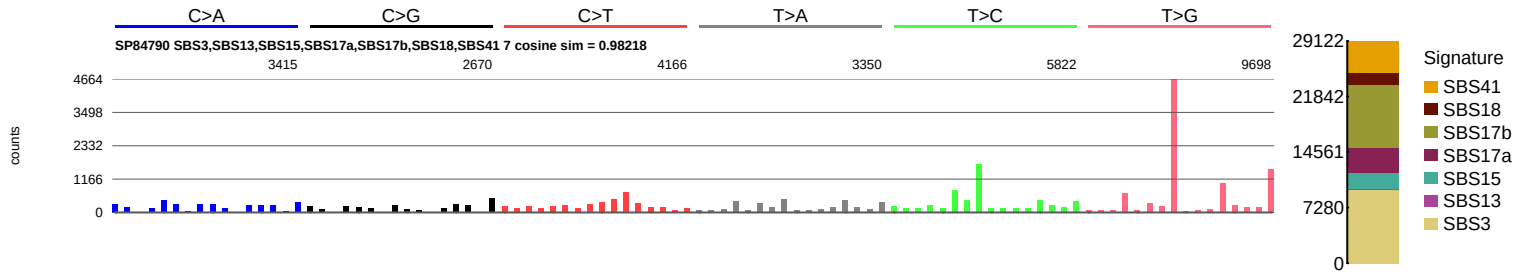

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS15,SBS17a,SBS17b,SBS40,SBS41 5 cosine sim = 0.98232
3213
2167
5023
2895
5831
9992
counts
4636
3477
2318
1159
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15

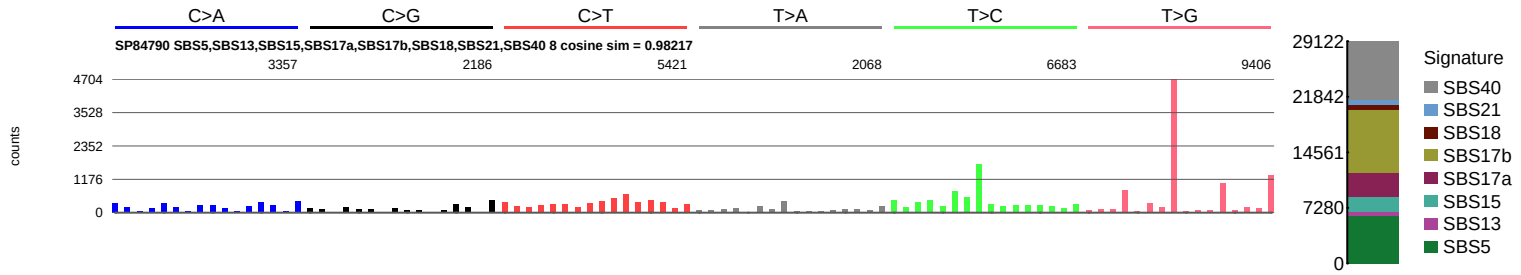

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS41 6 cosine sim = 0.9823
1809
2117
5542
2651
7404
9600
counts
4696
3522
2348
1174
0
29122
21842
14561
7280
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS5

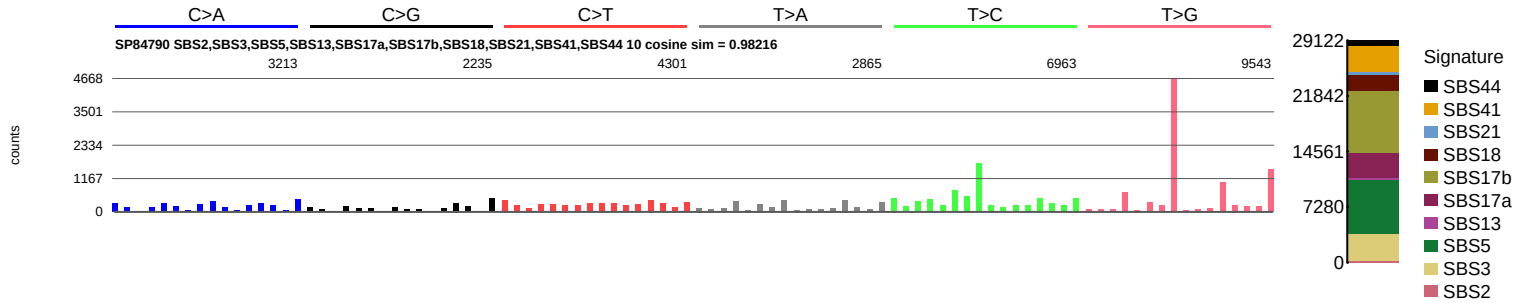

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS21 6 cosine sim = 0.98228
1832
1598
6712
1931
7805
9243
counts
4780
3585
2390
1195
0
29122
21842
14561
7280
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS1

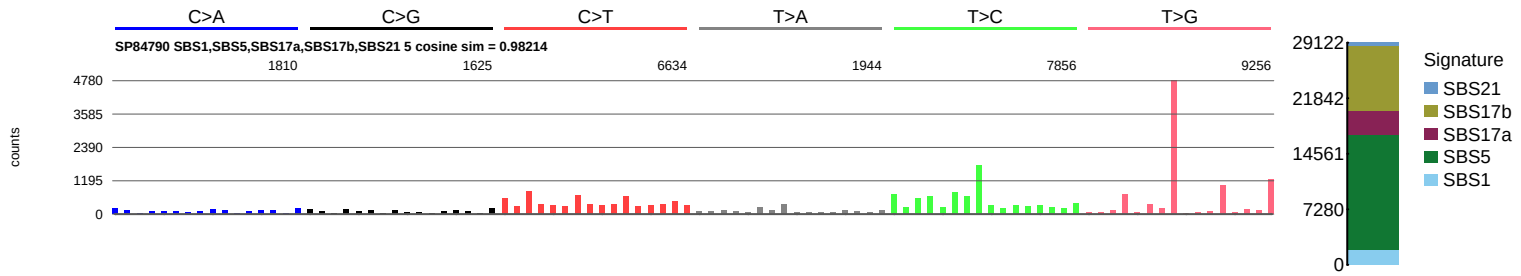

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.98225
2470
2805
4625
3303
6241
9676
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2

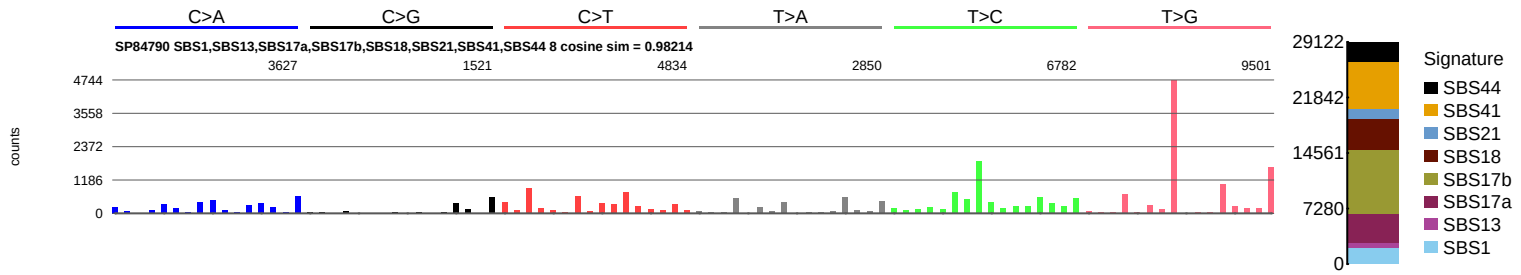

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b 6 cosine sim = 0.98223
1841
1624
6916
1944
7533
9262
counts
4784
3588
2392
1196
0
29122
21842
14561
7280
0
Signature
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

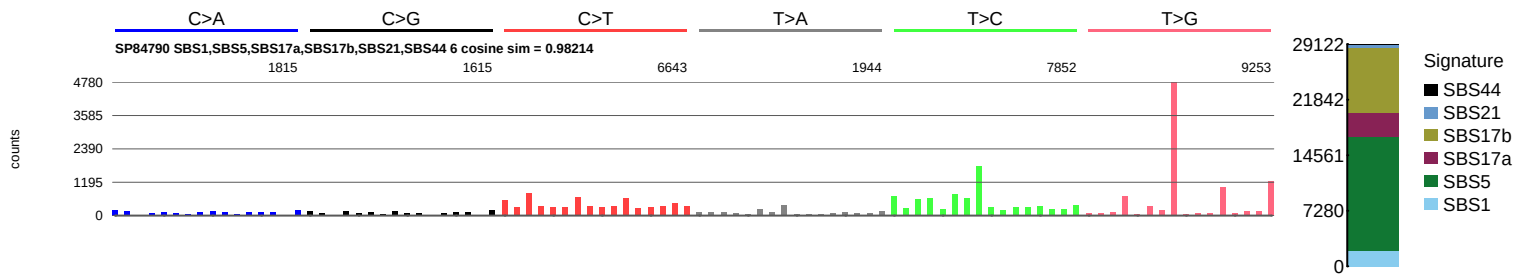

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98218
3415
2670
4166
3350
5822
9698
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3

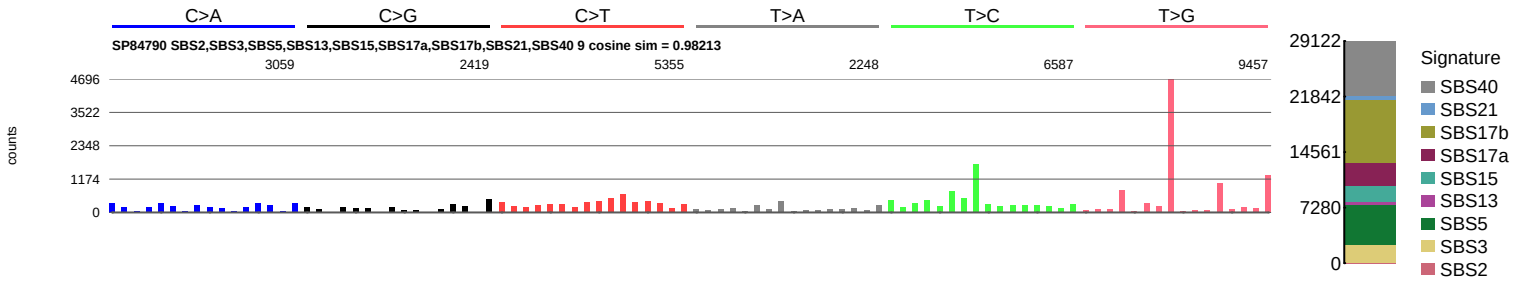

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98217
3357
2186
5421
2068
6683
9406
counts
4704
3528
2352
1176
0
29122
21842
14561
7280
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5

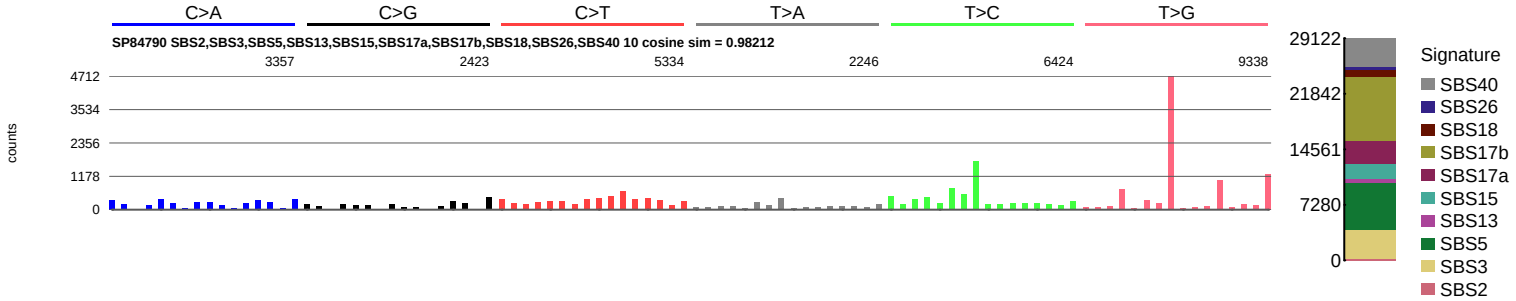

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98216
3213
2235
4301
2865
6963
9543
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2

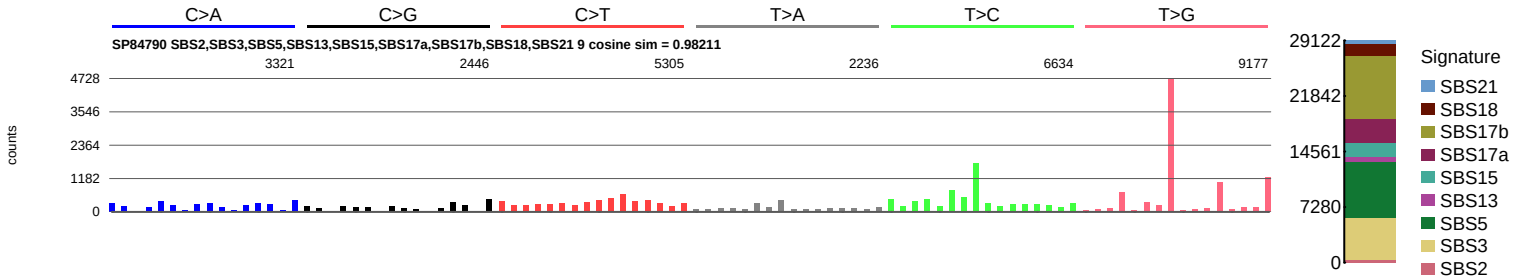

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS17a,SBS17b,SBS21 5 cosine sim = 0.98214
1810
1625
6634
1944
7856
9256
counts
4780
3585
2390
1195
0
29122
21842
14561
7280
0
Signature
SBS21
SBS17b
SBS17a
SBS5
SBS1

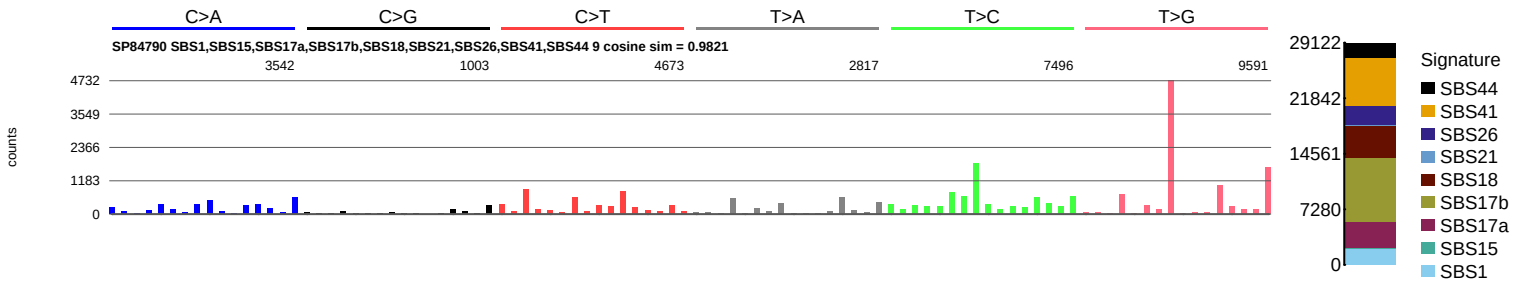

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98214
3627
1521
4834
2850
6782
9501
counts
4744
3558
2372
1186
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS1

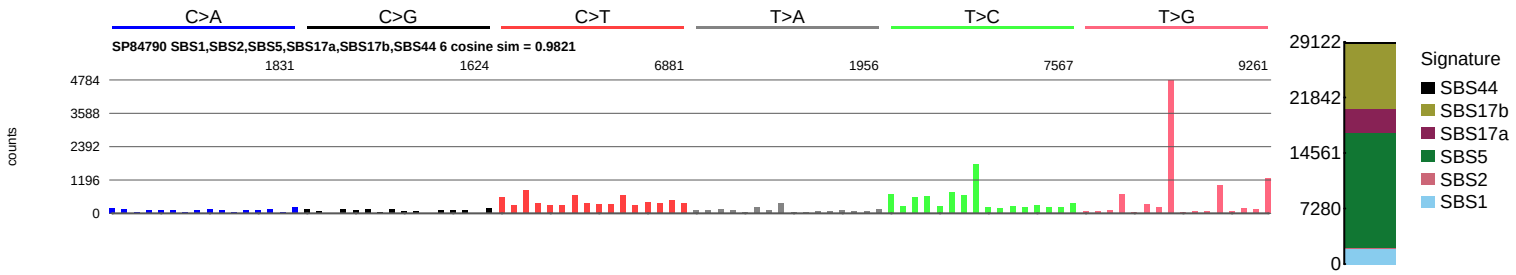

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS17a,SBS17b,SBS21,SBS44 6 cosine sim = 0.98214
1815
1615
6643
1944
7852
9253
counts
4780
3585
2390
1195
0
29122
21842
14561
7280
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS5
SBS1

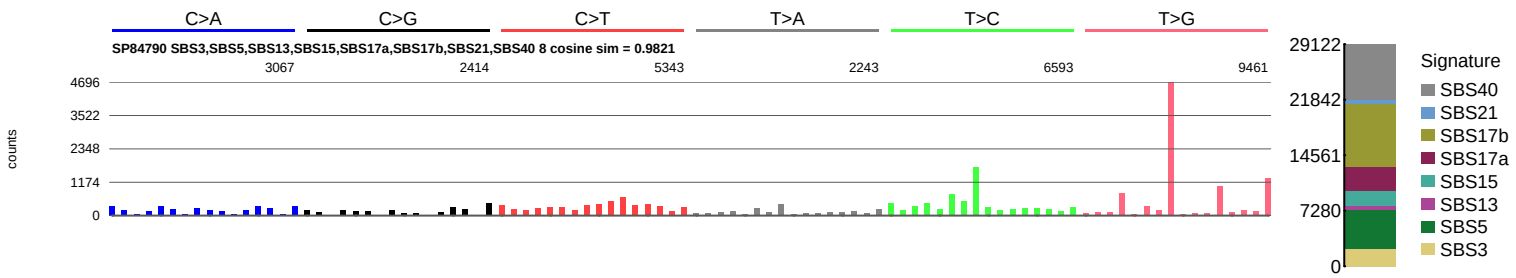

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40 9 cosine sim = 0.98213
3059
2419
5355
2248
6587
9457
counts
4696
3522
2348
1174
0
29122
21842
14561
7280
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2

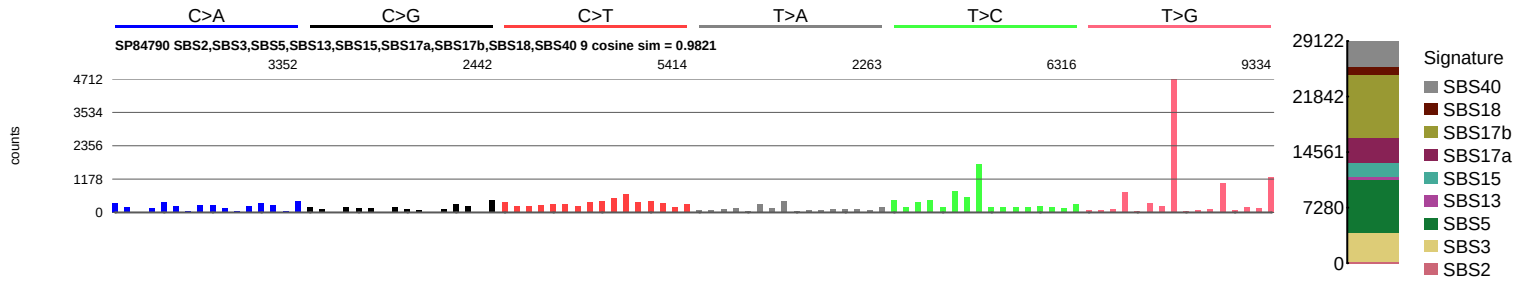

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 10 cosine sim = 0.98212
3357
2423
5334
2246
6424
9338
counts
4712
3534
2356
1178
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2

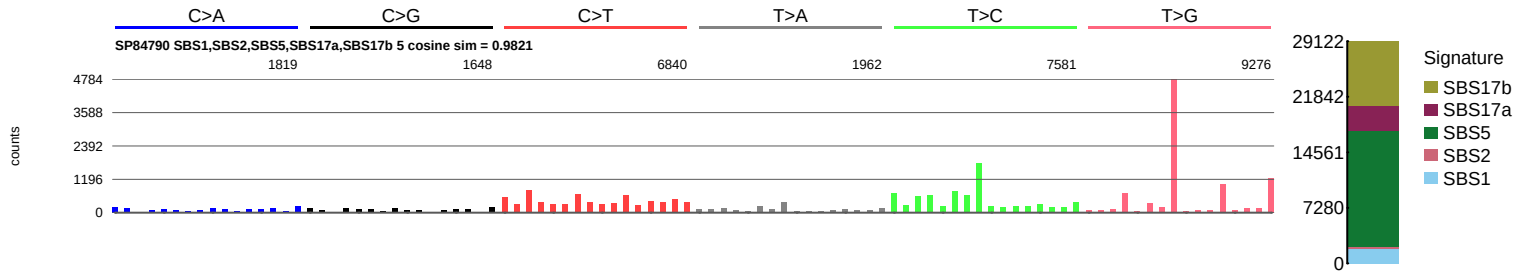

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98211
3321
2446
5305
2236
6634
9177
counts
4728
3546
2364
1182
0
29122
21842
14561
7280
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2

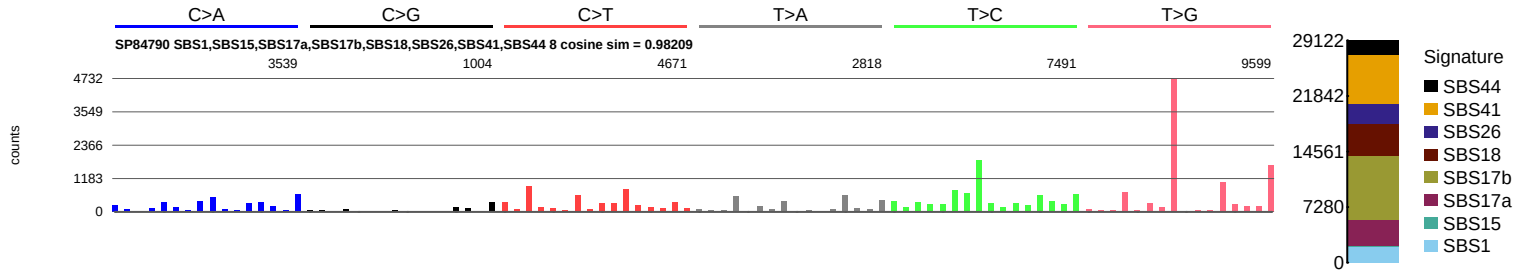

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 9 cosine sim = 0.9821
3542
1003
4673
2817
7496
9591
counts
4732
3549
2366
1183
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1

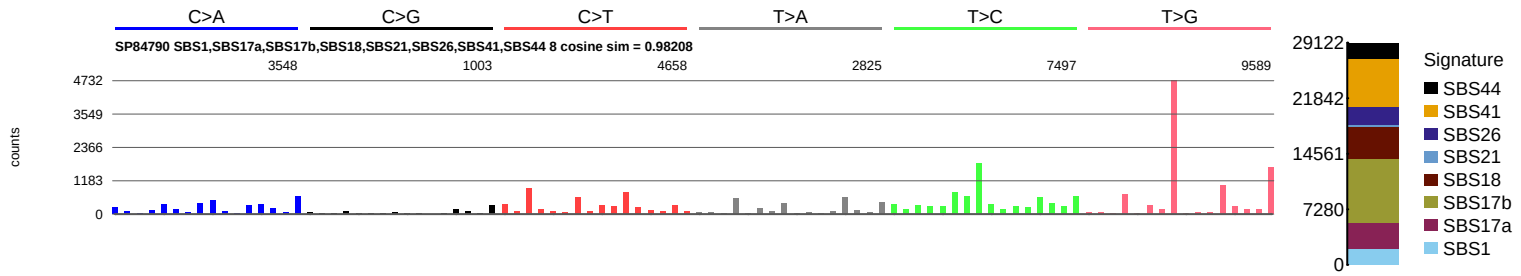

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS44 6 cosine sim = 0.9821
1831
1624
6881
1956
7567
9261
counts
4784
3588
2392
1196
0
29122
21842
14561
7280
0
Signature
SBS44
SBS17b
SBS17a
SBS5
SBS2
SBS1

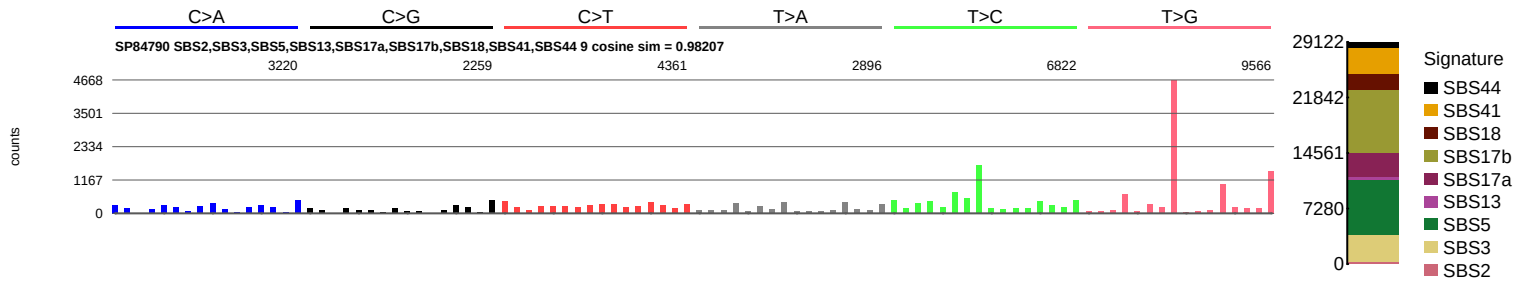

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40 8 cosine sim = 0.9821
3067
2414
5343
2243
6593
9461
counts
4696
3522
2348
1174
0
29122
21842
14561
7280
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3

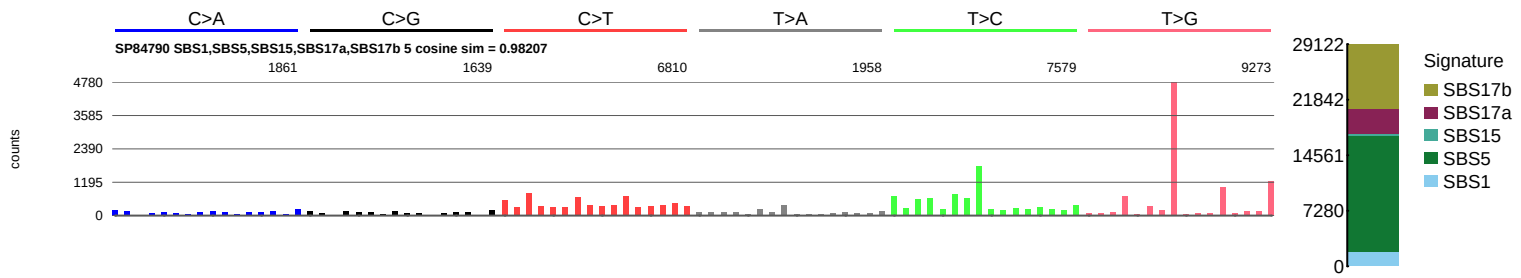

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 9 cosine sim = 0.9821
3352
2442
5414
2263
6316
9334
counts
4712
3534
2356
1178
0
29122
21842
14561
7280
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2

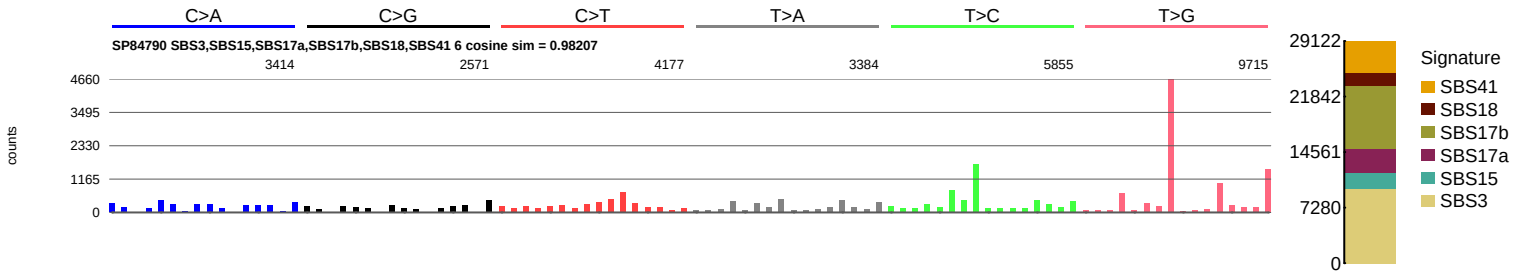

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS5,SBS17a,SBS17b 5 cosine sim = 0.9821
1819
1648
6840
1962
7581
9276
counts
4784
3588
2392
1196
0
29122
21842
14561
7280
0
Signature
SBS17b
SBS17a
SBS5
SBS2
SBS1

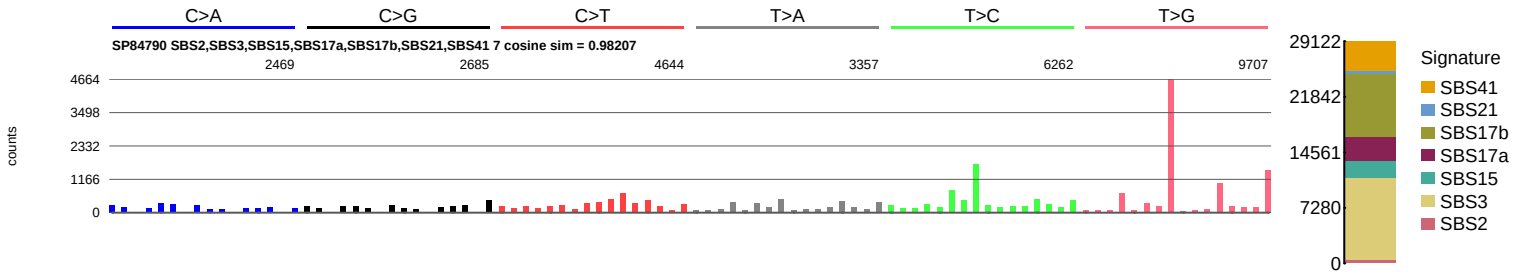

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98209
3539
1004
4671
2818
7491
9599
counts
4732
3549
2366
1183
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1

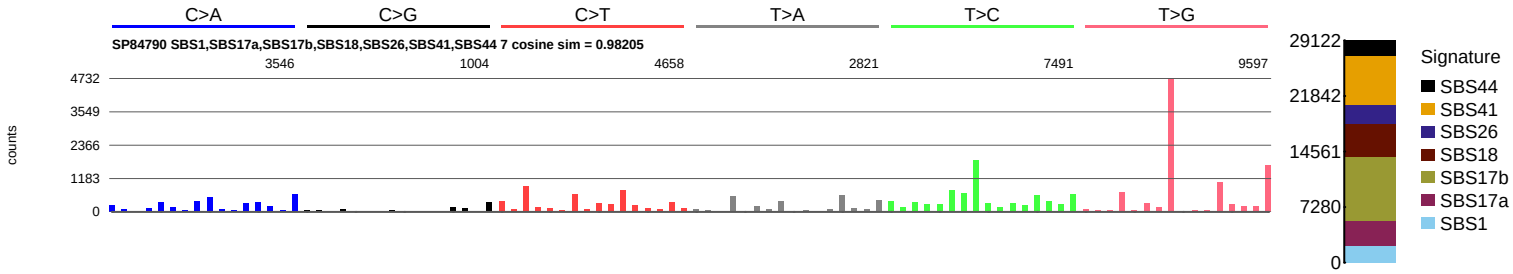

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 8 cosine sim = 0.98208
3548
1003
4658
2825
7497
9589
counts
4732
3549
2366
1183
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS1

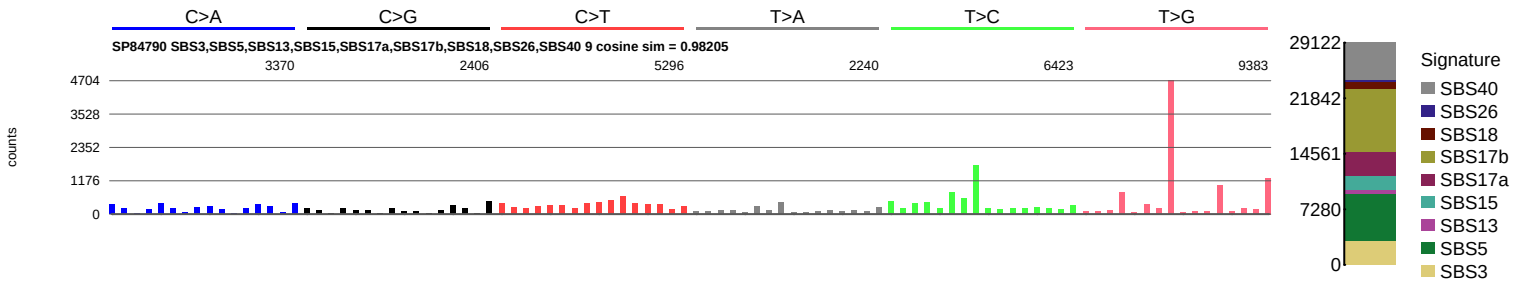

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98207
3220
2259
4361
2896
6822
9566
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2

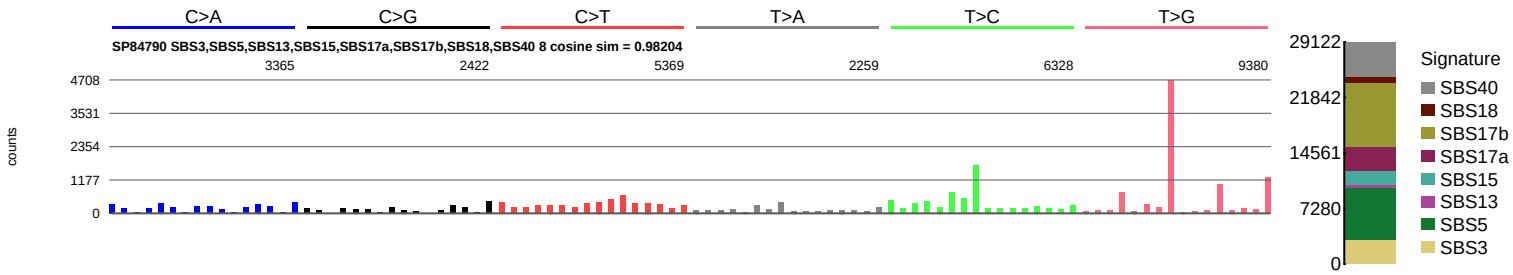

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS15,SBS17a,SBS17b 5 cosine sim = 0.98207
1861
1639
6810
1958
7579
9273
counts
4780
3585
2390
1195
0
29122
21842
14561
7280
0
Signature
SBS17b
SBS17a
SBS15
SBS5
SBS1

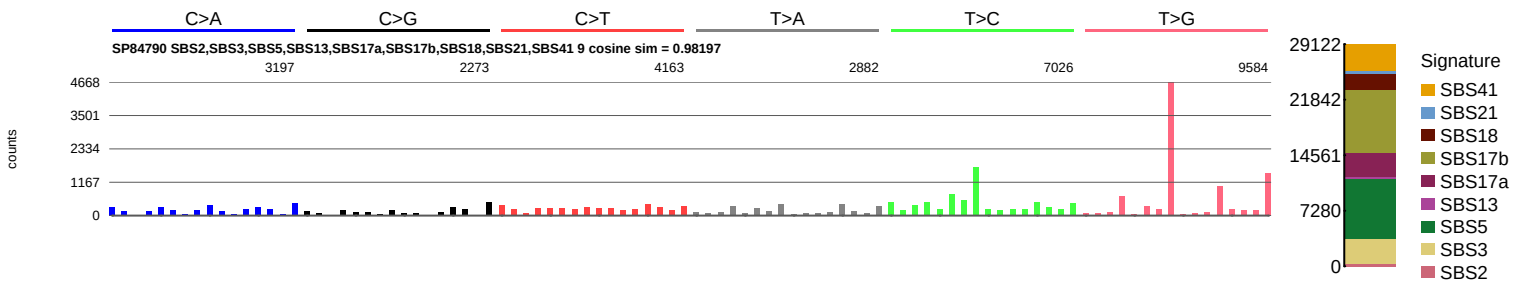

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98207
3414
2571
4177
3384
5855
9715
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3

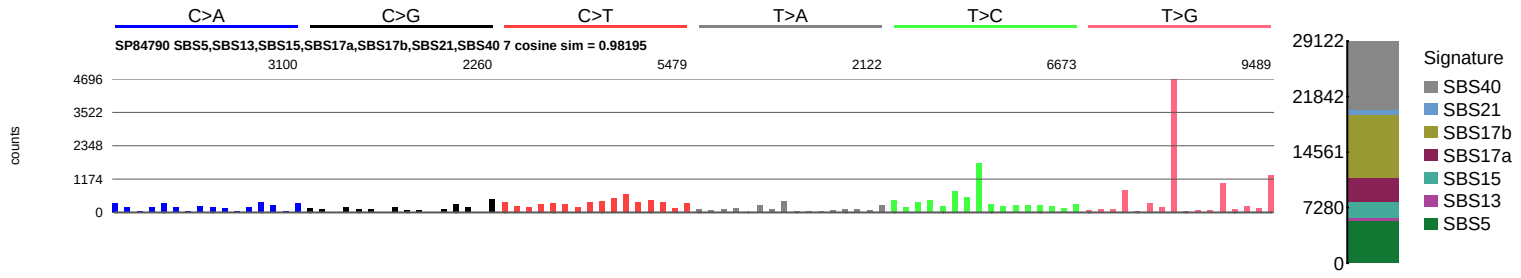

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98207
2469
2685
4644
3357
6262
9707
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2

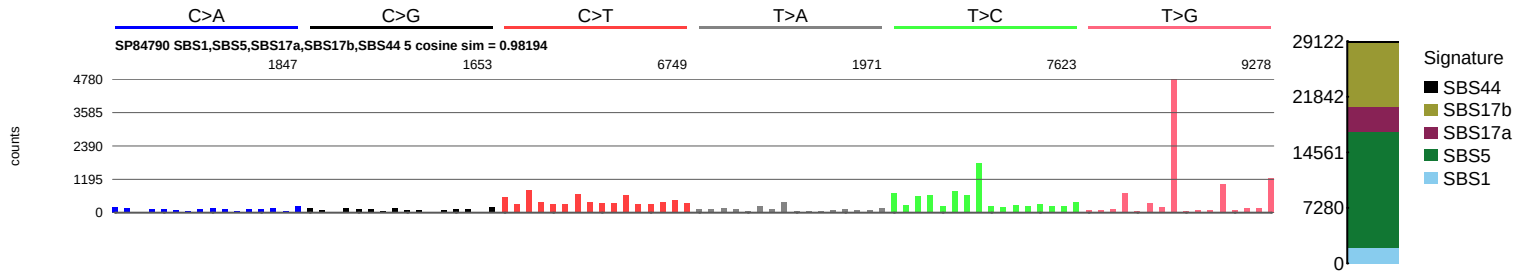

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 7 cosine sim = 0.98205
3546
1004
4658
2821
7491
9597
counts
4732
3549
2366
1183
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS1

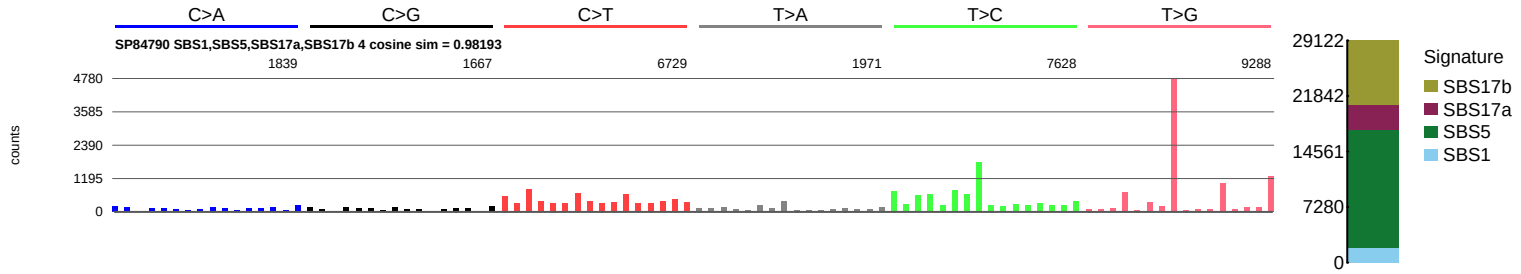

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98205
3370
2406
5296
2240
6423
9383
counts
4704
3528
2352
1176
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3

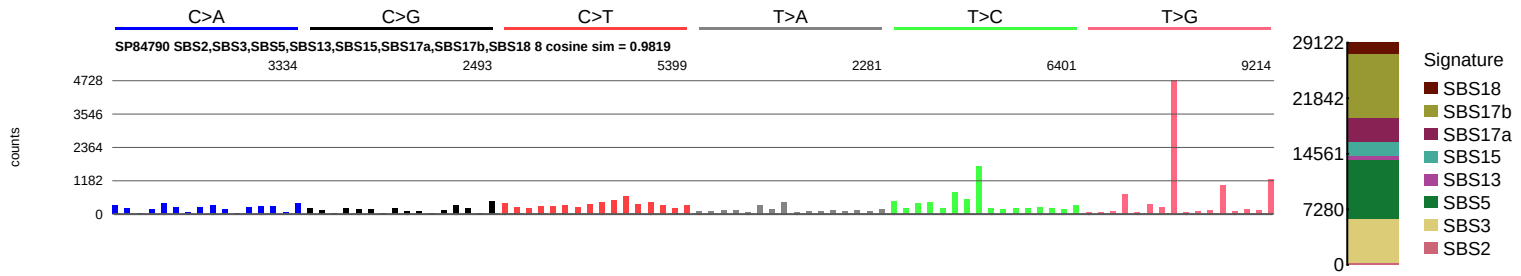

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98204
3365
2422
5369
2259
6328
9380
counts
4708
3531
2354
1177
0
29122
21842
14561
7280
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3

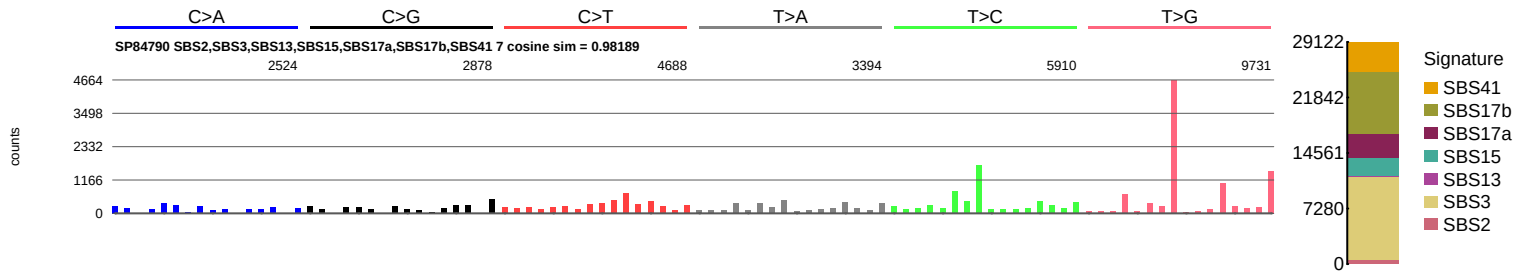

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98197
3197
2273
4163
2882
7026
9584
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2

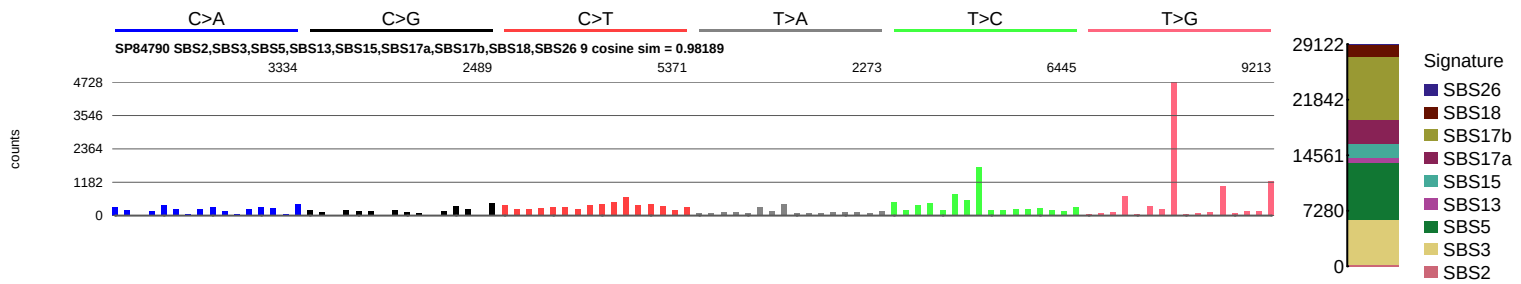

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40 7 cosine sim = 0.98195
3100
2260
5479
2122
6673
9489
counts
4696
3522
2348
1174
0
29122
21842
14561
7280
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5

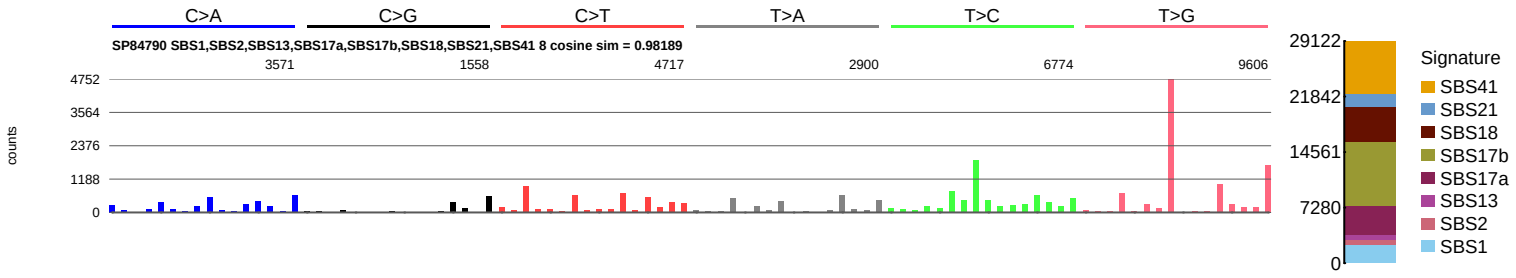

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS17a,SBS17b,SBS44 5 cosine sim = 0.98194
1847
1653
6749
1971
7623
9278
counts
4780
3585
2390
1195
0
29122
21842
14561
7280
0
Signature
SBS44
SBS17b
SBS17a
SBS5
SBS1

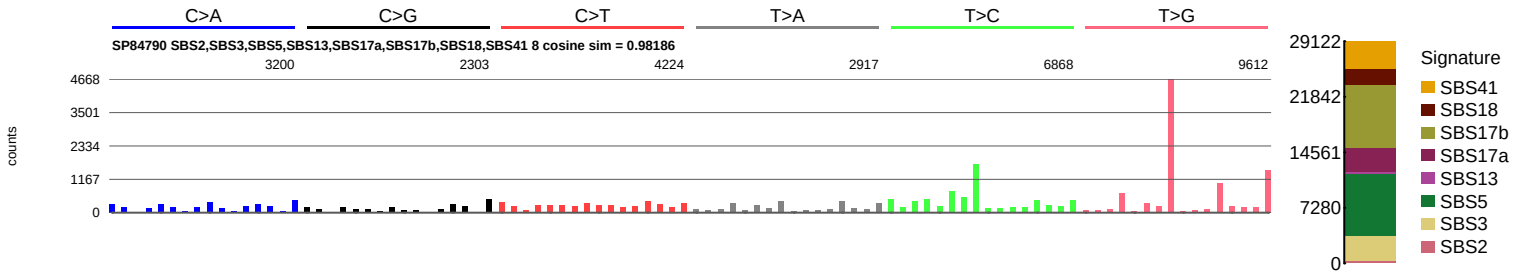

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS5,SBS17a,SBS17b 4 cosine sim = 0.98193
1839
1667
6729
1971
7628
9288
counts
4780
3585
2390
1195
0
29122
21842
14561
7280
0
Signature
SBS17b
SBS17a
SBS5
SBS1

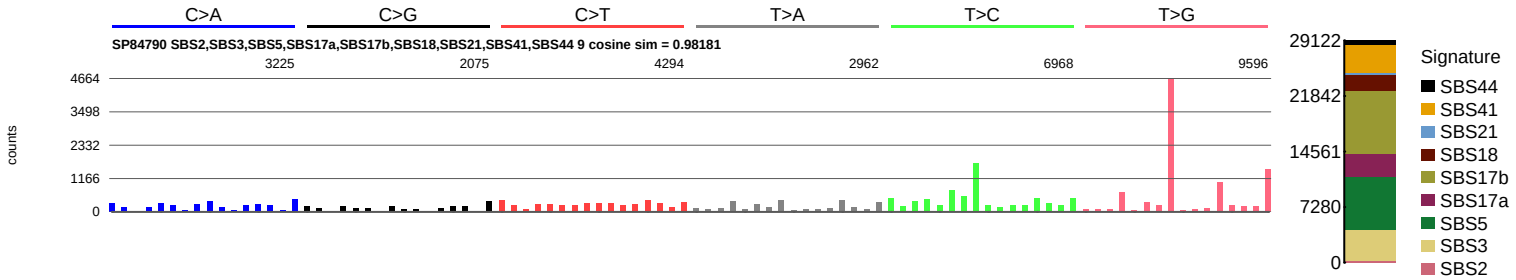

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.9819
3334
2493
5399
2281
6401
9214
counts
4728
3546
2364
1182
0
29122
21842
14561
7280
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2

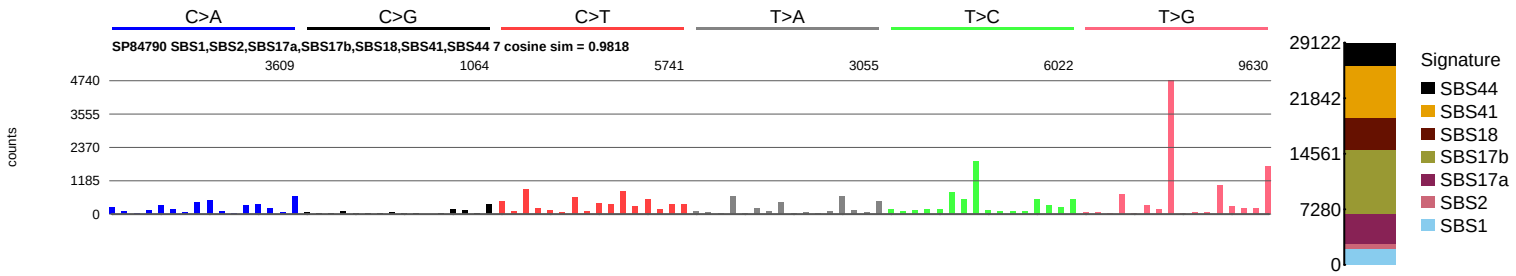

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98189
2524
2878
4688
3394
5910
9731
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2

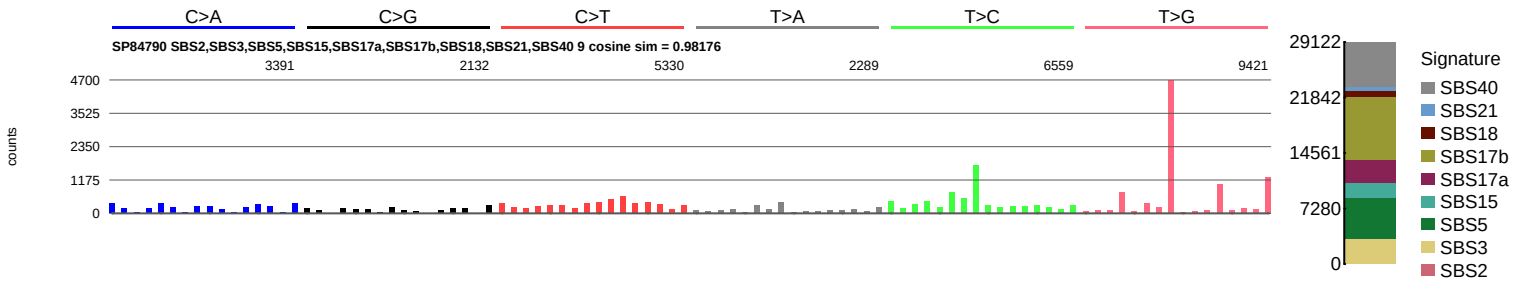

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 9 cosine sim = 0.98189
3334
2489
5371
2273
6445
9213
counts
4728
3546
2364
1182
0
29122
21842
14561
7280
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2

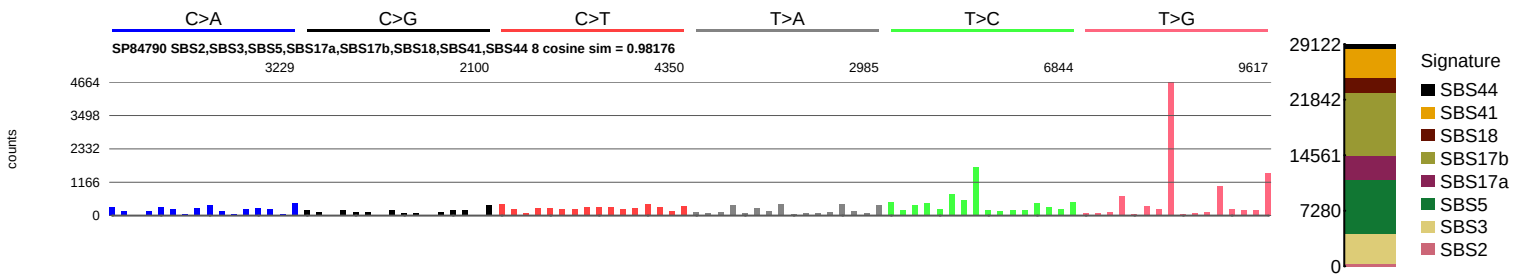

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98189
3571
1558
4717
2900
6774
9606
counts
4752
3564
2376
1188
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

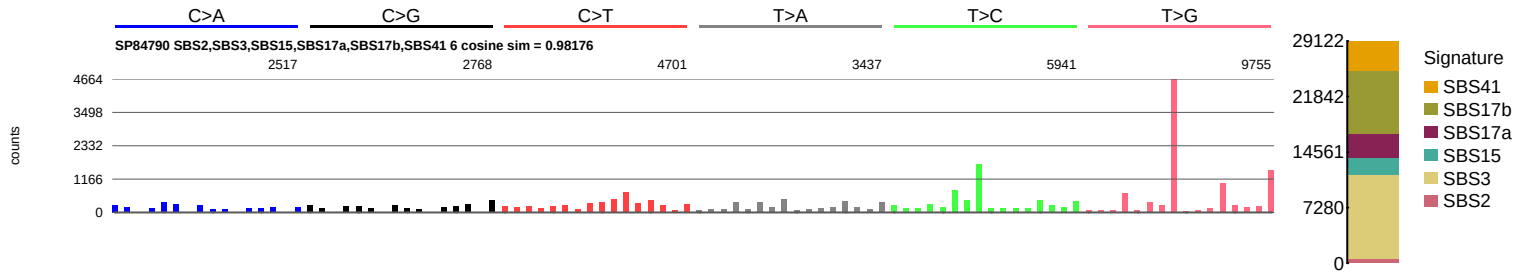

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98186
3200
2303
4224
2917
6868
9612
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2

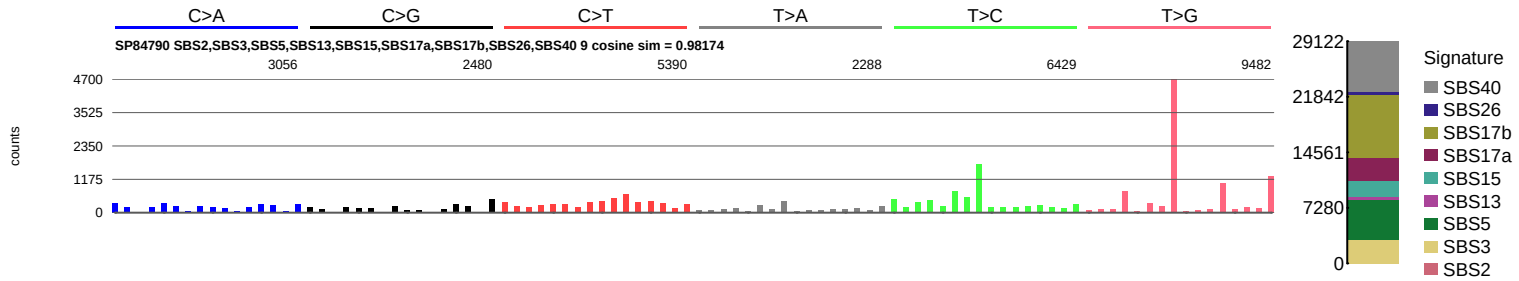

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98181
3225
2075
4294
2962
6968
9596
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2

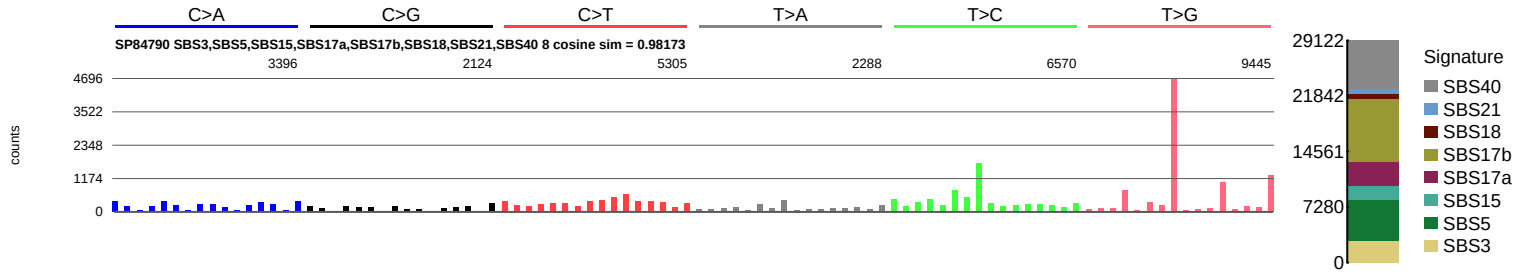

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.9818
3609
1064
5741
3055
6022
9630
counts
4740
3555
2370
1185
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS2
SBS1

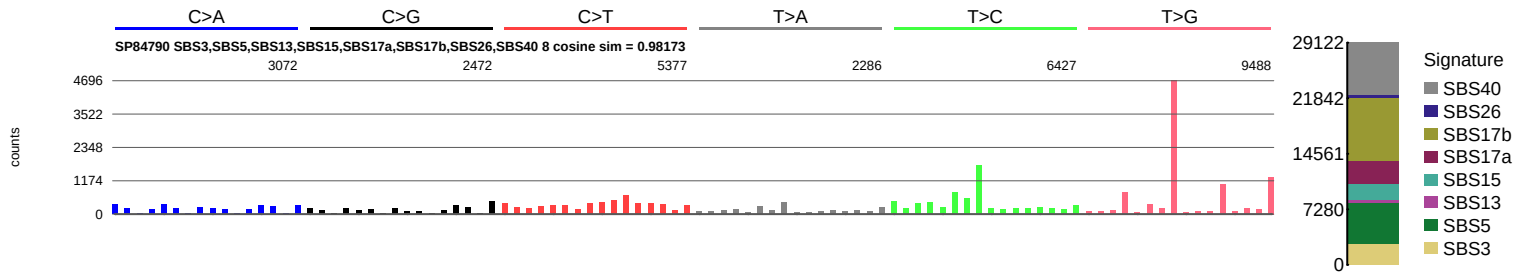

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98176
3391
2132
5330
2289
6559
9421
counts
4700
3525
2350
1175
0
29122
21842
14561
7280
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2

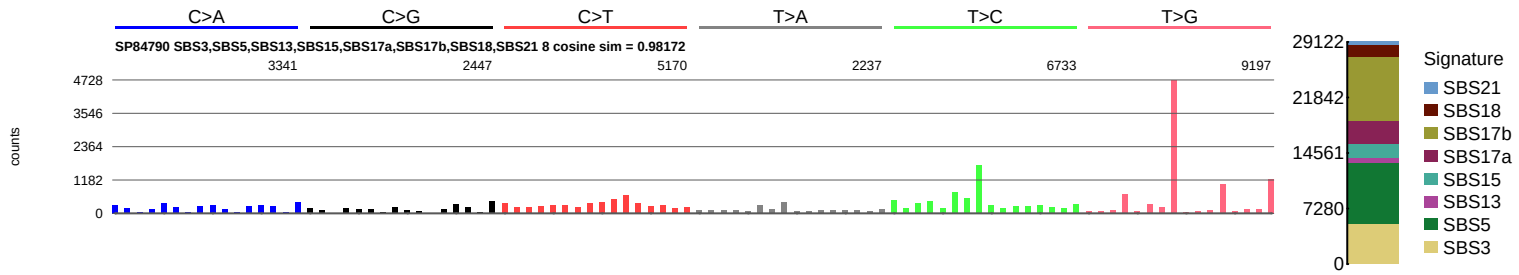

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98176
3229
2100
4350
2985
6844
9617
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2

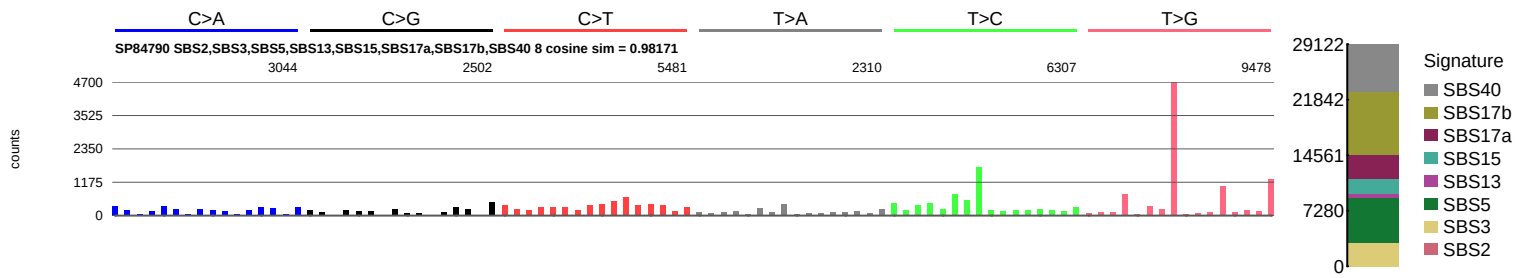

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98176
2517
2768
4701
3437
5941
9755
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS3
SBS2

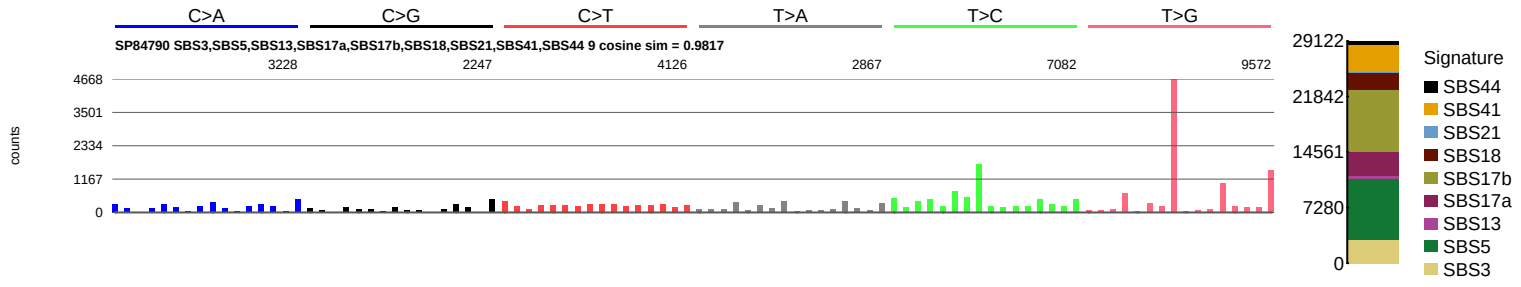

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40 9 cosine sim = 0.98174
3056
2480
5390
2288
6429
9482
counts
4700
3525
2350
1175
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2

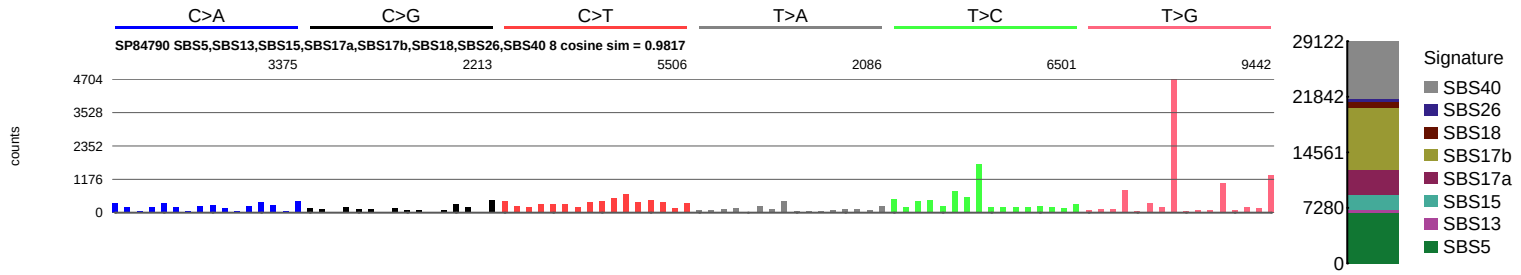

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98173
3396
2124
5305
2288
6570
9445
counts
4696
3522
2348
1174
0
29122
21842
14561
7280
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3

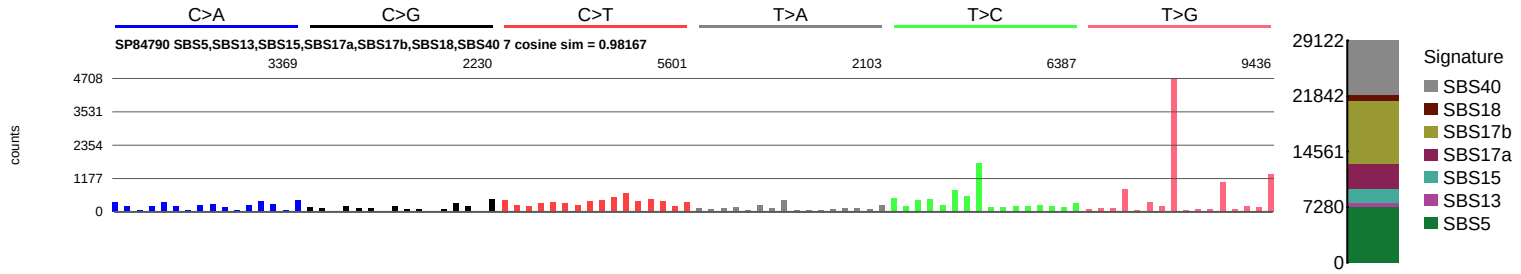

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40 8 cosine sim = 0.98173
3072
2472
5377
2286
6427
9488
counts
4696
3522
2348
1174
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3

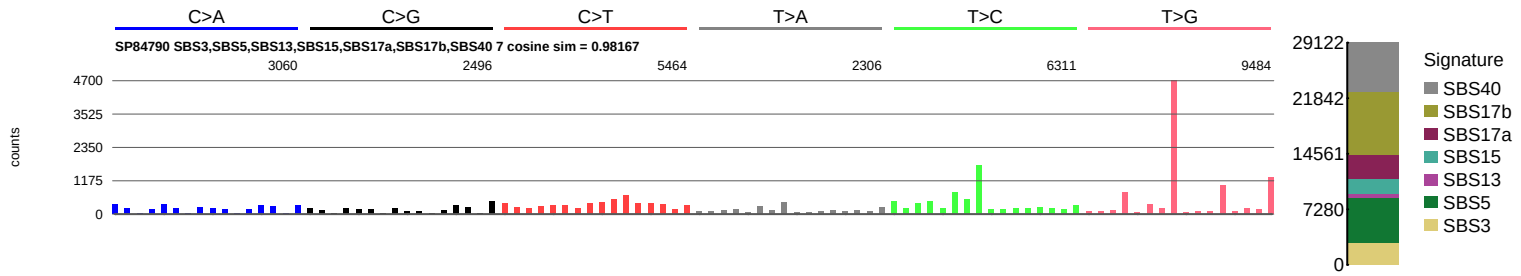

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98172
3341
2447
5170
2237
6733
9197
counts
4728
3546
2364
1182
0
29122
21842
14561
7280
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3

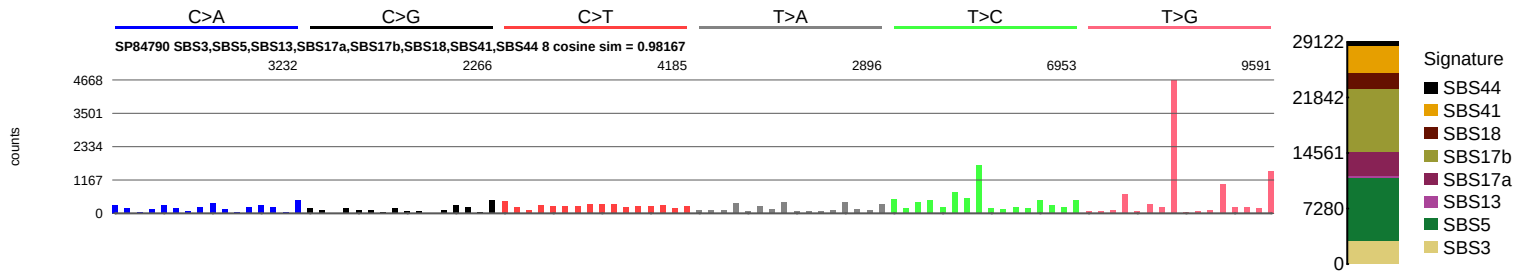

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40 8 cosine sim = 0.98171
3044
2502
5481
2310
6307
9478
counts
4700
3525
2350
1175
0
29122
21842
14561
7280
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2

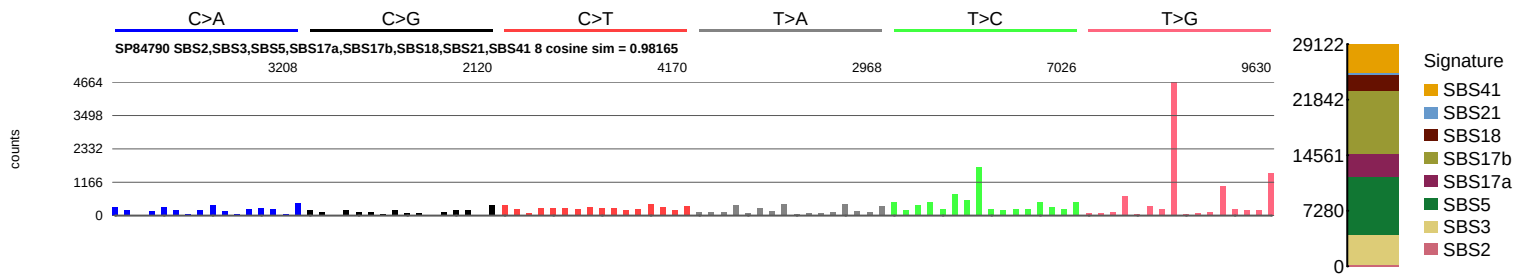

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.9817
3228
2247
4126
2867
7082
9572
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3

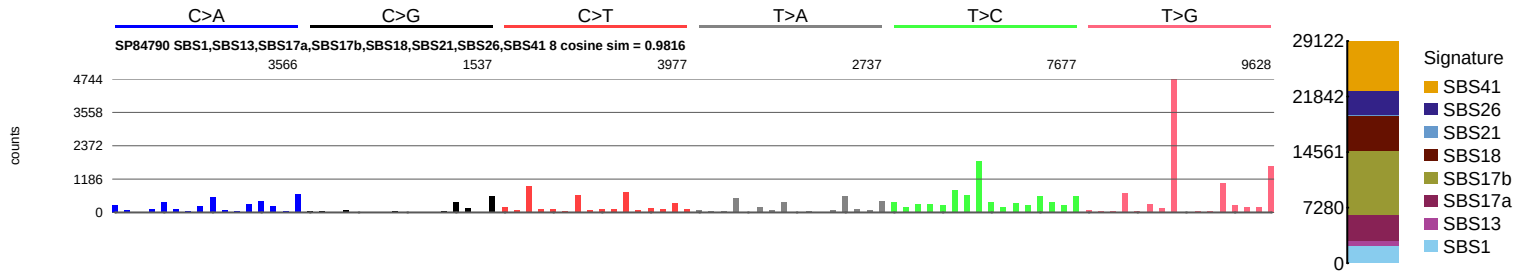

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.9817
3375
2213
5506
2086
6501
9442
counts
4704
3528
2352
1176
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5

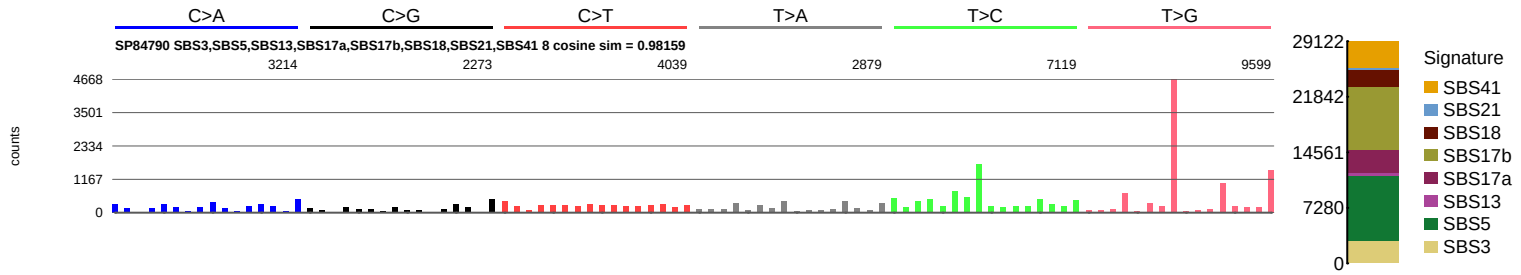

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98167
3369
2230
5601
2103
6387
9436
counts
4708
3531
2354
1177
0
29122
21842
14561
7280
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5

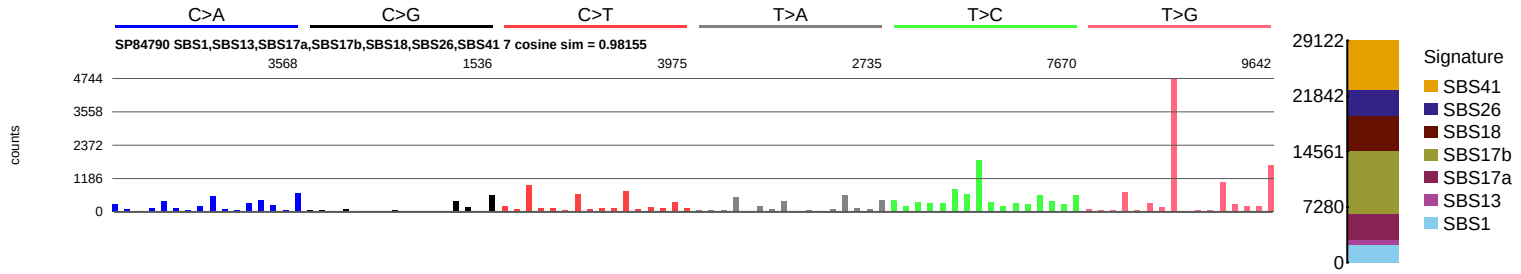

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40 7 cosine sim = 0.98167
3060
2496
5464
2306
6311
9484
counts
4700
3525
2350
1175
0
29122
21842
14561
7280
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3

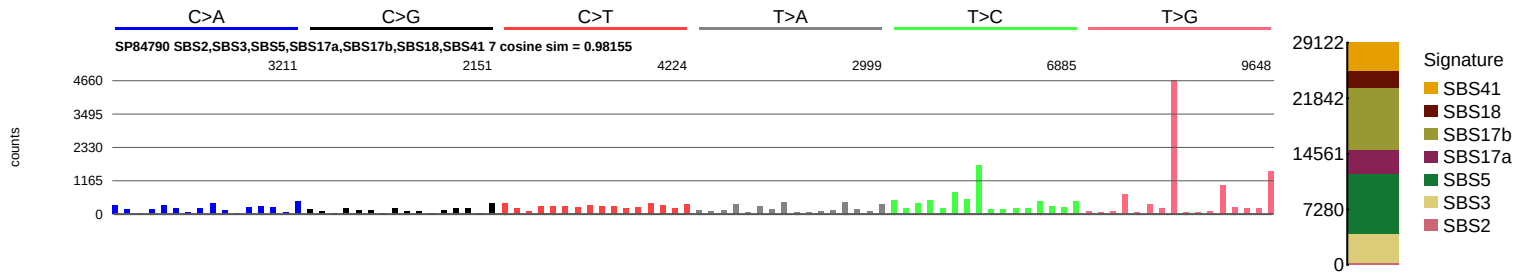

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98167
3232
2266
4185
2896
6953
9591
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3

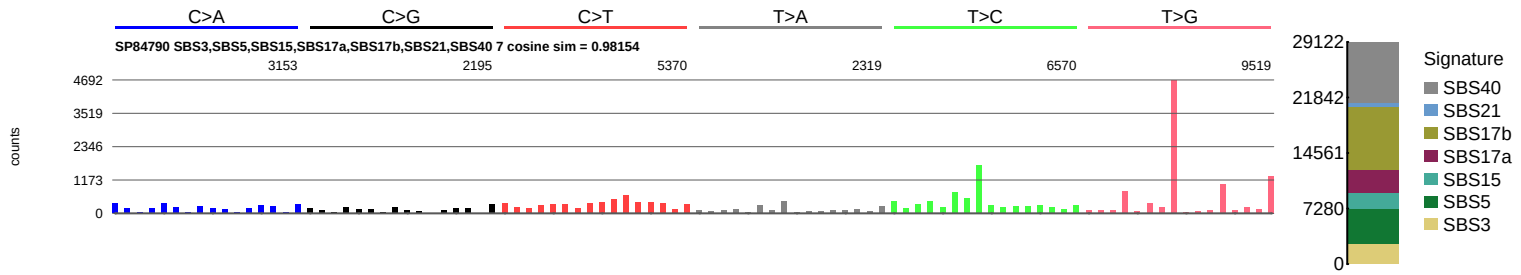

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98165
3208
2120
4170
2968
7026
9630
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2

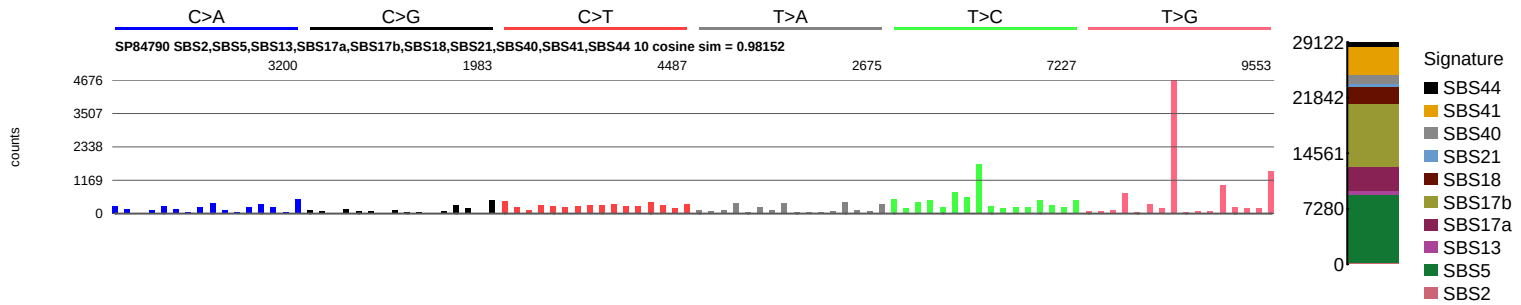

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 8 cosine sim = 0.9816
3566
1537
3977
2737
7677
9628
counts
4744
3558
2372
1186
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS1

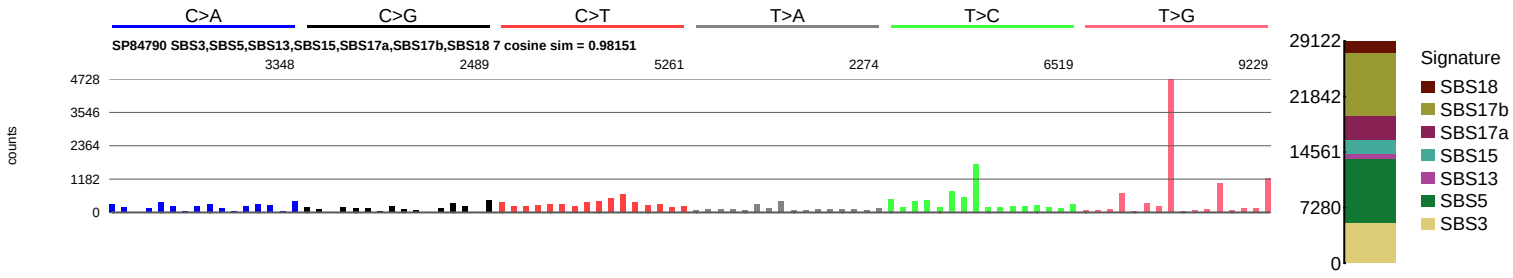

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98159
3214
2273
4039
2879
7119
9599
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3

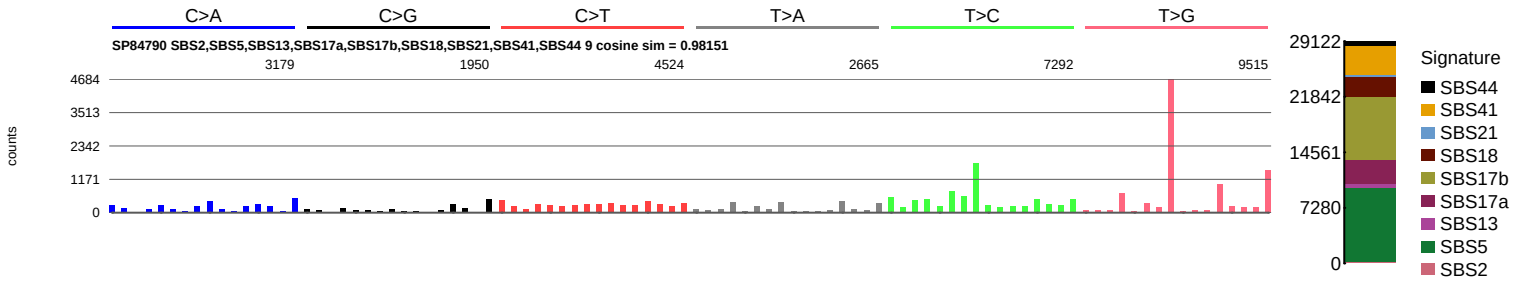

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.98155
3568
1536
3975
2735
7670
9642
counts
4744
3558
2372
1186
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS1

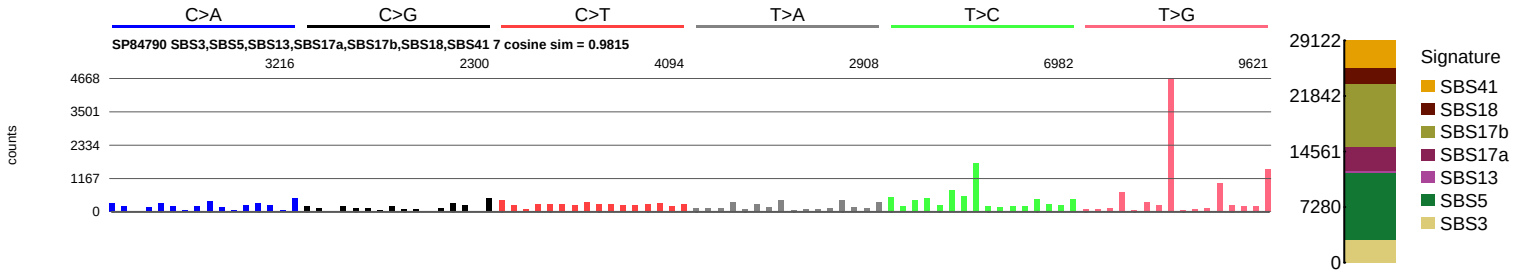

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98155
3211
2151
4224
2999
6885
9648
counts
4660
3495
2330
1165
0
29122
21842
14561
7280
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2

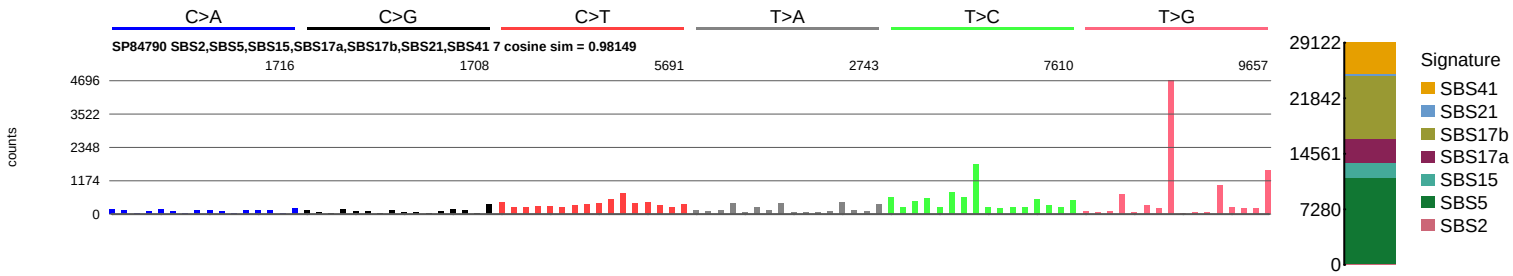

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40 7 cosine sim = 0.98154
3153
2195
5370
2319
6570
9519
counts
4692
3519
2346
1173
0
29122
21842
14561
7280
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3

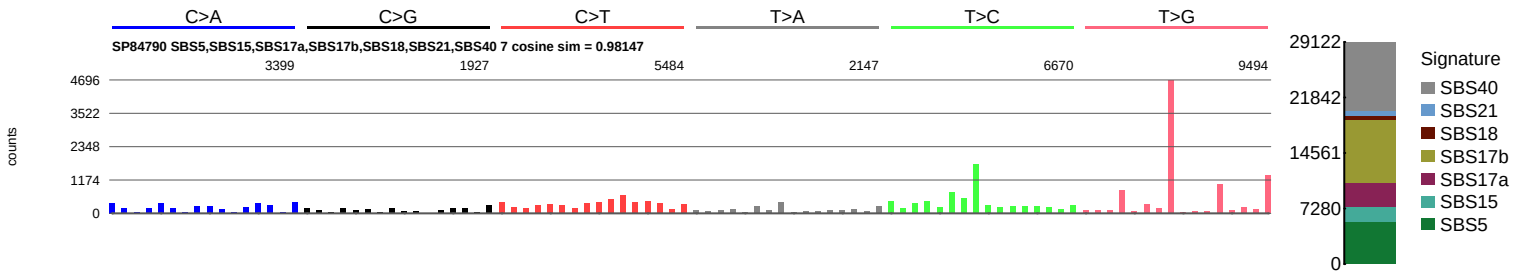

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98152
3200
1983
4487
2675
7227
9553
counts
4676
3507
2338
1169
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2

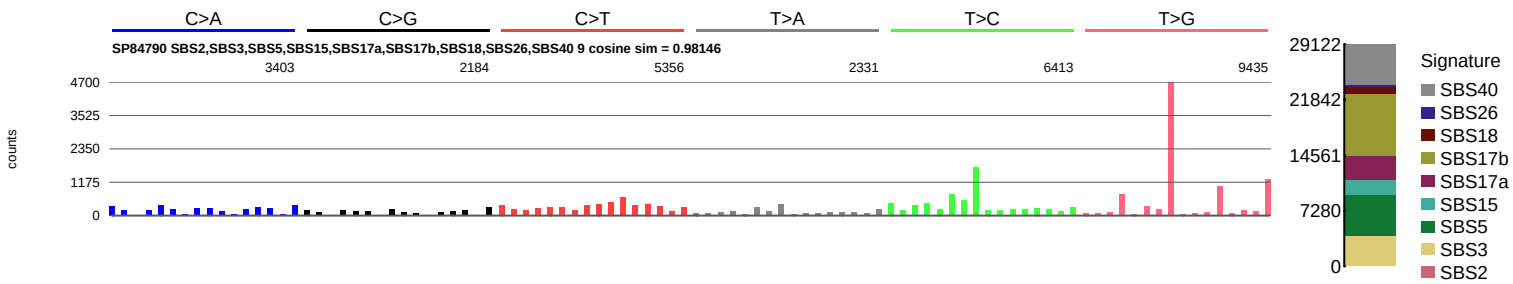

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98151
3348
2489
5261
2274
6519
9229
counts
4728
3546
2364
1182
0
29122
21842
14561
7280
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3

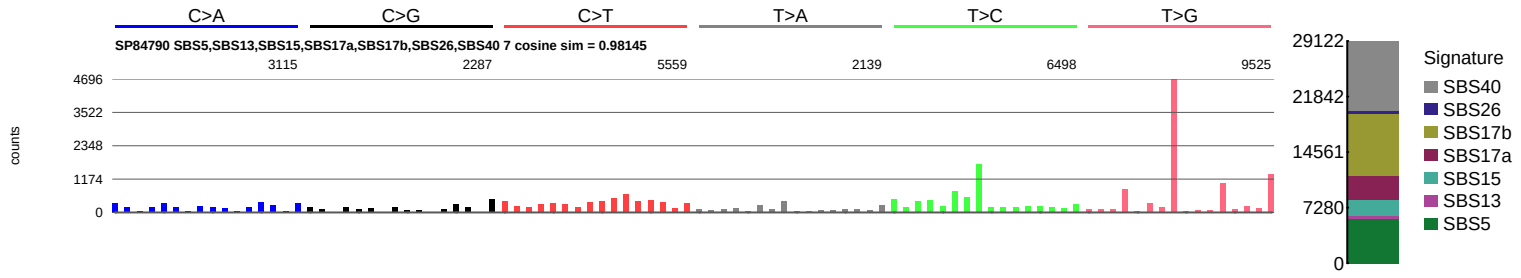

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98151
3179
1950
4524
2665
7292
9515
counts
4684
3513
2342
1171
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2

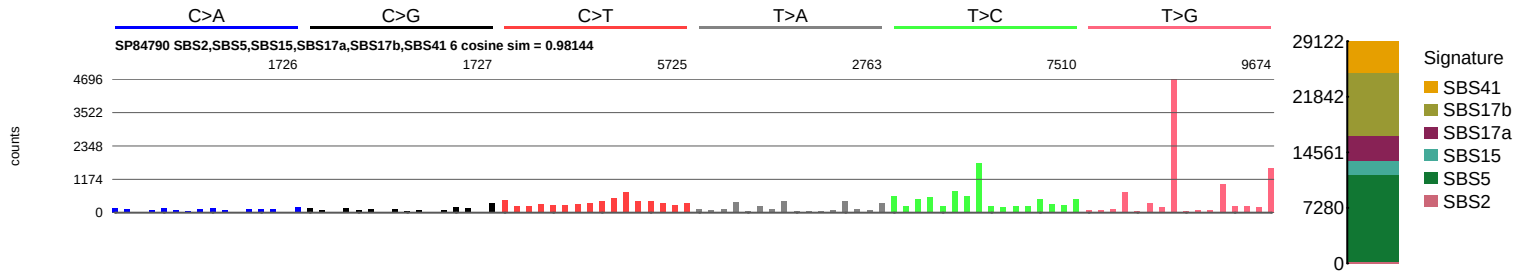

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.9815
3216
2300
4094
2908
6982
9621
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3

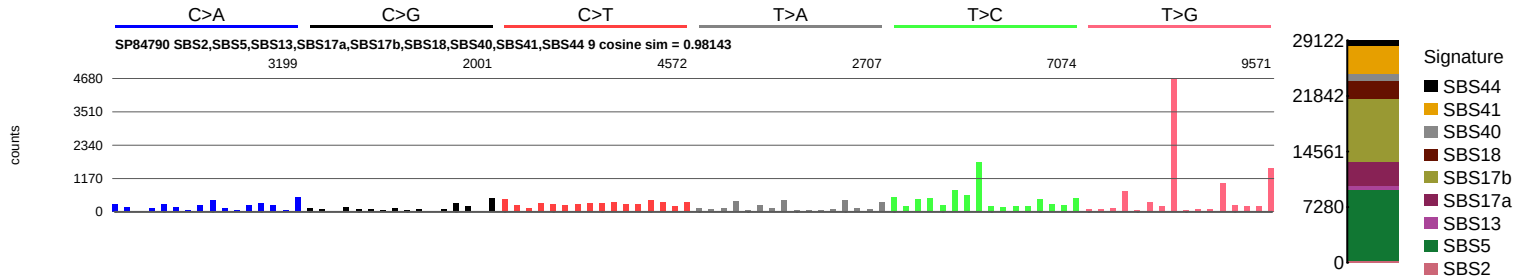

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98149
1716
1708
5691
2743
7610
9657
counts
4696
3522
2348
1174
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS2

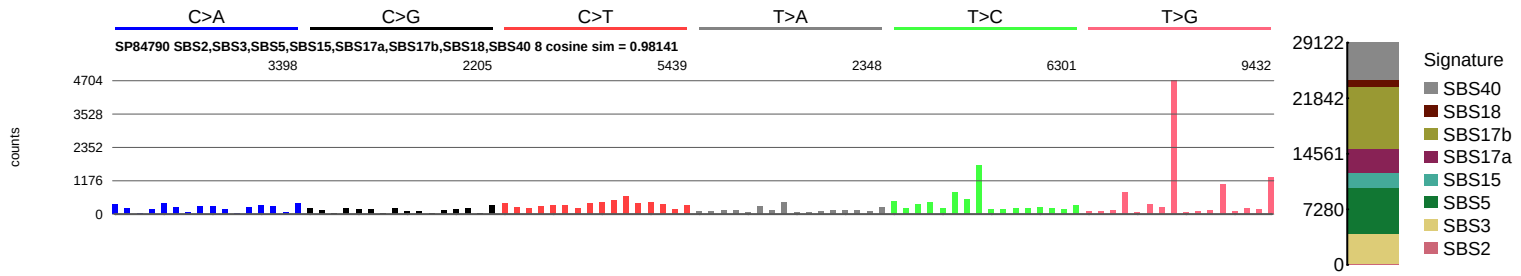

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98147
3399
1927
5484
2147
6670
9494
counts
4696
3522
2348
1174
0
29122
21842
14561
7280
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5

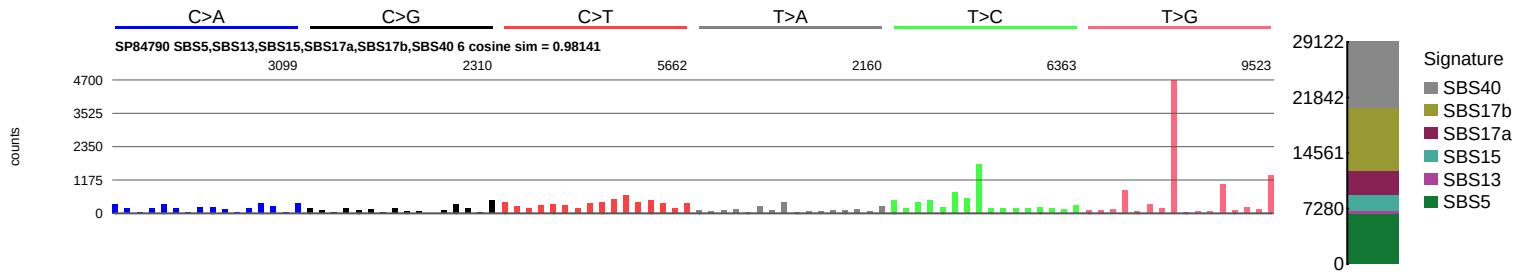

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98146
3403
2184
5356
2331
6413
9435
counts
4700
3525
2350
1175
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2

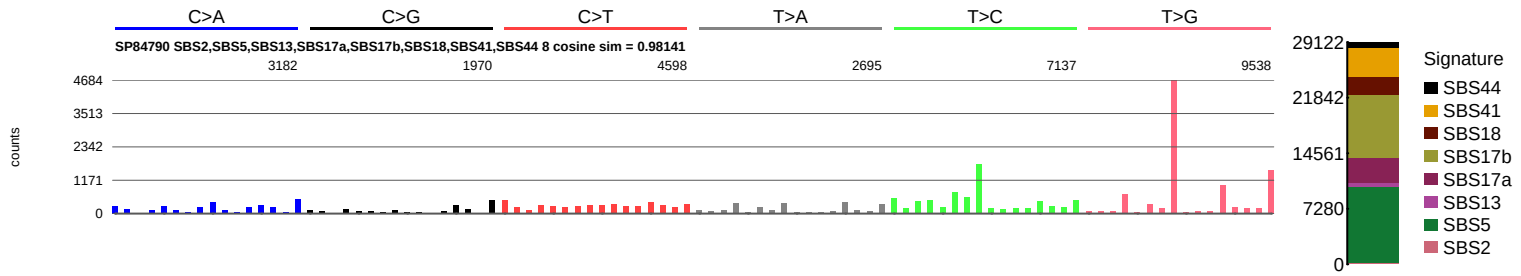

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40 7 cosine sim = 0.98145
3115
2287
5559
2139
6498
9525
counts
4696
3522
2348
1174
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5

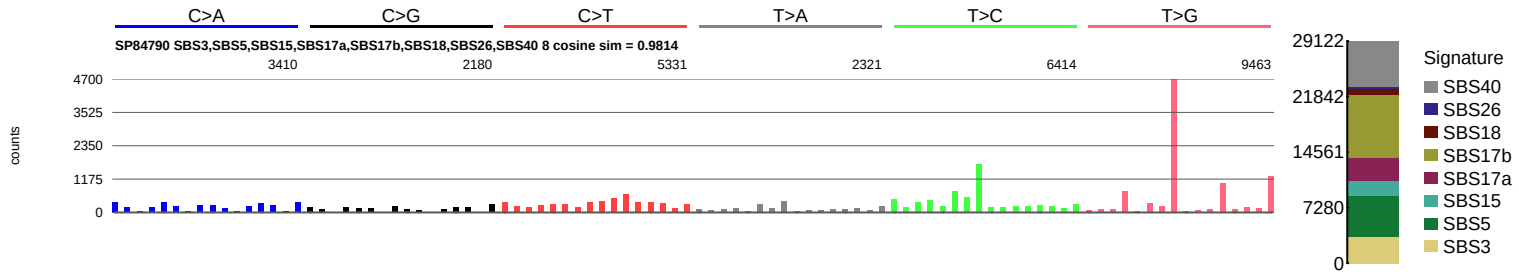

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98144
1726
1727
5725
2763
7510
9674
counts
4696
3522
2348
1174
0
29122
21842
14561
7280
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS2

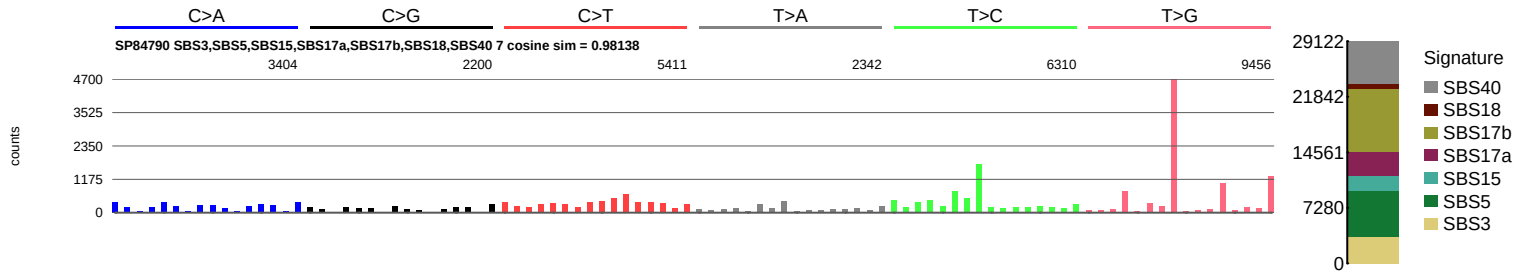

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98143
3199
2001
4572
2707
7074
9571
counts
4680
3510
2340
1170
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2

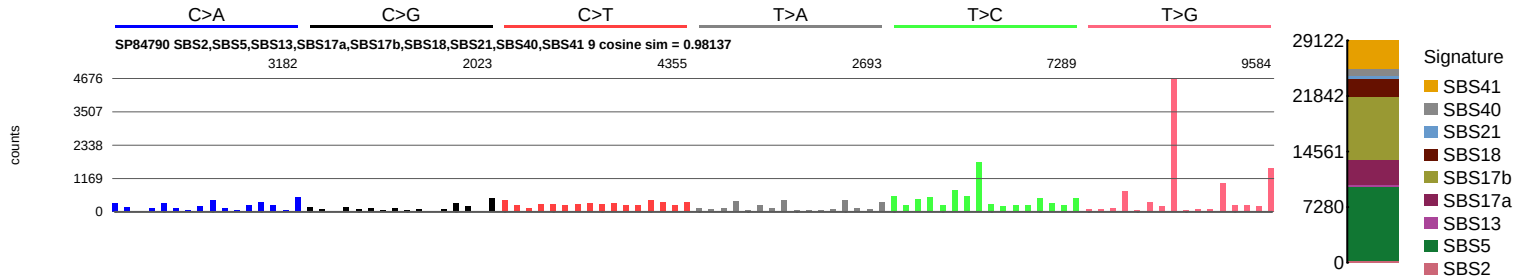

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98141
3398
2205
5439
2348
6301
9432
counts
4704
3528
2352
1176
0
29122
21842
14561
7280
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2

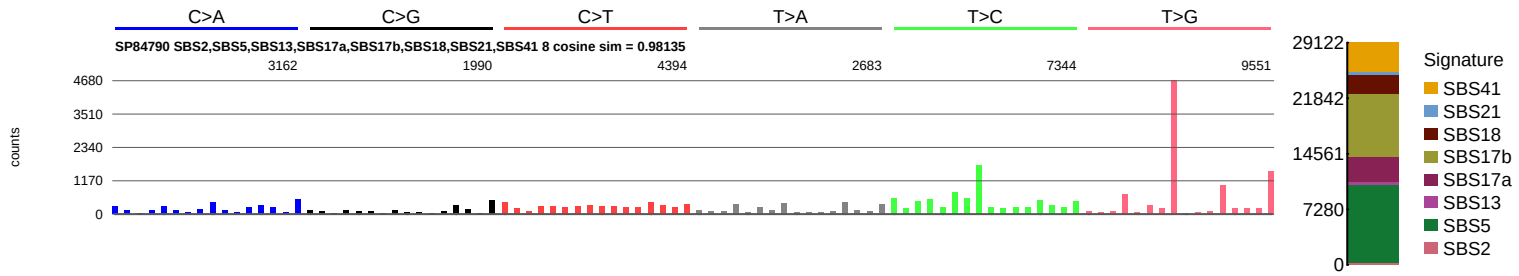

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40 6 cosine sim = 0.98141
3099
2310
5662
2160
6363
9523
counts
4700
3525
2350
1175
0
29122
21842
14561
7280
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS5

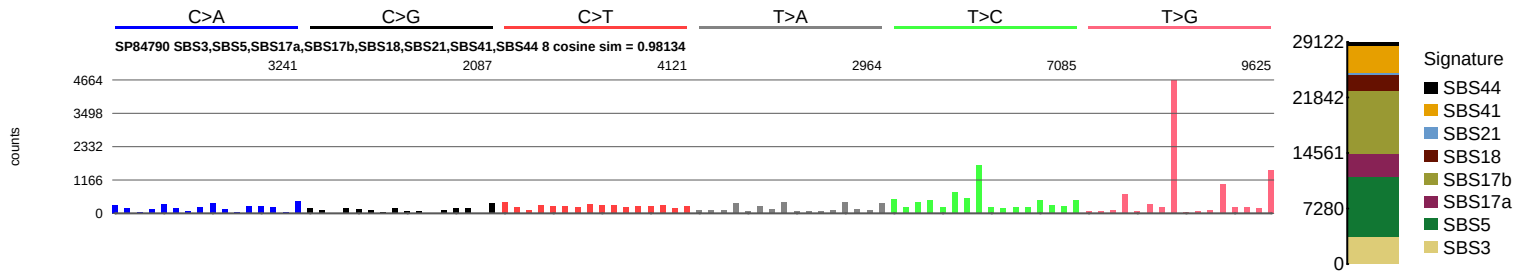

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98141
3182
1970
4598
2695
7137
9538
counts
4684
3513
2342
1171
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2

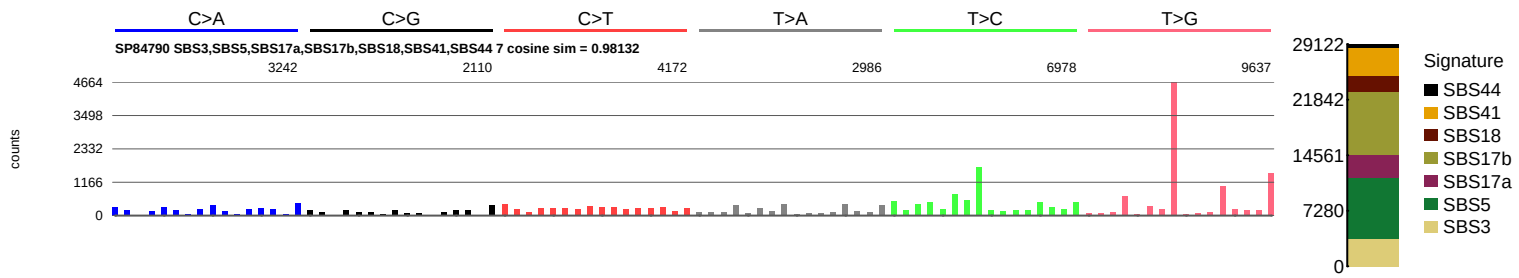

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.9814
3410
2180
5331
2321
6414
9463
counts
4700
3525
2350
1175
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3

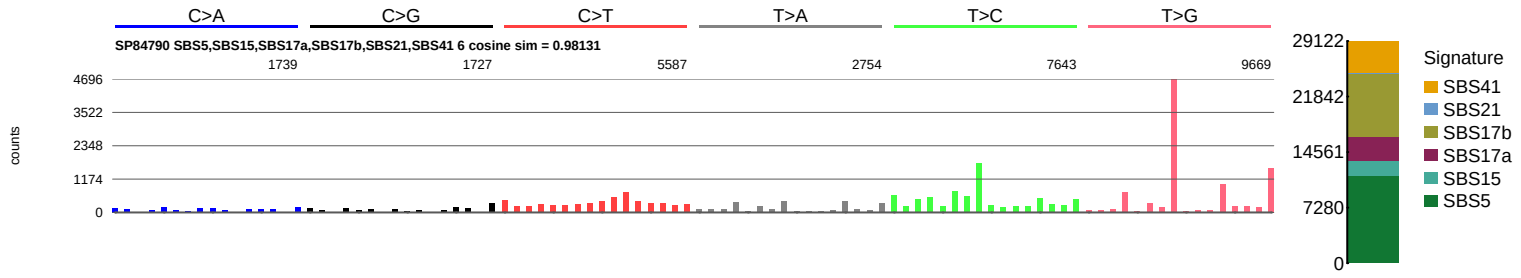

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98138
3404
2200
5411
2342
6310
9456
counts
4700
3525
2350
1175
0
29122
21842
14561
7280
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3

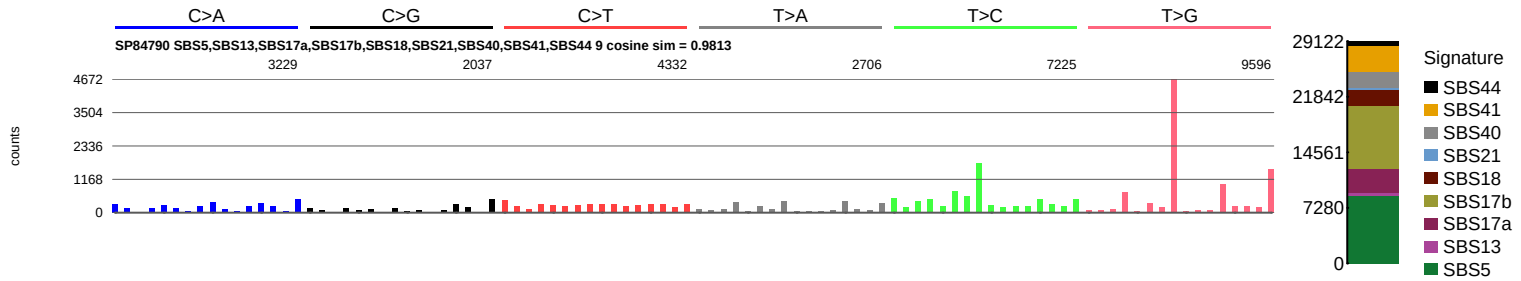

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98137
3182
2023
4355
2693
7289
9584
counts
4676
3507
2338
1169
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2

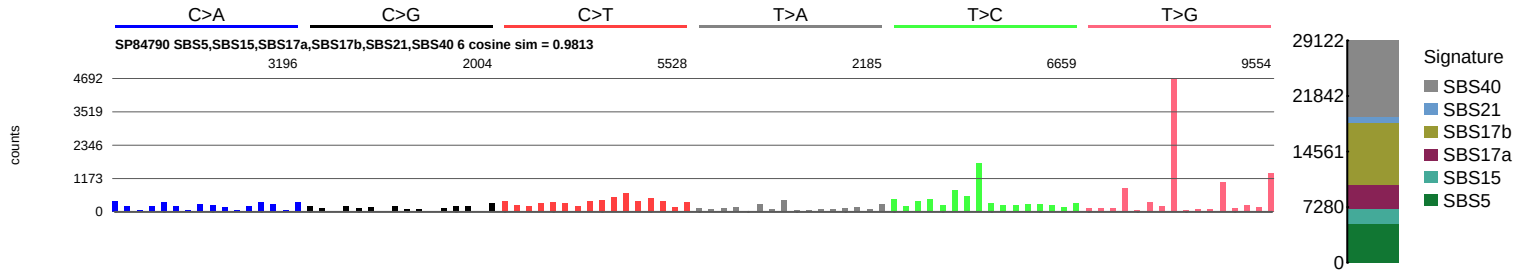

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98135
3162
1990
4394
2683
7344
9551
counts
4680
3510
2340
1170
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2

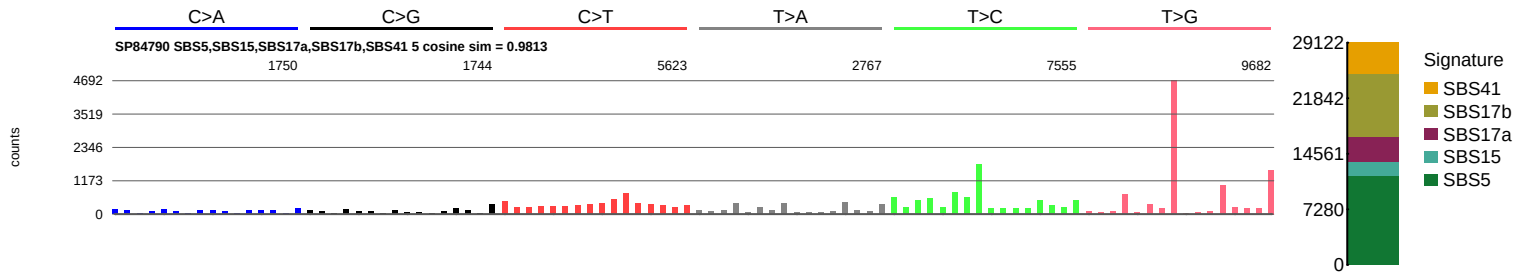

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98134
3241
2087
4121
2964
7085
9625
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3

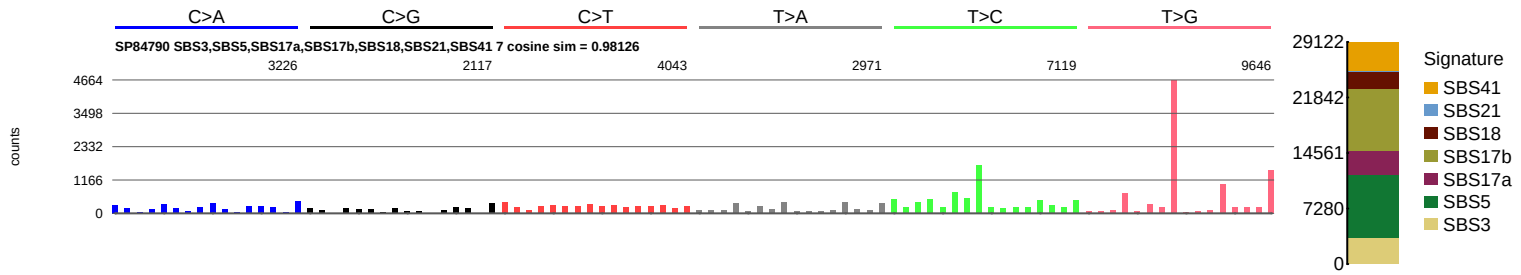

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.98132
3242
2110
4172
2986
6978
9637
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3

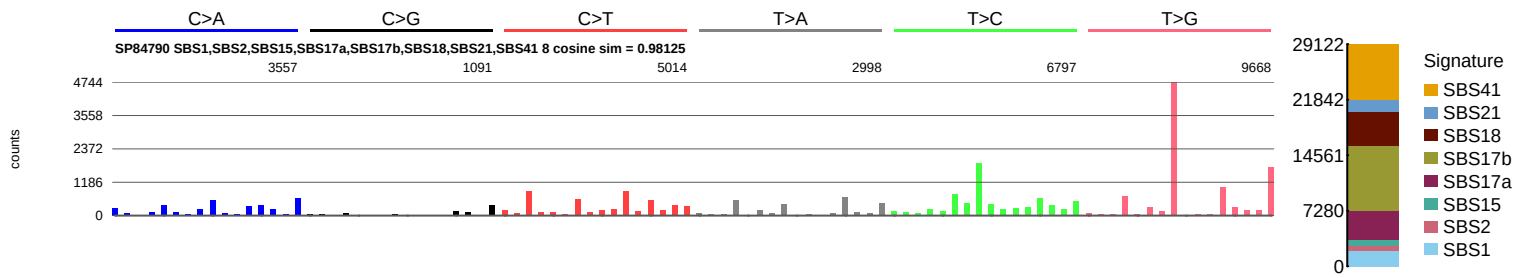

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.98131
1739
1727
5587
2754
7643
9669
counts
4696
3522
2348
1174
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5

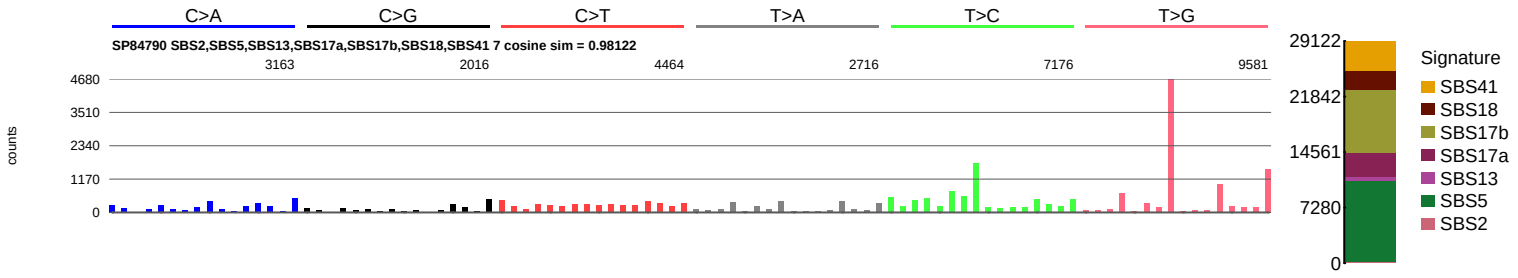

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.9813
3229
2037
4332
2706
7225
9596
counts
4672
3504
2336
1168
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5

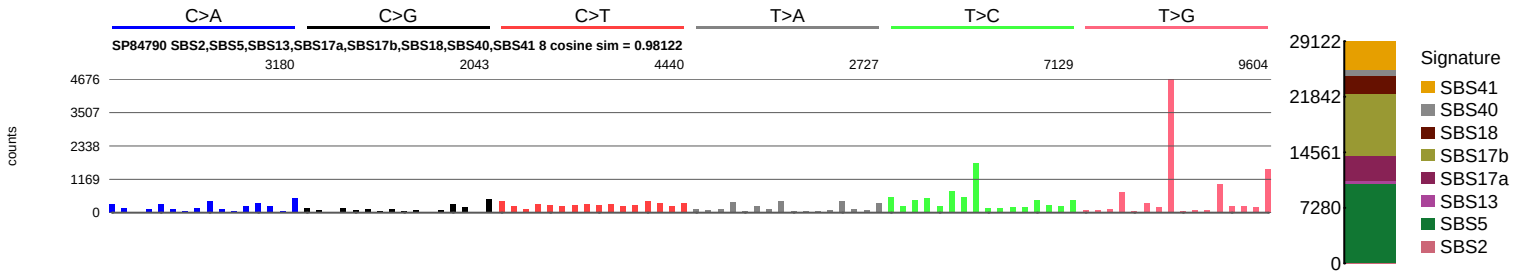

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40 6 cosine sim = 0.9813
3196
2004
5528
2185
6659
9554
counts
4692
3519
2346
1173
0
29122
21842
14561
7280
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5

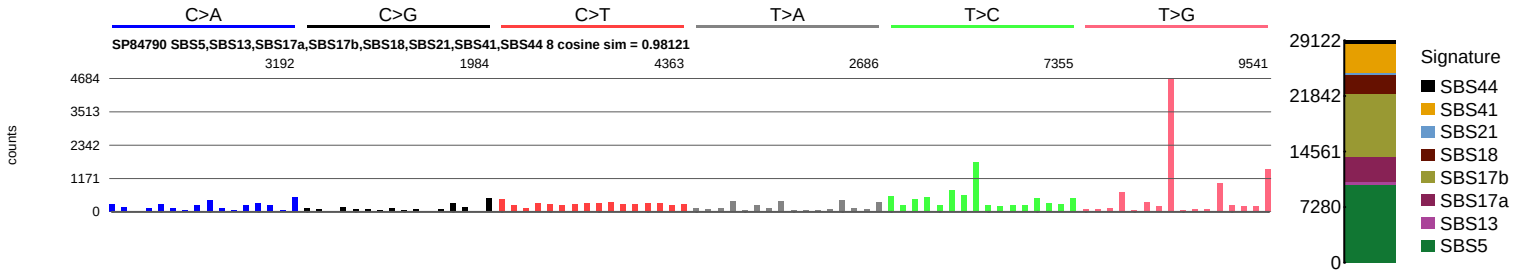

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS5,SBS15,SBS17a,SBS17b,SBS41 5 cosine sim = 0.9813
1750
1744
5623
2767
7555
9682
counts
4692
3519
2346
1173
0
29122
21842
14561
7280
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS5

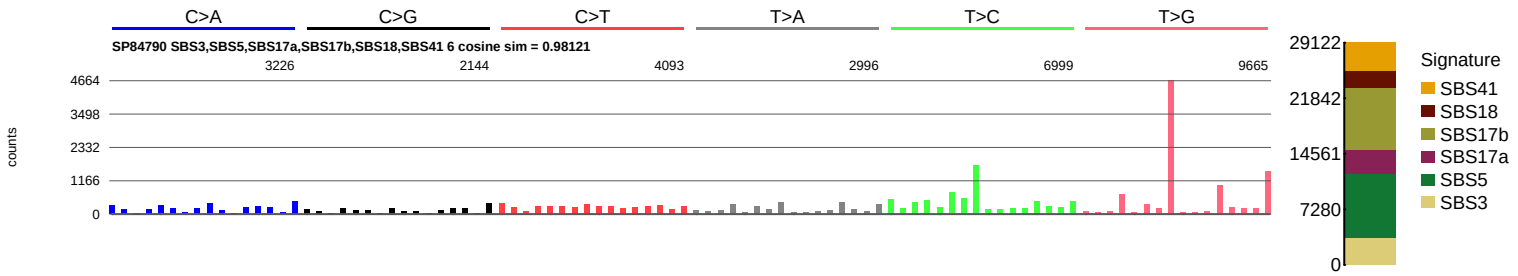

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98126
3226
2117
4043
2971
7119
9646
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3

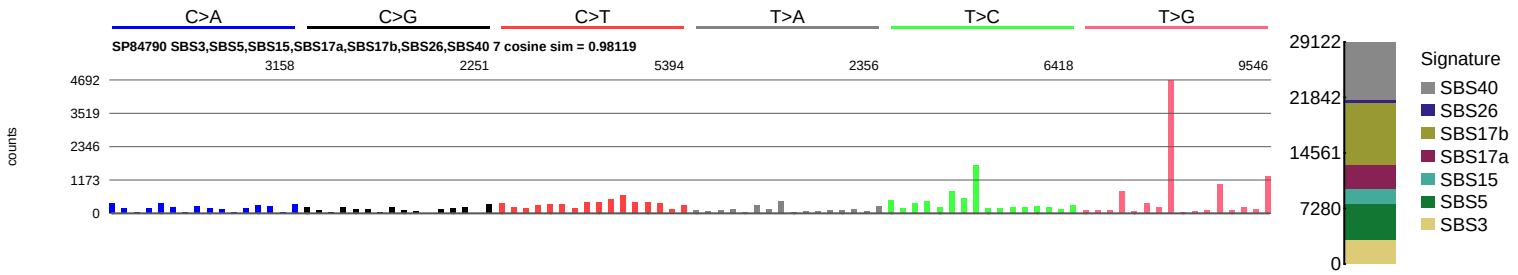

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98125
3557
1091
5014
2998
6797
9668
counts
4744
3558
2372
1186
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

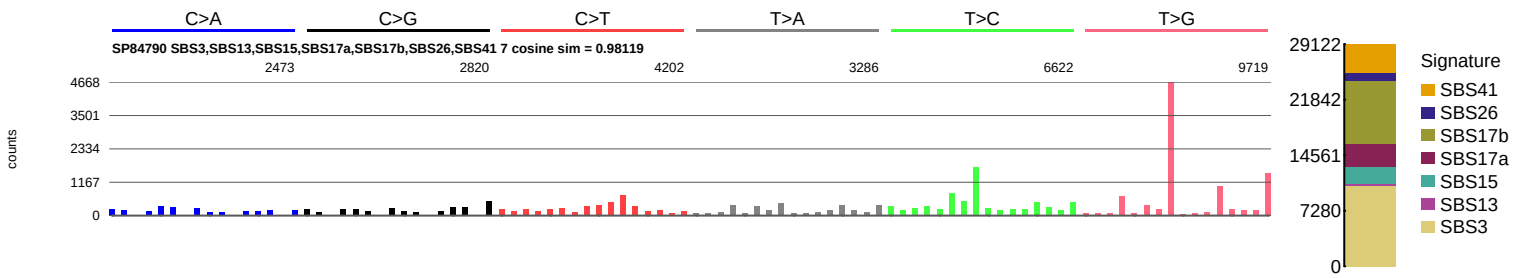

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98122
3163
2016
4464
2716
7176
9581
counts
4680
3510
2340
1170
0
29122
21842
14561
7280
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2

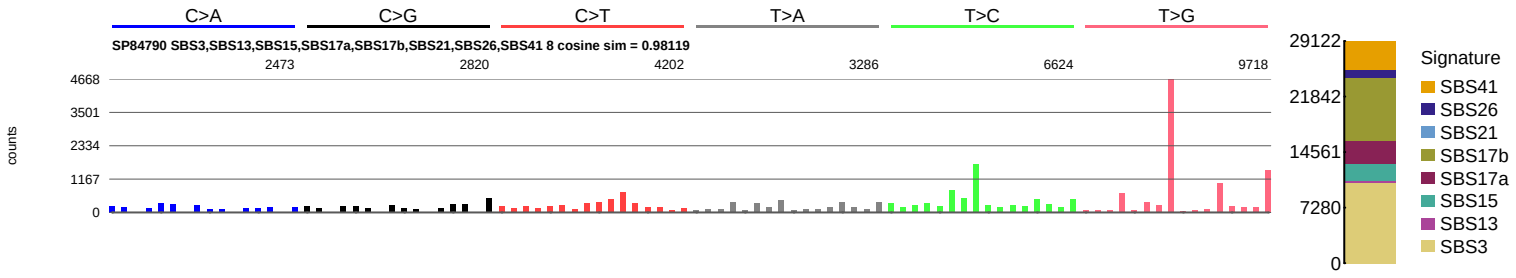

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98122
3180
2043
4440
2727
7129
9604
counts
4676
3507
2338
1169
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2

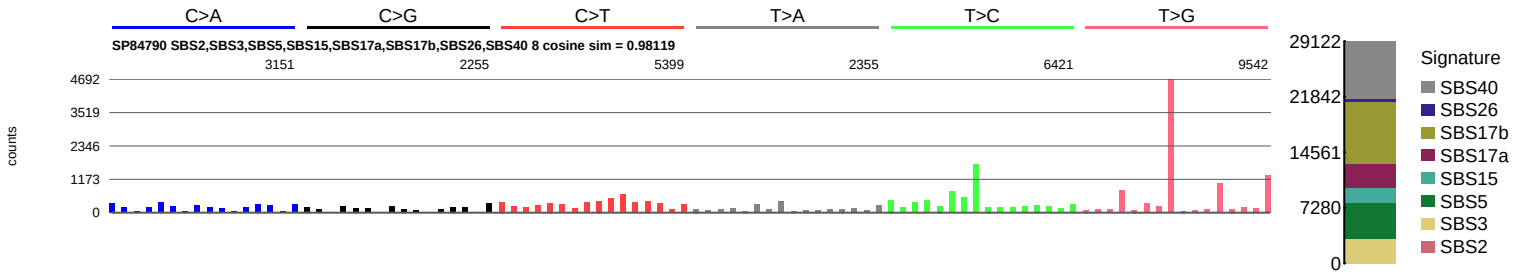

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98121
3192
1984
4363
2686
7355
9541
counts
4684
3513
2342
1171
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5

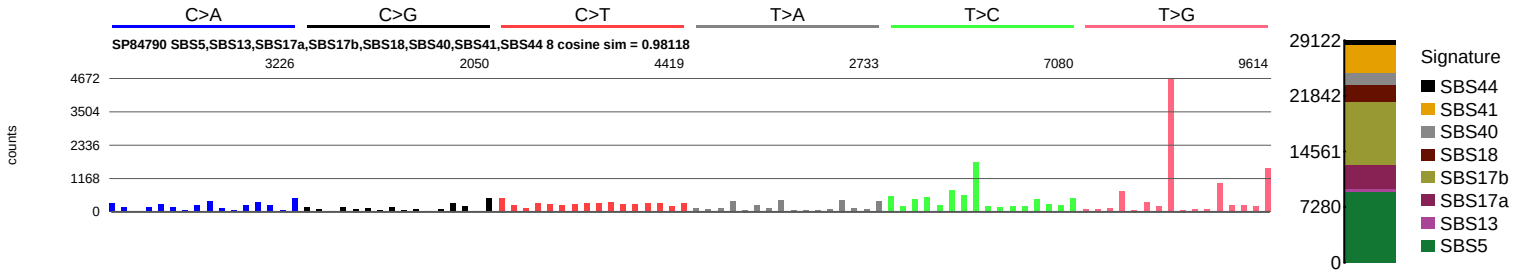

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98121
3226
2144
4093
2996
6999
9665
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3

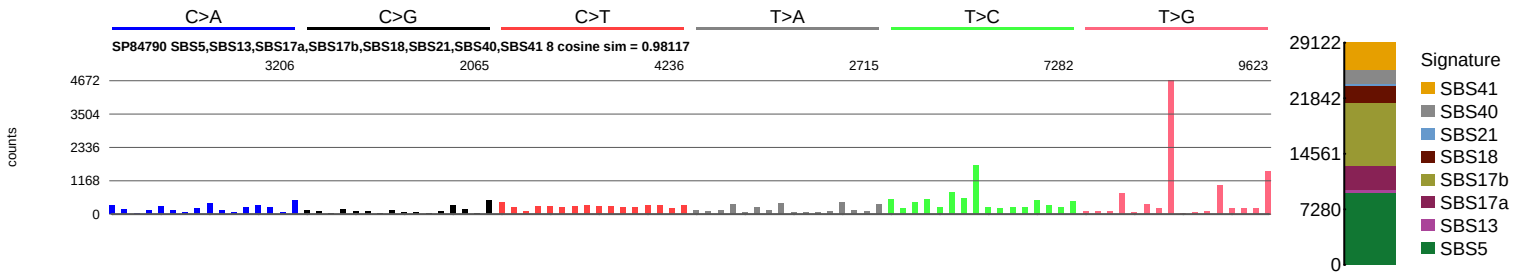

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS40 7 cosine sim = 0.98119
3158
2251
5394
2356
6418
9546
counts
4692
3519
2346
1173
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS3

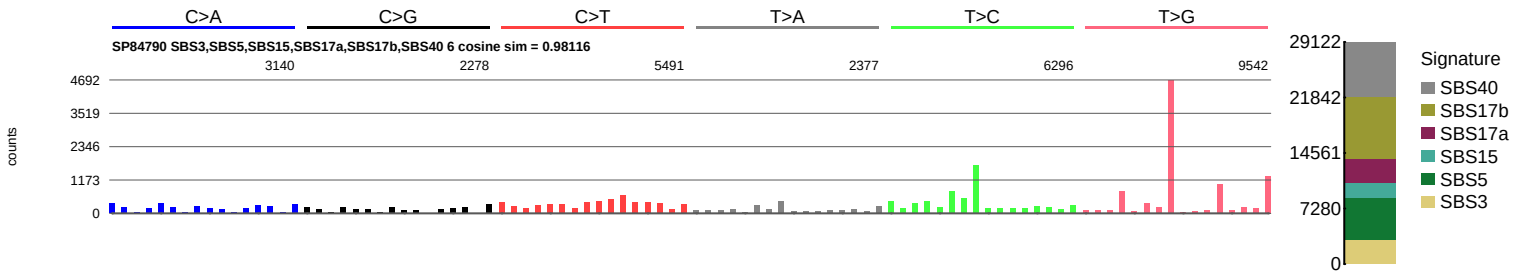

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41 7 cosine sim = 0.98119
2473
2820
4202
3286
6622
9719
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3

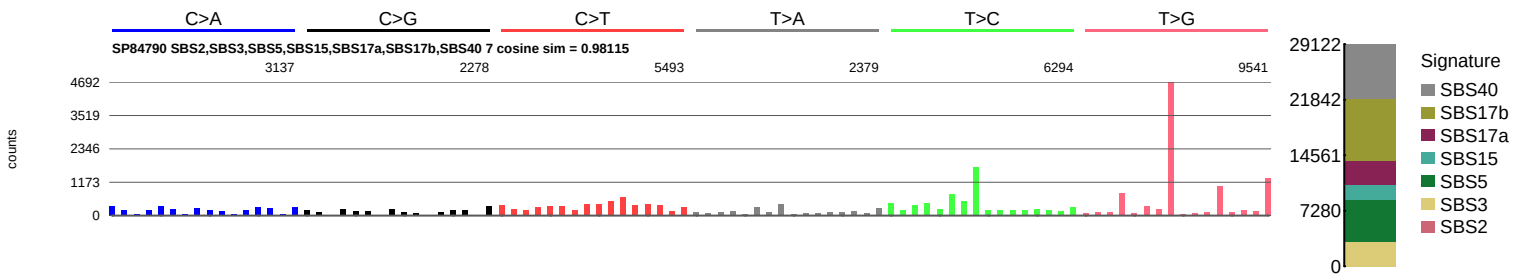

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41 8 cosine sim = 0.98119
2473
2820
4202
3286
6624
9718
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3

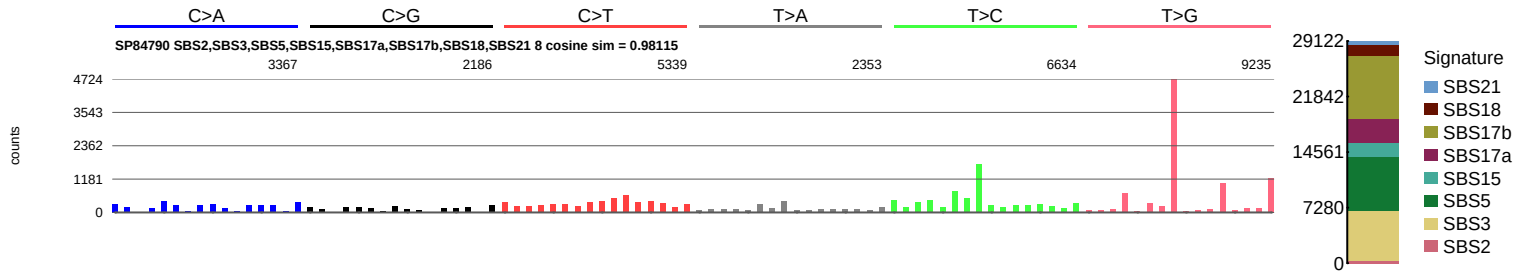

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS40 8 cosine sim = 0.98119
3151
2255
5399
2355
6421
9542
counts
4692
3519
2346
1173
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2

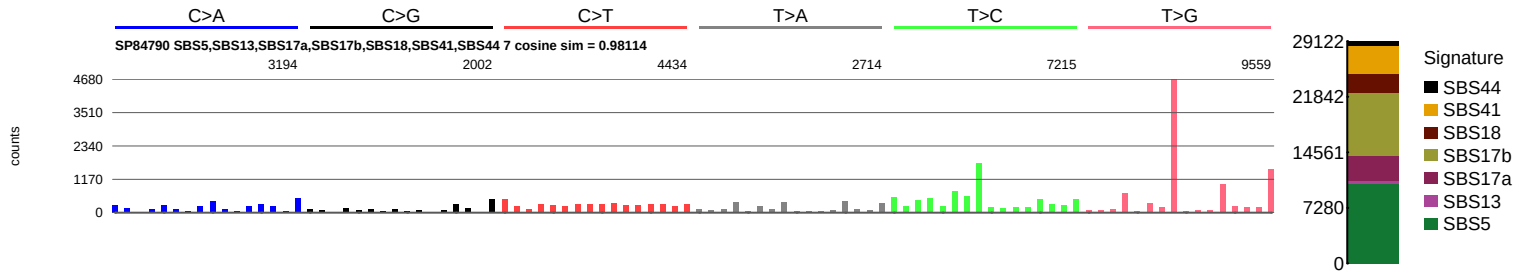

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98118
3226
2050
4419
2733
7080
9614
counts
4672
3504
2336
1168
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5

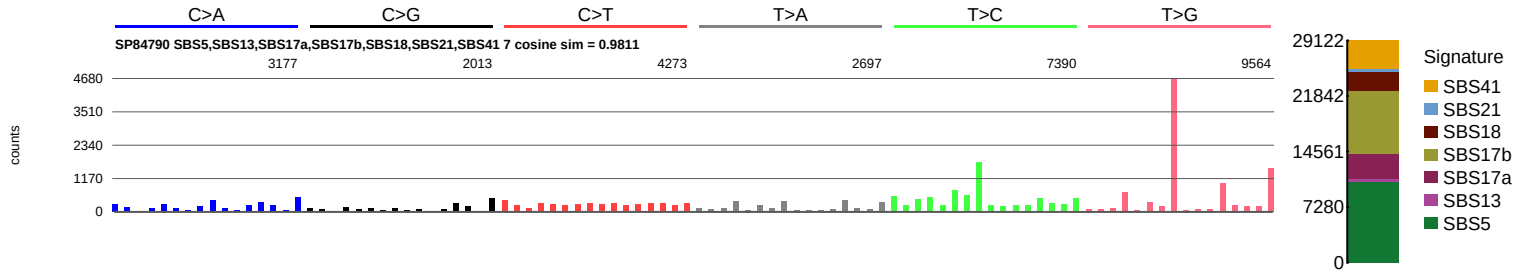

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98117
3206
2065
4236
2715
7282
9623
counts
4672
3504
2336
1168
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5

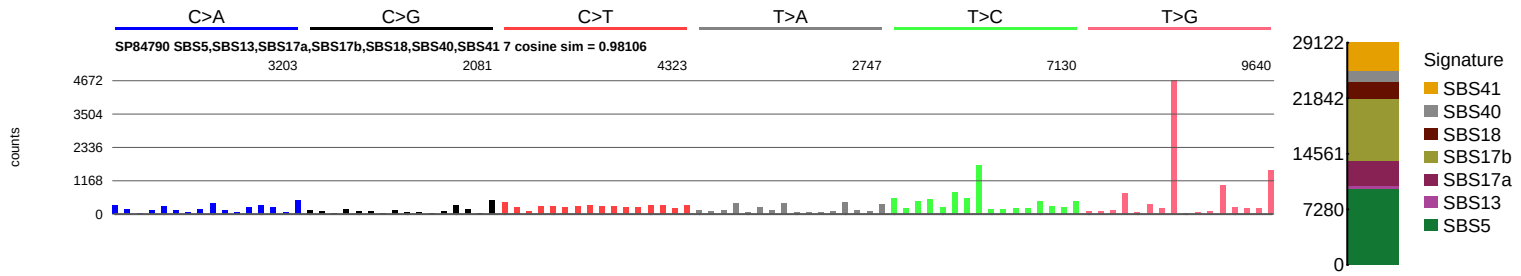

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS40 6 cosine sim = 0.98116
3140
2278
5491
2377
6296
9542
counts
4692
3519
2346
1173
0
29122
21842
14561
7280
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS3

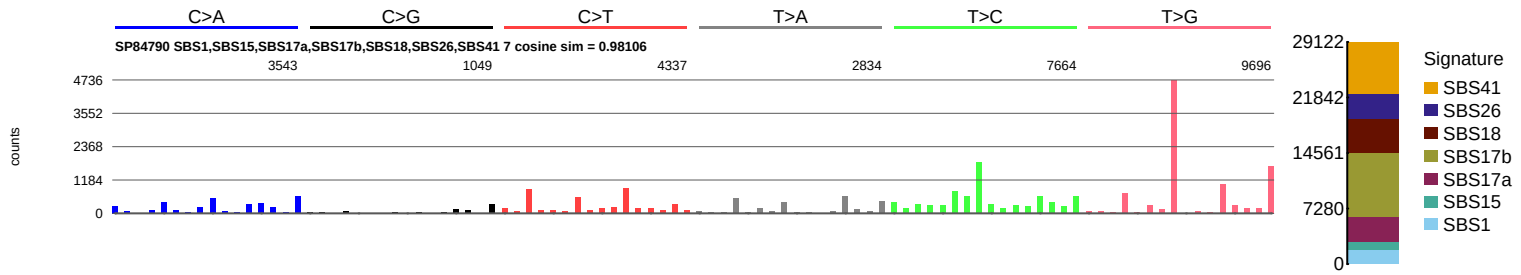

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS40 7 cosine sim = 0.98115
3137
2278
5493
2379
6294
9541
counts
4692
3519
2346
1173
0
29122
21842
14561
7280
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2

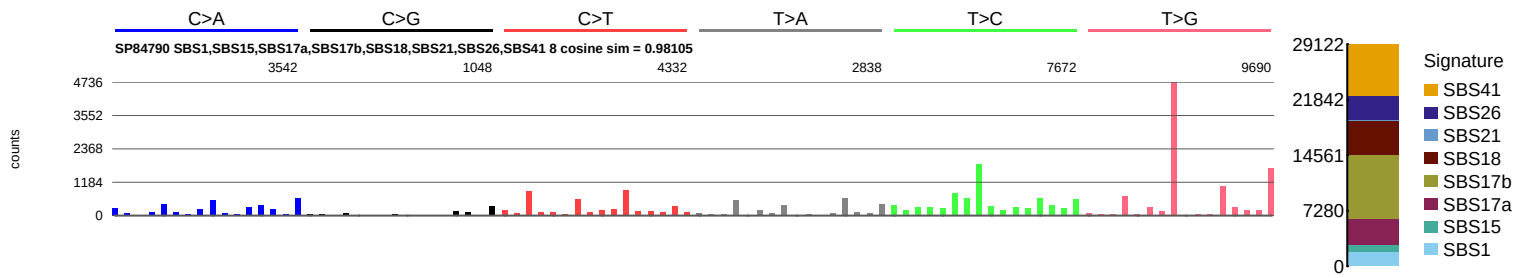

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98115
3367
2186
5339
2353
6634
9235
counts
4724
3543
2362
1181
0
29122
21842
14561
7280
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2

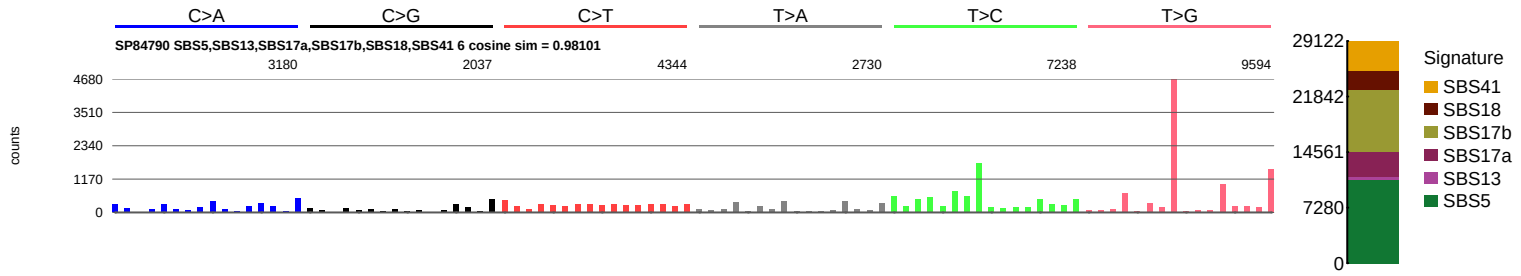

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.98114
3194
2002
4434
2714
7215
9559
counts
4680
3510
2340
1170
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5

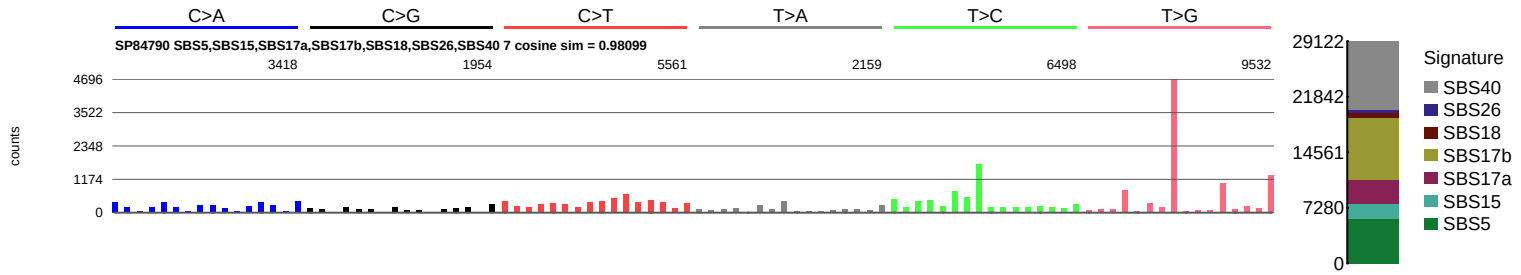

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.9811
3177
2013
4273
2697
7390
9564
counts
4680
3510
2340
1170
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5

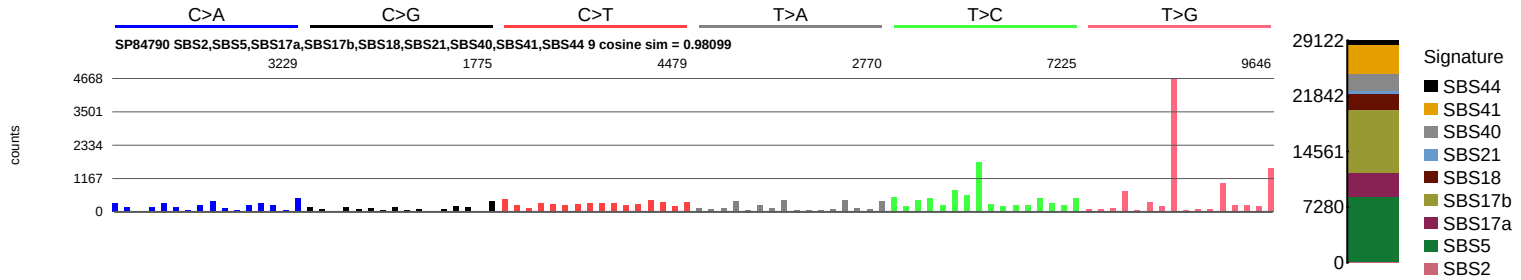

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98106
3203
2081
4323
2747
7130
9640
counts
4672
3504
2336
1168
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5

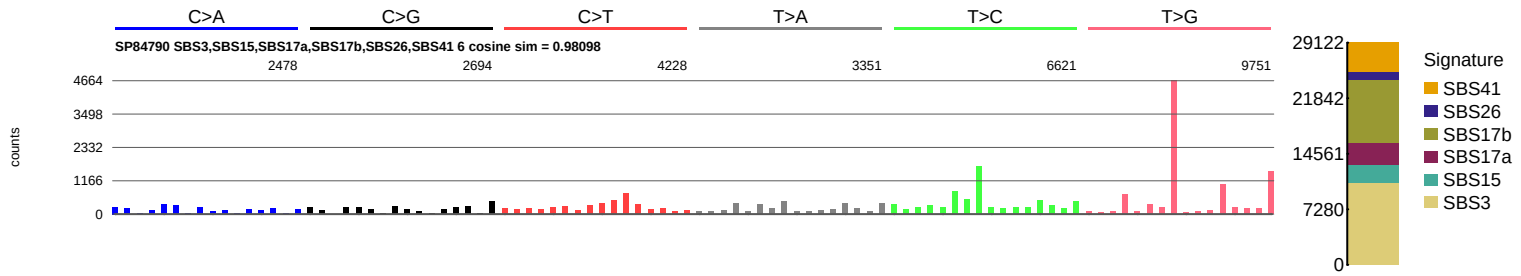

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.98106
3543
1049
4337
2834
7664
9696
counts
4736
3552
2368
1184
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1

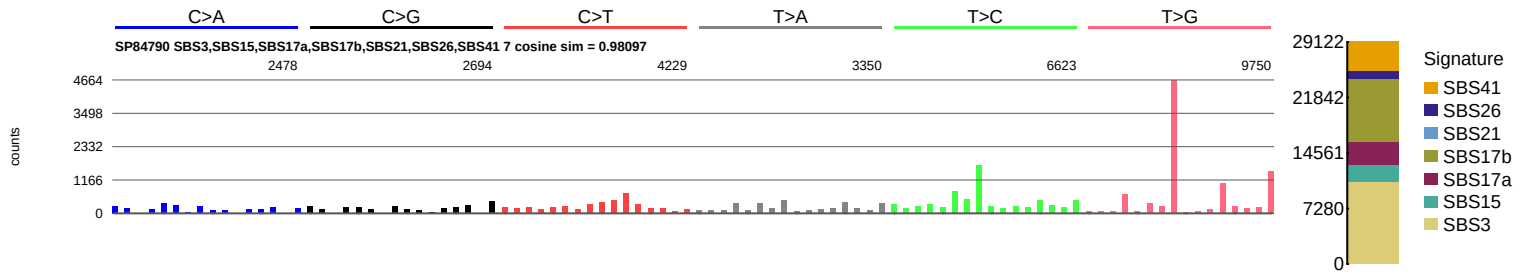

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 8 cosine sim = 0.98105
3542
1048
4332
2838
7672
9690
counts
4736
3552
2368
1184
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1

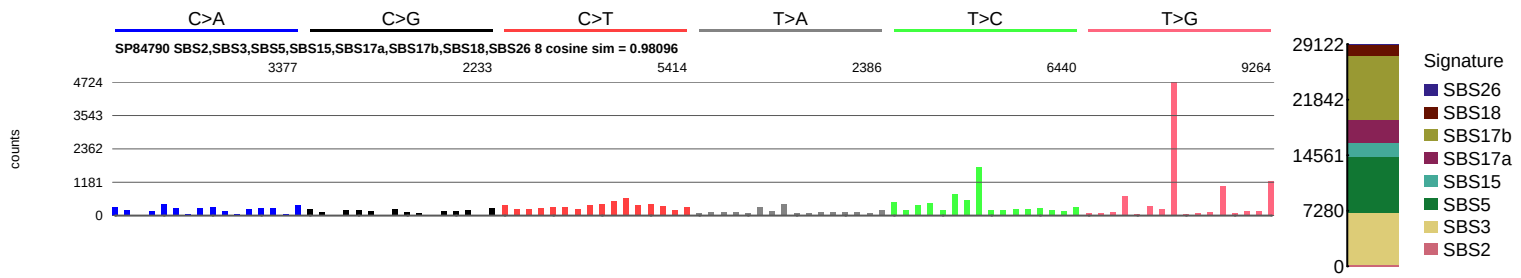

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98101
3180
2037
4344
2730
7238
9594
counts
4680
3510
2340
1170
0
29122
21842
14561
7280
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5

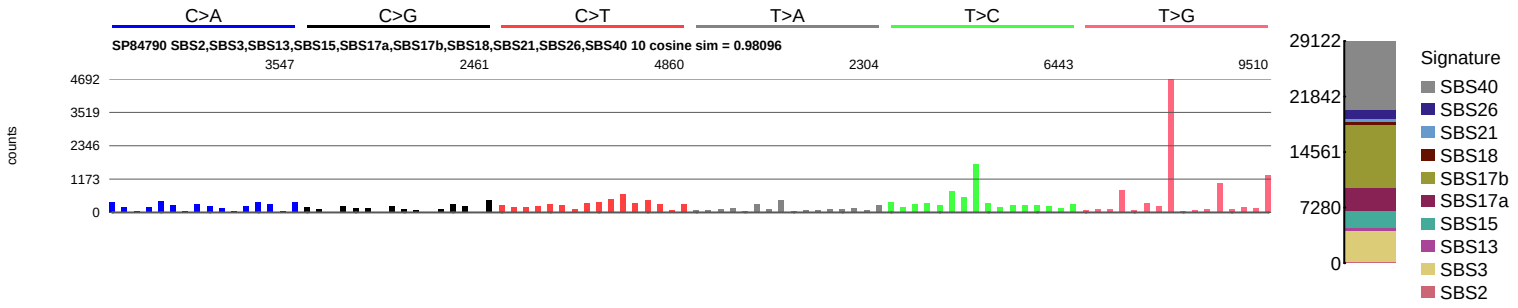

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98099
3418
1954
5561
2159
6498
9532
counts
4696
3522
2348
1174
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5

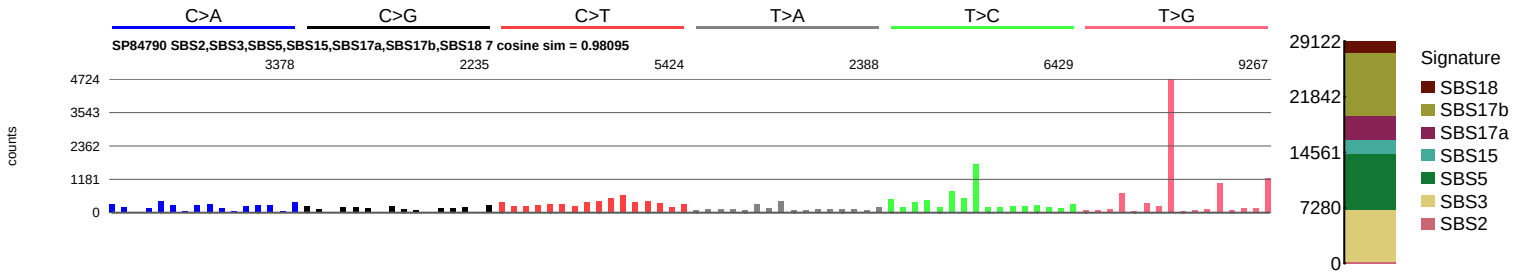

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98099
3229
1775
4479
2770
7225
9646
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2

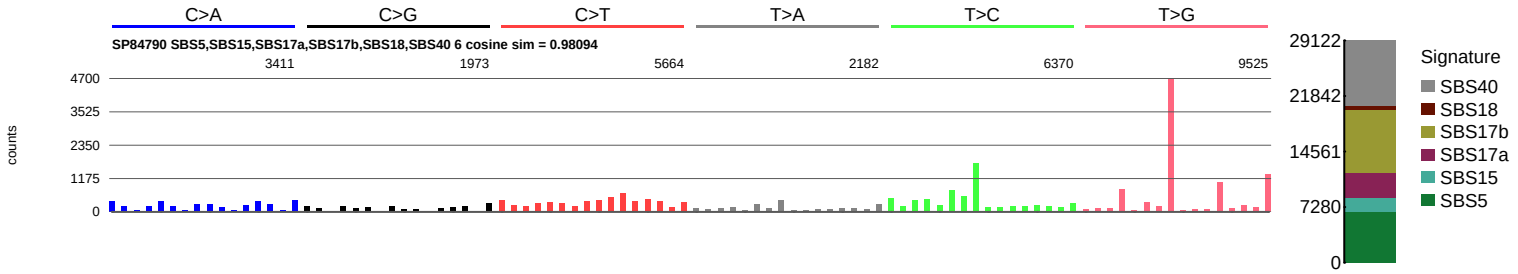

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS41 6 cosine sim = 0.98098
2478
2694
4228
3351
6621
9751
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS3

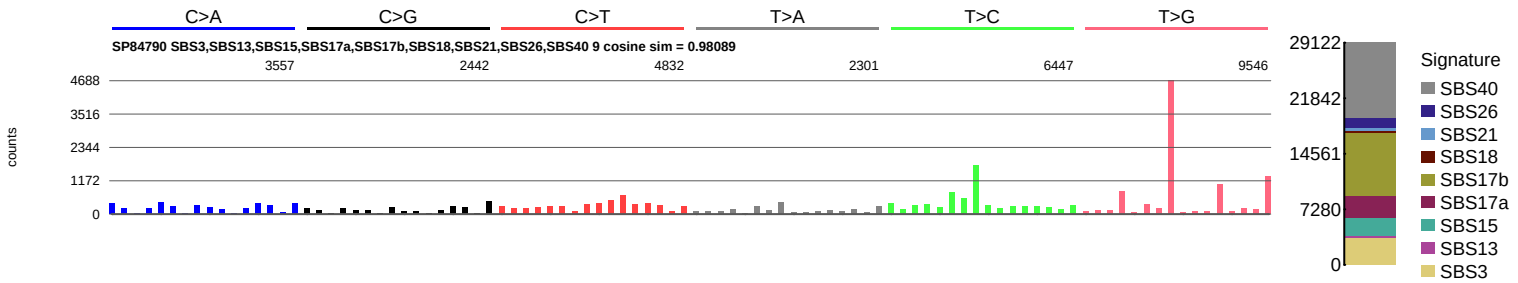

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41 7 cosine sim = 0.98097
2478
2694
4229
3350
6623
9750
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS3

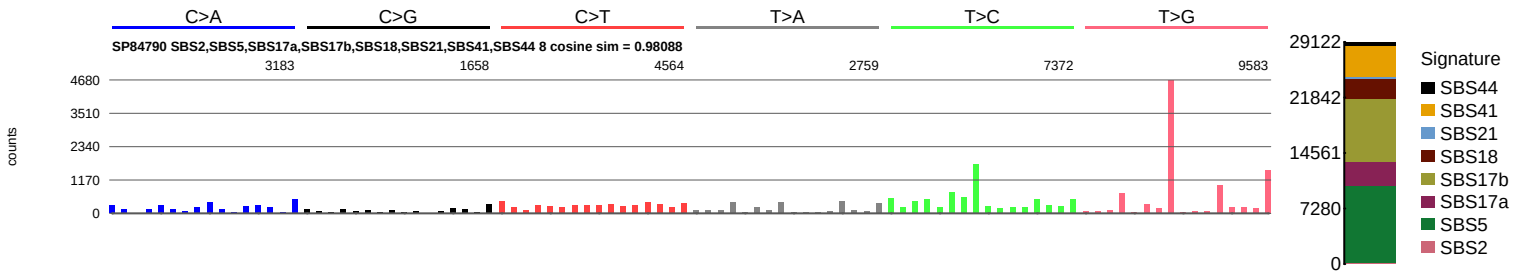

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98096
3377
2233
5414
2386
6440
9264
counts
4724
3543
2362
1181
0
29122
21842
14561
7280
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2

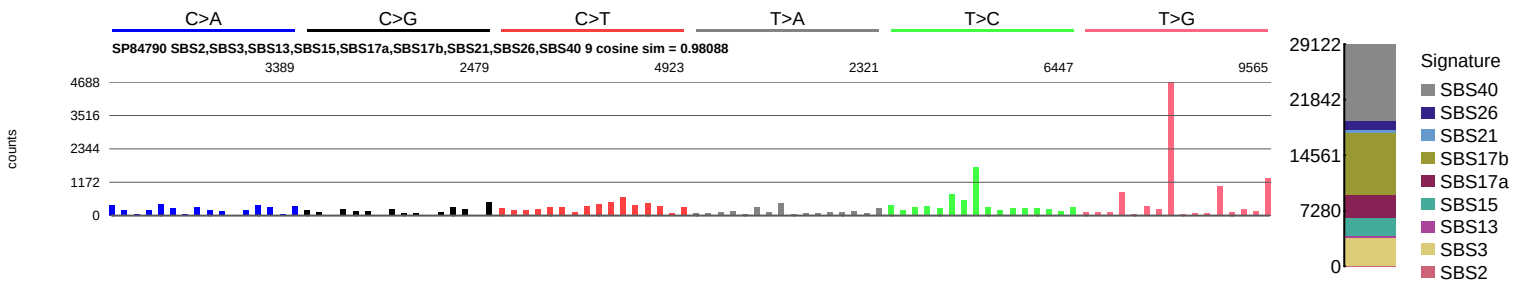

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 10 cosine sim = 0.98096
3547
2461
4860
2304
6443
9510
counts
4692
3519
2346
1173
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2

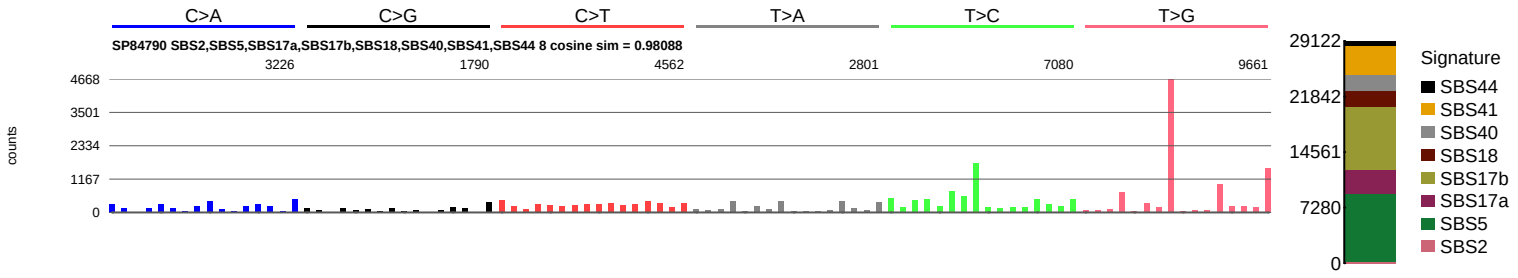

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98095
3378
2235
5424
2388
6429
9267
counts
4724
3543
2362
1181
0
29122
21842
14561
7280
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2

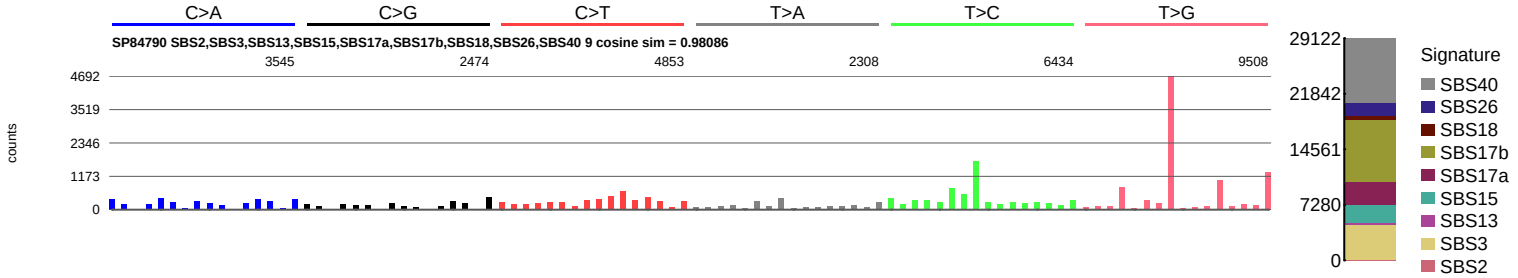

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40 6 cosine sim = 0.98094
3411
1973
5664
2182
6370
9525
counts
4700
3525
2350
1175
0
29122
21842
14561
7280
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5

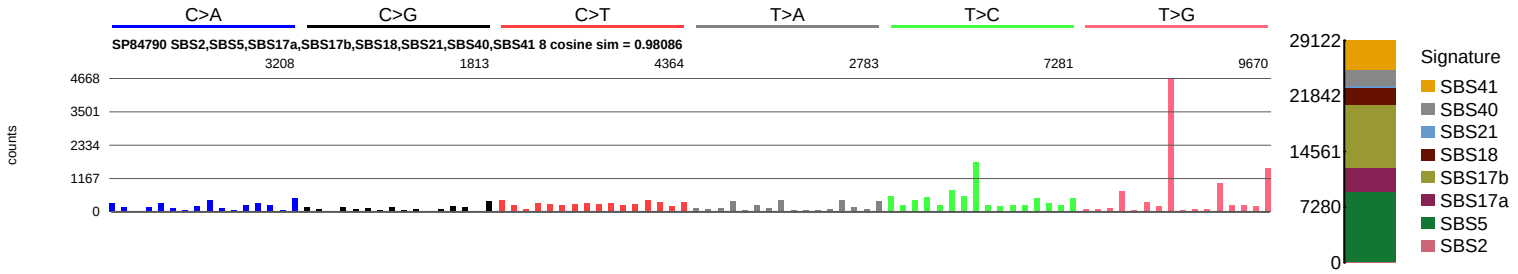

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98089
3557
2442
4832
2301
6447
9546
counts
4688
3516
2344
1172
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3

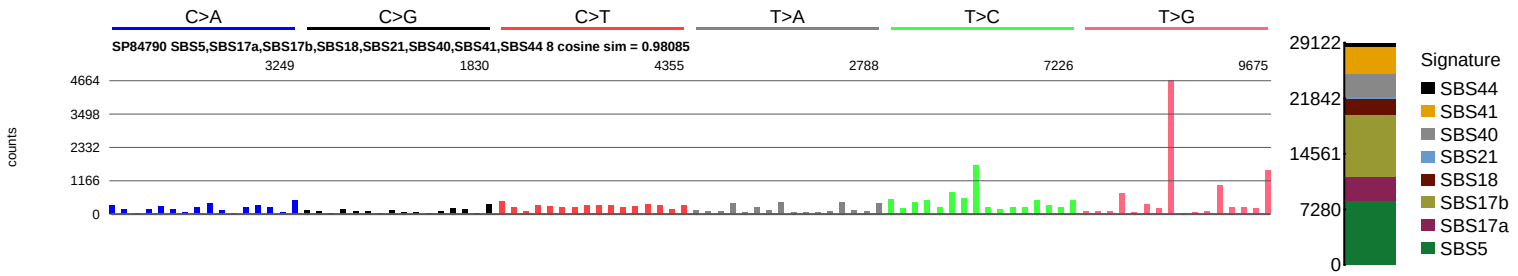

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98088
3183
1658
4564
2759
7372
9583
counts
4680
3510
2340
1170
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2

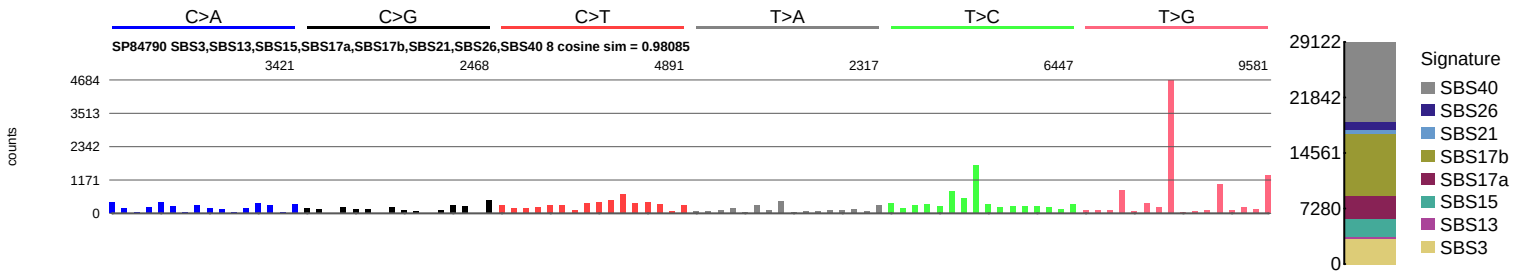

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40 9 cosine sim = 0.98088
3389
2479
4923
2321
6447
9565
counts
4688
3516
2344
1172
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2

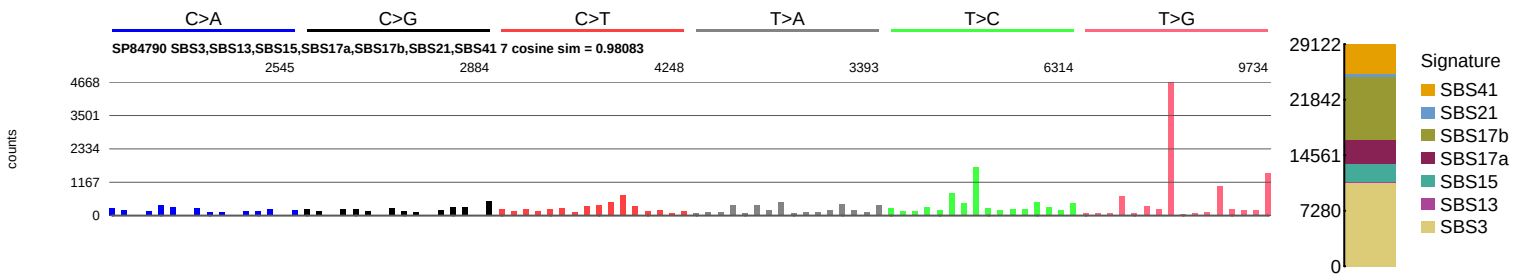

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98088
3226
1790
4562
2801
7080
9661
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2

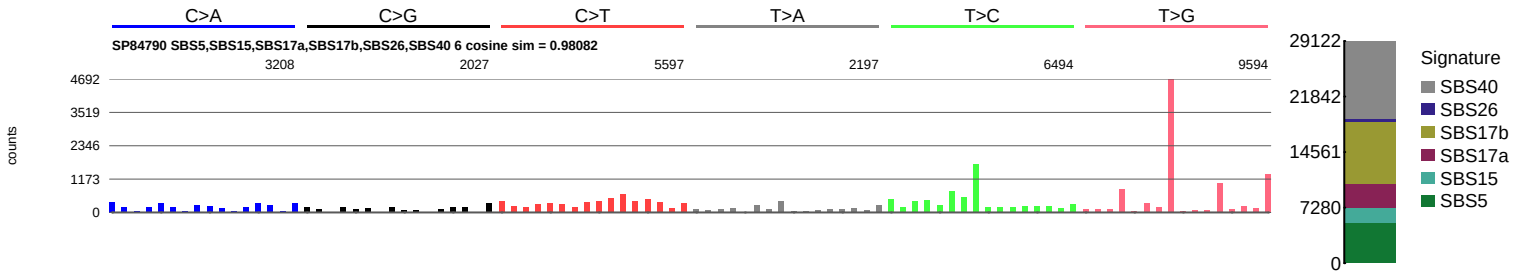

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98086
3545
2474
4853
2308
6434
9508
counts
4692
3519
2346
1173
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2

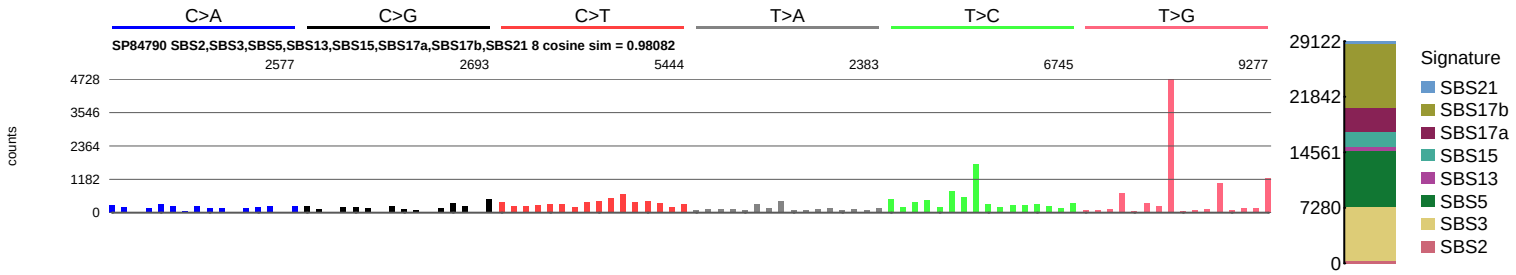

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98086
3208
1813
4364
2783
7281
9670
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2

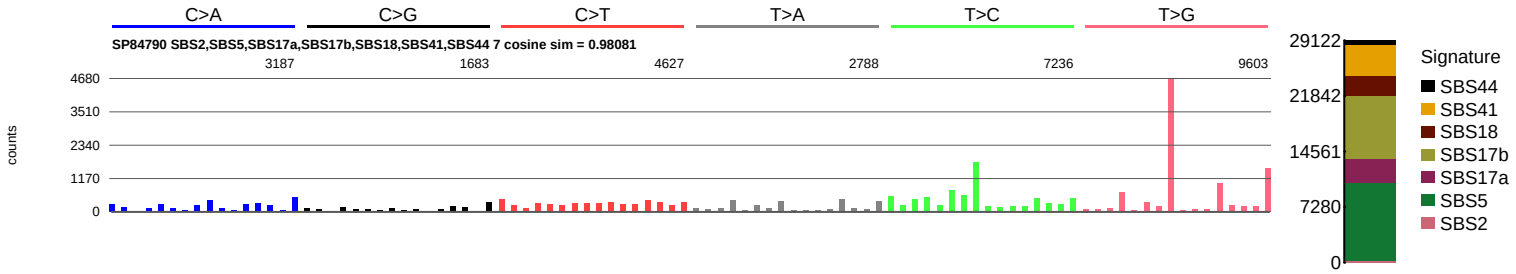

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98085
3249
1830
4355
2788
7226
9675
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5

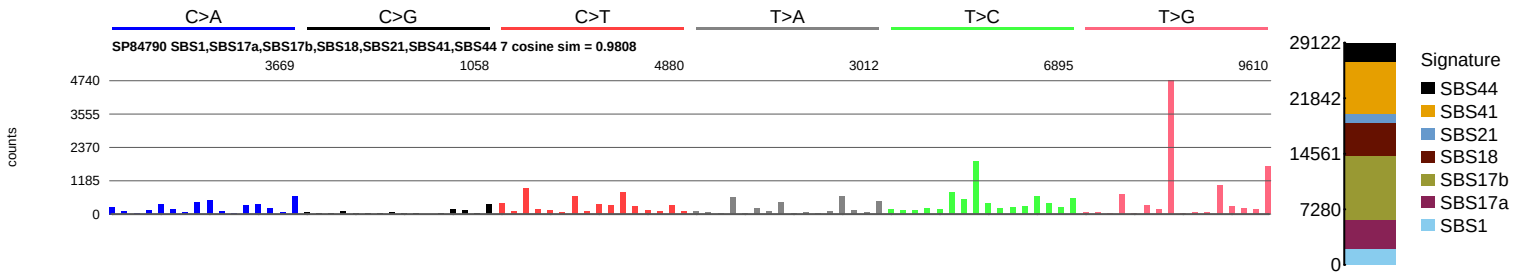

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40 8 cosine sim = 0.98085
3421
2468
4891
2317
6447
9581
counts
4684
3513
2342
1171
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3

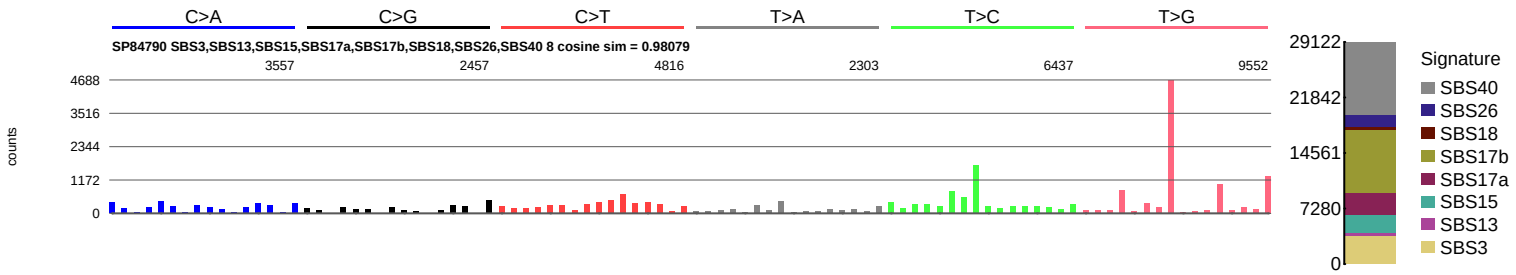

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98083
2545
2884
4248
3393
6314
9734
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3

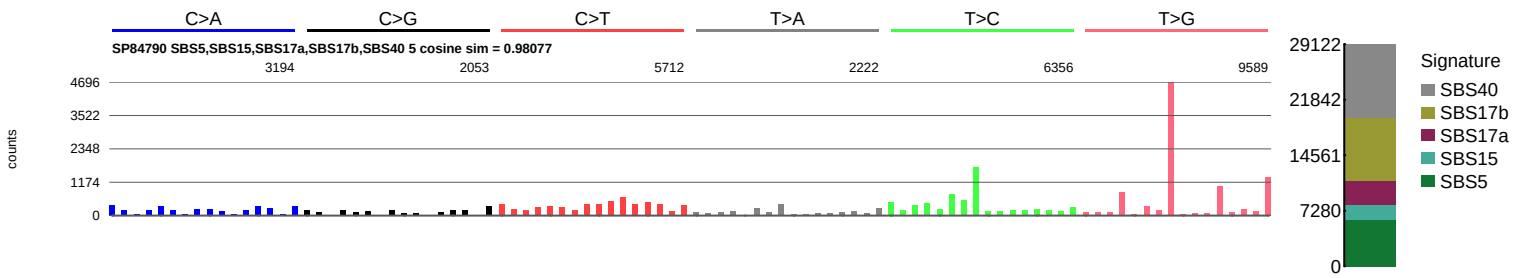

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS40 6 cosine sim = 0.98082
3208
2027
5597
2197
6494
9594
counts
4692
3519
2346
1173
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS5

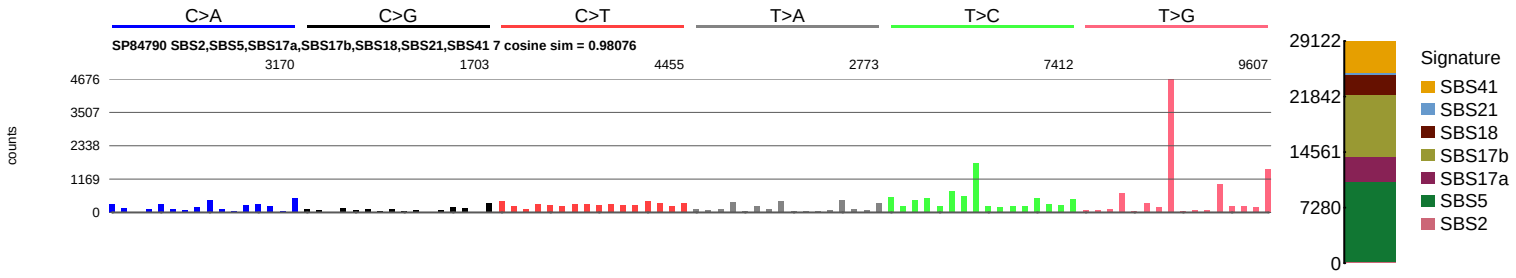

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21 8 cosine sim = 0.98082
2577
2693
5444
2383
6745
9277
counts
4728
3546
2364
1182
0
29122
21842
14561
7280
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2

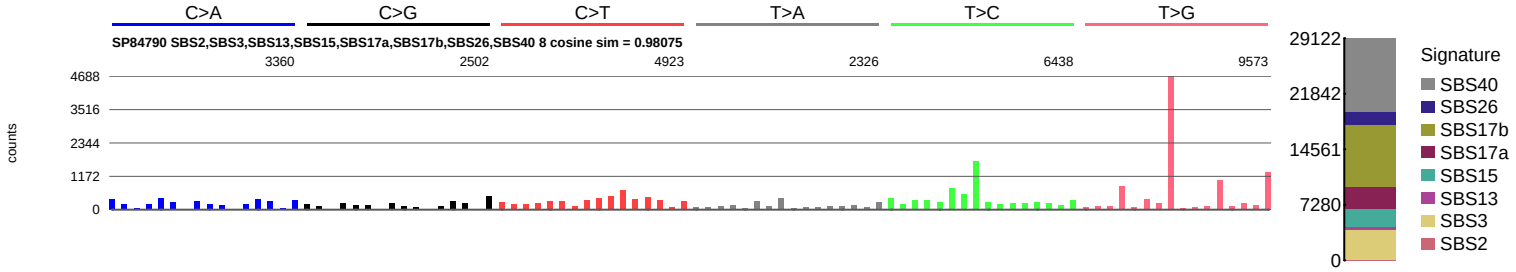

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.98081
3187
1683
4627
2788
7236
9603
counts
4680
3510
2340
1170
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS2

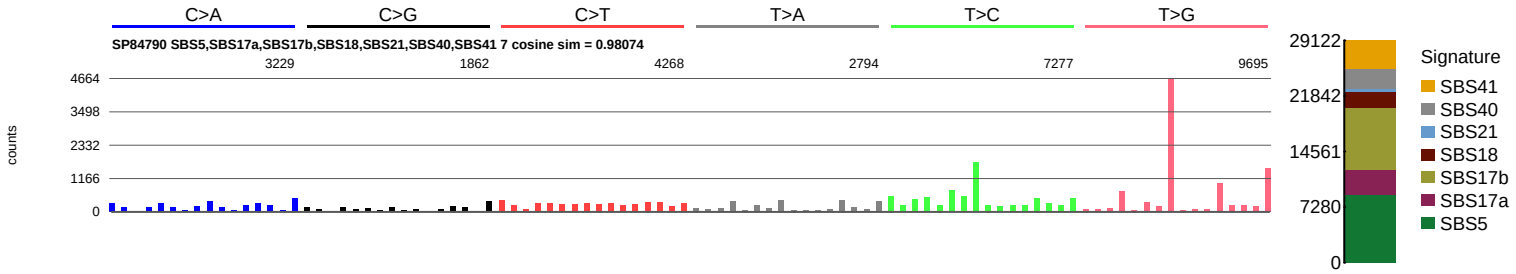

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 7 cosine sim = 0.9808
3669
1058
4880
3012
6895
9610
counts
4740
3555
2370
1185
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS1

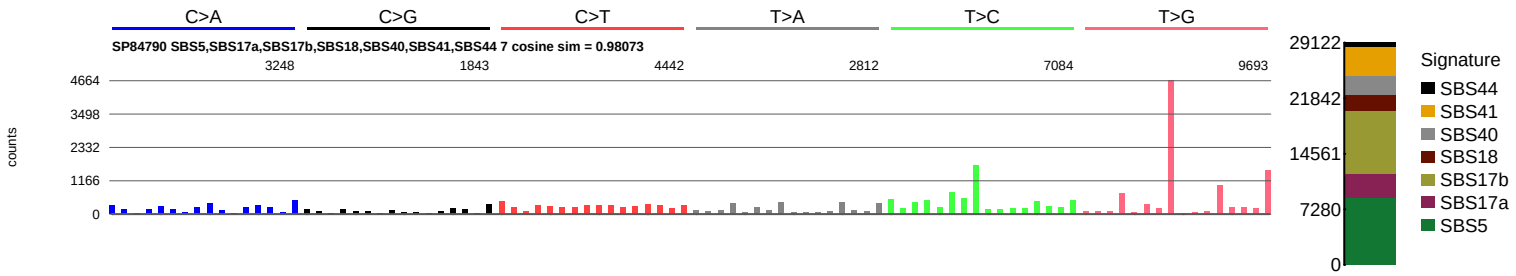

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98079
3557
2457
4816
2303
6437
9552
counts
4688
3516
2344
1172
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3

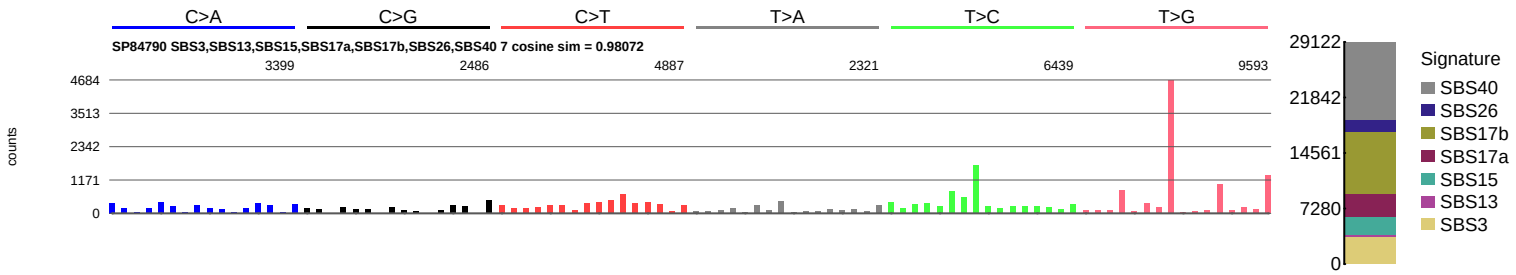

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS5,SBS15,SBS17a,SBS17b,SBS40 5 cosine sim = 0.98077
3194
2053
5712
2222
6356
9589
counts
4696
3522
2348
1174
0
29122
21842
14561
7280
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS5

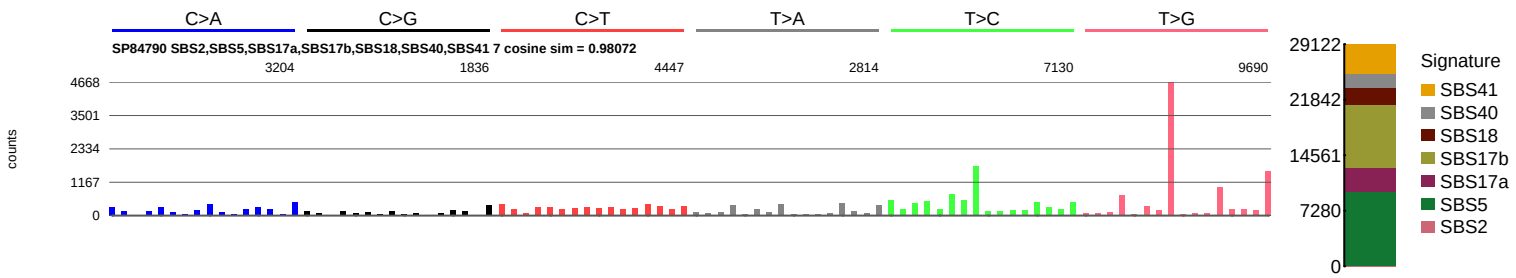

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98076
3170
1703
4455
2773
7412
9607
counts
4676
3507
2338
1169
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2

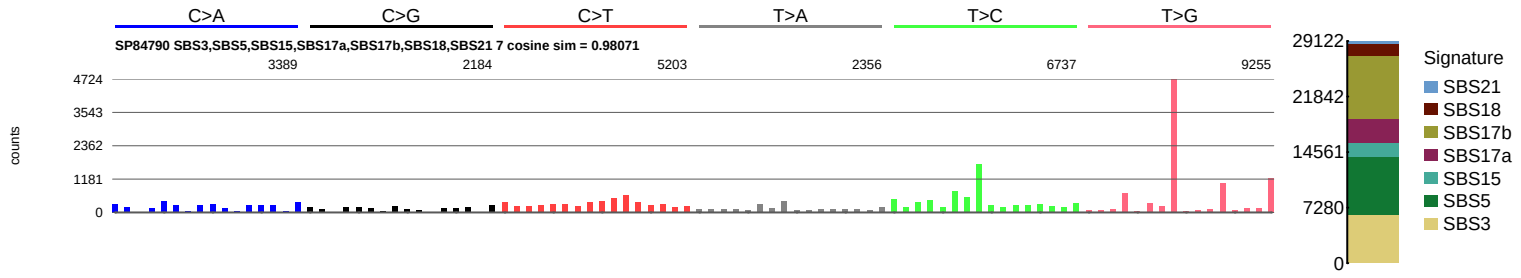

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40 8 cosine sim = 0.98075
3360
2502
4923
2326
6438
9573
counts
4688
3516
2344
1172
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2

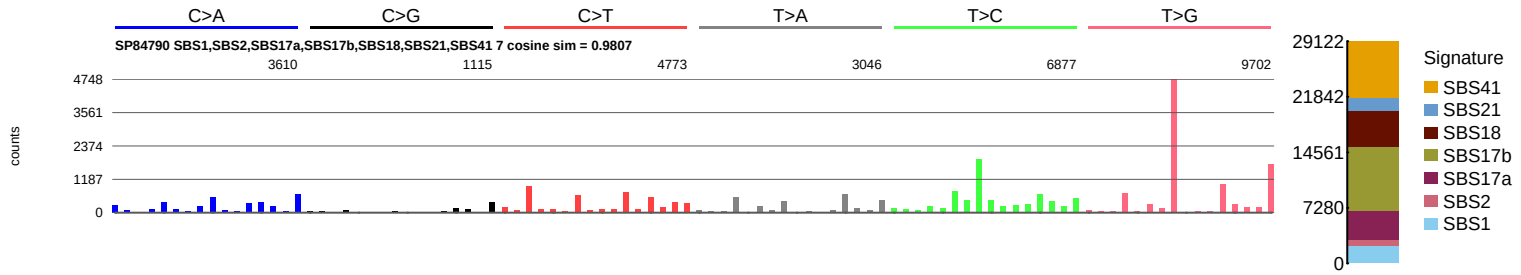

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.98074
3229
1862
4268
2794
7277
9695
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5

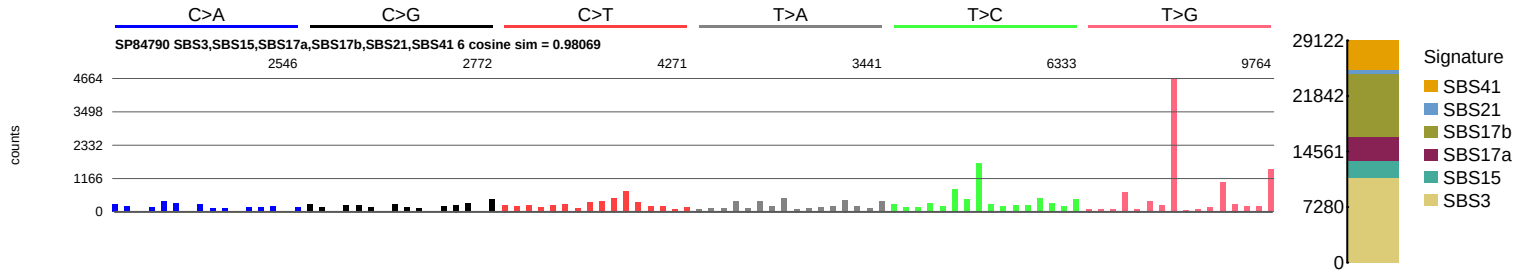

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 7 cosine sim = 0.98073
3248
1843
4442
2812
7084
9693
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5

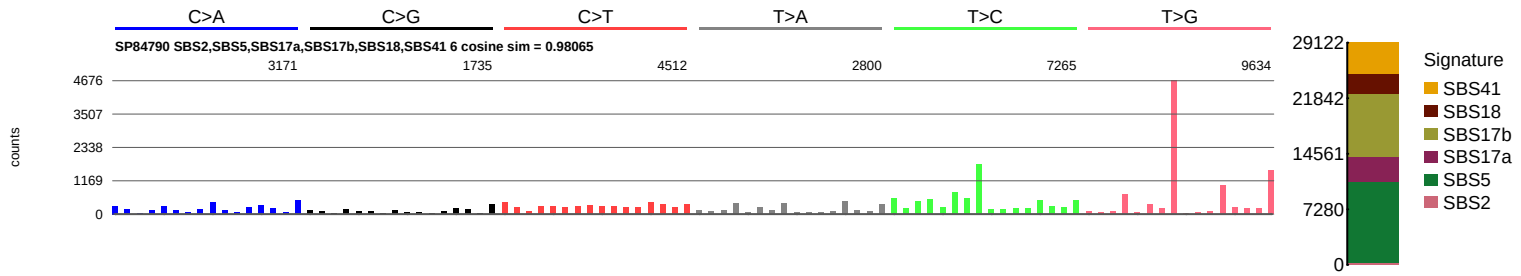

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40 7 cosine sim = 0.98072
3399
2486
4887
2321
6439
9593
counts
4684
3513
2342
1171
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3

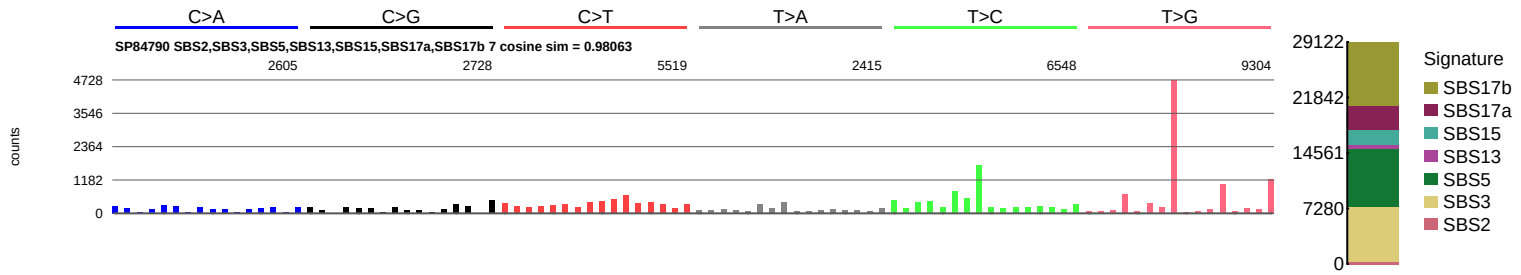

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98072
3204
1836
4447
2814
7130
9690
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2

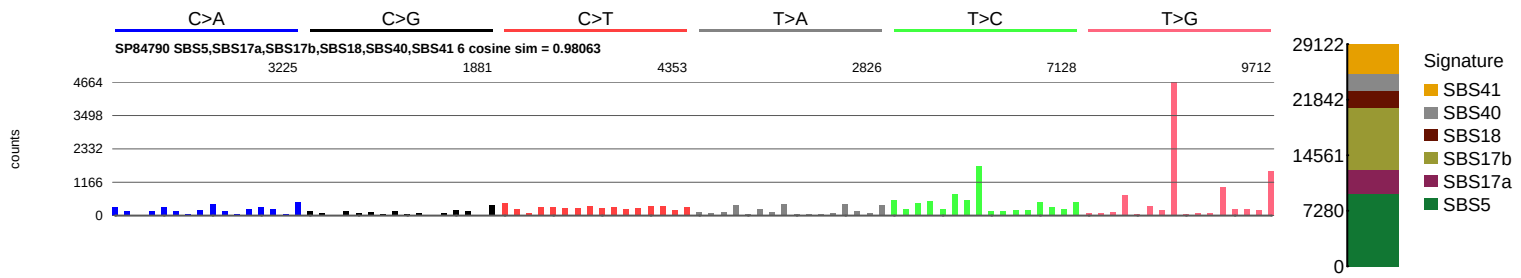

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98071
3389
2184
5203
2356
6737
9255
counts
4724
3543
2362
1181
0
29122
21842
14561
7280
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3

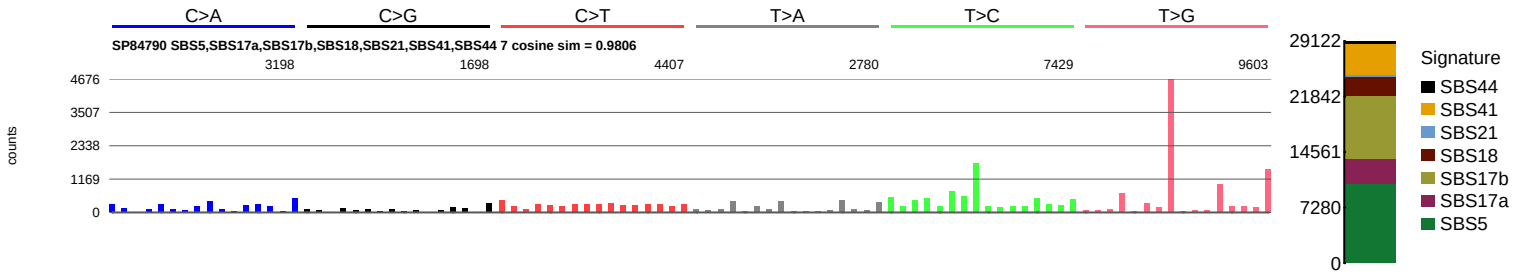

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.9807
3610
1115
4773
3046
6877
9702
counts
4748
3561
2374
1187
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1

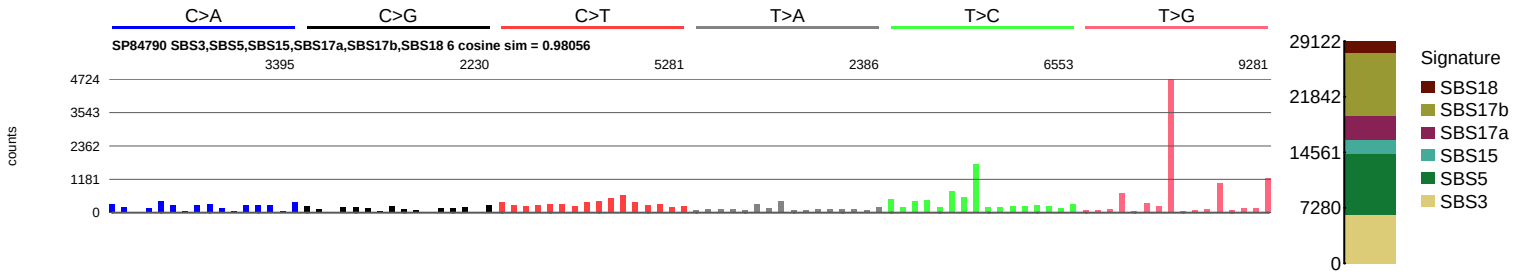

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.98069
2546
2772
4271
3441
6333
9764
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS3

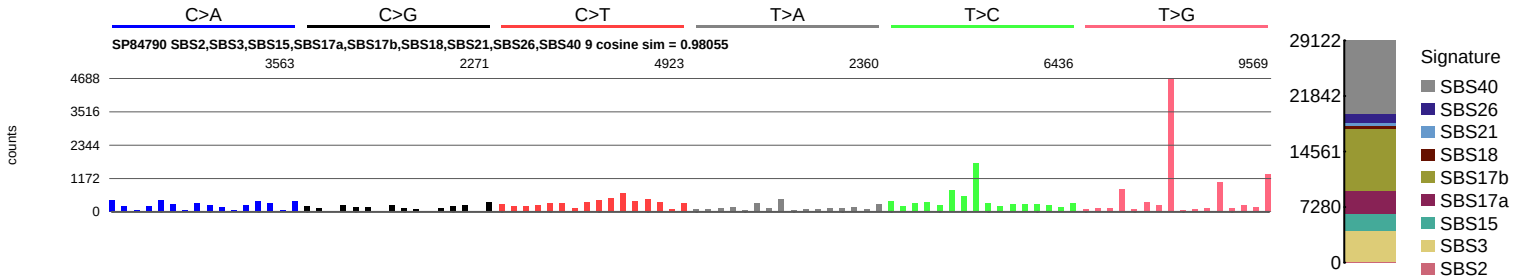

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98065
3171
1735
4512
2800
7265
9634
counts
4676
3507
2338
1169
0
29122
21842
14561
7280
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS2

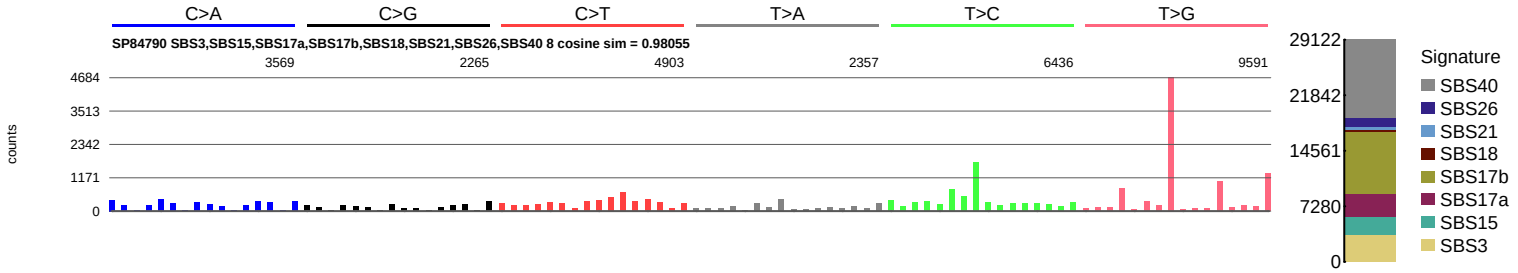

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b 7 cosine sim = 0.98063
2605
2728
5519
2415
6548
9304
counts
4728
3546
2364
1182
0
29122
21842
14561
7280
0
Signature
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2

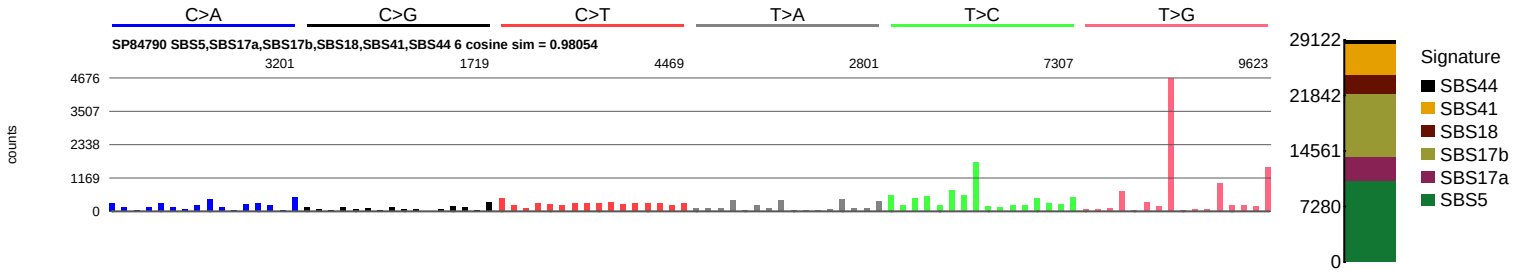

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 6 cosine sim = 0.98063
3225
1881
4353
2826
7128
9712
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5

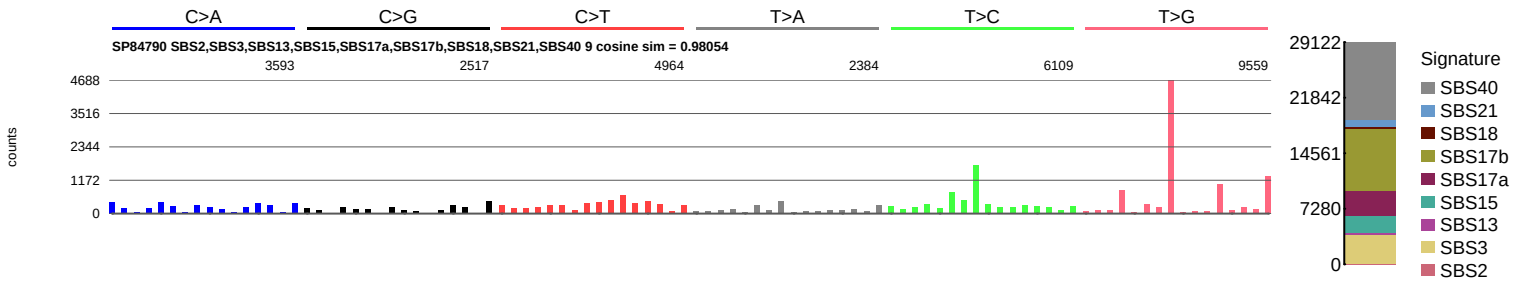

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 7 cosine sim = 0.9806
3198
1698
4407
2780
7429
9603
counts
4676
3507
2338
1169
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5

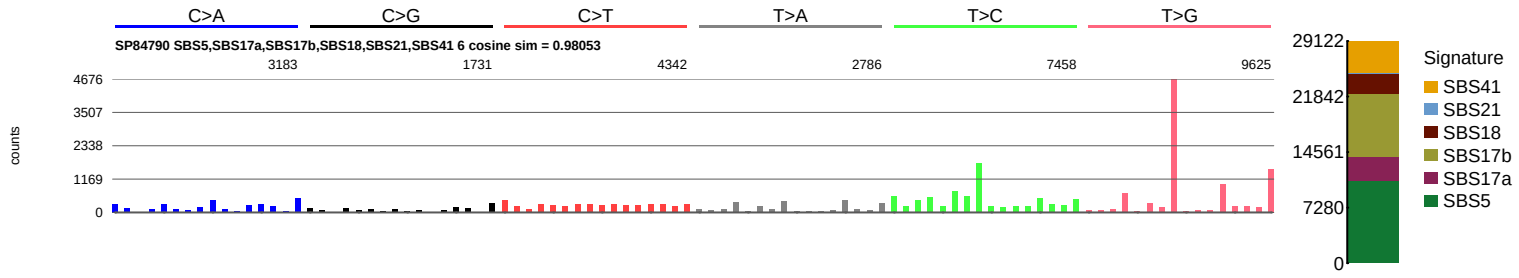

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98056
3395
2230
5281
2386
6553
9281
counts
4724
3543
2362
1181
0
29122
21842
14561
7280
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3

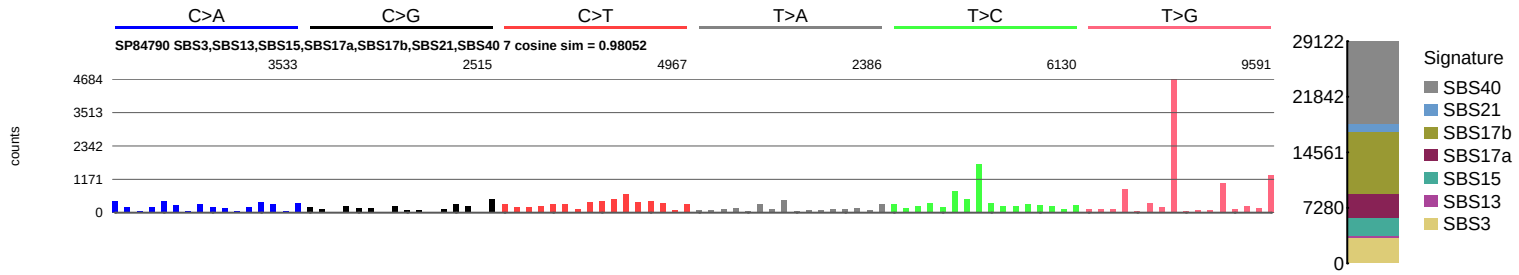

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98055
3563
2271
4923
2360
6436
9569
counts
4688
3516
2344
1172
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2

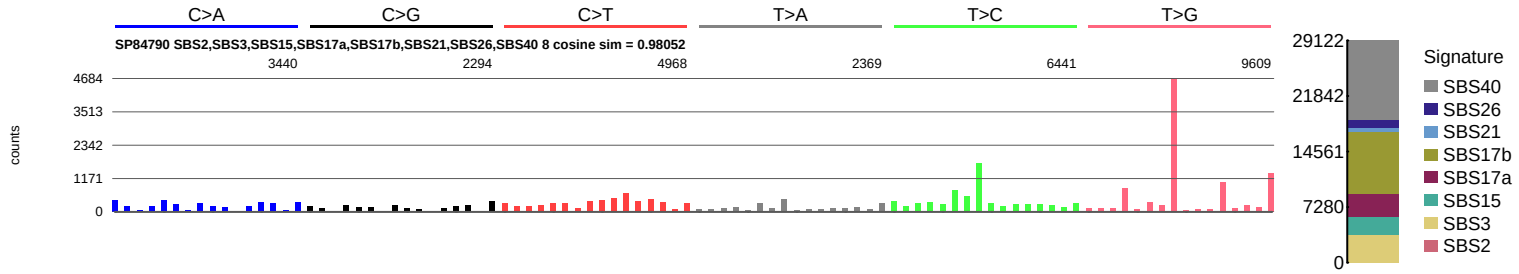

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98055
3569
2265
4903
2357
6436
9591
counts
4684
3513
2342
1171
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3

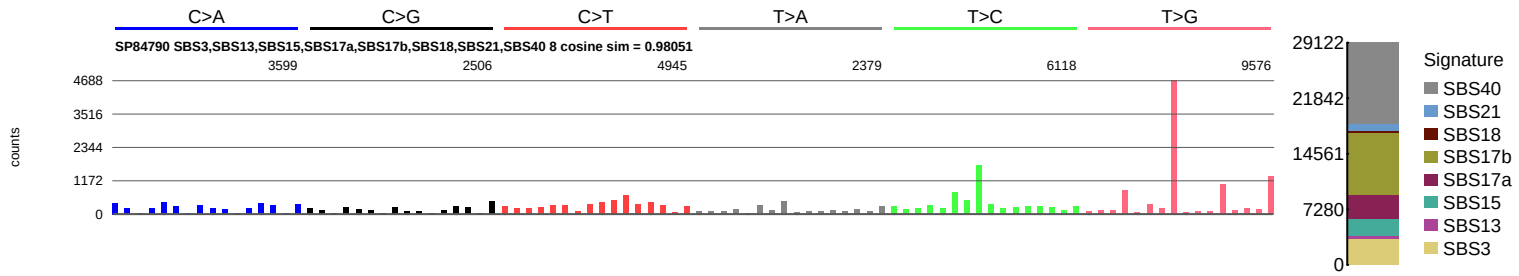

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS5,SBS17a,SBS17b,SBS18,SBS41,SBS44 6 cosine sim = 0.98054
3201
1719
4469
2801
7307
9623
counts
4676
3507
2338
1169
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS5

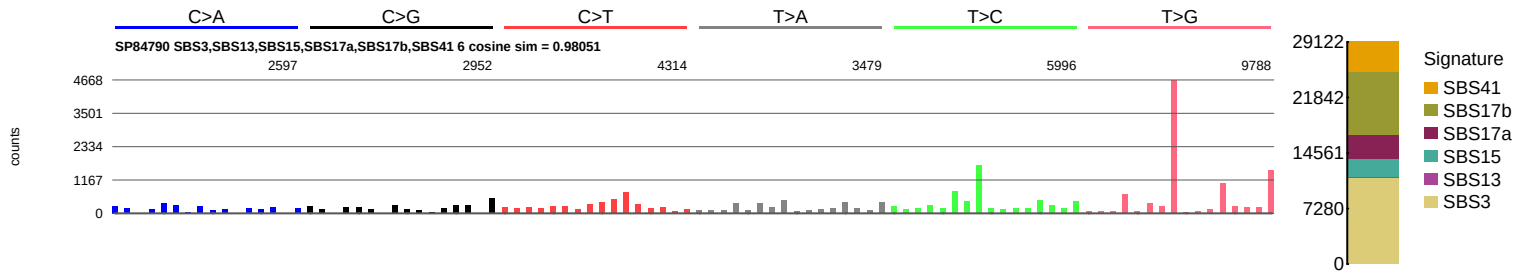

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98054
3593
2517
4964
2384
6109
9559
counts
4688
3516
2344
1172
0
29122
21842
14561
7280
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2

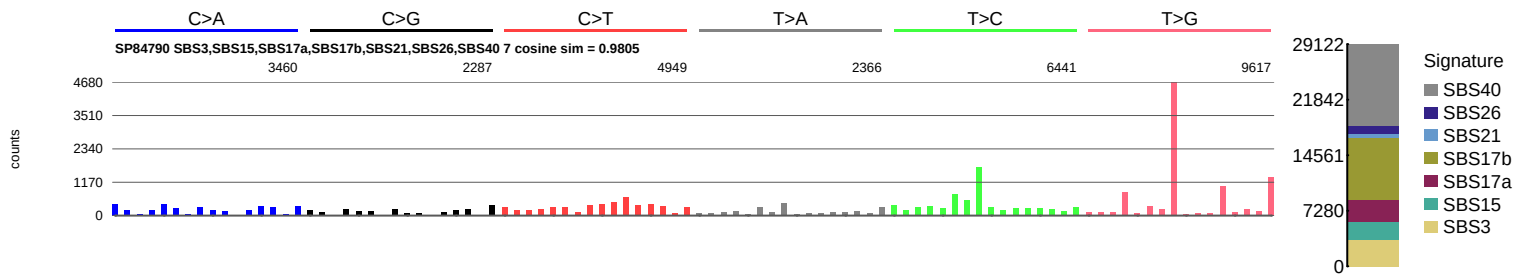

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 6 cosine sim = 0.98053
3183
1731
4342
2786
7458
9625
counts
4676
3507
2338
1169
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5

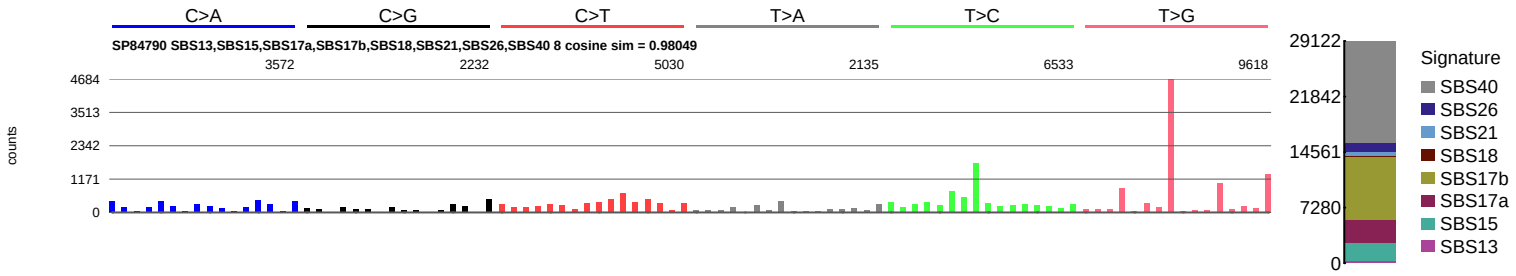

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40 7 cosine sim = 0.98052
3533
2515
4967
2386
6130
9591
counts
4684
3513
2342
1171
0
29122
21842
14561
7280
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3

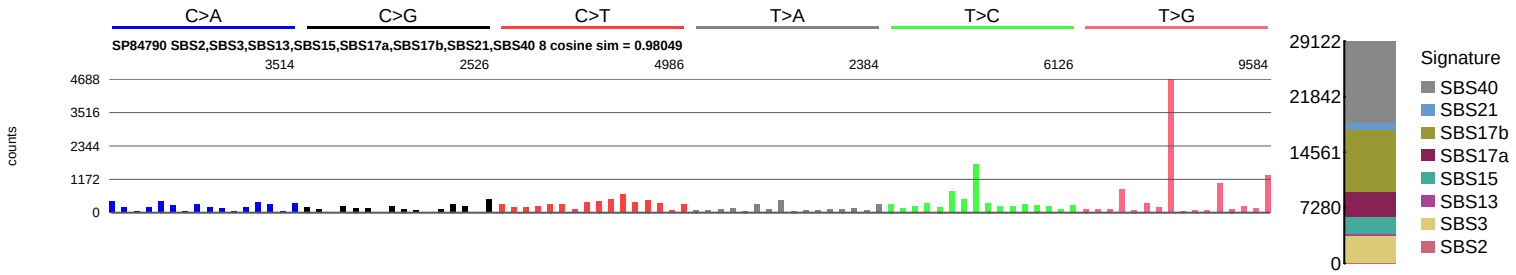

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40 8 cosine sim = 0.98052
3440
2294
4968
2369
6441
9609
counts
4684
3513
2342
1171
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2

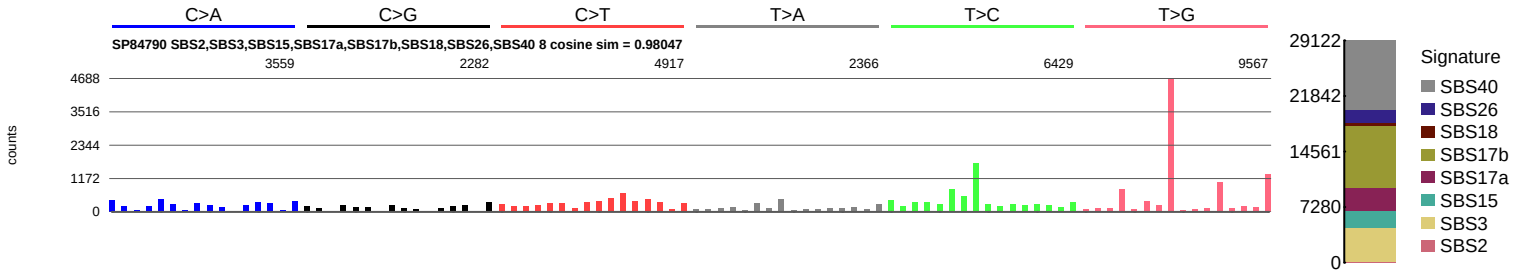

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98051
3599
2506
4945
2379
6118
9576
counts
4688
3516
2344
1172
0
29122
21842
14561
7280
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3

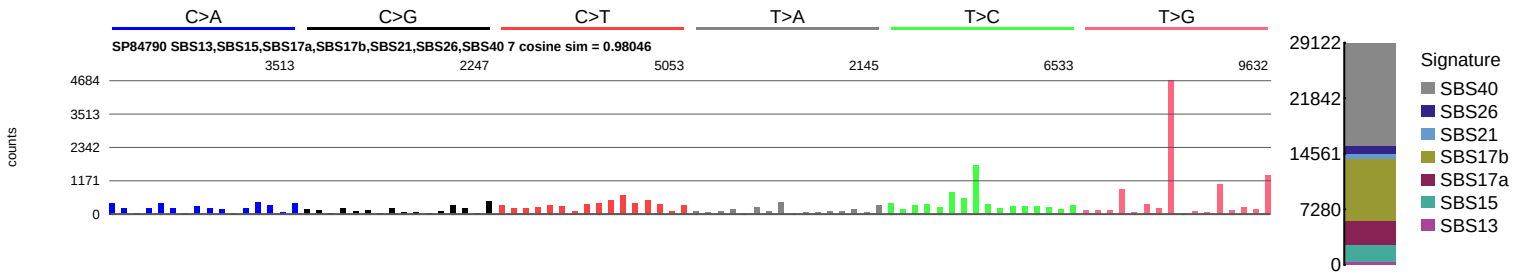

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98051
2597
2952
4314
3479
5996
9788
counts
4668
3501
2334
1167
0
29122
21842
14561
7280
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS3

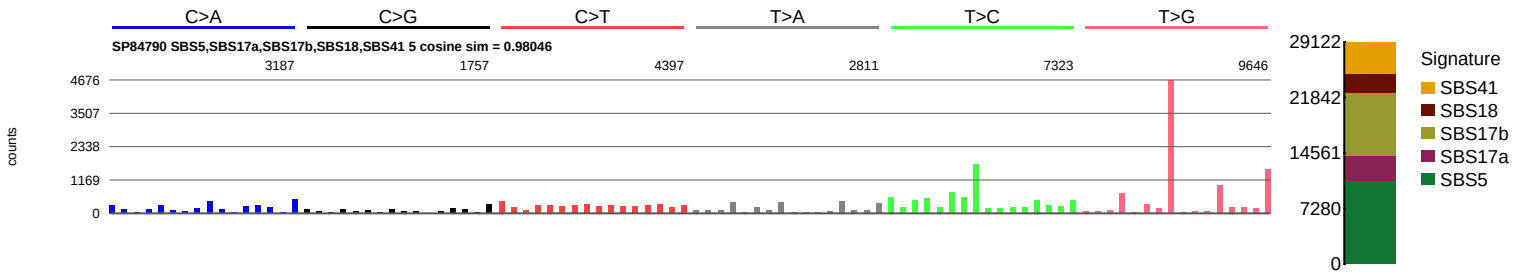

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40 7 cosine sim = 0.9805
3460
2287
4949
2366
6441
9617
counts
4680
3510
2340
1170
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS3

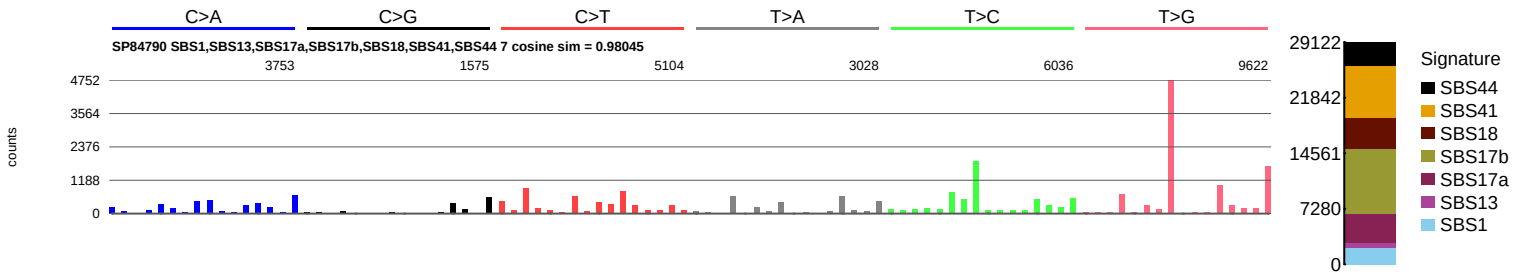

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98049
3572
2232
5030
2135
6533
9618
counts
4684
3513
2342
1171
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13

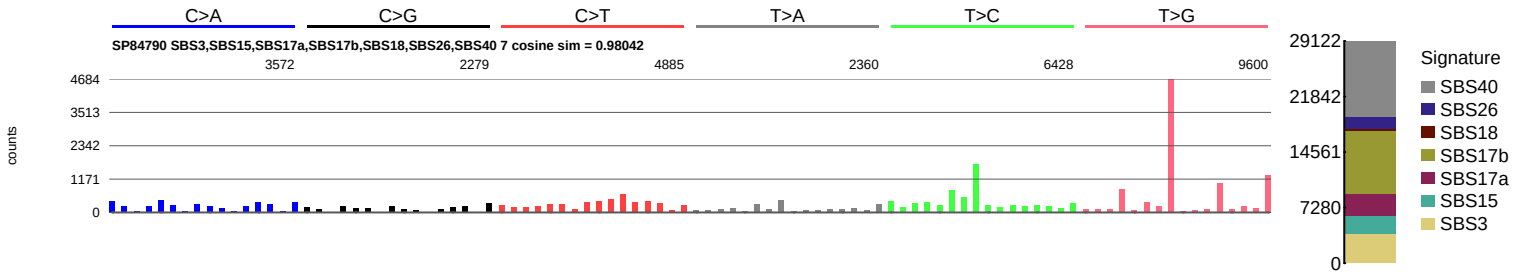

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40 8 cosine sim = 0.98049
3514
2526
4986
2384
6126
9584
counts
4688
3516
2344
1172
0
29122
21842
14561
7280
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2

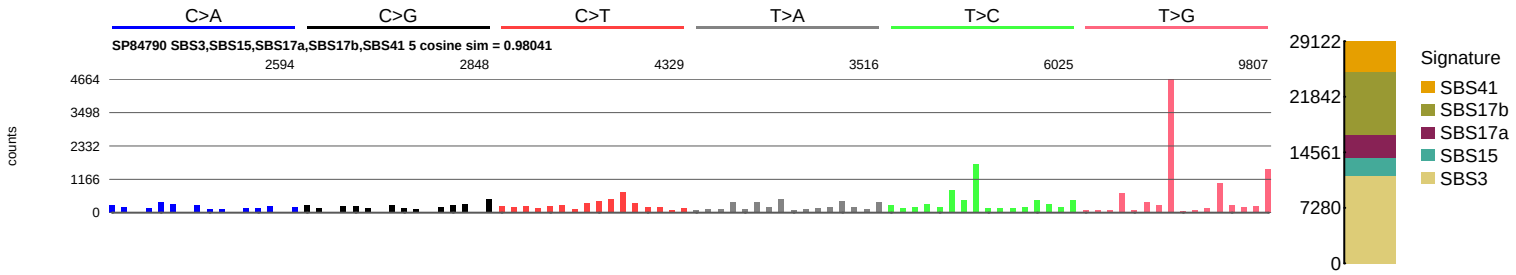

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98047
3559
2282
4917
2366
6429
9567
counts
4688
3516
2344
1172
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2

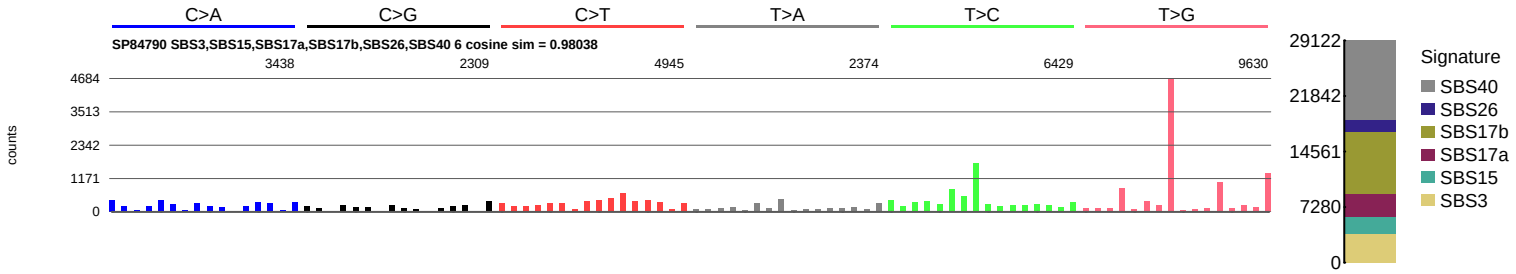

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40 7 cosine sim = 0.98046
3513
2247
5053
2145
6533
9632
counts
4684
3513
2342
1171
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13

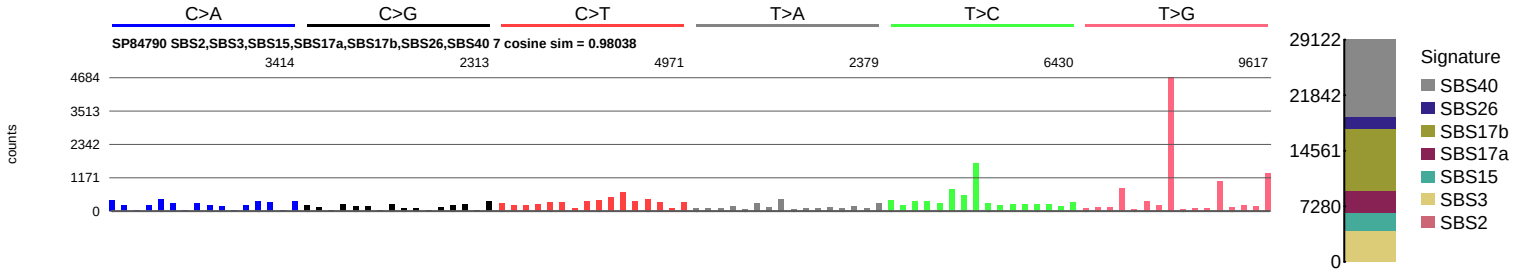

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS5,SBS17a,SBS17b,SBS18,SBS41 5 cosine sim = 0.98046
3187
1757
4397
2811
7323
9646
counts
4676
3507
2338
1169
0
29122
21842
14561
7280
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5

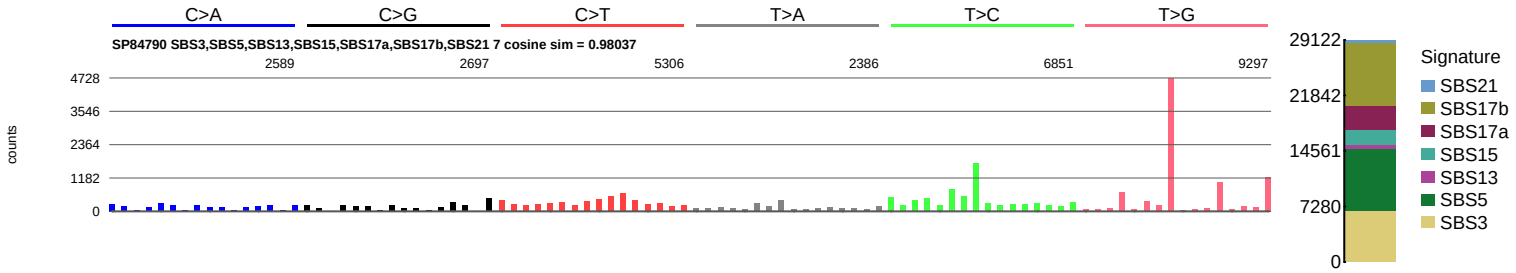

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.98045
3753
1575
5104
3028
6036
9622
counts
4752
3564
2376
1188
0
29122
21842
14561
7280
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS1

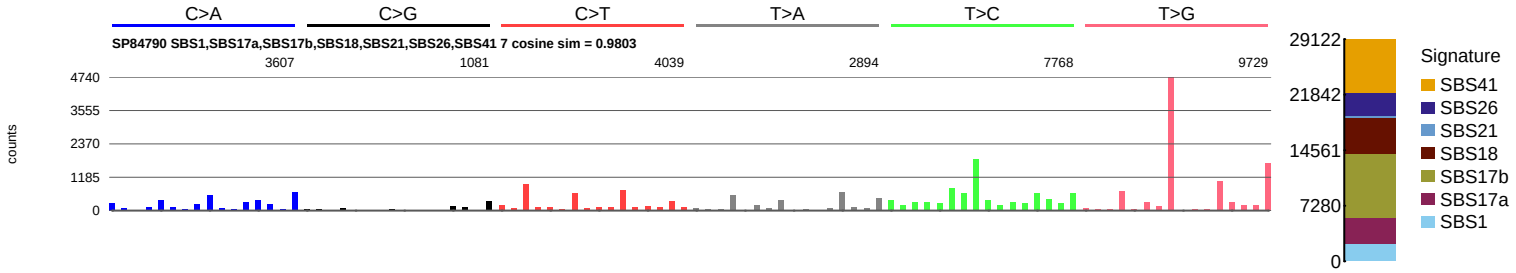

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98042
3572
2279
4885
2360
6428
9600
counts
4684
3513
2342
1171
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3

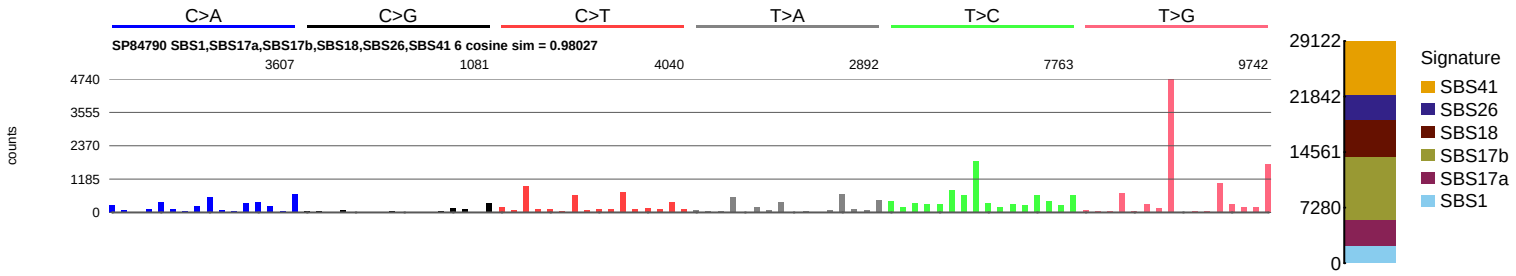

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS15,SBS17a,SBS17b,SBS41 5 cosine sim = 0.98041
2594
2848
4329
3516
6025
9807
counts
4664
3498
2332
1166
0
29122
21842
14561
7280
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS3

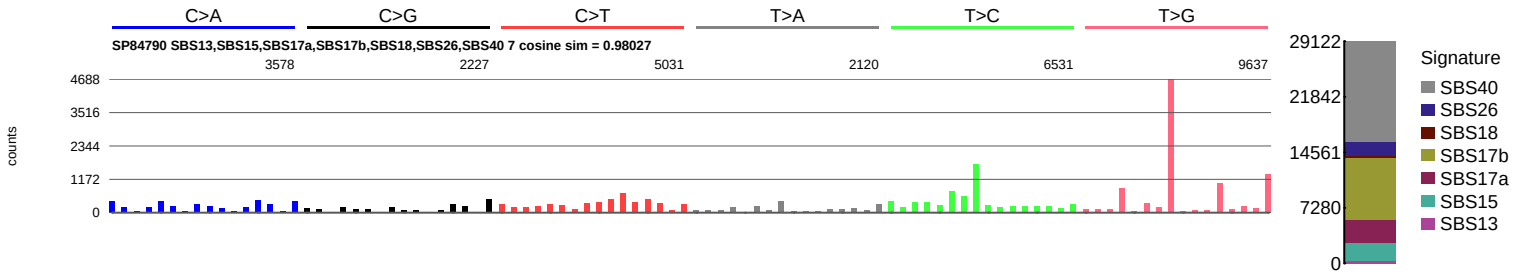

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS40 6 cosine sim = 0.98038
3438
2309
4945
2374
6429
9630
counts
4684
3513
2342
1171
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS3

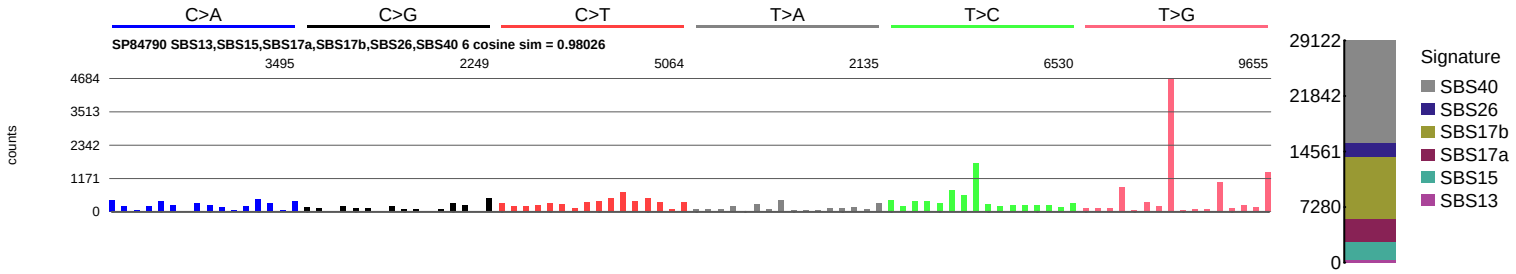

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS40 7 cosine sim = 0.98038
3414
2313
4971
2379
6430
9617
counts
4684
3513
2342
1171
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS2

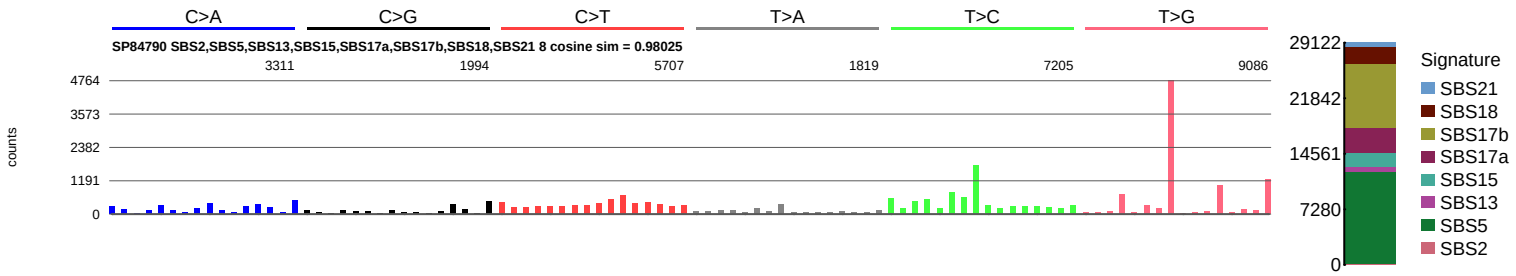

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98037
2589
2697
5306
2386
6851
9297
counts
4728
3546
2364
1182
0
29122
21842
14561
7280
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3

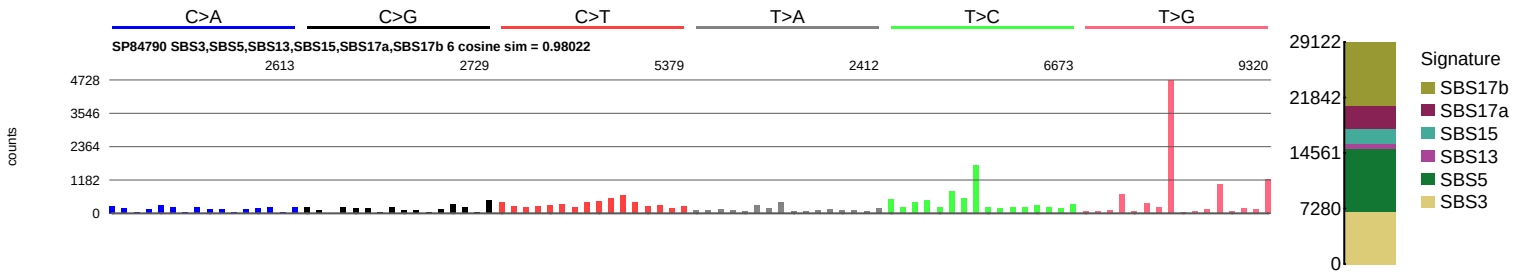

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 7 cosine sim = 0.9803
3607
1081
4039
2894
7768
9729
counts
4740
3555
2370
1185
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS1

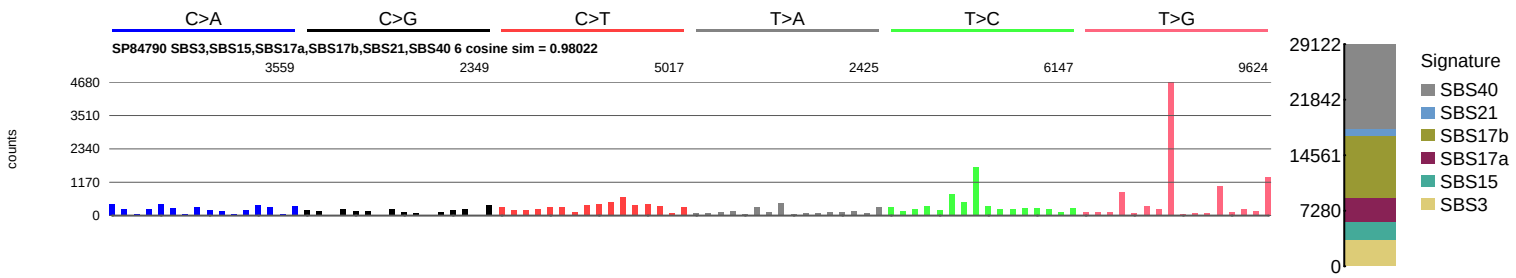

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS41 6 cosine sim = 0.98027
3607
1081
4040
2892
7763
9742
counts
4740
3555
2370
1185
0
29122
21842
14561
7280
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS1

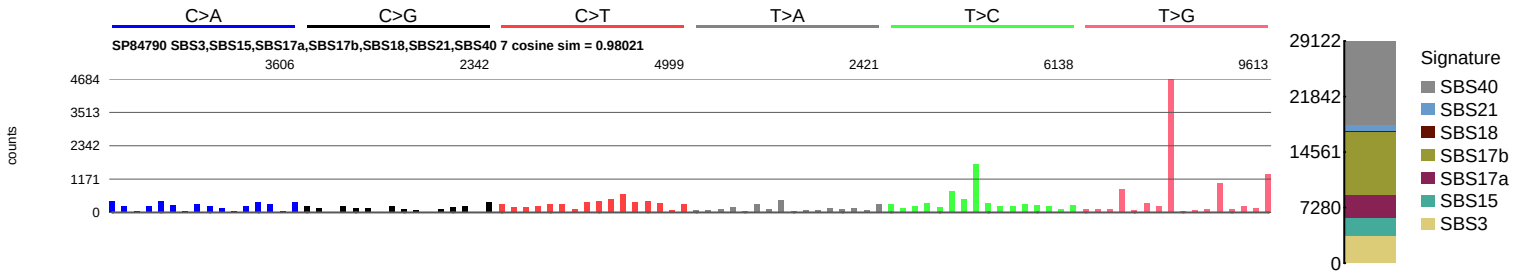

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98027
3578
2227
5031
2120
6531
9637
counts
4688
3516
2344
1172
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13

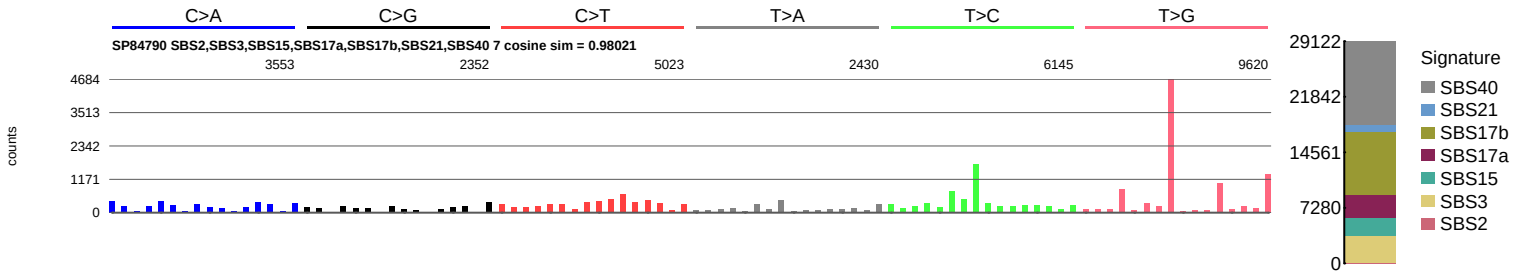

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40 6 cosine sim = 0.98026
3495
2249
5064
2135
6530
9655
counts
4684
3513
2342
1171
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13

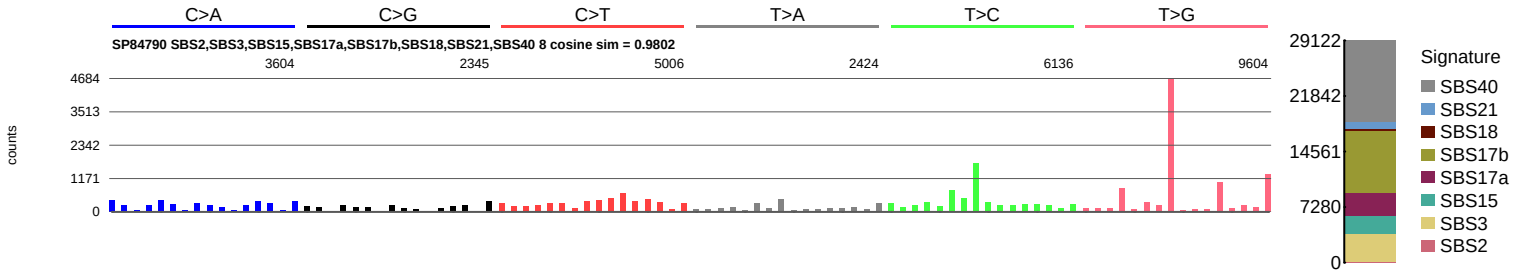

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98025
3311
1994
5707
1819
7205
9086
counts
4764
3573
2382
1191
0
29122
21842
14561
7280
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2

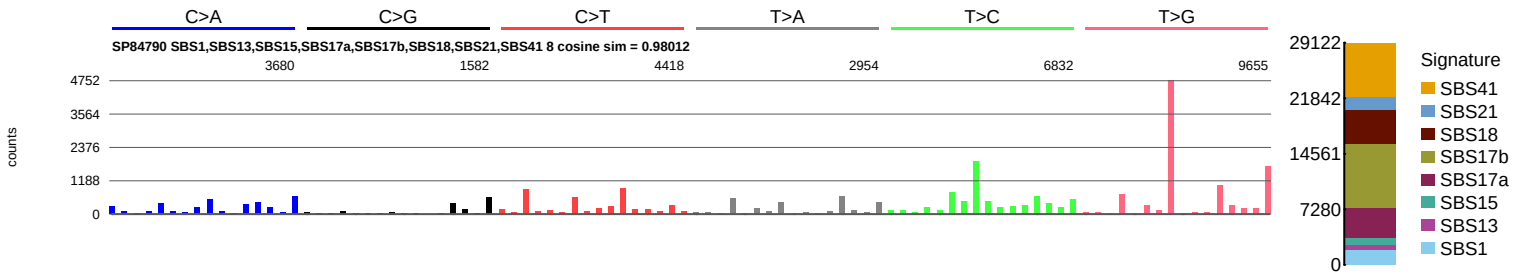

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b 6 cosine sim = 0.98022
2613
2729
5379
2412
6673
9320
counts
4728
3546
2364
1182
0
29122
21842
14561
7280
0
Signature
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3

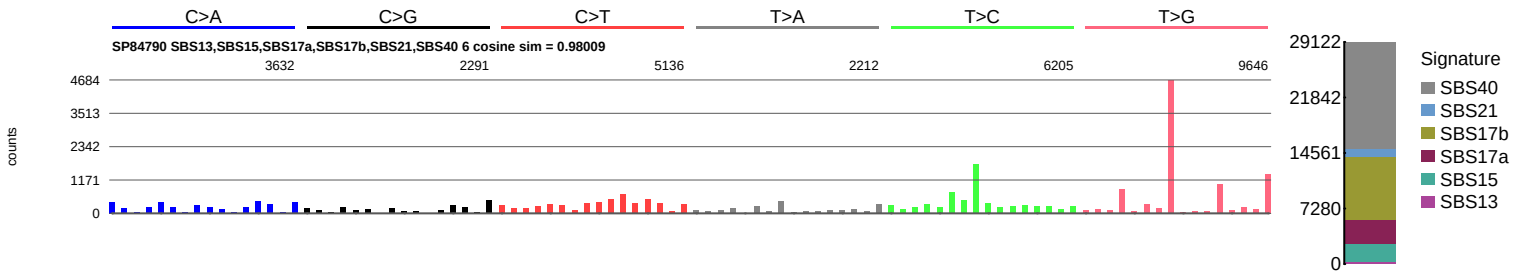

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS40 6 cosine sim = 0.98022
3559
2349
5017
2425
6147
9624
counts
4680
3510
2340
1170
0
29122
21842
14561
7280
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS3

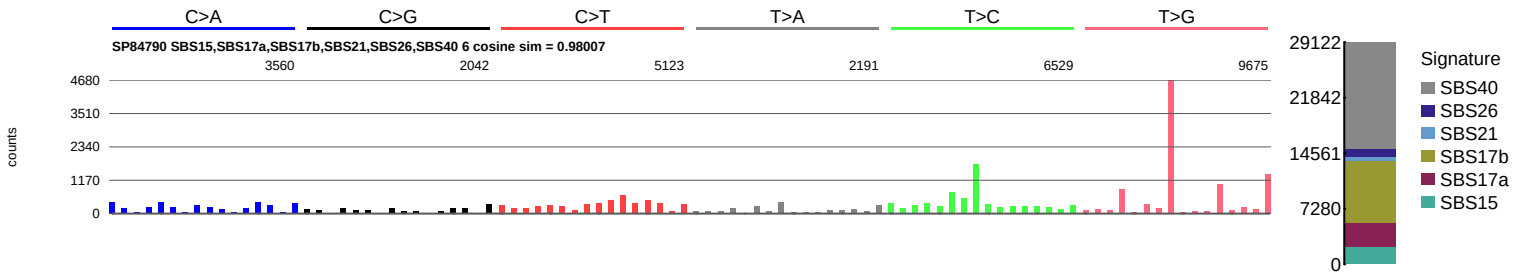

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98021
3606
2342
4999
2421
6138
9613
counts
4684
3513
2342
1171
0
29122
21842
14561
7280
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3

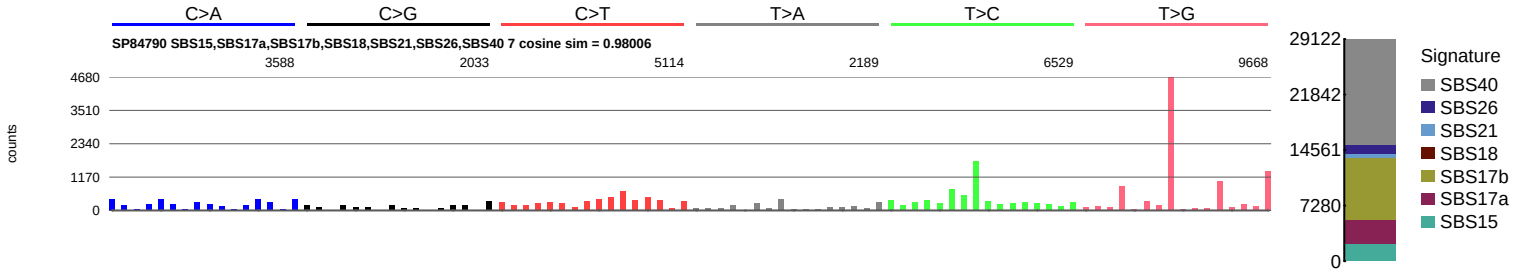

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS40 7 cosine sim = 0.98021
3553
2352
5023
2430
6145
9620
counts
4684
3513
2342
1171
0
29122
21842
14561
7280
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2

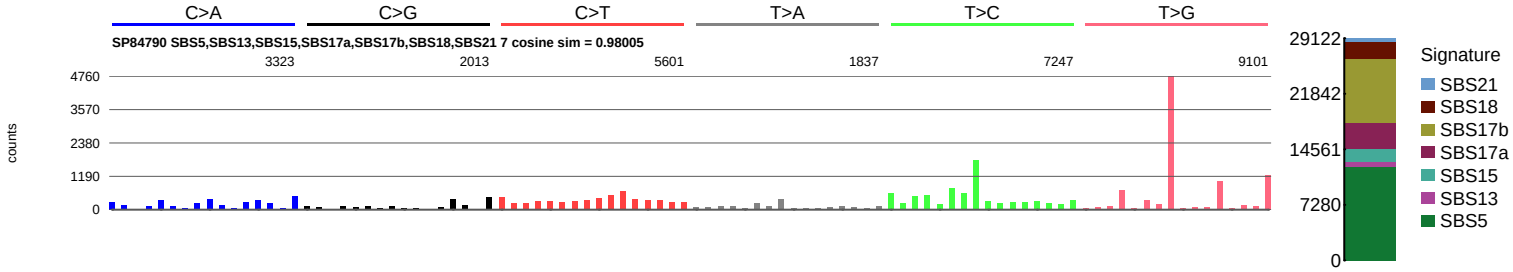

C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.9802
3604
2345
5006
2424
6136
9604
counts
4684
3513
2342
1171
0
29122
21842
14561
7280
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98012
3680
1582
4418
2954
6832
9655
counts
4752
3564
2376
1188
0
29122
21842
14561
7280
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40 6 cosine sim = 0.98009
3632
2291
5136
2212
6205
9646
counts
4684
3513
2342
1171
0
29122
21842
14561
7280
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13


C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40 6 cosine sim = 0.98007
3560
2042
5123
2191
6529
9675
counts
4680
3510
2340
1170
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15


C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 7 cosine sim = 0.98006
3588
2033
5114
2189
6529
9668
counts
4680
3510
2340
1170
0
29122
21842
14561
7280
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15


C>A
C>G
C>T
T>A
T>C
T>G
SP84790 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98005
3323
2013
5601
1837
7247
9101
counts
4760
3570
2380
1190
0
29122
21842
14561
7280
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5